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Francisco Facchinei
Jong-Shi Pang

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Preface

The finite-dimensional nonlinear complementarity problem (NCP) is a system of finitely many nonlinear inequalities in finitely many nonnegative variables along with a special equation that expresses the complementary relationship between the variables and corresponding inequalities. This complementarity condition is the key feature distinguishing the NCP from a general inequality system, lies at the heart of all constrained optimization problems in finite dimensions, provides a powerful framework for the modeling of equilibria of many kinds, and exhibits a natural link between smooth and nonsmooth mathematics. The finite-dimensional variational inequality (VI), which is a generalization of the NCP, provides a broad unifying setting for the study of optimization and equilibrium problems and serves as the main computational framework for the practical solution of a host of continuum problems in the mathematical sciences.

The systematic study of the finite-dimensional NCP and VI began in the mid-1960s; in a span of four decades, the subject has developed into a very fruitful discipline in the field of mathematical programming. The developments include a rich mathematical theory, a host of effective solution algorithms, a multitude of interesting connections to numerous disciplines, and a wide range of important applications in engineering and economics. As a result of their broad associations, the literature of the VI/CP has benefited from contributions made by mathematicians (pure, applied, and computational), computer scientists, engineers of many kinds (civil, chemical, electrical, mechanical, and systems), and economists of diverse expertise (agricultural, computational, energy, financial, and spatial). There are many surveys and special volumes, [54, 185, 191, 192, 226, 257, 429, 440], to name a few.

Written for novice and expert researchers and advanced graduate students in a wide range of disciplines, this two-volume monograph presents a comprehensive, state-of-the-art treatment of the finite-dimensional variational inequality and complementarity problem, covering the basic theory, iterative algorithms, and important applications. The materials presented

herein represent the work of many researchers worldwide. In undertaking this ambitious project, we have attempted to include every major aspect of the VI/CP, beginning with the fundamental question of existence and uniqueness of solutions, presenting the latest algorithms and results, extending into selected neighboring topics, summarizing many classical source problems, and including novel application domains. Despite our efforts, there are omissions of topics, due partly to our biases and partly to the scope of the presentation. Some omitted topics are mentioned in the notes and comments.

A Bird's-Eye View of the Subject

The subject of variational inequalities has its origin in the calculus of variations associated with the minimization of infinite-dimensional functionals. The systematic study of the subject began in the early 1960s with the seminal work of the Italian mathematician Guido Stampacchia and his collaborators, who used the variational inequality as an analytic tool for studying free boundary problems defined by nonlinear partial differential operators arising from unilateral problems in elasticity and plasticity theory and in mechanics. Some of the earliest papers on variational inequalities are [260, 373, 379, 542, 543]. In particular, the first theorem of existence and uniqueness of the solution of VIs was proved in [542]. The books by Baiocchi and Capelo [36] and Kinderlehrer and Stampacchia [324] provide a thorough introduction to the application of variational inequalities in infinite-dimensional function spaces; see also [40]. The lecture notes [276] treat complementarity problems in abstract spaces. The book by Glowinski, Lions, and Trémolière [238] is among the earliest references to give a detailed numerical treatment of such VIs. There is a huge literature on the subject of infinite-dimensional variational inequalities and related problems. Since a VI in an abstract space is in many respects quite distinct from the finite-dimensional VI and since the former problem is not the main concern of this book, in this section we focus our introduction on the latter problem only.

The development of the finite-dimensional variational inequality and nonlinear complementarity problem also began in the early 1960s but followed a different path. Indeed, the NCP was first identified in the 1964 Ph.D. thesis of Richard W. Cottle [111], who studied under the supervision of the eminent George B. Dantzig, "father of linear programming." Thus, unlike its infinite-dimensional counterpart, which was conceived in the area of partial differential systems, the finite-dimensional VI/CP was

born in the domain of mathematical programming. This origin has had a heavy influence on the subsequent evolution of the field; a brief account of the history prior to 1990 can be found in the introduction of the survey paper [257]; see also Section 1.2 in [256]. In what follows, we give a more detailed account of the evolutionary process of the field, covering four decades of major events and notable highlights.

In the 1960s, largely as a result of the celebrated almost complementary pivoting algorithm of Lemke and Howson for solving a bimatrix game formulated as a linear complementarity problem (LCP) [364] and the subsequent extension by Lemke to a general LCP [363], much focus was devoted to the study of the latter problem. Cottle, Pang, and Stone presented a comprehensive treatment of the LCP in the 1992 monograph [114]. Among other things, this monograph contains an extensive bibliography of the LCP up to 1990 and also detailed notes, comments, and historical accounts about this fundamental problem. Today, research on the LCP remains active and new applications continue to be uncovered. Since much of the pre-1990 details about the LCP are already documented in the cited monograph, we rely on the latter for most of the background results for the LCP and will touch on the more contemporary developments of this problem where appropriate.

In 1967, Scarf [514] developed the first constructive iterative method for approximating a fixed point of a continuous mapping. Scarf's seminal work led to the development of the entire family of fixed-point methods and of the piecewise homotopy approach to the computation of economic equilibria. The field of *equilibrium programming* was thus born. In essence, the term "equilibrium programming" broadly refers to the modeling, analysis, and computation of equilibria of various kinds via the methodology of mathematical programming. Since the infant days of linear programming, it was clear that complementarity problems have much to do with equilibrium programs. For instance, the primal-dual relation of a linear program provides clear evidence of the interplay between complementarity and equilibrium. Indeed, all the equilibrium problems that were amenable to solution by the fixed-point methods, including the renowned Walrasian problem in general equilibrium theory and variations of this problem [515, 580, 602], were in fact VIs/CPs.

The early research in equilibrium programming was to a large extent a consequence of the landmark discoveries of Lemke and Scarf. In particular, the subject of fixed-point computations via piecewise homotopies dominated much of the research agenda of equilibrium programming in the 1970s. A major theoretical advantage of the family of fixed-point ho-

motopy methods is their global convergence. Attracted by this advantage and the novelty of the methods, many well-known researchers including Eaves, Garcia, Gould, Kojima, Megiddo, Saigal, Todd, and Zangwill all made fundamental contributions to the subject. The flurry of research activities in this area continued for more than a decade, until the occurrence of several significant events that provided clear evidence of the practical inadequacy of this family of methods for solving realistic equilibrium problems. These events, to be mentioned momentarily, marked a turning point whereby the fixed-point/homotopy approach to the computation of equilibria gave way to an alternative set of methods that constitute what one may call a contemporary variational inequality approach to equilibrium programming. For completeness, we mention several prominent publications that contain important works on the subject of fixed-point computation via the homotopy approach and its applications [9, 10, 35, 148, 149, 150, 153, 209, 210, 235, 317, 334, 496, 515, 579, 627]. For a recent paper on this approach, see [640].

In the same period and in contrast to the aforementioned algorithmic research, Karamardian, in a series of papers [312, 313, 314, 315, 316], developed an extensive existence theory for the NCP and its cone generalization. In particular, the basic connection between the CP and the VI, Proposition **1.1.3**, appeared in [314]. The 1970s were a period when many fundamental articles on the VI/CP first appeared. These include the paper by Eaves [147] where the natural map $\mathbf{F}_K^{\rm nat}$ was used to prove a basic theorem of complementarity, important studies by Moré [413, 414] and Moré and Rheinboldt [416], which studied several distinguished classes of nonlinear functions and their roles in complementarity problems, and the individual and joint work of Kojima and Megiddo [335, 400, 401, 403], which investigated the existence and uniqueness of solutions to the NCP.

Although the initial developments of infinite-dimensional variational inequalities and finite-dimensional complementarity problems had followed different paths, there were attempts to bring the two fields more closely together, with the International School of Mathematics held in Summer 1978 in Erice, Italy, being the most prominent one. The proceedings of this conference were published in [113]. The paper [112] is among the earliest that describes some physical applications of VIs in infinite dimensions solvable by LCP methods.

One could argue that the final years of the 1970s marked the beginning of the contemporary chapter on the finite-dimensional VI/CP. During that time, the U.S. Department of Energy was employing a market equilibrium system known as the Project Independent Evaluation System (PIES) [271,

272] for energy policy studies. This system is a large-scale variational inequality that was solved on a routine basis by a special iterative algorithm known as the PIES algorithm, yielding remarkably good computational experience. For a detailed account of the PIES model, see the monograph by Ahn [5], who showed that the PIES algorithm was a generalization of the classical Jacobi iterative method for solving system of nonlinear equations [422]. For the convergence analysis of the PIES algorithm, see Ahn and Hogan [7]; for a recent update of the PIES model, which has become the National Energy Modeling System (NEMS), see [230].

The original PIES model provided a real-life economic model for which the fixed-point methods mentioned earlier were proved to be ineffective. This experience along with several related events inspired a new wave of research into iterative methods for solving VIs/CPs arising from various applied equilibrium contexts. One of these events is an important algorithmic advance, namely, the introduction of Newton's method for solving generalized equations (see below).

At about the same time as the PIES model appeared, Smith [525] and Dafermos [118] formulated the traffic equilibrium problem as a variational inequality. Parallel to the VI formulation, Aashitiani and Magnanti [1] introduced a complementarity formulation for Wardrop's user equilibrium principle [604] and established existence and uniqueness results of traffic equilibria using fixed-point theorems; see also [23, 211]. Computationally, the PIES algorithm had served as a model approach for the design of iterative methods for solving the traffic equilibrium problem [2, 212, 214]. More broadly, the variational inequality approach has had a significant impact on the contemporary point of view of this problem and the closely related spatial price equilibrium problem.

In two important papers [395, 396], Mathiesen reported computational results on the application of a sequential linear complementarity (SLCP) approach to the solution of economic equilibrium problems. These results firmly established the potential of this approach and generated substantial interest among many computational economists, including Manne and his (then Ph.D.) students, most notably, Preckel, Rutherford, and Stone. The volume edited by Manne [385] contains the papers [465, 548], which give further evidence of the computational efficiency of the SLCP approach for solving economic equilibrium problems; see also [397].

The SLCP method, as it was called in the aforementioned papers, turned out to be Newton's method developed and studied several years earlier by Josephy [293, 294, 295]; see also the later papers by Eaves [151, 152]. While the results obtained by the computational economists

clearly established the practical effectiveness of Newton's method through sheer numerical experience, Josephy's work provided a sound theoretical foundation for the fast convergence of the method. In turn, Josephy's results were based on the seminal research of Robinson, who in several landmark papers [495, 497, 498, 499] introduced the *generalized equations* as a unifying mathematical framework for optimization problems, complementarity problems, variational inequalities, and related problems. As we explain below, in addition to providing the foundation for the convergence theory of Newton's method, Robinson's work greatly influenced the modern development of sensitivity analysis of mathematical programs.

While Josephy's contributions marked a breakthrough in algorithmic advances of the field, they left many questions unanswered. From a computational perspective, Rutherford [513] recognized early on the lack of robustness in Newton's method applied to some of the most challenging economic equilibrium problems. Although ad hoc remedies and specialized treatments had lessened the numerical difficulty in solving these problems, the heuristic aids employed were far from satisfactory in resolving the practical deficiency of the method, which was caused by the lack of a suitable stabilizing strategy for global convergence. Motivated by the need for a computationally robust Newton method with guaranteed global convergence, Pang [425] developed the B-differentiable Newton method with a line search and established that the method is globally convergent and locally superlinearly convergent. While this is arguably the first work on global Newton methods for solving nonsmooth equations, Pang's method suffers from a theoretical drawback in that its convergence requires a Fréchet differentiability assumption at a limit point of the produced sequence.

Newton's method for solving nondifferentiable equations had been investigated before Pang's work. Kojima and Shindo [346] discussed such a method for PC^1 functions. Kummer [352] studied this method for general nondifferentiable functions. Both papers dealt with the local convergence but did not address the globalization of the method. Generalizing the class of semismooth functions of one variable defined by Mifflin [404], Qi and Sun [485] introduced the class of vector semismooth functions and established the local convergence of Newton's method for this class of functions. The latter result of Qi and Sun is actually a special case of the general theory of Kummer. Since its introduction, the class of vector semismooth functions has played a central role throughout the subsequent algorithmic developments of the field. Although focused mainly on the smooth case, the two recent papers [234, 626] present an enlightening summary of the historical developments of the convergence theory of Newton's method.

As an alternative to Pang's line search globalization strategy, Ralph [488] presented a path search algorithm that was implemented by Dirkse and Ferris in their highly successful PATH solver [136], which was awarded the 1997 Beale-Orchard-Hays Prize for excellence in computational mathematical programming; the accompanying paper [135] contains an extensive collection of MiCP test problems. In an important paper that dealt with an optimization problem [198], Fischer proposed the use of what is now called the Fischer-Burmeister function to reformulate the Karush-Kuhn-Tucker conditions arising from an inequality constrained optimization problem as a system of nonsmooth equations. Collectively, these works paved the way for an outburst of activities that started with De Luca, Facchinei, and Kanzow [123]. The latter paper discussed the application of a globally convergent semismooth Newton method to the Fischer-Burmeister reformulation of the nonlinear complementarity problem; the algorithm described therein provided a model approach for many algorithms that followed. The semismooth Newton approach led to algorithms that are conceptually and practically simpler than the B-differentiable Newton method and the path Newton method, and have, at the same time, better convergence properties.

The attractive theoretical properties of the semismooth methods and their good performance in practice spurred much research to investigate further this class of methods and inspired much of the subsequent studies. In the second half of the 1990s, a large number of papers was devoted to the improvement, extension, and numerical testing of semismooth algorithms, bringing these algorithms to a high level of sophistication. Among other things, these developments made it clear that the B-differentiable Newton method is intimately related to the semismooth Newton method applied to the min reformulation of the complementarity problem, thus confirming the breadth of the new approach.

The above overview gives a general perspective on the evolution of the VI/CP and documents several major events that have propelled this subject to its modern status as a fruitful and exciting discipline within mathematical programming. There are many other interesting developments, such as sensitivity and stability analysis, piecewise smooth functions, error bounds, interior point methods, smoothing methods, methods of the projection family, and regularization, as well as the connections with new applications and other mathematical disciplines, all of which add to the richness and vitality of the field and form the main topics in our work. The notes and comments of these developments are contained at the end of each chapter.

A Synopsis of the Book

Divided into two volumes, the book contains twelve main chapters, followed by an extensive bibliography, a summary of main results and key algorithms, and a subject index. The first volume consists of the first six chapters, which present the basic theory of VIs and CPs. The second volume consists of the remaining six chapters, which present algorithms of various kinds for solving VIs and CPs. Besides the main text, each chapter contains (a) an extensive set of exercises, many of which are drawn from published papers that supplement the materials in the text, and (b) a set of notes and comments that document historical accounts, give the sources for the results in the main text, and provide discussions and references on related topics and extensions. The bibliography contains more than 1,300 publications in the literature up to June 2002. This bibliography serves two purposes: one purpose is to give the source of the results in the chapters, wherever applicable; the other purpose is to give a documentation of papers written on the VI/CP and related topics.

Due to its comprehensiveness, each chapter of the book is by itself quite lengthy. Among the first six sections in Chapter **1**, Sections **1.1**, **1.2**, **1.3**, and **1.5** make up the basic introduction to the VI/CP. The source problems in Section **1.4** are of very diverse nature; they fall into several general categories: mathematical programming, economics, engineering, and finance. Depending on an individual's background, a reader can safely skip those subsections that are outside his/her interests; for instance, an economist can omit the subsection on frictional contact problems, a contact mechanician can omit the subsection on Nash-Cournot production models. Section **1.6** mainly gives the definition of several extended problems; except for (1.6.1), which is re-introduced and employed in Chapter **11**, this section can be omitted at first reading.

Chapters **2** and **3** contain the basic theory of existence and multiplicity of solutions. Several sections contain review materials of well-known topics; these are included for the benefit of those readers who are unfamiliar with the background for the theory. Section **2.1** contains the review of degree theory, which is a basic mathematical tool that we employ throughout the book; due to its powerful implications, we recommend this to a reader who is interested in the theoretical part of the subject. Sections **2.2**, **2.3** (except Subsection **2.3.2**), **2.4**, and **2.5** (except Subsection **2.5.3**) contain fundamental results. While Sections **2.6** and **2.8** can be skipped at first reading, Section **2.7** contains very specialized results for the discrete frictional contact problem and is included herein only to illustrate the application of

the theory developed in the chapter to an important class of mechanical problems.

Section **3.1** in Chapter **3** introduces the class of B-differentiable functions that plays a fundamental role throughout the book. With the exception of the nonstandard SBCQ, Section **3.2** is a review of various well-known CQs in NLP. Except for the last two subsections in Section **3.3** and Subsection **3.5.1**, which may be omitted at first reading, the remainder of this chapter contains important properties of solutions to the VI/CP.

Chapter **4** serves two purposes: One, it is a technical precursor to the next chapter; and two, it introduces the important classes of PA functions (Section **4.2**) and PC^1 functions (Section **4.6**). Readers who are not interested in the sensitivity and stability theory of the VI/CP can skip most of this and the next chapter. Nevertheless, in order to appreciate the class of semismooth functions, which lies at the heart of the contemporary algorithms for solving VIs/CPs, and the regularity conditions, which are key to the convergence of these algorithms, the reader is advised to become familiar with certain developments in this chapter, such as the basic notion of coherent orientation of PA maps (Definition **4.2.3**) and its matrix-theoretic characterizations for the special maps $\mathbf{M}_K^{\mathrm{nor}}$ and $\mathbf{M}_K^{\mathrm{nat}}$ (Proposition **4.2.7**) as well as the fundamental role of this notion in the globally unique solvability of AVIs (Theorem **4.3.2**). The inverse function Theorem **4.6.5** for PC^1 functions is of fundamental importance in nonsmooth analysis. Subsections **4.1.1** and **4.3.1** are interesting in their own right; but they are not needed in the remainder of the book.

Chapter **5** focuses on the single topic of sensitivity and stability of the VI/CP. While stability is the cornerstone to the fast convergence of Newton's method, readers who are not interested in this specialized topic or in the advanced convergence theory of the mentioned method can skip this entire chapter. Notwithstanding this suggestion, Section **5.3** is of classical importance and contains the most basic results concerning the local analysis of an isolated solution.

Chapter **6** contains another significant yet specialized topic that can be omitted at first reading. From a computational point of view, an important goal of this chapter is to establish a sound basis for understanding the connection between the exact solutions to a given problem and the computed solutions of iterative methods under prescribed termination criteria used in practical implementation. As evidenced throughout the chapter and also in Section **12.6**, the theory of error bounds has far-reaching consequences that extend beyond this goal. For instance, since the publication of the paper [171], which is the subject of discussion in Section **6.7**, there has been

an increasing use of error bounds in designing algorithms that can identify active constraints accurately, resulting in enhanced theoretical properties of algorithms and holding promise for superior computational efficiency.

Of independent interest, Chapters **7** and **8** contain the preparatory materials for the two subsequent chapters. While Sections **7.1** and **7.4** are both concerned with the fundamentals of nonsmooth functions, the former pertains to general properties of nonsmooth functions, whereas the latter focuses on the semismooth functions. As far as specific algorithms go, Algorithms **7.3.1** and **7.5.1** in Sections **7.3** and **7.5**, respectively, are the most basic and strongly recommended for anyone interested in the subsequent developments. The convergence of the former algorithm depends on the (strong) stability theory in Chapter **5**, whereas that of the latter is rather simple, provided that one has a good understanding of semismoothness. In contrast to the previous two algorithms, Algorithm **7.2.17** is closest to a straightforward application of the classical Newton method for smooth systems of equations to the NCP.

The path search Newton method **8.1.9** is the earliest algorithm to be coded in the highly successful PATH computer software [136]. Readers who are already familiar with the line search and/or trust region methods in standard nonlinear programming may wish to peruse Subsection **8.3.3** and skip the rest of Chapter **8** in order to proceed directly to the next chapter. When specialized to C^1 optimization problems, as is the focus in Chapters **9** and **10**, much of the material in Sections **8.3** and **8.4** is classical; these two sections basically offer a systematic treatment of known techniques and results and present them in a way that accommodates nonsmooth objective functions.

The last four chapters are the core of the algorithmic part of this book. While Chapter **9** focuses on the NCP, Chapter **10** is devoted to the VI. The first section of the former chapter presents a detailed exposition of algorithms based on the FB merit function and their convergence theory. The most basic algorithm, **9.1.10**, is described in Subsection **9.1.1** and is accompanied by a comprehensive analysis. Algorithm **9.2.3**, which combines the min function and the FB merit function in a line search method, is representative of a mixture of algorithms in one overall scheme. Example **9.3.3** contains several C-functions that can be used in place of the FB C-function. The box-constrained VI in Subsection **9.4.3** unifies the generalized problems in Section **9.4**.

The development in Section **10.1** is very similar to that in Section **9.1.1**; the only difference is that the analysis of the first section in Chapter **10** is tailored to the KKT system of a finitely representable VI. The other

major development in this chapter is the D-gap function in Section **10.3**, which is preceded by the preparatory discussion of the regularized gap function in Subsection **10.2.1**. The implicit Lagrangian function presented in Subsection **10.3.1** is an important merit function for the NCP.

Chapter **11** presents interior and smoothing methods for solving CPs of different kinds, including KKT systems. Developed in the abstract setting of constrained equations, the basic interior point method for the implicit MiCP, Algorithm **11.5.1**, is presented in Section **11.5**. An extensive theoretical study of the latter problem is the subject of the previous Section **11.4**; in which the important mixed P_0 property is introduced (see Definition **11.4.1**). A Newton smoothing method is outlined in Subsection **11.8.1**; this method is applicable to smoothed reformulations of CPs using the smoothing functions discussed in Subsection **11.8.2**, particularly those in Example **11.8.11**.

The twelveth and last chapter discusses various specialized methods that are applicable principally to (pseudo) monotone VIs and NCPs of the P_0 type. The first four sections of the chapter contain the basic methods and their convergence theories. The theory of maximal monotone operators in Subsection **12.3.1** plays a central role in the proximal point method that is the subject of Subsection **12.3.2**. Bregman-based methods in Subsection **12.7.2** are well researched in the literature, whereas the interior/barrier methods in Subsection **12.7.4** are recent entrants to the field.

Acknowledgments

Writing a book on this subject has been the goal of the second author since he and Harker published their survey paper [257] in 1990. This goal was not accomplished and ended with Harker giving a lecture series at the Université Catholique de Louvain in 1992 that was followed by the lecture notes [256]. The second author gratefully acknowledges Harker for the fruitful collaboration and for his keen interest during the formative stage of this book project.

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and many colleagues on various parts of the book. We thank them sincerely. Michael Ferris and Stefan Scholtes have provided useful comments on a preliminary version of the book that help to shape its final form. We wish to thank our Series Editor, Achi Dosanjh, the Production Editor, Louise Farkas, and the staff members at Springer-New York, for their skillful editorial assistance.

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Finally, the text of this monograph was typeset by the authors using LaTeX, a document preparation system based on Knuth's TeX program. We have used the document style files of the book [114] that were prepared by Richard Cottle and Richard Stone, and based on the LaTeX book style.

Rome, Italy — Francisco Facchinei
Baltimore, Maryland, U.S.A. — Jong-Shi Pang
December 9, 2002

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Acronyms

The numbers refer to the pages where the acronyms first appear.

AVI, 7	Affine Variational Inequality
B-function, 869	Box-function
CC, 1017	Coerciveness in the Complementary variables
CE, 989	Constrained Equation
C-function, 72	Complementarity function
CP, 4	Complementarity Problem
CQ, 17	Constraint Qualification
C^r, 13	Continuously differentiable of order r
$\mathrm{C}^{1,1}$, 529	$= \mathrm{LC}^1$
CRCQ, 262	Constant Rank Constraint Qualification
ESSC, 896	Extended Strong Stability condition
GUS, 122	Globally Uniquely Solvable
FOA, 443	First-Order Approximation
IP, 989	Interior Point
KKT, 9	Karush-Kuhn-Tucker
LC^1, 719	C^1 functions with Lipschitz continuous gradients
LCP, 8	Linear Complementarity Problem
LICQ, 253	Linear Independence Constraint Qualification
LP, 6	Linear Program
MFCQ, 252	Mangasarian-Fromovitz Constraint Qualification
MiCP, 7	Mixed Complementarity Problem
MLCP, 7	Mixed Linear Complementarity Problem
MPEC, 65	Mathematical Program with Equilibrium Constraints
MPS, 581	Minimum Principle Sufficiency
NCP, 6	Nonlinear Complementarity Problem
NLP, 13	Nonlinear Program
PA, 344	Piecewise Affine
PC^r, 384	Piecewise smooth of order r (mainly $r = 1$)

PL, 344	Piecewise Linear
QP, 15	Quadratic Program
QVI, 16	Quasi-Variational Inequality
SBCQ, 262	Sequentially Bounded Constraint Qualification
SC^1, 686	C^1 functions with Semismooth gradients
SCOC, 490	Strong Coherent Orientation Condition
SLCP, v	Sequential Linear Complementarity Problem
SMFCQ 253	Strict Mangasarian-Fromovitz Constraint Qualification
SPSD, 67	Symmetric Positive Semidefinite
SQP, 718	Sequential Quadratic Programming
VI, 2	Variational Inequality
WMPS, 1202	Weak Minimum Principle Sufficiency

Glossary of Notation

Spaces

$\mathcal{M}^n$	the subspace of symmetric matrices in $\mathbb{R}^{n\times n}$
$\mathcal{M}^n_+$	the cone of SPSD matrices of order n
$\mathcal{M}^n_{++}$	the cone of positive definite matrices in $\mathcal{M}^n$
$\mathbb{R}^n$	the real n-dimensional space
$\mathbb{R}^n_+$	the nonnegative orthant of $\mathbb{R}^n$
$\mathbb{R}^n_{++}$	the positive orthant of $\mathbb{R}^n$
$\mathbb{R}^{n\times m}$	the space of $n \times m$ real matrices

Matrices

A	$\equiv (a_{ij})$; a matrix with entries a_{ij}
$\det A$	the determinant of a matrix A
$\operatorname{tr} A$	the trace of a matrix A
A^T	the transpose of a matrix A
A^{-1}	the inverse of a matrix A
M/A	the Schur complement of A in M
A_s	$\equiv \frac{1}{2}(A + A^T)$; the symmetric part of a matrix A
$\lambda_{\max}(A)$	the largest eigenvalue of a matrix $A \in \mathcal{M}^n$
$\lambda_{\min}(A)$	the smallest eigenvalue of a matrix $A \in \mathcal{M}^n$
$\|A\|$	$\equiv \sqrt{\lambda_{\max}(A^T A)}$; the Euclidean norm of $A \in \mathbb{R}^{n\times n}$
$A \bullet B$	the Frobenius product of two matrices A and B in $\mathbb{R}^{n\times n}$
$\|A\|_F$	$\equiv \sqrt{A \bullet A}$; the Frobenius norm of $A \in \mathbb{R}^{n\times n}$
$A_{\cdot\alpha}$	the columns of A indexed by α
$A_{\alpha\cdot}$	the rows of A indexed by α
$A_{\alpha\beta}$	submatrix of A with rows and columns indexed by α and β, respectively
$\mathbf{I}_k$	identity matrix of order k (subscript often omitted)
$\operatorname{diag}(a)$	the diagonal matrix with diagonal elements equal to the components of the vector a

Scalars

$\mathbb{R}$	the real line
$\operatorname{sgn} t$	the sign, $1, -1, 0$, of a positive, negative, or zero scalar t
t_+	$\equiv \max(0, t)$; the nonnegative part of a scalar
t_-	$\equiv \max(0, -t)$; the nonpositive part of a scalar

Vectors

x^T	$\equiv (x_1, \ldots, x_n)$; the transpose of a vector x with components x_i
x^{-1}	$\equiv (1/x_i)_{i=1}^n$ for $x > 0$
x^+	$\equiv \max(0, x)$; the nonnegative part of a vector x
x^-	$\equiv \max(0, -x)$; the nonpositive part of a vector x
x_α	subvector of x with components indexed by α
$\{x^k\}$	a sequence of vectors x^1, x^2, x^3, ...
$x^T y$	the standard inner product of vectors in $\mathbb{R}^n$
$\|x\|_p$	$\equiv \left(\sum_{i=1}^n \lvert x_i \rvert^p \right)^{1/p}$; the ℓ_p-norm of a vector $x \in \mathbb{R}^n$
$\|x\|$	the ℓ_2-norm of $x \in \mathbb{R}^n$, unless otherwise specified
$\|x\|_\infty$	$\equiv \max_{1 \le i \le n} \lvert x_i \rvert$; the ℓ_∞-norm of $x \in \mathbb{R}^n$
$\|x\|_A$	$\equiv \sqrt{x^T A x}$; the A-norm of $x \in \mathbb{R}^n$ for $A \in \mathcal{M}_{++}^n$
$x \ge y$	the (usual) partial ordering: $x_i \ge y_i$, $i = 1, \ldots n$
$x \gneqq y$	$x \ge y$ and $x \ne y$
$x > y$	the strict ordering: $x_i > y_i$, $i = 1, \ldots, n$
$\min(x, y)$	the vector whose i-th component is $\min(x_i, y_i)$
$\max(x, y)$	the vector whose i-th component is $\max(x_i, y_i)$
$x \circ y$	$\equiv (x_i y_i)_{i=1}^n$; the Hadamard product of x and y
$x \perp y$	x and y are perpendicular
$\mathbf{1}_k$	k-vector of all ones (subscript often omitted)

Functions

$F : \mathcal{D} \to \mathcal{R}$	a mapping with domain $\mathcal{D}$ and range $\mathcal{R}$
$F\|_\Omega$	the restriction of the mapping F to the set Ω
$F \circ G$	composition of two functions F and G
F^{-1}	the inverse of a mapping F
$F'(\cdot\,;\cdot)$	directional derivative of the mapping F
JF	$\equiv \left(\frac{\partial F_i}{\partial x_j} \right)$; the $m \times n$ Jacobian of a mapping $F : \mathbb{R}^n \to \mathbb{R}^m$ $(m \ge 2)$
$J_\beta F_\alpha$	$\equiv (JF)_{\alpha\beta}$; a submatrix of JF
$J_y F(x, y)$	the partial Jacobian matrix of F with respect to y

Functions (continued)

$\nabla\theta$	$\equiv \left(\dfrac{\partial\theta}{\partial x_j} \right)$; the gradient of a function $\theta : \mathbb{R}^n \to \mathbb{R}$
$\nabla^2\theta$	$\equiv \left(\dfrac{\partial^2\theta}{\partial x_i x_j} \right)$; the Hessian matrix of $\theta : \mathbb{R}^n \to \mathbb{R}$
$\theta^D(\cdot;\cdot)$	Dini directional derivative of $\theta : \mathbb{R}^n \to \mathbb{R}$
$\operatorname{Jac} F = \partial_B F$	the limiting Jacobian or B-subdifferential of $F : \mathbb{R}^n \to \mathbb{R}^m$
∂F	$\equiv \operatorname{conv} \operatorname{Jac} F$; the Clarke Jacobian of $F : \mathbb{R}^n \to \mathbb{R}^m$
$\partial_C F$	$\equiv (\partial F_1(x) \times \partial F_2(x) \times \partial F_n(x))^T$
$\partial^2\theta$	$\equiv \partial\nabla\theta$; the generalized Hessian of an LC^1 function $\theta : \mathbb{R}^n \to \mathbb{R}$
$\varphi^*(y)$	the conjugate of a convex function $\varphi(x)$
$\varphi_\infty(d)$	the recession function of a convex function $\varphi(x)$
$o(t)$	any function such that $\lim\limits_{t\downarrow 0} \dfrac{o(t)}{t} = 0$
$O(t)$	any function such that $\limsup\limits_{t\downarrow 0} \dfrac{\lvert O(t)\rvert}{t} < \infty$
$\deg(\Phi, \Omega, p)$	the degree of Φ at p relative to Ω
$\deg(\Phi, \Omega)$	$\equiv \deg(\Phi, \Omega, 0)$
$\operatorname{ind}(\Phi, x)$	the index of Φ at x
$\Pi_K(x)$	the Euclidean projection of x on the set K
$\Pi_{K,A}(x)$	skewed projection of x on K under the A-norm
Π_K^A	$\equiv \Pi_{K,A} \circ A^{-1}$
$\operatorname{mid}(a, b; x)$	$\equiv \Pi_{[a,b]}(x)$; the mid function for given $a, b \in \mathbb{R}^n$
$\inf_S \theta(x)$	the infimum of the function θ on S
$\sup_S \theta(x)$	the supremum of the function θ on S
$\operatorname{dist}(x, W)$	Euclidean distance function from vector x to set W
$\operatorname{dist}_\infty(x, W)$	ℓ_∞-distance function from vector x to set W
J_Φ	the resolvent of the multifunction Φ
$D_f(x, y)$	$\equiv f(x) - f(y) - \nabla f(y)^T(x - y)$; Bregman distance induced by the strictly convex function f

Sets

$\in$, $\notin$	element membership, non-membership in a set
$\emptyset$, $\subseteq$, $\subset$	the empty set, set inclusion, proper set inclusion
$\cup, \cap, \times$	union, intersection, Cartesian product
$\prod S_i$	Cartesian product of sets S_i
$S_1 \setminus S_2$	the difference of two sets S_1 and S_2
$S_1 + S_2$	the vector sum of two sets S_1 and S_2

Sets (continued)

$\|S\|$	the cardinality of a finite set S
$\operatorname{aff} S$, $\operatorname{lin} S$	the affine, linear hull of a set S, respectively
$\operatorname{bd} S = \partial S$	the topological boundary of a set S
$\operatorname{cl} S$, $\operatorname{int} S$	the topological closure, interior of a set S, respectively
$\operatorname{conv} S$	the convex hull of a set S
$\operatorname{pos} A$	the conical hull of the columns of $A \in \mathbb{R}^{m\times n}$
$\operatorname{ri} S$	the relative interior of a set S
S^*, S_∞	the dual cone of a set S, the recession cone of S
$S^\perp$	the orthogonal complement of a set S
$\operatorname{dom} \Phi$	the domain of a (multi)function Φ
$\operatorname{gph} \Phi$	the graph of a (multi)function Φ
$\operatorname{ran} \Phi$	the range of a (multi)function Φ
$\mathbb{B}(x, \delta)$	the open ball with center at x and radius δ (a neighborhood $\mathcal{N}$ of x)
$\mathbb{B}(H; \varepsilon, S)$	ε-neighborhood of the function H restricted to the set S, comprising all continuous functions G such that $\|G - H\|_S \equiv \sup_{y\in S} \|G(y) - H(y)\| < \varepsilon$
$K_{\mathcal{N}}$	$\equiv K \cap \operatorname{cl} \mathcal{N}$
$\operatorname{argmax}_S \theta(x)$	the set of constrained maximizers of θ on S
$\operatorname{argmin}_S \theta(x)$	the set of constrained minimizers of θ on S
$\operatorname{supp}(x)$	the support of a vector x
$\mathcal{L}(x; S)$	the linearization cone of the set S at a point $x \in S$
$\mathcal{N}(x; S)$	the normal cone of the set S at a point $x \in S$
$\mathcal{T}(x; S)$	the tangent cone of the set S at a point $x \in S$
$\mathcal{C}(x; K, F)$	critical cone of the pair (K, F) at $x \in \operatorname{SOL}(K, F)$
$\mathcal{C}_\pi(x; K)$	$\equiv \mathcal{C}(\Pi_K(x); K, \mathbf{I} - x)$; critical cone of K at $x \in \mathbb{R}^n$
$\mathcal{I}(x)$	the index set of active constraints at x
$\mathcal{M}(x)$	the set of KKT multipliers at $x \in \operatorname{SOL}(K, F)$
$\mathcal{M}_\pi(x)$	the set of KKT multipliers at $\Pi_K(x)$
$\mathcal{M}^e(x)$	the (finite) set of extreme KKT multipliers in $\mathcal{M}(x)$
$P(A, b)$	$\{y \in \mathbb{R}^n : Ay \le b\}$; a polyhedron
$\mathcal{I}(A, b)$	family of index sets identifying the faces of $P(A, b)$
$\mathcal{B}_{\text{bas}}(A, b)$	normal family of basis matrix of $P(A, b)$
$[x, y]$	the closed line segment joining x and y in $\mathbb{R}^n$
(x, y)	the open line segment joining x and y in $\mathbb{R}^n$
$x^\perp$	the orthogonal complement of the vector x
$\operatorname{epi} \varphi$	the epigragph of a convex function φ
H_{++}	$\equiv H(\mathbb{R}^{2n}_{++} \times \mathbb{R}^m)$, used in IP theory
H_{+}	$\equiv H(\mathbb{R}^{2n}_{+} \times \mathbb{R}^m)$, used in IP theory

Problem Classes and Fundamental Objects

AVI (K, q, M)	AVI defined by the polyhedron K, vector q, and matrix M
CE (G, X)	constrained equation defined by the function G and the set X
CP (F, G)	vertical CP defined by two functions F and G
CP (K, F)	CP defined by the cone K and the mapping F
CP (K, q, M)	$\equiv$ CP (K, F) with $F(x) \equiv q + Mx$
$\mathcal{D}(K, M)$	VI domain of the pair (K, M)
FEA(K, F)	the feasible region of the CP (K, F)
$\mathcal{K}(K, M)$	VI kernel of the pair (K, M)
LCP (q, M)	LCP defined by the vector q and matrix M
NCP (F)	NCP defined by the function $F : \mathbb{R}^n \to \mathbb{R}^n$
$\mathcal{R}(K, M)$	VI range of the pair (K, M)
SOL(K, F)	solution set of the VI (K, F)
SOL(K, G, A, b)	solution set of the VI (K, G, A, b)
SOL(K, q, M)	solution set of the VI (K, q, M)
SOL(q, M)	solution set of the LCP (q, M)
VI (K, F)	VI defined by the set K and the mapping F
VI (K, G, A, b)	$\equiv$ VI (K, F) with $F(x) \equiv A^T G(Ax) + b$
VI (K, q, M)	$\equiv$ VI (K, F) with $F(x) \equiv q + Mx$

Matrix Classes

column sufficient	matrices M for which $x \circ Mx \leq 0 \Rightarrow x \circ Mx = 0$
copositive	matrices M such that $x^T Mx \geq 0$ for all $x \geq 0$
nondegenerate	matrices with nonzero principal minors
positive definite	matrices M such that $x^T Mx > 0$ for all $x \neq 0$
positive semidefinite	matrices M such that $x^T Mx \geq 0$ for all x
positive semi-definite plus	positive semidefinite + $[x^T Mx = 0 \Rightarrow Mx = 0]$
row sufficient	matrices whose transpose are column sufficient
semicopositive	matrices M for which $\forall\ x \gneqq 0\ \exists\ i$ such that $x_i(Mx)_i \geq 0$ and $x_i \neq 0$
strictly copositive	matrices M such that $x^T Mx > 0$ for all $x \gneqq 0$
strictly semi-copositive	matrices M for which $\forall\ x \gneqq 0\ \exists\ i$ such that $x_i(Mx)_i > 0$
P_0	matrices M for which $\forall\ x \neq 0\ \exists\ i$ such that $x_i(Mx)_i \geq 0$ and $x_i \neq 0$
P	matrices M for which $\forall\ x \neq 0\ \exists\ i$ such that $x_i(Mx)_i > 0$

Matrix Classes (continued)

R_0	matrices M such that $\mathrm{SOL}(0, M) = \{0\}$
S_0	matrices M such that $Mx \geq 0$ for some $x \gneqq 0$
S	matrices M such that $Mx > 0$ for some $x \geq 0$

CP Functions

$\psi_{\mathrm{CCK}}(a, b)$	$\equiv \psi_{\mathrm{FB}}(a, b) - \tau \max(0, a) \max(0, b)$; the Chen-Chen-Kanzow C-function
$\psi_{\mathrm{FB}}(a, b)$	$\equiv \sqrt{a^2 + b^2} - a - b$; the Fischer-Burmeister C-function
$\psi_{\mathrm{FB}_\mu}(a, b)$	$\equiv \sqrt{a^2 + b^2 + 2\mu} - a - b$; the smoothed Fischer-Burmeister function
$\psi_{\mathrm{CHKS}_\varepsilon}(a, b)$	$\equiv \sqrt{(a-b)^2 + 4\varepsilon} - (a + b)$; the smoothed Chen-Harker-Kanzow-Smale function
$\psi_{\mathrm{KK}}(a, b)$	$\equiv \dfrac{\sqrt{(a-b)^2 + 2qab} - (a+b)}{2-q}$, $q \in [0, 2)$; the Kanzow-Kleinmichel C-function
$\psi_{\mathrm{LT}}(a, b)$	$\equiv \|(a, b)\|_q - a - b$, $q > 1$; the Luo-Tseng C-function
$\psi_{\mathrm{LTKYF}}(a, b)$	$\equiv \phi_1(ab) + \phi_2(-a, -b)$, the Luo-Tseng-Kanzow-Yamashita-Fukushima family of C-functions
$\psi_{\mathrm{Man}}(a, b)$	$\equiv \zeta(\|a - b\|) - \zeta(b) - \zeta(a)$; Mangasarian's family of C-functions, includes the min function
$\psi_{\mathrm{U}}(a, b)$	Ulbrich's C-function; see Exercise **1.8.21**
$\psi_{\mathrm{YYF}}(a, b)$	$\equiv \frac{\eta}{2}\left((ab)_+\right)^2 + \frac{1}{2}\psi_{\mathrm{FB}}(a, b)^2$, $\eta > 0$; the Yamada-Yamashita-Fukushima C-function
$\mathbf{F}_\psi(x)$	$\equiv (\psi(x_i, F_i(x)))_{i=1}^n$; the reformulation function of the NCP (F) for a given C-function ψ
$\theta_\psi(x)$	$\equiv \frac{1}{2}\displaystyle\sum_{i=1}^{n} \psi^2(x_i, F_i(x))$; merit function induced by the C-function ψ
$\psi_{ab}(u, v)$	$\equiv \left(\dfrac{1}{2a} - \dfrac{1}{2b}\right) v^2 + \dfrac{1}{2b} \max(\,0, v - b\,u\,)^2 - \dfrac{1}{2a} \max(\,0, v - a\,u\,)^2$; for $b > a > 0$
$\theta_{ab}^{\mathrm{ncp}}(x)$	$\equiv \displaystyle\sum_{i=1}^{n} \psi_{ab}(x_i, F_i(x))$; the implicit Lagrangian function for the NCP (F)
$\phi_{\mathrm{Q}}(\tau, \tau'; r, s)$	Qi's B-function; see (9.4.7)
$\mathbf{H}_{\mathrm{IP}}(x, y, z)$	the IP function for implicit MiCP; see (11.1.4)
$\mathbf{H}_{\mathrm{CHKS}}(u, x, y, z)$	the IP function for the CHKS smoothing of the min function; see Exercise **11.9.8**

KKT Functions

$\mathbf{L}(x,\mu,\lambda)$ $\equiv F(x)+\sum_{j=1}^{\ell}\mu_j\,\nabla h_j(x)+\sum_{i=1}^{m}\lambda_i\,\nabla g_i(x)$; the vector Lagrangian function of the VI (K,F) with a finitely represented K

$\Phi_{\rm FB}(x,\mu,\lambda)$ $\equiv \begin{pmatrix} \mathbf{L}(x,\mu,\lambda) \\ -h(x) \\ \psi_{\rm FB}(-g_i(x),\lambda_i) : 1\le i\le m \end{pmatrix}$; the FB reformulation of the KKT system of a VI

$\theta_{\rm FB}(x,\mu,\lambda)$ $\equiv \frac{1}{2}\,\Phi_{\rm FB}(x,\mu,\lambda)^T\Phi_{\rm FB}(x,\mu,\lambda)$; the FB merit function of the KKT system of a VI

$\Phi_{\min}(x,\mu,\lambda)$ $\equiv \begin{pmatrix} \mathbf{L}(x,\mu,\lambda) \\ -h(x) \\ \min(-g(x),\lambda) \end{pmatrix}$; the min reformulation of the KKT system of a VI

$\theta_{\min}(x,\mu,\lambda)$ $\equiv \frac{1}{2}\,\Phi_{\min}(x,\mu,\lambda)^T\Phi_{\min}(x,\mu,\lambda)$; the min merit function of the KKT system of a VI

VI Functions

$\mathbf{F}_K^{\rm nat}$ the natural map associated with the pair (K,F)

$\mathbf{F}_K^{\rm nor}$ the normal map associated with the pair (K,F)

$\mathbf{F}_{K,D}^{\rm nat}(x)$ $\equiv x-\Pi_{K,D}(x-D^{-1}F(x))$; the skewed natural map associated with the pair (K,F) using $\Pi_{K,D}$

$\mathbf{F}_{K,\tau}^{\rm nat}$ the natural map associated with the VI $(K,\tau F)$

$\mathbf{M}_K^{\rm nat}$ $\equiv \mathbf{F}_K^{\rm nat}$ for $F(x)\equiv Mx$

$\mathbf{M}_K^{\rm nor}$ $\equiv \mathbf{F}_K^{\rm nor}$ for $F(x)\equiv Mx$

$\theta_{\rm gap}$ the gap function of a VI

$\theta_{\rm dual}$ the dual gap function of a VI

θ_c the regularized gap function with parameter $c>0$

$\theta_c^{\rm lin}$ the linearized gap function with parameter $c>0$

θ_{ab} $\equiv \theta_a-\theta_b$; the D-gap function for $b>a>0$

$y_c(x)$ unique maximizer in $\theta_c(x)$

$y_c^{\rm lin}(x)$ unique maximizer in $\theta_c^{\rm lin}(x)$

$T_c(x;K,F)$ $\equiv \mathcal{T}(x;K)\cap(-\mathcal{T}(y_c(x);K))\cap(\,-F(x)\,)^*$

$T_{ab}(x;K,F)$ $\equiv \mathcal{T}(y_b(x);K)\cap(-\mathcal{T}(y_a(x);K))\cap(\,-F(x)\,)^*$

Selected Function Classes and Properties

co-coercive on K	functions F for which $\exists\ \eta > 0$ such that $(x-y)^T(F(x)-F(y)) \geq \eta\|F(x)-F(y)\|^2$ for all x, y in K
monotone composite	functions $F(x) \equiv A^T G(Ax) + b$, where G is monotone
monotone plus	monotone and $(F(x)-F(y))^T(x-y) = 0 \Rightarrow F(x) = F(y)$
nonexpansive	functions F for which $\|F(x)-F(y)\| \leq \|x-y\|$ $\forall\ x$ and y
norm-coercive on X	functions F for which $\lim_{\substack{x\in X \\ \|x\|\to\infty}} \|F(x)\| = \infty$
P_0, P, $P_*(\sigma)$	see Definition **3.5.8**
(pseudo) monotone	see Definition **2.3.1**
(pseudo) monotone plus	see Definition **2.3.9**
strictly, strongly monotone	see Definition **2.3.1**
S	see Exercise **2.9.5**
strongly S	functions F such that $\forall\ q \in \mathbb{R}^n$, $\exists\ x \geq 0$ satisfying $F(x) > q$
symmetric	= gradient map; differentiable $F : \mathbb{R}^n \to \mathbb{R}^n$ with symmetric JF
uniformly P	see Definition **3.5.8**
univalent	= continuous plus injective
weakly univalent	uniform limit of univalent functions

Numbering System

The chapters of the book are numbered from 1 to 12; the sections are denoted by decimal numbers of the type **2.3** (meaning Section 3 of Chapter 2). Many sections are further divided into subsections; most subsections are numbered, some are not. The numbered subsections are by decimal numbers following the section numbers; e.g., Subsection **1.3.1** means Chapter 1, Section 3, Subsection 1.

All definitions, results, and miscellaneous items are numbered consecutively within each section in the form **1.3.5, 1.3.6**, meaning Items 5 and 6 in Section 3 of Chapter 1. All items are also identified by their types, for example, **1.4.1 Proposition.**, **1.4.2 Remark.** When an item is referred to in the text, it is called out as Algorithm **5.2.1**, Theorem **4.1.7**, and so forth. Equations are numbered consecutively and identified by chapter, section, and equation. Thus (3.1.4) means Equation (4) in Section 1 of Chapter 3.

Chapter 7

Local Methods for Nonsmooth Equations

This is the first of two chapters in which we develop numerical methods for the solution of systems of nonsmooth equations of the form

$$G(x) = 0, \tag{7.0.1}$$

where $G : \Omega \subseteq \mathbb{R}^n \to \mathbb{R}^n$ is locally Lipschitz on the open set Ω. This chapter focuses on locally convergent Newton-type methods, and the next one deals with the globalization of these methods. An algorithm is locally convergent if the starting iterate is required to belong to a suitably chosen neighborhood of the desired solution in order to guarantee the convergence of the algorithm. In contrast, an algorithm is globally convergent if such a nearness requirement can be removed. This chapter treats Newton methods for solving nonsmooth equations and studies their convergence rates. Nonsmooth systems are particularly important in the solution of VIs and CPs, since, as we saw in Chapter **1**, a VI/CP can be reformulated as a system of nonsmooth equations. In addition, nonsmooth equations are of independent interest and arise quite naturally in many disciplines. Specializations of the developments in this and the next chapter to various nonsmooth equation reformulations of VIs and CPs are the subjects of Chapters **9** and **10**, respectively.

In some cases it is appropriate to consider a closely related problem, namely, that of finding a solution of (7.0.1) that belongs to a given set X:

$$G(x) = 0, \qquad x \in X, \tag{7.0.2}$$

where $X \subseteq \Omega$ is a closed set. We call (7.0.2) a "constrained equation" (CE). In most of the cases of interest X either coincides with the whole

$\mathbb{R}^n$, in which case (7.0.2) reduces to (7.0.1), or is a polyhedral set, e.g. $\mathbb{R}^n_+$. There are at least two reasons for introducing the set X. The first one is that when considering the VI (K, F) the function F might be undefined outside (an open neighborhood containing) K and this feature is inherited by an equation reformulation of the VI. The second reason is that when considering an equation reformulation of VI (K, F) we know a priori that any solution must belong to the set K, and so it seems sensible to try to put to algorithmic use this information by taking $X = K$. Indeed, this strategy lies at the heart of some highly successful algorithms to be presented subsequently. In particular, the CE (7.0.2) provides a unified framework for the presentation and analysis of the family of "interior-point" methods, which is a vast subject by itself; see Chapter **11**.

Although many of the principal ideas underlying the methods presented in this chapter have their origin in similar ideas in methods for smooth equations, the nondifferentiability of G gives rise to a lot of complications that invalidate the classical methods. Thus, new tools need to be introduced and known methods need to be revised. For the most part, the analysis in this chapter is somewhat abstract; only basic schemes are considered so as to clarify as much as possible the main technical issues arising from nonsmooth equations.

7.1 Nonsmooth Analysis I: Clarke's Calculus

The aim of this section is to familiarize the reader with some extensions of classical results in real analysis for smooth functions to nonsmooth functions. More specifically, we consider locally Lipschitz functions and present Clarke's generalization of the concept of gradient and Jacobian and the corresponding calculus rules. We also briefly touch upon some slightly more complex topics like the theorem of the mean and the implicit function theorem. All the results of this section are given without proofs, since they are intended only as a summary of well-known, even if advanced, results, that are included here for completeness and convenience.

Until now the main tool we have used in the study of nonsmooth functions is the directional derivative. However, when designing algorithms we would encounter many difficulties if we would stick solely to this concept. In this section we introduce the "generalized Jacobian" of a locally Lipschitz function and explore some of its applications. We should warn the reader, however, that also this theory has its drawbacks, as we shall indicate later in this section, so that subsequently we will develop sharper results for classes of functions that are more amenable to our needs. Nev-

ertheless, the results of this section form the basis on which we can build practical and effective methods for the solution of VIs and CPs.

In Section **4.6**, we already introduced the concept of the limiting Jacobian that we recall (and slightly extend) here for the reader's convenience. Let $G : \Omega \subseteq \mathbb{R}^n \to \mathbb{R}^m$, with Ω open, be locally Lipschitz continuous at a vector $x \in \Omega$. Similar to Definition **4.6.2**, where we have $m = n$, we define the limiting Jacobian of G at $\bar{x}$ by

$$\begin{aligned} \operatorname{Jac} G(\bar{x}) \;=\; \partial_B G(\bar{x}) \;\equiv\; \{\, H \;\in\; \mathbb{R}^{m\times n} \;:\; H \;=\; \lim_{k\to\infty} JG(x^k), \\ \text{for some sequence } \{x^k\} \;\to\; \bar{x}, x^k \;\notin\; N_G \,\}, \end{aligned} \tag{7.1.1}$$

where we denote by N_G the negligible set of points at which G is not F-differentiable. Although this concept has been employed successfully in the study of PC^1 functions (see e.g. Section **4.6**), we cannot deny the fact that in general the limiting Jacobian is a difficult object to deal with. In particular, for an arbitrary nonsmooth function, it is not easy to calculate and manipulate the limiting Jacobian. In the context of nonsmooth optimization problems, this Jacobian is not particularly useful because it does not allow us to obtain optimality conditions. From the calculus point of view, we can not obtain mean-value theorems or other useful results based solely on the limiting Jacobian. To (partially) circumvent these difficulties (and also for other reasons that are beyond the scope of this section) we introduce the following definition.

7.1.1 Definition. Let $G : \Omega \subseteq \mathbb{R}^n \to \mathbb{R}^m$, with Ω open, be locally Lipschitz at a vector $\bar{x} \in \Omega$. The Clarke *generalized Jacobian* of G at $\bar{x}$ is:

$$\partial G(\bar{x}) \;\equiv\; \operatorname{conv} \operatorname{Jac} G(\bar{x}). \tag{7.1.2}$$

When $m = 1$, that is when G is a real-valued function $g : \mathbb{R}^n \to \mathbb{R}$, $\partial g(\bar{x})$ is called the *generalized gradient* of g at $\bar{x}$. Furthermore, in this case, consistently with the notation of the gradient of a smooth function, the elements of $\partial g(\bar{x})$ are viewed as column vectors. □

We often refer to the Clarke generalized Jacobian simply as the generalized Jacobian of G. When $m = 1$ there is a (traditional) notational problem, in that the notion of the (generalized) gradient is not consistent with that of the (generalized) Jacobian because of a transposition operation. Hopefully this won't cause any confusion.

We can illustrate these definitions with the simple function $|x|$. This function is globally Lipschitz continuous with a Lipschitz constant $L = 1$, and it is continuously differentiable everywhere except at the origin. It

is easy to check that at this point we have $\operatorname{Jac}|0| = \{-1, 1\}$, so that the generalized gradient is simply $\partial|0| = [-1, 1]$. In general, the calculation of generalized Jacobians can be simplified by the fact that the generalized Jacobian is "blind" to sets of zero measure, in the sense that if N_0 is any set of measure zero in $\mathbb{R}^n$, then

$$\partial G(\bar{x}) \equiv \operatorname{conv}\left\{H : H = \lim_{k\to\infty} JG(x^k),\ \{x^k\} \to \bar{x}, x^k \notin N_G \cup N_0\right\}.$$

We can use this result to illustrate the calculation of the generalized Jacobian of a slightly more complex function than the absolute-value function considered above.

7.1.2 Example. Consider the function in Example **4.6.4**, which is given by:

$$G(x, y) \equiv \begin{pmatrix} \min(x, y) \\ |x|^3 - y \end{pmatrix}.$$

Although the previous calculation of $\operatorname{Jac} G(0, 0)$ in Example **4.6.4** readily yields $\partial G(0, 0)$, we use the aforementioned result to produce $\partial G(0, 0)$ directly. To this end consider the four closed regions

$$\begin{aligned} R_1 &\equiv \{(x, y) : y \geq x,\ x \geq 0\}, \\ R_2 &\equiv \{(x, y) : y \geq x,\ x \leq 0\}, \\ R_3 &\equiv \{(x, y) : y \leq x,\ x \leq 0\}, \\ R_4 &\equiv \{(x, y) : y \leq x,\ x \geq 0\}. \end{aligned}$$

It is clear that the union of these four regions gives the whole space and that the union $\mathcal{B}$ of their boundaries is a set of measure zero. Note that the set $\mathcal{B}$ contains all the points where G is nondifferentiable but also points where G is continuously differentiable (for example the positive y semiaxis). By the result mentioned just before this example it is then easy to see that the generalized Jacobian of G at $(0, 0)$ is given by the convex hull of the limits of the Jacobians of G calculated for sequences converging to zero from the interior of each of the four regions R_i:

$$\partial G(0, 0) = \operatorname{conv}\left(\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & -1 \end{bmatrix}\right),$$

thus verifying the expression (7.1.2) in Definition **7.1.1**. □

Another interesting example in which we can easily calculate the generalized gradient of a function on the basis of the definition is the Euclidean norm.

7.1.3 Example. Let $g(x) = \| x \|_2$ be the Euclidean norm function on $\mathbb{R}^n$. This function is continuously differentiable everywhere except at the origin. If $x \neq 0$, we have

$$\nabla \| x \|_2 \; = \; \frac{x}{\| x \|_2},$$

so that the (Euclidean) norm of the gradient of the norm function is equal to one at every nonzero vector. From this calculation, it is then clear that $\operatorname{Jac} g(0)$ consists of all vectors with Euclidean norm of exactly one. The generalized gradient at the origin is the convex hull of this set and therefore we get $\partial \| 0 \| = \operatorname{cl} \mathbb{B}(0, 1)$.

Applying the above calculation to the Fischer-Burmeister C-function:

$$\psi_{\mathrm{FB}}(a, b) \; \equiv \; \sqrt{a^2 + b^2} - (\, a + b \,), \quad \forall \, (\, a, b \,) \; \in \; \mathbb{R}^2,$$

we obtain

$$\operatorname{Jac} \; \psi_{\mathrm{FB}}(0, 0) \; = \; \operatorname{bd} \mathbb{B} \left(\begin{pmatrix} -1 \\ -1 \end{pmatrix}, 1 \right)$$

and

$$\partial \psi_{\mathrm{FB}}(0, 0) \; = \; \operatorname{cl} \mathbb{B} \left(\begin{pmatrix} -1 \\ -1 \end{pmatrix}, 1 \right).$$

We note that $\operatorname{Jac} \psi_{\mathrm{FB}}(0, 0)$ is a compact but nonconvex set. □

From the definition of the generalized Jacobian, it is clear that if the real-valued function g is continuously differentiable at x, then $\partial g(x)$ is equal to the singleton $\{\nabla g(x)\}$. It can also be shown that if g is Lipschitz at x and Gâteaux differentiable there, then $\nabla g(x) \in \partial g(x)$, but the generalized gradient could contain other elements different from $\nabla g(x)$. In fact, the generalized gradient is equal to the singleton $\{\nabla g(x)\}$ if and only if g has a strong F-derivative at x. This is the first manifestation of the drawbacks of the generalized Jacobian. We could say that the term generalized Jacobian itself is inappropriate, since the generalized Jacobian does not necessarily reduce to the usual Jacobian if a function is Gâteaux differentiable at a point; in many situations the generalized Jacobian turns out to be "too large a set". For this reason, in the study of the VI/CP, we often consider special classes of locally Lipschitz continuous functions for which the Clarke generalized Jacobian is an appropriate tool; see e.g. Section **7.4**.

The generalized Jacobian can be viewed as a multifunction from Ω into subsets of $\mathbb{R}^{n \times m}$:

$$\partial G \; : \; x \; \in \; \Omega \; \mapsto \; \partial G(x) \; \subset \; \mathbb{R}^{n \times m}.$$

Below we report some basic properties of this multifunction at a vector x where G is locally Lipschitz; these properties are rather intuitive on the basis of what we have seen so far.

7.1.4 Proposition. Let a function $G : \Omega \subseteq \mathbb{R}^n \to \mathbb{R}^m$ be given, with G locally Lipschitz on the open set Ω. The following statements are valid for any $x \in \Omega$:

(a) $\partial G(x)$ is nonempty, convex and compact;

(b) the mapping ∂G is upper semicontinuous at x; thus for every $\varepsilon > 0$ there is a $\delta > 0$ such that, for all $y \in \mathbb{B}(x, \delta)$,

$$\partial G(y) \subseteq \partial G(x) + \mathbb{B}(0, \varepsilon).$$

Therefore ∂G is closed at x, that is, if $\{x^k\} \to x$, $H^k \in \partial G(x^k)$ and $H^k \to H$, then $H \in \partial G(x)$. □

For a real-valued function, the Clarke generalized gradient can be characterized by a certain kind of directional derivative that we define below.

7.1.5 Definition. Let a function $g : \Omega \subseteq \mathbb{R}^n \to \mathbb{R}$ be given, with g locally Lipschitz on the open set Ω. We define the *(Clarke) generalized directional derivative* of g at x in the direction d, denoted by $g^\circ(x; d)$, as

$$g^\circ(x; d) \equiv \limsup_{\substack{y \to x \\ t \downarrow 0}} \frac{g(y + td) - g(y)}{t}.$$

Since the lim sup is involved in the definition of the generalized directional derivative, it is obvious that $g^\circ(x; d)$ is well defined for every function. When g is locally Lipschitz at x, as in our case, then it is easy to see that the generalized directional derivative $g^\circ(x; d)$ is finite for every d. It turns out that $g^\circ(x; d)$ and $\partial g(x)$ are closely related; in fact, it holds that:

$$\partial g(x) = \{\xi \in \mathbb{R}^n : \xi^T d \leq g^\circ(x; d), \ \forall\, d \in \mathbb{R}^n\}. \tag{7.1.3}$$

Further properties of the Clarke generalized directional derivative are summarized in the following proposition.

7.1.6 Proposition. Let a function $g : \Omega \subseteq \mathbb{R}^n \to \mathbb{R}$ be locally Lipschitz continuous on the open set Ω. The following three statements hold.

(a) For every $x \in \Omega$, $g^\circ(x; \cdot)$ is Lipschitz continuous, positively homogeneous, and sublinear; sublinearity means

$$g^\circ(x; d + d') \leq g^\circ(x; d) + g^\circ(x; d')$$

for any two vectors d and d' in $\mathbb{R}^n$.

(b) For every $(x, d) \in \Omega \times \mathbb{R}^n$,

$$g^\circ(x; d) \;=\; \max\{\, \xi^T d \;:\; \xi \;\in\; \partial g(x)\,\}.$$

(c) As a function in (x, d), $g^\circ : \Omega \times \mathbb{R}^n \to \mathbb{R}$ is upper semicontinuous. □

By definition we have

$$g'(x; d) \;\leq\; g^\circ(x; d), \tag{7.1.4}$$

when the left-hand directional derivative exists. As we shall see shortly, in many situations it is important to ascertain whether equality actually holds in (7.1.4).

7.1.7 Definition. Let a function $g : \Omega \subseteq \mathbb{R}^n \to \mathbb{R}$ be given, with g being locally Lipschitz continuous on the open set Ω. We say that g is *C-regular* (C for Clarke) at $x \in \Omega$ if:

(a) g is directionally differentiable at x, and

(b) $g^\circ(x; d) = g'(x; d)$ for all directions $d \in \mathbb{R}^n$.

Whenever there is no confusion, we simply say that a function g is regular if it is C-regular. □

It can be shown that convex functions and continuously differentiable functions at x are C-regular and so are nonnegative combinations of regular functions at the same x. Moreover, if g is a C-regular function, then we have

$$g'(x; d) = \max\{\, \xi^T d \;:\; \xi \;\in\; \partial g(x)\},$$

which generalizes a well-known result in convex analysis. Thus the class of C-regular functions is rather large. However, simple functions exist that are not C-regular. Here is an example.

7.1.8 Example. Consider the one-variable function $g(x) \equiv -|x|$ and its (generalized) directional derivative at 0 in the direction $d = 1$. The function g is obviously directionally differentiable and $g'(0; 1) = -1$. On the other hand it is also easy to see that $g^\circ(0; 1) = 1$, so that the function is not C-regular. In a similar fashion, the min function of two arguments is also not regular. □

The previous example, which is phrased in terms of the generalized directional derivative, is another manifestation of the fact that the generalized Jacobian can be too large. We know that the (ordinary) directional

derivative of a locally Lipschitz continuous g, when it exists, gives an accurate first-order approximation of g in the direction; i.e.,

$$g(x+d) \;=\; g(x) + g'(x;d) + o(\,\|\,d\,\|\,).$$

Example **7.1.8** shows that the generalized directional derivative may fail to have this approximation property. The latter property is essential to the family of Newton methods for solving the equation (7.0.2) to be presented subsequently. Therefore, if one wants to use the generalized Jacobians to define algorithms for the solution of systems of nonsmooth equations, one needs to narrow the class of functions that one deals with. The definition of regularity can then be seen as a first attempt to define such a class of functions for which the generalized Jacobian may be fruitfully employed in a computational context.

In order to be able to compute the generalized gradient of a function, a number of calculus rules are available, which mimic the corresponding rules for differentiable functions.

7.1.9 Proposition. Let $g_i : \mathbb{R}^n \to \mathbb{R}$, $i = 1, \ldots m$, be a family of locally Lipschitz continuous functions at x.

(a) For any coefficients a_i, one has

$$\partial \left(\sum_{i=1}^{m} a_i\, g_i \right)(x) \;\subseteq\; \sum_{i=1}^{m} a_i \partial g_i(x),$$

with equality holding if all but at most one of the g_i are continuously differentiable at x or if all the functions are C-regular at x and each a_i is nonnegative. In particular, $\partial a g(x) = a \partial g(x)$ for all locally Lipschitz continuous functions g and all constants a.

(b) $\partial(g_1 g_2)(x) \subseteq g_2(x)\partial g_1(x) + g_1(x)\partial g_2(x)$, with equality holding if g_1 and g_2 are C-regular at x, $g_1(x) \geq 0$, and $g_2(x) \geq 0$.

(c) If $g_2(x) \neq 0$, then

$$\partial \left(\frac{g_1}{g_2} \right)(x) \;\subseteq\; \frac{g_2(x)\partial g_1(x) - g_1(x)\partial g_2(x)}{g_2^2(x)},$$

with equality holding if g_1 and $-g_2$ are C-regular at x, $g_1(x) \geq 0$ and $g_2(x) > 0$.

(d) For the pointwise maximum function:

$$g(x) \;\equiv\; \max\{g_i(x),\, i = 1, \ldots, m\},$$

it holds that

$$\partial g(x) \subseteq \operatorname{conv}\{\, \partial g_i(x) : i \in \mathcal{I}(x) \,\},$$

where $\mathcal{I}(x) \equiv \{i : g_i(x) = g(x)\}$. Moreover, equality holds in the above inclusion if all the g_i are C-regular at x. □

The point (d) is particularly significant because it provides a very important source of nondifferentiable functions (the maximum of a finite number of functions is not everywhere differentiable, even if all the functions g_i are differentiable) and has no counterpart in the smooth case. The fact that the calculus rules only give an inclusion unless more stringent conditions are met poses serious limits on the possibility to calculate generalized gradients easily. The following is a simple example showing the dramatic difference that can exist between the left-hand and right-hand sets in the inclusions of Proposition **7.1.9**.

7.1.10 Example. Let $g_i(x) = |x|$ and $g_2(x) = -|x|$. These two functions are locally Lipschitz continuous everywhere and their sum

$$g(x) \equiv g_1(x) + g_2(x)$$

is identically zero. The generalized gradient of g is therefore 0 everywhere. If we try to estimate the generalized gradient of g at zero by using Proposition **7.1.9**(b), we get the set $[-1, 1] + [-1, 1] = [-2, 2]$, which is obviously very different from $\partial g(0)$ although, as prescribed by the proposition, $\partial g(0) \subseteq \partial g_1(0) + \partial g_2(0)$. Note that g_2 is not C-regular at zero, as shown in Example **7.1.8**. □

The results listed in the previous proposition can be viewed as particular cases of the following result about composite functions.

7.1.11 Proposition. Let $f = g \circ G$, where $G : \mathbb{R}^n \to \mathbb{R}^m$ is locally Lipschitz continuous at x and where $g : \mathbb{R}^m \to \mathbb{R}$ is locally Lipschitz at $G(x)$. Then f is locally Lipschitz continuous at x and

$$\partial f(x) \subseteq \operatorname{conv}\{\, \xi = H^T \zeta : H \in \partial G(x),\ \zeta \in \partial g(G(x)) \,\}.$$

If in addition either one of the following two conditions is satisfied, then equality holds and the conv is superfluous.

(a) g is continuously differentiable at $G(x)$;

(b) g is C-regular at $G(x)$ and G is continuously differentiable at x (in this case, f is C-regular at x). □

Analogously to the smooth case we can consider *partial generalized gradients*. Suppose that a function $g : \mathbb{R}^p \times \mathbb{R}^q \to \mathbb{R}$ is given. We denote by $\partial_x g(x,y)$ the generalized gradient of $g(\cdot, y)$ at x, and by $\partial_y g(x,y)$ the generalized gradient of $g(x,\cdot)$ at y. In general neither $\partial g(x,y)$ is contained in $\partial_x g(x,y) \times \partial_y g(x,y)$ nor vice versa. However, it always holds that

$$\partial_x g(x,y) \subseteq \Pi_x \partial g(x,y) \quad \text{and} \quad \partial_y g(x,y) \subseteq \Pi_y \partial g(x,y),$$

where Π_x (Π_y) denotes the canonical projection on the subspace of the x (y) variables. Furthermore, if g is regular at (x,y) then it holds that

$$\partial g(x,y) \subseteq \partial_x g(x,y) \times \partial_y g(x,y).$$

Generalized gradients allow one to extend some well-known classical results in the smooth case.

7.1.12 Proposition. Suppose that $g : \mathbb{R}^n \to \mathbb{R}$ is Lipschitz continuous in a neighborhood of an (unconstrained) local minimum x of g. Then

$$0 \in \partial g(x). \tag{7.1.5}$$

If g is convex, the condition (7.1.5) is necessary and sufficient for x to be a global (unconstrained) minimum point of g. □

We call a vector x satisfying (7.1.5) a *C-stationary point* of g, where C stands for Clarke. Thus every local minimum of a locally Lipchitz continuous function a C-stationary point. We can also give mean-value type formulas, first for a scalar-valued function.

7.1.13 Proposition. Let a function $g : \mathbb{R}^n \to \mathbb{R}$ be locally Lipschitz on an open set containing the line segment $[x,y]$. There exists a point z in (x,y) such that

$$g(y) = g(x) + \xi^T (\, y - x \,), \tag{7.1.6}$$

for some ξ belonging to $\partial g(z)$. □

In the above results we confined ourselves to real-valued functions. A rich set of results, extending partially some of the ones we just listed, is available also for vector-valued functions. We first present a result that relates the generalized Jacobian of a vector-valued function to the generalized gradients of its component functions.

7.1.14 Proposition. Let $G : \Omega \subseteq \mathbb{R}^n \to \mathbb{R}^m$ be a locally Lipschitz continuous function on the open set Ω. If $x \in \Omega$, then

$$\partial G(x) \subseteq (\, \partial G_1(x) \times \partial G_2(x) \times \cdots \times \partial G_m(x) \,)^T.$$

Basically the above inclusion is an equality if the nondifferentiability of the various components of G are "unrelated", while the inclusion is usually strict otherwise. We illustrate this by two simple examples.

7.1.15 Example. Consider the function

$$F(x_1, x_2) \equiv \begin{pmatrix} |\, x_1 \,| - x_2 \\ x_1 + |\, x_2 \,| \end{pmatrix}.$$

We have

$$\partial F(0) = \left\{ \begin{pmatrix} \alpha & -1 \\ 1 & \beta \end{pmatrix}, \quad \forall \alpha \in [-1,1],\ \forall\, \beta \in [-1,1] \right\}.$$

In this case the inclusion in Proposition **7.1.14** is actually an equality.

Consider another function

$$F(x_1, x_2) \equiv \begin{pmatrix} \sin x_1 - |\, x_2 \,| \\ x_1^2 + |\, x_2 \,| \end{pmatrix}.$$

We have

$$\partial F(0) = \left\{ \begin{pmatrix} 1 & -\alpha \\ 0 & \alpha \end{pmatrix}, \quad \forall\, \alpha \in [-1,1] \right\},$$

while

$$\partial F_1(0) \times \partial F_2(0) = \left\{ \begin{pmatrix} 1 & \alpha \\ 0 & \beta \end{pmatrix}, \quad \forall\, \alpha \in [-1,1], \forall\, \beta \in [-1,1] \right\}.$$

In this case the inclusion in Proposition **7.1.14** is proper. □

The following result extends Proposition **7.1.13** to vector functions.

7.1.16 Proposition. Let a function $G : \Omega \subseteq \mathbb{R}^n \to \mathbb{R}^m$ be Lipschitz continuous on an open set Ω containing the segment $[x, y]$. There exist m points z^i in (x, y) and m scalars $\alpha_i \geq 0$ summing to unity such that

$$G(y) = G(x) + \sum_{i=1}^{m} \alpha_i\, H_i(y - x), \tag{7.1.7}$$

where, for each i, H_i belongs to $\partial G(z^i)$. □

The classical form of the mean value theorem for C^1 vector functions is usually stated in an integral form; even in the smooth case the alternative form stated above, which involves a simple summation, is not widely

known. It is possible to give an integral version of this theorem also in the nonsmooth case; however, for our purpose, the form presented above is easier and clearer. We present an application of Proposition **7.1.16** by establishing a result that gives an important connection between the directional derivative and Clarke's generalized Jacobian.

7.1.17 Proposition. Let a function $G : \Omega \subseteq \mathbb{R}^n \to \mathbb{R}^m$, with Ω open, be B-differentiable at a point x in Ω. For every vector $d \in \mathbb{R}^n$, there exists $H \in \partial G(x)$ such that $G'(x; d) = Hd$.

Proof. Let $\{\tau_k\}$ be an arbitrary sequence of positive scalars converging to zero. For every k, we can write

$$G(x + \tau_k d) - G(x) \;=\; \sum_{i=1}^{m} \tau_k \, \alpha_{i,k} \, H_i^k d \tag{7.1.8}$$

for some scalars $\alpha_{i,k}$ satisfying

$$\sum_{i=1}^{m} \alpha_{i,k} \;=\; 1, \qquad \alpha_{i,k} \;\geq\; 0$$

and some matrices $H_i^k \in \partial G(x + \tau'_{i,k} d)$, where $\tau'_{i,k} \in (0, \tau_k)$. By Proposition **7.1.4**, the sequence $\{H_i^k\}$ is bounded for every $i = 1, \ldots, m$. Without loss of generality, we may assume that each sequence $\{H_i^k\}$ converges to a limiting matrix H_i^∞, which must belong to $\partial G(x)$, by the closedness of the Clarke generalized Jacobian, also by Proposition **7.1.4**. We may further assume that each sequence $\{\alpha_{i,k}\}$ of scalars, for $i = 1, \ldots, m$, converges to a nonnegative scalar $\alpha_{i,\infty}$. Clearly, we have

$$\sum_{i=1}^{m} \alpha_{i,\infty} \;=\; 1.$$

Thus, dividing (7.1.8) by τ_k and letting $k \to \infty$, we deduce

$$G'(x; d) \;=\; \sum_{i=1}^{m} \alpha_{i,\infty} \, H_i^\infty d,$$

which shows that $G'(x; d) = Hd$, where

$$H \;\equiv\; \sum_{i=1}^{m} \alpha_{i,\infty} \, H_i^\infty$$

belongs to $\partial G(x)$, by the convexity of the generalized Jacobian. □

We present an implicit function theorem for a locally Lipschitz function. In what follows, for a function $G : \mathbb{R}^n \times \mathbb{R}^p \to \mathbb{R}^n$ of two variables $x \in \mathbb{R}^n$ and $y \in \mathbb{R}^p$, we denote by $\Pi_x \partial G(x, y)$ the set of $n \times n$ matrices M for which a $n \times p$ matrix N exists such that $[M \; N]$ belongs to $\partial G(x, y)$.

7.1.18 Proposition. Let $G : \mathbb{R}^n \times \mathbb{R}^p \to \mathbb{R}^n$ be Lipschitz continuous in a neighborhood of a point $(\bar{x}, \bar{y}) \in \mathbb{R}^n \times \mathbb{R}^p$ for which $G(\bar{x}, \bar{y}) = 0$. Assume that all matrices in $\Pi_x \partial G(\bar{x}, \bar{y})$ are nonsingular. There exist open neighborhoods U and V of $\bar{x}$ and $\bar{y}$ respectively such that, for every $y \in V$, the equation $G(x, y) = 0$ has a unique solution $x \equiv F(y) \in U$, $F(\bar{y}) = \bar{x}$, and the map $F : V \to U$ is Lipschitz continuous. □

A related important result is the nonsmooth inverse function theorem, which provides a sufficient condition for a locally Lipschitz function to be a locally Lipschitz homeomorphism. We say that the generalized Jacobian $\partial G(x)$ is *nonsingular* if all matrices in this Jacobian are nonsingular.

7.1.19 Proposition. Let $G : \Omega \subseteq \mathbb{R}^n \to \mathbb{R}^n$ be locally Lipschitz at $x \in \Omega$. If the generalized Jacobian $\partial G(x)$ is nonsingular, then G is a locally Lipschitz homeomorphism at x. □

We caution the reader that while the nonsingularity of the Jacobian is a necessary and sufficient condition for a continuously differentiable mapping to be a locally Lipschitz homeomorphism (see Proposition **5.2.9**), Proposition **7.1.19** only gives a sufficient condition. As shown by the example below, the converse of Proposition **7.1.19** is not true. In fact, this condition is often too broad to be necessary.

7.1.20 Example. We construct a piecewise linear function that is a global homeomorphism but whose generalized Jacobian contains a singular matrix. Define six vectors in the plane:

$$v^1 \equiv \begin{pmatrix} \cos 0 \\ \sin 0 \end{pmatrix}, \quad v^2 \equiv \begin{pmatrix} \cos \pi/4 \\ \sin \pi/4 \end{pmatrix}, \quad v^3 \equiv \begin{pmatrix} \cos 3\pi/8 \\ \sin 3\pi/8 \end{pmatrix},$$

$$v^4 \equiv \begin{pmatrix} \cos \pi/2 \\ \sin \pi/2 \end{pmatrix}, \quad v^5 \equiv \begin{pmatrix} \cos 3\pi/4 \\ \sin 3\pi/4 \end{pmatrix}, \quad v^6 \equiv \begin{pmatrix} \cos 5\pi/4 \\ \sin 5\pi/4 \end{pmatrix},$$

and let $v^7 \equiv v^1$. Define the six 2×2 matrices:

$$A^i \equiv \begin{bmatrix} v^i & v^{i+1} \end{bmatrix}, \quad \text{for } i = 1, \ldots, 6.$$

It is trivial to see that each A^i is a nonsingular matrix. Define another six vectors as follows:

$$u^1 \equiv v^1, \quad u^2 \equiv v^2, \quad u^3 \equiv \begin{pmatrix} \cos \pi \\ \sin \pi \end{pmatrix},$$

$$u^4 \equiv -v^4, \quad u^5 \equiv -v^5, \quad u^6 \equiv \begin{pmatrix} \cos 15\pi/8 \\ \sin 15\pi/8 \end{pmatrix},$$

and let $u^7 \equiv v^7$. Define the six 2×2 matrices:

$$B^i \equiv \begin{bmatrix} u^i & u^{i+1} \end{bmatrix}, \quad \text{for } i = 1, \ldots, 6.$$

For $i = 1, \ldots, 6$, let

$$K^i \equiv \text{pos}(\, v^i, v^{i+1}\,) \quad \text{and} \quad P^i \equiv \text{pos}(\, u^i, u^{i+1}\,).$$

The interiors and boundaries of the six cones $\{K^i : i = 1, \ldots, 6\}$ partition the plane; so do the other six cones $\{P^i : i = 1, \ldots, 6\}$. Define the PL map:

$$F(x) \equiv B^i\,(\,A^i\,)^{-1}x, \quad \text{if } x \in K^i,$$

which maps each K^i homeomorphically onto P^i. Therefore F is a global homeomorphism from $\mathbb{R}^2$ onto itself. Obviously all the matrices $B^i(A^i)^{-1}$ belong to $\partial F(0)$. Since

$$B^1\,(\,A^1\,)^{-1} = \mathbf{I}_2 \quad \text{and} \quad B^4\,(\,A^4\,)^{-1} = -\mathbf{I}_2,$$

we deduce that $\partial F(0)$ contains the zero matrix. Consequently $\partial F(0)$ is not nonsingular. □

We refer the reader to Subsection **2.1.2** for necessary and sufficient conditions for a locally Lipschitz continuous function to be a locally Lipschitz homeomorphism. For the subclass of PC^1 functions, see Theorem **4.6.5** for several such conditions; for the subclass of semismooth functions, see Exercise **7.6.18**.

7.2 Basic Newton-type Methods

Let a function $G : \Omega \subseteq \mathbb{R}^n \to \mathbb{R}^n$ be given. The development of Newton-type methods for solving the equation (7.0.1) when G is nonsmooth is motivated by the classical Newton algorithm for a continuously differentiable G. The latter algorithm is the prototype of many local, fast algorithms for solving smooth equations. Such algorithms have excellent convergence rates in a neighborhood of a zero of G, but may fail to converge if the starting point is far from the desired zero. The key idea in a general Newton-type method is to replace the function G by an approximation depending on the current iterate, resulting in an approximated problem that can be solved more easily. The solution of this approximation is then taken as a new iterate and the process is repeated.

If the function G is continuously differentiable there is a natural approximation at hand, namely, the linearization of the function at the current iterate. Specifically, given an iterate x^k, we can form the linear approximation

$$G(x^k) + JG(x^k)(x - x^k) \tag{7.2.1}$$

and calculate x^{k+1} as the zero of this linear approximation. We can give a rough argument to explain the fast convergence of the classical Newton method. As the first step in the analysis, suppose that the method is initiated at a vector x^0 that is sufficiently close to a zero x^* of G with $JG(x^*)$ being nonsingular. In general, given the iterate x^k sufficiently close to x^*, we can write by Taylor's expansion,

$$0 \;=\; G(x^*) \;=\; G(x^k) + JG(x^k)(x^* - x^k) + o_{x^k}(\| x^k - x^* \|),$$

where we write o_{x^k} to underline that the "small o" in Taylor's expansion depends, in principle, on the point x^k. If we premultiply the above expression by $JG(x^k)^{-1}$ (by continuity the Jacobian of G is invertible in a neighborhood of x^*) we get

$$x^k - x^* - JG(x^k)^{-1}G(x^k) \;=\; o_{x^k}(\| x^k - x^* \|).$$

If we can ensure that the "small o" function is actually independent of x^k the last equation immediately gives

$$(x^k - JG(x^k)^{-1}G(x^k)) - x^* \;=\; x^{k+1} - x^* \;=\; o(\| x^k - x^* \|)$$

This expression is the cornerstone to establish the well-definedness of the Newton sequence $\{x^k\}$ and its convergence to x^* as well as its Q-superlinear convergence rate (see Definition **7.2.1**). It turns out that the "small o" in Taylor's expansion is indeed independent of x^k (see Proposition **7.2.9**). Figure 7.1 illustrates the above iteration.

In the convergence analysis of iterative algorithms, several concepts of convergence rates play an important role. We formally define them as follows.

7.2.1 Definition. Let $\{x^k\} \subset \mathbb{R}^n$ be a sequence of vectors tending to the limit $x^\infty \neq x^k$ for all k. The convergence rate is said to be (at least)

(a) *Q-linear* if

$$\limsup_{k\to\infty} \frac{\| x^{k+1} - x^\infty \|}{\| x^k - x^\infty \|} \;<\; \infty;$$

(b) *Q-superlinear* if

$$\lim_{k\to\infty} \frac{\| x^{k+1} - x^\infty \|}{\| x^k - x^\infty \|} \;=\; 0;$$

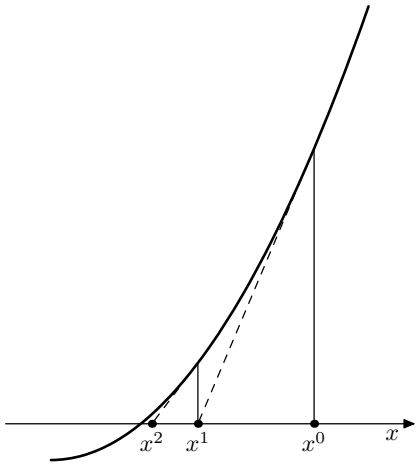


Figure 7.1: Smooth Newton's method illustrated.

(c) *Q-quadratic* if

$$\limsup_{k\to\infty} \frac{\| x^{k+1} - x^\infty \|}{\| x^k - x^\infty \|^2} < \infty;$$

(d) *R-linear* if

$$0 < \limsup_{k\to\infty} (\| x^k - x^\infty \|)^{1/k} < 1.$$

In each case, we say that $\{x^k\}$ converges to x^∞ (at least) Q-linearly, Q-superlinearly, Q-quadratically, and R-linearly, respectively. □

Returning to the above Newton process, we point out that there are two properties that make the process work: the linear model (7.2.1) provides a "good" approximation of G near x^k and the resulting linear equation is solvable because $JG(x^k)$ is invertible. These two properties hinge on the continuous differentiability of G and on the nonsingularity of $JG(x^*)$. When G is nondifferentiable, however, things become cumbersome and it is not immediately clear what properties are reasonable and useful to be imposed on a local model in order to develop an nonsmooth analog of the smooth Newton method.

In this section we give some general conditions that capture the essence of a Newton method and that are applicable for nondifferentiable functions. To this end we first rewrite (7.2.1) by setting $d = x - x^k$, so that (7.2.1) becomes

$$G(x^k) + JG(x^k)d \tag{7.2.2}$$

and x^{k+1} is obtained as $x^k + d^k$, where d^k is a zero of (7.2.2). Therefore d^k represents the shift we make from x^k to x^{k+1}; this is only a formal

change that we make for ease of notation. Note also that $JG(x^k)d$ is a "good" approximation of $G(x^k + d) - G(x^k)$, at least for small d. When G is nondifferentiable, the Jacobian of G is not even guaranteed to exist at x^k, and even if it exists, there is no reason to expect that we can use it to obtain a good approximation of the value of G at nearby points. Generalizing (7.2.2), we then assume that the Newton model is given by

$$G(x^k) + A(x^k, d), \tag{7.2.3}$$

where $A(x^k, d)$ is taken as a model of $G(x^k + d) - G(x^k)$ around $d = 0$, so that (7.2.3) should be regarded as an approximation to $G(x^k+d)$. Actually, to be more general and faithful to real situations, we may assume that there is a family of approximations $\mathcal{A}(x)$, where each element $A(x, \cdot) \in \mathcal{A}(x)$ is a function from $\mathbb{R}^n$ to $\mathbb{R}^n$ that can be used in (7.2.3). This may be surprising at first sight, but this is actually the more common situation in the nonsmooth case. For example we may wish to consider (and we will actually often consider) the linear model given by $G(x^k) + Hd$, where H is any element in the generalized Jacobian $\partial G(x)$. In this case, $\mathcal{A} = \partial G$. Later in this chapter we also consider other cases in which it is convenient to assume that at each step we have more than one model. If we want to use these models to develop local Newton methods for nonsmooth equations we obviously have to impose some adequate conditions. The next definition contains the essential features that we need.

7.2.2 Definition. Let G be a locally Lipschitz function from an open subset Ω of $\mathbb{R}^n$ to $\mathbb{R}^m$. We say that G has a *Newton approximation* at a point $\bar{x} \in \Omega$ if there exist a neighborhood $\Omega' \subseteq \Omega$ and a function $\Delta : (0, \infty) \to [0, \infty)$ with

$$\lim_{t \downarrow 0} \Delta(t) = 0, \tag{7.2.4}$$

such that for every point x in Ω' there is a family $\mathcal{A}(x)$ of functions each mapping $\mathbb{R}^n$ into $\mathbb{R}^m$ and satisfying the following two properties:

(a) $A(x, 0) = 0$ for every $A(x, \cdot) \in \mathcal{A}(x)$;

(b) for any $x \in \Omega'$ different from $\bar{x}$ and for any $A(x, \cdot) \in \mathcal{A}(x)$

$$\frac{\|G(x) + A(x, \bar{x} - x) - G(\bar{x})\|}{\| x - \bar{x} \|} \leq \Delta(\| x - \bar{x} \|). \tag{7.2.5}$$

We call $\mathcal{A}$ a *(Newton) approximation scheme* for G at $\bar{x}$. If the requirement (b) is strengthened to

(b') there exists a positive constant L' such that, for each $x \in \Omega'$ different from $\bar{x}$ and for every $A(x,\cdot) \in \mathcal{A}(x)$,

$$\frac{\| G(x) + A(x, \bar{x} - x) - G(\bar{x}) \|}{\| x - \bar{x} \|^2} \leq L', \tag{7.2.6}$$

then we say that F has a *strong Newton approximation* at $\bar{x}$ and that $\mathcal{A}$ is a *strong (Newton) approximation scheme*. Furthermore, if the following additional condition is met:

(c) ($m = n$ and) $\mathcal{A}$ is a family of *uniformly Lipschitz homeomorphisms* on Ω', by which we mean that there exist positive constants $L_{\mathcal{A}}$ and $\varepsilon_{\mathcal{A}}$ such that for each x in Ω' and for each $A(x,\cdot) \in \mathcal{A}(x)$, there are two open sets U_x and V_x, both containing $\mathbb{B}(0, \varepsilon_{\mathcal{A}})$, such that $A(x,\cdot)$ is a Lipschitz homeomorphism mapping U_x onto V_x with $L_{\mathcal{A}}$ being the Lipschitz modulus of the inverse of the restricted map $A(x,\cdot)|_{U_x}$,

we say that the (strong) Newton approximation is *nonsingular* and that the (strong) approximation scheme $\mathcal{A}$ is nonsingular. If $\mathcal{A}$ contains only one element for every x in Ω', we say that G admits a single-valued (strong) Newton approximation at $\bar{x}$ and that the approximate scheme $\mathcal{A}$ is single valued. □

Note that the requirement (c) can be met only if $m = n$, but the definition of Newton approximation is given also for the case $m \neq n$. In this book, we are mainly interested in the case $m = n$; but in establishing some results for composite functions, it is useful to have the concept of Newton approximation defined also for $m \neq n$.

The condition (a) tell us that the model (7.2.2) agrees with $G(x)$ for $d = 0$ and seems very natural. Condition (b) (and (b')) stipulate that the model (7.2.2) approximates well the value of $G(\bar{x})$. Note that this is a very weak requirement and in particular it does not require that the model (7.2.2) be a good approximation of $G(x)$ on a whole neighborhood of the current point. This condition will play the same role as Taylor's theorem does for smooth functions. Finally, condition (c) is a kind of nonsingularity condition and is basically necessary to guarantee that the Newton equation is solvable in a neighborhood of $\bar{x}$ and to ensure the Q-superlinear convergence rate. This condition is akin to the nonsingularity of the Jacobian of G, when G is continuously differentiable. In fact, it is easy to see that in the latter standard case the Newton models (7.2.1) are uniform *global* linear homeomorphisms.

7.2.3 Remarks. Conditions (b) and (b') can equivalently be stated as

$$\lim_{\substack{x\to\bar{x}\\ A(x,\cdot)\in\mathcal{A}(x)}} \frac{\|G(x)+A(x,\bar{x}-x)-G(\bar{x})\|}{\|\,x-\bar{x}\,\|} \;=\; 0$$

and

$$\limsup_{\substack{x\to\bar{x}\\ A(x,\cdot)\in\mathcal{A}(x)}} \frac{\|\,G(x)+A(x,\bar{x}-x)-G(\bar{x})\,\|}{\|\,x-\bar{x}\,\|^2} \;<\; \infty$$

respectively. This way of phrasing conditions (b) and (b') is probably more expressive; however the statement adopted in Definition **7.2.2** is more amenable to some of the subsequent developments. In particular, for all practical purposes, the only relevant property of the function Δ is the limit condition (7.2.4).

For notational simplicity, we denote the inverse of the function $A(x,\cdot)|_{\mathcal{U}_x}$ by $A^{-1}(x,\cdot)$ and omit the domain ($\mathcal{V}_x$) and range ($\mathcal{U}_x$) of this inverse. Thus for every $y\in\mathcal{V}_x$,

$$A^{-1}(x,y) \;=\; (\,A(x,\cdot)|_{\mathcal{U}_x}\,)^{-1}\,(y).$$

This notation is natural and should not give rise to any confusion. □

With the above definition we present the following natural extension of the smooth Newton method for solving nonsmooth equations and show subsequently that the extended method retains the main feature that characterizes the original method, namely its fast local convergence.

Nonsmooth Newton Method (NNM)

7.2.4 Algorithm.

Data: $x^0\in\mathbb{R}^n$ and $\varepsilon>0$.

Step 1: Set $k=0$.

Step 2: If $G(x^k)=0$, stop.

Step 3: Select an element $A(x^k,\cdot)$ in $\mathcal{A}(x)$ and find a vector d^k in $\mathbb{B}(0,\varepsilon)$ such that

$$G(x^k)+A(x^k,d^k) \;=\; 0. \tag{7.2.7}$$

Step 4: Set $x^{k+1}\equiv x^k+d^k$ and $k\leftarrow k+1$; go to Step 2.

An important word should be said about the termination check in Step 2. As stated above, Algorithm **7.2.4** stops in a finite number of iterations only if it arrives at an exact zero of G. In practice, this kind

of finite termination never happens. A practical stopping criterion is as follows:

$$\| G(x^k) \| \leq \text{ a prescribed tolerance.}$$

When this holds, the iterate x^k at termination is accepted as a satisfactory approximate zero of G. This stopping rule raises some important theoretical and practical issues that need to be addressed. These issues fall into the general domain of error bounds, which is the main subject of discussion in Chapter **6**.

The above remark about termination rule applies to all algorithms presented in the book; namely, all iterative algorithms are terminated in practical implementation according to a prescribed stopping criterion. In the statements of these algorithms, we continue to use the exact solution criterion as the check for termination. The main objective of a convergence analysis is then to demonstrate that such a criterion is satisfied asymptotically and at a reasonably fast rate by an infinite sequence of iterates generated by an algorithm. In such an analysis, the finite termination is of no interest.

The only formal difference between the nonsmooth and the classical Newton method is that the Newton equation (7.2.7) in Step 3 is not required to have a unique solution. As such, we have to specify some additional requirement on the solution of (7.2.7) we pick, namely that it belongs to a suitable small neighborhood of the origin. Even in this restricted neighborhood, the equation (7.2.7) may have more than one solution; see e.g. Theorem **7.3.5**. Nevertheless in many important situations, the local models $A(x^k, \cdot)$ are actually global homeomorphisms and therefore each Newton equation has one and only one solution. In situations where $A(x^k, \cdot)$ fails to be a global homeomorphism and the unique solvability of (7.2.7) is at risk, the restriction to a small neighborhood $\mathbb{B}(0, \varepsilon)$ is essential for the convergence of the overall algorithm.

Until now we have not explicitly addressed the issue of the difficulty of solving the Newton equation (7.2.7), but it is understood that this (sub)equation should be easier to solve than the original equation (7.0.2); otherwise the whole Newton process is of no interest.

The theorem below establishes the key theoretical properties of Algorithm **7.2.4**; the proof is a natural extension of the argument outlined at the beginning of this section for the classical Newton method. It is important to stress that the theorem is a local convergence result, meaning that there is the postulate that the initial iterate x^0 is chosen from a suitable neighborhood of a desired but unknown solution. This postulate is also needed in the extended Theorem **7.2.8**, which pertains to an "inex-

act" version of the method. We recall that $\varepsilon_{\mathcal{A}}$ is the positive constant introduced in Definition **7.2.2**.

7.2.5 Theorem. Let $G : \Omega \subseteq \mathbb{R}^n \to \mathbb{R}^n$, with Ω open, be a locally Lipschitz function in a neighborhood of $x^* \in \Omega$ satisfying $G(x^*) = 0$. Assume that G admits a nonsingular Newton approximation $\mathcal{A}$ at x^*. For every $\varepsilon \in (0, \varepsilon_{\mathcal{A}}]$, there exists a neighborhood $\mathbb{B}(x^*, \delta)$ of x^* such that if x^0 belongs to $\mathbb{B}(x^*, \delta)$ then the Newton Algorithm **7.2.4** generates a unique sequence $\{x^k\}$ that converges Q-superlinearly to x^*. If the Newton approximation $\mathcal{A}$ is strong, the convergence rate is Q-quadratic.

Proof. Let ε be any fixed constant in the interval $(0, \varepsilon_{\mathcal{A}}]$. We first note that by (7.2.5) we can find a positive δ such that for every $x \in \mathbb{B}(x^*, \delta)$ and for every $A(x, \cdot) \in \mathcal{A}(x)$ we can write

$$\begin{aligned} \| G(x) + A(x, x^* - x) \| &= \| G(x) + A(x, x^* - x) - G(x^*) \| \\ &\leq (2L_{\mathcal{A}})^{-1} \| x - x^* \|, \end{aligned} \tag{7.2.8}$$

where $L_{\mathcal{A}}$ is the Lipschitz constant in condition (c) of Definition **7.2.2**. Thanks to the Lipschitz continuity of G we can choose $\delta > 0$ to be less than $\min(2\varepsilon/3, \varepsilon/L)$ and also such that for all $x \in \mathbb{B}(x^*, \delta)$,

$$\| G(x) \| = \| G(x) - G(x^*) \| \leq L \| x - x^* \| \leq L\,\delta < \varepsilon, \tag{7.2.9}$$

where L is a Lipschitz constant of G around x^*. By property (c) in Definition **7.2.2**, (7.2.9) guarantees that the Newton equation (7.2.7) has a unique solution in $\mathcal{U}_{x^k}$, provided that x^k is in $\mathbb{B}(x^*, \delta)$. This solution is given by $d^k = A^{-1}(x^k, -G(x^k))$. Subsequently, we show that d^k belongs to the smaller ball $\mathbb{B}(0, \varepsilon)$.

Taking into account the definition of Algorithm **7.2.4**, property (c) in Definition **7.2.2**, (7.2.8), (7.2.9) and the definition of δ, we have, with $x^k \in \mathbb{B}(x^*, \delta)$,

$$\begin{aligned} \| x^{k+1} - x^* \| &= \| x^k - x^* + A^{-1}(x^k, -G(x^k)) \| \\ &= \| A^{-1}(x^k, -G(x^k)) - A^{-1}(x^k, A(x^k, x^* - x^k)) \| \\ &\leq L_{\mathcal{A}} \| G(x^k) + A(x^k, x^* - x^k) \| \leq \tfrac{1}{2} \| x^k - x^* \|. \end{aligned}$$

This implies

$$\| x^{k+1} - x^k \| \leq 1.5 \| x^k - x^* \| < \varepsilon,$$

where the last inequality follows from the choice of δ. Thus the vector d^k belongs to $\mathbb{B}(0, \varepsilon)$. Hence there exists a unique d^k satisfying Step 3 of Algorithm **7.2.4**.

In summary, the derivation so far shows that if x^0 belongs to $\mathbb{B}(x^*, \delta)$ then every vector x^k produced by Algorithm **7.2.4** also belongs to $\mathbb{B}(x^*, \delta)$ and the sequence $\{x^k\}$ is unique and converges to x^*; convergent because the sequence is contracting to x^*. Finally, the above chain of inequalities together with (7.2.5) gives

$$\begin{aligned} \| x^{k+1} - x^* \| &\leq L_{\mathcal{A}} \| G(x^k) + A(x^k, x^* - x^k) \| \\ &\leq L_{\mathcal{A}} \| x^k - x^* \| \, \Delta(\| x^k - x^* \|), \end{aligned}$$

thus establishing the Q-superlinear convergence. If (7.2.6) holds in place of (7.2.5) we can similarly prove that the convergence rate is Q-quadratic. □

In practical implementations it may be computationally very expensive to solve the Newton equation (7.2.7) exactly. This is true also if the model $A(x, \cdot)$ is linear (as in the smooth case) but very large. It is then useful to consider solving the equation (7.2.7) inexactly. It turns out that all the properties we have established for the (exact) nonsmooth Newton Algorithm **7.2.4** remain valid if we solve the Newton equation (7.2.7) in a suitably accurate way as stipulated by (7.2.10) and (7.2.11). These two conditions provide a unified framework for all inexact nonsmooth Newton schemes. (Traditionally, the latter methods are also known as *truncated* Newton methods, a terminology which is not used herein.)

Inexact Nonsmooth Newton Method (INNM)

7.2.6 Algorithm.

Data: $x^0 \in \mathbb{R}^n$, $\varepsilon > 0$ and a sequence of nonnegative scalars $\{\eta_k\}$.

Step 1: Set $k = 0$.

Step 2: If $G(x^k) = 0$, stop.

Step 3: Select an element $A(x^k, \cdot)$ in $\mathcal{A}(x)$ and find a direction d^k in $\mathbb{B}(0, \varepsilon)$ such that

$$G(x^k) + A(x^k, d^k) = r^k \tag{7.2.10}$$

where r^k is a vector satisfying

$$\| r^k \| \leq \eta_k \| G(x^k) \|. \tag{7.2.11}$$

Step 4: Set $x^{k+1} \equiv x^k + d^k$ and $k \leftarrow k + 1$; go to Step 2.

The inexact rule (7.2.11) is worth some further discussion. Roughly speaking, this rule stipulates that the inexactness of the vector d^k as an approximate solution of the Newton equation is proportional to the residual $\|G(x^k)\|$ of the current iterate x^k. This rule is essential for the fast convergence of the sequence $\{x^k\}$ produced by the above inexact Newton method; see Theorem **7.2.8** below. Before stating this theorem, we give a lemma that guarantees the existence of a direction d^k in Step 3 of Algorithm **7.2.6** for every given vector r^k satisfying (7.2.11).

7.2.7 Lemma. Let $G : \Omega \subseteq \mathbb{R}^n \to \mathbb{R}^n$, with Ω open, be a locally Lipschitz function in a neighborhood of $x^* \in \Omega$ satisfying $G(x^*) = 0$. Assume that G admits a nonsingular Newton approximation $\mathcal{A}$ at x^*. For every $\varepsilon \in (0, \varepsilon_{\mathcal{A}}]$ and every $\bar{\eta} > 0$, a neighborhood $\mathbb{B}(x^*, \delta)$ of x^* exists such that for every vector $x^k \in \mathbb{B}(x^*, \delta)$, every scalar $\eta_k \in (0, \bar{\eta}]$, and every vector r^k satisfying (7.2.11), the equation (7.2.10) has a unique solution d^k in $\mathbb{B}(0, \varepsilon)$.

Proof. By property (c) in Definition **7.2.2**, we can choose $\delta > 0$ such that for every $x^k \in \mathbb{B}(x^*, \delta)$,

$$\max(1, L_{\mathcal{A}})\,(1 + \bar{\eta})\,\|\,G(x^k)\,\| \;<\; \varepsilon,$$

and $A(x^k, \cdot)$ is a Lipschitz homeomorphism on $\mathbb{B}(0, \varepsilon)$ with $L_{\mathcal{A}}$ being the Lipschitz modulus of the inverse of $A(x^k, \cdot)|_{\mathbb{B}(0,\varepsilon)}$. The equation (7.2.10) is equivalent to

$$A(x^k, d^k) \;=\; r^k - G(x^k);$$

since the right-hand vector has norm not exceeding ε, this equation has a unique solution d^k satisfying

$$\|\,d^k\,\| \;\leq\; L_{\mathcal{A}}\,(1 + \bar{\eta})\,\|\,G(x^k)\,\|.$$

Thus d^k belongs to $\mathbb{B}(0, \varepsilon)$. □

The above lemma is conceptually useful. But its assumption is contrary to the intention of Algorithm **7.2.6**. Indeed, in the implementation of this inexact algorithm, the residual vector r^k is not given *a priori*; instead, Step 3 of the algorithm computes a suitable direction d^k so that the associated residual vector $r^k \equiv G(x^k) + A(x^k, d^k)$ satisfies (7.2.11). For instance, one could employ an iterative method to the equation $0 = G(x^k) + A(x^k, d)$ and terminate the method after finitely many (inner) iterations with (7.2.11) as the termination criterion. In general, there are many vectors d^k that can be picked by Step 3 of Algorithm **7.2.6**. This is in contrast to Algorithm **7.2.4** where the vector d^k is unique, under the

setting of Theorem **7.2.5**. The next result deals with an arbitrary sequence $\{x^k\}$ generated by Algorithm **7.2.6**. The focus of this result is neither the existence nor uniqueness of such a sequence, but rather its convergence.

7.2.8 Theorem. Let $G : \Omega \subseteq \mathbb{R}^n \to \mathbb{R}^n$, with Ω open, be a locally Lipschitz function in a neighborhood of $x^* \in \Omega$ satisfying $G(x^*) = 0$. Assume that G admits a nonsingular Newton approximation $\mathcal{A}$ at x^*. There exists a positive number $\bar{\eta}$ such that if $\eta_k \leq \bar{\eta}$ for every k then for every $\varepsilon \in (0, \varepsilon_{\mathcal{A}}]$ a neighborhood $\mathbb{B}(x^*, \delta)$ of x^* exists such that if x^0 belongs to $\mathbb{B}(x^*, \delta)$, the inexact Newton method **7.2.6** is well defined and every sequence $\{x^k\}$ generated by the method converges Q-linearly to x^*. Furthermore, if the sequence $\{\eta_k\} \to 0$, the convergence rate of $\{x^k\}$ to x^* is Q-superlinear. Finally, if the Newton approximation $\mathcal{A}$ is strong and for some $\tilde{\eta} > 0$, $\eta_k \leq \tilde{\eta}\|G(x^k)\|$ for all k, then the convergence rate is Q-quadratic.

Proof. Unlike the proof of Theorem **7.2.5**, we are somewhat loose in the specification of δ (and also $\bar{\eta}$) in the following proof. By (7.2.5) there exists a function Δ with $\lim_{t \downarrow 0} \Delta(t) = 0$ such that, for every x sufficiently near x^* and $A(x, \cdot) \in \mathcal{A}(x)$, we have

$$\begin{aligned} \| G(x) + A(x, x^* - x) - G(x^*) \| &= \| G(x) + A(x, x^* - x) \| \\ &\leq \| x - x^* \| \, \Delta(\|x - x^*\|). \end{aligned}$$

Let $\varepsilon \in (0, \varepsilon_{\mathcal{A}}]$ be given and suppose that $\eta_k \leq \bar{\eta}$ for every k. We can pick a $\delta > 0$ such that the following holds for every $x^k \in \mathbb{B}(x^*, \delta)$,

$$\begin{aligned} \| -G(x^k) + r^k \| &\leq \| G(x^k) \| + \eta_k \| G(x^k) \| \\ &\leq (1 + \bar{\eta}) \| G(x^k) \| \leq (1 + \bar{\eta}) L \| x^k - x^* \| \\ &\leq (1 + \bar{\eta}) L \delta < \varepsilon, \end{aligned}$$

where L is a Lipschitz constant of G around x^*. By the uniform Lipschitz homeomorphism property of $A^{-1}(x^k, \cdot)$ with constant $L_{\mathcal{A}}$, we have

$$\begin{aligned} \| x^{k+1} - x^* \| &= \| x^k - x^* + A^{-1}(x^k, -G(x^k) + r^k) \| \\ &= \| A^{-1}(x^k, -G(x^k) + r^k) - A^{-1}(x^k, A(x^k, x^* - x^k)) \| \\ &\leq L_{\mathcal{A}} \| G(x^k) + A(x^k, x^* - x^k) - r^k \| \\ &\leq L_{\mathcal{A}} [\| x^k - x^* \| \Delta(\| x^k - x^* \|) + \| r^k \|] \\ &\leq L_{\mathcal{A}} [\| x^k - x^* \| \Delta(\| x^k - x^* \|) + \eta_k \| G(x^k) \|] \\ &\leq L_{\mathcal{A}} [\| x^k - x^* \| \Delta(\| x^k - x^* \|) + \bar{\eta} L \| x^k - x^* \|]. \end{aligned}$$

It is now obvious that if $\bar{\eta}$ and δ are both chosen to be sufficiently small, then

$$\| x^{k+1} - x^* \| \le \tfrac{1}{2} \| x^k - x^* \|,$$

so that by induction we easily see that $\{x^k\}$ converges at least Q-linearly to x^*. If $\{\eta_k\}$ converges to zero, then the inequality established above:

$$\| x^{k+1} - x^* \| \le L_{\mathcal{A}} [\| x^k - x^* \| \Delta(\| x^k - x^* \|) + \eta_k L \| x^k - x^* \|]$$

clearly shows that the convergence rate is Q-superlinear.

Finally suppose that the approximation $\mathcal{A}$ is strong and η_k also satisfies the condition $\eta_k \le \tilde{\eta} \| G(x^k) \|$ for some positive $\tilde{\eta}$. In this case, reasoning in the same way as above, we obtain the following estimate

$$\| x^{k+1} - x^* \| \le L_{\mathcal{A}} (L' \| x - x^* \|^2 + \tilde{\eta} L^2 \| x - x^* \|^2),$$

from which the Q-quadratic convergence rate easily follows. □

Every function G admits an obvious (single-valued) strong Newton approximation scheme, namely the one given by $A(x, d) \equiv G(x + d) - G(x)$. However, finding a zero of (7.2.3) is in this case equivalent to solving the original problem. If G is B-differentiable at x^*, then $A(x, d) \equiv -G'(x^*, -d)$ is a Newton approximation scheme of G at x^*. For these two choices of $A(x, d)$, the resulting nonsmooth Newton method and its inexact version are practically useless, the reason with the former choice is as mentioned and that with the latter choice is because x^* is presumably an unknown zero of G. Nevertheless, as a theoretical consideration, these two choices of $A(x, d)$ suggest that the Newton approximation has an important role in the local theory of various classes of nonsmooth functions. For instance, the nonsingularity of the scheme $A(x, d) \equiv -G'(x^*; -d)$ amounts to the Lipschitz homeomorphism property of $G'(x^*; \cdot)$ and implies the isolatedness of x^* as a zero of G; see Theorem **7.2.10**.

We next consider the classical case where G is continuously differentiable near a zero x^*; we let $A(x^k, d)$ be the model given by (7.2.2). Algorithm **7.2.4** then reduces to the smooth Newton method. Furthermore, Theorem **7.2.5** easily yields the well-known convergence results for such a Newton method. To see this, we verify the properties (a), (b), (b') and (c) under the continuous differentiability assumption of G. The first property (a) is trivially satisfied. Assume now that $JG(x^*)$ is nonsingular. Then there exists a neighborhood Ω of x^* and a positive constant L such that

$$\| JG(x) \| \le L \quad \text{and} \quad \| JG(x)^{-1} \| \le L,$$

for every x in Ω. It is then clear that for every x^k in Ω both $A(x^k, d)$ and its inverse are globally Lipschitz homeomorphisms with a common Lipschitz constant given by L, hence condition (c) of Definition **7.2.2** is clearly satisfied. So we are only left with the verification of (b) and (b') of Definition **7.2.2**. This reduces to an elementary application of the mean-value theorem **7.1.16** for vector-valued functions. We formally state this classical result in a slightly more general form in the following proposition, which will be invoked several times subsequently.

7.2.9 Proposition. Let $G : \mathbb{R}^n \to \mathbb{R}^m$ be continuously differentiable in a neighborhood Ω of $\bar{x}$. A nondecreasing function $\Delta : (0, \infty) \to [0, \infty)$ with

$$\lim_{t \downarrow 0} \Delta(t) \;=\; 0$$

exists such that

$$\| G(x) + JG(x)(z - x) - G(z) \| \;\leq\; \| x - z \| \, \Delta(\| x - z \|)$$

for all z and x in Ω. Furthermore, if JG is Lipschitz continuous in a neighborhood of $\bar{x}$, then a subneighborhood $\Omega' \subseteq \Omega$ of $\bar{x}$ and a positive constant L' exist so that

$$\| G(x) + JG(x)(z - x) - G(z) \| \;\leq\; L' \, \| x - z \|^2$$

for all z and x in Ω'.

Proof. By the mean-value theorem for vector-valued functions, we can write, for a given $z \in \Omega$ and any $x \in \Omega$:

$$G(x) \;=\; G(z) + \sum_{i=1}^{m} \alpha_i(x) \, JG(y^i(x))(x - z),$$

where $\alpha_i(x)$ and $y^i(x)$ are such that

$$\alpha_i(x) \;\geq\; 0 \;\; \forall\, i, \quad \sum_{i=1}^{m} \alpha_i(x) \;=\; 1, \qquad y^i(x) \;\in\; [x, z] \;\; \forall\, i. \tag{7.2.12}$$

But then we can write

$$\begin{aligned} &\| G(x) + JG(x)(z - x) - G(z) \| \\ &\quad = \left\| \left[\sum_{i=1}^{m} \alpha_i(x) \, G(y^i(x)) - JG(x) \right] (x - z) \right\| \\ &\quad \leq \sum_{i=1}^{m} \alpha_i(x) \, \| JG(y^i(x)) - JG(x) \| \, \| x - z \|, \end{aligned}$$

where in the last inequality we have used the first two relations in (7.2.12). Furthermore, by using the last relation in (7.2.12) and the continuity of JG we see that, for every i,

$$\lim_{x \to z} \| JG(y^i(x)) - JG(x) \| = 0.$$

It is then easy to see that for the first assertion of the proposition it suffices to take

$$\Delta(t) \equiv \sup_{\substack{x,y \in \Omega \\ \|x-y\| \le t}} \| JG(y) - JG(x) \|$$

By the continuity of JG and the boundedness of Ω, $\Delta(t)$ is clearly finite and goes to zero when t tends to zero. If JG is locally Lipschitz continuous at $\bar{x}$, with Lipschitz constant L, we can write, by possibly restricting z and x to a suitable subneighborhood $\Omega' \subseteq \Omega$,

$$\| JG(y^i(x)) - JG(x) \| \le L \| y^i(x) - x \| \le L \| x - z \|,$$

which yields

$$\| G(x) + JG(x)(z - x) - G(z) \| \le L \| x - z \|^2$$

as we have asserted. □

Before giving examples of nonsmooth equations for which fast locally convergent algorithms can be obtained by the Newton schemes described so far, we give a few additional results that help to understand the nature of the assumptions. The first result shows that if G admits a nonsingular Newton approximation at a solution x^*, then x^* is an isolated solution. This parallels the classical result that if at a solution of a differentiable system of equations the Jacobian is nonsingular then the solution is isolated. The following theorem extends Lemma **5.2.1** to a general nonsmooth function in view of the fact that $A(x, d) \equiv -G'(x; -d)$ is a single-valued Newton approximation for every B-differentiable function G.

7.2.10 Theorem. Let $G : \Omega \subseteq \mathbb{R}^n \to \mathbb{R}^n$, with Ω open, be locally Lipschitz continuous at a zero $x^* \in \Omega$ of G. If G has a Newton approximation $\mathcal{A}(x)$ at x^* for which there exist a constant $c > 0$ and a neighborhood $\mathcal{N}$ of x^* and a neighborhood $\mathcal{U}$ of the origin such that, for every $x \in \mathcal{N}$, an element $A(x) \in \mathcal{A}(x)$ exists satisfying

$$\| A(x, d) \| \ge c \| d \|, \quad \forall d \in \mathcal{U},$$

then a neighborhood $\mathcal{N}'$ of x^* and a positive constant c' exist such that

$$\| x - x^* \| \le c' \| G(x) \|, \quad \forall x \in \mathcal{N}';$$

in particular, x^* is a locally unique zero of G.

Proof. Assume for the sake of contradiction that no such neighborhood $\mathcal{N}'$ and constant c' exist. Then there exists a sequence of vectors $\{x^k\}$ converging to x^* such that $x^k \neq x^*$ for every k and

$$\lim_{k\to\infty} \frac{G(x^k)}{\| x^k - x^* \|} = 0.$$

By condition (b) in Definition **7.2.2**, we have

$$0 = \lim_{k\to\infty} \frac{G(x^k) + A(x^k, x^* - x^k) - G(x^*)}{\| x^k - x^* \|} = \lim_{k\to\infty} \frac{A(x^k, x^* - x^k)}{\| x^k - x^* \|}.$$

But this contradicts the assumption on the Newton approximation because x^k belongs to the given neighborhood $\mathcal{N}$ of x^* and $x^k - x^*$ belongs to $\mathcal{U}$ for all k sufficiently large so that the last limit in the above expression is positive. □

7.2.11 Remark. The assumption of Theorem **7.2.10** is obviously valid if the Newton approximation $\mathcal{A}$ is nonsingular. Thus the existence of a nonsingular Newton approximation implies a pointwise error bound. □

Usually the hardest point to verify when defining a Newton approximation scheme is its nonsingularity. Theorem **7.2.13** shows that under an appropriate assumption we can deduce the nonsingularity of a Newton approximation from a pointwise nonsingularity at the base vector. The lemma below, which is a generalization of the classical finite-dimensional Banach perturbation lemma, is the cornerstone for proving the theorem.

7.2.12 Lemma. Let A and A' be two functions from a subset U of $\mathbb{R}^n$ into $\mathbb{R}^n$ such that A is a Lipschitz homeomorphism and $A - A'$ is Lipschitz continuous on U. Let $L > 0$ and $L' > 0$ be, respectively, the Lipschitz moduli of A^{-1} and $A - A'$. If $d^* \in U$ and $\delta > 0$ are such that:

(a) $\operatorname{cl} \mathbb{B}(d^*, \delta) \subseteq U$,

(b) $\operatorname{cl} \mathbb{B}(A(d^*), \delta/L) \subseteq A(U)$

(c) $LL' < 1$,

then A' is a Lipschitz homeomorphism from $\mathcal{U}$ onto $A'(U)$ and its inverse $(A')^{-1}$ has Lipschitz modulus

$$\frac{L}{1 - LL'} > 0;$$

moreover

$$\mathbb{B}\left(A'(d^*), \frac{(1 - LL')\delta}{L}\right) \subseteq A'(U). \tag{7.2.13}$$

Proof. Since $A' = (A' - A) + A$ and both $A' - A$ and A are Lipschitz continuous on U, thus so is A'. Define the perturbation function

$$P(\cdot) \equiv A'(\cdot) - A(\cdot) - (A'(d^*) - A(d^*)).$$

By the Lipschitz property of A^{-1}, we get

$$\begin{aligned} L &\geq \sup\left\{\frac{\|A^{-1}(y) - A^{-1}(y')\|}{\|y - y'\|} : y, y' \in A(U),\ y \neq y'\right\} \\ &= \left(\inf\left\{\frac{\|A(d) - A(d')\|}{\|d - d'\|} : d, d' \in U,\ d \neq d'\right\}\right)^{-1} \\ &\equiv 1/q. \end{aligned}$$

We have

$$\begin{aligned} &\inf\left\{\frac{\|(A+P)(d) - (A+P)(d')\|}{\|d - d'\|} : d, d' \in U,\ d \neq d'\right\} \\ &\geq q - \sup\left\{\frac{\|P(d) - P(d')\|}{\|d - d'\|} : d, d' \in U,\ d \neq d'\right\} \\ &\geq \frac{1}{L} - L'. \end{aligned}$$

Therefore $A+P$ is invertible on U and its inverse is Lipschitz with modulus

$$\frac{1}{(1/L) - L'} = \frac{L}{1 - LL'}.$$

We next show that

$$\mathbb{B}\left(A(d^*), \frac{(1 - LL')\delta}{L}\right) \subseteq (A+P)(U). \tag{7.2.14}$$

For $y \in \mathbb{B}\left(A(d^*), \frac{(1-LL')\delta}{L}\right)$ and $d \in \mathrm{cl}\,\mathbb{B}(d^*, \delta)$, let

$$T_y(d) \equiv A^{-1}(y - P(d)) \cap U.$$

Note that, since $P(d^*) = 0$, we have

$$\begin{aligned} \|y - P(d) - A(d^*)\| &\leq \|y - A(d^*)\| + \|P(d) - P(d^*)\| \\ &\leq \frac{(1 - LL')\delta}{L} + L'\delta = \frac{\delta}{L}, \end{aligned}$$

so that the set $T_y(d)$ is nonempty by (b); in fact it consists of a single point because $q > 0$. Therefore we can regard T_y as a single-valued function defined on $\mathrm{cl}\,\mathbb{B}(d^*, \delta)$. We have

$$T_y(\mathrm{cl}\,\mathbb{B}(d^*, \delta)) \subseteq \mathrm{cl}\,\mathbb{B}(d^*, \delta)$$

because for each d in $\mathrm{cl}\,\mathbb{B}(d^*, \delta)$,

$$\| T_y(d) - d^* \| = \| A^{-1}(y - P(d)) - A^{-1}(A(d^*)) \| \leq \delta.$$

Therefore T_y is a self-mapping from $\mathrm{cl}\,\mathbb{B}(d^*, \delta)$ into itself. Furthermore, if d and d' are any two elements in $\mathrm{cl}\,\mathbb{B}(d^*, \delta)$, then

$$\begin{aligned} \| T_y(d) - T_y(d') \| &= \| A^{-1}(y - P(d)) - A^{-1}(y - P(d')) \| \\ &\leq LL' \| d - d' \|, \end{aligned}$$

and since $LL' < 1$ it follows that T_y is a contraction and therefore has a unique fixed point in $\mathrm{cl}\,\mathbb{B}(d^*, \delta)$ (by Theorem **2.1.21**), which we denote $d(y)$. Clearly $(A+P)(d(y)) = y$ so that (7.2.14) follows. From this relation, the claimed homeomorphism properties of A' and the inclusion (7.2.13) can be easily established because $A'(U) = (A + P)(U) + A'(d^*) - A(d^*)$. □

The following theorem gives a sufficient condition for a Newton approximation to be nonsingular. As mentioned above, the key step in the proof of the theorem is to ensure the applicability of Lemma **7.2.12**.

7.2.13 Theorem. Let $G : \Omega \subseteq \mathbb{R}^n \to \mathbb{R}^n$ be a locally Lipschitz continuous function on the open set Ω, which contains a vector $\bar{x}$. Assume that $\mathcal{A}$ is a Newton approximation scheme of G at $\bar{x}$ for which there exist three positive constants ε_1, ε_2, and L satisfying

(A) for each $A(\bar{x}, \cdot) \in \mathcal{A}(\bar{x})$ there are two sets U and V containing $\mathbb{B}(0, \varepsilon_1)$ and $\mathbb{B}(0, \varepsilon_2)$ respectively, such that $A(\bar{x}, \cdot)$ is a Lipschitz homeomorphism from U to V and $A^{-1}(\bar{x}, \cdot)$ has Lipschitz modulus L.

Assume further that a function $L' : (0, \infty) \to [0, \infty)$ with $\lim_{t \downarrow 0} L'(t) = 0$ and a neighborhood $\mathcal{N}$ of $\bar{x}$ exist such that either one of the following two conditions holds:

(a) for every $x \in \mathcal{N}$ and every $A(x, \cdot) \in \mathcal{A}(x)$, there exists a member $A(\bar{x}, \cdot)$ in $\mathcal{A}(\bar{x})$ such that $A(x, \cdot) - A(\bar{x}, \cdot)$ is Lipschitz continuous with modulus $L'(\| x - \bar{x} \|)$ on U; or

(b) for every $x \in \mathcal{N}$, $\mathcal{A}(x) = \{A(x, \cdot)\}$ is single valued and $\tilde{A}(x, \cdot) - A(\bar{x}, \cdot)$ is Lipschitz continuous with modulus $L'(\| x - \bar{x} \|)$ on U, where

$$\tilde{A}(x, d) \equiv A(x, \bar{x} - x + d).$$

Then the approximation scheme $\mathcal{A}$ is nonsingular.

Proof. We need to show that property (c) of Definition **7.2.2** holds. First assume condition (a). Pick any $A(x,\cdot)$ from $\mathcal{A}(x)$ and let $A(\bar{x},\cdot) \in \mathcal{A}(\bar{x})$ be the function described in condition (a). Then $A(x,\cdot) - A(\bar{x},\cdot)$ is a Lipschitz function with modulus $L'(\| x - \bar{x} \|)$. We are now in the position to apply Lemma **7.2.12**. Choose a neighborhood $\mathbb{B}(\bar{x},\varepsilon)$ small enough so that for every x in this neighborhood,

$$L\,L'(\| x - \bar{x} \|) \;<\; \tfrac{1}{2}.$$

Assume further, and without loss of generality, that L is greater than 1, and that $\varepsilon_1 < \varepsilon_2$. Then we can apply Lemma **7.2.12** with $d^* = 0$ and conclude, recalling $A(x,0) = 0 = A(\bar{x},0)$, that $A(x,\cdot)$ is a Lipschitz homeomorphism from U to $A(x,U)$ and $A^{-1}(x,\cdot)$ has Lipschitz modulus

$$\frac{L}{1 - L\,L'(\| x - \bar{x} \|)} \;<\; 2L$$

and that

$$\mathbb{B}\left(0, \frac{\varepsilon_1}{2L}\right) \;\subseteq\; A(x,U).$$

This establishes the desired property (c) of Definition **7.2.2**.

Assume condition (b). Reasoning as above, we can similarly show that for some positive ε and for each $x \in \mathbb{B}(\bar{x},\varepsilon)$, $\tilde{A}(x,\cdot)$ is a Lipschitz homeomorphism from U to $\tilde{A}(x,U)$ and $\tilde{A}^{-1}(x,\cdot)$ has Lipschitz modulus $2L$; moreover

$$\mathbb{B}\left(0, \frac{\varepsilon_1}{2L}\right) \;\subseteq\; \tilde{A}(x,U).$$

Since by (7.2.5) we have

$$\lim_{x \to \bar{x}} A(x, \bar{x} - x) \;=\; \lim_{x \to \bar{x}} \tilde{A}(x,0) \;=\; 0;$$

we can assume, without loss of generality, that ε is small enough to guarantee

$$\| A(x, \bar{x} - x) \| \;=\; \| \tilde{A}(x,0) \| \;\leq\; \frac{\varepsilon_1}{4L} \quad \text{and} \quad \| \bar{x} - x \| \;\leq\; \tfrac{1}{2}\,\varepsilon_1.$$

Let $U'(x) \equiv U + \bar{x} - x$. The desired property (c) of Definition **7.2.2** can be established by the following argument: since $A(x,d) = \tilde{A}(x, d - (\bar{x} - x))$, $A(x,\cdot)$ is a Lipschitz homeomorphism from $U'(x)$ to $A(x,U'(x))$ and its inverse has Lipschitz modulus $2L$, and

$$\mathbb{B}\left(0, \tfrac{1}{2}\varepsilon_1\right) \;\subseteq\; U'(x), \qquad \mathbb{B}\left(0, \frac{\varepsilon_1}{4L}\right) \;\subseteq\; A(x,U'(x)).$$

Consequently, $\mathcal{A}$ is a nonsingular Newton approximation scheme under either condition (a) or (b). □

In the next two subsections we illustrate the theory developed so far by considering two simple, but important applications. Further developments are given in Section **7.4** and in the next chapter.

7.2.1 Piecewise smooth functions

The first application of the theory of nonsmooth Newton methods is to the family of piecewise smooth functions. This application results in a family of locally convergent Newton methods for solving smooth complementarity problems formulated as systems of PC^1 equations using the min function.

Let $G : \mathbb{R}^n \to \mathbb{R}^n$ be a PC^1 function. Suppose that $x^* \in \mathbb{R}^n$ is a zero of G. Let $\{G^1(x), \ldots, G^k(x)\}$ be the C^1 pieces of G at x^*. We recall that the set of active constraints $\mathcal{P}(x)$ at a point x is given by

$$\mathcal{P}(x) \;=\; \{\, i \;:\; G(x) \;=\; G^i(x) \,\}.$$

A natural Newton approximation scheme of such a function G is given by:

$$\mathcal{A}(x) \;\equiv\; \{\, JG^i(x) \;:\; i \;\in\; \mathcal{P}(x) \,\}. \tag{7.2.15}$$

Note that all $A(x, \cdot)$ are in this case linear so that the Newton equations are easy to solve; furthermore we will make an assumption that implies the nonsingularity of the $JG^i(x)$ for all $i \in \mathcal{P}(x)$, so that the Newton equations have unique solutions. With this approximation scheme, the nonsmooth Newton algorithm **7.2.4** becomes the following.

Piecewise Smooth Newton Method (PCNM)

7.2.14 Algorithm.

Data: $x^0 \in \mathbb{R}^n$

Step 1: Set $k = 0$.

Step 2: If $G(x^k) = 0$, stop.

Step 3: Select an i_k in $\mathcal{P}(x^k)$ and find a direction d^k such that

$$G(x^k) + JG^{i_k}(x^k)d^k = 0 \tag{7.2.16}$$

Step 4: Set $x^{k+1} \equiv x^k + d^k$ and $k \leftarrow k + 1$; go to Step 2.

The convergence of this algorithm follows easily from Theorem **7.2.5**. See Figure 7.2 for an illustration of the algorithm.

7.2.15 Theorem. Let $G : \Omega \subseteq \mathbb{R}^n \to \mathbb{R}^n$, with Ω open, be a PC^1 function with C^1 pieces $\{G^i : i = 1, \ldots, k\}$. Let $x^* \in \Omega$ be a zero of G. Suppose that the matrices $JG^i(x^*)$ are nonsingular for all $i \in \mathcal{P}(x^*)$. There exists a neighborhood $\mathbb{B}(x^*, \delta)$ of x^* such that if x^0 belongs to this neighborhood, Algorithm **7.2.14** generates a unique sequence $\{x^k\}$ that converges

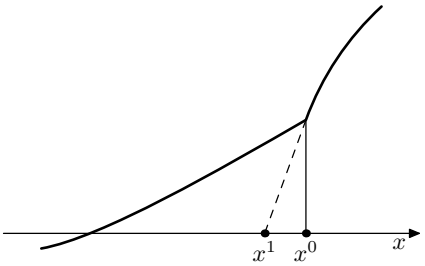


Figure 7.2: Illustration of Algorithm **7.2.14**.

Q-superlinearly to x^*. If the Jacobians of all the active pieces of G at points near x^* are locally Lipschitz at x^*, then the convergence rate is Q-quadratic.

Proof. We first observe that, by simple continuity arguments, if x is sufficiently close to x^* then

$$\mathcal{P}(x) \subseteq \mathcal{P}(x^*). \tag{7.2.17}$$

We now show that condition (a), (b) and (c) of Definition **7.2.2** are satisfied, thus verifying that the family $\mathcal{A}$ defined by (7.2.15) is a nonsingular Newton approximation scheme. Condition (a) is trivial, since all the $A(x,\cdot)$ are linear. To show (b) we have to prove that, for every sequence $\{y^k\}$ converging to x^* with $y^k \neq x^*$ for all k and for every $i_k \in \mathcal{P}(y^k)$,

$$\lim_{k\to\infty} \frac{\| G(y^k) + JG^{i_k}(y^k)(x^* - y^k) - G(x^*) \|}{\| y^k - x^* \|} = 0.$$

By (7.2.17), we have $G(x^*) = G^{i_k}(x^*)$ for all k sufficiently large. Since we obviously have $G(y^k) = G^{i_k}(y^k)$ by the choice of i_k, the left-hand limit in the above expression is equal to

$$\lim_{k\to\infty} \frac{\| G^{i_k}(y^k) + JG^{i_k}(y^k)(x^* - y^k) - G^{i_k}(x^*) \|}{\| y^k - x^* \|}.$$

Since there are finitely many indices i_k and each piece G^{i_k} is a C^1 function, the above limit is clearly equal to zero; cf. Proposition **7.2.9**. Consequently condition (b) of Definition **7.2.2** holds. Condition (c) holds by the following reasoning: each $JG^i(x^*)$ is nonsingular for all $i \in \mathcal{P}(x^*)$, the inclusion (7.2.17) is valid for all x sufficiently near x^*, for any such x and for every $A(x,\cdot) \in \mathcal{A}(x)$, there exists an index $i \in \mathcal{P}(x^*)$ such that $A(x,d) = JG^i(x)d$ for all $d \in \mathbb{R}^n$.

If the Jacobians of the active pieces of G at points near x^* are Lipschitz continuous around x^*, we can easily establish, by reasoning in the same way as we did to prove (b), that there exists a constant $L' > 0$ such that

$$\limsup_{k\to\infty} \frac{\| G(y^k) + JG^{i_k}(y^k)(x^* - y^k) - G(x^*) \|}{\| y^k - x^* \|^2} \leq L'$$

for every sequence $\{y^k\}$ converging to x^* with $y^k \neq x^*$ for every k and for every index $i_k \in \mathcal{P}(y^k)$. This shows that in this case, condition (b') in Definition **7.2.2** also holds, thus proving the desired Q-quadratic convergence rate. □

By specializing Algorithm **7.2.6**, we can easily obtain an inexact Newton method for PC^1 equations. Instead of repeating the details of such an inexact method, we present an application of an exact nonsmooth Newton method to the solution of a system of equations defined by the min function.

7.2.16 Example. Consider the nonsmooth equation $H(x) = 0$, where

$$H(x) = \min(F(x), G(x)), \quad \forall\, x \in \mathbb{R}^n,$$

with F and G being two continuously differentiable maps from $\mathbb{R}^n$ into itself. For an arbitrary vector x, let the $n \times n$ matrix $A(x)$ be defined row-wise as

$$A_{i\cdot}(x) \equiv \begin{cases} \nabla F_i(x)^T & \text{if } F_i(x) < G_i(x) \\ \text{either } \nabla F_i(x)^T \text{ or } \nabla G_i(x)^T & \text{if } F_i(x) = G_i(x) \\ \nabla G_i(x)^T & \text{if } G_i(x) < F_i(x) \end{cases}$$

There are $2^{|\beta(x)|}$ matrices of this form for every x, where

$$\beta(x) \equiv \{\, i : F_i(x) = G_i(x) \,\}.$$

These matrices are the Jacobian matrices of the C^1 pieces of the function H at x. When applied to the equation $H(x) = 0$, Algorithm **7.2.14** generates a sequence $\{x^k\}$ in the following way: for each k with x^k given, the next iterate x^{k+1} is defined as $x^k + d^k$, where d^k is a solution of the system of linear equations:

$$H(x^k) + \tilde{A}(x^k)d^k = 0,$$

where $\tilde{A}(x^k)$ is any one of the $2^{|\beta(x^k)|}$ matrices defined above. The local convergence of such a Newton sequence $\{x^k\}$ to a zero x^* of H is guaranteed by Theorem **7.2.15**. In order for this theorem to be applicable, we

postulate that every matrix $A(x^*)$ is nonsingular. Recalling the concept of a representative matrix, (cf. Definition **3.3.23**) we see that this postulate is equivalent to the condition that every row representative matrix M of the pair $(JF(x^*), JG(x^*))$ satisfying

$$M_{i\cdot} = \begin{cases} JF_i(x^*) & \forall\, i \text{ such that } F_i(x^*) = 0 < G_i(x^*) \\ JG_i(x^*) & \forall\, i \text{ such that } F_i(x^*) > 0 = G_i(x^*), \end{cases} \tag{7.2.18}$$

is nonsingular. This is the same nonsingularity assumption already used in Corollary **3.3.24**, which in turn reduces to the b-regularity of the solution x^* of the NCP (F) in the case where G is the identity function.

In summary, we see that for the CP (F, G), the same sufficient condition that guarantees the local uniqueness of a solution x^* is also sufficient for the locally fast convergence of the Newton scheme of computing this solution. The noteworthy point about such a scheme is that the major task at each iteration lies in solving a system of linear equations. □

The algorithm sketched in the above example provides a locally convergent method for solving the CP (F, G). Toward the end of Section **8.3**, we present a strategy to globalize the convergence of this method that allows for an arbitrary starting point; also in Section **9.2**, we present an alternate globalization strategy for solving the NCP (F) that is based on a combination of the min function and the FB function. For ease of later reference and also for the sake of clarity, we present a formal description of the local Newton method in Example **7.2.16** in terms of the set-valued map $\mathcal{A}_{\min} : \mathbb{R}^n \to \mathbb{R}^{n\times n}$, where for each $x \in \mathbb{R}^n$, $\mathcal{A}_{\min}(x)$ consists of all $n \times n$ matrices $A(x)$ defined in this example.

min **based Newton Method** (MBNM)

7.2.17 Algorithm.

Data: $x^0 \in \mathbb{R}^n$.

Step 1: Set $k = 0$.

Step 2: If $\min(F(x^k), G(x^k)) = 0$, stop.

Step 3: Select an element A^k in $\mathcal{A}_{\min}(x^k)$ and find a solution d^k of the system of linear equations:

$$\min(\, F(x^k), G(x^k)\,) + A^k d = 0. \tag{7.2.19}$$

Step 4: Set $x^{k+1} = x^k + d^k$ and $k \leftarrow k + 1$. Go to Step 2.

In addition to the convergence properties described in Example **7.2.16**, the above algorithm is actually finitely convergent when F and G are both affine functions. For completeness, we formally state the convergence properties of Algorithm **7.2.17** in the nonlinear case and also the finite-convergence result in the linear case.

7.2.18 Theorem. Let $F, G : \Omega \subseteq \mathbb{R}^n \to \mathbb{R}^n$ be continuously differentiable functions defined on the open set Ω. Let x^* be a solution of the CP (F, G) such that every row representative matrix M of the pair $(JF(x^*), JG(x^*))$ satisfying (7.2.18) is nonsingular. The following two statements hold.

(a) There exists a neighborhood $\mathbb{B}(x^0, \delta)$ of x^* such that if x^0 belongs to $\mathbb{B}(x^0, \delta)$, Algorithm **7.2.17** generates a unique sequence $\{x^k\}$ converging to x^* Q-superlinearly. If JF and JG are locally Lipschitz at x^*, then the convergence rate is Q-quadratic.

(b) If F and G are affine functions, then there exists a $\bar{k}$ such that $x^{\bar{k}} = x^*$.

Proof. It suffices to show statement (b). Let $\bar{k}$ be such that for all $k \geq \bar{k}$,

$$\begin{aligned} 0 = F_i(x^*) < G_i(x^*) &\Rightarrow F_i(x^k) < G_i(x^k) \\ 0 = G_i(x^*) < F_i(x^*) &\Rightarrow G_i(x^k) < F_i(x^k). \end{aligned}$$

For such an index k, we have

$$0 = F_i(x^k) + \nabla F_i(x^k)^T d^k = F_i(x^k + d^k) = F_i(x^{k+1})$$

for all i such that $0 = F_i(x^*) < G_i(x^*)$, and

$$0 = G_i(x^k) + \nabla G_i(x^k)^T d^k = G_i(x^k + d^k) = G_i(x^{k+1})$$

for all i such that $0 = G_i(x^*) < F_i(x^*)$. Moreover, for an index i such that $F_i(x^*) = G_i(x^*) = 0$, we have either $F_i(x^{k+1}) = 0$ or $G_i(x^{k+1}) = 0$. Since x^* also satisfies all these equations; by the uniqueness of x^{k+1}, we must have $x^* = x^{k+1}$. Thus in a finite number of iterations, Algorithm **7.2.17** terminates with an iterate that coincides with x^*. □

It is useful to consider the specialization of the equation (7.2.19) in the case where G is the identity map. Given x^k, we choose an index set $\beta' \subseteq \beta(x^k)$ and we solve the following system of linear equations for d^k:

$$\begin{aligned} F_i(x^k) + \nabla F_i(x^k)^T d^k &= 0 \quad \forall\, i \in \gamma(x^k) \cup \beta' \\ x_i^k + d_i^k &= 0 \quad \forall\, i \in \alpha(x^k) \cup (\, \beta(x^k) \setminus \beta' \,), \end{aligned} \tag{7.2.20}$$

where

$$\gamma(x^k) \equiv \{\, i : F_i(x^k) < x_i^k \,\} \quad \text{and} \quad \alpha(x^k) \equiv \{\, i : F_i(x^k) > x_i^k \,\}.$$

The noteworthy point about the system (7.2.20) is that the variables d_i^k for $i \in \alpha(x^k) \cup (\beta(x^k) \setminus \beta')$ are all trivially determined, which can then be substituted into the first equation in (7.2.20). The resulting system is of order $|\gamma(x^k) \cup \beta'|$, which could be significantly less than the original dimension n of the problem. This feature is a desirable advantage of Algorithm **7.2.17** applied to the NCP (F), and could become especially pronounced if the solution x^* being computed contains many zero components.

7.2.2 Composite maps

As another application of Algorithm **7.2.4**, we consider a function G that is the composition of a smooth map with a Lipschitz continuous map. Specifically, let G be of the form

$$G(x) \equiv S \circ N(x),$$

with

$$N : \mathbb{R}^n \to \Omega \subseteq \mathbb{R}^m \quad \text{and} \quad S : \Omega \to \mathbb{R}^n,$$

where Ω is an open set and where S is a C^1 function on Ω and N is locally Lipschitz continuous on $\mathbb{R}^n$. Composite functions of this kind are often encountered in applications and an important example is the normal map $\mathbf{F}_K^{\mathrm{nor}}$ of the VI (K, F). In the next subsection, we develop a Newton method for a variational inequality based on this model.

We consider a single-valued approximation scheme of G where for each point $x \in \mathbb{R}^n$ the function $A(x, \cdot)$ is given by

$$A(x, d) = JS(N(x))[N(x + d) - N(x)], \quad \forall\, d \in \mathbb{R}^n. \tag{7.2.21}$$

In this model, the smooth function S is replaced by its Jacobian JS (which is a first-order approximation) whereas the nonsmooth function N is essentially not changed. Thus G is "partially linearized".

At first sight, a model of this kind seems rather useless because the main difficulty associated with the nonsmooth function N has not been removed. However, we will see that in some practical cases the resulting Newton equation using $A(x, d)$ given by (7.2.21) is actually much simpler to solve and easier to analyze than the original problem, so that this approximation scheme may be desirable in these cases. The resulting Newton scheme is as follows.

Nonsmooth Newton Method for Composite Functions
(NNMCF)

7.2.19 Algorithm.

Data: $x^0 \in \mathbb{R}^n$ and $\varepsilon > 0$.

Step 1: Set $k = 0$.

Step 2: If $G(x^k) = 0$, stop.

Step 3: Find a direction $d^k \in \mathbb{B}(0, \varepsilon)$ such that

$$S(N(x^k)) + JS(N(x^k))[N(x^k + d^k) - N(x^k)] \;=\; 0. \qquad (7.2.22)$$

Step 4: Set $x^{k+1} \equiv x^k + d^k$ and $k \leftarrow k + 1$; go to Step 2.

It is easy to show that (7.2.21) defines an approximation scheme at any point $\bar{x}$ in Ω. In fact it is trivial to check that $A(x, 0) = 0$. Furthermore, taking into account the continuous differentiability of S around $N(\bar{x})$ and denoting by L' the Lipschitz constant of N around $\bar{x}$, we have

$$\begin{aligned}
&\limsup_{x \to \bar{x}} \frac{\| G(x) + A(x, \bar{x} - x) - G(\bar{x}) \|}{\| x - \bar{x} \|} \\
&\quad = \limsup_{x \to \bar{x}} \frac{\| S(N(x)) + JS(N(x))[N(x) - N(\bar{x})] - S(N(\bar{x})) \|}{\| x - \bar{x} \|} \\
&\quad \le \limsup_{x \to \bar{x}} \frac{\| S(N(\bar{x})) + o(\| N(x) - N(\bar{x}) \|) - S(N(\bar{x})) \|}{\| x - \bar{x} \|} \\
&\quad = \limsup_{x \to \bar{x}} L' \frac{o(\| x - \bar{x} \|)}{\| x - \bar{x} \|} \;=\; 0,
\end{aligned}$$

which establishes property (b) of Definition **7.2.2**. Property (b') can be proved in a similar way if the Jacobian of S is Lipschitz continuous around $N(\bar{x})$.

Our next task is to establish the nonsingularity of the approximation scheme. We employ Theorem **7.2.13** to show it is enough to postulate that $A(\bar{x}, \cdot)$ defined by (7.2.21) is a locally Lipschitz homeomorphism around the origin. Assuming this condition, we first show that there are a function $L' : (0, \infty) \to [0, \infty)$ satisfying

$$\lim_{t \downarrow 0} L'(t) \;=\; 0$$

and a neighborhood $\mathcal{N}$ of $\bar{x}$ such that $A(x, (\bar{x} - x) + \cdot) - A(\bar{x}, \cdot)$ is Lipschitz continuous with modulus $L'(\| x - \bar{x} \|)$ for every $x \in \mathcal{N}$. For any two vectors

d and d', a direct calculation shows that

$$\begin{aligned}
&\| \, (A(x, (\bar{x} - x) + d) - A(\bar{x}, d)) - (A(x, (\bar{x} - x) + d') - A(\bar{x}, d')) \, \| \\
&= \| \, JS(N(x))[\, N(\bar{x} + d) - N(x) \,] - JS(N(\bar{x}))[\, N(\bar{x} + d) - N(\bar{x}) \,] \\
&\quad - JS(N(x))[\, N(\bar{x} + d') - N(x) \,] + JS(N(\bar{x}))[\, N(\bar{x} + d') - N(\bar{x}) \,] \, \| \\
&= \| \, (JS(N(x)) - JS(N(\bar{x})))[\, N(\bar{x} + d) - N(\bar{x} + d') \,] \, \| \\
&\leq \| \, JS(N(x)) - JS(N(\bar{x})) \, \| \, \| \, N(\bar{x} + d) - N(\bar{x} + d') \, \| \\
&\leq L \, \| \, JS(N(x)) - JS(N(\bar{x})) \, \| \, \| \, d - d' \, \|.
\end{aligned}$$

So it is clear that we can take

$$L'(t) \equiv L \sup_{x \in \mathbb{B}(\bar{x}, t)} \| \, JS(N(x)) - JS(N(\bar{x})) \, \|.$$

By Theorem **7.2.13**, it follows that the approximation scheme $A(x, \cdot)$ is nonsingular. Summarizing the above discussion, we state the following result, which follows easily from Theorem **7.2.5**; no further proof is required.

7.2.20 Theorem. Let $G : \Omega \subseteq \mathbb{R}^n \to \mathbb{R}^n$, with Ω open, be a composite function as described above, and let $x^* \in \Omega$ be such that $G(x^*) = 0$. Suppose that $JS(N(x^*))[N(x^* + d) - N(x^*)]$ is a locally Lipschitz homeomorphism near $d = 0$. There exists a positive scalar $\bar{\varepsilon}$ such that for every $\varepsilon \in (0, \bar{\varepsilon}]$ a $\delta > 0$ exists such that if x^0 belongs to the neighborhood $\mathbb{B}(x^*, \delta)$ of x^*, Algorithm **7.2.19** generates a unique sequence $\{x^k\}$ that converges Q-superlinearly to x^*. If the Jacobian of S is Lipschitz continuous near $N(x^*)$, then the convergence rate is Q-quadratic.

If $N(x)$ is a PC^1 function, Theorem **4.6.5** provides various conditions for $JS(N(x^*))N(x^* + d)$, or equivalently $JS(N(x^*))[N(x^* + d) - N(x^*)]$, to be a locally Lipschitz homeomorphism near $d = 0$. By specializing Algorithm **7.2.6**, we could easily give an inexact Newton method for solving a composite nonsmooth equation; again we leave the details to the reader.

7.3 A Newton Method for VIs

Based on the Newton method for a composite equation, Algorithm **7.2.19**, we introduce a Newton method for solving the VI (K, F), where K is a closed convex set in $\mathbb{R}^n$ and F is a continuously differentiable function defined on an open set containing K. We accomplish this by using the normal map $\mathbf{F}_K^{\mathrm{nor}}$. Since this map is defined on a transformed z-space, it

is therefore also useful to rephrase the resulting algorithm in terms of the original variable x of the VI.

Throughout this discussion, we let x^* be a solution of the VI (K, F). We recall that the normal map $\mathbf{F}_K^{\rm nor}(z)$ associated with VI (K, F) is defined as

$$\mathbf{F}_K^{\rm nor}(z) \equiv F(\Pi_K(z)) + z - \Pi_K(z), \quad \forall\, z \in \mathbb{R}^n$$

and that $z^* \equiv x^* - F(x^*)$ is a zero of $\mathbf{F}_K^{\rm nor}$. In terms of z^*, we have $x^* \equiv \Pi_K(z^*)$. In order to apply Algorithm **7.2.19** to the normal equation $\mathbf{F}_K^{\rm nor}(z) = 0$, we need to exhibit a Newton approximation for the normal map and provide adequate conditions for this approximation to be nonsingular. As mentioned in Subsection **5.2.2**, the normal map $\mathbf{F}_K^{\rm nor}$ is a composite function $\mathbf{F}_K^{\rm nor} = S \circ N$, where the F-differentiable function $S : \mathbb{R}^{2n} \to \mathbb{R}^n$ is given by $S(a, b) \equiv F(a) + b$, while the Lipschitz continuous function $N : \mathbb{R}^n \to \mathbb{R}^{2n}$ is given by

$$N(z) \equiv \begin{pmatrix} \Pi_K(z) \\ \\ z - \Pi_K(z) \end{pmatrix},$$

which is not smooth due to the projector Π_K. Therefore we can use the approximation scheme (7.2.21), which results in the following single-valued approximation scheme of the normal map $\mathbf{F}_K^{\rm nor}$:

$$A(z, d) = (\, JF(\Pi_K(z)) - \mathbf{I}\,)\,(\,\Pi_K(z + d) - \Pi_K(z)\,) + d. \tag{7.3.1}$$

Let us examine the Newton equation (7.2.7) in this context. For a given vector z^k, the Newton equation is:

$$\mathbf{F}_K^{\rm nor}(z^k) + A(z^k, d^k) = 0,$$

By substitution and letting $z^{k+1} \equiv z^k + d^k$ and $x^k \equiv \Pi_K(z^k)$, the above equation reduces to

$$F(x^k) + JF(x^k)(\,\Pi_K(z^{k+1}) - x^k\,) + z^{k+1} - \Pi_K(z^{k+1}) = 0. \tag{7.3.2}$$

With x^k given, we easily recognize that the left-hand side of the above equation is the normal map of the VI (K, F^k) evaluated at the vector z^{k+1}, where

$$F^k(x) \equiv F(x^k) + JF(x^k)(\,x - x^k\,)$$

is the linearization of the function F at the vector x^k; the equation (7.3.2) then says that the vector $x^{k+1} \equiv \Pi_K(z^{k+1})$ is a solution of the VI (K, F^k). Note that if $\|z^{k+1} - z^k\| \le \varepsilon$, then by the nonexpansiveness of the Euclidean projector, we have

$$\|\, x^{k+1} - x^k \,\| \le \|\, z^{k+1} - z^k \,\| \le \varepsilon.$$

Consequently, we can describe the application of Algorithm **7.2.19** for solving the VI (K, F) in two equivalent ways. The first way is via the straightforward description in terms of the normal map $\mathbf{F}_K^{\rm nor}$, whereby a sequence of vectors $\{z^k\}$ is generated by iteratively solving the semi-linearized normal equation (7.3.2). (The term "semi-linearized" was previously used at the end of Section **5.2**.) The second way is via the conversion back to the original VI bypassing the transformation of variables, whereby a sequence of vectors $\{x^k\}$ is generated by iteratively solving the semi-linearized VI (K, F^k). In what follows, we present the Newton method for solving the VI (K, F) using the latter description, keeping in mind the former description in terms of the normal map $\mathbf{F}_K^{\rm nor}$ and the auxiliary sequence $\{z^k\}$.

Josephy-Newton Method for the VI (JNMVI)

7.3.1 Algorithm.

Data: $x^0 \in K$ and $\varepsilon > 0$.

Step 1: Set $k = 0$.

Step 2: If $x^k \in \mathrm{SOL}(K, F)$, stop.

Step 3: Let x^{k+1} be any solution of the VI (K, F^k) such that x^{k+1} belongs to $\mathbb{B}(x^k, \varepsilon)$.

Step 4: Set $k \leftarrow k + 1$ and go to Step 2.

In order to establish the local convergence of this algorithm, the first step is to ensure that the approximation scheme (7.3.1) of $\mathbf{F}_K^{\rm nor}$ is nonsingular at a zero. In turn, this step, and indeed the entire convergence analysis, very much hinges on some familiar limit properties of the error function:

$$\mathbf{e}_F(x) \equiv F(x) - F(x^*) - JF(x^*)(x - x^*), \tag{7.3.3}$$

which is the residual of the first-order Taylor expansion of F at x relative to x^*. If F is strongly F-differentiable at x^*, the error function satisfies:

$$\lim_{\substack{x^1 \neq x^2 \\ (x^1, x^2) \to (x^*, x^*)}} \frac{\mathbf{e}_F(x^1) - \mathbf{e}_F(x^2)}{\| x^1 - x^2 \|} = 0; \tag{7.3.4}$$

and if JF is Lipschitz continuous in a neighborhood of x^*, then

$$\limsup_{x^* \neq x \to x^*} \frac{\| \mathbf{e}_F(x) \|}{\| x - x^* \|^2} < \infty; \tag{7.3.5}$$

see Proposition **7.2.9**. Using the first limit (7.3.4) in its proof, the following

lemma provides a sufficient condition for the approximation scheme (7.3.1) to be nonsingular.

7.3.2 Lemma. Let K be a closed convex set and let F be continuously differentiable. Suppose that x^* is a strongly stable solution of VI (K, F). The approximation scheme (7.3.1) of $\mathbf{F}_K^{\mathrm{nor}}$ is nonsingular at z^*.

Proof. By Theorem **7.2.20** we only need to show that $A(z^*, \cdot)$ is a locally Lipschitz homeomorphism at the origin. Since x^* is a strongly stable solution of VI (K, F), we have, by Proposition **5.3.6**, that z^* is a strongly stable zero of $\mathbf{F}_K^{\mathrm{nor}}$. Since $A(z^*, \cdot)$ is clearly globally Lipschitz continuous, by Proposition **5.2.15** and Theorem **5.2.8**, it suffices to show that $A(z^*, z - z^*)$ is a strong FOA of $\mathbf{F}_K^{\mathrm{nor}}(z)$ at z^* (see Remark **5.2.16**); that is we need to show that with

$$\mathbf{e}(z) \equiv A(z^*, z - z^*) - \mathbf{F}_K^{\mathrm{nor}}(z),$$

denoting the error function of the approximation, the following limit holds:

$$\lim_{\substack{z^1 \neq z^2 \\ (z^1, z^2) \to (z^*, z^*)}} \frac{\mathbf{e}(z^1) - \mathbf{e}(z^2)}{\| z^1 - z^2 \|} = 0. \tag{7.3.6}$$

By substitution, we have (recalling $x^* = \Pi_K(z^*)$),

$$\begin{aligned} \mathbf{e}(z) &= JF(x^*)(\Pi_K(z) - x^*) - z^* + x^* - F(\Pi_K(z)) \\ &= -F(x^*) - z^* + x^* - \mathbf{e}_F(\Pi_K(z)). \end{aligned}$$

Since for $z^1 \neq z^2$,

$$\begin{aligned} &\frac{\mathbf{e}_F(\Pi_K(z^1)) - \mathbf{e}_F(\Pi_K(z^2))}{\| z^1 - z^2 \|} \\ &= \frac{\mathbf{e}_F(\Pi_K(z^1)) - \mathbf{e}_F(\Pi_K(z^2))}{\| \Pi_K(z^1) - \Pi_K(z^2) \|} \frac{\| \Pi_K(z^1) - \Pi_K(z^2) \|}{\| z^1 - z^2 \|}, \end{aligned}$$

the desired limit (7.3.6) follows readily from (7.3.4) and the nonexpansiveness of the projector Π_K. □

There is a subtle difference between Algorithm **7.3.1** and its "equivalent" formulation using the normal map $\mathbf{F}_K^{\mathrm{nor}}$. This has to do with the choice of initial iterates in the two formulations: $x^0 \in K$ in the former and $z^0 \in \mathbb{R}^n$ in the latter. When we specialize Theorem **7.2.20** to the normal equation $\mathbf{F}_K^{\mathrm{nor}}(z) = 0$, we may conclude that for any initial vector z^0 sufficiently close to a solution z^* of this equation, the Newton sequence

$\{z^k\}$ will have the desirable convergence properties. Translating this conclusion to Algorithm **7.3.1**, we can expect to establish that for any initial vector $x^0 \in K$ for which there exists a vector z^0 sufficiently close to z^* such that $\Pi_K(z^0) = x^0$, the sequence $\{x^k\}$ generated by Algorithm **7.3.1** will have similar convergence properties. Ideally, we wish to establish the latter properties for every $x^0 \in K$ that is sufficiently close to $x^* = \Pi_K(z^*)$. This entails us, given such a x^0, to construct an auxiliary vector z^0 with two properties: z^0 is close to z^* and $\Pi_K(z^0) = x^0$. This turns out to be not so easy. Thus, in the proof of the following convergence result, we circumvent the task of directly constructing z^0 by showing that given $x^0 \in K$ sufficiently close to x^*, the first semi-linearized VI (K, F^0) has a solution x^1 that remains close to x^*; moreover, the induced variable $z^1 \equiv x^1 - F(x^1)$ will be close to z^*. Since we now have $x^1 = \Pi_K(z^1)$, we can then complete the proof of the result by the argument outlined above with z^1 as the initial iterate. We formally state the following theorem pertaining to the convergence of the Newton Algorithm **7.3.1** for solving the VI (K, F).

7.3.3 Theorem. Suppose that x^* is a strongly stable solution of the VI (K, F), with F continuously differentiable around x^*. There exists a positive scalar $\bar{\varepsilon}$ such that for every $\varepsilon \in (0, \bar{\varepsilon}]$ a $\delta > 0$ exists such that for every $x^0 \in K \cap \mathbb{B}(x^*, \delta)$, Algorithm **7.3.1** generates a unique sequence $\{x^k\}$ that converges Q-superlinearly to x^*. If JF is Lipschitz continuous near x^*, the convergence rate is Q-quadratic.

Proof. By Lemma **7.3.2** and Theorem **7.2.20**, there exists a positive scalar $\bar{\varepsilon}$ such that for every $\varepsilon \in (0, \bar{\varepsilon}]$ a scalar δ_z exists such that if z^1 is chosen from $\mathbb{B}(z^*, \delta_z)$, a sequence $\{z^k\}$, where z^{k+1} satisfies (7.3.2) and belongs to $\mathbb{B}(z^k, \varepsilon)$ for every k, is uniquely defined and converges Q-superlinearly to z^*; moreover, the convergence rate is Q-quadratic if JF is Lipschitz continuous near x^*.

Let $\varepsilon \in (0, \bar{\varepsilon}]$ be given. As explained above, we need to show the existence of a positive scalar δ such that if x^0 belongs to $K \cap \mathbb{B}(x^*, \delta)$, Algorithm **7.3.1** with the given ε generates a unique vector x^1 as described above. By the continuous differentiability of F near x^*, the affine function

$$F^*(x) \equiv F(x^*) + JF(x^*)(x - x^*) \tag{7.3.7}$$

is a strong FOA of F at x^*. Since x^* is a strongly stable solution of the VI (K, F), it follows that x^* is a strongly stable solution of the VI (K, F^*). Thus there exist three positive constants η, δ_x, and c with $\delta_x < \varepsilon/2$ such that for every function $G \in \mathbb{B}(F^*; \eta, K_{\mathcal{N}})$, where $\mathcal{N} \equiv \mathbb{B}(x^*, \delta_x)$, it holds

that $\mathrm{SOL}(K,G) \cap \mathcal{N}$ is a singleton; moreover, if x_G is the unique element in $\mathrm{SOL}(K,G) \cap \mathcal{N}$, then

$$\| x_G - x^* \| \leq c \| e(x_G) \|,$$

where $e(x) \equiv F^*(x) - G(x)$. We have

$$\begin{aligned} & F^*(x) - F^0(x) \\ &= F(x^*) - F(x^0) + JF(x^*)(x - x^*) - JF(x^0)(x - x^0) \\ &= -\mathbf{e}_F(x^0) + (JF(x^*) - JF(x^0)) (x - x^0). \end{aligned}$$

Since JF is continuous near x^*, there exists $\delta \in (0, \varepsilon/2)$ such that with $x^0 \in \mathbb{B}(x^*, \delta)$, we have $F^0 \in \mathbb{B}(F^*; \eta, K_{\mathcal{N}})$. Thus $\mathrm{SOL}(K, F^0) \cap \mathbb{B}(x^*, \delta_x)$ is a singleton; let x^1 be the unique element in this set. We have

$$\| x^1 - x^0 \| \leq \| x^1 - x^* \| + \| x^0 - x^* \| \leq \delta_x + \delta < \varepsilon.$$

Furthermore,

$$\| x^1 - x^* \| \leq c [\varepsilon \| JF(x^*) - JF(x^0) \| + \| \mathbf{e}_F(x^0) \|]$$

Letting $z^1 \equiv x^1 - F(x^1)$, we have $x^1 = \Pi_K(z^1)$ and

$$\| z^1 - z^* \| \leq \| x^1 - x^* \| + \| F(x^1) - F(x^*) \| \leq (1 + L) \| x^1 - x^* \|,$$

where $L > 0$ is the locally Lipschitz modulus of F near x^*. Consequently, by shrinking δ if necessary, we can make

$$\| JF(x^*) - JF(x^0) \| \quad \text{and} \quad \| \mathbf{e}_F(x^0) \|$$

arbitrarily small and ensure that z^1 belongs to $\mathbb{B}(z^*, \delta_z)$. From this point on, Algorithm **7.3.1** becomes a straightforward reformulation of Algorithm **7.2.19**. What remains to be shown is that the convergence properties of $\{z^k\}$ to z^* translate into analogous properties of $\{x^k\}$ to x^*. Since $x^{k+1} = \Pi_K(z^{k+1})$, we have

$$\| x^{k+1} - x^* \| \leq \| z^{k+1} - z^* \|,$$

which implies that $\{x^k\}$ converges to x^*. Since $z^k = x^k - F(x^k)$ for $k \geq 1$, we have

$$\| z^k - z^* \| \leq (1 + L) \| x^k - x^* \|,$$

Combining the above two inequalities, we deduce for all k sufficiently large,

$$\frac{\| x^{k+1} - x^* \|}{\| x^k - x^* \|} \leq (1 + L)^{-1} \frac{\| z^{k+1} - z^* \|}{\| z^k - z^* \|},$$

which shows that $\{x^k\}$ converges to x^* Q-superlinearly because $\{z^k\}$ converges to z^* Q-superlinearly. If $\{z^k\}$ converges to z^* Q-quadratically, then a similar inequality shows that $\{x^k\}$ converges to x^* Q-quadratically. $\square$

7.3.4 Remark. In the previous discussion we have required x^k to belong to K for every k. However, it is not difficult to check that x^0 does not need to be in K in Algorithm **7.3.1**. In any case all the iterates x^k for $k \geq 1$ will be feasible and all the properties established so far still hold even if x^0 does not belong to K. This means that we can start the algorithm with any x^0 sufficiently close to x^*, a fact that will be used in Chapter **10**. □

A consequence of the strong stability of x^* is that the sequence $\{x^k\}$ in Algorithm **7.3.1** is uniquely defined. It turns out that by assuming only the stability of x^* as a solution of the semi-linearized VI (K, F^*), Algorithm **7.3.1** remains well defined and generates a (possibly nonunique) sequence $\{x^k\}$ that still converges to x^* with the same convergence rate. We formally state this generalized result below.

7.3.5 Theorem. Let F be continuously differentiable. Suppose that x^* is a stable solution of the VI (K, F^*), where F^* is defined by (7.3.7). For every $\varepsilon > 0$, there exists a $\delta > 0$ such that for every $x^0 \in K \cap \mathbb{B}(x^*, \delta)$, Algorithm **7.3.1** generates a well-defined sequence $\{x^k\}$ in $\mathbb{B}(x^*, \delta)$, and every such sequence converges Q-superlinearly to x^*. If JF is Lipschitz continuous near x^*, the convergence rate is Q-quadratic.

Proof. Since we can no longer apply the nonsmooth equation theory, the proof below does not rely on the auxiliary sequence $\{z^k\}$. Instead, the entire proof is given in the x-space.

By the stability of the VI (K, F^*) at x^*, there exist three positive constants η, δ_x, and c such that for every function $G \in \mathbb{B}(F^*; \eta, K_{\mathcal{N}})$, where $\mathcal{N} \equiv \mathbb{B}(x^*, \delta_x)$, it holds that $\mathrm{SOL}(K, G) \cap \mathcal{N}$ is nonempty; moreover, for every element x in $\mathrm{SOL}(K, G) \cap \mathcal{N}$, we have

$$\| x - x^* \| \leq c \, \| e(x) \|$$

where $e(x) \equiv F^*(x) - G(x)$. By the proof of Theorem **7.3.3**, it follows that, for all $\delta \in (0, \delta_x)$ sufficiently small, if x^0 belongs to $\mathbb{B}(x^*, \delta)$, then the VI (K, F^0) has a solution x^1 that belongs to the neighborhood $\mathcal{N}$; moreover, any such solution satisfies:

$$\| x^1 - x^* \| \leq c \, [\, \| JF(x^*) - JF(x^0) \| \, \| x^1 - x^0 \| + \| \mathbf{e}_F(x^0) \| \,]$$

which implies

$$\| x^1 - x^* \| \leq \frac{c \, [\, \| JF(x^*) - JF(x^0) \| \, \| x^* - x^0 \| + \| \mathbf{e}_F(x^0) \| \,]}{1 - c \, \| JF(x^*) - JF(x^0) \|}$$

and

$$\| x^0 - x^1 \| \leq \frac{\| x^0 - x^* \| + c \| \mathbf{e}_F(x^0) \|}{1 - c \| JF(x^*) - JF(x^0) \|}.$$

Consequently, for every $\varepsilon > 0$, we can choose a $\delta > 0$ such that for any x^0 belonging to $\mathbb{B}(x^*, \delta)$, $\mathrm{SOL}(K, F^0) \cap \mathcal{N}$ is nonempty,

$$\frac{c [\| JF(x^*) - JF(x^0) \| \| x^* - x^0 \| + \| \mathbf{e}_F(x^0) \|]}{1 - c \| JF(x^*) - JF(x^0) \|} \leq \tfrac{1}{2} \| x^0 - x^* \|$$

and

$$\frac{\| x^0 - x^* \| + c \| \mathbf{e}_F(x^0) \|}{1 - c \| JF(x^*) - JF(x^0) \|} < \varepsilon.$$

Hence, any $x^1 \in \mathrm{SOL}(K, F^0) \in \mathcal{N}$ must satisfy

$$\| x^1 - x^* \| \leq \tfrac{1}{2} \| x^0 - x^* \| \quad \text{and} \quad \| x^1 - x^0 \| < \varepsilon.$$

In particular $x^1 \in \mathbb{B}(x^*, \delta)$. We may repeat the argument with x^1 replacing x^0. Inductively, we can therefore establish that Algorithm **7.3.1** generates a well-defined sequence $\{x^k\}$ in $\mathbb{B}(x^*, \delta)$ and every such sequence converges to x^*. Moreover, for any such sequence $\{x^k\}$, we can show that for every k,

$$\| x^{k+1} - x^* \| \leq \frac{c [\| JF(x^*) - JF(x^k) \| \| x^* - x^k \| + \| \mathbf{e}_F(x^k) \|]}{1 - c \| JF(x^*) - JF(x^k) \|},$$

which easily establishes the Q-superlinear convergence of the sequence $\{x^k\}$. If JF is Lipschitz continuous near x^*, the Q-quadratic convergence rate also follows from the same inequality with the aid of (7.3.5). □

7.3.6 Remark. Similar to Remark **7.3.4**, we note that the proof of Theorem **7.3.5** does not need the initial vector x^0 to belong to K. □

It is possible to introduce an inexact Newton method for solving a given VI by allowing the semi-linearized sub-VI to be solved inexactly. Central to any such method is a measure of the inexactness of an approximate solution to the sub-VIs. There are various such measures in general. For the purpose of defining an inexact version of Algorithm **7.3.1**, a measure based on the residual $\| \mathbf{F}_K^{k,\mathrm{nat}}(x^{k+1}) \|$ seems to be an obvious choice, where

$$\mathbf{F}_K^{k,\mathrm{nat}}(x) \equiv x - \Pi_K(x - F(x^k) - JF(x^k)(x - x^k))$$

is the natural map of the sub-VI (K, F^k). The resulting inexact rule (7.3.8) is similar in spirit to (7.2.10). We formally state the following inexact algorithm for solving the VI (K, F).

Inexact Josephy-Newton Method for the VI (IJNMVI)

7.3.7 Algorithm.

Data: $x^0 \in K$, $\varepsilon > 0$ and a sequence of nonnegative scalars $\{\eta_k\}$.

Step 1: Set $k = 0$.

Step 2: If $x^k \in \mathrm{SOL}(K, F)$, stop.

Step 3: Let x^{k+1} be any vector belonging to $\mathbb{B}(x^k, \varepsilon)$ that satisfies

$$\| \mathbf{F}_K^{k,\mathrm{nat}}(x^{k+1}) \| \leq \eta_k \| \mathbf{F}_K^{\mathrm{nat}}(x^k) \|. \tag{7.3.8}$$

Step 4: Set $k \leftarrow k + 1$ and go to Step 2.

The following is the convergence result of Algorithm **7.3.7** under a stability condition.

7.3.8 Theorem. Let F be continuously differentiable. Assume that x^* is a stable solution of the VI (K, F^*), where F^* is defined by (7.3.7). There exists $\bar{\eta} > 0$ such that if $\eta_k \leq \bar{\eta}$ for every k, then for every $\varepsilon > 0$, a neighborhood $\mathbb{B}(x^*, \delta)$ of x^* exists such that if x^0 belongs to $\mathbb{B}(x^*, \delta)$, the inexact Newton method **7.3.7** generates a well-defined sequence $\{x^k\}$ in $\mathbb{B}(x^*, \delta)$, and every such sequence converges Q-linearly to x^*. Furthermore, if $\{\eta_k\} \to 0$, then the convergence rate of $\{x^k\}$ to x^* is Q-superlinear. Finally, if JF is Lipschitz continuous near x^* and for some $\tilde{\eta} > 0$, $\eta_k \leq \tilde{\eta} \| \mathbf{F}_K^{\mathrm{nat}}(x^k) \|$ for all k, then the convergence rate is Q-quadratic.

Proof. The proof is similar to the proof of the previous theorem and Theorem **7.2.8**. By the stability of the VI (K, F^*) at x^*, there exist three positive constants η, δ_x, and c such that for every function $G \in \mathbb{B}(F^*; \eta, K_{\mathcal{N}})$, where $\mathcal{N} \equiv \mathbb{B}(x^*, \delta_x)$, it holds that $\mathrm{SOL}(K, G) \cap \mathcal{N}$ is nonempty; moreover, for every element x in $\mathrm{SOL}(K, G) \cap \mathcal{N}$, we have

$$\| x - x^* \| \leq c \| e(x) \|$$

where $e(x) \equiv F^*(x) - G(x)$. We focus on the convergence of a sequence $\{x^k\}$ generated by Algorithm **7.3.1** that lies in $\mathcal{N}$. Write $r^{k+1} \equiv \mathbf{F}_K^{k,\mathrm{nat}}(x^{k+1})$. We have

$$0 = x^{k+1} - r^{k+1} - \Pi_K(x^{k+1} - F^k(x^{k+1})).$$

Thus $y^{k+1} \equiv x^{k+1} - r^{k+1}$ is an exact solution of the VI $(K, \tilde{F}^k)$, where

$$\begin{aligned} \tilde{F}^k(x) &\equiv F^k(x + r^{k+1}) - r^{k+1} \\ &= F(x^k) + JF(x^k)(x + r^{k+1} - x^k) - r^{k+1}. \end{aligned}$$

This function remains a perturbation of the function F^*, provided that $\|r^{k+1}\|$ is sufficiently small. More precisely, we have

$$\begin{aligned}\| F^*(x) - \tilde{F}^k(x) \| \;\le\; & \| \mathbf{e}_F(x^k) \| + \\ & \| JF(x^*) - JF(x^k) \| \, \| x - x^k \| + (1 + \| JF(x^k) \|) \, \| r^{k+1} \|.\end{aligned}$$

By the inexact rule (7.3.8) and noting $x^* = \Pi_K(x^* - F(x^*))$, we have

$$\begin{aligned}\| r^{k+1} \| \;&\le\; \eta_k \, \| x^k - \Pi_K(x^k - F(x^k)) \| \\ &\le\; \eta_k \, [\, \| x^k - x^* \| + \| \Pi_K(x^* - F(x^*)) - \Pi_K(x^k - F(x^k)) \| \,] \\ &\le\; \eta_k \, (2 + L) \, \| x^k - x^* \|,\end{aligned}$$

where L is the local Lipschitz modulus of F near x^*. Consequently, we may choose a $\delta \in (0, \delta_x)$ sufficiently small such that if x^k belongs to $\mathbb{B}(x^*, \delta)$, the function $\tilde{F}^k$ belongs to $\mathbb{B}(F^*; \eta, K_{\mathcal{N}})$. Hence any solution y of the VI $(K, \tilde{F}^k)$ that lies in $\mathcal{N}$ must satisfy

$$\| y - x^* \| \;\le\; c \, \| F^*(y) - \tilde{F}^k(y) \|.$$

Thus, we have

$$\begin{aligned}\| y^{k+1} - x^* \| \;\le\; & \; c \, [\, \| JF(x^*) - JF(x^k) \| \, \| y^{k+1} - x^k \| + \\ & \| \mathbf{e}_F(x^k) \| + \eta_k \, (2 + L) \, (1 + \| JF(x^k) \|) \, \| x^k - x^* \| \,],\end{aligned}$$

which implies

$$\begin{aligned}\| y^{k+1} - x^* \| \;\le\; & \frac{c \, \| \mathbf{e}_F(x^k) \|}{1 - c \, \| JF(x^*) - JF(x^k) \|} + \\ & \frac{c \, [\, \| JF(x^*) - JF(x^k) \| + \eta_k \, (2 + L) \, (1 + \| JF(x^k) \|) \,] \, \| x^k - x^* \|}{1 - c \, \| JF(x^*) - JF(x^k) \|}.\end{aligned}$$

Consequently,

$$\begin{aligned}\| x^{k+1} - x^* \| \;\le\; & \| r^{k+1} \| + \frac{c \, \| \mathbf{e}_F(x^k) \|}{1 - c \, \| JF(x^*) - JF(x^k) \|} + \\ & \frac{c \, [\, \| JF(x^*) - JF(x^k) \| + \eta_k \, (2 + L) \, (1 + \| JF(x^k) \|) \,] \, \| x^k - x^* \|}{1 - c \, \| JF(x^*) - JF(x^k) \|}.\end{aligned}$$

By an argument similar to that in the two previous theorems mentioned at the beginning of the proof, we can easily establish the desired convergence properties of any inexact Newton sequence $\{x^k\}$ that lies in $\mathbb{B}(x^*, \delta)$. □

The specialization of Algorithms **7.3.1** and **7.3.7** to a MiCP is fairly straightforward. The specialized algorithms solve a MiCP by solving a

sequence of sub-MLCPs, exactly or inexactly. It would be interesting to consider the case of a MiCP that arises from the KKT system of the VI (K, F), where K is finitely representable given by

$$K \equiv \{\, x \in \mathbb{R}^n : h(x) = 0,\, g(x) \leq 0 \,\},$$

where $h : \mathbb{R}^n \to \mathbb{R}^\ell$ and $g : \mathbb{R}^n \to \mathbb{R}^m$ are twice continuously differentiable and $F : \mathbb{R}^n \to \mathbb{R}^n$ is continuously differentiable. Specifically, consider the system:

$$\begin{aligned} \mathbf{L}(x, \mu, \lambda) &= 0 \\ h(x) &= 0 \\ 0 \leq \lambda \perp g(x) &\leq 0, \end{aligned}$$

where

$$\mathbf{L}(x, \mu, \lambda) \equiv F(x) + \sum_{j=1}^{\ell} \mu_j \nabla h_j(x) + \sum_{i=1}^{m} \lambda_i \nabla g_i(x)$$

is the (vector) Lagrangian function of the VI (K, F). The above KKT system is equivalent to the VI $(\mathbb{R}^{n+\ell} \times \mathbb{R}^m_+, \mathbf{F})$, where

$$\mathbf{F}(x, \mu, \lambda) = \begin{pmatrix} \nabla_x L(x, \mu, \lambda) \\ -h(x) \\ -g(x) \end{pmatrix}.$$

Given a triple (x^k, μ^k, λ^k), we examine the associated Newton subproblem. Writing $z \equiv (x, \mu, \lambda)$, we have

$$\begin{aligned} \mathbf{F}^k(z) &\equiv \mathbf{F}(z^k) + J\mathbf{F}(z^k)(\, z - z^k \,) \\ &= \begin{pmatrix} \mathbf{L}(z^k) \\ -h(x^k) \\ -g(x^k) \end{pmatrix} + \begin{bmatrix} \nabla_x \mathbf{L}(z^k) & Jh(x^k)^T & Jg(x^k)^T \\ -Jh(x^k) & 0 & 0 \\ -Jg(x^k) & 0 & 0 \end{bmatrix} \begin{pmatrix} x - x^k \\ \mu - \mu^k \\ \lambda - \lambda^k \end{pmatrix} \\ &= \begin{pmatrix} F(x^k) \\ -h(x^k) \\ -g(x^k) \end{pmatrix} + \begin{bmatrix} \nabla_x \mathbf{L}(x^k) & Jh(x^k)^T & Jg(x^k)^T \\ -Jh(x^k) & 0 & 0 \\ -Jg(x^k) & 0 & 0 \end{bmatrix} \begin{pmatrix} x - x^k \\ \mu \\ \lambda \end{pmatrix}. \end{aligned}$$

We easily recognize the sub-VI $(\mathbb{R}^{n+\ell} \times \mathbb{R}^m_+, \mathbf{F}^k)$ as an MLCP, which in turn is the KKT system of the AVI $(K(x^k), q^k, \nabla_x \mathbf{L}(z^k))$, where

$$\begin{aligned} K(x^k) \equiv \{\, x \in \mathbb{R}^n : \quad & h(x^k) + Jh(x^k)(x - x^k) = 0, \\ & g(x^k) + Jg(x^k)(x - x^k) \leq 0 \,\}, \end{aligned}$$

and

$$q^k \equiv F(x^k) - \nabla_x \mathbf{L}(z^k)x^k.$$

The salient features of the latter AVI are as follows. Its constraint set $K(x^k)$ is a polyhedron formed by linearizing the constraints of the original VI (K, F); its defining function is a special linearization of the original function F. Instead of using the Jacobian matrix of F alone, the linearlization involves the second-order derivatives of the constraint functions h and g and the current estimates of the multipliers; that is,

$$F(x) \approx F(x^k) + \left(JF(x^k) + \sum_{j=1}^{\ell} \mu_j^k \nabla^2 h_j(x^k) + \sum_{i=1}^{m} \lambda_i^k \nabla^2 g_i(x^k) \right) (x - x^k).$$

Thus the overall Newton method solves the KKT system of the VI (K, F) by solving a sequence of AVIs, each formed in the way described above. The local convergence of the resulting algorithm to a stable KKT triple follows easily from Theorem **7.3.5**.

7.4 Nonsmooth Analysis II: Semismooth Functions

Motivated by the theory developed in Section **7.2**, we now pause to introduce a very important subclass of locally Lipschitz continuous functions; namely, the "semismooth" functions. The generalized Jacobian of Clarke is a powerful tool that allows us to extend to locally Lipschitz functions many classical results valid for smooth functions. However, a straightforward extension of the Newton method for smooth equations to general nonsmooth equations using the Clarke generalized Jacobian $\partial G(x)$ is doomed to failure, because in general a linear model defined by the generalized Jacobian matrices $H \in \partial G(x)$ does not define a Newton approximation scheme of G; that is, the limit condition

$$\lim_{\substack{\bar{x} \neq x \to \bar{x} \\ H \in \partial G(x)}} \frac{G(x) + H(\bar{x} - x) - G(\bar{x})}{\| x - \bar{x} \|} = 0.$$

does not generally hold for any locally Lipschitz functions G. This is illustrated by the following example.

7.4.1 Example. We build a real-valued function of one variable f from $\mathbb{R}$ into itself such that $f(0) = 0$ in the following way. For any integer $n \geq 2$, let $I_n \equiv [1/n, 1/(n-1)]$ and let m_n and m_{2n} be the midpoints of I_n and I_{2n} respectively. Furthermore define

$$a_n \equiv \frac{2n}{4n-1} \quad \text{and} \quad b_n \equiv \frac{8n-4}{4n-3};$$

note that $a_n \leq b_n$. For each $n \geq 2$, define two linear functions:

$$f_n^1(x) \equiv a_n(x + m_n) \quad \text{and} \quad f_n^2(x) \equiv b_n(x - m_{2n}).$$

These functions satisfy

$$f_n^1\left(\frac{1}{n-1}\right) = \frac{1}{n-1}, \qquad f_n^2\left(\frac{1}{n}\right) = \frac{1}{n}, \qquad f_n^1(m_n) < f_n^2(m_n).$$

By the last inequality we see that the point y_n defined by the equation $f_n^1(y_n) = f_n^2(y_n)$ belongs to the open interval $(1/n, m_n)$. We are now ready to define the function f:

$$f(x) = \begin{cases} 0 & \text{if} \quad x = 0 \\ f_n^2(x) & \text{if} \quad x \in [1/n, y_n] \\ f_n^1(x) & \text{if} \quad x \in [y_n, 1/(n-1)] \\ f_2^1(x) & \text{if} \quad x \geq 1 \\ -f(-x) & \text{if} \quad x < 0. \end{cases}$$

Note that since $a_n \to 1/2$ and $b_n \to 2$ the function f is locally Lipschitz continuous on the entire real line $\mathbb{R}$. Moreover, $\partial f(0) = [1/2, 2]$; thus all elements of the Clarke Jacobian $\partial f(0)$ are nonzero. It is also easy to see that $\bar{x} = 0$ is the only solution to $f(x) = 0$. To show that $f(x) + \xi(\bar{x} - x)$, with $\xi \in \partial f(x)$ is not a Newton approximation, consider the sequence $\{x^k\}$ converging to $\bar{x} = 0$, where x^k is defined as the midpoint of the interval $[1/k, y_k]$. Taking into account that the function f is differentiable at each x^k and its derivative there is b_k and that $x^k \leq m_k$, we have

$$\begin{aligned} \lim_{k\to\infty} \frac{f(x^k) + f'(x^k)(\bar{x} - x^k) - f(\bar{x})}{|\, x^k - \bar{x}\,|} &= \lim_{k\to\infty} \frac{b_k x^k + b_k m_{2k} - b_k x^k}{x^k} \\ &\geq \lim_{k\to\infty} \frac{b_k m_{2k}}{m_k} = \frac{2}{3}. \end{aligned}$$

Thus, the definition of a Newton approximation is not met.

Consider now the Newton process defined by $x^{k+1} \equiv x^k + d^k$ where d^k is the solution to

$$f(x^k) + \xi^k\, d = 0, \qquad \xi^k \in \partial f(x^k).$$

If we take $x^0 \in (1/n, y_n)$ the function f is differentiable at x^0 and it is easy to check that the sequence of points produced is

$$x^1 = -m_{2n}, \quad x^2 = m_{2n}, \quad x^3 = -m_{2n}, \cdots$$

so that the algorithm does not converge to 0. The same phenomenon occurs if we take $x^0 \in (y_n, 1/(n-1))$; the sequence generated is

$$x^1 = -m_n, \quad x^2 = m_n, \quad x^3 = -m_n \cdots$$

We leave to the reader to check that the same kind of behavior occurs if $x^0 > 1$ or for the corresponding negative values of x^0. See Figure 7.3 for an illustration of the above discussion. □

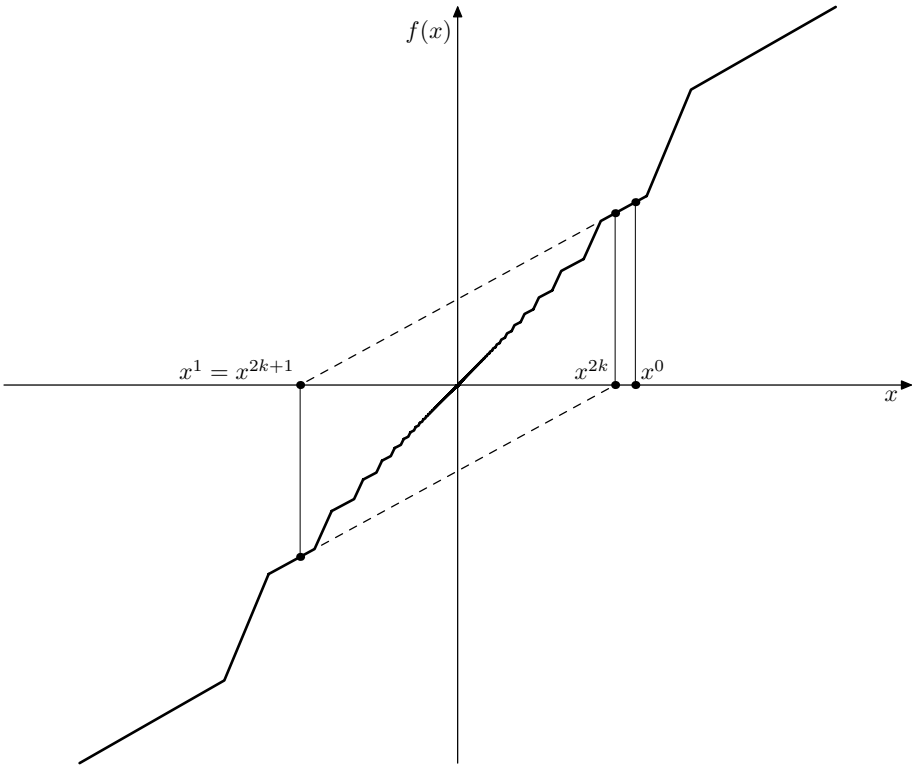


Figure 7.3: Non-convergence of nonsmooth Newton method with Clarke's generalized Jacobian.

Semismooth functions are locally Lipschitz continuous functions for which the Clarke generalized Jacobians define a legitimate Newton approximation scheme. This class of functions turns out to be broad enough to include most of the functions we need to deal with; more importantly, the Newton method described in the previous section becomes an extremely simple but powerful tool that lies at the heart of many highly successful methods for solving VIs and CPs. This section is therefore devoted to the

study of semismooth functions and some of its generalizations. The results obtained herein are key to the analysis of the Newton method for solving semismooth equations to be presented in the next section. In the following definition, the domain and range of the function G can be in different Euclidean spaces.

7.4.2 Definition. Let $G : \Omega \subseteq \mathbb{R}^n \to \mathbb{R}^m$, with Ω open, be a locally Lipschitz continuous function on Ω. We say that G is *semismooth* at a point $\bar{x} \in \Omega$ if G is directionally differentiable near $\bar{x}$ and there exist a neighborhood $\Omega' \subseteq \Omega$ of $\bar{x}$ and a function $\Delta : (0, \infty) \to [0, \infty)$ with

$$\lim_{t \downarrow 0} \Delta(t) \;=\; 0,$$

such that for any $x \in \Omega'$ different from $\bar{x}$,

$$\frac{\| G'(x; x - \bar{x}) - G'(\bar{x}; x - \bar{x}) \|}{\| x - \bar{x} \|} \;\leq\; \Delta(\| x - \bar{x} \|). \tag{7.4.1}$$

If the above requirement is strengthened to

$$\limsup_{\bar{x} \neq x \to \bar{x}} \frac{\| G'(x; x - \bar{x}) - G'(\bar{x}; x - \bar{x}) \|}{\| x - \bar{x} \|^2} \;<\; \infty, \tag{7.4.2}$$

we say that G is *strongly semismooth* at $\bar{x}$. If G is (strongly) semismooth at each point of Ω, then we say that G is (strongly) semismooth on Ω. □

It is easy to see that G is semismooth at $\bar{x}$ if and only if G is B-differentiable near $\bar{x}$ and the following limit holds:

$$\lim_{\bar{x} \neq x \to \bar{x}} \frac{\| G'(x; x - \bar{x}) - G'(\bar{x}; x - \bar{x}) \|}{\| x - \bar{x} \|} \;=\; 0.$$

Indeed, (7.4.1) clearly implies the above limit. Conversely, if the above limit holds, then it suffices to define

$$\Delta(t) \;\equiv\; \sup_{x \in \mathbb{B}(\bar{x}, t) \setminus \{\bar{x}\}} \frac{\| G'(x; x - \bar{x}) - G'(\bar{x}; x - \bar{x}) \|}{\| x - \bar{x} \|}, \quad \forall\, t \;>\; 0.$$

The B-differentiability of G implies that $\Delta(t)$ is finite for all $t > 0$ sufficiently small and tends to zero as t approaches zero. (For all practical purposes, $\Delta(t)$ for t outside a neighborhood of zero is not of interest.)

By definition, if G is semismooth at $\bar{x}$, then it is B-differentiable in a neighborhood of $\bar{x}$. In particular, the directional derivative $G'(\bar{x}; x - \bar{x})$ provides a good approximation to $G(x)$ for all x sufficiently close to $\bar{x}$.

What we have in addition, in the semismooth case, is that the directional derivative $G'(\bar{x}; x-\bar{x})$ can be approximated with a good degree of precision by using any element of the generalized Jacobian of G at x. This assertion is contained in the theorem below, which gives several equivalent ways of describing semismoothness. The theorem provides the basis for the use of generalized Jacobians to build linear Newton approximations to G at $\bar{x}$.

7.4.3 Theorem. Let $G : \Omega \subseteq \mathbb{R}^n \to \mathbb{R}^m$, with Ω open, be B-differentiable near $\bar{x} \in \Omega$. The following three statements are equivalent:

(a) G is semismooth at $\bar{x}$;

(b) the following limit holds:

$$\lim_{\substack{\bar{x} \neq x \to \bar{x} \\ H \in \partial G(x)}} \frac{G'(\bar{x}; x-\bar{x}) - H(x-\bar{x})}{\| x-\bar{x} \|} = 0; \tag{7.4.3}$$

(c) the following limit holds:

$$\lim_{\substack{\bar{x} \neq x \to \bar{x} \\ H \in \partial G(x)}} \frac{G(x) + H(\bar{x}-x) - G(\bar{x})}{\| x-\bar{x} \|} = 0. \tag{7.4.4}$$

If G is strongly semismooth at $\bar{x}$, then

$$\limsup_{\bar{x} \neq x \to \bar{x}} \frac{G(x) - G(\bar{x}) - G'(\bar{x}, x-\bar{x})}{\| x-\bar{x} \|^2} < \infty, \tag{7.4.5}$$

$$\limsup_{\substack{\bar{x} \neq x \to \bar{x} \\ H \in \partial G(x)}} \frac{G(x) + H(\bar{x}-x) - G(\bar{x})}{\| x-\bar{x} \|^2} < \infty. \tag{7.4.6}$$

Proof. (a) $\Rightarrow$ (b). Suppose that G is semismooth at $\bar{x}$. The limit (7.4.3) is clearly equivalent to

$$\lim_{\substack{0 \neq d \to 0 \\ H \in \partial G(\bar{x}+d)}} \frac{G'(\bar{x}; d) - Hd}{\| d \|} = 0.$$

By Carathéodory's theorem applied to the convex compact set $\partial G(\bar{x}+d)$, it follows that for every $d \in \mathbb{R}^n$ and every element $H \in \partial G(\bar{x}+d)$, there exist scalars α_i for $i = 1, \ldots, m+1$ and sequences of vectors $\{d^{i,k}\}$ such that

$$\sum_{i=1}^{m+1} \alpha_i = 1, \quad \alpha_i \geq 0$$

$$\lim_{k \to \infty} d^{i,k} = d, \quad \forall\, i = 1, \ldots, m+1,$$

G is F-differentiable at $\bar{x} + d^{i,k}$, and

$$H = \sum_{i=1}^{m+1} \alpha_i \lim_{k \to \infty} JG(\bar{x} + d^{i,k}).$$

Thus we have

$$\begin{aligned} & G'(\bar{x};d) - Hd \\ &= \sum_{i=1}^{m+1} \alpha_i \lim_{k\to\infty} [\,(\,G'(\bar{x};d^{i,k}) - G'(\bar{x}+d^{i,k};d)\,] \\ &= \sum_{i=1}^{m+1} \alpha_i \lim_{k\to\infty} [\,(\,G'(\bar{x};d^{i,k}) - G'(\bar{x}+d^{i,k};d^{i,k}) + \\ & \qquad G'(\bar{x}+d^{i,k};d^{i,k}) - G'(\bar{x}+d^{i,k};d)\,]. \end{aligned}$$

By the semismoothness of G at $\bar{x}$, it follows that for every $\varepsilon > 0$, there exists $\delta > 0$ such that for every vector d' satisfying $\|d'\| \le \delta$,

$$\|\,G'(\bar{x};d') - G'(\bar{x}+d';d')\,\| \;<\; \varepsilon\,\|\,d'\,\|.$$

Moreover, by the locally Lipschitz continuity of G at $\bar{x}$, it follows that there exists a neighborhood $\mathcal{N}$ of $\bar{x}$ and a constant $L > 0$ such that for all vectors $y \in \mathcal{N}$ and u and $v \in \mathbb{R}^n$,

$$\|\,G'(y;u) - G'(y;v)\,\| \;\le\; L\,\|\,u-v\,\|.$$

Thus

$$\lim_{\substack{y\to x \\ \|u-v\|\to 0}} (\,G'(y;u) - G'(y;v)\,) \;=\; 0.$$

Consequently, we deduce that for all d with $\|d\|$ sufficiently small,

$$\|\,G'(\bar{x};d) - Hd\,\| \;\le\; \varepsilon\,\|\,d\,\|.$$

Hence (7.4.3) holds.

(b) $\Leftrightarrow$ (c). By the B-differentiability of G at $\bar{x}$, we have

$$\lim_{\bar{x}\neq x\to\bar{x}} \frac{G(x) - G(\bar{x}) - G'(\bar{x};x-\bar{x})}{\|\,x-\bar{x}\,\|} \;=\; 0;$$

cf. Proposition **3.1.3**. This limit clearly shows that (7.4.3) and (7.4.4) are equivalent.

(b) $\Rightarrow$ (a). By Proposition **7.1.17**, for every x sufficiently close to $\bar{x}$, there exists $H \in \partial G(x)$ such that $G'(x;x-\bar{x}) = H(x-\bar{x})$. Thus (a) follows from (b) readily.

Assume now that G is strongly semismooth at $\bar{x}$. For any given vector d, let $\Gamma(t) \equiv G(\bar{x}+td)$. It is clear that Γ is locally Lipschitz continuous, and hence differentiable almost everywhere on $[0,1]$ Therefore, for all d

with $\|d\|$ sufficiently small, we can write

$$\begin{aligned} G(\bar{x}+d) - G(\bar{x}) &= \Gamma(1) - \Gamma(0) = \int_0^1 \Gamma'(t)\,dt \\ &= \int_0^1 G'(\bar{x}+td;d)\,dt \\ &= \int_0^1 \left[\, G'(\bar{x};d) + t\,O(\|\,d\,\|^2)\,\right]\,dt \\ &= G'(\bar{x};d) + O(\|\,d\,\|^2), \end{aligned}$$

where the fourth equality follows from (7.4.2). By taking $d \equiv x - \bar{x}$, this chain of equalities establishes (7.4.5). Finally, using (7.4.5) and following the above proof of the equivalence of statements (a), (b) and (c), we can easily establish (7.4.6). □

Theorem **7.4.3** justifies our interest in the class of semismooth functions. Indeed when $m = n$, the limit (7.4.4) says that $\mathcal{A}(x) \equiv \partial G(x)$ is a Newton approximation scheme of G at $\bar{x}$ if G is semismooth at $\bar{x}$. If G is strongly semismooth at $\bar{x}$, then, because of (7.4.6), the approximation scheme is strong. Note that in general, $\mathcal{A}(x)$ contains an infinite number of members.

An important way to obtain semismooth functions is through composition. The next result makes this statement precise. This result implies in particular that the sum and difference of two (strongly) semismooth functions are (strongly) semismooth.

7.4.4 Proposition. Let a function $F : \Omega_F \subseteq \mathbb{R}^n \to \mathbb{R}^m$, with Ω_F open, a point $\bar{x}$ belonging to Ω_F, and a function $g : \Omega_g \subseteq \mathbb{R}^m \to \mathbb{R}$, with Ω_g being a neighborhood of $F(\bar{x})$, be given. If F and g are (strongly) semismooth at $\bar{x}$ and $F(\bar{x})$ respectively, then the composite function $g \circ F$ is (strongly) semismooth at $\bar{x}$.

Proof. We only consider the semismooth case; the strongly semismooth case can be proved in a similar way. Since g is assumed to be a real-valued function, elements in $\partial(g \circ F)(x)$ are column vectors. By Proposition **3.1.6**, we know that $g \circ F$ is B-differentiable $\bar{x}$ and that

$$(\,g \circ F)'(\bar{x};x-\bar{x}) = g'(F(\bar{x});F'(\bar{x};x-\bar{x})). \tag{7.4.7}$$

We need to show

$$\lim_{\substack{\bar{x}\neq x\to\bar{x} \\ \xi\in\partial(g\circ F)(x)}} \frac{(\,g\circ F\,)'(\bar{x};x-\bar{x}) - \xi^T(x-\bar{x})}{\|\,x-\bar{x}\,\|} = 0.$$

By Proposition **7.1.9**(e) we can write, for every x sufficiently close to $\bar{x}$:

$$\partial(g \circ F)(x) \subseteq \operatorname{conv} S(x),$$

where $S(x) \equiv \partial g(F(x))\partial F(x)$. Therefore we get

$$\begin{aligned} &\max_{\xi \in \partial(g\circ F)(x)} \| \xi^T(x-\bar{x}) - F'(\bar{x}, x-\bar{x}) \| \\ &\quad \leq \max_{\xi \in \operatorname{conv} S(x)} \| \xi^T(x-\bar{x}) - F'(\bar{x}; x-\bar{x}) \|, \end{aligned} \tag{7.4.8}$$

where we can write the maximum in the above formula because both the sets $\partial(g \circ F)(x)$ and $S(x)$ are (nonempty and) compact. Let us denote by $r : \mathbb{R}^n \to [0, \infty)$ the function

$$r(\xi) \equiv \| \xi^T(x-\bar{x}) - (g \circ F)'(\bar{x}; x-\bar{x}) \|.$$

Obviously, r is a convex function on $\mathbb{R}^n$; thus the maximum of $r(\xi)$ over $\operatorname{conv} S(x)$ is attained at a point $\bar{\xi}$ in $S(x)$. Let then $\bar{\xi}$ be a point in $S(x)$ where r achieves the maximum. By the definition of $S(x)$ we can find a V in $\partial F(x)$ and a ζ in $\partial g(F(x))$ such that $\bar{\xi} = V\zeta$. Therefore, writing

$$d = x - \bar{x} \quad \text{and} \quad F_d = F(x) - F(\bar{x}),$$

we have, for every x sufficiently close to $\bar{x}$,

$$\begin{aligned} &\max_{\xi \in \partial(g\circ F)(x)} \| \xi^T d - (g \circ F)'(\bar{x}; d) \| \\ &\leq \| \bar{\xi}^T d - (g \circ F)'(\bar{x}; d) \| && \text{by (7.4.8)} \\ &= \| \zeta^T V^T d - g'(F(x); F'(\bar{x}; d)) \| && \text{by (7.4.7)} \\ &\leq \| \zeta^T V^T d - g'(F(x); F_d) \| + o(\| d \|) && \text{by (7.4.3) and the} \\ &&& \text{Lip. continuity of } g'(F(x); \cdot) \\ &\leq \| \zeta^T V F_d - g'(F(x); F_d) \| + o(\| d \|) && \text{by (7.4.4)} \\ &\leq \max_{\xi \in \partial g(F(x))} \| \xi^T V F_d - g'(F(x); F_d) \| \\ &\qquad + o(\| d \|) && \text{because } \zeta \in \partial g(F(x)) \\ &\leq o(\| F_d \|) + o(\| d \|) && \text{by the semismoothness of } g \\ &= o(\| d \|) && \text{by the Lip. continuity of } F. \end{aligned}$$

This chain of inequalities obviously completes the proof. □

By Definition **7.4.2**, it is easy to check that a vector-valued function is (strongly) semismooth if and only if each of its component functions is (strongly) semismooth. Thus Proposition **7.4.4** implies that the composition of two vector semismooth functions is semismooth. In the rest of this section, we present several results that identify various subclasses of semismooth functions. These results show that the class of semismooth functions is indeed very broad. We begin with a result for real-valued functions.

7.4.5 Proposition. Let $f : \Omega \subseteq \mathbb{R}^n \to \mathbb{R}$, with Ω open, and a point $\bar{x}$ belonging to Ω be given.

(a) If f is continuously differentiable in a neighborhood of $\bar{x}$, then f is semismooth at $\bar{x}$.

(b) If f is continuously differentiable with a Lipschitz continuous gradient in a neighborhood of $\bar{x}$, then f is strongly semismooth at $\bar{x}$.

(c) If f is convex on a neighborhood of $\bar{x}$, then f is semismooth at $\bar{x}$.

Proof. If f is continuously differentiable in an open neighborhood of $\bar{x}$, then $\partial f(x) = \{\nabla f(x)\}$ in the same neighborhood. With this observation, the semismoothness of f follows from Proposition **7.2.9**; so does the strong semismoothness if ∇f is Lipschitz continuous near $\bar{x}$.

To prove (c), we have to check that for every sequence $\{x^k\}$ converging to $\bar{x}$ and every sequence $\{\xi^k\}$, with $x^k \neq \bar{x}$ and $\xi^k \in \partial f(x^k)$ for every k, we have

$$\lim_{k\to\infty} f'(\bar{x}; d^k) = \lim_{k\to\infty} (\xi^k)^T d^k, \tag{7.4.9}$$

where

$$d^k \equiv \frac{x^k - \bar{x}}{\| x^k - \bar{x} \|}.$$

Without loss of generality, we may assume that

$$\lim_{k\to\infty} d^k = \bar{d} \quad \text{and} \quad \lim_{k\to\infty} \xi^k = \bar{\xi} \in \partial f(\bar{x}).$$

Since the left-hand limit and right-hand limit in (7.4.9) are equal to $f'(\bar{x}; \bar{d})$ and $\bar{\xi}^T \bar{d}$, respectively, it remains to show that

$$\bar{\xi}^T \bar{d} = f'(\bar{x}; \bar{d}).$$

Since f is convex and ξ^k is a subgradient (in the sense of classical convex analysis) of f at x^k, we have

$$f(\bar{x}) - f(x^k) \geq (\xi^k)^T (\bar{x} - x^k);$$

dividing by $\|x^k - \bar{x}\|$ and letting $k \to \infty$, we deduce

$$\bar{\xi}^T \bar{d} \geq f'(\bar{x}; \bar{d}).$$

But since $\bar{\xi} \in \partial f(\bar{x})$, we also have

$$f'(\bar{x}; \bar{d}) \geq \bar{\xi}^T \bar{d}.$$

Thus equality holds. □

Extending the class of PC^1 functions, we say that a continuous function G is *piecewise semismooth* near a vector x in the domain of G if G satisfies the conditions in Definition **4.5.1** except that the pieces $\{G^i : i = 1, \ldots, k\}$ are semismooth functions near x. At first glance, it may seem that the class of piecewise semismooth functions is broader than the class of semismooth functions. It turns out that this is not true; indeed, every piecewise semismooth function is semismooth. In particular, PC^1 functions are semismooth.

7.4.6 Proposition. Let $G : \Omega \subseteq \mathbb{R}^n \to \mathbb{R}^m$, with Ω open, be piecewise semismooth near the vector $\bar{x}$ in Ω. Then G is semismooth at $\bar{x}$.

Proof. By Lemma **4.6.1**, G is B-differentiable near $\bar{x}$. Let $\{x^\nu\}$ be an arbitrary sequence converging to $\bar{x}$ with $x^\nu \neq \bar{x}$ for all ν. Write

$$d^\nu \equiv \frac{x^\nu - \bar{x}}{\| x^\nu - \bar{x} \|}.$$

To show that G is semismooth at $\bar{x}$, it suffices to show that

$$\lim_{\nu \to \infty} \frac{G'(\bar{x}; x^\nu - \bar{x}) - G'(x^\nu; x^\nu - \bar{x})}{\| x^\nu - \bar{x} \|} = 0. \tag{7.4.10}$$

Without loss of generality, we may assume that

$$\lim_{\nu \to \infty} d^\nu = \bar{d}.$$

Let $\{G^1, G^2, \cdots, G^k\}$ be the locally active semismooth pieces of G at $\bar{x}$, so that we have $G(\bar{x}) = G^i(\bar{x})$ for all $i = 1, \ldots, k$. For every ν sufficiently large, there exist $i_\nu \in \{1, \ldots, k\}$ and $H^{i_\nu} \in \partial G^{i_\nu}(x^\nu)$ such that

$$G'(x^\nu; x^\nu - \bar{x}) = (G^{i_\nu})'(x^\nu; x^\nu - \bar{x}) = H^{i_\nu}(x^\nu - \bar{x}),$$

$$G^{i_\nu}(x^\nu) = G(x^\nu) \quad \text{and} \quad G^{i_\nu}(\bar{x}) = G(\bar{x}).$$

Furthermore, we have

$$\begin{aligned} G'(\bar{x}; x^\nu - \bar{x}) &= G(x^\nu) - G(\bar{x}) + o(\| x^\nu - \bar{x} \|) \\ &= G^{i_\nu}(x^\nu) - G^{i_\nu}(\bar{x}) + o(\| x^\nu - \bar{x} \|). \end{aligned}$$

Therefore,

$$\frac{G'(\bar{x}; x^\nu - \bar{x}) - G'(x^\nu; x^\nu - \bar{x})}{\| x^\nu - \bar{x} \|} =$$

$$\frac{G^{i_\nu}(x^\nu) - G^{i_\nu}(\bar{x}) - H^{i_\nu}(x^\nu - \bar{x})}{\| x^\nu - \bar{x} \|} + \frac{o(\| x^\nu - \bar{x} \|)}{\| x^\nu - \bar{x} \|}.$$

Since there are only finitely many indices i_ν, by the semismoothness of each piece G^{i_ν} and a simple argument via contradiction, we can easily deduce that the limit (7.4.10) holds. □

It follows from Proposition **7.4.6** that if G^1 and G^2 are two semismooth functions at the same vector $\bar{x}$, then $\min(G^1, G^2)$ and $\max(G^1, G^2)$, being piecewise semismooth functions, are both semismooth at $\bar{x}$. Moreover, if G^1 and G^2 are two affine functions, then $\min(G^1, G^2)$ and $\max(G^1, G^2)$ are both strongly semismooth. The reader can easily verify the last assertion by a direct calculation of the directional derivatives of the min and max functions. More generally, the pointwise minimum and maximum of a finite family of affine functions are strongly semismooth. Summarizing this discussion, we state the following result which identifies a broad class of strongly semismooth functions.

7.4.7 Proposition. Every PA map from $\mathbb{R}^n$ into $\mathbb{R}^m$ is strongly semismooth.

Proof. It suffices to prove that every real-valued PA map is strongly semismooth. From the discussion preceeding Proposition **4.2.1**, we know that every real-valued PA has a maxmin representation in terms of a finite family of affine functions. This respresentation and the aforegoing remarks easily establish the proposition. □

Every norm function, being convex, is semismooth. The next proposition shows that the ℓ_p norms are strongly semismooth.

7.4.8 Proposition. The norm function $\| \cdot \|_p : \mathbb{R}^n \to \mathbb{R}_+$ is strongly semismooth for every $p \in [1, \infty]$.

Proof. Since

$$\begin{aligned} \| \cdot \|_1 &= |x_1| + \ldots + |x_n| \\ \| \cdot \|_\infty &= \max(|x_1|, \ldots, |x_n|), \end{aligned}$$

the strong semismoothness of the ℓ_1 and ℓ_∞ norms follows from Proposition **7.4.4** and the above remark about the strong semismoothness of the maximum of a finite family of affine functions.

Consider the case $p \in (1, \infty)$. In this case $\|\cdot\|_p$ is infinitely many times differentiable at every point except at the origin. By Proposition **7.4.5**(b), we only need to consider the strong semismoothnes of $\|\cdot\|_p$ at the origin. For simplicity we set $f \equiv \|\cdot\|_p$. At any $x \neq 0$ we have

$$\frac{\partial f(x)}{\partial x_i} \;=\; \frac{\mathrm{sign}(x_i)\,|\,x_i\,|^{p-1}}{\|\,x\,\|_p^{p-1}},$$

which implies

$$\nabla f(x)^T x \;=\; f(x), \quad \forall\, x \;\neq\; 0. \tag{7.4.11}$$

Since the norm function is positively homogeneous, we have

$$f'(0;d) \;=\; \lim_{t \downarrow 0} \frac{f(td) - f(0)}{t} \;=\; \lim_{t \downarrow 0} \frac{tf(d)}{t} \;=\; f(d). \tag{7.4.12}$$

By (7.4.11) and (7.4.12) it follows that for all $d \neq 0$,

$$f'(0;d) - f'(d;d) \;=\; 0,$$

which establishes the strong semismoothness of f at the origin. □

For a finitely representable convex set K, Theorem **4.5.2** shows that the Euclidean projector Π_K is a PC^1 function near a vector x where the projected vector $\bar{x} \equiv \Pi_K(x)$ satisfies the CRCQ. Thus, both the normal map $\mathbf{F}_K^{\mathrm{nor}}$ and the natural map $\mathbf{F}_K^{\mathrm{nat}}$ for the VI (K, F) with a polyhedral K and a C^1 function F are semismooth. The Fischer-Burmeister function:

$$\psi_{\mathrm{FB}}(a,b) \;=\; \sqrt{a^2 + b^2} - (\,a + b\,), \quad (\,a, b\,) \;\in\; \mathbb{R}^2$$

is strongly semismooth. In fact, for any scalar $\mu \in (0, 4)$, the modified FB function:

$$\psi_{\mathrm{FB}_\mu}(a,b) \;=\; \sqrt{(\,a - b\,)^2 + \mu\, a\, b} - (\,a + b\,), \quad (\,a, b\,) \;\in\; \mathbb{R}^2$$

is also strongly semismooth. One way to see this is by observing that the square root term defines an elliptic norm whose strong semismoothness can be shown similarly to the Euclidean norm. The function ψ_{FB_μ} is a C-function; it includes both the min function (when $\mu \to 0+$) and the FB function (when $\mu = 2$).

For any strongly semismooth vector function F and any $p \in [1, \infty]$, Propositions **7.4.4** and **7.4.8** imply that $\|F\|_p$ is strongly semismooth. The reader can easily verify that the function $\Phi(r, x_1, x_2)$ in Example **3.1.5** is semismooth.

Summarizing the various classes of nonsmooth functions that have been introduced, we present the following chain of implications that shows the relations between these classes.

$$\mathrm{PC}^1 \Rightarrow \text{semismooth} \Rightarrow \text{B-diff} \Leftrightarrow [\ \text{loc. Lip.} + \text{dir. diff}\ ].$$

7.4.1 SC^1 functions

Semismooth vector functions can be used to define an important class of real-valued functions. Specifically, we say that $\theta : \Omega \subseteq \mathbb{R}^n \to \mathbb{R}$, with Ω open, is *semismoothly differentiable*, or SC^1 in short, at a point $x \in \Omega$ if θ is continuously differentiable in a neighborhood of x and $\nabla\theta$ is semismooth at x. We say that θ is SC^1 on Ω if it is SC^1 at every point in Ω. Consistent with the smooth case, we call the generalized Jacobian of $\nabla\theta$ the *generalized Hessian* of the SC^1 function θ; we write

$$\partial^2\theta \equiv \partial\nabla\theta = \operatorname{conv}\partial_B\nabla\theta,$$

where the second equality expresses the generalized Hessian of θ as the convex hull of the B-subdifferential of $\nabla\theta$. The set $\partial^2\theta$ is well defined because $\nabla\theta$ is a Lipschitz continuous function. Given a vector $d \in \mathbb{R}^n$ with unit norm, we define a special subset of the generalized Hessian of θ that can viewed as "arising" from d. Denoted by $\partial_d^2\theta(x)$, this subset of generalized Hessian matrices is formally defined as follows:

$$\begin{aligned}\partial_d^2\theta(x) \equiv \{ \quad & H \in \partial^2\theta(x) : \\ & \exists\, \{ x^k \} \overset{d}{\to} x, \text{ and } \exists\, \{ H^k \} \to H \text{ with } H^k \in \partial^2\theta(x^k) \},\end{aligned}$$

where the notation $\{x^k\} \overset{d}{\to} x$ means that $\{x^k\}$ converges to x *in the direction* d, that is, $x^k \neq x$ eventually,

$$\lim_{k\to\infty} x^k = x, \quad \text{and} \quad \lim_{k\to\infty} \frac{x^k - x}{\| x^k - x \|} = d.$$

The set $\partial_d^2\theta(x)$ is called the *directional generalized Hessian* of θ at x along the direction d.

Semimsmoothly differentiable functions abound in applications. Twice continuously differentiable functions are certainly semismoothly differentiable. If G is any integrable semismooth vector function and $G = \nabla\theta$ for some real-valued function θ, then θ is SC^1. If G is any C^2 vector function, then $\theta(x) \equiv \max(0, G(x))^T \max(0, G(x))$ is SC^1. Exercise **7.6.8** gives a broad family of SC^1 functions that generalizes the latter function θ. Exercise **7.6.9** shows how a general constrained optimization problem can be converted into the unconstrained minimization of an SC^1 function. Many merit functions for the VI/CP are SC^1; see the next chapter for details on such functions. In what follows, we verify that the squared Fischer-Burmeister function ψ_{FB}^2 is (strongly) SC^1.

7.4.9 Example. The function $\theta(a,b) \equiv \psi^2_{\rm FB}(a,b)$ is continuously differentiable with gradient given by

$$\nabla\theta(a,b) \;=\; \left(\begin{array}{c} \dfrac{a}{\sqrt{a^2+b^2}} - 1 \\ \\ \dfrac{b}{\sqrt{a^2+b^2}} - 1 \end{array} \right) \psi_{\rm FB}(a,b), \quad \forall\,(\,a,b\,) \,\in\, \mathbb{R}^2.$$

This expression includes the case where $(a,b) = (0,0)$, which yields

$$\nabla\theta(0,0) \;=\; (0,0).$$

This is justified because $\psi_{\rm FB}(0,0) = 0$ and the multiplicative vector in front of $\psi_{\rm FB}(a,b)$ in the above expression of $\nabla\theta(a,b)$ is bounded. There are several similar expressions in the following analysis, which can all be justified in the same way. To show that $\nabla\theta(a,b)$ is strongly semismooth on the plane, it suffices to show that the function

$$(\,a,b\,) \,\in\, \mathbb{R}^2 \mapsto \frac{a}{\sqrt{a^2+b^2}}\,\psi_{\rm FB}(a,b), \quad \forall\,(\,a,b\,) \,\in\, \mathbb{R}^2$$

is everywhere strongly semismooth. In turn, substituting the definition of $\psi_{\rm FB}(a,b)$, we see that it suffices to show that the function

$$\varphi(a,b) \,\equiv\, \frac{a(a+b)}{\sqrt{a^2+b^2}}, \quad \forall\,(\,a,b\,) \,\in\, \mathbb{R}^2$$

is strongly semismooth at the origin $(0,0)$, where $\varphi(a,b) = 0$. We first show that this function is globally Lipschitz continuous on the plane. Thus we need to establish the existence of a constant $L > 0$ such that for any two pairs (a_i,b_i) for $i = 1,2$,

$$|\,\varphi(a_1,b_1) - \varphi(a_2,b_2)\,| \;\le\; L\,\sqrt{(a_1-a_2)^2 + (b_1-b_2)^2}. \tag{7.4.13}$$

This inequality is clearly valid with $L = 1$ if either pair (a_1,b_1) or (a_2,b_2) is zero. Suppose that both pairs are nonzero. We have

$$\begin{aligned} \varphi(a_1,b_1) - \varphi(a_2,b_2) \;=\; & \left(\frac{a_1}{\sqrt{a_1^2+b_1^2}} + \frac{a_2}{\sqrt{a_2^2+b_2^2}} \right) (\,a_1 - a_2\,) + \\ & \frac{a_1}{\sqrt{a_1^2+b_1^2}}\,\frac{a_2}{\sqrt{a_2^2+b_2^2}}\,(\,\sqrt{a_2^2+b_2^2} - \sqrt{a_1^2+b_1^2}\,). \end{aligned}$$

This identity easily shows that the inequality (7.4.13) holds with $L = 2$. We leave it as an exercise for the reader to verify that φ is directionally

differentiable everywhere in the plane. By the definition of strong semismoothness, it remains to show

$$\limsup_{(0,0)\neq(a,b)\to(0,0)} \frac{\varphi'((a,b);(a,b)) - \varphi'((0,0);(a,b))}{a^2+b^2} < \infty.$$

The function $\varphi(a,b)$ is positively homogeneous; hence,

$$\varphi'((0,0);(a,b)) = \varphi(a,b), \quad \forall\,(a,b) \in \mathbb{R}^2.$$

Moreover, for $(a,b) \neq (0,0)$, we can easily verify

$$\varphi'((a,b);(a,b)) = \varphi(a,b).$$

Consequently the numerator in the above limit is identically equal to zero. □

Using the basic properties of a semismooth vector function, we establish a second-order limit result for an SC^1 function, which extends the second-order Taylor theorem for a C^2 function.

7.4.10 Proposition. Let $\theta : \Omega \to \mathbb{R}$, with Ω open, be SC^1 at $x \in \Omega$. It holds that

$$\lim_{\substack{\|d\|\to 0 \\ H\in\partial\partial^2\theta(x+d)}} \frac{\theta(x+d) - \theta(x) - \nabla\theta(x)^T d - \frac{1}{2}\, d^T H d}{\|\, d\,\|^2} = 0.$$

Proof. By Theorem **7.4.3**, it suffices to show

$$\lim_{\|d\|\to 0} \frac{\theta(x+d) - \theta(x) - \nabla\theta(x)^T d - \frac{1}{2}\, d^T G'(x;d)}{\|\, d\,\|^2} = 0, \qquad (7.4.14)$$

where $G \equiv \nabla\theta$. By the B-differentiability of G, for every $\varepsilon > 0$, there exists $\delta > 0$ such that

$$\|\, d\,\| \le \delta \;\Rightarrow\; \|\, G(x+d) - G(x) - G'(x;d)\,\| \le \varepsilon\,\|\, d\,\|.$$

Fix an arbitrary vector d satisfying $\|d\| \le \delta$ and define the real-valued function $\psi : [0,1] \to \mathbb{R}$ by

$$\psi(t) \equiv \theta(x+td) - \theta(x) - t\,\nabla\theta(x)^T d, \quad t \in [0,1].$$

Thus $\psi(1) = \theta(x+d) - \theta(x) - \nabla\theta(x)^T d$ and $\psi(0) = 0$; moreover, for all $t \in [0,1]$, $\psi'(t)$ exists and

$$\begin{aligned} \psi'(t) &= (\,\nabla\theta(x+td) - \nabla\theta(x)\,)^T d \\ &= t\,G'(x;d)^T d + (\,G(x+td) - G(x) - G'(x;td)\,)^T d. \end{aligned}$$

Since $\psi(1) - \psi(0) = \int_0^1 \psi'(t)dt$, we deduce

$$\left| \theta(x+d) - \theta(x) - \nabla\theta(x)^T d - \tfrac{1}{2}\, d^T G'(x;d) \right| \le \varepsilon \,\|\, d \,\|^2.$$

Consequently, the desired limit (7.4.14) follows. □

Given an SC^1 function θ, the B-subdifferential $\partial_B \nabla\theta(x)$ of the gradient function $\nabla\theta$ at a vector x plays an important role in the local property of θ around x. To explain this role, we collect several properties of the B-subdifferential of a vector function, considered as a multifunction from $\mathbb{R}^n$ into the set of $n \times n$ matrices. These properties are similar to those of the generalized Clarke Jacobian.

7.4.11 Proposition. Let $G : \Omega \subseteq \mathbb{R}^n \to \mathbb{R}^n$, with Ω open, be locally Lipschitz continuous on Ω. The multifunction $\partial_B G : \Omega \to \mathbb{R}^{n\times n}$ is nonempty, compact valued and upper semicontinuous on Ω.

Proof. The B-subdifferential is nonempty because its convex hull yields the generalized Jacobian, which is nonempty. Moreover, $\partial_B G$ is locally bounded by the locally Lipschitz continuity of G. The closedness of the set $\partial_B G(x)$ follows by a standard "diagonal" argument. Suppose that $H^k \in \partial_B G(x)$ and $\{H^k\}$ converges to H^∞. There are sequences $\{x_\nu^k\}$ each converging to x and such that

$$\lim_{\nu\to\infty} JG(x_\nu^k) = H^k.$$

for every k. It then follows that

$$\lim_{\nu\to\infty} x_\nu^\nu = x \quad \text{and} \quad \lim_{\nu\to\infty} JG(x_\nu^\nu) = H^\infty,$$

thus establishing that $H^\infty \in \partial_B G(x)$. Hence $\partial_B G$ is compact valued.

We are left with the verification of the upper semicontinuity. Since $\partial_B G$ is compact valued and locally bounded, the upper semicontinuity is equivalent to its closedness. Therefore, assume that we have a sequence $\{x^k\}$ converging to x and a sequence $\{H^k\}$ converging to H^∞ with H^k in $\partial_B G(x^k)$ for every k. We need to show that $H \in \partial_B G(x)$. Again this follows by a simple diagonal argument; we leave the details to the reader. □

Consider the minimization problem

$$\begin{aligned} &\text{minimize} && \theta(x) \\ &\text{subject to} && x \in X, \end{aligned} \tag{7.4.15}$$

where X is a given subset of $\mathbb{R}^n$. In Subsection **3.3.4**, the concept of a strong local minimum of this problem is defined; it is established there that if θ is twice continuously differentiable and X is finitely representable and satisfies a CQ, then the strict copositivity of the Hessian of the Lagrangian function of (7.4.15) provides a sufficient condition for a KKT point of this problem to be a strong local minimum; see Corollary **3.3.20**. Proposition **7.4.12** below, which uses $\partial_d^2\theta$, complements this corollary; in the proposition, we assume that θ is only SC^1 and X is convex.

7.4.12 Proposition. Let $\theta : \mathbb{R}^n \to \mathbb{R}$ be an SC^1 function and X be a closed convex subset of $\mathbb{R}^n$. Let x^* be a stationary point of (7.4.15). Consider the following three statements.

(a) All matrices in $\partial_B \nabla\theta(x^*)$ are strictly copositive on the critical cone of the pair $(X, \nabla\theta)$ at x^*.

(b) For every d in the critical cone of the pair $(X, \nabla\theta)$ at x^* such that $\|d\| = 1$,
$$d^T H d > 0, \qquad \forall\, H \in \partial_d^2\theta(x^*). \tag{7.4.16}$$

(c) x^* is a strong local minimum of θ on X, i.e. a positive constant c and a neighborhood $\mathcal{N}$ of x^* exist such that
$$\theta(x) \geq \theta(x^*) + c\,\|x - x^*\|^2, \quad \forall\, x \in X \cap \mathcal{N}. \tag{7.4.17}$$

It holds that (a) $\Rightarrow$ (b) $\Rightarrow$ (c); furthermore, if X is polyhedral then (b) and (c) are equivalent.

Proof. Since $\partial_d^2\theta(x^*) \subseteq \partial^2\theta(x^*)$ and $\partial^2\theta(x^*) = \operatorname{conv} \partial_B \nabla\theta(x^*)$, it is clear that (a) implies (b). To prove that (b) implies (c), we assume for the sake of contradiction that this is false. There exists a sequence of feasible vectors $\{x^k\}$ converging to x^* such that for every k, $x^k \neq x$ and
$$\theta(x^k) < \theta(x^*) + k^{-1}\,\|\,x^k - x^*\,\|^2.$$
By Proposition **7.4.10**, we can write
$$\begin{aligned}\theta(x^k) \;&=\; \theta(x^*) + \nabla\theta(x^*)^T(\,x^k - x^*\,) + \\ &\tfrac{1}{2}\,(\,x^k - x^*\,)^T H^k\,(\,x^k - x^*\,) + o(\|\,x^k - x^*\,\|^2),\end{aligned}$$
where $H^k \in \partial_B \nabla\theta(x^k)$. Combining the above two expressions, we obtain
$$\begin{aligned}k^{-1}\,\|\,x^k - x^*\,\|^2 \;&>\; \nabla\theta(x^*)^T(\,x^k - x^*\,) + \\ &\tfrac{1}{2}\,(\,x^k - x^*\,)^T H^k\,(\,x^k - x^*\,) + o(\|\,x^k - x^*\,\|^2).\end{aligned} \tag{7.4.18}$$

Since X is convex, the stationarity of x^* means that x^* is a solution of the VI $(X, \nabla\theta)$. Since $x^k \in X$, we have

$$(\, x^k - x^* \,)^T \nabla\theta(x^*) \,\geq\, 0. \tag{7.4.19}$$

We may assume without loss of generality that the normalized sequence $\{(x^k - x^*)/\|x^k - x^*\|\}$ converges to a limit d^∞, which must necessarily be a nonzero tangent vector to X at x^*. From (7.4.19), we clearly have $\nabla\theta(x^*)^T d^\infty \geq 0$. Dividing by $\|x^k - x^*\|$ and letting $k \to \infty$ in (7.4.18) yield $\nabla\theta(x^*)^T d^\infty \leq 0$; thus equality holds. Hence d^∞ belongs the critical cone of the pair $(X, \nabla\theta)$ at x^*. Combining the two inequalities (7.4.18) and (7.4.19), we deduce

$$k^{-1} \,\|\, x^k - x^* \,\|^2 \,>\, \tfrac{1}{2} \,(\, x^k - x^* \,)^T H^k \,(\, x^k - x^* \,) + o(\|\, x^k - x^* \,\|^2). \tag{7.4.20}$$

By the upper semicontinuity of the generalized Hessian, we may assume without loss of generality that the sequence of matrices $\{H^k\}$ converges to some limit H^∞, which must necessarily belong to $\partial_d^2\theta(x^*)$. Since (b) holds, we have

$$(\, d^\infty \,)^T H^\infty d^\infty \,>\, 0$$

because d^∞ is nonzero. Dividing (7.4.20) by $\|x^k - x^*\|^2$ and letting $k \to \infty$, we obtain

$$0 \,\geq\, (\, d^\infty \,)^T H^\infty d^\infty,$$

which is a contradiction. Therefore, x^* is a strong local minimum of (7.4.15).

We now prove that (c) implies (b) if X is polyhedral. Suppose that there exist a positive constant c and a neighborhood $\mathcal{N}$ of x^* such that (7.4.17) holds. Assume for the sake of contradiction that (b) does not hold. There exists a direction d in the critical cone of $(X, \nabla\theta)$ at x^* and a matrix $\bar{H}$ in $\partial_d^2\theta(x^*)$ such that

$$d^T \bar{H} d \,\leq\, 0.$$

Since $\nabla\theta$ is semismooth at x^*, an easy application of Proposition **7.4.10** yields

$$d^T H d \,\leq\, 0, \qquad \forall\, H \,\in\, \partial_d^2\theta(x^*). \tag{7.4.21}$$

Consider the sequence defined by $x^k \equiv x^* + k^{-1}d$. Since X is polyhedral, Lemma **3.3.6** shows that x^k is feasible for every k large enough. Furthermore, the sequence $\{x^k\}$ clearly converges to x^* in the direction d. By Proposition **7.4.10**, we can write

$$\begin{aligned} \theta(x^k) - \theta(x^*) \,&=\, \nabla\theta(x^*)^T (\, x^k - x^* \,) + \\ &\tfrac{1}{2} \,(\, x^k - x^* \,)^T H^k \,(\, x^k - x^* \,) + o(\|\, x^k - x^* \,\|^2), \end{aligned}$$

where $H^k \in \partial^2\theta(x^k)$. Combining this equality with (7.4.17) and taking into account that $\nabla\theta(x^*)^T(x^k - x^*) = k^{-1}\nabla\theta(x^*)^T d = 0$, we have

$$c\,\|\,x^k - x^*\,\|^2 \;\leq\; \tfrac{1}{2}\,(\,x^k - x^*\,)^T H^k\,(\,x^k - x^*\,) + o(\|\,x^k - x^*\,\|^2).$$

Dividing by $\|x^k - x^*\|^2$ and passing to the limit we get

$$0 \;<\; c \;\leq\; \tfrac{1}{2}\,d^T H^\infty d,$$

where $H^\infty \in \partial_d^2\theta(x^*)$. This obviously contradicts (7.4.21) and concludes the proof. □

7.5 Semismooth Newton Methods

In this section we study Newton methods for solving semismooth systems of equations. Due to their importance, we perform an in-depth analysis of these methods and consider a variation that solves a linear least-squares subproblem at every iteration. Specifically, in addition to specializing the Newton and inexact Newton methods described in Section **7.2** to semismooth systems, we fully characterize the Q-superlinear rate of convergence of sequences of points to a solution and apply such a characterization to a Levenberg-Marquardt-type algorithm. One of the main advantages of the Newton-type methods we consider in this section is that they are very similar to the classical methods for smooth equations and retain the main characteristics of the latter methods while being applicable to nonsmooth systems. This permits an easy extension of all the technical and numerical tools that are available for smooth systems.

Consider the system (7.0.1) where G is (strongly) semismooth. We saw in the previous section that the Clarke generalized Jacobian $\partial G(x)$ is a (strong) Newton approximation scheme of G at x. Thus we can expect that, under a suitable nonsingularity condition, the following algorithm is locally well defined and Q-superlinearly convergent.

Semismooth Newton Method (SNM)

7.5.1 Algorithm.

Data: $x^0 \in \mathbb{R}^n$.

Step 1: Set $k = 0$.

Step 2: If $G(x^k) = 0$, stop.

Step 3: Select an element $H^k \in \partial G(x^k)$. Find a direction $d^k \in \mathbb{R}^n$ such

that

$$G(x^k) + H^k d^k \;=\; 0. \tag{7.5.1}$$

Step 4: Set $x^{k+1} \equiv x^k + d^k$ and $k \leftarrow k+1$; go to Step 2.

Algorithm **7.5.1** is particularly attractive because (7.5.1) is a system of linear equations that has a unique solution, provided that H^k is nonsingular. The next lemma essentially guarantees this nonsingularity property by postulating that all the generalized Jacobians of G at a zero x^* are nonsingular and that x^k is sufficiently close to x^*. We present this lemma in a form that is slightly more general than necessary at this stage, but that is useful for subsequent developments.

7.5.2 Lemma. Let $J : \Omega \subseteq \mathbb{R}^n \to \mathbb{R}^{n\times n}$, with Ω open, be a compact-valued, upper semicontinuous set-valued mapping. Suppose that at a point $\bar{x} \in \Omega$ all the matrices in $J(\bar{x})$ are nonsingular. There exist positive constants κ and δ such that

$$\sup_{\substack{x\in\mathbb{B}(\bar{x},\delta)\\ H\in J(x)}} \max\{\,\|H\|,\,\|H^{-1}\|\,\} \;\le\; \kappa.$$

In particular, all the matrices $H \in J(x)$ for $x \in \mathbb{B}(\bar{x},\delta)$ are nonsingular.

Proof. By the upper semicontinuity and compactness assumption on J, for every positive ε, we can find a positive δ such that $J(x)$ is contained in $J(\bar{x}) + \mathbb{B}(0,\varepsilon)$ for every $x \in \mathbb{B}(\bar{x},\delta)$. The lemma follows readily from this observation and from the continuity of the determinant. □

We state and prove the main convergence result of Algorithm **7.5.1**.

7.5.3 Theorem. Let $G : \Omega \subseteq \mathbb{R}^n \to \mathbb{R}^n$, with Ω open, be semismooth at $x^* \in \Omega$ satisfying $G(x^*) = 0$. If $\partial G(x^*)$ is nonsingular, then there exists a $\delta > 0$ such that, if $x^0 \in \mathbb{B}(x^*,\delta)$, the sequence $\{x^k\}$ generated by Algorithm **7.5.1** is well defined and converges Q-superlinearly to x^*. Furthermore, if G is strongly semismooth at x^*, then the convergence rate is Q-quadratic.

Proof. We only need to show that under the stated nonsingluarity assumption on the elements of $\partial G(x^*)$ the approximation scheme given by $\partial G(x)$ is nonsingular. Once this is done the theorem follows trivially from Theorem **7.2.5**. By assumptions and Lemma **7.5.2**, the quantity

$$L \;\equiv\; \sup_{H\in\partial G(x^*)} \{\|H\|, \|H^{-1}\|\},$$

is finite. We then see that all the elements in $\partial G(x^*)$ are global (linear) homeomorphism with the common Lipschitz constant L. By the upper semicontinuity of the generalized Jacobian mapping and by the compactness of $\partial G(x)$, it follows that for every positive ε, a positive δ exists such that if $x \in \mathbb{B}(x^*, \delta)$ and H belongs to $\partial G(x)$, there exists an element H^* in $\partial G(x^*)$ satisfying $\|H - H^*\| \leq \varepsilon$. We can invoke Theorem **7.2.13**(a) to complete the proof of the nonsingularity of the approximation scheme defined by $\partial G(x)$. □

Next we consider an inexact version of Algorithm **7.5.1**, corresponding to Algorithm **7.2.6**.

Semismooth Inexact Newton Method (SINM)

7.5.4 Algorithm.

Data: $x^0 \in \mathbb{R}^n$ and a sequence of nonnegative scalars $\{\eta_k\}$.

Step 1: Set $k = 0$.

Step 2: If $G(x^k) = 0$, stop.

Step 3: Select an element $H^k \in \partial G(x^k)$ find a direction $d^k \in \mathbb{R}^n$ such that

$$G(x^k) + H^k d^k = r^k,$$

where $r^k \in \mathbb{R}^n$ is a vector satisfying

$$\| r^k \| \leq \eta_k \|G(x^k)\|.$$

and η_k is a nonnegative number.

Step 4: Set $x^{k+1} \equiv x^k + d^k$ and $k \leftarrow k + 1$; go to Step 2.

We have shown in the proof of Theorem **7.5.3** that the nonsingularity of all the elements in $\partial G(x^*)$ induces the nonsingularity of the approximation scheme defined by the generalized Jacobian $\partial G(x)$. Hence the following theorem is an immediate consequence of Theorem **7.2.8** and requires no proof.

7.5.5 Theorem. Let $G : \Omega \subseteq \mathbb{R}^n \to \mathbb{R}^n$, with Ω open, be semismooth at $x^* \in \Omega$ satisfying $G(x^*) = 0$. If $\partial G(x^*)$ is nonsingular, then there exist a positive number $\bar{\eta}$ such that, if $\eta_k \leq \bar{\eta}$ for every k, there is a neighborhood $\mathbb{B}(x^*, \delta)$ of x^* such that if x^0 belongs to $\mathbb{B}(x^*, \delta)$, the inexact Newton method **7.5.4** is well defined and the sequence $\{x^k\}$ so generated

converges Q-linearly to x^*. Furthermore, if $\eta_k \to 0$ the convergence rate is Q-superlinear. Finally, if G is strongly semismooth at x^* and for some $\tilde{\eta} > 0$, $\eta_k \le \tilde{\eta}\|G(x^k)\|$ for all k, then the convergence rate is Q-quadratic. □

Again, we have to remark that the above inexact Newton method is very similar to its counterpart for smooth equations. The one difference is that in the smooth case we can take $\bar{\eta} \in (0,1)$ arbitrarily in order to obtain the local Q-linear convergence; whereas in the nonsmooth case, Theorem **7.5.5** just states that there exists an $\bar{\eta} > 0$ such that the sequence $\{x^k\}$ converges Q-linearly if $\eta_k \le \bar{\eta}$ holds for all k. This difference apart, all the efficient numerical techniques developed to solve linear systems inexactly can be used in the practical implementation of Algorithm **7.5.4**.

7.5.6 Example. Consider the function $F(x_1, x_2)$ of two variables defined by: for $x = (x_1, x_2)$,

$$F_1(x) = \begin{cases} 2x_1 & \text{if } x_1 \ge 0 \\ x_1 & \text{if } x_1 < 0 \end{cases} \quad \text{and} \quad F_2(x) = \begin{cases} x_2 & \text{if } x_2 \ge 0 \\ 2x_2 & \text{if } x_2 < 0. \end{cases}$$

The unique solution of the equation $F(x_1, x_2) = 0$ is $x^* = 0$, where F is not smooth. It is easy to check that all matrices in $\partial F(x^*)$ are nonsingular. For any given $\varepsilon > 0$, define the sequence $\{x^k\}$ as follows:

$$x^{2k} = (\varepsilon, \varepsilon/2), \quad \text{and} \quad x^{2k+1} = (-\varepsilon/2, -\varepsilon).$$

It is not hard to see that F is F-differentiable at every x^k; moreover,

$$F(x^{2k}) = (2\varepsilon, \varepsilon/2) \quad \text{and} \quad F(x^{2k+1}) = (-\varepsilon/2, -2\varepsilon).$$

By an easy calculation, we can show that

$$JF(x^k)^T(x^{k+1} - x^k) + F(x^k) = \begin{cases} (-\varepsilon, -\varepsilon) & \text{if } k \text{ is even} \\ (\varepsilon, \varepsilon) & \text{if } k \text{ is odd} \end{cases}$$

Thus, for all k,

$$\| JF(x^k)^T(x^{k+1} - x^k) + F(x^k) \| = \sqrt{2}\,\varepsilon$$

whereas

$$\| F(x^k) \| = (\sqrt{17}/2)\,\varepsilon.$$

Thus, provided that each η_k is not smaller than $2\sqrt{2}/\sqrt{17}$, $\{x^k\}$ is an acceptable inexact Newton sequence for all $\varepsilon > 0$, which is a constant. This sequence shows that we can choose x^0 arbitrarily close to x^*, yet Algorithm **7.5.4** may fail to converge if $\sup \eta_k$ is not small enough. □

Characterization of Q-superlinear convergence

Another special feature of the class of semismooth functions is that we can obtain a characterization of the Q-superlinear rate of convergence of a sequence of iterates to a zero of such a function, which is analogous to the classical Dennis-Moré condition for smooth functions. We first define an important concept and establish a simple lemma.

For a given convergent sequence $\{x^k\}$ with limit x^*, we say that a sequence $\{d^k\}$ is *superlinearly convergent* with respect to $\{x^k\}$ if the following limit holds:

$$\lim_{k\to\infty} \frac{\| x^k + d^k - x^* \|}{\| x^k - x^* \|} = 0. \tag{7.5.2}$$

Clearly, this limit implies that for some scalar $\delta > 0$,

$$(1-\delta)\,\| x^k - x^* \| \le \| d^k \| \le (1+\delta)\,\| x^k - x^* \|,$$

for all k sufficiently large. Thus, if $\{d^k\}$ is superlinearly convergent with respect to $\{x^k\}$, then $\{\|d^k\|\}$ is of the same order as $\{\|x^k - x^*\|\}$ in the sense that the above bounds hold for some constant $\delta > 0$. The following result is a refinement of this observation.

7.5.7 Lemma. Suppose that $\{x^k\}$ is a sequence of points in $\mathbb{R}^n$ converging to x^*. If $\{d^k\}$ is a superlinearly convergence sequence of vectors with respect to $\{x^k\}$, then the following limit holds:

$$\lim_{k\to\infty} \frac{\| d^k \|}{\| x^k - x^* \|} = 1. \tag{7.5.3}$$

Proof. By (7.5.2) we have that, for every positive ε we can find an index $k(\varepsilon)$ such that, for every $k \ge k(\varepsilon)$,

$$\varepsilon > \left\| \frac{x^k - x^*}{\| x^k - x^* \|} + \frac{d^k}{\| x^k - x^* \|} \right\| \ge \left| 1 - \frac{\| d^k \|}{\| x^k - x^* \|} \right|,$$

from which the limit (7.5.3) follows easily since ε is arbitrary. □

The next result provides a necessary and sufficient condition for the rate of convergence of a sequence of iterates to a zero of a semismooth function to be Q-superlinear.

7.5.8 Theorem. Let $G : \mathbb{R}^n \to \mathbb{R}^n$ be a locally Lipschitz function on the open convex set $\Omega \subseteq \mathbb{R}^n$. Let $\{x^k\}$ be any sequence in Ω converging to $x^* \in \Omega$ with $x^k \ne x^*$ for all k. Assume that all the matrices in $\partial G(x^*)$ are nonsingular. The following two statements hold.

(a) If G is semismooth at x^*, then $\{x^k\}$ converges Q-superlinearly to x^* and $G(x^*) = 0$ if and only if there exists a sequence $\{H^k\}$, where $H^k \in \partial G(x^k)$ for every k, such that

$$\lim_{k\to\infty} \frac{\| G(x^k) + H^k d^k \|}{\| d^k \|} = 0, \tag{7.5.4}$$

where $d^k \equiv x^{k+1} - x^k$.

(b) If G is strongly semismooth at x^*, then $\{x^k\}$ converges Q-quadratically to x^* and $G(x^*) = 0$ if and only if there exists a sequence $\{H^k\}$, where $H^k \in \partial G(x^k)$ for every k, such that

$$\limsup_{k\to\infty} \frac{\| G(x^k) + H^k d^k \|}{\| d^k \|^2} < \infty. \tag{7.5.5}$$

Proof. To prove (a), assume (7.5.4). Without loss of generality, we may assume that $x^{k+1} \neq x^k$ for every k. Let $e^k \equiv x^k - x^*$ denote the error vector at the k-th iterate. Then $d^k = e^{k+1} - e^k$ and both sequences $\{e^k\}$ and $\{d^k\}$ converge to 0. We can write

$$G(x^*) = [G(x^k) + H^k d^k] - [G(x^k) - G(x^*) - H^k e^k] - H^k e^{k+1}. \tag{7.5.6}$$

The semismoothness of G at x^* implies that the term in the second square bracket converges to zero as k tends to infinity; moreover, since $\{H^k\}$ is bounded by Lemma **7.5.2** and $e^{k+1} \to 0$, the last term in (7.5.6) also tends to zero. By (7.5.4), the term in the first square bracket also tends to 0; hence $G(x^*) = 0$. To establish the Q-superlinear convergence rate, we first need to show that

$$\limsup_{k\to\infty} \frac{\| d^k \|}{\| e^k \|} < \infty. \tag{7.5.7}$$

By the boundedness of the sequence $\{\|(H^k)^{-1}\|\}$, (7.5.4) easily yields

$$\| d^k \| = O(\| G(x^k) \|).$$

Since $G(x^*) = 0$ and G is Lipschitz continuous near x^*, the above expression implies that for suitable constants c_2 and L:

$$\limsup_{k\to\infty} \frac{\| d^k \|}{\| e^k \|} \leq c_2 \limsup_{k\to\infty} \frac{\| G(x^k) - G(x^*) \|}{\| e^k \|} \leq c_2 L,$$

which proves (7.5.7). From (7.5.6), we deduce

$$e^{k+1} = (H^k)^{-1}[(G(x^k) + H^k d^k) - (G(x^k) - G(x^*) - H^k e^k)]. \tag{7.5.8}$$

Taking norms and dividing by $\|e^k\|$ on both sides, using again the boundedness of $\{\|(H^k)^{-1}\|\}$, and (7.5.7), we can conclude that

$$\lim_{k\to\infty} \frac{\| e^{k+1} \|}{\| e^k \|} = 0, \tag{7.5.9}$$

which establishes the Q-superlinear convergence rate of the sequence $\{x^k\}$.

Conversely, suppose that $G(x^*) = 0$ and (7.5.9) holds. By reversing the above arguments we can easily deduce that

$$\lim_{k\to\infty} \frac{\| G(x^k) + H^k d^k \|}{\| e^k \|} = 0.$$

By Lemma **7.5.7**, (7.5.4) follows readily.

To prove part (b), assume first that (7.5.5) holds. This clearly implies that

$$\lim_{k\to\infty} \frac{\| G(x^k) + H^k d^k \|}{\| d^k \|} = 0.$$

By part (a), this implies in turn $G(x^*) = 0$ and $\{x^k\} \to x^*$ Q-superlinearly. Taking norms in (7.5.8), letting $c > 0$ be the bound for $\|(H^k)^{-1}\|$, and dividing by $\| e^k \|^2$, we obtain

$$\frac{\| e^{k+1} \|}{\| e^k \|^2} \le c \left(\frac{\| G(x^k) + H^k d^k \|}{\| d^k \|^2} \frac{\| d^k \|^2}{\| e^k \|^2} + \frac{\| G(x^k) - G(x^*) - H^k e^k \|}{\| e^k \|^2} \right).$$

Since we already know that $x^k \to x^*$ Q-superlinearly, we have

$$\lim_{k\to\infty} \frac{\| d^k \|}{\| e^k \|} = 1$$

by Lemma **7.5.7**. Combining this limit with (7.5.5) and (7.4.6), we immediately obtain

$$\limsup_{k\to\infty} \frac{\| e^{k+1} \|}{\| e^k \|^2} < \infty,$$

i.e., $\{x^k\}$ converges Q-quadratically to x^*.

Conversely, if $G(x^*) = 0$ and $\{x^*\}$ converge Q-quadratically to x^*, we can show that (7.5.5) holds by simply reversing the above arguments; we omit the details here. □

The Q-superlinear and Q-quadratic convergence of Algorithms **7.5.1** and **7.5.4** are immediate consequences of Theorem **7.5.8**, provided that the sequence $\{x^k\}$ generated by these algorithms can be shown to converge. The reason is obvious: the numerator in (7.5.2) and (7.5.3) is equal to zero identically. Theorem **7.5.8** has other uses. For instance, consider a sequence $\{x^k\}$ that satisfies

$$\| G(x^k) + G'(x^k; x^{k+1} - x^k) \| \le \eta_k \| G(x^k) \|,$$

where $\{\eta_k\}$ is a prescribed sequence of positive scalars. Provided that the sequence $\{x^k\}$ converges to a zero x^* of G where G is semismooth and $\{\eta_k\}$ converges to zero, then $\{x^k\}$ converges to x^* Q-superlinearly. This follows easily from Proposition **7.1.17** and Theorem **7.5.8**. If G is strongly semismooth at x^* and $\eta_k \leq \tilde{\eta}\|G(x^k)\|$ for all k, then $\{x^k\}$ converges to x^* Q-quadratically. Another use of Theorem **7.5.8** is to show that the conditions on the sequence of scalars $\{\eta_k\}$ given in Theorem **7.5.5** to guarantee the Q-superlinear and Q-quadratic convergence of the semismooth inexact Newton method are not only sufficient, but also necessary (see Exercise **7.6.15**). A further use of Theorem **7.5.8** is to establish the convergence property of algorithms that do not fall directly under the general framework of Section **7.2**. In what follows, we consider one such algorithm.

Semismooth Inexact LM Newton Method (SILMNM)

7.5.9 Algorithm.

Data: $x^0 \in \mathbb{R}^n$ and two sequences $\{\sigma_k\}$ and $\{\eta_k\}$ of nonnegative scalars.

Step 1: Set $k = 0$.

Step 2: If $G(x^k) = 0$, stop.

Step 3: Select an element $H^k \in \partial G(x^k)$ and find a direction $d^k \in \mathbb{R}^n$ such that

$$(\,(H^k)^T H^k + \sigma_k \mathbf{I}\,)d^k \;=\; -(H^k)^T G(x^k) + r^k, \tag{7.5.10}$$

where $r^k \in \mathbb{R}^n$ is a vector satisfying

$$\|\, r^k \,\| \;\leq\; \eta_k \,\|\, (H^k)^T G(x^k) \,\|. \tag{7.5.11}$$

Step 4: Set $x^{k+1} \equiv x^k + d^k$ and $k \leftarrow k+1$; go to Step 2.

7.5.10 Remark. We can also consider an "exact" version of the Algorithm **7.5.9** which has $\eta_k = 0$ for all k. We have presented directly the inexact version for brevity. In any event, the properties of the exact version can be easily derived from the analysis that follows. □

The motivation to consider the Levenberg-Marquardt method is mainly for computational purposes. Specifically, Algorithms **7.5.1** and **7.5.4** require at each iteration the solution (exactly or inexactly) of a linear system defined by the matrix $H^k \in \partial G(x^k)$ that is typically not symmetric. Although this matrix is guaranteed to be nonsingular in theory (by the non-

singularity of the matrices in the generalized Jacobian $\partial G(x^*)$, in practical implementation of the algorithms, this guarantee is not always effective because the iterate x^k is not necessarily close enough to the desired solution x^*, which is the object of the computations. Thus we could run into the situation of a singular system (7.5.1); when this happens, the two mentioned algorithms can not be continued. Another more likely situation is that the matrix H^k is nonsingular but ill-conditioned, thereby causing numerical instabilities and/or other difficulties when solving the equation (7.5.1) exactly or inexactly. There is yet another consideration when the equation (7.5.1) is solved. In large-scale problems, the matrix H^k is very large and highly sparse. In order to profitably exploit these special structures, one often applies an iterative solver (such as the conjugate gradient method) that does not destroy the sparsity of the matrix. For both practical and theoretical reasons, such an iterative solver is most effective when applied to symmetric systems of linear equations. The modified system (7.5.10) is symmetric. When H^k is nonsingular, the system (7.5.10) with $\sigma_k = 0$ and $r^k = 0$ is equivalent to (7.5.1). In general, the system

$$(H^k)^T H^k d \;=\; -(H^k)^T G(x^k) \tag{7.5.12}$$

is equivalent to the least-squares problem:

$$\text{minimize} \quad (\,G(x^k) + H^k d\,)^T (\,G(x^k) + H^k d\,).$$

The matrix $(H^k)^T H^k$ is symmetric positive semidefinite, and definite if and only if H^k is nonsingular. The latter least-squares problem, and thus its equivalent equation (7.5.12), always has a solution, regardless of whether H^k is nonsingular or not. The scalar σ_k, when chosen positive, ensures that the resulting matrix $(H^k)^T H^k + \sigma_k \mathbf{I}$ is positive definite. The latter property is the key for the successful application of such iterative linear-equation solvers as the conjugate gradient method and its many variations. In addition, this positive definiteness property also helps to maintain the numerical stability of the linear-equation solvers.

The modified system (7.5.10) is not without drawbacks. For one thing, the matrix $(H^k)^T H^k$ is likely to be completely dense even for a highly sparse H^k. Also the condition number of the product matrix is the square of that of H^k. Nevertheless, effective numerical solvers exist that can successfully deal with these two concerns. For example, to deal with the sparsity issue, these solvers can be implemented using only vector-matrix multiplications and without actually forming the product $(H^k)^T H^k$; thus any sparsity patterns that H^k might have can be exploited. The positive scalar σ_k helps to alleviate the issue of the squared condition number. The

choice of σ_k is a delicate matter in practice. We do not discuss this topic further, which is beyond the scope of this book. Suffices it to say that a main purpose of this scalar is to ensure the numerical stability of solving the system of Newton equations (7.5.10).

The following result is the main convergence of Algorithm **7.5.9**. We provide a direct proof of this result; the reader is asked to use Theorem **7.5.8** to prove parts (b) and (c).

7.5.11 Theorem. Let $G : \Omega \subseteq \mathbb{R}^n \to \mathbb{R}^n$, with Ω open, be semismooth at $x^* \in \Omega$ satisfying $G(x^*) = 0$. Assume that $\partial G(x^*)$ is nonsingular. There exist positive scalars $\bar{\eta}$ and $\bar{\sigma}$ and a neighborhood $\mathbb{B}(x^*, \delta)$ of x^* such that if $\eta_k \leq \bar{\eta}$ and $\sigma_k \leq \bar{\sigma}$ for every k, the following three statements hold:

(a) if x^0 belongs to $\mathbb{B}(x^*, \delta)$ then the inexact Levenberg-Marquardt Newton method **7.5.4** is well defined and every sequence $\{x^k\}$ so generated converges Q-linearly to x^*;

(b) if in addition
$$\lim_{k\to\infty} \eta_k = \lim_{k\to\infty} \sigma_k = 0,$$
the convergence rate is Q-superlinear;

(c) finally, if G is strongly semismooth at x^* and for some positive $\tilde{\eta}$ and $\tilde{\sigma}$, $\eta_k \leq \tilde{\eta}\|(H^k)^T G(x^k)\|$ and $\sigma_k \leq \tilde{\sigma}\|(H^k)^T G(x^k)\|$ for all k, then the convergence rate is Q-quadratic.

Proof. By Lemma **7.5.2**, there exist positive constants c and σ' and a neighborhood $\mathbb{B}(x^*, \delta_1)$ of x^* such that for all $x \in \mathbb{B}(x^*, \delta_1)$, $\sigma \in [0, \sigma']$, and $H \in \partial G(x^*)$, the matrix $H^T H + \sigma \mathbf{I}$ is nonsingular, and

$$\| (H^T H + \sigma \mathbf{I})^{-1} \| \leq c. \tag{7.5.13}$$

Without loss of generality, we may assume that the neighborhood $\mathbb{B}(x^*, \delta_1)$ is where G is Lipschitz continuous with modulus $L > 0$ and (in view of the upper semicontinuity of the generalized Jacobian) for some constant $\kappa > 0$,

$$\| H^T \| \leq \kappa, \quad \forall H \in \partial G(x), \tag{7.5.14}$$

for all $x \in \mathbb{B}(x^*, \delta_1)$. Choose positive scalars β, $\bar{\sigma}$ and $\bar{\eta}$ such that $\bar{\sigma} \leq \sigma'$ and

$$\tau \equiv c(\kappa\beta + \bar{\sigma} + \bar{\eta}\kappa L) < 1.$$

Corresponding to the scalar β, by Theorem **7.4.3**, there exists a neighborhood $\mathbb{B}(x^*, \delta_2)$ of x^* such that for all x in this neighborhood and all $H \in \partial G(x)$,

$$\| G(x) - G(x^*) - H(x - x^*) \| \leq \beta \| x - x^* \|. \tag{7.5.15}$$

Let $\delta \equiv \min(\delta_1, \delta_2)$. For all $x \in \mathbb{B}(x^*, \delta)$ and all $H \in \partial G(x)$, we have

$$\| H^T G(x) \| \leq \| H^T \| \, \| G(x^k) - G(x^*) \| \leq \kappa L \| x^k - x^* \|. \tag{7.5.16}$$

Let x^0 be an arbitrary vector in the neighborhood $\mathbb{B}(x^*, \delta)$. The vector x^1 is well defined because the matrix $H_0^T H_0 + \sigma_0 \mathbf{I}$ is nonsingular. From the steps of Algorithm **7.5.9**, we deduce

$$\begin{aligned}
&[\, H_0^T H_0 + \sigma_0 \mathbf{I} \,] \, (x^1 - x^*) \\
&\quad = [\, H_0^T H_0 + \sigma_0 \mathbf{I} \,] \, (x^k - x^*) - H_0^T G(x^0) + r^0 \\
&\quad = H_0^T [\, G(x^*) - G(x^0) + H_0 (x^0 - x^*) \,] + \sigma_0 \, (x^0 - x^*) + r^0.
\end{aligned}$$

Thus, we have

$$\begin{aligned}
\| x^1 - x^* \| &\leq c \, [\, \| H_0^T \| \, \| G(x^0) - G(x^*) - H_0 (x^0 - x^*) \| \\
&\qquad + \sigma_0 \| x^0 - x^* \| + \eta_0 \| H_0^T G(x^0) \| \,] \\
&\leq c \, [\, \kappa \beta \| x^0 - x^* \| + \bar{\sigma} \| x^0 - x^* \| + \bar{\eta} \kappa L \| x^0 - x^* \| \,] \\
&= \tau \| x^0 - x^* \|,
\end{aligned}$$

where we have used the inequalities (7.5.11)–(7.5.16). Since τ is less than one, it follows that x^1 belongs to the neighborhood $\mathbb{B}(x^*, \delta)$. We may repeat the above derivation, with x^1 replacing x^0 and x^2 replacing x^1. By an inductive argument, we can therefore establish that the entire sequence $\{x^k\}$ is well defined and satisfies:

$$\| x^{k+1} - x^* \| \leq \tau \| x^k - x^* \|, \quad \forall \, k.$$

Consequently, the sequence $\{x^k\}$ converges to x^* Q-linearly. Moreover, the above derivation also shows that

$$\begin{aligned}
\| x^{k+1} - x^* \| \leq \; & c \, \| (H^k)^T \| \, \| G(x^k) - G(x^*) - H^k (x^k - x^*) \| + \\
& c \, (\sigma_k + \eta_k L) \, \| x^k - x^* \|.
\end{aligned}$$

Consequently, if $\{\eta_k\}$ and $\{\sigma_k\}$ both converge to zero, the convergence of the sequence $\{x^k\}$ is Q-superlinear. The Q-quadratic rate follows readily under the strong semismoothness assumption and the further restrictions on η_k and σ_k. □

7.5.12 Remark. Part (a) of Theorem **7.5.11** remains valid if condition (7.5.11) is replaced by $\|r^k\| \leq \eta_k \|G(x^k)\|$. This is due to the boundedness of the sequence $\{H^k\}$. Under this modified inexact rule, part (b) of the

same theorem remains valid too. Part (c) remains valid if $\sigma_k = O(\|G(x^k)\|)$ and $\|r^k\| = O(\|G(x^k)\|^2)$. These are alternative inexact rules that do not affect the convergence properties of Algorithm **7.5.4**. □

By the proofs of Theorems **7.5.8** and **7.5.11**, we can establish the following. If $\{x^k\}$ is any convergent sequence (no matter how it is generated) with limit x^*, and if $\partial G(x^*)$ is nonsingular, then for any sequence of vectors $\{d^k\}$ satisfying (7.5.10) for every k, where $\sigma_k = O(\|G(x^k)\|)$ and $\|r^k\| = O(\|G(x^k)\|^2)$, it holds that

$$\lim_{k\to\infty} \frac{\| x^k + d^k - x^* \|}{\| x^k - x^* \|} = 0; \tag{7.5.17}$$

that is, such a sequence $\{d^k\}$ must be superlinearly convergent with respect to $\{x^k\}$. This observation is useful for instance in the context of a line search method (see Section **8.3**), which generates a sequence $\{x^k\}$ of iterates along with a sequence $\{d^k\}$ of search directions. If it can be shown that $\{x^k\}$ converges, then the above result can be used to establish the Q-superlinear convergence of $\{x^k\}$ to its limit by verifying that d^k satisfies (7.5.10); for details, see the mentioned section.

7.5.1 Linear Newton approximation schemes

As noted previously, the semismooth Newton methods are very appealing for several reasons. However, the choice to use elements of the generalized Jacobians of ∂G to define these methods is not without drawbacks. In fact, the calculation of generalized Jacobians is not always an easy task and, in some cases, it may even turn out to be too difficult a task to accomplish. Another point to consider is that to ensure a nonsingular approximation scheme we have to postulate that all the matrices in $\partial G(x^*)$ are nonsingular. This condition may turn out to be too strong and not necessary for convergence. A simple example such as the absolute value function of one variable easily illustrates this situation. The function $|x|$ has a unique zero at $x^* = 0$; but $\partial|0|$ is equal to the interval $[-1, 1]$, which contains the singular element 0. It is trivial to check that starting from any nonzero scalar, the exact version of Algorithm **7.5.4** with $\sigma_k = 0$ converges in one step.

To overcome the above drawback, we note that in the development of the semismooth Newton methods, we have relied on two special properties of the generalized Jacobian:

(a) as a multifunction, the generalized Jacobian is nonempty valued, compact valued, and upper semicontinuous at x^*, and its elements are matrices;

(b) the generalized Jacobian defines a (strong) Newton approximation scheme of G at x^* if G is (strongly) semismooth at x^*.

In particular, these properties allow us to show that the nonsingularity of all elements in $\partial G(x^*)$ is sufficient for the nonsingularity of the approximation scheme $\partial G(x)$ at x^* (cf. Lemma **7.5.2**). This discussion motivates the following definition.

7.5.13 Definition. Let $G : \Omega \subseteq \mathbb{R}^n \to \mathbb{R}^m$, with Ω open, be locally Lipschitz on Ω. We say that G admits a *linear Newton approximation* at a vector $\bar{x} \in \Omega$ if there exists a multifunction $T : \Omega \to \mathbb{R}^{n\times n}$ such that $T(x)$ is a Newton approximation scheme of G at $\bar{x}$ and T has nonempty compact images and is upper semicontinuous at $\bar{x}$. If T is a strong Newton approximation scheme, then we say the G admits a *strong linear Newton approximation* at $\bar{x}$. We also say that T is a (strong) linear Newton approximation scheme of G. □

This definition is an attempt to capture when a function G has a Newton approximation scheme defined by linear functions such that the nonsingularity of the scheme is the immediate consequence of the nonsingularity of a set of distinguished matrices. With this definition it is clear that (strongly) semismooth functions are a class of functions for which the generalized Jacobian defines a (strong) linear Newton approximation scheme. More interestingly, semismooth functions can have linear Newton approximations other than the generalized Jacobian (some examples are given later in this section). As one can expect, functions that are not semismooth can admit linear Newton approximations also; see the next chapter. In all these cases we can similarly define Newton, inexact Newton, Levenberg-Marquardt methods and characterize their convergence rate as done in the previous section. It is not particularly useful to repeat all these developments. As an illustration we just describe the basic Newton method and its convergence properties; the other extensions can be obtained along similar lines.

Linear Newton Method (LNM)

7.5.14 Algorithm.

Data: Let $x^0 \in \mathbb{R}^n$.

Step 1: Set $k = 0$.

Step 2: If $G(x^k) = 0$, stop.

Step 3: Select an element $H^k \in T(x^k)$. Find a direction $d^k \in \mathbb{R}^n$ such

that

$$G(x^k) + H^k d^k \;=\; 0.$$

Step 4: Set $x^{k+1} \equiv x^k + d^k$ and $k \leftarrow k+1$; go to Step 2.

The following is the convergence result for the above method. No proof is required.

7.5.15 Theorem. Assume that $G : \Omega \subseteq \mathbb{R}^n \to \mathbb{R}^n$, with Ω open, is locally Lipschitz continuous on Ω and has a linear Newton approximation scheme T at $x^* \in \Omega$, which satisfies $G(x^*) = 0$. Suppose that all the matrices H belonging to $T(x^*)$ are nonsingular. There exists a $\delta > 0$ such that, if $x^0 \in \mathbb{B}(x^*, \delta)$, the sequence $\{x^k\}$ generated by Algorithm **7.5.14** is well defined and converges Q-superlinearly to the solution x^*. Furthermore, if the linear approximation scheme T is strong, then the convergence rate is Q-quadratic. □

In the remaining part of this section we look at possible linear approximation schemes of nonsmooth functions. There is in general a trade-off: on the one hand, the larger the set $T(x)$ is, the easier it is to calculate an element H belonging to $T(x)$. On the other hand, the larger the set $T(x^*)$ is, the more unlikely it is for all the elements in $T(x^*)$ to be nonsingular. It is therefore useful to have many choices of such schemes to pick from in order to meet practical needs.

An important choice is to take $T(x)$ to be the B-subdifferential $\partial_B G(x)$. Recalling Proposition **7.4.11**, we can prove that the B-subdifferential defines a (strong) linear Newton approximation scheme, if G is (strongly) semismooth. This follows easily from Theorem **7.4.3** and the fact that $\partial_B G(x)$ is a subset of $\partial G(x)$.

7.5.16 Proposition. Assume that $G : \Omega \subseteq \mathbb{R}^n \to \mathbb{R}^n$ is Lipschitz continuous in a neighborhood of $\bar{x} \in \Omega$ and (strongly) semismooth at $\bar{x}$. Then $\partial_B G$ defines a (strong) linear Newton approximation scheme of G at $\bar{x}$. □

The use of the B-subdifferential to define a linear Newton approximation facilitates the verification of the nonsingularity of such an approximation, since the B-subdifferential is contained in the generalized Jacobian. An example is the absolute value function whose B-subdifferential at the origin is $\partial_B f(0) = \{-1, 1\}$. On the flip side, the B-subdifferential may be too small. In particular, to make the calculation of an element in $T(x)$ easier, we may want to consider a larger set than $\partial G(x)$. An example of the kind

of mapping T that can be used in practical situations is

$$\partial_C G(x) \equiv (\,\partial G_1(x) \times \partial G_2(x) \times \partial G_n(x)\,)^T\,, \tag{7.5.18}$$

which contains $\partial G(x)$, by Proposition **7.1.14**. The subscript "C" signifies the Cartesian product structure of the set on the right-hand side. The fact that this map $\partial_C G$ is a linear Newton approximation can be proved by a straightforward verification of the defining conditions of such an approximation. Instead of proving this fact, we present the next theorem, where a systematic way of generating linear Newton approximations of a composite map is considered. For simplicity, we restrict to maps defined on the whole space, but obvious extensions can be made for maps with arbitrary open domains.

7.5.17 Theorem. Suppose that $G : \mathbb{R}^n \to \mathbb{R}^n$ is a locally Lipschitz continuous map given by the composition of two maps: $G \equiv A \circ B$, where $B : \mathbb{R}^n \to \mathbb{R}^m$ and $A : \mathbb{R}^m \to \mathbb{R}^n$ are both locally Lipschitz. Suppose that T_A and T_B are (strong) linear Newton approximation schemes of A and B at $B(\bar{x})$ and $\bar{x}$ respectively. Then

$$T(x) \equiv \{\, VW \,:\, V \in T_A(B(x)),\, W \in T_B(x)\,\},$$

is a (strong) linear Newton approximation scheme of G at $\bar{x}$.

Proof. All the elements in $T(x)$ are obviously linear maps; furthermore the point-to-set map T has nonempty compact images and is upper semicontinuous. It remains to show that T is a (strong) Newton approximation scheme of G. Let $x \in \mathbb{R}^n$ be sufficiently close to $\bar{x}$ and set $\bar{y} = B(\bar{x})$. Let H belong to $T(x)$, so that there are $V \in T_A(y)$ and $W \in T_B(x)$, where $y = B(x)$, with $H = VW$. Then we have

$$B(x) + W(\bar{x} - x) - B(\bar{x}) \;=\; o(\|\, x - \bar{x}\,\|), \tag{7.5.19}$$

which implies

$$\|\, y - \bar{y}\,\| \;=\; O(\|\, x - \bar{x}\,\|) \tag{7.5.20}$$

by the local compactness and upper semicontinuity of T_B. Then we have

$$\begin{aligned}
G(x) \;&=\; A(B(x)) \;=\; A(y) \\
&=\; A(\bar{y}) - V(\bar{y} - y) + o(\|\, y - \bar{y}\,\|) && \text{because } V \in T_A(y) \\
&=\; A(\bar{y}) - V(W(\bar{x} - x) + o(\|\, x - \bar{x}\,\|)) \\
&\quad + o(\|\, x - \bar{x}\,\|) && \text{by (7.5.19) and (7.5.20)} \\
&=\; A(B(\bar{x})) - H(\bar{x} - x) + o(\|\, x - \bar{x}\,\|) && \text{by local boundedness of } T_A.
\end{aligned}$$

This proves that T is a linear Newton approximation at $\bar{x}$. The proof for the strong case is similar and the details are left to the reader. □

We point out the importance of this theorem by a simple example. Suppose that $G = G_1 + G_2$ and assume that both G_1 and G_2 are semismooth at $\bar{x}$. Then ∂G_1 and ∂G_2 linear Newton approximations of G_1 and G_2, respectively. Obviously, ∂G is also a linear Newton approximation of G. However, it is not easy to calculate ∂G from ∂G_1 and ∂G_2, because although the inclusion

$$\partial G(x) \subseteq \partial G_1(x) + \partial G_2(x)$$

holds in general, by the calculus rules of the generalized Jacobian (see Proposition **7.1.9**), equality does not necessarily hold, unless stringent requirements are made on G_1 and G_2. What Theorem **7.5.17** tells us is that although $T(x) \equiv \partial G_1(x) + \partial G_2(x)$ may not coincide with the generalized Jacobian of G, $T(x)$ is still a linear Newton approximation of G. We can generalize this observation to many other important cases as stated in the following corollary, which is presented for real-valued functions but some of whose parts can easily be extended to vector functions.

7.5.18 Corollary. Suppose that $g_1 : \mathbb{R}^n \to \mathbb{R}$ and $g_2 : \mathbb{R}^n \to \mathbb{R}$ are two locally Lipschitz continuous functions. If T_1 and T_2 are (strong) linear Newton approximations of g_1 and g_2 at $\bar{x}$, respectively, then the following statements hold.

(a) $T_1 + T_2$ is a (strong) linear Newton approximation of $g_1 + g_2$ at $\bar{x}$.

(b) For every scalar c, cT_1 is a (strong) linear Newton approximation of cg_1 at $\bar{x}$

(c) $g_1T_2 + g_2T_1$ is a (strong) linear Newton approximation of g_1g_2 at $\bar{x}$.

(d) If $g_2(\bar{x}) \neq 0$, then

$$\frac{g_2T_1 - g_1T_2}{g_2}$$

is a (strong) linear Newton approximation of g_1/g_2 at $\bar{x}$.

(e) $T_1 \times T_2$ is a (strong) linear Newton approximation of (g_1, g_2) at $\bar{x}$.

(f) T, defined as

$$T(x) \equiv \begin{cases} T_1(x) & \text{if } g_1(x) < g_2(x), \\ T_2(x) & \text{if } g_1(x) > g_2(x), \\ T_1(x) \cup T_2(x) & \text{if } g_1(x) = g_2(x), \end{cases}$$

is a (strong) linear Newton approximation of $\min\{g_1(x), g_2(x)\}$ at $\bar{x}$.

Proof. The results from (a) to (e) are all immediate consequences of Theorem **7.5.17** and simple calculus rules for the derivatives of sums, products and so on. Only part (f) requires some additional words. The rule for the min function is derived also from Theorem **7.5.17** once we observe that a linear Newton approximation scheme S of the function $\min\{x_1, x_2\}$ is given by

$$S(x) \equiv \begin{cases} \{\,(1,0)\,\} & \text{if} \quad x_1 < x_2, \\ \{\,(0,1)\,\} & \text{if} \quad x_1 > x_2, \\ \{\,(1,0),\,(0,1)\,\} & \text{if} \quad x_1 = x_2; \end{cases}$$

the verification of the latter fact is left as an exercise for the reader. □

7.6 Exercises

7.6.1 Give an example of a univariate, Lipschitz continuous, monotone function that is not directionally differentiable at the origin.

7.6.2 Let $F : \mathbb{R} \to \mathbb{R}$ be thrice continuously differentiable. Show that if x^* is a zero of F and $F'(x^*) \neq 0$ while $F''(x^*) = 0$, Newton method (locally) converges at least Q-cubically; that is,

$$\limsup_{k \to \infty} \frac{\| x^{k+1} - x^* \|}{\| x^k - x^* \|^3} < \infty.$$

7.6.3 Compute the limiting Jacobian of the C-function $\psi_{\mathrm{U}}(a, b)$ given in Exercise **1.8.21**.

7.6.4 Generalize Example **7.1.3** to the ℓ_p-norm for any $p \geq 1$. Specifically, show that $\partial \|x\|_p$ at $x = 0$ is equal to the set

$$\{\, x \in \mathbb{R}^n \,:\, \| x \|_q \leq 1 \,\},$$

where q satisfies $1/p + 1/q = 1$.

7.6.5 Let $f : \mathbb{R} \to \mathbb{R}$ be the univariate function in Example **7.4.1** and define $g : \mathbb{R}^2 \to \mathbb{R}$ by $g(x_1, x_2) \equiv f(\min(|x_1|, |x_2|))$. Show that g is locally Lipschitz continuous on $\mathbb{R}^2$ but g is not directionally differentiable at the origin.

7.6.6 Let $G : \mathbb{R}^n \to \mathbb{R}^m$ be B-differentiable near the origin. Show that if G is positively homogeneous, then G is semismooth at the origin. (Hint: show that $Vx = G(x)$ for every $V \in \partial G(x)$ for every x.)

7.6.7 Use Exercise **1.8.28** to show that the Euclidean projector Π_K onto the Lorentz cone is semismooth. According to Exercise **4.8.3**, Π_K is everywhere B-differentiable.

7.6.8 Let K be a closed convex set in $\mathbb{R}^n$.

(a) Let $G : \Omega \to \mathbb{R}^n$ be a continuously differentiable function on the open subset Ω of $\mathbb{R}^k$. Show that the composite squared distance function $\rho(y) \equiv \frac{1}{2}\,\mathrm{dist}(G(y), K)^2$ is C^1 on Ω with

$$\nabla\rho(y) \;=\; JG(y)^T(\,G(y) - \Pi_K(G(y))\,).$$

It follows that if Π_K and JG are semismooth, then ρ is SC^1.

(b) Let $G(y) \equiv r + Ey$ for some affine pair (r, E) and let $K \equiv P(A, b)$. Recall the family $\mathcal{I}(A, b)$ of index sets (4.1.5) that define the normal manifold induced by the polyhedron $P(A, b)$. Show that

$$\partial_B\nabla\rho(y) \subseteq \{\, E^TB^T(BB^T)^{-1}BE \;:\; \text{rows of } B$$
$$\text{form a basis of the rows of } A_{I\cdot} \text{ for some } I \in \mathcal{I}(A, b)\,\}.$$

Note that the matrices in the right-hand set are symmetric positive semidefinite. This is consistent with Exercise **2.9.13**, which shows that ρ is convex if G is affine; see also Exercise **7.6.12**.

(c) Let $\theta(y) \equiv \|\max(0, r + Ey)\|^2$. Verify that

$$\partial^2\theta(y) \;\subseteq\; \mathrm{conv}\{\, E^TD(J)E \;:\; J \;\subseteq\; \mathcal{P}(y)\,\}$$

where $\mathcal{P}(y)$ is the active index set at y; i.e.,

$$\mathcal{P}(y) \;\equiv\; \{\, i \;:\; r_i + E_{i\cdot}y \;\geq\; 0\,\},$$

and $D(J)$ is the diagonal matrix whose diagonal entries are given by:

$$D(J)_{jj} \;\equiv\; \begin{cases} 1 & \text{if } j \in J \\ 0 & \text{otherwise.} \end{cases}$$

7.6.9 Let K be a closed convex set in $\mathbb{R}^n$ and let $f : \mathbb{R}^n \to \mathbb{R}$ be a twice continuously differentiable function such that

$$\sup_{x\in\mathbb{R}^n} \lambda_{\max}(\nabla^2 f(x)) < \infty.$$

Let $\tau > 0$ be such that $\mathbf{I} - \tau\nabla^2 f(x)$ is positive definite for all $x \in \mathbb{R}^n$. Consider the nonlinear program

$$\begin{aligned} &\text{minimize} && f(x) \\ &\text{subject to} && x \in K. \end{aligned}$$

(a) Show that a vector is a stationary point of the above NLP if and only if the vector is an unconstrained stationary point of the C^1 function:

$$\psi(x) \equiv \tau f(x) - \tfrac{1}{2}\tau^2 \nabla f(x)^T \nabla f(x) + \tfrac{1}{2} \operatorname{dist}(x - \tau \nabla f(x), K)^2.$$

Show further that

$$\nabla \psi(x) = (\mathbf{I} - \tau \nabla^2 f(x))(x - \Pi_K(x - \tau \nabla f(x))).$$

Thus ψ is SC^1 if Π_K and $\nabla^2 f$ are semismooth.

(b) Let $f(x) \equiv q^T x + \frac{1}{2} x^T M x$, where q is arbitrary and M is symmetric positive (semi)definite (K remains the same). Suppose τ is a scalar satisfying $0 < \tau < 1/\lambda_{\max}(M)$. Show that $M - \tau M^2$ is positive (semi)definite and ψ is (strongly) convex.

(c) The following special quadratic program arises from the support vector machine: find $w \in \mathbb{R}^n$, $y \in \mathbb{R}^m$, and $t \in \mathbb{R}$ to

$$\begin{array}{ll} \text{minimize} & q^T y + \frac{1}{2} w^T w \\ \text{subject to} & Aw - tp + y \geq b \\ \text{and} & y \geq 0. \end{array}$$

Show that this quadratic program is equivalent to the quadratic program in the variable $v \in \mathbb{R}^m$:

$$\begin{array}{ll} \text{minimize} & f(v) \equiv \frac{1}{2} v^T A A^T v - b^T v \\ \text{subject to} & v \in K \equiv \{ v \in [0, q] : p^T v = 0 \}. \end{array}$$

Note that the set K is the intersection of a rectangle and a halfspace; as such the projection Π_K is easy to compute. By the first two parts, the above quadratic program is further equivalent to the unconstrained minimization of a convex piecewise quadratic function.

7.6.10 This exercise aims at deriving a second-order Taylor expansion with remainder term for an LC^1 function, which is a scalar-valued C^1 function with a locally Lipschitz continuous gradient function.

(a) Let $F : \Omega \to \mathbb{R}^m$ be a locally Lipschitz continuous function on the open convex subset Ω of $\mathbb{R}^n$. Let $x \in \Omega$ and $d \in \mathbb{R}^n$ be two given n-vectors. Let $\Omega_{x,d}$ be an open interval consisting of scalars t such that $x + td \in \Omega$. Define $G(t) \equiv F(x + td)$ for $t \in \Omega_{x,d}$. Show that for every $t \in \Omega_{x,d}$,

$$\partial G(t) \subseteq \partial F(x + td) d \equiv \{ Ad : A \in \partial F(x + td) \}.$$

Give an example to show that the inclusion is proper.

(b) Let $m = n$ in (a) and define $\varphi(t) \equiv F(x+td)^T d$ for $t \in \Omega_{x,d}$. Show that

$$\partial\varphi(t) \subseteq d^T \partial F(x+td) d = \{ d^T A d : A \in \partial F(x+td) \}.$$

(c) Let $\varphi : \mathbb{R} \to \mathbb{R}$ be an LC^1 function on an open set containing $[0,1]$. By using Proposition **7.1.12** extend classical arguments to show that

$$\varphi(1) - \varphi(0) - \varphi'(0) \in \tfrac{1}{2}\, \partial\varphi'(t)$$

for some $t \in [0,1]$.

(d) Let $\theta : \Omega \to \mathbb{R}$ be LC^1 on the open convex subset Ω of $\mathbb{R}^n$. Use classical arguments and the above part (c) with $\varphi(t) \equiv \theta(x+t(y-x))$ to show that for any two vectors x and y in Ω there exist a vector $z \in (x,y)$ and a matrix $A \in \partial^2\theta(z)$ satisfying

$$\theta(y) - \theta(x) - \nabla\theta(x)^T (\, y - x\,) = \tfrac{1}{2}\, (\, y - x\,)^T A (\, y - x\,).$$

Thus,

$$|\, \theta(y) - \theta(x) - \nabla\theta(x)^T (\, y - x\,)\,| \leq \sup_{\substack{A \in \partial^2\theta(z) \\ z \in [x,y]}} \tfrac{1}{2}\, \|\, A \,\|\, \|\, y - x \,\|^2.$$

7.6.11 Let $H : \mathbb{R}^{2n+m} \to \mathbb{R}^{n+m}$ be continuously differentiable. Define the function

$$\mathbf{H}_{\mathrm{FB}}(x,y,z) = \begin{pmatrix} \psi_{\mathrm{FB}}(x_1, y_1) \\ \vdots \\ \psi_{\mathrm{FB}}(x_n, y_n) \\ H(x,y,z) \end{pmatrix},$$

where

$$\psi_{\mathrm{FB}}(a,b) = \sqrt{a^2 + b^2} - a - b$$

is the FB C-function. Show that a linear Newton approximation of $\mathbf{H}_{\mathrm{FB}}$ at (x,y,z) is provided by the family of matrices:

$$T(x,y,z) \equiv \left\{ \begin{pmatrix} D_x & D_y & 0 \\ J_x H(x,y,z) & J_y H(x,y,z) & J_z H(x,y,z) \end{pmatrix} \right\},$$

where D_x and D_y are diagonal matrices whose diagonal entries are given by: for $i = 1, \ldots, n$,

$$(\,(D_x)_{ii}, (D_y)_{ii}\,) \begin{cases} = \dfrac{(\, x_i, y_i\,)}{\sqrt{x_i^2 + y_i^2}} - (1,1) & \text{if } (\, x_i, y_i\,) \neq 0 \\ \in \mathrm{cl}\, \mathbb{B}(0,1) - (1,1) & \text{otherwise.} \end{cases}$$

Show further that $\partial \mathbf{H}_{\mathrm{FB}}(x,y,z) = T(x,y,z)$.

7.6.12 Let $G : \Omega \subseteq \mathbb{R}^n \to \mathbb{R}^n$, with Ω open, be locally Lipschitz continuous and monotone on Ω. Show that every matrix in $\partial G(x)$ is positive semidefinite for all $x \in \Omega$.

7.6.13 Let $x^* \in X$ be a zero of $H : \mathbb{R}^n \to \mathbb{R}^n$, and assume that X is closed and convex. Suppose that M is a superlinearly convergent algorithmic map for the solution of the equation $H(x) = 0$. By this we mean that there exist a sufficiently small neighborhood $\mathbb{B}(x^*, \rho)$ of x^* and a function $\eta : \mathbb{B}(x^*, \rho) \to \mathbb{R}_+$ such that η is continuous at x^*, $\eta(x^*) = 0$, and

$$\| M(x) - x^* \| \leq \eta(x) \| x - x^* \|, \quad \forall x \in \mathbb{B}(x^*, \rho).$$

This implies that if x^0 belongs to $\mathbb{B}(x^*, \rho)$, the sequence $\{x^k\}$ generated by setting $x^{k+1} = M(x^k)$ converges Q-superlinearly to x^*. Note that x^{k+1} need not belong to X. Show that for every $x^0 \in \mathbb{B}(x^*, \rho)$, the sequence $\{\tilde{x}^k\} \subset X$ generated by setting $\tilde{x}^{k+1} \equiv \Pi_K \circ M(\tilde{x}^k)$ converges Q-superlinearly to x^*.

7.6.14 Give a direct proof of Theorem **7.2.15** using results from the classical Newton method for smooth equations and the fact that all the functions G_i for $i \in \mathcal{P}(x^*)$ have a zero at x^*. Develop a Newton algorithm for a "piecewise strongly semismooth" equation that is analogous to Algorithm **7.2.14** for a PC^1 equation; generalize the above proof to establish the Q-quadratic convergence of the resulting Newton algorithm under the assumption that all matrices in $\partial_B G_i(x^*)$ are nonsingular for all $i \in \mathcal{P}(x^*)$.

7.6.15 Use Theorem **7.5.8** and Lemma **7.5.7** to show that if the sequence $\{x^k\}$ produced by Algorithm **7.5.4** converges Q-superlinearly to a zero x^* of a locally Lipschitz continuous function G, where all the matrices in $\partial G(x^*)$ are nonsingular, then

$$\lim_{k \to \infty} \frac{r^k}{\| G(x^k) \|} = 0.$$

This shows that the conditions on the sequence of scalars $\{\eta_k\}$ given in Theorem **7.5.5** to guarantee the Q-superlinear convergence of the semismooth inexact Newton Algorithm **7.5.4** are not only sufficient, but also necessary.

7.6.16 Suppose that the sequence $\{x^k\}$ defined by $x^{k+1} \equiv x^k + d^k$ for all $k \geq 0$ converges to x^*. Show that $\{x^k\}$ converges to x^* Q-superlinearly if and only if (7.5.3) holds and for every infinite subset κ of $\{0, 1, 2, \ldots\}$ for which $\{d^k / \|d^k\| : k \in \kappa\}$ converges to a vector d,

$$\lim_{k(\in\kappa) \to \infty} \frac{x^* - x^k}{\| x^k - x^* \|} = d.$$

Roughly speaking, this shows that $\{x^k\}$ converges to x^* Q-superlinearly if and only if $d^k \approx x^* - x^k$.

The following sequence $\{x^k\}$ of vectors in the plane obviously does not converge to the origin Q-superlinearly: for $k = 0, 1, 2, \ldots$,

$$x^{4k+i} \equiv \begin{cases} (\, 0.9^k, 0.9^k \,) & \text{if } i = 1 \\ (\, -0.9^k, 0.9^k \,) & \text{if } i = 2 \\ (\, -0.9^k, -0.9^k \,) & \text{if } i = 3 \\ (\, 0.9^k, -0.9^k \,) & \text{if } i = 4. \end{cases}$$

Use the above characterization of Q-superlinear convergence to identify a direction d that violates the condition.

7.6.17 Consider the VI (K, F) with a compact convex set K and a continuously differentiable function F. By Theorem **10.2.1**, the gap function $\theta_{\rm gap}$ is directionally differentiable and

$$\theta'_{\rm gap}(x; d) \;=\; F(x)^T d + \max_{y \in M(x)} (\, x - y \,)^T JF(x) d,$$

where $M(x)$ is the argmax of $\theta_{\rm gap}(x)$; i.e.,

$$M(x) \;\equiv\; \{\, y \;\in\; K \;:\; \theta_{\rm gap}(x) \;=\; (\, x - y \,)^T F(x) \,\}.$$

(a) Suppose that $x^k \in K$ and $JF(x^k)$ is copositive on $\mathcal{T}(x^k; K)$. Assume further that $x^k \notin \mathrm{SOL}(K, F)$. Show that if x^{k+1} is a solution of the semi-linearlized VI (K, F^k), then $x^{k+1} - x^k$, which must be nonzero, is a descent direction of the gap function $\theta_{\rm gap}$ at x^k in the sense that $\theta'_{\rm gap}(x^k; x^{k+1} - x^k) < 0$.

(b) Let $x \in K$ be such that $JF(x)$ is copositive on $\mathcal{T}(x; K)$. Use part (a) to show that $x \in \mathrm{SOL}(K, F)$ if and only if (i) the semi-linearized VI (K, F^x), where

$$F^x(z) \;\equiv\; F(x) + JF(x)(\, z - x \,) \quad \forall\, z \;\in\; K,$$

has a solution and (ii) x is a constrained stationary point of the gap function $\theta_{\rm gap}$ on K; i.e., $\theta'_{\rm gap}(x; y - x) \geq 0$ for all $y \in K$.

7.6.18 Let $\Phi : \mathbb{R}^n \to \mathbb{R}^n$ be Lipschitz continuous in a neighborhood of a vector $x^* \in \Phi^{-1}(0)$. Show that statement (a) below implies statement (b).

(a) Φ is a locally Lipschitz homeomorphism near x^*;

(b) for every $V \in \partial_B \Phi(x^*)$,

$$\mathrm{sgn} \det V \;=\; \mathrm{ind}(\Phi, x^*) \;=\; \pm 1.$$

Assume in addition that Φ is directionally differentiable near x^*. Consider the following two additional statements:

(c) $\Psi \equiv \Phi'(x^*;\cdot)$ is a globally Lipschitz homeomorphism;

(d) for every $V \in \partial_B \Psi(0)$,

$$\operatorname{sgn}\det V \;=\; \operatorname{ind}(\Psi, 0) \;=\; \operatorname{ind}(\Phi, x^*) \;=\; \pm 1.$$

Show that (a) $\Rightarrow$ (c) $\Rightarrow$ (d). Show further that if Φ is semismooth near x^*, then (a) $\Leftrightarrow$ (b). In this case, the local inverse of Φ near x^* is semismooth at the origin. (Hint: use Exercise **3.7.6**.) Finally, show that if Φ is semismooth near x^* and

$$\partial_B \Phi(x^*) \;\subseteq\; \partial_B \Psi(0),$$

then the four statements (a), (b), (c), and (d) are equivalent.

7.6.19 This exercise extends Theorem **7.5.8** to a locally Lipschitz continuous function $G : \Omega \to \mathbb{R}^n$, where Ω is open convex set in $\mathbb{R}^n$. Let T be a linear Newton approximation of G at a point $x^* \in \Omega$. Let $\{x^k\}$ be any sequence in Ω converging to x^* with $x^k \neq x^*$ for all k. Assume that the Newton approximation T is nonsingular at x^*. Show the following two statements.

(a) The sequence $\{x^k\}$ converges Q-superlinearly to x^* and $G(x^*) = 0$ if and only if there exists a sequence $\{H^k\}$, where $H^k \in T(x^k)$ for every k, such that

$$\lim_{k\to\infty} \frac{\| G(x^k) + H^k d^k \|}{\| d^k \|} \;=\; 0,$$

where $d^k \equiv x^{k+1} - x^k$.

(b) If the approximation T is strong at x^*, then the sequence $\{x^k\}$ converges Q-quadratically to x^* and $G(x^*) = 0$ if and only if there exists a sequence $\{H^k\}$, where $H^k \in T(x^k)$ for every k, such that

$$\limsup_{k\to\infty} \frac{\| G(x^k) + H^k d^k \|}{\| d^k \|^2} \;<\; \infty.$$

7.6.20 This exercise shows what can happen if in a Newton method for the solution of a system of equations, we allow for inexact evaluation of the elements in the linear Newton approximation. Let $G : \Omega \to \mathbb{R}^n$ be a locally Lipschitz function on the open convex set Ω in $\mathbb{R}^n$ and let T be a nonsingular linear Newton approximation of G at the zero $x^* \in \Omega$. Starting at an $x^0 \in \Omega$, define the sequence $\{x^k\}$ by the iteration

$$x^{k+1} \;\equiv\; x^k - (\,V^k\,)^{-1} G(x^k),$$

where $V^k \in \mathbb{R}^{n\times n}$ is such that $\operatorname{dist}(V^k, T(x^k)) \leq \eta$ and η is a nonnegative constant. Prove that if x^0 is chosen sufficiently close to x^* and η is sufficiently small, the sequence $\{x^k\}$ is well defined and converges at least Q-linearly to x^k. Show further that $\{x^k\}$ converges Q-superlinearly to x^* if and only if there exists a sequence $\{H^k\}$, where $H^k \in T(x^k)$ for every k, such that

$$\lim_{k\to\infty} \frac{\| (H^k - V^k)(x^{k+1} - x^k) \|}{\| (x^{k+1} - x^k) \|} = 0.$$

7.6.21 Let $G : \Omega \to \mathbb{R}^n$ be a locally Lipschitz continuous function on the open convex set Ω in $\mathbb{R}^n$ and let T be a nonsingular, strong, linear Newton approximation of G at the zero $x^* \in \Omega$. Consider the sequence $\{x^k\}$ generated by the iteration

$$x^{k+1} \equiv x^k - (H^k)^{-1}[G(x^k) + G(x^k - (H^k)^{-1}G(x^k))],$$

where, for each k, $H^k \in T(x^k)$. Show that if x^0 is sufficiently close to x^*, the sequence $\{x^k\}$ is well defined and converges at least Q-cubically to x^*. Note that to obtain the point x^{k+1} from x^k, we have to construct one element H^k in $T(x^k)$ and solve two systems of linear equations involving the same matrix H^k. Therefore, the cost of this iteration is lower than that of two Newton iterations.

7.7 Notes and Comments

Newton methods for smooth systems of equations have a long history and is very well researched. The reader can find ample accounts of both the theoretical and practical behavior of the method in classical references such as [132, 422]. The review papers [234, 626], already mentioned in the Preface, give a good historical perspective with an eye towards the infinite-dimensional case. In order to generalize the Newton methods to nonsmooth equations, some appropriate analytic tools are necessary. The basis of our development is Clarke's original approach to nonsmooth calculus, as detailed in his book [108]. Essentially all of Section **7.1** is from Clarke's monograph, to which we refer the reader for original sources and further discussions. An impressive *tour de force*, the treatise of Rockafellar and Wets [510] provides a more up-to-date and integrated account of these nonsmooth tools. Proposition **7.1.17** is from [485], while Example **7.1.20** is from [353].

The question of what a "good" approximation to a nonsmooth function should be and, more importantly, what kind of approximation is suitable for defining a Newton method has no simple answer. Early attempts to define

a Newton method for structured nonsmooth problems (VIs, for example) consisted of replacing any smooth functions by their linearizations while retaining the nonsmooth components of the problems. This was done for example in [151, 293, 294, 495, 497]. What is needed, however, is a broader approach that can be applied to general, non-structured, problems. This was provided by several researchers: Gwinner [250], Kummer [352, 355, 356], Pang [425], and Robinson [501] all proposed definitions of suitable approximations to nonsmooth functions that can be used in generalized Newton methods, and, with the exception of Pang, all considered infinite-dimensional settings. Robinson defined the influential concept of "point based approximation" (PBA) in his paper [501], whose preprint version was widely circulated since 1988. Specifically, a function $A : \Omega \times \Omega \to \mathbb{R}^n$ is a PBA approximation for $G : \Omega \to \mathbb{R}^n$, where Ω is an open subset of $\mathbb{R}^n$, if there exists a positive constant ρ such that

$$\| G(x) - A(y, x) \| \leq \rho \| x - y \|^2, \quad \forall\, x, y \in \Omega,$$

and the function $A(y, \cdot) - A(x, \cdot)$ is Lipschitz continuous on Ω with modulus $\rho\|x - y\|$. It is clear that every PBA approximation to G readily provides a strong Newton approximation to G, but not vice versa because of the Lipschitz requirement on $A(y, \cdot) - A(x, \cdot)$. Based on this definition, and assuming an additional "nonsingularity" condition, Robinson was able to establish convergence of a generalized Newton method along with its quadratic convergence rate. Gwinner's earlier approach shares many common points with that of Robinson, but has not received much attention, possibly because it was published in an edited volume, the writing was dense and very technical, and practically no examples illustrating the applicability of the method were provided. The same reasons may also explain the little influence of Kummer's papers [352, 355, 356], whose results were partially and independently rediscovered at the beginning of the 1990s and whose work forms the basis of our development in this chapter. In particular, Kummer [352] defines a concept very close to that of a linear Newton approximation and establishes local superlinear convergence under some additional conditions that are not easy to verify. One feature of Kummer's method that distinguishes it from the other approaches described above is that it uses families of approximations, as we also do. Motivated by the desire to use various kinds of generalized derivatives as approximations in a Newton process, Kummer [355, 356] considers some schemes that are close to those based on Definition **7.2.2**. Albeit phrased in quite different terms, our definition of a (nonsingular) Newton approximation can be seen as a variant of the definition used in these papers of Kummer. Specifically,

Kummer considers subproblems of the form $0 \in G(x^k) + D(x^k, d)$, where D is a set-valued map that associates a nonempty subset of $\mathbb{R}^n$ to every (x, d) and such that $D(x, 0) = \{0\}$ for every x. Assuming that d^k is a solution of the aforementioned inclusion, the Newton process is then obtained by setting $x^{k+1} = x^k + d^k$. Two conditions: (CA) and (CI) are needed for this method to work. Condition (CA) is essentially equivalent to (7.2.5). Condition (CI) is a global uniform injectivity condition on $D(x, \cdot)$ that is similar to the condition used in Theorem **7.2.10**, which, however, is local in nature. Condition (c) in Definition **7.2.2** takes the place of (CI).

It is not too difficult to construct approximations to a function that can formally be used in a nonsmooth Newton Method. The challenge is to construct such approximations that lead to practically implementable and computationally efficient algorithms. This chapter provided evidence of the breadth and wide applicability of Definition **7.2.2**. Theorem **7.2.5** shows convergence of Algorithm **7.2.4** assuming that the iterative process is initiated in a suitable neighborhood of a desired solution and is therefore an extension of the well-known domain-of-attraction theorem for Newton's method in the smooth case. In this book we do not consider Kantorovich-type (or "semi-local") results, where the existence of a solution of the system of equations is a consequence of the assumptions. Robinson [501] presents a result of the latter kind. The Inexact Nonsmooth Newton Method, Algorithm **7.2.6**, is a natural and simple extension of the basic Algorithm **7.2.4** and parallels the development of inexact Newton methods in the smooth case [125]. Inexact versions of nonsmooth Newton methods were described already in [355, 356], but without the superlinear convergence results. Lemma **7.2.12** is a restatement of a result in [500].

The Piecewise Smooth Newton Method, Algorithm **7.2.14**, was first analyzed by Kojima and Shindo in [346]. Their original convergence proof is very simple and, roughly speaking, is based on the observation that if x^* is a solution of a piecewise function, then x^* is a zero of all the active pieces at x^*, to which a Newton method can be applied. An important motivation of Kojima and Shindo to solve piecewise smooth equations was to apply the method to the solution of a nonsmooth equation reformulation of the NCP. Kummer showed that the Piecewise Smooth Newton Method is also a particular case of his framework [352].

The particular Newton approximation (7.2.21) for the composite map $S \circ N$ illustrates what was said before about structured problems: here we linearize the smooth part S while leaving the nonsmooth part N intact. This kind of Newton approximation appears in [501, 622]. The main application of such a composite Newton scheme has so far been restricted to

the VI (K, F) as described in Section **7.3**. Applications to other problems await to be explored.

Minus the name, the Josephy-Newton method for VIs, Algorithm **7.3.1**, first appeared in a 1976 paper by Bakushinskii [37]. In this paper the author established the local superlinear convergence of the algorithm under very restrictive assumptions: F is monotone with bounded Jacobian matrices that are uniformly positive definite. Although Bakushinskii relaxed the latter condition by a regularization scheme, the monotonicity assumption remained. Based on Robinson's generalized equation formalism whereby the VI (K, F) is represented by the inclusion $0 \in F(x) + \mathcal{N}(x; K)$ and relying on the theory of strong regularity [497], Josephy established Theorem **7.3.3**. Josephy also considered quasi-Newton versions of the method [294] and applied these methods to the solution of the PIES model [295].

The inexact version of the Josephy-Newton method, Algorithm **7.3.7**, appears in Pang [424], where the convergence is established under Robinson's strong regularity assumption. The issue of whether the strong stability assumption can be relaxed was discussed in the paper by Pang [428] for the exact Josephy-Newton method. Focusing on linearly constrained VIs and relying on some sensitivity results, Pang showed that it is possible to relax the strong stability assumption to a weaker condition at the solution, called "pseudo regularity". In essence, pseudo regularity corresponds to the stability of the semi-linearized problem at the solution. Theorem **7.3.5** improves on the results in [427] by allowing the set K to be non-polyhedral; Theorem **7.3.8** further extends the former theorem by treating inexact methods. Under the assumed stability condition in these theorems, it is possible to show that the Newton process still has a local superlinear/quadratic convergence rate; the main difference is that the sub-VIs no longer have unique solutions. Bonnans [58, 59] also recognized that semiregularity suffices for the convergence of the Josephy-Newton method. Dontchev, in a series of papers [140, 142, 143, 144], studies the convergence of Newton methods for generalized equations under Robinson's strong regularity and Lipschitzian stability. The paper [436] discussed the parallel implementation of the Josephy-Newton method for solving NCPs.

The Josephy-Newton method is preceded by the family of locally convergent *Sequential Quadratic Programming* (SQP) algorithms for nonlinear programs first introduced in Wilson's Ph.D. thesis [610]. Indeed, when K is finitely representable and F is the gradient map of the scalar function θ, the specialization of Algorithms **7.3.1** and **7.3.7** to the MiCP corresponding to the KKT system of the VI (K, F) leads to, respectively, an exact

and an inexact SQP method for solving the NLP:

$$\begin{aligned} &\text{minimize} && \theta(x) \\ &\text{subject to} && x \in K. \end{aligned}$$

Robinson [494] gave a convergence proof for Wilson's method. There is a large number of papers devoted to the theoretical analysis and practical implementation of SQP methods for NLPs; it is outside our scope to attempt even a summary of these contributions. We only mention that there remains some interest in establishing desirable convergence properties of SQP methods under the weakest possible assumptions. For instance, in the recent references [19, 205, 251, 613], by exploiting properties of the NLPs, improved results are established that are finer than those obtained from a direct specialization of the results presented in this chapter for the KKT system of a VI. Moreover, there has been in recent years a growing amount of research in applying SQP methods to MPECs; see [20, 213, 223].

With the aim of identifying a favorable class of functions in nonsmooth convex minimization to which the bundle methods could be applied, Mifflin [404] introduced the semismoothness property of a scalar function in several variables. This property was later extended to vector functions by Qi and Sun in [485], who had a different objective in mind, namely, to define a Newton method for nonsmooth equations that circumvents the drawback of Pang's B-differentiable Newton method [425]. We caution the reader that our Definition **7.4.2** of semismoothness is slightly different from that usually found in the literature. In fact, we require the function F to be directionally differentiable in a neighborhood of the point x at which semismoothness is defined, while in the usual definition the directional differentiability is required only at x. We adopted this slightly more restrictive definition because in the context of this book there is no difference between the two definitions; moreover, the new definition makes the analytical developments more elegant and easier.

The main source for the properties of a semismooth function presented in Section **7.4** is the papers by Qi and Sun [485] and Qi [472]. Example **7.4.1** is from [352]. Proposition **7.4.4** was proved by Mifflin for a scalar function and by Fischer [202] in the strongly semismooth case. Exercise **7.6.18**, which is an inverse function theorem for semismooth functions, is based on [435].

The observation that the FB merit function θ_{FB} for an NCP is an SC^1 function was first formally made in [180]. (Since $\psi_{\mathrm{FB}}(a,b) = \sqrt{a^2+b^2}-a-b$ is convex, the semismoothness of ψ_{FB} follows from Mifflin's original paper [404].) Scalar functions with a Lipschitz gradient (LC^1 functions) were

widely discussed, for example, in [270, 330, 331, 353, 354] in connection with optimality conditions and sensitivity results for optimization problems. In particular, in [270], the concept of a generalized Hessian was introduced; it was established that if a function θ is LC^1 in a neighborhood of a point x, then, for every y sufficiently close to x, we can write

$$\theta(y) \;=\; \theta(x) + \nabla\theta(x)^T(y-x) + \tfrac{1}{2}\,(y-x)^T H(y-x),$$

for some H belonging to the generalized Hessian of θ calculated at a point in the open segment (x, y). Proposition **7.4.10** can be viewed as an extension of this result to the SC^1 case.

The class of SC^1 functions has been a subject of interest in relation to the development of minimization algorithms; see [166, 434, 473]. We will discuss more about the first two of these papers in Section **8.6**. Optimization problems with SC^1 objective functions arise, for example, in stochastic programming and control theory; also augmented Lagrangian functions and exact penalty functions are SC^1 under standard assumptions. For an interesting application of SC^1 optimization to convex best interpolation, see [145]. Hiriart-Urruty [268] was arguably the first to note the properties of the B-subdifferential of a vector function reported in Proposition **7.4.11**.

The idea of constructing a set of generalized Hessian matrices "arising" from a certain direction, as done in the definition of $\partial_d^2\theta(x)$, has its origin in the work of Chaney [82] and is also related to the notion of "generalized Jacobian relative to a set" used in [267, 268]. The authors of the latter papers also consider using these notions to give sufficient optimality conditions for nonlinear programs, as we do in Proposition **7.4.12**.

Semismooth Newton methods, and in particular Algorithm **7.5.1**, were introduced and popularized by the work of Qi and Sun [485]. At first, it was not immediately recognized that the algorithm is just a particular instance of the general scheme proposed by Kummer in [352]. In view of the classical work [125] for smooth problems, it is natural to consider an inexact version of the semismooth Newton method. The convergence of such an inexact method was analyzed by Martínez and Qi [394], who also pointed out the difference from the smooth case highlighted in Example **7.5.6**. The refined results on the quadratic convergence rate in the case of strongly semismooth functions presented in Theorem **7.5.5** are from [168]. Sun [563] applies a nonsmooth Newton method to piecewise quadratic programs that arise from SQP, trust region, and penalty methods.

In the smooth case, the Dennis-Moré characterization of superlinear convergence of the iterates [130] plays a fundamental role in the design and analysis of algorithms for the solution of systems of equations and optimiza-

tion problems; such a role persists in the nonsmooth case. Theorem **7.5.8** is taken from [433]. Incidentally, the latter paper is among one of the earliest to study algorithms for solving general nonsmooth equations in a systematic way. Levenberg-Marquardt type algorithms are among the most popular methods for the solution of (smooth) systems of equations; even a partial list of the literature on this topic is beyond the scope of these notes. In the case of semismooth equations the Levenberg-Marquardt method was first considered in [174], which is the source for Theorem **7.5.11**. See [97] for a suggestion to deal with singular smooth and nonsmooth equations using adaptive outer inverses.

Linear Newton approximations were introduced in [475] under the name "C-differentials". Although developed independently, the resulting methods are also a particular realization of Kummer's general method. The motivation to consider the concept of C-differential came from the need to enhance the applicability of Newton methods for semismooth systems of equations when the calculation of generalized Jacobians or of B-subdifferentials is difficult. This inspired Sun and Han [559] to use the object $\partial_C G$ (7.5.18) to develop a Newton method for nonsmooth equations. The general theory given in Subsection **7.5.1** permits one to analyze a host of different algorithms in a unified framework.

The classic paper on quasi-Newton methods for smooth equations is [131]. In this book we do not consider these methods for general nonsmooth equations. The paper [462] discusses secant methods for semismooth equations. In general, quasi-Newton methods are neither intensively studied nor very much used for solving nonsmooth equations. This can be partially explained as follows. It is widely accepted by researchers in the field that, in order to obtain a local fast convergence rate for a nonsmooth system, differentiability of the function at the solution x^* is unavoidable. This problem was evidenced for the first time by Ip and Kyparisis [275], who proved a local convergence result by assuming some kind of "radially Lipschitz continuity" of the directional derivative of G. Their assumption implies, in particular, the strong differentiability of G at x^*. Although this assumption was somewhat relaxed by Qi [476], differentiability of G at the solution seems to be an essential requirement in order to achieve a superlinear convergence rate. More interesting results can be obtained when the function G has some kind of structure that can be exploited, such as in the reformulations of the VIs/CPs as systems of equations; see the notes and comments in the next two chapters for references and further discussion.

As remarked at the beginning of the section, the developments in this chapter have been based principally on Clarke's approach to nonsmooth

functions and related developments. We should mention that Demyanov and Rubinov [128] proposed a "quasidifferentiable calculus". A (not necessarily differentiable) function $g : \mathbb{R}^n \to \mathbb{R}$ is said to be quasidifferentiable at a point $x \in \mathbb{R}^n$ if it is directionally differentiable at x and there exist two convex and compact subsets A and B of $\mathbb{R}^n$ such that for every direction $d \in \mathbb{R}^n$ we can write

$$g'(x;d) = \max_{a \in A} a^T d + \min_{b \in B} b^T d.$$

Based on this definition, it is possible to define "quasidifferential" Newton methods [126, 647]; however, in this case the corresponding theory does not yield results as interesting as those discussed in this chapter. In particular, the only case where the quasidifferential approach seems capable of providing practically useful methods is that in which the functions involved are semismooth; as a result, there seems to be little gain in adopting a quasidifferential point of view.

Chapter 8

Global Methods for Nonsmooth Equations

The methods described in the previous chapter are all locally convergent in that the starting iterate is required to be sufficiently close to a desired but unknown zero of the function under consideration. If this requirement is not met, the convergence of the algorithms is in jeopardy. Such non-convergence is well known in the case of smooth equations. Therefore, in order to deal with the situation where a good iterate is not available to initiate a locally convergent algorithm, we have to face the challenge of introducing globally convergent algorithms that allow arbitrary starting vectors, which could be far from a zero of the system of equations under consideration. The principal objective of the this chapter is to present some such global methods for solving nonsmooth equations.

Associated with the constrained equation

$$G(x) = 0, \qquad x \in X, \tag{8.0.1}$$

we often consider a merit function $\theta : X \to \mathbb{R}_+$ with the property that

$$\theta(x) = 0 \quad \Leftrightarrow \quad G(x) = 0.$$

A preliminary discussion of the role of merit functions in the study of the VI/CP is presented in Subsection **1.5.3**. Merit functions are key to the design of globally convergent algorithms for the solution of the CE (8.0.1) because they allow us to recast (8.0.1) as the *(global) minimization problem*

$$\begin{array}{ll} \text{minimize} & \theta(x) \\ \text{subject to} & x \in X. \end{array} \tag{8.0.2}$$

This reformulation offers a useful algorithmic point of view on the CE. Some natural examples of merit functions are $\theta(x) \equiv \| G(x) \|$ for a suitable norm

$\|\cdot\|$ and $\theta(x) \equiv (1/2)G(x)^T G(x)$. In the application to a VI/CP, there are additional merit functions that are of a different type, e.g., the gap function $\theta_{\rm gap}$, which are not norm functions derived from an underlying equation reformulation of the problem. Thus a detailed algorithmic study of (8.0.2) with a general nonsmooth function θ is of interest.

8.1 Path Search Algorithms

A common approach to globalize the convergence of the local methods in the smooth case is via a line search on a certain merit function. In the case of nonsmooth equations, this is by no means the only approach, nor is it the most natural. To motivate the method to be described in this section, let us review the damped Newton method for solving the unconstrained system $G(x) = 0$ where $G : \mathbb{R}^n \to \mathbb{R}^n$ is C^1. Let x^k be the current iterate at the beginning of an iteration. We compute the direction d^k by solving the system of linear equations:

$$G(x^k) + JG(x^k)d \;=\; 0.$$

In the locally convergent Newton method, we set $x^{k+1} \equiv x^k + d^k$ and the iteration terminates. In the damped Newton method, we check if $x^k + d^k$ is satisfactory according to some criterion. If so, we accept it as the next iterate x^{k+1}; otherwise, we backtrack from $x^k + d^k$ and search a suitable new point along the line segment joining x^k and $x^k + d^k$. Let $p^k(\tau)$ be such a Newton path (which is a line segment); that is,

$$p^k(\tau) \;\equiv\; x^k + \tau\, d^k, \quad \forall\, \tau \;\in\; [0,1].$$

Since G is smooth by assumption and $d^k = -JG(x^k)^{-1}G(x^k)$, we have

$$G(p^k(\tau)) \;=\; (\,1 - \tau\,)G(x^k) + o(\tau)$$

for all $\tau > 0$ sufficiently small. Consequently, if $G(x^k) \neq 0$, we are guaranteed to find a step size τ_k in $(0,1]$ such that $\|G(x^{k+1})\|$ is "sufficiently" less than $\|G(x^k)\|$, where

$$x^{k+1} \;\equiv\; p^k(\tau_k) \;=\; x^k + \tau_k\, d^k.$$

The determination of the step size $\tau_k \in (0,1]$ is the "damping" process. Under suitable conditions, it can be shown that the sequence $\{x^k\}$ so generated has all the desirable convergence properties as in the local method, for any initial vector $x^0 \in \mathbb{R}^n$. The latter is the *globally convergent* aspect of the damped Newton method.

When we consider nonsmooth equations, things become more complicated. First of all we saw in Section **7.2** that there is no longer a single natural model for the Newton methods; furthermore it is not clear that we can find a new point along the line segment $p^k(\tau)$ such that $\|G(p^k(\tau))\|$ is less than $\|G(x^k)\|$ for $\tau > 0$. In order to introduce a globally convergent Newton method for solving nonsmooth equations, our first order of business is to introduce a concept of uniform approximation of G that globalizes the local Newton approximation concept introduced in Definition **7.2.2**. Such a uniform approximation serves two important purposes in the global Newton method to be defined subsequently; it guarantees that the method is well defined from any initial iterate in $\mathbb{R}^n$, and, more importantly, the resulting method still possesses the same desirable convergence properties as the local methods.

8.1.1 Definition. Let G be a function from an open subset Ω of $\mathbb{R}^n$ into $\mathbb{R}^n$. We say that G has a *nonsingular uniform Newton approximation* on Ω if there exist positive scalars ε and L and a function $\Delta : (0, \infty) \to [0, \infty)$, with

$$\lim_{t \downarrow 0} \Delta(t) \;=\; 0,$$

such that for every point x in Ω there is a family $\mathcal{A}(x)$ of functions each mapping $\mathbb{R}^n$ into itself and satisfying the following three properties:

(a) $A(x, 0) = 0$ for every $A(x, \cdot) \in \mathcal{A}(x)$;

(b) for any two distinct vectors x and x' in Ω and for any $A(x, \cdot) \in \mathcal{A}(x)$,

$$\frac{\| G(x) + A(x, x' - x) - G(x') \|}{\| x - x' \|} \;\leq\; \Delta(\| x - x' \|); \tag{8.1.1}$$

(c) $\mathcal{A}(x)$ is a family of *uniformly locally Lipschitz homeomorphisms* with modulus L on Ω, by which we mean that for each $A(x, \cdot) \in \mathcal{A}(x)$, there are two open sets U_x and V_x, both containing $\mathbb{B}(0, \varepsilon)$, such that $A(x, \cdot)$ is a Lipschitz homeomorphism mapping U_x onto V_x with L being the Lipschitz modulus of the inverse of the restricted map $A(x, \cdot)|_{U_x}$.

If the requirement (b) is strengthened to

(b') there exists a positive constant L' such that, for any two distinct vectors x and x' in Ω and for any $A(x, \cdot) \in \mathcal{A}(x)$,

$$\frac{\| G(x) + A(x, x' - x) - G(x') \|}{\| x - x' \|^2} \;\leq\; L', \tag{8.1.2}$$

then we say that G has a *nonsingular uniform strong Newton approximation* on Ω. □

Note that a nonsingular uniform Newton approximation on an open set Ω involves a much stronger requirement than a nonsingular Newton approximation at every point x of the set Ω; the additional requirement in the former approximation is that the function Δ and the constants L and ε in Definition **8.1.1** do not depend on x. Another distinguished aspect of the latter definition is that x and x' are two arbitrary vectors in Ω (as opposed to x and a fixed $\bar{x}$ in Definition **7.2.2**). These combined aspects allow us to show that if a continuous function defined on $\mathbb{R}^n$ has a nonsingular uniform Newton approximation on $\mathbb{R}^n$, then the function is a global homeomorphism. This result is an extension of the classical Hadamard theorem for a smooth function. The proof of this result makes use of the continuation property defined below.

8.1.2 Definition. The mapping $G : \Omega \subseteq \mathbb{R}^n \to \mathbb{R}^n$ has the *continuation property* for a given continuous function $q : [0,1] \to \mathbb{R}^n$ if the existence of a continuous function $p : [0,a) \to \Omega$, $a \in (0,1]$ such that $G(p(t)) = q(t)$ for all $t \in [0,a)$ implies that

$$\lim_{t \uparrow a} p(t) \;=\; p(a) \tag{8.1.3}$$

exists with $p(a) \in \Omega$ and $G(p(a)) = q(a)$. (Note: if G is continuous, the last equality is obvious, provided that the limit (8.1.3) exists.) □

The continuation property has an important role in the global bijectivity of a locally homeomorphic function. The following result is classical; it summarizes various consequences of the continuation property.

8.1.3 Proposition. Let $G : \Omega \subseteq \mathbb{R}^n \to \mathbb{R}^n$ be a local homeomorphism at each point of the open set Ω. The following four statements are valid.

(a) If G has the continuation property for all linear functions q mapping $[0,1]$ into $\mathbb{R}^n$, then $G(\Omega) = \mathbb{R}^n$.

(b) Let $q : [0,1] \times [0,1] \to \mathbb{R}^n$ and $r : [0,1] \to \Omega$ be continuous functions such that $G(r(s)) = q(s,0)$ for all $s \in [0,1]$. If for each $s \in [0,1]$, G has the continuation property for $q(s,\cdot)$, then there exists a unique continuous function $p : [0,1] \times [0,1] \to \Omega$ such that $p(s,0) = r(s)$ and $G(p(s,t)) = q(s,t)$ for all s, $t \in [0,1]$. Moreover, if

$$q(s,1) \;=\; q(0,t) = q(1,t)$$

for all s, $t \in [0,]$, then $r(0) = r(1)$.

(c) If Ω is in addition path-connected, then G maps Ω homeomorphically onto $\mathbb{R}^n$ if and only if G has the continuation property for all continuous functions $q : [0,1] \to \mathbb{R}^n$.

(d) If Ω is in addition convex, then G maps Ω homeomorphically onto $\mathbb{R}^n$ if and only if G has the continuation property for all linear functions $q : [0,1] \to \mathbb{R}^n$. □

Using the above proposition, we state and prove the aforementioned global homeomorphism property of a continuous function that has a nonsingular uniform Newton approximation.

8.1.4 Theorem. Let G be a continuous function from $\mathbb{R}^n$ into $\mathbb{R}^n$. If G has a nonsingular uniform Newton approximation on $\mathbb{R}^n$, then G is a homeomorphism from $\mathbb{R}^n$ onto itself.

Proof. We first show that for every $x \in \mathbb{R}^n$, there exists a $\delta > 0$ such that G is injective in $\mathbb{B}(x, \delta)$. Assume for the sake of contradiction that no such δ exists for some x. There exists sequences $\{y^k\}$ and $\{z^k\}$ such that $y^k \neq z^k$ and $G(y^k) = G(z^k)$ for every k, and

$$\lim_{k\to\infty} \| y^k - z^k \| = 0.$$

Hence for all k sufficiently large,

$$\| A(y^k, z^k - y^k) \| \leq \| y^k - z^k \| \, \Delta(\| y^k - z^k \|).$$

Let L and ε be the two positive scalars prescribed in Definition **8.1.1**. For all k sufficiently large, we have

$$\max(\, \| y^k - z^k \|, \, \| A(y^k, z^k - y^k) \| \,) < \varepsilon.$$

For all such k, it follows by conditions (a) and (c) in Definition **8.1.1** that

$$\| y^k - z^k \| \leq L \, \| A(y^k, z^k - y^k) \|.$$

Hence, we deduce

$$L^{-1} \leq \Delta(\| y^k - z^k \|),$$

which is a contradiction because the right-hand side converges to zero as $k \to \infty$ while the left-hand side remains a positive constant.

Consequently, for every $x \in \mathbb{R}^n$, there exists a $\delta > 0$ such that G is injective in $\mathbb{B}(x, \delta)$. By Proposition **2.1.12**, it follows that G is a local homeomorphism on $\mathbb{R}^n$. We next show that G has the continuation property for any linear path

$$q(t) = (1 - t)y^0 + ty^1, \quad t \in [0,1], \; y^0, y^1 \in \mathbb{R}^n;$$

that is, for any continuous function $p : [0, a) \to \mathbb{R}^n$, $a \in (0, 1]$ such that, $G(p(t)) = q(t)$ for all t in $[0, a)$, the limit (8.1.3) exists. For any pair (t_1, t_2) of scalars in $[0, a)$ we have

$$\begin{aligned} &\| G(p(t_1)) + A(p(t_1), p(t_2) - p(t_1)) - G(p(t_2)) \| \\ &\quad \leq \| p(t_1) - p(t_2) \| \, \Delta(\| p(t_1) - p(t_2) \|). \end{aligned}$$

Provided that $\|p(t_1) - p(t_2)\|$ is sufficiently small, we have

$$\| p(t_2) - p(t_1) \| \leq L \| A(p(t_1), p(t_2) - p(t_1)) \|.$$

Thus

$$\begin{aligned} &\| p(t_2) - p(t_1) \| \\ &\quad \leq L [\| q(t_1) - q(t_2) \| + \| p(t_1) - p(t_2) \| \, \Delta(\| p(t_1) - p(t_2) \|)] \end{aligned}$$

which implies

$$\| p(t_2) - p(t_1) \| \leq \frac{L \| y^0 - y^1 \|}{1 - L \, \Delta(\| p(t_1) - p(t_2) \|)} \, | t_1 - t_2 |.$$

This derivation shows that the function $p(t)$ is locally Lipschitz continuous on $[0, a)$. More importantly, there is a common Lipschitz constant for all $t \in [0, a)$. Specifically, a constant $L_p > 0$ exists such that for each $t \in (0, a)$ there is a $\delta_t > 0$ such that for all t_1 and t_2 belonging to the open interval $\mathcal{N}_t \equiv (t - \delta_t, t + \delta_t)$,

$$\| p(t_2) - p(t_1) \| \leq L_p \, | t_1 - t_2 |. \tag{8.1.4}$$

By a compactness argument, this local property can be extended to a uniform property on the entire open interval $(0, a)$. That is, (8.1.4) holds for any two points $t_2 < t_1$ in $(0, a)$. Indeed, the union

$$\bigcup_{t \in [t_1, t_2]} \mathcal{N}_t$$

is an open covering of the compact interval $[t_1, t_2]$. Thus there exists a finite subcovering, which implies the existence of a partition

$$t_1 \equiv \tau_0 < \tau_1 < \ldots < \tau_k \equiv t_2$$

such that for every $i = 0, 1, \ldots, k - 1$,

$$\| p(\tau_i) - p(\tau_{i+1}) \| \leq L_p \, (\, \tau_{i+1} - \tau_i \,).$$

Adding these k inequalities yields (8.1.4) readily.

Consequently, if $\{t_k\}$ is an arbitrary sequence in $[0, a)$ converging to a, then $\{p(t_k)\}$ is a Cauchy sequence and hence converges. Moreover, if $\{t'_k\}$ is another such sequence converging to a, then by (8.1.4), the two sequences $\{p(t_k)\}$ and $\{p(t'_k)\}$ must have the same limit. Consequently, the limit of $p(t)$ as $t \uparrow a$ exists. Hence G has the desired continuation property. By Proposition **8.1.3** (d), it follows that G is a global homeomorphism from $\mathbb{R}^n$ onto itself. □

8.1.5 Remark. The above partitioning argument is very similar to the one used in the proof of Proposition **4.2.2**(c), which shows that every PA map is globally Lipschitz continuous. More generally, this argument also shows that every locally Lipschitz continuous function $G : \Omega \to \mathbb{R}^m$ defined on a convex set $\Omega \subseteq \mathbb{R}^n$ with a common Lipschitz constant at every point must be globally Lipschitz on Ω. □

Central to the global Newton method in this section is the notion of a Newton path that generalizes that of a Newton segment discussed previously. Formally, a *path* in an open subset Ω of $\mathbb{R}^n$ is a continuous function p from the interval $[0, \bar{\tau}]$ or $[0, \bar{\tau})$ into Ω, for some positive scalar $\bar{\tau} \in (0, 1]$. The following proposition identifies the kind of paths we are going to use in the global Newton algorithm.

8.1.6 Proposition. Let $A : U \subseteq \mathbb{R}^n \to V \subseteq \mathbb{R}^n$ be a Lipschitz homeomorphism between two open sets U and V. Let x be a point in U such that $A(x) \neq 0$, and assume that $A(x) + \mathbb{B}(0, \varepsilon) \subset V$ for some positive scalar ε. Let $\bar{\tau} \equiv \min(\varepsilon/\|A(x)\|, 1)$. The unique path in U with domain $[0, \bar{\tau}]$ such that

$$p(0) = x \quad \text{and} \quad A(p(\tau)) = (1 - \tau)A(x), \quad \forall\, \tau \in [0, \bar{\tau}], \tag{8.1.5}$$

is given by

$$p(\tau) = A^{-1}((1 - \tau)A(x)), \quad \forall\, \tau \in [0, \bar{\tau}]. \tag{8.1.6}$$

Furthermore, p is Lipschitz continuous on $[0, \bar{\tau}]$ and A is a local homeomorphism near $p(\bar{\tau})$.

Proof. By the definition of $\bar{\tau}$, the line segment $[A(x), (1 - \bar{\tau})A(x)]$ is contained entirely in V. By the assumptions made, the path considered in (8.1.6) is a well-defined continuous function and satisfies (8.1.5). So we have to show that this is the only path in U that satisfies (8.1.5). But this trivially holds because A is a bijection between U and V. Furthermore, by its definition and by the assumption that A is a Lipschitz homeomorphism, it follows that p is Lipschitz continuous on $[0, \bar{\tau}]$. Finally, since U is open,

$p(\bar{\tau})$ belongs to its interior, it is easily seen that A is a local homeomorphism around this point. □

We next establish a technical lemma that lies at the heart of the path search Newton method and its convergence proof. Since this lemma refers to a given vector x, it does not require the uniformity of the nonsingular Newton approximation.

8.1.7 Lemma. Let G be a Lipschitz continuous function from an open subset Ω of $\mathbb{R}^n$ into $\mathbb{R}^n$. Suppose that G admits a nonsingular Newton approximation at a vector x in Ω, where $G(x) \neq 0$. For any $A(x,\cdot) \in \mathcal{A}(x)$, there exists a unique Lipschitz continuous path $p : I \to \Omega$ with largest domain such that the following conditions hold:

(a) $p(0) = x$;

(b) either $I = [0,1]$ or $I = [0,\bar{\tau})$ for some $\bar{\tau} \in (0,1]$;

(c) for each $\tau \in I$, $G(x) + A(x, p(\tau) - x) = (1-\tau)G(x)$;

(d) $G(x) + A(x, \cdot - x)$ is a local homeomorphism near every point on the path.

Moreover, for every scalar $\gamma \in (0,1)$, there exists a positive $\tau' \in I$ such that

$$\| G(p(\tau)) \| \leq (1 - \gamma\tau) \| G(x) \| \quad \forall \tau \in [0, \tau'].$$

Proof. Let $\varepsilon_{\mathcal{A}}$, $L_{\mathcal{A}}$, U_x, and V_x be the positive scalars and open sets given by Definition **7.2.2**, all associated with the given point x and family $\mathcal{A}(x)$. Since U_x contains $\mathbb{B}(0,\varepsilon)$, we may take $\mathcal{U}$ to be an open neighborhood of x such that for some scalar $\varepsilon_1 \in (0, \varepsilon_{\mathcal{A}})$,

$$\mathbb{B}(x, \varepsilon_1) \subseteq \mathcal{U} \subseteq \Omega \cap (x + U_x).$$

We claim that with any scalar ε_2 in the interval $(0, \min(\varepsilon_{\mathcal{A}}, \varepsilon_1/L_{\mathcal{A}}))$, the set

$$\mathcal{V} \equiv G(x) + A(x, \mathcal{U} - x),$$

which is open because $A(x,\cdot)$ is a homeomorphism between U_x and V_x and $\mathcal{U}$ is an open subset of $x + U_x$, must satisfy

$$\mathbb{B}(G(x), \varepsilon_2) \subseteq \mathcal{V} \subseteq G(x) + V_x.$$

The second inclusion is obvious. To see the first inclusion, let ζ belong to $\mathbb{B}(G(x), \varepsilon_2)$. Then $\zeta - G(x)$ belongs to $\mathbb{B}(0, \varepsilon_2)$ which is a subset of V_x. Let z be the unique vector in U_x such that

$$A(x, z) = \zeta - G(x).$$

By the definition of $L_{\mathcal{A}}$, we have

$$\| z \| \leq L_{\mathcal{A}} \| \zeta - G(x) \| < \varepsilon_1,$$

which implies $x + z \in \mathbb{B}(x, \varepsilon_1) \subseteq \mathcal{U}$. Since

$$\zeta = G(x) + A(x, z),$$

it follows that $\zeta \in \mathcal{V}$ as claimed.

The map $G(x)+A(x, \cdot - x)$ restricted to $\mathcal{U}$ is a Lipschitz homeomorphism mapping the open set $\mathcal{U}$ onto $\mathcal{V}$. By Proposition **8.1.6**, we deduce the existence of a path p in $\mathcal{U}$ with domain $I \equiv [0, \bar{\tau}]$, where

$$\bar{\tau} \equiv \min \left(\frac{\varepsilon_2}{\|G(x)\|}, 1 \right),$$

such that (a) and (c) are satisfied and $G(x) + A(x, \cdot - x)$ is a local homeomorphism at every point on the path p, including $p(\bar{\tau})$. If $\bar{\tau} = 1$ there is nothing else to prove. Assume that $\bar{\tau} < 1$. Since $G(x) + A(x, \cdot - x)$ is still a homeomorphism around $p(\bar{\tau})$ (and $G(x) + A(x, p(\bar{\tau}) - x)) \neq 0$) we can reapply Proposition **8.1.6** and enlarge the domain I of p to $[0, \bar{\tau}\,']$, with $\bar{\tau}\,' > \bar{\tau}$. Continuing this argument, we arrive at one of two cases. Either after a finite number of enlargements, the domain I is equal to the entire interval $[0, 1]$, so that (b) also holds, or this process continues on indefinitely and we obtain the largest domain with respect to the conditions (a), (b) and (c).

Let $\gamma \in (0, 1)$ be given. If the desired τ' fails to exists, then there exists a sequence of positive scalars $\{\tau_k\}$ converging to zero such that for every k,

$$\| G(p(\tau_k)) \| > (1 - \gamma \tau_k) \|G(x)\| \tag{8.1.7}$$

Since $p(0) = x$, by property (b) in Definition **7.2.2**, we have

$$\frac{\| G(p(\tau_k)) - G(x) - A(x, p(\tau_k) - x) \|}{\| p(\tau_k) - x \|} \leq \Delta(\| p(\tau_k) - x \|).$$

Since the left-hand fraction is equal to

$$\frac{\| G(p(\tau_k)) - (1 - \tau_k) G(x) \|}{\| p(\tau_k) - x \|},$$

it follows that

$$\lim_{k \to \infty} \frac{\| G(p(\tau_k)) - (1 - \tau_k) G(x) \|}{\| p(\tau_k) - x \|} = 0.$$

Since the path p is Lipschitz continuous, the above limit implies

$$\lim_{k\to\infty} \frac{\| G(p(\tau_k)) - (1-\tau_k) G(x) \|}{\tau_k} = 0.$$

Hence,

$$\| G(p(\tau_k)) \| \leq (1-\tau_k) \| G(x) \| + o(\tau_k).$$

By (8.1.7), it follows that

$$(1-\gamma)\tau_k \| G(x) \| \leq o(\tau_k)$$

which implies $G(x) = 0$ because γ is less than 1. This is a contradiction. □

8.1.8 Remark. If $A(x,\cdot)$ is a globally Lipschitz homeomorphism from $\mathbb{R}^n$ onto itself and Ω is equal to $\mathbb{R}^n$, then the domain of the path p is the entire unit interval $[0,1]$. □

In what follows, we present the *path Newton algorithm*. With the exception of dealing with paths instead of line segments, the algorithm is formally very similar to the classical damped Newton method for smooth equations. The setting of the path Newton algorithm is as follows. A function G is defined on an open subset $\Omega \subseteq \mathbb{R}^n$ and G has a nonsingular uniform Newton approximation on Ω. For any $x \in \Omega$ such that $G(x) \neq 0$, let p_x be the path in Ω prescribed by Lemma **8.1.7** and let I_x be its domain. According to this lemma, I_x is equal to either $[0, \bar{\tau}_x]$, with $\bar{\tau}_x = 1$, or $[0, \bar{\tau}_x)$ with $\bar{\tau}_x \in (0,1]$.

Path Newton Method (PNM)

8.1.9 Algorithm.

Data: $x^0 \in \mathbb{R}^n$ and $\gamma \in (0,1)$.

Step 1: Set $k = 0$.

Step 2: If $G(x^k) = 0$, stop.

Step 3: Select an element $A(x^k,\cdot)$ in $\mathcal{A}(x^k)$ and consider the corresponding path $p^k \equiv p_{x^k}$ with domain $I_k \equiv I_{x^k}$ defined by the scalar $\bar{\tau}_k$. Find the smallest nonnegative integer i_k such that with $i \equiv i_k$, $2^{-i}\bar{\tau}_k \in I_k$ and

$$\| G(p^k(2^{-i}\bar{\tau}_k)) \| \leq (1 - \gamma\, 2^{-i}\, \bar{\tau}_k) \| G(x^k) \|. \tag{8.1.8}$$

Step 4: Set $\tau_k \equiv 2^{-i_k}\bar{\tau}_k$, $x^{k+1} \equiv p^k(\tau_k)$, and $k \leftarrow k+1$; go to Step 2.

By the last assertion in Lemma **8.1.7**, the step size τ_k can be determined in a finite number of trials starting with $i = 0$ and increasing i by 1 after each failure of the criterion (8.1.8). This step-size procedure is the well-known *Armijo step-size rule* in nonlinear programming algorithms. Instead of using 1/2 as the "backtracking factor", we could use any scalar $\rho \in (0, 1)$ to carry out this rule. With such a backtracking factor ρ, (8.1.8) becomes

$$\| G(p^k(\rho^i \bar{\tau}_k)) \| \leq (1 - \gamma \rho^i \bar{\tau}_k) \| G(x^k) \|, \tag{8.1.9}$$

and the step size $\tau_k \equiv \rho^{i_k} \bar{\tau}_k$, where i_k is the smallest nonnegative integer i satisfying the above inequality. If the integer i_k is nonzero, the Armijo rule implies the following two inequalities:

$$\| G(x^{k+1}) \| \leq (1 - \gamma \rho^{i_k} \bar{\tau}_k) \| G(x^k) \|,$$

and

$$\| G(p^k(\rho^{i_k - 1} \bar{\tau}_k)) \| > (1 - \gamma \rho^{i_k - 1} \bar{\tau}_k) \| G(x^k) \|,$$

where the latter inequality is the result of the failure of the rule (8.1.9) at the next-to-last trial $i_k - 1$. Throughout the rest of this section, we fix the backtracking factor to be 1/2, as specified in Algorithm **8.1.9**.

The sequence $\{x^k\}$ in Algorithm **8.1.9** is thus well defined. The convergence properties of this sequence are summarized in Theorem **8.1.10** below. In proving this result, there are some subtle technical details when the domain Ω is a proper subset of $\mathbb{R}^n$. Consequently, we state and prove the theorem assuming that the set Ω is equal to the entire space $\mathbb{R}^n$. For any positive scalar $c > 0$, define the set

$$S_c \equiv \{ x \in \mathbb{R}^n : \| G(x) \| \leq c \}.$$

It is clear that this set is bounded for all $c > 0$ if and only if G is norm-coercive on $\mathbb{R}^n$. Moreover, if G is a global homeomorphism from $\mathbb{R}^n$ onto itself, then the set S_c must be bounded for all $c > 0$. The reason is quite simple. For every $c > 0$, S_c is contained in $G^{-1}(\mathrm{cl}\, \mathbb{B}(0, c))$; the latter set is compact because G^{-1} is continuous.

8.1.10 Theorem. Let $G : \mathbb{R}^n \to \mathbb{R}^n$ be locally Lipschitz on $\mathbb{R}^n$. Assume that G has a nonsingular uniform Newton approximation $\mathcal{A}$ on $\mathbb{R}^n$. For every $x^0 \in \mathbb{R}^n$, the sequence $\{x^k\}$ of vectors produced by Algorithm **8.1.9** converges Q-superlinearly to the unique zero of G. Moreover, if $\mathcal{A}$ is a nonsingular uniform strong approximation of G, then the convergence rate of the sequence $\{x^k\}$ is Q-quadratic.

Proof. The nonnegative sequence $\{\|G(x^k)\|\}$ is monotonically decreasing; it thus has a limit. We want to derive a contradiction if this limit is positive. For this purpose, assume that there exists a constant $\eta > 0$ such that

$$\| G(x^k) \| \geq \eta, \quad \forall k.$$

By the construction of the path p^k at each iteration, there exists a constant $\varepsilon > 0$ such that for all k,

$$\bar{\tau}_k \geq \min(\varepsilon / \| G(x^k) \|, 1).$$

Since $\|G(x^k)\| < \|G(x^0)\|$ for all k, we obtain,

$$\bar{\tau}_k \geq \min(\varepsilon / \| G(x^0) \|, 1) \equiv \eta' > 0, \quad \forall k.$$

By the properties of the function Δ, we can choose a positive t_* small enough so that

$$\Delta(t) \leq \frac{1-\gamma}{L_{\mathcal{A}}}, \qquad \forall t \in (0, t_*].$$

We have

$$\frac{t_*}{L_{\mathcal{A}} \| G(x^k) \|} \geq \frac{t_*}{L_{\mathcal{A}} \| G(x^0) \|} \equiv \beta > 0, \quad \forall k.$$

Define the positive scalar

$$\xi \equiv \min(\eta', \beta) > 0.$$

Note that $\bar{\tau}_k \geq \xi$ for all k. By the definition of the path p^k, we have, for every $\tau \in [0, \bar{\tau}_k]$,

$$A(x^k, p^k(\tau) - x^k) = -\tau G(x^k);$$

by the (uniform) Lipschitz continuity of the inverse of $A(x^k, \cdot)$, it follows that for every $\tau \in [0, \xi]$,

$$\| p^k(\tau) - x^k \| \leq \tau L_{\mathcal{A}} \|G(x^k)\| \leq t^*.$$

So, for the same values of τ,

$$\begin{aligned} \| p^k(\tau) - x^k \| \Delta(\|p^k(\tau) - x^k\|) &\leq \| p^k(\tau) - x^k \| \frac{1-\gamma}{L_{\mathcal{A}}} \\ &\leq \tau (1-\gamma) \| G(x^k) \|. \end{aligned}$$

Hence, by condition (b) in Definition **8.1.1**,

$$\begin{aligned} &\| G(p^k(\tau)) \| \\ &\leq \| A(x^k, p^k(\tau) - x^k) + G(x^k) \| + \| p^k(\tau) - x^k \| \Delta(\|p^k(\tau) - x^k\|) \\ &\leq (1-\tau) \| G(x^k) \| + \tau (1-\gamma) \| G(x^k) \| \\ &= (1 - \tau\gamma) \| G(x^k) \|. \end{aligned}$$

Consequently,

$$\| G(p^k(\tau)) \| \leq (1 - \tau\gamma) \| G(x^k) \|, \quad \forall \tau \in [0, \xi]. \tag{8.1.10}$$

By the definition of the step size τ_k, it follows that there exists $\xi' \in (0, \xi]$ such that $\tau_k \geq \xi'$ for all k. Indeed, if no such ξ' exists, then $\{\tau_k\}$ converges to zero. This implies that the sequence of integers $\{i_k\}$ is unbounded. Consequently, by the definition of i_k, we have, for all k sufficiently large,

$$\| G(p^k((1/2)^{i_k - 1}\bar{\tau}_k)) \| > (1 - \gamma (1/2)^{i_k - 1} \bar{\tau}_k) \| G(x^k) \|;$$

but this contradicts (8.1.10) with $\tau \equiv (1/2)^{i_k - 1}\bar{\tau}_k$. Consequently, the desired ξ' exists.

The inequality (8.1.10) implies that

$$\| G(x^{k+1}) \| \leq (1 - \tau_k \gamma) \| G(x^k) \|.$$

Passing to the limit $k \to \infty$, we deduce a contradiction because

$$\lim_{k \to \infty} \| G(x^k) \| \geq \eta > 0$$

and the sequence $\{\tau_k\}$ is bounded away from zero. This contradiction establishes that the sequence $\{G(x^k)\}$ converges to zero.

By Theorem **8.1.4**, G is a homeomorphism from $\mathbb{R}^n$ onto itself. Thus the set $S_{\|G(x^0)\|}$ is bounded. Since the sequence $\{x^k\}$ is contained in this bounded set, it is therefore bounded and thus has at least one accumulation point; moreover, every such point is a zero of G. Since G is a global homeomorphism, it has a unique zero to which $\{x^k\}$ must converge. To conclude the proof we show that eventually the algorithm sets $\tau_k = 1$ for all k sufficiently large. For this purpose, we recall that, as indicated at the beginning of the proof,

$$\bar{\tau}_k \geq \min(\varepsilon / \| G(x^k) \|, 1).$$

Since $\|G(x^k)\|$ goes to 0, this implies that eventually $\bar{\tau}_k = 1$. Reasoning in the same way as before, we see that the inequality

$$\| G(p^k(\tau)) \| \leq (1 - \tau\gamma) \| G(x^k) \|$$

holds for every τ satisfying

$$0 < \tau \leq \min \left(\frac{t_*}{L_{\mathcal{A}} \| G(x^k) \|}, \frac{\varepsilon}{\| G(x^k) \|} \right).$$

Since $\|G(x^k)\|$ goes to 0, the right-hand quantity goes to infinity; hence $\tau_k = 1$ eventually. Consequently, after a finite number of iterations, Algorithm **8.1.9** reduces to the local Newton method **7.2.4**. Since the present

assumptions imply the assumptions needed to establish Theorem **7.2.5**, the asserted convergence rates of the sequence $\{x^k\}$ follow immediately from Theorem **7.2.5**. □

As we have noted, the condition of a nonsingular uniform Newton approximation for a function is fairly restrictive. Algorithm **8.1.9** and its convergence result, Theorem **8.1.10**, are both valid in a relaxed setting. Specifically, the pointwise existence of a nonsingular Newton approximation is sufficient for the well-definedness of Algorithm **8.1.9**, by Lemma **8.1.7**. It turns out that the same convergence properties remain valid for the sequence $\{x^k\}$ produced by this algorithm, provided that a uniform inverse Lipschitz continuity property is imposed on the map $A(x, \cdot)$ for all x belonging to the level set $S_{\|G(x^0)\|}$, which we denote S_0 in the following theorem.

8.1.11 Theorem. Let $G : \mathbb{R}^n \to \mathbb{R}^n$ be locally Lipschitz on $\mathbb{R}^n$. Let x^0 be given. Assume that G has a Newton approximation $\mathcal{A}$ at every point in the set S_0. Suppose that there exist positive constants ε and L such that for every $x \in S_0$ and for each $A(x, \cdot) \in \mathcal{A}(x)$, there are two open sets U_x and V_x, both containing $\mathbb{B}(0, \varepsilon)$, such that $A(x, \cdot)$ is a Lipschitz homeomorphism mapping U_x onto V_x with L being the Lipschitz modulus of the inverse of the restricted map $A(x, \cdot)|_{U_x}$. The following two statements hold for the sequence $\{x^k\}$ produced by Algorithm **8.1.9**.

(a) The sequence $\{\|G(x^k)\|\}$ converges to zero.

(b) If S_0 is bounded, then the conclusions of Theorem **8.1.10** are all valid.

Proof. The proof of Theorem **8.1.10** can be applied to establish the two statements (a) and (b) under the present assumptions. In particular, the proof of (a) does not require the boundedness of the sequence $\{x^k\}$. □

Although Theorem **8.1.11** is a substantial improvement over Theorem **8.1.10** with the relaxation of the uniformity of the nonsingular Newton approximation, a major deficiency of the path method persists, namely, the method requires a nonsingular Newton approximation at each of the iterates. It is plausible that further study could lead to the removal or relaxation of such a stringent requirement, no such refinement of the path method exists at the time this book is being written.

8.2 Dini Stationarity

The path Newton method **8.1.9** provides a way of globalizing the convergence of the local Newton Algorithm **7.2.4** via a path search. While

theoretically elegant, the method is restrictive in terms of its convergence properties. In the remainder of this chapter, we present two alternative approaches to globalize the locally convergent Newton method that circumvent the nonsingularity requirement of the path method. Both of these approaches are based on applying a numerical routine to minimize a nonnegative merit function $\theta(x)$ that satisfies:

$$\theta(x) = 0 \quad \Leftrightarrow \quad G(x) = 0.$$

The nonnegativity of θ and the fact that we are seeking its zero are important features that distinguish some algorithms presented herein from other algorithms for solving general minimization problems that arise from different contexts.

In order to have a clear understanding of the object of computation of a minimization algorithm, we discuss below various concepts of stationarity for the constrained optimization problem:

$$\begin{array}{ll} \text{minimize} & \theta(x) \\ \text{subject to} & x \in X, \end{array} \tag{8.2.1}$$

where X is a closed set in $\mathbb{R}^n$ and θ is a nonsmooth function defined on an open set containing X. This problem provides the basic setting for the development of the algorithms throughout this chapter. In Subsection **1.5.3**, we have defined a stationary point for (8.2.1) with a directionally differentiable θ as a point $x \in X$ that satisfies (1.5.21), which we rephrase as follows:

$$\theta'(x; d) \geq 0, \quad \forall\, d \in \mathcal{T}(x; X), \tag{8.2.2}$$

where $\mathcal{T}(x; X)$ is the tangent cone of X at x. If the set X is convex, the condition (8.2.2) is equivalent to

$$\theta'(x; y - x) \geq 0, \quad \forall\, y \in X,$$

provided that θ is B-differentiable at x. If θ is not directionally differentiable, it is still possible to define a stationarity concept by replacing the directional derivative $\theta'(x; d)$, which is now not guaranteed to exist, by the *upper Dini directional derivative* defined as:

$$\theta^D(x; d) \equiv \limsup_{t \downarrow 0} \frac{\theta(x + td) - \theta(x)}{t}.$$

Note that because a lim sup is involved in its definition, $\theta^D(x; d)$ is well defined (but possibly equal to plus infinity) for every function θ and for

every pair (x, d). If θ is locally Lipschitz near x, then $\theta^D(x; d)$ is finite for all d; moreover, for every sequence $\{d^k\}$ converging to a limit d^∞, we have

$$\limsup_{k\to\infty} \theta^D(x; d^k) \leq \theta^D(x; d^\infty). \tag{8.2.3}$$

It is also easy to see that if θ is directionally differentiable at x, then $\theta^D(x; d) = \theta'(x; d)$ for all $d \in \mathbb{R}^n$. We call a feasible point $x^* \in X$ a *D-stationary point* ("D" for Dini) of the problem (8.2.1) if the upper Dini directional derivative of θ at x in the direction d is nonnegative for every $d \in \mathcal{T}(x; X)$; that is,

$$\theta^D(x; d) \geq 0, \qquad \forall\, d \in \mathcal{T}(x; X). \tag{8.2.4}$$

If X is convex and θ is locally Lipschitz near x, (8.2.4) is equivalent to

$$\theta^D(x; d) \geq 0, \qquad \forall\, d \in X - x. \tag{8.2.5}$$

Indeed, (8.2.4) clearly implies (8.2.5) because $X - x$ is a subset of $\mathcal{T}(x; X)$ by the convexity of X; the converse implication follows from (8.2.3) and the fact that the tangent cone $\mathcal{T}(x : X)$ is the closure of the cone generated by the set $X - x$.

The next result summarizes the relation between a D-stationary point and a local minimum of the constrained minimization problem (8.2.1). The second part of this result complements Proposition **7.4.12**, which pertains to an SC^1 minimization problem; unlike this previous result, the proposition below deals with first-order conditions for a minimization problem with a nondifferentiable objective function.

8.2.1 Proposition. Let $\theta : X \to \mathbb{R}$ be a real-valued function and X a closed subset of $\mathbb{R}^n$. Let $x^* \in X$ be given. If x^* is local minimum of (8.2.1), then (8.2.4) holds. Conversely, if θ is B-differentiable at x^* and $\theta'(x^*; d) > 0$ for every nonzero d belonging to $\mathcal{T}(x^*; X)$, then x^* is a strict local minimum of (8.2.1).

Proof. The first assertion is obvious. For the sake of contradiction, we assume that the second assertion is false. There exists a sequence of vectors $\{x^k\} \subset X$ converging to x^* such that $x^k \neq x^*$ and $\theta(x^*) \geq \theta(x^k)$ for all k. Without loss of generality, by letting

$$d^k \equiv \frac{x^k - x^*}{\| x^k - x^*\|},$$

we may assume that the sequence $\{d^k\}$ converges to a vector d^∞, which must belong to $\mathcal{T}(x^*; X)$. With $\tau_k \equiv \|x^k - x^*\|$, we may write

$$\theta(x^k) = \theta(x^* + \tau_k d^k) = \theta(x^* + \tau_k d^\infty) + [\,\theta(x^* + \tau_k d^k) - \theta(x^* + \tau_k d^\infty)\,].$$

Hence

$$\theta(x^* + \tau_k d^\infty) - \theta(x^*) \leq \theta(x^* + \tau_k d^k) - \theta(x^* + \tau_k d^\infty).$$

Dividing by τ_k, letting k tend to infinity, and utilizing the locally Lipschitz continuity of θ at x^*, we deduce

$$\theta'(x^*; d^\infty) \leq 0,$$

which is a contradiction. □

Based on Proposition **7.1.12**, we have defined an (unconstrained) C-stationary point of a locally Lipschitz function θ to be a vector x such that $0 \in \partial\theta(x)$. Recalling the equality (7.1.3), i.e.,

$$\partial\theta(x) = \{\xi \in \mathbb{R}^n : \xi^T d \leq \theta^\circ(x, d), \ \forall d \in \mathbb{R}^n\},$$

we see that a vector x^* is an (unconstrained) C-stationary point of θ if and only if

$$\theta^\circ(x^*; d) \geq 0, \quad \forall\, d \in \mathbb{R}^n. \tag{8.2.6}$$

We clearly have

$$\theta^D(x^*; d) \leq \theta^\circ(x^*; d), \quad \forall\, d \in \mathbb{R}^n.$$

Thus every D-stationary point must be C-stationary. Recalling Definition **7.1.7**, we see that if θ is C-regular at x^*, then D-stationarity and C-stationarity are the same concept, and both become the elementary stationarity concept (8.2.2) with $X = \mathbb{R}^n$, which is in terms of the directional derivative $\theta'(x^*; \cdot)$.

8.3 Line Search Methods

In this section we consider line search methods for solving the constrained minimization problem (8.2.1), and apply them to a least-squares objective function $\theta(x) \equiv G(x)^T G(x)/2$, which is closely related to solving the constrained system of equations (7.0.2). In order to get a better understanding of the line search methods and of the kind of requirements to establish their convergence, we assume initially that the objective function θ is continuously differentiable and $X = \mathbb{R}^n$. This is the classical unconstrained optimization problem of minimizing a C^1 function. Thanks to several differentiable optimization formulations of the VI/CP, such as those based on the differentiable C-functions for the NCP (see Subsection **1.5.1**), the line search methods for smooth functions can be employed to define a large

family of efficient descent methods for solving the VI/CP. Nevertheless, since other merit functions of the VI/CP are nonsmooth functions, it is therefore useful to consider line search methods for minimizing nonsmooth functions.

Consider first the problem of minimizing a C^1 function $\theta(x)$ on $\mathbb{R}^n$. Initiated at a given vector x^0, a line search algorithm generates a sequence $\{x^k\}$ that has the form

$$x^{k+1} \;=\; x^k + \tau_k\, d^k, \qquad k \;=\; 0, 1, \ldots,$$

where $d^k \neq 0$ is the *search direction* and τ_k is the *step size*. The precise definition of d^k and τ_k at each iteration distinguishes a particular algorithm from the others. Needless to say, we want the sequence of objective values $\{\theta(x^k)\}$ to be decreasing. Nevertheless, it is well known that a mere decrease of these values at each iteration is not sufficient to ensure the overall "convergence" of the generated sequence $\{x^k\}$. Care must be taken in choosing d^k and τ_k in order for the algorithm to possess good convergence properties and be practically efficient. The common goal is for the entire sequence $\{x^k\}$ to converge to a stationary point of θ. Under some second-order sufficiency conditions, such a point is a local minimizer of θ. For a nonconvex function θ, this is the best we can hope for with a line search method. In the important case where θ is the norm of a vector function G, an appropriate "regularity" condition at a stationary point of θ will imply that such a point will be a zero of G; see Proposition **1.5.13**.

Let x^k be a given nonstationary point; that is $\nabla\theta(x^k) \neq 0$. For a nonzero vector d^k, consider the vector $x^k + \tau d^k$ for $\tau \in (0, 1]$. We have, by a first-order expansion of θ at x^k,

$$\theta(x^k + \tau d^k) \;=\; \theta(x^k) + \tau\, \nabla\theta(x^k)^T d^k + o(\tau)$$

Therefore, if d^k satisfies the *descent condition*:

$$\nabla\theta(x^k)^T d^k \;<\; 0, \tag{8.3.1}$$

then for any constant $\gamma \in (0, 1)$, there exists a $\bar{\tau}_k \in (0, 1]$ such that for all $\tau \in [0, \bar{\tau}_k]$,

$$\theta(x^k + \tau d^k) \;\leq\; \theta(x^k) + \gamma\, \tau\, \nabla\theta(x^k)^T d^k;$$

see Lemma **8.3.1** below. In particular, with τ_k chosen to be one such τ, we deduce

$$\theta(x^{k+1}) \;\leq\; \theta(x^k) - \gamma\, \tau_k\, \sigma(x^k, d^k), \tag{8.3.2}$$

where we have let

$$\sigma(x^k, d^k) \;\equiv\; -\nabla\theta(x^k)^T d^k. \tag{8.3.3}$$

By the descent condition (8.3.1), $\sigma(x^k, d^k)$ is a positive number. There are many choices of the direction d^k that satisfies (8.3.1). A broad family of such directions is given by

$$d^k \equiv -H^k \nabla\theta(x^k),$$

for any symmetric positive definite matrix H^k. Letting H^k be the identity matrix yields the classical "steepest descent direction" $d_s^k \equiv -\nabla\theta(x^k)$. If the Hessian $\nabla^2\theta(x^k)$ exists and is positive definite, it is a legitimate choice for H^k too. The latter choice defines the "Newton direction"

$$d_N^k \equiv -(\nabla^2\theta(x^k))^{-1}\nabla\theta(x^k)$$

for a twice continuously differentiable function θ. The direction d_N^k coincides with the vector obtained by applying one iteration of Newton's method to the gradient equation $\nabla\theta(x) = 0$ at the iterate x^k, thus the name "Newton direction". For our purpose in this book, we seldom have the luxury of dealing with a C^2 function; thus the Newton direction d_N^k is of no use to us. Nevertheless, this consideration is very useful for the class of SC^1 objective functions θ (recall that this is the subclass of C^1 functions whose gradients are semismooth vector functions; see Subsection **7.4.1**).

Repeating the above general process we see that unless we land at a stationary point of θ after a finite number of iterations (at which time the process terminates), we obtain a sequence $\{x^k\}$ of vectors such that (8.3.2) holds for all k. In particular, the sequence of functional values $\{\theta(x^k)\}$ is strictly decreasing. Suppose that θ is bounded from below. (This condition is trivially satisfied if θ is the squared Euclidean norm of a vector function G.) The sequence $\{\theta(x^k)\}$ converges. By (8.3.2), this implies that

$$\lim_{k\to\infty} \tau_k \, \sigma(x^k, d^k) = 0. \tag{8.3.4}$$

Suppose that κ is an infinite subset of $\{1, 2, \ldots\}$ such that

$$\liminf_{k(\in\kappa)\to\infty} \tau_k > 0.$$

By (8.3.4), we have

$$\lim_{k(\in\kappa)\to\infty} \sigma(x^k, d^k) = 0.$$

Suppose further that there exist positive constants c and ζ such that

$$\sigma(x^k, d^k) \geq c \, \| \nabla\theta(x^k) \|^\zeta, \quad \forall\, k.$$

(Note: with d^k being the steepest descent direction and $\sigma(x^k, d^k)$ defined by (8.3.3), the above inequality clearly holds as an equality with $c = 1$ and

$\zeta = 2$.) We then have

$$\lim_{k(\in\kappa)\to\infty} \nabla\theta(x^k) \;=\; 0.$$

Thus the subsequence $\{x^k : k \in \kappa\}$ is "asymptotically stationary". Notice that this property does not require the subsequence on hand to be bounded; if $\{x^k : k \in \kappa\}$ is bounded, then every accumulation point of this (sub)sequence is a stationary point of θ.

The above discussion provides the basis for extension to a nonsmooth θ. In essence, we need at every iteration a direction along which we can decrease the objective function adequately according to a criterion such as (8.3.2). This criterion is defined in terms of a "forcing function" whose principal role is to ensure that the accumulation points of the generated sequence of iterates, if they exist, are stationary points of some sort. When the feasible set X is a proper subset of $\mathbb{R}^n$, we need to deal with the feasibility issue as well. If X is convex, this can be easily dealt with by requiring the search direction d^k to belong to $X - x^k$. With x^k in X to start with, the convexity of X then implies that $x^k + \tau d^k$ belongs to X for all $\tau \in [0, 1]$.

Throughout the following discussion, X is a convex subset of $\mathbb{R}^n$ and θ is a locally Lipschitz continuous function on X. We introduce an abstract multifunction Ξ from X into $\mathbb{R}^n$ to describe the set of admissible directions to be employed in each iteration of a line search algorithm. Specifically, for any vector $x \in X$, consider the set of all vectors d satisfying the following feasible descent condition:

$$d \in X - x \quad \text{and} \quad \theta^D(x; d) < 0. \tag{8.3.5}$$

Obviously, if no such vector d exists, then x is a D-stationary point of θ on X; in this case, $\Xi(x)$ is defined to be the empty set; otherwise, $\Xi(x)$ is taken to be a nonempty subset of the above set of vectors d satisfying (8.3.5). Subsequently, we consider specific algorithms where elements of $\Xi(x)$ are further specified. For now, elements of $\Xi(x)$ are required to satisfy only (8.3.5). As a multifunction, the effective domain of Ξ is the set of non-Dini stationary points of the optimization problem (8.2.1). The next lemma justifies why (8.3.5) is a reasonable descent condition for a nonsmooth function θ and identifies a key condition for the forcing function $\sigma(x, d)$ to be used in the generation of the line-search iterates. This lemma, which we call the *descent lemma*, is the analog of the last assertion of Lemma **8.1.7**, which dealt with a path method.

8.3.1 Lemma. Let X be a convex subset of $\mathbb{R}^n$ and θ be a locally Lipschitz continuous function. Let $(x, d) \in \operatorname{gph} \Xi$ be given. For every constant $\gamma \in (0, 1)$ and positive scalar $\sigma(x, d)$ satisfying

$$\sigma(x, d) \leq -\theta^D(x; d),$$

there exists $\bar{\tau} \in (0, 1]$ such that for all $\tau \in [0, \bar{\tau}]$,

$$\theta(x + \tau d) \leq \theta(x) - \gamma \tau \sigma(x, d).$$

Proof. Assume for the sake of contradiction that no such scalar $\bar{\tau}$ exists. There exists a sequence $\{\tau_\nu\}$ of positive scalars converging to zero such that

$$\theta(x + \tau_\nu d) - \theta(x) > -\gamma \tau_\nu \sigma(x, d), \quad \forall \nu.$$

Dividing by τ_ν and letting ν tend to ∞, we deduce

$$\theta^D(x; d) \geq -\gamma \sigma(x, d).$$

By assumption, the left-hand quantity is not greater than $-\sigma(x, d)$, which is a negative number. Since $\gamma \in (0, 1)$, we obtain a contradiction. □

Based on the above setting, we are ready to present the general form of a line search algorithm for minimizing a locally Lipschitz function θ on a convex set X. Each realization of this algorithm requires the specification of the line search map Ξ and of the forcing function σ.

General Line Search Algorithm (GLSA)

8.3.2 Algorithm.

Data: $x^0 \in X$, $\gamma \in (0, 1)$, and a function $\sigma : \mathbb{R}^{2n} \to \mathbb{R}_+$ satisfying

$$0 < \sigma(x, d) \leq -\theta^D(x; d) \quad \forall (x, d) \in \operatorname{gph} \Xi.$$

Step 1: Set $k = 0$.

Step 2: If x^k is a D-stationary point of θ on X, stop.

Step 3: Choose a vector d^k from the set $\Xi(x^k)$.

Step 4: Find the smallest nonnegative integer i_k such that with $i = i_k$,

$$\theta(x^k + 2^{-i} d^k) \leq \theta(x^k) - \gamma 2^{-i} \sigma(x^k, d^k);$$

set $\tau_k \equiv 2^{-i_k}$.

Step 5: Set $x^{k+1} = x^k + \tau_k d^k$ and $k \leftarrow k + 1$; go to Step 2.

Step 4 implements the Armijo step-size rule; Lemma **8.3.1** justifies that the step size τ_k can be obtained after a finite number of trials. Similar to the path method **8.1.9**, we could use any backtracking factor $\rho \in (0,1)$ other than one-half. The convexity of X ensures that each iterate x^{k+1} belongs to X. Thus, the above algorithm generates a well-defined sequence of feasible vectors $\{x^k\} \subset X$ such that

$$\theta(x^{k+1}) - \theta(x^k) \;\leq\; -\gamma\,\tau_k\,\sigma(x^k, d^k), \quad \forall\, k, \tag{8.3.6}$$

and if $i_k \geq 1$,

$$\theta(x^k + (1/2)^{i_k-1} d^k) - \theta(x^k) \;>\; -\gamma\,(1/2)^{i_k-1}\,\sigma(x^k, d^k). \tag{8.3.7}$$

In order to investigate the convergence properties of the sequence $\{x^k\}$, we need to impose a technical condition on the forcing sequence $\{\sigma(x^k, d^k)\}$. Generalizing the requirement that $\sigma(x,d) \leq -\theta^D(x,d)$, which is equivalent to

$$-\sigma(x,d) \;\geq \limsup_{\tau \to 0+} \frac{\theta(x+\tau d) - \theta(x)}{\tau},$$

we postulate a key assumption, labeled (LS) in Theorem **8.3.3** below, that can be satisfied fairly easily in the smooth case. The following is a convergence result of Algorithm **8.3.2** under this postulate.

8.3.3 Theorem. Let X be a convex subset of $\mathbb{R}^n$ and let θ be a locally Lipschitz function on X. Let $\{x^k\} \subset X$ be an infinite sequence of vectors generated by Algorithm **8.3.2**. If $\{x^k : k \in \kappa\}$ is a subsequence of $\{x^k\}$ satisfying the following two conditions:

(BD_θ) the objective sequence $\{\theta(x^k) : k \in \kappa\}$ is bounded below;

(LS) for every sequence $\{t_k : k \in \kappa\}$ of positive scalars converging to zero,

$$\limsup_{k(\in\kappa)\to\infty} \frac{\theta(x^k + t_k d^k) - \theta(x^k) + t_k\,\sigma(x^k, d^k)}{t_k} \;\leq\; 0,$$

then

$$\lim_{k(\in\kappa)\to\infty} \sigma(x^k, d^k) \;=\; 0.$$

Proof. The sequence $\{\theta(x^k)\}$ is strictly decreasing. Since $\{\theta(x^k) : k \in \kappa\}$ is bounded below by (BD_θ), it follows that this subsequence $\{\theta(x^k) : k \in \kappa\}$, and thus the entire sequence $\{\theta(x^k)\}$, converges. Hence

$$\lim_{k\to\infty} \left(\, \theta(x^{k+1}) - \theta(x^k) \,\right) \;=\; 0.$$

From (8.3.6), we deduce

$$\lim_{k\to\infty} \tau_k \, \sigma(x^k, d^k) \;=\; 0. \tag{8.3.8}$$

Assume for the sake of contradiction that $\{\sigma(x^k, d^k) : k \in \kappa\}$ does not converge to zero. There exists an infinite subset κ' of κ such that

$$\liminf_{k(\in\kappa')\to\infty} \sigma(x^k, d^k) \;>\; 0,$$

By (8.3.8), it follows that

$$\lim_{k(\in\kappa')\to\infty} \tau_k \;=\; 0.$$

This implies

$$\lim_{k(\in\kappa')\to\infty} i_k \;=\; \infty.$$

Thus (8.3.7) implies that, for all $k \in \kappa'$ sufficiently large,

$$\frac{\theta(x^k + \tau_k' d^k) - \theta(x^k)}{\tau_k'} \;>\; -\gamma \, \sigma(x^k, d^k),$$

where $\tau_k' \equiv (1/2)^{i_k - 1}$. Thus

$$\frac{\theta(x^k + \tau_k' d^k) - \theta(x^k) + \tau_k' \, \sigma(x^k, d^k)}{\tau_k'} \;>\; (\,1 - \gamma\,) \, \sigma(x^k, d^k),$$

Letting $k(\in \kappa') \to \infty$, we obtain, by (LS),

$$(\,1 - \gamma\,) \, \limsup_{k(\in\kappa)\to\infty} \sigma(x^k, d^k) \;\leq\; 0$$

Since $\gamma < 1$ and $\sigma(x^k, d^k)$ is nonnegative, the above inequality yields

$$\limsup_{k(\in\kappa')\to\infty} \sigma(x^k, d^k) \;=\; 0,$$

which is a contradiction. □

Assumption (BD_θ) obviously holds if either $\{x^k : k \in \kappa\}$ is bounded or θ is bounded below on X. Notice that Theorem **8.3.3** requires neither one of these two conditions to hold; in particular, the theorem does not assume that the sequence $\{x^k\}$ is bounded. Since the sequence $\{x^k\}$ is obviously contained in the level set

$$\{\, x \in X \,:\, \theta(x) \;\leq\; \theta(x^0)\,\},$$

a sufficient condition for $\{x^k\}$ to be bounded is that the above set is bounded. In turn, if θ is coercive on X, that is, if

$$\lim_{\substack{x \in X \\ \|x\| \to \infty}} \theta(x) = \infty,$$

then all level sets of θ are bounded. As will be shown below, condition (LS) holds rather easily when θ is continuously differentiable.

From the proof of Theorem **8.3.3**, it is apparent that we can actually choose the step size τ_k according to more relaxed rules and still retain all the convergence properties of the GLSA. In particular, it is not necessary to halve the step size until a suitable one is found. If a certain step size τ fails the Armijo test in Step 4, we can next try any value in the interval $[\beta_{\min}\tau, \beta_{\max}\tau]$, where $\beta_{\min} \leq \beta_{\max} < 1$ are two positive constants. More generally, it is easy to check that any step-size rule, which at each iteration gives a larger reduction of the objective value than the Armijo rule (or any of its variants), inherits the convergence properties stated in Theorem **8.3.3**. The only important remark to be made with respect to the various step-size rules is that if one wants to obtain a locally fast convergence rate for the resulting algorithm, then it is important that eventually the Armijo test is satisfied by a unit step size; the reasons for this are discussed later in this section.

The use of the step-size variants can certainly be important in the practical implementation of the GLSA. However, since they do not add much to the main theoretical issues we are dealing with and since their explicit consideration would complicate considerably the notation and the exposition, we prefer to stick to the simplest scheme of the GLSA as described in Algorithm **8.3.2**.

In what follows, we consider several realizations of Algorithm **8.3.2**. The focus of the discussion is on the generation of the search directions and the verification of the condition (LS). In all cases, we aim at establishing that every accumulation point of the generated sequence of iterates is a stationary point of the minimization problem. This kind of conclusion is referred to as the *subsequential convergence* of the method. Subsequently, we discuss *sequential convergence* under appropriate conditions; the latter kind of convergence asserts that the entire sequence of iterates converges to a stationary point of the problem in question.

We begin with a straightforward generalization of the descent methods described at the opening of this section. Specifically, we consider the constrained problem (8.2.1), where X is a closed convex subset of $\mathbb{R}^n$ and θ is a B-differentiable function. For a given vector $x \in X$ and a symmetric

positive definite matrix H, consider the following minimization problem in the variable d:

$$\begin{array}{ll} \text{minimize} & \theta'(x;d) + \frac{1}{2}\, d^T H d \\ \text{subject to} & d \in X - x. \end{array} \tag{8.3.9}$$

In general, if θ is not smooth, the objective function of (8.3.9) is not convex. Nevertheless, this program has several important properties, which we summarize in the result below.

8.3.4 Proposition. Let X be a closed convex subset of $\mathbb{R}^n$, $x \in X$, and H be a symmetric positive definite matrix. Let θ be a B-differentiable function. The following two statements are valid.

(a) The program (8.3.9) attains a finite optimum solution.

(b) The optimum objective value of (8.3.9) is nonpositive; it is equal to zero if and only if x is a stationary point of (8.2.1).

Proof. The objective function of (8.3.9) is coercive, that is

$$\lim_{\|d\| \to \infty} [\, \theta'(x;d) + \tfrac{1}{2}\, d^T H d \,] \;=\; \infty.$$

Since it is also continuous, statement (a) follows readily. Since $d = 0$ is a feasible solution of (8.3.9), the optimum objective value of this program is clearly nonpositive. If x is a stationary point of (8.3.9), then $\theta'(x;d) \geq 0$ for all d in $X - x$. Thus (8.3.9) must have a zero optimum objective value. Conversely, if the optimum objective value of (8.3.9) is zero, then for all $d \in X - x$, we have

$$\theta'(x;d) + \tfrac{1}{2}\, d^T H d \;\geq\; 0.$$

For a vector $y \in X$, the vector $x + \tau(y - x)$ belongs to X for all $\tau \in [0,1]$ by convexity of X. Thus $\tau(y - x)$ belongs to $X - x$; hence

$$\tau\, \theta'(x; y - x) + \frac{\tau^2}{2}\, (\, y - x\,)^T H (\, y - x\,) \;\geq\; 0.$$

Dividing by $\tau > 0$ and letting $\tau \to 0$, we deduce

$$\theta'(x; y - x) \;\geq\; 0, \quad \forall\, y \;\in\; X.$$

In other words, x is a stationary point of (8.2.1). □

For a nonstationary point x of (8.2.1), let $\Xi(x, H)$ be the set of optimal solutions of (8.3.9) corresponding to a prescribed symmetric positive definite matrix H. The next lemma shows that the union of these solutions is bounded if the vector x stays in a bounded set and the matrix H is

"uniformly positive definite"; see condition (8.3.10). The notation $\lambda_{\min}(A)$ in the displayed condition denotes the smallest eigenvalue of a symmetric matrix A.

8.3.5 Lemma. Let X be a closed convex subset of $\mathbb{R}^n$ and let θ be a B-differentiable function. For every convergent sequence $\{x^\nu\}$ of nonstationary points of (8.2.1) and every sequence of symmetric positive definite matrix $\{H^\nu\}$ such that

$$0 < \inf_\nu \lambda_{\min}(H^\nu) \tag{8.3.10}$$

the union

$$\bigcup_\nu \Xi(x^\nu, H^\nu)$$

is bounded.

Proof. Since θ is locally Lipschitz and the sequence $\{x^\nu\}$ converges, it follows that there exists a constant $L > 0$ such that

$$|\,\theta'(x^\nu; d)\,| \le L\,\|\,d\,\|, \quad \forall\,\nu \text{ and } \forall\, d \in \mathbb{R}^n.$$

For any vector $d \in \Xi(x^\nu, H^\nu)$, we have

$$\theta'(x^\nu; d) + \tfrac{1}{2}\, d^T H^k d \le 0.$$

Letting $c > 0$ be a common positive lower bound of the smallest eigenvalues of the matrices H^ν for all ν, we easily deduce $\|d\| \le 2L/c$. Therefore all such optimal solutions d are bounded in norm by a constant independent of ν. □

Letting $\sigma(x, d) \equiv -\theta'(x; d)$, we obtain a first specialization of Algorithm **8.3.2** for minimizing a B-differentiable function θ on a convex set X; this specialized algorithm is certainly applicable when θ is continuously differentiable.

B-Differentiable Line Search Algorithm (BDLSA)

8.3.6 Algorithm.

Data: $x^0 \in X$, $\gamma \in (0, 1)$ and a sequence of symmetric positive definite matrices $\{H^k\}$.

Step 1: Set $k = 0$.

Step 2: If x^k is a stationary point of θ on X, stop.

Step 3: Choose a vector d^k from the set $\Xi(x^k, H^k)$.

Step 4: Find the smallest nonnegative integer i_k such that with $i = i_k$,

$$\theta(x^k + 2^{-i} d^k) \;\le\; \theta(x^k) + \gamma\, 2^{-i}\, \theta'(x^k; d^k);$$

set $\tau_k \equiv 2^{-i_k}$.

Step 5: Set $x^{k+1} = x^k + \tau_k d^k$ and $k \leftarrow k+1$; go to Step 2.

Proposition **8.3.4** and Lemma **8.3.1** justify that the above algorithm is well defined. We have the following convergence result for this algorithm.

8.3.7 Proposition. Let X be a closed convex subset of $\mathbb{R}^n$ and let θ be a B-differentiable function on X. Let $\{x^k\} \subset X$ be an infinite sequence of vectors generated by Algorithm **8.3.6**. If κ is an infinite subset of $\{0, 1, 2, \ldots\}$ such that

(a) there exist positive scalars c_1 and c_2 such that for every $k \in \kappa$,

$$c_1\, y^T y \;\le\; y^T H^k y \;\le\; c_2\, y^T y, \quad \forall\, y \in \mathbb{R}^n, \tag{8.3.11}$$

(b) the subsequence $\{x^k : k \in \kappa\}$ converges to a vector x^*,

(c) θ has a strong F-derivative at x^*;

then x^* is a stationary point of (8.2.1).

Proof. By Exercise **3.7.2**, the strong F-differentiability of θ at x^* implies that the directional derivative θ' is continuous at (x^*, v) for every $v \in \mathbb{R}^n$.

Clearly, the vector x^* belongs to X. We need to verify that the subsequence $\{x^k : k \in \kappa\}$ satisfies the condition (LS). By Lemma **8.3.5**, the sequence $\{d^k : k \in \kappa\}$ is bounded. Without loss of generality, we may assume that $\{d^k : k \in \kappa\}$ converges to a vector d^∞. Let $\{t_k : k \in \kappa\}$ be a sequence of positive scalars converging to zero. By the strong F-differentiability of θ at x^*, we can write, for all $k \in \kappa$ sufficiently large,

$$\theta(x^k + t_k d^k) - \theta(x^k) \;=\; t_k\, \nabla\theta(x^*)^T d^k + o(t_k);$$

thus

$$\lim_{k(\in\kappa)\to\infty} \frac{\theta(x^k + t_k d^k) - \theta(x^k)}{t_k} \;=\; \nabla\theta(x^*)^T d^\infty.$$

Since θ' is continuous at (x^*, d^∞), we deduce

$$\lim_{k(\in\kappa)\to\infty} \sigma(x^k, d^k) \;=\; -\lim_{k(\in\kappa)\to\infty} \theta'(x^k; d^k) \;=\; -\nabla\theta(x^*)^T d^\infty.$$

Consequently (LS) holds. By Theorem **8.3.3**, we obtain $\nabla\theta(x^*)^T d^\infty = 0$. Let y be an arbitrary vector in X. We need to show that

$$\nabla\theta(x^*)^T (y - x^*) \;\ge\; 0.$$

Without loss of generality, we may assume that the sequence of matrices $\{H^k : k \in \kappa\}$ converges to a limit H^∞. For every k and every $t \in [0,1]$, we have $t(y - x^k) \in X - x^k$; hence

$$t\,\theta'(x^k; y - x^k) + \tfrac{t^2}{2}\,(y - x^k)^T H^k (y - x^k) \geq$$

$$\theta'(x^k; d^k) + \tfrac{1}{2}\,(d^k)^T H^k d^k \geq \theta'(x^k; d^k).$$

Letting $k(\in \kappa) \to \infty$ and using the continuity of θ' at (x^*, v) for any vector v, we deduce

$$t\,\theta'(x^*; y - x^*) + \frac{t^2}{2}\,(y - x^*)^T H^\infty (y - x^*) \geq 0.$$

Since this holds for all $t \in (0,1]$, dividing by $t > 0$ and letting $t \to 0$, we obtain $\theta'(x^*; y - x^*) \geq 0$ as desired. □

From the proof of Proposition **8.3.7**, we see that the postulate (LS) holds easily if (i) the sequence $\{(x^k, d^k) : k \in \kappa\}$ is convergent and (ii) θ is continuously differentiable. The latter smooth case is the principal source of application for Algorithm **8.3.6** and its convergence result, Proposition **8.3.7**. Therefore, it follows that for a C^1 function θ, if we choose the sequence $\{H^k\}$ such that for some positive constants c_1 and c_2 (8.3.11) holds for all k, then every accumulation point of the sequence $\{x^k\}$ generated by Algorithm **8.3.6** is a stationary point of (8.2.1); that is, the algorithm is subsequentially convergent. Moreover, in this case, the objective function of the directional subprogram (8.3.9) is quadratic and strictly convex, which we can write as

$$\begin{array}{ll} \text{minimize} & \nabla\theta(x)^T d + \frac{1}{2}\, d^T H d \\ \text{subject to} & d \in X - x. \end{array} \tag{8.3.12}$$

The optimal solution set $\Xi(x, H)$ of the above program is a singleton. If in addition X is polyhedral, then (8.3.12) is a strictly convex quadratic program.

We next investigate Algorithm **8.3.2** applied to minimize the function $\theta(x) \equiv \frac{1}{2} G(x)^T G(x)$ on $X = \mathbb{R}^n$, where G is a given vector function from $\mathbb{R}^n$ into itself. The two procedures discussed below exploit the special definition of θ in generating the search directions. We first consider the smooth case where G is continuously differentiable. Specifically, we study a modification of the well-known Gauss-Newton method for solving the equation $G(x) = 0$. The method is as follows. Given an iterate x^k that is not a solution of the latter equation, we compute the search direction d^k

by solving the system of linear equations:

$$(\, JG(x^k)^T JG(x^k) + \| G(x^k) \| \, \mathbf{I} \,) d \;=\; -JG(x^k)^T G(x^k). \tag{8.3.13}$$

Since $G(x^k)$ is nonzero, the matrix on the left-hand side of (8.3.13) is symmetric positive definite; thus (8.3.13) always has a unique solution. With $\theta(x) = \frac{1}{2} G(x)^T G(x)$, we have

$$\nabla\theta(x) \;=\; JG(x)^T G(x) \;=\; \sum_{i=1}^{n} G_i(x) \, \nabla G_i(x).$$

Thus the equation (8.3.13) is equivalent to the minimization of the strictly convex quadratic function

$$\nabla\theta(x^k)^T d + \tfrac{1}{2} \, d^T H^k d,$$

where

$$H^k \;\equiv\; JG(x^k)^T JG(x^k) + \| G(x^k) \| \, \mathbf{I}. \tag{8.3.14}$$

In the notation introduced above, the unique solution d^k of (8.3.13) is the single element of the set $\Xi(x^k, H^k)$. As in Algorithm **8.3.6**, let

$$\sigma(x^k, d^k) \;\equiv\; -\theta'(x^k; d^k).$$

By the F-differentiability of θ, we have

$$\sigma(x^k, d^k) \;=\; -\nabla\theta(x^k)^T d^k \;=\; -(\, JG(x^k)^T G(x^k) \,)^T d^k,$$

which implies, by (8.3.13),

$$\sigma(x^k, d^k) \;=\; \| \, JG(x^k) \, d^k \, \|^2 + \|G(x^k)\| \, \|d^k\|^2. \tag{8.3.15}$$

We have the following convergence result for the sequence $\{x^k\}$ generated by this procedure, which we refer to as a *modified Gauss-Newton method*.

8.3.8 Proposition. Let $G : \mathbb{R}^n \to \mathbb{R}^n$ be continuously differentiable. Every accumulation point of the sequence $\{x^k\}$ generated by the modified Gauss-Newton method is an (unconstrained) stationary point of $\frac{1}{2} G^T G$.

Proof. Let $\{x^k : k \in \kappa\}$ be an arbitrary subsequence converging to a vector x^*. If $G(x^*) = 0$, then x^* is a global minimum, and hence a stationary point of θ. Thus we may assume that $G(x^*) \neq 0$. With the identification (8.3.14), if follows that the subsequence $\{x^k : k \in \kappa\}$ and its limit x^* satisfy the conditions (a), (b), and (c) in Proposition **8.3.7**. By this proposition, the desired conclusion follows readily. □

It goes without saying that if x is a stationary point of θ and if the Jacobian matrix $JG(x)$ is nonsingular, then x is a desired zero of G. By means of the regularized equation (8.3.13), the procedure presented above generates a well-defined sequence $\{x^k\}$ without requiring the individual matrices $JG(x^k)$ to be nonsingular. The nonsingularity of the Jacobian matrix at a limit point of the generated sequence is needed only as a theoretical guarantee that such a point is indeed a zero of the equation $G(x) = 0$ under consideration.

We next consider the case where G is not C^1 but θ is continuously differentiable. In the next chapter, we see that this case includes many nonsmooth systems $G(x) = 0$, where G is a semismooth vector function and $G^T G$ is C^1 (for instance, recall the squared Fischer-Burmesiter function of an NCP; see Example **7.4.9**).

In what follows, let G be a locally Lipschitz function on $\mathbb{R}^n$ such that $\theta(x) \equiv \frac{1}{2} G(x)^T G(x)$ is C^1. Let T be a Newton approximation scheme for G defined on $\mathbb{R}^n$. (The following discussion does not require T to be nonsingular.) To define the search directions we proceed as follows. Let two positive constants $\rho > 0$ and $p > 1$ be given. Select an element $H^k \in T(x^k)$ and find a solution d^k of the system of linear equations:

$$G(x^k) + H^k d \;=\; 0. \tag{8.3.16}$$

If the system (8.3.16) is not solvable or if the condition

$$\nabla\theta(x^k)^T d^k \;\leq\; -\rho \,\|\, d^k \,\|^p \tag{8.3.17}$$

is not met, set $d^k = -\nabla\theta(x^k)$. We also let

$$\sigma(x^k, d^k) \equiv -\nabla\theta(x^k)^T d^k.$$

We demonstrate that this procedure is subsequentially convergent. Let $\{x^k : k \in \kappa\}$ be a convergent subsequence with limit x^*. By the definition of d^k, we have

$$\|\, d^k \,\| \;\leq\; \max\{\, \|\, \nabla\theta(x^k) \,\|,\, (\, \rho^{-1} \,\|\, \nabla\theta(x^k) \,\|\,)^{1/(p-1)} \,\}.$$

Thus $\{d^k : k \in \kappa\}$ is bounded. As in the proof of Proposition **8.3.8**, we deduce

$$\lim_{k(\in\kappa)\to\infty} \sigma(x^k, d^k) \;=\; 0.$$

If $d^k = -\nabla\theta(x^k)$ for infinitely many $k \in \kappa$, then, $\nabla\theta(x^*) = 0$; hence x^* is a stationary point of θ. If (8.3.16) and (8.3.17) hold for all but finitely many $k \in \kappa$, then (8.3.17) implies that $\{d^k : k \in \kappa\}$ converges to zero.

Since $G(x^k) + H^k d^k = 0$ and $\{H^k : k \in \kappa\}$ is bounded by the upper semicontinuity of T, it follows that $G(x^*) = 0$. Therefore, in either case, we have established that every accumulation point of the sequence $\{x^k\}$ is a stationary point of θ.

We still need to address the question of when a stationary point of θ is a zero of G. Generalizing the C^1 case, we can easily show that for a B-differentiable vector function G, if x is a stationary point of θ and the directional derivative $G'(x;\cdot)$ is a surjective map, then x must be zero of G. Indeed, the surjectivity of $G'(x;\cdot)$ implies the existence of a vector d satisfying

$$G(x) + G'(x;d) = 0.$$

Since x is a stationary point of θ, we have

$$0 = \nabla\theta(x)^T d = \sum_{i=1}^{n} G_i(x) G_i'(x;d) = -\| G(x) \|^2.$$

This simple proof is a specialization of the proof of Proposition **1.5.13**, which pertains to a function θ with no special structure.

Except for the upper semicontinuity property, the fact that T is a Newton approximation scheme of G has not been exploited in the above procedure. The approximation property of T is important when it comes to study the rate of convergence of the method. In essence, the above procedure provides a constructive approach to globalize the convergence of the linear Newton method **7.5.14**. Under a suitable condition, the global method eventually becomes the local method; when that happens, we can then deduce the fast local convergence rate of the former procedure based on the theory established in Subsection **7.5.1**.

8.3.1 Sequential convergence

We next turn our attention to the issue of sequential convergence. We state below a well-known classical result due to Ostrowski that describes an important property of the set of accumulation points of any bounded sequence satisfying a limit condition, which we call the *Ostrowski condition*; see (8.3.18) below.

8.3.9 Theorem. Let $\{x^\nu\}$ be a bounded sequence of vectors in $\mathbb{R}^n$ satisfying

$$\lim_{\nu\to\infty} \| x^{\nu+1} - x^\nu \| = 0. \tag{8.3.18}$$

The set of accumulation points of $\{x^\nu\}$ is nonempty, closed, and connected. □

There are many consequences of this theorem. As an illustration, Theorem **8.3.9** yields the following corollary. A bounded sequence satisfying the Ostrowski condition (8.3.18) that has finitely many accumulation points must converge. For our purpose, we want to establish a criterion for sequential convergence without assuming *a priori* the boundedness of the sequence. We say that an accumulation point of a sequence is *isolated* if there exists a neighborhood of this point within which no other accumulation point exists. In addition to not making the boundedness assumption, the following result shows that sometimes it is possible to prove sequential convergence under an assumption other than the Ostrowski condition; see statement (b) below.

8.3.10 Proposition. Let $\{x^\nu\}$ be a sequence of vectors in $\mathbb{R}^n$ with an isolated accumulation point x^∞. The following three statements are equivalent.

(a) Ostrowski's condition (8.3.18) holds.

(b) For every subsequence $\{x^\nu : \nu \in \kappa\}$ converging to x^∞, it holds that

$$\lim_{\nu(\in\kappa)\to\infty} \| x^{\nu+1} - x^\nu \| = 0. \tag{8.3.19}$$

(c) The sequence $\{x^\nu\}$ converges to x^∞.

Proof. (a) $\Rightarrow$ (b). This is obvious.

(b) $\Rightarrow$ (c). Let $\varepsilon > 0$ be a scalar such that no other accumulation point exists in $\mathbb{B}(x^\infty, \varepsilon)$. For the sake of contradiction, we assume that $\{x^\nu\}$ does not converge to x^∞. For each positive integer ν, let $p(\nu)$ be the first integer greater than ν such that

$$\| x^{p(\nu)} - x^\infty \| > \varepsilon.$$

Such an integer $p(\nu)$ is well defined because $\{x^\nu\}$ does not converge to x^∞ and x^∞ is the only accumulation point in $\mathbb{B}(x^\infty, \varepsilon)$. Except for possibly finitely many members, the sequence $\{x^{p(\nu)-1}\}$ is contained in the compact ball $\operatorname{cl} \mathbb{B}(x^\infty, \varepsilon)$. Therefore the sequence $\{x^{p(\nu)-1}\}$ must converge to x^∞. By (8.3.19), $\{x^{p(\nu)} : \nu \in \kappa\}$ also converges to x^∞. But this is impossible by the definition of $p(\nu)$. □

If the sequence $\{x^k\}$ of iterates is generated by Algorithm **8.3.2**, the satisfaction of condition (b) in Proposition **8.3.10** usually requires only an appropriate nonsingularity condition on an accumulation point. The significance of this proposition is that knowing the existence of an isolated accumulation point x^∞ of the sequence $\{x^k\}$ that satisfies condition (b)

is sufficient to guarantee the convergence of the entire sequence to that point. This is in contrast to Ostrowski's original theorem (and its corollary mentioned above), which requires the entire sequence $\{x^k\}$ to be bounded in the first place and also the sequential limit (8.3.18) to hold. Proposition **8.3.11** provides a simple condition for the subsequential limit (8.3.19) to hold in a line search method; the condition is a linkage between the sequence $\{d^k : k \in \kappa\}$ of directions and the sequence $\{\sigma(x^k, d^k) : k \in \kappa\}$ of forcing quantities. Notice that condition (LS) is not needed in the following result.

8.3.11 Proposition. Let X be a convex subset of $\mathbb{R}^n$ and let θ be a locally Lipschitz function on X. Let $\{x^k\} \subset X$ be an infinite sequence of vectors generated by Algorithm **8.3.2**. Suppose that $\{x^k : k \in \kappa\}$ is a subsequence of $\{x^k\}$ satisfying condition (BD_θ) in Theorem **8.3.3**. If there exist constants c and $\zeta \in (0, 1)$ such that

$$\| d^k \| \le c \max(\sigma(x^k, d^k), \sigma(x^k, d^k)^\zeta), \quad \forall\, k \in \kappa, \tag{8.3.20}$$

then the subsequential limit (8.3.19) holds.

Proof. By the proof of Theorem **8.3.3**, the limit (8.3.8) holds; that is

$$\lim_{k\to\infty} \tau_k \, \sigma(x^k, d^k) = 0.$$

(This proof does not require condition (LS).) Since the sequence $\{\tau_k\}$ of step sizes is bounded, the above limit and (8.3.20) easily imply

$$\lim_{k(\in\kappa)\to\infty} \tau_k \| d^k \| = 0,$$

Since $x^{k+1} - x^k = \tau_k d^k$, the limit (8.3.19) follows readily. □

Applied to the entire sequence $\{x^k\}$, Proposition **8.3.11** provides a sufficient condition for Ostrowski's sequential limit (8.3.18) to hold. We illustrate the satisfaction of the latter limit in two instances. First consider Algorithm **8.3.6** and assume that some subsequence of $\{\theta(x^k)\}$ is bounded below. Since

$$\theta'(x^k; d^k) + \tfrac{1}{2} (d^k)^T H^k d^k \le 0$$

and $\sigma(x^k, d^k) \equiv -\theta'(x^k; d^k)$, provided that

$$c_1 \equiv \inf_k \lambda_{\min}(H^k) > 0,$$

we obtain

$$\| d^k \| \le c_1^{-1/2} \sqrt{\sigma(x^k, d^k)}, \quad \forall\, k.$$

Consequently (8.3.18) follows. Combining this derivation with Proposition **8.3.10**, we immediately obtain the following corollary of Proposition **8.3.7**, which does not require further proof.

8.3.12 Corollary. In the setting of Proposition **8.3.7**, if the sequence $\{x^k\}$ produced by Algorithm **8.3.6** has an isolated accumulation point where θ is strongly F-differentiable, then $\{x^k\}$ converges to that point, which must be a stationary point of (8.2.1). □

Consider next the problem of solving the nonsmooth equation $G(x) = 0$ with a C^1 norm function $\theta(x) \equiv \frac{1}{2} G(x)^T G(x)$. Let $\{x^k\}$ be the sequence of iterates generated by the line search method where the sequence $\{d^k\}$ of search directions is such that, for every k, either (8.3.16) and (8.3.17) hold or $d^k = -\nabla\theta(x^k)$. With $\sigma(x^k, d^k) = -\nabla\theta(x^k)^T d^k$, it is easy to verify that for every k,

$$\| d^k \| \leq \max \left\{ \sqrt{\sigma(x^k, d^k)},\ (\rho^{-1}\, \sigma(x^k, d^k))^{1/p} \right\}.$$

Again (8.3.18) follows. Therefore if the sequence $\{x^k\}$ has an isolated accumulation point, then $\{x^k\}$ converges to this point, which must necessarily be a stationary point of θ.

An example of a line search method for which it is easier to verify the subsequential condition (8.3.19) than the sequential condition (8.3.18) is the modified Gauss-Newton method for computing a zero of a continuously differentiable vector function. Let G be a C^1 function from $\mathbb{R}^n$ into itself. Let $\{x^k\}$ be the sequence of iterates generated by the modified Gauss-Newton method, where the sequence $\{d^k\}$ of search directions satisfies (8.3.13) for every k. Suppose that $\{x^k\}$ has an accumulation point x^* where $JG(x^*)$ is nonsingular. Let $\{x^k : k \in \kappa\}$ be any subsequence of $\{x^k\}$ converging to x^*. By (8.3.15), we have

$$\sigma(x^k, d^k) = \| JG(x^k)\, d^k \|^2 + \|G(x^k)\| \|d^k\|^2 \geq \| JG(x^k)\, d^k \|^2.$$

Since $JG(x^*)$ is nonsingular, it follows easily that there exists a constant $c > 0$ such that

$$\| d^k \| \leq c \sqrt{\sigma(x^k, d^k)}, \quad \forall\, k \in \kappa.$$

Thus (8.3.19) holds. Based on this derivation, we can establish the following convergence result for the modified Gauss-Newton method.

8.3.13 Corollary. Let $G : \mathbb{R}^n \to \mathbb{R}^n$ be continuously differentiable. If the sequence $\{x^k\}$ produced by the modified Gauss-Newton method has an accumulation point x^* such that $JG(x^*)$ is nonsingular, then $\{x^k\}$ converges to x^*.

Proof. Continuing the argument started above, we claim that $G(x^*) = 0$ and x^* is an isolated accumulation point of G. By Proposition **8.3.8**, x^* is a stationary point of $G^T G$. Since $JG(x^*)$ is nonsingular, this implies that x^* is an isolated zero of G. Suppose that x^* is not an isolated accumulation point of $\{x^k\}$; there must exist another accumulation point x^∞, which can be chosen to be arbitrarily close to but distinct from x^*. For any such point x^∞, $JG(x^\infty)$ is nonsingular, by the continuity of the Jacobian matrix JG. Since $G(x^\infty) = 0$ also, we obtain a contradiction to the isolated zero property of x^*. The sequential convergence of $\{x^k\}$ to x^* follows readily from Proposition **8.3.10**. □

8.3.2 Q-superlinear convergence

Having analyzed the convergence properties of the GLSA and its specializations, we proceed to study when a sequence of vectors produced by a line search method converges Q-superlinearly. For this purpose, we recall the concept of a superlinearly convergent sequence of directions with respect to a convergent sequence of iterates, defined just before Lemma **7.5.7**; see the limit (7.5.17). Taking into account the way the Armijo step-size rule is designed (see condition (ii) below), we state and prove the following simple result, which is related to Theorem **7.5.8**.

8.3.14 Proposition. Let $\{x^\nu\}$ and $\{d^\nu\}$ two sequences of vectors and $\{\tau_\nu\}$ be a sequence of scalars in $(0, 1]$ satisfying the following two conditions:

(i) for every ν, $d^\nu \neq 0$ and $x^{\nu+1} = x^\nu + \tau_\nu d^\nu$;

(ii) there exists a constant $\rho \in (0, 1)$ such that for all ν, either $\tau_\nu = 1$ or $\tau_\nu \leq \rho$.

Suppose that $\{x^\nu\}$ converges to x^∞. Any two of the following three statements imply the third.

(a) $\{d^\nu\}$ is superlinearly convergent with respect to $\{x^\nu\}$.

(b) There exists $\nu_0 > 0$ such that for all $\nu \geq \nu_0$, $\tau_\nu = 1$.

(c) $\{x^\nu\}$ converges Q-superlinearly to x^∞.

Proof. We have

$$\frac{x^{\nu+1} - x^\infty}{\| x^\nu - x^\infty \|} = \tau_\nu \frac{x^\nu + d^\nu - x^\infty}{\| x^\nu - x^\infty \|} + (1 - \tau_\nu) \frac{x^{\nu+1} - x^\infty}{\| x^\nu - x^\infty \|}.$$

Clearly if (b) holds, then (a) and (c) are equivalent. Suppose (a) and (c) hold but τ_ν is less than one for infinitely many ν. For these ν, we have

$\tau_\nu \leq \rho$ so that $1-\tau_\nu \geq 1-\rho > 0$. But this contradicts (a) and (c) combined. □

The above proposition suggests that there are two things to be verified when it comes to establish the Q-superlinear convergence of a convergent line search method. The first is the verification that the sequence $\{d^k\}$ of search directions is superlinearly convergent with respect to the (convergent) sequence of iterates $\{x^k\}$. Once this is done, it then follows that the convergence rate of $\{x^k\}$ is Q-superlinear if and only if a unit step size is attained after a finite number of iterations. We illustrate this argument with a line search method for minimizing a continuously differentiable θ function on a closed convex set X. With θ being C^1, we know that if a sequence $\{x^k\}$ produced by Algorithm **8.3.6** has an isolated accumulation point, then $\{x^k\}$ converges to this point. The following result is essentially a formal statement of the aforementioned ideas applied to this particular problem.

8.3.15 Theorem. Let $X \subset \mathbb{R}^n$ be closed convex and let $\theta : \mathbb{R}^n \to \mathbb{R}$ be continuously differentiable. Let $\{x^k\}$ be a sequence of vectors generated by Algorithm **8.3.6** with a sequence $\{H^k\}$ of symmetric positive definite matrices for which there exist positive constants c_1 and c_2 such that (8.3.11) holds for all k. Suppose that $\{x^k\}$ has an isolated accumulation point x^*. The following statements are valid.

(a) x^* is a stationary point of (8.2.1).

(b) $\{x^k\}$ converges to x^*.

(c) If the limit holds:

$$\lim_{k\to\infty} \frac{\nabla\theta(x^k) - \nabla\theta(x^*) - H^k(x^k - x^*)}{\| x^k - x^* \|} = 0, \tag{8.3.21}$$

then $\{x^k\}$ converges to x^* Q-superlinearly if and only if the step size τ_k is equal to one after a finite number of iterations.

Proof. Only part (c) requires a proof. Since θ is assumed to be continuously differentiable, d^k is the unique optimal solution of the strictly convex program:

$$\begin{aligned} &\text{minimize} && \nabla\theta(x^k)^T d + \tfrac{1}{2}\, d^T H^k d \\ &\text{subject to} && d \in X - x^k; \end{aligned}$$

cf. (8.3.12). By the variational principle of this program, we deduce

$$0 \leq (\, y - x^k - d^k \,)\, (\, \nabla\theta(x^k) + H^k d^k \,), \quad \forall\, y \in X.$$

In particular, with $y = x^*$, we have

$$\begin{aligned} 0 \leq \quad & (\, x^* - x^k - d^k \,)^T (\, \nabla\theta(x^k) - \nabla\theta(x^*) - H^k(x^k - x^*) \,) + \\ & (\, x^* - x^k - d^k \,)^T [\, \nabla\theta(x^*) + H^k(x^k - x^* + d^k) \,]. \end{aligned}$$

Since $x^k + d^k \in X$ and x^* is a stationary point of (8.2.1), we have

$$0 \leq (\, x^k + d^k - x^* \,)^T \nabla\theta(x^*).$$

Adding the two inequalities and rearranging terms, we obtain

$$\begin{aligned} & (\, x^k + d^k - x^* \,) H^k (\, x^k + d^k - x^* \,) \\ & \quad \leq (\, x^* - x^k - d^k \,) (\, \nabla\theta(x^k) - \nabla\theta(x^*) - H^k(x^k - x^*) \,). \end{aligned}$$

By (8.3.11), we deduce

$$\| \, x^k + d^k - x^* \, \| \leq c_1^{-1} \, \| \, \nabla\theta(x^k) - \nabla\theta(x^*) - H^k(x^k - x^*) \, \|.$$

From this inequality and the limit (8.3.21), it follows that $\{d^k\}$ is a superlinearly convergence sequence of directions with respect to $\{x^k\}$. Thus (c) follows readily from Proposition **8.3.14**. □

The next thing to do in a convergence rate analysis is to verify the ultimate attainment of a unit step size. Invariably, this entails some additional assumption on the smoothness of the objective function θ (cf. the SC^1 requirement in Theorem **8.3.19**) and some kind of "regularity" condition of the stationary point (cf. conditions (a) and (b) in the result below). In what follows, we establish a result that lies at the heart of all superlinearly convergent line search methods for solving systems of equations.

8.3.16 Proposition. Let $G : \mathbb{R}^n \to \mathbb{R}^n$ be locally Lipschitz. Suppose that a sequence $\{x^\nu\}$ converges to a zero x^∞ of G, with $G(x^\nu) \neq 0$ for all ν. Let $\{d^\nu\}$ be a superlinearly convergent sequence with respect to $\{x^\nu\}$. Consider the following three statements.

(a) G has a nonsingular Newton approximation scheme at x^∞.

(b) There exist positive constants c and δ such that

$$\| \, G(x) \, \| \geq c \, \| \, x - x^\infty \, \|, \quad \forall \, x \in \mathbb{B}(x^\infty, \delta).$$

(c) For any positive constant c', there exists $\nu_0 > 0$ such that for all $\nu \geq \nu_0$,

$$\| \, G(x^\nu + d^\nu) \, \| \leq c' \, \| \, G(x^\nu) \, \|.$$

It holds that (a) ⇒ (b) ⇒ (c).

Proof. That (a) implies (b) follows from Theorem **7.2.10**. It remains to show (b) $\Rightarrow$ (c). By (b) and the locally Lipschitz assumption there exist positive scalars L and δ' such that

$$c \,\|\, x - x^* \,\| \;\leq\; \|\, G(x) \,\| \;=\; \|\, G(x) - G(x^*) \,\| \;\leq\; L \,\|\, x - x^* \,\|,$$

for all $x \in \mathbb{B}(x^*, \delta')$. Therefore for all ν sufficiently large we can write

$$\begin{aligned} \frac{\|\, G(x^\nu + d^\nu) \,\|}{\|\, G(x^\nu) \,\|} &\leq L \,\frac{\|\, x^\nu + d^\nu - x^* \,\|}{\|\, G(x^\nu) \,\|} \\ &\leq \frac{L}{c}\,\frac{\|\, x^\nu + d^\nu - x^* \,\|}{\|\, x^\nu - x^* \,\|}. \end{aligned}$$

Since $\{d^\nu\}$ is superlinearly convergent, the last fraction in the above inequalities converges to 0 as $\nu \to \infty$. Consequently (c) follows readily. $\square$

The above proposition suggests that when it comes to solving the system of equations $G(x) = 0$ via the minimization of a merit function such as the familiar $\theta \equiv \frac{1}{2} G^T G$, one way to ensure the ultimate acceptance of a unit step size is to employ a rule of the following form: for a given constant $\gamma \in (0, 1)$,

$$\theta(x^k + 2^{-i} d^k) \;\leq\; \gamma\, \theta(x^k). \tag{8.3.22}$$

Under the assumptions of Proposition **8.3.16**, a unit step size will eventually be attained. Although (8.3.22) is also a descent rule, its drawback is that it is not always easily satisfied. There are several possible causes for the failure of such a rule, most prominently, when x^k is far from a zero of G or when d^k is not defined to facilitate the easy satisfaction of (8.3.22). Since it is invariably the case that d^k is a descent direction of θ at x^k, we want to apply the Armijo rule with a forcing function $\sigma(x^k, d^k)$ in order to ensure a reasonable step size and a sufficient decrease of θ. This consideration along with the rule (8.3.22) suggests the following modification of Step 4 of the GLSA.

Modified Step 4 of the GLSA

Step 4′: Find the smallest nonnegative integer i_k such that with $i = i_k$,

$$\theta(x^k + 2^{-i} d^k) \leq \theta(x^k) - \min[\gamma 2^{-i} \sigma(x^k, d^k), (1 - \gamma)\theta(x^k)] \tag{8.3.23}$$

and set $\tau_k \equiv 2^{-i_k}$.

It is not difficult to the establish the following analog of Theorem **8.3.3** for the GLSA with the above modified Armijo step-size procedure. The following result requires that θ be a nonnegative function.

8.3.17 Theorem. Let X be a convex subset of $\mathbb{R}^n$ and let θ be a locally Lipschitz continuous, nonnegative function on X. Let $\{x^k\} \subset X$ be an infinite sequence of vectors generated by Algorithm **8.3.2** with the modified Step 4′. If $\{x^k : k \in \kappa\}$ is a subsequence of $\{x^k\}$ satisfying the conditions (BD_θ) and (LS), then

$$\lim_{k(\in\kappa)\to\infty} \min(\, \theta(x^k),\, \sigma(x^k, d^k)\,) \;=\; 0. \tag{8.3.24}$$

Proof. If

$$\theta(x^{k+1}) \;\leq\; \gamma\, \theta(x^k) \tag{8.3.25}$$

holds for infinitely many $k \in \kappa$, then, since $\gamma \in (0,1)$ and $\{\theta(x^k)\}$ is a decreasing sequence of nonnegative scalars, it follows that $\{\theta(x^k) : k \in \kappa\}$ converges to zero. This clearly implies the desired limit (8.3.24) because each $\sigma(x^k, d^k)$ is nonnegative. If (8.3.25) holds for only finitely many $k \in \kappa$, then we are back to the earlier situation of Theorem **8.3.3**; in this case $\{\sigma(x^k, d^k) : k \in \kappa\}$ converges to zero. This also implies (8.3.24). □

Other algorithms can be similarly modified with their convergence preserved. Furthermore, the modified step-size procedure offers the additional benefit that under the assumptions of Proposition **8.3.16** a unit step size is ultimately accepted, thereby guaranteeing the Q-superlinear convergence of the sequence of iterates. Incidentally, the rule (8.3.23) is somewhat unusual and fully exploits the fact that we are minimizing a merit function whose minimum value is expected to be zero. This is actually the line search acceptance rule that we will use quite frequently in the next chapter.

Sometimes, it may be beneficial to consider the following variant of the above modified step-size procedure. Given x^k and d^k, first check if (8.3.25) holds. If so, then set $\tau_k = 1$ and $x^{k+1} = x^k + d^k$. Otherwise, apply the standard Armijo rule (i.e. the original Step 4 of Algorithm **8.3.2**) to determine τ_k and set x^{k+1} accordingly.

An alternative and simpler situation in which we can show that a unit step size is eventually attained is when the function θ is SC^1, $X = \mathbb{R}^n$ and an additional mild condition is satisfied. In particular, a unit step size is eventually accepted if the second-order sufficiency condition for optimality (7.4.16) holds. This is shown in the next proposition, which is different from the previous analysis in that the function θ is a general SC^1 function and is not required to be the squared norm of a system of equations. Moreover, the sufficiency condition (7.4.16) is phrased in a simplified form in view of the fact that $X = \mathbb{R}^n$; cf. condition (b) below.

8.3.18 Proposition. Let $\theta : \mathbb{R}^n \to \mathbb{R}$ be SC^1 near a zero x^* of $\nabla\theta$. The following two statements, (a) and (b), are equivalent:

(a) x^* is a strong, unconstrained local minimum of θ, i.e., a positive constant c and a neighborhood $\mathcal{N}$ of x^* exist such that

$$\theta(x) \geq \theta(x^*) + c\,\|x - x^*\|^2, \quad \forall\, x \in \mathcal{N}; \tag{8.3.26}$$

(b) $d^T H d > 0$ for all $d \neq 0$ and $H \in \partial_d^2 \theta(x^*)$.

Suppose that a sequence $\{x^\nu\}$ converges to x^*, with $x^\nu \neq x^*$ for all ν. Let $\{d^\nu\}$ be a superlinearly convergent sequence with respect to $\{x^\nu\}$. Consider the following two statements.

(c) A positive constant ρ exists such that eventually

$$\nabla\theta(x^\nu)^T d^\nu \leq -\rho\,\|\,d^\nu\,\|^2; \tag{8.3.27}$$

(d) For every $\gamma \in (0, 1/2)$, eventually

$$\theta(x^\nu + d^\nu) \leq \theta(x^\nu) + \gamma\,\nabla\theta(x^\nu)^T d^\nu. \tag{8.3.28}$$

It holds that (b) $\Rightarrow$ (c) $\Rightarrow$ (d).

Proof. The equivalence of (a) and (b) follows immediately from Proposition **7.4.12**. Assume that (b) holds. By Lemma **7.5.7** and the assumption that $\{d^\nu\}$ is superlinearly convergent with respect to $\{x^\nu\}$, we have

$$\|\,x^\nu + d^\nu - x^*\,\| = o(\|\,d^\nu\,\|). \tag{8.3.29}$$

Assume, without loss of generality, that $\{d^\nu/\|d^\nu\|\}$ converges to some vector d^∞ with unit norm. We can write

$$\begin{aligned}
\lim_{\nu\to\infty} \frac{x^\nu - x^*}{\|\,x^\nu - x^*\,\|} &= \lim_{\nu\to\infty} \frac{x^\nu - x^* - (x^\nu + d^\nu - x^*)}{\|\,x^\nu - x^*\,\|} \\
&= \lim_{\nu\to\infty} \frac{-d^\nu}{\|\,x^\nu - x^*\,\|} \\
&= \lim_{\nu\to\infty} \frac{-d^\nu}{\|\,x^\nu - x^*\,\|} \frac{\|\,x^\nu - x^*\,\|}{\|\,d^\nu\,\|} \\
&= \lim_{\nu\to\infty} \frac{-d^\nu}{\|\,d^\nu\,\|} \\
&= -d^\infty,
\end{aligned}$$

showing that $\{x^\nu\}$ converges to x^* in the direction $-d^\infty$. Therefore, by condition (b), it follows that for any sequence $\{H^\nu\}$ with $H^\nu \in \partial^2\theta(x^\nu)$ for every ν,

$$\liminf_{\nu\to\infty} \frac{(\,d^\nu\,)^T H^\nu d^\nu}{\|\,d^\nu\,\|^2} > 0. \tag{8.3.30}$$

We now proceed in a way that is very close to the proof of Theorem **7.5.8**. By the boundedness of $\{H^\nu\}$ and by (8.3.29), we have

$$\lim_{\nu\to\infty} \frac{\| H^\nu(x^\nu + d^\nu - x^*)\|}{\| d^\nu \|} = 0,$$

while the semismoothness of $\nabla\theta$ and Lemma **7.5.7** imply that

$$\lim_{\nu\to\infty} \frac{\| \nabla\theta(x^\nu) - \nabla\theta(x^*) - H^\nu(x^\nu - x^*) \|}{\| d^\nu \|} = 0.$$

The last two equations and the fact that $\nabla\theta(x^*) = 0$ give

$$\lim_{\nu\to\infty} \frac{\| \nabla\theta(x^\nu) + H^\nu d^\nu \|}{\| d^\nu \|} = 0.$$

In turn this implies

$$\begin{aligned} 0 &= \lim_{\nu\to\infty} \frac{\| \nabla\theta(x^\nu) + H^\nu d^\nu \|}{\| d^\nu \|} \\ &= \lim_{\nu\to\infty} \frac{\| d^\nu \| \| \nabla\theta(x^\nu) + H^\nu d^\nu \|}{\| d^\nu \|^2} \\ &\geq \lim_{\nu\to\infty} \frac{| \nabla\theta(x^\nu)^T d^\nu + (d^\nu)^T H^\nu d^\nu |}{\| d^\nu \|^2}. \end{aligned}$$

Hence,

$$\lim_{\nu\to\infty} \frac{| \nabla\theta(x^\nu)^T d^\nu + (d^\nu)^T H^\nu d^\nu |}{\| d^\nu \|^2} = 0.$$

This last equation together with (8.3.30) proves that (c) holds.

It remains to show the implication (c) $\Rightarrow$ (d). By Proposition **7.4.10**, we can write

$$\begin{aligned} \theta(x^\nu + d^\nu) &= \theta(x^*) + \nabla\theta(x^*)^T(x^\nu + d^\nu - x^*) \\ &\quad + \tfrac{1}{2} (x^\nu + d^\nu - x^*)^T H^\nu(x^\nu + d^\nu - x^*) \\ &\quad + o(\| x^\nu + d^\nu - x^* \|^2) \end{aligned}$$

and

$$\begin{aligned} \theta(x^\nu) &= \theta(x^*) + \nabla\theta(x^*)^T(x^\nu - x^*) \\ &\quad + \tfrac{1}{2} (x^\nu - x^*)^T R^\nu(x^\nu - x^*) + o(\| x^\nu - x^* \|^2), \end{aligned}$$

for any $H^\nu \in \partial^2\theta(x^\nu + d^\nu)$ and $R^\nu \in \partial^2\theta(x^\nu)$. Subtracting the second equation from the first one, taking into account that $\nabla\theta(x^*) = 0$, and recalling (8.3.29), we have

$$\theta(x^\nu + d^\nu) - \theta(x^\nu) = -\tfrac{1}{2} (x^\nu - x^*)^T R^\nu(x^\nu - x^*) + o(\| d^\nu \|^2);$$

subtracting $\frac{1}{2}\nabla\theta(x^\nu)^T d^\nu$ from both sides, we get, also taking into account Lemmas **7.5.7** and (8.3.29),

$$\begin{aligned}
&\theta(x^\nu + d^\nu) - \theta(x^\nu) - \tfrac{1}{2}\,\nabla\theta(x^\nu)^T d^\nu \\
&\quad = -\tfrac{1}{2}\,(\,x^\nu - x^*\,)^T R^\nu(\,x^\nu - x^*\,) - \tfrac{1}{2}\,\nabla\theta(x^\nu)^T d^\nu + o(\|\,d^\nu\,\|^2) \\
&\quad = -\tfrac{1}{2}\,(\,x^\nu + d^\nu - x^*\,)^T R^\nu(\,x^\nu - x^*\,) + \tfrac{1}{2}\,(\,d^\nu\,)^T R^\nu(\,x^\nu - x^*\,) \\
&\qquad -\tfrac{1}{2}\,\nabla\theta(x^\nu)^T d^\nu + o(\|\,d^\nu\,\|^2) \\
&\quad = -\tfrac{1}{2}\,(\,d^\nu\,)^T[\,\nabla\theta(x^\nu) - R^\nu(\,x^\nu - x^*\,)\,] + o(\|\,d^\nu\,\|^2) \\
&\quad = o(\|\,d^\nu\,\|^2).
\end{aligned}$$

From this, taking into account that $\gamma \in (0, 1/2)$, we can write:

$$\begin{aligned}
\theta(x^\nu + d^\nu) - \theta(x^\nu) &= \tfrac{1}{2}\,\nabla\theta(x^\nu)^T d^\nu + o(\|\,d^\nu\,\|^2) \\
&= (\,\tfrac{1}{2} - \gamma\,)\nabla\theta(x^\nu)^T d^\nu + \gamma\,\nabla\theta(x^\nu)^T d^\nu + o(\|\,d^\nu\,\|^2) \\
&\le \gamma\,\nabla\theta(x^\nu)^T d^\nu + \big[\,-\rho\,(\tfrac{1}{2} - \gamma)\,\|\,d^\nu\,\|^2 + o(\|\,d^\nu\,\|^2)\,\big] \\
&\le \gamma\,\nabla\theta(x^\nu)^T d^\nu, \quad \forall\,\nu \text{ sufficiently large,}
\end{aligned}$$

which shows that (d) holds. □

8.3.3 SC^1 minimization

In the remainder of this section, we consider two specific applications of the material developed so far. The first application is a line search method for minimizing a convex SC^1 function θ on a closed convex set X. The convexity of θ implies that every matrix $H \in \partial\nabla\theta(x)$ is symmetric positive semidefinite for all x. This can be easily seen because any such matrix H is the convex combination of finitely many matrices, each of which is the limit of a sequence of Hessian matrices $\nabla^2\theta(x^k)$, where $\{x^k\}$ is a sequence of F-differentiable points of $\nabla\theta$ converging to x. Since each $\nabla^2\theta(x^k)$ is symmetric, it follows that H is symmetric. Moreover, for every F-differentiable point y of $\nabla\theta$ and for every vector d, we have

$$d^T\nabla^2\theta(y)d = \lim_{\tau\to 0+} \frac{\theta(y + \tau d) - \theta(y) - \tau\,\nabla\theta(y)^T d}{\tau^2}.$$

By the gradient inequality of a convex function, the right-hand side is nonnegative; thus $\nabla^2\theta(y)$ is positive semidefinite. Consequently so is H.

Consider a sequence $\{x^k\}$ generated by the GLSA as follows. Let each search direction d^k be the unique optimal solution of the strictly convex

program:

$$\begin{array}{ll} \text{minimize} & \nabla\theta(x^k)^T d + \frac{1}{2} d^T (V^k + \varepsilon_k \mathbf{I}) d \\ \text{subject to} & d \in X - x^k, \end{array}$$

where $V^k \in \partial\nabla\theta(x^k)$ and $\varepsilon_k \geq 0$ is such that $V^k + \varepsilon_k \mathbf{I}$ is positive definite. (For instance, if V^k is positive definite, we may let $\varepsilon_k = 0$.) As with many line search methods, let $\sigma(x^k, d^k) \equiv -\nabla\theta(x^k)^T d^k$. The resulting method, which is basically Algorithm **8.3.6** with $H^k \equiv V^k + \varepsilon_k \mathbf{I}$, is a Newton-type method with line search for solving a convex SC^1 minimization problem. (The reader can see that the method is of the Newton type by considering the case where θ is twice continuously differentiable and $X = \mathbb{R}^n$.) By restricting each matrix H^k to be chosen from the B-subdifferential $\partial_B\nabla\theta(x^k)$ and postulating that the sequence $\{\varepsilon_k\}$ of scalars converges to zero and the sequence $\{x^k\}$ of iterates has an accumulation point that satisfies a certain nonsingularity condition, the next result establishes that $\{x^k\}$ converges Q-superlinearly to this point, which turns out to be the unique global minimum of (8.2.1). The main part of the proof consists of the demonstration that a unit step size is attained after a finite number of iterations.

8.3.19 Theorem. Let $X \subseteq \mathbb{R}^n$ be closed convex and let $\theta : \mathbb{R}^n \to \mathbb{R}^n$ be a convex SC^1 function. Let $\{\varepsilon_k\}$ be a sequence of positive scalars converging to zero. Let $\{x^k\}$ be a sequence of iterates generated by Algorithm **8.3.6** with the constant $\gamma < 1/2$ and $H^k \equiv V^k + \varepsilon_k \mathbf{I}$ for all k, where V^k belongs to $\partial_B\nabla\theta(x^k)$. Suppose that $\{x^k\}$ has an accumulation point x^* such that all matrices in $\partial_B\nabla\theta(x^*)$ are nonsingular, then $\{x^k\}$ converges Q-superlinearly to x^*, which must necessarily be the unique global minimum of θ on X.

Proof. We first show that $\{x^k\}$ converges to x^*. By Proposition **8.3.7**, we know that x^* is a stationary point, thus a global minimum of (8.2.1). Since every matrix V in $\partial_B\nabla\theta(x^*)$ is symmetric positive semidefinite and nonsingular, V must be positive definite. By Proposition **7.4.12** and the convexity of (8.2.1), it follows that x^* must be the unique global minimum of this program. This uniqueness shows that x^* is the only accumulation point of $\{x^k\}$, because any other accumulation point of $\{x^k\}$ must be a global minimum of $\{x^k\}$. Therefore, $\{x^k\}$ converges to x^*.

It remains to establish the ultimate attainment of a unit step size. Since every matrix in $\partial_B\nabla\theta(x^*)$ is symmetric positive definite and since $\partial_B\nabla\theta$ is upper semicontinuous, it follows that there exist positive constants c_1 and c_2 such that (8.3.11) holds for all k. Thus, by the proof of Theorem **8.3.15**, the sequence $\{d^k\}$ is superlinearly convergent with respect to $\{x^k\}$. By Proposition **8.3.18**, a unit step size is accepted eventually. □

The above proof of convergence (but not the rate of convergence) is made easy by the convexity assumptions. In fact, the proof has bypassed much of the theory developed previously. Nevertheless, the proof of the Q-superlinear convergence rate is based entirely on Theorem **8.3.15**. The convex SC^1 program has many applications. For instance, in Exercise **7.6.9**, we have seen that a general convex optimization problem with a quadratic objective function can be converted into an equivalent convex SC^1 minimization problem to which the above algorithm can be applied. Exercise **8.5.4** and **1.8.6** show that under appropriate conditions a convex-concave saddle problems is also an equivalent to a convex SC^1 minimization problem. Detailed discussion of these and other applications of SC^1 programs is beyond the scope of this book.

8.3.4 Application to a complementarity problem

When θ is a nonsmooth function, the directional subprogram (8.3.9) is a nonconvex program. In order to circumvent the computational difficulty of solving such a program, it would be reasonable to consider replacing $\theta'(x;\cdot)$ by a convex majorant $a(x,\cdot)$. That is, let $a(x,\cdot)$ be a convex function in the second argument such that $a(x,\cdot) \geq \theta'(x;\cdot)$ for all fixed but arbitrary x. Consider the following modified directional subprogram:

$$\begin{array}{ll} \text{minimize} & a(x,d) + \frac{1}{2}\, d^T H d \\ \text{subject to} & d \in X - x. \end{array} \tag{8.3.31}$$

This is a convex program in d with a strongly convex objective function and a nonempty closed convex feasible set; thus it has a unique optimal solution. Define the forcing function $\sigma(x,d) \equiv -a(x,d)$; it is then possible to develop a convergence theory for the resulting line search method that is based on (8.3.31) and this modified forcing function. Instead of repeating the development, we present a specific application of the above idea, namely, for solving the complementarity problem with constraint: find $x \in X$ such that

$$0 \leq F(x) \perp G(x) \geq 0 \tag{8.3.32}$$

where $F, G : \mathbb{R}^n \to \mathbb{R}^n$ are continuously differentiable, via the minimization of the B-differentiable function:

$$\theta(x) \equiv \tfrac{1}{2} \| \min(F(x),\, G(x)\,) \|^2, \quad x \in \mathbb{R}^n$$

on the set X. Letting $H(x) \equiv \min(F(x), G(x))$, we have

$$\theta'(x;d) \equiv \sum_{i=1}^{n} H_i(x)\, H_i'(x;d), \quad \forall\, (\,x, d\,) \in \mathbb{R}^{2n},$$

where

$$H_i'(x;d) = \begin{cases} F_i'(x;d) & \text{if } F_i(x) < G_i(x) \\ G_i'(x;d) & \text{if } F_i(x) > G_i(x) \\ \min(\, F_i'(x;d),\, G_i'(x;d)\,) & \text{if } F_i(x) = G_i(x). \end{cases}$$

Let $\mathcal{I}_F(x)$, $\mathcal{I}_G(x)$, and $\mathcal{I}_=(x)$ denote, respectively, the index sets identified by the three cases in the above expression of $H_i'(x;d)$. In terms of these index sets, we can define a function $a(x,d)$ that can be easily seen to majorize $\theta'(x;d)$:

$$a(x,d) \equiv \sum_{i\in\mathcal{I}_F(x)} F_i(x)\,\nabla F_i(x)^T d + \sum_{i\in\mathcal{I}_G(x)} G_i(x)\,\nabla G_i(x)^T d + \sum_{i\in\mathcal{I}_=(x)} \max(\, F_i(x)\,\nabla F_i(x)^T d,\, G_i(x)\,\nabla G_i(x)^T d\,).$$

Clearly, for each fixed but arbitrary x, $a(x,\cdot)$ is a convex piecewise linear function of the second argument. There are other choices of such a majorizing function; for instance, another choice is obtained by replacing the term $\max(F_i(x)\nabla F_i(x)^T d, G_i(x)\nabla G_i(x)^T d)$ by the average of the two maximands for every i in $\mathcal{I}_=(x)$ with $F_i(x) > 0$. The convergence theory developed below can easily be modified to suit these different choices.

To prepare for the subsequent analysis, we first establish a preliminary result having to do with the limit condition (LS) in this context.

8.3.20 Proposition. Let $\{(x^k, d^k)\}$ be a convergent sequence. For every sequence $\{t_k\}$ of positive scalars converging to zero,

$$\limsup_{k\to\infty}\, a(x^k,d^k) \;\geq\; \limsup_{k\to\infty}\, \frac{\theta(x^k + t_k d^k) - \theta(x^k)}{t_k}. \tag{8.3.33}$$

Proof. Consider a component i and an index k for which $F_i(x^k) = G_i(x^k)$ and $H_i(x^k + t_k d^k) = F_i(x^k + t_k d^k)$. We have

$$\begin{aligned} H_i(x^k + t_k d^k)^2 - H_i(x^k)^2 &= F_i(x^k + t_k d^k)^2 - F_i(x^k)^2 \\ &= 2\, t_k\, F_i(x^k)\,\nabla F_i(x^k)^T d^k + o(t_k) \\ &\leq 2\, t_k\, a_i(x^k, d^k) + o(t_k). \end{aligned}$$

Similarly, we can show that

$$H_i(x^k + t_k d^k)^2 - H_i(x^k)^2 \;\leq\; 2\, t_k\, a_i(x^k, d^k) + o(t_k)$$

if $H_i(x^k + t_k d^k) = G_i(x^k + t_k d^k)$. Moreover, the above inequality holds as an equality if $F_i(x^k) \neq G_i(x^k)$. Dividing by $2t_k$, adding up these inequalities

and equalities for $i = 1, \ldots, n$, and letting $k \to \infty$, we obtain the desired inequality (8.3.33). □

Consider an infinite sequence $\{x^k\}$ generated in the following way. Let $\{H^k\}$ be a sequence of symmetric positive definite matrices such that for some positive constants c_1 and c_2, (8.3.11) holds for all k. Given a vector $x^k \in X$, compute the search direction d^k by solving the convex program:

$$\begin{array}{ll} \text{minimize} & a(x^k, d) + \frac{1}{2}\, d^T H^k d \\ \text{subject to} & d \in X - x^k. \end{array} \tag{8.3.34}$$

Let $\sigma(x^k, d^k) \equiv -a(x^k, d^k)$. If $a(x^k, d^k) = 0$, stop. Otherwise (we must have $a(x^k, d^k) < 0$), compute the next iterate x^{k+1} by the Armijo line search rule with the forcing function σ. Iterate until a prescribed termination rule is satisfied.

Suppose that a subsequence $\{x^k : k \in \kappa\}$ converges to a limit x^*, which must necessarily belong to X. Since

$$\| d^k \| \leq c_1^{-1/2} \sqrt{-a(x^k, d^k)}, \quad \forall\, k, \tag{8.3.35}$$

from the definition of $a(x^k, d^k)$, it follows that $\{d^k : k \in \kappa\}$ is bounded. Proposition **8.3.20** implies that the sequence $\{x^k : k \in \kappa\}$ satisfies condition (LS) in Theorem **8.3.3**. Hence, by this theorem, we have

$$\lim_{k(\in\kappa)\to\infty} a(x^k, d^k) = 0.$$

By (8.3.35), we deduce that $\{d^k : k \in \kappa\}$ converges to zero. By imposing a certain condition on x^*, we can demonstrate that $\theta(x^*) = 0$; that is, x^* is a desired solution of the constrained CP (F, G). The condition in question is essentially a realization of the assumptions in Proposition **1.5.13** specialized to this application.

8.3.21 Proposition. Let X be a closed convex set and let F and G be continuously differentiable functions from $\mathbb{R}^n$ into itself. Let x^* be the limit of the subsequence $\{x^k : k \in \kappa\}$. Then, x^* solves (8.3.32) if and only if for every partitioning of the index sets $\mathcal{I}_=(x^*)$ into three mutually disjoint subsets β_F, $\beta_=$, and β_G, there exists a vector $d^* \in X - x^*$ satisfying

$$\begin{aligned} F_i(x^*) + \nabla F_i(x^*)^T d^* &= 0, \quad \forall\, i \in \mathcal{I}_F(x^*) \cup \beta_F, \\ G_i(x^*) + \nabla G_i(x^*)^T d^* &= 0, \quad \forall\, i \in \mathcal{I}_G(x^*) \cup \beta_G, \end{aligned} \tag{8.3.36}$$

$$H_i(x^*)^2 + \max(F_i(x^*)\nabla F_i(x^*)^T d^*, G_i(x^*)\nabla G_i(x^*)^T d^*) \leq 0, \ \forall\, i \in \beta_=.$$

Proof. It suffices to show the "if" part. We may assume, by working with a suitable subsequence, that index sets $\mathcal{J}_F$, $\mathcal{J}_=$, and $\mathcal{J}_G$ exist such that

$$\mathcal{I}_F(x^k) = \mathcal{J}_F, \quad \mathcal{I}_=(x^k) = \mathcal{J}_=, \quad \text{and} \quad \mathcal{I}_G(x^k) = \mathcal{J}_G, \tag{8.3.37}$$

for all $k \in \kappa$. Clearly, we have

$$\mathcal{I}_F(x^*) \subseteq \mathcal{J}_F, \quad \mathcal{I}_=(x^*) \supseteq \mathcal{J}_=, \quad \text{and} \quad \mathcal{I}_G(x^*) \subseteq \mathcal{J}_G. \tag{8.3.38}$$

Let

$$\beta_F \equiv \mathcal{J}_F \setminus \mathcal{I}_F(x^*), \quad \beta_= \equiv \mathcal{J}_=, \quad \text{and} \quad \beta_G \equiv \mathcal{J}_G \setminus \mathcal{I}_G(x^*).$$

These three index sets β_F, $\beta_=$, and β_G partition $\mathcal{I}_=(x^*)$. Corresponding to this triple $(\beta_F, \beta_=, \beta_G)$ of index sets, let d^* be a vector in $X - x^*$ satisfying the displayed system (8.3.36) in the proposition. For every scalar $t \in (0, 1]$, the vector $d \equiv t(d^* + x^* - x^k)$ is feasible to (8.3.34). Thus, for all k,

$$\begin{aligned} & t\, a(x^k, d^* + x^* - x^k) + \frac{t^2}{2} (\, d^* + x^* - x^k \,)^T H^k (\, d^* + x^* - x^k \,) \\ & \quad \geq a(x^k, d^k) + \tfrac{1}{2} (\, d^k \,)^T H^k d^k. \end{aligned}$$

Let $k(\in \kappa) \to \infty$; the right-hand side converges to zero. Next, dividing by $t > 0$, letting $t \to 0$, and invoking (8.3.37), we obtain

$$\begin{aligned} & \sum_{i \in \mathcal{J}_F} F_i(x^*) \nabla F_i(x^*)^T d^* + \sum_{i \in \mathcal{J}_G} G_i(x^*) \nabla G_i(x^*)^T d^* + \\ & \sum_{i \in \mathcal{J}_=} \max(\, F_i(x^*) \nabla F_i(x^*)^T d^*,\, G_i(x^*) \nabla G_i(x^*)^T d^* \,) \geq 0. \end{aligned}$$

By (8.3.38) and the assumed property of d^*, we deduce that the left-hand side of the above inequality is less than or equal to $-2\theta(x^*)$. Consequently, we must have $\theta(x^*) = 0$ as desired. □

The system of inequalities (8.3.36) satisfied by d^* is worth further discussion. For simplicity, we take the set X to be the entire space $\mathbb{R}^n$. Let x be a given vector; let $\mathcal{I}$, $\mathcal{J}$, and $\mathcal{K}$ be three mutually disjoint index sets partitioning $\{1, \ldots, n\}$ such that

$$\mathcal{I} \supseteq \mathcal{I}_F(x), \quad \mathcal{J} \supseteq \mathcal{I}_G(x), \quad \text{and} \quad \mathcal{K} \subseteq \mathcal{I}_=(x).$$

Consider the system of linear inequalities in the variable d:

$$\begin{aligned} & F_i(x) + \nabla F_i(x)^T d = 0, \quad \forall\, i \in \mathcal{I}, \\ & G_i(x) + \nabla G_i(x)^T d = 0, \quad \forall\, i \in \mathcal{J}, \end{aligned} \tag{8.3.39}$$

$$H_i(x)^2 + \max(F_i(x)\nabla F_i(x)^T d, G_i(x)\nabla G_i(x)^T d) \leq 0, \ \forall\, i \in \mathcal{K}.$$

The last inequality is equivalent to a set of four inequalities:

$$\left.\begin{array}{r} F_i(x) + \nabla F_i(x)^T d \ \leq\ 0 \\ G_i(x) + \nabla G_i(x)^T d \ \leq\ 0 \end{array}\right\} \quad \forall\, i \in \mathcal{K} \text{ with } H_i(x) > 0, \tag{8.3.40}$$

$$\left.\begin{array}{r} F_i(x) + \nabla F_i(x)^T d \ \geq\ 0 \\ G_i(x) + \nabla G_i(x)^T d \ \geq\ 0 \end{array}\right\} \quad \forall\, i \in \mathcal{K} \text{ with } H_i(x) < 0. \tag{8.3.41}$$

Suppose that the following matrix is nonsingular:

$$\mathcal{B}(x) \equiv \left[\begin{array}{cc} JF(x)_{\mathcal{I}\mathcal{I}} & JF(x)_{\mathcal{I}\mathcal{J}} \\ JG(x)_{\mathcal{J}\mathcal{I}} & JG(x)_{\mathcal{J}\mathcal{J}} \end{array}\right].$$

We can then solve for the components $d_{\mathcal{I}}$ and $d_{\mathcal{J}}$ from the two equations in (8.3.39), obtaining

$$\left(\begin{array}{c} d_{\mathcal{I}} \\ d_{\mathcal{J}} \end{array}\right) = -\mathcal{B}(x)^{-1}\left\{\left(\begin{array}{c} F_{\mathcal{I}}(x) \\ G_{\mathcal{J}}(x) \end{array}\right) + \left[\begin{array}{c} JF(x)_{\mathcal{I}\mathcal{K}} \\ JG(x)_{\mathcal{J}\mathcal{K}} \end{array}\right] d_{\mathcal{K}}\right\}.$$

Substituting this expression into (8.3.40) and (8.3.41) results in a system of linear inequalities that involves only the component $d_{\mathcal{K}}$. Define the index sets

$$\mathcal{K}_+ \equiv \{\, i \in \mathcal{K} : H_i(x) > 0 \,\} \quad \text{and} \quad \mathcal{K}_- \equiv \{\, i \in \mathcal{K} : H_i(x) < 0 \,\}.$$

The system satisfied by $d_{\mathcal{K}}$ can be written very simply as:

$$b^+ + A^+ d_{\mathcal{K}} \ \leq\ 0 \quad \text{and} \quad b^- + A^- d_{\mathcal{K}} \ \geq\ 0, \tag{8.3.42}$$

where

$$b^{\pm} \equiv \left(\begin{array}{c} F_{\mathcal{K}_\pm}(x) \\ G_{\mathcal{K}_\pm}(x) \end{array}\right) - \left[\begin{array}{cc} JF(x)_{\mathcal{K}_\pm\mathcal{I}} & JF(x)_{\mathcal{K}_\pm\mathcal{J}} \\ JG(x)_{\mathcal{K}_\pm\mathcal{I}} & JG(x)_{\mathcal{K}_\pm\mathcal{J}} \end{array}\right] \mathcal{B}(x)^{-1} \left(\begin{array}{c} F_{\mathcal{I}}(x) \\ G_{\mathcal{J}}(x) \end{array}\right)$$

and

$$A^{\pm} \equiv \left[\begin{array}{c} JF(x)_{\mathcal{K}_\pm\mathcal{K}} \\ JG(x)_{\mathcal{K}_\pm\mathcal{K}} \end{array}\right] - \left[\begin{array}{cc} JF(x)_{\mathcal{K}_\pm\mathcal{I}} & JF(x)_{\mathcal{K}_\pm\mathcal{J}} \\ JG(x)_{\mathcal{K}_\pm\mathcal{I}} & JG(x)_{\mathcal{K}_\pm\mathcal{J}} \end{array}\right] \mathcal{B}(x)^{-1} \left[\begin{array}{c} JF(x)_{\mathcal{I}\mathcal{K}} \\ JG(x)_{\mathcal{J}\mathcal{K}} \end{array}\right].$$

A sufficient condition for the system (8.3.42) to have a solution is that the matrix

$$\tilde{A} \equiv \begin{bmatrix} A^+ \\ -A^- \end{bmatrix}$$

is an *S matrix* ("S" for Stiemke); i.e., there exists a vector $v > 0$ such that $\tilde{A}v > 0$. In general, checking whether a given matrix is an S matrix can be determined by linear programming; see Exercise **3.7.29** for the related class of S_0 matrices.

When G is the identity map, i.e. for the NCP (F), the above matrix-theoretic conditions simplify considerably, resulting in a set of reduced conditions that ensure the sequential and Q-superlinear convergence of the above line search method. Though simplified, the latter conditions are combinatorial in nature; more specifically, they have to hold for various triples of index sets $\mathcal{I}$, $\mathcal{J}$, and $\mathcal{K}$ that are derived from different partitions of the degenerate index set $\mathcal{I}_=(x^*)$. This kind of combinatorial conditions persist in all complementarity problems with degenerate solutions. See Subsection **9.1.1** for related discussion.

8.4 Trust Region Methods

In this section we consider a different approach to design a global method for the solution of a system of equations. As before, we first consider the minimization problem (8.2.1). At each iteration of a line search method, we compute a search direction by solving a "simplified" program, such as (8.3.9) or (8.3.31), followed by a step size determination. One way to think of the direction finding subproblem is that it is an approximation of the original problem, with the objective function $\theta(x)$ of the latter problem replaced by an approximation such as

$$\theta(x) \approx \theta(x^k) + \theta'(x^k; x - x^k) + \tfrac{1}{2}(x - x^k)^T H^k (x - x^k),$$

or, in terms of a more general surrogate derivative function $a(x, d)$ such that $a(x, 0) = 0$,

$$\theta(x) \approx \theta(x^k) + a(x^k, x - x^k) + \tfrac{1}{2}(x - x^k)^T H^k (x - x^k).$$

In a trust region approach, the function on the right-hand side is taken as a model function that we *trust* to be a good approximation of θ within a certain region $\mathbb{B}(x^k, \Delta)$, for some $\Delta > 0$. We therefore attempt to (approximately) minimize the model function within this region and take the solution so obtained as the new iteration x^{k+1}, provided that x^{k+1} satisfies

a certain prescribed criterion, which ensures a satisfactorily progress of the iteration. On the basis of the information gathered during the iteration we can also decide to shrink or to enlarge the radius Δ of the region where we believe the model function is a good approximation to the objective function θ.

Like the line search methods, we can motivate many of the features of the trust region methods by first considering the simple case of the unconstrained minimization of a C^2 function θ. In this case Taylor expansion shows that if $\|d\|$ is sufficiently small, we have

$$\theta(x^k + d) \approx \theta(x^k) + \nabla\theta(x^k)^T d + \frac{1}{2} d^T \nabla^2\theta(x^k) d,$$

which corresponds to taking $a(x,d) \equiv \nabla\theta(x)^T d$ and $H^k \equiv \nabla^2\theta(x^k)$. A natural idea is then to minimize the quadratic approximation in order to find a suitable direction d^k. However this simple idea has two deficiencies. First, the quadratic function could be unbounded below; second, its minimizer could lie outside the region where the model approximates well the function θ near x^k. So it seems sensible to consider the following direction finding subproblem:

$$\begin{array}{ll} \text{minimize} & \nabla\theta(x^k)^T d + \frac{1}{2}\, d^T \nabla^2\theta(x^k) d \\ \text{subject to} & \| d \| \leq \Delta, \end{array} \tag{8.4.1}$$

where Δ is a positive scalar that represents the radius of the region where we presume the quadratic model is accurate. Note that the problem (8.4.1) always has a finite optimal solution because its feasible region is nonempty and compact. Furthermore efficient procedures exists to compute the global minimum of this quadratic problem even in the case where $\nabla^2\theta(x^k)$ is indefinite. An added benefit of (8.4.1) is that unless x^k is an (unconstrained) stationary point of θ satisfying the second-order necessary condition, that is unless $\nabla\theta(x^k) = 0$ and $\nabla^2\theta(x^k)$ is positive semidefinite, it is always possible to improve the current value of θ, provided that Δ is sufficiently small. To see this, first suppose that x^k is a stationary point of θ but $\nabla^2\theta(x^k)$ is not positive semidefinite. There must exist a vector $\tilde{d}_n$ such that $\tilde{d}_n^T \nabla^2\theta(x^k)\tilde{d}_n < 0$. (Any such vector $\tilde{d}_n$ is called a *direction of negative curvature* of θ at x^k, thus the subscript "n".) Let

$$d_n \equiv \frac{\Delta}{\| \tilde{d}_n \|} \tilde{d}_n.$$

Provided that Δ is sufficiently small, we have (recalling that $\nabla\theta(x^k) = 0$)

$$\theta(x^k + d_n) = \theta(x^k) + \tfrac{1}{2}\, d_n^T \nabla^2\theta(x^k) d_n + o(\Delta^2) < \theta(x^k).$$

If x^k is not stationary, then we may set

$$d_c \;=\; -\frac{\Delta}{\|\,\nabla\theta(x^k)\,\|}\,\nabla\theta(x^k),$$

which is a positive multiple of the steepest descent direction $-\nabla\theta(x^k)$. By the Taylor expansion, we have

$$\theta(x^k + d_c) \;=\; \theta(x^k) - \Delta\|\,\nabla\theta(x^k)\,\| + \frac{\Delta^2}{2}\,\frac{\nabla\theta(x^k)^T\nabla^2\theta(x^k)\nabla\theta(x^k)}{\|\,\nabla\theta(x^k)\,\|^2} + o(\Delta^2),$$

provided that Δ is sufficiently small. Since $\nabla\theta(x^k)$ is nonzero, the term $-\Delta\|\nabla\theta(x^k)\|$ dominates the Δ^2 terms in the right-hand side; hence

$$\theta(x^k + d_c) \;<\; \theta(x^k).$$

Combining these two cases, we conclude that if x^k is not a "second-order stationary point" of θ (i.e., either $\nabla\theta(x^k) \neq 0$ or $\nabla^2\theta(x^k)$ is not positive semidefinite), then we can always reduce the objective function $\theta(x^k)$ by searching for a minimizer of (8.4.1) for a sufficiently small Δ. Obviously many details are missing. Two immediate questions can be asked. How should we choose the trust region radius Δ? How much should we try to reduce the objective function in order to guarantee the overall convergence of the generated iterates to a stationary point?

In what follows we answer the above questions in the case of a nondifferentiable θ. In this case we cannot use the gradient or the Hessian of the objective function because they are no longer guaranteed to exist. We therefore have to use some substitutes as suggested at the beginning of this section. Unlike the smooth case where we can expect a trust region algorithm to successfully compute a sequence of iterates whose accumulation points are second-order stationary points of the minimization problem, we are content in the nonsmooth case to be able to compute a (first-order) stationary point, since looking for some kind of a generalized second-order stationary point may prove to be an elusive task.

Let the feasible set X be a closed convex, proper subset of $\mathbb{R}^n$. For a given vector x in X, a positive scalar Δ, and a matrix H, we compute a vector d by solving, possibly inexactly, the following trust region subproblem:

$$\begin{aligned} \text{minimize} \quad & a(x,d) + \tfrac{1}{2}\,d^T H d \\ \text{subject to} \quad & \|\,d\,\| \;\leq\; \Delta \\ & d \;\in\; X - x \end{aligned} \tag{8.4.2}$$

Let $\mathrm{TR}(x, H, \Delta)$ denote the problem (8.4.2) corresponding to a given triple (x, H, Δ). The only difference between the trust-region subproblem (8.4.2)

and the line search subproblem (8.3.31) is the presence of the trust-region constraint $\|d\| \leq \Delta$ in the former. In theory, any vector norm $\|\cdot\|$ can be used to define such a region; nevertheless, in practice, the most often used norms are the maximum norm and the Euclidean norm. As we see below, there is a major operational difference between a trust region method and a line search method.

Suppose that the function $a(x, \cdot)$ is lower semicontinuous in the second argument. The trust region subproblem (8.4.2) always has an optimal solution because its feasible region is nonempty (contains the zero vector) and compact. Let d^* denote any such optimal solution. Since $d = 0$ is feasible, the optimal objective value of (8.4.2) is always nonpositive. In the trust region algorithm described below, we apply an inexact rule to compute a direction d as follows. For a given scalar $\rho \in (0, 1]$, let $\Xi_\rho(x, H, \Delta)$ denote the set of vectors $d \in X - x$ satisfying $\|d\| \leq \Delta$ and

$$a(x, d) + \tfrac{1}{2}\, d^T H d \;\leq\; \rho \,\left[\, a(x, d^*) + \tfrac{1}{2}\,(\, d^*\,)^T H d^* \,\right] \tag{8.4.3}$$

Note that the right-hand quantity is nonpositive. The above rule accepts any feasible vector of (8.4.2) with an objective value not exceeding a prescribed fraction of the optimal objective value.

The following is a formal description of the trust region method for solving the minimization problem (8.2.1). For any pair of vectors (x, d), let $\sigma(x, d)$ denote the negative of the objective value of (8.4.2) evaluated at d; let $\sigma(x, H, \Delta)$ denote the negative of the optimal objective value of this trust-region subproblem. Thus, we have $\sigma(x, H, \Delta) = \sigma(x, d^*)$, where d^* is an optimal solution of the TR(x, H, Δ). Moreover, (8.4.3) becomes: $\sigma(x, d) \geq \rho\sigma(x, H, \Delta)$.

General Trust Region Algorithm (GTRA)

8.4.1 Algorithm.

Data: $x^0 \in X$, $0 < \gamma_1 < \gamma_2 < 1$, $\Delta_0 > 0$, $\Delta_{\min} > 0$, $\rho \in (0, 1]$, and a sequence of symmetric matrices $\{H^k\}$.

Step 1: Set $k = 0$.

Step 2: Compute a vector $d^k \in \Xi_\rho(x^k, H^k, \Delta_k)$.

Step 3: If $\sigma(x^k, d^k) = 0$ stop.

Step 4: Otherwise, if

$$\theta(x^k + d^k) - \theta(x^k) \;\leq\; -\gamma_1\, \sigma(x^k, d^k), \tag{8.4.4}$$

then set $x^{k+1} = x^k + d^k$ and

$$\Delta_{k+1} = \begin{cases} \max(2\Delta_k, \Delta_{\min}) & \text{if } \theta(x^k + d^k) - \theta(x^k) \le -\gamma_2 \sigma(x^k, d^k) \\ \max(\Delta_k, \Delta_{\min}) & \text{otherwise.} \end{cases}$$

Set $k \leftarrow k + 1$ and go to Step 2. Otherwise, if

$$\theta(x^k + d^k) - \theta(x^k) > -\gamma_1 \sigma(x^k, d^k),$$

set $x^{k+1} = x^k$, $\Delta_{k+1} = \frac{1}{2}\Delta_k$, and $k \leftarrow k + 1$; go to Step 2.

We distinguish two kinds of iterations. The *outer iterations* are those in which the test (8.4.4) is satisfied and x^k is updated; the *inner iterations* are those in which the test (8.4.4) is not passed and the trust region radius is reduced while x^k is kept fixed. After each outer iteration, the objective value is decreased; moreover the trust region radius is adjusted in one of several possible ways: it stays the same, it expands, or it is set to equal the lower bound $\Delta_{\min}$. In any event, the trust region radius does not decrease after each outer iteration; the region is enlarged ($\Delta_{k+1} = 2\Delta_k$) only if the decrease of θ is more than that prescribed by the test (8.4.4), since $\gamma_2 > \gamma_1$. In contrast, after each inner iteration, the objective value stays the same since there is no change in the iterate, and the trust region radius is halved.

It is useful to compare Algorithms **8.3.2** and **8.4.1**. Both algorithms generate a sequence of feasible vectors $\{x^k\}$ with nonincreasing objective values $\{\theta(x^k)\}$. The two algorithms distinguish themselves in the way they produce the iterates. In the former algorithm, given an iterate $x^k \in X$, a search direction d^k is first identified, and a step size τ_k is then determined. In the latter algorithm, this two-stage procedure is reversed. Indeed, a positive scalar Δ_k is first decided (from the previous iteration), which can be thought of as the maximum step size to be taken in the current iteration; a direction d^k is then computed with norm not exceeding Δ_k. Presumably, the direction step is easier to implement in the line search method than in the trust region method, due to the presence of the direction norm constraint in $\mathrm{TR}(x^k, H^k, \Delta_k)$. Nevertheless, as explained above, any added computation in solving the latter directional subproblem is compensated by a stronger property of the limit point(s) obtained in a trust region method, at least in the smooth case. Thus both algorithms have their own strengths and imperfections.

In Subsection **8.3.4**, we have specified $a(x, \cdot)$ to be a majorant of the directional derivative $\theta'(x; \cdot)$ for a B-differentiable function θ. In what fol-

lows, we present a convergence theory of Algorithm **8.4.1** without explicitly specifying the surrogate derivative function a. Instead, we postulate some general conditions that this function must satisfy. These conditions are fairly mild and are inspired by the properties of the directional derivatives of a locally Lipschitz continuous function.

(TR1a) For every $x \in X$, $a(x,0) = 0$ and $a(x,\cdot)$ is lower semicontinuous.

(TR1b) For every $(x,d) \in X \times \mathbb{R}^n$ and for all $t \in [0,1]$, it holds that $a(x,td) \leq ta(x,d)$.

(TR1c) For every bounded sequence $\{x^\nu\}$, there exist positive scalars Δ and c such that $\|a(x^\nu,d)\| \leq c$ for all ν and all $d \in \mathrm{cl}\,\mathbb{B}(0,\Delta)$.

In order to understand the asymptotic properties of Algorithm **8.4.1**, let us first examine the situation where this algorithm terminates finitely in Step 3 with an iterate x^k satisfying $\sigma(x^k,d^k) = 0$. Since

$$0 \;=\; \sigma(x^k,d^k) \;\geq\; \rho\,\sigma(x^k,H^k,\Delta_k) \;\geq\; 0,$$

it follows that

$$\sigma(x^k,H^k,\Delta_k) \;=\; 0,$$

which implies

$$a(x^k,d) + \tfrac{1}{2}\,d^T H^k d \;\geq\; 0,$$

for all $d \in X - x^k$ satisfying $\|d\| \leq \Delta_k$. Since x^k belongs to X, which is convex by assumption, we see that

$$d \;\in\; X - x^k \;\Rightarrow \tau\, d \;\in\; X - x^k \;\forall\, \tau \;\in\; [0,1].$$

Consequently, by a standard limiting argument, we can deduce

$$a(x^k,d) \;\geq\; 0, \quad \forall\, d \;\in\; X - x^k,\; \|\,d\,\| \;\leq\; \Delta_k.$$

Our goal is to extend this finite-termination result to the case where the algorithm generates an infinite sequence $\{x^k\}$ and to establish that the above expression holds "asymptotically". For this purpose, we define the following quantity that provides a kind of surrogate measure of stationarity:

$$\tilde{\sigma}(x,\Delta) \;\equiv\; \max\{-a(x,d) \;:\; \|\,d\,\| \;\leq\; \Delta, x + d \;\in\; X\}.$$

By (TR1a), the maximum on the right-hand side is attained. The quantity $\tilde{\sigma}(x,\Delta)$ is not calculated during the GTRA. In terms of this quantity, the result established above for the case when Algorithm **8.4.1** terminates finitely at the iterate x^k can be rephrased simply as $\tilde{\sigma}(x^k,\Delta_k) = 0$. The

next proposition collects some important properties of the function $\tilde{\sigma}$. In this result, H is an arbitrary matrix, not necessarily semidefinite; moreover, $H = 0$ is allowed.

8.4.2 Proposition. Let X be a closed convex subset of $\mathbb{R}^n$ and let $x \in X$ be arbitrary. Let $\rho \in (0,1)$ and $H \in \mathbb{R}^{n\times n}$ be arbitrary. Assume conditions (TR1a), (TR1b), and (TR1c).

(a) The function $\tilde{\sigma}(x,\cdot)$ is nonnegative and nondecreasing.

(b) For every scalar $\Delta \geq 0$,

$$\tilde{\sigma}(x,\Delta) \;\geq\; \min(\Delta,1)\,\tilde{\sigma}(x,1).$$

(c) For every scalar $\Delta > 0$ and every vector $d \in \Xi_\rho(x,H,\Delta)$,

$$\sigma(x,d) \;\geq\; \frac{\rho}{2}\,\tilde{\sigma}(x,\Delta)\,\min\left\{1,\,\frac{\tilde{\sigma}(x,\Delta)}{\|\,H\,\|\,\Delta^2}\right\}.$$

(d) For every scalar $\Delta > 0$ and every vector $d \in X - x$ satisfying $\|d\| \leq \Delta$,

$$\sigma(x,d) \;\leq\; \tilde{\sigma}(x,\Delta) + \frac{\Delta^2}{2}\,\|\,H\,\|.$$

Proof. Part (a) follows readily from (TR1a) and the definition of $\tilde{\sigma}(x,\Delta)$. We next turn to part (b). In view of the fact that $\tilde{\sigma}(x,\cdot)$ is nondecreasing it suffices to consider the case where $\Delta \in [0,1)$. Let $\tilde{d}$, with $\|\tilde{d}\| \leq 1$, be a vector for which $\tilde{\sigma}(x,1) = -a(x,\tilde{d})$. Then $x + \Delta\tilde{d}$ belongs to X, by the convexity of X. By (TR1b) we get

$$\tilde{\sigma}(x,\Delta) \;\geq\; -a(x,\Delta\tilde{d}) \;\geq\; -\Delta\,a(x,\tilde{d}) \;=\; \Delta\,\tilde{\sigma}(x,1).$$

For a $\Delta > 0$, let $\tilde{d}_\Delta$, with $\|\tilde{d}_\Delta\| \leq \Delta$, denote a vector for which $\tilde{\sigma}(x,\Delta) = -a(x,\tilde{d}_\Delta)$. Taking into account (8.4.3) and (TR1b), we can write for every $d \in \Xi_\rho(x,H,\Delta)$,

$$\begin{aligned}
\sigma(x,d) &\geq \rho\,[\,-a(x,d^*) - \tfrac{1}{2}\,(\,d^*\,)^T H d^*\,] \\
&\geq \rho\,\max_{t\in[0,1]}\,\{\,-a(x,t\tilde{d}_\Delta) - \tfrac{1}{2}\,t^2\,(\,\tilde{d}_\Delta\,)^T H \tilde{d}_\Delta\,\} \\
&\geq \rho\,\max_{t\in[0,1]}\,\{\,t\,\tilde{\sigma}(x,\Delta) - \tfrac{1}{2}\,t^2\,\|\,H\,\|\,\Delta^2\,\} \\
&\geq \rho\,\min\left\{\tilde{\sigma}(x,\Delta),\,\frac{\tilde{\sigma}(x,\Delta)^2}{2\,\|\,H\,\|\,\Delta^2}\right\} \\
&\geq \frac{\rho}{2}\,\tilde{\sigma}(x,\Delta)\,\min\left\{1,\,\frac{\tilde{\sigma}(x,\Delta)}{\|\,H\,\|\,\Delta^2}\right\}.
\end{aligned}$$

This establishes the part (c). It remains to show part (d). But this follows easily from the observation that

$$-a(x,d) \;\leq\; -a(x,\tilde{d}_\Delta) \;=\; \tilde{\sigma}(x,\Delta)$$

and from the definition of σ,

$$\sigma(x,d) \;\equiv\; -a(x,d) - \tfrac{1}{2}\, d^T H d.$$

Combining these two expressions immediately yields the remaining inequality in part (c). □

It follows from parts (a) and (b) of Proposition **8.4.2** that $\tilde{\sigma}(x,\Delta) = 0$ for some $\Delta > 0$ if and only if $\tilde{\sigma}(x,1) = 0$. Moreover, $\tilde{\sigma}(x,\Delta) = 0$ for some $\Delta > 0$ means that

$$a(x,d) \;\geq\; 0, \quad \forall\, d \;\in\; X - x,\; \|\, d \,\| \;\leq\; \Delta,$$

which, as noted above, coincides with the condition at the finite termination of Algorithm **8.4.1**. Therefore, our goal in the asymptotic analysis of this algorithm is to show that the sequence of scalars $\{\tilde{\sigma}(x^k,1)\}$ converges to zero, at least subsequentially. In other words, we aim at establishing a result for the GTRA that is analogous to Theorem **8.3.3**, which pertains to the GLSA.

The first task to accomplish is to show that if the GTRA does not terminate finitely, then it is not possible for any iterate to stay constant indefinitely. Since an iterate must stay constant during an inner iteration, our claim amounts to saying that after a finite number of consecutive reductions of the trust region radius Δ_k, we must arrive at a subsequent direction d^ℓ that passes the test (8.4.4), thereby producing an iterate $x^{\ell+1}$ that is distinct from the current x^ℓ. Consequently, the algorithm cannot generate an infinite sequence of inner iterations. The following result is a formal statement of this claim.

8.4.3 Proposition. Let $x \in X$ be such that $\tilde{\sigma}(x,1) > 0$. Let $\rho \in (0,1)$ and $H \in \mathbb{R}^{n\times n}$ be given. Suppose that for every sequence $\{d^\nu\}$ of vectors and every sequence $\{t_\nu\}$ of positive scalars converging to zero such that $\|d^\nu\| \leq t_\nu$ and d^ν belongs to $X - x$ for every ν, we have

$$\limsup_{\nu\to\infty} \; \frac{\theta(x+d^\nu) - \theta(x) + \tilde{\sigma}(x,t_\nu)}{t_\nu} \;\leq\; 0. \tag{8.4.5}$$

For any scalar $\gamma \in (0,1)$, there exists a positive number $\bar{\Delta}$ such that

$$\theta(x+d) - \theta(x) \;\leq\; -\gamma\, \sigma(x,d),$$

for every $\Delta \in (0,\bar{\Delta}]$ and every $d \in X - x$ satisfying $\|d\| \leq \Delta$.

Proof. The proof is by contradiction. Assume that for a given $\gamma \in (0,1)$ no such $\bar{\Delta}$ exists. There exist a sequence of positive scalars $\{\Delta_\nu\}$ converging to zero and a sequence $\{d^\nu\}$ of vectors such that for each ν, $d^\nu \in X - x$, $\|d^\nu\| \le \Delta_\nu$, and

$$\theta(x + d^\nu) - \theta(x) \;>\; -\gamma\, \sigma(x, d^\nu).$$

By part (d) of Proposition **8.4.2**, we deduce

$$\theta(x + d^\nu) - \theta(x) \;>\; -\gamma \left(\tilde{\sigma}(x, \Delta_\nu) + \frac{\Delta_\nu^2}{2} \| H \| \right).$$

Adding $\tilde{\sigma}(x, \Delta_\nu)$ to both sides, taking into account $\gamma \in (0,1)$ and Proposition **8.4.2** (a) and (b), we get, for all ν sufficiently large such that $\Delta_\nu < 1$,

$$\begin{aligned} \theta(x + d^\nu) - \theta(x) + \tilde{\sigma}(x, \Delta_\nu) \quad &> \quad (1 - \gamma)\, \tilde{\sigma}(x, \Delta_\nu) - \gamma\, \frac{\Delta_\nu^2}{2} \| H \| \\ &\ge \quad (1 - \gamma)\, \Delta_\nu \tilde{\sigma}(x, 1) - \gamma\, \frac{\Delta_\nu^2}{2} \| H \|. \end{aligned}$$

Dividing the inequality by Δ_ν, taking the lim sup, and using the limit condition (8.4.5), we deduce $(1-\gamma)\tilde{\sigma}(x,1) \le 0$. Since $\tilde{\sigma}(x,1)$ is nonnegative it follows that $\tilde{\sigma}(x,1) = 0$, which is a contradiction. □

We are ready to state and prove the main convergence result for the GTRA. Similar to Theorem **8.3.3**, we need an assumption that is the analog of condition (LS) for the GLSA. This assumption, labeled (TR2) below, is the sequential version of the pointwise limit condition in Proposition **8.4.3**. As in the previous theorem, the following theorem does not assume the boundedness of the trust region sequence. Also, the only assumption needed for the sequence $\{H^k\}$ is its boundedness.

8.4.4 Theorem. Let X be a closed convex subset of $\mathbb{R}^n$ and let θ be a locally Lipschitz function on X. Let $\{H^k\}$ be a bounded sequence of symmetric matrices. Assume conditions (TR1a), (TR1b) and (TR1c). Let $\{x^k\} \subset X$ be an infinite sequence of vectors generated by Algorithm **8.4.1**. If $\{x^k : k \in \kappa\}$ is a subsequence of $\{x^k\}$ satisfying condition (BD_θ) and

(TR2) for every sequence $\{\tilde{d}^k\}$ of vectors and every sequence $\{t_k : k \in \kappa\}$ of positive scalars converging to zero such that $\|\tilde{d}\| \le t_k$ and $\tilde{d}^k$ belongs to $X - x^k$ for every $k \in \kappa$, we have

$$\limsup_{k(\in\kappa)\to\infty} \frac{\theta(x^k + \tilde{d}^k) - \theta(x^k) + \tilde{\sigma}(x^k, t_k)}{t_k} \;\le\; 0,$$

then

$$\lim_{k(\in\kappa)\to\infty} \tilde{\sigma}(x^k, 1) \;=\; 0. \tag{8.4.6}$$

Proof. Let κ' be the infinite subset of κ such that the test (8.4.4) is satisfied for each $k \in \kappa'$. Clearly,

$$\lim_{k(\in\kappa')\to\infty} \tilde{\sigma}(x^k, 1) \;=\; \lim_{k(\in\kappa)\to\infty} \tilde{\sigma}(x^k, 1).$$

Therefore, if the left-hand limit is equal to zero, then so is the right-hand limit. In essence, the set κ' contains the counters of the outer iterations where distinct iterates are generated; the set $\kappa \setminus \kappa'$ contains the counters of the inner iterations where suitable trust region radii are computed.

Assume for the sake of contradiction that

$$\liminf_{k\in(\kappa')\to\infty} \tilde{\sigma}(x^k, 1) \;>\; 0. \tag{8.4.7}$$

Since the sequence $\{\theta(x^k)\}$ is nonincreasing and a subsequence is bounded below, it follows that $\{\theta(x^k)\}$ converges. Thus,

$$\lim_{k(\in\kappa')\to\infty} \sigma(x^k, d^k) \;=\; 0.$$

By Proposition **8.4.2** (c), we have

$$\sigma(x^k, d^k) \;\geq\; \frac{\rho}{2}\, \tilde{\sigma}(x^k, \Delta_k)\, \min\left\{1, \frac{\tilde{\sigma}(x^k, \Delta_k)}{\| H^k \| \Delta_k^2}\right\}.$$

Since the left-hand side converges to zero as $k(\in \kappa)$ tends to infinity, and the right-hand side is nonnegative, we deduce

$$0 \;=\; \lim_{k(\in\kappa')\to\infty} \tilde{\sigma}(x^k, \Delta_k)\, \min\left\{1, \frac{\tilde{\sigma}(x^k, \Delta_k)}{\| H^k \| \Delta_k^2}\right\}.$$

By Proposition **8.4.2** (b), (8.4.7), and the boundedness of $\{H^k\}$, we can easily establish that

$$\lim_{k(\in\kappa')\to\infty} \Delta_k \;=\; 0.$$

Since the trust region radius is halved during every inner iteration, the above limit implies

$$\lim_{k(\in\kappa)\to\infty} \Delta_k \;=\; 0.$$

Since at the beginning of each cycle of inner iterations Δ_k is always greater or equal to $\Delta_{\min}$, we conclude that there must exist a sequence of directions $\{\hat{d}^k : k \in \hat{\kappa}\}$, where $\hat{\kappa}$ is an infinite subset of $\kappa \setminus \kappa'$, such that for each $k \in \hat{\kappa}$, $\hat{d}^k$ belongs to $X - x^k$, $\|d^k\| \leq 2\Delta_k$, and

$$\begin{aligned} \theta(x^k + \hat{d}^k) - \theta(x^k) \;&>\; -\gamma_1\, \sigma(x^k, \hat{d}_k) \\ &\geq\; -\gamma_1\, \tilde{\sigma}(x^k, 2\Delta_k) - \gamma_1\, \rho\, \Delta_k^2 \,\| H^k \|, \end{aligned}$$

where the last inequality is by Proposition **8.4.2** (d). For all $k \in \kappa$ sufficiently large, we have $\Delta_k \leq 1/2$. By Proposition **8.4.2** (b) we can then write,

$$\begin{aligned}
&\theta(x^k + \hat{d}^k) - \theta(x^k) + \tilde{\sigma}(x^k, 2\Delta_k) \\
&\quad \geq (1 - \gamma_1)\, \tilde{\sigma}(x^k, 2\Delta_k) - 2\gamma_1 \rho\, \Delta_k^2 \,\|\, H^k \,\| \\
&\quad \geq 2(1 - \gamma_1)\, \Delta_k \tilde{\sigma}(x^k, 1) - 2\gamma_1 \rho\, \Delta_k^2 \,\|\, H^k \,\|.
\end{aligned}$$

Dividing by $2\Delta_k$, taking the lim sup (with k in $\hat{\kappa}$), and noting that the sequences $\{\hat{d}^k\}$ and $\{2\Delta_k\}$ satisfy the conditions needed to invoke (TR2), we get

$$\limsup_{k(\in\kappa)\to\infty} (1 - \gamma_1)\, \tilde{\sigma}(x^k, 1) \leq 0.$$

Since $\sigma(x^k, 1)$ is nonnegative this implies

$$\lim_{k(\in\kappa)\to\infty} \tilde{\sigma}(x^k, 1) = 0,$$

which contradicts (8.4.7). □

At this point, the reader can develop further convergence results for the trust region algorithm by properly specifying the function $a(x, d)$. In particular, much of the considerations presented subsequent to Theorem **8.3.3** for the line search methods can be repeated almost verbatim for Algorithm **8.4.1** and its specializations. Such considerations include the issue of a superlinear convergence rate. For this purpose, we need to ensure that the sequence of directions $\{d^k\}$ is properly calculated so that it is superlinearly convergent with respect to the sequence of iterates $\{x^k\}$. A necessary condition for the Q-superlinear convergence of $\{x^k\}$ is then that the trust region radius Δ_k is eventually never reduced.

In what follows, we introduce a modification of Step 4 of the trust region Algorithm **8.4.1** that is similar to the modification of the same step of the line search Algorithm **8.3.2**. As before, the purpose of this modification is to induce superlinear convergence of the iterates in the case of a nonnegative function θ, such as $\theta \equiv \frac{1}{2} G^T G$ for a vector function G.

Modified Step 4 of the GTRA

Step 4′: If

$$\theta(x^k + d^k) - \theta(x^k) \leq \min(-\gamma_1 \sigma(x^k, d^k), (1 - \gamma_1)\, \theta(x^k))$$

then set $x^{k+1} = x^k + d^k$ and

$$\Delta_{k+1} = \begin{cases} \max(2\Delta_k, \Delta_{\min}) & \text{if } \theta(x^k + d^k) - \theta(x^k) \leq -\gamma_2 \sigma(x^k, d^k) \\ \max(\Delta_k, \Delta_{\min}) & \text{otherwise.} \end{cases}$$

Set $k = k + 1$ and go to Step 2.

Otherwise, if

$$\theta(x^k + d^k) - \theta(x^k) > -\gamma_1 \sigma(x^k, d^k),$$

set $x^{k+1} = x^k$, $\Delta_{k+1} = \frac{1}{2}\Delta_k$, and $k = k + 1$; go to Step 2.

Similar to Theorem **8.3.17**, we can establish a convergence result for the GTRA under the above modified Step 4. We omit the details.

The rest of this section addresses an important issue in the trust region Algorithm **8.4.1**, namely, the calculation of the direction d^k in each iteration. In the original statement of the algorithm, this calculation is left unspecified; in practice, without solving the trust region subproblem $\mathrm{TR}(x^k, H^k, \Delta_k)$, it is not easy to compute such a direction because we need to ensure that it gives a fixed fraction of the optimal objective value $\sigma(x^k, H^k, \Delta_k)$, which is generally not known. In what follows, we present an alternative way to generate the direction d^k in order to alleviate this computational bottleneck.

Assume that for every fixed $x \in X$, $a(x, \cdot)$ is locally Lipschitz continuous and C-regular. This is trivially true if $a(x, \cdot)$ is linear in the second argument as in the case where θ is continuously differentiable and $a(x, d) \equiv \nabla\theta(x)^T d$. More generally, this assumption is also satisfied if $a(x, \cdot)$ is a convex function in the second argument as in the case of the vertical CP (8.3.32), where we take $a(x, \cdot)$ to be a convex, piecewise linear majorant of the directional derivative of the function $\frac{1}{2}\| \min(F(x), G(x))\|^2$

We introduce some special notation. Omitting the subscript "d", we write $\partial a(x, \cdot)$ to denote the Clarke generalized gradient of the function $a(x, \cdot)$ at d; thus

$$\partial a(x, d) = \partial a(x, \cdot)(d) = \partial_d a(x, d).$$

Similarly, we write $a'(x, d)$ to denote the directional derivative of the function $a(x, \cdot)$ at d; thus $a'(x, d) = a(x, \cdot)'(d; \cdot)$. For a fixed pair (x, d), $a'(x, d)$ is a Lipschitz continuous, positively homogeneous function from $\mathbb{R}^n$ into itself. By Proposition **7.1.6** and the C-regularity of $a(x, \cdot)$, we have

$$a'(x, d)(v) = \max\{ \zeta^T v : \zeta \in \partial a(x, d) \}.$$

Define the *normalized steepest descent direction* $d_s(x)$ of the function $a(x, \cdot)$

at the origin to be the negative of the normalized least-norm vector in $\partial a(x,0)$; that is,

$$d_s(x) \equiv \frac{-\xi(x)}{\| \xi(x) \|}, \quad \text{with } \xi(x) \equiv \operatorname{argmin} \{ \| \zeta \| : \zeta \in \partial a(x,0) \},$$

where, by convention, $d_s(x) \equiv 0$ if $0 \in \partial a(x,0)$. Since $\partial a(x,0)$ is a compact convex set in $\mathbb{R}^n$, $\xi(x)$ exists and is unique; thus $d_s(x)$ is well defined for every x. By the variational characterization of $\xi(x)$, we have

$$(\zeta - \xi(x))^T \xi(x) \geq 0, \quad \forall \zeta \in \partial a(x,0). \tag{8.4.8}$$

To prepare for some important properties of $d_s(x)$, we establish a technical lemma that has to do with a function of a single variable. Although easy, the proof of the lemma is not entirely trivial.

8.4.5 Lemma. Let $h : [t_1, t_2] \subset \mathbb{R} \to \mathbb{R}$ be continuous and right differentiable at every $t \in [t_1, t_2)$. If $h'(t;1) < 0$ for every $t \in [t_1, t_2)$, then h is decreasing in $[t_1, t_2]$.

Proof. Let ε be a fixed but arbitrary positive scalar and let I denote the subset of $[t_1, t_2]$ defined by

$$I \equiv \{ t \in [t_1, t_2] : h(s) - h(t_1) \leq \varepsilon (s - a), \ \forall s \in [t_1, t] \}.$$

We show that I is a nonempty closed interval. In fact t_1 obviously belongs to I and it is also easy to see that if $t \in I$ then $[t_1, t) \subseteq I$ so that I is an interval. To show I is closed it is sufficient to show that its right end-point, t_3, belongs to the interval. But this is easy because by definition

$$h(s) - h(t_1) \leq \varepsilon (s - t_1), \quad \forall s \in [t_1, t_3);$$

simply let s tend to t_3 and we obtain $t_3 \in I$. Therefore we can write $I = [t_1, t_3]$. We claim that $t_3 = t_2$. Assume by contradiction that $t_3 < t_2$. There exists a sequence $\{\tau_k\}$ of positive scalars converging to zero such that for every k

$$h(t_3 + \tau_k) - h(t_1) > \varepsilon (t_3 + \tau_k - t_1).$$

Since $h(t_3) - h(t_1) \leq \varepsilon(t_3 - t_1)$, we deduce $h(t_3 + \tau_k) - h(t_3) > \varepsilon \tau_k$. Dividing by τ_k and passing to the limit, we get $h'(t_3;1) \geq \varepsilon > 0$, a contradiction to the negativity of the right derivative of h at t_3. Therefore $t_2 = t_3$ and

$$h(t_2) - h(t_1) \leq \varepsilon (t_2 - t_1).$$

Since ε is arbitrary, it follows that $h(t_2) - h(t_1) \leq 0$. Repeating the above proof for any two points $t_1' < t_2'$ in the interval (t_1, t_2), we conclude that the function h is nonincreasing in the interval $[t_1, t_2]$. To complete the proof, we need to show that the function h is actually decreasing in $[t_1, t_2]$. Assume for the sake of contradiction that there are two distinct points, $t_4 < t_5$, belonging to $[t_1, t_2]$ such that $h(t_4) = h(t_5)$. Since h is nonincreasing we have $h(t_4) = h(t)$ for every $t \in [t_4, t_5]$. But this implies that $h'(t_4; 1) = 0$ thus contradicting the assumption that this directional derivative is negative. □

The following proposition identifies several properties of the direction $d_s(x)$, which justify the terminology "steepest descent" in this abstract setting. These properties are natural extensions of those in the case where $a(x, d) \equiv \nabla\theta(x)^T d$ for a smooth function θ.

8.4.6 Proposition. Assume that $a(x, \cdot)$ is locally Lipschitz continuous and C-regular.

(a) The normalized steepest descent direction $d_s(x)$ is a minimizer of the directional derivative $a'(x, 0)(d)$ on the closed Euclidean unit ball; that is,

$$d_s(x) \in \operatorname{argmin} \{ a'(x, 0)(d) : d \in \operatorname{cl} \mathbb{B}(0, 1) \}.$$

(b) $d_s(x) = 0$ if and only if $a'(x, 0)(d) \geq 0$ for all $d \in \mathbb{R}^n$.

(c) If $d_s(x) \neq 0$, then $d_s(x)$ is the unique minimizer of $a'(x, 0)(d)$ on $\operatorname{cl} \mathbb{B}(0, 1)$.

(d) If $d_s(x) \neq 0$, there exists a positive scalar $\bar{t}$ such that the function $h(t) \equiv a(x, t d_s(x))$ is decreasing in $[0, \bar{t}]$.

Proof. For every $d \in \mathbb{R}^n$, we have

$$a'(x, 0)(d) = \max\{ \zeta^T d : \zeta \in \partial a(x, 0) \}.$$

Since both $\operatorname{cl} \mathbb{B}(0, 1)$ and $\partial a(x, 0)$ are compact convex sets, by Corollary **2.2.10**, we deduce

$$\begin{aligned} \min_{d \in \operatorname{cl} \mathbb{B}(0,1)} a'(x, 0)(d) &= \min_{d \in \operatorname{cl} \mathbb{B}(0,1)} \max_{\zeta \in \partial a(x,0)} \zeta^T d \\ &= \max_{\zeta \in \partial a(x,0)} \min_{d \in \operatorname{cl} \mathbb{B}(0,1)} \zeta^T d \\ &= \max_{\zeta \in \partial a(x,0)} -\| \zeta \| = -\| \xi(x) \|. \end{aligned}$$

By (8.4.8), we have

$$\zeta^T \xi(x) \geq \xi(x)^T \xi(x), \quad \forall \zeta \in \partial a(x, 0),$$

which implies

$$\zeta^T d_s(x) \leq -\| \xi(x) \|, \quad \forall \zeta \in \partial(a,0).$$

Hence, $a'(x,0)(d_s(x)) \leq -\|\xi(x)\|$, showing that $d_s(x)$ minimizes $a'(x,0)(d)$ on $\mathrm{cl}\,\mathbb{B}(0,1)$. This establishes (a).

If $d_s(x) = 0$, then clearly $a'(x,0)(d)$ is nonnegative for all $d \in \mathbb{R}^n$, by the positive homogeneity of the directional derivative $a'(x,0)(\cdot)$. Conversely, if $a'(x,0)(d)$ is nonnegative for all $d \in \mathbb{R}^n$, then $d = 0$ is a minimizer of this directional derivative. By Proposition **7.1.12**, it follows that $0 \in \partial a(x,0)$; hence $d_s(x) = 0$. This establishes (b).

Suppose $d_s(x) \neq 0$. Then $\xi(x) \neq 0$. If $d \in \mathrm{cl}\,\mathbb{B}(0,1)$ satisfies

$$a'(x,0)(d) = -\| \xi(x) \|,$$

then for every $\zeta \in \partial a(x,0)$, we have

$$\zeta^T d \leq -\| \xi(x) \|.$$

In particular, with $\zeta \equiv \xi(x)$, we deduce

$$-\| \xi(x) \| \leq \xi(x)^T d \leq -\| \xi(x) \|.$$

Thus equality holds throughout. This implies that d must be equal to $d_s(x)$. Hence (c) follows.

In order to prove (d) we observe that the function $h(t)$ is continuous and right differentiable at every nonnegative t sufficiently small and that for such t

$$h'(t;1) = a'(x, td_s(x))(d_s(x)) = a(x,\cdot)'(td_s(x); d_s(x))$$

by the chain rule for directional derivatives; see Proposition **3.1.6**. By Proposition **7.1.6** and the assumed regularity we know that the directional derivative $a(x,\cdot)'(\cdot,\cdot)$ is an upper semicontinuous function of both arguments. Since $h'(0;1)$ is negative, a positive $\bar{t}$ must exist such that $h'(t;1) < 0$ for every $t \in [0,\bar{t}]$. We are in a position to apply Lemma **8.4.5** to complete the proof. □

Let a vector $x \in X$ be given such that the associated normalized steepest descent direction $d_s(x)$ is nonzero. For a positive radius $\Delta > 0$, consider the simplified trust region subproblem in the step size variable t:

$$\begin{array}{ll} \text{minimize} & a(x, td_s(x)) + \dfrac{t^2}{2} d_s(x)^T H d_s(x) \\ \text{subject to} & \| x + td_s(x) \| \leq \Delta \\ & x + td_s(x) \in X, \quad t \geq 0. \end{array} \tag{8.4.9}$$

This problem is a restriction of the trust region problem $\mathrm{TR}(x, H, \Delta)$ where the feasible set is constrained to be a bounded ray emanating from x in the direction $d_s(x)$. We define a *Cauchy point* of $\mathrm{TR}(x, H, \Delta)$ to be a vector

$$d_c(x) \;=\; t_c(x)\, d_s(x),$$

where t_c is any minimizer of (8.4.9). Since we are only interested in the value of the objective function of $\mathrm{TR}(x, H, \Delta)$ at a Cauchy point, and not in the Cauchy point itself, there is no need to further specify which of the possibly many Cauchy points we use. Based on such a Cauchy point, we can modify the trust region Algorithm **8.4.1** by requiring that the search direction d^k at iteration k gives a prescribed fraction of reduction of the objective value of $\mathrm{TR}(x^k, H^k, \Delta_k)$ calculated at a Cauchy point, instead of the same fraction of reduction calculated at the minimum point of this trust region subproblem. The rationale behind this modification is that it seems reasonable to expect the calculation of a direction satisfying (8.4.10) below to be easier than the calculation of a direction satisfying the previous (8.4.3).

Trust Region Algorithm Using a Cauchy Point (TRAUCP)

This is identical to Algorithm **8.4.1** except that at Step 2, the direction d^k is required to satisfy, instead of (8.4.3),

$$a(x^k, d^k) + \tfrac{1}{2}\,(d^k)^{\,T} H^k d^k \;\le\; \rho\,\big[\, a(x^k, d_c^k) + \tfrac{1}{2}\,(d_c^k)^{\,T} H^k d_c^k \,\big]\,, \qquad (8.4.10)$$

where d_c^k is a Cauchy point of the $\mathrm{TR}(x^k, H^k, \Delta_k)$.

One possible implementation of the above algorithm is to take $\rho = 1$. In this case, we simply accept the Cauchy point d_c^k and proceed with the rest of the calculations. Analogously to the convergence theory developed for the original GTRA, we can develop a similar theory for the above modified trust region algorithm. We omit the details.

8.5 Exercises

8.5.1 Give an example of a function $\theta : \mathbb{R} \to \mathbb{R}$ such that θ is not directionally differentiable at $x = 0$ and such that $\theta^D(0; 1) < \theta^\circ(0; 1)$.

8.5.2 Prove the assertion made after Theorem **8.3.3**; that is, show that a modification of the General Line Search Algorithm where at each iteration τ_k is chosen such that $x^{k+1} \equiv x^k + \tau_k d^k$ gives an objective function reduction that is larger than the one provided by the choice at Steps 4 and 5 of

the General Line Search Algorithm, inherits all the convergence properties of the GLSA.

8.5.3 The setting of this exercise is as in Exercise **7.6.9**. Let K be a closed convex set in $\mathbb{R}^n$ and let $f : \mathbb{R}^n \to \mathbb{R}$ be a twice continuously differentiable function such that

$$-\infty < \inf_{x \in \mathbb{R}^n} \lambda_{\min}(\nabla^2 f(x)) \leq \sup_{x \in \mathbb{R}^n} \lambda_{\max}(\nabla^2 f(x)) < \infty.$$

Let $\tau > 0$ be such that

$$0 < \inf_{x \in \mathbb{R}^n} \lambda_{\min}(\mathbf{I} - \tau \nabla^2 f(x)) \leq \sup_{x \in \mathbb{R}^n} \lambda_{\max}(\mathbf{I} - \tau \nabla^2 f(x)) < \infty.$$

Let $x^0 \in \mathbb{R}^n$ be arbitrary. Consider a sequence $\{x^k\}$ generated as follows. Let $d^k \equiv x^k - \Pi_K(x^k - \tau \nabla f(x^k))$. Let $x^{k+1} \equiv x^k - t_k d^k$, where t_k is obtained by the Armijo line search applied to the function

$$\psi(x) \equiv \tau f(x) - \tfrac{1}{2} \tau^2 \nabla f(x)^T \nabla f(x) + \tfrac{1}{2} \operatorname{dist}(x - \tau \nabla f(x), K)^2.$$

at the vector x^k along the direction d^k and with $\sigma(x^k, d^k) \equiv -\nabla \psi(x^k)^T d^k$. Show that every accumulation point of $\{x^k\}$ is a constrained stationary point of f on K. Moreover if x^* is such an accumulation point with $\nabla^2 f(x^*)$ strictly copositive on the critical cone $\mathcal{C}(x^*; K, \nabla f)$, then $\{x^k\}$ converges to x^*.

8.5.4 This exercise is an application of Danskin's Theorem **10.2.1** and also of some sensitivity results of Chapter **5**. Let $L : \mathbb{R}^{n+m} \to \mathbb{R}$ be a convex-concave saddle function on $\mathbb{R}^{n+m}$, and let $X \subset \mathbb{R}^n$ and $Y \subset \mathbb{R}^m$ be two compact, convex sets. Consider the minimax problem:

$$\min_{x \in X} \max_{y \in Y} L(x, y).$$

By introducing the function

$$\theta(x) \equiv \max_{y \in Y} L(x, y),$$

the minimax problem can be equivalently rewritten as

$$\min_{x \in X} \theta(x).$$

(a) Show that θ is a convex function on $\mathbb{R}^n$ and finite everywhere.

(b) Assume that L is LC^1 and that for some positive constant c the following condition holds: for all x in X and all y^1 and y^2 in Y,

$$(y^1 - y^2)^T (\nabla_y L(x, y^2) - \nabla_y L(x, y^1)) \leq c \| y^1 - y^2 \|^2.$$

This means that $L(x, \cdot)$ is strongly concave in y with the same strong concavity constant for all x in X. Prove that θ is LC^1 on X with $\nabla\theta(x) = \nabla_x L(x, y(x))$, where $y(x)$ is the unique maximizer of $L(x, \cdot)$ over Y. (Hint: use Theorem **10.2.1**).

(c) Prove that if in addition Y is defined by a finite number of twice continuously differentiable convex inequalities satisfying the CRCQ at every point in Y, then θ is actually SC^1 on X. (Hint: characterize $y(x)$ in a suitable way and then use Theorem **4.5.2** and Proposition **7.4.6**.)

8.6 Notes and Comments

In this chapter we concentrated on the study of numerical approaches to the solution of nonsmooth equations that are "natural" globalizations of a local Newton method; these include path search, line search, and trust region algorithms. Other approaches to the minimization of a nonsmooth (merit) function are possible, for example bundle and subgradient methods [325, 521]; but these will not be discussed here. In fact, the following notes and comments are confined to methods for nonsmooth problems that are directly relevant to the VI/CP.

Path search methods were pioneered by Ralph [488], who introduced the idea of searching along a suitable path instead of a line segment as a natural way to circumvent the difficulties associated with nonsmooth equations. The convergence theory of the resulting Algorithm **8.1.9** is rather restrictive in its assumptions, which nevertheless guarantee the existence of a solution *a priori*. This special feature was already noted by Ralph and is clearly highlighted in Theorem **8.1.4**, which is new. In order to prove this theorem we used a classical result on homeomorphisms (Definition **8.1.2** and Proposition **8.1.3**) that can be found in [422]. The rest of the analysis in Section **8.1** is based on [488], which also addresses applications to the normal map of VIs and NCPs. A vital point in the implementation of the path search algorithm is the calculation of a suitable point along the path at each iteration. The use of Lemke's method in the case of the normal map of a complementarity problem is discussed in [136, 488].

A further point discussed in [488] that we did not deal with is the use of "nonmonotone" strategies. In all the algorithms in this chapter, the value of some merit function is strictly decreasing in each iteration. In a nonmonotone strategy, this is no longer imposed, and the possibility exists that the merit function value may actually increase, in a controlled way, from one iteration to the next. An example of how this can be achieved is

to substitute the test at Step 3 in the Path Newton Method with

$$\| G(p^k(2^{-i}\bar{\tau}_k)) \| \leq (1 - \gamma\, 2^{-i}\, \bar{\tau}_k) \max_{1 \leq j \leq \min(r,k+1)} \{ \| G(x^{k+1-j}) \| \},$$

where r is a positive integer. If r is equal to 1, we obtain the test in Step 3; otherwise we have a relaxation of the monotonicity requirements on $\{\|G(x^k)\|\}$. The nonmonotone strategies have been shown experimentally to yield superior computational results to the more standard monotone counterparts. Although there are some scattered results in the literature, in the context of unconstrained optimization, nonmonotone methods were discussed in detail and popularized by Grippo, Lampariello, and Lucidi [243, 244]. The techniques in these papers were successfully applied in many different contexts, including a significant extension (other than the path algorithm) to the solution of nondifferentiable equations [184]. Even if we don't mention this further, most of the algorithms in this and in the next two chapters can be easily modified to accommodate nonmonotone strategies. Indeed, this is done in most of the practical implementations of the algorithms with good effects.

Line search methods are a more common and more reliable alternative to the path search algorithm. Central to line search (and trust region) globalization strategies is the use of merit functions that reduce the problem of solving a system of nonsmooth equations to that of finding an (unconstrained or constrained) minimum of a nondifferentiable function. From this point of view, methods for nonsmooth optimization are clearly relevant to the analysis in Sections **8.3** and **8.4**. Actually, the contents of these two sections basically revolve around the description of nonsmooth optimization methods.

The use of Dini or Clarke's derivatives to define suitable stationarity concepts is rather standard, even if in most of the cases we deal with in the book some sharper definition of stationarity can be employed. Pang [425] described one of the earlier attempts to define a globally convergent modification of Newton's method to solve systems of B-differentiable equations. In essence his approach consisted in the damping of a Newton direction obtained by solving the (nonlinear) system

$$G(x^k) + G'(x^k; d^k) = 0. \tag{8.6.1}$$

However, since the directional derivative $G'(\cdot;\cdot)$ is not continuous in the first argument, differentiability of G at an accumulation point of the sequence generated by the algorithm has to be assumed in order to prove convergence to a solution of the system $G(x) = 0$. This approach is closely

related to the B-differentiable Line Search Algorithm **8.3.6**, and in fact in Proposition **8.3.7** we see that the existence of the strong F-derivative of θ at x^* is required in order to show convergence to a stationary point. This requirement was dropped in several specific classes of equations in [426]. Building on the latter paper, Pang, Han, and Rangaraj developed a general line search framework for the minimization of a locally Lipschitz function [432] and of a system of Lipschitz continuous equations [254]. Central to these two papers is the use of an "iteration function" $\phi(x^k; d)$ to replace directional derivatives. In the case of equations, this means defining the search direction subproblem by $G(x^k) + \phi(x^k; d^k) = 0$ instead of (8.6.1). Under adequate conditions on the iteration function, it is possible to show convergence. A somewhat more general approach very similar to that of [432] was developed in [184], where no explicit rules for calculating the search direction were given; it was only postulated that some suitable relations between the essential elements used in the algorithm (such as the directions and forcing functions) hold. Our General Line Search Algorithm **8.3.2** and the related analysis are the product of these collective efforts and in particular of [184]. The concrete applications of the general scheme are, instead, derived from the Pang-Han-Rangaraj iteration function approach. Some other related developments can be found in [472, 478]. Poliquin and Qi [461] studied in detail iteration functions for general Lipschitz functions, exhibited very broad classes of Lipschitz functions for which iteration functions can be built, and showed that a necessary condition for a Lipschitz function to admit an iteration function is that Clarke's directional derivative be equal to the upper Dini directional derivative.

Ostrowski's Theorem **8.3.9** is classical, and its proof can be found in [423]. The easily derived, but practically very useful, Proposition **8.3.10** was given in this form in [165], even if the key implication (b) $\Rightarrow$ (c) was already noted in [417]. The applications of this proposition in the rest of Subsection **8.3.1** are immediate extensions of standard results in smooth optimization. Proposition **8.3.15** is a well-known result; in fact, this result pertains to the analysis of the behavior of general sequences of iterates $\{x^k\}$ and of search directions $\{d^k\}$; in particular, it makes no assumption on how the latter sequence is generated. Apparently, the idea of using the modified line search procedure (8.3.23) (more precisely, of the variant described after Theorem **8.3.17**) was employed for the first time, in the context of algorithms for the solution of nonsmooth problems, by Qi [472] and, independently, in the context of algorithms for the solution of VIs/CPs, by Facchinei and Soares [180]. This simple "trick" was successfully used to simplify the analysis of the convergence rate of algorithms for VIs and CPs.

The class of SC^1 optimization problems is very broad and important because, as seen in Section **7.7**, SC^1 functions arise in many contexts. The equivalence between (a) and (b) in Proposition **8.3.18** was shown by Qi [471], while the implications (b) $\Rightarrow$ (c) $\Rightarrow$ (d) in the same proposition were proved by Facchinei [166]. The latter study was motivated by the investigation of the convergence rate of algorithms that use differentiable, exact penalty functions for the solution of constrained optimization problems. In fact, it turns out that regardless of the continuity properties of the original constrained problem, differentiable, exact penalty functions cannot be expected to be more than SC^1 in general. The algorithm for the constrained minimization of a convex SC^1 in Subsection **8.3.3** is from [434] where, however, only the case of a polyhedral set X was analyzed. The application to a complementarity problem in Subsection **8.3.4** is an extension of the NE/SQP method of Pang and Gabriel [431] for the NCP; for the inexact version of the latter method, see [232].

Conn, Gould and Toint [110] wrote the definitive reference on trust region methods. Their book is an exhaustive source of information for the history of these methods, for the numerical issues connected to the solution of trust region subproblems, and for all kinds of applications of the trust region approach. Trust region methods for the minimization of nonsmooth functions have been considered since the 1980s under the assumption of specific structures of the objective functions, such as those arising from the use of nondifferentiable penalty functions for constrained optimization. For a discussion on these algorithms tailored to specific structures, see [110].

When considering more general nondifferentiable problems, i.e., less structured problems, one can trace a research line that originates from the seminal work of Dennis, Li, and Tapia [129]. In this work the authors considered the unconstrained minimization of a general Lipschitz continuous function that is C-regular. In our terminology, their main assumptions are as follows. In (8.4.2), the function $a(x, \cdot)$ is both Lipschtiz continuous and C-regular, a is continuous in x for every fixed d, and the directional derivatives of θ at x and $a(x, \cdot)$ (at the origin) are the same. Using a condition similar to (8.4.3) for the accuracy in the solution of the trust region subproblems, Dennis, Li, and Tapia proved convergence to a C-stationary point of θ. They also showed that their analysis subsumed that of many previous authors who had analyzed structured nonsmooth problems.

An important advance along these lines was made by Scholtes and Stöhr [516], who, in the case of composite nonsmooth functions, which include penalty functions for MPECs, could relax the C-regularity assumption on the objective function and assume only the directional differentiability of θ.

Another significant improvement was given in [110], where the algorithm of Dennis, Li, and Tapia is modified in order to permit a more relaxed inexactness criterion in the solution of the subproblem (8.4.2). In all these papers, relatively strong assumptions are imposed on the function θ and on a. A possible remedy to these restrictions is to mix trust region methods with bundle methods. Again, we refer to the discussion in [110] for more details on these very specialized developments.

A somewhat different line of research, also aimed at relaxing as much as possible the assumptions on the objective function θ and on the model a, was initiated by Qi and Sun [486]. In this reference, the iteration function approach, discussed above in connection to a line search algorithm, is modified and adapted to trust region methods for the unconstrained minimization of a nonsmooth function. Under appropriate assumptions on the iteration function, convergence to stationary points of the sequence generated by the algorithm is proved. Another noteworthy feature of [486] is the use of a function similar to our $\tilde{\sigma}$ in order to gauge the criticality of the points generated by the algorithm. This same route was also explored by Gabriel and Pang [233], who, however, used a different iteration function and focused on linearly constrained problems. A proposal in [233] that we maintain in our General Trust Region Algorithm is the imposition of a lower bound $\Delta_{\min}$ on the trust region radius Δ at the beginning of each outer iteration. This strategy has been successfully followed in the design of many trust region algorithms for the solution of VIs and CPs, but is not present in the other papers that we have discussed until now. It is not clear to us whether this small requirement on Δ can be removed in our very general setting. In both [233] and [486], the selected approximate solution d^k to (8.4.2) is required to satisfy a condition of the type (8.4.3).

The approach in Section **8.4** departs from [233, 486] in that we use a still different type of iteration function and allow for the presence of any kind of convex constraint set X. The results concerning the use of the Cauchy point to develop more easily implementable algorithms parallel those in [110]. A different approach to this problem is discussed by Martínez and Moretti [393], who propose to accept an inexact solution d^k of (8.4.2) if d^k gives an objective function of (8.4.2) that is at least a fixed fraction of the same objective function calculated at a point obtained by solving a trust region subproblem with a "simplified" objective function.

Chapter 9

Equation-Based Algorithms for CPs

This chapter is devoted to the exposition of iterative algorithms for the solution of CPs based on the theory developed in the two previous chapters. The methods presented herein are applicable to general classes of the complementarity problem and are all based on various equivalent equation/merit function reformulations of the CP, most of which we have introduced in Section **1.5**. In order to apply the theory developed in Chapters **7** and **8**, we need to first analyze in detail the relevant properties of these reformulations.

Basically we can attempt to apply any of the methods considered in Chapters **7** and **8** to any of the reformulations considered in Section **1.5**, or to some of their variants to be discussed later. It is neither beneficial nor possible to give a detailed treatment of all these possibilities. Instead, we focus on some selected cases to illustrate the kind of analysis that can be performed, and encourage the readers to develop similar arguments also for other cases that we have left out, and which may better suit their specific needs. Obviously, the choice of topics and the relative emphasis we place on different issues reflect our understanding and beliefs in the applicability and importance of these algorithms. We also extend the approach considered for CPs to other problems that have a similar structure and that can be dealt with in a similar fashion, including the mixed complementarity problems and variational inequalities with box constraints. In the next chapter, we consider general variational inequalities.

9.1 Nonlinear Complementarity Problems

In this section we consider algorithms for the nonlinear complementarity problem:

$$0 \leq x \perp F(x) \geq 0,$$

where we assume that F is a continuously differentiable function defined on $\mathbb{R}^n$. (We caution the reader that if the domain of definition or differentiability of F is a proper subset of $\mathbb{R}^n$, one must be careful in applying the algorithms and results developed herein; in some extreme cases, it might even be necessary to abandon the general approach and seek alternative approaches to deal with the restricted domains. In the present and subsequent chapters, F-differentiability at all relevant iterates is a minimum requirement of the input functions that we deal with.) The NCP is certainly one of the simplest and most basic kind of variational inequalities. The algorithms and analysis we develop for this class of problems are the paradigm for the more complex problems we consider subsequently. In this chapter the reader has the opportunity to become familiar with many basic approaches and techniques that can be extended to other types of CPs, such as the CP involving two functions F and G: $0 \leq F(x) \perp G(x) \geq 0$.

In Section **1.5** we already encountered several equivalent reformulations of the NCP (F) as smooth systems of equations. A natural question is then: why are we still interested in a nonsmooth reformulation such as that based on a nondifferentiable C-function? At first sight, such a nonsmooth reformulation seems to have little value and thus is not worthy of further consideration. This is not quite so. There are different reasons why a nonsmooth equation reformulation may be preferable. The first reason is that a smooth equation reformulation can often fail to provide a sound basis for the development of fast local methods. This is clarified in the following proposition.

9.1.1 Proposition. Suppose that $F : \mathbb{R}^n \to \mathbb{R}^n$ is continuously differentiable. Let ψ be a continuously differentiable C-function and let

$$\mathbf{F}_\psi(x) \equiv \begin{pmatrix} \psi(x_1, F_1(x)) \\ \vdots \\ \psi(x_n, F_n(x)) \end{pmatrix}, \quad \forall\, x \in \mathbb{R}^n.$$

If x^* is a *degenerate* solution of the NCP (F), then $J\mathbf{F}_\psi(x^*)$ is singular.

Proof. Let $i \in \{1, \ldots, n\}$ be an index for which $x_i^* = 0 = F_i(x^*)$. By the formula for the derivative of composite functions, the i-th row of the

Jacobian of $\mathbf{F}_\psi(x^*)$ is given by

$$\frac{\partial\psi(0,0)}{\partial a}\,(\,e^i\,)^T + \frac{\partial\psi(0,0)}{\partial b}\,\nabla F_i(x^*)^T,$$

where $\partial\psi/\partial a$ and $\partial\psi/\partial b$ denote, respectively, the partial derivative of ψ with respect to the first and second argument, and where e^i denotes the i-th coordinate vector. By this expression, it is clear that if we can show that $\nabla\psi(0,0)$ is equal to zero, then the i-th row of the Jacobian $J\mathbf{F}_\psi(x^*)$ is also zero; and thus the Jacobian $J\mathbf{F}_\psi(x^*)$ is singular. By the differentiability of ψ and the fact that $\psi(0,0) = 0$, we have

$$\frac{\partial\psi(0,0)}{\partial a} = \lim_{a\downarrow 0}\frac{\psi(a,0)}{a} = 0,$$

where the second equality holds because ψ is a C-function. Similarly, we can show that the partial derivative of ψ with respect to its second argument is also equal to zero at $(0,0)$. □

The above proposition clearly shows that we cannot expect to be able to develop locally fast methods based on smooth C-functions for the solution of the NCP (F) if the computed solution happens to be degenerate. Furthermore, practical experience also suggests that nonsmooth equation based methods are often more efficient. This explains why, following the common practice in the field, we give more emphasis to nonsmooth equation methods. Clearly, nonsmooth equation methods have their own drawbacks, the most notable one being the difficulty in the design of globally convergent algorithms. Therefore, an important task of this chapter is to develop remedies to overcome the latter difficulty, with the goal of obtaining algorithms for solving the NCP (F) that are globally convergent with locally fast convergence rate. In designing such algorithms, the following are important considerations that require particular attention.

1. It is preferable to use nonsmooth equation reformulations that are (strongly) semismooth and whose associated merit functions are smooth.

2. It is desirable to use linear Newton approximations to the nonsmooth equation so that systems of linear equations are solved at each iteration.

3. In addition to being globally convergent, the resulting methods should generate iterates all of whose limit points are solutions of the complementarity problem.

4. The merit functions should have bounded level sets so that at least one limit point of the iterates exists.

5. We want to ensure that the linear Newton approximations are nonsingular at a solution, so that a superlinear convergence rate can be achieved.

6. Although difficult, it is useful to be able to obtain some asymptotic convergence results if the merit functions have unbounded level sets.

As we know, the Fischer-Burmeister C-function gives rise to an equation formulation of the NCP (F) that satisfies the requirement in Point 1. We will see more formulations of this kind later in this chapter. Restricted to such equations, linear Newton approximation schemes can be designed without too much difficulty with the use of the generalized Jacobian or one of its restrictions or enlargements considered in Section **7.5.1**. Furthermore, since the merit functions are smooth, globalization of these Newton schemes can be easily carried out via either a line search or a trust region procedure; locally superlinear convergence can be readily ensured, provided that the schemes are nonsingular at a computed solution. So far, we have already prepared the groundwork for all these tasks and it remains to specialize the developments from the previous chapters. Toward this end, a detailed study of Points 3, 4, and 5 is in order. Specifically, we need to understand what classes of functions F will ensure the merit functions being used to have bounded level sets (Point 4) and what solution properties will guarantee the nonsingularity of the Newton approximations (Point 5). The crux of Point 3 is the question of when a stationary point of a merit function is a solution to the NCP. We have addressed this issue several times already (see for instance Propositions **1.5.13** and **8.3.21**); but we need to undertake a more thorough analysis for a host of merit functions not dealt with before. Such an analysis will enable us to compare different merit functions and understand more about their relative strengths and weaknesses as a computational tool for solving the CPs. We illustrate this comparison with an example of an NCP and using two merit functions.

9.1.2 Example. Consider the NCP (F) where $F : \mathbb{R} \to \mathbb{R}$ is given by

$$F(x) \;=\; (x-3)^3 + 1.$$

The unique solution of this NCP is $x^* = 2$. Note that F is strictly monotone and that $JF(x)$ is positive definite everywhere except at $x = 3$, where it is positive semidefinite.

Consider two different unconstrained minimization reformulations of this complementarity problem via the following merit functions:

$$\begin{aligned}
\theta_{\mathrm{FB}}(x) &= x^2 + F(x)^2 + xF(x) - (x + F(x))\sqrt{x^2 + F(x)^2};\\
\theta_{\mathrm{MS}}(x) &= xF(x) + \frac{1}{4}\,[\,(x - 2F(x))_+^2 - x^2 + (F(x) - 2x)_+^2 - F(x)^2\,].
\end{aligned}$$

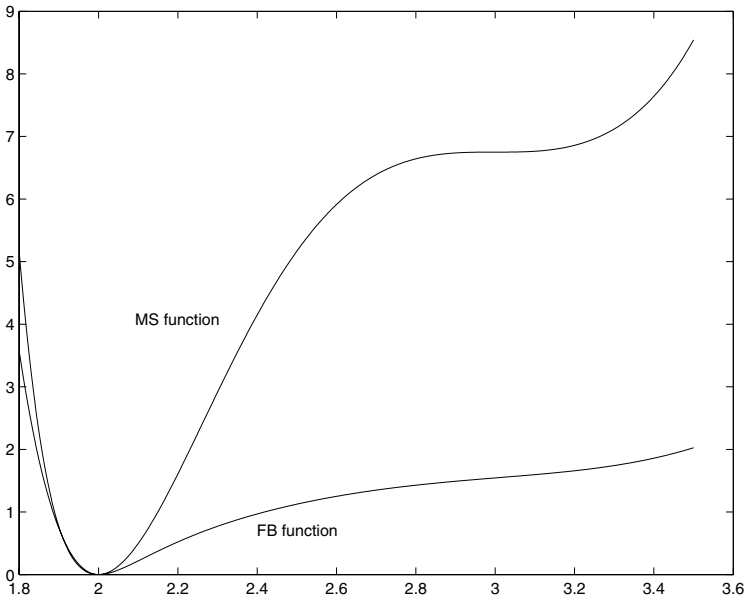


Figure 9.1: $\theta_{\rm MS}$ versus $\theta_{\rm FB}$.

The first function $\theta_{\rm FB}$ is equal to $\frac{1}{2}(\mathbf{F}_{\rm FB})^2$, which is the merit function corresponding to the FB function; the second function $\theta_{\rm MS}$ is derived from the "implicit Lagrangian" introduced by Mangasarian and Solodov. The latter function is a particular case of a more general merit function for variational inequalities that will be studied in the next chapter, to which we refer the reader for a proof of the fact that $\theta_{\rm MS}$ is also an unconstrained merit function for the NCP (F); see Subsection **10.3.1**.

Note that both merit functions are nonnegative and continuously differentiable (this can be checked directly in this simple case or the reader can refer to Proposition **1.5.3** for $\theta_{\rm FB}$ and to the analysis in the next chapter for $\theta_{\rm MS}$). Moreover, it is a matter of straightforward computations to verify that $\nabla\theta_{\rm FB}(x) = 0$ for $x = 2$ only, while $\nabla\theta_{\rm MS}(x) = 0$ for $x = 2$ and also for $\tilde{x} = 3$. Thus $\theta_{\rm FB}$ has a unique stationary point, which coincides with the unique solution of the NCP (F). $\theta_{\rm MS}$ has two stationary points; one of these is the solution of the same NCP and the other is a singular point of the derivative of $\theta_{\rm MS}$ and is not a solution of the NCP (F). The functions $\theta_{\rm FB}$ and $\theta_{\rm MS}$ are plotted in Figure **9.1**, which clearly shows the different behavior of the two functions. If we employ a standard unconstrained minimization technique to minimize $\theta_{\rm FB}$, we are sure that every

limit point of the sequence so produced, being a stationary point of $\theta_{\rm FB}$, is the solution of the complementarity problem. The same thing cannot be said, however, for the implicit Lagrangian, since the sequence of iterates may be converging to the wrong stationary point $\tilde{x} = 3$. It is clear that in this case the merit function $\theta_{\rm FB}$ is preferred to $\theta_{\rm MS}$. □

Conditions to ensure the nonsingularity of the Newton approximation at a solution (Point 5) are obviously important. We will see that even though almost all the Newton approximation schemes we consider in this chapter turn out to be nonsingular at a strongly stable solution of the CP, differences with regard to this issue exist among various reformulations.

Point 6 pertains to an advanced convergence analysis. The goal of such an analysis is to extend the usual results that concern the limit points of the iterates to the case where such limit points can not be guaranteed to exist (for example, when the merit functions have unbounded level sets). To date, results of such an unbounded asymptotic analysis are scarce. The difficulty lies in the lack of proper tools to deal with unbounded sequences for nonconvex/nonmonotone problems.

9.1.1 Algorithms based on the FB function

The first family of iterative methods presented in this chapter is based on the equation $\mathbf{F}_{\rm FB}(x) = 0$, where

$$\mathbf{F}_{\rm FB}(x) \equiv \begin{pmatrix} \psi_{\rm FB}(x_1, F_1(x)) \\ \cdots \\ \psi_{\rm FB}(x_n, F_n(x)) \end{pmatrix},$$

with $\psi_{\rm FB}$ being the FB C-function:

$$\psi_{\rm FB}(a,b) \equiv \sqrt{a^2+b^2} - (a+b), \quad (a,b) \in \mathbb{R}^2.$$

The naturally associated merit function is the squared Euclidean norm of $\mathbf{F}_{\rm FB}(x)$; i.e.,

$$\theta_{\rm FB}(x) \equiv \tfrac{1}{2}\, \mathbf{F}_{\rm FB}(x)^T \mathbf{F}_{\rm FB}(x).$$

The following lemma establishes some useful bounds for $\psi_{\rm FB}(a,b)$ in terms of the min function. These bounds are very useful in understanding the growth properties of $\psi_{\rm FB}(a,b)$.

9.1.3 Lemma. For any two scalars a and b, it holds that

$$\frac{2}{2+\sqrt{2}}\, |\, \min(a,b)\, | \;\le\; |\, \psi_{\rm FB}(a,b)\, | \;\le\; (2+\sqrt{2})\, |\, \min(a,b)\, |. \qquad (9.1.1)$$

Proof. Without loss of generality assume $a \leq b$, so that $\min(a,b) = a$. If $a + b > 0$ we have that $b \neq 0$ and

$$\begin{aligned} |\,\psi_{\rm FB}(a,b)\,| &= |\,\sqrt{a^2+b^2} - (a+b)\,| \\ &= \left| \frac{[\sqrt{a^2+b^2} - (a+b)][\sqrt{a^2+b^2} + (a+b)]}{\sqrt{a^2+b^2} + (a+b)} \right| \\ &= \frac{|\,2ab\,|}{\sqrt{a^2+b^2} + (a+b)} \\ &= \frac{2\,|\,a\,|}{\sqrt{(a/b)^2+1} + (a/|b|) + 1} \\ &= \frac{2}{\sqrt{(a/b)^2+1} + (a/|b|) + 1}\,|\,\min(a,b)\,|. \end{aligned}$$

Since $a \leq b$, so that $b \geq |a|$, we also have

$$1 \;\leq\; \sqrt{(a/b)^2+1} + (a/|b|) + 1 \;\leq\; 2 + \sqrt{2},$$

which combined with the above equation easily gives the inequalities in (9.1.1). Suppose that $a + b \leq 0$. We have

$$|\,\psi_{\rm FB}(a,b)\,| \;=\; |\,\sqrt{a^2+b^2} - (a+b)\,| \;\geq\; \sqrt{a^2+b^2} \;\geq\; |\,a\,| \;=\; |\,\min(a,b)\,|.$$

Since $a \leq b$, so that $a \leq -|b|$, we also have $a^2 \geq b^2$, so that

$$\begin{aligned} |\,\psi_{\rm FB}(a,b)\,| &= \sqrt{a^2+b^2} - (\,a+b\,) \\ &\leq \sqrt{a^2+b^2} - 2a \\ &= \sqrt{2}\,|\,a\,| + 2\,|\,a\,| \;=\; (2+\sqrt{2})\,|\,\min(a,b)\,|, \end{aligned}$$

which gives the right inequality in (9.1.1). □

Since $|\min(a,b)| \leq \sqrt{a^2+b^2}$, we obtain, from the right-hand inequality in (9.1.1),

$$|\,\psi_{\rm FB}(a,b)\,| \;\leq\; (\,2+\sqrt{2}\,)\,\sqrt{a^2+b^2}.$$

It follows trivially from (9.1.1) that, for an arbitrary sequence $\{(a_k,b_k)\}$,

$$\lim_{k\to\infty} |\,\psi_{\rm FB}(a_k,b_k)\,| \;=\; \infty \;\Leftrightarrow\; \lim_{k\to\infty} |\,\min(a_k,b_k)\,| \;=\; \infty.$$

Since limits of these sort are clearly related to the issue of the boundedness of the level sets of the merit function $\theta_{\rm FB}(x)$, one can expect that Lemma **9.1.3** has an important role to play when we come to investigate the latter boundedness issue.

In addition to being easy to deal with, the merit function $\theta_{\rm FB}(x)$ possesses several favorable properties that make it particularly attractive from the computational point of view. In particular, although $\mathbf{F}_{\rm FB}$ is nonsmooth, the associated merit function $\theta_{\rm FB}$ is continuously differentiable if F is; see Proposition **1.5.3**. In the next proposition we establish several other differentiability properties of $\mathbf{F}_{\rm FB}$ and $\theta_{\rm FB}$.

9.1.4 Proposition. Assume that $F : \Omega \subseteq \mathbb{R}^n \to \mathbb{R}^n$ is continuously differentiable on the open set Ω. The following statements hold.

(a) The generalized Jacobian of $\mathbf{F}_{\rm FB}$ satisfies

$$\partial \mathbf{F}_{\rm FB}(x) \subseteq \mathcal{D}_a(x) + \mathcal{D}_b(x) JF(x), \tag{9.1.2}$$

where $\mathcal{D}_a(x)$ and $\mathcal{D}_b(x)$ are the sets of $n \times n$ diagonal matrices $\mathrm{diag}(a_1(x), \ldots, a_n(x))$ and $\mathrm{diag}(b_1(x), \ldots, b_n(x))$ respectively, with

$$(a_i(x), b_i(x)) \begin{cases} \equiv \dfrac{(x_i, F_i(x))}{\sqrt{x_i 2 + F_i(x) 2}} - (1,1) & \text{if } (x_i, F_i(x)) \neq 0 \\ \in \mathrm{cl}\, \mathbb{B}(0,1) - (1,1) & \text{if } (x_i, F_i(x)) = 0. \end{cases}$$

(b) $\mathbf{F}_{\rm FB}$ is semismooth on Ω.

(c) $\theta_{\rm FB}$ is continuously differentiable on Ω and its gradient $\nabla \theta_{\rm FB}(x)$ is equal to $H^T \mathbf{F}_{\rm FB}(x)$ for every H in $\partial \mathbf{F}_{\rm FB}(x)$.

(d) If the Jacobian $JF(x)$ is locally Lipschitz on Ω, then $\mathbf{F}_{\rm FB}$ is strongly semismooth on Ω.

Proof. As a composition of Lipschitz functions, the function $\mathbf{F}_{\rm FB}$ is locally Lipschitz continuous. By Proposition **7.1.14**, we have

$$\partial \mathbf{F}_{\rm FB}(x)^T \subseteq \partial (\mathbf{F}_{\rm FB})_1(x) \times \cdots \times \partial (\mathbf{F}_{\rm FB})_n(x).$$

If i is such that $(x_i, F_i(x)) \neq (0,0)$, then it is easy to check that $(\mathbf{F}_{\rm FB})_i$ is differentiable at x and, with e^i denoting the i-th coordinate vector,

$$\nabla (\mathbf{F}_{\rm FB})_i(x) = \left(\frac{x_i}{\sqrt{x_i^2 + F_i(x)^2}} - 1 \right) e^i + \left(\frac{F_i(x)}{\sqrt{x_i^2 + F_i(x)^2}} - 1 \right) \nabla F_i(x).$$

If $(x_i, F_i(x)) = (0,0)$, by using the theorem on the generalized gradient of a composite function and recalling that $\partial \|(0,0)\| = \mathrm{cl}\, \mathbb{B}(0,1)$, we get

$$\partial (\mathbf{F}_{\rm FB})_i(x) = \{ (\xi - 1) e^i + (\rho - 1) \nabla F_i(x) : (\xi, \rho) \in \mathrm{cl}\, \mathbb{B}(0,1) \}.$$

From these equalities (a) follows.

To prove (b) it is sufficient to note that each component $(\mathbf{F}_{\mathrm{FB}})_i$, being the composition of the strongly semismooth function ψ_{FB} and the smooth function $x \mapsto (x_i, F_i(x))$, is semismooth, by Proposition **7.4.4**. This observation also establishes (d).

To prove (c) it suffices to establish the expression of $\nabla\theta_{\mathrm{FB}}(x)$. But this follows easily from Proposition **7.1.11** on the generalized Jacobian of composite functions and from the fact that $\partial\theta_{\mathrm{FB}}(x) = \{\nabla\theta_{\mathrm{FB}}(x)\}$ by the continuous differentiability of θ_{FB}. □

9.1.5 Remark. Part (c) of Proposition **9.1.4** implies that $H^T\mathbf{F}_{\mathrm{FB}}(x)$ is independent of H, as long as H belongs to $\partial\mathbf{F}_{\mathrm{FB}}(x)$. We further note that if a point x is such that $(x_i, F_i(x)) \neq (0,0)$ for all i, then $\mathbf{F}_{\mathrm{FB}}$ is continuously differentiable in a neighborhood of x. □

In Section **9.3** we will see that $\mathcal{D}_a(x) + \mathcal{D}_b(x)JF(x)$ is actually a linear Newton approximation of $\mathbf{F}_{\mathrm{FB}}$ at x. The inclusion (9.1.2) does not provide an effective way to compute a matrix in the generalized Jacobian $\partial\mathbf{F}_{\mathrm{FB}}(x)$. Nevertheless, this inclusion identifies plausible candidates for such a matrix. We will give shortly a simple procedure to calculate elements in the limiting Jacobian of $\mathbf{F}_{\mathrm{FB}}$.

We denote the (diagonal) matrices in the two sets $\mathcal{D}_a(x)$ and $\mathcal{D}_b(x)$ by $D_a(x)$ and $D_b(x)$, respectively. The diagonal elements of these matrices are of the form:

$$a_i(x) \equiv \xi_i - 1 \quad \text{and} \quad b_i(x) \equiv \rho_i - 1, \tag{9.1.3}$$

for some $(\xi_i, \rho_i) \in \mathbb{R}^2$ satisfying $\xi_i^2 + \rho_i^2 \leq 1$. As such, these entries have several obvious properties that we record in the next result. The second part of this result is especially useful in the subsequent analysis.

9.1.6 Proposition. Let $(a_i(x), b_i(x))$ be given by (9.1.3). The following three properties hold:

(a) both $a_i(x)$ and $b_i(x)$ are nonpositive;

(b) $\max(\,|a_i(x)|, |b_i(x)|\,) \leq 2$;

(c) $a_i(x)^2 + b_i(x)^2 \geq 3 - 2\sqrt{2} > 0$.

Let $D_a(x)$ and $D_b(x)$ be any two (diagonal) matrices in $\mathcal{D}_a(x)$ and $\mathcal{D}_b(x)$ respectively. Define

$$v \equiv D_a(x)\mathbf{F}_{\mathrm{FB}}(x) \quad \text{and} \quad z \equiv D_b(x)\mathbf{F}_{\mathrm{FB}}(x).$$

The following four properties are valid for each i:

$$\begin{aligned}
&z_i > 0 \;\Rightarrow\; v_i > 0 \;\Rightarrow\; \psi_{\rm FB}(x_i, F_i(x)) < 0,\\
&z_i = 0 \;\Rightarrow\; v_i = 0 \;\Rightarrow\; \psi_{\rm FB}(x_i, F_i(x)) = 0,\\
&z_i < 0 \;\Rightarrow\; v_i < 0 \;\Rightarrow\; \psi_{\rm FB}(x_i, F_i(x)) > 0,\\
&z_i^2 + v_i^2 \geq (3 - 2\sqrt{2})\psi_{\rm FB}(x_i, F_i(x))^2.
\end{aligned}$$

Proof. Statements (a) and (b) are fairly obvious. By considering the simple optimization problem:

$$\begin{aligned}
&\text{minimize} \quad (\xi - 1)^2 + (\rho - 1)^2\\
&\text{subject to} \quad \xi^2 + \rho^2 \leq 1,
\end{aligned}$$

whose minimum is attained at $\xi = \rho = 1/\sqrt{2}$, we easily deduce the third property of $(a_i(x), b_i(x))$. For each i, we have

$$v_i \equiv a_i(x)\psi_{\rm FB}(x_i, F_i(x)) \quad \text{and} \quad z_i \equiv b_i(x)\psi_{\rm FB}(x_i, F_i(x)).$$

The four properties of the pair (v_i, z_i) follow easily from the three properties of the pair $(a_i(x), b_i(x))$. □

For the subsequent convergence analysis in the absence of a boundedness assumption, the uniform continuity of the gradient of the merit function becomes an essential requirement. To prepare for such an analysis, we provide sufficient conditions for $\nabla\theta_{\rm FB}$ to be uniformly continuous. We first give a definition. We say that a function $G : \mathbb{R}^n \to \mathbb{R}^m$ is *uniformly continuous near a sequence* $\{x^k\} \subset \mathbb{R}^n$ if for every $\varepsilon > 0$ there exists a $\delta > 0$ such that for all k and all y,

$$\|\, x^k - y \,\| \leq \delta \;\Rightarrow\; \|\, G(x^k) - G(y) \,\| \leq \varepsilon.$$

This definition is clearly applicable to a matrix-valued function. In particular, we can speak about the uniform continuity of JG if G is F-differentiable. In general, if G is uniformly continuous on $\mathbb{R}^n$, then G is uniformly continuous near any (bounded or unbounded) sequence of vectors in $\mathbb{R}^n$. In particular, the latter property holds if G is affine.

Following are two technical results pertaining to the uniform continuity property near a sequence. The first result concerns a general vector-valued function; the second result is specific to the FB functional $\psi_{\rm FB}$ and the associated vector function $\mathbf{F}_{\rm FB}$.

9.1.7 Lemma. If $F : \mathbb{R}^n \to \mathbb{R}^n$ is F-differentiable and the Fréchet derivative $JF : \mathbb{R}^n \to \mathbb{R}^{n\times n}$ is uniformly continuous near a sequence $\{x^k\}$ and $\{JF(x^k)\}$ is bounded, then F is uniformly continuous near $\{x^k\}$.

Proof. By the mean value theorem in integral form, we may write

$$\begin{aligned} & F(y) - F(x^k) \\ &= \int_0^1 JF(x^k + t(y - x^k))(\, y - x^k \,)\, dt \\ &= \int_0^1 [\, JF(x^k + t(y - x^k)) - JF(x^k) \,](\, y - x^k \,)\, dt + JF(x^k)(\, y - x^k \,). \end{aligned}$$

Since $\{JF(x^k)\}$ is bounded and JF is uniformly continuous near $\{x^k\}$, the uniform continuity of F near $\{x^k\}$ follows readily. □

9.1.8 Lemma. Let $F : \mathbb{R}^n \to \mathbb{R}^n$ be uniformly continuous near a sequence $\{x^k\} \subset \mathbb{R}^n$. The vector function $\mathbf{F}_{\mathrm{FB}}$ is uniformly continuous near $\{x^k\}$. If in addition $\{\mathbf{F}_{\mathrm{FB}}(x^k)\}$ is bounded then, for every $\varepsilon > 0$, there exists a $\delta > 0$ such that for all k and all matrices $D_a(y)$ in $\mathcal{D}_a(y)$ and $D_a(x^k)$ in $\mathcal{D}_a(x^k)$,

$$\| \, y - x^k \, \| \le \delta \;\Rightarrow\; \| \, [\, D_a(y) - D_a(x^k) \,]\, \mathbf{F}_{\mathrm{FB}}(y) \, \| \le \varepsilon.$$

Proof. Since ψ_{FB} is a globally Lipschitz function on the plane, the uniform continuity of $\mathbf{F}_{\mathrm{FB}}$ near $\{x^k\}$ is obvious. Hence, for every $\varepsilon > 0$, there exists a $\delta > 0$ such that, for all k and i,

$$\| \, y - x^k \, \| \le \delta \;\Rightarrow\; | \, \psi_{\mathrm{FB}}(y_i, F_i(y)) - \psi_{\mathrm{FB}}(x_i^k, F_i(x^k)) \, | \le \frac{\varepsilon}{8}. \tag{9.1.4}$$

For simplicity in the proof, we assume that the sequence $\{\mathbf{F}_{\mathrm{FB}}(x^k)\}$ converges to a limit $\mathbf{F}^\infty$. Consider an arbitrary component $\mathbf{F}_i^\infty$. If $\mathbf{F}_i^\infty = 0$, then for all sufficiently large k,

$$| \, (\, \mathbf{F}_{\mathrm{FB}}(x^k) \,)_i \, | \le \frac{\varepsilon}{8};$$

moreover, we have

$$\begin{aligned} & | \, \psi_{\mathrm{FB}}(y_i, F_i(y)) \, | \\ & \quad \le | \, \psi_{\mathrm{FB}}(x_i^k, F_i(x^k)) \, | + | \, \psi_{\mathrm{FB}}(y_i, F_i(y)) - \psi_{\mathrm{FB}}(x_i^k, F_i(x^k) \, | \,] \\ & \quad \le \varepsilon/4; \end{aligned}$$

hence, by Proposition **9.1.6**(b),

$$| \, [\, a_i(y) - a_i(x^k) \,]\, \psi_{\mathrm{FB}}(y_i, F_i(y)) \, | \le \varepsilon.$$

If $|\mathbf{F}_i^\infty| > 0$ then, by adjusting δ if necessary, we deduce from the uniform continuity of $\mathbf{F}_{\mathrm{FB}}$ near $\{x^k\}$ that for some constant $c > 0$, for all k sufficiently large and all y satisfying $\|y - x^k\| \le \delta$, we have $|\psi_{\mathrm{FB}}(y_i, F_i(y)| \ge c$, which implies by the inequality in Lemma **9.1.3**,

$$\sqrt{y_i^2 + F_i(y)^2} \ge \frac{c}{2 + \sqrt{2}}\,.$$

Hence we deduce

$$a_i(y) - a_i(x^k) \;=\; \frac{y_i}{\sqrt{y_i^2 + F_i(y)^2}} - \frac{x_i^k}{\sqrt{(\,x_i^k\,)^2 + F_i(x^k)^2}}.$$

Since the two denominators in the right-hand side are bounded away from zero and F_i is uniformly continuous near $\{x^k\}$, it follows that $|a_i(y)-a_i(x^k)|$ can be made arbitrarily small uniformly for all k, whenever y is sufficiently close to x^k. Hence the same is true for $|a_i(y) - a_i(x^k)||\psi_{\rm FB}(y_i, F_i(y))|$ because $\psi_{\rm FB}(y_i, F_i(y))$ is bounded by (9.1.4). Since i is arbitrary, the desired conclusion of the lemma follows readily. □

Using the above results, we are ready to establish a uniform continuity property of $\nabla\theta_{\rm FB}$ assuming such a condition on JF.

9.1.9 Proposition. Let $F : \mathbb{R}^n \to \mathbb{R}^n$ be a continuously differentiable function and $\{x^k\}$ be an arbitrary sequence with $\{\theta_{\rm FB}(x^k)\}$ and $\{JF(x^k)\}$ bounded. If JF is uniformly continuous near $\{x^k\}$, then $\nabla\theta_{\rm FB}$ is uniformly continuous near $\{x^k\}$.

Proof. There exist matrices $D_a(y)$ and $D_a(x^k)$ in $\mathcal{D}_a(y)$ and $\mathcal{D}_a(x^k)$ respectively, and matrices $D_b(y)$ and $D_b(x^k)$ in $\mathcal{D}_b(y)$ and $\mathcal{D}_b(x^k)$ respectively such that

$$\begin{aligned}\nabla\theta_{\rm FB}(y) - \nabla\theta_{\rm FB}(x^k) \;=\; &[\,D_a(y)\mathbf{F}_{\rm FB}(y) - D_a(x^k)\mathbf{F}_{\rm FB}(x^k)\,]+\\ &[\,JF(y)^{\,T} D_b(y)\mathbf{F}_{\rm FB}(y) - JF(x^k)^{\,T} D_b(x^k)\mathbf{F}_{\rm FB}(x^k)\,].\end{aligned}$$

The term within the first square bracket in the right-hand side is equal to

$$D_a(x^k)\,[\,\mathbf{F}_{\rm FB}(y) - \mathbf{F}_{\rm FB}(x^k)\,] + [\,D_a(y) - D_a(x^k)\,]\,\mathbf{F}_{\rm FB}(y).$$

By Lemma **9.1.7**, it follows that F is uniformly continuous near $\{x^k\}$; this in turn implies that $\mathbf{F}_{\rm FB}$ is uniformly continuous near $\{x^k\}$. Thus the first summand in the first expression can be made arbitrarily small in norm uniformly for all k whenever y is sufficiently close to x^k; the same conclusion holds for the second summand by Lemma **9.1.8** because the boundedness of $\{\theta_{\rm FB}(x^k)\}$ is equivalent to that of $\{\mathbf{F}_{\rm FB}(x^k)\}$, which is required by the lemma. By a similar argument, it can be shown that the term

$$JF(y)^{\,T} D_b(y)\mathbf{F}_{\rm FB}(y) - JF(x^k)^{\,T} D_b(x^k)\mathbf{F}_{\rm FB}(x^k)$$

can also be made arbitrarily small in norm uniformly for all k whenever y is sufficiently close to x^k. This is enough to establish that $\nabla\theta_{\rm FB}$ is uniformly continuous near $\{x^k\}$. □

The differentiability of $\theta_{\rm FB}$ lies at the heart of many global methods we are going to present. In what follows, we use the FB reformulation of the NCP to illustrate in detail the practical application of all the algorithms described in the previous chapters. In addition to presenting some specific algorithms, we obtain through the discussion many useful insights about the kind of issues we have to tackle in order to be able to apply the general algorithmic theory successfully.

The first algorithm we present is a line search method based on the Linear Newton Method **7.5.14**. In the following description, T is a linear Newton approximation scheme of $\mathbf{F}_{\rm FB}$ on $\mathbb{R}^n$. In the general form presented below, the matrix H^k in Step 3 of the algorithm is not required to be a generalized Jacobian matrix of $\mathbf{F}_{\rm FB}$ at the iterate x^k. The main convergence result of the algorithm, Theorem **9.1.11**, accommodates this generality.

FB Line Search Algorithm (FBLSA)

9.1.10 Algorithm.

Data: $x^0 \in \mathbb{R}^n$, $\rho > 0$, $p > 1$, and $\gamma \in (0, 1)$.

Step 1: Set $k = 0$.

Step 2: If x^k is a stationary point of $\theta_{\rm FB}$ stop.

Step 3: Select an element H^k in $T(x^k)$ and find a solution d^k of the system

$$\mathbf{F}_{\rm FB}(x^k) + H^k d \;=\; 0. \tag{9.1.5}$$

If the system (9.1.5) is not solvable or if the condition

$$\nabla\theta_{\rm FB}(x^k)^{\,T} d^k \;\leq\; -\rho \,\|\, d^k \,\|^p \tag{9.1.6}$$

is not satisfied, (re)set $d^k \equiv -\nabla\theta_{\rm FB}(x^k)$.

Step 4: Find the smallest nonnegative integer i_k such that, with $i = i_k$,

$$\theta_{\rm FB}(x^k + 2^{-i} d^k) \;\leq\; \theta_{\rm FB}(x^k) + \gamma\, 2^{-i}\, \nabla\theta_{\rm FB}(x^k)^{\,T} d^k; \tag{9.1.7}$$

set $\tau_k \equiv 2^{-i_k}$.

Step 5: Set $x^{k+1} \equiv x^k + \tau_k d^k$ and $k \leftarrow k + 1$; go to Step 2.

By the analysis in Section **8.3**, particularly Theorem **8.3.3** and the subsequent discussion, we can establish the following convergence result

for the above algorithm. This result has two parts. The first part pertains to the limit points of a sequence $\{x^k\}$ generated by Algorithm **9.1.10**, if any such point exists. The second part pertains to the case where the sequence $\{x^k\}$ may not have accumulation points. In the latter case, the result relies on a uniform continuity assumption of the Jacobian of F and a reasonable choice of the matrices H^k.

9.1.11 Theorem. Let $F : \mathbb{R}^n \to \mathbb{R}^n$ be continuously differentiable. Let $\{x^k\}$ be an arbitrary sequence produced by Algorithm **9.1.10** with T being a linear Newton approximation scheme of $\mathbf{F}_{\mathrm{FB}}$. The following two statements hold.

(a) Every limit point x^* of $\{x^k\}$ satisfies $\nabla\theta_{\mathrm{FB}}(x^*) = 0$.

(b) If JF is uniformly continuous near a subsequence $\{x^k : k \in \kappa\}$, $\{JF(x^k) : k \in \kappa\}$ is bounded and $H^k \in \mathcal{D}_a(x^k) + \mathcal{D}_b(x^k)JF(x^k)$ for each $k \in \kappa$, then

$$\lim_{k(\in\kappa)\to\infty} \nabla\theta_{\mathrm{FB}}(x^k) \;=\; 0. \tag{9.1.8}$$

Proof. The following inequality holds for all k:

$$\| d^k \| \;\leq\; \max\Big\{ \| \nabla\theta_{\mathrm{FB}}(x^k) \|, \, (\rho^{-1} \| \nabla\theta_{\mathrm{FB}}(x^k) \|)^{1/(p-1)} \Big\}. \tag{9.1.9}$$

This inequality implies that if $\{\nabla\theta_{\mathrm{FB}}(x^k) : k \in \kappa\}$ is bounded, then so is $\{d^k : k \in \kappa\}$.

To prove (a), let $\{x^k : k \in \kappa\}$ be a subsequence converging to x^*. It then follows that the two conditions (BD) and (LS) in Theorem **8.3.3** are satisfied. By this theorem, we have

$$\lim_{k(\in\kappa)\to\infty} \nabla\theta_{\mathrm{FB}}(x^k)^{\,T} d^k \;=\; 0. \tag{9.1.10}$$

Arguing as in Section **8.3**, we can establish that x^* is a stationary point of θ_{FB}.

To prove (b), we claim that the sequence $\{\nabla\theta_{\mathrm{FB}}(x^k) : k \in \kappa\}$ is bounded. Indeed, for every k, there exist matrices $\tilde{D}_a(x^k)$ and $\tilde{D}_b(x^k)$ belonging to $\mathcal{D}_a(x^k)$ and $\mathcal{D}_b(x^k)$, respectively, such that

$$\nabla\theta_{\mathrm{FB}}(x^k) \;=\; (\tilde{D}_a(x^k) + JF(x^k)^{\,T} \tilde{D}_b(x^k) \,) \, \mathbf{F}_{\mathrm{FB}}(x^k).$$

By Proposition **9.1.6**, the sequences $\{\tilde{D}_a(x^k)\}$ and $\{\tilde{D}_b(x^k)\}$ are bounded; so is the sequence $\{JF(x^k) : x^k \in \kappa\}$ by assumption. Moreover, $\{\mathbf{F}_{\mathrm{FB}}(x^k)\}$ is bounded because $\{\theta_{\mathrm{FB}}(x^k)\}$ is bounded. Consequently, our claim is established. It remains to show that $\{\nabla\theta_{\mathrm{FB}}(x^k) : k \in \kappa\}$ converges to zero.

We first verify that condition (LS) in Theorem **8.3.3** holds. By the remark made at the opening of the proof, the sequence $\{d^k : k \in \kappa\}$ is bounded. Let $\{t_k : k \in \kappa\}$ be any sequence of positive scalars converging to zero. We have

$$\begin{aligned}\theta_{\mathrm{FB}}(x^k + t_k d^k) - \theta_{\mathrm{FB}}(x^k) - t_k \nabla\theta_{\mathrm{FB}}(x^k)^T d^k \\ = t_k \left(\nabla\theta_{\mathrm{FB}}(x^k + t'_k d^k) - \nabla\theta_{\mathrm{FB}}(x^k)\right)^T d^k,\end{aligned}$$

for some $t'_k \in (0, t_k)$. By Proposition **9.1.9**, $\nabla\theta_{\mathrm{FB}}$ is uniformly continuous near $\{x^k : k \in \kappa\}$. This uniform continuity and the boundedness of d^k for k in κ imply that

$$\lim_{k(\in\kappa)\to\infty} \frac{\theta_{\mathrm{FB}}(x^k + t_k d^k) - \theta_{\mathrm{FB}}(x^k) - t_k \nabla\theta_{\mathrm{FB}}(x^k)^T d^k}{t_k} = 0;$$

thus (LS) holds. By Theorem **8.3.3**, (9.1.10) holds. If $d^k = -\nabla\theta_{\mathrm{FB}}(x^k)$ for infinitely many k in κ, then the theorem is proved. So suppose that for all but finitely many k in κ, d^k satisfies (9.1.5) and (9.1.6). From (9.1.6) and (9.1.10), we deduce that $\{d^k : k \in \kappa\}$ converges to zero. By assumption, $\{JF(x^k) : k \in \kappa\}$ is bounded, it follows from the choice of H^k that $\{H^k : k \in \kappa\}$ is bounded. Consequently,

$$\lim_{k(\in\kappa)\to\infty} \mathbf{F}_{\mathrm{FB}}(x^k) = 0. \tag{9.1.11}$$

Moreover, since

$$\bigcup_{k\in\kappa} \left(\mathcal{D}_a(x^k) + \mathcal{D}_b(x^k) JF(x^k)\right)$$

is bounded, so is

$$\bigcup_{k\in\kappa} \partial\mathbf{F}_{\mathrm{FB}}(x^k).$$

By Proposition **9.1.4**(c) and (9.1.11), we deduce that $\{\nabla\theta_{\mathrm{FB}}(x^k) : k \in \kappa\}$ converges to zero. □

The two statements (a) and (b) in the above theorem differ in their respective assumptions. Statement (a) is only meaningful when the sequence $\{x^k\}$ has at least one accumulation point; this requires implicitly the boundedness of a subsequence. Statement (b) has no such requirement; instead it assumes the boundedness of the Jacobian (sub)sequence $\{JF(x^k) : k \in \kappa\}$ and a uniform continuity of the Jacobian JF. Thus (b) is a more general result. Statement (a) of the theorem does not assert that x^* is a solution of the NCP (F); this issue is precisely Point 3 mentioned earlier and will be treated fully later. Here, we make a preliminary observation. Namely, if $\{x^k : k \in \kappa\}$ is a subsequence that converges to a point x^* and the corresponding sequence $\{d^k : k \in \kappa\}$ of directions satisfies

(9.1.6) infinitely often, then x^* must be a solution of the NCP (F). The justification of this observation is very similar to the last part of the proof of statement (b). Indeed, by (9.1.9), a subsequence of $\{d^k : k \in \kappa\}$ tends to zero. In turn this implies, by (9.1.5) and the boundedness of $\{H^k : k \in \kappa\}$ because T is upper semicontinuous on $\mathbb{R}^n$, that $\{\mathbf{F}_{\mathrm{FB}}(x^k) : k \in \kappa\}$, which is convergent because $\{x^k : k \in \kappa\}$ converges, goes to zero. Consequently, x^* is a solution of the NCP (F). Put in another way, if a subsequence of iterates generated by Algorithm **9.1.10** converges to a stationary point of θ_{FB} and this point is not a solution of the NCP, then, on this subsequence, only gradient steps are taken, eventually.

Before addressing the boundedness of the sequence $\{x^k\}$ and the question of whether an accumulation point of such a sequence is a solution, we describe a simple procedure to compute a matrix H in $\operatorname{Jac}\mathbf{F}_{\mathrm{FB}}(x)$. In terms of ease of computation and other practical reasons, the procedure yields a good choice for the linear Newton approximation T needed in Algorithm **9.1.10**. Furthermore, the computed element H belongs to $\mathcal{D}_a(x) + \mathcal{D}_b(x)JF(x)$; in particular, the assumption on H^k in Theorem **9.1.11** is fulfilled.

Procedure to calculate an element H in $\operatorname{Jac}\mathbf{F}_{\mathrm{FB}}(x)$

Step 1: Set $\beta \equiv \{i : x_i = 0 = F_i(x)\}$.

Step 2: Choose $z \in \mathbb{R}^n$ such that $z_i \neq 0$ for all i belonging to β.

Step 3: For each $i \notin \beta$ set the i-th column of H^T equal to

$$\left(\frac{x_i}{\sqrt{x_i^2 + F_i(x)^2}} - 1\right) e^i + \left(\frac{F_i(x)}{\sqrt{x_i^2 + F_i(x)^2}} - 1\right) \nabla F_i(x).$$

Step 4: For each $i \in \beta$ set the i-th column of H^T equal to

$$\left(\frac{z_i}{\sqrt{z_i^2 + (\nabla F_i(x)^T z)^2}} - 1\right) e^i + \left(\frac{\nabla F_i(x)^T z}{\sqrt{z_i^2 + (\nabla F_i(x)^T z)^2}} - 1\right) \nabla F_i(x).$$

9.1.12 Proposition. The matrix H calculated by the above procedure is an element of $\operatorname{Jac}\mathbf{F}_{\mathrm{FB}}(x)$.

Proof. It suffices to build a sequence of points $\{y^k\}$ converging to x such that $\mathbf{F}_{\mathrm{FB}}$ is F-differentiable at each y^k and $\{J\mathbf{F}_{\mathrm{FB}}(y^k)\}$ tends to H. Let $y^k \equiv x + \varepsilon_k z$, where z is the vector given in Step 2 of the procedure and $\{\varepsilon_k\}$ is a sequence of positive numbers converging to 0. By definition, if $i \notin \beta$, then either $x_i \neq 0$ or $F_i(x) \neq 0$; moreover $z_i \neq 0$ for all $i \in \beta$. Thus

we can assume, by continuity, that ε_k is small enough so that, for each i, either $y_i^k \neq 0$ or $F_i(y^k) \neq 0$; $\mathbf{F}_{\mathrm{FB}}$ is therefore F-differentiable at y^k.

If i does not belong to β, it is obvious, by continuity, that the i-th row of $J\mathbf{F}_{\mathrm{FB}}(y^k)$ tends to the i-th row of H; so the only case of concern is when i belongs to β. We recall that, according to Proposition **9.1.4**, the i-th row of $J\mathbf{F}_{\mathrm{FB}}(y^k)$ is given by

$$(\, a_i(y^k) - 1\,)\,(\, e^i\,)^T + (\, b_i(y^k) - 1\,)\,\nabla F_i(y^k)^T, \tag{9.1.12}$$

where

$$a_i(y^k) \equiv \frac{\varepsilon_k\, z_i}{\sqrt{\varepsilon_k^2\, z_i^2 + F_i(y^k)^2}} \quad \text{and} \quad b_i(y^k) \equiv \frac{F_i(y^k)}{\sqrt{\varepsilon_k^2\, z_i^2 + F_i(y^k)^2}}.$$

By the Taylor expansion, we can write, for each $i \in \beta$,

$$F_i(y^k) \;=\; F_i(x) + \varepsilon_k \nabla F_i(\zeta^k)^T z \;=\; \varepsilon_k \nabla F_i(\zeta^k)^T z,$$

for some ζ^k on the line segment joining y^k and x. Hence

$$(\, a_i(y^k), b_i(y^k)\,) \;=\; \frac{1}{\sqrt{z_i^2 + (\,\nabla F_i(\zeta^k)^T z\,)^2}}\,(\, z_i, \nabla F_i(\zeta^k)^T z\,). \tag{9.1.13}$$

Clearly $\{\zeta^k\}$ converges to x. Substituting (9.1.13) in (9.1.12), passing to the limit, and taking into account the continuity of JF, we deduce that the rows of $J\mathbf{F}_{\mathrm{FB}}(y^k)$ corresponding to indices in β also tend to the corresponding rows of H defined in Step 4. □

Changing the z in the procedure we may expect to get a different element in $\mathrm{Jac}\,\mathbf{F}_{\mathrm{FB}}(x)$. The simplest choice is $z_i = 0$ if $i \notin \beta$ and $z_i = 1$ if $i \in \beta$. In any event, the computation of an element of $\mathrm{Jac}\,\mathbf{F}_{\mathrm{FB}}(x)$ is trivial, provided that $JF(x)$ is readily available.

9.1.2 Pointwise FB regularity

Having given a concrete example for the linear Newton approximation scheme T we next examine the important issue of when a stationary point of θ_{FB} is a solution of the NCP (F), or equivalently, when such a point is a global minimizer of θ_{FB} with value zero. By Proposition **9.1.4** (c), it follows that if x is stationary point of θ_{FB} and $\partial\mathbf{F}_{\mathrm{FB}}(x)$ contains a nonsingular matrix, then x must satisfy $\mathbf{F}_{\mathrm{FB}}(x) = 0$ and thus is a solution of the NCP (F). The discussion below is therefore most relevant when the generalized Jacobian of $\mathbf{F}_{\mathrm{FB}}$ at a stationary point of θ_{FB} contains only singular matrices; we call such a point a *singular stationary point* of θ_{FB}.

To this end we introduce the important notion of a "FB regular" vector. For simplicity in the following discussion, we sometimes suppress the dependence on x in our notation. For example, we write the gradient of the differentiable function $\theta_{\rm FB}$ as:

$$\begin{aligned}\nabla\theta_{\rm FB}(x) &= D_a(x)\mathbf{F}_{\rm FB}(x) + JF(x)^T D_b(x)\mathbf{F}_{\rm FB}(x)\\ &= D_a\mathbf{F}_{\rm FB} + JF^T D_b\mathbf{F}_{\rm FB},\end{aligned}$$

where we have suppressed the vector x in the last expression. The signs of the components of the vectors $D_a\mathbf{F}_{\rm FB}$ and $D_b\mathbf{F}_{\rm FB}$ play an important role in our analysis. To highlight such a role, we introduce several index sets associated with a vector $x \in \mathbb{R}^n$:

$$\begin{aligned}&\mathcal{C} \equiv \{i:\ x_i \geq 0, F_i(x) \geq 0, x_i F_i(x) = 0\} && (\underline{\rm c}\text{omplementary indices}),\\ &\mathcal{R} \equiv \{1,\ldots n\} \setminus \mathcal{C} && (\underline{\rm r}\text{esidual indices}),\\ &\mathcal{P} \equiv \{i \in \mathcal{R}:\ x_i > 0, F_i(x) > 0\} && (\underline{\rm p}\text{ositive indices}),\\ &\mathcal{N} \equiv \mathcal{R} \setminus \mathcal{P} && (\underline{\rm n}\text{egative indices}).\end{aligned}$$

Notice that $i \in \mathcal{N}$ if and only if either $x_i < 0$ or $F_i(x) < 0$. The above index sets all depend on x, but the notation does not reflect this dependence. However, this should not cause any confusion because the reference vector x will always be clear from the context. Note also that x is a solution of the NCP (F) if and only if $\mathcal{R}$ is empty.

The notation $\mathcal{P}$ and $\mathcal{N}$ of the above index sets is motivated by the following simple relations, which are refinements of the first three properties of the vectors $v \equiv D_a\mathbf{F}_{\rm FB}$ and $z \equiv D_b\mathbf{F}_{\rm FB}$ in Proposition **9.1.6**:

$$\begin{aligned}&(D_a\mathbf{F}_{\rm FB})_i > 0 \ \Leftrightarrow\ (D_b\mathbf{F}_{\rm FB})_i > 0 \ \Leftrightarrow\ i \in \mathcal{P},\\ &(D_a\mathbf{F}_{\rm FB})_i = 0 \ \Leftrightarrow\ (D_b\mathbf{F}_{\rm FB})_i = 0 \ \Leftrightarrow\ i \in \mathcal{C},\\ &(D_a\mathbf{F}_{\rm FB})_i < 0 \ \Leftrightarrow\ (D_b\mathbf{F}_{\rm FB})_i < 0 \ \Leftrightarrow\ i \in \mathcal{N};\end{aligned} \tag{9.1.14}$$

in particular, the signs of the corresponding components of $D_a\mathbf{F}_{\rm FB}$ and $D_b\mathbf{F}_{\rm FB}$ are the same.

The FB regularity of a point x is defined by a property of the Jacobian matrix of F at x. As suggested by its name, this property is very much tailored to the FB function of the NCP.

9.1.13 Definition. A point $x \in \mathbb{R}^n$ is called *FB regular* if for every vector $z \neq 0$ such that (the index sets below are all defined with respect to x)

$$z_{\mathcal{C}} = 0, \quad z_{\mathcal{P}} > 0, \quad z_{\mathcal{N}} < 0, \tag{9.1.15}$$

there exists a nonzero vector $y \in \mathbb{R}^n$ such that

$$y_{\mathcal{C}} = 0, \quad y_{\mathcal{P}} \geq 0, \quad y_{\mathcal{N}} \leq 0,$$

and

$$z^T JF(x)y \geq 0. \tag{9.1.16}$$

The following result relates stationary points of θ_{FB}, FB regular points, and solutions of the NCP (F).

9.1.14 Theorem. Suppose that $F : \mathbb{R}^n \to \mathbb{R}^n$ is continuously differentiable. If $x \in \mathbb{R}^n$ is a stationary point of θ_{FB}, then x is a solution of the NCP (F) if and only if x is an FB regular point of θ_{FB}.

Proof. Assume first that $x \in \mathbb{R}^n$ is a solution of NCP (F). It then follows that x is a global minimum of θ_{FB} and hence a stationary point of θ_{FB}. Moreover, $\mathcal{P} = \mathcal{N} = \emptyset$ in this case; therefore the FB regularity of x holds vacuously since $z = z_{\mathcal{C}}$, and there exists no nonzero vector z satisfying conditions (9.1.15).

Conversely, suppose that x is FB regular and that $\nabla\theta_{\mathrm{FB}}(x) = 0$. The stationary condition can be written as

$$D_a\mathbf{F}_{\mathrm{FB}} + JF(x)^T D_b\mathbf{F}_{\mathrm{FB}} = 0.$$

Consequently, we have, for any $y \in \mathbb{R}^n$,

$$y^T D_a\mathbf{F}_{\mathrm{FB}} + y^T JF(x)^T D_b\mathbf{F}_{\mathrm{FB}} = 0. \tag{9.1.17}$$

Assume that x is not a solution of NCP (F). The set $\mathcal{R}$ is then nonempty and hence, by (9.1.14), $z \equiv D_b\mathbf{F}_{\mathrm{FB}}$ is a nonzero vector with

$$z_{\mathcal{C}} = 0, \qquad z_{\mathcal{P}} > 0, \qquad z_{\mathcal{N}} < 0.$$

Recalling that the components of $D_a\mathbf{F}_{\mathrm{FB}}$ and $z = D_b\mathbf{F}_{\mathrm{FB}}$ have the same signs, and taking y from the definition of the FB regularity of x, we have

$$y^T(D_a\mathbf{F}_{\mathrm{FB}}) = y_{\mathcal{C}}^T(D_a\mathbf{F}_{\mathrm{FB}})_{\mathcal{C}} + y_{\mathcal{P}}^T(D_a\mathbf{F}_{\mathrm{FB}})_{\mathcal{P}} + y_{\mathcal{N}}^T(D_a\mathbf{F}_{\mathrm{FB}})_{\mathcal{N}} > 0 \tag{9.1.18}$$

(since $y_{\mathcal{R}} \neq 0$), and

$$y^T JF(x)^T(D_b\mathbf{F}_{\mathrm{FB}}) = y^T JF(x)^T z \geq 0. \tag{9.1.19}$$

The inequalities (9.1.18) and (9.1.19) together, however, contradict condition (9.1.17). Hence $\mathcal{R} = \emptyset$. This means that x is a solution of NCP (F). □

If the Jacobian of F at x is positive semidefinite, then x is a FB regular point. This can be seen by taking $y = z$ in Definition **9.1.13**. It turns out x is FB regular under a a much broader condition. To motivate this condition, we note that (9.1.16) can be written as

$$\sum_{i \in \mathcal{P}} z_i \, (\, JF(x)y \,)_i + \sum_{i \in \mathcal{N}} z_i \, (\, JF(x)y \,)_i \, \geq \, 0.$$

Taking into account the signs of the components $z_{\mathcal{P}}$ and $z_{\mathcal{N}}$, we see that the above inequality holds if

$$\begin{aligned} \nabla F_i(x)^T y \, &\geq 0, \quad \forall \, i \, \in \, \mathcal{P} \\ \nabla F_i(x)^T y \, &\leq 0, \quad \forall \, i \, \in \, \mathcal{N}. \end{aligned}$$

Consequently, a sufficient condition for x to an FB regular point is that there exists a nonzero vector y satisfying $y_{\mathcal{C}} = 0$, $y_{\mathcal{P}} \geq 0$, $y_{\mathcal{N}} \leq 0$ and the above two inequalities. In turn, this condition is equivalent to the existence of a nonzero vector $(u_{\mathcal{P}}, u_{\mathcal{N}})$ satisfying

$$\left[\begin{array}{cc} J_{\mathcal{P}} F_{\mathcal{P}}(x) & -J_{\mathcal{N}} F_{\mathcal{P}}(x) \\ -J_{\mathcal{P}} F_{\mathcal{N}}(x) & J_{\mathcal{N}} F_{\mathcal{N}}(x) \end{array} \right] \left(\begin{array}{c} u_{\mathcal{P}} \\ u_{\mathcal{N}} \end{array} \right) \geq 0$$

$$u_{\mathcal{P}}, u_{\mathcal{N}} \, \geq \, 0.$$

We recall from Exercise **3.7.29** that a matrix $M \in \mathbb{R}^{n \times m}$ for which there exists a nonzero vector $u \in \mathbb{R}^m_+$ such that $Mu \geq 0$ is called an S_0 matrix. This class of matrices is larger than the class of S matrices introduced at the end of Section **8.3**; that is, every S matrix must be S_0. In terms of this concept, we have therefore shown that a sufficient condition for a vector x to be an FB regular point is that the square matrix:

$$\left[\begin{array}{cc} J_{\mathcal{P}} F_{\mathcal{P}}(x) & -J_{\mathcal{N}} F_{\mathcal{P}}(x) \\ -J_{\mathcal{P}} F_{\mathcal{N}}(x) & J_{\mathcal{N}} F_{\mathcal{N}}(x) \end{array} \right] \tag{9.1.20}$$

is an S_0 matrix. Computationally, the latter sufficient condition has the advantage over the FB regularity condition because the S_0 property of the matrix (9.1.20) can be verified by linear programming, which is a finite procedure; whereas the verification of FB regularity can not be accomplished by a finite procedure in general.

To relate the matrix (9.1.20) to the Jacobian matrix $JF(x)$, let us define the sign matrix $\Lambda(x)$ as the diagonal matrix whose diagonal entries λ_i, $i = 1, \ldots, n$ satisfy:

$$\lambda_i \, \equiv \, \left\{ \begin{array}{rl} 1 & \text{if } i \in \mathcal{P} \\ -1 & \text{if } i \in \mathcal{N} \\ 0 & \text{if } i \in \mathcal{C}. \end{array} \right.$$

It is easy to see that the matrix (9.1.20) is the principal submatrix of the matrix

$$\Xi(x) \equiv \Lambda(x)JF(x)\Lambda(x)$$

with rows and same columns corresponding to the indices in the residual set $\mathcal{R} = \mathcal{P} \cup \mathcal{N}$ of the vector x. This observation leads to the following definition.

9.1.15 Definition. Let $F : \mathbb{R}^n \to \mathbb{R}^n$ be continuously differentiable. We say that the Jacobian matrix $JF(x)$ is a *signed S_0 matrix* if $\Xi(x)_{\mathcal{R}\mathcal{R}}$ is an S_0 matrix. We say that F has the *differentiable signed S_0 property* at x if $JF(x)$ is a signed S_0 matrix. If F has the differentiable signed S_0 property at every point in its domain, then we simply say that F is a signed S_0 function. □

Consequently, the signed S_0 property of $JF(x)$ is a sufficient condition for x to be an FB regular point. Combining this conclusion with Theorem **9.1.14**, we obtain the following important corollary, which provides a sufficient condition that guarantees even the singular stationary points of $\theta_{\rm FB}$ are solutions of the NCP (F).

9.1.16 Corollary. Suppose that $F : \mathbb{R}^n \to \mathbb{R}^n$ is continuously differentiable. If F has the differentiable signed S_0 property at every singular stationary point of $\theta_{\rm FB}$, then every stationary point of the merit function $\theta_{\rm FB}$ is a solution of NCP (F). In this case, every accumulation point of a sequence of iterates produced by Algorithm **9.1.10** solves this NCP.

Proof. If x is a stationary point of $\theta_{\rm FB}$, by Proposition **9.1.4**(c) it follows that

$$H^T \mathbf{F}_{\rm FB}(x) = 0$$

for every $H \in \partial \mathbf{F}_{\rm FB}(x)$. If $\mathbf{F}_{\rm FB}(x)$ is nonzero, then every matrix in $\partial \mathbf{F}_{\rm FB}(x)$ is singular. By assumption, F has the differentiable signed S_0 property at x. This implies that x is an FB regular point; thus by Theorem **9.1.14**, x is a solution of the NCP (F). This is a contradiction. The last assertion of the corollary follows from Theorem **9.1.11**(a). □

The class of S_0 matrices, on which the differentiable signed S_0 property of a function is based, is very broad, as can be seen from Proposition **9.1.17** below. To prove the proposition, we note that by Ville's theorem of the alternatives, a matrix M is not an S_0 matrix if and only if $-M^T$ is an S

matrix; that is, the following equivalences hold:

$$\begin{aligned}&[\,\nexists 0 \neq x \geq 0 \text{ such that } Mx \geq 0\,]\\ &\quad\Leftrightarrow\ [\,\exists y \geq 0 \text{ such that } M^T y < 0\,]\\ &\quad\Leftrightarrow\ [\,\exists y' > 0 \text{ such that } M^T y' < 0\,].\end{aligned}$$

9.1.17 Proposition. Let M be an $n \times n$ matrix. Consider the following statements.

(a) M is positive semidefinite.

(a') M is positive definite.

(b) M is P_0.

(b') M is P.

(c) M is semicopositive.

(c') M is strictly semicopositive

(d) M and all its principal submatrices are S_0.

(d') M and all its principal submatrices are S.

It holds that

$$\text{(a)} \Rightarrow \text{(b)} \Rightarrow \text{(c)} \Leftrightarrow \text{(d)};$$

and

$$\text{(a')} \Rightarrow \text{(b')} \Rightarrow \text{(c')} \Leftrightarrow \text{(d')}.$$

Proof. The implications (a) $\Rightarrow$ (b) $\Rightarrow$ (c) are well-known. So are the parallel implications (a') $\Rightarrow$ (b') $\Rightarrow$ (c'). We show that (c) $\Rightarrow$ (d). Since every principal submatrix of a semicopositive matrix is clearly semicopositive, it suffices to show that every semicopositive matrix is an S_0 matrix. We show this by induction. By an inductive hypothesis, we assume that every semicopositive matrix of order less than n is an S_0 matrix. Let M be a semicopositive matrix of order n. It follows that for every index subset α of $\{1, \ldots, n\}$, the system

$$M_{\alpha\alpha} x_\alpha < 0, \quad x_\alpha \geq 0$$

has no solution because $M_{\alpha\alpha}$ is semicopositive; hence, $-M_{\alpha\alpha}$ is not an S matrix. By the remark made above, it follows that

$$M^T_{\alpha\alpha} \equiv (M^T)_{\alpha\alpha} = (M_{\alpha\alpha})^T$$

is an S_0 matrix. Thus, M^T and all its principal submatrices are S_0 matrices. In particular, there exists a nonzero vector $\bar{x} \geq 0$ satisfying $M^T\bar{x} \geq 0$.

Assume for the sake of contradiction that M is not an S_0 matrix. There exists a vector $u > 0$ such that $M^T u < 0$. Clearly, there exists a scalar $\lambda > 0$ such that $u - \lambda\bar{x}$ is nonnegative with at least one zero component and $M^T(u - \bar{x}) < 0$. This shows that some proper principal submatrix of M is not an S_0 matrix. But this is a contradiction because such a submatrix is semicopositive and thus S_0 by the induction hypothesis. Consequently, (c) implies (d).

The above proof also establishes the following equivalence: M and all its principal submatrices are S_0 matrices if and only if M^T and all its principal submatrices are S_0 matrices. Employing this auxiliary result, we show that (d) implies (c). Suppose that (d) holds but M is not semicopositive. There exists a nonzero vector $x \geq 0$ such that

$$x_i > 0 \;\Rightarrow\; (Mx)_i < 0.$$

Let α be the subset of $\{1, \ldots, n\}$ corresponding to the positive components of x. Thus,

$$x_\alpha > 0, \quad M_{\alpha\alpha}x_\alpha < 0.$$

This implies that $M_{\alpha\alpha}^T$ is not an S_0 matrix. Since (d) holds, the observation made above about M and its transpose implies that $M_{\alpha\alpha}^T$ is an S_0 matrix. This is a contradiction. Finally, the proof of (c') $\Leftrightarrow$ (d') is similar and left as an exercise for the reader. □

By Proposition **3.5.9**, if $F : \mathbb{R}^n \to \mathbb{R}^n$ is a continuously differentiable P_0 function, then $JF(x)$ is a P_0 matrix for all $x \in \mathbb{R}^n$. Hence, all differentiable P_0 functions defined on $\mathbb{R}^n$ are signed S_0 functions; thus so are the monotone functions. Consequently, by Corollary **9.1.16**, for a differentiable P_0 function F, every stationary point of the merit function θ_{FB} is a solution of NCP (F). In what follows, we give an example of a signed S_0 function that is not P_0; thus the class of signed S_0 functions is larger than the class of P_0 functions.

9.1.18 Example. Consider the function:

$$F(x, y) \equiv \begin{pmatrix} -y^2\, e^{-x} \\ -(\min(0, x))^2\, e^{-y} \end{pmatrix}, \quad (x, y) \in \mathbb{R}^2.$$

We show that this function is a signed S_0 function on $\mathbb{R}^2$. Let (x, y) be an arbitrary vector. Note that $F(x, y)$ is nonpositive. Hence the set $\mathcal{P}$ is

empty. It suffices to show that $J_{\mathcal{N}}F_{\mathcal{N}}(x,y)$ is an S_0 matrix. There are three cases to consider.

1. $\mathcal{N} = \{1\}$. Since $\partial F_1(x,y)/\partial x$ is a nonnegative scalar, it follows that $J_{\mathcal{N}}F_{\mathcal{N}}(x,y)$ is an S_0 matrix.

2. $\mathcal{N} = \{2\}$. This is similar to the previous case because $\partial F_2(x,y)/\partial y$ is also nonnegative.

3. $\mathcal{N} = \{1,2\}$. We have

$$JF(x,y) \;=\; \begin{bmatrix} y^2e^{-x} & -2ye^{-x} \\ -2\min(0,x)\,e^{-y} & (\,\min(0,x)\,)^2e^{-y} \end{bmatrix}.$$

If $y \leq 0$, then $JF(x,y)$ is a nonnegative matrix and is thus an S_0 matrix. So suppose $y > 0$. Since 2 is an element of the set $\mathcal{N}$, we must have $x < 0$. Thus the second row of $JF(x,y)$ is positive. Clearly,

$$JF(x,y)\begin{pmatrix} 2/y \\ 1 \end{pmatrix}$$

is a nonnegative vector. Hence $JF(x,y)$ is also an S_0 matrix in this case.

In summary, we have therefore proved that F is a differentiable signed S_0 function on $\mathbb{R}^2$. But F is not a P_0 function because $JF(-\varepsilon,-\varepsilon)$, which has a negative determinant, is not a P_0 matrix. □

9.1.3 Sequential FB regularity

An interesting consequence of Corollary **9.1.16** is that if an NCP with a differentiable signed S_0 function has no solution, then the FB merit function θ_{FB} has no stationary points. The last statement of this corollary is not meaningful when a sequence of iterates produced by Algorithm **9.1.10** has no accumulation point. In this case, under the assumptions in part (b) of Theorem **9.1.11** and by refining the above analysis, we could hope to obtain a limit result of the form:

$$\lim_{k(\in\kappa)\to\infty} \theta_{\mathrm{FB}}(x^k) \;=\; \inf_{x\in\mathbb{R}^n} \theta_{\mathrm{FB}}(x).$$

This shows that the (sub)sequence of iterates $\{x^k : k \in \kappa\}$ is an asymptotically minimizing sequence of θ_{FB} in the sense that the limit of the corresponding sequence of functional values $\{\theta_{\mathrm{FB}}(x^k) : k \in \kappa\}$ is equal to the infimum of $\theta_{\mathrm{FB}}(x)$. In the absence of any accumulation point of the sequence $\{x^k\}$, this kind of an asymptotically minimizing property is the best possible conclusion one can hope for.

In Exercise **6.9.7**, we have formally defined the concept of an asymptotically minimizing sequence and that of an asymptotically stationary sequence and explored the connection between these two concepts. In Exercise **6.9.8**, we have given sufficient conditions under which an asymptotically minimizing sequence of the FB merit function $\theta_{\rm FB}$ must be asymptotically stationary. For our purpose here, we are interested in the reverse implication. It turns out that by extending the pointwise FB regularity to a sequence of vectors, we can establish the sequentially, asymptotically minimizing property of Algorithm **9.1.10**.

Specifically, we say that F is *(sequentially) FB regular* at the sequence $\{x^k\}$ of non-solutions of the NCP (F) if for some triple of mutually disjoint index sets $\mathcal{C}$, $\mathcal{P}$, and $\mathcal{N}$ that partition $\{1, \cdots, n\}$,

$$\begin{aligned} \{\, i : 0 \le x_i^k \perp F_i(x^k) \ge 0 \,\} &= \mathcal{C}, \quad \forall\, k \\ \{\, i : x_i^k > 0 \text{ and } F_i(x^k) > 0 \,\} &= \mathcal{P}, \quad \forall\, k \\ \{\, i : x_i^k < 0 \text{ or } F_i(x^k) < 0 \,\} &= \mathcal{N}, \quad \forall\, k, \end{aligned} \tag{9.1.21}$$

and for any two convergent sequences $\{z^k\}$ and $\{v^k\}$ satisfying the following conditions (9.1.22)–(9.1.24):

$$\left.\begin{array}{llll} z_{\mathcal{C}}^k = 0, & z_{\mathcal{P}}^k > 0, & z_{\mathcal{N}}^k < 0 \\ v_{\mathcal{C}}^k = 0, & v_{\mathcal{P}}^k > 0, & v_{\mathcal{N}}^k < 0 \end{array}\right\} \quad \forall\, k, \tag{9.1.22}$$

$$\limsup_{k\to\infty} [\, z^k \circ (\, JF(x^k)^T z^k \,) \,] \le 0, \tag{9.1.23}$$

and

$$\liminf_{k\to\infty} (\, \| v^k \| + \| z^k \|) > 0, \tag{9.1.24}$$

there exists an index i such that

$$\limsup_{k\to\infty} |\, (\, v^k + JF(x^k)^T z^k \,)_i \,| > 0.$$

We say that F is *asymptotically FB regular* on a set W if F is FB regular at every sequence of non-solutions in W.

The sequential FB regularity yields two important consequences. The first consequence is the following corollary of Theorem **9.1.11**.

9.1.19 Theorem. Let $F : \mathbb{R}^n \to \mathbb{R}^n$ be continuously differentiable. Let $\{x^k\}$ be an arbitrary sequence produced by Algorithm **9.1.10** with T being a linear Newton approximation scheme of $\mathbf{F}_{\rm FB}$. If JF is uniformly continuous near an infinite subsequence $\{x^k : k \in \kappa\}$ at which F is FB regular,

$\{JF(x^k) : k \in \kappa\}$ is bounded and $H^k \in \mathcal{D}_a(x^k) + \mathcal{D}_b(x^k)JF(x^k)$ for each $k \in \kappa$, then

$$\lim_{k(\in\kappa)\to\infty} \theta_{\rm FB}(x^k) \;=\; 0.$$

Proof. By Theorem **9.1.11**, we have

$$\lim_{k(\in\kappa)\to\infty} \nabla\theta_{\rm FB}(x^k) \;=\; 0.$$

Assume for the sake of contradiction and without loss of generality that for some constant $\delta > 0$, $\theta_{\rm FB}(x^k) \geq \delta$ for all $k \in \kappa$ and that for some triple of mutually disjoint index sets $\mathcal{C}$, $\mathcal{P}$, and $\mathcal{N}$ that partition $\{1, \cdots, n\}$, (9.1.21) holds for all $k \in \kappa$. By Proposition **9.1.4**, we can write, for each k,

$$\nabla\theta_{\rm FB}(x^k) \;=\; v^k + JF(x^k)^T z^k,$$

where

$$v^k \;\equiv\; D_a(x^k)\mathbf{F}_{\rm FB}(x^k) \quad \text{and} \quad z^k \;\equiv\; D_b(x^k)\mathbf{F}_{\rm FB}(x^k)$$

for some diagonal matrices $D_a(x^k) \in \mathcal{D}_a(x^k)$ and $D_b(x^k) \in \mathcal{D}_b(x^k)$. The sequences $\{v^k\}$ and $\{z^k\}$ satisfy (9.1.22), by the properties of the latter matrices; see Proposition **9.1.6**. Moreover, by the boundedness of these matrices and the boundedness of the sequence $\{\mathbf{F}_{\rm FB}(x^k)\}$, we may assume that the sequences $\{v^k\}$ and $\{z^k\}$ are convergent. It then follows that

$$z^k \circ JF(x^k)^T z^k \;=\; z^k \circ \nabla\theta_{\rm FB}(x^k) - z^k \circ v^k \;\leq\; z^k \circ \nabla\theta_{\rm FB}(x^k),$$

which implies

$$\limsup_{k(\in\kappa)\to\infty} \left[\, z^k \circ (\, JF(x^k)^T z^k \,) \,\right] \;\leq\; 0.$$

For each $i = 1, \ldots, n$, we have

$$(\, v_i^k \,)^2 + (\, z_i^k \,)^2 \;\geq\; (\, 3 - 2\sqrt{2}\,)\, \psi_{\rm FB}(x_i^k, F_i(x^k))^2.$$

Hence, for all $k \in \kappa$,

$$\| \, v^k \, \|^2 + \| \, z^k \, \|^2 \;\geq\; (\, 3 - 2\sqrt{2}\,) \, \| \, \mathbf{F}_{\rm FB}(x^k) \, \|^2 \;\geq\; 2\,(\, 3 - 2\sqrt{2}\,)\, \delta.$$

This establishes (9.1.24). Consequently, by the sequential FB regularity of F at $\{x^k : k \in \kappa\}$, there exists an index i such that

$$\limsup_{k(\in\kappa)\to\infty} \; |\, (\, v^k + JF(x^k)^T z^k \,)_i \,| \;>\; 0.$$

But this is a contradiction. □

Another consequence of the sequential FB regularity is a global Lipschitzian error bound for the NCP with the FB residual. Since the min residual is equivalent to the FB residual, by Lemma **9.1.3**, we can obtain a similar global error bound with the min residual.

9.1.20 Proposition. Let $F : \mathbb{R}^n \to \mathbb{R}^n$ be a continuously differentiable function such that the NCP (F) has a solution. Suppose there exist a subset W of $\mathbb{R}^n$ and a constant $c > 0$ such that F is asymptotically FB regular on W and

$$\operatorname{dist}(x, S) \leq c^{-1} \| \mathbf{F}_{\mathrm{FB}}(x) \|, \quad \forall x \notin W,$$

where S is the solution set of the NCP (F). There exists a constant $c' > 0$, such that

$$\operatorname{dist}(x, S) \leq c' \| \mathbf{F}_{\mathrm{FB}}(x) \|, \quad \forall x \in \mathbb{R}^n.$$

Proof. The proof is an application of Proposition **6.5.5** to the function $G \equiv \mathbf{F}_{\mathrm{FB}}$ and is very similar to the proof of Theorem **9.1.19**. According to the cited proposition, it suffices to show the existence of a scalar $\delta > 0$ such that for every vector $x \in W$ that is not a solution of the NCP (F),

$$\| D_a(x)\mathbf{F}_{\mathrm{FB}}(x) + JF(x)^T D_b(x)\mathbf{F}_{\mathrm{FB}}(x) \| \geq \delta \| \mathbf{F}_{\mathrm{FB}}(x) \|.$$

Assume for the sake of contradiction that no such δ exists. There exists a sequence of vectors $\{x^k\} \subset W$ such that $\mathbf{F}_{\mathrm{FB}}(x^k) \neq 0$ for every k and

$$\lim_{k\to\infty} \frac{D_a(x^k)\mathbf{F}_{\mathrm{FB}}(x^k) + JF(x^k)^T D_b(x^k)\mathbf{F}_{\mathrm{FB}}(x^k)}{\| \mathbf{F}_{\mathrm{FB}}(x^k) \|} = 0.$$

Without loos of generality, we may assume that the index sets

$$\{ i : 0 \leq x_i^k \perp F_i(x^k) \geq 0 \}$$
$$\{ i : x_i^k > 0 \text{ and } F_i(x^k) > 0 \}$$
$$\{ i : x_i^k < 0 \text{ or } F_i(x^k) < 0 \},$$

are constant sets, which we call $\mathcal{C}$, $\mathcal{P}$, and $\mathcal{N}$, respectively. Let

$$z^k \equiv D_b(x^k) \frac{\mathbf{F}_{\mathrm{FB}}(x^k)}{\| \mathbf{F}_{\mathrm{FB}}(x^k) \|} \quad \text{and} \quad v^k \equiv D_a(x^k) \frac{\mathbf{F}_{\mathrm{FB}}(x^k)}{\| \mathbf{F}_{\mathrm{FB}}(x^k) \|}.$$

We have

$$\lim_{k\to\infty} (v^k + JF(x^k)^T z^k) = 0. \tag{9.1.25}$$

Since $\{D_a(x^k)\}$ and $\{D_b(x^k)\}$ are bounded diagonal matrices, we may assume without loss of generality that $\{z^k\}$ and $\{v^k\}$ are convergent sequences. By Proposition **9.1.6**, we have, for every k,

$$\| z^k \|^2 + \| v^k \|^2 \geq 3 - 2\sqrt{2}.$$

Furthermore,

$$z_{\mathcal{C}}^k = 0, \quad z_{\mathcal{P}}^k > 0, \quad z_{\mathcal{N}}^k < 0,$$

and

$$v^k_{\mathcal{C}} \,=\, 0, \quad v^k_{\mathcal{P}} \,>\, 0, \quad v^k_{\mathcal{N}} \,<\, 0.$$

Since

$$z^k \,\circ\, (\, JF(x^k)^{\,T} z^k \,) \,=\, z^k \,\circ\, (\, v^k + JF(x^k)^{\,T} z^k \,) - z^k \circ v^k,$$

it follows easily from the boundedness of $\{z^k\}$ and (9.1.25) that

$$\limsup_{k\to\infty} \, z^k \circ (\, JF(x^k)^{\,T} z^k \,) \,\leq\, 0.$$

Therefore there exists an index i such that

$$\limsup_{k\to\infty} \, |\, (\, v^k + JF(x^k)^{\,T} z^k \,)_i \,| \,>\, 0.$$

But this contradicts (9.1.25). □

Clearly, if

$$\inf_{x\in\mathbb{R}^n} \max_{z\in \mathrm{bd}\, \mathbb{B}(0,1)} \max_{1\leq i\leq n} z_i (\, JF(x)^{\,T} z \,)_i \,>\, 0,$$

then F has the asymptotic FB regularity property on $\mathbb{R}^n$. In particular, if F is affine and $JF(x)$ is a P matrix, then this property holds. In this case, Proposition **9.1.20** recovers the global error bound for an LCP with a P matrix; cf. Proposition **6.3.1**. In what follows, we present an example of a non-P matrix M and a vector q for which a global error bound holds for the LCP (q, M). This example also illustrates the role of the set W in Proposition **9.1.20**.

9.1.21 Example. Consider the 2×2 matrix

$$M \,\equiv\, \begin{bmatrix} 8/3 & -1 \\ -2 & 3/4 \end{bmatrix}.$$

This is a singular P_0 matrix that is not positive semidefinite. The two rows of M are negative multiples of each other. Let $q \equiv (0, 0)$. Note that

$$\mathrm{SOL}(0, M) \,\equiv\, \{\, (\, x_1, x_2 \,) \,\in\, \mathbb{R}^2_+ \,:\, x_1 \,=\, 3x_2/8 \,\}$$

is an unbounded ray; hence M is not an R_0 matrix. Let W consist of all vectors (x_1, x_2) such that either $x_1 > 0$ or $x_2 > 0$. We first show that if $x \notin W$, then

$$\mathrm{dist}(x, \mathrm{SOL}(0, M)) \,\leq\, \|\, \mathbf{F}_{\mathrm{FB}}(x) \,\|.$$

Write $F(x) \equiv Mx$. If $x \notin W$, then $x_1 \leq 0$ and $x_2 \leq 0$. Hence, for $i = 1, 2$,

$$\begin{aligned} \psi_{\mathrm{FB}}(x_i, F_i(x)) &= \sqrt{x_i^2 + F_i(x)^2} - x_i - F_i(x) \\ &\geq -x_i = |\, x_i \,|. \end{aligned}$$

Consequently,

$$\operatorname{dist}(x, \mathrm{SOL}(0, M)) \leq \| x \| \leq \| \mathbf{F}_{\mathrm{FB}}(x) \|,$$

as claimed. We next verify that F has the asymptotic FB regularity property on W. Let $\{x^k\}$ be a sequence in $W \setminus \mathrm{SOL}(0, M)$. Let $\mathcal{C}$, $\mathcal{P}$, and $\mathcal{N}$ satisfy (9.1.21). We claim that $\mathcal{P} \neq \{1, 2\}$ and $\mathcal{N} \neq \{1, 2\}$. Indeed, if $\mathcal{P} = \{1, 2\}$ then $Mx^k > 0$; but this is impossible because the two rows of M are negative multiples of each other. If $\mathcal{N} = \{1, 2\}$ then, for every k, there are four possibilities:

(a) $x_1^k < 0$ and $F_2(x^k) < 0$;

(b) $x_1^k < 0$ and $x_2^k < 0$;

(c) $F_1(x^k) < 0$ and $x_2^k < 0$;

(d) $F_1(x^k) < 0$ and $F_2(x^k) < 0$.

Case (d) is impossible for the same reason as the case $\mathcal{P} = \{1, 2\}$ is impossible. Case (b) is not possible because $x^k \in W$, which means that either x_1^k or x_2^k is positive. Consider case (a). We must have $x_1^k < 0$ and $x_2^k > 0$, which implies $F_2(x^k) > 0$. Thus (a) is also not possible. Similarly, (c) is not possible. Consequently $\mathcal{N} \neq \{1, 2\}$.

Let $\{z^k\}$ and $\{v^k\}$ be two convergent sequences satisfying the conditions (9.1.22), (9.1.23), and (9.1.24). Let z^∞ and v^∞ be the limits of $\{z^k\}$ and $\{v^k\}$ respectively. We have

$$\begin{gathered} z_{\mathcal{C}}^\infty = 0, \quad z_{\mathcal{P}}^\infty \geq 0, \quad z_{\mathcal{N}}^\infty \leq 0; \\ v_{\mathcal{C}}^\infty = 0, \quad v_{\mathcal{P}}^\infty \geq 0, \quad v_{\mathcal{N}}^\infty \leq 0; \\ z^\infty \circ Mz^\infty \leq 0; \end{gathered}$$

and $(z^\infty, v^\infty) \neq 0$. We need to show that $v^\infty + Mz^\infty$ is nonzero. This is certainly true if $Mz^\infty = 0$. We therefore assume that Mz^∞ is nonzero. Suppose that $(Mz^\infty)_1$ is nonzero. Without loss of generality, assume that $(Mz^\infty)_1$ is positive. It follows that $z_1^\infty \leq 0$. If $1 \in \mathcal{C} \cup \mathcal{P}$, then $v_1^\infty \geq 0$; hence $(v^\infty + Mz^\infty)_1 > 0$. If $1 \in \mathcal{N}$, then $2 \in \mathcal{C} \cup \mathcal{P}$; thus $z_2^\infty \geq 0$. This implies $(Mz^\infty)_1 \leq 0$, which is a contradiction. Thus, F has the asymptotic FB regularity property on W; consequently, the LCP $(0, M)$ admits a global Lipschitzian error bound in terms of the FB function, and hence the min residual. □

9.1.4 Nonsingularity of Newton approximation

For all practical purposes, the linear Newton approximation T in Algorithm **9.1.10** should be chosen to satisfy

$$T(x) \subseteq \mathcal{D}_a(x) + \mathcal{D}_b(x)JF(x)$$

for all vectors x of relevance to the algorithm (see Theorem **9.1.29** (d)). For this reason, we are interested in deriving sufficient conditions for all matrices in the right-hand set to be nonsingular. This kind of analysis serves a dual purpose. On the one hand, the nonsingularity of all elements in $T(x^k)$ ensures the unique solvability of the system (9.1.5), albeit the solution d^k does not necessarily satisfy (9.1.6). On the other hand, the nonsingularity of all elements in $\mathcal{D}_a(x) + \mathcal{D}_b(x)JF(x)$ will ensure the applicability of Theorem **7.5.15**; this in turn is essential to establish a fast convergence rate of Algorithm **9.1.10**.

Having motivated the analysis to follow, we should immediately point out that such an analysis is necessarily different from the kind of stationarity analysis that we just completed. For one thing, to ensure the nonsingularity of all matrices in $\mathcal{D}_a(x)+\mathcal{D}_b(x)JF(x)$, invariably, we have to impose some additional property on the function F, which is not needed in the previous analysis. Nevertheless, the additional functional assumption allows us to establish the locally fast convergence of Algorithm **9.1.10**, among other benefits. For another thing, the main issue in the previous analysis pertains to the singular stationary points of $\theta_{\rm FB}$. By definition, if x is such a point, then a large subset (namely $\partial\mathbf{F}_{\rm FB}(x)$) of $\mathcal{D}_a(x) + \mathcal{D}_b(x)JF(x)$ consists of singular matrices. Indeed, if a sequence produced by Algorithm **9.1.10** converges to a singular solution of the NCP, chances are the convergence rate will not be Q-superlinear. To date, there is no known rigorous study of the latter issue.

The following lemma is the first step toward the ultimate goal of establishing the nonsingularity of all matrices in the set $\mathcal{D}_a(x) + \mathcal{D}_b(x)JF(x)$.

9.1.22 Lemma. Let $M \in \mathbb{R}^{n\times n}$ be a given matrix. The following two statements are equivalent.

(a) M is a P_0 matrix.

(b) Every matrix of the form $D_a + D_bM$ is nonsingular for all nonnegative (nonpositive) diagonal matrices D_a and D_b with $(D_a)_{ii} > (<)0$ for all $i = 1, \ldots, n$.

Proof. (a) $\Rightarrow$ (b). We only consider (b) in which the two matrices D_a and D_b are nonnegative (thus D_a has positive diagonal entries); the other case

is analogous. Let M be a P_0 matrix and

$$D_a \equiv \mathrm{diag}(a_1, \ldots, a_n) \quad \text{and} \quad D_b \equiv \mathrm{diag}(b_1, \ldots, b_n)$$

be such that $a_i > 0$ for all i. Assume that $(D_a + D_b M)q = 0$ for some $q \in \mathbb{R}^n$ so that $D_a q = -D_b M q$; we want to show that $q = 0$. If $b_i = 0$ for some i then $q_i = 0$ because $a_i > 0$. Taking into account the fact that a principal submatrix of a P_0 matrix is still a P_0 matrix, we may assume without loss of generality that D_b is positive definite. Thus we obtain $Mq = -D_b^{-1} D_a q$, which implies

$$q \circ Mq \;=\; -q \circ (\, D_b^{-1} D_a q \,).$$

Since $D_b^{-1} D_a$ is a diagonal matrix with positive diagonals, we conclude that M reverses the sign of q; that is, for all $i = 1, \ldots, n$,

$$q_i \neq 0 \;\Rightarrow\; q_i (\, Mq \,)_i \;<\; 0.$$

By the P_0 property of M, it follows that q must be the zero vector. Thus (b) holds.

(b) $\Rightarrow$ (a). Suppose that M is not a P_0 matrix. We derive a contradiction by assuming (b) outside the parentheses. The proof under the assumption inside the parentheses is similar. There exists a nonzero vector x such that for all $i = 1, \ldots, n$,

$$x_i \neq 0 \;\Rightarrow\; x_i (\, Mx \,)_i \;<\; 0.$$

Let

$$\begin{aligned} \alpha_+ &\equiv \{\, i \,:\, x_i \;>\; 0 \} \\ \alpha_0 &\equiv \{\, i \,:\, x_i \;=\; 0 \} \\ \alpha_- &\equiv \{\, i \,:\, x_i \;<\; 0 \}. \end{aligned}$$

Since x is nonzero, the union $\alpha_+ \cup \alpha_-$ is not the empty set. Let D_a be the identity matrix and define the diagonal entries of the diagonal matrix D_b as follows:

$$(\, D_b \,)_{ii} \equiv \begin{cases} -x_i/(Mx)_i & \text{if } x_i \neq 0 \\ 0 & \text{if } x_i = 0. \end{cases}$$

Since M reverses the sign of the vector x, all diagonal entries of D_b are therefore nonnegative. Clearly we have $(D_a + D_b M)x = 0$. □

We introduce several index sets associated with a given vector $x \in \mathbb{R}^n$. Suppressing the dependence on x, let

$$\alpha \quad \equiv \quad \{\, i \,:\, x_i \;=\; 0 \;<\; F_i(x) \,\},$$

$$\begin{aligned}\beta &\equiv \{\, i : x_i = 0 = F_i(x) \,\},\\ \gamma &\equiv \{\, i : x_i > 0 = F_i(x) \,\},\\ \delta &\equiv \{1, \ldots, n\} \setminus (\, \alpha \cup \beta \cup \gamma \,).\end{aligned}$$

If the point x under consideration is a solution of the NCP (F), then $\delta = \emptyset$ and the sets α, β and γ coincide with the respective sets already introduced in Section **3.3**. In terms of the set $\mathcal{C}$ of complementary indices, we see that the index sets α, β and γ form a partition of $\mathcal{C}$, whereas δ is equal to the set $\mathcal{R}$ of residual indices. (We changed the notation of the latter set for the sake of notational uniformity.) Note that β denotes the set of degenerate indices of x. If $\beta = \emptyset$, then $\mathbf{F}_{\mathrm{FB}}$ is differentiable at x.

The following is the main theorem that provides a sufficient condition for the nonsingularity of the matrices in $\mathcal{D}_a(x) + \mathcal{D}_b(x)JF(x)$.

9.1.23 Theorem. Suppose that $F : \mathbb{R}^n \to \mathbb{R}^n$ is continuously differentiable in a neighborhood of a given vector $x \in \mathbb{R}^n$. Let $M \equiv JF(x)$; also let $\bar{\alpha} \equiv \gamma \cup \beta \cup \delta$ be the complement of α in $\{1, \ldots, n\}$. Assume that

(a) the submatrices $M_{\tilde{\gamma}\tilde{\gamma}}$ are nonsingular for all $\tilde{\gamma}$ satisfying

$$\gamma \subseteq \tilde{\gamma} \subseteq \gamma \cup \beta,$$

(b) the Schur complement of $M_{\gamma\gamma}$ in $M_{\bar{\alpha}\bar{\alpha}}$ is a P_0 matrix.

All matrices in $\mathcal{D}_a(x) + \mathcal{D}_b(x)JF(x)$ are nonsingular. In particular, this is true if $JF(x)_{\bar{\alpha}\bar{\alpha}}$ is a P matrix.

Proof. Let $H \equiv D_a + D_b M$, where D_a and D_b are diagonal matrices with diagonal elements a_i and b_i respectively, which are given by

$$(a_i, b_i) \begin{cases} \equiv \dfrac{(x_i, F_i(x))}{\sqrt{x_i^2 + F_i(x)^2}} - (1,1) & \text{if } (x_i, F_i(x)) \neq 0 \\ \in \operatorname{cl} \mathbb{B}(0,1) - (1,1) & \text{if } (x_i, F_i(x)) = 0. \end{cases}$$

We can partition the index set β into the following three subsets:

$$\begin{aligned}\beta_a &\equiv \{\, i \in \beta : a_i = 0 > b_i \,\},\\ \beta_b &\equiv \{\, i \in \beta : a_i < 0 = b_i \,\},\\ \beta_n &\equiv \{\, i \in \beta : a_i \neq 0 \text{ and } b_i \neq 0 \,\},\end{aligned}$$

and define

$$\tilde{\alpha} \equiv \alpha \cup \beta_b, \quad \tilde{\gamma} \equiv \gamma \cup \beta_a \quad \text{and} \quad \tilde{\delta} \equiv \delta \cup \beta_n.$$

These index sets partition $\{1, \ldots, n\}$. Notice that

$$(\, D_a \,)_{\tilde{\gamma}\tilde{\gamma}} \quad \text{and} \quad (\, D_b \,)_{\tilde{\alpha}\tilde{\alpha}}$$

are both zero matrices; moreover,

$$(\, D_a \,)_{\tilde{\alpha}\tilde{\alpha}} \quad \text{and} \quad (\, D_b \,)_{\tilde{\gamma}\tilde{\gamma}}$$

are nonsingular diagonal matrices; furthermore, it is easy to see that

$$(\, D_a \,)_{\tilde{\delta}} \equiv (\, D_a \,)_{\tilde{\delta}\tilde{\delta}} \quad \text{and} \quad (\, D_b \,)_{\tilde{\delta}} \equiv (\, D_b \,)_{\tilde{\delta}\tilde{\delta}}$$

are both negative definite diagonal matrices. By assumption (a), $M_{\tilde{\gamma}\tilde{\gamma}}$ is nonsingular.

Let $q \in \mathbb{R}^n$ be such that $(D_a + D_b M)q = 0$. We claim that $q = 0$ necessarily, thus establishing the nonsingularity of H. We write q and M in partitioned form according to the three index sets $\tilde{\alpha}$, $\tilde{\gamma}$, and $\tilde{\delta}$,

$$M = \begin{pmatrix} M_{\tilde{\gamma}\tilde{\gamma}} & M_{\tilde{\gamma}\tilde{\alpha}} & M_{\tilde{\gamma}\tilde{\delta}} \\ M_{\tilde{\alpha}\tilde{\gamma}} & M_{\tilde{\alpha}\tilde{\alpha}} & M_{\tilde{\alpha}\tilde{\delta}} \\ M_{\tilde{\delta}\tilde{\gamma}} & M_{\tilde{\delta}\tilde{\alpha}} & M_{\tilde{\delta}\tilde{\delta}} \end{pmatrix} \quad \text{and} \quad q = \begin{pmatrix} q_{\tilde{\gamma}} \\ q_{\tilde{\alpha}} \\ q_{\tilde{\delta}} \end{pmatrix}.$$

The equation $(D_a + D_b M)q = 0$ implies $q_{\tilde{\alpha}} = 0$ and

$$M_{\tilde{\gamma}\tilde{\gamma}} q_{\tilde{\gamma}} + M_{\tilde{\gamma}\tilde{\alpha}} q_{\tilde{\alpha}} + M_{\tilde{\gamma}\tilde{\delta}} q_{\tilde{\delta}} = 0 \tag{9.1.26}$$

$$(\, D_a \,)_{\tilde{\delta}} q_{\tilde{\delta}} + (\, D_b \,)_{\tilde{\delta}} \, (\, M_{\tilde{\delta}\tilde{\gamma}} q_{\tilde{\gamma}} + M_{\tilde{\delta}\tilde{\alpha}} q_{\tilde{\alpha}} + M_{\tilde{\delta}\tilde{\delta}} q_{\tilde{\delta}} \,) = 0. \tag{9.1.27}$$

Since $M_{\tilde{\gamma}\tilde{\gamma}}$ is nonsingular, (9.1.26) implies

$$q_{\tilde{\gamma}} = -(\, M_{\tilde{\gamma},\tilde{\gamma}} \,)^{-1} M_{\tilde{\gamma},\tilde{\delta}} q_{\tilde{\delta}}. \tag{9.1.28}$$

Substituting this into (9.1.27) and rearranging terms, we obtain

$$\left[(\, D_a \,)_{\tilde{\delta}} + (\, D_b \,)_{\tilde{\delta}} \left(M_{\tilde{\delta}\tilde{\delta}} - M_{\tilde{\delta}\tilde{\gamma}} (\, M_{\tilde{\gamma}\tilde{\gamma}} \,)^{-1} M_{\tilde{\gamma}\tilde{\delta}} \right) \right] q_{\tilde{\delta}} = 0. \tag{9.1.29}$$

We claim that the matrix

$$M_{\tilde{\delta}\tilde{\delta}} - M_{\tilde{\delta}\tilde{\gamma}} (\, M_{\tilde{\gamma}\tilde{\gamma}} \,)^{-1} M_{\tilde{\gamma}\tilde{\delta}} \tag{9.1.30}$$

is of class P_0. Thus we need to show that for any subset κ of $\tilde{\delta}$, the matrix

$$M_{\kappa\kappa} - M_{\kappa\tilde{\gamma}} (\, M_{\tilde{\gamma}\tilde{\gamma}} \,)^{-1} M_{\tilde{\gamma}\kappa} \tag{9.1.31}$$

has nonnegative determinant. This matrix is the Schur complement of $M_{\tilde{\gamma}\tilde{\gamma}}$ in the matrix

$$\begin{bmatrix} M_{\kappa\kappa} & M_{\kappa\tilde{\gamma}} \\ M_{\kappa\tilde{\gamma}} & M_{\tilde{\gamma}\tilde{\gamma}} \end{bmatrix}; \tag{9.1.32}$$

hence by the Schur determinantal formula, the determinant of the matrix (9.1.31) is equal to the quotient of the determinant of the matrix (9.1.32) divided by the determinant of $M_{\tilde{\gamma}\tilde{\gamma}}$. Since

$$M_{\tilde{\gamma}\tilde{\gamma}} \;=\; \begin{bmatrix} M_{\beta_a\beta_a} & M_{\beta_a\gamma} \\ M_{\gamma\beta_a} & M_{\gamma\gamma} \end{bmatrix},$$

the determinant of $M_{\tilde{\gamma}\tilde{\gamma}}$ is equal to the product of the determinant of $M_{\gamma\gamma}$ and the determinant of the Schur complement of $M_{\gamma\gamma}$ in the matrix $M_{\tilde{\gamma}\tilde{\gamma}}$. By assumption (b), the latter Schur complement has a nonnegative, thus positive, determinantal sign; hence, it follows that the determinantal sign of $M_{\tilde{\gamma}\tilde{\gamma}}$ is equal to the determinantal sign of $M_{\gamma\gamma}$. Similarly, we can show that the determinant of the matrix (9.1.32), if nonzero, must have the same sign as $\det M_{\gamma\gamma}$. Consequently, it follows that the determinant of the matrix (9.1.31), if nonzero, must be positive. This shows that $M_{\tilde{\delta}\tilde{\delta}}$ is a P_0 matrix.

Since $(D_a)_{\tilde{\delta}}$ and $(D_b)_{\tilde{\delta}}$ are negative definite matrices, by Lemma **9.1.22**, the matrix within the square bracket in equation (9.1.29) is nonsingular. Consequently $q_{\tilde{\delta}} = 0$. This, in turn, implies that $q_{\tilde{\gamma}} = 0$ by (9.1.28). Since $q_{\tilde{\alpha}} = 0$, we therefore have $q = 0$. This establishes the first assertion of the theorem. Since every principal submatrix of a P matrix is nonsingular and the Schur complement of every principal submatrix of a P matrix is a P matrix, the second assertion of the theorem follows readily. □

When x is a solution of the NCP (F) so that the index set δ is empty, condition (a) in Theorem **9.1.23** reduces to the b-regularity of x; hence by the proof of Proposition **5.3.21**, under condition (a) in Theorem **9.1.23**, condition (b) is equivalent to the Schur complement of $M_{\gamma\gamma}$ in $M_{\bar{\alpha}\bar{\alpha}}$ being a P matrix. Consequently, recalling Corollary **5.3.20**, we deduce that if x is strongly b-regular, or equivalently, a strongly stable solution of the NCP (F), then all matrices in $\mathcal{D}_a(x) + \mathcal{D}_b(x)JF(x)$ are nonsingular. See also Proposition **5.3.21**. We summarize this discussion in the corollary below.

9.1.24 Corollary. Suppose that $F : \mathbb{R}^n \to \mathbb{R}^n$ is continuously differentiable in a neighborhood of a given vector $x \in \mathbb{R}^n$. If x is a strongly stable solution of the NCP (F), then all matrices in $\mathcal{D}_a(x) + \mathcal{D}_b(x)JF(x)$ are nonsingular. □

9.1.5 Boundedness of level sets

We next consider the issue of the boundedness of the level sets of the merit function θ_{FB}; that is, the boundedness of sets of the form:

$$\{\, x \in \mathbb{R}^n \,:\, \| \mathbf{F}_{\mathrm{FB}}(x) \| \,\leq\, c \,\}$$

for given constants $c \geq 0$. Since the solutions of the NCP (F) are the global minima with value zero of θ_{FB}, an obvious necessary condition for the level sets of θ_{FB} to be bounded is that the solution set of the NCP be bounded. However, this is not a sufficient condition, as shown by the example below.

9.1.25 Example. Consider the NCP (F), where $F : \mathbb{R} \to \mathbb{R}$ is defined by $F(x) \equiv 1$. The unique solution of this problem is $x^* = 0$ and the solution set is therefore bounded. It is easy to verify that

$$\mathbf{F}_{\mathrm{FB}}(x) \;=\; \sqrt{1 + x^2} - (\,1 + x\,), \quad \forall\, x \,\in\, \mathbb{R},$$

which obviously has unbounded level sets. Indeed, it is not difficult to show that

$$\lim_{x \to \infty} \theta_{\mathrm{FB}}(x) \;=\; 1,$$

which shows that θ_{FB} is not coercive. □

By Lemma **9.1.3**, all level sets of $\theta_{\mathrm{FB}}(x)$ are bounded if and only if the function

$$\theta_{\min}(x) \;\equiv\; \tfrac{1}{2} \,\|\, \min(\, x, F(x)\,) \,\|^2$$

has the same property; this is further equivalent to the function $\theta_{\min}(x)$, or, equivalently, $\theta_{\mathrm{FB}}(x)$, being coercive on $\mathbb{R}^n$; i.e.,

$$\lim_{\|x\| \to \infty} \theta_{\min}(x) \;=\; \infty, \quad \text{or, equivalently,} \quad \lim_{\|x\| \to \infty} \theta_{\mathrm{FB}}(x) \;=\; \infty.$$

If F is an affine function, say $F(x) \equiv q + Mx$ for some vector-matrix pair q and M, this issue has been dealt with in Proposition **2.6.5**. Specializing this proposition to the LCP, we give several sufficient conditions for $\theta_{\min}(x)$ to be coercive on $\mathbb{R}^n$ in terms of some special properties of M.

9.1.26 Proposition. Let $F(x) \equiv q + Mx$ for some vector $q \in \mathbb{R}^n$ and matrix $M \in \mathbb{R}^{n \times n}$. The function $\theta_{\min}(x)$, thus $\theta_{\mathrm{FB}}(x)$, is coercive on $\mathbb{R}^n$ if and only if M is an R_0 matrix. In turn, this is true if any of the following three conditions holds:

(a) M is nondegenerate;

(b) M is strictly semicopositive;

(c) for every nonempty subset α of $\{1, \ldots, n\}$, there exists a vector y_α such that

$$(\, y_\alpha \,)^T M_{\alpha\alpha} \;>\; 0.$$

Proof. By Gordan's theorem of the alternatives, condition (c) is equivalent to the following condition: for every nonempty subset α of $\{1, \ldots, n\}$, the system

$$M_{\alpha\alpha} x_\alpha \;=\; 0, \quad x_\alpha \;\geq\; 0$$

has $x_\alpha = 0$ as the only solution. Based on this observation, both conditions (a) and (b) are clearly special cases of (c). Furthermore, it is easy to see that if (c) holds, then M must be an R_0 matrix. Thus the proposition follows from Proposition **2.6.5**. □

Extending the R_0 property to a nonlinear map F, we obtain the following result, which provides a necessary and sufficient condition for the merit functions $\theta_{\min}(x)$ and $\theta_{\mathrm{FB}}(x)$ to have bounded level sets.

9.1.27 Proposition. Let $F : \mathbb{R}^n \to \mathbb{R}^n$ be a continuous function. The function $\theta_{\min}(x)$, or equivalently $\theta_{\mathrm{FB}}(x)$, is coercive on $\mathbb{R}^n$ if and only if the following implication holds for every infinite sequence $\{x^k\}$,

$$\left.\begin{array}{r} \lim\limits_{k\to\infty} \|\, x^k \,\| \;=\; \infty \\ \limsup\limits_{k\to\infty} \|\, (\, -x^k \,)_+ \,\| \;<\; \infty \\ \limsup\limits_{k\to\infty} \|\, (\, -F(x^k) \,)_+ \,\| \;<\; \infty \end{array}\right\}$$
$$\Rightarrow\; \exists\, i \text{ such that } \limsup_{k\to\infty} \min(\, x_i^k, F_i(x^k) \,) \;=\; \infty.$$

Proof. Suppose that the implication holds for every infinite sequence $\{x^k\}$. We claim that $\theta_{\min}(x)$ is coercive. Assume for the sake of contradiction that there exists a sequence $\{x^k\}$ such that $\{\min(x^k, F(x^k))\}$ is bounded but

$$\lim_{k\to\infty} \|\, x^k \,\| \;=\; \infty.$$

We claim that $\{x^k\}$ satisfies the other two conditions in the left-hand side of the displayed implication. If

$$\limsup_{k\to\infty} \|\, (\, -x^k \,)_+ \,\| \;=\; \infty,$$

then there must exists an index i such that

$$\liminf_{k\to\infty} x_i^k \;=\; -\infty,$$

which implies

$$\liminf_{k\to\infty} \min(\, x_i^k, F_i(x^k) \,) \;=\; -\infty;$$

this is a contradiction. Similarly, we can show that

$$\limsup_{k\to\infty} \| \, (-F(x^k))_+ \, \| \, < \, \infty.$$

Thus by assumption, we deduce

$$\limsup_{k\to\infty} \min(\, x_i^k, F_i(x^k) \,) \, = \, \infty,$$

which is a contradiction. This establishes the "if" statement of the proposition.

To prove the "only if" statement, let $\{x^k\}$ be an infinite sequence satisfying the three left-hand conditions in the displayed implication. The coercivity of $\theta_{\min}(x)$ implies that an index i exists satisfying

$$\limsup_{k\to\infty} \, | \, \min(\, x_i^k, F_i(x^k) \,) \, | \, = \, \infty.$$

But the second and third condition on $\{x^k\}$ imply that we can remove the absolute value in the left-hand min term. Consequently, the desired implication holds. □

Unlike the affine case, it is not easy to give some simple sufficient conditions that imply the coercivity of $\theta_{\mathrm{FB}}(x)$. Nevertheless, we note that if $\min(x, F(x))$ is a global homeomorphism from $\mathbb{R}^n$ onto itself, as in the case when F is a continuous uniformly P function on $\mathbb{R}^n$, then $\min(x, F(x))$ must be norm-coercive on $\mathbb{R}^n$. Thus the following result holds readily. Nevertheless, for the sake of illustrating Proposition **9.1.27**, we present a proof of the corollary by verifying the conditions in this proposition. For a generalization of the corollary, see Exercise **9.5.5**.

9.1.28 Corollary. Let $F : \mathbb{R}^n \to \mathbb{R}^n$ be a continuous uniformly P function. All level sets of the function $\theta_{\mathrm{FB}}(x)$ are bounded.

Proof. Let $\{x^k\}$ be an arbitrary sequence satisfying the three left-hand conditions in the displayed implication in Proposition **9.1.27**. We need to show that the right-hand conclusion holds. Define the index set

$$\alpha \, \equiv \, \{ \, i \, : \, \{x_i^k\} \text{ is unbounded} \, \}.$$

Since $\{x^k\}$ is unbounded, $\alpha \neq \emptyset$. Let $\{z^k\}$ denote a sequence defined in the following way

$$z_i^k \, \equiv \, \begin{cases} 0 & \text{if } i \, \in \, \alpha \\ x_i^k & \text{if } i \, \notin \, \alpha. \end{cases}$$

The sequence $\{z^k\}$ is clearly bounded. Let $\mu > 0$ be a constant associated with the uniformly P property of F. We have

$$\begin{aligned}
\mu \sum_{i\in\alpha} (\, x_i^k \,)^2 &= \mu \,\|\, x^k - z^k \,\|^2 \\
&\leq \max_{i\in\{1,\ldots,n\}} \left(x_i^k - z_i^k \right) \left(F_i(x^k) - F_i(z^k) \right) \\
&= \max_{i\in\alpha} x_i^k \left(F_i(x^k) - F_i(z^k) \right) \qquad (9.1.33) \\
&= x_j^k \left(F_j(x^k) - F_j(z^k) \right) \\
&= |\, x_j^k \,| \, \left| F_j(x^k) - F_j(z^k) \right|,
\end{aligned}$$

where j is one of the indices for which the maximum is attained. Without loss of generality, we may assume that such a maximizing index is independent of k. Since $j \in \alpha$, we can assume, without loss of generality,

$$\lim_{k\to\infty} |\, x_j^k \,| = \infty, \qquad (9.1.34)$$

so that, dividing by $|x_j^k|$ in (9.1.33) gives

$$\mu \,|\, x_j^k \,| \leq \left| F_j(x^k) - F_j(z^k) \right|,$$

which in turn, since $F_j(z^k)$ is bounded, implies

$$\lim_{k\to\infty} \left| F_j(x^k) \right| = \infty. \qquad (9.1.35)$$

The second and third condition on $\{x^k\}$ and the two limits (9.1.34) and (9.1.35) imply

$$\limsup_{k\to\infty} \min(\, x_j^k, F_j(x^k) \,) = \infty,$$

which is the desired limit. □

We have all the elements to give a complete description of the convergence properties of Algorithm **9.1.10**. The proof of these properties follows easily from the general theory developed in the previous chapter and the special properties of the merit function θ_{FB}.

9.1.29 Theorem. Let $F : \mathbb{R}^n \to \mathbb{R}^n$ be continuously differentiable and let T be a linear Newton approximation scheme of $\mathbf{F}_{\mathrm{FB}}$. Let $\{x^k\}$ be an infinite sequence generated by Algorithm **9.1.10**.

(a) Every accumulation point of $\{x^k\}$ is a stationary point of the merit function θ_{FB}.

(b) If x^* is an accumulation point of $\{x^k\}$ such that x^* is FB-regular, then x^* is a solution of the NCP (F).

(c) If $\{x^k\}$ has an isolated limit point, then the whole sequence $\{x^k\}$ converges to that point.

(d) Suppose that x^* is a limit point of $\{x^k\}$ and a solution of NCP (F). Assume that $T(x) \subseteq \partial \mathbf{F}_{\mathrm{FB}}(x)$ for every x in a neighborhood of x^* and that all the matrices belonging to $T(x^*)$ are nonsingular. The whole sequence $\{x^k\}$ converges to x^*; furthermore, if $p > 2$ and $\gamma < 1/2$ in Algorithm **9.1.10**, the following statements hold:

  (i) eventually d^k is always the solution of system (9.1.5);

  (ii) eventually a unit step size is accepted so that $x^{k+1} = x^k + d^k$;

  (iii) the convergence rate is Q-superlinear; furthermore, if the Jacobian $JF(x)$ is Lipschitz continuous in a neighborhood of x^*, the convergence rate is Q-quadratic.

Proof. Statement (a) follows from Theorem **9.1.11**. Statement (b) follows from (a) and Theorem **9.1.14**. Statement (c) follows from Propositions **8.3.10** and **8.3.11** because, with $\sigma(x^k, d^k) \equiv -\nabla\theta_{\mathrm{FB}}(x^k)^T d^k$, we have

$$\| d^k \| \leq \max \left\{ \sqrt{\sigma(x^k, d^k)}, (\rho^{-1} \sigma(x^k, d^k))^{1/p} \right\}.$$

To prove (d), let x^* satisfy the given assumptions. By Lemma **7.5.2** and Theorem **7.2.10**, it follows that x^* is a locally unique solution of NCP (F). Lemma **7.5.2** also implies that eventually the matrices H^k used in the system (9.1.5) are nonsingular and both sequences $\{H^k\}$ and $\{(H^k)^{-1}\}$ are bounded. By the nonsingularity of H^k it follows that eventually the system (9.1.5) has a unique solution d^k. We still need to show that this d^k satisfies (9.1.6). To this end it is sufficient to prove that d^k satisfies, for some positive ρ_1 independent of k, the condition

$$\nabla\theta_{\mathrm{FB}}(x^k)^T d^k \leq -\rho_1 \|d^k\|^2. \tag{9.1.36}$$

Since d^k is the solution of system (9.1.5), we have

$$\| d^k \| \leq c \| \mathbf{F}_{\mathrm{FB}}(x^k) \|,$$

where c is an upper bound on $\|(H^k)^{-1}\|$. This shows that $\{d^k\}$ converges to zero. By assumption, $T(x^k)$ is contained in $\partial \mathbf{F}_{\mathrm{FB}}(x^k)$ for all k sufficiently large; hence $\nabla\theta_{\mathrm{FB}}(x^k)$ is equal to $(H^k)^T \mathbf{F}_{\mathrm{FB}}(x^k)$. We have

$$\nabla\theta_{\mathrm{FB}}(x^k)^T d^k = -\|\mathbf{F}_{\mathrm{FB}}(x^k)\|^2 \leq -\frac{\|d^k\|^2}{c^2}. \tag{9.1.37}$$

Thus (9.1.36) follows from (9.1.37) by taking $\rho_1 \leq 1/c^2$. Since $\{\|d^k\|\}$ converges to zero, (9.1.36) implies that eventually (9.1.6) holds for any $p > 2$ and any positive ρ.

We next show that eventually the step size determined by the Armijo test (9.1.7) is one. By the given assumptions, the sequence $\{d^k\}$ is superlinearly convergent with respect to $\{x^k\}$, by Theorem **7.5.15**. Thus by Proposition **8.3.16**, we have

$$\theta_{\mathrm{FB}}(x^k + d^k) \leq (1 - 2\gamma)\,\theta_{\mathrm{FB}}(x^k)$$

for all k sufficiently large, where γ denotes the constant in the Armijo test (9.1.7). (Note: $\gamma < 1/2$ is needed here.) Therefore we deduce, for all k sufficiently large,

$$\begin{aligned}
\theta_{\mathrm{FB}}(x^k + d^k) &\leq \theta_{\mathrm{FB}}(x^k) - \gamma\, \mathbf{F}_{\mathrm{FB}}(x^k)^T \mathbf{F}_{\mathrm{FB}}(x^k) \\
&= \theta_{\mathrm{FB}}(x^k) + \gamma\, (d^k)^T (H^k)^T \mathbf{F}_{\mathrm{FB}}(x^k) \\
&= \theta_{\mathrm{FB}}(x^k) + \gamma\, (d^k)^T \nabla \theta_{\mathrm{FB}}(x^k),
\end{aligned}$$

where the first equality holds because d^k is the solution of the linear equation (9.1.5) and the second equality holds because $\nabla\theta_{\mathrm{FB}}(x^k)$ is equal to $(H^k)^T \mathbf{F}_{\mathrm{FB}}(x^k)$ as we have noted above. This is enough to establish that eventually a unit step size is attained.

Finally, to complete the proof of the theorem, we need to establish the convergence rates in part (iii) of (d). By Theorem **7.5.15**, we obtain the Q-superlinear convergence readily. For the quadratic rate, we argue as follows. By the Lipschitz property of $JF(x)$ near x^*, Proposition **9.1.4** (d) implies that $\mathbf{F}_{\mathrm{FB}}$ is strongly semismooth near x^*. In turn, Theorem **7.4.3** implies that T is a strong Newton approximation of $\mathbf{F}_{\mathrm{FB}}$ at x^*. The Q-quadratic rate then follows from Theorem **7.5.15** also. □

9.1.30 Remarks. Although we have spent considerable effort investigating the coercivity of θ_{FB}, since the sequence $\{\theta_{\mathrm{FB}}(x^k)\}$ is decreasing, a sufficient condition for $\{x^k\}$ to have an accumulation point is that the following level set

$$\{\, x \in \mathbb{R}^n : \theta_{\mathrm{FB}}(x) \leq \theta_{\mathrm{FB}}(x^0) \,\}$$

is bounded.

In part (d) of Theorem **9.1.29**, which deals with the rate of convergence of Algorithm **9.1.10**, we require $p > 2$ and $\gamma < 1/2$. This is the only part of the theorem where these restrictions are needed. □

As a corollary of the convergence theory developed so far, we give an existence result for an NCP with a P_0 function. It is interesting to point out that although this result is an immediate consequence of the constructive approach presented above, the existence claim of the result does not seem easily provable by the analytic theory of Section **3.5**.

9.1.31 Corollary. Let $F : \mathbb{R}^n \to \mathbb{R}^n$ be a continuous P_0 function on $\mathbb{R}^n$. If $\min(x, F(x))$ is norm-coercive on $\mathbb{R}^n$, then the NCP (F) has a nonempty bounded solution set.

Proof. The norm-coercivity of $\min(x, F(x))$ implies that any sequence produced by Algorithm **9.1.10** is bounded and thus has at least one accumulation point. The P_0 property of F ensures that such a point must be a solution of the NCP (F). The boundedness of the solution set follows readily from the norm-coerciveness of the min function. □

9.1.6 Some modifications

In proving part (d) of Theorem **9.1.29**, we assumed that $T(x)$ is contained in $\partial \mathbf{F}_{\mathrm{FB}}(x)$ for all x in a neighborhood of x^* and in effect all matrices in the generalized Jacobian $\partial \mathbf{F}_{\mathrm{FB}}(x^*)$ are nonsingular. While the latter condition is very natural (in order to obtain the fast convergence rate), the former condition deserves some comments. First of all we underline that the most natural choice we can make for T is the limiting Jacobian $\operatorname{Jac} \mathbf{F}_{\mathrm{FB}}$ of $\mathbf{F}_{\mathrm{FB}}$, because we know how to calculate easily elements of this latter set. From the proof of the theorem, we see that the assumption $T(x) \subseteq \partial \mathbf{F}_{\mathrm{FB}}(x)$ is needed only to ensure the ultimate attainment of a unit step size. Thus if we could ensure such a step by another means, then we would not need $T(x) \subseteq \partial \mathbf{F}_{\mathrm{FB}}(x)$. To remove this requirement, the idea is to perform a further test in Step 3 of Algorithm **9.1.10**, just after the calculation of the search direction. If system (9.1.5) has a solution d^k, we can immediately check whether

$$\| \mathbf{F}_{\mathrm{FB}}(x^k + d^k) \| \leq \gamma' \| \mathbf{F}_{\mathrm{FB}}(x^k) \|,$$

where $\gamma' \in (0, 1)$ is a fixed constant. If this test is passed we set x^{k+1} equal to $x^k + d^k$ and $k \leftarrow k + 1$ and we return to Step 2. Using the same arguments already employed in Section **8.3**, we can show that Algorithm FBLSA modified in this way preserves all the properties described in Theorem **9.1.29** without requiring $T(x) \subseteq \partial \mathbf{F}_{\mathrm{FB}}(x)$. The steps of the modified algorithm are the same as those of the FBLSA except for the search direction calculation in Step 3, which is substituted by Step 3′ as described below.

Variant of the FB Line Search Algorithm (VFBLSA)

Let γ' be a constant in $(0, 1)$. Replace Step 3 in FBLSA by the following.

Step 3′: Select an element H^k in $T(x^k)$ and find a solution of the system

$$\mathbf{F}_{\mathrm{FB}}(x^k) + H^k d \;=\; 0. \tag{9.1.38}$$

If system (9.1.38) is solvable and the solution d^k satisfies

$$\| \mathbf{F}_{\mathrm{FB}}(x^k + d^k) \| \;\leq\; \gamma' \| \mathbf{F}_{\mathrm{FB}}(x^k) \|, \tag{9.1.39}$$

set $\tau_k = 1$ and go to Step 5. Otherwise, if the system (9.1.38) is not solvable or if the condition

$$\nabla\theta_{\mathrm{FB}}(x^k)^T d^k \;\leq\; -\rho \| d^k \|^p$$

is not satisfied, set $d^k = -\nabla\theta_{\mathrm{FB}}(x^k)$.

Note that this is just a variant of the strategy (8.3.23). The idea behind this simple variant is that it may be preferable to check whether the Newton direction is acceptable according to the criterion (9.1.39) before checking the sufficient descent condition (9.1.7)

The key idea of both the FBLSA and its variant, the VFBLSA, is very simple: perform a standard line search along the Newton direction (9.1.5) if this is judged sufficiently good; otherwise, revert to the steepest descent direction of the merit function. This simple strategy is possible because the merit function is differentiable; however, it is by no means the only possible choice. A potential drawback of this strategy is that if gradient steps occur too often, this can slow down convergence.

Alternative globalization strategies can easily be envisaged. One possibility is described below and is based on the idea of a "smooth" transition between the Newton direction and the steepest descent direction. To this end we first note that the latter direction is trivially the solution of the system

$$\nabla\theta_{\mathrm{FB}}(x^k) + d \;=\; 0.$$

Thus, combining this equation with (9.1.38), we can then look for a search direction that is the solution of the following system:

$$(\, m\lambda \nabla\theta_{\mathrm{FB}}(x^k) + \mathbf{F}_{\mathrm{FB}}(x^k) \,) + (\, m\lambda \mathbf{I} + H^k \,) d \;=\; 0, \tag{9.1.40}$$

where λ is a positive constant and m is a nonnegative integer. When $m = 0$

the system (9.1.40) reduces to the Newton system (9.1.38); when m tends to infinity the solution of (9.1.40) approaches the steepest descent direction of $\theta_{\rm FB}$ at x^k. (This can be justified by a rigorous proof.) The solution of the system (9.1.40) for increasing values of m can then be seen as an attempt to combine in a weighted way the Newton direction and the steepest descent direction. This consideration leads to the following natural modification of Algorithm **9.1.10** where all steps remain the same except for Step 3.

Modified FB Line Search Algorithm (MFBLSA)

Let λ and c_1 and c_2 be positive constants with both c_1 and c_2 less than one. Substitute Step 3 in FBLSA by the following.

Step 3″: Select an element H^k in $T(x^k)$ and compute the smallest nonnegative integer m such that system (9.1.40) admits a solution d^k that satisfies the following condition:

$$\nabla\theta_{\rm FB}(x^k)^T d^k \;\le\; -c_1 \min[\,\|\nabla\theta_{\rm FB}(x^k)\|^2, \|\nabla\theta_{\rm FB}(x^k)\|^3\,] \tag{9.1.41}$$

and

$$\nabla\theta_{\rm FB}(x^k)^T d^k \;\le\; -c_2 \min[\,\|d^k\|^2, \|d^k\|^3\,]. \tag{9.1.42}$$

At first sight, the two conditions (9.1.41) and (9.1.42) seem somewhat unusual. It turns out that these conditions are key to establishing the locally fast convergence of the MFBLSA. Before addressing the convergence of this modified algorithm, we first state and prove a lemma that establishes the existence of a nonnegative integer m satisfying the three conditions (9.1.40), (9.1.41) and (9.1.42), with m depending on x^k.

9.1.32 Lemma. Let positive scalars λ, c_1 and c_2 be given with c_1 and c_2 both less than one. Let x^k be a given non-stationary point of $\theta_{\rm FB}$. For every compact set $T(x^k) \subset \mathbb{R}^{n\times n}$, there exists a positive integer m_k such that for every integer $m \ge m_k$ and every matrix $H^k \in T(x^k)$, a solution d^k to (9.1.40) exists that satisfies (9.1.41) and (9.1.42).

Proof. To simplify the notation, we drop the counter k in the following proof. Since $T(x)$ is a compact set of matrices, it follows that there must exist a positive integer $\bar{m}$ such that for every integer $m \ge \bar{m}$, the matrix $m\lambda\mathbf{I} + H$ is nonsingular for every H belonging to $T(x)$ and

$$m\,\lambda\,\nabla\theta_{\rm FB}(x) + \mathbf{F}_{\rm FB}(x) \;\neq\; 0.$$

The latter property implies that d is nonzero. The equation (9.1.40) yields

$$\left(1 - \frac{H}{m\lambda}\right) \| d \| \leq \| \mathbf{F}_{\mathrm{FB}}(x) \| + \frac{\| \nabla\theta_{\mathrm{FB}}(x) \|}{m\lambda}.$$

Let c_3 be a positive scalar satisfying

$$c_3 \geq \max\left\{ \| \nabla\theta_{\mathrm{FB}(x)} \|, \| \mathbf{F}_{\mathrm{FB}}(x) \|, \max_{H \in T(x)} \|H\| \right\},$$

we obtain

$$\| d \| \leq 6c_3,$$

for every $m \geq m_1 \equiv \max(\bar{m}, 1/\lambda, 2c_3/\lambda)$, every $H \in T(x)$ and corresponding solution d. The equation (9.1.40) also implies that

$$d = -\nabla\theta_{\mathrm{FB}}(x) - \frac{1}{m\lambda}\left(Hd + \mathbf{F}_{\mathrm{FB}}(x)\right).$$

Premultiplying this equation by $\nabla\theta_{\mathrm{FB}}(x)^T$ and by taking into account the bound on $\|d\|$, we get

$$\nabla\theta_{\mathrm{FB}}(x)^T d \leq -\| \nabla\theta_{\mathrm{FB}}(x) \|^2 + \frac{36c_3^3 + 6c_3^2}{m\lambda}.$$

Thus given the constants λ and c_1, we can clearly find an integer $m_2 \geq m_1$ such that for every integer $m \geq m_2$ and every matrix $H \in T(x)$, the solution d of the equation (9.1.40) also satisfies

$$\nabla\theta_{\mathrm{FB}}(x)^T d \leq -c_1 \| \nabla\theta_{\mathrm{FB}}(x) \|^2.$$

This inequality clearly implies (9.1.41). Similarly, we can also deduce that

$$-\| d \|^2 = \nabla\theta_{\mathrm{FB}}(x)^T d + \frac{d^T H d + d^T \mathbf{F}_{\mathrm{FB}}(x)}{m\lambda},$$

which implies, in view of the boundedness of $\|H\|$ and $\|d\|$, that for all m sufficiently large, we have

$$\nabla\theta_{\mathrm{FB}}(x)^T d \leq -c_2 \| d \|^2,$$

which yields (9.1.42) readily. □

Among other things, Lemma **9.1.32** implies that d^k is a descent direction of θ_{FB} at x^k. Thus the MFBSLA generates a well-defined sequence $\{x^k\}$. The following theorem asserts that such a sequence possesses the same properties as the one produced by the basic FBLSA. The proof of the theorem reveals the important role played by the two tests (9.1.41) and (9.1.42).

9.1.33 Theorem. Let $F : \mathbb{R}^n \to \mathbb{R}^n$ be continuously differentiable and let T be a linear Newton approximation scheme of $\mathbf{F}_{\mathrm{FB}}$. Statements (a)–(d) in Theorem **9.1.29** remain valid for any infinite sequence $\{x^k\}$ of iterates generated by the MFBLSA.

Proof. The proof of statements (a)–(c) is identical to that of Theorem **9.1.29**. So we prove only statement (d). To this end, observe that in a neighborhood of the solution x^* within which T provides a nonsingular Newton approximation of $\mathbf{F}_{\mathrm{FB}}$, the system (9.1.40) is certainly solvable for $m = 0$. Moreover, with d^k denoting the solution of (9.1.40) corresponding to $m = 0$, i.e., $\mathbf{F}_{\mathrm{FB}}(x^k) + H^k d^k = 0$, arguing as in the proof of Theorem **9.1.29**, we can deduce that for all k sufficiently large,

$$\nabla\theta_{\mathrm{FB}}(x^k)^T d^k \leq -c_1 \| \nabla\theta_{\mathrm{FB}}(x^k) \|^3$$

and

$$\nabla\theta_{\mathrm{FB}}(x^k)^T d^k \leq -c_2 \| d^k \|^3$$

Clearly these two inequalities imply (9.1.41) and (9.1.42), respectively. Consequently $m = 0$ is accepted eventually and thus the modified FBLSA reduces to the basic FBSLA with $p = 3$ eventually. □

9.1.34 Remark. Similar to the variant of the FBLSA, one can easily design a variant of the MFBLSA whereby the requirement that $T(x)$ be contained in $\partial\mathbf{F}_{\mathrm{FB}}(x)$ for all x in a neighborhood of a solution x^* is no longer needed and yet the superlinear convergence of the resulting algorithm is preserved. The details are not repeated. □

The advantage of using Step 3″ to calculate the search direction is that we try to make use of the Newton direction as much as possible by not completely switching to the steepest descent direction. The price we have to pay is that the calculation of the search direction becomes more involved and may require the solution of several systems of linear equations. However these equations are strongly related because at each iteration k, the defining matrices in two consecutive trials (from m to $m+1$) differ only by the positive multiple λ of the identity matrix. This special feature can be profitably exploited to greatly reduce the cost of solving these equations in practical implementation.

At this stage it should be clear that all the algorithms considered so far follow a similar pattern. We consider a locally convergent Newton algorithm for the solution of the system $\mathbf{F}_{\mathrm{FB}}(x) = 0$ and then, if necessary, we modify the corresponding direction so that it satisfies the conditions for the

global convergence of the line search algorithms described in Section **8.3**. In the case of the FB reformulation of complementarity problems this is a very easy task because the merit function θ_{FB} is continuously differentiable. In order to conciliate the global convergence and a fast local convergence rate, we then have to show that locally the algorithm reduces to the basic Newton method we started with. Again, under natural assumptions, showing that this actually happens is easy because of the continuous differentiability of the merit function. The local algorithm on which FBLSA is based is the NNM described in Section **7.2**; however there is no difficulty in considering different local algorithms. The most obvious choice, especially effective for large problems, is the inexact version of NNM, the INNM described in Section **7.2**. The resulting algorithm is presented below.

Inexact FB Line Search Algorithm (IFBLSA)

The steps of the algorithm are the same of those of the FBLSA except for the search direction calculation in Step 3, which is substituted by the following one. Let $\{\eta_k\}$ be a sequence of nonnegative scalars.

Step 3‴: Select an element H^k in $T(x^k)$ and calculate a vector d^k such that

$$\mathbf{F}_{\mathrm{FB}}(x^k) + H^k d \;=\; r^k \tag{9.1.43}$$

where r^k is a vector satisfying

$$\| r^k \| \;\leq\; \eta_k \| \mathbf{F}_{\mathrm{FB}}(x^k) \|. \tag{9.1.44}$$

If the condition

$$\nabla\theta_{\mathrm{FB}}(x^k)^T d^k \;\leq\; -\rho \| d^k \|^p$$

is not satisfied, set $d^k = -\nabla\theta_{\mathrm{FB}}(x^k)$.

An alternative is the Levenberg-Marquardt version of the algorithm, along the line presented in Section **7.5**.

Inexact LM FB Line Search Algorithm (ILMFBLSA)

Let $\{\eta_k\}$ and $\{\sigma_k\}$ be two sequences of nonnegative scalars. The steps of the algorithm are the same of those of the IFBLSA except for the equation

(9.1.43) and the inexact rule (9.1.44), which are revised as follows:

$$(\,(\,H^k\,)^T H^k + \sigma_k \mathbf{I}\,)d \;=\; -(\,H^k\,)^T \mathbf{F}_{\mathrm{FB}}(x^k) + r^k, \tag{9.1.45}$$

where $r^k \in \mathbb{R}^n$ is a vector satisfying

$$\|\, r^k \,\| \leq \eta_k \|\,(\,H^k\,)^T \mathbf{F}_{\mathrm{FB}}(x^k)\,\|, \tag{9.1.46}$$

The convergence properties of the above two algorithms can easily be derived from Theorems **7.2.8** and **7.5.11**, respectively. In essence, under the assumptions of Theorem **9.1.29**, the resulting sequence $\{x^k\}$ produced by the IFBLSA enjoys all the limiting properties as the sequence produced by the basic FBLSA, provided that the sequence of scalars $\{\eta_k\}$ satisfies the requirements in Theorem **7.2.8**. Specifically, if T is a linear Newton approximation scheme of $\mathbf{F}_{\mathrm{FB}}$, and if $\bar{\eta} \geq 0$ exists such that $\eta_k \leq \bar{\eta}$ for every k, then statements (a)–(c) of Theorem **9.1.29** hold for $\{x^k\}$ generated by the IFBLSA. Moreover, if the sequence $\{\eta_k\}$ tends to zero, then with the exception of the quadratic rate of convergence, statement (d) and its subparts remain valid for the inexact sequence of iterates. Finally, if $\tilde{\eta}$ exists such that $\eta_k \leq \tilde{\eta}\|\mathbf{F}_{\mathrm{FB}}(x^k)\|$ for all k, then the Q-quadratic rate also holds. Similar conclusions are valid for the sequence produced by the ILMFBLSA, provided that in addition the sequence of scalars $\{\sigma_k\}$ satisfies similar assumptions as described in Theorem **7.5.11**. The details are omitted.

Obviously, further refinements of these inexact algorithms are possible. For instance, we can apply the variant of the basic FBLSA that led to the VFBLSA, thereby removing the restriction $T(x) \subseteq \partial \mathbf{F}_{\mathrm{FB}}(x)$. Moreover, we could also consider inexact versions of the MFBLSA. All these refinements are fairly straightforward and do not present any technical difficulty in analysis.

9.1.7 A trust region approach

Up to now we have considered in detail algorithms based on the line search framework. However, we can obviously study algorithms based on the other general paradigm considered in Chapter **8**, namely that of a trust region method. In what follows, we consider a basic version of the trust region algorithm that is a straightforward adaptation of Algorithm **8.4.1** applied to minimize the merit function $\theta_{\mathrm{FB}}(x)$ with no restriction on the variable x; thus $X = \mathbb{R}^n$. The reader can easily derive many variants of this scheme by following the development in the line search approach.

Given x^k, a major step of the trust region algorithm is to compute an approximate optimal solution of the convex program in the variable d:

$$\begin{array}{ll} \text{minimize} & \nabla\theta_{\rm FB}(x^k)^T d + \frac{1}{2}\, d^T (H^k)^T H^k d \\ \text{subject to} & \| d \| \leq \Delta, \end{array} \tag{9.1.47}$$

where Δ is a positive scalar and $H^k \in T(x^k)$ with T being a linear Newton approximation of $\mathbf{F}_{\rm FB}$. In the notation of Section **8.4**, the above problem is $\mathrm{TR}(x^k, \tilde{H}^k, \Delta)$, where $\tilde{H}^k \equiv (H^k)^T H^k$. Consistent with the previous notation, let $\sigma(x^k, \tilde{H}^k, \Delta)$ denote the negative of the optimal objective value of this subprogram. Furthermore, let $\sigma(x^k, d)$ denote the negative of the objective function of (9.1.47) evaluated at any vector d.

FB Trust Region Algorithm (FBTRA)

9.1.35 Algorithm.

Data: $x^0 \in \mathbb{R}^n$, $0 < \gamma_1 < \gamma_2 < 1$, $\Delta_0 > 0$, $\Delta_{\min} > 0$, and $\rho \in (0, 1]$.

Step 1: Set $k = 0$.

Step 2: Select an element H^k in $T(x^k)$ and compute a feasible solution d^k of the trust region subproblem $\mathrm{TR}(x^k, \tilde{H}^k, \Delta_k)$ such that

$$\sigma(x^k, d^k) \;\geq\; \rho\, \sigma(x^k, \tilde{H}^k, \Delta_k). \tag{9.1.48}$$

Step 3: If $\sigma(x^k, d^k) = 0$ stop.

Step 4: If

$$\theta_{\rm FB}(x^k + d^k) - \theta_{\rm FB}(x^k) \;\leq\; -\gamma_1\, \sigma(x^k, d^k), \tag{9.1.49}$$

then set $x^{k+1} \equiv x^k + d^k$, and

$$\Delta_{k+1} \equiv \begin{cases} \max(2\Delta_k, \Delta_{\min}) & \text{if } \theta_{\rm FB}(x^k + d^k) - \theta_{\rm FB}(x^k) \leq -\gamma_2 \sigma(x^k, d^k) \\ \max(\Delta_k, \Delta_{\min}) & \text{otherwise.} \end{cases}$$

Set $k \leftarrow k + 1$ and go to Step 2. Otherwise, if

$$\theta_{\rm FB}(x^k + d^k) - \theta_{\rm FB}(x^k) \;>\; -\gamma_1\, \sigma(x^k, d^k),$$

set $x^{k+1} = x^k$, $\Delta_{k+1} = \frac{1}{2}\Delta_k$, and $k \leftarrow k + 1$; go to Step 2.

The objective function of (9.1.47) is quadratic and convex and actually strictly convex if H^k is nonsingular. Since the feasible region of (9.1.47) is obviously bounded, this problem has at least one optimal solution. With the norm on d being the Euclidean norm, (9.1.47) is a convex quadratic minimization problem with one single convex quadratic constraint. As such, there are numerically efficient ways for solving such a problem; we refer the interested reader to Section **9.6** for bibliographical references.

Algorithm **9.1.35** enjoys properties similar to those of its line search counterpart, the FBLSA. Based on the analysis of Section **8.4** and the continuous differentiability of θ_{FB}, we can easily establish these properties, which we summarize in the following theorem.

9.1.36 Theorem. Let $F : \mathbb{R}^n \to \mathbb{R}^n$ be continuously differentiable and let T be a linear Newton approximation scheme of $\mathbf{F}_{\mathrm{FB}}$. Let $\{x^k\}$ be an infinite sequence of iterates generated by Algorithm **9.1.35**.

(a) Every accumulation point of $\{x^k\}$ is a stationary point of the merit function θ_{FB}. If such a point is FB regular then it is a solution of NCP (F);

(b) Suppose that x^* is a limit point of $\{x^k\}$ and a solution of NCP (F). Assume further that T is nonsingular at x^* that $T(x) \subseteq \partial \mathbf{F}_{\mathrm{FB}}(x)$ for every x in a neighborhood of x^* and that $\rho = 1$. The whole sequence $\{x^k\}$ converges to x^* and

  (i) eventually the vector d^k coincides with the unique unconstrained global minimizer of $-\sigma(x^k, \cdot)$;

  (ii) eventually the test (9.1.49) is always successful;

  (iii) the convergence rate is Q-superlinear; furthermore, if the linear approximation T is strong at x^*, the convergence rate is Q-quadratic.

Proof. With $a(x^k, d) \equiv \nabla \theta_{\mathrm{FB}}(x^k)^T d$, it is easy to check that the assumptions TR1a, TR1b, TR1c and TR2 in Section **8.4** are all satisfied. Thus the first assertion in (a) follows from Theorem **8.4.4**. The second assertion is clear.

To prove (b), let $\{x^k : k \in \kappa\}$ be an arbitrary subsequence of $\{x^k\}$ converges to x^*. We show that

$$d^k = -(H^k)^{-1}\mathbf{F}_{\mathrm{FB}}(x^k), \qquad \forall k \in \kappa \text{ sufficiently large.} \tag{9.1.50}$$

In fact, for every such k, H^k is nonsingular by the assumptions made on T. Therefore the objective function of (9.1.47) is strictly convex and its

unconstrained minimizer is

$$d_u^k \equiv -(H^k)^{-1}\mathbf{F}_{\rm FB}(x^k).$$

Since $\mathbf{F}_{\rm FB}(x^*) = 0$, we have

$$\lim_{k(\in\kappa)\to\infty} (H^k)^{-1}\mathbf{F}_{\rm FB}(x^k) = 0.$$

Consequently, for all $k \in \kappa$ sufficiently large,

$$\| d_u^k \| = \| (H^k)^{-1}\mathbf{F}_{\rm FB}(x^k) \| \leq \Delta_{\min}.$$

Since at the beginning of a series of inner iterations $\Delta_k \geq \Delta_{\min}$, in order to show that (9.1.50) holds we only need to show that (9.1.49) is satisfied by d_u^k. Recalling that $\nabla\theta_{\rm FB}(x^k) = (H^k)^T\mathbf{F}_{\rm FB}(x^k)$, we have

$$\sigma(x^k, d_u^k) = \theta_{\rm FB}(x^k).$$

The test (9.1.49) then becomes

$$\theta_{\rm FB}(x^k + d_u^k) \leq (1 - \gamma_1)\,\theta_{\rm FB}(x^k),$$

which must holds eventually, because $\{d_u^k : k \in \kappa\}$ is a superlinearly convergent sequence of directions with respect to $\{x^k : k \in \kappa\}$. Since x^* is an isolated solution by Theorem **7.2.10**, the convergence of the whole sequence $\{x^k\}$ to x^* follows from Proposition **8.3.10**. The rest of the proof is by now standard. □

We briefly discuss a "subspace minimization" method that has no parallel in the line search algorithms considered previously. This method is nothing else than a specific way of calculating the approximate solution d^k of the trust region subproblem $\mathrm{TR}(x^k, \tilde{H}^k, \Delta_k)$ required in Step 2 of Algorithm **9.1.35**. The basic idea is that of computing an *approximate* solution to this subproblem by computing an *exact* solution to a lower dimensional trust region subproblem. To be more precise, assume for the time being that H^k is nonsingular. Consider the minimization problem in the vector d and scalars c_1 and c_2:

$$\begin{aligned} \text{minimize} \quad & \nabla\theta_{\rm FB}(x^k)^T d + \tfrac{1}{2}\, d^T (H^k)^T H^k d \\ \text{subject to} \quad & \| d \| \leq \Delta_k, \\ & d = c_1\,(-\nabla\theta_{\rm FB}(x^k)) + c_2\,(-(H^k)^{-1}\nabla\theta_{\rm FB}(x^k)). \end{aligned} \tag{9.1.51}$$

This problem is just the minimization of the usual objective function, with the standard trust region constraint, but with d restricted to lie in the

subspace determined by the negative of the gradient of θ_{FB} at x^k and by the Newton direction $-(H^k)^{-1}\nabla\theta_{\mathrm{FB}}(x^k)$. By substituting in the objective function and in the trust region constraint the expression of d defined by the equality constraint, we see that this problem is equivalent to a two-dimensional trust region subproblem, the variables being the two scalars c_1 and c_2.

With $a(x^k, d^k) \equiv \nabla\theta_{\mathrm{FB}}(x^k)^T d^k$, the exact solution of (9.1.51) satisfies (8.4.10) with $\rho = 1$, because the Cauchy point of $\mathrm{TR}(x^k, \tilde{H}^k, \Delta_k)$ belongs to the feasible region of the current subproblem. Naturally we also hope that the presence of the Newton direction will induce a fast local convergence rate under some assumptions. It is important to stress that the calculation of d^k consists of the solution of a system of linear equations (to obtain the Newton direction) and the solution of a *two dimensional* trust region subproblem; thus the cost of calculating d^k is usually much lower than that of solving the full dimensional trust region problem $\mathrm{TR}(x^k, \tilde{H}^k, \Delta_k)$.

Having motivated the approach, we formally introduce a more realistic direction generation routine that does not require H^k to be invertible.

Subspace FB Trust Region Algorithm (SFBTRA)

The steps of the algorithm are the same of those of FBTRA except that Step 2 is substituted by the following one.

Step 2′: Select a subspace V^k containing $-\nabla\theta_{\mathrm{FB}}(x^k)$ and compute an exact solution d^k of the trust region subproblem $\mathrm{TR}(\Delta_k, V^k)$ defined by

$$\begin{array}{ll} \text{minimize} & \nabla\theta_{\mathrm{FB}}(x^k)^T d + \frac{1}{2}\, d^T (H^k)^T H^k d \\ \text{subject to} & \|\, d \,\| \le \Delta_k, \\ \text{and} & d \in V^k. \end{array} \tag{9.1.52}$$

In contrast to the ususal trust region method, the present trust region subproblem (9.1.52) includes the additional requirement that d^k belongs to an appropriate subspace V^k. If the dimension of this subspace is p and a basis is known for this subspace, subproblem (9.1.52) is then equivalent to a standard trust region subproblem in p variables. The convergence of the SFBTRA is summarized below.

9.1.37 Theorem. Let $F : \mathbb{R}^n \to \mathbb{R}^n$ be continuously differentiable and let T be a linear Newton approximation scheme of $\mathbf{F}_{\mathrm{FB}}$. Let $\{x^k\}$ be an

infinite sequence of iterates generated by the Subspace FB Trust Region Algorithm.

(a) Every accumulation point of $\{x^k\}$ is a stationary point of the merit function $\theta_{\rm FB}$. If such a point is regular then it is a solution of NCP (F).

(b) Suppose that x^* is a limit point of the sequence $\{x^k\}$ and a solution of the NCP (F), that T is nonsingular at x^* and that $T(x) \subseteq \partial \mathbf{F}_{\rm FB}(x)$ for every x in a neighborhood of x^*. Assume further that eventually V^k contains the vector $-(H^k)^{-1}\mathbf{F}_{\rm FB}(x^k)$. The whole sequence $\{x^k\}$ converges to x^* and the three statements (i), (ii), and (iii) in Theorem **9.1.36** remain valid for the sequence $\{x^k\}$.

Proof. Since V^k contains $-\nabla\theta_{\rm FB}(x^k)$, d^k satisfies (8.4.3), so that the global convergence follows from the general theory of Section **8.4**. Under the assumptions in (b), we also have that the Newton direction belongs to V^k eventually, so that the remaining assertions of the theorem can be proved exactly as in the proof of Theorem **9.1.36**. □

9.1.8 Constrained methods

The algorithms considered so far all assume that the function F is defined on the whole space. In some applications, however, this may not be the case, and the function F may be well defined only in (an open neighborhood containing) $\mathbb{R}^n_+$. Although heuristics can be designed to tackle this complication and the methods described above can be adjusted to deal with the restricted domain of definition of F, it is important to be able to design algorithms that are theoretically sound and heuristic-free. To this end the most natural approach is to consider a constrained optimization reformulation of the NCP:

$$\begin{array}{ll} \text{minimize} & \theta_{\rm FB}(x) \\ \text{subject to} & x \in \mathbb{R}^n_+, \end{array} \tag{9.1.53}$$

and develop algorithms that solve (9.1.53) while maintaining the nonnegativity of the iterates. This constrained approach is particularly beneficial for those problems where the function F, which may be defined everywhere, has good properties only on $\mathbb{R}^n_+$ and has no distinguished properties outside of this orthant. For example the function F may fail to be differentiable outside of the nonnegative orthant.

In essence, the methods developed in Chapter **8** can easily be applied to (9.1.53) with the simple identification of the feasible set X being the

nonnegative orthant. One important issue that requires attention is the question of when a constrained stationary point of (9.1.53) is actually a solution of the NCP (F). We note that a vector x is a stationary point of (9.1.53) if and only if x satisfies:

$$0 \leq x \perp \nabla\theta_{\mathrm{FB}}(x) \geq 0,$$

or equivalently,

$$\min(\, x,\, \nabla\theta_{\mathrm{FB}}(x)\,) \,=\, 0.$$

The following result shows that the same FB regularity condition that we used in the unconstrained case to deal with the issue on hand also works in the present constrained setting.

9.1.38 Theorem. Suppose that $F : \Omega \supset \mathbb{R}^n_+ \to \mathbb{R}^n$ be continuously differentiable on the open set Ω and that x is a constrained stationary point of the problem (9.1.53). Then x is a solution of the NCP (F) if and only if x is an FB regular point of θ_{FB}.

Proof. The proof of the "only if" statement is the same as that of Theorem **9.1.14**. To prove the "if" statement, suppose that x is an FB regular constrained stationary point of (9.1.53) but is not a solution of NCP (F). Let us denote by $I_0 \equiv \{i : x_i = 0\}$ the set of active constraints at x, and by $I_+ \equiv \{i : x_i > 0\}$ the set of the positive components of x. The stationarity condition can be written as

$$\begin{aligned} (\nabla\theta_{\mathrm{FB}}(x))_i &= 0, \quad \forall\, i \in I_+ \\ (\nabla\theta_{\mathrm{FB}}(x))_i &\geq 0, \quad \forall\, i \in I_0. \end{aligned}$$

Let $z \equiv D_b\mathbf{F}_{\mathrm{FB}}(x)$, where D_b is an arbitrary diagonal matrix in $\mathcal{D}_b(x)$. Since x is not a solution of the NCP (F), $z \neq 0$. Take y to be the vector in Definition **9.1.13** associated with the FB regularity of x. We note that

$$i \in I_0 \;\Rightarrow\; x_i = 0 \;\Rightarrow\; i \in \mathcal{C} \cup \mathcal{N} \;\Rightarrow\; y_i \leq 0.$$

Therefore

$$y^T\nabla\theta_{\mathrm{FB}}(x) \,=\, y_{I_0}^T\nabla_{I_0}\theta_{\mathrm{FB}}(x) \,\leq\, 0. \tag{9.1.54}$$

On the other hand we also have $y^T JF(x)^T z \geq 0$; and similar to (9.1.18), $y^T D_a\mathbf{F}_{\mathrm{FB}}(x) > 0$. But the last two inequalities clearly contradict (9.1.54), thus concluding the proof. □

Following the recipe for the unconstrained methods, we can exploit the continuous differentiabilty of θ_{FB} to derive some globally and locally

superlinearly convergent methods for solving the NCP (F) based on the constrained FB reformulation (9.1.53). For this purpose, we need to start with a fast local method and modify it for global convergence. For such a method, let us consider Algorithm **7.3.1**, which when applied to the NCP (F) solves a sequence of LCPs. Specifically, given an iterate x^k, this algorithm computes the next iterate by solving the LCP $(q^k, JF(x^k))$, where

$$q^k \equiv F(x^k) - JF(x^k)x^k.$$

In order to globalize the convergence of this locally convergent algorithm, we make use of the FB merit function $\theta_{\rm FB}$ and supplement the above calculation with a line search based on a constrained steepest descent direction. For simplicity, we consider a method for computing the latter direction that is of the gradient projection type. Specifically, given $x^k \geq 0$, we define a search direction $\hat{d}^k$ as the unique solution of the following simple constrained optimization problem:

$$\begin{array}{ll} \text{minimize} & \nabla\theta_{\rm FB}(x^k)^T d + \frac{1}{2}\, d^T d \\ \text{subject to} & x^k + d \in \mathbb{R}^n_+. \end{array} \tag{9.1.55}$$

It is easy to see that

$$\hat{d}^k = \max(-x^k, -\nabla\theta_{\rm FB}(x^k)) = -\min(x^k, \nabla\theta_{\rm FB}(x^k))$$

Hence, $\hat{d}^k = 0$ if and only if $\min(x^k, \nabla\theta_{\rm FB}(x^k)) = 0$. Thus if x^k is not a stationary point of (9.1.53), then $\hat{d}^k$ is a nonzero descent direction of $\theta_{\rm FB}$ at x^k; in this case, we have

$$\nabla\theta_{\rm FB}(x^k)^T \hat{d}^k + \tfrac{1}{2}(\hat{d}^k)^T \hat{d}^k < 0,$$

which implies

$$\|\hat{d}^k\| < \sqrt{-2\,\nabla\theta_{\rm FB}(x^k)^T \hat{d}^k}. \tag{9.1.56}$$

Thus $\hat{d}^k$ can be used as a "fall-back" direction for the minimization of $\theta_{\rm FB}$. Combined with Algorithm **7.3.1**, we obtain the following constrained line search algorithm for solving the NCP (F).

Constrained FB Line Search Algorithm I (CFBLSA)

9.1.39 Algorithm.

Data: $x^0 \in \mathbb{R}^n_+$, $\rho > 0$, $p > 1$, and $\gamma \in (0, 1)$.

Step 1: Set $k = 0$.

Step 2: If x^k is a stationary point of problem (9.1.53), stop.

Step 3: Find a solution y^{k+1} of the LCP $(q^k, JF(x^k))$ and set

$$d^k \equiv y^{k+1} - x^k.$$

If this LCP is not solvable or if the condition

$$\nabla\theta_{\mathrm{FB}}(x^k)^T d^k \leq -\rho \| d^k \|^p \tag{9.1.57}$$

is not satisfied, set $d^k = \hat{d}^k$.

Step 4: Find the smallest nonnegative integer i_k such that with $i = i_k$,

$$\begin{aligned} \theta_{\mathrm{FB}}(x^k + 2^{-i} d^k) \leq\ & \theta_{\mathrm{FB}}(x^k) - \\ & \min\left\{ -2^{-i}\gamma \nabla\theta_{\mathrm{FB}}(x^k)^T d^k,\ (1-\gamma)\,\theta_{\mathrm{FB}}(x^k) \right\} \end{aligned} \tag{9.1.58}$$

and set $\tau_k \equiv 2^{-i_k}$.

Step 5: Set $x^{k+1} \equiv x^k + \tau_k d^k$ and $k \leftarrow k+1$; go to Step 2.

Clearly, any sequence $\{x^k\}$ generated by the above algorithm is nonnegative. The asymptotic properties of such a sequence are summarized in Theorem **9.1.40** below. The line of proof of this theorem is by now fairly familiar. The proof is based on the results in Section **8.3** and Section **7.3**, particularly Proposition **8.3.7**, Corollary **8.3.12**, and Theorem **7.3.3**, the latter for the convergence rate of the algorithm.

9.1.40 Theorem. Let $F : \Omega \supset \mathbb{R}^n_+ \to \mathbb{R}^n$ be continuously differentiable on the open set Ω. Let $\{x^k\}$ be the sequence generated by Algorithm **9.1.39**.

(a) Every accumulation point of the sequence $\{x^k\}$ is a stationary point of the problem (9.1.53). Any such point that is FB regular is a solution of NCP (F).

(b) If $\{x^k\}$ has an isolated accumulation point, then $\{x^k\}$ converges to that point.

(c) Suppose that x^* is a limit point of $\{x^k\}$ and a strongly stable solution of the NCP (F). The whole sequence $\{x^k\}$ converges to x^*; moreover if $p > 2$ and $\gamma < 1/2$, then the following statements hold:

(i) eventually d^k is always a solution of the LCP $(q^k, JF(x^k))$;

(ii) eventually a unit step size is always accepted; hence for all k sufficiently large, $x^{k+1} = x^k + d^k$;

(iii) the convergence rate is superlinear, furthermore, if $JF(x)$ is Lipschitz continuous at x^*, the convergence rate is quadratic.

Proof. With $\sigma(x^k, d^k) \equiv -\nabla\theta_{\mathrm{FB}}(x^k)^T d^k$, Theorem **8.3.3** implies that

$$\lim_{k(\in\kappa)\to\infty} \nabla\theta_{\mathrm{FB}}(x^k)^T d^k \;=\; 0,$$

for every convergent subsequence $\{x^k : k \in \kappa\}$, whose limit we denote x^*. There are two cases to consider. Suppose first that $d^k = \hat{d}^k$ for infinitely many $k \in \kappa$. By (9.1.56) and the above limit, we deduce that for some infinite subset κ' of κ,

$$0 \;=\; \lim_{k(\in\kappa')\to\infty} \hat{d}^k,$$

which implies

$$0 \;=\; \min(\, x^*, \nabla\theta_{\mathrm{FB}}(x^*)\,).$$

Hence x^* is a stationary point of (9.1.53).

Suppose next that $d^k = y^{k+1} - x^k$ and (9.1.57) holds for all but finitely many $k \in \kappa$, where y^{k+1} solves the LCP $(q^k, JF(x^k))$. Thus, we have, for all $k \in \kappa$ sufficiently large,

$$0 \;\le\; y^{k+1} \;\perp\; F(x^k) + JF(x^k)d^k \;\ge\; 0 \tag{9.1.59}$$

and

$$\|\, d^k \,\|^p \;\le\; -\rho^{-1}\, \nabla\theta_{\mathrm{FB}}(x^k)^T d^k.$$

Since the right-hand side converges to zero as $k(\in\kappa) \to \infty$, it follows that

$$\lim_{k(\in\kappa)\to\infty} d^k \;=\; 0.$$

Consequently, passing to the limit $k(\in\kappa) \to \infty$ in (9.1.59), we deduce that x^* solves the NCP (F). Thus x^* must be a stationary point of (9.1.53). This establishes statement (a).

To prove (b), we note that, for every k, by (9.1.56) and (9.1.57) and the definition of d^k,

$$\|\, d^k \,\| \;\le\; \max\left\{ \sqrt{-2\,\nabla\theta_{\mathrm{FB}}(x^k)^T d^k},\; (\,-\rho^{-1}\,\nabla\theta_{\mathrm{FB}}(x^k)^T d^k\,)^{1/p} \right\}.$$

Since $\sigma(x^k, d^k) \equiv -\nabla\theta_{\mathrm{FB}}(x^k)^T d^k$, (b) follows from Propositions **8.3.10** and **8.3.11**.

Finally, to prove (c), it suffices to note that by Theorem **7.3.3** the sequence $\{y^{k+1} - x^k\}$ is superlinearly convergent with respect to $\{x^k\}$. The rest of the proof is a verbatim repetition of the proof of part (d) of Theorem **9.1.29**. □

9.1.41 Remark. The boundedness of the sequence $\{x^k\}$ of iterates produced by Algorithm **9.1.39**, and thus the existence of an accumulation point of such a sequence, is ensured by the boundedness of the level set

$$\{\, x \in \mathbb{R}^n_+ \,:\, \|\, \mathbf{F}_{\mathrm{FB}}(x) \,\| \,\leq\, \|\, \mathbf{F}_{\mathrm{FB}}(x^0) \,\|\},$$

which is a subset of

$$\{\, x \in \mathbb{R}^n \,:\, \|\, \mathbf{F}_{\mathrm{FB}}(x) \,\| \,\leq\, \|\, \mathbf{F}_{\mathrm{FB}}(x^0) \,\|\,\}$$

whose boundedness is needed for an unconstrained method to generate a bounded sequence of iterates. □

The algorithm we present next is an attempt to alleviate the burden of solving an LCP in each iteration of Algorithm **9.1.39**. Specifically, the local algorithm we employ is the (exact) semismooth Levenberg-Marquardt method, i.e., Algorithm **7.5.9** with $\eta_k = 0$ for all k, augmented by a line search as in the modified Gauss-Newton method presented in Section **8.3**. As in Algorithms **9.1.10** and **9.1.39**, the continuous differentiability of θ_{FB} makes the line search in the next algorithm possible, which also takes into account the nonnegativity constraint on the iterates. Although the modified Gauss-Newton method was presented for an unconstrained system of smooth equations, the extension of this method to the constrained nonsmooth system:

$$\mathbf{F}_{\mathrm{FB}}(x) \,=\, 0, \quad x \,\geq\, 0.$$

is straightforward. Omitting the details, we can present the resulting algorithm, which employs, in addition to the linear Newton approximation scheme T of $\mathbf{F}_{\mathrm{FB}}$, a continuous function $\rho : \mathbb{R}^n_+ \to \mathbb{R}^n_+$ such that $\rho(t) = 0$ if and only if $t = 0$. With the iterate $x^k \geq 0$ given at the beginning of the k-th iteration, we choose a matrix $H^k \in T(x^k)$ and solve the following convex quadratic program: with $\tilde{H}^k \equiv (H^k)^{\,T} H^k + \rho(\theta_{\mathrm{FB}}(x^k))\mathbf{I}$,

$$\begin{array}{ll} \text{minimize} & \nabla\theta_{\mathrm{FB}}(x^k)^{\,T} d + \frac{1}{2}\, d^{\,T} \tilde{H}^k d \\ \text{subject to} & x^k + d \,\geq\, 0, \end{array} \tag{9.1.60}$$

By the definition of the function ρ, the matrix $\tilde{H}^k$ is (symmetric) positive definite, provided that x^k is not a solution of the NCP (F). Since the feasible set of the quadratic program (9.1.60) is obviously nonempty, problem (9.1.60) always admits a unique solution d^k, which must satisfy:

$$\nabla\theta_{\mathrm{FB}}(x^k)^{\,T} d^k + \tfrac{1}{2}\,(\,d^k\,)^{\,T} \tilde{H}^k d^k \,\leq\, 0 \tag{9.1.61}$$

because $d = 0$ is a feasible solution of (9.1.60). By Proposition **8.3.4**, equality holds in (9.1.61) if and only if x^k is a stationary point of (9.1.53).

Constrained LM FB Line Search Algorithm (CLMFBLSA)

9.1.42 Algorithm.

Data: $x^0 \in \mathbb{R}^n$ and $\gamma \in (0, 1)$.

Step 1: Set $k = 0$.

Step 2: If x^k is a stationary point of (9.1.53) stop.

Step 3: Select an element $H^k \in T(x^k)$. Let d^k be the unique solution of the quadratic program (9.1.60).

Step 4: If

$$\| \theta_{\rm FB}(x^k + d^k) \| \leq \gamma \| \theta_{\rm FB}(x^k) \| \tag{9.1.62}$$

set $i_k = 0$. Otherwise, let i_k be the smallest nonnegative integer such that with $i = i_k$,

$$\theta_{\rm FB}(x^k + 2^{-i} d^k) \leq \theta_{\rm FB}(x^k) + \gamma\, 2^{-i}\, \nabla\theta_{\rm FB}(x^k)^T d^k. \tag{9.1.63}$$

Let $\tau_k \equiv 2^{-i_k}$.

Step 5: Set $x^{k+1} \equiv x^k + \tau_k d^k$ and $k \leftarrow k + 1$; go to Step 2.

The following is the convergence result for the above algorithm.

9.1.43 Theorem. Let $F : \Omega \supset \mathbb{R}^n_+ \to \mathbb{R}^n$ be continuously differentiable on the open set Ω. Let T be a linear Newton approximation scheme of $\mathbf{F}_{\rm FB}$. Let $\{x^k\}$ be an infinite sequence generated by Algorithm **9.1.42**. The following statements hold.

(a) Every limit point of $\{x^k\}$ is a stationary point of (9.1.53).

(b) If a limit point x^* of the sequence $\{x^k\}$ is a solution of the NCP (F) and $T(x^*)$ is nonsingular, then

(i) the whole sequence $\{x^k\}$ converges to x^*;

(ii) the convergence rate is Q-superlinear;

(iii) if the Newton scheme T is strong at x^* and

$$\rho(\theta_{\rm FB}(x)) = O(\theta_{\rm FB}(x))$$

for all x sufficiently near x^*, then the convergence rate is Q-quadratic.

Proof. Let $\{x^k : k \in \kappa\}$ be a subsequence converging to x^*. If x^* is a solution of the NCP (F), then x^* is obviously a stationary point of (9.1.53). So assume that x^* is not a solution of NCP (F). The sequence of scalars $\{\rho(\theta_{\rm FB}(x^k)) : k \in \kappa\}$ is bounded below by a positive scalar. Since T is upper semicontinuous, the assumptions of Proposition **8.3.7** are all satisfied; the first assertion (a) therefore follows immediately from this proposition.

Assume the conditions in (b). We know from Theorem **7.2.10** that x^* is an isolated solution of NCP (F). To prove (bi), it suffices to show, by Proposition **8.3.10**, that the sequence of directions $\{d^k : k \in \kappa\}$ converges to zero, where $\{x^k : k \in \kappa\}$ is any subsequence converging to x^*. Since $T(x^*)$ is nonsingular, there exists a positive constant c_1 such that for every $k \in \kappa$ sufficiently large,

$$\| (H^k)^{-1} \| \le c_1, \tag{9.1.64}$$

which implies

$$\| d^k \| \le c_1 \| H^k d^k \|. \tag{9.1.65}$$

By (9.1.61) and similar to the proof of Lemma **8.3.5**, we have

$$\begin{aligned} 0 &\ge \nabla\theta_{\rm FB}(x^k)^T d^k + \tfrac{1}{2}(d^k)^T[(H^k)^T H^k + \rho(\theta_{\rm FB}(x^k))\mathbf{I}]d^k \\ &\ge \nabla\theta_{\rm FB}(x^k)^T d^k + \tfrac{1}{2}(d^k)^T (H^k)^T H^k d^k \\ &\ge \tfrac{1}{2}\| H^k d^k \|^2 - \| \mathbf{F}_{\rm FB}(x^k) \| \| H^k d^k \|, \end{aligned}$$

where the last inequality is due to $\nabla\theta_{\rm FB}(x^k) = (H^k)^T \mathbf{F}_{\rm FB}(x^k)$. Hence

$$\| H^k d^k \| \le 2 \| \mathbf{F}_{\rm FB}(x^k) \|. \tag{9.1.66}$$

Combining (9.1.65) and (9.1.66) we obtain, for some positive constant c_2,

$$\|d^k\| \le c_2 \| \mathbf{F}_{\rm FB}(x^k) \|$$

for every $k \in \kappa$ such that x^k is sufficiently close to x^*. This is enough to yield (bi).

In order to investigate the convergence rate we next show that the sequence $\{d^k\}$ is superlinearly convergent with respect to $\{x^k\}$; that is,

$$\lim_{k\to\infty} \frac{\| x^k + d^k - x^* \|}{\| x^k - x^* \|} = 0; \tag{9.1.67}$$

moreover, if the assumptions stated in (b3) hold we even have

$$\limsup_{k\to\infty} \frac{\| x^k + d^k - x^* \|}{\| x^k - x^* \|^2} < \infty. \tag{9.1.68}$$

These estimates are sufficient to complete the proof of the theorem. By property (b) of Definition **7.2.2** of a Newton approximation, we have

$$\lim_{k\to\infty} \frac{\| \mathbf{F}_{\mathrm{FB}}(x^k) - \mathbf{F}_{\mathrm{FB}}(x^*) - H^k(x^k - x^*) \|}{\| x^k - x^* \|} = 0, \tag{9.1.69}$$

or, if the Newton approximation scheme T is strong,

$$\limsup_{k\to\infty} \frac{\| \mathbf{F}_{\mathrm{FB}}(x^k) - \mathbf{F}_{\mathrm{FB}}(x^*) - H^k(x^k - x^*) \|}{\| x^k - x^* \|^2} < \infty. \tag{9.1.70}$$

By the nonsingularity of $T(x^*)$ and the proof of Theorem **8.3.15**, we deduce the existence of a constant $c' > 0$ such that for all k sufficiently large,

$$\| x^k + d^k - x^* \| \leq c' \| \nabla\theta(x^k) - \nabla\theta(x^*) - \tilde{H}^k(x^k - x^*) \|,$$

where $\tilde{H}^k \equiv (H^k)^T H^k + \rho(\theta_{\mathrm{FB}}(x^k))\mathbf{I}$. We have

$$\begin{aligned} &\nabla\theta(x^k) - \nabla\theta(x^*) - \tilde{H}^k(x^k - x^*) \\ &= (H^k)^T [\, \mathbf{F}_{\mathrm{FB}}(x^k) - \mathbf{F}_{\mathrm{FB}}(x^*) - H^k(x^k - x^*) \,] - \rho(\theta_{\mathrm{FB}}(x^k))\, (\, x^k - x^* \,). \end{aligned}$$

Since the sequence $\{H^k\}$ is bounded, the limit (9.1.67) follows easily from the limit (9.1.69) and the last two expressions. Similarly, (9.1.68) follows from (9.1.70) and the assumption on ρ because the latter assumption implies that

$$\rho(\theta_{\mathrm{FB}}(x^k)) = O(\theta_{\mathrm{FB}}(x^k)) = O(\| x^k - x^* \|)$$

by the locally Lipschitz continuity of θ_{FB} near x^*. □

9.2 Global Algorithms Based on the min Function

In this section we consider algorithms that are based on the min function reformulation of the NCP (F); namely:

$$\mathbf{F}_{\min}(x) \equiv \min(\, x, F(x) \,) = 0.$$

Along with the FB reformulation, this is certainly among the simplest and most interesting equation reformulation of the NCP. As we saw in Example **7.2.16** and the subsequent development, the min reformulation allows us to define a locally fast algorithm that requires solving a system of linear equations of reduced dimension at each iteration; cf. Algorithm **7.2.17** with G being the identity map. The convergence of this algorithm can be established under weaker assumptions than those employed in the FB reformulation; moreover, the algorithm can be shown to be finitely convergent

for the LCP. Indeed, the b-regularity of a solution x^* being computed is a sufficient condition for the convergence of Algorithm **7.2.17** applied to the NCP (F); whereas for the local convergence of a Newton method applied to the equation $\mathbf{F}_{\mathrm{FB}}(x) = 0$, we need all matrices in $\mathcal{D}_a(x^*) + \mathcal{D}_b(x^*)JF(x^*)$ to be nonsingular. In turn, a sufficient condition for the latter requirement to hold is given in Theorem **9.1.23**, which requires a condition of consistent determinantal signs (the P_0 property) in addition to the b-regularity. Furthermore, we have finite convergence in the case of the LCP using the min formulation, which is not possible using the FB formulation.

In what follows, we give an example to show that the gap between the requirements for the local convergence of a Newton algorithm based on the min formulation and the FB formulation is significant. The example is an LCP with a unique, totally degenerate solution x^* that is b-regular but not strongly stable. More importantly, $\operatorname{Jac}\mathbf{F}_{\mathrm{FB}}(x^*)$ contains a singular matrix.

9.2.1 Example. Let $n = 2$ and $F : \mathbb{R}^2 \to \mathbb{R}^2$ be defined by

$$F(x) \equiv \begin{pmatrix} -x_1 + x_2 \\ -x_2 \end{pmatrix}, \quad x = (x_1, x_2) \in \mathbb{R}^2.$$

It is easy to see that the NCP (F) has the unique solution $x^* = (0, 0)$, which is b-regular (because $JF(x^*)$ is a nondegenerate matrix) but not strongly stable (because condition (b) of Corollary **5.3.20** is not satisfied). We have

$$\mathbf{F}_{\mathrm{FB}}(x_1, x_2) \equiv \begin{pmatrix} \sqrt{x_1^2 + (x_2 - x_1)^2} - x_2 \\ \sqrt{2}\, x_2 \end{pmatrix}.$$

Now consider the sequence $\{x^k\}$ defined by

$$x^k \equiv \begin{pmatrix} 1/k \\ 2/k \end{pmatrix}, \quad k = 1, 2, \ldots.$$

The sequence $\{x^k\}$ clearly converges to x^* and $\mathbf{F}_{\mathrm{FB}}$ is continuously differentiable at all x^k with Jacobian

$$J\mathbf{F}_{\mathrm{FB}}(x^k) = \begin{pmatrix} 0 & \frac{1}{\sqrt{2}} - 1 \\ 0 & \sqrt{2} \end{pmatrix}$$

for all k. The sequence $\{J\mathbf{F}_{\mathrm{FB}}(x^k)\}$ consists of a constant singular matrix, which therefore must belong to $\operatorname{Jac}\mathbf{F}_{\mathrm{FB}}(x^*)$. We conclude that any linear Newton approximation scheme T such that $\operatorname{Jac}\mathbf{F}_{\mathrm{FB}}(x^*) \subseteq T(x^*)$, and in particular the Newton scheme $T(x) \equiv \operatorname{Jac}\mathbf{F}_{\mathrm{FB}}(x)$, is singular at x^*. Consequently, the local convergence of a Newton method based on the function

$\mathbf{F}_{\mathrm{FB}}$ for solving this LCP is in jeopardy. Theorem **7.2.18** guarantees, however, that a Newton method based on the function $\mathbf{F}_{\min}$ for solving the same LCP is finitely convergent. □

We consider two strategies for designing a globally convergent and locally superlinearly convergent method for solving the NCP (F). The first strategy is based on the merit function derived from the min function:

$$\theta_{\min}(x) \equiv \tfrac{1}{2}\, \mathbf{F}_{\min}(x)^T \mathbf{F}_{\min}(x).$$

A resulting line search algorithm applicable for solving the NCP (F) was described toward the end of Section **8.3**. The subsequential convergence of such an algorithm was established in Proposition **8.3.21**; however, the sequential convergence and the convergence rate were not addressed. In what follows, we pick up from we left off in the previous subsection and show that a variant of the algorithm possesses the familiar sequential convergence properties as the algorithms that are based on the FB function.

For ease of reference, we state the complete global algorithm for solving the NCP (F) that is based on the min function. We continue to make use of the one-dimensional function $\rho : \mathbb{R}_+ \to \mathbb{R}_+$ that has the property: $\rho(t) = 0$ if and only if $t = 0$. Instead of a convex majorant of the directional derivative of $\theta_{\min}(x; \cdot)$, we directly define the quadratic objective function in the subproblem for calculating the search direction:

$$\phi(x, d) \equiv$$

$$\sum_{i:x_i > F_i(x)} (\, F_i(x) + \nabla F_i(x)^T d\,)^2 + \sum_{i:x_i \le F_i(x)} (\, x_i + d_i\,)^2 + \frac{\rho(\theta_{\min}(x))}{2}\, d^T d.$$

Let the forcing function be

$$\sigma(x, d) \equiv \sum_{i:x_i > F_i(x)} F_i(x) \nabla F_i(x)^T d + \sum_{i:x_i \le F_i(x)} x_i d_i.$$

We specialize the corresponding index sets $\mathcal{I}_F(x)$, $\mathcal{I}_=(x)$ and $\mathcal{I}_G(x)$ by setting

$$\begin{aligned} \mathcal{I}_F(x) &\equiv \{\, i \,:\, F_i(x) < x_i\}, \\ \mathcal{I}_=(x) &\equiv \{\, i \,:\, F_i(x) = x_i\}, \\ \mathcal{I}_x(x) &\equiv \{\, i \,:\, F_i(x) > x_i\}. \end{aligned}$$

With the above preparation, we present the promised line search algorithm for solving the NCP (F) that is based on the merit function $\theta_{\min}$ and the direction search function ϕ.

min-Based Line Search Algorithm (minLSA)

9.2.2 Algorithm.

Data: $x^0 \in \mathbb{R}^n_+$ and $\gamma \in (0,1)$.

Step 1: Set $k = 0$.

Step 2: Let d^k be an optimal solution of the convex quadratic program:

$$\begin{array}{ll} \text{minimize} & \phi(x^k, d) \\ \text{subject to} & x^k + d \geq 0. \end{array} \tag{9.2.1}$$

Step 3: If $d^k = 0$ stop.

Step 4: Find the smallest nonnegative integer i_k such that with $i = i_k$,

$$\theta_{\min}(x^k + 2^{-i} d^k) \leq \theta_{\min}(x^k) - \gamma\, 2^{-i}\, \sigma(x^k, d^k). \tag{9.2.2}$$

Step 5: Let $\tau_k \equiv 2^{-i_k}$, $x^{k+1} = x^k + \tau_k d^k$, and $k \leftarrow k+1$; go to Step 2.

A major theoretical drawback of Algorithm **9.2.2** is that at the termination Step 3, we can not establish that x^k is a stationary point of the merit function $\theta_{\min}$, without assuming a suitable regularity condition on x^k. Similarly, in an asymptotic analysis of this algorithm, we cannot prove that every accumulation point of an infinite sequence $\{x^k\}$ is a stationary point of $\theta_{\min}$. This is quite different from the previous algorithms based on the FB merit function $\theta_{\rm FB}$, where such a stationarity result is an easy consequence of the continuous differentiability of $\theta_{\rm FB}$ without any additional assumption. Instead, what we can prove for Algorithm **9.2.2** is a result similar to Proposition **8.3.21**, which provides a necessary and sufficient condition for an accumulation point to be a solution of the NCP (F).

As a remedy, we next turn to a different way of globalizing the local min-based Newton direction (see (9.2.3) below) that combines the advantages of this direction with a smooth merit function based on the FB functional. Let $T_{\min}$ be a linear Newton approximation scheme of $\mathbf{F}_{\min}$.

min-FB Line Search Algorithm (minFBLSA)

9.2.3 Algorithm.

Data: $x^0 \in \mathbb{R}^n$, $\varepsilon > 0$, $\rho > 0$, $p > 1$, and $\gamma \in (0,1)$.

Step 1: Set $k = 0$.

Step 2: If x^k is a stationary point of θ_{FB} stop.

Step 3: Select an element H^k in $T_{\min}(x^k)$ and find a solution of the system

$$\mathbf{F}_{\min}(x^k) + H^k d \;=\; 0. \tag{9.2.3}$$

If system (9.2.3) is solvable and the solution d^k satisfies

$$\| \mathbf{F}_{\mathrm{FB}}(x^k + d^k) \| \;\leq\; \gamma \| \mathbf{F}_{\mathrm{FB}}(x^k) \|, \tag{9.2.4}$$

set $\tau_k = 1$ and go to Step 5. Otherwise, if the system (9.2.3) is not solvable or if the condition

$$\nabla\theta_{\mathrm{FB}}(x^k)^{\,T} d^k \;\leq\; -\rho \,\| \, d^k \, \|^p \tag{9.2.5}$$

is not satisfied, set $d^k = -\nabla\theta_{\mathrm{FB}}(x^k)$.

Step 4: Find the smallest nonnegative integer i_k such that with $i = i_k$,

$$\theta_{\mathrm{FB}}(x^k + 2^{-i} d^k) \leq \theta_{\mathrm{FB}}(x^k) - \gamma\, 2^{-i}\, \nabla\theta_{\mathrm{FB}}(x^k)^{\,T} d^k \tag{9.2.6}$$

and set $\tau_k = 2^{-i_k}$.

Step 5: Set $x^{k+1} = x^k + \tau_k d^k$ and $k \leftarrow k + 1$; go to Step 2.

The only difference between this algorithm and the VFBLSA is in the way the Newton direction is calculated, cf. equation (9.2.3) and (9.1.38), respectively. In the present case the calculation is based on the min function reformulation, whereas in the case of the VFBLSA the Newton direction is based on the FB function reformulation. It is easy to prove that Algorithm **9.2.3** inherits the global convergence properties of Algorithmalg:FBLSA and the local properties of Algorithm **7.2.17**. Roughly speaking, the global convergence can be proved as we did for the former algorithm, while the local convergence properties can be proved as we did for VBLSA, taking into account Theorem **7.2.18**. We leave the details to the reader.

9.2.4 Theorem. Let $F : \mathbb{R}^n \to \mathbb{R}^n$ be continuously differentiable and let $T_{\min}$ be a linear Newton approximation scheme of $\mathbf{F}_{\min}$. Let $\{x^k\}$ be an infinite sequence of iterates generated by Algorithm **9.2.3**.

(a) Every accumulation point of the sequence $\{x^k\}$ is a stationary point of the merit function θ_{FB}. Let x^* be such a limit point; if x^* is FB regular then it is a solution of NCP (F).

(b) If $\{x^k\}$ has an isolated accumulation point, then the whole sequence $\{x^k\}$ converges to x^*.

(c) Suppose that x^* is a limit point of $\{x^k\}$ and a solution of NCP (F). If $T_{\min}$ is nonsingular at x^* then the whole sequence $\{x^k\}$ converges to x^*; moreover, if $p > 2$ and $\gamma < 1/2$, then

(i) eventually d^k is always the solution of system (9.1.5);

(ii) eventually a unit step size is always accepted; hence for all k sufficiently large, $x^{k+1} = x^k + d^k$;

(iii) the convergence rate is Q-superlinear; furthermore, if the linear approximation $T_{\min}$ is strong at x^* and $JF(x)$ is Lipschitz at x^*, the convergence rate is Q-quadratic. □

9.3 More C-Functions

In this section we present some more C-functions that can be used to reformulate the nonlinear complementarity problem. We do not develop detailed algorithms, because this can easily be done along the line of the previous sections. We are mainly interested in those C-functions that give rise to differentiable merit functions since they generally lead to much simpler and usually more effective algorithms. When presenting these merit functions we concentrate on several main characteristics that are likely to affect the key behavior of algorithms:

- the (strong) semismoothnes of the equation reformulation and the degree of smoothness of the corresponding merit function;
- the definition of Newton approximation schemes and their nonsingularity at a solution;
- conditions which guarantee that a stationary point of the associated optimization problem is a solution of the complementarity problem; and
- conditions that ensure the boundedness of the level sets of the optimization problem.

In fact the analysis of the previous sections showed that these are the key characteristics we should pay attention to.

The development based on the FB function is a prototype for similar developments based on other C-functions. To avoid repetitions, we outline here a general framework that extends the theory developed for the FB

function. This will eventually allow us to obtain, in a simple and unified fashion, algorithms and results for many different equation reformulations of the NCP.

The starting point of our discussion is a semismooth C-function ψ; i.e., ψ is a function of two arguments satisfying

$$\psi(a,b) \;=\; 0 \;\Leftrightarrow\; [\,(\,a,b\,) \;\geq\; 0 \text{ and } ab \;=\; 0.\,]$$

The results of Section **7.4** are very useful for verifying the (strong) semismoothness of a C-function. Given such a function ψ, the NCP (F) is equivalent to the equation:

$$0 \;=\; \mathbf{F}_\psi(x) \;=\; \begin{pmatrix} \psi(x_1, F_1(x)) \\ \cdots \\ \psi(x_n, F_n(x)) \end{pmatrix}.$$

The corresponding merit function is given by

$$\theta_\psi(x) \;\equiv\; \tfrac{1}{2}\,\mathbf{F}_\psi(x)^T\mathbf{F}_\psi(x).$$

The next two propositions can be seen as partial extensions of the results given in Proposition **9.1.4**. In particular, Proposition **9.3.1** gives some insights into the structure of the generalized Jacobian of $\mathbf{F}_\psi$ and shows how to construct a Newton approximation scheme of $\mathbf{F}_\psi$.

9.3.1 Proposition. Let $F : \mathbb{R}^n \to \mathbb{R}^n$ be continuously differentiable, and let $\psi : \mathbb{R}^2 \to \mathbb{R}$ be a semismooth C-function. Let $g_\psi : \mathbb{R}^2 \to \mathbb{R}^2$ be a linear Newton approximation of ψ such that

$$g_\psi(a,b) \;\subseteq\; \partial\psi(a,b), \quad \forall\,(\,a,b\,) \;\in\; \mathbb{R}^2.$$

Define

$$\mathcal{A}(x) \;\equiv\; \{\, A \;\in\; \mathbb{R}^{n\times n} \;:\; A \;=\; D_a + D_b JF(x)\,\}, \quad \forall\, x \;\in\; \mathbb{R}^n, \qquad (9.3.1)$$

where D_a and D_b are both diagonal matrices whose i-th diagonal entries are the first and second component of an element $\xi^i \in g_\psi(x_i, F_i(x))$, respectively. The following three statements are valid.

(a) $\mathcal{A}$ is a linear Newton approximation scheme of $\mathbf{F}_\psi$ at every $x \in \mathbb{R}^n$.

(b) If ψ is strongly semismooth at x and F has a locally Lipschitz Jacobian at x, then the linear Newton approximation $\mathcal{A}$ is strong at x.

(c) If $g_\psi(a,b) = \partial\psi(a,b)$, then

$$\partial\mathbf{F}_\psi(x)^T \;\subseteq\; \partial(\mathbf{F}_\psi)_1(x) \times \cdots \times \partial(\mathbf{F}_\psi)_n(x) \;\subseteq\; \mathcal{A}^T(x).$$

Proof. The assertions (a) and (b) follow from Theorem **7.5.17**. The statement (c) can be proved by applying standard rules for the calculation of generalized Jacobians in pretty much the same way as in the proof of Proposition **9.1.4**(a). □

The next proposition gives an easily verifiable condition guaranteeing that the merit function $\theta_\psi(x)$ is continuously differentiable.

9.3.2 Proposition. Let $F : \mathbb{R}^n \to \mathbb{R}^n$ be continuously differentiable, and let ψ be a locally Lipschitz continuous C-function. If

$$[\,\psi \text{ is not differentiable at } (a,b)\,] \;\Rightarrow\; \psi(a,b) = 0,$$

then θ_ψ is continuously differentiable and

$$\nabla\theta_\psi(x) = A^T\mathbf{F}_\psi(x), \quad \forall\, A \in \mathcal{A}(x),$$

where $\mathcal{A}(x)$ is defined by (9.3.1) with $g_\psi \equiv \partial\psi$.

Proof. This is an immediate generalization of Proposition **9.1.4**(c) and can be proved in exactly the same way. □

Before considering further results we discuss three examples of merit functions that fit into the scheme being developed here. These functions are modifications of the FB function, which are aimed at improving some of its features and at offering a greater flexibility to algorithm design. One of these functions, $\psi_{\rm CCK}$, was first described in Chapter **1**; see the discussion following Proposition **1.5.3**.

9.3.3 Example. The following are all C-functions: for all $(a,b) \in \mathbb{R}^2$,

- $\psi_{\rm LT}(a,b) \equiv \|(a,b)\|_q - (a+b)$, $q > 1$;
- $\psi_{\rm KK}(a,b) \equiv \dfrac{\sqrt{(a-b)^2 + 2qab} - (a+b)}{2-q}$, $q \in [0,2)$;
- $\psi_{\rm CCK}(a,b) \equiv \psi_{\rm FB}(a,b) - qa_+b_+$, $q \geq 0$,

where $\|\cdot\|_q$ denotes the ℓ_q-norm of vectors; that is,

$$\|\,(a,b)\,\|_q \;\equiv\; (\,a^q + b^q\,)^{1/q}.$$

We refer to Proposition **1.5.3** for a proof that $\psi_{\rm CCK}(a,b)$ is a C-function. The proof that $\psi_{\rm LT}(a,b)$ and $\psi_{\rm KK}(a,b)$ are also C-functions is left for the reader. The above definitions actually yield families of C-functions, with each admissible value of the parameter q giving a C-function in each family.

Furthermore these classes all include the FB function, which corresponds to q equal to 2, 1 and 0 in ψ_{LT}, ψ_{KK}, and ψ_{CCK}, respectively.

When $q = 0$, $\psi_{\mathrm{KK}}(a,b) = -\min(a,b)$. Therefore, the function ψ_{KK} represents an attempt to unify the min function and the FB function. Since

$$(\,a-b\,)^2 + 2qab \;=\; (\,a+(q-1)b\,)^2 + q(2-q)b^2,$$

the restriction $q \in [0,2)$ is needed to ensure that the above quantity is always nonnegative.

Clearly, ψ_{FB} majorizes ψ_{CCK}; moreover, these two functions differ only in the positive orthant, where both functions are negative. Consequently, we have

$$|\,\psi_{\mathrm{CCK}}(a,b)\,| \;\geq\; |\,\psi_{\mathrm{FB}}(a,b)\,|, \quad \forall\,(a,b) \in \mathbb{R}^2.$$

The function ψ_{CCK} is an attempt to make the level sets of the resulting merit function compact under weaker assumptions.

All three functions ψ_{LT}, ψ_{KK} and ψ_{CCK} are strongly semismooth C-functions. They are nondifferentiable only at points (a,b) such that $a \geq 0$, $b \geq 0$ and $ab = 0$, where the functions all have value zero. More precisely, ψ_{LTKYF} and ψ_{KK} are differentiable everywhere except at the origin, while ψ_{CCK} is differentiable everywhere except on the boundary of the nonnegative orthant. In spite of the seemingly less favorable differentiability characteristics of ψ_{CCK} (it is nondifferentiable at a multitude of points) we will see shortly that it is ψ_{CCK} that enjoys the best properties among these C-functions. If $F : \mathbb{R}^n \to \mathbb{R}^n$ is a continuously differentiable function, it follows from Proposition **9.3.2** that θ_{LTKYF}, θ_{KK} and θ_{CCK} are continuously differentiable everywhere in $\mathbb{R}^n$. □

In order to further analyze the above C-functions, it is convenient to give a natural choice for $g_\psi(a,b)$ required in Proposition **9.3.1**; for such a choice, we use the limiting gradient of each of these C-functions:

$$\mathrm{Jac}\,\psi_{\mathrm{LT}}(a,b) \;\equiv$$

$$\begin{cases} \begin{pmatrix} 1-\mathrm{sgn}(a)\left(\dfrac{|\,a\,|}{\|\,(a,b)\,\|}\right)^{q-1} \\[2ex] 1-\mathrm{sgn}(b)\left(\dfrac{|\,b\,|}{\|\,(a,b)\,\|}\right)^{q-1} \end{pmatrix} & \text{if } (a,b) \neq 0, \\[2ex] \left\{ \begin{pmatrix} c \\ d \end{pmatrix} : (c+1)^{\frac{q}{q-1}} + (d+1)^{\frac{q}{q-1}} = 1 \right\} & \text{if } (a,b) = 0, \end{cases}$$

$$\operatorname{Jac}\psi_{\mathrm{KK}}(a,b) \equiv$$

$$\begin{cases} \dfrac{1}{2-q}\begin{pmatrix} \dfrac{a+(q-1)b}{\sqrt{(a-b)^2+2qab}}-1 \\ \dfrac{b+(q-1)a}{\sqrt{(a-b)^2+2qab}}-1 \end{pmatrix} & \text{if } (a,b)\neq 0, \\ \left\{ \begin{pmatrix} c \\ d \end{pmatrix} : \left\| \begin{pmatrix} c+1 \\ d+1 \end{pmatrix} \right\| = \sqrt{\dfrac{q^2-2q+4}{2(2-q)}} \right\} & \text{if } (a,b)=0, \end{cases}$$

$$\operatorname{Jac}\psi_{\mathrm{CCK}}(a,b) \equiv \operatorname{Jac}\psi_{\mathrm{FB}}(a,b) - q\,(\,b_+\ \operatorname{Jac}\ a_+ + a_+\ \operatorname{Jac}\ b_+\,),$$

and $\operatorname{Jac}\psi_{\mathrm{FB}}(a,b)$ can be obtained from either $\operatorname{Jac}\psi_{\mathrm{LT}}(a,b)$ with $q=2$ or $\operatorname{Jac}\psi_{\mathrm{KK}}(a,b)$ with $q=1$; see also Example **7.1.3** for $\operatorname{Jac}\psi_{\mathrm{FB}}(0,0)$; while for a scalar τ,

$$\operatorname{Jac}\ \tau_+ \equiv \begin{cases} 0 & \text{if } \tau<0 \\ \{0,1\} & \text{if } \tau=0 \\ 1 & \text{if } \tau>0. \end{cases}$$

In each of the above formulas, the proof that the limiting gradient (thus the usual gradient) at a differentiable point is equal to the respective displayed singleton is straightforward. Similarly, so is the proof that the limiting gradient at a nondifferentiable point is contained in the respective displayed set; the proof for the reverse inclusion, and thus the equality, in the latter case follows from a proof similar to that of Proposition **9.1.12**.

Associated with the above limiting gradients, let $\mathcal{A}_{\mathrm{LT}}$, $\mathcal{A}_{\mathrm{KK}}$ and $\mathcal{A}_{\mathrm{CCK}}$ denote, respectively, the linear Newton approximations of the functions $\mathbf{F}_{\mathrm{LT}}$, $\mathbf{F}_{\mathrm{KK}}$ and $\mathbf{F}_{\mathrm{CCK}}$ as defined in Proposition **9.3.1**. We examine the nonsingularity issue of these Newton approximations. We recall some index sets defined for an arbitrary vector $x\in\mathbb{R}^n$:

$$\begin{aligned} \alpha &\equiv \{i : x_i = 0 < F_i(x)\}, \\ \beta &\equiv \{i : x_i = 0 = F_i(x)\}, \\ \gamma &\equiv \{i : x_i > 0 = F_i(x)\}, \\ \delta &\equiv \{1,\ldots,n\} \setminus (\,\alpha\cup\beta\cup\gamma\,). \end{aligned}$$

The following result is a direct extension of Theorem **9.1.23**, phrased in terms of the nonsingularity of the Newton schemes.

9.3.4 Theorem. Suppose that $F:\mathbb{R}^n\to\mathbb{R}^n$ is continuously differentiable in a neighborhood of a given vector $x\in\mathbb{R}^n$. Let $M\equiv JF(x)$; also let $\bar\alpha\equiv\gamma\cup\beta\cup\delta$ be the complement of α in $\{1,\ldots,n\}$. Assume that

(a) the submatrices $M_{\tilde{\gamma}\tilde{\gamma}}$ are nonsingular for all $\tilde{\gamma}$ satisfying

$$\gamma \subseteq \tilde{\gamma} \subseteq \gamma \cup \beta,$$

(b) the Schur complement of $M_{\gamma\gamma}$ in $M_{\bar{\alpha}\bar{\alpha}}$ is a P_0 matrix.

The Newton schemes $\mathcal{A}_{\mathrm{LT}}$, $\mathcal{A}_{\mathrm{KK}}$ and $\mathcal{A}_{\mathrm{CCK}}$ are nonsingular at x.

Proof. We outline the proof for $\mathcal{A}_{\mathrm{LT}}$, leaving that for the other two Newton schemes to the reader. We want to show that all the matrices in $\mathcal{A}_{\mathrm{LT}}(x)$ are nonsingular. By definition all these matrices have the form $D_a + D_b M$, where the elements of the diagonal matrices D_a and D_b are given, respectively, by the first and second component of an element in $\operatorname{Jac}\psi_{\mathrm{LT}}(x_i, F_i(x))$. But then the proof is identical to the proof of Theorem **9.1.23**. Indeed, it is easy to see that the actual values of the diagonal entries of D_a and D_b are irrelevant for the validity of the proof of the latter theorem; only the following properties are essential:

- $(D_a)_{\tilde{\gamma}} = 0$ and $(D_b)_{\tilde{\alpha}} = 0$;
- $(D_b)_{\tilde{\gamma}}$ is nonsingular;
- $(D_a)_{\tilde{\delta}}$ and $(D_b)_{\tilde{\delta}}$ are negative definite,

where we employ the same notation as in the previous proof. But it can be checked directly that these properties still hold in the current setting; hence the same proof is applicable. □

The next result deals with the issue of when a stationary point of the merit functions θ_{LT}, θ_{KK}, and θ_{CCK} is a solution of the NCP (F). See Exercise **9.5.16** for a unified treatment of this issue.

9.3.5 Theorem. Let $F : \mathbb{R}^n \to \mathbb{R}^n$ be continuously differentiable. If $x \in \mathbb{R}^n$ is a stationary point of θ_{LT} (θ_{KK}, θ_{CCK}), then x is a solution of NCP (F) if and only if x is an FB regular point as in Definition **9.1.13**.

Proof. It is easy to check that ψ_{LT}, ψ_{KK} and ψ_{CCK} are negative in the interior of the first orthant and positive in the second, third and fourth orthant except the nonnegative axis. It can further be checked that the diagonal elements a_i and b_i of the matrices D_a and D_b are always nonpositive and can possibly be zero only if $x_i \geq 0, F_i(x) \geq 0$ and $x_i F_i(x) = 0$. Therefore the relations in (9.1.14) still hold for the functions $\mathbf{F}_{\mathrm{LT}}$, $\mathbf{F}_{\mathrm{KK}}$ and $\mathbf{F}_{\mathrm{CCK}}$. Thus the proof of the present theorem can be carried out identically to that of Theorem **9.1.14**. □

As a consequence of Theorem **9.3.5** and in view of the sufficient conditions developed in Section **9.1.1**, we obtain finitely verifiable conditions that will ensure the stationary points of the merit functions θ_{LT}, θ_{KK} and θ_{CCK} to be solutions of the NCP (F).

The last issue we consider is the boundedness of the level sets of the merit functions, or equivalently, the coercivity of these functions. We leave it as an exercise for the reader to treat the merit function θ_{LT}. As for θ_{KK}, it is not difficult to show, similar to Lemma **9.1.3**, that this merit function is of the same order as the FB function θ_{FB}, via the min function $\theta_{\min}$. Thus all level sets of any one of the functions θ_{KK}, θ_{FB}, and $\theta_{\min}$ are bounded if and only if all level sets of all these functions are bounded; in other words, if any one of these functions is coercive on $\mathbb{R}^n$, then all three functions are coercive on $\mathbb{R}^n$. See Proposition **9.1.27** for a sufficient condition that ensures this property. Since θ_{CCK} majorizes θ_{FB}, it follows that this condition also ensures the coercivity of θ_{CCK}. In what follows, we provide a necessary and sufficient condition for θ_{CCK} to be coercive.

9.3.6 Theorem. Let $F : \mathbb{R}^n \to \mathbb{R}^n$ be continuous and $q > 0$ be given. The function θ_{CCK} is coercive if and only if the following implication holds for any infinite sequence $\{x^k\}$:

$$\left.\begin{array}{r} \lim\limits_{k\to\infty} \| x^k \| = \infty \\ \limsup\limits_{k\to\infty} \| (-x^k)_+ \| < \infty \\ \limsup\limits_{k\to\infty} \| (-F(x^k))_+ \| < \infty \end{array}\right\} \quad (9.3.2)$$
$$\Rightarrow \exists\, i \text{ such that } \limsup_{k\to\infty} (x_i^k)_+ (F_i(x^k))_+ = \infty.$$

In particular, this holds under either one of the following two conditions:

(a) F is a uniform P function;

(b) F is monotone and the solution set of the NCP (F) is nonempty and bounded.

Proof. Suppose that $\{x^k\}$ is a sequence satisfying

$$\lim_{k\to\infty} \| x^k \| = \infty \quad \text{and} \quad \limsup_{k\to\infty} \theta_{\mathrm{CCK}}(x^k) < \infty.$$

It follows from the second limit that $\{\theta_{\mathrm{FB}}(x^k)\}$ is bounded and that there is no index i such that $\{x_i^k\} \to -\infty$ or $\{F_i(x^k)\} \to -\infty$. Therefore, by assumption, an index, say j, exists such that $\{(x_j^k)_+(F_j(x^k))_+\} \to \infty$

(at least on a subsequence). But this easily implies that $\{\theta_{\rm CCK}(x^k)\}$ is unbounded, because $q > 0$, a contradiction.

Conversely, suppose that $\theta_{\rm CCK}$ is coercive. Let $\{x^k\}$ be an infinite sequence satisfying the left-hand side of (9.3.2) but

$$\forall\, i \quad \limsup_{k\to\infty} (\, x_i^k \,)_+ (\, F_i(x^k) \,)_+ \,<\, \infty. \tag{9.3.3}$$

By the coercivity of $\theta_{\rm CCK}$, we have

$$\lim_{k\to\infty} \theta_{\rm CCK}(x^k) \,=\, \infty.$$

Hence there exists an index j such that for some infinite subset κ of $\{1, 2, \ldots\}$,

$$\lim_{k(\in\kappa)\to\infty} |\, \psi_{\rm FB}(x_j^k, F_j(x^k)) \,| \,=\, \infty.$$

By Lemma **9.1.3**, it follows that

$$\lim_{k(\in\kappa)\to\infty} |\, \min(x_j^k, F_j(x^k)) \,| \,=\, \infty.$$

In turn this implies that

$$\lim_{k(\in\kappa)\to\infty} \min(x_j^k, F_j(x^k)) \,=\, \infty,$$

which contradicts (9.3.3).

If F is a uniformly P function, Corollary **9.1.28** implies that $\theta_{\rm FB}$ is coercive, thus so is $\theta_{\rm CCK}$. Suppose that F is monotone and the NCP (F) has a nonempty bounded solution set. By Theorem **2.4.4**, there exists a point $\hat{x}$ such that $\hat{x} > 0$ and $F(\hat{x}) > 0$. Let $\{x^k\}$ be a sequence satisfying the left-hand conditions in (9.3.2). By working with an appropriate subsequence if necessary, we may assume that $\{x_i^k\} \to \infty$ for all indices i for which $\{x_i^k\}$ is unbounded and that $\{F_j(x^k)\} \to \infty$ for all indices j for which $\{F_j(x^k)\}$ is unbounded. Since F is monotone we have

$$(\, x^k \,)^T F(\hat{x}) + \hat{x}^T F(x^k) \,\le\, (\, x^k \,)^T F(x^k) + \hat{x}^T F(\hat{x}).$$

It follows that $(x^k)^T F(x^k) \to \infty$. By the left-hand conditions in (9.3.2), this implies in turn that for some index j,

$$\limsup_{k\to\infty} (\, x_j^k \,)_+ (\, F_j(x^k) \,)_+ \,=\, \infty,$$

thus establishing the desired implication (9.3.2). □

The boundedness of the solution set of the NCP (F) is obviously a necessary condition for the boundedness of the level sets of any one of

the merit functions. The above result shows that for a monotone solvable NCP this condition turns out to be necessary and sufficient when the merit function is $\theta_{\rm CCK}$. This result does not hold for the FB merit function $\theta_{\rm FB}$, as shown by the following example.

9.3.7 Example. Consider the NCP (F) where F is the constant univariate function equal to one. This NCP has zero as the unique solution. From Example **9.1.25**, we know that $\theta_{\rm FB}$ is not coercive. Nevertheless, $\theta_{\rm CCK}(x) = \frac{1}{2}(\sqrt{1+x^2} - (1+x) - qx_+)^2$ differs from $\theta_{\rm FB}(x)$ only for positive values of x. We know from Theorem **9.3.6** that the level sets of $\theta_{\rm CCK}(x)$ are bounded; indeed, the reader can easily check that

$$\lim_{x \to \pm\infty} \theta_{\rm CCK}(x) \;=\; \infty,$$

for all positive q. □

9.4 Extensions

In this section we briefly outline how the results presented so far in this chapter can be extended to several classes of problems more general than the NCPs but that, nevertheless, share some common features. Specifically, we consider mixed complementarity problems and variational inequalities whose feasible region K is defined by lower *or* upper bounds or by lower *and* upper bounds on the variables. For all these problems we can define equation reformulations and corresponding merit functions that extend those introduced for the NCP. As in the previous section, we do not consider specific algorithms, since these are basically identical to those considered in Section **9.1** once the merit functions have been defined. In fact, we even omit much of the properties of the equations and the merit functions. Furthermore, to avoid unnecessary repetitions, we only consider the extension of the FB-based functions.

9.4.1 Finite lower (or upper) bounds only

Consider the VI (K, F) in which the set K is defined by finite lower bounds:

$$K \;\equiv\; \{\, x \in \mathbb{R}^n \,:\, x \geq a \,\},$$

where $a \in \mathbb{R}^n$ is a given vector. This problem can easily be converted into an NCP by the simple change of variables $y = x - a$. Alternatively, we can deal with this problem directly without such a conversion. Observe that a point $x \in K$ is a solution of VI (K, F) if and only if, for every i,

$$[\, x_i \;=\; a_i \;\Rightarrow\; F_i(x) \;\geq\; 0 \,] \quad \text{and} \quad [\, x_i \;>\; a_i \;\Rightarrow\; F_i(x) \;=\; 0 \,].$$

Thus, an equation reformulation for this VI is given by

$$0 \;=\; \mathbf{F}^{\ell}_{\mathrm{FB}}(x) \;\equiv\; \begin{pmatrix} \psi_{\mathrm{FB}}(x_1 - a_1, F_1(x)) \\ \cdots \\ \psi_{\mathrm{FB}}(x_n - a_n, F_n(x)) \end{pmatrix},$$

which immediately suggests the merit function

$$\theta^{\ell}_{\mathrm{FB}}(x) \;\equiv\; \tfrac{1}{2}\, \mathbf{F}^{\ell}_{\mathrm{FB}}(x)^{T} \mathbf{F}^{\ell}_{\mathrm{FB}}(x).$$

If there are upper bounds instead of lower bounds, i.e. if

$$K \;\equiv\; \{\, x \in \mathbb{R}^n \;:\; x \leq b \,\},$$

where $b \in \mathbb{R}^n$, we can reason in a similar way. In this case a point $x \in K$ is a solution of the VI (K, F) if and only if

$$[\, x_i \;=\; b_i \;\Rightarrow\; F_i(x) \leq 0 \,] \quad \text{and} \quad [\, x_i \;<\; b_i \;\Rightarrow\; F_i(x) \;=\; 0 \,].$$

This immediately leads to the equation reformulation

$$0 \;=\; \mathbf{F}^{u}_{\mathrm{FB}}(x) \;\equiv\; \begin{pmatrix} \psi_{\mathrm{FB}}(b_1 - x_1, -F_1(x)) \\ \cdots \\ \psi_{\mathrm{FB}}(b_n - x_n, -F_n(x)) \end{pmatrix},$$

with the natural merit function

$$\theta^{u}_{\mathrm{FB}}(x) \;\equiv\; \tfrac{1}{2}\, \mathbf{F}^{u}_{\mathrm{FB}}(x)^{T} \mathbf{F}^{u}_{\mathrm{FB}}(x).$$

The functions $\mathbf{F}^{\ell}_{\mathrm{FB}}$, $\mathbf{F}^{u}_{\mathrm{FB}}$, $\theta^{\ell}_{\mathrm{FB}}$ and θ^{u}_{FB} enjoy properties similar to those of their respective counterparts $\mathbf{F}_{\mathrm{FB}}$ and θ_{FB}, which we do not repeat.

9.4.2 Mixed complementarity problems

Consider the MiCP (G, H) defined by the two mappings G and H mappings from $\mathbb{R}^n$ into $\mathbb{R}^{n_1}$ and $\mathbb{R}^{n_2}_{+}$, respectively, where $n_1 + n_2 = n$. This problem aims at finding a pair of vectors (u, v) in $\mathbb{R}^{n_1} \times \mathbb{R}^{n_2}$ such that

$$G(u, v) \;=\; 0, \quad u \text{ free}$$
$$0 \leq v \;\perp\; H(u, v) \geq 0.$$

The equivalent FB-based equation formulation for this problem is:

$$0 \;=\; \mathbf{F}^{\mathrm{mcp}}_{\mathrm{FB}}(u, v) \;\equiv\; \begin{pmatrix} G(u, v) \\ \psi_{\mathrm{FB}}(v_1, H_1(u, v)) \\ \cdots \\ \psi_{\mathrm{FB}}(v_{n_2}, H_{n_2}(u, v)) \end{pmatrix},$$

with the associated merit function:

$$\theta_{\rm FB}^{\rm mcp}(u,v) \equiv \tfrac{1}{2}\mathbf{F}_{\rm FB}^{\rm mcp}(u,v)^T\mathbf{F}_{\rm FB}^{\rm mcp}(u,v).$$

In what follows, we summarize, without proof, the key properties of the functions $\mathbf{F}_{\rm FB}^{\rm mcp}$ and $\theta_{\rm FB}^{\rm mcp}$.

9.4.1 Proposition. Let $G : \mathbb{R}^n \to \mathbb{R}^{n_1}$ and $H : \mathbb{R}^n \to \mathbb{R}^{n_2}$ be continuously differentiable. The following two statements are valid.

(a) $\mathbf{F}_{\rm FB}^{\rm mcp}$ is semismooth and $\theta_{\rm FB}^{\rm mcp}$ is continuously differentiable. If G and H have Lipschitz continuous derivatives, then $\mathbf{F}_{\rm FB}^{\rm mcp}$ is strongly semismooth.

(b) If $g_{\rm FB}(a,b)$ is a linear Newton approximation of $\psi_{\rm FB}(a,b)$, then a linear Newton approximation scheme of $\mathbf{F}_{\rm FB}^{\rm mcp}$ is given by

$$\mathcal{A}(u,v) \equiv \left\{ \begin{pmatrix} J_uG(u,v) & J_vG(u,v) \\ D_bJ_uH(u,v) & D_a + D_bJ_vH(u,v) \end{pmatrix} \right\}, \qquad (9.4.1)$$

where D_a and D_b are $n_2 \times n_2$ diagonal matrices whose i-th diagonal entries are the first and second component respectively of an element $\xi_i \in g_{\rm FB}(v_i, H_i(u,v))$. If $g_{\rm FB}(a,b) \subseteq \partial\psi_{\rm FB}(a,b)$, then

$$\nabla\theta_{\rm FB}^{\rm mcp}(u,v) = A^T\mathbf{F}_{\rm FB}^{\rm mcp}(u,v), \quad \forall\, A \in \mathcal{A}(u,v).$$

To examine further the properties of the functions $\mathbf{F}_{\rm FB}^{\rm mcp}$ and $\theta_{\rm FB}^{\rm mcp}$, we introduce some index sets that are the natural extension of those used in Section **9.1**. Specifically, let

$$\begin{aligned} \alpha &\equiv \{\, i : v_i = 0 < H_i(u,v) \,\}, \\ \beta &\equiv \{\, i : v_i = 0 = H_i(u,v) \,\}, \\ \gamma &\equiv \{\, i : v_i > 0 = H_i(u,v) \,\}, \\ \delta &\equiv \{\, 1, \ldots, n_2 \,\} \setminus (\, \alpha \cup \beta \cup \gamma \,). \end{aligned}$$

9.4.2 Proposition. Let $G : \mathbb{R}^n \to \mathbb{R}^{n_1}$ and $H : \mathbb{R}^n \to \mathbb{R}^{n_2}$ be continuously differentiable and let $(u,v) \in \mathbb{R}^{n_1+n_2}$ be given. Also let $\bar{\alpha} \equiv \gamma \cup \beta \cup \delta$ be the complement of α in $\{1, \ldots, n_2\}$. Assume that

(a) the submatrices

$$\begin{bmatrix} J_uG(u,v) & J_{v_{\tilde{\gamma}}}G(u,v) \\ J_uH_{\tilde{\gamma}}(u,v) & J_{v_{\tilde{\gamma}}}H_{v_{\tilde{\gamma}}}(u,v) \end{bmatrix}$$

are nonsingular for all $\tilde{\gamma}$ satisfying

$$\gamma \subseteq \tilde{\gamma} \subseteq \gamma \cup \beta,$$

(b) the Schur complement of

$$\begin{bmatrix} J_u G(u,v) & J_{v_\gamma} G(u,v) \\ J_u H_{v_\gamma}(u,v) & J_{v_\gamma} H_{v_\gamma}(u,v) \end{bmatrix}$$

in

$$\begin{bmatrix} J_u G(u,v) & J_{v_{\bar\alpha}} G(u,v) \\ J_u H_{v_{\bar\alpha}}(u,v) & J_{v_{\bar\alpha}} H_{v_{\bar\alpha}}(u,v) \end{bmatrix}$$

is a P_0 matrix.

The linear Newton scheme (9.4.1) is nonsingular at (u,v). □

In an analogous fashion we can extend the notion of an FB regular point for the MiCP (G,H) by taking into account the presence of the equation $G(u,v)=0$. Analogously to the index sets $\mathcal{C}$, $\mathcal{P}$ and $\mathcal{N}$ introduced before Definition **9.1.13** for the NCP, we define

$$\begin{aligned} \mathcal{C} &\equiv \{\, i : v_i \geq 0,\, H_i(u,v) \geq 0,\, v_i H_i(u,v) = 0 \,\} \\ \mathcal{P} &\equiv \{\, i : v_i > 0,\, H_i(u,v) > 0 \,\} \\ \mathcal{N} &\equiv \{\, 1, \ldots n_2 \} \setminus (\, \mathcal{C} \cup \mathcal{P} \,). \end{aligned}$$

The following definition extends Definition **9.1.13**.

9.4.3 Definition. A point $(u,v) \in \mathbb{R}^{n_1+n_2}$ is called *FB regular* (for the MiCP (G,H) or the merit function $\theta_{\mathrm{FB}}^{\mathrm{mcp}}$) if $J_u G(u,v)$ is nonsingular and if for every nonzero vector $z \in \mathbb{R}^{n_2}$ such that

$$z_{\mathcal{C}} = 0, \quad z_{\mathcal{P}} > 0, \quad z_{\mathcal{N}} < 0, \tag{9.4.2}$$

there exists a nonzero vector $y \in \mathbb{R}^{n_2}$ such that

$$y_{\mathcal{C}} = 0, \quad y_{\mathcal{P}} \geq 0, \quad y_{\mathcal{N}} \leq 0,$$

and

$$z^T (M(u,v)/J_u G(u,v)) y \geq 0, \tag{9.4.3}$$

where

$$M(u,v) \equiv \begin{bmatrix} JG(u,v) \\ JH(u,v) \end{bmatrix} \in \mathbb{R}^{n \times n}$$

and $M(u,v)/J_u G(u,v)$ is the Schur complement of $J_u G(u,v)$ in $M(u,v)$. □

Omitting the details, we note that if $J_u G(u,v)$ is nonsingular and the Schur complement $M(u,v)/J_u G(u,v)$ has the "signed S_0 property with respect to the v-variable", then (u,v) is an FB regular point for the MiCP (G,H). By Definition **9.4.3** and the formula for $\nabla\theta_{\rm FB}^{\rm mcp}(u,v)$ given in Proposition **9.4.1**, and similar to Theorem **9.1.14**, we can prove the following theorem.

9.4.4 Theorem. Let $G : \mathbb{R}^n \to \mathbb{R}^{n_1}$ and $H : \mathbb{R}^n \to \mathbb{R}^{n_2}$ be continuously differentiable and let $(u,v) \in \mathbb{R}^{n_1+n_2}$ be given. Assume that $(u,v) \in \mathbb{R}^n$ is a stationary point of $\theta_{\rm FB}^{\rm mcp}$. Then (u,v) is a solution of the MiCP (G,H) if and only if (u,v) is an FB regular point of $\theta_{\rm FB}^{\rm mcp}$. □

9.4.3 Box constrained VIs

Consider the box constrained VI (K,F) with the set K given by (1.1.7), which we restate below:

$$K \equiv \{\, x \in \mathbb{R}^n : a_i \le x_i \le b_i,\ i = 1, \ldots n \,\}, \tag{9.4.4}$$

where for each i, $-\infty \le a_i < b_i \le \infty$. This extended framework includes as particular cases all the complementarity problems discussed in this chapter. Similar to the case of one-sided bounds, we observe that a point x in K is a solution of a box constrained VI (K,F) if and only if (1.2.4) holds for every i; that is,

$$\begin{aligned} x_i = a_i \quad &\Rightarrow \quad F_i(x) \ge 0, \\ a_i < x_i < b_i \quad &\Rightarrow \quad F_i(x) = 0, \\ x_i = b_i \quad &\Rightarrow \quad F_i(x) \le 0. \end{aligned} \tag{9.4.5}$$

We are interested in defining a generalization of a C-function that captures the implications in (9.4.5). Specifically, given a pair of extended scalars τ (possibly equal to $-\infty$) and τ' (possibly equal to ∞) with $\tau < \tau'$, we call a function $\phi(\tau,\tau';\cdot,\cdot) : \mathbb{R}^2 \to \mathbb{R}$ a *B-function* (B for Box) if $\phi(\tau,\tau';r,s) = 0$ if and only if $\tau \le r \le \tau'$ and (r,s) satisfies

$$\begin{aligned} r = \tau \quad &\Rightarrow \quad s \ge 0, \\ \tau < r < \tau' \quad &\Rightarrow \quad s = 0, \\ r = \tau' \quad &\Rightarrow \quad s \le 0. \end{aligned}$$

Note that a B-function of the type $\phi(0,\infty;\cdot,\cdot)$ is just a C-function. Conversely, given any C-function ψ that satisfies the following sign reversal property:

$$\psi(a,b) \ge 0 \ \Rightarrow\ ab \le 0, \tag{9.4.6}$$

and any pair of extended scalars τ and τ' satisfying

$$-\infty \leq \tau < \tau' \leq \infty,$$

the function $\phi(\tau, \tau'; \cdot, \cdot)$ defined by

$$\phi(\tau, \tau'; r, s) \equiv \begin{cases} \psi(\tau' - r, -s) & \text{if } -\infty = \tau \text{ and } \tau' < \infty \\ \psi(r - \tau, \psi(\tau' - r, -s)) & \text{if } -\infty < \tau \text{ and } \tau' < \infty \\ \psi(r - \tau, s) & \text{if } -\infty < \tau \text{ and } \tau' = \infty \\ s & \text{if } -\infty = \tau \text{ and } \tau' = \infty \end{cases}$$

is a B-function. To verify this, it suffices to consider the case where both τ and τ' are finite. In this case, we have $\phi(\tau, \tau'; r, s) = 0$ if and only if $0 \leq r - \tau \perp \psi(\tau' - r, -s) \geq 0$. If $\tau < r$, then $\psi(\tau' - r, -s) = 0$, which implies $0 \leq \tau' - r \perp s \leq 0$. If $\tau' > r$, then $s = 0$. If $\tau' = r$, then $s \leq 0$. If $\tau = r$, then $\psi(\tau' - r, -s) \geq 0$ and $\tau' - r > 0$. Hence by the assumed sign reversal property of ψ, we deduce $s \geq 0$. Consequently, $\phi(\tau, \tau'; \cdot, \cdot)$ is a B-function. Incidentally, the C-functions ψ_{FB}, ψ_{KK} and ψ_{CCK} all satisfy (9.4.6); but the min function does not satisfy this sign property.

Given the box constrained VI (K, F) with K being the rectangle (9.4.4) and given any B-function $\phi(a_i, b_i; \cdot, \cdot)$ associated with the bound (a_i, b_i) for $i = 1, \ldots, n$, the system of equations:

$$0 = \mathbf{F}_\phi^{\text{box}}(x) \equiv \begin{pmatrix} \phi(a_1, b_1; x_i, F_i(x)) \\ \vdots \\ \phi(a_n, b_n; x_n, F_n(x)) \end{pmatrix}$$

is clearly equivalent to the VI (K, F). In turn, corresponding to this reformulation of the VI, we may define the merit function

$$\theta_\phi^{\text{box}}(x) \equiv \tfrac{1}{2} (\mathbf{F}_\phi^{\text{box}}(x))^T \mathbf{F}_\phi^{\text{box}}(x),$$

which can be used as the basis for the design of iterative descent methods for solving the box constrained VI (K, F).

The above construction of a B-function starting from a C-function is not completely satisfactory from a computational point of view. Indeed the nesting of C-functions very easily gives rise to a highly nonlinear function. For example, the nesting of two FB functions involves the square root of a term that in turn contains a square root. Therefore we are interested in developing other equation reformulations for a box constrained VI that take into account directly the structure of the problem and that avoid the

drawback just mentioned. For simplicity, we assume in the rest of the section that all lower and upper bounds are finite. The results below can easily be modified to deal with mixed finite and infinite bounds.

Consider a B-function derived from the mid function. For a given pair of (finite) scalars $\tau < \tau'$, $\mathrm{mid}(\tau, \tau'; \cdot) : \mathbb{R} \to \mathbb{R}$ is defined by:

$$\mathrm{mid}(\tau, \tau', y) \equiv \Pi_{[\tau,\tau']}(y) = \begin{cases} \tau & \text{if } \tau > y \\ y & \text{if } y \in [\tau, \tau'] \\ \tau' & \text{if } \tau' < y. \end{cases}$$

It is easy to verify that the function $\phi(\tau, \tau'; \cdot, \cdot)$ defined as

$$\phi(\tau, \tau'; r, s) \equiv r - \mathrm{mid}(\tau, \tau'; r - s)$$

is a piecewise affine B-function. Similar to the min function, the mid function naturally leads to a locally convergent Newton method for solving the box constrained VI; it suffices to follow the steps in Example **7.2.16**. Like the function $\theta_{\min}$, the merit function for the box constrained VI obtained from the mid reformulation is nonsmooth and hence it is difficult to globalize the resulting local Newton method directly.

In what follows, we define an alternative piecewise function that has more desirable properties than the mid function when it comes to designing globally convergent methods for solving the box constrained VI. For one thing, the defined function will be differentiable everywhere except at points that correspond to the upper and lower bounds. This feature is consistent with the FB function, which is differentiable everywhere except at points that correspond to the lower bounds (namely, zero). In addition, the differentiability property of the FB merit function remains valid for the corresponding merit function defined below.

Specifically, given an arbitrary pair of (finite) scalars $\tau < \tau'$, define

$$\phi_{\mathrm{Q}}(\tau, \tau'; r, s) \equiv$$
$$\begin{cases} r - \tau & \text{if } r \le \tau \text{ and } s \ge 0 \\ r - \tau + s - \sqrt{(r-\tau)^2 + s^2} & \text{if } \tau \le r \le \tau' \text{ and } s \ge 0 \\ r - \tau - \sqrt{(r-\tau)^2 + s^2} + \sqrt{(r-\tau')^2 + s^2} & \text{if } r \ge \tau' \text{ and } s \ge 0 \\ r - \tau' - \sqrt{(r-\tau)^2 + s^2} + \sqrt{(r-\tau')^2 + s^2} & \text{if } r \le \tau \text{ and } s \le 0 \\ r - \tau' + s + \sqrt{(r-\tau')^2 + s^2} & \text{if } \tau \le r \le \tau' \text{ and } s \le 0 \\ r - \tau' & \text{if } r \ge \tau' \text{ and } s \le 0. \end{cases} \tag{9.4.7}$$

It is easy to check that this function is continuous and is a B-function. The pieces of $\phi_{\mathrm{Q}}(\tau, \tau'; \cdot, \cdot)$ are rather simple, but they are not all differentiable. Thus $\phi_{\mathrm{Q}}(\tau, \tau'; \cdot, \cdot)$ is not a PC^1 function. Nevertheless, the following proposition asserts that this function is strongly semismooth on $\mathbb{R}^2$.

9.4.5 Proposition. For any two scalars $\tau < \tau'$, $\phi_{\mathrm{Q}}(\tau, \tau'; \cdot, \cdot)$ is a strongly semismooth B-function. Thus, if $F : \mathbb{R}^n \to \mathbb{R}^n$ is continuously differentiable then for any two vectors $a \leq b$, the function

$$\mathbf{F}_{\mathrm{Q}}^{\mathrm{box}}(x) \equiv \begin{pmatrix} \phi_{\mathrm{Q}}(a_1, b_1; x_i, F_i(x)) \\ \vdots \\ \phi_{\mathrm{Q}}(a_n, b_n; x_n, F_n(x)) \end{pmatrix}$$

is semismooth on $\mathbb{R}^n$; moreover, if the Jacobian of F is locally Lipschitz continuous then $\mathbf{F}_{\mathrm{Q}}^{\mathrm{box}}$ is strongly semismooth.

Proof. It suffices to show that $\phi_{\mathrm{Q}}(\tau, \tau'; \cdot, \cdot)$, which we write as ϕ for simplicity, is strongly semismooth; the remaining assertions then follow from Theorem **7.5.17**. It is straightforward, albeit lengthy, to check that ϕ is continuously differentiable with a locally Lipschitz Jacobian at every point except $(\tau, 0)$ and $(\tau', 0)$. We now prove that ϕ is strongly semismooth at $(\tau, 0)$, the proof for the point $(\tau', 0)$ is similar. Consider the function

$$\phi_1(r, s) \equiv \begin{cases} 0 & \text{if } s \geq 0 \\ r - \tau' + \sqrt{(r - \tau')^2 + s^2} & \text{if } s \leq 0 \end{cases}$$

and observe that it is continuously differentiable with a Lipschitz Jacobian near $(\tau, 0)$ and therefore strongly semismooth there. Define the function

$$\phi_2(r, s) \equiv \phi_{\mathrm{Q}}(r, s) - \phi_1(r, s),$$

and observe that it is continuous near $(\tau, 0)$ and C^1 there except possibly at $(\tau, 0)$. Furthermore $\phi_2(r, s)$ is locally homogeneous at $(\tau, 0)$, i.e., there is a neighborhood of $(\tau, 0)$, such that if both $(\tau + r, s)$ and $(\tau + tr, ts)$ belong to this neighborhood, then $\phi_2(\tau + tr, ts) = t\phi_2(\tau + r, s)$. Since ϕ_1 is strongly semismooth, we only need to check the strong semismoothness of ϕ_2. Since ϕ_2 is the difference of two Lipschitz contiuous functions near $(\tau, 0)$, it is Lipschitz continuous there. Furthermore, since ϕ_2 is C^1 near $(\tau, 0)$ except, possibly at $(\tau, 0)$ itself, and locally homogeneous at $(\tau, 0)$, it is directionally differentiable at $(\tau, 0)$; moreover

$$\phi_2'((\tau, 0); (r, s)) = \phi_2(\tau + r, s) = \nabla\phi_2(\tau + r, s)^T \begin{pmatrix} r \\ s \end{pmatrix},$$

provided that $\|(r, s)\|$ is sufficiently small. But this shows that ϕ_2 is strongly semismooth at $(\tau, 0)$ and concludes the proof. □

Although the function $\phi_{\mathrm{Q}}(\tau, \tau'; \cdot, \cdot)$ is defined for finite τ and τ', it is natural to ask what happens to this function when $\tau = 0$ and τ' approaches ∞. It is easy to see that $\lim_{\tau' \to \infty} \phi_{\mathrm{Q}}(0, \tau'; r, s) = \psi_{\mathrm{Q}}^{\infty}(r, s)$, where

$$\psi_{\mathrm{Q}}^{\infty}(r, s) \equiv \begin{cases} r & \text{if } r \leq 0 \text{ and } s \geq 0, \\ r + s - \sqrt{r^2 + s^2} & \text{if } (r, s) \geq 0, \\ -\sqrt{r^2 + s^2} & \text{if } (r, s) \leq 0, \\ s & \text{if } 0 \leq r \text{ and } s \leq 0. \end{cases}$$

The function $\psi_{\mathrm{Q}}^{\infty}(r, s)$ is a C-function; it is equal to $\min(r, s)$ in the second and fourth orthant and equal to $-\psi_{\mathrm{FB}}(r, s)$ in the first orthant. Thus $\psi_{\mathrm{Q}}^{\infty}$ can be viewed as a certain mixture of the min function and the FB function.

It is useful to write down explicitly the gradient of $\phi_{\mathrm{Q}}(\tau, \tau'; \cdot, \cdot)$, which, as noted in the above proof, exists everywhere except at $(\tau, 0)$ and $(\tau', 0)$. For notational simplicity, we write this function as ϕ_{Q}, omitting its dependence on the pair (τ, τ'):

$$\nabla \phi_{\mathrm{Q}}(r, s) =$$

$$\begin{cases} \begin{pmatrix} 1 \\ 0 \end{pmatrix} & \begin{cases} \text{if } r \leq \tau, s \geq 0 \\ \text{but } (r, s) \neq (\tau, 0) \end{cases} \\ \begin{pmatrix} 1 \\ 0 \end{pmatrix} & \begin{cases} \text{if } r \geq \tau', s \leq 0 \\ \text{but } (r, s) \neq (\tau', 0) \end{cases} \\ \begin{pmatrix} 1 - \dfrac{r - \tau}{\sqrt{(r - \tau)^2 + s^2}} \\ 1 - \dfrac{s}{\sqrt{(r - \tau)^2 + s^2}} \end{pmatrix} & \begin{cases} \text{if } \tau < r \leq \tau' \\ \text{and } s \geq 0 \\ \text{but } (r, s) \neq (\tau', 0) \end{cases} \\ \begin{pmatrix} 1 - \dfrac{r - \tau}{\sqrt{(r - \tau)^2 + s^2}} + \dfrac{r - \tau'}{\sqrt{(r - \tau')^2 + s^2}} \\ -\dfrac{s}{\sqrt{(r - \tau)^2 + s^2}} + \dfrac{s}{\sqrt{(r - \tau')^2 + s^2}} \end{pmatrix} & \begin{cases} \text{if } r > \tau', s > 0 \\ \text{or } r < \tau, s < 0 \end{cases} \\ \begin{pmatrix} 1 + \dfrac{r - \tau'}{\sqrt{(r - \tau')^2 + s^2}} \\ 1 + \dfrac{s}{\sqrt{(r - \tau')^2 + s^2}} \end{pmatrix} & \begin{cases} \text{if } \tau \leq r < \tau' \\ \text{and } s \leq 0 \\ \text{but } (r, s) \neq (\tau, 0). \end{cases} \end{cases}$$

Note that the above expression excludes the gradient of ϕ_{Q} at the non-

differentiable points $(\tau, 0)$ and $(\tau', 0)$. Wherever the gradient exists, its (Euclidean) norm is bounded above by the constant 3; this confirms that $\mathrm{Jac}\,\phi_{\mathrm{Q}}$ is well defined everywhere. Contrary to the case of a PC^1 function however, $\mathrm{Jac}\,\phi_{\mathrm{Q}}(\tau, 0)$ and $\mathrm{Jac}\,\phi_{\mathrm{Q}}(\tau', 0)$ is easily seen to consist of an infinite number of elements. Indeed, letting

$$\partial\mathbb{B}^n_+ \equiv \{\, x \in \mathbb{R}^n_+ : \|\, x \,\|_2 = 1 \,\}$$

be the intersection of the unit Euclidean ball and the nonnegative orthant, we can derive from the expression of $\nabla\phi_{\mathrm{Q}}$ that

$$\mathrm{Jac}\,\phi_{\mathrm{Q}}(\tau, 0) = \{\, (1-\rho, 1-\xi) : (\rho, \xi) \in \partial\mathbb{B}^2_+ \,\} \cup \partial\mathbb{B}^2_+.$$

A similar expression holds also at the point $(\tau', 0)$.

By applying Theorem **7.5.17**, we obtain a linear Newton approximation scheme of $\mathbf{F}^{\mathrm{box}}_{\mathrm{Q}}$ as follows:

$$\mathcal{A}_{\mathrm{Q}}(x) \equiv \{\, A \in \mathbb{R}^{n\times n} : A = D_a(x) + D_b(x)JF(x) \,\},$$

where D_a and D_b are $n \times n$ diagonal matrices whose i-th diagonal elements are, respectively, the first and second component of an element in $\mathrm{Jac}\,\phi_{\mathrm{Q}}(x_i, F_i(x))$. By observing that ϕ_{Q} is continuously differentiable except at points where it is zero and arguing exactly in the way as in the proof of Proposition **9.1.4**(c) (see also Proposition **9.3.2**), we deduce that the merit function

$$\theta^{\mathrm{box}}_{\mathrm{Q}} \equiv \tfrac{1}{2}\,(\,\mathbf{F}^{\mathrm{box}}_{\mathrm{Q}}\,)^T(\,\mathbf{F}^{\mathrm{box}}_{\mathrm{Q}}\,)$$

is continuously differentiable everywhere on $\mathbb{R}^n$ and

$$\nabla\theta^{\mathrm{box}}_{\mathrm{Q}}(x) = A^T\mathbf{F}^{\mathrm{box}}_{\mathrm{Q}}(x), \qquad \forall\, A \in \mathcal{A}_{\mathrm{Q}}(x).$$

As Proposition **9.4.6** shows, the merit function $\theta^{\mathrm{box}}_{\mathrm{Q}}$ is always coercive, regardless of any property of F. This is a remarkable difference from the merit function θ_{FB}, which is the counterpart of $\theta^{\mathrm{box}}_{\mathrm{Q}}$ for the NCP. One could relate the coercivity of $\theta^{\mathrm{box}}_{\mathrm{Q}}$ to the boundedness of the box K. In particular, as a consequence of the coercivity, the function $\theta^{\mathrm{box}}_{\mathrm{Q}}(x)$ always has an unconstrained minimum (provided that F is a continuous function), which must necessarily be a solution of the box constrained VI (K, F). The latter statement follows easily from the fact that the box constrained VI must have a solution and from the equivalent reformulation $\mathbf{F}^{\mathrm{box}}_{\mathrm{Q}}(x) = 0$ of this VI. We formally state these properties in the following result.

9.4.6 Proposition. Let K be a compact rectangle in $\mathbb{R}^n$ and F be a continuous mapping from $\mathbb{R}^n$ into itself. The function $\theta_{\mathrm{Q}}^{\mathrm{box}}$ is coercive; that is,

$$\lim_{\|x\|\to\infty} \theta_{\mathrm{Q}}^{\mathrm{box}}(x) = \infty.$$

Thus $\theta_{\mathrm{Q}}^{\mathrm{box}}(x)$ always has an unconstrained minimum, which must necessarily be a solution of the box constrained VI (K, F).

Proof. Let $\{x^k\}$ be an arbitrary unbounded sequence. We want to show that

$$\lim_{k\to\infty} \theta_{\mathrm{Q}}^{\mathrm{box}}(x^k) = \infty.$$

In turn, it is sufficient to show that, for at least one i, $\{(\mathbf{F}_{\mathrm{Q}}^{\mathrm{box}})_i(x^k)\}$ is unbounded. Since $\{x^k\}$ is unbounded we may assume without loss of generality that there exists an index i for which either $\{x_i^k\} \to -\infty$ or $\{x_i^k\} \to \infty$. In either case, for all k sufficiently large, the function $(\mathbf{F}_{\mathrm{Q}}^{\mathrm{box}})_i(x^k)$ is equal to one of two different expressions according to whether $F_i(x^k)$ is positive or nonpositive. It is then easy to check, using the definition of $\phi_{\mathrm{Q}}(a_i, b_i; \cdot, \cdot)$, that $\{(\mathbf{F}_{\mathrm{Q}}^{\mathrm{box}})_i(x^k)\}$ is unbounded. The last statement of the proposition does not require a proof. □

It is now easy to continue along the same line as in Section **9.3** to obtain conditions for the nonsingularity of the above Newton scheme. In fact once again the structure of the elements in the Newton approximation scheme is the same as before. We only need to extend in a suitable way the definition of the index sets given before Theorem **9.3.4**; namely,

$$\begin{aligned}
\alpha &\equiv \{\, i : x_i = a_i,\ 0 < F_i(x) \,\} \cup \{\, i : x_i = b_i,\ F_i(x) < 0 \,\},\\
\beta &\equiv \{\, i : x_i = a_i,\ 0 = F_i(x) \,\} \cup \{\, i : x_i = b_i,\ F_i(x) = 0 \,\},\\
\gamma &\equiv \{\, i : a_i < x_i < b_i,\ 0 = F_i(x) \,\},\\
\delta &\equiv \{\, 1, \ldots, n \,\} \setminus (\, \alpha \cup \beta \cup \gamma \,).
\end{aligned}$$

The following theorem can be proved in the same way as Theorem **9.1.23**.

9.4.7 Theorem. Suppose that $F : \mathbb{R}^n \to \mathbb{R}^n$ is continuously differentiable in a neighborhood of a given vector $x \in \mathbb{R}^n$. Let $M \equiv JF(x)$; also let $\bar{\alpha} \equiv \gamma \cup \beta \cup \delta$ be the complement of α in $\{1, \ldots, n\}$. Assume that

(a) the submatrices $M_{\tilde{\gamma}\tilde{\gamma}}$ are nonsingular for all $\tilde{\gamma}$ satisfying

$$\gamma \subseteq \tilde{\gamma} \subseteq \gamma \cup \beta,$$

(b) the Schur complement of $M_{\gamma\gamma}$ in $M_{\bar{\alpha}\bar{\alpha}}$ is a P_0 matrix.

The Newton schemes $\mathcal{A}_Q$ is nonsingular at x. □

Similar to Corollary **9.1.24** for the case of the NCP (F), we can show that if x is a solution of the box constrained VI (K, F), then the conditions (a) and (b) of Theorem **9.4.7** are implied by the strong regularity of x. The proof of this assertion is left as an exercise.

Thanks to the structural similarities between the merit function $\theta_Q^{\rm box}$ and the FB merit function $\theta_{\rm FB}$ for the NCP, we can naturally extend the notion of a regular point to obtain a necessary and sufficient condition for a stationary point of $\theta_Q^{\rm box}$ to be a solution of VI (K, F). Specifically, we define the following index sets:

$$\begin{aligned}\mathcal{C} &\equiv \{i : x_i = l_i,\, F_i(x) \geq 0\} \cup \{i : l_i \leq x_i \leq u_i,\, F_i(x) = 0\} \cup \\ &\qquad \{i : x_i = u_i,\, F_i(x) \leq 0\}, \\ \mathcal{R} &\equiv \{1, \ldots n\} \setminus \mathcal{C}.\end{aligned}$$

We further partition the index set $\mathcal{R}$ as follows:

$$\begin{aligned}\mathcal{P} &= \{i : l_i < x_i \leq u_i, F_i(x) > 0\} \cup \{i : u_i < x_i\} \\ \mathcal{N} &= \{i : l_i \leq x_i < u_i, F_i(x) < 0\} \cup \{i : x_i < l_i\}.\end{aligned}$$

Similar to (9.1.14), we have the following implications:

$$\begin{aligned}(D_a \mathbf{F}_{\rm FB})_i > 0 \quad &\Leftarrow \quad i \in \mathcal{P} \quad \Rightarrow \quad (D_b \mathbf{F}_{\rm FB})_i \geq 0, \\ (D_a \mathbf{F}_{\rm FB})_i = 0 \quad &\Leftarrow \quad i \in \mathcal{C} \quad \Rightarrow \quad (D_b \mathbf{F}_{\rm FB})_i = 0, \\ (D_a \mathbf{F}_{\rm FB})_i < 0 \quad &\Leftarrow \quad i \in \mathcal{N} \quad \Rightarrow \quad (D_b \mathbf{F}_{\rm FB})_i \leq 0.\end{aligned} \tag{9.4.8}$$

Note that x is a solution of the VI (K, F) if and only if $\mathcal{R}$ is empty.

9.4.8 Definition. A point $x \in \mathbb{R}^n$ is said to be *box-regular* if for every vector $z \neq 0$ such that

$$z_{\mathcal{C}} = 0, \quad z_{\mathcal{P}} > 0, \quad z_{\mathcal{N}} < 0,$$

there exists a nonzero vector $y \in \mathbb{R}^n$ such that

$$y_{\mathcal{C}} = 0, \quad y_{\mathcal{P}} \geq 0, \quad y_{\mathcal{N}} \leq 0,$$

and

$$z^T JF(x) y \geq 0.$$

Using this definition and the implications (9.4.8) we can prove the following result by the same argument used to prove Theorem **9.1.14**.

9.4.9 Theorem. Let K be a compact rectangle in $\mathbb{R}^n$ and F be a continuous differentiable mapping from $\mathbb{R}^n$ into itself. Let $x \in \mathbb{R}^n$ be a stationary point of θ_Q^{box}. Then x is a solution of the VI (K, F) if and only if x is a box-regular point. □

Finally, we can establish a result simillar to Corollary **9.1.16** that gives a sufficient condition for every stationary point of θ_Q^{box} to be a solution of the box constrained VI (K, F). The details are not repeated. In particular, it follows that every stationary point of θ_Q^{box} relative to a P_0 box constrained VI (K, F) with F continuously differentiable is a solution of the VI.

9.5 Exercises

9.5.1 In this exercise we analyze some differentiability properties of the C-functions ψ_{Man}, whose definition we recall here for convenience: Let $\zeta : \mathbb{R} \to \mathbb{R}$ be any strictly increasing function with $\zeta(0) = 0$. The C-function ψ_{Man} is defined by

$$\psi_{\text{Man}}(a, b) \equiv \zeta(|a - b|) - \zeta(b) - \zeta(a), \quad (a, b) \in \mathbb{R}^2.$$

Consider an NCP (F) with F continuously differentiable. Prove the following assertions.

(a) If ζ is continuously differentiable and $\zeta'(0) = 0$, $\mathbf{F}_{\text{Man}}$ is continuously differentiable.

(b) Assume in addition that $\zeta'(t) > 0$ for every positive t. If x^* is a nondegenerate solution of the NCP (F) and $JF(x^*)$ is a nondegenerate matrix, $J\mathbf{F}_{\text{Man}}(x^*)$ is nonsingular.

9.5.2 Let $F : \mathbb{R}^n \to \mathbb{R}^n$ be continuous and consider the NCP (F). Define the function $\theta_{\text{LTKYF}} : \mathbb{R}^n \to \mathbb{R}_+$ by

$$\theta_{\text{LTKYF}}(x) \equiv \phi_0(x^T F(x)) + \sum_{i=1}^{n} \phi_i(-x_i, -F_i(x)),$$

where $\phi_0 \in \mathbf{\Phi}_1$ and $\phi_i \in \mathbf{\Phi}_2$ for all i (see Proposition **1.5.4** for the definition of $\mathbf{\Phi}_1$ and $\mathbf{\Phi}_2$).

(a) Show that θ_{LTKYF} is an unconstrained merit function for the NCP (F).

(b) Assume that $F(x) = Mx + q$, with M positive semidefinite and that the ϕ_i, $i = 0, \ldots, n$, are convex. Prove that θ_{LTKYF} is also convex.

9.5.3 Let $\sigma : \mathbb{R}^2 \to \mathbb{R}$ be a positively homogeneous convex function whose level curve $\{(a,b) : \sigma(a,b) = 1\}$ intersects the line $\{(a,b) : a+b = 1\}$ only at the points $(1,0)$ and $(0,1)$. Show that $\phi(a,b) \equiv \sigma(a,b) - (a+b)$ is a C-function. Prove furthermore that $\psi(a,b) \equiv (\phi(-a,-b))_+)^p$, with $p \geq 1$ is convex and belongs to $\mathbf{\Phi}_2$.

9.5.4 Let $F : \mathbb{R}^n_+ \to \mathbb{R}^n$ be locally Lipschitz and monotone. Show that $0 \in \partial\theta_{\rm FB}(x)$ if and only if x is a solution of the NCP (F). (Hint: use Exercise **7.6.12**.)

9.5.5 Let F be a continuous mapping from $\mathbb{R}^n$ into itself. By extending the proof of Corollary **9.1.28**, show that if for all $y \in \mathbb{R}^n$,

$$\lim_{\|x\| \to \infty} \max_{1 \leq i \leq n} \frac{(\, x_i - y_i \,)\,(\, F_i(x) - F_i(y) \,)}{\| \, x - y \, \|} \;=\; \infty,$$

then

$$\lim_{\|x\| \to \infty} \| \, \min(x, F(x)) \, \| \;=\; \infty.$$

Hence it follows that

$$\lim_{\|x\| \to \infty} \theta_{\rm CCK}(x) \;=\; \infty.$$

See Exercise **10.5.12** for an extension to a VI.

9.5.6 Let $F : \mathbb{R}^n \to \mathbb{R}^n$ be a strongly monotone function with strong monotonicity modulus $c > 0$. Consider the NCP (F). Given a point $x \in \mathbb{R}^n$, define a direction $d(x) \in \mathbb{R}^n$, by

$$d_i(x) \;\equiv\; -\frac{\partial \psi_{\rm FB}}{\partial b}(x_i, F_i(x)).$$

Show that

(a) $\nabla\theta_{\rm FB}(x)^T d(x) \;\leq\; -c \| \, d(x) \, \|^2$;

(b) $\nabla\theta_{\rm FB}(x)^T d(x) \;=\; 0$ if and only if x is the solution of the NCP (F).

(Hint: For (a), show preliminarily that

$$\frac{\partial \psi_{\rm FB}}{\partial a}(a,b)\, \frac{\partial \psi_{\rm FB}}{\partial b}(a,b) \;\geq\; 0, \quad \forall\, (a,b) \;\in\; \mathbb{R}^2,$$

and for (b), that

$$\frac{\partial \psi_{\rm FB}}{\partial a}(a,b)\, \frac{\partial \psi_{\rm FB}}{\partial b}(a,b) \;=\; 0$$

if and only if $\psi_{\rm FB}(a,b) = 0$.)

9.5.7 In the previous exercise we saw that if F is strongly monotone it is possible to build a descent direction for $\theta_{\rm FB}$ without using the derivatives of F. It is natural then to consider an algorithm for the minimization of $\theta_{\rm FB}$ that uses such a direction. To this end consider the following iteration. Given a point x^k, set $d^k \equiv d(x^k)$. Find the smallest nonnegative integer i_k such that with $i = i_k$,

$$\theta_{\rm FB}(x^k + 2^{-i} d^k) \;\leq\; \theta_{\rm FB}(x^k) + \gamma\, 2^{-2i}\, \nabla\theta_{\rm FB}(x^k)^T d^k;$$

set $x^{k+1} \equiv x^k + 2^{-2i} d^k$. Show that the corresponding sequence $\{x^k\}$ generated starting from any $x^0 \in \mathbb{R}^n$ converges to the unique solution of the NCP (F). (Hint: revisit the proof of Theorem **9.1.10** and take into account Remark **9.1.30**.)

9.5.8 Consider the algorithm described in Exercise **9.5.7**. Prove that if $x^0 \in \mathbb{R}^n_+$, then $x^k \in \mathbb{R}^n_+$ for every k. This means that if $x^0 \in \mathbb{R}^n_+$ the algorithm can be viewed as a (very simple) feasible algorithm for the minimization of $\theta_{\rm FB}$ on the nonnegative orthant $\mathbb{R}^n_+$. (Hint: show first that if $a \geq 0$ then $a - \partial\psi_{\rm FB}/\partial b \geq 0$. Using also this fact, prove then that $x^k + d^k \geq 0$ for all k.)

9.5.9 Consider the nonnegatively constrained minimization problem

$$\begin{array}{ll} \text{minimize} & \theta(x) \\ \text{subject to} & x \in \mathbb{R}^n_+, \end{array}$$

where the function θ is assumed to be strongly convex and continuously differentiable. Define a minimization algorithm for this problem by the following iterative procedure. Given a feasible point x^k, set $d^k = d(x^k)$, where $d(x^k)$ is the direction defined in Exercise **9.5.6** with reference to the NCP $(\nabla\theta)$. Find the smallest nonnegative integer i_k such that, with $i = i_k$,

$$\theta(x^k + 2^{-i} d^k) \;\leq\; \theta(x^k) + \gamma\, 2^{-2i}\, \nabla\theta_{\rm FB}(x^k)^T d^k;$$

set $x^{k+1} \equiv x^k + 2^{-2i} d^k$.

(a) Show that $\nabla\theta(x^k)^T d^k \leq 0$ and that $\nabla\theta(x^k)^T d^k = 0$ if and only if x^k is a solution of the constrained problem. (Hint: prove and then use the following result: if $a \geq 0$, then $b(\partial\psi_{\rm FB}(a,b)/\partial b) \geq 0$ for every b and equality holds if and only if $b \geq 0$ and $ab = 0$.)

(b) Show that $\{x^k\}$ is feasible and converges to the unique solution of the minimization problem. (Hint: use (a), Exercise **9.5.8**, the continuity of $d(x)$, and standard arguments.)

9.5.10 Let $F : \mathbb{R}^n \to \mathbb{R}^n$ be given. Consider the function

$$\theta_{\rm S}(x) \equiv \sum_{i=1}^{n} \psi_{\rm S}(x_i, F_i(x)),$$

where

$$\psi_{\rm S}(a, b) \equiv a(b_+)^2 + ((-b)_+)^2.$$

Show that $\theta_{\rm S}$ is a merit function on $\mathbb{R}^n_+$ for the NCP (F). Show further that if x^* is a nondegenerate solution of the NCP (F), then x^* is a KKT point of the problem

$$\begin{array}{ll} \text{minimize} & \theta_{\rm S}(x) \\ \text{subject to} & x \in \mathbb{R}^n_+, \end{array}$$

at which strict complementarity holds. (Hint: to prove the second assertion, show that $\frac{\partial \psi_{\rm S}}{\partial b}(x_i^*, F_i(x^*)) = 0$ for every i.) Note that this behavior is sharply different from that of the FB merit function, for which a solution x^* is always a "totally degenerate" KKT point of

$$\begin{array}{ll} \text{minimize} & \theta_{\rm FB}(x) \\ \text{subject to} & x \in \mathbb{R}^n_+ \end{array}$$

in the sense that $\nabla \theta_{\rm FB}(x) = 0$.

9.5.11 Consider the function $\theta_{\rm S}$ introduced in Exercise **9.5.10**. Prove that if F is continuous and monotone and the NCP (F) is strictly feasible, then the level sets

$$\{\, x \in \mathbb{R}^n_+ \,:\, \theta_{\rm S}(x) \leq \eta \,\}$$

are bounded for every η. (Hint: argue by contradiction and assume there is an unbounded sequence $\{x^k\}$ such that $\theta_{\rm S}(x^k)$ is bounded. Prove preliminarily that $\psi_{\rm S}(a, b)$ goes to infinity if either $b \to -\infty$ or $ab \to \infty$. Next use the monotonicity of F and the strict feasibility to show $\{(x^k)^T F(x^k)\}$ goes to infinity.)

9.5.12 Consider the uni-dimensional NCP (F) with $F(x) = -x$. Determine its unique solution x^*. Prove that x^* is not strongly regular and that $\partial \mathbf{F}_{\rm FB}(x^*)$ contains singular matrices while $\operatorname{Jac} \mathbf{F}_{\rm FB}(x^*)$ does not. This shows that if we apply Theorem **9.1.29** to this problem we cannot prove superlinear convergence if we take $T(x) = \partial \mathbf{F}_{\rm FB}(x)$, but superlinear convergence is guaranteed for the choice $T(x) = \operatorname{Jac} \mathbf{F}_{\rm FB}(x)$.

9.5.13 Show that a matrix $M \in \mathbb{R}^{n \times n}$ is strictly semicopositive if and only if M and all its principal submatrices are S.

9.5.14 Let $F : \mathbb{R}^n \to \mathbb{R}^n$ be continuously differentiable. Show that if x is an unconstrained stationary point of the merit function $\theta_{\min}$, then x is a solution of the NCP (F) if and only if the following MLCP has a solution in the variable d:

$$\left.\begin{aligned} x_i + d_i &= 0, \quad \forall\, i \in \mathcal{I}_x \\ x_i + d_i &\geq 0 \\ F_i(x) + \nabla F_i(x)^T d &\geq 0 \\ (\, x_i + d_i \,)\,(\, F_i(x) + \nabla F_i(x)^T d \,) &= 0 \end{aligned}\right\} \forall\, i \in \mathcal{I}_=$$

$$F_i(x) + \nabla F_i(x)^T d = 0, \quad \forall\, i \in \mathcal{I}_F,$$

where

$$\begin{aligned} \mathcal{I}_x &\equiv \{\, i : x_i < F_i(x) \,\} \\ \mathcal{I}_= &\equiv \{\, i : x_i = F_i(x) \,\} \\ \mathcal{I}_F &\equiv \{\, i : x_i > F_i(x) \,\}. \end{aligned}$$

9.5.15 Consider the 2-dimensional NCP (F) with

$$F(x_1, x_2) \equiv \begin{pmatrix} 1 \\ \dfrac{1}{(\, x_1 + x_2 - 1 \,)^2 + 1} - 1 \end{pmatrix}, \quad (\, x_1, x_2 \,) \in \mathbb{R}^2.$$

This NCP has a unique solution $(0, 1)$. Let $\{x^k\}$ be the sequence defined by

$$x^k = (\, x_1^k, x_2^k \,) \equiv (\, 0, k \,), \quad \forall\, k.$$

Show that $\nabla \theta_{\mathrm{FB}}(x^k) \to 0$ but $\theta_{\mathrm{FB}}(x^k) \to \frac{1}{2}$ as $k \to \infty$. Give two sequences $\{z^k\}$ and $\{v^k\}$ satisfying (9.1.22)–(9.1.24) but for which

$$\lim_{k \to \infty} (\, v^k + JF(x^k) z^k \,) = 0.$$

9.5.16 Let $F : \mathbb{R}^n \to \mathbb{R}^n$ be continuously differentiable and let $\psi(a, b)$ be a semismooth C-function on $\mathbb{R}^2$ such that the non-differentiable points of ψ are a subset of the zeros of ψ. Let x be an unconstrained stationary point of the merit function θ_ψ. Suppose that for some pair of diagonal matrices (D_a, D_b) as specified in Proposition **9.3.1**,

$$(\, D_a \,)_{ii}\, (\, D_b \,)_{ii} \geq 0, \quad \forall\, i = 1, \ldots, n,$$

and

$$(\, D_a \,)_{ii}\, (\, D_b \,)_{ii} = 0 \;\Rightarrow\; \psi(x_i, F_i(x)) = 0.$$

Show that x solves the NCP (F) if and only if x is FB regular.

9.5.17 Let $F : \mathbb{R}^n \to \mathbb{R}^n$ be a continuous P_0 function. Suppose that the NCP (F) has a nonempty bounded solution set. Show that there exists a constant $\eta > 0$ such that the level set

$$\{ x \in \mathbb{R}^n : \theta_{\mathrm{FB}}(x) \leq \eta \}$$

is bounded. Thus every asymptotically minimizing sequence of θ_{FB} must be bounded.

9.5.18 Let $F(x) \equiv A^T G(Ax) + b$ be a monotone composite function, where $G : \mathbb{R}^m \to \mathbb{R}^m$ is strongly monotone and continuously differentiable with a Lipschitz continuous Jacobian, $A \in \mathbb{R}^{m \times n}$, and $b \in \mathbb{R}^n$. Suppose that the NCP (F) has a nonempty bounded solution set, which we denote by S. Let $\{x^k\}$ be an arbitrary sequence produced by Algorithm **9.1.10** with T being a linear Newton approximation scheme of $\mathbf{F}_{\mathrm{FB}}$ such that $T(x^k)$ is contained in $\mathcal{D}_a(x^k) + \mathcal{D}_b(x^k)JF(x^k)$ for all k. Suppose that $\{JF(x^k)\}$ is bounded. Show that $\{x^k : k \in \kappa\}$ is bounded and the sequence $\{F(x^k)\}$ converges to the unique element in $F(S)$.

9.6 Notes and Comments

The main aim of this chapter has been the development of global methods for the NCP through its reformulation as a system of equations or as a minimization problem. The history of equation/merit function reformulations of the NCP can be traced to the early equation reformulations of the KKT conditions of an NLP in the 1970s (see Section **10.6** for more details); but the focus at that time was not on development of Newton methods. The latter begins in early 1990. The first paper devoted principally to the reformulation of an NCP as a system of (differentiable) equations is a 1976 paper by Mangasarian [382], where the family of C-functions ψ_{Man} was introduced (See also Section **1.9** for more notes.) Exercise **9.5.1** is from this source. In spite of the importance of Mangasarian's contribution, and with the exception of [605], sustained developments of Newton methods occurred much later.

Unconstrained, smooth reformulations are at the basis of the methods developed in [184, 296, 297, 383, 552, 553], which, however, suffer from the drawback evidenced in Proposition **9.1.1** and thus require a nondegeneracy assumption to achieve superlinear convergence. So researchers turned very quickly to nonsmooth reformulations. Actually, the first attempts [232, 258, 426, 431], based on (variants or refinements of) the B-differentiable Newton method of Pang [425], even predate the aforementioned smooth approaches.

But these early nonsmooth Newton methods call for the solution of either a linear mixed complementarity problem or a convex, quadratic subproblem at each iteration in order to calculate a search direction. Moreover, it is not even possible to prove that a limit point is at least a stationary point of the merit function without assuming some regularity conditions; an example of this is Algorithm minLSA, which coincides with the NE/SQP algorithm proposed in [431].

An alternative approach may be obtained by applying the method described in Section **8.1** to the (nonsmooth) normal map of the NCP. This was done by Dirkse and Ferris in [136], who employed the box constrained VI as the basic framework. The resulting algorithm was the basis for the original version of the GAMS-based [66] computer software known as PATH, which utilized Lemke's algorithm for solving linear complementarity subproblems in order to generate a path along which a new iterate is computed. In spite of the good performance of the computer code, the original PATH method had some theoretical drawbacks in that the very strong assumptions described in Section **8.1** must be satisfied by the nonsmooth equation reformulation in order to guarantee convergence. Over the years, Ferris and his collaborators have maintained and upgraded the PATH solver continuously; it is to date indisputably the best available software for solving MiCPs. More material related to the implementation of the PATH code and to its many enhancements can be found in [51, 137, 138, 139, 186, 187, 188, 189]. Other articles that report the practical experience for solving realistic MiCPs include [124, 135, 194, 195, 419]. Besides GAMS, the programming language AMPL [216] also contains a complementarity command that facilitates the programming description of MiCP applications for computer solution.

It was not until semismooth Newton methods for nonsmooth equations were developed that the equation/merit function reformulation approach gained momentum. As already discussed in the Preface, the introduction of the Fischer-Burmeister function ψ_{FB} in [198] in order to reformulate the KKT conditions of an inequality constrained optimization problem triggered immediate interest. Although the potentials of the function ψ_{FB} were immediately appreciated [199, 200, 201, 236], it was only in [123, 180] that it was fully recognized that the semismoothness of ψ_{FB} and the continuous differentiability of the corresponding merit function θ_{FB} could be combined to form a simple and efficient algorithm for the solution of complementarity problems (see also [286, 288] for some closely related results). The approach of [123] set a model for many of the algorithmic developments to follow, and paved the way to the simple extension of practically

any standard technique for the solution of smooth systems of equations to complementarity problems.

Proposition **9.1.1** was proved in [306]. It is interesting to observe that some most recent developments by Izmailov and Solodov [283] suggest the possibility to define efficient solution algorithms for an NCP based on its smooth reformulations, which could overcome the likely singularity of the Jacobian of the equation reformulation at a solution. However, this approach has not yet been fully explored.

The differential properties of $\mathbf{F}_{\rm FB}$ described in Proposition **9.1.4**, the procedure to calculate an element in $\operatorname{Jac}\mathbf{F}_{\rm FB}$, and Algorithm **9.1.10** are all from [123, 180]. Originated from the matrix-theoretic characterization of strong regularity of a solution to an NCP (see Corollary **5.3.20**), Pang and Gabriel [232, 431] introduced the concept of pointwise s-regularity ("s" for stationarity) for the NCP (defined in Exercise **9.5.14**) and showed that if x^* is a s-regular accumulation point of a sequence of iterates produced by the NE/SQP algorithm, then x^* must be a solution of the NCP. This regularity concept was refined by Moré [415], who presented a similar condition in connection with a bound constrained reformulation of the NCP. Without using the term "FB regularity", Definition **9.1.13** was formally introduced in [123]. From a theoretical point of view, FB regularity is not more or less desirable than the original s-regularity; it is just that the FB C-function has more desirable properties than the min C-function in the design solution algorithms for the NCP. The matrix-theoretic relations in Proposition **9.1.17** are well known in LCP theory; see [114]. See [204] for the connection between merit functions and stability for CPs. Lemma **9.1.3** is proved by Tseng [587].

If each stationary point of $\theta_{\rm FB}$ is FB regular, then every stationary point of $\theta_{\rm FB}$ is a global minimizer. Another condition that yields the same conclusion is the convexity of $\theta_{\rm FB}$. However, the structure of the FB C-function makes very unlikely that $\theta_{\rm FB}$ is convex, even under very strong assumptions on F. It is therefore interesting to note that convexity may hold for other merit functions. This is shown by Exercise **9.5.2**, which is based on [376]. Other results on the convexity of merit functions (in some cases with reference to general variational inequalities) are given in [310, 358, 447, 631].

The analysis of the behavior of Algorithm **9.1.10** when an unbounded sequences is produced (see Theorem **9.1.11**) follows the approach in [104, 225]. The results in Subsection **9.1.3**, including the sequential FB regularity, are new. The results on the nonsingularity of the Newton approximation, Subsection **9.1.4**, are again from [123].

In the case of monotone problems, it is possible to perturb the (possibly singular) Newton equation so as to permit the calculation of a useful descent direction and guarantee convergence, see [200, 632]. A function satisfying the condition in Proposition **9.1.27** has been called an R_0-function in, for example, [83, 89, 202]. As a matter of fact, Chen [83] has a refined categorization of such a function; see Section **6.10** for further discussion of Chen's work. In [202] it was shown that the R_0 condition implies the boundedness of the level sets of θ_{FB}. Results in this vein, based on slight variants of the R_0 condition and in connection with the merit function θ_{LTKYF} introduced in Exercise **9.5.2**, were obtained in [376]. However, the necessity part in Proposition **9.1.27** seems new. Corollary **9.1.28** was proved in [180] and in [288], while in [236] the boundedness of the level sets of θ_{FB} was obtained under a strong monotonicity assumption on F.

Corollary **9.1.31**, when specialized to the LCP defined by a P_0 matrix M, implies that the LCP (q, M) has a nonempty bounded solution set for all vectors q if and only if M is R_0. This result was first proved by Aganagic and Cottle [4] by resorting to yet another class of matrices in connection with Lemke's algorithm [363].

The modifications of the basic algorithm described in Subsection **9.1.6** should be considered as just a small sample of many possible variants. Algorithm VFBLSA was proposed by Jiang and Qi in [288], which considered only the case of uniformly P functions. Algorithm MFBLSA, instead, is derived from an approach proposed by Ferris and Lucidi in the report version of [184]. The Inexact FB Line Search Algorithm was presented in [168], while its more computationally efficient Levenberg-Marquardt variant was described in the subsequent [174].

Pang [425] initiated the use of the merit function $\theta_{\min}$ in an attempt to globalize the convergence of a Newton method for solving the equation $\min(x, F(x)) = 0$. This proposal leads to the B-differentiable Newton method for solving the NCP (F); see Example **7.2.16**. In spite of its theoretical deficiencies, this method is very effective for solving a host of discrete frictional contact problems; see [105, 106, 107, 549, 550]. Led by Klarbring, this group at Linköping University pioneered the use of the Josephy-Newton SLCP method [57, 326, 328, 329] and is at the forefront of using nonsmooth Newton methods for solving contact problems [290, 291]. In the mechanics literature, the B-differentiable Newton method was independently discovered by Alart and Curnier [8, 117] and was called the augmented Lagrangian method; see also [327].

The use of the Fischer-Burmeister merit function to globalize the Newton algorithm for the min reformulation of the NCP (Algorithm minF-

BLSA) was first described in [180], where the motivation for the "fast" direction was rather different from the one given here. In this reference it is also observed that if one takes $T_{\min}(x)$ to be equal to the set $\mathcal{A}_{\min}(x)$ introduced in Example **7.2.16**, Algorithm minFBLSA converges finitely when F is affine and one of the limit points of the sequence generated by the algorithm is a b-regular solution of the LCP; see also [208] for a similar result.

Algorithm minFBLSA and Algorithm VFBLSA are essentially identical, the only difference being that in the former algorithm the "fast" direction is the Newton direction for the system $\mathbf{F}_{\min}(x) = 0$, while in the latter the "fast" direction is the Newton direction for the system $\mathbf{F}_{\mathrm{FB}}(x) = 0$. Actually, with the exception of Algorithm MFBLSA, all the algorithms described so far can be seen as an attempt to combine a globally convergent "first-order algorithm" with a fast, but only locally convergent method in such a way as to obtain global convergence and a fast local convergence rate. This is a well-known strategy in optimization and can be analyzed at various levels of abstraction. We have given concrete examples rather than a general theory. The reader interested in a more abstract approach can find material and bibliographical references in [124, 289, 460]; the first of these papers also contains extensive numerical testing and comparisons.

Trust region methods for the solution of NCPs have been investigated less extensively than line search methods. For a general overview on trust region methods and all relevant numerical issues we refer the reader to the excellent and comprehensive monograph by Conn, Gould, and Toint [110]. The first application of trust region methods to the solution of a complementarity problem (and to some of its generalizations) is to be found in [233], where the authors considered a constrained reformulation involving the min function. While basically all subsequent trust region proposals employed smooth merit functions, the cited reference introduced the key feature that the trust region radius should never be smaller than a prescribed constant $\Delta_{\min}$ at the beginning of each iteration (see Step 4 in Algorithm **9.1.35**, for example). Although it is shown in [110] that in some cases it is possible to dispense with this (small) restriction, this feature has been maintained by all subsequent methods. Algorithm **9.1.35** and Theorem **9.1.36** can be considered as refinements of the results in [287] in that inexact solutions of the trust region subproblem are allowed and stationarity of every limit point of the sequence generated by the algorithm is proved. The Subspace Trust Region Algorithm is taken from [311]. Trust region methods match well with constrained reformulations of the NCP, since (simple) constraints can be taken into account in the

trust region subproblems, which are constrained by their own nature. In fact, the paper [233] already considered (simply) constrained optimization reformulations for NCPs; further results along this line can be found in [415] and [596].

Constrained reformulations of the NCP have been the subject of much interest, since in many ways they may appear as the most natural kind of reformulation. It is obvious that when considering constrained reformulations of complementarity problems one is mainly interested in algorithms that maintain feasibility throughout the process, since otherwise, many of the benefits of such reformulations would be lost. One should distinguish two main (and related) issues: the analytical properties of the reformulations and the development of (constrained) algorithms for their solution. The first constrained reformulations of the NCP are probably those employing the implicit Lagrangian [383] (the implicit Lagrangian is discussed more in detail in the next chapter). Constrained reformulations using the function θ_{FB} were first analyzed in [203], where necessary and sufficient conditions for a constrained stationary point to be a solution of the complementarity problem are given. Theorem **9.1.38** shows that such a condition is just FB regularity; the simple proof employed here is patterned after an analogous proof for the implicit Lagrangian given in [173]. Apart from the already mentioned work [415], (necessary and) sufficient conditions for a stationary point to be a solution of the NCP were given, with reference to different reformulations, in [14, 193, 217, 410, 529, 595, 628].

Regarding feasible algorithms for the solution of constrained reformulations, there are almost endless possibilities; our aim in the main text was only to illustrate some of them showing the typical considerations one encounters in such an approach. Algorithm CFBLSA as such does not seem to have been considered in the literature, while Algorithm CLMAFB is derived from [172], where the solution of KKT systems is considered. Algorithm CFBLSA exemplifies the approach, already described in connection to unconstrained reformulations, in which two methods with different convergence properties are combined with the aim of preserving the good properties of both. Instead, Algorithm CLMAFB is an example of a method in which a search direction is defined that when far from a solution behaves like a "first order direction" while becoming closer and closer to a Newton type direction when approaching a desired solution. In this sense (and only in this sense) it may be considered close in spirit to Algorithm MFBLSA. A variant of Algorithm CLMAFB that uses the inexact solution of a quadratic problem is considered in [299]. Both Algorithm CFBLSA and Algorithm CLMAFB (and its variant [299]) are computationally more

expensive than their unconstrained counterparts. One line of research has then aimed at reducing these costs by studying algorithms that require only the solution of linear systems at each iteration. Results in this direction can be found in [301, 302, 309], where active set strategies similar to those considered in Section **6.7** are proposed that identify the zero variables at the solution and basically reduce the constrained problem to an unconstrained one. More generally, since we are minimizing a smooth objective function, a large array of standard methods is available; one only has to combine in an appropriate way these methods with a nonsmooth local Newton method to obtain a suitable feasible algorithm. A general framework in which it is possible to combine two algorithms with different properties is given in [183] and used there in connection with the PATH solver in order to increase its robustness.

The C-function ψ_{KK} and the corresponding merit function were introduced in [307]; for the other C-functions considered in Section **9.3** see the discussion in Section **1.9**. Among the C-functions considered in Section **9.3**, the more interesting one is certainly ψ_{CCK}. This function shares all the favorable properties of the FB function and has some additional theoretical advantages as described in Theorem **9.3.6**, which, incidentally, improves on some results in [86]. The function θ_{CCK} is used in an algorithm similar to Algorithm minFBLSA and proves to be extremely effective [86]. These findings are confirmed also by the results in [419], where extensive numerical results on a sophisticated implementation of semismooth algorithms are reported.

Other merit functions have been proposed in the literature. For example, the function θ_{S}, studied in Exercise **9.5.10**, was introduced by Solodov in [529]; other proposals can be found in [625, 560]. The review paper of Fischer and Jiang [207] is a valuable survey that summarizes the large amount of research on merit functions for complementarity and related problems.

The extensions considered in Section **9.4** are for the most part rather straightforward and discussed in some of the aforementioned papers. The composition of two C-functions in order to obtain a B-function was first considered in Billups' Ph.D. dissertation [48]. The function $\phi_{\mathrm{Q}}(\tau, \tau'; r, s)$ was proposed by Qi in [477]. Extensions and refinements of many of the results described in this chapter to box constrained problems and to further generalizations, more numerical results and some related approaches can be found in [15, 17, 49, 53, 168, 169, 179, 190, 193, 228, 300, 303, 304, 562, 597].

There are some topics that we did not discuss in the text and that we want to mention briefly here. The first is quasi-Newton and finite-difference methods. The interest in these kinds of algorithms seems not to

be great, due probably to the availability of automatic differentiation tools and to practical experience indicating that often large problems are well structured and first-order derivatives are readily available. Nevertheless, valuable results have been obtained in this area, see [253, 368, 369, 457, 512, 559]. A far more interesting topic is the extension of the semismooth Newton methods to CPs in SPSD matrices. Tseng [590] and Yamashita and Fukushima [633] show that it is possible to extend some of the merit functions discussed in this chapter to these CPs; see also [101].

We have always assumed throughout our treatment that the function F defining the NCP is at least continuously differentiable. If this is not the case, meaningful results can still be obtained. The interested reader is referred to [202, 206, 285]; in particular, the first reference discusses some problems that give rise to NCPs with nondifferentiable functions. In Exercises **9.5.6**–**9.5.9** we saw that merit functions can be used to define, under suitable monotonicity conditions, simple methods that do not use the derivative of F. Exercises **9.5.6** and **9.5.7** are based on results by Geiger and Kanzow [236]. This kind of methods (usually referred to as "derivative-free" methods) was first proposed by Yamashita and Fukushima in [630] in connection with the implicit Lagrangian. In the latter two papers strong monotonicity of F is required. In [285, 310, 376] the authors replaced this condition by a mere monotonicity assumption, while Fischer [202] further extended the approach to nondifferentiable functions. In [384, 625] it is shown that, under a strong monotonicity assumption, a linear convergence rate for two derivative-free algorithms can be obtained. Exercise **9.5.8** is based on [448].

We have already mentioned Watson's paper [605] in which a homotopy method is employed to solve a smooth equation reformulation of the NCP based on ψ_{Man}. Of course, the homotopy approach is not new in the complementarity field; the classic Lemke's algorithm [114, 363] for the LCP is well recognized as a homotopy method. In a nutshell, in a homotopy method to solve a system of equations $G(x) = 0$, a homotopy function $H(x, t)$ is constructed, where $H(\cdot, 1) = G$ while $H(\cdot, 0)$ is a function with a known zero $x(0)$. Starting at $x(0)$, an attempt is made to follow the curve $H^{-1}(0)$ in the (x, t) space to reach a solution of $G(x) = 0$. For a general introduction to this family of methods, we refer to [9, 10]. An important advantage of the homotopy methods is their robustness; but they tend to be rather complex to implement and usually involve substantial computations that could be very slow on problems of large size. Some recent advances in this field can be found in [50, 55, 56, 517, 518] (the last two of these papers actually refer to normal map reformulations of a general

VI). In particular Billups [50] tries to cross borders and describes a semismooth Newton method that resorts to a homotopy method if the process seems to be stalling around a local but non-global minimum point of the norm of an equation reformulation of the underlying problem. Some homotopy methods are also very close to path-following algorithms discussed in Chapter **11**, to which we refer for further discussion and references. We finally mention that Ferris and co-workers have set up what can be considered the standard library of test problems for NCP and box constrained VI algorithms; descriptions of this library and of the related tools are given in [135, 139, 182, 186, 194].

Chapter 10

Algorithms for VIs

In this chapter we consider algorithms for the solution of general VIs. The methods considered here can be viewed as the natural extension of those presented in the previous chapters and are therefore applicable to non-monotone problems.

We explore two basic approaches, one in which we attempt to solve the KKT conditions of the VI (K, F) and another one in which we try to handle the problem directly. The KKT conditions offer a very convenient approach from several points of view. First of all, these conditions can easily be reformulated as a mixed complementarity problem and so the methods of Chapter **9** can be readily applied; furthermore, sharp convergence results can be established by exploiting the particular structure of the KKT system. The KKT approach is not without drawbacks; for one thing, it inherently requires the function F to be defined everywhere; furthermore, the resulting merit function, which is defined in the joint space of the primary variable and the multipliers, is likely to have unbounded level sets. More generally, this approach may not be able to easily and fully exploit the good properties that a problem might possess (monotonicity, for example). In spite of this limitation, the simplicity of the algorithms based on the KKT conditions make them very attractive and it is safe to say that if the constraints of the VI are not particularly simple, solving the KKT conditions is the preferred approach for solving the VI in practice.

The second alternative we consider is based on the definition of suitable merit functions for the VI defined in the space of the primary variable only. While very attractive theoretically, this approach suffers from the severe drawback that the mere evaluation of such merit functions is typically a nontrivial task by itself. For instance, consider the gap function defined in (1.5.15). To evaluate this function at a vector, we need to solve an

optimization problem. Although some remedies can be envisaged, methods based on merit functions of this kind are likely to be practically effective only in the cases where the set K is relatively simple.

10.1 KKT Conditions Based Methods

In this section we analyze several approaches that aim at finding a KKT triple of the VI (K, F). Consequently we assume that the set K is finitely representable as

$$K \,=\, \{\, x \,\in\, \mathbb{R}^n \,:\, h(x) \,=\, 0,\, g(x) \,\leq\, 0\,\}, \tag{10.1.1}$$

where $h : \mathbb{R}^n \to \mathbb{R}^\ell$ and $g : \mathbb{R}^n \to \mathbb{R}^m$ are continuously differentiable functions. The KKT conditions of the VI (K, F) are

$$\begin{array}{c} F(x) + Jh(x)^T\mu + Jg(x)^T\lambda \,=\, 0 \\ h(x) \,=\, 0 \\ 0 \,\geq\, g(x) \,\perp\, \lambda \,\geq\, 0. \end{array} \tag{10.1.2}$$

For the most part in this section, we treat (10.1.2) as a MiCP without regards to whether K is convex or not. Thus g_i is not assumed convex and h is not assumed affine. The blanket differentiability assumption made throughout the chapter is that F is continuously differentiable and g and h are both twice continuously differentiable on $\mathbb{R}^n$.

10.1.1 Using the FB function

We first consider a reformulation of the KKT system that uses the FB function. We therefore rewrite (10.1.2) as:

$$0 \,=\, \Phi_{\rm FB}(x,\mu,\lambda) \,\equiv\, \left(\begin{array}{c} \mathbf{L}(x,\mu,\lambda) \\ -h(x) \\ \psi_{\rm FB}(-g_1(x),\lambda_1) \\ \vdots \\ \psi_{\rm FB}(-g_m(x),\lambda_m) \end{array}\right), \tag{10.1.3}$$

where

$$\mathbf{L}(x,\mu,\lambda) \,\equiv\, F(x) + Jh(x)^T\mu + Jg(x)^T\lambda,$$

is the VI Lagrangian function. The natural merit function associated with the system (10.1.3) is

$$\theta_{\rm FB}(x,\mu,\lambda) \,\equiv\, \tfrac{1}{2}\,\Phi_{\rm FB}(x,\mu,\lambda)^T\Phi_{\rm FB}(x,\mu,\lambda). \tag{10.1.4}$$

By the results in the previous chapter we know that $\Phi_{\rm FB}$ is semismooth and $\theta_{\rm FB}$ is continuously differentiable on $\mathbb{R}^{n+\ell+m}$. Furthermore, if JF and each $\nabla^2 g_i$ and $\nabla^2 h_j$ are locally Lipschitz continuous, then $\Phi_{\rm FB}$ is strongly semismooth.

We begin our study by considering conditions that guarantee the nonsingularity of all elements in the generalized Jacobian $\partial\Phi_{\rm FB}(x,\mu,\lambda)$, or more generally, in a suitably defined linear Newton approximation scheme of $\Phi_{\rm FB}$. We also relate these conditions to the strong regularity of the KKT system (10.1.2). In principle, we could apply here the results of Section **9.4**; but the resulting conditions would not be easily related to known properties of the VI (K,F). Therefore, we prefer to proceed in a different way, and in so doing we will obtain sharper and more transparent results.

The following result, which is analogous to Proposition **9.1.4**, identifies the structure of the matrices in the generalized Jacobian $\partial\Phi_{\rm FB}(x,\mu,\lambda)$.

10.1.1 Proposition. The generalized Jacobian $\partial\Phi_{\rm FB}(x,\mu,\lambda)$ is contained in the following family of matrices:

$$\mathcal{A}(x,\mu,\lambda) \equiv \left\{ \begin{bmatrix} J_x\mathbf{L}(x,\mu,\lambda) & Jh(x)^T & Jg(x)^T \\ -Jh(x) & 0 & 0 \\ -D_g(x,\lambda)Jg(x) & 0 & D_\lambda(x,\lambda) \end{bmatrix} \right\}, \quad (10.1.5)$$

where

$$D_\lambda(x,\lambda) \equiv \operatorname{diag}(\, a_1(x,\lambda),\ldots,a_m(x,\lambda)\,)$$

and

$$D_g(x,\lambda) \equiv \operatorname{diag}(\, b_1(x,\lambda),\ldots,b_m(x,\lambda)\,)$$

are $m\times m$ diagonal matrices whose diagonal elements are given by:

$$(a_i(x,\lambda),b_i(x,\lambda)) \begin{cases} \equiv \dfrac{(\,\lambda_i,-g_i(x)\,)}{\sqrt{g_i(x)^2+\lambda_i^2}} - (\,1,1\,) & \text{if } (g_i(x),\lambda_i)\neq 0 \\ \in \operatorname{cl}\mathbb{B}(0,1) - (\,1,1\,) & \text{if } (g_i(x),\lambda_i)=0. \end{cases}$$

Moreover, $\mathcal{A}(x,\mu,\lambda)$ is a linear Newton approximation of $\Phi_{\rm FB}$ at (x,μ,λ), which is strong if JF and each $\nabla^2 g_i$ and $\nabla^2 h_j$ are locally Lipschitz continuous at x.

Proof. The containment of $\partial\Phi_{\rm FB}(x,\mu,\lambda)$ in $\mathcal{A}(x,\mu,\lambda)$ follows easily from Propositions **7.1.14** and **7.1.11**. The assertion about the Newton approximation scheme follows immediately from Theorem **7.5.17** and trivial considerations. □

We introduce several index sets associated with a given pair (x, λ) in $\mathbb{R}^{n+m}$. All these index sets are subsets of $I \equiv \{1, \ldots, m\}$. Specifically, let

$$\begin{aligned} I_0 &\equiv \{\, i \in I : g_i(x) = 0 \leq \lambda_i \,\} \\ I_< &\equiv \{\, i \in I : g_i(x) < 0 = \lambda_i \,\} \\ I_R &\equiv I \setminus (\, I_0 \cup I_< \,). \end{aligned}$$

Note that $i \in I_R$ if and only if one of three things holds: $g_i(x) > 0$, $\lambda_i < 0$, or $(-g_i(x), \lambda_i) > 0$. If $g(x) \leq 0 \leq \lambda$, then I_0 is equal to the index set $\mathcal{I}(x)$ of binding (inequality) constraints at x. If $0 \geq g(x) \perp \lambda \geq 0$, then I_R is empty. More generally, we have

$$\psi_{\mathrm{FB}}(-g_i(x), \lambda_i) = 0 \;\Leftrightarrow\; i \in I_0 \cup I_<.$$

We define two subsets of I_0:

$$I_{00} \equiv \{\, i \in I_0 : \lambda_i = 0 \,\} \quad \text{and} \quad I_+ \equiv \{\, i \in I_0 : \lambda_i > 0 \,\}.$$

For a given matrix H in the generalized Jacobian $\partial \Phi_{\mathrm{FB}}(x, \mu, \lambda)$ with particular elements $a_i(x, \lambda)$ and $b_i(x, \lambda)$, cf. Proposition **10.1.1**, we further refine the index set I_{00}:

$$\begin{aligned} I_{01} &\equiv \{\, i \in I_{00} : a_i(x, \lambda) = 0 \,\} \\ I_{02} &\equiv \{\, i \in I_{00} : \max(a_i(x, \lambda), b_i(x, \lambda)) < 0 \,\} \\ I_{03} &\equiv \{\, i \in I_{00} : b_i(x, \lambda) = 0 \,\}; \end{aligned}$$

let

$$I_{R2} \equiv I_R \cup I_{02}.$$

The latter four index sets depend on the matrix H; since this dependence will be obvious in the context where the index sets will be employed, the dependence is not explicitly included in the above definition of the index sets.

The following relationships between these index sets hold:

$$I = I_0 \cup I_< \cup I_R, \qquad I_0 = I_{00} \cup I_+, \qquad I_{00} = I_{01} \cup I_{02} \cup I_{03}.$$

Hence,

$$I = I_+ \cup I_{01} \cup I_{R2} \cup I_{03} \cup I_<.$$

Moreover, we have

$$(\, D_g \,)_{ii} = 0 \quad \text{and} \quad (\, D_\lambda \,)_{ii} = -1, \quad \forall\, i \in I_< \cup I_{03}; \tag{10.1.6}$$

$$(D_g)_{ii} = -1 \quad \text{and} \quad (D_\lambda)_{ii} = 0, \quad \forall\, i \in I_+ \cup I_{01}; \tag{10.1.7}$$

$$\max((D_\lambda)_{ii}, (D_g)_{ii}) < 0, \quad \forall\, i \in I_{R2} = I_R \cup I_{02}. \tag{10.1.8}$$

It would be useful to define a family of matrices that are closely related to the strong stability condition. Specifically, for any subset $\mathcal{J}$ of $I_{00} \cup I_R$, let

$$M_{\rm FB}(\mathcal{J}) \equiv \begin{pmatrix} J_x\mathbf{L} & Jh^T & Jg_+^T & Jg_{\mathcal{J}}^T \\ -Jh & 0 & 0 & 0 \\ -Jg_+ & 0 & 0 & 0 \\ -Jg_{\mathcal{J}} & 0 & 0 & 0 \end{pmatrix},$$

With such a matrix, we are in a position to present an equivalent formulation of strong stability that better suites our needs here.

10.1.2 Lemma. A KKT triple $(x^*, \mu^*, \lambda^*) \in \mathbb{R}^{n+\ell+m}$ of the VI (K, F) is strongly stable if and only if for all subsets $\mathcal{J} \subseteq I_{00}$, the determinants of the matrices $M_{\rm FB}(\mathcal{J})$ have the same nonzero sign.

Proof. Suppose that (x^*, μ^*, λ^*) is a strongly stable KKT triple. By Corollary **5.3.22**, we know that the matrix

$$\mathbf{B} \equiv \begin{bmatrix} J_x\mathbf{L}(x^*, \mu^*, \lambda^*) & Jh(x^*)^T & Jg_+(x^*)^T \\ -Jh(x^*) & 0 & 0 \\ -Jg_+(x^*) & 0 & 0 \end{bmatrix}$$

is nonsingular and the Schur complement

$$\mathbf{C} \equiv \begin{bmatrix} Jg_{00}(x^*) & 0 & 0 \end{bmatrix} \mathbf{B}^{-1} \begin{bmatrix} Jg_{00}(x^*)^T \\ 0 \\ 0 \end{bmatrix}$$

is a P matrix. Let $\mathcal{J}$ be an arbitrary subset of I_{00}. By the Schur determinantal formula, we have

$$\det M_{\rm FB}(\mathcal{J}) = \det \mathbf{B} \, \det M_{\rm FB}(\mathcal{J})/\mathbf{B}. \tag{10.1.9}$$

The Schur complement $M_{\rm FB}(\mathcal{J})/\mathbf{B}$ is a principal submatrix of $\mathbf{C}$; hence $\det M_{\rm FB}(\mathcal{J})/\mathbf{B}$ is positive. Therefore, the sign of $\det M_{\rm FB}(\mathcal{J})$ is nonzero and equal to the sign of $\det \mathbf{B}$, which is independent of $\mathcal{J}$. For the converse, it suffices to note that $\mathbf{B} = M_{\rm FB}(\emptyset)$ and reverse the above argument. □

The above lemma suggests a definition for an arbitrary triple (x, μ, λ) in $\mathbb{R}^{n+\ell+m}$. If this happens to be a KKT triple, the definition is equivalent

to the strong stability of this triple. Thus the ESSC defined below can be viewed as an *algebraic extension* of the concept of strong stability to a non-KKT triple.

10.1.3 Definition. The *Extended Strong Stability condition* (ESSC) is said to hold at a triple (x, μ, λ) in $\mathbb{R}^{n+\ell+m}$ if

(a) $\operatorname{sgn}\det M_{\mathrm{FB}}(\mathcal{J})$ is a nonzero constant for all index sets $\mathcal{J} \subseteq I_{00}$;

(b) $\det M_{\mathrm{FB}}(\mathcal{J}) \det M(\mathcal{J}') \geq 0$ for any two index sets $\mathcal{J}$ and $\mathcal{J}'$ that are subsets of $I_{00} \cup I_R$. □

Equivalently, the ESSC holds at (x, μ, λ) if and only if (i) the matrix

$$\mathbf{B} \equiv \begin{bmatrix} J_x \mathbf{L}(x, \mu, \lambda) & Jh(x)^T & Jg_+(x)^T \\ -Jh(x) & 0 & 0 \\ -Jg_+(x) & 0 & 0 \end{bmatrix} = M_{\mathrm{FB}}(\emptyset)$$

is nonsingular, (ii) $M_{\mathrm{FB}}(\mathcal{J})$ is nonsingular for all nonempty subset $\mathcal{J}$ of I_{00}, and (iii) for all subset $\mathcal{J}$ of $\mathcal{I}_{00} \cup I_R$ for which $M_{\mathrm{FB}}(\mathcal{J})$ is nonsingular, $\operatorname{sgn}\det M_{\mathrm{FB}}(\mathcal{J}) = \operatorname{sgn}\det \mathbf{B}$. Clearly, by considering the nonsingularity of the matrix $M_{\mathrm{FB}}(I_{00})$, it follows easily that if the ESSC holds at (x, μ, λ), then the gradients

$$\{ \nabla h_j(x) : j = 1, \ldots, \ell \} \cup \{ \nabla g_i(x) : i \in I_0 \}$$

are linearly independent. In Proposition **10.1.7**, we show that this linear independence condition plus another condition together imply the ESSC.

The central role of the ESSC is described in Theorem **10.1.4** below. To prove this theorem, we recall the following determinantal formula: for any square matrix A of order r and any diagonal matrix D of the same order,

$$\det(D + A) = \sum_{\alpha} \det D_{\alpha\alpha} \det A_{\bar{\alpha}\bar{\alpha}}, \tag{10.1.10}$$

where the summation ranges over all subsets α of $\{1, \ldots, r\}$ with complement $\bar{\alpha}$. Included in this summation are the two extreme cases corresponding to α being the empty set and the full set; the convention for these cases is that the determinant of the empty matrix is set equal to one.

10.1.4 Theorem. Suppose that the ESSC holds at the triple (x, μ, λ) in $\mathbb{R}^{n+\ell+m}$. All elements in $\mathcal{A}(x, \mu, \lambda)$ (and therefore all matrices in the generalized Jacobian $\partial\Phi_{\mathrm{FB}}(x, \mu, \lambda)$) are nonsingular.

Proof. Consider an arbitrary matrix H in $\mathcal{A}(x, \mu, \lambda)$. By definition (cf. Proposition **10.1.1**) and by (10.1.6), (10.1.7), and (10.1.8), such a matrix has the following structure (the dependence on (x, μ, λ) is suppressed for simplicity):

$$H = \begin{bmatrix} J_x\mathbf{L} & Jh^T & Jg_+^T & Jg_{01}^T & Jg_{R2}^T & Jg_{03}^T & Jg_<^T \\ -Jh & 0 & 0 & 0 & 0 & 0 & 0 \\ -Jg_+ & 0 & 0 & 0 & 0 & 0 & 0 \\ -Jg_{01} & 0 & 0 & 0 & 0 & 0 & 0 \\ -(D_g)_{R2}Jg_{R2} & 0 & 0 & 0 & (D_\lambda)_{R2} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\mathbf{I} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\mathbf{I} \end{bmatrix},$$

where $(D_g)_{R2}$ and $(D_\lambda)_{R2}$ are negative definite diagonal matrices. Note that we have abbreviated g_{I_+} etc. by g_+ etc. in the matrix H. Clearly, H is nonsingular if and only if the following matrix is nonsingular:

$$\tilde{H} = \begin{bmatrix} J_x\mathbf{L}^T & Jh^T & Jg_+^T & Jg_{01}^T & Jg_{R2}^T \\ -Jh & 0 & 0 & 0 & 0 \\ -Jg_+ & 0 & 0 & 0 & 0 \\ -Jg_{01} & 0 & 0 & 0 & 0 \\ -Jg_{R2} & 0 & 0 & 0 & ((D_g)_{R2})^{-1}(D_\lambda)_{R2} \end{bmatrix}.$$

The matrix on the right-hand side is equal to the sum of the matrix $M_{\mathrm{FB}}(\mathcal{J})$ with $\mathcal{J} \equiv I_{01} \cup I_{R2}$ and a block diagonal matrix D, all of whose blocks are zero except for the last diagonal block, which is equal to the positive definite diagonal matrix $((D_g)_{R2})^{-1}(D_\lambda)_{R2}$. By the formula (10.1.10), we deduce that

$$\det \tilde{H} = \sum_{\alpha \subseteq I_{R2}} \det(((D_g)_{R2})^{-1}(D_\lambda)_{R2})_{\alpha\alpha} \det M_{\mathrm{FB}}(\mathcal{J})_{\alpha'\alpha'},$$

where the summation ranges over all subsets α of I_{R2} and $\alpha' \equiv \mathcal{J} \setminus \alpha$. Since this summation includes $\alpha = I_{R2}$, it is easily seen that $\det \tilde{H}$ is positive, by the ESSC. □

We see that the ESSC is a sufficient condition for the nonsingularity of all the elements in the generalized Jacobian of Φ_{FB} at (x, μ, λ). If the triple at hand satisfies the KKT conditions, then the ESSC is also necessary.

10.1.5 Theorem. Let $(x^*, \mu^*, \lambda^*) \in \mathbb{R}^{n+\ell+m}$ be a KKT triple of VI (K, F). The following three statements are equivalent.

(a) The ESSC holds at (x^*, μ^*, λ^*).

(b) All matrices in $\mathcal{A}(x^*, \mu^*, \lambda^*)$ are nonsingular.

(c) All matrices in $\partial\Phi_{\rm FB}(x^*, \mu^*, \lambda^*)$ are nonsingular.

Proof. It suffices to show that (c) implies (a). So we assume that all matrices in $\partial\Phi_{\rm FB}(x^*, \mu^*, \lambda^*)$ are nonsingular. We claim that the gradients

$$\{\, \nabla g_i(x^*) \,:\, i \in I_{00} \,\} \tag{10.1.11}$$

are linearly independent. Define a sequence $\{\lambda^k\}$ as follows. For each k, let

$$\lambda_i^k \equiv \begin{cases} \lambda_i^* & \text{if } i \in I \setminus I_{00} \\ 1/k & \text{if } i \in I_{00} \end{cases}$$

The sequence $\{\lambda^k\}$ converges to λ^*; moreover, for each k, $\Phi_{\rm FB}$ is continuously differentiable at (x^*, μ^*, λ^k). The sequence of Jacobian matrices $\{J\Phi_{\rm FB}(x^*, \mu^*, \lambda^k)\}$ converges to a matrix H in $\partial\Phi_{\rm FB}(x^*, \mu^*, \lambda^*)$ with $I_{01} = I_{00}$ and $I_{02} = I_{03} = \emptyset$. By assumption, H is nonsingular, the linear independence of the gradients (10.1.11) follows from the structure of H; see the proof of Theorem **10.1.4**.

We next show that for any subset $\mathcal{J}$ of I_{00}, there exists an element in $\partial\Phi_{\rm FB}(x^*, \mu^*, \lambda^*)$ such that $I_{01} = \mathcal{J}$, $I_{02} = \emptyset$ and $I_{03} = I_{00} \setminus \mathcal{J}$. Since the gradients (10.1.11) are linearly independent, the classical implicit function theorem implies that there is a sequence $\{x^k\}$ converging to x^* such that $g_i(x^k) > 0$ for all $i \in I_{00}$ and all k. For each k, define

$$\lambda_i^k \equiv \begin{cases} \lambda_i^* & \text{if } i \in I \setminus I_{00} \\ \sqrt{g_i(x^k)} & \text{if } i \in \mathcal{J} \\ g_i(x^k)^2 & \text{if } i \in I_{00} \setminus \mathcal{J}. \end{cases}$$

Clearly, the sequence $\{(x^k, \mu^*, \lambda^k)\}$ converges to (x^*, μ^*, λ^*); moreover, $\Phi_{\rm FB}$ is continuously differentiable at (x^k, μ^*, λ^k) for every k because there is no index i such that $g_i(x^k) = \lambda_i^k = 0$. Furthermore, by using Proposition **10.1.1** and the continuity of the functions involved, it easily follows that $\{J\Phi_{\rm FB}(x^k, \mu^*, \lambda^k)\}$ converges to an element H with the desired properties.

Since all matrices in $\partial\Phi_{\rm FB}(x^*, \mu^*, \lambda^*)$, which is a convex set, are nonsingular, it follows that all such matrices have the same nonzero determinantal

sign. Otherwise, if H_1 and H_2 are two matrices in $\partial\Phi_{\rm FB}(x^*,\mu^*,\lambda^*)$ such that $\det H_1 < 0$ and $\det H_2 > 0$, since the determinant is a continuous function of its arguments, a convex combination $\bar{H}$ of H_1 and H_2 exists such that $\det \bar{H} = 0$, contradicting the nonsingularity of all elements in the generalized Jacobian $\partial\Phi_{\rm FB}(x^*,\mu^*,\lambda^*)$.

To complete the proof of the theorem, it remains to show that the matrices $M_{\rm FB}(\mathcal{J})$ have the same nonzero determinantal sign for all $\mathcal{J} \subseteq I_{00}$. For each such index set $\mathcal{J}$, let $H(\mathcal{J}) \in \partial\Phi_{\rm FB}(x^*,\mu^*,\lambda^*)$ be such that $\mathcal{J} = I_{01}$ and $I_{03} = I_{00} \setminus I_{01}$. From the structure of $H(\mathcal{J})$ (cf. the proof of Theorem **10.1.4**), we have

$$\begin{aligned} \operatorname{sgn}\det H(\mathcal{J}) &= (-1)^m \operatorname{sgn}\det \begin{bmatrix} J_x\mathbf{L} & Jh^T & Jg_+^T & Jg_{\mathcal{J}}^T \\ -Jh & 0 & 0 & 0 \\ -Jg_+ & 0 & 0 & 0 \\ -Jg_{\mathcal{J}} & 0 & 0 & 0 \end{bmatrix} \\ &= (-1)^m \operatorname{sgn}\det M_{\rm FB}(\mathcal{J}). \end{aligned}$$

Since all matrices in $\partial\Phi_{\rm FB}(x^*,\mu^*,\lambda^*)$ have determinants with the same nonzero sign, it follows that the ESSC holds. □

As we already observed, if (x^*,μ^*,λ^*) is a KKT triple, then the ESSC is equivalent to the strong stability of this triple. We therefore get as an immediate consequence of Theorem **10.1.5** the following characterization of a strongly stable KKT triple.

10.1.6 Corollary. A KKT triple (x^*,μ^*,λ^*) is strongly stable if and only if all matrices in the generalized Jacobian $\partial\Phi_{\rm FB}(x^*,\mu^*,\lambda^*)$ are nonsingular. □

It is interesting to note that a parallel result does not hold for the FB reformulation of an NCP, where the strong stability of a solution x^* is only a sufficient condition for the nonsingularity of all the elements in $\partial\mathbf{F}_{\rm FB}(x^*)$; cf. Corollary **9.1.24**. When the function $\Phi_{\rm FB}$ is specialized to the NCP (F), it becomes

$$\Phi_{\rm FB}(x,\lambda) = \begin{pmatrix} F(x) - \lambda \\ \psi_{\rm FB}(x_1,\lambda_1) \\ \vdots \\ \psi_{\rm FB}(x_n,\lambda_n) \end{pmatrix},$$

which can be compared to

$$\mathbf{F}_{\rm FB}(x) \equiv \begin{pmatrix} \psi_{\rm FB}(x_1, F_1(x)) \\ \vdots \\ \psi_{\rm FB}(x_n, F_n(x)), \end{pmatrix}.$$

These two functions and their generalized Jacobians have much in common; yet what distinguishes them is the presence of the auxiliary variable λ in $\Phi_{\rm FB}(x, \lambda)$. This distinguished feature is key to the proof of Theorem **10.1.5**, where by choosing the sequence $\{\lambda^k\}$ suitably, we can establish a much needed element H in $\partial\Phi_{\rm FB}(x^*, \mu^*, \lambda^*)$ that corresponds to a given index subset $\mathcal{J}$ of I_{00}. This freedom of choosing the auxiliary sequence $\{\lambda^k\}$ is absent in the function $\mathbf{F}_{\rm FB}(x)$ and accounts for the failure of using $\partial\mathbf{F}_{\rm FB}(x^*)$ as a characterization of the strong stability of x^* as a solution of the NCP (F). In this sense, $\partial\mathbf{F}_{\rm FB}(x^*)$ is not as rich as $\partial\Phi_{\rm FB}(x^*, \lambda^*)$ for the NCP (F).

In the following proposition we give a sufficient condition for the ESSC to be satisfied at an arbitrary triple. When this is a KKT triple, this condition is similar to the strong second-order sufficiency condition for optimality in constrained optimization and the result can be deduced easily from Corollary **5.3.23**. The proof for the case of a non-KKT triple is an easy consequence of Exercise **1.8.9**.

10.1.7 Proposition. Let $(x, \mu, \lambda) \in \mathbb{R}^{n+\ell+m}$ be any given triple. If

(a) the gradients

$$\{\, \nabla h_j(x) \,:\, j = 1, \ldots, \ell \,\} \cup \{\, \nabla g_i(x) \,:\, i \in I_0 \,\}$$

are linearly independent;

(b) $J_x\mathbf{L}(x, \mu, \lambda)$ is positive definite on the null space of the gradients

$$\{\, \nabla h_j(x) \,:\, j = 1, \ldots, \ell \,\} \cup \{\, \nabla g_i(x) \,:\, i \in I_+ \,\}, \tag{10.1.12}$$

then the ESSC holds at (x, μ, λ).

Proof. Assumption (b) implies that $J_x\mathbf{L}(x, \mu, \lambda)$ is positive definite on the null space of the gradients:

$$\{\, \nabla h_j(x) \,:\, j = 1, \ldots, \ell \,\} \cup \{\, \nabla g_i(x) \,:\, i \in I_+ \cup \mathcal{J} \,\}, \tag{10.1.13}$$

for any subset $\mathcal{J}$ of $I_{00} \cup I_R$. Clearly, the latter property is equivalent to the positive definiteness of the matrix $Z^T J_x\mathbf{L}(x, \mu, \lambda)Z$, where Z is any

matrix whose columns form an orthonormal basis of the null space of the gradients (10.1.13).

The proposition will be established if we can show three things: (i) if $\mathcal{J}$ is a subset of I_{00}, then $\det M_{\rm FB}(\mathcal{J})$ is positive; (ii) if $\mathcal{J}$ is a subset of $I_{00} \cup I_R$ such that the gradients (10.1.13) are linearly dependent, then $M_{\rm FB}(\mathcal{J})$ is singular; and (iii) if $\mathcal{J}$ is a subset of $I_{00} \cup I_R$ such that the gradients (10.1.13) are linearly independent, then $\det M_{\rm FB}(\mathcal{J})$ is positive. All these follow rather easily from the determinantal formula in part (ii) of Exercise **1.8.9**, the assumptions (a) and (b) herein, and the above remark. □

We illustrate the results obtained so far with an example that pertains to two LCPs.

10.1.8 Example. Let $n = 1$, $\ell = 0$, $m = 1$, and $g_1(x) \equiv -x$. Thus the VI is a 1-dimensional LCP (q, M). Consider first the case where $q = -1$ and $M = 1$; i.e., $F(x) \equiv x - 1$. We have

$$\Phi_{\rm FB}(x, \lambda) = \begin{pmatrix} x - 1 - \lambda \\ \sqrt{x^2 + \lambda^2} - x - \lambda \end{pmatrix}.$$

Consider the KKT pair $(x, \lambda) = (0, 0)$; we have $I_{00} = \{1\}$ and $I_+ = I_R = \emptyset$. Moreover, $M_{\rm FB}(\emptyset) = 1$ and

$$M_{\rm FB}(\{1\}) = \begin{bmatrix} 1 & 1 \\ -1 & 0 \end{bmatrix};$$

hence the ESSC is satisfied. From the expression of $\Phi_{\rm FB}$ it can be easily seen that the elements of $\partial \Phi_{\rm FB}(0, 0)$ are given by

$$\begin{bmatrix} 1 & -1 \\ a - 1 & b - 1 \end{bmatrix},$$

where $a^2 + b^2 \leq 1$. As expected, all these matrices are nonsingular; in fact, their determinants are all negative.

Next, consider the case where $q = 0$ and $M = -1$; i.e., $F(x) \equiv -x$. We have

$$\Phi_{\rm FB}(x, \lambda) = \begin{pmatrix} -x - \lambda \\ \sqrt{x^2 + \lambda^2} - x - \lambda \end{pmatrix}.$$

Consider the KKT pair $(x, \lambda) = (0, 0)$; we have $I_{00} = \{1\}$ and $I_+ = I_R = \emptyset$. Moreover, $M_{\rm FB}(\emptyset) = -1$ and, as before,

$$M_{\rm FB}(\{1\}) = \begin{bmatrix} -1 & 1 \\ -1 & 0 \end{bmatrix};$$

hence the ESSC is not satisfied. From the expression of $\Phi_{\rm FB}$ it can be easily seen that the elements of $\partial\Phi_{\rm FB}(0,0)$ are given by

$$\begin{bmatrix} -1 & -1 \\ a-1 & b-1 \end{bmatrix},$$

where $a^2 + b^2 \leq 1$. Take for example $a = b = 1/\sqrt{2}$; we see that the corresponding generalized Jacobian matrix is singular. □

We next examine other crucial issues, which we are by now accustomed to. We begin by giving conditions under which every stationary point of $\theta_{\rm FB}(x,\mu,\lambda)$ is a KKT triple of the VI (K,F). As a prerequisite result, we identify the structure of the gradient of $\theta_{\rm FB}(x,\mu,\lambda)$ in a result that is similar to Proposition **9.1.4**. Since the proof follows a similar pattern, it is therefore omitted.

10.1.9 Proposition. If F is a C^1 function and g and h are C^2 functions, then $\theta_{\rm FB}(x,\mu,\lambda)$ is continuously differentiable and

$$\nabla\theta_{\rm FB}(x,\mu,\lambda) \;=\; H^T\Phi_{\rm FB}(x,\mu,\lambda)$$

for every $H \in \partial\Phi_{\rm FB}(x,\mu,\lambda)$. □

Obviously, if (x,μ,λ) is a stationary point of $\theta_{\rm FB}$ and some matrix H in $\partial\Phi_{\rm FB}(x,\mu,\lambda)$ is nonsingular, then (x,μ,λ) is a KKT triple of the VI (K,F). In particular, the latter conclusion is valid if (x,μ,λ) is a stationary point satisfying the ESSC. The ESSC requires certain gradients of the constraint functions to be linearly independent; see condition (a) in Proposition **10.1.7**. In the next result, we provide an alternative condition for a stationary point of $\theta_{\rm FB}$ to be a KKT triple under a positive semidefiniteness assumption on $J_x\mathbf{L}(x,\mu,\lambda)$.

10.1.10 Theorem. Let $(x^*,\mu^*,\lambda^*) \in \mathbb{R}^{n+\ell+m}$ be a stationary point of $\theta_{\rm FB}(x,\mu,\lambda)$. Assume that

(a) $J_x\mathbf{L}(x^*,\mu^*,\lambda^*)$ is positive semidefinite on $\mathbb{R}^n$;

(b) $J_x\mathbf{L}(x^*,\mu^*,\lambda^*)$ is positive definite on the null space of the gradients

$$\{\,\nabla h_j(x) \,:\, j \,=\, 1,\ldots,\ell\,\} \cup \{\,\nabla g_i(x) \,:\, i \,=\, 1,\ldots,m\,\}; \quad (10.1.14)$$

and either one of the following two conditions holds:

(c1) $Jh(x^*)$ has full row rank; or

(c2) h is an affine function and there exists x satisfying $h(x) = 0$.

It holds that $\theta_{\rm FB}(x^*, \mu^*, \lambda^*) = 0$; i.e., (x^*, μ^*, λ^*) is a KKT triple of the VI (K, F).

Proof. We need to show

$$\mathbf{L}^* \equiv \mathbf{L}(x^*, \mu^*, \lambda^*) = 0, \quad h^* \equiv h(x^*) = 0$$

and

$$\psi_i^* \equiv \psi_{\rm FB}(-g_i(x^*), \lambda^*) = 0, \quad \forall\, i = 1, \ldots, m.$$

We have noted that $\psi_i^* = 0$ for all $i \in I_0 \cup I_<$. By Proposition **10.1.9**,

$$0 = \theta_{\rm FB}(x^*, \mu^*, \lambda^*) = H^T \Phi_{\rm FB}(x^*, \mu^*, \lambda^*)$$

for every matrix H in $\partial \Psi_{\rm FB}(x^*, \mu^*, \lambda^*)$. Substituting the form of H as given in Propositions **10.1.1**, we obtain

$$(\, J_x \mathbf{L}(x^*, \mu^*, \lambda^*)\,)^T \mathbf{L}^* - Jh(x^*)^T h^* - Jg_R(x^*)^T (\, D_g^*)_R \psi_R^* = 0 \quad (10.1.15)$$

$$\left.\begin{array}{r} Jh(x^*)\mathbf{L}^* = 0 \\ \nabla g_i(x^*)^T \mathbf{L}^* = 0, \quad \forall\, i \in I_0 \cup I_< \end{array}\right\} \quad (10.1.16)$$

$$Jg_R(x^*)\mathbf{L}^* + (\, D_\lambda^*)_R \psi_R^* = 0, \quad (10.1.17)$$

where $(D_g^*)_R$ and $(D_\lambda^*)_R$ are two negative definite diagonal matrices. Solving for ψ_R^* in (10.1.17) and substituting into (10.1.15), premultiplying (10.1.15) by $(\mathbf{L}^*)^T$ and using (10.1.16), we obtain

$$(\, \mathbf{L}^*)^T \left[(\, J_x \mathbf{L}^*\,)^T + Jg_R(x^*)^T D Jg_R(x^*) \right] \mathbf{L}^* = 0, \quad (10.1.18)$$

where D is the positive definite diagonal matrix $(D_g^*)_R((D_\lambda^*)_R)^{-1}$ and $J_x \mathbf{L}^*$ is the shorthand for $J_x \mathbf{L}(x^*, \mu^*, \lambda^*)$. By condition (a), the matrix in the square bracket is positive semidefinite definite. Hence,

$$(\, \mathbf{L}^*)^T (\, J_x \mathbf{L}^*\,)^T \mathbf{L}^* = 0 \quad \text{and} \quad Jg_R(x^*)\mathbf{L}^* = 0.$$

Taking into account the equations in (10.1.16), we deduce that $\mathbf{L}^*$ belongs to the null space of the gradients (10.1.14). By condition (b), it follows that $\mathbf{L}^* = 0$. By (10.1.17), we obtain $\psi_R^* = 0$. Hence $\psi_i^* = 0$ for all $i = 1, \ldots, m$. The equation (10.1.15) therefore becomes

$$Jh(x^*)^T h^* = 0.$$

It remains to show that $h^* = 0$. This clearly holds under (c1). Noting that $Jh(x^*)^T h^*$ is the gradient at x^* of the norm function $\frac{1}{2} h^T h$, which is convex because h is affine, we easily deduce that $h(x^*) = 0$ under (c2). $\square$

A particularly interesting case is when K is polyhedral, i.e.,

$$h(x) \equiv Ax - b, \quad \text{and} \quad g(x) \equiv Cx - d,$$

with $A \in \mathbb{R}^{\ell \times n}$, $b \in \mathbb{R}^{\ell}$, $C \in \mathbb{R}^{m \times n}$ and $d \in \mathbb{R}^{m}$. This case corresponds to the linearly constrained VI for which the KKT conditions are both necessary and sufficient for a vector x to be a solution of the VI. With h and g both being affine functions, the null space of the gradients (10.1.14) is the intersection of the null space of A and the null space of C, which is the lineality space of the set K. Another simplification is that $J_x \mathbf{L}(x, \mu, \lambda) = JF(x)$. Putting all these remarks together, we obtain the following consequence of Theorem **10.1.10** for a linearly constrained VI.

10.1.11 Corollary. Let $(x^*, \mu^*, \lambda^*) \in \mathbb{R}^{n+\ell+m}$ be a stationary point of $\theta_{\rm FB}(x, \mu, \lambda)$. Assume that K is polyhedral and $JF(x^*)$ is positive semidefinite. If any one of the following three conditions holds:

(a) $JF(x^*)$ is positive definite on the lineality space of K,

(b) the set K is bounded, or

(c) K is contained in the nonnegative orthant $\mathbb{R}^n_+$,

then x^* solves the linearly constrained VI (K, F).

Proof. The aforementioned observations establish the corollary under (a). If either (b) or (c) holds, then the set K contains no lines; hence the lineality space of K is the singleton $\{0\}$. Hence (a) holds. □

Another noteworthy case is $K = K_1 \cap K_2$, where K_1 is finitely represented and K_2 is polyhedral. If the set K_2 satisfies the assumptions of Corollary **10.1.11** or, more generally, if its lineality space is $\{0\}$, then condition (b) in Theorem **10.1.10** becomes superfluous. As an illustration of this case, suppose that we have a VI (K, F) with K contained, for example, in the nonnegative orthant. In order to guarantee that a stationary point $\theta_{\rm FB}$ is a zero of $\Psi_{\rm FB}$, we need conditions (a) and (b) of Theorem **10.1.10**. However we can consider the VI $(K \cap \{x : x \geq 0\}, F)$, which is obviously equivalent to the original VI (K, F); with this equivalent problem, requirement (b) of Theorem **10.1.10** is now superfluous.

With the results established so far we can now extend all the algorithms given in Section **9.1** for the unconstrained minimization of $\theta_{\rm FB}(x, \mu, \lambda)$. Formally there is no need for any major change, except for the equation and merit function involved. We do not repeat the details; instead we highlight two disadvantages hidden in this application. The first point

concerns the boundedness of the level sets of $\theta_{\rm FB}(x,\mu,\lambda)$. Specifically, we generally cannot expect $\theta_{\rm FB}(x,\mu,\lambda)$ to have bounded level sets, no matter what assumptions we make on F. The following example identifies what the problem is.

10.1.12 Example. Let $n = 2$ and $m = \ell = 1$, and set

$$F(x) \equiv \begin{pmatrix} x_1 \\ x_2 \end{pmatrix},$$

which is a strongly monotone function, and

$$h(x) \equiv x_1 - x_2, \qquad g(x) \equiv e^{-x_1}.$$

Let

$$x^k \equiv (k,k), \quad \mu_k \equiv k, \quad \lambda_k \equiv 2k/e^{-k}.$$

It is easy to check that $\Phi_{\rm FB}(x^k,\mu_k,\lambda_k)$ tends to zero, so that $\theta_{\rm FB}(x,\mu,\lambda)$ does not have bounded level sets. Note that the two gradients $\nabla h(x)$ and $\nabla g(x)$ are linearly independent for all x in $\mathbb{R}^2$. □

Due to the very strong properties of the function F and the linear independence of the gradients of the constraint functions in the above example, it is not very likely that reasonable conditions exist that will imply the boundedness of the level sets of $\theta_{\rm FB}(x,\mu,\lambda)$. The main reason for this undesirable feature is that the primary variable x and the multiplier variables μ and λ are, in a sense, "decoupled".

The second point of concern is the following. The conditions given in Theorem **10.1.10** to ensure that stationary points of the merit function are KKT triples, although relatively weak, include the assumption that the partial Jacobian of the Lagrangian with respect to the x-variable,

$$J_x\mathbf{L}(x,\mu,\lambda) \;=\; JF(x) + \sum_{j=1}^{\ell} \mu_j\, \nabla^2 h_j(x) + \sum_{i=1}^{m} \lambda_i\, \nabla^2 g_i(x),$$

is positive semidefinite. This condition is satisfied for all triples (x,μ,λ) if, for instance, F is monotone and the constraints are affine so that $JF(x)$ is positive semidefinite and the Hessians $\nabla^2 h_j(x)$ and $\nabla^2 g_i(x)$ all vanish. However, if one considers the most natural extension of this case, where F is monotone, h is affine, and g is convex (but nonlinear), it is easy to see that, since the matrices $\nabla^2 g_i(x)$ are positive semidefinite, if λ_i is negative and large enough, $J_x\mathbf{L}(x,\mu,\lambda)$ is not likely to be positive semidefinite. Note also that this conclusion is independent of the structure of F or h. We illustrate this point by the following example.

10.1.13 Example. Let $n = m = 1$ and $\ell = 0$ and set

$$F(x) \equiv \tfrac{1}{2}\,x - 5, \qquad g(x) \equiv \tfrac{1}{2}\,x^2 - x,$$

so that F is strongly monotone and g is strongly convex. It is easy to compute that $\nabla\theta_{\rm FB}(x,\lambda) = 0$ both for $(x,\lambda) = (0,-1)$ and $(x,\lambda) = (2,4)$. But while the latter stationary point satisfies the KKT conditions (10.1.2), the former point does not. In fact, it is easy to check that

$$\theta_{\rm FB}(0,-1) \;\neq\; 0 \;=\; \theta_{\rm FB}(2,4),$$

so that $(2,4)$ is a solution of (10.1.3) but $(0,-1)$ is not. □

The feature highlighted in the above example is due to the special structure of the KKT system and is rather disturbing, since it implies that, even if we solve a strongly monotone VI over a closed convex set defined by nonlinear inequalities, we cannot ensure convergence to the unique solution of the VI. Since the problem clearly arises because of the negative values of some variable λ_i that we know *a priori* has to be nonnegative at a solution of system (10.1.3), we are naturally led to consider the following constrained version of problem (10.1.3):

$$\begin{array}{ll} \text{minimize} & \theta_{\rm FB}(x,\mu,\lambda) \\ \text{subject to} & \lambda \;\geq\; 0. \end{array} \tag{10.1.19}$$

A triple (x,μ,λ) is a stationary point of (10.1.19) if and only if

$$\begin{aligned} \nabla_x\theta_{\rm FB}(x,\mu,\lambda) \;&=\; 0 \\ \nabla_\mu\theta_{\rm FB}(x,\mu,\lambda) \;&=\; 0 \\ 0 \;\leq\; \lambda \;\perp\; \nabla_\lambda\theta_{\rm FB}(x,\mu,\lambda) \;&=\; 0. \end{aligned}$$

Usually, the main reason to consider constrained reformulations of a CP is to avoid points where the function F might be not defined; in this case we consider the simply constrained reformulation (10.1.19) in order to be able to give reasonable conditions under which every stationary point of the minimization problem is a desired KKT triple of the VI (K,F). In the next theorem we show that because of the presence of the nonnegativity constraint in (10.1.19), we can actually achieve our goal. The difference between the theorem below and the previous Theorem **10.1.10** is condition (b).

10.1.14 Theorem. Let $(x^*,\mu^*,\lambda^*) \in \mathbb{R}^{n+\ell+m}$ be a stationary point of (10.1.19). It holds that (x^*,μ^*,λ^*) is a solution of the KKT system (10.1.2) under conditions (a), (b), and [either (c1) or (c2)] stated below:

(a) $J_x\mathbf{L}(x^*,\mu^*,\lambda^*)$ is positive semidefinite on $\mathbb{R}^n$;

(b) $J_x\mathbf{L}(x^*,\mu^*,\lambda^*)$ is strictly copositive on the cone

$$\begin{aligned}\mathcal{C}(x^*,\lambda^*) \equiv \{ \quad & v \in \mathbb{R}^n : Jh(x^*)v = 0,\\ & \nabla g_i(x^*)^T v \geq 0, \quad \forall\, i \in I_{00} \cup I_<\\ & \nabla g_i(x^*)^T v = 0, \quad \forall\, i \in I_+ \cup I_R \,\};\end{aligned}$$

(c1) $Jh(x^*)$ has full row rank;

(c2) h is an affine function.

Proof. The proof resembles that of Theorem **10.1.10**. As in the previous proof, the stationarity of (x^*,μ^*,λ^*) implies

$$(\, J_x\mathbf{L}^*(x^*,\mu^*,\lambda^*)\,)^T\mathbf{L}^* - Jh(x^*)^T h^* - Jg_R(x^*)^T(\, D_g^*\,)_R\psi_R^* = 0 \tag{10.1.20}$$

$$\left.\begin{aligned} Jh(x^*)\mathbf{L}^* &= 0\\ 0 \leq \lambda_i^* \perp \nabla g_i(x^*)^T\mathbf{L}^* &\geq 0, \quad \forall\, i \in I_0 \cup I_< \end{aligned}\right\} \tag{10.1.21}$$

$$0 \leq \lambda_R^* \perp Jg_R(x^*)\mathbf{L}^* + (\, D_\lambda^*\,)_R\psi_R^* \geq 0, \tag{10.1.22}$$

where $(D_g^*)_R$ and $(D_\lambda^*)_R$ are two negative definite diagonal matrices. For each index $i \in I_R$ such that $\lambda_i^* > 0$, we have by (10.1.22),

$$\nabla g_i(x^*)^T\mathbf{L}^* + (\, D_\lambda^*\,)_{ii}\psi_i^* = 0.$$

For each index $i \in I_R$ such that $\lambda_i^* = 0$, we must have $g_i(x^*) > 0$, which implies $\psi_i^* = 2g_i(x^*) > 0$. Thus $(D_\lambda^*)_{ii}\psi_i^* < 0$; hence $\nabla g_i(x^*)^T\mathbf{L}^* > 0$. Premultiplying (10.1.20) by $(\mathbf{L}^*)^T$ and utilizing (10.1.21), we deduce

$$\begin{aligned}&(\,\mathbf{L}^*)^T J_x\mathbf{L}(x^*,\mu^*,\lambda^*)\mathbf{L}^* =\\ &\quad -\sum_{i\in I_R:\lambda_i^*>0} d_{ii}\,(\,\psi_i^*\,)^2 + \sum_{i\in I_R:\lambda_i^*=0} (\,\nabla g_i(x^*)^T\mathbf{L}^*\,)\,(D_g^*)_{ii}\,\psi_i^*,\end{aligned}$$

where $d_{ii} \equiv (D_g^*)_{ii}(D_\lambda^*)_{ii}$ is a positive scalar. By assumption (a), the left-hand side is nonnegative, whereas the right-hand side is nonpositive. Hence we must have

$$(\,\mathbf{L}^*)^T(\, J_x\mathbf{L}(x^*,\mu^*,\lambda^*)\,)^T\mathbf{L}^* = 0,$$

$$\psi_i^* = 0, \quad \forall\, i \in I_R \text{ such that } \lambda_i^* > 0,$$

$$\nabla g_i(x^*)^T\mathbf{L}^* = 0, \quad \forall\, i \in I_R \text{ such that } \lambda_i^* = 0.$$

Consequently $Jg_R(x^*)\mathbf{L}^* = 0$. Taking into account (10.1.21), we deduce that $\mathbf{L}^*$ belongs to the cone $\mathcal{C}(x^*,\lambda^*)$. By (b), we deduce $\mathbf{L}^* = 0$. This

implies that $\psi_i^* = 0$ for all $i = 1, \ldots, m$. The proof that $h^* = 0$ is the same as in Theorem **10.1.10**. □

The next corollary follows easily from Theorem **10.1.14** and does not require a proof.

10.1.15 Corollary. Let $(x^*, \mu^*, \lambda^*) \in \mathbb{R}^{n+\ell+m}$ be a stationary point of (10.1.19). If

(a) F is monotone, h is affine and each g_i is convex;

(b) $J_x\mathbf{L}(x^*, \mu^*, \lambda^*)$ is strictly copositive on the cone $\mathcal{C}(x^*, \lambda^*)$;

then x^* solves the VI (K, F). □

Assumption (a) of Corollary **10.1.15** clearly holds for monotone VIs and for convex optimization problems. Condition (b) is certainly satisfied by a monotone VI (K, F) with a strongly monotone function F. The two assumptions of Corollary **10.1.15** are obviously satisfied by the problem in Example **10.1.13**.

Iterative algorithms for the minimization of $\theta_{\mathrm{FB}}(x, \mu, \lambda)$ that maintain the nonnegativity of λ throughout the process can be developed easily in several ways as described in Chapter **8**. Since this is fairly straightforward, we omit the details. Instead, we briefly discuss the solution of the Newton equation associated with a given triple (x, μ, λ) and a matrix H in the Newton approximation $\mathcal{A}(x, \mu, \lambda)$. The equation is:

$$\Phi_{\mathrm{FB}}(x, \mu, \lambda) + H \begin{pmatrix} dx \\ d\mu \\ d\lambda \end{pmatrix} = 0$$

in which we are Solving for the triple $(dx, d\mu, d\lambda)$. Writing out the equation, we have

$$\mathbf{L}(x, \mu, \lambda) + J_x\mathbf{L}(x, \mu, \lambda)dx + \sum_{j=1}^{\ell} d\mu_j \nabla h_j(x) + \sum_{i=1}^{m} d\lambda_i \nabla g_i(x) = 0$$

$$h_j(x) + \nabla h_j(x)dx = 0, \quad j = 1, \ldots, \ell$$

$$\nabla g_i(x)^T dx = 0, \quad i \in I_+ \cup I_{01}$$

$$\psi_{\mathrm{FB}}(-g_i(x), \lambda_i) - (D_g)_{ii} \nabla g_i(x)^T dx + (D_\lambda)_{ii} d\lambda_{ii} = 0, \quad i \in I_R$$

$$-(D_g)_{ii} \nabla g_i(x)^T dx + (D_\lambda)_{ii} d\lambda_{ii} = 0, \quad i \in I_{02}$$

$$d\lambda_i = 0, \quad i \in I_{03} \cup I_< .$$

We may use the last three equations to solve for $d\lambda_i$ for i in $I_{R2} \cup I_{03} \cup I_<$ and then substitute into the first equation, obtaining

$$\tilde{r}(x,\lambda) + \tilde{M}dx + \sum_{j=1}^{\ell} d\mu_j \nabla h_j(x) + \sum_{i \in I_+ \cup I_{01}} d\lambda_i \nabla g_i(x) \;=\; 0,$$

where

$$\tilde{M} \;\equiv\; J_x \mathbf{L}(x,\mu,\lambda) + \sum_{i \in I_{R2}} (D_g)_{ii} (D_\lambda)_{ii}^{-1} \nabla g_i(x) \nabla g_i(x)^T$$

$$\tilde{r}(x,\lambda) \;\equiv\; \mathbf{L}(x,\mu,\lambda) - \sum_{i \in I_R} (D_\lambda)_{ii}^{-1} \psi_{\rm FB}(-g_i(x),\lambda_i) \nabla g_i(x).$$

Thus the resulting system of linear equations that needs to be solved is of the form:

$$\begin{pmatrix} \tilde{r}(x,\lambda) \\ h(x) \\ 0 \end{pmatrix} + \begin{bmatrix} \tilde{M} & Jh^T & Jg_{I_+ \cup I_{01}}^T \\ Jh & 0 & 0 \\ Jg_{I_+ \cup I_{01}} & 0 & \end{bmatrix} \begin{pmatrix} dx \\ d\mu \\ d\lambda_{I_+ \cup I_{01}} \end{pmatrix} \;=\; 0;$$

the latter system is of the order $n + \ell + |I_+| + |I_{01}|$, which is less than the order $(n + \ell + m)$ of the full system.

10.1.2 Using the min function

We next consider another reformulation of the KKT conditions: the one using the min function. We expect this reformulation to be in some sense better than the one based on the FB function, but less amenable to globalization. We briefly elaborate on this point in the remaining part of this section. First of all we rewrite the KKT conditions as:

$$\Phi_{\min}(x,\mu,\lambda) \;\equiv\; \begin{pmatrix} \mathbf{L}(x,\mu,\lambda) \\ -h(x) \\ \min(-g(x),\lambda) \end{pmatrix}, \quad (\,x,\mu,\lambda\,) \in \mathbb{R}^{n+\ell+m}. \quad (10.1.23)$$

We know that $\Phi_{\min}$ is semismooth and actually strongly semismooth if F has locally Lipschitz continuous derivatives and g and h have locally Lipschitz continuous second derivatives. However, the natural merit function

$$\theta_{\min}(x,\mu,\lambda) \;\equiv\; \tfrac{1}{2}\, \Phi_{\min}(x,\mu,\lambda)^T \Phi_{\min}(x,\mu,\lambda)$$

is not differentiable in general. The following proposition shows that it is possible to give an exact expression for the limiting Jacobian of $\Phi_{\min}$. For

this purpose, we introduce three index sets:

$$\begin{aligned}\alpha(x,\lambda) &\equiv \{\, i \in I : -g_i(x) < \lambda_i \,\}\\ \beta(x,\lambda) &\equiv \{\, i \in I : -g_i(x) = \lambda_i \,\}\\ \gamma(x,\lambda) &\equiv \{\, i \in I : -g_i(x) > \lambda_i \,\}.\end{aligned}$$

10.1.16 Proposition. For an arbitrary triple (x,μ,λ) in $\mathbb{R}^{n+\ell+m}$, the limiting Jacobian $\operatorname{Jac}\Phi_{\min}(x,\mu,\lambda)$ is equal to the family of matrices

$$\begin{bmatrix} J_x\mathbf{L}(x,\mu,\lambda) & Jh(x)^T & Jg(x)^T \\ -Jh(x) & 0 & 0 \\ -D_a(x,\lambda)Jg(x) & 0 & D_b(x,\lambda) \end{bmatrix},$$

where

$$D_a(x,\lambda) \equiv \operatorname{diag}\left(a_1(x,\lambda),\ldots,a_m(x,\lambda)\right),$$

is the diagonal matrix whose diagonal elements are given by

$$a_i(x,\lambda) \equiv \begin{cases} 1 & \text{if } i \in \alpha(x,\lambda) \\ 0 \text{ or } 1 & \text{if } i \in \beta(x,\lambda) \\ 0 & \text{if } i \in \gamma(x,\lambda), \end{cases}$$

and $D_b(x,\lambda) \equiv \mathbf{I}_m - D_a(x,\lambda)$.

Proof. This follows immediately from the definition of $\Phi_{\min}(x,\mu,\lambda)$ and the independence of the variables x and λ. □

It is natural at this stage to consider the Newton approximation given by the limiting Jacobian and to consider the nonsingularity of this Newton scheme. In the case of $\Phi_{\mathrm{FB}}(x,\mu,\lambda)$ we saw that the family of matrices $M_{\mathrm{FB}}(\mathcal{J})$ played an important role. For $\Phi_{\min}(x,\mu,\lambda)$, a related family of matrices plays a similar role. The following definition extends the b-regularity concept for the NCP to the KKT system. In this definition, (x,μ,λ) is not required to be a KKT triple.

10.1.17 Definition. A triple $(x,\mu,\lambda) \in \mathbb{R}^{n+\ell+m}$ is said to be *quasi-regular* if the matrices

$$M_{\min}(\mathcal{J}) \equiv \begin{bmatrix} J_x\mathbf{L}(x,\mu,\lambda) & Jh(x)^T & Jg_\alpha(x)^T & Jg_{\mathcal{J}}(x)^T \\ -Jh(x) & 0 & 0 & 0 \\ -Jg_\alpha(x) & 0 & 0 & 0 \\ -Jg_{\mathcal{J}}(x) & 0 & 0 & 0 \end{bmatrix}$$

are nonsingular for all subsets $\mathcal{J}$ of $\beta(x,\lambda)$. □

If (x^*, μ^*, λ^*) is a KKT triple, then this triple is quasi-regular if and only if it is b-regular. As such, every quasi-regular KKT triple must be locally unique. (This conclusion also follows from Theorems **10.1.18** and **7.2.10**.) Moreover, by taking $\mathcal{J} = \beta(x^*, \lambda^*)$, it follows that if (x^*, μ^*, λ^*) is a quasi-regular KKT triple, then the LICQ must hold at x^*; i.e., the gradients of the active constraints

$$\{ \nabla h_j(x^*) : j = 1, \ldots, \ell \} \cup \{ \nabla g_i(x^*) : i \in I_0 \}$$

must be linearly independent. Clearly, every strongly stable KKT stable must be quasi-regular. The fundamental role of quasi-regularity is explained in the next result.

10.1.18 Theorem. A triple $(x, \mu, \lambda) \in \mathbb{R}^{n+\ell+m}$ is quasi-regular if and only if all matrices in $\operatorname{Jac} \Phi_{\min}(x, \mu, \lambda)$ are nonsingular.

Proof. In view of Proposition **10.1.16**, every matrix $H \in \operatorname{Jac} \Phi_{\min}(x, \mu, \lambda)$ is associated with an index set $\mathcal{J} \subseteq \beta(x, \lambda)$ with complement $\mathcal{I} \equiv \beta(x, \lambda) \backslash \mathcal{J}$ such that

$$H = \begin{bmatrix} J_x L & Jh^T & Jg_\alpha^T & Jg_{\mathcal{J}}^T & Jg_{\mathcal{I}}^T & Jg_\gamma^T \\ -Jh & 0 & 0 & 0 & 0 & 0 \\ -Jg_\alpha & 0 & 0 & 0 & 0 & 0 \\ -Jg_{\mathcal{J}} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \mathbf{I}_{|\mathcal{I}|} & 0 \\ 0 & 0 & 0 & 0 & 0 & \mathbf{I}_{|\gamma|} \end{bmatrix}.$$

Conversely, every pair of complementary index subsets $\mathcal{J}$ and $\mathcal{I}$ of $\beta(x, \lambda)$ determines an element of the limiting Jacobian $\operatorname{Jac} \Phi_{\min}(x, \mu, \lambda)$. Clearly, the above matrix H is nonsingular if and only if $M_{\min}(\mathcal{J})$ is nonsingular. □

As in the case of the min reformulation for CPs, it is easy to see that the Newton method based on the limiting Jacobian is superlinearly (or quadratically) convergent under an assumption which is weaker than the assumption needed for the FB reformulation of the KKT conditions. Furthermore, the structure of the elements in the limiting Jacobian $\operatorname{Jac} \Phi_{\min}$ also results in a reduction in size of the linear equations that need to be solved at each iteration.

10.2 Merit Functions for VIs

In this section we define both constrained and unconstrained smooth merit functions for the solution of a VI (K, F) in general form, without assuming that K is finitely representable. These merit functions will be the basis for the development of iterative algorithms for the solution of VIs to be examined in the next section.

In Subsection **1.5.3**, we already considered the merit function

$$\theta_{\rm gap}(x) \equiv \sup_{y \in K} F(x)^T (x - y),$$

and observed that it is nonnegative for all $x \in K$; moreover, a zero of $\theta_{\rm gap}$ that lies in K must solve the VI (K, F). The difficulty with the function $\theta_{\rm gap}$ is that it is nondifferentiable and extended-valued in general. The latter drawback is not present if the set K is bounded; but even in such a favorable situation, $\theta_{\rm gap}$ may still be nondifferentiable. To better understand the reasons for the nondifferentiability of $\theta_{\rm gap}$, we study first the differentiability properties of functions given by the supremum of a family of real-valued functions. This will suggest a modification of the function $\theta_{\rm gap}$ that will allow us to overcome the nondifferentiability of $\theta_{\rm gap}$.

10.2.1 Theorem. Let $K \subseteq \mathbb{R}^m$ be a nonempty, closed set and let $\Omega \subseteq \mathbb{R}^n$ be a nonempty, open set. Assume that the function $f : \Omega \times K \to \mathbb{R}$ is continuous on $\Omega \times K$ and that $\nabla_x f(x, y)$ exists and is continuous on $\Omega \times K$. Define the function $g : \Omega \to \mathbb{R} \cup \{\infty\}$ by

$$g(x) \equiv \sup_{y \in K} f(x, y), \quad x \in \Omega$$

and set

$$M(x) \equiv \{\, y \in K : g(x) = f(x, y) \,\}.$$

Let $x \in \Omega$ be a given vector. Suppose that a neighborhood $\mathcal{N} \subseteq \Omega$ of x exists such that $M(x')$ is nonempty for all $x' \in \mathcal{N}$ and the set

$$\bigcup_{x' \in \mathcal{N}} M(x') \tag{10.2.1}$$

is bounded. The following two statements (a) and (b) are valid.

(a) The function g is directionally differentiable at x and

$$g'(x; d) = \sup_{y \in M(x)} \nabla_x f(x, y)^T d.$$

(b) If $M(x)$ reduces to a singleton, say $M(x) = \{y(x)\}$, then g is Gâteaux differentiable at x and

$$\nabla g(x) \;=\; \nabla_x f(x, y(x)).$$

Proof. Let $d \in \mathbb{R}^n$ be given and suppose that $\bar{t} > 0$ is small enough so that $x + td \in \mathcal{N}$ for every $t \in (0, \bar{t}]$. By the definition of g one can write, for every $y \in M(x)$,

$$g(x + td) - g(x) \;\geq\; f(x + td, y) - f(x, y),$$

so that by dividing by t and passing to the limit we have

$$\liminf_{t \downarrow 0} \frac{g(x + td) - g(x)}{t} \;\geq\; \sup_{y \in M(x)} \nabla_x f(x, y)^T d. \tag{10.2.2}$$

Similarly, for every $y_t \in M(x + td)$ we have

$$g(x + td) - g(x) \;\leq\; f(x + td, y_t) - f(x, y_t) \;=\; t \nabla_x f(\bar{x}_t, y_t)^T d$$

for some suitable $\bar{x}_t \in (x, x + td)$. This implies

$$\frac{g(x + td) - g(x)}{t} \;\leq\; \nabla_x f(\bar{x}_t, y_t)^T d, \qquad \forall\, y_t \in M(x + td).$$

By the boundedness of the union (10.2.1),it is easy to show that for every sequence $\{t_k\}$ of positive numbers converging to zero, if $\{y^k\}$ is any sequence of vectors such that $y^k \in M(x + t_k d)$ for every k, then $\{y^k\}$ is bounded and every limit point of this sequence belongs to $M(x)$. Consequently, it follows from the above inequality that

$$\limsup_{t \downarrow 0} \frac{g(x + td) - g(x)}{t} \;\leq\; \sup_{y \in M(x)} \nabla_x f(x, y)^T d. \tag{10.2.3}$$

From equations (10.2.2) and (10.2.3), part (a) follows easily. Part (b) is a trivial consequence of (a). □

10.2.1 The regularized gap function

Theorem **10.2.1** shows clearly the reason why θ_{gap} is not necessarily differentiable. Namely, it has to do with the likely existence of multiple optimal solutions to the maximization problem that defines θ_{gap}. There is a simple way to remedy this situation that will accomplish a dual purpose. Specifically, the objective function $y \mapsto F(x)^T(x - y)$ of the maximization problem that defines $\theta_{\text{gap}}(x)$ is linear in y. By adding a strongly concave function to this linear objective function, we can ensure that the resulting

maximization problem has a unique optimal solution, which is continuously differentiable in x. Based on this consideration, we introduce the following definition in which we add a strictly concave quadratic function to "regularize" the gap function.

10.2.2 Definition. Let the VI (K, F) be given, with F defined on an open set Ω containing K. Let c be a positive constant and let G be a symmetric positive definite matrix. The *regularized gap function* of the VI (K, F) is defined as

$$\theta_c(x) \equiv \sup_{y \in K} \left\{ F(x)^T(x - y) - \frac{c}{2}(x - y)^T G(x - y) \right\},$$

for all x in Ω. □

The above notation stresses the role of the scalar c and not that of the matrix G. We adopted this notation for simplicity, because usually G is fixed (for most practical purposes, G can be taken to be the identify matrix); as we see below, there is a need to consider the regularized gap function for distinct values of c. We note immediately that

$$\theta_b(x) \leq \theta_a(x), \quad \forall\, x \in \Omega,$$

for any two scalars $b > a > 0$. Similar to the gap function $\theta_{\text{gap}}(x)$, we obviously have

$$\theta_c(x) \geq 0, \quad \forall\, x \in K.$$

Defining the function

$$\psi_c(x, y) \equiv F(x)^T(x - y) - \frac{c}{2}(x - y)^T G(x - y), \quad \forall\, (x, y) \in \Omega \times \mathbb{R}^n$$

we see that

$$\theta_c(x) \equiv \sup_{y \in K} \psi_c(x, y).$$

Clearly, with K being a nonempty closed set, for each $x \in \Omega$, there exists a unique vector, denoted $y_c(x)$, in K that maximizes $\psi_c(x, \cdot)$ on K; thus the supremum in $\theta_c(x)$ can be replaced by a maximum. Moreover, with F being continuously differentiable on Ω, the function $\theta_c(x)$ is continuously differentiable on Ω, by Theorem **10.2.1**. The key to the continuous differentiability of $\theta_c(x)$ is the continuity of $y_c(x)$. In turn, the latter continuity follows from the result below, which gives an explicit representation of the maximizer $y_c(x)$ in terms of the skewed projector $\Pi_{K,G}$ defined in (1.5.9). The result also summarizes various important properties of the regularized gap function and its relation to the VI (K, F).

10.2.3 Theorem. Let $K \subseteq \mathbb{R}^n$ be closed convex and $F : \Omega \supset K \to \mathbb{R}^n$ be continuous on the open set Ω. Let c be a positive scalar and let G be a symmetric positive definite matrix. The following four statements are valid.

(a) For every $x \in \Omega$, $y_c(x) = \Pi_{K,G}(x - c^{-1}G^{-1}F(x))$.

(b) $\theta_c(x)$ is continuous on Ω and nonnegative on K.

(c) $[\ \theta_c(x) = 0,\ x \in K\ ]$ if and only if $x \in \mathrm{SOL}(K, F)$.

(d) If F is continuously differentiable on Ω, then so is θ_c; moreover,

$$\nabla\theta_c(x) \ = \ F(x) + (\, JF(x) - c\,G\,)^T(\, x - y_c(x)\,). \tag{10.2.4}$$

Proof. Since

$$\begin{aligned}\psi_c(x,y) \ &= \ (2c)^{-1}\, F(x)^T G^{-1} F(x) - \\ &\quad \frac{c}{2}\,(\, x - c^{-1}\, G^{-1}F(x) - y\,)G(\, x - c^{-1}\, G^{-1}F(x) - y\,),\end{aligned}$$

the expression for $y_c(x)$ in (a) follows immediately from the definition of the skewed projector. This expression shows that y_c is a continuous function on Ω; hence so is θ_c. This establishes (b). In terms of $y_c(x)$, we have

$$\theta_c(x) \ = \ F(x)^T(\, x - y_c(x)\,) - \frac{c}{2}\,(\, x - y_c(x)\,)^T G(\, x - y_c(x)\,).$$

Suppose $\theta_c(x) = 0$ and $x \in K$. On the one hand,

$$F(x)^T(\, x - y_c(x)\,) \ = \ \frac{c}{2}\,(\, x - y_c(x)\,)^T G(\, x - y_c(x)\,) \ \geq \ 0. \tag{10.2.5}$$

On the other hand, by the variational principle of the maximization problem that defines $\theta_c(x)$, we have

$$(\, z - y_c(x)\,)^T\,[\, F(x) + c\,G(\, y_c(x) - x\,)\,] \ \geq \ 0, \quad \forall\, z \ \in \ K.$$

Substituting $z = x$, we obtain

$$(\, x - y_c(x)\,)^T\,[\, F(x) + c\,G(\, y_c(x) - x\,)\,] \ \geq \ 0,$$

which together with (10.2.5) yields $y_c(x) = x$. By (a), we therefore have $x \ = \ \Pi_{K,G}(x - c^{-1}G^{-1}F(x))$, which shows that x solves the VI (K, F). Conversely, if $x \in \mathrm{SOL}(K, F)$, then clearly $\theta_c(x) \leq 0$, which easily implies $\theta_c(x) \ = \ 0$. This establishes (c). Finally, the gradient formula (10.2.4) follows easily from Theorem **10.2.1**. By part (b) of this theorem, we have

$$\nabla\theta_c(x) \ = \ \nabla_x\psi_c(x, y_c(x)).$$

From the definition of ψ_c, (10.2.4) follows readily. □

10.2.4 Remark. The convexity of the set K is not needed in proving the "if" part in statement (c) of Theorem **10.2.3**. □

It is important to note that regardless of the domain of definition Ω of the function F, the nonnegativity of the regularized gap function θ_c is valid only on K; moreover, in order for a zero of θ_c to be a solution of the VI (K, F), it is essential that such a zero belongs to K; see part (c) of Theorem **10.2.3**. Therefore, θ_c is a valid merit function for the VI (K, F) only on the set K, irrespective of the domain Ω, which could be the entire space $\mathbb{R}^n$.

When K is polyhedral (or more generally, finitely representable satisfying CQs), the projection $\Pi_{K,G}$ is piecewise linear (piecewise smooth); hence y_c a is PC^1 function, provided that F is continuously differentiable. In this case, the regularized gap function θ_c is a semismooth function.

Using the regularized gap function, we define the following constrained minimization problem, called the *regularized gap program*:

$$\begin{array}{ll} \text{minimize} & \theta_c(x) \\ \text{subject to} & x \in K. \end{array} \tag{10.2.6}$$

The constraint K is needed in order to ensure that a global minimizer of the program with zero objective value solves the VI (K, F). As usual, we are interested in necessary and sufficient conditions for a stationary point of this program to be a solution of the VI (K, F). Theorem **10.2.5** below identifies two such conditions on a certain pair consisting of a closed convex cone and the transpose of the Jacobian matrix of F. In order to present the result, we introduce the cone in question. The notation $(-F(x))^*$ denotes the dual cone of the singleton $\{-F(x)\}$; i.e.,

$$(\,-F(x)\,)^* \equiv \{\, d \in \mathbb{R}^n \,:\, d^T F(x) \leq 0 \,\}.$$

For a vector $x \in K$, we write

$$T_c(x; K) \equiv \mathcal{T}(x; K) \cap (\,-\mathcal{T}(y_c(x); K)\,)$$

and

$$T_c(x; K, F) \equiv T_c(x; K) \cap (\,-F(x)\,)^*.$$

Since K is closed and convex, $T_c(x; K)$ is a closed convex cone containing the vector $y_c(x) - x$; $T_c(x; K, F)$ is a also a closed convex cone. At first glance, the latter cone seems like a novel object that has not been encountered until now. In what follows, we show that $T_c(x; K, F)$ reduces to a familiar space when x is a solution of the VI (K, F). For this purpose,

assume that $x \in \mathrm{SOL}(K, F)$. It follows that $\theta_c(x) = 0$, and thus $x = y_c(x)$ because $y_c(x)$ is the unique vector in K that satisfies

$$\theta_c(x) \;=\; F(x)^T(\,x - y_c(x)\,) + \frac{c}{2}\,(\,y_c(x) - x\,)^T G(\,y_c(x) - x\,).$$

Therefore, $T_c(x; K) = \mathcal{T}(x; K) \cap (-\mathcal{T}(x; K))$, which is equal to the lineality space of the tangent cone of K at x. Moreover,

$$\mathcal{T}(x; K) \,\cap\, (\,-F(x)\,)^* \;=\; \mathcal{T}(x; K) \,\cap\, F(x)^{\perp},$$

which is equal the critical cone $\mathcal{C}(x; K, F)$. Hence

$$T_c(x; K) \,\cap\, (\,-F(x)\,)^* \;=\; \mathcal{C}(x; K, F) \,\cap\, (\,-\mathcal{C}(x; K, F)\,).$$

Consequently, we have shown that if $x \in \mathrm{SOL}(K, F)$, then $T_c(x; K, F)$ is equal to the lineality space of the critical cone $\mathcal{C}(x; K, F)$ of the VI (K, F) at x. In particular, $T_c(x; K, F)$ is a linear subspace if $x \in \mathrm{SOL}(K, F)$. In the following result, we need the cone $T_c(x; K, F)$ for a vector $x \in K$ that is not (yet) known to be a solution of the VI (K, F).

10.2.5 Theorem. Let $K \subseteq \mathbb{R}^n$ be closed convex and $F : \Omega \supset K \to \mathbb{R}^n$ be continuously differentiable on the open set Ω. Let c be a positive scalar and let G be a symmetric positive definite matrix. Suppose that x is a stationary point of (10.2.6). The following three statements are equivalent.

(a) x solves the VI (K, F).

(b) $T_c(x; K, F)$ is contained in $F(x)^{\perp}$.

(c) The implication below holds:

$$\left.\begin{array}{rl} d \;\in\; T_c(x; K, F) \\ JF(x)^T d \;\in\; -T_c(x; K, F)^* \end{array}\right\} \;\Rightarrow\; d^T F(x) \;=\; 0. \qquad (10.2.7)$$

Proof. If $x \in \mathrm{SOL}(K, F)$, then by the above observations, it follows that $T_c(x; K, F)$ is contained in $F(x)^{\perp}$. Thus (a) implies (b). Clearly, (b) implies (c). It remains to show that (c) implies (a). Assume that (10.2.7) holds. We have by the definition of $y_c(x)$,

$$(\,z - y_c(x)\,)^T\,[\,F(x) + c\,G(\,y_c(x) - x\,)\,] \;\geq\; 0, \quad \forall\, z \;\in\; K. \qquad (10.2.8)$$

As noted above, the vector $dx \equiv y_c(x) - x$ belongs to $T_c(x; K)$. Moreover, the substitution of $z = x$ in (10.2.8) yields

$$(\,x - y_c(x)\,)^T F(x) \;\geq\; 0;$$

thus $dx \in (-F(x))^*$. Hence dx belongs to $T_c(x; K, F)$. We next verify that $JF(x)^T dx$ belongs to $-T_c(x; K, F)^*$. Let d be an arbitrary vector in the cone $T_c(x; K)$. Since x is a stationary point of (10.2.6), we have

$$(z - x)^T \nabla\theta_c(x) \geq 0, \quad \forall z \in K.$$

Substituting the expression for $\nabla\theta_c(x)$ from part (d) of Theorem **10.2.3**, we obtain

$$(z - x)^T [F(x) + (JF(x)^T - cG)(x - y_c(x))] \geq 0.$$

Therefore, since $d \in \mathcal{T}(x; K)$, it follows that

$$d^T [F(x) + (JF(x)^T - cG)(x - y_c(x))] \geq 0.$$

Moreover, since $-d \in \mathcal{T}(y_c(x); K)$, (10.2.8) yields

$$-d^T [F(x) + cG(y_c(x) - x)] \geq 0$$

Adding the last two inequalities, we deduce $-d^T JF(x)^T dx \geq 0$. Thus $JF(x)^T dx$ belongs to $-T_c(x; K)^*$. Since $T_c(x; K, F)^*$ contains $T_c(x; K)^*$, it follows that $JF(x)^T dx$ also belongs to the larger cone $-T_c(x; K, F)^*$. Therefore $dx^T F(x) = 0$ by assumption. From (10.2.8) with $z = x$, we deduce $y_c(x) = x$. Hence x solves the VI (K, F). □

Theorem **10.2.5** can be equivalently stated in a form that provides a characterization of a solution of the VI (K, F). We phrase this equivalent statement in the following corollary, which does not require a proof.

10.2.6 Corollary. Let $K \subseteq \mathbb{R}^n$ be closed convex and $F : \Omega \supset K \to \mathbb{R}^n$ be continuously differentiable on the open set Ω. Let c be a positive scalar and let G be a symmetric positive definite matrix. A vector $x \in K$ solves the VI (K, F) if and only if x is a stationary point of (10.2.6) and the implication (10.2.7) holds. □

In what follows, we derive an important special case of Theorem **10.2.5**, which assumes that $JF(x)$ is copositive on $T_c(x; K, F)$. (A similar assumption was made in Exercise **7.6.17** that pertained to the gap program of minimizing θ_{gap} on K.) In this case, something is already known about the set of vectors d satisfying the left-hand conditions in (10.2.7). In fact, in Section **2.5**, we have introduced for an arbitrary pair (C, M), where C is a closed convex cone in $\mathbb{R}^n$ and M is a matrix in $\mathbb{R}^{n \times n}$, a set of exactly this kind. Moreover, we have shown in Proposition **2.5.2** that if M is copositive on C, such a set is contained in the CP kernel $\mathcal{K}(C, M)$ and equal to

the latter kernel if and only if M is copositive-star on C. Based on this observation, we obtain easily the following corollary of Theorem **10.2.5**. Contained in the corollary is a sufficient condition for a stationary point x of (10.2.6) to be a solution of the VI (K, F) without involving the vector $y_c(x)$.

10.2.7 Corollary. Let $K \subseteq \mathbb{R}^n$ be closed convex and $F : \Omega \supset K \to \mathbb{R}^n$ be continuously differentiable on the open set Ω. Let c be a positive scalar and let G be a symmetric positive definite matrix. Suppose that x is a stationary point of (10.2.6).

(a) If $JF(x)$ is copositive on $T_c(x; K, F)$, then x is a solution of the VI (K, F) if and only if $\mathcal{K}(T_c(x; K, F), JF(x))$ is contained in $F(x)^\perp$.

(b) If $JF(x)$ is copositive on $T_c(x; K, F)$ and $(T_c(x; K, F), JF(x))$ is a R_0 pair, then x is a solution of the VI (K, F).

(c) If $JF(x)$ is strictly copositive on $\mathcal{T}(x; K) \cap (-F(x))^*$, then x is an isolated solution of the VI (K, F).

Proof. Part (a) follows from the aforementioned remarks. Part (b) follows immediately from (a). If $JF(x)$ is strictly copositive on $\mathcal{T}(x; K) \cap (-F(x))^*$, then $JF(x)$ is strictly copositive on $T_c(x; K, F)$; hence x is a solution of the VI (K, F). Therefore it remains to show that in this case x is isolated. But this follows easily from Remark **3.3.5** because $\mathcal{T}(x; K) \cap (-F(x))^*$ becomes the critical cone $\mathcal{C}(x; K, F)$ of the VI (K, F) at the solution x. □

10.2.8 Remark. In general, since $T_c(x; K, F)$ is a subset of $(-F(x))^*$, we have

$$\mathcal{K}(T_c(x; K, F), JF(x)) \subseteq F(x)^\perp \Leftrightarrow F(x) \in (\mathcal{K}(T_c(x; K, F), JF(x)))^*.$$

Therefore, part (a) in Corollary **10.2.7** is equivalent to the dual statement: if $JF(x)$ is copositive on $T_c(x; K, F)$, then x is a solution of the VI (K, F) if and only if $F(x)$ is an element of $(\mathcal{K}(T_c(x; K, F), JF(x)))^*$. □

In the next result we provide a sufficient condition in terms of a growth property of F on K in order for θ_c to be coercive on K. This result implies that for a strongly monotone VI (K, F), the minimization problem (10.2.6) always has an optimal solution for $c > 0$ sufficiently small.

10.2.9 Proposition. Let $K \subseteq \mathbb{R}^n$ be closed convex and $F : \Omega \supset K \to \mathbb{R}^n$ be continuous on the open set Ω. Suppose a vector $x^0 \in K$ exists such that

$$\eta \equiv \liminf_{\substack{x \in K \\ \|x\| \to \infty}} \frac{(F(x) - F(x^0))^T (x - x^0)}{\|x - x^0\|^2} > 0.$$

Let G be a symmetric positive definite matrix and c be a positive scalar satisfying $c\|G\| < 2\eta$. The function θ_c is coercive on K, that is,

$$\lim_{\substack{x \in K \\ \|x\| \to \infty}} \theta_c(x) = \infty. \tag{10.2.9}$$

Proof. By the definition of θ_c and the fact that $x^0 \in K$ and

$$(x^0 - x)^T G(x^0 - x) \leq \|G\| \|x - x^0\|^2,$$

we deduce, for all $x \in K$,

$$\begin{aligned}
&\theta_c(x) \\
&\geq F(x)^T(x - x^0) - \frac{c}{2}(x^0 - x)^T G(x^0 - x) \\
&\geq F(x^0)^T(x - x^0) + (F(x) - F(x^0))^T(x - x^0) - \frac{c}{2}\|G\| \|x - x^0\|^2 \\
&\geq \|x - x^0\|^2 \left[\frac{(F(x) - F(x^0))^T(x - x^0)}{\|x - x^0\|^2} - \frac{c}{2}\|G\| - \frac{\|F(x^0)\|}{\|x - x^0\|} \right].
\end{aligned}$$

Since $c\|G\| < 2\eta$, the expression within the square bracket is positive as $\|x\|$ tends to infinity with $x \in K$. Hence the coerciveness of θ_c on K follows readily. □

If the set K is bounded, the coerciveness of θ_c on K holds vacuously. The main utility of Proposition **10.2.9** is for the case where K is unbounded, such as that of the NCP. For a detailed treatment of the latter problem, see Subsection **10.3.1**.

In the development so far, we have employed a strictly concave quadratic function to regularize the gap function. The quadratic function is a popular choice (especially with G being the identity matrix) because the resulting problem (10.2.6) is a strictly concave quadratic maximization program when K is polyhedral; thus many efficient algorithms can be used to evaluate the function $\theta_c(x)$. Setting aside this computational consideration, we note that from a theoretical point of view, there is no need to restrict the regularizing function to be quadratic. Thus we may consider a more general definition of the regularization. Specifically, let $q : \mathbb{R}^{2n} \to \mathbb{R}$ be a function of two arguments $(x, y) \in \mathbb{R}^{2n}$ satisfying the following conditions:

(c1) q is continuously differentiable and nonnegative on $\mathbb{R}^{2n}$;

(c2) $q(x, \cdot)$ is strongly convex, uniformly in x, that is, there exists a constant $\rho > 0$ such that, for any $x \in \mathbb{R}^n$ and for any y, $y' \in \mathbb{R}^n$,

$$q(x, y) - q(x, y') \geq \nabla_y q(x, y')^T(y - y') + \rho\|y - y'\|^2;$$

(c3) $q(x, y) = 0$ if and only if $x = y$;

(c4) for any x and y in $\mathbb{R}^n$, $\nabla_x q(x, y) = -\nabla_y q(x, y)$.

A large family of functions q satisfying the above conditions is obtained by taking a continuously differentiable, strongly convex function $r : \mathbb{R}^n \to \mathbb{R}$ with $r(0) = 0$ and by defining

$$q(x, y) \equiv r(x - y), \quad x, y \in \mathbb{R}^n.$$

In general, with a function q satisfying (c1)–(c4), we may define a generalized gap function by:

$$\theta_c^q(x) \equiv \max_{y \in K} \left\{ F(x)^T (x - y) - c\, q(x, y) \right\}, \quad x \in \Omega.$$

It is possible to check that all the results established in this section still hold. In particular, θ_c^q is a merit function for the VI (K, F) on K and is continuously differentiable on Ω (if F is so); moreover, a stationary point of the problem

$$\begin{array}{ll} \text{minimize} & \theta_c^q(x) \\ \text{subject to} & x \in K \end{array}$$

is a solution of VI (K, F) under sufficient conditions on the extended pair $(T_c^q(x; K, F), JF(x))$, where $T_c^q(x; K, F)$ is defined similarly to the previous $T_c(x; K, F)$ with the vector $y_c(x)$ replaced by the optimal solution $y_c^q(x)$ of the maximization problem defining $\theta_c^q(x)$. Results similar to Theorem **10.2.5** and its Collorary **10.2.7** can be obtained in this broader context.

Obviously, the practical choice of the function q should be such that it is easy to evaluate the function $\theta_c^q(x)$. It is conceivable that some nonquadratic choice of q could offer some theoretical and/or numerical advantages for solving the VI (K, F). Nevertheless, there has been no computational study in this general direction that offers guidelines for choosing q most profitably.

10.2.2 The linearized gap function

Before pursuing further the minimization approach of solving the VI (K, F) based on the regularized gap program (10.2.6), we digress to introduce a related merit function giving rise to an equivalent reformulation of the VI that offers a distinct computational advantage over (10.2.6). As noted above, one of the main drawbacks of the regularized gap function is that its evaluation requires the calculation of a projection on the set K, a task

that, unless K is polyhedral, cannot be accomplished by a finite procedure, in general. In the case of a finitely representable set K, a plausible idea is to replace K by its pointwise linearization. More precisely, suppose that

$$K \equiv \{\, x \in \mathbb{R}^n : g_i(x) \leq 0,\, i = 1, \ldots m \,\}, \tag{10.2.10}$$

where each g_i is convex and continuously differentiable on the open set Ω containing K. As before, Ω is the domain of definition of the function F in the VI (K, F). Equality constraints can be included without difficulty, but are omitted for simplicity. We define

$$L(x) \equiv \{\, y \in \mathbb{R}^n : g_i(x) + \nabla g_i(x)^T (y - x) \leq 0,\, i = 1, \ldots, m \,\},$$

and define a *linearized gap function* $\theta_c^{\rm lin}(x)$ by

$$\theta_c^{\rm lin}(x) \equiv \sup_{y \in L(x)} \left\{ F(x)^T (x - y) - \frac{c}{2} (x - y)^T G (x - y) \right\}. \tag{10.2.11}$$

Lemma **10.2.10** below shows that the set $L(x)$ is nonempty for all x in Ω. Thus the maximization problem in the right-hand side of the above expression has a unique optimal solution, which we denote $y_c^{\rm lin}(x)$. Clearly,

$$\theta_c^{\rm lin}(x) = F(x)^T (\, x - y_c^{\rm lin}(x) \,) - \frac{c}{2} \| x - y_c^{\rm lin}(x) \|_G^2 . \tag{10.2.12}$$

Similar to Theorem **10.2.3**, we can show

$$y_c^{\rm lin}(x) \equiv \Pi_{L(x),G}(x - c^{-1} G^{-1} F(x)). \tag{10.2.13}$$

Note that the calculation of $y_c^{\rm lin}(x)$, and therefore of $\theta_c^{\rm lin}(x)$, only involves the projection onto the polyhedral set $L(x)$, that is the solution of a strictly convex quadratic problem. Moreover $y_c^{\rm lin}(x)$ is characterized by the variational inequality:

$$(\, y - y_c^{\rm lin}(x) \,)^T [\, F(x) + c\, G(y_c^{\rm lin}(x) - x) \,] \geq 0, \quad \forall\, y \in L(x).$$

It turns out that $\theta_c^{\rm lin}(x)$ is also a merit function for the VI (K, F). The simple lemma below shows that if K is convex, the linearized gap function $\theta_c^{\rm lin}$ is well defined and majorizes the regularized gap function $\theta_c(x)$ on Ω.

10.2.10 Lemma. Let K be defined by (10.2.10), where each g_i is convex and continuously differentiable on the open set Ω containing K. Let $c > 0$ be a given scalar and G be a given symmetric positive definite matrix. For every $x \in \Omega$, $K \subseteq L(x)$; thus $\theta_c^{\rm lin}(x) \geq \theta_c(x)$;

Proof. Let $z \in K$. By convexity we have, for every $x \in \Omega$,

$$0 \geq g(z) \geq g(x) + Jg(x)(z - x),$$

which shows that K is a subset of $L(x)$. From the definition of $\theta_c^{\text{lin}}(x)$ and $\theta_c(x)$, it is obvious that the former value is no less than the latter value. □

The above lemma implies that θ_c^{lin} is a nonnegative function on K. The next result shows that, under the Abadie constraint qualification, θ_c^{lin} is a merit function for the VI (K, F) on K.

10.2.11 Proposition. Let K be defined by (10.2.10), where each g_i is convex and continuously differentiable on the open set Ω containing K. Let $c > 0$ be a given scalar and G be a given symmetric positive definite matrix. If $x \in K$ and $\theta_c^{\text{lin}}(x) = 0$, then $x \in \text{SOL}(K, F)$ Conversely, if $x \in \text{SOL}(K, F)$ and the Abadie CQ holds at x, then $\theta_c^{\text{lin}}(x) = 0$.

Proof. If $x \in K$ and $\theta_c^{\text{lin}}(x) = 0$, then $\theta_c(x) = 0$, by Lemma **10.2.10**. Hence x solves the VI (K, F), by part (c) of Theorem **10.2.3**.

Conversely, suppose that x solves VI (K, F) and the Abadie CQ holds at x. Since K is convex, we have

$$F(x) \in -\mathcal{N}(x; K) = \mathcal{T}(x; K)^*.$$

Since the Abadie's constraint qualification holds at x we also have

$$\mathcal{T}(x; K) = \{\, d \in \mathbb{R}^n : \nabla g_i(x)^T d \leq 0, \quad \forall\, i \in \mathcal{I}(x) \,\}.$$

Thus $L(x) \subseteq x + \mathcal{T}(x; K)$. Hence

$$(\, y - x \,)^T F(x) \geq 0, \quad \forall\, y \in L(x).$$

This shows that $x = y_c^{\text{lin}}(x)$; therefore $\theta_c^{\text{lin}}(x) = 0$. □

Although the linearized gap function $\theta_c^{\text{lin}}(x)$ is much easier to compute than the regularized gap function $\theta_c(x)$, θ_c^{lin} does have its own drawbacks, the main one being that in general it is not continuously differentiable. In fact, up to this point, we have not even claimed that θ_c^{lin} is continuous. The difficulty of these issues is due to the fact that the maximization problem in $\theta_c^{\text{lin}}(x)$ is constrained by the set $L(x)$, which varies with x. In particular, Theorem **10.2.1** is not applicable. Instead, we recognize that the maximization problem in $\theta_c^{\text{lin}}(x)$ is a parametric optimization problem to which the sensitivity analysis of Section **5.4** are relevant. Our immediate goal is to show that under the Slater constraint qualification for the convex set

K, $\theta_c^{\rm lin}$ is directionally differentiable on Ω. To prepare for this differentiability result, we first consider in the greater depth some properties of the function $y_c^{\rm lin}$. By definition, $y_c^{\rm lin}(x)$ is the unique optimal solution of the following strictly convex quadratic program in the variable y with x as the parameter:

$$\begin{array}{ll} \text{minimize} & F(x)^T(y-x) + \dfrac{c}{2}(y-x)^T G(y-x) \\ \text{subject to} & g_i(x) + \nabla g_i(x)^T(y-x) \leq 0, \quad i = 1,\ldots,m; \end{array}$$

equivalently, $y_c^{\rm lin}(x)$ is the unique solution of the AVI $(L(x), q(x), cG)$, where

$$q(x) \equiv F(x) - cGx.$$

Considering this as a parametric AVI with y as the primary variable and x as the parameter, and letting λ be the vector of multipliers for the constraints in $L(x)$, we may write, in the notation of Section **5.4**, the (vector) Lagrangian function of the AVI $(L(x), q(x), cG)$ as:

$$\mathbf{L}_c^{\rm lin}(y, \lambda; x) \equiv F(x) + cG(y-x) + \sum_{i=1}^{m} \lambda_i \nabla g_i(x).$$

Let $\mathcal{M}_c^{\rm lin}(x)$ denote the set of multipliers satisfying the KKT system of the AVI $(L(x), q(x), cG)$ associated with the solution $y_c^{\rm lin}(x)$; thus $\lambda \in \mathcal{M}_c^{\rm lin}(x)$ if and only if

$$\begin{array}{c} \mathbf{L}_c^{\rm lin}(y_c^{\rm lin}(x), \lambda; x) = 0, \\ 0 \leq \lambda \perp g(x) + Jg(x)(y_c^{\rm lin}(x) - x) \leq 0. \end{array}$$

The following result establishes in particular the continuity of $y_c^{\rm lin}$ and $\theta_c^{\rm lin}$.

10.2.12 Lemma. Let $F : \Omega \supset K \to \mathbb{R}^n$ be continuous on the open set Ω. Let K be defined by (10.2.10), where each g_i is convex and continuously differentiable on Ω. Let $c > 0$ be a given scalar and G be a given symmetric positive definite matrix. If the Slater CQ holds for K, then for every $x \in \Omega$, there exist a neighborhood $\mathcal{N} \subset \Omega$ of x and a constant $c > 0$ such that

$$\| y_c^{\rm lin}(x') - y_c^{\rm lin}(x) \| \leq c \| x - x' \|$$

for all $x' \in \mathcal{N}$. Thus both $y_c^{\rm lin}$ and $\theta_c^{\rm lin}$ are continuous functions on Ω.

Proof. Since each g_i is a convex function and the Slater CQ holds for K, it follows that for every x in Ω, the same CQ holds for the set $L(x)$.

Since $J_y \mathbf{L}_c^{\text{lin}}(y, \lambda; x) = cG$ is a constant positive definite matrix for all λ in $\mathcal{M}_c^{\text{lin}}(x)$ and all x and y, the lemma follows easily from Theorem **5.4.4**. □

Toward the demonstration of the directional differentiability of θ_c^{lin}, we proceed as in Proposition **5.4.8**. Let $\{x^k\}$ be a sequence such that

$$\lim_{k\to\infty} x^k = x^* \quad \text{and} \quad \lim_{k\to\infty} \frac{x^k - x^*}{\| x^k - x^* \|} = dx.$$

Write $y^k \equiv y_c^{\text{lin}}(x^k)$. By continuity, the sequence $\{y^k\}$ converges to the vector $y^* \equiv y_c^{\text{lin}}(x^*)$; moreover, the sequence

$$\left\{ \frac{y^k - y^*}{\| x^k - x^* \|} \right\}$$

is bounded. Assume that

$$\lim_{k\to\infty} \frac{y^k - y^*}{\| x^k - x^* \|} = dy$$

for some vector dy. Using (10.2.12) and some algebraic manipulation similar to the derivation of the product rule of differentiation, we can show that

$$\begin{aligned}
&\lim_{k\to\infty} \frac{\theta_c^{\text{lin}}(x^k) - \theta_c^{\text{lin}}(x^*)}{\| x^k - x^* \|} \\
&= F(x^*)^T (\, dx - dy \,) \\
&\qquad + (\, JF(x^*) dx \,)^T (\, x^* - y^* \,) - c \,(\, x^* - y^* \,)^T G (\, dx - dy \,) \\
&= [\, F(x^*) + (\, JF(x^*) - c\, G \,)^T (\, x^* - y^* \,) \,]^T dx \\
&\qquad - [\, F(x^*) + c\, G (\, y^* - x^* \,) \,]^T dy \\
&= T_1(dx) - T_2(dy),
\end{aligned}$$

where

$$T_1(dx) \equiv [\, F(x^*) + (\, JF(x^*) - c\, G \,)^T (\, x^* - y^* \,) \,]^T dx$$

is independent of dy, whereas the second term

$$T_2(dy) \equiv [\, F(x^*) + c\, G (\, y^* - x^* \,) \,]^T dy$$

depends on dy. In what follows, we focus on the evaluation of the latter term $T_2(dy)$.

Corresponding to the given pair (x^*, y^*) and the vector dx, consider the following linear program in the variable λ:

$$\begin{aligned}
&\text{maximize} \quad \sum_{i=1}^{m} \lambda_i \,(\, y^* - x^* \,)^T \nabla^2 g_i(x^*)\, dx \\
&\text{subject to} \quad \lambda \in \mathcal{M}_c^{\text{lin}}(x^*).
\end{aligned} \tag{10.2.14}$$

In essence this is the linear program (5.4.15) in Proposition **5.4.7** that defines the nonempty directional critical set of the AVI $(L(x^*), q(x^*), cG)$ at the solution y^* in the direction dx. In order to obtain a better description of such critical elements, let $\mathcal{D}_c^{\rm lin}(x^*; dx)$ denote the set of dual optimal solutions of (10.2.14); we further let $\mathcal{I}_c^{\rm lin}(x^*)$ denote the set of binding constraints in $L(x^*)$ at its feasible vector y^*, i.e.,

$$\mathcal{I}_c^{\rm lin}(x^*) \equiv \{\, i \,:\, g_i(x^*) + \nabla g_i(x^*)^T (\, y^* - x^* \,) \,=\, 0 \,\}.$$

The set $\mathcal{M}_c^{\rm lin}(x^*)$ consists of all vectors λ satisfying the following system of linear inequalities:

$$\begin{aligned} F(x^*) + c\, G(\, y^* - x^* \,) + \sum_{i \in \mathcal{I}_c^{\rm lin}(x^*)} \lambda_i \, \nabla g_i(x^*) \;&=\; 0, \\ \lambda_i \;\geq\; 0, \quad \forall\, i \;&\in\; \mathcal{I}_c^{\rm lin}(x^*), \\ \lambda_i \;=\; 0, \quad \forall\, i \;&\notin\; \mathcal{I}_c^{\rm lin}(x^*). \end{aligned}$$

Thus $\mathcal{D}_c^{\rm lin}(x^*; dx)$ is the set of optimal solutions to the linear program in the variable v:

$$\begin{aligned} &\text{minimize} \quad v^T [\, F(x^*) + c\, G(\, y^* - x^* \,) \,] \\ &\text{subject to} \\ &(\, y^* - x^* \,)^T \nabla^2 g_i(x^*) dx + \nabla g_i(x^*)^T v \leq 0, \quad \forall\, i \in \mathcal{I}_c^{\rm lin}(x^*). \end{aligned} \tag{10.2.15}$$

In essence, this is the dual linear program (5.4.16) specialized to the present context. Note that the term $T_2(dy)$ is the objective value of the dual program (10.2.15) evaluated at $v = dy$.

With the above preparation, we can establish the directional differentiability of $\theta_c^{\rm lin}$.

10.2.13 Theorem. Let $F : \Omega \supset K \to \mathbb{R}^n$ be continuously differentiable on the open set Ω. Let K be defined by (10.2.10), where each g_i is convex and twice continuously differentiable on Ω. Let $c > 0$ be a given scalar and G be a given symmetric positive definite matrix. If the Slater CQ holds for K, then $\theta_c^{\rm lin}$ is directionally differentiable on Ω; moreover, for every x^* in Ω and dx in $\mathbb{R}^n$,

$$(\, \theta_c^{\rm lin} \,)'(x^*; dx) \;=\; T_1(dx) - T_2^*(dx),$$

where $T_2^*(dx)$ is the optimal objective value of (10.2.15).

Proof. As noted in the proof of Lemma **10.2.12**, the Slater CQ continues to hold for the set $L(x^*)$. Let $\{\tau_k\}$ be an arbitrary sequence of positive

scalars converging to zero. Set $x^k \equiv x^* + \tau_k dx$. Continuing the analysis that began above, we deduce from Proposition **5.4.8** that the vector dy must belong to the set $\mathcal{D}_c^{\rm lin}(x^*; dx)$. This implies that the term $T_2(dy)$ is equal to the optimal objective value of the linear program (10.2.15), which depends only on x^* and dx and is independent of dy. Consequently, the limit

$$\lim_{k\to\infty} \frac{\theta_c^{\rm lin}(x^* + \tau_k dx) - \theta_c^{\rm lin}(x^*)}{\tau_k}$$

exists and is equal to $T_1(dx) - T_2^*(dx)$. □

A special case of Theorem **10.2.13** is worth mentioning. Suppose x^* is such that $\mathcal{M}_c^{\rm lin}(x^*)$ is a singleton. In this case, for all $dx \in \mathbb{R}^n$,

$$T_2^*(dx) \;=\; \sum_{i=1}^{m} \lambda_i \,(\, y^* - x^* \,)^{\,T} \nabla^2 g_i(x^*)\, dx,$$

where λ is the unique vector in $\mathcal{M}_c^{\rm lin}(x^*)$. More importantly, λ is independent of dx; hence $T_2^*(dx)$ is a linear function in dx. Since $T_1(dx)$ is obviously linear in dx also, we have established the following corollary.

10.2.14 Corollary. Assume the setting of Theorem **10.2.13**. The function $\theta_c^{\rm lin}$ is G-differentiable at a vector $x^* \in \Omega$ if $\mathcal{M}_c^{\rm lin}(x^*)$ is a singleton. □

Consider the linearized gap program:

$$\begin{array}{ll} \text{minimize} & \theta_c^{\rm lin}(x) \\ \text{subject to} & x \,\in\, K. \end{array} \tag{10.2.16}$$

A stationary point of this program is a vector x^* in K such that

$$(\,\theta_c^{\rm lin}\,)'(x^*; dx) \;\geq\; 0, \quad \forall\, dx \,\in\, x^* + \mathcal{T}(x^*; K).$$

The following lemma is instrumental in showing that such a point x^* is a solution of the VI (K, F); it also plays a central role later on, when defining an algorithm for solving (10.2.16).

10.2.15 Lemma. Assume the setting of Theorem **10.2.13**. For every x in Ω, with $dx \equiv y_c^{\rm lin}(x) - x$,

$$\begin{aligned} (\,\theta_c^{\rm lin}\,)'(x; dx) \;\leq\; & -dx^{\,T} JF(x) dx + \\ & \min_{\lambda \in \mathcal{M}_c^{\rm lin}(x)} \left\{ \sum_{\mathcal{I}_+(x)} \lambda_i\, g_i(x) - \sum_{i=1}^{m} \lambda_i\, dx^{\,T} \nabla^2 g_i(x) dx \right\}, \end{aligned} \tag{10.2.17}$$

where $\mathcal{I}_+(x) \equiv \{i : g_i(x) > 0\}$.

Proof. We have

$$(\,\theta_c^{\rm lin}\,)'(x;dx) \;=\; T_1 - T_2,$$

where

$$T_1 \;\equiv\; [\,F(x) - (\,JF(x) - c\,G\,)^{\,T} dx\,]^{\,T} dx,$$

and, by linear programming duality, T_2 is the optimal objective value of the linear program:

$$\begin{array}{ll} \text{maximize} & \displaystyle\sum_{i=1}^{m} \lambda_i \, dx^{\,T} \nabla^2 g_i(x)\, dx \\ \text{subject to} & \lambda \,\in\, \mathcal{M}_c^{\rm lin}(x). \end{array}$$

Consequently, for any $\lambda \in \mathcal{M}_c^{\rm lin}(x)$, we have

$$T_2 \;\geq\; \sum_{i=1}^{m} \lambda_i \, dx^{\,T} \nabla^2 g_i(x) dx.$$

Furthermore,

$$\begin{array}{rcl} T_1 & = & -dx^{\,T} JF(x) dx - \displaystyle\sum_{i \in \mathcal{I}_c^{\rm lin}(x)} \lambda_i \, \nabla g_i(x)^{\,T} dx \\ & = & -dx^{\,T} JF(x) dx + \displaystyle\sum_{i \in \mathcal{I}_c^{\rm lin}(x)} \lambda_i \, g_i(x) \\ & \leq & -dx^{\,T} JF(x) dx + \displaystyle\sum_{i \in \mathcal{I}_+(x)} \lambda_i \, g_i(x). \end{array}$$

Consequently, for any $\lambda \in \mathcal{M}_c^{\rm lin}(x)$,

$$T_1 - T_2 \;\leq\; -dx^{\,T} JF(x) dx + \sum_{i \in \mathcal{I}_+(x)} \lambda_i \, g_i(x) - \sum_{i=1}^{m} \lambda_i \, dx^{\,T} \nabla^2 g_i(x) dx,$$

from which the inequality (10.2.17) follows readily. □

The next result shows that if x is a stationary point of the constrained minimization problem (10.2.16) such that $J_x \mathbf{L}(x, \lambda)$, where

$$\mathbf{L}(x, \lambda) \;=\; F(x) + \sum_{i=1}^{m} \lambda_i \, \nabla g_i(x)$$

is the Lagrangian function of the VI (K, F), is strictly copositive on the cone $\mathcal{T}(x; K) \cap (-F(x))^*$ for some $\lambda \in \mathcal{M}_c^{\rm lin}(x)$, then x must solve the VI (K, F). We recall that a similar strict copositivity assumption is also used in part (c) of Corollary **10.2.7**.

10.2.16 Theorem. Under the assumptions of Theorem **10.2.13**, if x is a stationary point of (10.2.16) such that $J_x\mathbf{L}(x,\lambda)$ is strictly copositive on the cone $\mathcal{T}(x;K)\cap(-F(x))^*$ for some $\lambda\in\mathcal{M}_c^{\rm lin}(x)$, then x solves the VI (K,F).

Proof. Since K is convex, the stationarity of x means

$$(\,\theta_c^{\rm lin}\,)'(x;y-x)\;\geq\;0,\quad\forall\,y\;\in\;K.$$

In particular, substituting $y=y_c^{\rm lin}(x)$ and letting $dx\equiv y_c^{\rm lin}(x)-x$, we deduce,

$$\begin{aligned}0\;&\leq\;(\,\theta_c^{\rm lin}\,)'(x;dx)\\ &\leq\;-dx^{\,T}JF(x)dx-\max\left\{\sum_{i=1}^{m}\lambda_i\,dx^{\,T}\nabla^2 g_i(x)dx\;:\;\lambda\;\in\;\mathcal{M}_c^{\rm lin}(x)\right\},\end{aligned}$$

where we have used Lemma **10.2.15** and the fact that $\mathcal{I}_+(x)$ is the empty set because x belongs to K. Thus,

$$dx^{\,T}J_x\mathbf{L}(x,\lambda)dx\;\leq\;0,\quad\forall\,\lambda\;\in\;\mathcal{M}_c^{\rm lin}(x).\tag{10.2.18}$$

Since the Slater CQ holds, we know that the tangent vectors of K at x consists of all vectors that make a non-acute angle with the gradients of the binding constraints of K at x. From this fact, it is easy to see that the vector dx must belong to $\mathcal{T}(x;K)$. Moreover, since x belongs to K, x must be an element of $L(x)$ also. Hence we have

$$(\,x-y_c^{\rm lin}(x)\,)[\,F(x)+c\,G(\,y_c^{\rm lin}(x)-x\,)\,]\;\geq\;0,$$

which easily implies that dx belongs to $(-F(x))^*$. By assumption, there exists a $\bar\lambda\in\mathcal{M}_c^{\rm lin}(x)$ such that $J_x\mathbf{L}(x,\bar\lambda)$ is strictly copositive on the cone $\mathcal{T}(x;K)\cap(-F(x))^*$. Thus $dx^{\,T}J_x\mathbf{L}(x,\bar\lambda)dx\;>\;0$ unless $dx\;=\;0$. By (10.2.18), the latter must hold; thus $y_c^{\rm lin}(x)=x$, which implies that $x\in{\rm SOL}(K,F)$. □

One important distinction between the regularized gap function $\theta_c(x)$ and the linearized gap function $\theta_c^{\rm lin}(x)$ is that the maximizing vector $y_c(x)$ of the former must be an element of the set K, whereas the maximizing vector $y_c^{\rm lin}(x)$ of the latter is an element of the set $L(x)$ but is not necessarily an element of K. This distinction is the source for the difference in the theoretical results: Theorem **10.2.5** and Corollary **10.2.7** for the regularized gap program (10.2.6) versus Theorem **10.2.16** for the linearized gap program (10.2.16). The latter theorem, which takes advantage of the finite representation of K, is analogous to the former corollary; but the former theorem has no analog for (10.2.16).

10.3 The D-Gap Merit Function

In this section, we return to the case of a general closed convex set K without assuming that it is finitely representable. The domain of definition of the regularized gap function θ_c coincides with that of F. In particular, if F is defined on the entire space $\mathbb{R}^n$, then so is θ_c. Nevertheless, even in this case, the regularized gap program (10.2.6) is a constrained minimization problem. A question arises as to whether there is an equivalent unconstrained minimization formulation of the VI (K, F) when F is defined everywhere on $\mathbb{R}^n$. Interestingly, the question has a very simple, affirmative answer. Indeed, a family of such unconstrained merit functions is given by the following definition. Throughout this subsection, F is assumed to be defined on the entire space $\mathbb{R}^n$.

10.3.1 Definition. Let $F : \mathbb{R}^n \to \mathbb{R}^n$ be a given mapping and K a closed convex set in $\mathbb{R}^n$. Let a and b be given scalars satisfying $b > a > 0$. The *D-gap function* of the VI (K, F) is defined as

$$\theta_{ab}(x) \equiv \theta_a(x) - \theta_b(x), \quad \forall\, x \in \mathbb{R}^n,$$

where D stands for Difference. □

The following lemma is very useful for the study of the D-gap function.

10.3.2 Lemma. For every $x \in \mathbb{R}^n$ it holds that

$$\frac{b-a}{2} \| x - y_b(x) \|_G^2 \leq \theta_{ab}(x). \tag{10.3.1}$$

Proof. By the definition of the D-gap function, Theorem **10.2.3**, and simple majorizations, we have

$$\begin{aligned}
\theta_{ab}(x) &= \sup_{y \in K} \left\{ F(x)^T(x-y) - \frac{a}{2}(x-y)^T G(x-y) \right\} \\
&\quad - \sup_{y \in K} \left\{ F(x)^T(x-y) - \frac{b}{2}(x-y)^T G(x-y) \right\} \\
&\geq F(x)^T(x - y_b(x)) - \frac{a}{2}(x - y_b(x))^T G(x - y_b(x)) \\
&\quad - F(x)^T(x - y_b(x)) + \frac{b}{2}(x - y_b(x))^T G(x - y_b(x)) \\
&= \frac{b-a}{2} \| x - y_b(x) \|_G^2,
\end{aligned}$$

establishing the desired inequality (10.3.1). □

The following theorem shows that the D-gap function is indeed an unconstrained merit function of the VI (K, F).

10.3.3 Theorem. Let $F : \mathbb{R}^n \to \mathbb{R}^n$ be continuous and K be a closed convex subset of $\mathbb{R}^n$. For $b > a > 0$, the D-gap function θ_{ab} is continuous on $\mathbb{R}^n$ and

(a) $\theta_{ab}(x) \geq 0$ for all x in $\mathbb{R}^n$;

(b) $\theta_{ab}(x) = 0$ if and only if $x \in \mathrm{SOL}(K, F)$;

(c) if F is continuously differentiable on $\mathbb{R}^n$, then so is the D-gap function θ_{ab} and

$$\begin{aligned} \nabla\theta_{ab}(x) \;=\; & JF(x)^T(\, y_b(x) - y_a(x)\,) + \\ & a\,G(\, y_a(x) - x\,) - b\,G(\, y_b(x) - x\,). \end{aligned} \tag{10.3.2}$$

Proof. Statements (a) and (c) and the sufficiency part of (b) need no further proof. To complete the proof of (b), assume that $\theta_{ab}(x) = 0$. By Lemma **10.3.2**, we have $x = y_b(x)$. This implies that $x \in K$ and $\theta_b(x) = 0$. Hence x solves the VI (K, F), by part (c) of Theorem **10.2.3**. □

Having established that θ_{ab} is an unconstrained merit function for the VI (K, F), we proceed to investigate the question of when an unconstrained stationary point x of θ_{ab} is a solution of the VI. In providing a sufficient condition for this to hold, we need to be aware of the fact that such a point x is not automatically an element of K; indeed, the membership of x in K has to be a consequence of the assumption to be made. This is in contrast to the situations in Theorems **10.2.5** and **10.2.16**, which deal with a constrained stationary point of (10.2.6) and (10.2.16), respectively. For an arbitrary vector $x \in \mathbb{R}^n$, define

$$T_{ab}(x; K) \;\equiv\; \mathcal{T}(y_b(x); K) \,\cap\, (\, -\mathcal{T}(y_a(x); K)\,)$$

and

$$T_{ab}(x; K, F) \;\equiv\; T_{ab}(x; K) \,\cap\, (\, -F(x)\,)^*.$$

Since K is closed and convex, $T_{ab}(x; K)$ is a closed convex cone containing the vector $y_a(x) - y_b(x)$; moreover, $T_{ab}(x; K, F)$ is also a closed convex cone that reduces to the lineality space of the critical cone of the VI (K, F) at x if the latter is a solution of the VI.

10.3.4 Theorem. Let $F : \mathbb{R}^n \to \mathbb{R}^n$ be continuous and K be a closed convex subset of $\mathbb{R}^n$. Let a and b be given scalars satisfying $b > a > 0$ and let G be a symmetric positive definite matrix. Suppose that x is a stationary point of θ_{ab} on $\mathbb{R}^n$. The following three statements are equivalent.

(a) x solves the VI (K, F).

(b) $T_{ab}(x;K,F)$ is contained in $F(x)^{\perp}$.

(c) The implication below holds:

$$\left.\begin{array}{rl} d & \in\ T_{ab}(x;K,F) \\ JF(x)^T d & \in\ -T_{ab}(x;K,F)^* \end{array}\right\} \ \Rightarrow\ d^T F(x) \ =\ 0. \qquad (10.3.3)$$

Moreover the two statements (a) and (b) in Corollary **10.2.7** hold with $T_c(x;K,F)$ replaced by $T_{ab}(x;K,F)$.

Proof. If $x \in \mathrm{SOL}(K,F)$, then $T_{ab}(x;K,F)$ becomes the lineality space of the critical cone $\mathcal{C}(x;K,F)$. Hence (a) implies (b), which in turn implies (c). Assume that (10.3.3) holds. By the variational principle of the maximization problem that defines $\theta_a(x)$, we have

$$(\, z - y_a(x)\,)^T\,[\, F(x) + a\,G(\, y_a(x) - x\,)\,]\ \geq\ 0, \quad \forall\, z\ \in\ K.$$

Substituting $z = y_b(x)$, we obtain

$$(\, y_b(x) - y_a(x)\,)^T\,[\, F(x) + a\,G(\, y_a(x) - x\,)\,]\ \geq\ 0. \qquad (10.3.4)$$

Similarly, we have

$$(\, y_a(x) - y_b(x)\,)^T\,[\, F(x) + b\,G(\, y_b(x) - x\,)\,]\ \geq\ 0. \qquad (10.3.5)$$

Adding the last two inequalities, we deduce

$$dx^T[\, b\,G(\, y_b(x) - x\,) - a\,G(\, y_a(x) - x\,)\,]\ \geq\ 0,$$

where $dx \equiv y_a(x) - y_b(x)$ belongs to $T_{ab}(x;K)$. We claim that $dx^T F(x) \leq 0$. We have

$$\begin{array}{rcl} 0 & \leq & -dx^T[\, F(x) + a\,G(\, y_a(x) - x\,)\,] \\ & = & -dx^T\left[\, F(x) + a\,Gdx + \dfrac{a}{b}\,b\,G(\, y_b(x) - x\,)\,\right] \\ & \leq & -\left(1 - \dfrac{a}{b}\right)\,dx^T F(x), \end{array}$$

which establishes the claim because $b > a > 0$. Consequently, dx belongs to $T_{ab}(x;K,F)$. Since

$$F(x) + a\,G(\, y_a(x) - x\,)\ \in\ \mathcal{T}(y_a(x);K)^*$$

and

$$F(x) + b\,G(\, y_b(x) - x\,)\ \in\ \mathcal{T}(y_b(x);K)^*,$$

by subtracting the former expression from the latter expression, we deduce

$$b\,G(\,y_b(x)-x\,)-a\,G(\,y_a(x)-x\,)\;\in\;\mathcal{T}(y_b(x);K)^*-\mathcal{T}(y_a(x);K)^*.$$

It is easy to see that

$$\mathcal{T}(y_b(x);K)^*-\mathcal{T}(y_a(x);K)^*\;\subseteq\;T_{ab}(x;K)^*\;\subseteq\;T_{ab}(x;K,F)^*;$$

thus

$$b\,G(\,y_b(x)-x\,)-a\,G(\,y_a(x)-x\,)\;\in\;T_{ab}(x;K,F)^*.$$

Since $\nabla\theta_{ab}(x)=0$, by (10.3.2), we deduce

$$JF(x)^T(\,y_b(x)-y_a(x)\,)+a\,G(\,y_a(x)-x\,)-b\,G(\,y_b(x)-x\,)\;=\;0, \quad (10.3.6)$$

which implies $JF(x)^T dx\;\in\;-T_{ab}(x;K,F)^*$. By (10.3.3), it follows that $dx^T F(x)=0$. From (10.3.4) and (10.3.5), we have

$$0\;\leq\;(\,y_b(x)-y_a(x)\,)^T G(\,y_a(x)-x\,)$$

and

$$0\;\leq\;(\,y_a(x)-y_b(x)\,)^T G(\,y_b(x)-x\,).$$

Adding the two inequalities, we obtain

$$0\;\leq\;(\,y_a(x)-y_b(x)\,)^T G(\,y_b(x)-y_a(x)\,).$$

Since G is positive definite, we have $y_a(x)=y_b(x)$. Thus (10.3.6) yields $(a-b)G(y_a(x)-x)=0$. Since $b>a$ and G is nonsingular, it follows that $x=y_a(x)$. Consequently, x solves the VI (K,F). The proof of the last assertion of the theorem is the same as in Corollary **10.2.7**. □

When K is the Cartesian product of closed convex sets of lower dimensions, the cone $T_{ab}(x;K,F)$ can be redefined to take advantage of this Cartesian product. Specifically, let K be given by (3.5.1); i.e.,

$$K\;=\;\prod_{\nu=1}^{N}K_\nu, \quad (10.3.7)$$

where N is a positive integer and each K_ν is a subset of $\mathbb{R}^{n_\nu}$ with

$$\sum_{\nu=1}^{N}n_\nu\;=\;n.$$

For $x\in K$, define

$$T_{ab}^{\circ}(x;K,F)\;\equiv\;T_{ab}(x;K)\,\cap\,(\,-F(x)\,)^{\circ},$$

where v° is the *Hadamard cone* of the vector v; i.e.,

$$v^\circ \equiv \{ d \in \mathbb{R}^n : d_\nu^T v_\nu \geq 0, \forall \nu = 1, \ldots, N \}.$$

Clearly, $T^\circ_{ab}(x; K, F)$ is a subcone of $T_{ab}(x; K, F)$. By part (d) of Exercise **1.8.2**, we have

$$T_{ab}(x; K) = \prod_{\nu=1}^{N} \{ \mathcal{T}((y_b(x))_\nu; K_\nu) \cap (-\mathcal{T}((y_a(x))_\nu; K_\nu)) \}.$$

Consequently, $T^\circ_{ab}(x; K, F)$ has the Cartesian product structure:

$$T^\circ_{ab}(x; K, F) = \prod_{\nu=1}^{N} \{ \mathcal{T}((y_b(x))_\nu; K_\nu) \cap (-\mathcal{T}((y_a(x))_\nu; K_\nu)) \cap (-F_\nu(x))^* \}.$$

We have the following analog of Theorem **10.3.4** for a VI defined over a Cartesian product of sets.

10.3.5 Theorem. Let $F : \mathbb{R}^n \to \mathbb{R}^n$ be continuous and K be given by (10.3.7), where each K_ν is closed and convex. Let a and b be given scalars satisfying $b > a > 0$ and let $G \equiv \mathrm{diag}(G_1, \ldots, G_N)$, where each G_ν is a symmetric positive definite matrix of order n_ν. Suppose that x is a stationary point of θ_{ab} on $\mathbb{R}^n$. The following three statements are equivalent.

(a) x solves the VI (K, F).

(b) $T^\circ_{ab}(x; K, F)$ is contained in $F(x)^\perp$.

(c) The implication below holds:

$$\left.\begin{aligned} d &\in T^\circ_{ab}(x; K, F) \\ JF(x)^T d &\in -(T^\circ_{ab}(x; K, F))^* \end{aligned}\right\} \Rightarrow d^T F(x) = 0. \qquad (10.3.8)$$

Proof. If $x \in \mathrm{SOL}(K, F)$, then $T_{ab}(x; K, F)$ is a subset of $F(x)^\perp$ by Theorem **10.3.4**. Since $T^\circ_{ab}(x; K, F)$ is a subset of $T_{ab}(x; K, F)$, it follows that (b) holds. Clearly (b) implies (c). To prove that (c) implies (a), it suffices to refine the proof of Theorem **10.3.4** by taking advantage of the Cartesian product of K in order to show that $y_a(x) - y_b(x)$ belongs to $(-F(x))^\circ$. The rest of the proof applies verbatim. □

In what follows, we derive a set of inequalities that relate the D-gap function $\theta_{ab}(x)$ to various merit functions of the VI (K, F) that are residuals of equivalent equation formulations of the VI; see Proposition **10.3.7**. These inequalities show that as a residual function for the VI, the D-gap

function is equivalent to the other residuals, including the natural residual, i.e., $\|\mathbf{F}_K^{\rm nat}(x)\|$. To simplify the notation somewhat, we take G to be the identity matrix from here on. With this simplification, we have

$$y_a(x) \;=\; \Pi_K(x - a^{-1}F(x)).$$

Define

$$\mathbf{F}_a^{\rm nat}(x) \;\equiv\; x - y_a(x), \quad \forall\, x \,\in\, \mathbb{R}^n.$$

Notice that $\mathbf{F}_1^{\rm nat}$ is just the natural map $\mathbf{F}_K^{\rm nat}$. In terms of $\mathbf{F}_a^{\rm nat}$, Exercise **10.5.4** shows that

$$\theta_a(x) \;=\; F(x)^{\,T}\mathbf{F}_a^{\rm nat}(x) - \frac{a}{2}\,\rho_a(x)^2, \tag{10.3.9}$$

where

$$\rho_a(x) \;\equiv\; \|\,\mathbf{F}_a^{\rm nat}(x)\,\|$$

is also a merit function for the VI (K, F). The following lemma shows that $\rho_a(x)$ is nonincreasing in a and $a\rho_a(x)$ is nondecreasing in a. This lemma extends Exercise **4.8.4** that compares $\rho_a(x)$ with $\rho_1(x) = \|\mathbf{F}_K^{\rm nat}(x)\|$.

10.3.6 Proposition. For any two scalars $b > a > 0$ and any $x \in \mathbb{R}^n$,

$$\rho_b(x) \;\leq\; \rho_a(x) \quad \text{and} \quad b\,\rho_b(x) \;\geq\; a\,\rho_a(x).$$

Proof. By the definition of $\mathbf{F}_a^{\rm nat}(x)$, we have

$$(\,y - x + \mathbf{F}_a^{\rm nat}(x)\,)^{\,T}(\,F(x) - a\,\mathbf{F}_a^{\rm nat}(x)\,) \;\geq\; 0, \quad \forall\, y \,\in\, K.$$

In particular, with $y = \Pi_K(x - b^{-1}F(x)) = x - \mathbf{F}_b^{\rm nat}(x)$, we obtain

$$(\,\mathbf{F}_a^{\rm nat}(x) - \mathbf{F}_b^{\rm nat}(x)\,)^{\,T}(\,F(x) - a\,\mathbf{F}_a^{\rm nat}(x)\,) \;\geq\; 0. \tag{10.3.10}$$

Similarly, we have

$$(\,\mathbf{F}_b^{\rm nat}(x) - \mathbf{F}_a^{\rm nat}(x)\,)^{\,T}(\,F(x) - b\,\mathbf{F}_b^{\rm nat}(x)\,) \;\geq\; 0. \tag{10.3.11}$$

Adding the last two inequalities, we obtain

$$0 \;\geq\; b\,\rho_b(x)^2 - (\,a + b\,)\,\mathbf{F}_b^{\rm nat}(x)^{\,T}\mathbf{F}_a^{\rm nat}(x) + a\,\rho_a(x)^2,$$

which yields

$$0 \;\geq\; (\,b\rho_b(x) - a\rho_a(x)\,)\,(\,\rho_b(x) - \rho_a(x)\,).$$

Since $b > a > 0$, we must have $\rho_a(x) \geq \rho_b(x)$ and $b\rho_b(x) \geq a\rho_a(x)$. □

Proposition **10.3.6** shows that the residuals in the family $\{\rho_a : a > 0\}$, which includes the natural residual, are all equivalent. Using this proposition and a simple manipulation, we can extend the equivalence to include the D-gap function θ_{ab}.

10.3.7 Proposition. For any two scalars $b > a > 0$ and any $x \in \mathbb{R}^n$,

$$\frac{b-a}{2}\,\rho_b(x)^2 \;\le\; \theta_{ab}(x) \;\le\; \frac{b-a}{2}\,\rho_a(x)^2 \tag{10.3.12}$$

and

$$\frac{a\,(b-a)}{2\,b}\,\rho_a(x)^2 \;\le\; \theta_{ab}(x) \;\le\; \frac{b\,(b-a)}{2\,a}\,\rho_b(x)^2. \tag{10.3.13}$$

Hence,

$$\theta_{ab}(x) \;\ge\; \frac{b-a}{2}\,\|\,x - \Pi_K(x)\,\|^2, \quad \forall\, x \,\in\, \mathbb{R}^n. \tag{10.3.14}$$

Proof. The left-hand inequality in (10.3.12) is simply Lemma **10.3.2**. To prove the right-hand inequality in (10.3.12), we proceed as follows. By the definition of $\theta_{ab}(x)$, we have

$$\begin{aligned}
\theta_{ab}(x) \;&=\; \sup_{y\in K}\left\{\, F(x)^T(\,x-y\,) - \frac{a}{2}\,(\,x-y\,)^T(\,x-y\,)\,\right\} \\
&\quad\; - \sup_{y\in K}\left\{\, F(x)^T(\,x-y\,) - \frac{b}{2}\,(\,x-y\,)^T(\,x-y\,)\,\right\} \\
&\le\; F(x)^T(\,x-y_a(x)\,) - \frac{a}{2}\,(\,x-y_a(x)\,)^T(\,x-y_a(x)\,) \\
&\quad\; -F(x)^T(\,x-y_a(x)\,) + \frac{b}{2}\,(\,x-y_a(x)\,)^T(\,x-y_a(x)\,) \\
&=\; \frac{b-a}{2}\,\rho_a(x)^2.
\end{aligned}$$

To prove the right-hand inequality in (10.3.13), we have

$$\theta_{ab}(x) \;=\; F(x)^T(\,\mathbf{F}_a^{\rm nat}(x) - \mathbf{F}_b^{\rm nat}(x)\,) - \frac{a}{2}\,\rho_a(x)^2 + \frac{b}{2}\,\rho_b(x)^2.$$

By (10.3.11) and Proposition **10.3.7**, we deduce

$$\begin{aligned}
\theta_{ab}(x) \;&\le\; b\,\mathbf{F}_b^{\rm nat}(x)^T(\,\mathbf{F}_a^{\rm nat}(x) - \mathbf{F}_b^{\rm nat}(x)\,) - \frac{a}{2}\,\rho_a(x)^2 + \frac{b}{2}\,\rho_b(x)^2 \\
&\le\; b\,\rho_a(x)\,\rho_b(x) \;- \frac{a}{2}\,\rho_a(x)^2 - \frac{b}{2}\,\rho_b(x)^2 \\
&\le\; \left(\frac{b^2}{a} - \frac{b^2}{2\,a} - \frac{b}{2}\right)\rho_b(x)^2 \;=\; \frac{b\,(b-a)}{2\,a}\,\rho_b(x)^2.
\end{aligned}$$

To prove the left-hand inequality in (10.3.13), we use (10.3.10) to obtain

$$\begin{aligned}
\theta_{ab}(x) \;&\ge\; a\,\mathbf{F}_a^{\rm nat}(x)^T(\,\mathbf{F}_a^{\rm nat}(x) - \mathbf{F}_b^{\rm nat}(x)\,) - \frac{a}{2}\,\rho_a(x)^2 + \frac{b}{2}\,\rho_b(x)^2 \\
&\ge\; \frac{a}{2}\,\rho_a(x)^2 - a\,\rho_a(x)\,\rho_b(x) + \frac{b}{2}\,\rho_b(x)^2 \\
&\ge\; \left(\frac{a}{2} - \frac{a^2}{b} + \frac{a^2}{2\,b}\right)\rho_a(x)^2 \;=\; \frac{a\,(b-a)}{2\,b}\,\rho_a(x)^2.
\end{aligned}$$

Hence both inequalities (10.3.12) and (10.3.13) hold. Finally, since $y_a(x)$ belongs to K, we have

$$\rho_a(x) \;\geq\; \| \, x - \Pi_K(x) \, \|, \quad \forall \, x \;\in\; \mathbb{R}^n;$$

thus (10.3.14) follows readily from (10.3.12). □

10.3.8 Remark. It is fairly easy to show that

$$\theta_a(x) \;\geq\; \frac{a}{2} \, \rho_a(x)^2, \quad \forall \, x \;\in\; K;$$

nevertheless, it does not seem obvious in general how to bound $\theta_a(x)$ in terms of $\rho_a(x)$ from above. □

From Proposition **10.3.7**, it follows immediately that if for some $\bar{a} > 0$, the function $\rho_{\bar{a}}(x)$ is coercive on $\mathbb{R}^n$, then so are $\rho_b(x)$ for all $b > 0$ and $\theta_{ab}(x)$ for all $b > a > 0$. In particular, this is true if $\|\mathbf{F}_K^{\mathrm{nat}}\|$ is coercive on $\mathbb{R}^n$. The next result shows that the latter property holds if either F is ξ-monotone for some scalar $\xi > 1$ or K is bounded; in the latter case, no condition is needed on F. See also Exercise **10.5.12** for a further result.

10.3.9 Proposition. Let $K \subseteq \mathbb{R}^n$ be closed convex and $F : \mathbb{R}^n \to \mathbb{R}^n$ be continuous. Under either one of the following two conditions:

(a) for some scalar $\xi > 1$, F is ξ-monotone on $\mathbb{R}^n$; i.e., a constant $c > 0$ exists such that

$$(\, F(x) - F(x') \,)^T (\, x - x' \,) \;\geq\; c \, \| \, x - x' \, \|^{\xi}, \quad \forall \, x, x' \;\in\; \mathbb{R}^n,$$

(b) K is bounded,

the functions ρ_a and θ_{ab} are coercive on $\mathbb{R}^n$ for all $b > a > 0$.

Proof. As noted above, it suffices to show that

$$\lim_{\|x\| \to \infty} \| \, \mathbf{F}_K^{\mathrm{nat}}(x) \, \| \;=\; \infty. \tag{10.3.15}$$

Write $r \equiv \mathbf{F}_K^{\mathrm{nat}}(x)$; thus $x = r + \Pi_K(x - F(x))$. Therefore, if K is bounded, then clearly the limit (10.3.15) holds. Suppose that F is ξ-monotone on $\mathbb{R}^n$. By the variational principle of the Euclidean projector, we have, for all y in K,

$$\begin{aligned} 0 \;&\leq\; (\, y - x + r \,)^T (\, F(x) - r \,) \\ &=\; (\, y - x + r \,)^T (\, F(x) - F(y + r) \,) + (\, y - x + r \,)^T (\, F(y + r) - r \,). \end{aligned}$$

Thus dividing by $\|y - x + r\|^{\xi}$, we obtain

$$0 \leq \frac{(\, y - x + r\,)^T (\, F(x) - F(y + r)\,)}{\|\, y - x + r\,\|^{\xi}} + \frac{\|\, F(y + r) - r\,\|}{\|\, y - x + r\,\|^{\xi - 1}},$$

which yields

$$c\,\|\, y - x + r\,\|^{\xi - 1} \;\leq\; \|\, F(y + r) - r\,\|.$$

Letting y be a fixed vector in K, we easily deduce the limit (10.3.15) from the above inequality. □.

Combined with Proposition **5.3.7**, Proposition **10.3.7** yields two very important properties of the D-gap function at a semistable solution of the VI.

10.3.10 Proposition. Let $K \subseteq \mathbb{R}^n$ be closed convex and $F : \mathbb{R}^n \to \mathbb{R}^n$ be Lipschitz continuous in a neighborhood of a solution x^* of the VI (K, F). If x^* is semistable, there exist a constant $\eta > 0$ and a neighborhood $\mathcal{N}$ of x^* such that for every $x \in \mathcal{N}$,

$$\|\, x - x^*\,\| \;\leq\; \eta\, \sqrt{\theta_{ab}(x)}.$$

Consequently, $d^T H d > 0$ for all $d \neq 0$ and $H \in \partial_d^2 \theta_{ab}(x^*)$.

Proof. By Proposition **5.3.7**, a constant $\eta' > 0$ and a neighborhood $\mathcal{N}$ of x^* exist such that for every $x \in \mathcal{N}$,

$$\|\, x - x^*\,\| \;\leq\; \eta'\, \|\, \mathbf{F}_K^{\mathrm{nat}}(x)\,\|.$$

Since $\|\mathbf{F}_K^{\mathrm{nat}}(x)\|$ and $\sqrt{\theta_{ab}(x)}$ are equivalent by Proposition **10.3.7**, the above inequality easily implies a similar inequality with the upper bound replaced by $\eta\sqrt{\theta_{ab}(x)}$ for some constant η. The last assertion of the proposition is an immediate consequence of Proposition **8.3.18**. □

Combining Propositions **6.3.1** and **10.3.7**, we obtain the following global square-root error bound with the D-gap residual for a VI under the setting in the former proposition.

10.3.11 Proposition. Let K be given by (6.3.1), i.e.,

$$K \;=\; \prod_{\nu=1}^{N} K_\nu,$$

where N is a positive integer and each K_ν is a closed convex subset of $\mathbb{R}^{n_\nu}$ with

$$\sum_{\nu=1}^{N} n_\nu \;=\; n.$$

Let $F : \mathbb{R}^n \to \mathbb{R}^n$ be a continuous uniformly P function on $\mathbb{R}^n$. For any $b > a > 0$, there exists a constant $\eta > 0$ such that

$$\| x - x^* \| \leq \frac{\eta}{\sqrt{b-a}} \sqrt{2\,\theta_{ab}(x)}, \quad \forall\, x \in \mathbb{R}^n,$$

where x^* is the unique solution of the VI (K, F).

Proof. Since $x - y_b(x)$ is the natural map of the VI $(K, b^{-1}F)$, there exists a constant $\eta > 0$ such that

$$\| x - x^* \| \leq \eta \| x - y_b(x) \|.$$

The desired error bound in terms of $\theta_{ab}(x)$ follows easily from Proposition **10.3.7**. □

Notice that we need the square root of the D-gap residual to bound $\|x - x^*\|$, whereas the natural residual itself provides the desired bound for $\|x - x^*\|$. Analogously, we can obtain a result similar to Proposition **6.3.3** in terms of the D-gap residual. We omit the details.

It would be tempting at this stage to introduce a "linearized D-gap function" for a finitely representable set K, which is similar to the function θ_c^{lin} for the regularized gap function. In other words, we could in principle consider the function

$$\theta_{ab}^{\text{lin}}(x) \equiv \theta_a^{\text{lin}}(x) - \theta_b^{\text{lin}}(x),$$

which remains directionally differentiable under the Slater CQ on K. However, it is not easy to establish a result relating the stationary points of θ_{ab}^{lin} to the solutions of the VI (K, F). In particular it is difficult to establish an analog of Theorem **10.2.16**. The difficulty lies in the fact that the directional derivative $(\theta_{ab}^{\text{lin}})'(x; dx)$ would depend on two different sets of multipliers, one relative to the calculation of $y_a^{\text{lin}}(x)$ and the other to that of $y_b^{\text{lin}}(x)$, and it is not easy to manipulate the terms containing these multipliers. Therefore, we do not pursue this idea further.

10.3.1 The implicit Lagrangian for the NCP

It is both instructive and interesting to specialize the D-gap function to the NCP (F). With an understanding of how the nonnegative orthant of the NCP can be exploited in the specialization, it is then easy to extend the development herein to the box constrained VI (K, F), where the set K is defined by lower and upper bounds only. Such an extended development is not going to add much new insight to the theory, and the notation will be quite messy. For these reasons, we choose to focus on the NCP.

With $K = \mathbb{R}^n_+$ and G taken to be the identity matrix, the regularized gap function for the NCP (F) has an explicit form. By part (a) of Theorem **10.2.3**, we have

$$y_c(x) \;=\; \max(\,0, x - c^{-1}F(x)\,),$$

which implies

$$x - y_c(x) \;=\; c^{-1}\,[\,F(x) - \max(\,0, F(x) - c\,x\,)\,].$$

Consequently, a regularized gap function for the NCP (F) is given by

$$\theta_c^{\rm ncp}(x) \;=\; \frac{1}{2c}\,\left[\,\|\,F(x)\,\|^2 - \|\,\max(\,0, F(x) - c\,x\,)\,\|^2\,\right].$$

For two positive scalars $b > a$, the D-gap function for the NCP (F) is therefore equal to

$$\theta_{ab}^{\rm ncp}(x) \;=\; \left(\,\frac{1}{2a} - \frac{1}{2b}\,\right)\,\|\,F(x)\,\|^2$$

$$+\frac{1}{2b}\,\|\,\max(\,0, F(x) - b\,x\,)\,\|^2 - \frac{1}{2a}\,\|\,\max(\,0, F(x) - a\,x\,)\,\|^2.$$

When $a \equiv 1/b$ for some scalar $b > 1$, the function $\theta_{ab}^{\rm ncp}(x)$ has a symmetric form in x and $F(x)$, which we denote by $\theta_b^{\rm MS}(x) \equiv \theta_{\frac{1}{b}b}^{\rm ncp}(x)$. Omitting an easy algebraic verification, we obtain,

$$\theta_b^{\rm MS}(x) \;=\; x^{\,T}F(x) + \frac{1}{2b}\quad \begin{array}{l}[\,\|\,\max(\,0, x - b\,F(x)\,)\,\|^2 - x^{\,T}x \\ \qquad +\|\,\max(\,0, F(x) - b\,x\,)\,\|^2 - F(x)^{\,T}F(x)\,]\end{array}$$

(We have seen an instance of $\theta_2^{\rm MS}(x)$ in Example **9.1.2**.) The special merit function $\theta_b^{\rm MS}(x)$ is connected to the augmented Lagrangian formulation of the natural optimization formulation of the NCP (F); cf. (1.5.13):

$$\begin{array}{ll}\text{minimize} & x^{\,T}F(x) \\ \text{subject to} & x \;\geq\; 0 \text{ and } F(x) \;\geq\; 0.\end{array} \tag{10.3.16}$$

Indeed, the standard Lagrangian function of this program is:

$$L(x, u, v) \;\equiv\; x^{\,T}F(x) - u^{\,T}x - v^{\,T}F(x), \quad (\,x, u, v\,) \;\in\; \mathbb{R}^{3n}.$$

The augmented Lagrangian function of the same program is: for a scalar $b > 0$,

$$L_b(x, u, v) \;\equiv\; x^{\,T}F(x) + \frac{1}{2b}\quad \begin{array}{l}[\,\|\,\max(\,0, u - b\,F(x)\,)\,\|^2 - u^{\,T}u \\ \qquad +\|\,\max(\,0, v - b\,x\,)\,\|^2 - v^{\,T}v\,].\end{array}$$

Since for the NCP, the pair $(x, F(x))$ serves as the "optimal multipliers" of (10.3.16), we may replace (u, v) by $(x, F(x))$; this replacement results in the merit function $\theta_b^{\mathrm{MS}}(x)$. Due to this (informal) connection with the augmented Lagrangian theory of NLP, and extending the consideration to an arbitrary pair of scalars a and b satisfying $b > a > 0$, we call the merit function $\theta_{ab}^{\mathrm{ncp}}(x)$ the *implicit Lagrangian* (merit) function of the NCP (F).

With the explicit expression of $\theta_{ab}^{\mathrm{ncp}}(x)$, we may derive a necessary and sufficient condition, i.e., a regularity property on x, for the implication

$$\nabla\theta_{ab}^{\mathrm{ncp}}(x) \,=\, 0 \;\Rightarrow\; \theta_{ab}^{\mathrm{ncp}}(x) \,=\, 0 \tag{10.3.17}$$

to hold. There are two ways to proceed. One is to apply Theorem **10.3.5** and impose a condition on a stationary point of $\theta_{ab}^{\mathrm{ncp}}$ so that the implication (10.3.8), and hence (10.3.17), is valid. The other way is to follow the analysis of the FB function θ_{FB} in Subsection **9.1.1**. Although the resulting conditions are somewhat different for a non-solution of the NCP, they both become necessary if the given point x is a solution of the NCP (F); see Theorems **10.3.12** and **10.3.15**. In order to provide an illustration of Theorem **10.3.5**, we begin with the investigation of the implication (10.3.8) in the present context.

We need to look at the tangent cone $\mathcal{T}(y_c(x); K)$. It is easy to see that a vector d belongs to this cone if and only if for each $i = 1, \ldots, n$,

$$d_i \begin{cases} \geq 0 & \text{if } c\,x_i \leq F_i(x) \\ \text{free} & \text{if } c\,x_i > F_i(x). \end{cases}$$

Based on this observation, we can verify that a vector d belongs to the cone $T_{ab}^{\circ}(x; \mathbb{R}^n_+, F)$ if and only if

$$d_i \begin{cases} = 0 & \text{if } 0 < b\,x_i \leq F_i(x) \\ \leq 0 & \text{if } 0 < a\,x_i \leq F_i(x) < b\,x_i \\ \leq 0 & \text{if } 0 < F_i(x) < a\,x_i \\ \geq 0 & \text{if } F_i(x) < 0 < x_i \\ \text{free} & \text{if } F_i(x) = 0 < x_i \\ = 0 & \text{if } x_i = 0 \leq F_i(x) \\ \geq 0 & \text{if } F_i(x) < 0 = x_i \\ = 0 & \text{if } x_i < 0 \text{ and } a\,x_i \leq F_i(x) \\ \geq 0 & \text{if } b\,x_i \leq F_i(x) < a\,x_i < 0 \\ \geq 0 & \text{if } F_i(x) < b\,x_i < 0. \end{cases}$$

According to the conditions that describe the components d_i, we may partition the index set $\{1,\ldots,n\}$ into four mutually disjoint index subsets:

$$\{\,1,\ldots,n\,\} \;=\; \mathcal{I}^{+}_{ab} \cup \mathcal{I}^{0}_{ab} \cup \mathcal{I}^{-}_{ab} \cup \mathcal{I}^{f}_{ab},$$

such that

$$T^{\circ}_{ab}(x;\mathbb{R}^n_+,F) \;=\; \mathbb{R}^{n_+}_{+} \times \{\,0_{n_0}\,\} \times \mathbb{R}^{n_-}_{-} \times \mathbb{R}^{n_f},$$

where

$$n_+ \equiv |\,\mathcal{I}^{+}_{ab}\,| \quad n_0 \equiv |\,\mathcal{I}^{0}_{ab}\,| \quad n_- \equiv |\,\mathcal{I}^{-}_{ab}\,| \quad n_f \equiv |\,\mathcal{I}^{f}_{ab}\,|.$$

Specifically,

$$\begin{aligned}
\mathcal{I}^{+}_{ab} &= \{\,i \,:\, F_i(x) < \max(\,ax_i, bx_i\,) \text{ and } F_i(x) < 0\,\};\\
\mathcal{I}^{0}_{ab} &= \{\,i \,:\, F_i(x) \geq \max(\,ax_i, bx_i\,)\,\};\\
\mathcal{I}^{-}_{ab} &= \{\,i \,:\, 0 < F_i(x) < bx_i\,\};\\
\mathcal{I}^{f}_{ab} &= \{\,i \,:\, 0 = F_i(x) < x_i\,\}.
\end{aligned}$$

The dual of $T^{\circ}_{ab}(x;\mathbb{R}^n_+,F)$ is therefore,

$$(\,T^{\circ}_{ab}(x;\mathbb{R}^n_+,F)\,)^{*} \;=\; \mathbb{R}^{n_+}_{+} \times \mathbb{R}^{n_0} \times \mathbb{R}^{n_-}_{-} \times \{\,0_{n_f}\,\}.$$

We may partition the matrix $JF(x)$ accordingly:

$$JF(x) \;=\; \begin{bmatrix}
J_{\mathcal{I}^{+}_{ab}}F_{\mathcal{I}^{+}_{ab}}(x) & J_{\mathcal{I}^{0}_{ab}}F_{\mathcal{I}^{+}_{ab}}(x) & J_{\mathcal{I}^{-}_{ab}}F_{\mathcal{I}^{+}_{ab}}(x) & J_{\mathcal{I}^{f}_{ab}}F_{\mathcal{I}^{+}_{ab}}(x)\\
J_{\mathcal{I}^{+}_{ab}}F_{\mathcal{I}^{0}_{ab}}(x) & J_{\mathcal{I}^{0}_{ab}}F_{\mathcal{I}^{0}_{ab}}(x) & J_{\mathcal{I}^{-}_{ab}}F_{\mathcal{I}^{0}_{ab}}(x) & J_{\mathcal{I}^{f}_{ab}}F_{\mathcal{I}^{0}_{ab}}(x)\\
J_{\mathcal{I}^{+}_{ab}}F_{\mathcal{I}^{-}_{ab}}(x) & J_{\mathcal{I}^{0}_{ab}}F_{\mathcal{I}^{-}_{ab}}(x) & J_{\mathcal{I}^{-}_{ab}}F_{\mathcal{I}^{-}_{ab}}(x) & J_{\mathcal{I}^{f}_{ab}}F_{\mathcal{I}^{-}_{ab}}(x)\\
J_{\mathcal{I}^{+}_{ab}}F_{\mathcal{I}^{f}_{ab}}(x) & J_{\mathcal{I}^{0}_{ab}}F_{\mathcal{I}^{f}_{ab}}(x) & J_{\mathcal{I}^{-}_{ab}}F_{\mathcal{I}^{f}_{ab}}(x) & J_{\mathcal{I}^{f}_{ab}}F_{\mathcal{I}^{f}_{ab}}(x)
\end{bmatrix}.$$

Thus the implication (10.3.3) becomes

$$\left.\begin{aligned}
J_{\mathcal{I}^{+}_{ab}}F_{\mathcal{I}^{+}_{ab}}(x)^{T}d_{\mathcal{I}^{+}_{ab}} + J_{\mathcal{I}^{+}_{ab}}F_{\mathcal{I}^{-}_{ab}}(x)^{T}d_{\mathcal{I}^{-}_{ab}} + J_{\mathcal{I}^{+}_{ab}}F_{\mathcal{I}^{f}_{ab}}(x)^{T}d_{\mathcal{I}^{f}_{ab}} &\leq 0\\
J_{\mathcal{I}^{-}_{ab}}F_{\mathcal{I}^{+}_{ab}}(x)^{T}d_{\mathcal{I}^{+}_{ab}} + J_{\mathcal{I}^{-}_{ab}}F_{\mathcal{I}^{-}_{ab}}(x)^{T}d_{\mathcal{I}^{-}_{ab}} + J_{\mathcal{I}^{-}_{ab}}F_{\mathcal{I}^{f}_{ab}}(x)^{T}d_{\mathcal{I}^{f}_{ab}} &\geq 0\\
J_{\mathcal{I}^{f}_{ab}}F_{\mathcal{I}^{+}_{ab}}(x)^{T}d_{\mathcal{I}^{+}_{ab}} + J_{\mathcal{I}^{f}_{ab}}F_{\mathcal{I}^{-}_{ab}}(x)^{T}d_{\mathcal{I}^{-}_{ab}} + J_{\mathcal{I}^{f}_{ab}}F_{\mathcal{I}^{f}_{ab}}(x)^{T}d_{\mathcal{I}^{f}_{ab}} &= 0\\
d_{\mathcal{I}^{+}_{ab}} \geq 0, \quad d_{\mathcal{I}^{-}_{ab}} \leq 0, \quad d_{\mathcal{I}^{f}_{ab}} &\text{ free}
\end{aligned}\right\}$$

$$\Rightarrow\; F(x)^{T}_{\mathcal{I}^{+}_{ab}}d_{\mathcal{I}^{+}_{ab}} + F(x)^{T}_{\mathcal{I}^{-}_{ab}}d_{\mathcal{I}^{-}_{ab}} \;=\; 0.$$

By an obvious change in the sign of $d_{\mathcal{I}_{ab}^-}$ and a corresponding change in the middle inequality and by the sign of $F_i(x)$ for i in $\mathcal{I}_{ab}^+$ and $\mathcal{I}_{ab}^-$, the above implication is clearly equivalent to:

$$\left.\begin{aligned} J_{\mathcal{I}_{ab}^+}F_{\mathcal{I}_{ab}^+}(x)^T d_{\mathcal{I}_{ab}^+} - J_{\mathcal{I}_{ab}^+}F_{\mathcal{I}_{ab}^-}(x)^T d_{\mathcal{I}_{ab}^-} + J_{\mathcal{I}_{ab}^+}F_{\mathcal{I}_{ab}^f}(x)^T d_{\mathcal{I}_{ab}^f} &\leq 0 \\ -J_{\mathcal{I}_{ab}^-}F_{\mathcal{I}_{ab}^+}(x)^T d_{\mathcal{I}_{ab}^+} + J_{\mathcal{I}_{ab}^-}F_{\mathcal{I}_{ab}^-}(x)^T d_{\mathcal{I}_{ab}^-} - J_{\mathcal{I}_{ab}^-}F_{\mathcal{I}_{ab}^f}(x)^T d_{\mathcal{I}_{ab}^f} &\leq 0 \\ J_{\mathcal{I}_{ab}^f}F_{\mathcal{I}_{ab}^+}(x)^T d_{\mathcal{I}_{ab}^+} - J_{\mathcal{I}_{ab}^f}F_{\mathcal{I}_{ab}^-}(x)^T d_{\mathcal{I}_{ab}^-} + J_{\mathcal{I}_{ab}^f}F_{\mathcal{I}_{ab}^f}(x)^T d_{\mathcal{I}_{ab}^f} &= 0 \\ (\, d_{\mathcal{I}_{ab}^+}\, d_{\mathcal{I}_{ab}^-}\,) \geq 0, \quad d_{\mathcal{I}_{ab}^f} \text{ free} & \end{aligned}\right\}$$

$$\Rightarrow\ d_{\mathcal{I}_{ab}^+} = 0 \text{ and } d_{\mathcal{I}_{ab}^-} = 0.$$

By Motzkin's Theorem of the Alternative, the above implication is equivalent to the existence of a triple $(v_{\mathcal{I}_{ab}^+}, v_{\mathcal{I}_{ab}^-}, v_{\mathcal{I}_{ab}^f})$ such that

$$\begin{aligned} J_{\mathcal{I}_{ab}^+}F_{\mathcal{I}_{ab}^+}(x)v_{\mathcal{I}_{ab}^+} - J_{\mathcal{I}_{ab}^-}F_{\mathcal{I}_{ab}^+}(x)v_{\mathcal{I}_{ab}^-} + J_{\mathcal{I}_{ab}^f}F_{\mathcal{I}_{ab}^+}(x)v_{\mathcal{I}_{ab}^f} &> 0 \\ -J_{\mathcal{I}_{ab}^+}F_{\mathcal{I}_{ab}^-}(x)v_{\mathcal{I}_{ab}^+} + J_{\mathcal{I}_{ab}^-}F_{\mathcal{I}_{ab}^-}(x)v_{\mathcal{I}_{ab}^-} - J_{\mathcal{I}_{ab}^f}F_{\mathcal{I}_{ab}^-}(x)v_{\mathcal{I}_{ab}^f} &> 0 \\ J_{\mathcal{I}_{ab}^+}F_{\mathcal{I}_{ab}^f}(x)v_{\mathcal{I}_{ab}^+} - J_{\mathcal{I}_{ab}^-}F_{\mathcal{I}_{ab}^f}(x)v_{\mathcal{I}_{ab}^-} + J_{\mathcal{I}_{ab}^f}F_{\mathcal{I}_{ab}^f}(x)v_{\mathcal{I}_{ab}^f} &= 0 \\ (\, v_{\mathcal{I}_{ab}^+}\, v_{\mathcal{I}_{ab}^-}\,) \geq 0, \quad v_{\mathcal{I}_{ab}^f} \text{ free.} & \end{aligned} \tag{10.3.18}$$

If the principal submatrix $J_{\mathcal{I}_{ab}^f}F_{\mathcal{I}_{ab}^f}(x)$ is nonsingular, the existence of the triple $(v_{\mathcal{I}_{ab}^+}, v_{\mathcal{I}_{ab}^-}, v_{\mathcal{I}_{ab}^f})$ is in turn equivalent to the Schur complement of $J_{\mathcal{I}_{ab}^f}F_{\mathcal{I}_{ab}^f}(x)$ in

$$\begin{bmatrix} J_{\mathcal{I}_{ab}^+}F_{\mathcal{I}_{ab}^+}(x) & -J_{\mathcal{I}_{ab}^-}F_{\mathcal{I}_{ab}^+}(x) & J_{\mathcal{I}_{ab}^f}F_{\mathcal{I}_{ab}^+}(x) \\ -J_{\mathcal{I}_{ab}^+}F_{\mathcal{I}_{ab}^-}(x) & J_{\mathcal{I}_{ab}^-}F_{\mathcal{I}_{ab}^-}(x) & -J_{\mathcal{I}_{ab}^f}F_{\mathcal{I}_{ab}^-}(x) \\ J_{\mathcal{I}_{ab}^+}F_{\mathcal{I}_{ab}^f}(x) & -J_{\mathcal{I}_{ab}^-}F_{\mathcal{I}_{ab}^f}(x) & J_{\mathcal{I}_{ab}^f}F_{\mathcal{I}_{ab}^f}(x) \end{bmatrix}$$

being an S matrix. Summarizing the above discussion, we obtain the following result, which gives a sufficient condition for (10.3.17) to hold.

10.3.12 Theorem. Let $F : \mathbb{R}^n \to \mathbb{R}^n$ be continuously differentiable and let $b > a > 0$ be given scalars. Assume that x is an unconstrained stationary point of θ_{ab}^{ncp}. The system (10.3.18) is consistent if and only if x is a solution of the NCP (F).

Proof. This follows immediately from Theorem **10.3.5**. A direct argument can also be given to establish the "if" assertion. Indeed if x is a solution of the NCP (F), then the index sets $\mathcal{I}_{ab}^+$ and $\mathcal{I}_{ab}^-$ are empty. The system (10.3.18) is clearly consistent because $v_{\mathcal{I}_{ab}^f} = 0$ is obviously a solution. □

The above derivation is based on the specialization of a result for a VI on a general Cartesian product of sets. In what follows, we adopt an alternative approach more like that of Theorem **9.1.14**, where it was shown that FB regularity provides a necessary and sufficient condition for a stationary point of the function $\theta_{\rm FB}$ to be a solution of the NCP. We begin by defining the scalar-valued function: for $(u, v) \in \mathbb{R}^2$,

$$\psi_{ab}(u, v) \equiv \left(\frac{1}{2a} - \frac{1}{2b}\right) v^2 + \frac{1}{2b} \max(\,0, v - b\,u\,)^2 - \frac{1}{2a} \max(\,0, v - a\,u\,)^2.$$

We note that

$$\theta_{ab}^{\rm ncp}(x) \;=\; \sum_{i=1}^{n} \psi_{ab}(x_i, F_i(x)),$$

which implies,

$$\nabla\theta_{ab}^{\rm ncp}(x) \;=\; \varphi_u(x) + JF(x)^T \varphi_v(x), \tag{10.3.19}$$

where $\varphi_u(x)$ and $\varphi_v(x)$ are vectors with components:

$$(\,\varphi_u(x)\,)_i \;=\; \frac{\partial\psi_{ab}(x_i, F_i(x))}{\partial u} \quad \text{and} \quad (\,\varphi_v(x)\,)_i \;=\; \frac{\partial\psi_{ab}(x_i, F_i(x))}{\partial v}$$

The partial derivatives of $\psi_{ab}(u, v)$ are given by

$$\frac{\partial\psi_{ab}(u, v)}{\partial u} \;=\; -\max(\,0, v - bu\,) + \max(\,0, v - au\,)$$

and

$$\frac{\partial\psi_{ab}(u, v)}{\partial v} \;=\; \left(\frac{1}{a} - \frac{1}{b}\right) v + \frac{1}{b} \max(\,0, v - bu\,) - \frac{1}{a} \max(\,0, v - au\,).$$

Several key properties of $\psi_{ab}(u, v)$ and its partial derivatives are summarized below. The reader is asked to prove these properties in Exercise **10.5.5**.

10.3.13 Proposition. Let $b > a > 0$ be given scalars.

(a) $\psi_{ab}(u, v)$ is a C-function; moreover, $\psi_{ab}(u, v) = 0 \Leftrightarrow \nabla\psi_{ab}(u, v) = 0$.

(b) $\dfrac{\partial\psi_{ab}(u, v)}{\partial u}\dfrac{\partial\psi_{ab}(u, v)}{\partial v} \geq 0$.

(c) $\dfrac{\partial\psi_{ab}(u, v)}{\partial v} \geq 0$ for all $(u, v) \geq 0$.

(d) $\dfrac{\partial\psi_{ab}(0, v)}{\partial u} = 0 \geq \dfrac{\partial\psi_{ab}(0, v)}{\partial v}$. □

Based on Proposition **10.3.13**(a), we can established the following preliminary result.

10.3.14 Lemma. Let $F : \mathbb{R}^n \to \mathbb{R}^n$ be continuously differentiable and let $b > a > 0$ be given scalars. An unconstrained stationary point x of $\theta_{ab}^{\mathrm{ncp}}$ is a solution of the NCP (F) if and only if

$$\frac{\partial \psi_{ab}(x_i, F_i(x))}{\partial v} = 0, \quad \forall\, i = 1, \ldots, n. \tag{10.3.20}$$

Proof. Suppose (10.3.20) holds. Since x is a stationary point of $\theta_{ab}^{\mathrm{ncp}}$, it follows from (10.3.19) that $\varphi_u(x) = 0$. Therefore we have

$$\frac{\partial \psi_{ab}(x_i, F_i(x))}{\partial v} = 0 = \frac{\partial \psi_{ab}(x_i, F_i(x))}{\partial v}, \quad \forall\, i = 1, \ldots, n.$$

By Proposition **10.3.13**(a), we deduce $\psi_{ab}(x_i, F_i(x)) = 0$ for all i. Thus x solves the NCP (F). The converse follows easily from the same proposition. □

Similar to the definition of FB regularity, we define three index sets associated with an arbitrary vector $x \in \mathbb{R}^n$:

$$\mathcal{C} = \left\{ i : \frac{\partial \psi_{ab}(x_i, F_i(x))}{\partial v} = 0 \right\} \quad \text{(complementary indices)}$$

$$\mathcal{P} = \left\{ i : \frac{\partial \psi_{ab}(x_i, F_i(x))}{\partial v} > 0 \right\} \quad \text{(positive indices)}$$

$$\mathcal{N} = \left\{ i : \frac{\partial \psi_{ab}(x_i, F_i(x))}{\partial v} < 0 \right\} \quad \text{(negative indices)}.$$

Incidentally, the relations between these three index sets and the four index sets $\mathcal{I}_{ab}^{+}$, $\mathcal{I}_{ab}^{0}$, $\mathcal{I}_{ab}^{-}$, $\mathcal{I}_{ab}^{f}$ are as follows:

$$\mathcal{C} = \mathcal{I}_{ab}^{0} \cup \mathcal{I}_{ab}^{f}, \quad \mathcal{P} = \mathcal{I}_{ab}^{-} \quad \text{and} \quad \mathcal{N} = \mathcal{I}_{ab}^{+}.$$

In principle, we can formally introduce a regularity property at a stationary point of $\theta_{ab}^{\mathrm{ncp}}$ and show that this property is sufficient for the stationary point to be a solution of the NCP. Instead of such a formal definition, we embed it in the following result, which is the analog of Theorem **9.1.14**.

10.3.15 Theorem. Let $F : \mathbb{R}^n \to \mathbb{R}^n$ be continuously differentiable and let $b > a > 0$ be given scalars. An unconstrained stationary point x of $\theta_{ab}^{\mathrm{ncp}}$ is a solution of the NCP (F) if and only if for every vector $z \neq 0$ such that

$$z_{\mathcal{C}} = 0, \quad z_{\mathcal{P}} > 0, \quad z_{\mathcal{N}} < 0 \tag{10.3.21}$$

(these index sets are defined with respect to x and the vector $\varphi_v(x)$), there exists a nonzero vector $y \in \mathbb{R}^n$ such that

$$y_{\mathcal{C}} = 0, \quad y_{\mathcal{P}} \geq 0, \quad y_{\mathcal{N}} \leq 0,$$

and

$$z^T JF(x)y > 0. \tag{10.3.22}$$

In particular, if the matrix

$$\begin{bmatrix} J_{\mathcal{P}}F_{\mathcal{P}}(x) & -J_{\mathcal{N}}F_{\mathcal{P}}(x) \\ -J_{\mathcal{P}}F_{\mathcal{N}}(x) & J_{\mathcal{N}}F_{\mathcal{N}}(x) \end{bmatrix} \tag{10.3.23}$$

is an S matrix, then (10.3.17) holds.

Proof. The regularity property is clearly necessary because if x is a solution of the NCP (F), then $\mathcal{P}$ and $\mathcal{N}$ are both empty and the property therefore holds vacuously. Conversely, suppose that this property holds and let x be stationary. We have

$$\varphi_u(x) + JF(x)^T \varphi_v(x) = 0.$$

Let $z \equiv \varphi_v(x)$. Clearly z is nonzero. Let y be the associated vector satisfying the sign conditions and (10.3.22). By Proposition **10.3.13**(b), we have

$$(\varphi_u(x))_i \geq 0, \quad \forall\, i \in \mathcal{P}$$

and

$$(\varphi_u(x))_i \leq 0, \quad \forall\, i \in \mathcal{N}.$$

Consequently, we deduce

$$0 = y^T \varphi_u(x) + y^T JF(x)z \geq y^T JF(x)z,$$

which is a contradiction. The last assertion of the theorem follows easily as in the analysis of the FB function. □

Except for a different definition of the index sets $\mathcal{P}$ and $\mathcal{N}$, the matrix (10.3.23) is the same as the matrix (9.1.20). As to the properties of these two matrices, we need an S property for the former matrix whereas an S_0 property suffices for the latter. In terms of this requirement, the FB merit function has a slight theoretical advantage over the implicit Lagrangian function; see also Exercise **10.5.6**.

We complete the discussion of the implicit Lagrangian function by considering the issue of boundedness of the level sets of this function. Combining Propositions **10.3.7** and **10.3.6**, we deduce the existence of positive constants η' and η such that

$$\eta' \,\|\, \min(x, F(x)) \,\| \leq \theta_{ab}^{\mathrm{ncp}}(x) \leq \eta \,\|\, \min(x, F(x)) \,\|, \quad \forall\, x \in \mathbb{R}^n.$$

Consequently,

$$\lim_{\|x\|\to\infty} \theta_{ab}^{\rm ncp}(x) \,=\, \infty \;\Leftrightarrow\; \lim_{\|x\|\to\infty} \|\, \min(\, x, F(x)\,)\,\| \,=\, \infty.$$

By invoking Proposition **9.1.27**, we obtain a necessary and sufficient condition for $\theta_{ab}^{\rm ncp}(x)$ to have bounded level sets, which is the same as the one for $\theta_{\rm FB}$. For more properties of $\theta_{ab}^{\rm ncp}$, see Exercises **10.5.6** and **10.5.7**.

10.4 Merit Function Based Algorithms

With the differentiable merit functions for the VI (K, F) introduced in Section **10.2**, it is natural to try to follow the development in Chapter **9** in order to obtain some iterative descent algorithms for solving the VI. In so doing, we immediately have to face several problems that did not appear in the complementarity case. We list them below.

• The evaluation of $\theta_a(x)$ and of its gradient $\nabla\theta_a(x)$ requires the calculation of a Euclidean projection on the set K.

• The evaluation of $\theta_{ab}(x)$ and of its gradient $\nabla\theta_{ab}(x)$ requires the calculation of two projections on the set K.

• Neither the function θ_a nor θ_{ab} is naturally associated with a system of semismooth equations; i.e., these merit functions are not the norm functions of some semismooth equation formulation of the VI.

The first two points have important computational implications. In particular, unless the set K is polyhedral, in which case the Euclidean projector can be computed by solving a strictly convex quadratic program, in general the evaluation of the merit functions requires an infinite iterative procedure. The third point suggests that it is not easy to define a locally fast algorithm for the solution of VI (K, F) based on the merit functions θ_a or θ_{ab}. In what follows, we consider ways of lessening these difficulties. Throughout the discussion, we take the matrix G to be the identify matrix.

10.4.1 Algorithms based on the D-gap function

We first consider iterative algorithms for the unconstrained minimization of the function θ_{ab}. The first thing we do is to introduce a local, superlinearly convergent method for the minimization of this D-gap function. By part (c) of Theorem **10.3.3**, if F is continuously differentiable, then

$$\nabla\theta_{ab}(x) \,=\, JF(x)^{\,T}(\, y_b(x) - y_a(x)\,) + b\,(\, x - y_b(x)\,) - a\,(\, x - y_a(x)\,),$$

where

$$y_a(x) \,=\, \Pi_K(x - a^{-1}F(x)), \quad \text{and} \quad y_b(x) \,=\, \Pi_K(x - b^{-1}F(x)).$$

In order to define a locally fast method we attempt to define a Newton approximation scheme of $\nabla\theta_{ab}$. Suppose for a moment that the projector Π_K is continuously differentiable in a neighborhood of $x^* - a^{-1}F(x^*)$ and $x^* - b^{-1}F(x^*)$ for a given vector x^*. If F is twice continuously differentiable around x^*, then θ_{ab} is also twice continuously differentiable around x^*. However the calculation of the Hessian of $\theta_{ab}(x)$ would require on the one hand the second derivatives of $F(x)$ and on the other hand the Jacobian of the Euclidean projector. It is clear that such an approach would be computationally unattractive. Therefore we want to define a Newton approximation scheme of $\nabla\theta_{ab}$, which allows us to bypass these computational requirements.

In order to illustrate the above comment about the second derivatives of $F(x)$, consider the simple case where K is the entire space $\mathbb{R}^n$. In this case, we have $y_a(x) = x - a^{-1}F(x)$ and

$$\nabla\theta_{ab}(x) \,=\, (\,a^{-1} - b^{-1}\,)\,JF(x)^{\,T}F(x) \,=\, (\,a^{-1} - b^{-1}\,)\,\sum_{i=1}^{n} F_i(x)\nabla F_i(x).$$

Therefore,

$$\nabla^2\theta_{ab}(x) \,=\, (\,a^{-1} - b^{-1}\,)\,\sum_{i=1}^{n}\,[\,\nabla F_i(x)\nabla F_i(x)^{\,T} + F_i(x)\nabla^2 F_i(x)\,],$$

which contains the Hessian of F_i. This case corresponds to solving the system of nonlinear equations $F(x) = 0$. We recall the Gauss-Newton method in which the matrix

$$\sum_{i=1}^{n} \nabla F_i(x)\nabla F_i(x)^{\,T} \,=\, JF(x)^{\,T}JF(x),$$

which is equal to $\nabla^2\theta_{ab}(x)$ without the $\nabla^2 F_i(x)$ terms, is employed to define the Gauss-Newton directions. In particular, if $F(x) = 0$, then $\nabla^2\theta_{ab}(x) = (a^{-1} - b^{-1})JF(x)^{\,T}JF(x)$. In essence, the development below is an extension of the idea of "dropping the Hessian of the functions F_i", which is justified when x is near a solution of the VI (K, F), i.e., when $x - y_a(x)$ and $x - y_b(x)$ are sufficiently small in norm.

In order to obtain a Newton approximation of the projector Π_K, we assume that K is defined by a finite number of convex inequalities. More precisely, let

$$K \,=\, \{\,x \,\in\, \mathbb{R}^n \,:\, g_i(x) \,\leq\, 0,\, i \,=\, 1, \ldots, m\,\}, \tag{10.4.1}$$

where each $g_i : \mathbb{R}^n \to \mathbb{R}$ is a twice continuously differentiable convex function on $\mathbb{R}^n$. Our immediate goal is to define a Newton approximation scheme for the projector Π_K. Section **4.4** already considered some detailed differentiability properties of the projector Π_K. For ease of reference, we recall herein some notation and results we need. For each vector x in $\mathbb{R}^n$, the projected vector $\bar{x} \equiv \Pi_K(x)$ is the unique solution of the convex program:

$$\begin{aligned} &\text{minimize} && \tfrac{1}{2}\,(\,v - x\,)^T(\,v - x\,) \\ &\text{subject to} && g(v) \,\leq\, 0. \end{aligned}$$

Suppose that the CRCQ holds at $\bar{x}$. This CQ continues to hold at all vectors in K that are sufficiently close to $\bar{x}$. Let $\mathcal{M}_\pi(x)$ denote the set of multiplier λ satisfying the KKT conditions for the above problem and let $\mathcal{B}(x)$ be the collection of index subsets $\mathcal{J} \subseteq \mathcal{I}(\bar{x})$ such that the family of gradient vectors

$$\{\, \nabla g_i(\bar{x}) \,:\, i \,\in\, \mathcal{J}\,\} \tag{10.4.2}$$

is linearly independent, where $\mathcal{I}(\bar{x})$ denotes the set of active constraints at $\bar{x}$. Denote by $\mathcal{B}'(x)$ the subcollection of $\mathcal{B}(x)$ consisting of index sets $\mathcal{J}$ in $\mathcal{B}(x)$ for which there exists a vector $\lambda \in \mathcal{M}_\pi(x)$ such that $\mathrm{supp}(\lambda)$ is contained in $\mathcal{J}$. Such a multiplier is necessarily unique and we denote it by $\lambda(\mathcal{J}, x)$. This notation reflects the dependence of this multiplier on the base vector x (which will vary in the discussion to follow) and the index set $\mathcal{J}$.

To motivate the definition of the desired linear Newton approximation of Π_K, we recall that in the proof of Theorem **4.5.2**, we considered, for each $\mathcal{J} \in \mathcal{B}'(x)$, the function

$$\Phi_{\mathcal{J}} \,:\, (\,v, u, \eta_{\mathcal{J}}\,) \,\in\, \mathbb{R}^{2n+|\mathcal{J}|} \,\mapsto\, \begin{pmatrix} v - u + \displaystyle\sum_{j\in\mathcal{J}} \eta_j\, \nabla g_j(v) \\ \\ -g_{\mathcal{J}}(v) \end{pmatrix} \,\in\, \mathbb{R}^{n+|\mathcal{J}|},$$

which vanishes at the triple $(\bar{x}, x, \lambda_{\mathcal{J}})$, where $\lambda \equiv \lambda(\mathcal{J}, x)$. The partial Jacobian matrix of $\Phi_{\mathcal{J}}$ with respect to the pair $(v, \eta_{\mathcal{J}})$, at such a triple is nonsingular and equal to

$$J_{v,\eta_{\mathcal{J}}}\Phi_{\mathcal{J}}(\bar{x}, x, \lambda_{\mathcal{J}}) \,=\, \begin{bmatrix} \mathbf{I}_n + \displaystyle\sum_{j\in\mathcal{J}} \lambda_j\, \nabla^2 g_j(\bar{x}) & Jg_{\mathcal{J}}(\bar{x})^T \\ \\ -Jg_{\mathcal{J}}(\bar{x}) & 0 \end{bmatrix}.$$

By the classical implicit function theorem applied to $\Phi_{\mathcal{J}}$ at the triple $(\bar{x}, x, \lambda_{\mathcal{J}})$, where $\lambda = \lambda(\mathcal{J}, x)$, with $(v, \eta_{\mathcal{J}})$ as the primary variable and

u as the parameter, there exist open neighborhoods $\mathcal{V}(\mathcal{J})$ of $\bar{x}$, $\mathcal{N}(\mathcal{J})$ of x, and $\mathcal{U}(\mathcal{J})$ of $\lambda_{\mathcal{J}}$ and a continuously differentiable function

$$z : \mathcal{N}(\mathcal{J}) \to \mathcal{V}(\mathcal{J}) \times \mathcal{U}(\mathcal{J})$$

such that for every vector u in the neighborhood $\mathcal{N}(\mathcal{J})$, $z(u)$ is the unique pair $(y, \eta_{\mathcal{J}})$ in $\mathcal{V}(\mathcal{J}) \times \mathcal{U}(\mathcal{J})$ satisfying

$$\Phi_{\mathcal{J}}(y, u, \eta_{\mathcal{J}}) = 0.$$

Furthermore, the Jacobian matrix of z at x is equal to

$$Jz(x) = -J_{v,\eta_{\mathcal{J}}}\Phi_{\mathcal{J}}(\bar{x}, x, \lambda_{\mathcal{J}})^{-1} J_u \Phi_{\mathcal{J}}(\bar{x}, x, \lambda_{\mathcal{J}}).$$

Let $z_{\mathcal{J}}$ be the y-part of the function z. By Theorem **4.5.2**, Π_K is a PC^1 function near x with C^1 pieces given by

$$\{\, z_{\mathcal{J}} : \mathcal{J} \in \mathcal{B}'(x) \,\}.$$

At this point, we could in principle invoke the proof of Theorem **7.2.15** to construct a linear Newton approximation of Π_K at x. Specifically, for each vector x' sufficiently near x, let $\mathcal{P}(x')$ be the family of index sets $\mathcal{J}$ in $\mathcal{B}'(x)$ such that

$$\Pi_K(x') = z_{\mathcal{J}}(x')$$

and define the family

$$\{\, Jz_{\mathcal{J}}(x') : \mathcal{J} \in \mathcal{P}(x') \,\}.$$

The proof of Theorem **7.2.15** shows that this family provides a linear Newton approximation of Π_K near x.

The above family is conceptually acceptable but practically not very useful because to compute a Newton matrix at x', we need to know the family $\mathcal{P}(x')$ and the implicit function $z_{\mathcal{J}}$, both of which depend on the vector x. Ideally, we wish to define the Newton matrices at x' directly from this vector itself. To derive the alternative linear Newton approximation, we first show that the function $z_{\mathcal{J}}$ is needed only as a technical tool and can be bypassed in defining the Newton matrices. For this purpose, we let

$$G_{\mathcal{J}}(\bar{x}) \equiv \mathbf{I}_n + \sum_{i \in \mathcal{J}} \lambda_i \nabla^2 g_i(\bar{x}),$$

where $\lambda = \lambda(\mathcal{J}, x)$. (A word of clarification about the notation: Corresponding to a given pair $(\mathcal{J}, x)$, the multiplier $\lambda(\mathcal{J}, x)$ is uniquely determined; this multiplier is used to define the matrix $G_{\mathcal{J}}(\bar{x})$ but is not included in the notation of the matrix.)

Writing $G_{\mathcal{J}}$ for $G_{\mathcal{J}}(\bar{x})$, we can easily show that

$$J_{v,\eta_{\mathcal{J}}}\Phi_{\mathcal{J}}(\bar{x},x,\lambda_{\mathcal{J}})^{-1} = \begin{bmatrix} G_{\mathcal{J}}^{-1} - G_{\mathcal{J}}^{-1}Jg_{\mathcal{J}}(\bar{x})^T Q^{-1} Jg_{\mathcal{J}}(\bar{x}) G_{\mathcal{J}}^{-1} & -G_{\mathcal{J}}^{-1}Jg_{\mathcal{J}}(\bar{x})^T Q^{-1} \\ Q^{-1}Jg_{\mathcal{J}}(\bar{x})G_{\mathcal{J}}^{-1} & Q^{-1} \end{bmatrix},$$

where Q is a shorthand for

$$Q_{\mathcal{J}}(\bar{x}) \equiv Jg_{\mathcal{J}}(\bar{x})\, G_{\mathcal{J}}(\bar{x})^{-1}\, Jg_{\mathcal{J}}(\bar{x})^T,$$

which is a symmetric positive definite matrix, by the linear independence of the gradients (10.4.2) and the positive definiteness of the matrix $G_{\mathcal{J}}(\bar{x})$. It is straightforward to check that

$$Jz_{\mathcal{J}}(x) = G_{\mathcal{J}}^{-1} - G_{\mathcal{J}}^{-1}Jg_{\mathcal{J}}(\bar{x})^T Q^{-1} Jg_{\mathcal{J}}(\bar{x}) G_{\mathcal{J}}^{-1}.$$

Notice that the matrix on the right-hand side is completely determined by the pair $(x,\mathcal{J})$ and does not depend on the knowledge of the implicit function $z_{\mathcal{J}}$. To highlight the independence from the latter function, we write

$$A(x,\mathcal{J}) \equiv G_{\mathcal{J}}(\bar{x})^{-1} - G_{\mathcal{J}}(\bar{x})^{-1}Jg_{\mathcal{J}}(\bar{x})^T Q_{\mathcal{J}}(\bar{x})^{-1} Jg_{\mathcal{J}}(\bar{x}) G_{\mathcal{J}}(\bar{x})^{-1}.$$

In what follows, we show that the family

$$\mathcal{A}_\pi(x') \equiv \{\, A(x',\mathcal{J}) : \mathcal{J} \in \mathcal{B}'(x') \,\}, \tag{10.4.3}$$

defines a linear Newton approximation scheme for Π_K at x' that is near x. Before showing this, we remark that it is easy to compute an element in $\mathcal{A}_\pi(x')$ once $\bar{x}' = \Pi_K(x')$ is known. Indeed, simply choose an extreme point $\bar{\lambda}$ of the multiplier set $\mathcal{M}_\pi(x')$ and let $\mathcal{J} \equiv \mathrm{supp}(\bar{\lambda})$. This index set $\mathcal{J}$ then determines an element in $\mathcal{A}_\pi(x')$.

Toward the demonstration of the desired Newton approximation property of $\mathcal{A}_\pi$, we establish a key lemma.

10.4.1 Lemma. Let K be given by (10.4.1), where each g_i is a twice continuously differentiable convex function on $\mathbb{R}^n$. Assume that $\bar{x} \equiv \Pi_K(x)$ satisfies the CRCQ. There exist a neighborhood $\mathcal{N}$ of x and a constant $\eta > 0$ such that

$$\mathcal{B}'(x') \subseteq \mathcal{B}'(x)$$

and

$$\| \lambda(\mathcal{J},x') - \lambda(\mathcal{J},x) \| \leq \eta \| x - x' \|,$$

for all $\mathcal{J} \in \mathcal{B}'(x')$ and all $x' \in \mathcal{N}$.

Proof. Assume for the sake of contradiction that there exists a sequence $\{x^k\}$ of vectors converging to x such that for each k, an index set $\mathcal{J}_k$ exists, which belongs to $\mathcal{B}'(x^k)$ but not to $\mathcal{B}'(x)$. Without loss of generality, we may assume that $\mathcal{J}_k$ are all equal. Denote this common index set by $\mathcal{J}$. Thus

$$\mathcal{J} \in \mathcal{B}'(x') \setminus \mathcal{B}'(x), \quad \forall\, k.$$

We may further assume that $\mathcal{I}(\bar{x}^k) \subseteq \mathcal{I}(\bar{x})$ for all k, where $\bar{x}^k \equiv \Pi_K(x^k)$. Thus, $\mathcal{J} \subseteq \mathcal{I}(\bar{x})$. For each k, the gradients

$$\{\, \nabla g_i(\bar{x}^k) \,:\, k \in \mathcal{J} \,\}$$

are linearly independent; by the CRCQ, so are the gradients

$$\{\, \nabla g_i(\bar{x}) \,:\, k \in \mathcal{J} \,\}.$$

Hence $\mathcal{J} \in \mathcal{B}(x)$. For each k, let $\lambda^k \in \mathcal{M}_\pi(x^k)$ be such that $\mathrm{supp}(\lambda^k) \subseteq \mathcal{J}$. The sequence $\{\lambda^k\}$ must be bounded; moreover, every accumulation point of this sequence must be a member of $\mathcal{M}_\pi(x)$. Consequently, $\mathcal{J}$ belongs to $\mathcal{B}'(x)$; a contradiction. This establishes the first assertion of the lemma.

To prove the second assertion, we note that for each vector $x' \in \mathbb{R}^n$ and each index set $\mathcal{J}$ in $\mathcal{B}'(x')$, we have

$$\lambda'_{\mathcal{J}} \;=\; (\, Jg_{\mathcal{J}}(\bar{x}')Jg_{\mathcal{J}}(\bar{x}')^{\,T}\,)^{-1}Jg_{\mathcal{J}}(\bar{x}')\,(\,x' - \Pi_K(x')\,),$$

where $\lambda' = \lambda(\mathcal{J}, x')$. Provided that x' belongs to the neighborhood $\mathcal{N}$, such an index set $\mathcal{J}$ must belong to the family $\mathcal{B}'(x)$. Since $Jg_{\mathcal{J}}(\bar{x})^{\,T}$ has linearly independent columns and there are only finitely many such index sets $\mathcal{J}$, the existence of the desired constant $\eta > 0$ satisfying the second assertion of the lemma follows readily from the above expression of the multipliers. □

Next, we wish to obtain an upper bound for

$$\|\, A(x, \mathcal{J}) - A(x', \mathcal{J})\,\|$$

in terms of $\|x - x'\|$ for x' sufficiently close to x, where $\mathcal{J} \in \mathcal{B}'(x')$, which is a subset of $\mathcal{B}'(x)$. We proceed as follows. Write $\bar{x}' \equiv \Pi_K(x')$ and

$$E(x', \mathcal{J}) \;\equiv\; G_{\mathcal{J}}(\bar{x}')^{-1}Jg_{\mathcal{J}}(\bar{x}').$$

In what follows, we write $\lambda = \lambda(\mathcal{J}, x)$ and $\lambda' = \lambda(\mathcal{J}, x')$. We have

$$A(x, \mathcal{J}) - A(x', \mathcal{J}) \;=\; T_1 - T_2,$$

where

$$T_1 \equiv G_{\mathcal{J}}(\bar{x})^{-1} - G_{\mathcal{J}}(\bar{x}')^{-1}$$

and

$$T_2 \equiv E(x,\mathcal{J})Q_{\mathcal{J}}(\bar{x})^{-1}E(x,\mathcal{J})^T - E(x',\mathcal{J})Q_{\mathcal{J}}(\bar{x}')^{-1}E(x',\mathcal{J})^T.$$

To bound $\|T_1\|$, we first note that since each λ_i is nonnegative and $\nabla^2 g_i(\cdot)$ is symmetric positive semidefinite, by the convexity of g_i, it is easy to deduce that, for all pairs $(\bar{x}', \mathcal{J})$,

$$\| G_{\mathcal{J}}(\bar{x}')^{-1} \|_2 \leq 1. \tag{10.4.4}$$

Moreover, we can write

$$\begin{aligned} & G_{\mathcal{J}}(\bar{x}) - G_{\mathcal{J}}(\bar{x}') \\ & \quad = \sum_{i\in\mathcal{J}} \left\{ \lambda_i [\nabla^2 g_i(\bar{x}) - \nabla^2 g_i(\bar{x}')] + (\lambda_i - \lambda_i')\nabla g_i(\bar{x}') \right\}. \end{aligned}$$

We note the following simple inequality:

$$\| A^{-1} - B^{-1} \| \leq \| A^{-1} \| \| A - B \| \| B^{-1} \|,$$

which holds for any two invertible matrices A and B of the same order. Consequently, by (10.4.4), we obtain

$$\begin{aligned} & \| G_{\mathcal{J}}(\bar{x})^{-1} - G_{\mathcal{J}}(\bar{x}')^{-1} \| \\ & \quad \leq \sum_{i\in\mathcal{J}} \left\{ \lambda_i \| \nabla^2 g_i(\bar{x}) - \nabla^2 g_i(\bar{x}') \| + |\lambda_i - \lambda_i'| \| \nabla g_i(\bar{x}') \| \right\}. \end{aligned}$$

To bound $\|T_2\|$, we write

$$\begin{aligned} T_2 = \; & E(x,\mathcal{J})[Q_{\mathcal{J}}(\bar{x})^{-1} - Q_{\mathcal{J}}(\bar{x}')^{-1}]E(x,\mathcal{J})^T + \\ & (E(x,\mathcal{J}) - E(x',\mathcal{J}))Q_{\mathcal{J}}(\bar{x}')^{-1}(E(x,\mathcal{J}) - E(x',\mathcal{J}))^T + \\ & 2(E(x,\mathcal{J}) - E(x',\mathcal{J}))Q_{\mathcal{J}}(\bar{x}')^{-1}E(x',\mathcal{J})^T. \end{aligned}$$

In turn,

$$\begin{aligned} & E(x,\mathcal{J}) - E(x',\mathcal{J}) \\ & \quad = G_{\mathcal{J}}(\bar{x})^{-1}Jg_{\mathcal{J}}(\bar{x}) - G_{\mathcal{J}}(\bar{x}')^{-1}Jg_{\mathcal{J}}(\bar{x}') \\ & \quad = [G_{\mathcal{J}}(\bar{x})^{-1} - G_{\mathcal{J}}(\bar{x}')^{-1}]Jg_{\mathcal{J}}(\bar{x}) + G_{\mathcal{J}}(\bar{x}')^{-1}(Jg_{\mathcal{J}}(\bar{x}) - Jg_{\mathcal{J}}(\bar{x}')). \end{aligned}$$

Finally, we need to evaluate $Q_{\mathcal{J}}(\bar{x})^{-1} - Q_{\mathcal{J}}(\bar{x}')^{-1}$. We can write

$$\begin{aligned} & Q_{\mathcal{J}}(\bar{x}) - Q_{\mathcal{J}}(\bar{x}') \\ & \quad = Jg_{\mathcal{J}}(\bar{x})[G_{\mathcal{J}}(\bar{x})^{-1} - G_{\mathcal{J}}(\bar{x}')^{-1}]Jg_{\mathcal{J}}(\bar{x})^T + \\ & \quad (Jg_{\mathcal{J}}(\bar{x}) - Jg_{\mathcal{J}}(\bar{x}'))G_{\mathcal{J}}(\bar{x}')^{-1}(Jg_{\mathcal{J}}(\bar{x}) - Jg_{\mathcal{J}}(\bar{x}'))^T + \\ & \quad 2(Jg_{\mathcal{J}}(\bar{x}) - Jg_{\mathcal{J}}(\bar{x}'))G_{\mathcal{J}}(\bar{x}')^{-1}Jg_{\mathcal{J}}(x')^T. \end{aligned}$$

Moreover,

$$\| Q_{\mathcal{J}}(\bar{x})^{-1} - Q_{\mathcal{J}}(\bar{x}')^{-1} \| \leq \| Q_{\mathcal{J}}(\bar{x})^{-1} \| \| Q_{\mathcal{J}}(\bar{x}) - Q_{\mathcal{J}}(\bar{x}') \| \| Q_{\mathcal{J}}(\bar{x}')^{-1} \|.$$

Based on the above derivation and Lemma **10.4.1**, we can establish the following result.

10.4.2 Proposition. Let K be given by (10.4.1), where each g_i is a twice continuously differentiable convex function on $\mathbb{R}^n$. Assume that the CRCQ holds at $\bar{x} \equiv \Pi_K(x)$. The family $\mathcal{A}_\pi$ is a linear Newton approximation scheme for Π_K at x. If each $\nabla^2 g_i$ is Lipschitz continuous near $\Pi_K(x)$, then the approximation scheme is strong.

Proof. We recall that $\mathcal{B}'(x)$ is a finite set; hence so is

$$\bigcup_{\mathcal{J} \in \mathcal{B}'(x)} \lambda(\mathcal{J}, x).$$

By Lemma **10.4.1** and the above derivation, it follows that for every $\varepsilon > 0$, there exists a neighborhood $\mathcal{N}$ of x such that for any $x' \in \mathcal{N}$ and any $\mathcal{J} \in \mathcal{B}'(x')$,

$$\| A(x', \mathcal{J}) - A(x, \mathcal{J}) \| \leq \varepsilon.$$

Moreover, if each $\nabla^2 g_i$ is Lipschitz continuous near $\bar{x}$, then a constant $\eta > 0$ exists such that

$$\| A(x', \mathcal{J}) - A(x, \mathcal{J}) \| \leq \eta \| x - x' \|.$$

With $z_{\mathcal{J}}$ denoting the implicit function based on the pair $(\mathcal{J}, x)$, we have for all x' sufficiently near x and any $\mathcal{J} \in \mathcal{B}'(x')$,

$$\begin{aligned} &\Pi_K(x') + A(x', \mathcal{J})(x - x') \\ &\quad = z_{\mathcal{J}}(x') + Jz_{\mathcal{J}}(x)(x - x') + [A(x', \mathcal{J}) - A(x, \mathcal{J})](x - x') \\ &\quad = z_{\mathcal{J}}(x) + o(\| x - x' \|) + [A(x', \mathcal{J}) - A(x, \mathcal{J})](x - x'). \end{aligned}$$

Thus,

$$\begin{aligned} &\Pi_K(x') + A(x', \mathcal{J})(x - x') - \Pi_K(x) \\ &\quad = [A(x', \mathcal{J}) - A(x, \mathcal{J})](x - x') + o(\| x - x' \|). \end{aligned}$$

From the above expression, condition (b) in Definition **7.2.2** holds easily, establishing that $\mathcal{A}_\pi$ is a linear Newton approximation of Π_K at x. Moreover, this approximation scheme is strong if each $\nabla^2 g_i$ is Lipschitz continuous near $\bar{x}$. □

According to Exercise **4.8.7**, each $A(x, \mathcal{J})$ is the matrix representation of the operator Π_C^G, where C is the null space of the gradients (10.4.2) and $G \equiv G_{\mathcal{J}}(\bar{x})$. As such the next proposition follows easily from the cited exercise.

10.4.3 Proposition. All matrices $A(x, \mathcal{J})$ in $\mathcal{A}_\pi(x)$ are symmetric and positive semidefinite and $\|A(x, \mathcal{J})\|_2 \leq 1$. Thus $\mathbf{I} - A(x, \mathcal{J})$ is positive semidefinite. □

Using Proposition **10.4.2**, we can define a linear Newton approximation scheme for various functions involving the projector. Consider the regularized gap function $\theta_c(x)$ with

$$\nabla\theta_c(x) \;=\; F(x) + (\, JF(x)^T - c^{-1}\mathbf{I}_n \,)(\, x - y_c(x) \,),$$

where $y_c(x) = \Pi_K(x - c^{-1}F(x))$. Since the function y_c is the composite $\Pi_K \circ (\mathbf{I} - c^{-1}F)$ of the Euclidean projector with the differentiable function $\mathbf{I} - c^{-1}F$, it follows from Proposition **10.4.2** and Theorem **7.5.17** that if the CRCQ holds at the vector $y_c(x)$, then a linear Newton approximation for y_c at x is given by the family of product matrices:

$$\mathcal{Y}_c(x') \;\equiv\; \{\, A(\,\mathbf{I} - c^{-1}JF(x')\,) \;:\; A \,\in\, \mathcal{A}_\pi(x' - c^{-1}F(x'))\,\}$$

for vectors x' near x. Moreover, this approximation is strong if each $\nabla^2 g_i$ is Lipschitz continuous near $y_c(x)$ and JF is Lipschitz continuous near x. Since

$$\mathbf{F}_c^{\mathrm{nat}}(x) \;\equiv\; x - y_c(x),$$

Corollary **7.5.18** immediately yields that

$$\mathcal{A}_c^{\mathrm{nat}}(x') \;\equiv\; \mathbf{I} - \mathcal{Y}_c(x')$$

provides a (strong) linear Newton approximation of $\mathbf{F}_c^{\mathrm{nat}}$ at x. Extending these approximations to the function $\nabla\theta_c$, we have the following result.

10.4.4 Proposition. Let K be given by (10.4.1), where each g_i is a twice continuously differentiable convex function on $\mathbb{R}^n$. Let F be continuously differentiable. Suppose that the CRCQ holds at a solution x^* of the VI (K, F). For any scalar $c > 0$, the family of matrices:

$$\mathcal{A}_c(x) \;\equiv\; \{\, JF(x) + (JF(x)^T - c^{-1}\mathbf{I})(\mathbf{I} - Y\,) \;:\; Y \,\in\, \mathcal{Y}_c(x)\,\} \qquad (10.4.5)$$

is a linear Newton approximation of $\nabla\theta_c$ at x^*. Thus, for $b > a > 0$,

$$\mathcal{A}_{ab}(x) \;\equiv\; \{\, A_a - A_b \;:\; A_a \,\in\, \mathcal{A}_a(x),\; A_b \,\in\, \mathcal{A}_b(x)\,\} \qquad (10.4.6)$$

is a linear Newton approximation scheme for $\nabla\theta_{ab}$ at x^*. Furthermore,

$$\mathrm{Jac}(\nabla\theta_c(x^*)) \subseteq \mathcal{A}_c(x^*) \quad \text{and} \quad \mathrm{Jac}(\nabla\theta_{ab}(x^*)) \subseteq \mathcal{A}_{ab}(x^*).$$

Finally, if in addition $\nabla^2 g_i$ and JF are Lipschitz continuous at x^*, both $\mathcal{A}_c$ and $\mathcal{A}_{ab}$ are strong linear Newton approximation schemes of θ_c and θ_{ab} at x^*, respectively.

Proof. We give the proof only for the regularized gap function θ_c. The proof for the D-gap function θ_{ab} follows from Corollary **7.5.18**. In turn, by this corollary and the special form of $\nabla\theta_c$, in order to show that $\mathcal{A}_c$ provides a linear Newton approximation of θ_c at x^*, it suffices to show that a linear Newton approximation of the function

$$H : x \mapsto JF(x)^T(\, x - y_c(x)\,)$$

at x^* is provided by the family of matrices

$$\{\, JF(x)^T(\, \mathbf{I} - Y\,) : Y \in \mathcal{Y}_c(x)\,\}.$$

Since x^* solves the VI (K, F), we have $x^* = y_c(x^*)$. Thus, $H(x^*) = 0$. Hence, for any $Y \in \mathcal{Y}_c(x)$, we have

$$H(x) + Y(x^* - x) - H(x^*) = -JF(x)^T(\, y_c(x) + Y(x - x^*) - y_c(x^*)\,)$$

Since $\mathcal{Y}_c$ is a linear Newton approximation of y_c at x^*, the right-hand side is $o(\|x - x^*\|)$ ($O(\|x - x^*\|^2$ if each $\nabla^2 g_i$ is Lipschitz continuous at x^*). The establishes the approximation property of $\mathcal{A}_c$ and the strong approximation property, provided that JF and each $\nabla^2 g_i$ are Lipschitz continuous at x^*.

Finally, to show that the limiting Jacobian of θ_c at x^* is contained in $\mathcal{A}_c(x^*)$, we note that $\nabla\theta_c$ is a PC^1 function at x^* with C^1 pieces given by the implicit functions $z_{\mathcal{J}}$ for $\mathcal{J} \in \mathcal{B}'(x^*)$. Thus the desired containment is an immediate consequence of Lemma **4.6.3**. □

We illustrate the above approximation schemes with the NCP.

10.4.5 Example. Consider the NCP (F) with F being continuously differentiable, which corresponds to the above setting with $g(x) = -x$. We illustrate the family $\mathbf{I} - \mathcal{Y}_1(x)$, which provides a linear Newton approximation for the map

$$\mathbf{F}_1^{\mathrm{nat}}(x) = x - \Pi_{\mathbb{R}^n_+}(x - F(x)) = \min(x, F(x)).$$

It is easy to see that $\mathcal{M}_\pi(x)$ is equal to the singleton $\{-\min(0, x)\}$. Therefore, since the LICQ holds for this problem, we easily see that

$$\mathcal{B}'(x) = \{\, \mathcal{J} : \mathcal{I}_<(x) \subseteq \mathcal{J} \subseteq \mathcal{I}_\leq(x)\,\},$$

where

$$\mathcal{I}_<(x) \equiv \{\, i : x_i < 0 \,\} \quad \text{and} \quad \mathcal{I}_\leq(x) \equiv \{\, i : x_i \leq 0 \,\}.$$

For an index set $\mathcal{J} \in \mathcal{B}'(x)$, we have

$$A(x, \mathcal{J}) = \mathbf{I}_n - \begin{bmatrix} \mathbf{I}_{\mathcal{J}\mathcal{J}} & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & \mathbf{I}_{\bar{\mathcal{J}}\bar{\mathcal{J}}} \end{bmatrix},$$

where $\bar{\mathcal{J}}$ is the complement of $\mathcal{J}$ in $\{1, \ldots, n\}$. A direct calculation yields

$$\mathcal{A}_1^{\rm nat}(x) = \left\{ \begin{bmatrix} \mathbf{I}_{\mathcal{J}\mathcal{J}} & 0 \\ JF(x)_{\bar{\mathcal{J}}\mathcal{J}} & JF(x)_{\bar{\mathcal{J}}\bar{\mathcal{J}}} \end{bmatrix}, \right.$$

$$\left. \forall\, \mathcal{J} \text{ such that } \{i : x_i < F_i(x)\} \subseteq \mathcal{J} \subseteq \{i : x_i \leq F_i(x)\} \right\}.$$

We easily recognize that the family $\mathcal{A}_1^{\rm nat}(x)$ coincides with the family of matrices obtained in Example **7.2.16** specialized to the case $G(x) \equiv x$. □

By a simple rearrangement, we see that a matrix in $\mathcal{A}_a(x)$ has the form

$$JF(x) + JF(x)^T - a\,\mathbf{I} + a^{-1}\,(\,JF(x) - a\,\mathbf{I}\,)^T A'_a(\,JF(x) - a\,\mathbf{I}\,), \quad (10.4.7)$$

where the matrix $A'_a \in \mathcal{A}_\pi(x - a^{-1}F(x))$ is positive semidefinite. Thus,

$$\begin{aligned} \mathcal{A}_{ab}(x) = \{\, &(\,b - a\,)\,\mathbf{I}_n - P_b + P_a \;: \\ P_a \in\; &a^{-1}\,(\,a\mathbf{I}_n - JF(x)^T\,)\mathcal{A}_\pi(x - a^{-1}F(x))(\,a\mathbf{I}_n - JF(x)\,), \\ P_b \in\; &b^{-1}\,(\,b\mathbf{I}_n - JF(x)^T\,)\mathcal{A}_\pi(x - b^{-1}F(x))(\,b\mathbf{I}_n - JF(x)\,)\,\}. \end{aligned} \quad (10.4.8)$$

The latter expression is useful for proving the next proposition, which provides a sufficient condition for the matrices in the family $\mathcal{A}_{ab}(x)$ to be nonsingular.

10.4.6 Proposition. Let K be given by (10.4.1), where each g_i is a twice continuously differentiable convex function on $\mathbb{R}^n$. Let F be continuously differentiable. Let $b > a > 0$ be given scalars and $x^* \in \mathbb{R}^n$ be a given vector. Suppose that the CRCQ holds at $y_a(x^*)$ and $y_b(x^*)$. The following four statements hold.

(a) If

$$\lambda_{\min}(JF(x^*) + JF(x^*)^T) > a, \quad (10.4.9)$$

all the matrices in $\mathcal{A}_a(x^*)$ are positive definite.

(b) If

$$\lambda_{\min}(JF(x^*) + JF(x^*)^T) > a + b^{-1}\|JF(x^*)\|^2, \tag{10.4.10}$$

all the matrices in $\mathcal{A}_{ab}(x^*)$ are positive definite.

(c) If x^* is a strongly nondegenerate solution of the VI (K, F) and $JF(x^*)$ is positive definite, then $\mathcal{A}_{ab}(x^*)$ consists of a single symmetric positive definite matrix.

(d) If $x^* \in \mathrm{SOL}(K, F)$ and all matrices in $\mathcal{A}_{ab}(x^*)$ are positive definite, then $d^T H d > 0$ for all $d \neq 0$ and $H \in \partial_d^2 \theta_{ab}(x^*)$.

Proof. Every matrix in $\mathcal{A}_a(x^*)$ has the form (10.4.7):

$$JF(x^*) + JF(x^*)^T - a\,\mathbf{I} + a^{-1}\,(\,JF(x^*) - a\,\mathbf{I}\,)^T A'_a(\,JF(x^*) - a\,\mathbf{I}\,)$$

for some positive semidefinite matrix A'_a. From this representation, the assertion (a) follows readily.

By Proposition **10.4.3** the Euclidean norm of all matrices in $\mathcal{A}_\pi(x')$ is not greater than one, for any x'. By (10.4.8), any $A \in \mathcal{A}_{ab}(x^*)$ can be written as the difference $P_a - P_b$ for some matrices $P_a \in \mathcal{A}_a(x^*)$ and $P_b \in \mathcal{A}_b(x^*)$. Thus,

$$\begin{aligned} d^T A d &= (\,b - a\,)\,\|d\|^2 - d^T P_b d + d^T P_a d \\ &\geq (\,b - a\,)\,\|d\|^2 - b^{-1}\,\|\,(\,b\,\mathbf{I} - JF(x^*)\,)^T d\,\|^2 \\ &= d^T(\,JF(x^*) + JF(x^*)^T\,)d - a\,\|d\|^2 \\ &\quad -b^{-1}\,d^T(\,JF(x^*)JF(x^*)^T\,)d. \end{aligned}$$

It is easy to see that if (10.4.10) is satisfied, then $d^T A d$ is positive, provided that $d \neq 0$. Thus (b) holds.

If x^* is a strongly nondegenerate solution of the VI (K, F), then for every scalar $c > 0$, $y_c(x^*) = x^*$; moreover, there is only one index set $\mathcal{J} \subseteq \mathcal{I}(x^*)$ such that $\mathrm{supp}(\lambda^*) \subseteq \mathcal{J}$, where $\{\lambda^*\} = \mathcal{M}(x^*)$. Hence $\mathcal{B}'(x^* - c^{-1}F(x^*))$ is a singleton equal to $\{\mathcal{I}(x^*)\}$. Therefore, $\mathcal{A}_\pi(x^* - c^{-1}F(x^*))$ consists of a single symmetric positive semidefinite matrix A, which is independent of c, such that $\mathbf{I} - A$ is also positive semidefinite. It is then easy to verify that $\mathcal{A}_{ab}(x^*)$ also consists of a single matrix that is given by

$$\begin{aligned} &(\,b - a\,)\mathbf{I} + a^{-1}\,(\,JF(x^*) - a\,\mathbf{I}\,)^T A(\,JF(x^*) - a\,\mathbf{I}\,) \\ &\qquad -b^{-1}\,(\,JF(x^*) - b\,\mathbf{I}\,)^T A(\,JF(x^*) - b\,\mathbf{I}\,) \\ &= (\,b - a\,)\,(\,\mathbf{I} - A\,) + (\,a^{-1} - b^{-1}\,)\,JF(x^*)^T A JF(x^*). \end{aligned}$$

The above matrix is clearly symmetric positive semidefinite. If it is not positive definite, then there exists a nonzero vector $v \in \mathbb{R}^n$ such that

$$(\mathbf{I} - A)v = JF(x^*)^T AJF(x^*)v = 0.$$

Since A is symmetric positive semidefinite, it follows that $0 = AJF(x^*)v$, which yields $0 = v^T AJF(x^*)v = v^T JF(x^*)v$. Since $JF(x^*)$ is positive definite, we deduce $v = 0$, which is a contradiction. Hence (c) holds.

Finally, if $x^* \in \mathrm{SOL}(K, F)$ and every matrix in $\mathcal{A}_{ab}(x^*)$ is positive definite, then so is every matrix in $\partial^2\theta_{ab}(x^*)$. This is because every matrix in $\partial^2\theta_{ab}(x^*)$ is a convex combination of finitely many matrices in $\mathrm{Jac}\,\nabla\theta_{ab}(x^*)$, each of which is contained in $\mathcal{A}_{ab}(x^*)$ by Proposition **10.4.4**, and is thus positive definite. Since $\partial_d^2\theta_{ab}(x^*)$ is a subset of $\partial^2\theta_{ab}(x^*)$, (d) follows. □

10.4.7 Remark. Conditions (10.4.9) and (10.4.10) both imply that the matrix $JF(x^*)$ is positive definite. Therefore, on the one hand, if x^* is a solution of the VI (K, F) and (10.4.10) holds, then x^* is a strongly stable, thus semistable, solution of the VI (K, F), by Corollary **5.1.8**. Hence part (d) of Proposition **10.4.6** also follows from Proposition **10.3.10** in this case. On the other hand, if x^* is a strongly nondegenerate solution of the VI (K, F), then $\mathrm{Jac}\,\nabla\theta_{ab}(x^*) = \partial^2\theta_{ab}(x^*)$ consists of a single symmetric matrix, which must be positive definite if $JF(x^*)$ is nonsingular. Part (d) of Proposition **10.4.6** is then obvious because $\partial_d^2\theta_{ab}(x^*)$ must equal to $\partial^2\theta_{ab}(x^*)$. Moreover, in this case θ_{ab} is twice continuously differentiable in a neighborhood of x^*, which is a strongly stable zero of $\nabla\theta_{ab}$ and a strongly stable solution of the VI (K, F). This raises a question. Suppose x^* is a strongly stable solution of the VI (K, F), does it follow that all matrices in $\mathcal{A}_{ab}(x^*)$ must be positive definite? Neither a proof nor a counterexample exists at this time. □

With the above background results in place, we can present algorithms for the solution of VI (K, F) based on the minimization of the D-gap function θ_{ab}. The algorithm below is the analog of Algorithm **9.1.10** that pertains to the NCP. As before, the algorithm is a line search method based on the Linear Newton Method **7.5.14**.

A D-gap Line Search Algorithm (DGLSA I)

10.4.8 Algorithm.

Data: $x^0 \in \mathbb{R}^n$, $b > a > 0$, $\rho > 0$, $p > 1$, and $\gamma \in (0, 1)$.

Step 1: Set $k = 0$.

Step 2: If x^k is a stationary point of θ_{ab} stop.

Step 3: Select an element H^k in $\mathcal{A}_{ab}(x^k)$ and find a solution d^k of the system

$$\nabla\theta_{ab}(x^k) + H^k d = 0. \tag{10.4.11}$$

If the system (10.4.11) is not solvable or if the condition

$$\nabla\theta_{ab}(x^k)^T d^k \leq -\rho \| d^k \|^p \tag{10.4.12}$$

is not satisfied, (re)set $d^k \equiv -\nabla\theta_{ab}(x^k)$.

Step 4: Find the smallest nonnegative integer i_k such that with $i = i_k$,

$$\theta_{ab}(x^k + 2^{-i} d^k) \leq \theta_{ab}(x^k) + \gamma\, 2^{-i}\, \nabla\theta_{ab}(x^k)^T d^k; \tag{10.4.13}$$

set $\tau_k \equiv 2^{-i_k}$.

Step 5: Set $x^{k+1} \equiv x^k + \tau_k d^k$ and $k \leftarrow k+1$; go to Step 2.

The analysis of the algorithm is similar to that of Algorithm **9.1.10**, with the appropriate conditions imposed to ensure the applicability of the convergence theory for Algorithm **7.5.14** specialized to the present context. The following theorem is the analog of Theorem **9.1.29**, which pertains to the NCP.

10.4.9 Theorem. Let K be given by (10.4.1), where each g_i is a twice continuously differentiable convex function on $\mathbb{R}^n$. Let F be continuously differentiable. Let $b > a > 0$ be given scalars. Let $\{x^k\}$ be an infinite sequence generated by Algorithm **10.4.8**.

(a) Every accumulation point of $\{x^k\}$ is a stationary point of the merit function θ_{ab}.

(b) If x^* is an accumulation point of $\{x^k\}$ such that (10.3.3) holds, then x^* is a solution of the VI (K, F).

(c) If $\{x^k\}$ has an isolated limit point, then the whole sequence $\{x^k\}$ converges to that point.

(d) Suppose that x^* is a limit point of $\{x^k\}$ and a solution of the VI (K, F). Assume that the CRCQ holds at x^* and (10.4.10) holds. The whole sequence $\{x^k\}$ converges to x^*; furthermore, if the scalars $p > 2$ and $\gamma < 1/2$ in Algorithm **10.4.8**, the following statements hold:

(i) eventually d^k is always the solution of system (10.4.11);

(ii) eventually a unit step size is accepted so that $x^{k+1} = x^k + d^k$;

(iii) the convergence rate is Q-superlinear; furthermore, if the Jacobian $JF(x)$ is Lipschitz continuous in a neighborhood of x^*, the convergence rate is Q-quadratic.

Proof. Statement (a) follows from Theorem **9.1.11**. Statement (b) follows from (a) and Theorem **10.3.4**. Statement (c) follows from Propositions **8.3.10** and **8.3.11** because with $\sigma(x^k, d^k) \equiv -\nabla\theta_{ab}(x^k)^T d^k$, we have

$$\| d^k \| \leq \max \left\{ \sqrt{\sigma(x^k, d^k)}, \, (\rho^{-1} \sigma(x^k, d^k))^{1/p} \right\}.$$

To prove (d), let x^* satisfy the given assumptions. Obviously x^* is a locally unique solution of VI (K, F). By part (b) of Proposition **10.4.6**, the Newton scheme $\mathcal{A}_{ab}$ is nonsingular at x^*. Therefore the systems (10.4.11) are solvable eventually, and define a superlinearly convergent sequence of directions $\{d^k\}$ with respect to $\{x^k\}$. Assertions (i) and (ii) of part (d) follow from part (c) of Proposition **10.4.6** and Proposition **8.3.18**. Finally, the Q-quadratic rate in assertion (iii) can be proved in the same way as in the corresponding result in Theorem **9.1.29**. □

The obvious problem with Algorithm **10.4.8** is that usually the two scalars a and b are not known in advance for condition (10.4.10) to hold, which in turn is responsible for the superlinear convergence of the algorithm. The one special case in which we may give beforehand correct values to these two parameters is when $F(x) = Mx + q$ (but K is not required to be polyhedral). In this case in fact, we have the following obvious result.

10.4.10 Corollary. Assume the same setting of Theorem **10.4.9** Suppose that $F(x) = Mx + q$ with M positive definite. Let $\{x^k\}$ be an infinite sequence generated by Algorithm **10.4.8**. If $\lambda_{\min}(M + M^T) > a + b^{-1}\|M\|^2$, then $\{x^k\}$ converges quadratically to the unique solution x^* of the VI (K, F). □

If F is not affine we can still try to adaptively vary a and b to obtain superlinear convergence. This can be achieved by a simple modification of Algorithm **10.4.8**.

D-gap Line Search Algorithm II (DGLSA II)

10.4.11 Algorithm. The steps of the algorithm are the same as those of Algorithm **10.4.8** except for Step 5, which is modified as follows. Let $c > 0$ and $\varepsilon > 0$ be given scalars.

Step 5: If

$$\theta_{ab}(x^k) \leq c \tag{10.4.14}$$

$$\lambda_{\min}(JF(x^k) + JF(x^k)^T) > \varepsilon\, c \tag{10.4.15}$$

and

$$\lambda_{\min}(JF(x^k) + JF(x^k)^T) \leq 2(a + b^{-1}\|JF(x^k)\|^2), \tag{10.4.16}$$

set $a \leftarrow \frac{1}{2}a$, $b \leftarrow 2b$ and $c \leftarrow \frac{1}{2}c$. Set $x^{k+1} = x^k + \tau_k d^k$ and $k \leftarrow k+1$; go to Step 2.

The idea behind this modification is that if the algorithm is converging to a solution of the variational inequality in the sense that $\theta_{ab}(x^k)$ is sufficiently small, (to be precise, smaller than the prescribed constant c), and the smallest eigenvalue of $JF(x^k) + JF(x^k)^T$ is sufficiently positive (greater than the positive tolerance factor εc), we adjust a and b to facilitate the satisfaction of condition (10.4.10) in Proposition **10.4.6** (b). This modification ensures superlinear convergence.

10.4.12 Theorem. Assume the same setting of Theorem **10.4.9**. Let $\{x^k\}$ be an infinite sequence generated by Algorithm **10.4.11**.

(a) If a and b are updated an infinite number of times, then every limit point of the sequence $\{x^k\}$ is a solution of VI (K, F).

(b) If a and b are updated a finite number of times, then the assertions (a), (b) and (c) of Theorem **10.4.9** hold. Furthermore, if x^* is a limit point of $\{x^k\}$ such that $JF(x^*)$ is positive definite and the CRCQ holds at x^*, then $\{x^k\}$ converges superlinearly to x^*; quadratically if JF is locally Lipschitz continuous near x^*.

Proof. We first note that if a_0, b_0, a and b are positive numbers such that $a \leq a_0 < b_0 \leq b$, then $\theta_{a_0 b_0}(x) \leq \theta_{ab}(x)$ for every x. This follows easily from the definition of the D-gap function. Let a_0 and b_0 be, respectively, the initial values of a and b at the beginning of Algorithm **10.4.11**. By the opening observation and the fact that at each iteration the value of the D-gap function is reduced at the new point x^{k+1}, we see that the sequence $\{\theta_{a_0 b_0}(x^k)\}$ is decreasing. Therefore, if a and b are updated infinitely often, and if κ denotes the subset of indices at which these parameters are updated, it follows, by the instructions of the modified Step 5, that $\{\theta_{ab}(x^k) : k \in \kappa\} \to 0$, which yields $\{\theta_{a_0 b_0}(x^k)\} \to 0$. This implies (a).

Suppose that a and b are updated a finite number of times only. Algorithm **10.4.11** eventually reduces to Algorithm **10.4.8**, so that (a), (b) and (c) of Theorem **10.4.9** hold. Furthermore, if x^* is a limit point of $\{x^k\}$ and $JF(x^*)$ is positive definite, then $\{x^k\}$ converges to x^* and x^* is a solution of VI (K, F). But then the tests (10.4.14) and (10.4.15) are satisfied eventually. Since a and b are no longer updated after a finite number of iterations, this means that (10.4.16) is never passed, so that by Proposition **10.4.6** and by simple continuity arguments we deduce that the Newton approximation scheme $\mathcal{A}_{ab}$ is nonsingular at x^*. From this observation, all the remaining assertions follow readily. □

From a theoretical point of view, the conclusions of Algorithm **10.4.11** are rather favorable. In fact, if a and b are updated only a finite number of times, the algorithm behaves like Algorithm **10.4.8** with the important difference that if one of the limit points is a solution of the VI (K, F), then the whole sequence converges to that point with a superlinear convergence rate, under no additional assumptions. On the other hand, if a and b are updated an infinite number of times, all the limit points of the sequence generated are solutions of the VI (K, F). There is however a severe computational concern in the latter case. In fact it is obvious that handling the function θ_{ab} with a approaching zero and b converging to infinity will certainly cause significant numerical difficulties. In practice, we could set a lower bound on a and an upper bound on b and stop updating these scalars when the bounds are reached. In this way we loose, in principle, some of the theoretical properties of the algorithm. For practical purposes, the choice of the bounds on a and b has to take into account the balance between fast convergence and numerical stability.

The next result is an important specialization of Theorem **10.4.12**, in which the conclusions are facilitated by the strong assumption on the function F.

10.4.13 Corollary. Assume the same setting of Theorem **10.4.9** and that F is strongly monotone and continuously differentiable. Any infinite sequence generated by Algorithm **10.4.11** converges to the unique solution x^* of the VI (K, F); moreover, the scalars a and b are updated only a finite number of times. If the CRCQ holds at x^* the convergence rate is superlinear; if JF is locally Lipschitz near x^*, the convergence rate is Q-quadratic.

Proof. Let $\{x^k\}$ be an infinite sequence generated by Algorithm **10.4.11**. We claim that a and b are updated a finite number of times only. First observe that $\{x^k\}$ is bounded. In fact, suppose for the sake of contradiction

that $\{x^k : k \in \kappa\}$ is unbounded for some subsequence indexed by κ. Let a_0 and b_0 be, respectively, the initial values of a_0 and b_0; then

$$\lim_{k(\in\kappa)\to\infty} \theta_{a_0 b_0}(x^k) = \infty$$

because $\theta_{a_0 b_0}$ is coercive, by Proposition **10.3.9**. But reasoning as in the proof of Theorem **10.4.12**, we see that this implies

$$\lim_{k(\in\kappa)\to\infty} \theta_{ab}(x^k) = \infty,$$

which is impossible because, as we also showed in Theorem **10.4.12**, the sequence $\{\theta_{ab}(x^k)\}$ is decreasing. Hence $\{x^k\}$ is bounded. By Theorem **10.4.12** (a), every limit point of $\{x^k\}$ is a solution of the VI (K, F), which must be unique by the strong monotonicity of F. Thus $\{x^k\}$ converges to the unique solution x^* of the VI. Moreover, since $JF(x^*)$ is positive definite, eventually the test (10.4.16) is never passed; therefore a and b is no longer updated after a finite number of iterations. From this point on, the corollary follows from Theorem **10.4.12** (c). □

It should be noted that while the locally fast convergence of Algorithm **10.4.11** requires a tuning of the scalars a and b, the global convergence of Algorithm **10.4.8** is valid for fixed values of a and b. As noted above, the tuning of these scalars has some computational disadvantages and could lead to numerical difficulties if theoretically fast convergence is insisted. To alleviate such difficulties, we may apply the original Algorithm **10.4.8**, where a and b are not adjusted, and attempt to speed up the convergence by resorting to a different locally fast method, such as the Newton Algorithm **7.3.1** whereby a sequence of semi-linearized sub-VIs (K, F^k) is solved, where

$$F^k(x) \equiv F(x^k) + JF(x^k)(x - x^k), \quad \forall x \in \mathbb{R}^n.$$

The next algorithm is derived from this idea; it combines a globally convergent method based on the D-gap function with a locally fast algorithm based on the semi-linearized subproblems.

D-gap Line Search Algorithm III (DGLSA III)

10.4.14 Algorithm.

Data: $x^0 \in \mathbb{R}^n$, $0 < a < b$, $\rho > 0$, $p > 1$, and $\gamma \in (0, 1)$.

Step 1: Set $k = 0$.

Step 2: If x^k is a stationary point of θ_{ab} stop.

Step 3: Find a solution $x^{k+1/2}$ to the semi-linearized sub-VI (K, F^k). If no such solution exists or if $d^k \equiv x^{k+1/2} - x^k$ does not satisfies the condition

$$\nabla\theta_{ab}(x^k)^T d^k \;\leq\; -\rho \,\|\, d^k \,\|^p, \tag{10.4.17}$$

set $d^k = -\nabla\theta_{ab}(x^k)$.

Step 4: Find the smallest nonnegative integer i_k such that with $i = i_k$,

$$\theta_{ab}(x^k + 2^{-i}d^k) \;\leq\; \theta_{ab}(x^k) + \gamma\, 2^{-i}\, \nabla\theta_{ab}(x^k)^T d^k; \tag{10.4.18}$$

set $\tau_k \equiv 2^{-i_k}$.

Step 5: Set $x^{k+1} \equiv x^k + \tau_k d^k$ and $k \leftarrow k+1$; go to Step 2.

The only difference between Algorithms **10.4.8** and **10.4.14** is in Step 3, where the system of linear equations (10.4.11) is replaced by the semi-linearized sub-VI (K, F^k). Needless to say, solving the latter sub-VI is computationally much more intensive than solving the former system of linear equations. The benefit of Algorithm **10.4.14** is that the scalars a and b are fixed throughout the algorithm. More importantly, Algorithm **10.4.14** can be shown to converge superlinearly under a much weaker assumption than the positive definiteness of the Jacobian matrix.

10.4.15 Theorem. Let K be a closed convex set in $\mathbb{R}^n$. Let F be continuously differentiable. Let $b > a > 0$ be given scalars. Let $\{x^k\}$ be an infinite sequence generated by Algorithm **10.4.14**.

(a) Every accumulation point of $\{x^k\}$ is a stationary point of the merit function θ_{ab}.

(b) If x^* is an accumulation point of $\{x^k\}$ such that (10.3.3) holds, then x^* is a solution of the VI (K, F).

(c) If $\{x^k\}$ has an isolated limit point, then the whole sequence $\{x^k\}$ converges to that point.

(d) Suppose that x^* is a limit point of $\{x^k\}$ and a stable solution of the VI (K, F). The whole sequence $\{x^k\}$ converges to x^*; furthermore, if the scalars $p > 2$ and $\gamma < 1/2$ in Algorithm **10.4.14**, the following statements hold:

(i) eventually d^k is always given by $x^{k+1/2} - x^k$;

(ii) eventually a unit step size is accepted so that $x^{k+1} = x^k + d^k$;

(iii) the convergence rate is Q-superlinear; furthermore, if the Jacobian $JF(x)$ is Lipschitz continuous in a neighborhood of x^*, the convergence rate is Q-quadratic.

Proof. The proof of parts (a), (b), and (c) of this theorem is identical to that of Theorem **10.4.9**. The only difference is the proof of part (d). Assume that x^* is a limit point of $\{x^k\}$ and a stable solution of the VI (K, F). Since x^* is a stable solution of the VI (K, F), it is isolated. Therefore, part (c) implies that $\{x^k\}$ converges to x^*. Moreover, by Theorem **7.3.5** and Remark **7.3.6**, the sequence $\{x^{k+1/2} - x^k\}$ is superlinearly convergent with respect to $\{x^k\}$. By Propositions **10.3.10** and **8.3.18**, it follows that each direction $d^k \equiv x^{k+1/2} - x^k$ generated by the solution of the semi-linearized sub-VI will eventually pass the test (10.4.17) and the unit step size will be accepted in a neighborhood of the limit vector x^*. With these observations, the theorem follows readily. □

10.4.2 The case of affine constraints

All the results and algorithms considered so far in this section can be improved considerably in the case where K is polyhedral, i.e., when the functions g_i defining K are all affine. The first noteworthy point about this case is that the Euclidean projection onto K can be accomplished by solving a convex quadratic program; consequently, the calculation of the D-gap function and its gradient at a given point can be carried out fairly efficiently.

To prepare for the presentation of the improved results we first establish several preliminary facts and introduce some more terminology. Let L be a subspace in $\mathbb{R}^n$ with $\dim L \geq 1$. We may decompose every vector $x \in \mathbb{R}^n$ in a unique way as $x = x_L + x_{L^\perp}$, where x_L and $x_{L^\perp}$ are, respectively, the Euclidean projection of x on L and $L^\perp$. We say that a matrix $A \in \mathbb{R}^{n\times n}$ is nonsingular on a certain subspace L if

$$0 \neq x \in L \;\Rightarrow\; (Ax)_L \neq 0. \tag{10.4.19}$$

Let P_L be the $n \times n$ matrix representing the Euclidean projector onto the subspace L; by Exercise **1.8.14**, P_L is symmetric positive semidefinite and satisfies $P^2 = P$. If the subspace L is represented as the solution set of a homogeneous system of linear equations, say,

$$L \equiv \{\, x \in \mathbb{R}^n \,:\, Dx = 0 \,\},$$

for some $p \times n$ matrix D with full row rank, we have

$$P_L = \mathbf{I} - D^T (DD^T)^{-1} D,$$

while

$$P_{L^\perp} = \mathbf{I} - P_L = D^T(DD^T)^{-1}D.$$

The nonsingularity of A on L is equivalent to

$$P_L v \neq 0 \Rightarrow P_L A P_L v \neq 0.$$

Alternatively, if Z is an $n \times k$ matrix whose columns form an orthonormal basis of L, then A is nonsingular on L if and only if the $k \times k$ matrix $Z^T AZ$ is nonsingular.

If x^* is a solution of the VI (K, F), then $\Pi_K(x^* - a^{-1}F(x^*)) = x^*$ for all scalars $a > 0$. Thus the family $\mathcal{B}'(x^* - a^{-1}F(x^*))$ of index sets is clearly independent of a. We denote this common family of index sets by $\mathcal{B}'_*$, omitting the argument $x^* - a^{-1}F(x^*)$. Based on this observation, the following result is easy to prove.

10.4.16 Lemma. Let K be a polyhedron given by

$$K = \{x \in \mathbb{R}^n : Bx \leq d\},$$

for some $m \times n$ matrix B and m-vector d. Let F be continuously differentiable around a solution x^* of the (linearly constrained) VI (K, F). The family of matrices $\mathcal{A}_\pi(x^* - a^{-1}F(x^*))$ is independent of a; indeed,

$$\mathcal{A}_\pi(x^* - a^{-1}F(x^*)) = \{\mathbf{I} - (B_{\mathcal{J}.})^T(B_{\mathcal{J}.}(B_{\mathcal{J}.})^T)^{-1}B_{\mathcal{J}.} : \mathcal{J} \in \mathcal{B}'_*\}.$$

Proof. This follows easily from the remark made above and the polyhedrality of K, which implies that the matrix $G_{\mathcal{J}}(x^*)$ is equal to the identity for all index sets $\mathcal{J}$ in the family $\mathcal{B}'_*$. □

Besides its computational simplifications, the polyhedrality of the set K offers a major theoretical benefit; namely, the nonsingularity of the linear Newton approximation scheme $\mathcal{A}_a^{\mathrm{nat}}(x^*)$ of the function $\mathbf{F}_a^{\mathrm{nat}}$ can be established under a nonsingularity condition of the Jacobian matrix $JF(x^*)$ on certain subspaces. The latter nonsingularity condition can be contrasted to the positive definiteness condition (10.4.10), which is needed for a non-polyhedral set K.

10.4.17 Proposition. Let K be a polyhedron given by

$$K = \{x \in \mathbb{R}^n : Bx \leq d\},$$

for some $m \times n$ matrix B and m-vector d. Let F be continuously differentiable around a solution x^* of the (linearly constrained) VI (K, F). Assume that $JF(x^*)$ is nonsingular on the subspace

$$L_{\mathcal{J}} \equiv \{x \in \mathbb{R}^n : B_{\mathcal{J}.}x = 0\}$$

for every $\mathcal{J} \in \mathcal{B}'_*$. The following two statements hold.

(a) For every $a > 0$, every matrix in the linear Newton approximation $\mathcal{A}^{\rm nat}_a(x^*)$ is nonsingular.

(b) x^* is a semistable solution of the VI (K, F); consequently, $v^T H v > 0$ for all $v \neq 0$ and $H \in \partial^2_v \theta_{ab}(x^*)$.

Proof. By Lemma **10.4.16**, every matrix in $\mathcal{A}^{\rm nat}_a(x^*)$ is of the form: for some index set $\mathcal{J}$ in $\mathcal{B}'_*$,

$$\begin{aligned} \mathbf{I} - P_{L_\mathcal{J}} \left(\mathbf{I} - a^{-1} JF(x^*) \right) &= \mathbf{I} - P_{L_\mathcal{J}} + a^{-1} P_{L_\mathcal{J}} JF(x^*) \\ &= P_{L^\perp_\mathcal{J}} + a^{-1} P_{L_\mathcal{J}} JF(x^*) \end{aligned}$$

Suppose a vector $v \neq 0$ is such that

$$[\, \mathbf{I} - P_{L_\mathcal{J}} \left(\mathbf{I} - a^{-1} JF(x^*) \right)] v \, = \, 0.$$

With $y \equiv P_{L^\perp_\mathcal{J}} v$ and $z \equiv P_{L_\mathcal{J}} JF(x^*) v$, we have $y + a^{-1} z = 0$. Since $P_{L_\mathcal{J}} P_{L^\perp_\mathcal{J}} = 0$, it follows that $y^T z = 0$. Hence $y = z = 0$. Therefore $v \in L_\mathcal{J}$; the nonsingularity of $JF(x^*)$ on $L_\mathcal{J}$ implies $P_{L_\mathcal{J}} JF(x^*) v \neq 0$, which is a contradiction. Consequently, every matrix in $\mathcal{A}^{\rm nat}_a(x^*)$ must be nonsingular. Hence (a) follows. This implies that $\mathcal{A}^{\rm nat}_a(x^*)$ is a nonsingular linear Newton approximation of $\mathbf{F}^{\rm nat}_a$ at x^*. By Theorem **7.2.10**, it follows that there exist a neighborhood $\mathcal{N}$ of x^* and a constant $\eta > 0$ such that

$$\| \, x - x^* \, \| \, \leq \, \eta \, \| \, x - \Pi_K (x - a^{-1} F(x) \, \|, \quad \forall \, x \, \in \, \mathcal{N}.$$

In turn, by Proposition **10.3.6**, $\|x - \Pi_K(x - a^{-1}F(x))\|$ and $\|\mathbf{F}^{\rm nat}_K(x)\|$ are equivalent; hence, for some constant $\eta' > 0$,

$$\| \, x - x^* \, \| \, \leq \, \eta' \, \| \, \mathbf{F}^{\rm nat}_K(x) \, \|, \quad \forall \, x \, \in \, \mathcal{N}.$$

By Proposition **5.3.7**, x^* is a semistable solution of the VI (K, F). The last assertion of the proposition follows from Proposition **10.3.10**. □

Proposition **10.4.17** prompts us to consider a simple modification of Algorithm **10.4.8**, valid for problems with linear constraints, in which the fast direction is calculated based on the Newton approximation $\mathcal{A}^{\rm nat}_a$.

D-gap Line Search Algorithm IV (DGLSA IV)

10.4.18 Algorithm. The steps of the algorithm are the same as those of Algorithm **10.4.8** except for Step 3, where the matrix H^k is taken to be an element of $\mathcal{A}^{\rm nat}_a(x^k)$.

The convergence of Algorithm **10.4.18** is exactly the same as that of Algorithm **10.4.18**, except for a simple change in part (d) of Theorem **10.4.9**. Namely, instead of the positive definiteness condition (10.4.10), it suffices to assume that $JF(x^*)$ is nonsingular on the linear subspace $L_{\mathcal{J}}$ for every $\mathcal{J} \in \mathcal{B}'_* \equiv \mathcal{B}'(x^* - F(x^*))$. Under this much simpler assumption, it can be shown that eventually d^k is always the solution arising from the solution of the Newton system $H^k d = -\nabla\theta_{ab}(x^k)$ and the other conclusions of Theorem **10.4.9** remain valid for Algorithm **10.4.18**. The details are omitted.

The linearly constrained VI is a special case of the general VI for which the D-gap function can be effectively computed by a finite algorithm, and Algorithm **10.4.18** is the only algorithm considered up to now for solving the general VI that is fully implementable in practice. This algorithm can be considered an extension of Algorithm **9.2.3** for the NCP in that both algorithms combine a smooth merit function with a local method based on a nonsmooth reformulation of the respective problem, which is different from the one naturally associated with the merit function. More specifically, for the linearly constrained VI, the merit function is θ_{ab} and the local method is based on $\mathbf{F}_a^{\text{nat}}$; whereas for the NCP, the merit function is θ_{FB} and the local method is based on $\mathbf{F}_{\min}$. As we saw in Example **10.4.5**, the local Newton method in the case of the NCP reduces to the local method considered in Algorithm **9.2.3**. The convergence of Algorithm **10.4.18** specialized to the NCP requires assumptions that are comparable to those needed for Algorithm **9.2.3**.

10.4.3 The case of a bounded K

In this subsection, we consider the VI (K, F), where K is a compact convex set. We present an algorithm, which generates a bounded sequence of iterates each of whose limit points is a solution of the VI. The basic idea of the algorithm is as follows. Use any descent algorithm to minimize the D-gap function θ_{ab} for given values of a and b. At each iteration of the minimization algorithm, we compare the norm of the gradient $\|\nabla\theta_{ab}(x)\|$ with the value $\theta_{ab}(x)$. More specifically, we compare the two quantities

$$(\, b - a \,) \,\|\, \nabla\theta_{ab}(x) \,\| \quad \text{and} \quad c\,\theta_{ab}(x),$$

where $c > 0$ is a given constant. If at any iteration, the inequality

$$(\, b - a \,) \,\|\, \nabla\theta_{ab}(x) \,\| \;\geq\; c\,\theta_{ab}(x), \tag{10.4.20}$$

is violated, then we terminate the minimization of the function θ_{ab} and adjust the scalars a and b. We then resume the minimization of θ_{ab} using

the adjusted values of a and b. The benefit of maintaining the inequality (10.4.20) is obvious. Namely, if $\{x^k\}$ is a sequence of iterates satisfying (10.4.20) for a fixed pair a and b, then

$$\lim_{k\to\infty} \nabla\theta_{ab}(x^k) = 0 \;\Rightarrow\; \lim_{k\to\infty} \theta_{ab}(x^k) = 0.$$

Consequently, if we could drive the gradient of the D-gap function to zero, then the D-gap function itself would be driven to zero too. If (10.4.20) is violated, we adjust a and b and repeat the process. The algorithm below implements this strategy for solving the VI (K, F).

A D-gap Algorithm for a VI with a Bounded Set (DGVIB)

10.4.19 Algorithm.

Data: $x^0 \in \mathbb{R}^n$, $0 \le a_0 < b_0$, $\rho > 0$, $c > 0$, and a sequence $\{\eta_k\}$ of positive numbers.

Step 1: Set $r = 0$, $k = 0$, and $x_r^k = x^0$.

Step 2: If x^k is a solution of VI (K, F), stop.

Step 3: Apply an iterative descent method $\mathcal{M}$ to the minimization of $\theta_{a_k b_k}$ starting with x_0^k, generating a sequence $\{x_r^k\}$. If at any step r_k of method $\mathcal{M}$,

$$(\, b_k - a_k \,) \,\|\, \nabla\theta_{a_k b_k}(x_{r_k}^k) \,\| \;\le\; c\, \theta_{a_k b_k}(x_{r_k}^k), \tag{10.4.21}$$

stop the application of algorithm $\mathcal{M}$. Select two values a_{k+1} and b_{k+1} such that $0 < a_{k+1} \le \frac{1}{2} a_k$, $b_{k+1} \ge 2b_k$, and

$$\frac{\theta_{a_{k+1} b_{k+1}}(x_{r_k}^k)}{b_{k+1} - a_{k+1}} \;\le\; \frac{\theta_{a_k b_k}(x^k)}{b_k - a_k} + \eta_k. \tag{10.4.22}$$

Go to Step 4. If (10.4.21) is not satisfied during the complete execution of the method $\mathcal{M}$ for the minimization of $\theta_{b_k a_k}$, stop the algorithm at the termination of method $\mathcal{M}$, which presumably is determined by a prescribed termination rule.

Step 4: Set $x^{k+1} \equiv x_{r_k}^k \equiv x_0^{k+1}$, $a_{k+1} \equiv \frac{1}{2}\alpha_k$, $b_{k+1} \equiv 2b_k$, $k \leftarrow k+1$, and $r \leftarrow 0$; go to Step 2.

Observe that either the pair (a_k, b_k) is updated at every iteration, which implies

$$\|\, \nabla\theta_{a_k b_k}(x^{k+1}) \,\| \;\le\; \frac{c}{b_k - a_k}\, \theta_{a_k b_k}(x^{k+1}), \quad \forall\, k,$$

or the algorithm becomes the method $\mathcal{M}$ applied to the minimization of $\theta_{a_{\bar{k}}b_{\bar{k}}}$ for some finite iteration index $\bar{k}$. The following lemma shows that it is always possible to find suitable a_{k+1} and b_{k+1} satisfying (10.4.22) at Step 3. This is not an obvious fact, since both the numerator and the denominator on the left-hand side of (10.4.22) increase as a_{k+1} decreases and b_{k+1} increases.

10.4.20 Lemma. Let K be a compact convex subset of $\mathbb{R}^n$. For arbitrary positive scalars $\bar{a} < \bar{b}$ and an arbitrary vector $x \in \mathbb{R}^n$,

$$\limsup_{\substack{a\downarrow 0\\ b\to\infty}} \frac{\theta_{ab}(x)}{b-a} \le \frac{\theta_{\bar{a}\bar{b}}(x)}{\bar{b}-\bar{a}}.$$

Consequently, for every scalar $\eta > 0$, there exist positive scalars a and b satisfying $b \ge 2\bar{b}$ and $a \le \frac{1}{2}\bar{a}$ such that

$$\frac{\theta_{ab}(x)}{b-a} \le \frac{\theta_{\bar{a}\bar{b}}(x)}{\bar{b}-\bar{a}} + \eta.$$

Proof. Let $\bar{x} \equiv \Pi_K(x)$. By the definition of the D-gap function, we have, for all positive scalars a and b,

$$\frac{\theta_{ab}(x)}{b-a} = \frac{2\,F(x)^T(\,y_b(x)-y_a(x)\,) - a\,\|\,x-y_a(x)\,\|^2 + b\,\|\,x-y_b(x)\,\|^2}{2\,(\,b-a\,)}.$$

Since $y_b(x)$ and $y_a(x)$ are vectors in K, which is a bounded set, we see that

$$\limsup_{\substack{a\downarrow 0\\ b\to\infty}} \frac{\theta_{ab}(x)}{b-a} = \tfrac{1}{2}\,\|\,x-\bar{x}\,\|^2 \le \frac{\theta_{\bar{a}\bar{b}}(x)}{\bar{b}-\bar{a}}.$$

where the last inequality follows from (10.3.14). The last assertion of the lemma is obvious. □

The above lemma implies that for any given pair (a_k, b_k), one can find a_{k+1} and b_{k+1} such that $0 < a_{k+1} \le \frac{1}{2}a_k$, $b_{k+1} \ge 2b_k$, and

$$\frac{\theta_{a_{k+1}b_{k+1}}(x^k_{r_k})}{b_{k+1}-a_{k+1}} \le \frac{\theta_{a_kb_k}(x^k_{r_k})}{b_k-a_k} + \eta_k.$$

Since $\theta_{a_kb_k}(x^k_{r_k}) < \theta_{a_kb_k}(x^k)$, we obtain the deisred inequality (10.4.22).

The next lemma is frequently used in the convergence proof of inexact methods.

10.4.21 Lemma. Let $\{\delta_k\}$ and $\{\eta_k\}$ be two sequences of numbers such that $\delta_k \ge 0$ for all k,

$$\delta_{k+1} \le \delta_k + \eta_k \quad \forall\, k$$

and

$$\sum_{k=0}^{\infty} \eta_k \ < \ \infty. \tag{10.4.23}$$

The sequence $\{\delta_k\}$ converges.

Proof. For any two positive integers $k > s$, we have

$$\delta_k \ \leq \ \delta_s + \bar{\eta}_k, \tag{10.4.24}$$

where

$$\bar{\eta}_k \ \equiv \ \sum_{i=k-1}^{\infty} \eta_i.$$

By (10.4.23), it follows that the nonnegative sequence $\{\delta_k\}$ is bounded. Suppose for the sake of contradiction that two subsequences $\{\delta_k : k \in \kappa\}$ and $\{\delta_k : k \in \kappa'\}$ converge to two distinct limits: δ_∞ and δ'_∞, respectively. Note that $\{\bar{\eta}_k\}$ is a sequence converging to zero. By letting $s \in \kappa'$ and $s < k \in \kappa$, we deduce from (10.4.24) that $\delta_\infty \leq \delta'_\infty$. Reversing the role of k and s, we obtain the other inequality $\delta'_\infty \leq \delta_\infty$. Thus equality holds and the lemma follows. □

The following is the main convergence result of Algorithm **10.4.19** for solving a VI with a compact convex set.

10.4.22 Theorem. Let $K \subset \mathbb{R}^n$ be compact convex and $F : \mathbb{R}^n \to \mathbb{R}^n$ be continuously differentiable. Suppose that the method $\mathcal{M}$ applied to the minimization of $\theta_{a_k b_k}$ is a descent algorithm such that every limit point it generates is a stationary point of $\theta_{a_k b_k}$. Assume also that $\{\eta_k\}$ satisfies (10.4.23).

(a) If a_k and b_k are updated only a finite number of times, and $\bar{k} - 1$ is the last (outer) iteration where these scalars are updated, then the sequence $\{x_r^{\bar{k}}\}$ is bounded and and every limit point of the latter sequence is a solution of VI (K, F).

(b) If a_k and b_k are updated an infinite number of times, then the sequence $\{x^k\}$ is bounded, and every limit point x^* of this sequence for which $JF(x^*)$ is positive semidefinite is a solution of VI (K, F).

Proof. In case (a) the algorithm reduces to the application of the method $\mathcal{M}$ to the minimization of $\theta_{a_{\bar{k}} b_{\bar{k}}}$, which generates the sequence $\{x_r^{\bar{k}}\}$. Since K is bounded, $\theta_{a_{\bar{k}} b_{\bar{k}}}$ is coercive by Proposition **10.3.9**; by the descent property of $\mathcal{M}$, the sequence $\{x_r^{\bar{k}}\}$ is bounded. Furthermore, by the assumed stationarity property of $\mathcal{M}$, it follows that

$$\lim_{r \to \infty} \nabla \theta_{a_{\bar{k}} b_{\bar{k}}}(x_r^{\bar{k}}) \ = \ 0.$$

Since, for every iteration r,

$$(\, b_{\bar{k}} - a_{\bar{k}} \,) \,\|\, \nabla\theta_{a_{\bar{k}} b_{\bar{k}}}(x_r^{\bar{k}}) \,\|\, > \, c\, \theta_{a_{\bar{k}} b_{\bar{k}}}(x_r^{\bar{k}}),$$

we deduce

$$\lim_{r\to\infty} \theta_{a_{\bar{k}} b_{\bar{k}}}(x_r^{\bar{k}}) \, = \, 0.$$

This establishes (a).

To prove (b), assume that $\{a_k\}$ and $\{b_k\}$ are updated infinitely many times. Let

$$\delta_s \, \equiv \, \frac{\theta_{a_s b_s}(x^s)}{(b_s - a_s)}.$$

The inequality (10.4.22) and the descent property of the method $\mathcal{M}$ implies

$$\delta_{k+1} \, \leq \, \delta_k + \eta_k.$$

By Lemma **10.4.21**, the sequence $\{\delta_k\}$ converges. By Proposition **10.3.7** we have

$$\|\, x^k - y_{b_k}(x^k) \,\| \, \leq \, \sqrt{2\,\delta_k}, \tag{10.4.25}$$

this, together with the boundedness of K, implies that $\{x^k\}$ is bounded. Let δ_∞ be the limit of $\{\delta_k\}$. We claim that $\delta_\infty = 0$. Suppose the contrary; then for all k sufficiently large, $\delta_k \geq \delta_\infty/2 > 0$ or, equivalently,

$$\delta_\infty \, \leq \, \frac{2\, F(x^k)^{\,T}(y_{b_k}(x^k) - y_{a_k}(x^k)) + b_k \|x^k - y_{b_k}(x^k)\|^2 - a_k \|x^k - y_{a_k}(x^k)\|^2}{b_k - a_k}.$$

The boundedness of $\{x^k\}$ and the fact that $\{b_k\} \to \infty$ and $\{a_k\} \to 0$ give

$$\liminf_{k\to\infty} \, \|\, x^k - y_{b_k}(x^k) \| \, \geq \, \delta_\infty \, > \, 0.$$

Hence

$$\lim_{k\to\infty} \, b_k \,\|\, x^k - y_{b_k}(x^k) \,\| \, = \, \infty. \tag{10.4.26}$$

Since

$$\nabla\theta_{a_k b_k}(x^k) \, = \, JF(x^k)^{\,T}(\, y_{b_k}(x^k) - y_{a_k}(x^k) \,) + a_k \,(\, y_{a_k}(x^k) - x^k \,) - b_k \,(\, y_{b_k}(x^k) - x^k \,),$$

it follows that

$$\lim_{k\to\infty} \, \|\, \nabla\theta_{a_k b_k}(x^k) \,\| \, = \, \infty,$$

thus contradicting (10.4.21) and the boundedness of $\{x^k\}$. Hence $\delta_\infty = 0$. Consequently we deduce from (10.4.25) and the boundedness of $\{x^k\}$ that every limit point of $\{x^k\}$ belongs to K.

Let x^∞ be the limit of the subsequence $\{x^k : k \in \kappa\}$ such that $JF(x^\infty)$ is positive definite. Suppose for the moment that

$$\lim_{k(\in\kappa)\to\infty} \theta_{a_k}(x^k) \;=\; 0. \tag{10.4.27}$$

By the definition of θ_{a_k}, we can write for every $x \in K$,

$$\theta_{a_k}(x^k) \;\geq\; F(x^k)^T(\,x^k - x\,) - \frac{a_k}{2}\,\|\,x^k - x\,\|^2.$$

Passing to the limit $k(\in \kappa) \to \infty$ and using (10.4.27) and the fact that $\{a_k\} \to 0$, we obtain

$$F(x^\infty)^T(\,x^\infty - x\,) \;\leq\; 0,$$

thus showing that x^∞ is a solution of VI (K, F). To conclude the proof it therefore remains to show that (10.4.27) holds.

Since $\delta_\infty = 0$, (10.4.25) implies

$$\lim_{k\to\infty} \|\,x^k - y_{b_k}(x^k)\,\| \;=\; 0;$$

in turn, since

$$\theta_{a_k b_k}(x^k) \;=$$
$$F(x^k)^T(y_{b_k}(x^k) - y_{a_k}(x^k)) - \frac{a_k}{2}\,\|y_{a_k}(x^k) - x^k\|^2 + \frac{b_k}{2}\,\|y_{b_k}(x^k) - x^k\|^2,$$

it follows that

$$\lim_{k\to\infty} \frac{\theta_{a_k b_k}(x^k)}{b_k - a_k} \;=\; 0$$

because $\{x^k\}$, $\{y_{a_k}(x^k)\}$ and $\{y_{b_k}(x^k)\}$ are bounded sequences, $\{a_k\} \to 0$, and $\{b_k\} \to \infty$. Furthermore, since

$$\|\,\nabla\theta_{a_k b_k}(x^{k+1})\,\| \;\leq\; \frac{c}{b_k - a_k}\,\theta_{a_k b_k}(x^{k+1}), \quad \forall\, k,$$

we deduce

$$\lim_{k\to\infty} \nabla\theta_{a_k b_k}(x^k) \;=\; 0. \tag{10.4.28}$$

For every $x \in \mathbb{R}^n$ and every pair of positive scalars $b > a$, by the variational characterization of projection $\Pi_K(x - a^{-1}F(x))$, we have, since $y_b(x) \in K$,

$$0 \;\leq\; F(x)^T(\,y_b(x) - y_a(x)\,) + a\,(\,y_a(x) - x\,)^T(\,y_b(x) - y_a(x)\,).$$

Analogously, inverting the role of $y_a(x)$ and $y_b(x)$ we also get

$$0 \;\leq\; F(x)^T(\,y_a(x) - y_b(x)\,) + b\,(\,y_b(x) - x\,)^T(\,y_a(x) - y_b(x)\,).$$

Adding the last two inequalities, we obtain

$$[\, b\,(\, x - y_b(x)\,) - a\,(\, x - y_a(x)\,)\,]^T(\, y_b(x) - y_a(x)\,) \;\geq\; 0.$$

Therefore,

$$\begin{aligned}
&\nabla\theta_{ab}(x)^T(\, y_b(x) - y_a(x)\,) \\
&\quad = (\, y_b(x) - y_a(x)\,)^T[\, JF(x)^T(\, y_b(x) - y_a(x)\,) \\
&\qquad + b\,(\, x - y_b(x)\,) - a\,(\, x - y_a(x)\,)\,] \\
&\quad \geq (\, y_b(x) - y_a(x)\,)^T JF(x)(\, y_b(x) - y_a(x)\,).
\end{aligned}$$

By (10.4.28), it follows that

$$\lim_{k(\in\kappa)\to\infty} (\, y_{b_k}(x^k) - y_{a_k}(x^k)\,) \;=\; 0.$$

By (10.4.26), we deduce

$$\lim_{k(\in\kappa)\to\infty} y_{a_k}(x^k) \;=\; \lim_{k(\in\kappa)\to\infty} x^k \;=\; x^\infty.$$

Since

$$\theta_{a_k}(x^k) \;=\; F(x^k)^T(\, x^k - y_{a_k}(x^k)\,) - \frac{a_k}{2}\,\|\, x^k - y_{a_k}(x^k)\,\|^2,$$

(10.4.27) follows readily. □

10.4.4 Algorithms based on θ_c

In this subsection we briefly discuss some algorithms for solving the VI (K, F) based on the constrained minimization reformulation:

$$\begin{array}{ll} \text{minimize} & \theta_c(x) \\ \text{subject to} & x \in K, \end{array} \tag{10.4.29}$$

with the understanding that we are interested in feasible methods. The motivations to consider this approach are the same as those discussed in Subsection **9.1.8**. However, this constrained reformulation is less attractive than the corresponding one for complementarity problems since feasible, constrained methods for the resolution of (10.4.29) are more complex when K is a general convex set than when K is the nonnegative orthant.

Similar to what is done in Algorithm **9.1.39**, we can globalize a a fast local algorithm, for example, the Newton algorithm **7.3.1**, by combining it with a standard first-order feasible method, such as the gradient projection or the conditional gradient method. Since we already encountered several

concrete examples of this approach we take this time a slightly more general point of view. To this end, let us denote by $\mathcal{M} : K \to K$ a feasible algorithm for the minimization of θ_c. By this we mean that starting with any feasible point $x^0 \in K$, the sequence $\{x^k\}$ obtained by setting $x^{k+1} = \mathcal{M}(x^k)$ is feasible and each of its limit points is a stationary point of θ_c. We further assume that if x^* is an isolated limit point of the sequence $\{x^k\}$, then the whole sequence converges to x^*. This latter property is satisfied by practically all feasible, first-order methods. We recall that as usual VI (K, F^k) denotes the semi-linearized VI with

$$F^k(x) \equiv F(x^k) + JF(x^k)(x - x^k).$$

The numerical solution of the latter VI is the workhorse in the following algorithm.

Constrained Regularized Gap Algorithm (CRGA)

10.4.23 Algorithm.

Data: $x^0 \in K$, $\rho > 0$, $p > 1$, and $\gamma \in (0, 1)$.

Step 1: Set $k = 0$.

Step 2: If x^k is a stationary point of problem (10.4.29), stop.

Step 3: Find a solution $x^{k+1/2}$ of the semi-linearized sub-VI (K, F^k) and set

$$d^k \equiv x^{k+1/2} - x^k.$$

If the sub-VI is not solvable or if the condition

$$\nabla\theta_c(x^k)^T d^k \leq -\rho \| d^k \|^p$$

is not satisfied, set $x^{k+1} \equiv \mathcal{M}(x^k)$ and go to Step 5.

Step 4: Find the smallest nonnegative integer i_k such that with $i = i_k$,

$$\theta_c(x^k + 2^{-i}d^k) \leq \theta_c(x^k) - \min\left\{ -2^{-i}\gamma\nabla\theta_c(x^k)^T d^k,\ (1-\gamma)\,\theta_c(x^k) \right\}.$$

Set $\tau_k \equiv 2^{-i_k}$ and $x^{k+1} \equiv x^k + \tau_k d^k$.

Step 5: Set $k \leftarrow k + 1$ and go to Step 2.

Note also the parallelism with Algorithm **10.4.14**, which uses the same strategy but with reference to the unconstrained minimization of θ_{ab} and

with a concrete choice for $\mathcal{M}$. The following result is rather easy to prove along the lines of Theorem **10.4.15**. The reader is asked to supply the details in Exercise **10.5.14**.

10.4.24 Theorem. Let K be a closed convex set in $\mathbb{R}^n$. Let F be continuously differentiable. Let $c > 0$ be a given scalar. Let $\{x^k\}$ be an infinite sequence generated by Algorithm **10.4.23**.

(a) Every accumulation point of $\{x^k\}$ is a constrained stationary point of the merit function θ_c.

(b) If x^* is an accumulation point of $\{x^k\}$ such that (10.2.7) holds, then x^* is a solution of the VI (K, F).

(c) If $\{x^k\}$ has an isolated limit point, then the whole sequence $\{x^k\}$ converges to that point.

(d) Suppose that x^* is a limit point of $\{x^k\}$ that is a stable solution of the VI (K, F). The whole sequence $\{x^k\}$ converges to x^*; furthermore, if the scalars $p > 2$ and $\gamma < 1/2$ in Algorithm **10.4.23**, the following statements hold:

  (i) eventually d^k is always given by $x^{k+1/2} - x^k$;

  (ii) eventually a unit step size is accepted so that $x^{k+1} = x^k + d^k$;

  (iii) the convergence rate is Q-superlinear; furthermore, if the Jacobian $JF(x)$ is Lipschitz continuous in a neighborhood of x^*, the convergence rate is Q-quadratic. □

If one wants to avoid the solution of the semi-linearized sub-VI (K, F^k) in Algorithm **10.4.23**, one can use the linear Newton approximation $\mathcal{A}_c(x)$ introduced in Proposition **10.4.4**. In this case we must also assume that the set K is defined by a finite set of convex inequalities, as in (10.4.1). We may introduce an analog of Algorithm **9.1.42**. Given the iterate $x^k \in K$, we choose a matrix $H^k \in \mathcal{A}_c((x^k)$ and solve the convex quadratic program

$$\begin{array}{ll} \text{minimize} & \nabla\theta_c(x^k)^T d + \frac{1}{2}\, d^T [\,(H^k)^T H^k + \rho(\theta_c(x^k))\mathbf{I}\,] d \\ \text{subject to} & x^k + d \,\in\, K, \end{array} \tag{10.4.30}$$

in order to calculate a search direction d^k along which we perform a line search to obtain x^{k+1}. To ensure a superlinear rate when the sequence $\{x^k\}$ so generated converges to solution of VI (K, F) where the CRCQ holds and the Jacobian of F is positive definite, one has to assume that the value of c is sufficiently small. All the strategies we have illustrated in the case of the D-gap function to adjust the parameters a and b can be easily be adopted

to the adjustment of c. The same holds for the enhancements introduced in the case in which the set K is polyhedral or compact. We leave these issues to the reader.

There exist feasible algorithms for problem (10.4.29) that only require the solution of systems of linear equations. Some recent extensions also indicate the possibility to make these algorithms superlinearly convergent in the case of functions that are not twice continuously differentiable. Given the great technical complexity of these methods and the fact that their detailed exposition would not add much to the comprehension of the problems we are studying, we prefer to direct the reader to the comments and references in Section **10.6**.

10.5 Exercises

10.5.1 Consider the optimization problem

$$\begin{array}{ll} \text{minimize} & \theta(x) \\ \text{subject to} & g(x) \in -K, \end{array}$$

where the objective function $\theta : \mathbb{R}^n \to \mathbb{R}$ and $g : \mathbb{R}^n \to \mathbb{R}^m$ are continuously differentiable and $K \subseteq \mathbb{R}^m$ is a closed convex cone. This is a generalization of the classical NLP, where $K = \mathbb{R}^{m_1}_+ \times \mathbb{R}^{m_2}$. It is known that under appropriate constraint qualifications (see Exercise **3.7.8**) a necessary condition for a point x to be a solution of this problem is that there exist a multiplier $\lambda \in \mathbb{R}^m$ such that the following generalized KKT system is satisfied:

$$\begin{aligned} \nabla\theta(x) + Jg(x)^T\lambda &= 0 \\ -K \ni g(x) \perp \lambda &\in K^*. \end{aligned}$$

Show that the three relations in the second line of the above generalized KKT system are equivalent to either of the following equations:

$$\begin{aligned} \Pi_{K^*}(g(x) + \lambda) &= \lambda, \\ \Pi_{-K}(g(x) + \lambda) &= g(x). \end{aligned}$$

This shows that the generalized KKT system can be reformulated as a system of equations. (Hint: use the known fact that any element $y \in \mathbb{R}^m$ can be uniquely decomposed into the sum of two orthogonal elements y^1 and y^2 belonging to $-K$ and K^* and that $y^1 = \Pi_{-K}(y)$ and $y^2 = \Pi_{K^*}(y)$.)

10.5.2 Consider the following minimization problem:

$$\begin{aligned} \text{minimize} \quad & x_1^2 + x_2^2 + 4x_1x_2 \\ \text{subject to} \quad & x_1 \geq 0, \\ & x_2 \geq 0. \end{aligned}$$

Check that its global minimizer is $x^* = (0, 0)$ and find the corresponding Lagrange multipliers λ^*. Show that the point (x^*, λ^*) is quasi-regular but not strongly stable. Verify that all the matrices in $\operatorname{Jac} \Phi_{\min}(x^*, \lambda^*)$ are nonsingular while $\partial \Phi_{\mathrm{FB}}(x^*, \lambda^*)$ contains singular elements

10.5.3 Recall the dual gap function defined for a pseudo monotone VI (K, F), where F is pseudo monotone on K; see (2.3.13):

$$\theta_{\mathrm{dual}}(x) \equiv \inf_{y \in K} F(y)^T(y - x), \quad x \in \mathbb{R}^n.$$

Unlike the primal gap function θ, it is not easy to regularize θ_{dual}. Nevertheless if K is a compact convex set, the following properties of θ_{dual} hold:

(a) θ_{dual} is everywhere finite, concave, and B-differentiable on $\mathbb{R}^n$;

(b) if F is pseudo monotone plus on K and $\mathrm{SOL}(K, F) \neq \emptyset$, then θ_{dual} is G-differentiable at every vector $x^* \in \mathrm{SOL}(K, F)$; moreover $\nabla \theta_{\mathrm{dual}}$ is a constant on $\mathrm{SOL}(K, F)$.

10.5.4 Verify the identity (10.5.4).

10.5.5 Prove Proposition **10.3.13**.

10.5.6 This exercise contrasts the implicit Lagrangian function $\theta_{ab}^{\mathrm{ncp}}$ with the FB merit function θ_{FB} for the NCP. Let $F : \mathbb{R}^n \to \mathbb{R}^n$ be continuously differentiable and let $b > a > 0$ be given scalars. Suppose that $JF(x)$ is a row sufficient matrix. Show that if x is an unconstrained stationary point of $\theta_{ab}^{\mathrm{ncp}}$ but not a solution of the NCP (F), then $JF(x)_{\bar{\mathcal{C}}\bar{\mathcal{C}}}$ is singular, where $\bar{\mathcal{C}} \equiv \mathcal{N} \cup \mathcal{P}$ and $\mathcal{N}$ and $\mathcal{P}$ are defined at x with respect to $\theta_{ab}^{\mathrm{ncp}}$.

Since a row sufficient matrix must be P_0, it follows from Proposition **9.1.17** that $JF(x)$ must be a signed S_0 matrix and thus x must be a FB regular point. Consequently, if x is an unconstrained stationary point of θ_{FB}, then x must be a solution of the NCP (F).

10.5.7 Let $F : \mathbb{R}^n \to \mathbb{R}^n$ be continuously differentiable and let $b > a > 0$ be given scalars. Suppose that x is a constrained stationary point of $\theta_{ab}^{\mathrm{ncp}}$ on the nonnegative orthant. Show that x is a solution of the NCP (F) if and only if x satisfies the regularity property in Theorem **10.3.15**.

10.5.8 Let $F : K \to \mathbb{R}^n$ be continuous and strongly monotone on the closed convex set $K \subseteq \mathbb{R}^n$. Recall the generalized gap function $\tilde{\theta}_{\rm gap}(x)$ from Section **2.10** and Exercise **2.9.22**. Assume that the function f in the definition of $\tilde{\theta}_{\rm gap}(x)$ has a Lipschitz continuous gradient. Show that there exists a constant η such that

$$\| x - x^* \| \leq \eta \sqrt{\tilde{\theta}_{\rm gap}(x)}, \quad \forall\, x \in K,$$

where x^* is the unique solution of the VI (K, F).

10.5.9 Let K be a closed convex subset of $\mathbb{R}^n$ and let $F(x) \equiv Mx + q$ with M positive definite. Show that if c is sufficiently small, the function θ_c is convex on K.

10.5.10 Let K be a bounded polyhedron and denote by V the set of its vertices. Consider the VI (K, F), where $F : K \to \mathbb{R}^n$. We can define a nonnegative function in the following way:

$$\theta_{\rm S}(x) \equiv \sum_{i \in V} [\, (\, F(x)^T (x - x^i)\,)_+ \,]^p, \quad \forall\, x \in K,$$

where $p \geq 1$. Prove that $\theta_{\rm S}$ is a merit function on K. Show further that if $p \geq 2$ then $\theta_{\rm S}$ is continuously differentiable if F is also continuously differentiable. (Hint: recall that all points in a bounded polyhedron can be represented as a convex combination of the vertices).

10.5.11 Let K be a closed convex subset of $\mathbb{R}^n$ and F be a continuously differentiable mapping from $\mathbb{R}^n$ into itself. Let $b > a > 0$ be given scalars. Suppose that there exists a constant $\delta > 0$ such that

$$(\, y_a(x) - y_b(x)\,)^T JF(x)(\, y_a(x) - y_b(x)\,) \geq \delta \,\| \, y_a(x) - y_b(x) \,\|^2, \quad \forall\, x \in \mathbb{R}^n.$$

Show that there exists a constant $c > 0$ satisfying

$$\mathrm{dist}(x, \mathrm{SOL}(K, F)) \leq c \sqrt{\theta_{ab}(x)}, \quad \forall\, x \in \mathbb{R}^n$$

if and only if there exists a constant $c' > 0$ such that

$$\mathrm{dist}(x, \mathrm{SOL}(K, F)) \leq c' \sqrt{\theta_{ab}(x)},$$

for all x satisfying $\|y_a(x) - y_b(x)\| < \sqrt{\theta_{ab}(x)}$.

10.5.12 Let K be a closed convex set in $\mathbb{R}^n$ and let F be a continuous mapping from $\mathbb{R}^n$ into itself.

(a) Show that if there exists $x^{\rm ref}$ in K such that

$$\lim_{\substack{x \in K \\ \|x\| \to \infty}} \frac{(\, x - x^{\rm ref}\,)^T F(x)}{\| \, x - x^{\rm ref} \, \|} \; = \; \infty,$$

then $\mathbf{F}_K^{\rm nor}$ is norm-coercive on $\mathbb{R}^n$.

(b) Show that if for all $y \in \mathbb{R}^n$,

$$\lim_{\|x\| \to \infty} \frac{(\, x - y\,)^T F(x)}{\| \, x - y \, \|} \; = \; \infty, \tag{10.5.1}$$

then $\mathbf{F}_K^{\rm nat}$ is norm-coercive on $\mathbb{R}^n$.

In contrast to (a), the sufficient condition in (b) is a kind of *strong coercivity* of F because the limit is required to hold for all vectors y in $\mathbb{R}^n$. A counterexample exists if such a limit holds only for some vectors y.

(c) Consider the function of two variables $F : \mathbb{R}^2 \to \mathbb{R}^2$ defined by

$$F(x_1, x_2) \; \equiv \; \begin{cases} \begin{pmatrix} x_2 \, (\, x_1^2 + x_2^2 \,) \\ -x_1 \, (\, x_1^2 + x_2^2 \,) + x_2 \end{pmatrix} & \text{if } x_1 \geq 0 \\ \begin{pmatrix} x_2 \, (\, x_1^2 + x_2^2 \,) + x_1^3 \\ -x_1 \, (\, x_1^2 + x_2^2 \,) - 2 x_1 x_2 + x_2 \end{pmatrix} & \text{if } x_1 < 0 \end{cases}$$

Let K be the nonnegative x_1-axis in the plane. Show that the limit (10.5.1) holds for $y = (0, 1)$ but $\mathbf{F}_K^{\rm nat}$ is not norm-coercive, by considering the unbounded sequence $\{(k, 0)\}$.

10.5.13 Let $F : \mathbb{R}^n \to \mathbb{R}^n$ be a continuous function satisfying (10.5.1) for all $y \in \mathbb{R}^n$. Let K be a closed convex set in $\mathbb{R}^n$. Suppose that the VI (K, F) has a semistable solution. Prove or disprove that there exists $\eta > 0$ such that

$$\mathrm{dist}(x, \mathrm{SOL}(K, F)) \; \leq \; \eta \, \| \, \mathbf{F}_K^{\rm nat}(x) \, \|, \quad \forall \, x \, \in \, \mathbb{R}^n.$$

10.5.14 Prove Theorem **10.4.24**.

10.6 Notes and Comments

Computing a KKT triple has long been the main aim of algorithms in nonlinear programming. It is not surprising then that equation reformulations of the KKT system have been considered extensively in the NLP community, even though the early focus was not on their numerical solution. One

of the earliest such reformulations was given by Mangasarian [380, 381], who showed that the gradient of the augmented Lagrangian of an NLP in fact formed an equation reformulation of the KKT system of the underlying NLP. The results in [382] can also be used to reformulate the KKT conditions both as smooth and nonsmooth systems of equations. Kojima [336] was the first to utilize a nonsmooth reformulation of the KKT system; Kojima's formulation has already been discussed in Section **5.7**, to which we refer the reader. Wierzbicki [607], on whose work Exercise **10.5.1** is based, gave still another nonsmooth reformulation, generalizing some of the results in [382]. Other smooth reformulations include [3, 333]. Although never stated explicitly in these terms, the gradient of any differentiable exact augmented Lagrangian constitutes an equation reformulation of the KKT condition of an NLP, provided that a penalty parameter is sufficiently small. This idea has led to the development of Newton methods. We refer to [133, 134] as entry points to the literature on this topic.

In early 1990s, fuelled by new advances in the solution of nonsmooth equations, there was a surge of interest in methods that would find a KKT triple of an NLP or of a VI by solving directly an equation reformulation of the KKT conditions. Actually, in its pure form, this approach seems more suited to VIs than to NLPs, since the KKT system of the latter problems does not distinguish a (local) minimum from a (local) maximum. The reasons for which nonsmooth reformulations were (and still are) preferred to smooth ones are similar to those already discussed in Section **9.1** for the NCP. The first contributions in this area are based on the application of Pang's B-differentiable Newton method [425] to two equation reformulations of the KKT conditions, using the min map [426] and the normal map [619, 620]. The resulting algorithms are rather complex and computationally expensive due to the nondifferentiability of the merit function used to enforce global convergence. The article [430] presented an SQP method for solving an NLP, using a merit function that involves a penalization of a nonsmooth (min) equation reformulation of the entire KKT system.

In Qi and Jiang [480] reformulations based on the FB C-function and on the min function were considered in connection with the semismooth Newton method. It was shown that if a KKT triple is strongly stable, then all the elements in the generalized Jacobian of both equation reformulations are nonsingular, thus paving the way to the development of locally convergent semismooth Newton (and quasi-Newton) methods for the KKT conditions and yielding the superlinear/quadratic convergence of the sequence of KKT iterates. An interesting result in the cited reference is that the same convergence rate also holds for the sequence of primal variables

alone; a similar result was obtained previously for an NLP in [177]. Results close to those in [480] were obtained independently by Sun and Han [559] with the use of a different linear Newton approximation from the one used by Qi and Jiang. The article [639] discussed a trust region method for solving the KKT system of a VI based on a semismooth reformulation of the system.

The presentation in Subsection **10.1.1** is largely based on the paper [170] by Facchinei, Fischer, and Kanzow, who, independently of [480], analyzed the FB-based equation reformulation of the KKT of a VI. The equivalence between the strong stability of a KKT triple and the nonsingularity of all elements in $\partial\Phi_{\mathrm{FB}}$ intuitively illustrates that strong stability is just a "form of nonsingularity". This observation was first noted by Jongen, Klatte, and Tammer [292] in terms of the Kojima map (i.e., the normal map) of the KKT system [336].

The analysis of the constrained formulation (10.1.19) is from [172]. Other constrained reformulations of the KKT conditions are studied in [13, 16, 218, 309, 444]. The analysis developed for the FB reformulation of the KKT conditions extends readily to the min function, as reported in Subsection **10.1.2**. The notion of quasi-regularity was introduced in [171] in connection with the problem of identifying active constraints.

Quasi-Newton methods for the solution of the nonsmooth reformulations of the KKT conditions were considered already in [480]. Rather standard conditions in NLP are extended in the case of a "structured" update where only the Jacobian of the Lagrangian is approximated by a quasi-Newton formula. Related material can be found in [370, 559, 638]. An interesting development was proposed recently by Izmailov and Solodov in [284]. By using the technique for the identification of active constraints described in Section **6.7** in conjunction with an error bound condition for 2-regular mappings (see [282] and Section **6.10**) Izmailov and Solodov define an active set algorithm for which a locally fast convergence rate can be guaranteed under conditions weaker than quasi-regularity.

The use of the gap function θ_{gap} in an iterative descent method for monotone VIs can be traced to the Russian school [127, 466, 658, 659]. Marcotte and Dussault [387, 390] considered monotone VIs over compact polyhedra and used the gap function θ_{gap} to monitor convergence; see also [359]. The methods described in these papers rely very much on the polyhedral structure of the feasible set. In [389], a monotone VI over a compact set is considered. At each iteration a semi-linearized VI is solved, yielding the Josephy-Newton direction along which a line search is performed to achieve a "sufficient" decrease of θ_{gap}; see Exercise **7.6.17**. Part of the

contribution of the authors of this reference is the demonstration that such a Newton direction is indeed a descent direction for the gap function of a monotone VI. The resulting algorithm is shown to be globally convergent in the sense that every limit point of the produced (bounded) sequence is a solution of the VI. Furthermore, if the problem is strongly monotone and K is a polyhedron and if an additional nondegeneracy assumption holds, the algorithm is shown to be quadratically convergent to the solution of the VI.

There are several obvious drawbacks to the use of the gap function as a merit function. First, it is restricted to a compact defining set K; thus it is not applicable to the NCP. Second, it is applicable only to monotone problems. In spite of these drawbacks, the idea of descent and its implementation persist throughout the subsequent work that follows a similar approach.

The question of whether it is possible to reformulate an asymmetric VI as a continuously differentiable optimization problem was an open issue until it was settled by Auchmuty [24] and Fukushima [220], who independently introduced the regularized gap function θ_c. The key Theorem **10.2.1** in this regard was proved by Danskin [122]; see also [25]. We have reviewed Auchmuty's approach in Section **2.10**; see in particular Exercise **2.9.22**. Our presentation in the main text of this chapter has followed the simpler route taken by Fukushima, who published an excellent survey [221] on merit functions for VIs and CPs up to the year 1996. To see the connection between the two approaches it is sufficient to note that the generalized primal gap function $\tilde{\theta}_{\rm gap}$ introduced in Section **2.10** coincides with θ_c if we take $f(x) \equiv (c/2)x^T Gx$ in the definition of $\tilde{\theta}_{\rm gap}$.

Theorem **10.2.3** is from [220]. The (necessary and) sufficient conditions in Theorem **10.2.5** and Corollary **10.2.6** for a stationary point of the regularized gap function to be a solution of the VI are new; these results are a significant extension of the special case where the Jacobian of F is positive definite at a stationary point, which is the sufficient condition originally considered in [220]. Proposition **10.2.9** extends a corresponding result in [575] where the case of a strongly monotone function was considered. Collectively, these new results along with Exercise **7.6.17** demonstrate for the first time that the gap function approach is applicable to nonmonotone VIs.

There has been much research on the regularized gap function. Taji, Fukushima, and Ibaraki [575] considered strongly monotone problems, and extended the results of [389] but without the compactness assumption on K. Zhu and Marcotte [655] extended a "derivative-free" algorithm, which

had been proposed earlier by Fukushima [220] in the strongly monotone case, to handle monotone problems over compact sets. The significance of the algorithm lies mainly in the fact that for the first time a dynamic strategy for the control of the parameter c that appears in the definition of θ_c is introduced. To be more precise, at each iteration the direction $d_c^k \equiv y_c(x^k) - x^k$ is considered. Based on a very simple test, either a line search is performed along d_c^k or the parameter c is reduced. The rationale behind the approach is that it is possible to show that for monotone problems, d_c^k will yield "sufficient" decrease in the regularized gap function, provided that c is sufficiently small. In a second paper [656], Zhu and Marcotte elaborated instead on [575] in order to tackle a larger class of merit functions. It is important to note that all the algorithms discussed thus far were proposed only for the solution of problems that are, at best, monotone. Our treatment in Section **10.4.4** pertains to general VIs, and the resulting algorithms achieve a fast convergence rate under weaker conditions than those previously known. Furthermore, some of the outlined variants are worthy of further exploration.

In Section **10.4.4** we mentioned that there exist algorithms for the solution of inequality constrained NLPs that maintain feasibility throughout the process and only solve equations. A recent reference is [470]. In general, the superlinear convergence of these methods requires the objective function to be twice continuously differentiable, while we know that the function θ_c cannot be expected to be more than once continuously differentiable. A possible remedy to this situation is envisaged in [176], where a feasible algorithm is studied that is locally superlinearly convergent under a reduced differentiability requirement on the objective function.

It was shown in [575] that $\sqrt{\theta_c(x)}$ gives an error bound for the VI (K, F) for all $x \in K$, provided that c is sufficiently small. Exercise **10.5.8**, taken from [656] (see also [440]), improves on this result. The extension in which non-quadratic functions are used to regularize the gap function is from [618]. As shown by Patriksson [440], this class of regularized gap functions is also closely related to the general approach of Auchmuty. A further variant along these lines, in which the function q can act as a barrier function for the set K, is discussed in [454], where a connection of the approach to the FB C-function is also highlighted.

Extending Auchmuty's approach [24], Larsson and Patriksson investigated differentiability and convexity properties of instances of Auchmuty's merit functions (see Exercise **10.5.9**) and showed how to generate descent directions under suitable monotonicity assumptions both in the case of differentiable and nondifferentiable merit functions. This line of research was

reported in [438, 439] and culminated in Patriksson's monograph [440], where a "cost approximation" scheme was introduced within which merit functions were derived in a unified way. This scheme subsumes also Cohen's auxiliary problem [109] and applies to a very general problem of the form

$$0 \in \mathcal{N}(x; K) + \partial u(x) + F(x),$$

where K is as usual a closed convex set, $u : \mathbb{R}^n \to \mathbb{R} \cup \{+\infty\}$ is a lower semicontinuous, proper convex function, and $F : K \cap \operatorname{dom} u \to \mathbb{R}^n$ is a continuous function. Incidentally, Smith's merit function for linearly constrained VIs [526, 527, 528] (see Exercise **10.5.10**) falls under Patriksson's framework as well.

The linearized gap function was investigated in Taji and Fukushima [574], which is the basis of our presentation in Subsection **10.2.2**. The sufficient condition for a stationary point of this function to be a solution of the VI given in Theorem **10.2.16** improves on the positive definiteness condition in the reference, where an iterative algorithm was described. Close to a classical SQP method for nonlinear programming, the latter algorithm uses a standard nondifferentiable penalty function to assess the acceptability of the iterates generated, which may be infeasible. Under some additional assumptions, it was shown that if the VI is strongly monotone, the algorithm converges to the unique solution of the VI. The complete algorithm is rather complicated, thus confirming the difficulties in the use of the linearized gap function. For related work, see [441, 572, 573].

Since the possibility to perform an unconstrained minimization of a smooth function to solve a VI seems much more attractive than a constrained approach, the next burst of activity in algorithmic developments came with the definition of the D-gap function. This was made possible by Peng [456, 445], who observed in the first paper that the implicit Lagrangian of an NCP is equal to the difference of two regularized gap functions on the nonnegative orthant and then formally extended the observation to a general VI. Peng's original definition of θ_{ab} used a single scalar $a \in (0, 1)$ and had $b \equiv 1/a$. Extending Peng's work, Yamashita, Taji, and Fukushima [637] employed two scalars $b > a > 0$ and formally coined the name "D-gap function" for θ_{ab}. The advantage of two parameters over one is flexibility.

Prior to the D-gap function, Yamashita and Fukushima [631] had already introduced a smooth, unconstrained merit function for the VI, by considering the Moreau-Yoshida regularization of θ_c; however this approach is computationally expensive, if not infeasible, since the mere calculation of the Moreau-Yoshida regularized merit function is prohibitive. Therefore,

the contribution of the D-gap function was that it provided the first unconstrained, differentiable, and, most importantly, readily computable merit function for the general VI on a closed convex set.

Yamashita, Taji, and Fukushima [637] showed that the level sets of θ_{ab} are bounded if F is strongly monotone and the set K is bounded. The results on the coercivity of the D-gap function in [445, 637] were further improved in [451], where it was proved that strong monotonicity alone, with no further assumptions, was sufficient for the coercivity of θ_{ab}. While the presentation in Section **10.3** follows the line in the mentioned references, our results are significant improvements over those in these references. Part (b) in Exercise **10.5.12**, which implies the coercivity of θ_{ab} under the strong coercivity of F on $\mathbb{R}^n$ defined therein, and also part (c) are due to recent work of Yamashita and Fukushima [635].

The implicit Lagrangian for the NCP (F) has already been referred to several times by now, and it was in fact introduced four years before the D-gap function. However, we have preferred to collect facts relevant to the implicit Lagrangian function after discussing the D-gap function, because this places the former function within the context of a broader perspective and brings out the connection of the two functions more clearly. Specifically, the implicit Lagrangian was introduced by Mangasarian and Solodov [383]; the name and the motivation for this function came from its connection with an augmented Lagrangian for the minimization problem (10.3.16) as explained in Subsection **10.3.1**. In the cited reference, a so-called "restricted" implicit Lagrangian is also introduced, which is a merit function for the NCP (F) on the nonnegative orthant. It turns out that the restricted implicit Lagrangian is nothing else than the regularized gap function θ_c for the VI $(\mathbb{R}^n_+, F)$. Extensions of the implicit Lagrangian to generalized CPs can be found in [595].

The question of when a stationary point of the implicit Lagrangian is a solution of the underlying NCP was first addressed in [630], where it was shown that a stationary point is a solution of the NCP if the Jacobian of F is positive definite at the point. The issue was finally settled by Facchinei and Kanzow in [173], where Theorem **10.3.15** was proved. Exercise **10.5.7** (with $b = 1/a$) is also from the latter source. Kanzow and Fukushima [304] investigated in detail the D-gap function for the VI with box constraints; these two authors were the first to prove the particular instances of Propositions **10.3.9** and **10.3.11** that correspond to the box constrained VI. Furthermore, Kanzow and Fukushima also showed that if F is a continuous uniformly P function, every stationary point of the D-gap function for a box constrained VI is a solution of the problem.

The linear Newton approximations $\mathcal{A}_c$ and $\mathcal{A}_{ab}$ of the gradients of θ_c and θ_{ab}, respectively, are introduced in [558]. Similar Newton approximations have also been used, in the case of box constraints, in [304]. As discussed in Section **10.4**, these Newton approximations, based on a deep understanding of the differential properties of the projector, are an important tool in the development of efficient algorithms for the solution of a general VI. Their use, however, requires the fine tuning of the parameters a and b. Various schemes to accomplish this tuning are discussed in Solodov and Tseng [539], who elaborate on and generalize the work of Zhu and Marcotte [655] regarding the D-gap function. Further methods using the D-gap function that are close to those just described can be found in [303, 451, 452]. In particular, in [303, 452] the case of the box constrained VI is considered; by using the D-gap function, Newton methods based on the natural residual or on the Josephy-Newton direction are globalized. In [451] instead, the globalization of the Josephy-Newton method is considered for a general VI. It is shown that every limit point of the sequence produced by the algorithm is a stationary point of the D-gap function and that if one of the limit points is a point where the Jacobian of F is positive definite then the whole sequence converges to this point and the convergence rate is at least superlinear. Subsections **10.4.1**, **10.4.2**, and **10.4.3** are built on all the mentioned results.

There are some open questions that we would like to mention. We saw that in our analysis there is a discrepancy between the conditions for fast convergence in the case of affine constraints, where the Jacobian of F is required to be *nonsingular* on a certain subspace, and the case of general nonlinear constraints, where we need the Jacobian of F to be *positive definite*. We expect that, in analogy to what is known both for the affine constraints and for the Josephy-Newton method, it is possible to obtain the superlinear convergence of the type of Newton methods considered in Subsection **10.4.1** under a much weaker assumption than positive definiteness. This could be achieved through a more refined analysis of either the linear Newton approximation $\mathcal{A}_c^{\mathrm{nat}}$ or $\mathcal{A}_{ab}$. A related issue is the definition of simpler or more powerful linear Newton approximations for the natural map or for the gradients of θ_c and θ_{ab} that would benefit algorithmic developments.

Chapter 11

Interior and Smoothing Methods

The main topic of this chapter is "interior point (IP) methods" for solving constrained equations (CEs), with emphasis on their applications to CPs of various kinds including KKT systems of VIs. We present a unified framework within which a basic IP method can be defined and develop an accompanying theory that analyzes its convergence. We do not treat rates of convergence for the IP methods, however.

Originated from the classical Sequential Unconstrained Minimization Technique for solving constrained minimization problems and made popular by their great success in solving linear programs, IP methods are highly effective also for solving monotone CPs and convex programs. The distinguished feature of these methods is the way they handle the constraints; namely, they generate iterates that satisfy these constraints strictly using iterative procedures for unconstrained problems.

The basic problem of this chapter is the following CE, which we denote by the pair (G, X):

$$G(x) = 0, \quad x \in X, \tag{11.0.1}$$

where $G : \Omega \supset X \to \mathbb{R}^n$ is a given mapping defined on the open set Ω that contains the constraint set X, which is a given closed set in $\mathbb{R}^n$ with a nonempty interior. The latter blanket property, which is assumed throughout this chapter, is natural and necessary for IP methods whose essential feature is to generate iterates that lie in $\operatorname{int} X \neq \emptyset$. Although constrained equations were also discussed in Chapter **8**, there was no particular emphasis there on the structure of X. In contrast, the development of this chapter is based on a set of postulates that link G and X very closely. Within the IP framework, a sequence of iterates in $\operatorname{int} X$ is generated that converges to a zero of G under appropriate conditions. The generation of such a sequence is via a modified Newton method applied to the equa-

tion $G(x) = 0$. The modification is essential in order to effectively handle the constraint set X. A general algorithmic framework for solving the CE (G, X) is presented in Section **11.3**.

In addition to the algorithms, we also present some general analytical results for the CE, including the existence of a solution, that are based on a blanket local homeomorphism assumption on G; see assumption (IP3) in Section **11.2**. These results cover not only the basic problem (11.0.1) but also treat various issues that are relevant to the IP algorithms. The local homeomorphism assumption is reasonable because this is also the requirement that underlies the convergence of a Newton method for solving a smooth, unconstrained system of equations. Based on the general theory in Section **11.2** and the algorithmic framework in Section **11.3**, a comprehensive treatment of the implicit MiCP is presented in Sections **11.4** and **11.5**; the former section presents the analytic results and the latter section presents the specialized IP algorithms.

Like all the algorithms presented in previous chapters, the algorithms in Sections **11.3** and **11.5** are iterative descent methods; specifically, the progress of these methods is monitored by a scalar merit function, which is reduced at every iteration of the algorithms. In the IP literature, such a function is called a potential function; the name "potential reduction algorithm" is therefore associated with an IP method of this kind. As an alternative to a potential reduction approach, a homotopy, or path-following approach can also be used in conjunction with smoothing. In Section **11.6**, we present an alternative IP method for solving the implicit MiCP that is not based on potential reduction.

The CE provides an effective framework for turning a CP, which is equivalent to a system of nonsmooth equations, into a smooth problem to which the IP methods are applied. Related methods for solving the CP can be developed based on the smoothing of various nonsmooth equation formulations of the CP, such as the min formulation and the FB formulation. Collectively, these methods constitute the broad family of smoothing methods for solving the CP, of which the IP methods are distinguished members. In Section **11.7**, we present a path-following method for solving the implicit MiCP that is based on a special smoothing of the FB function. Section **11.8** presents a general theory of smoothing for solving an unconstrained, locally Lipschitz equation and builds an iterative algorithm based on the application of a globally convergent Newton method to the smoothed problems.

11.1 Preliminary Discussion

The CE framework is broad enough to include CPs of various kinds. For instance, the standard NCP (F), where F is a mapping from $\mathbb{R}^n$ into itself, is easily seen to be equivalent to the CE (G, X), where

$$G(x,y) \equiv \begin{pmatrix} y \circ x \\ y - F(x) \end{pmatrix} \in \mathbb{R}^{2n}, \quad (x,y) \in \mathbb{R}^{2n}, \tag{11.1.1}$$

and $X \equiv \mathbb{R}^{2n}_+$. Similarly, the KKT system:

$$\begin{aligned} \mathbf{L}(x,\mu,\lambda) \equiv F(x) + \sum_{j=1}^{\ell} \mu_j \nabla h_j(x) + \sum_{i=1}^{m} \lambda_i \nabla g_i(x) &= 0 \\ h(x) &= 0 \\ 0 \le \lambda \perp g(x) &\le 0 \end{aligned}$$

is a special instance of the CE (G, X), where, for $(\lambda, w, x, \mu) \in \mathbb{R}^{2m+n+\ell}$,

$$G(\lambda,w,x,\mu) \equiv \begin{pmatrix} w \circ \lambda \\ w + g(x) \\ \mathbf{L}(x,\mu,\lambda) \\ h(x) \end{pmatrix} \in \mathbb{R}^{2m+n+\ell}, \tag{11.1.2}$$

and $X \equiv \mathbb{R}^{2m}_+ \times \mathbb{R}^{n+\ell}$. More generally, the implicit MiCP of the form:

$$\begin{aligned} H(x,y,z) &= 0 \\ 0 \le x \perp y &\ge 0, \end{aligned} \tag{11.1.3}$$

where $H : \mathbb{R}^{2n+m} \to \mathbb{R}^{n+m}$, can be written as the CE $(\mathbf{H}_{\rm IP}, X)$, where for $(x, y, z) \in \mathbb{R}^{2n+m}$,

$$\mathbf{H}_{\rm IP}(x,y,z) = \begin{pmatrix} x \circ y \\ H(x,y,z) \end{pmatrix} \in \mathbb{R}^{2n+m}, \tag{11.1.4}$$

and $X \equiv \mathbb{R}^{2n}_+ \times \mathbb{R}^m$ (we recall that the term "implicit" was introduced in Subsection **1.4.9** to describe this particular MiCP, see (1.4.58)). A particularly interesting special case of the implicit MiCP (11.1.3) is when H is affine:

$$H(x,y,z) \equiv q + Ax + By + Cz, \quad (x,y,z) \in \mathbb{R}^{2n+m},$$

for some vector $q \in \mathbb{R}^{n+m}$ and matrices $A,\ B \in \mathbb{R}^{(n+m)\times n}$ and C in $\mathbb{R}^{(n+m)\times m}$. The MiCP (11.1.3) includes many other kinds of CPs, such as the CP (F, G) defined by a pair of functions F and G; see the discussion at the end of Chapter **2**.

A box constrained VI (K, F), where K is the closed rectangle

$$K \equiv \{\, x \in \mathbb{R}^n \,:\, a \leq x \leq b \,\},$$

with $a \leq b$ being two given vectors in $\mathbb{R}^n$, is equivalent to the CE (G, X), where

$$G(y, v, x) \equiv \begin{pmatrix} y \circ (\, x - a \,) \\ v \circ (\, b - x \,) \\ y - F(x) - v \end{pmatrix} \in \mathbb{R}^{3n}, \quad (\, y, v, x \,) \in \mathbb{R}^{3n},$$

and $X \equiv \mathbb{R}^{2n}_+ \times K$. When some of the bounds are infinite (i.e., $a_i = -\infty$ or $b_j = \infty$ for some i, j), the VI (K, F) can similarly be reformulated as a CE with an obvious modification of the pair (G, X). Note that in the above CE formulations of the respective CPs, the complementarity condition is always written in terms of the Hadamard (componentwise) product so that the resulting map G is "square", that is, its domain and range are in the same Euclidean space; moreover, the set X is solid and quite simple.

For a CP in SPSD matrices, there are several equivalent ways of describing the complementarity condition, each leading to a different algorithm for solving such a CP. Specifically, let $F : \mathcal{M}^n \to \mathcal{M}^n$ be an operator from the set $\mathcal{M}^n$ of $n \times n$ symmetric matrices into itself and consider the CP in the space $\mathcal{M}^n$:

$$\mathcal{M}^n_+ \ni A \perp F(A) \in \mathcal{M}^n_+.$$

With $X \equiv \mathcal{M}^n_+ \times \mathcal{M}^n_+$, the following two maps G both lead to an equivalent CE formulation for this CP:

$$G(A, B) \equiv \begin{pmatrix} (\, AB + BA \,)/2 \\ B - F(A) \end{pmatrix} \in \mathcal{M}^n \times \mathcal{M}^n, \quad (\, A, B \,) \in \mathcal{M}^n \times \mathcal{M}^n;$$

$$G(A, B) \equiv \begin{pmatrix} AB \\ B - F(A) \end{pmatrix} \in \mathbb{R}^{n\times n} \times \mathcal{M}^n, \quad (\, A, B \,) \in \mathcal{M}^n \times \mathcal{M}^n.$$

The justification of these equivalent CE formulations is based on Proposition **1.4.9**. Again, X is solid with $\operatorname{int} X = \mathcal{M}^n_{++} \times \mathcal{M}^n_{++}$. Notice that the product matrix AB is not necessarily symmetric. This lack of symmetry distinguishes the above two mappings G.

11.1.1 The notion of centering

Our treatment of IP methods is very general and, at the same time, basically simple; the price we pay for the generality is the use of a rather abstract setting. To provide the proper motivation for the abstraction, we introduce the topic in an informal way so that the reader can more easily grasp the essential ideas of the methods and appreciate the role and the importance of the various elements that are analyzed in the subsequent sections. In particular, we describe the "central path" associated with the complementarity constraint, which plays a major role in IP methods.

Consider the CE reformulation (11.1.1) of the NCP (F). Assume that F is smooth; thus so is G. The Newton direction for the equation $G(x, y) = 0$ is the solution (dx, dy) of the system of linear equations

$$\begin{pmatrix} y \circ x \\ y - F(x) \end{pmatrix} + \begin{bmatrix} \operatorname{diag}(y) & \operatorname{diag}(x) \\ -JF(x) & \mathbf{I} \end{bmatrix} \begin{pmatrix} dx \\ dy \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix},$$

where for a given vector b, $\operatorname{diag}(b)$ is the diagonal matrix whose diagonal entries are equal to the components of b. A full step along this Newton direction will usually bring the iterates outside the set X. In IP methods we want to stay in the interior of X, and so we backtrack along the Newton direction in order to stay in $\operatorname{int} X$. Since we suppose that (x, y) itself belongs to the interior of X this is always possible. Experience shows, however, that often we can only take a very small step along the Newton direction before leaving the interior of X, thus resulting in very slow progress toward a solution. It may then be useful to "bend" the Newton direction toward the interior of X. To this end consider embedding the original CE reformulation of the NCP (F) into a family of parameterized CEs:

$$\begin{pmatrix} y \circ x \\ y - F(x) \end{pmatrix} = \begin{pmatrix} t\,\mathbf{1} \\ 0 \end{pmatrix}, \quad (x, y) \in X \equiv \mathbb{R}^{2n}_{+}, \tag{11.1.5}$$

where t is a nonnegative parameter. If $t = 0$ we get back the CE reformulation of the NCP (F) described above; if t is positive, instead, the solution of (11.1.5) is in the interior of X. Hence we may hope that the Newton direction for (11.1.5) will be biased toward the interior of X. Furthermore, under reasonable conditions, the solution $(x(t), y(t))$ of (11.1.5) forms a path that can be expected to approach a solution of the original CE when t tends to zero.

A natural strategy then emerges: use the bent Newton direction arising from system (11.1.5) and at the same time drive t to zero. We may represent

various possible directions as the solutions of the system

$$\begin{pmatrix} y \circ x \\ y - F(x) \end{pmatrix} + \begin{bmatrix} \operatorname{diag}(y) & \operatorname{diag}(x) \\ -JF(x) & \mathbf{I} \end{bmatrix} \begin{pmatrix} dx \\ dy \end{pmatrix} = \begin{pmatrix} \sigma\, t\, \mathbf{1} \\ 0 \end{pmatrix}, \quad (11.1.6)$$

where σ is a number in $[0, 1]$. If $\sigma = 0$ we get the pure, original Newton direction; if $\sigma = 1$ we obtain the bent Newton direction arising from (11.1.5). Increasing, intermediate values of σ give rise to increasingly biased directions. Using directions bent toward the interior of X we expect that longer steps may be taken before leaving $\operatorname{int} X$. Furthermore these directions will usually bring the iterates "well within" the set X, so that if in subsequent iterations we want to use the pure Newton direction we may expect that longer steps can be taken. Note that underlying the whole reasoning above is the idea that the Newton direction is good and we should avoid backtracking as much as possible; this idea is well founded on practical experience. Figure 11.1 illustrates the pure, intermediate, and bent, Newton direction.

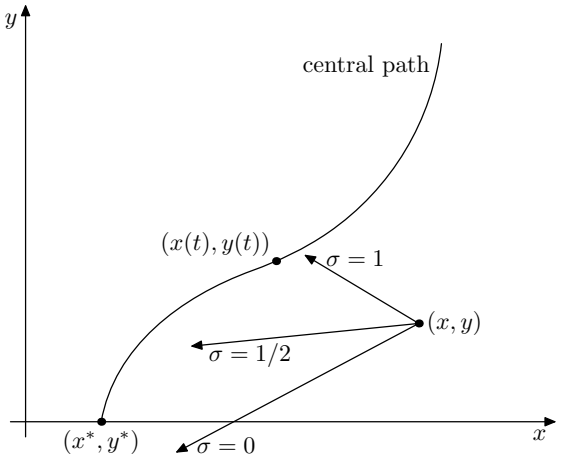


Figure 11.1: Central path and several directions for different values of σ.

The path described by the solution pair $(x(t), y(t))$ of (11.1.6) is called the *central path* of the NCP (F) and has traditionally played a key role in standard IP methods for linear programs and monotone LCPs. Roughly speaking, a possible approach to solving the latter problems by interior point methods is to trace the central path by driving t to zero. The individual algorithms differ in the numerical tracing of this path. In our treatment of the abstract CE (G, X), the role of the central path is, for the most part, no longer apparent; instead we pay more attention to the role

of the related *centering parameter* σ and retain the key idea of suitably bending the Newton direction.

A further question that must be addressed in this framework is: when backtracking, how do we assess the quality of a point along the bent Newton direction? The usual way this task is accomplished in previous chapters is by gauging the natural merit function $\frac{1}{2}\|G(x,y)\|^2$. However, because of the simple-minded strategy we are adopting to deal with the constraint set X, this is no longer a feasible approach. The reader can check that the theory of Chapter **8** simply does not allow us to prove any useful result in this case. We then have to resort to a different merit function, the *potential function*, that should in some sense contains information about the set X itself to make up for the otherwise poor treatment of this set.

At first sight, some potential functions may look rather odd; for example, we will show that for the CE reformulation of the NCP (F) a suitable potential function is: for $(x,y) \in \mathbb{R}^{2n}_{++}$,

$$\psi(x,y) \equiv n \log(\| x \circ y \|^2 + \| y - F(x) \|^2) - \sum_{i=1}^{n} \log(x_i y_i). \qquad (11.1.7)$$

For the considerations to follow, it is profitable to view the function ψ as the composition of two functions: $\psi \equiv p \circ G$, where G is defined in (11.1.1) and $p : \operatorname{int} S \to \mathbb{R}$ is given by

$$p(u,v) \equiv n \log(\| u \|^2 + \| v \|^2) - \sum_{i=1}^{n} \log u_i, \quad (u, v) \in \operatorname{int} S,$$

with $S \equiv \mathbb{R}^n_+ \times \mathbb{R}^n$. The function p should be seen as acting on $G(\operatorname{int} X)$. Note that p is not defined on the boundary of S. Actually, for any sequence $\{(u^k, v^k)\}$ converging to a nonzero vector on the boundary of S, $p(u^k, v^k)$ goes to infinity. This implies that if $\{(x^k, y^k)\}$ is a sequence in $\operatorname{int} X$ such that $\{G(x^k, y^k)\}$ converges to the boundary of S, either $\{\psi(x^k, y^k)\}$ goes to infinity or $\{G(x^k, y^k)\}$ converges to zero. An important consequence of the boundary behavior of ψ is the following. Suppose that at each iteration, the bent Newton direction (dx^k, dy^k) computed from (11.1.6) is a descent direction for ψ at $(x^k, y^k) \in \operatorname{int} X$. By searching along this direction from the current point, we can find a new point (x^{k+1}, y^{k+1}) in $\operatorname{int} X$ satisfying $\psi(x^{k+1}, y^{k+1}) < \psi(x^k, y^k)$. If, at the same time, we can also guarantee that $G(x^k, y^k)$ is approaching the boundary of S (we can achieve this by suitably decreasing t so that we will follow the central path toward $(x(0), y(0))$) then we must be converging to a zero of G. This is the principal strategy adopted by the IP methods. See Figure 11.2 for an illustration of the function $p(u,v)$ for $n = 1$.

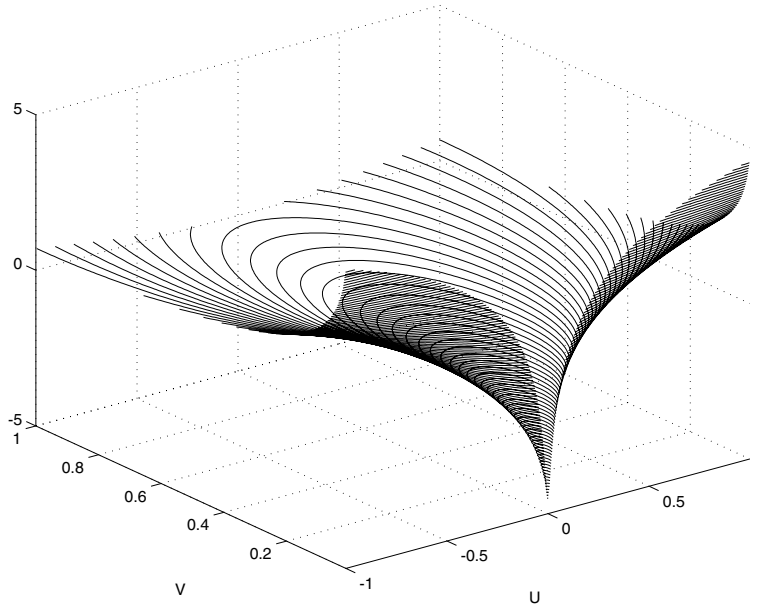


Figure 11.2: Representation of the function $p(u, v)$ for $n = 1$.

The discussion so far raises several questions:

(a) When is the central path (or some generalizations of it) a well-defined and well-behaved object?

(b) In particular, when does a solution to the CE exist?

(c) What are the properties of the function p (or, equivalently, of the function ψ) needed to ensure the convergence of the IP trajectory?

(d) How can we put all these elements together to form a working algorithm?

In the following sections we give a full answer to all these questions. Specifically, in the next section we analyze (a) and (b), while in Section **11.3** we consider (c) and (d) and present a very broad algorithmic scheme. Subsequent sections will be built upon these results to deal with specialized situations.

11.2 An Existence Theory

Existence results for CPs of various kinds were already considered in Chapters **2** and **3**. Those results readily extend to the CE reformulations of com-

plementarity problems. The developments in this section, however, apply to general CEs that do not necessarily arise from the reformulation of a CP; furthermore they have a somewhat different flavor that makes them suitable for the subsequent algorithmic developments.

In order to better explain the kind of existence issues we deal with, consider the pair $(\mathbf{H}_{\rm IP}, X)$ corresponding to the implicit MiCP (11.1.3), where $\mathbf{H}_{\rm IP}$ is defined by (11.1.4) and $X \equiv \mathbb{R}^{2n}_{+} \times \mathbb{R}^{m}$. This problem is more general than the NCP and is the main formulation to which we apply the IP methods.

We obviously have

$$\mathbf{H}_{\rm IP}(\operatorname{int} X) \subseteq \mathbb{R}^{n}_{++} \times H(\operatorname{int} X). \tag{11.2.1}$$

It is natural to ask when equality will hold in this inclusion. This issue is closely related to the questions (a) and (b) raised at the end of the previous section. Specifically, let $a \in \mathbb{R}^{n}_{++}$ and $b \in \mathbb{R}^{n+m}$ be two given vectors. Consider the parametric system:

$$\begin{aligned} x \circ y &= ta \\ H(x, y, z) &= tb \\ (x, y) &\geq 0, \end{aligned} \tag{11.2.2}$$

which yields the problem (11.1.3) when $t = 0$. If $H(x, y, z) = y - F(x)$, that is if the MiCP reduces to the NCP (F), the central path is obtained with $a = \mathbf{1}$ and $b = 0$; thus the solution trajectory to (11.2.2) can be considered a generalized central path for the MiCP. A fundamental question is: Does the system (11.2.2) have a solution for $t > 0$? Likewise, the asymptotic properties, such as continuity and boundedness, of the solution trajectory as t goes to zero are also important concerns. If equality holds in (11.2.1), then the solvability question has an affirmative answer provided that $tb \in H(\operatorname{int} X)$ for all $t > 0$. In particular, with $b = 0$, the latter condition becomes $0 \in H(\operatorname{int} X)$, which turns out to be a key condition in all IP methods for solving the MiCP. The continuity and the boundedness of the trajectory will be guaranteed if we can further show that $\mathbf{H}_{\rm IP}$ is a proper, local homeomorphism (see below for the definition of the properness property).

The background result to establish the aforementioned properties is best stated in terms of mappings between metric spaces. Part of the reason for working in this abstract framework is that we do not need to be concerned with the topological properties of the sets of interest, which we simply treat as metric spaces. We start by giving a few definitions. Assume that M

and N are two metric spaces and that $G : M \to N$ is a map between these two spaces. The map G is said to be *proper* with respect to a set $E \subseteq N$ if $G^{-1}(K) \subseteq M$ is compact for every compact set $K \subseteq E$. If G is proper with respect to N, we will simply say that G is proper. For $D \subseteq M$ and $E \subseteq N$ such that $G(D) \subseteq E$, the restricted map $\tilde{G} : D \to E$ defined by $\tilde{G}(u) \equiv G(u)$ for all $u \in D$ is denoted by $G|_{(D,E)}$; if $E = N$ then we write this $\tilde{G}$ simply as $G|_D$. We will also refer to $G|_{(D,E)}$ as "G restricted to the pair (D, E)", and to $G|_D$ as "G restricted to D". A metric space M is said to be *connected* if there exists no partition $(\mathcal{O}_1, \mathcal{O}_2)$ of M for which both $\mathcal{O}_1$ and $\mathcal{O}_2$ are nonempty and open. A metric space M is said to be *path-connected* if there is a path joining any two points u^0 and u^1 in M; that is, there exists a continuous function $p : [0, 1] \to M$ such that $p(0) = u^0$ and $p(1) = u^1$.

The following result from nonlinear analysis is the cornerstone of the theory presented herein. It does not require the mapping G to be differentiable

11.2.1 Theorem. Let M and N be two metric spaces and $G : M \to N$ be a continuous map. Let $M_0 \subseteq M$ and $N_0 \subseteq N$ be given sets satisfying the following conditions:

(a) $G|_{M_0}$ is a local homeomorphism, and

(b) $\emptyset \neq G^{-1}(N_0) \subseteq M_0$.

If G is proper with respect to a set E such that $N_0 \subseteq E \subseteq N$, then the map G restricted to the pair $(G^{-1}(N_0), N_0)$ is a proper local homeomorphism. If, in addition, N_0 is connected, then $G(M_0) \supseteq N_0$ and $G(\operatorname{cl} M_0) \supseteq E \cap \operatorname{cl} N_0$.

Proof. We first prove the following statement.

(c) Let $\tilde{G} : \tilde{M} \to \tilde{N}$ be a proper local homeomorphism from the metric space $\tilde{M}$ into the connected metric space $\tilde{N}$. The map $\tilde{G}$ is onto.

To prove this statement, for every $y \in \tilde{N}$, denote by $[y]$ the cardinality of the set $\tilde{G}^{-1}(y)$. We first show that

(d) for every $\bar{y} \in \tilde{N}$, $[\bar{y}]$ is finite and there exists a neighborhood W of $\bar{y}$ such that $[y]$ is constant for all $y \in W$.

The finiteness of $[\bar{y}]$ follows from the fact that $\tilde{G}^{-1}(\bar{y})$, being the preimage of the singleton $\{\bar{y}\}$, is compact if nonempty (by the properness of $\tilde{G}$), which then implies that $\tilde{G}^{-1}(\bar{y})$ is a discrete set (by the local homeomorphism assumption). To continue the proof of (d), consider first the case where

$\tilde{G}^{-1}(\bar{y})$ is empty. We need to show that a neighborhood of $\bar{y}$ exists such that $\tilde{G}^{-1}(y)$ is empty for all y in the neighborhood. Assume for contradiction that no such neighborhood exists. There exists then a sequence $\{y^k\} \subset \tilde{N}$ converging to $\bar{y}$ such that $\tilde{G}^{-1}(y^k) \neq \emptyset$ for all k. The union of these preimages,

$$\bigcup_k \tilde{G}^{-1}(y^k),$$

is compact by the properness of $\tilde{G}$. Thus there exists a sequence $\{x^{k_\nu}\} \subset \tilde{M}$ with a limit x^∞. Clearly, by the continuity of $\tilde{G}$, we must have $\tilde{G}(x^\infty) = \bar{y}$, which is a contradiction.

Next assume that $\tilde{G}^{-1}(\bar{y}) = \{\bar{x}^1, \bar{x}^2, \ldots, \bar{x}^m\}$ for some positive integer m. We can find a neighborhood $\mathcal{X}_i$ of x^i for $i = 1, 2, \ldots, m$ and a neighborhood $\mathcal{Y}$ of $\bar{y}$ such that all the $\tilde{G}|_{(\mathcal{X}_i, \mathcal{Y})}$ are homeomorphisms. Therefore, for every $y \in \mathcal{Y}$, there is one and only one point $x^i \in \mathcal{X}_i$ such that $\tilde{G}(x^i) = y$. We then see that $[y] \geq [\bar{y}]$ for every $y \in \mathcal{Y}$. We claim that there exists a subneighborhood $\bar{\mathcal{Y}}$ of $\bar{y}$ contained in $\mathcal{Y}$, such that $[y] = [\bar{y}]$ for every $y \in \bar{\mathcal{Y}}$. Assume for contradiction that this is not true. We can then find a sequence $\{y^k\}$ converging to $\bar{y}$ such that $[y^k] > [\bar{y}]$ for every k. Hence we can also find a sequence $\{z^k\}$ such that, for every k, $\tilde{G}(z^k) = y^k$ and $z^k \notin \mathcal{X}_i$ for all $i = 1, \ldots, m$. Since $\tilde{G}$ is proper, the set $\tilde{G}^{-1}(\{y^k\} \cup \{\bar{y}\}) \supseteq \{z^k\}$ is compact. Thus $\{z^k\}$ has a limit point $\bar{z}$ not belonging to $\mathcal{X}_i$ for every i and such that $\tilde{G}(\bar{z}) = \bar{y}$. This is a contradiction and assertion (c) is proved.

We are ready to show that $\tilde{G}$ is onto, which is the claim of statement (c). Fix a $\bar{y} \in \tilde{N}$ with $[\bar{y}] > 0$. Consider the set

$$\mathcal{O} \equiv \{\, y \in \tilde{N} : [y] = [\bar{y}] \,\},$$

which is open by (d). It is also closed by (d) too. Since $\mathcal{O}$ is nonempty, the connectedness of $\tilde{N}$ implies that $\tilde{N} = \mathcal{O}$, which establishes the surjectivity of $\tilde{G}$.

Finally, we use (c) to complete the proof of the theorem. Let $\tilde{G}$ be $G|_{(G^{-1}(N_0), N_0)}$. Note that the domain of $\tilde{G}$ is nonempty by (b) and that $\tilde{G}$ is a local homeomorphism by (a). We claim that $\tilde{G}$ is also proper. Indeed, let $K \subseteq N_0$ be a compact set. Using (b) we easily see that $\tilde{G}^{-1}(K) = G^{-1}(K)$. By the properness of G with respect to E it follows that $\tilde{G}^{-1}(K) = G^{-1}(K)$ is compact, so that $\tilde{G}$ is proper. Assume now that N_0 is connected. By (c), it follows that $\tilde{G}$ is onto and so $G(M_0) \supseteq N_0$. To prove the final inclusion $G(\operatorname{cl} M_0) \supseteq E \cap \operatorname{cl} N_0$, take a $\bar{y} \in E \cap \operatorname{cl} N_0$. We can find a sequence $\{y^k\} \subset N_0$ converging to $\bar{y}$. Since $G(M_0) \supseteq N_0$, there exists a sequence $\{x^k\} \subset M_0$ with $G(x^k) = y^k$ for every k. Since $\bar{y} \in E$ and $\{y^k\} \subset N_0 \subseteq E$,

the set $G^{-1}(\{y^k\} \cup \{\bar{y}\}) \supseteq \{x^k\}$ is compact. Therefore, $\{x^k\}$ has a limit point $\bar{x} \in M$. Clearly we have $G(\bar{x}) = \bar{y}$ and $\bar{x} \in \operatorname{cl} M_0$. □

A major application of the above theorem is to establish the surjectivity of a function defined on an open subset of $\mathbb{R}^n$ onto a desirable subset in the range space. This kind of "constrained surjectivity" is distinct from the kind of global surjectivity of functions defined on the entire space $\mathbb{R}^n$; see for example Proposition **2.1.8** which was established by degree theory. The theme of constrained surjectivity and the related property of "constrained homeomorphism" are the fundamental theoretical basis on which the interior point methods for solving the VI/CP are built. Theorem **11.2.1** is also the cornerstone for the class of Bregman-based methods for solving monotone VIs; see Subsection **12.7.2**, particularly, Lemma **12.7.9**.

11.2.1 Applications to CEs

In order to apply Theorem **11.2.1** to the CE (G, X), we introduce some blanket assumptions on this pair. Subsequently, these assumptions will be verified in the context of several applications of the CE to the CPs. The reader should bear in mind that, for the reasons explained before, existence results of the type presented herein also have strong algorithmic implications.

Specifically, we postulate the existence of a closed convex subset S that relates to the range of G and possesses certain special properties. Part of the generality of our framework stems from the freedom in the choice of S. By properly choosing S, we obtain various IP algorithms for solving CPs of different kinds. The assumptions are as follows.

(IP1) The closed set X has a nonempty interior.

(IP2) There exists a solid, closed, convex set $S \subset \mathbb{R}^n$ such that (see Figure 11.3)

(a) $0 \in S$;

(b) the (open) set $X_{\mathcal{I}} \equiv G^{-1}(\operatorname{int} S) \cap \operatorname{int} X$ is nonempty;

(c) the set $G^{-1}(\operatorname{int} S) \cap \operatorname{bd} X$ is empty.

(IP3) G is a local homeomorphism on $X_{\mathcal{I}}$.

By the inverse function theorem, (IP3) holds if

(IP3') G is continuously differentiable on $X_{\mathcal{I}}$, and $JG(x)$ is nonsingular for all $x \in X_{\mathcal{I}}$.

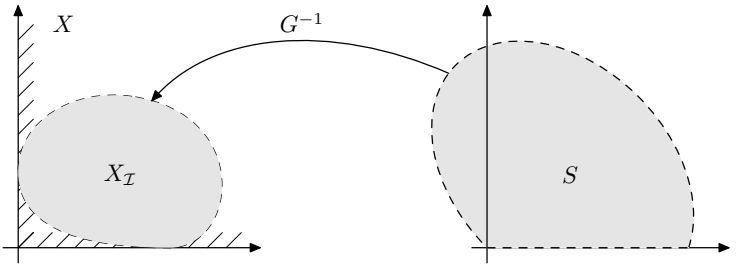


Figure 11.3: The set $X_{\mathcal{I}}$ and (IP2).

As mentioned before, assumption (IP1) is basic to the development of an interior point method. It is important to point out that for the kind of CEs solvable by such a method, such as the CPs, the solution being computed typically lies on the boundary of X. The interiority assumption facilitates the computation of such a solution. By restricting the method to operate within $\operatorname{int} X$, which is an open set, one can then treat the CE as if it were unconstrained. The sets S and $X_{\mathcal{I}}$ in assumption (IP2) contain the key elements of an IP method. If G is considered to be a mapping with domain X, conditions (b) and (c) in (IP2) are equivalent to the condition that

$$\emptyset \neq G^{-1}(\operatorname{int} S) \subseteq \operatorname{int} X.$$

Whereas S pertains to the range of G, $X_{\mathcal{I}}$ pertains to the domain. We give a preliminary explanation about the role of these two sets in an IP method. Initiated at a vector x^0 in $X_{\mathcal{I}}$, an IP algorithm generates a sequence of iterates $\{x^k\} \subset X_{\mathcal{I}}$ so that the sequence $\{G(x^k)\} \subset \operatorname{int} S$ will eventually converge to zero, thus accomplishing the goal of solving the CE (G, X), at least approximately. Assumption (IP3') facilitates the generation of the sequence $\{x^k\}$ via a Newton scheme. For the purpose of deriving the existence results for the CE (G, X), assumption (IP3), which does not require G to be differentiable, is sufficient. Moreover, in light of the Newton methods developed in Chapter **8** for solving nonsmooth equations, it is possible to generalize the methods presented in this chapter to deal with the case where G is nonsmooth, say semismooth. For simplicity reasons, we refrain from discussing the generalization and focus on the case where G is smooth when we discuss the IP methods.

11.2.2 Theorem. Assume that conditions (IP1)–(IP3) hold and there exists a convex set $E \subseteq S$ such that $0 \in E$, $E \cap G(X_{\mathcal{I}})$ is nonempty and $G : X \to \mathbb{R}^n$ is proper with respect to E. The following two statements hold:

(a) $E \subseteq G(X)$; in particular, CE (G, X) has a solution;

(b) G restricted to the pair $(\, X_{\mathcal{I}} \cap G^{-1}(E),\, E \cap \operatorname{int} S\,)$ is a proper local homeomorphism.

Proof. To apply Theorem **11.2.1**, we let $M \equiv X$, $N \equiv \mathbb{R}^n$, $M_0 \equiv X_{\mathcal{I}}$ and $N_0 \equiv E \cap \operatorname{int} S$. Using (IP2) and the assumption that $E \cap G(X_{\mathcal{I}}) \neq \emptyset$, we easily see that $\emptyset \neq G^{-1}(N_0) \subseteq M_0$. Moreover, $G|_{M_0}$ is a local homeomorphism by (IP3). Since G is proper with respect to E by assumption, it follows from Theorem **11.2.1** that

$$G(X) \supseteq G(\operatorname{cl} X_{\mathcal{I}}) = G(\operatorname{cl} M_0) \supseteq E \cap \operatorname{cl} N_0 = E \cap \operatorname{cl}(E \cap \operatorname{int} S) = E,$$

where the last equality follows from the expression:

$$\operatorname{cl}(\, E \cap \operatorname{int} S\,) = (\, \operatorname{cl} E\,) \cap \operatorname{cl}(\, \operatorname{int} S\,) = (\, \operatorname{cl} E\,) \cap S,$$

by elementary properties of convex sets. Hence (a) holds. It also follows from Theorem **11.2.1** that G restricted to the pair $(G^{-1}(N_0),\, N_0)$ is a proper local homeomorphism. By (IP2), we have

$$G^{-1}(N_0) = X \cap G^{-1}(E \cap \operatorname{int} S) = X_{\mathcal{I}} \cap G^{-1}(E);$$

consequently, (b) holds. □

11.2.3 Remark. Needless to say, the above Theorem **11.2.2** and the next Theorem **11.2.4** remain valid if instead of condition (IP3), we assume the stronger property (IP3'). □

The next result is a specialization of the previous theorem.

11.2.4 Theorem. Assume that conditions (IP1)–(IP3) hold and that G is proper with respect to S. The following two statements hold:

(a) $S \subseteq G(X)$; in particular, the CE (G, X) has a solution; and

(b) G restricted to $X_{\mathcal{I}}$ maps each path-connected component of $X_{\mathcal{I}}$ homeomorphically onto $\operatorname{int} S$.

Proof. Conclusion (a) follows immediately from Theorem **11.2.2**(a) with $E = S$. Using part (b) of this theorem with $E = S$, we conclude that G restricted to the pair $(X_{\mathcal{I}}, \operatorname{int} S)$ is a proper local homeomorphism. If $\mathcal{T} \subseteq X_{\mathcal{I}}$ is a path-connected component of $X_{\mathcal{I}}$ then G restricted to the pair $(\mathcal{T},\, \operatorname{int} S)$ is a proper local homeomorphism because $\mathcal{T}$ is both open and closed with respect to $X_{\mathcal{I}}$. Since every proper local homeomorphism from a path-connected set into a convex set is a homeomorphism, (b) follows. □

Compared to Theorem **11.2.4**, the usefulness of Theorem **11.2.2** lies in the set E being a (proper) subset of S. Unlike S, E is not required to be either solid or closed. Such a set E allows us to deal with functions G that are not necessarily proper with respect to the larger set S; cf. Theorem **11.4.16** and the subsequent discussion.

11.3 A General Algorithmic Framework

In this section we present the promised IP scheme for solving a CE. We begin by introducing three further assumptions that delimit the potential functions, followed by the description of the scheme. Applications, refinements and variants of this scheme will be discussed in the subsequent sections.

11.3.1 Assumptions on the potential function

The Newton scheme relies on a potential function for the set $X_{\mathcal{I}}$, which in turn is induced by another potential function defined on $\operatorname{int} S$. Specifically, we postulate the existence of a potential function $p : \operatorname{int} S \to \mathbb{R}$ with certain properties and use the function $\psi(x) \equiv p(G(x))$ to measure progress in the algorithm. We assume that the function p satisfies the following two properties, which do not involve the set X:

(IP4) for every sequence $\{u^k\} \subset \operatorname{int} S$ such that

$$\text{either } \lim_{k\to\infty} \|u^k\| = \infty \text{ or } \lim_{k\to\infty} u^k = \bar{u} \in \operatorname{bd} S \setminus \{0\},$$

we have

$$\lim_{k\to\infty} p(u^k) = \infty; \tag{11.3.1}$$

(IP5) p is continuously differentiable on its domain and $u^T \nabla p(u) > 0$ for all nonzero $u \in \operatorname{int} S$.

A useful consequence of (IP4) is that if an interior sequence in S with bounded (e.g. decreasing) p-values is approaching the boundary of S then the sequence must converge to the origin. Thus, we can rely on p to keep a sequence of interior vectors in S away from the boundary of S while leading the sequence toward the zero vector. Hence the role of p is slightly different from that of a standard "barrier function" used in nonlinear programming, which in contrast penalizes a vector in S when it gets close to the boundary of S. See Figure 11.4 for an illustration of (IP4).

An inf-compactness condition equivalent to (IP4) is stated in the following result. The reader is asked to supply a proof in Exercise **11.9.2**.

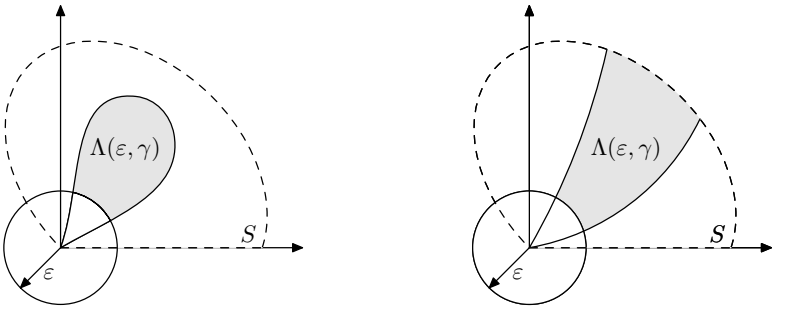


Figure 11.4: Left: (IP4) is satisfied. Right: (IP4) is violated.

11.3.1 Lemma. Condition (IP4) holds if and only if for all $\gamma \in \mathbb{R}$ and $\varepsilon > 0$, the set

$$\Lambda(\varepsilon, \gamma) \equiv \{ u \in \operatorname{int} S : p(u) \leq \gamma, \ \|u\| \geq \varepsilon \}$$

is compact. □

Recognizing that an important role of p is to drive an interior vector u in S toward the zero vector while keeping u away from the boundary of S, we can easily understand the role of (IP5). Namely, for any nonzero interior vector u of S, $-u$ is a descent direction of p at the vector u. Thus starting at u, we can move toward the origin and decrease p, thus accomplishing the goal of "staying away from $\operatorname{bd} S$".

We introduce the last assumption, which postulates the existence of an important vector a that is closely linked to the potential function p. Like (IP4) and (IP5), the following assumption (IP6) pertains only to S and p.

(IP6) There exists a pair $(a, \bar{\sigma}) \in \mathbb{R}^n \times (0, 1]$ such that

$$\|a\|^2 \left(u^T \nabla p(u) \right) \geq \bar{\sigma} \left(a^T u \right) \left(a^T \nabla p(u) \right), \quad \forall \ u \in \operatorname{int} S,$$

Trivially (IP6) holds with $a = 0$ and any $\bar{\sigma} \in (0, 1]$. It follows that the entire development in this chapter holds with $a = 0$. Nevertheless the interesting case is when $a \neq 0$. The purpose of (IP6) is to identify a broad class of such vectors a for which one can establish the convergence of a potential reduction algorithm developed in a later subsection.

In what follows, we give a preliminary illustration of these assumptions with three fundamental examples. The first example pertains to the most basic problem of solving a smooth system of unconstrained equations, which corresponds to $X = \mathbb{R}^n$. In this case, we may simply take S to be the entire space $\mathbb{R}^n$ (so that $\operatorname{bd} S = \emptyset$), $p(u)$ to be the function $\|u\|^2$, a to be any

vector and $\bar{\sigma} = 1$. It is then clear that (IP2) and (IP4)–(IP6) all hold easily, with (IP6) being an immediate consequence of the Cauchy-Schwarz inequality. (Remark: if $\mathrm{bd}\, S \setminus \{0\}$ is an empty set, such as in this example, then any coercive function p satisfies (IP4).)

Another fundamental case to illustrate assumptions (IP4)–(IP6) is when S is the nonnegative orthant $\mathbb{R}^n_+$, and here we expand the discussion in Subsection **11.1.1**. This case is natural for dealing with the NCP. Specifically, consider the function

$$p(u) \equiv \zeta \log u^T u - \sum_{i=1}^{n} \log u_i, \quad u > 0$$

and the pair $(a, \bar{\sigma}) = (\mathbf{1}_n, 1)$, where $\zeta > n/2$ is an arbitrary scalar. (Note: the ℓ_1 norm of u, instead of $u^T u$, could also be used in the first logarithmic term. The analysis remains the same with the constant ζ properly adjusted.) Clearly, p is coercive on $\mathbb{R}^n_{++}$, i.e.

$$\lim_{\substack{u>0 \\ \|u\|\to\infty}} p(u) = \infty,$$

because for $u > 0$,

$$\begin{aligned} p(u) &\geq \zeta \left(2 \log \left(\sum_{i=1}^{n} u_i \right) - \log n \right) - \sum_{i=1}^{n} \log u_i \\ &> (2\zeta - n) \log \left(\sum_{i=1}^{n} u_i \right) - (\zeta - n) \log n, \end{aligned}$$

where the first inequality follows from the fact that $\|u\|_1 \leq \sqrt{n}\|u\|$ and the second inequality is due to the inequality:

$$n \log \left(\sum_{i=1}^{n} u_i \right) - \sum_{i=1}^{n} \log u_i \geq n \log n.$$

Moreover, for any positive sequence $\{u^k\}$ converging to a nonzero nonnegative vector with at least one zero component, the limit (11.3.1) clearly holds. Thus (IP4) follows. For a positive vector u, denote by u^{-1} the vector whose i-th component is equal to $1/u_i$. Since

$$u^T \nabla p(u) = u^T \left(\frac{2\zeta}{\| u \|^2} u - u^{-1} \right) = 2\zeta - n > 0,$$

(IP5) holds. Observe that $u^T \nabla p(u)$ is a constant for all $u \in \mathrm{int}\, S$. This is a very important feature of the potential function p. Moreover, with

$(a, \bar{\sigma}) = (\mathbf{1}_n, 1)$, we claim that (IP6) also holds. Indeed, we have for $u > 0$,

$$a^T \nabla p(u) = \frac{2\,\zeta \sum_{i=1}^{n} u_i}{\sum_{i=1}^{n} u_i^2} - \sum_{i=1}^{n} u_i^{-1};$$

thus

$$\frac{(a^T \nabla p(u))\,(a^T u)}{\|a\|^2} = n^{-1} \left[\frac{2\,\zeta\,\|u\|_1^2}{\|u\|^2} - \left(\sum_{i=1}^{n} u_i^{-1} \right) \left(\sum_{i=1}^{n} u_i \right) \right]$$

$$\leq 2\,\zeta - n = u^T \nabla p(u),$$

where the last inequality follows from the fact that $\|u\|_1 \leq \sqrt{n}\,\|u\|$ and the arithmetic-geometric mean inequality.

A third case of interest is a combination of the unconstrained case and the case of the nonnegative orthant. Namely, $S = \mathbb{R}^{n_1}_+ \times \mathbb{R}^{n_2}$ for some positive integers n_1 and n_2. This case corresponds to solving mixed CPs such as the KKT system of a VI. With this S, we let, for a given $\zeta > n_1/2$,

$$p(u, v) \equiv \zeta \log(\|(u, v)\|^2) - \sum_{i=1}^{n_1} \log u_i, \quad (u, v) \in \mathbb{R}^{n_1}_{++} \times \mathbb{R}^{n_2}, \tag{11.3.2}$$

and $a \equiv (\mathbf{1}_{n_1}, 0)$ and $\bar{\sigma} = 1$. We leave it as an exercise for the reader to verify that conditions (IP4)–(IP6) are valid.

11.3.2 A potential reduction method for the CE

We return to the CE (G, X), which we assume satisfies the blanket assumptions (IP1), (IP2), (IP3'), and (IP4)–(IP6). In particular, G is continuously differentiable on $X_{\mathcal{I}}$. The algorithm for solving this CE is a modified, damped Newton method applied to the equation $G(x) = 0$. There are two major modifications; both are needed to facilitate the treatment of the constraint set X. The first modification is that the Newton equation to compute the search direction is augmented by the given vector a in condition (IP6); see (11.3.3). The second modification is that the merit function for the line search is given by

$$\psi(x) \equiv p(G(x)), \quad \forall\, x \in X_{\mathcal{I}}.$$

When $X = \mathbb{R}^n$ and $p(u) = \|u\|^2$, the function $\psi(x)$ becomes the squared (Euclidean) norm function of $G(x)$, which is the familiar merit function in

a damped Newton method for smooth or nonsmooth unconstrained equations; see Chapter **8**. When X is a proper subset of $\mathbb{R}^n$ (such as the nonnegative orthant), the function ψ could be quite different from any norm function of G. In general, by (IP3') and (IP5), the function ψ is continuously differentiable on $X_{\mathcal{I}}$ with gradient given by

$$\nabla\psi(x) \;=\; JG(x)^T\nabla p(G(x)), \quad \forall\, x \,\in\, X_{\mathcal{I}}.$$

With the above explanation, we give the full details of the modified Newton method for solving the CE (G, X). We call this a potential reduction algorithm because it reduces the potential (i.e., merit) function $\psi(x)$ at every iteration. If X is a proper subset of $\mathbb{R}^n$, the algorithm will not necessarily drive ψ to zero. This is a distinguishing feature of the algorithm versus for example the Gauss-Newton method discussed in Chapter **8**. A still more significant difference is that a solution x^* of the CE (G, X) is not necessarily a minimizer of the potential function which, in many cases, is not even well defined at x^*.

A Potential Reduction Algorithm for the CE (PRACE)

11.3.2 Algorithm.

Data: $x^0 \in X_{\mathcal{I}}$, $\gamma \in (0,1)$, and a sequence $\{\sigma_k\} \subset [0, \bar{\sigma})$.

Step 1: Set $k = 0$.

Step 2: If $G(x^k) = 0$, stop.

Step 3: Solve the system of linear equations

$$G(x^k) + JG(x^k)d \;=\; \sigma_k \,\frac{a^T G(x^k)}{\|\, a \,\|^2}\, a \tag{11.3.3}$$

to obtain the search direction d^k. (The right-hand side of the above equation is assumed to be zero if $a = 0$. This convention is assumed throughout this chapter.)

Step 4: Find the smallest nonnegative integer i_k such that with $i = i_k$, $x^k + 2^{-i}d^k \in X_{\mathcal{I}}$ and

$$\psi(x^k + 2^{-i}d^k) - \psi(x^k) \;\le\; \gamma\, 2^{-i}\, \nabla\psi(x^k)^T d^k;$$

set $\tau_k \equiv 2^{-i_k}$.

Step 5: Set $x^{k+1} \equiv x^k + \tau_k d^k$ and $k \leftarrow k + 1$; go to Step 2.

Since $x^k \in X_{\mathcal{I}}$, assumption (IP3') implies that the (modified) Newton equation (11.3.3) has a unique solution, which we have denoted by d^k. The following lemma guarantees that d^k is a descent direction for ψ at x^k. This descent property, along with the openness of $X_{\mathcal{I}}$, ensures that the integer i_k can be determined in a finite number of trials, starting with $i = 0$ and increasing i by one at each trial. Consequently, the next iterate x^{k+1} is well defined.

11.3.3 Lemma. Suppose that conditions (IP5) and (IP6) hold. Assume also that $x \in X_{\mathcal{I}}$, $d \in \mathbb{R}^n$, and $\sigma \in [0, \bar{\sigma})$ satisfy $G(x) \neq 0$ and

$$G(x) + JG(x)d \;=\; \sigma \,\frac{a^{\,T} G(x)}{\| a \|^2}\, a.$$

It holds that $\nabla\psi(G(x))^{\,T} d < 0$.

Proof. Write $u \equiv G(x)$. We have $0 \neq u \in \operatorname{int} S$. Furthermore,

$$\begin{aligned} \nabla\psi(x)^{\,T} d \;&=\; \nabla p(G(x))^{\,T} JG(x)d \;=\; \nabla p(u)^{\,T} \left(-u + \sigma \,\frac{a^{\,T} G(x)}{\| a \|^2}\, a \right) \\ &\leq\; -\nabla p(u)^{\,T} u \left(1 - \frac{\sigma}{\bar{\sigma}} \right) \;<\; 0, \end{aligned}$$

where the last two inequalities follow from (IP5), (IP6), and the choice of σ. □

In what follows, we state and prove a limiting property of an infinite sequence of iterates $\{x^k\}$ generated by the above algorithm. Before stating the theorem, we observe that such a sequence $\{x^k\}$ necessarily belongs to the set $X_{\mathcal{I}}$; thus $\{G(x^k)\} \subset \operatorname{int} S$. Since the sequence $\{x^k\}$ is infinite, we have $G(x^k) \neq 0$ for all k. There are four conclusions in the theorem below. The first three of these do not assert the boundedness of $\{x^k\}$; this boundedness is the consequence of a properness assumption on G, which is stated in part (d) in the theorem. An immediate consequence of statement (d) is that the CE (G, X) has a solution. A consequence of statement (c) in the theorem is

$$\inf \{ \| G(x) \| \,:\, x \,\in\, X \} \;=\; 0,$$

which is a kind of asymptotic solvability, as opposed to exact solvability, of the CE (G, X). In turn, this inf condition implies that the CE (G, X) has "ε-solutions" for every scalar $\varepsilon > 0$ in the sense that for any such ε, there exists a vector $x^\varepsilon \in X$ satisfying $\|G(x^\varepsilon)\| \leq \varepsilon$; moreover, x^ε can be computed by the potential reduction Newton Algorithm **11.3.2** starting at the given vector $x^0 \in X_{\mathcal{I}}$.

11.3.4 Theorem. Assume conditions (IP1), (IP2), (IP3') and (IP4)–(IP6) hold and that

$$\limsup_k \sigma_k < \bar{\sigma}.$$

Let $\{x^k\}$ be any infinite sequence produced by Algorithm **11.3.2**. The following statements are valid:

(a) the sequence $\{G(x^k)\}$ is bounded;

(b) any accumulation point of $\{x^k\}$, if it exists, solves the CE (G, X); in particular, if $\{x^k\}$ is bounded, then the CE (G, X) has a solution.

Moreover, for any closed subset E of S containing the sequence $\{G(x^k)\}$,

(c) if G is proper with respect to $E \cap \operatorname{int} S$, then

$$\lim_{k\to\infty} G(x^k) = 0;$$

(d) if G is proper with respect to E, then $\{x^k\}$ is bounded.

Proof. Let $\eta \equiv \psi(x^0)$ and write $u^k \equiv G(x^k) \in \operatorname{int} S$ for all k. Clearly,

$$p(u^k) = \psi(x^k) < \psi(x^0) = \eta.$$

Hence $\{u^k\}$ is contained in the set $\Lambda(\varepsilon, \eta) \cup \operatorname{cl} \mathbb{B}(0, \varepsilon)$, which is compact by Lemma **11.3.1**. This establishes (a).

To show (b), let x^∞ be an accumulation point of $\{x^k\}$. Clearly, x^∞ belongs to X because X is a closed set. Assume for contradiction that $u^\infty \equiv G(x^\infty) \neq 0$. Let $\{x^k : k \in \kappa\}$ be a subsequence converging to x^∞ and assume without loss of generality that $\{\sigma_k : k \in \kappa\}$ converges to some scalar σ_∞, which must satisfy $0 \leq \sigma_\infty < \bar{\sigma}$. Since

$$\lim_{k(\in\kappa)\to\infty} u^k = u^\infty \neq 0,$$

there exists $\varepsilon > 0$ such that the subsequence $\{u^k : k \in \kappa\}$ is contained in $\Lambda(\varepsilon, \eta)$. Since the latter set is compact, it follows that

$$u^\infty = G(x^\infty) \in \Lambda(\varepsilon, \eta) \subset \operatorname{int} S.$$

Hence $x^\infty \in X_{\mathcal{I}}$ because of condition (IP2c). By assumption (IP3'), $JG(x^\infty)^{-1}$ exists. This implies that the sequence $\{d^k : k \in \kappa\}$ converges to a vector d^∞ satisfying

$$G(x^\infty) + JG(x^\infty)d^\infty = \sigma_\infty \frac{a^T G(x^\infty)}{\| a \|^2}\, a.$$

By Lemma **11.3.3**, we have $\nabla p(x^\infty)^T d^\infty < 0$. Since $\{x^k : k \in \kappa\}$ converges to $x^\infty \in X_{\mathcal{I}}$, where ψ is continuous, it follows that $\{\psi(x^k) : k \in \kappa\}$ converges. This implies that the entire sequence $\{\psi(x^k)\}$ converges because it is monotonically decreasing. Using the relation

$$\psi(x^{k+1}) - \psi(x^k) = \psi(x^k + \tau_k d^k) - \psi(x^k) \leq \gamma\, \tau_k\, \nabla\psi(x^k)^T d^k < 0,$$

which holds for all k, we deduce that

$$\lim_{k\to\infty} \tau_k\, \nabla\psi(x^k)^T d^k = 0$$

and hence that

$$\lim_{k(\in\kappa)\to\infty} \tau_k = 0$$

because

$$\lim_{k(\in\kappa)\to\infty} \nabla\psi(x^k)^T d^k = \nabla\psi(x^\infty)^T d^\infty < 0.$$

Consequently,

$$\lim_{k(\in\kappa)\to\infty} i_k = \infty,$$

which implies in particular that $i_k \geq 2$ for all $k \in \kappa$ sufficiently large. Moreover the sequence $\{x^k + 2^{-i_k+1} d^k : k \in \kappa\}$ converges to x^∞. Since x^∞ belongs to $X_{\mathcal{I}}$, which is an open set, it follows that for all $k \in \kappa$ sufficiently large, the vector $x^k + 2^{-i_k+1} d^k$ belongs to $X_{\mathcal{I}}$. By the definition of i_k, we therefore deduce that for all $k \in \kappa$ sufficiently large,

$$\frac{\psi(x^k + 2^{-i_k+1} d^k) - \psi(x^k)}{2^{-i_k+1}} > \gamma\, \nabla\psi(x^k)^T d^k.$$

Letting $k(\in \kappa) \to \infty$ in the above expression, we deduce

$$\nabla\psi(x^\infty)^T d^\infty \geq \gamma\, \nabla\psi(x^\infty)^T d^\infty,$$

which is a contradiction because $\gamma < 1$ and $\nabla\psi(x^\infty)^T d^\infty < 0$. Consequently, we must have $G(x^\infty) = 0$ and statement (b) of the theorem is established.

To prove (c), assume for the sake of contradiction that for an infinite subset $\kappa \subset \{1, 2, \ldots\}$, we have

$$\liminf_{k(\in\kappa)\to\infty} \| u^k \| > 0.$$

By an argument similar to that employed above, we may deduce the existence of an $\varepsilon > 0$ such that

$$\{ u^k : k \in \kappa \} \subset \Lambda(\varepsilon, \eta) \cap E.$$

The right-hand set is a compact subset of $E \cap \operatorname{int} S$ because E is closed. By the properness assumption of G, it follows that $\{x^k : k \in \kappa\}$ is bounded. By (b), every accumulation point of this subsequence is a zero of G. This contradiction establishes (c). Finally, by (a) and the fact that E is closed, we deduce that $\{u^k\}$ is contained in a compact subset E_1 of E. Since G is proper with respect to E, it follows that $G^{-1}(E_1) \supset \{x^k\}$ is bounded. Hence (d) follows. □

The following is an important corollary of Theorem **11.3.4**. In essence, it gives a practical condition in terms of certain two-sided (one-sided) level sets of $\|G\|$ under which a compact set E can be identified easily so that G is proper with respect to $E \cap \operatorname{int} S$ (E). Since it is more convenient to give a direct proof of the corollary, we bypass using the theorem in the following proof.

11.3.5 Corollary. Assume that, in addition to the conditions in Theorem **11.3.4**, for every pair of positive scalars δ and η, the set

$$X(\delta, \eta) \equiv \{\, x \in X_{\mathcal{I}} : \delta \leq \| G(x) \| \leq \eta \,\}$$

is bounded. If $\{x^k\}$ is any infinite sequence produced by Algorithm **11.3.2**, then

$$\lim_{k \to \infty} G(x^k) = 0.$$

Alternatively, if for every positive scalar η, the set

$$X(\eta) \equiv \{\, x \in X_{\mathcal{I}} : \| G(x) \| \leq \eta \,\},$$

is bounded, then the sequence $\{x^k\}$ is bounded.

Proof. Assume for contradiction that the sequence $\{G(x^k)\}$ does not converge to zero. There exists an infinite subset $\kappa \subset \{1, 2, \ldots\}$ such that

$$\liminf_{k(\in\kappa) \to \infty} \| G(x^k) \| > 0.$$

By part (a) of Theorem **11.3.4**, the sequence $\{G(x^k)\}$ is bounded. Hence there exist positive scalars δ and η such that $\{x^k : k \in \kappa\}$ is contained in the set $X(\delta, \eta)$. By the boundedness of the latter set, it follows that $\{x^k : k \in \kappa\}$ is bounded and thus it has at least one accumulation point. By part (b) of Theorem **11.3.4**, every accumulation point of $\{x^k : k \in \kappa\}$ is a zero of G. Hence the subsequence $\{G(x^k) : k \in \kappa\}$ converges to zero. This contradiction establishes the first assertion of the corollary. The second assrtion of the corollary is trivial because $\{x^k\}$ is contained in the set $X(\eta)$ for some appropriate $\eta > 0$. □

It is useful to note the distinction between the two additional assumptions in the above corollary. Namely, the first assumption about the boundedness of the set $X(\delta, \eta)$ for all positive (δ, η) is weaker than the second assumption about the boundedness of the set $X(\eta)$ for all positive η. In particular, the sequence $\{x^k\}$ is not asserted to be bounded under the former assumption; nor is it asserted that the CE (G, X) has a solution. The second assumption, on the other hand, implies that the CE (G, X) must have a solution.

In principle, by assuming a further nonsingularity assumption on the Jacobian $JG(x^\infty)$ at a limit point x^∞ of the sequence $\{x^k\}$, we could establish the sequential convergence of the sequence and characterize the superlinear rate of convergence in terms of the ultimate attainment of a unit step size. However, in the application to CPs, the nonsingularity assumption tends to be fairly restrictive because it essentially corresponds to imposing a strict complementarity condition at the limit solution. There are tremendous efforts in circumventing such a restriction for CPs of various kinds, including the KKT systems of monotone VIs. Such an analysis is typically very tedious and highly technical and requires very tight bounds of the generated directions in terms of the system residuals. More details and references are given in Section **11.10**.

11.4 Analysis of the Implicit MiCP

This is the first of two sections devoted to an application of theory developed so far to the implicit MiCP (11.1.3). In this section, we discuss the application of Theorems **11.2.2** and **11.2.4**, in the next section we consider specialized algorithms for the same class of problems. Further applications of the results to the CP (F, G) defined by a pair of functions F and G are briefly mentioned.

As a starting point of the specialized theory to be developed herein, we introduce a basic property of a partitioned matrix of the following form:

$$Q \equiv [\, A \quad B \quad C \,], \tag{11.4.1}$$

where A and B are of order $(n + m) \times n$ and C is of order $(n + m) \times m$. Consideration of such a matrix Q stems from the mixed CP (11.1.3); indeed, the Jacobian matrix of H is given by

$$JH(x, y, z) = [\, J_x H(x, y, z) \quad J_y H(x, y, z) \quad J_z H(x, y, z) \,], \tag{11.4.2}$$

which is in the form of the partitioned matrix Q. The case $m = 0$ is permitted throughout the following discussion; in this case, the matrix C

is vacuous and both A and B are square matrices of order n. This case includes, for instance, the standard NCP.

11.4.1 Definition. The matrix Q given by (11.4.1) is said to have the *mixed* P_0 property if C has full column rank and

$$\left.\begin{array}{l} Au + Bv + Cw \;=\; 0 \\ (\,u, v\,) \;\neq\; 0 \end{array}\right\} \\ \Rightarrow\; u_i v_i \;\geq\; 0 \text{ for some } i \text{ such that } |u_i| + |v_i| > 0. \tag{11.4.3}$$

If the above implication holds with a vacuous C, we say that the (A, B) is a P_0 pair. □

The above definition is a generalization of the P_0 property of a square matrix M, which corresponds to the case where C is vacuous and either A or B is the identity matrix and the other one is equal to $-M$. The following proposition formalizes this connection.

11.4.2 Proposition. A matrix $M \in \mathbb{R}^{n\times n}$ is P_0 if and only if $(\mathbf{I}, -M)$ is a P_0 pair.

Proof. If M is a P_0-matrix, it can easily be shown that $(\mathbf{I}, -M)$ is a P_0 pair. The proof of the converse turns out to be not entirely trivial. Suppose that $(\mathbf{I}, -M)$ is a P_0 pair but M is not a P_0 matrix. There exists a nonzero vector $v \in \mathbb{R}^n$ such that $v_j(Mv)_j < 0$ for all $v_j \neq 0$. Let $\alpha \equiv \mathrm{supp}(v)$. Since $(\mathbf{I}, -M)$ has the P_0 property, α must be a proper subset of $\{1, \ldots, n\}$ and $M_{\bar{\alpha}\alpha} v_\alpha$ cannot equal zero, where $\bar{\alpha}$ is the complement of α in $\{1, \ldots, n\}$. Pick an index $i \in \bar{\alpha}$ such that $M_{i\alpha} v_\alpha \neq 0$ and choose a scalar w_i whose sign is opposite that of $M_{i\alpha} v_\alpha$. Define the vector $\tilde{v}$ as follows:

$$\tilde{v}_j \;\equiv\; \begin{cases} v_j & \text{if } j \neq i \\ \varepsilon\, w_i & \text{if } j = i. \end{cases}$$

For $\varepsilon > 0$ sufficiently small, the nonzero vector $\tilde{v}$ satisfies $\tilde{v}_j(M\tilde{v})_j < 0$ for all $\tilde{v}_j \neq 0$; moreover $\mathrm{supp}(\tilde{v})$ is equal to $\mathrm{supp}(v) \cup \{i\}$. Repeating the above argument with v replaced by $\tilde{v}$, we obtain either a contradiction or a superset of $\mathrm{supp}(\tilde{v})$. Since the expansion of the latter index set must eventually stop because there are only finitely many indices, we arrive at a contradiction. □

The fundamental role of the mixed P_0 property in the IP theory is elucidated in the following lemma.

11.4.3 Lemma. Let A and B be matrices of order $(n+m)\times n$ and C be a matrix of order $(n+m)\times m$. The matrix Q given by (11.4.1) has the mixed P_0 property if and only if for every pair of $n\times n$ diagonal matrices D_1 and D_2 both having positive diagonal entries, the $(2n+m)\times(2n+m)$ square matrix

$$\mathbf{M} \equiv \begin{bmatrix} D_1 & D_2 & 0 \\ A & B & C \end{bmatrix} \tag{11.4.4}$$

is nonsingular.

Proof. Suppose that Q has the mixed P_0 property. Let D_1 and D_2 be any two $n\times n$ diagonal matrices both with positive diagonals. Let (u,v,w) be such that

$$D_1u + D_2v = 0$$
$$Au + Bv + Cw = 0.$$

Suppose $(u,v)\neq 0$. An index i exists such that $u_iv_i \geq 0$ and $|u_i|+|v_i|>0$. The first displayed equation implies $u_i = -\tau v_i$ for some positive $\tau > 0$. This is impossible. Thus we must have $(u,v)=0$. Since C has full column rank, it follows that $w=0$. Thus $\mathbf{M}$ is nonsingular.

Conversely, suppose $\mathbf{M}$ is nonsingular for any two diagonal matrices D_1 and D_2 as specified. It is clear that C must have full column rank. Suppose (u,v,w) satisfies $(u,v)\neq 0$ and $Au+Bv+Cw=0$. Assume for the sake of contradiction that $u_iv_i<0$ for all i such that $|u_i|+|v_i|>0$. Let D_1 be the identity matrix of order n; define the diagonal matrix D_2 as follows: for each $i=1,\ldots,n$,

$$(D_2)_{ii} \equiv \begin{cases} |u_i|/|v_i| & \text{if } |u_i|+|v_i|>0 \\ 1 & \text{otherwise.} \end{cases}$$

We clearly have $D_1u+D_2v=0$, contradicting the nonsingularity of the matrix $\mathbf{M}$. □

A sufficient condition for the implication (11.4.3) to hold is:

$$Au + Bv + Cw = 0 \;\Rightarrow\; u^Tv \geq 0, \tag{11.4.5}$$

which is a generalization of the positive semidefiniteness property of a square matrix M (take $A=\mathbf{I}$, $B=-M$ and C vacuous). The following definition offers a formal terminology for a pair of matrices (A,B) satisfying the implication (11.4.5) with a vacuous C.

11.4.4 Definition. A pair of $n\times n$ matrices A and B is said to be *column monotone* if $Au+Bv=0 \Rightarrow u^Tv\geq 0$. □

Clearly, a square matrix M is positive semidefinite if and only if the pair $(\mathbf{I}, -M)$ is column monotone. It turns out that column monotonicity is even more closely related to positive semidefiniteness than what this observation shows. In order to state the following result, Proposition **11.4.5**, we recall from Definition **3.3.23** the concept of a column representative matrix of a pair of matrices. Specifically, given a pair of $n \times n$ matrices A and B, we say that an $n \times n$ matrix E is a column representative matrix of the pair (A, B) if the i-th column of E is an element of the set $\{A_{\cdot i}, B_{\cdot i}\}$ for all $i = 1, \ldots, n$. If E is a column representative matrix of (A, B), the *complement* of E is the column representative matrix $\bar{E}$ of (A, B) such that $\{E_{\cdot i}, \bar{E}_{\cdot i}\} = \{A_{\cdot i}, B_{\cdot i}\}$ for all $i = 1, \ldots, n$.

11.4.5 Proposition. A pair of $n \times n$ matrices A and B is column monotone if and only if there exists a nonsingular column representative matrix E of the pair (A, B) such that $-E^{-1}\bar{E}$ is positive semidefinite.

Proof. The sufficiency is obvious. We only need to prove the necessity. Let (A, B) be column monotone. In turn, it suffices to show the existence of a nonsingular column representative matrix E of (A, B) because once this is done, the positive semidefiniteness of $-E^{-1}\bar{E}$ is obvious. We show the existence of E by induction on n. The assertion clearly holds with $n = 1$. Suppose that every column monotone pair of $(n-1) \times (n-1)$ matrices must have a nonsingular column representative matrix (of order $n-1$). Let (A, B) be a column monotone pair of square matrices of order n. It is easy to see that for every $i = 1, \ldots, n$, either $A_{\cdot i}$ or $B_{\cdot i}$ must be nonzero. Suppose without loss of generality that the first column of A is nonzero. By permuting the rows of the matrix $[A \;\; B]$, we may assume that the first diagonal entry of A is nonzero. Such a permutation preserves the column monotonicity of the resulting pair of matrices. Pivoting on the nonzero entry a_{11}, we perform a row operation on the matrix $[A \;\; B]$ to zero out the entries in the first column of A below the first diagonal entry. Such a row operation preserves the column monotonicity of the resulting pair of matrices, which we denote $\bar{A}$ and $\bar{B}$. Let A' and B' be the $(n-1)\times(n-1)$ principal submatrices of $\bar{A}$ and $\bar{B}$, respectively, obtained by removing the first row and first column of $\bar{A}$ and $\bar{B}$. It is not difficult to show that the pair of matrices A' and B' is column monotone. Since these are matrices of order $n-1$, the induction hypothesis implies the existence of a nonsingular column representative matrix E', which is of order $n-1$, of the pair (A', B'). It is then easy to demonstrate that the $n \times n$ column representative matrix E of (A, B) obtained by taking $E_{\cdot 1}$ to be the first column of A and $E_{\cdot i}$ for $i = 2, \ldots, n-1$ to be either $A_{\cdot i}$ or $B_{\cdot i}$ depending

on the definition of the corresponding column of E' is nonsingular. This completes the inductive step. □

A consequence of the above proposition is that the horizontal LCP (see Exercise **4.8.15**):

$$Ax + By = c$$
$$0 \leq x \perp y \geq 0,$$

where the pair (A, B) is column monotone, is equivalent to a standard, monotone LCP (q, M) for a suitable vector q and a positive semidefinite matrix M. The proof of the proposition offers a construction for the conversion from the horizontal problem to the standard problem.

11.4.1 The differentiable case

With the above matrix-theoretic preparation, we are ready to investigate the implicit CP (11.1.3) and the associated CE $(\mathbf{H}_{\mathrm{IP}}, X)$, where $\mathbf{H}_{\mathrm{IP}}$ is given by (11.1.4) and $X \equiv \mathbb{R}^{2n}_{+} \times \mathbb{R}^m$. By choosing the set S appropriately, we obtain results for this CP under a broad set of assumptions on H. We first consider the case of a differentiable H satisfying the mixed P_0 property and a certain norm-coerciveness condition. Specifically, assume that H is continuously differentiable at every point $(x, y, z) \in \operatorname{int} X$ and that the Jacobian matrix $JH(x, y, z)$ in partitioned form (11.4.2) has the mixed P_0 property. A sufficient condition for this property to hold is that $J_z H(x, y, z)$ has full column rank for all $(x, y, z) \in \operatorname{int} X$ and that H is *differentiably monotone* on $\operatorname{int} X$. By this we mean that the Jacobian matrix $JH(x, y, z)$ in partitioned form satisfies the implication (11.4.5); i.e., for every $(x, y, z) \in \mathbb{R}^{2n}_{++} \times \mathbb{R}^m$,

$$J_x H(x, y, z)dx + J_y H(x, y, z)dy + J_z H(x, y, z)dz = 0 \;\Rightarrow\; dx^T dy \geq 0.$$

The Jacobian matrix of the mapping $\mathbf{H}_{\mathrm{IP}}$ is equal to

$$J\mathbf{H}_{\mathrm{IP}}(x, y, z) = \begin{bmatrix} \operatorname{diag}(y) & \operatorname{diag}(x) & 0 \\ J_x H(x, y, z) & J_y H(x, y, z) & J_z H(x, y, z) \end{bmatrix}.$$

Thus $J\mathbf{H}_{\mathrm{IP}}(x, y, z)$ is of the form (11.4.4); moreover, provided that x and y are positive vectors, Lemma **11.4.3** is applicable to infer that $J\mathbf{H}_{\mathrm{IP}}(x, y, z)$ is nonsingular for all $(x, y, z) \in \operatorname{int} X$. Consequently, $\mathbf{H}_{\mathrm{IP}}$ is a local homeomorphism at every point in $\operatorname{int} X$.

If H is norm-coercive on X, that is, if

$$\lim_{\substack{(x,y)\geq 0 \\ \|(x,y,z)\|\to\infty}} \| H(x, y, z) \| = \infty, \tag{11.4.6}$$

and if H is continuous, then $\mathbf{H}_{\rm IP} : X \to \mathbb{R}^n_+ \times \mathbb{R}^{n+m}$ is proper with respect to $\mathbb{R}^n_+ \times \mathbb{R}^{n+m}$. Unfortunately, the above norm-coerciveness of H is not a practical condition; indeed for $m = 0$ and H affine, say

$$H(x,y) \equiv q + Ax + By, \quad \forall\, (x,y) \in \mathbb{R}^{2n},$$

for some $n \times n$ matrices A and B, the limit condition (11.4.6) is equivalent to the implication:

$$[\, Ax + By = 0, \quad (x,y) \geq 0\,] \;\Rightarrow\; (x,y) = 0.$$

To see that the latter implication is indeed too restrictive, consider the case where B is the negative identity matrix, so that the implicit CP becomes a standard LCP. The above implication is then equivalent to:

$$[\, Ax \geq 0,\; x \geq 0\,] \;\Rightarrow\; x = 0,$$

which fails to hold when A is an S_0 matrix. Since the class of LCPs of the S_0 type is rather broad (it includes LCPs of the semicopositive, and thus P_0, type; see Exercise **3.7.29**), we need to seek an alternative condition on H to ensure the properness of $\mathbf{H}_{\rm IP}$ that is more realistic for interesting CPs. The following result identifies one such condition, which we term *coerciveness in the complementary variables*, and abbreviate as (CC).

11.4.6 Lemma. Suppose that $H : \mathbb{R}^{2n}_+ \times \mathbb{R}^m \to \mathbb{R}^{n+m}$ is continuous and satisfies the following condition:

(CC) for every sequence $\{(x^k, y^k, z^k)\} \subset \mathbb{R}^{2n}_+ \times \mathbb{R}^m$ and for every subset α of $\{1, \ldots, n\}$ with complement $\bar{\alpha}$,

$$\left.\begin{array}{r} \{\, (x^k_\alpha, y^k_{\bar{\alpha}})\,\} \text{ is bounded} \\ \lim\limits_{k\to\infty} \|\, (x^k_{\bar{\alpha}}, y^k_\alpha, z^k)\,\| = \infty \end{array}\right\} \;\Rightarrow\; \lim_{k\to\infty} \|\, H(x^k, y^k, z^k)\,\| = \infty.$$

For every compact subset S of $\mathbb{R}^n_+ \times \mathbb{R}^{n+m}$, the inverse image

$$\mathbf{H}^{-1}_{\rm IP}(S) \equiv \{\, (x,y,z)\, \mathbb{R}^{2n}_+ \times \mathbb{R}^m \,:\, \mathbf{H}_{\rm IP}(x,y,z) \in S\,\}$$

is a compact subset of $\mathbb{R}^{2n}_+ \times \mathbb{R}^m$.

Proof. The set $\mathbf{H}^{-1}_{\rm IP}(S)$ is obviously closed. Suppose $\{(x^k, y^k, z^k)\}$ is an infinite sequence in $\mathbf{H}^{-1}_{\rm IP}(S)$ such that

$$\lim_{k\to\infty} \|\, (x^k, y^k, z^k)\,\| = \infty. \tag{11.4.7}$$

Since S is bounded, we have that $\{x^k \circ y^k\}$ and $\{H(x^k, y^k, z^k)\}$ are bounded. It is easy to deduce the existence of a subset α of $\{1, \ldots, n\}$ (possibly $\alpha = \emptyset$

or $\alpha = \{1, \ldots, n\}$) and an infinite index set κ such that $\{x_\alpha^k : k \in \kappa\}$ is bounded and

$$\lim_{k(\in\kappa)\to\infty} x_i^k = \infty, \quad \forall\, i \notin \alpha.$$

Since $\{x^k \circ y^k\}$ is bounded, it follows that

$$\lim_{k(\in\kappa)\to\infty} y_i^k = 0, \quad \forall\, i \notin \alpha.$$

Hence the sequence $\{(x_\alpha^k, y_{\bar\alpha}^k) : k \in \kappa\}$ is bounded. This, (11.4.7), and (CC) together imply that $\{\|H(x^k, y^k, z^k)\|\}$ is unbounded, a contradiction. □

At first glance, the (CC) property may seem a bit artificial. As it turns out, this is not the case when we consider special instances of the function H. The next result shows that in the context of the NCP, the (CC) condition holds if and only if the min function is norm-coercive on the nonnegative orthant; thus (CC) holds for broad classes of CPs, including the NCP with a continuous uniformly P function and the affine CPs satisfying a generalized R_0 property; see Proposition **11.4.11**.

11.4.7 Proposition. Let $F : \mathbb{R}_+^n \to \mathbb{R}^n$ be continuous. The function $H(x, y) \equiv y - F(x)$ satisfies condition (CC) if and only if the function $\min(x, F(x))$ is norm-coercive on $\mathbb{R}_+^n$, i.e.,

$$\lim_{\substack{x\geq 0 \\ \|x\|\to\infty}} \| \min(x, F(x)) \| = \infty.$$

In particular, this holds if F is a continuous uniformly P function.

Proof. Assume that the min function is norm-coercive on $\mathbb{R}_+^n$ but H fails to satisfy condition (CC). Thus there exist a sequence $\{(x^k, y^k)\} \subset \mathbb{R}_+^{2n}$ and an index set $\alpha \subseteq \{1, \ldots, n\}$ such that $\{(x_\alpha^k, y_{\bar\alpha}^k)\}$ and $\{y^k - F(x^k)\}$ are both bounded and

$$\lim_{k\to\infty} \| (x_{\bar\alpha}^k, y_\alpha^k) \| = \infty.$$

If the sequence $\{x^k\}$ is bounded, then $\{y^k\}$ is unbounded. Hence, by the continuity of F, $\{y^k - F(x^k)\}$ is unbounded. Since this is not possible, $\{x^k\}$ is unbounded. By the norm coercivity of $\min(x, F(x))$ on $\mathbb{R}_+^n$, it follows that for some index i, the sequence $\{\min(x_i^k, F_i(x^k))\}$ is unbounded. If $\{x_i^k\}$ is bounded, then a subsequence of $\{F_i(x^k)\}$ must tend to $-\infty$ as $k \to \infty$. The latter implies, since y_i^k is nonnegative for all k, that a subsequence of $\{y_i^k - F_i(x^k)\}$ tends to ∞ as $k \to \infty$. Since this can not happen, the index i must belong to $\bar\alpha$. Hence $\{y_i^k\}$ is bounded; thus so is $\{F_i(x^k)\}$. But then $\{\min(x_i^k, F_i(x^k))\}$ must also be bounded because x_i^k is nonnegative for all k. This contradiction shows that (CC) holds for H.

Conversely, suppose that condition (CC) holds for H but there exists a sequence $\{x^k\}$ in $\mathbb{R}^n_+$ such that $\{\min(x^k, F(x^k))\}$ is bounded and

$$\lim_{k\to\infty} \| x^k \| = \infty.$$

By working with a suitable subsequence of $\{x^k\}$, we may assume without loss of generality that there is a proper subset α of $\{1, \ldots, n\}$ such that $\{x^k_i\}$ is bounded for every $i \in \alpha$ and

$$\lim_{k\to\infty} x^k_i = \infty$$

for every $i \notin \alpha$. Define

$$y^k \equiv F(x^k) - \min(\, x^k, F(x^k)\,).$$

The sequence $\{y^k\}$ is nonnegative; moreover, for every $i \in \bar{\alpha}$, $y^k_i = 0$ for all k sufficiently large because $\{\min(x^k, F(x^k)\}$ is bounded. Since the sequence $\{y^k - F(x^k) = -\min(x^k, F(x^k))\}$ is bounded, the sequence $\{(x^k, y^k)\}$ contradicts condition (CC) for H. This contradiction establishes the necessary and sufficient condition for the (CC) condition to hold. The last assertion of the proposition follows from Corollary **9.1.28** and Proposition **9.1.27**. □

To this end, we recall several results relevant to the norm-coercivity of the function $\min(x, F(x))$ on $\mathbb{R}^n$; two of these, Corollary **9.1.28** and Proposition **9.1.27** have been used in the above proof. In addition, Proposition **9.1.26** deals with the special case where F is affine; see also Proposition **2.6.5**.

A corollary of Lemma **11.4.6** is that if H is continuous and satisfies the (CC) condition, then the level set

$$\{\, (x, y, z) \in \mathbb{R}^{2n}_+ \times \mathbb{R}^m : \| H(x, y, z) \| \leq \delta,\ x \circ y \leq \eta\, \mathbf{1}_n \,\},$$

if nonempty, is compact for all nonnegative scalars δ and η. In particular, under the (CC) condition, the implicit MiCP (11.1.3) must have a (possibly empty) bounded solution set. Based on Theorem **11.2.4**, we present a first existence result for the implicit CP (11.1.3) under a differentiability assumption. For notational convenience, we write

$$H_{++} \equiv H(\mathbb{R}^{2n}_{++} \times \mathbb{R}^m) \quad \text{and} \quad H_+ \equiv H(\mathbb{R}^{2n}_+ \times \mathbb{R}^m).$$

Clearly, we have $H_{++} \subseteq H_+$. In general, H_+ is not necessarily closed and H_{++} is not necessary open. Under the assumptions in the following theorem, these two sets are equal and coincide with the whole space $\mathbb{R}^{n+m}$.

11.4.8 Theorem. Suppose that $H : \mathbb{R}^{2n}_{+} \times \mathbb{R}^m \to \mathbb{R}^{n+m}$ is continuous and satisfies the (CC) condition. Assume further that for every (x, y, z) in $\mathbb{R}^{2n}_{++} \times \mathbb{R}^m$, the Jacobian matrix $JH(x, y, z)$ exists, is continuous, and has the mixed P_0 property. The following three statements are valid:

(a) $\mathbf{H}_{\mathrm{IP}}$ maps $\mathbb{R}^{2n}_{++} \times \mathbb{R}^m$ homeomorphically onto $\mathbb{R}^{n}_{++} \times \mathbb{R}^{n+m}$;

(b) $H_{++} = H_{+} = \mathbb{R}^{n+m}$;

(c) $\mathbf{H}_{\mathrm{IP}}(\mathbb{R}^{2n}_{+} \times \mathbb{R}^m) = \mathbb{R}^{n}_{+} \times \mathbb{R}^{n+m}$; in particular, the implicit CP (11.1.3) has a solution.

Proof. Let $S \equiv \mathbb{R}^{n}_{+} \times \mathbb{R}^{n+m}$. With $X \equiv \mathbb{R}^{2n}_{+} \times \mathbb{R}^m$, the set $X_{\mathcal{I}}$ is clearly nonempty because it is equal to $\operatorname{int} X$. Thus conditions (IP1) and (IP2) hold. Moreover, (IP3) also holds by Lemma **11.4.3**. Since G is proper with respect to S, by Lemma **11.4.6**, Theorem **11.2.4** implies that S is a subset of $\mathbf{H}_{\mathrm{IP}}(X)$ and $\mathbf{H}_{\mathrm{IP}}$ maps $\operatorname{int} X$ homeomorphically onto $\operatorname{int} S$. Thus $\mathbf{H}_{\mathrm{IP}}(X) \subseteq S$; consequently, $\mathbf{H}_{\mathrm{IP}}(\mathbb{R}^{2n}_{+} \times \mathbb{R}^m) = \mathbb{R}^{n}_{+} \times \mathbb{R}^{n+m}$. Thus (a) and (c) follow readily. To prove (b), consider the inclusions:

$$\mathbf{H}_{\mathrm{IP}}(\mathbb{R}^{2n}_{++} \times \mathbb{R}^m) \subseteq \mathbb{R}^{n}_{++} \times H_{++} \subseteq \mathbb{R}^{n}_{++} \times H_{+} \subseteq \mathbb{R}^{n}_{++} \times \mathbb{R}^{n+m};$$

and the fact that the left-most set is equal to the right-most set, by (a). Thus equality holds throughout the above expression and (b) follows. □

As a consequence of the above result, we illustrate the behavior of a homotopy path of the system:

$$\begin{aligned} x \circ y &= a \\ H(x, y, z) &= 0 \\ (\, x, y \,) &\geq 0, \end{aligned} \tag{11.4.8}$$

where $a \in \mathbb{R}^{n}_{+}$. Note that when $a = 0$, the above system is equivalent to problem (11.1.3). Let $(x^0, y^0, z^0) \in \mathbb{R}^{2n}_{++} \times \mathbb{R}^m$ be a given vector and let $r : [0, 1] \to \mathbb{R}^{n}_{+}$ and $s : [0, 1] \to \mathbb{R}^{n+m}$ be paths satisfying

$$\begin{aligned} r(t) &\in \mathbb{R}^{n}_{++}, \quad \forall\, t \in (0, 1] \\ r(1) &= x^0 \circ y^0 > 0, \\ s(1) &= H(x^0, y^0, z^0), \\ s(0) &= 0, \quad r(0) = a. \end{aligned}$$

Under the assumptions of Theorem **11.4.8**, it follows that for each $t \in (0, 1]$, there exists a unique triple $(x(t), y(t), z(t))$ in $\mathbb{R}^{2n}_{++} \times \mathbb{R}^m$ such that

$$\begin{aligned} x(t) \circ y(t) &= r(t) \\ H(x(t), y(t), z(t)) &= s(t). \end{aligned}$$

Moreover, $(x(t), y(t), z(t))$ is continuous in t and

$$(\, x(1), y(1), z(1)\,) \;=\; (\, x^0, y^0, z^0\,);$$

hence, the path has one endpoint at the given triple (x^0, y^0, z^0). By Lemma **11.4.6**, it follows that

$$\{\, (\, x(t), y(t), z(t)\,) \;:\; t \;\in\; (0,1)\,\}$$

is bounded, and every accumulation point of the path $t \mapsto (x(t), y(t), z(t))$ as $t \to 0$ is a solution of the system (11.4.8). Hence if we follow this path starting at (x^0, y^0, z^0), we will eventually approach a solution of (11.4.8). A trivial way to construct the paths $r(t)$ and $s(t)$ is as follows:

$$\begin{aligned} r(t) \;&=\; (\,1-t\,)a + t\,x^0 \circ y^0, \quad \forall\, t \;\in\; [0,1] \\ s(t) \;&=\; t\,H(x^0, y^0, z^0), \qquad\quad \forall\, t \;\in\; [0,1]. \end{aligned}$$

As another consequence of Theorem **11.4.8**, we consider the case where H is affine:

$$H(x,y,z) \;\equiv\; q + Ax + By + Cz, \quad \forall\,(\,x,y,z\,) \;\in\; \mathbb{R}^{2n+m}, \qquad (11.4.9)$$

for some matrices A and B of order $(n+m) \times n$, a matrix C of order $(n+m) \times m$, and a vector $q \in \mathbb{R}^{n+m}$. In this case, the (CC) property of H can be satisfied by assuming a property on the triple (A, B, C) that involves the column representative matrices of the pair (A, B).

11.4.9 Corollary. Suppose that $[A \;\; B \;\; C]$ has the mixed P_0 property and that for every column representative matrix E of the pair (A, B),

$$\left.\begin{aligned} Eu + Cv \;&=\; 0 \\ u \;&\geq\; 0 \end{aligned}\right\} \;\Rightarrow\; (\,u, v\,) = 0. \qquad (11.4.10)$$

Theorem **11.4.8** holds for the affine mapping (11.4.9).

Proof. It suffices to note that H satisfies the (CC) condition. The proof of this property is rather easy by a standard normalization followed by a limiting argument. □

11.4.10 Remark. The implication (11.4.10) implies that C has full column rank. □

The condition on the column representative matrices of (A, B) is a generalization of the familiar R_0 property of a matrix. Indeed, when C is

vacuous and B is the negative identity matrix, this condition is equivalent to the R_0 property of A. As we recall, one characterization of an R_0 matrix A is that for all vectors q, the LCP (q, A) has a bounded (possibly empty) solution set; cf. Proposition **2.5.6** and the subsequent discussion. The following result generalizes this characterization to an arbitrary pair of matrices (A, B).

11.4.11 Proposition. The implication (11.4.10) holds for all column representative matrices E of the pair (A, B) if and only if for all vectors p in $\mathbb{R}^{n+m}$, the mixed, horizontal LCP:

$$\begin{aligned} p + Ax + By + Cz \;&=\; 0 \\ 0 \;\leq\; x \;\perp\; y \;&\geq\; 0 \end{aligned} \tag{11.4.11}$$

has a bounded (possible empty) solution set.

Proof. "Necessity." Suppose that for some vector $p \in \mathbb{R}^{n+m}$, there is an unbounded sequence $\{(x^k, y^k, z^k)\}$ of solutions to (11.4.11). By a standard normalization followed by a limiting argument, it is easy to exhibit a column representative matrix E of (A, B) and a nonzero pair (u, v) satisfying the left-hand condition in the implication (11.4.10). This is a contradiction.

"Sufficiency." Since every pair (u, v) satisfying the left-hand condition in (11.4.10) easily extends to a solution of (11.4.11) with $p = 0$, the assumption that the latter (homogeneous) problem has a bounded solution set, which must be a singleton because of homogeneity, implies that such a pair must be equal to zero. □

11.4.2 The monotone case

We next consider the same implicit CP (11.1.3) without assuming the differentiability of H. Moreover, instead of the mixed P_0 property of the Jacobian matrix of H, which no longer exists, we use the properties introduced below.

11.4.12 Definition. Let a set $U \subseteq X \times Y$ be given, where X and Y are two subsets of $\mathbb{R}^n$. A mapping $T : X \times Y \times \mathbb{R}^m \to \mathbb{R}^{n+m}$ is said to be

(a) (x, y) *equi-monotone* on U if for any (x, y, z) and (x', y', z') in $U \times \mathbb{R}^m$,

$$T(x, y, z) \;=\; T(x', y', z') \;\Rightarrow\; (\, x - x' \,)^T (\, y - y' \,) \;\geq\; 0;$$

(b) (x, y) *co-monotone* on U if there exist two continuous functions

$$h \,:\, U \times \mathbb{R}^m \to \mathbb{R}^{n+m} \quad \text{and} \quad c \,:\, \mathbb{R}^{2(n+m)} \to \mathbb{R}$$

with $c(v, v) = 0$ for all $v \in \mathbb{R}^{n+m}$ such that

$$\left.\begin{array}{l} T(x,y,z) \;=\; v \\ T(x',y',z') \;=\; v' \\ (x,y,z),(x',y',z') \in U \times \mathbb{R}^m \end{array}\right\} \Rightarrow$$

$$(x - x')^T(y - y') \;\geq\; (v - v')^T(h(x,y,z) - h(x',y',z')) + c(v,v');$$

(c) *z-injective* on U if for each $(x, y) \in U$, the function $T(x, y, \cdot)$ is injective on $\mathbb{R}^m$;

(d) *z-coercive* on U if for any sequence $\{(x^k, y^k, z^k)\} \subset U \times \mathbb{R}^m$

$$\left.\begin{array}{r} \{\,(\,x^k, y^k\,)\,\} \text{ bounded} \\ \lim\limits_{k\to\infty} \|\, z^k \,\| \;=\; \infty \end{array}\right\} \;\Rightarrow\; \lim_{k\to\infty} \|\, T(x^k, y^k, z^k) \,\| \;=\; \infty.$$

Clearly, if T is (x, y) co-monotone, then it must be (x, y) equi-monotone. Like the mixed P_0 property, the above definition of monotonicity may seem a little unusual at first glance. As we see subsequently, this definition is actually quite broad and includes the familiar monotonicity concept of a function of one argument and other matrix-theoretic properties; see part (a) of Proposition **11.4.18** and also Lemma **11.4.20**. For now, we make several elementary observations.

• If $m = 0$ and H is affine, say $H(x, y) \equiv Ax + By + q$ for some n by n matrices A and B, then the following three statements are equivalent:

(a) H is co-monotone on $\mathbb{R}^{2n}$;

(b) H is equi-monotone on $\mathbb{R}^{2n}$;

(c) the pair (A, B) is column monotone.

To prove that (c) $\Rightarrow$ (a), use the characterization of column monotonicity in Proposition **11.4.5**. Part (b) of Proposition **11.4.18** generalizes the implication (c) $\Rightarrow$ (a) to the case $m \neq 0$.

• If H is affine, say $H(x, y, z) \equiv Ax + By + Cz + q$ for some $(n + m) \times n$ matrices A and B and some $(n+m) \times n$ matrix C, then the following three statements are equivalent:

(a) H is z-injective on $\mathbb{R}^{2n}$;

(b) H is z-coercive on $\mathbb{R}^{2n}$;

(c) C has full column rank.

• If $H(x,y,z)$ satisfies the (CC) condition, then H is (x,y)-coercive on $\mathbb{R}^{2n}_+$.

The following simple lemma is stated without proof.

11.4.13 Lemma. Assume that (x,y) and $(\hat{x},\hat{y})$ are two points in $\mathbb{R}^n \times \mathbb{R}^n$ satisfying $(x-\hat{x})^T(y-\hat{y}) \geq 0$. If $(x,y) > 0$, $(\hat{x},\hat{y}) > 0$ and $x \circ y = \hat{x} \circ \hat{y}$ then $(x,y) = (\hat{x},\hat{y})$. □

In the next lemma, we present two important properties of the mapping $\mathbf{H}_{\mathrm{IP}}$ defined in (11.1.4) under the assumptions of H described in Definition **11.4.12**.

11.4.14 Lemma. Suppose that the mapping $H : \mathbb{R}^{2n}_+ \times \mathbb{R}^m \to \mathbb{R}^{n+m}$ is continuous, (x,y) equi-monotone on $\mathbb{R}^{2n}_+$, z-injective on $\mathbb{R}^{2n}_{++}$, and z-coercive on $\mathbb{R}^{2n}_+$. Then $\mathbf{H}_{\mathrm{IP}} : \mathbb{R}^{2n}_+ \times \mathbb{R}^m \to \mathbb{R}^n_+ \times \mathbb{R}^{n+m}$ is injective on $\mathbb{R}^{2n}_{++} \times \mathbb{R}^m$ and proper with respect to $\mathbb{R}^n_+ \times H_{++}$.

Proof. We first show that $\mathbf{H}_{\mathrm{IP}}$ is injective on $\mathbb{R}^{2n}_{++} \times \mathbb{R}^m$. Let (x,y,z) and (x',y',z') be two vectors belonging to $\mathbb{R}^{2n}_{++} \times \mathbb{R}^m$ such that $\mathbf{H}_{\mathrm{IP}}(x,y,z)$ is equal to $\mathbf{H}_{\mathrm{IP}}(x',y',z')$. We have

$$x \circ y \;=\; x' \circ y' \quad \text{and} \quad H(x,y,z) \;=\; H(x',y',z').$$

The equi-monotonicity of H then implies $(x-y)^T(x'-y') \geq 0$. By Lemma **11.4.13**, we obtain $x = x'$ and $y = y'$. The z-injectiveness of H on $\mathbb{R}^{2n}_{++}$ then implies $z = z'$. We have thus shown that $(x,y,z) = (x',y',z')$, which shows that G is injective on $\mathbb{R}^{2n}_{++} \times \mathbb{R}^m$. We next show that $\mathbf{H}_{\mathrm{IP}}$ is proper with respect to $\mathbb{R}^n_+ \times H_{++}$. Indeed, let S be a compact subset of $\mathbb{R}^n_+ \times H_{++}$. By continuity of $\mathbf{H}_{\mathrm{IP}}$, it follows that $\mathbf{H}^{-1}_{\mathrm{IP}}(S)$ is a closed set. We next show that $\mathbf{H}^{-1}_{\mathrm{IP}}(S)$ is a bounded set; this then implies that $\mathbf{H}^{-1}_{\mathrm{IP}}(S)$ is a compact set, and hence that $\mathbf{H}_{\mathrm{IP}}$ is proper with respect to $\mathbb{R}^n_+ \times H_{++}$. Suppose for contradiction that there exists a sequence $\{(x^k,y^k,z^k)\} \subset \mathbf{H}^{-1}_{\mathrm{IP}}(S)$ for which

$$\lim_{k\to\infty} \| (x^k,y^k,z^k) \| \;=\; \infty.$$

Since S is compact, we may assume without loss of generality that there exists a vector $H^\infty \in H_{++}$ such that

$$H^\infty \;=\; \lim_{k\to\infty} H(x^k,y^k,z^k).$$

By the definition of H_{++}, there exists $(x^\infty,y^\infty,z^\infty) \in \mathbb{R}^{2n}_{++} \times \mathbb{R}^m$ such that $H(x^\infty,y^\infty,z^\infty) = H^\infty$. Let

$$\varepsilon \;\equiv\; \tfrac{1}{2} \min\{ \min(x^\infty_i, y^\infty_i) \,:\, 1 \leq i \leq n \},$$

and let $\mathbb{B}^\infty$ be the open ball with center at (x^∞, y^∞) and radius ε. Clearly, $\mathbb{B}^\infty \times \mathbb{R}^m \subset \mathbb{R}^{2n}_{++} \times \mathbb{R}^m$. Moreover, since $\mathbf{H}_{\rm IP}$ is injective, it follows from the domain invariance theorem that $\mathbf{H}_{\rm IP}(\mathbb{B}^\infty \times \mathbb{R}^m)$ is an open set, and thus so is $H(\mathbb{B}^\infty \times \mathbb{R}^m)$. Consequently, for all k sufficiently large, $H(x^k, y^k, z^k)$ belongs to $H(\mathbb{B}^\infty \times \mathbb{R}^m)$. Hence there exists $(\tilde{x}^k, \tilde{y}^k, \tilde{z}^k)$ in $\mathbb{B}^\infty \times \mathbb{R}^m$ such that $H(x^k, y^k, z^k) = H(\tilde{x}^k, \tilde{y}^k, \tilde{z}^k)$. By the (x, y) equi-monotonicity of H, we deduce

$$(\, x^k - \tilde{x}^k \,)^T (\, y^k - \tilde{y}^k \,) \geq 0,$$

which yields

$$(x^k)^T \tilde{y}^k + (y^k)^T \tilde{x}^k \leq (\tilde{x}^k)^T \tilde{y}^k + (x^k)^T y^k. \tag{11.4.12}$$

Since $(x^k \circ y^k, H(x^k, y^k, z^k)) \in S$, it follows that $\{((x^k)^T y^k, H(x^k, y^k, z^k))\}$ is bounded. Moreover, we must have

$$\min(\tilde{x}^k_i, \tilde{y}^k_i) \leq \varepsilon \quad \text{and} \quad (\tilde{x}^k)^T \tilde{y}^k \leq (x^\infty)^T y^\infty + n\,\varepsilon.$$

for all k sufficiently large and all $i = 1, \ldots, n$. Consequently, by (11.4.12), it follows that $\{(x^k, y^k)\}$ is bounded. Thus, by the z-coerciveness condition and the fact that $\{H(x^k, y^k, z^k)\}$ is bounded, it follows that $\{z^k\}$ must also be bounded. But this is a contradiction. Consequently, $\mathbf{H}^{-1}_{\rm IP}(S)$ is bounded. □

Unlike the differentiable case where the (CC) condition was in place, the set H_{++} is not necessarily convex under the equi-monotonicity assumption stated above. This is illustrated by the following example.

11.4.15 Example. Let $n = 2$ and $m = 0$ and let H be given by

$$H(x, y) \equiv \begin{pmatrix} x_1 + x_2 \\ x_1 + x_2 + x_2^3 \end{pmatrix}, \quad \text{for } (x, y) \in \mathbb{R}^2 \times \mathbb{R}^2.$$

It is trivial to verify that $H(x, y) = H(x', y')$ implies $x = x'$. Thus H is equi-monotone on $\mathbb{R}^2 \times \mathbb{R}^2$. Nevertheless, the set

$$H(\mathbb{R}^4_{++}) = \{\, (\, r + s, r + s + s^3 \,) : (r, s) > 0 \,\}$$

is easily seen to be nonconvex. □

In spite of the possible nonconvexity of the set H_{++}, we can still establish the solvability of CE $(\mathbf{H}_{\rm IP}, X)$ under a "strict feasibility" condition. We note that the two conclusions in the next result, Theorem **11.4.16**,

are not as strong as the corresponding conclusions in Theorem **11.4.8**. The main reason why we can no longer retain the same conclusions is due to the restricted properness of $\mathbf{H}_{\rm IP}$ under the assumptions made in Theorem **11.4.16** below. Indeed, by Lemma **11.4.14**, $\mathbf{H}_{\rm IP}$ is proper with respect to $\mathbb{R}^n_+ \times H_{++}$, which is a subset of $\mathbb{R}^n_+ \times \mathbb{R}^{n+m}$. Moreover, since the equi-monotonicity of H is not enough to ensure the convexity of H_{++}, there is no obvious way of identifying a convex set S in the range space of $\mathbf{H}_{\rm IP}$ satisfying the assumptions (IP2) and (IP3) and such that $\mathbf{H}_{\rm IP}$ is proper with respect to a convex subset E of S. Thus, we can not directly apply the CE theory as presented in Section **11.2**, in particular, Theorem **11.2.4**, like we did in the previous case where differentiability was assumed. Instead, the proof of the next theorem resorts to the main result, Theorem **11.2.1**.

11.4.16 Theorem. Suppose that the mapping $H : \mathbb{R}^{2n}_+ \times \mathbb{R}^m \to \mathbb{R}^{n+m}$ is continuous, (x, y) equi-monotone on $\mathbb{R}^{2n}_+$, z-injective on $\mathbb{R}^{2n}_{++}$, and z-coercive on $\mathbb{R}^{2n}_+$. The following two statements hold:

(a) $\mathbf{H}_{\rm IP}$ maps $\mathbb{R}^{2n}_{++} \times \mathbb{R}^m$ onto $\mathbb{R}^n_{++} \times H_{++}$ homeomorphically;

(b) there holds

$$\mathbb{R}^n_+ \times H_{++} \subseteq \mathbf{H}_{\rm IP}(\mathbb{R}^{2n}_+ \times \mathbb{R}^m); \tag{11.4.13}$$

consequently, if $0 \in H_{++}$ the system (11.4.8) has a solution for every vector $a \in \mathbb{R}^n_+$; in particular, so does the problem (11.1.3).

Proof. By Lemma **11.4.14** and the domain invariance theorem, it follows that $\mathbf{H}_{\rm IP}$ maps $\mathbb{R}^{2n}_{++} \times \mathbb{R}^m$ homeomorphically onto $\mathbf{H}_{\rm IP}(\mathbb{R}^{2n}_{++} \times \mathbb{R}^m)$, which is clearly a subset of $\mathbb{R}^n_{++} \times H_{++}$. In particular, it follows that $\mathbf{H}_{\rm IP}$ restricted to the pair $(\mathbb{R}^{2n}_{++} \times \mathbb{R}^m, \mathbb{R}^n_{++} \times H_{++})$ is a local homeomorphism. Moreover, this restricted map is also proper due to Lemma **11.4.14**. Let

$$\begin{aligned} &M \equiv \mathbb{R}^{2n}_+ \times \mathbb{R}^m, \quad M_0 \equiv \mathbb{R}^{2n}_{++} \times \mathbb{R}^m, \quad N \equiv \mathbb{R}^n_+ \times H_+, \\ &N_0 \equiv \mathbb{R}^n_{++} \times H_{++}, \quad E \equiv \mathbb{R}^n_+ \times H_{++}, \end{aligned}$$

and the mapping F be the restriction of $\mathbf{H}_{\rm IP}$ to the pair (M, N). Note that N_0 is path-connected, thus connected. Hence by Theorem **11.2.1**, it follows that $\mathbb{R}^n_{++} \times H_{++}$ is contained in $\mathbf{H}_{\rm IP}(\mathbb{R}^{2n}_{++} \times \mathbb{R}^m)$. Thus equality must hold between these two sets, and statement (a) follows. Moreover, by the same theorem, we also have $\mathbb{R}^n_+ \times H_{++} \subseteq \mathbf{H}_{\rm IP}(\mathbb{R}^{2n}_+ \times \mathbb{R}^m)$, which clearly implies (11.4.13). The last statement of the theorem follows easily from (11.4.13). □

If the set H_{++} is convex and contains the origin, then Theorem **11.4.16** can be proved by applying Theorem **11.2.2** with the following identifica-

tions:

$$X \equiv \mathbb{R}^{2n}_{+} \times \mathbb{R}^{m}, \quad S \equiv \mathbb{R}^{n}_{+} \times \mathbb{R}^{n+m} \quad \text{and} \quad E \equiv \mathbb{R}^{n}_{+} \times H_{++}.$$

Note that $\mathbf{H}_{\mathrm{IP}}$ is not necessarily proper with respect to S, only with respect to E, which is not necessarily closed.

Part of the reason why the set H_{++} is not convex under the (x, y) equimonotonicity stated above is that this condition is rather weak; more specifically, it does not say anything about two triples (x, y, z) and (x', y', z') when $H(x, y, z) \neq H(x', y', z')$. The following result shows that H_{++} is indeed convex under the stronger assumption of co-monotonicity.

11.4.17 Proposition. If the mapping $H : \mathbb{R}^{2n}_{+} \times \mathbb{R}^{m} \to \mathbb{R}^{n+m}$ is continuous, co-monotone on $\mathbb{R}^{2n}_{+}$, z-injective on $\mathbb{R}^{2n}_{++}$, and z-coercive on $\mathbb{R}^{2n}_{+}$, then the set H_{++} is convex.

Proof. It suffices to show that the set $\mathbf{H}_{\mathrm{IP}}(\mathbb{R}^{2n}_{++} \times \mathbb{R}^{m})$ is convex. Let $M \equiv \mathbb{R}^{2n}_{++} \times \mathbb{R}^{m}$. The map $F \equiv (\mathbf{H}_{\mathrm{IP}})|_{\mathbb{R}^{2n}_{++} \times \mathbb{R}^{m}}$ is a homeomorphism due to Theorem **11.4.16**(a). We claim that for any two triples (x^i, y^i, z^i) in M, $i = 1, 2$, the set $F^{-1}(E)$ is compact, where E is the line segment $[F(x^1, y^1, z^1), F(x^2, y^2, z^2)]$. Let (x, y, z) be an arbitrary triple in the set $\mathbf{H}^{-1}_{\mathrm{IP}}(E) \supseteq F^{-1}(E)$. By the definition of $\mathbf{H}_{\mathrm{IP}}$, there exists $\tau \in [0, 1]$ such that

$$\begin{aligned} x \circ y &= \tau(x^1 \circ y^1) + (1 - \tau)(x^2 \circ y^2) \\ H(x, y, z) &= \tau H(x^1, y^1, z^1) + (1 - \tau) H(x^2, y^2, z^2). \end{aligned}$$

Using the first relation and the fact that both (x^1, y^1) and (x^2, y^2) belong to $\mathbb{R}^{2n}_{++}$, we easily see that $(x, y) \in \mathbb{R}^{2n}_{++}$. This implies that (x, y, z) belongs to $F^{-1}(E)$, and hence that $\mathbf{H}^{-1}_{\mathrm{IP}}(E) = F^{-1}(E)$. Extending this argument and using the continuity of $\mathbf{H}_{\mathrm{IP}}$, we can deduce that $\mathbf{H}^{-1}_{\mathrm{IP}}(E) = F^{-1}(E)$ is closed. Next, we show that $F^{-1}(E)$ is bounded. Let (x, y, z) be an arbitrary element of this set. Write $b^i \equiv H(x^i, y^i, z^i)$ for $i = 1, 2$, and $b(\tau) = \tau b^1 + (1 - \tau) b^2$. By the co-monotonicity of H, we have

$$\begin{aligned} &(x^1 - x)^T (y^1 - y) \\ &\quad \geq (1 - \tau)\, (b^1 - b^2)^T (h(x^1, y^1, z^1) - h(x, y, z)) + c(b^1, b(\tau)), \end{aligned}$$

which implies

$$x^T y^1 + y^T x^1 \leq c_1 + (1 - \tau)\, (b^1 - b^2)^T h(x, y, z) \tag{11.4.14}$$

where

$$\begin{aligned} c_1 \equiv\ & (x^1)^T y^1 + \max((x^1)^T y^1, (x^2)^T y^2) + \\ & \max_{\tau \in [0,1]} |\, c(b^1, b(\tau))\, | + |\, (b^1 - b^2)^T h(x^1, y^1, z^1)\, |. \end{aligned}$$

Similarly, we can deduce

$$x^T y^2 + y^T x^2 \leq c_2 + \tau\,(b^2 - b^1)^T h(x, y, z) \tag{11.4.15}$$

where

$$\begin{aligned} c_2 \equiv \quad & (x^2)^T y^2 + \max((x^1)^T y^1, (x^2)^T y^2) + \\ & \max_{\tau \in [0,1]} |\, c(b^2, b(\tau))\,| + |\,(b^1 - b^2)^T h(x^2, y^2, z^2)\,|. \end{aligned}$$

Multiplying (11.4.14) by τ and (11.4.15) by $1 - \tau$ and adding, we deduce

$$(\,\tau y^1 + (1-\tau) y^2\,)^T x + (\,\tau x^1 + (1-\tau) x^2\,)^T y \leq \max(C_1, C_2).$$

Thus (x, y) is bounded. By the z-coerciveness of H , so is the vector z.

Consequently, applying Theorem **11.2.1** with

$$M_0 \equiv M, \quad N \equiv \mathbb{R}^n_{++} \times H_{++} \quad \text{and} \quad N_0 \equiv E,$$

we deduce that $\mathbf{H}_{\mathrm{IP}}(\mathbb{R}^{2n}_{++} \times \mathbb{R}^m)$ contains the line segment joining the two vectors $\mathbf{H}_{\mathrm{IP}}(x^1, y^1, z^1)$ and $\mathbf{H}_{\mathrm{IP}}(x^2, y^2, z^2)$, for any two vectors (x^i, y^i, z^i) in $\mathbb{R}^{2n}_{++} \times \mathbb{R}^m$. This establishes the convexity of $\mathbf{H}_{\mathrm{IP}}(\mathbb{R}^{2n}_{++} \times \mathbb{R}^m)$. □

An upshot of the theory developed so far for the implicit MiCP (11.1.3) is that under the assumptions of Proposition **11.4.17**, the conditions (IP1)–(IP3) are satisfied and the CE mapping $\mathbf{H}_{\mathrm{IP}}$ is proper with respect to a convex set in the range space. Hence this CP is amenable to the IP methods under the the further assumption of differentiability of H. Details appear subsequently.

In what follows, we show that the co-monotonicity property holds in three familiar cases of the mapping H; these correspond to the monotone, mixed CP:

$$\begin{aligned} 0 \leq x \perp f(x, z) \geq 0 \\ g(x, z) = 0, \end{aligned} \tag{11.4.16}$$

where $(f, g) : \mathbb{R}^{n+m} \to \mathbb{R}^{n+m}$ is a monotone mapping; the vertical CP (F, G):

$$0 \leq F(x) \perp G(x) \geq 0 \tag{11.4.17}$$

defined by a pair of "jointly monotone" mappings F, $G : \mathbb{R}^n \to \mathbb{R}^n$; see (11.4.18); and the monotone, mixed, horizontal LCP:

$$\begin{aligned} 0 = q + Ax + By + Cz \\ 0 \leq x \perp y \geq 0, \end{aligned}$$

where the triple (A, B, C) satisfies a certain monotonicity property; see part (c) of the next proposition.

11.4.18 Proposition. The co-monotonicity property holds on X if H is of any one of the following three forms:

(a) $X \equiv \mathbb{R}^{2n}_+$ and

$$H(x,y,z) \equiv \begin{pmatrix} f(x,z) - y \\ g(x,z) \end{pmatrix},$$

where $f : \mathbb{R}^n_+ \times \mathbb{R}^m \to \mathbb{R}^n$ and $g : \mathbb{R}^n_+ \times \mathbb{R}^m \to \mathbb{R}^m$ satisfy the standard monotonicity assumption: for any two pairs (x,z) and (x',z') in $\mathbb{R}^n_+ \times \mathbb{R}^m$,

$$(x - x')^T(f(x,z) - f(x',z')) + (z - z')^T(g(x,z) - g(x',z')) \geq 0;$$

(b) $X \equiv \mathbb{R}^{2n}$ and

$$H(u,v,x) \equiv \begin{pmatrix} u - F(x) \\ v - G(x) \end{pmatrix},$$

where F and $G : \mathbb{R}^n \to \mathbb{R}^n$ are *jointly monotone* on $\mathbb{R}^n$; that is,

$$(F(x) - F(x'))^T(G(x) - G(x')) \geq 0, \quad \forall\, x, x' \in \mathbb{R}^m; \quad (11.4.18)$$

(c) $X \equiv \mathbb{R}^{2n}$ and

$$H(x,y,z) \equiv \begin{pmatrix} A_1x + B_1y + C_1z + q_1 \\ A_2x + B_2y + C_2z + q_2 \end{pmatrix},$$

where C_1 is a nonsingular matrix of order m and the pair

$$A_2' \equiv A_2 - C_2C_1^{-1}A_1, \quad \text{and} \quad B_2' \equiv B_2 - C_2C_1^{-1}B_1,$$

is column monotone.

Proof. To prove that the mapping H given in (a) is (x,y) co-monotone on $\mathbb{R}^{2n}_+$, define $h : \mathbb{R}^{2n}_+ \times \mathbb{R}^m \to \mathbb{R}^{n+m}$ and $c : \mathbb{R}^{2(n+m)} \to \mathbb{R}$ by

$$h(x,y,z) \equiv -(x,z) \quad \text{and} \quad c(x,y,z) \equiv 0.$$

Suppose $H(x^i,y^i,z^i) = (r^i,s^i)$ for $i = 1,2$. We have

$$\begin{aligned}&(x^1 - x^2)^T(f(x^1,z^1) - f(x^2,z^2)) + (z^1 - z^2)^T(g(x^1,z^1) - g(x^2,z^2)) \\ &= (x^1 - x^2)^T(y^1 - y^2 + r^1 - r^2) + (z^1 - z^2)^T(s^1 - s^2).\end{aligned}$$

By the monotonicity of (f,g), the left-hand expression is nonnegative. Hence, rearranging terms in the right-hand expression, we deduce

$$\begin{aligned}&(x^1 - x^2)^T(y^1 - y^2) \geq \\ &\quad (H(x^1,y^1,z^1) - H(x^2,y^2,z^2))^T(h(x^1,y^1,z^1) - h(x^2,y^2,z^2)).\end{aligned}$$

Next, we prove that the mapping H given in (b) is co-monotone on $\mathbb{R}^{2n}$. Let $H(u^i, v^i, x^i) \equiv (r^i, s^i)$ for $i = 1, 2$. We have

$$u^1 - u^2 \;=\; F(x^i) - F(x^2) + r^1 - r^2 \quad \text{and} \quad v^1 - v^2 \;=\; G(x^i) - G(x^2) + s^1 - s^2.$$

Thus,

$$\begin{aligned}
&(\, u^1 - u^2 \,)^T (\, v^1 - v^2 \,) \\
&= (\, u^1 - u^2 \,)^T [\, (\, G(x^i) - G(x^2) + s^1 - s^2 \,] \\
&= (\, F(x^i) - F(x^2) \,)^T (\, G(x^i) - G(x^2) \,) + (\, r^1 - r^2 \,)^T (\, G(x^i) - G(x^2) \,) \\
&\qquad + (\, s^1 - s^2 \,)^T (\, u^1 - u^2 \,).
\end{aligned}$$

Hence with

$$h(u, v, x) \;\equiv\; \begin{pmatrix} G(x) \\ u \end{pmatrix} \quad \text{and} \quad c(u, v, x) \;\equiv\; 0,$$

it follows that H is (u, v) co-monotone on $\mathbb{R}^{2n}$.

Finally, we prove that the mapping H given in (c) is (x, y) co-monotone on $\mathbb{R}^{2n}$. By Proposition **11.4.5**, we may assume without loss of generality that A_2' is nonsingular and that $-(A_2')^{-1}B_2'$ is positive semidefinite. As before, let $H(x^i, y^i, z^i) = (r^i, s^i)$ for $i = 1, 2$. By a simple manipulation, we can easily deduce that

$$x^1 - x^2 = -(A_2')^{-1}B_2'(y^1 - y^2) + (A_2')^{-1}[(s^1 - s^2) - C_2(C_1)^{-1}(r^1 - r^2)].$$

Hence by defining

$$h(x, y, z) \equiv \begin{pmatrix} (\,(\,A_2'\,)^T\,)^{-1} y \\ -(\,[\,(\,A_2'\,)^{-1} C_2 (\,C_1\,)^{-1}\,]^T\,)^{-1} y \end{pmatrix},$$

and $c(x, y, z) \equiv 0$, we can easily verify the desired co-monotonicity property. □

Note the two different ways of writing the mapping H in parts (a) and (b) of the above proposition. Specifically, in part (a), we write the condition $f(x, z) \geq 0$ (and similarly for $g(x, z) \geq 0$) as $[f(x, z) - y = 0$ and $y \geq 0]$ with a negative sign associated with the slack variable y in the equation. In part (b), we write the condition $F(x) \geq 0$ (and similarly for $G(x) \geq 0$) as $[\, u - F(x) = 0$ and $u \geq 0\,]$ with a positive sign in front of the slack variable u. The resulting mapping H is co-monotone in both forms.

Based on Theorem **11.4.16** and Propositions **11.4.17** and **11.4.18**, we can easily establish the following corollary that pertains to the two special CPs (11.4.16) and (11.4.17).

11.4.19 Corollary. The following two statements hold.

(a) Let $(f, g) : \mathbb{R}^n_+ \times \mathbb{R}^m \to \mathbb{R}^{n+m}$ be continuous and monotone. Suppose that for each $x \in \mathbb{R}^n_+$, the mapping

$$\begin{pmatrix} f(x,\cdot) \\ g(x,\cdot) \end{pmatrix} : \mathbb{R}^m \to \mathbb{R}^{n+m}$$

is injective and norm-coercive. Theorem **11.4.16** holds for the map H defined in part (a) of Proposition **11.4.18** and the set H_{++} is convex. In particular, if there exists a pair $(\tilde{x}, \tilde{z})$ with $\tilde{x} > 0$ such that $f(\tilde{x}, \tilde{z}) > 0$ and $g(\tilde{x}, \tilde{z}) = 0$, then for every vector $a \in \mathbb{R}^n_+$, the system

$$\begin{aligned} H(x, y, z) &= 0 \\ x \circ y &= a \\ (\, x, y \,) &\geq 0 \end{aligned}$$

has a solution.

(b) Let F, $G : \mathbb{R}^m \to \mathbb{R}^n$ be continuous and jointly monotone on $\mathbb{R}^n$. Suppose that the map

$$\begin{pmatrix} F \\ G \end{pmatrix} : \mathbb{R}^n \to \mathbb{R}^{2n}$$

is injective and norm coercive. Theorem **11.4.16** holds for the map H defined in part (b) of Proposition **11.4.18** and the set H_{++} is convex. In particular, if there exists $\tilde{x}$ such that $F(\tilde{x}) > 0$ and $G(\tilde{x}) > 0$, then for every vector $a \in \mathbb{R}^n_+$, the system

$$\begin{aligned} (\, F(x), G(x) \,) &\geq 0 \\ F(x) \circ G(x) &= a \end{aligned}$$

has a solution.

Proof. It suffices to note that the respective map H satisfies all the assumptions in Theorem **11.4.16** and Proposition **11.4.17**. □

11.4.3 The KKT map

We next discuss the the mapping G given by (11.1.2) that corresponds to the KKT system of a monotone VI (K, F) defined on the solution set K of a system of finitely many differentiable convex inequalities and by a

monotone mapping F; this discussion covers the case of a convex program. For the mapping (11.1.2), the point of the discussion is not to establish the existence of solutions to the CE (G, X); instead, the focus is on discovering properties of the corresponding mapping $H : \mathbb{R}^{2m}_+ \times \mathbb{R}^{n+\ell} \to \mathbb{R}^{m+n+\ell}$ defined by: for all $(\lambda, w, x, \mu) \in \mathbb{R}^{2m}_+ \times \mathbb{R}^{n+\ell}$,

$$H(\lambda, w, x, \mu) \equiv \begin{pmatrix} w + g(x) \\ \mathbf{L}(x, \mu, \lambda) \\ h(x) \end{pmatrix}, \tag{11.4.19}$$

that facilitate the application of the IP algorithm for solving the VI (K, F) via its KKT system. Although the treatment of the map H of (11.4.19) is very similar to that of the mixed CP in part (a) of Proposition **11.4.18**, due to the change of notation and the importance of the map (11.4.19) for a VI, we give a full consideration of this map and summarize a main result in Corollary **11.4.24**.

A word of caution about the association of variables is in order. The pair (λ, w) in (11.4.19) is the pair (x, y) in Definition **11.4.12**, and the pair (x, μ) in (11.4.19) is the z variable in Definition **11.4.12**. With this identification of variables, we first establish the (λ, w) co-monotonicity of H under suitable assumptions on F, g and h. The proof of the following lemma is very similar to that of parts (a) and (b) of Proposition **11.4.18**.

11.4.20 Lemma. Suppose that the function $F : \mathbb{R}^n \to \mathbb{R}^n$ is continuous and monotone, each $g_i : \mathbb{R}^n \to \mathbb{R}$ is continuously differentiable and convex, and $h : \mathbb{R}^n \to \mathbb{R}^\ell$ is affine. The mapping H defined by (11.4.19) is (λ, w) co-monotone on $\mathbb{R}^{2m}_+$.

Proof. Suppose

$$H(\lambda, w, x, \mu) = r \quad \text{and} \quad H(\lambda', w', x', \mu') = r'$$

for some (λ, w, x, μ) and (λ', w', x', μ') with (λ, w) and (λ', w') nonnegative. Write

$$r \equiv \begin{pmatrix} p \\ q \\ s \end{pmatrix} \quad \text{and} \quad r' \equiv \begin{pmatrix} p' \\ q' \\ s' \end{pmatrix}$$

in accordance with the partitioning of the components of H. Since h is affine, we have for all j,

$$\begin{aligned} s_j - s'_j &= h_j(x) - h_j(x') \\ &= \nabla h_j(x')^T (x - x') = \nabla h_j(x)^T (x - x'). \end{aligned}$$

We also have $w - w' = p - p' - g(x) + g(x')$, which implies

$$(\lambda' - \lambda)^T(w - w') = (\lambda - \lambda')^T(g(x) - g(x') - p + p').$$

Since

$$F(x) - F(x') + \sum_{i=1}^{m}(\lambda_i \nabla g_i(x) - \lambda_i' \nabla g_i(x')) + \sum_{j=1}^{\ell}(\mu_j - \mu_j')\nabla h_j(x) = q - q'$$

we deduce, by the last three expressions and a simple algebraic manipulation,

$$\begin{aligned}
&-(\lambda - \lambda')^T(p - p') + (x - x')^T(q - q') - (\mu - \mu')^T(s - s') \\
&= (x - x')^T(F(x) - F(x')) + (\lambda - \lambda')^T(g(x) - g(x') - p + p') + \\
&\quad \sum_{i=1}^{m} \lambda_i' \left[g_i(x) - g_i(x') - \nabla g_i(x')^T(x - x') \right] + \\
&\quad \sum_{i=1}^{m} \lambda_i \left[g_i(x') - g_i(x) - \nabla g_i(x)^T(x' - x) \right] \\
&\geq -(\lambda - \lambda')^T(w - w'),
\end{aligned}$$

where the last inequality follows from the monotonicity of F, the convexity of each g_i and the nonnegativity of λ_i and λ_i'. Consequently, with

$$h(\lambda, w, x, \mu) \equiv \begin{pmatrix} \lambda \\ x \\ \mu \end{pmatrix} \quad \text{and} \quad c(\lambda, w, x, \mu) \equiv 0,$$

it follows easily that H is (λ, w) co-monotone on $\mathbb{R}^{2m}_+$. □

The nonnegativity of λ plays an essential role in the above proof. In particular, we can not establish that H is (λ, w) co-monotone on the entire space $\mathbb{R}^{2m}$. The positivity of the pair (λ, μ) plays a similarly important role in the next result, which establishes the (x, μ)-injectiveness of H on $\mathbb{R}^{2m}_{++}$ under the strict monotonicity assumption of F (or the strict convexity of some g_i) and a full row rank assumption on the Jacobian matrix $Jh(x)$, the latter being a constant for all vectors x because of the affinity assumption of h.

11.4.21 Lemma. Suppose that the function $F : \mathbb{R}^n \to \mathbb{R}^n$ is continuous and monotone, each $g_i : \mathbb{R}^n \to \mathbb{R}$ is continuously differentiable and convex,

and $h : \mathbb{R}^n \to \mathbb{R}^\ell$ is affine. Assume that either F is strictly monotone or one of the functions g_i is strictly convex. Assume also that the vectors $\{\nabla h_j(x) : j = 1, \ldots, \ell\}$ are linearly independent. The mapping H of (11.4.19) is (x, μ)-injective on $\mathbb{R}^{2m}_{++}$.

Proof. Suppose $H(\lambda, w, x, \mu) = H(\lambda, w, x', \mu')$ for some $(\lambda, w) > 0$. By using either the strict monotonicity of F or the strict convexity of some g_i, the fact that $\lambda > 0$ and the proof of Lemma **11.4.20**, we can deduce that $x = x'$. Furthermore, since $\mathbf{L}(x, \mu, \lambda) = \mathbf{L}(x', \mu', \lambda)$, by the linear independence assumption of the vectors $\nabla h_j(x)$, we easily deduce $\mu = \mu'$. □

The next lemma establishes the (x, μ)-coerciveness of H under the same rank assumption on $Jh(x)$ and the assumption that the set

$$K \equiv \{ x \in \mathbb{R}^n : g(x) \leq 0,\ h(x) = 0 \} \tag{11.4.20}$$

is bounded.

11.4.22 Lemma. Suppose that the function $F : \mathbb{R}^n \to \mathbb{R}^n$ is continuous, $h : \mathbb{R}^n \to \mathbb{R}^\ell$ is affine, and each $g_i : \mathbb{R}^n \to \mathbb{R}$ is continuously differentiable and convex. Assume also that the vectors $\{\nabla h_j(x) : j = 1, \ldots, \ell\}$ are linearly independent and the set K defined in (11.4.20) is bounded and nonempty. The mapping H of (11.4.19) is (x, μ)-coercive on $\mathbb{R}^{2m}_{+}$.

Proof. By the convexity of each g_i and the affinity of h, it follows from convex analysis that for all vectors $c \in \mathbb{R}^m$ and nonnegative scalars d, the set

$$K_{c,d} \equiv \{ x \in \mathbb{R}^n : g(x) \leq c,\ \| h(x) \| \leq d \}$$

is bounded. Indeed, consider the convex function:

$$\varphi(x) \equiv \max \left\{ \| h(x) \|,\ \max_{1 \leq i \leq m} g_i(x) \right\}, \quad x \in \mathbb{R}^n,$$

whose level set

$$\{ x \in \mathbb{R}^n : \varphi(x) \leq 0 \} = K$$

is bounded and nonempty by assumption. A classical result of convex analysis states that if a convex function has a bounded level set, then all the level sets of that function are bounded. Thus, every level set of $\varphi(x)$ is bounded. Clearly, $K_{c,d}$ is one such level set. Hence $K_{c,d}$ is bounded.

Let $\{(\lambda^k, w^k, x^k, \mu^k)\}$ be a sequence such that each (λ^k, w^k) is nonnegative and the sequences $\{(\lambda^k, w^k)\}$ and $\{H(\lambda^k, w^k, x^k, \mu^k)\}$ are bounded. We need to show that $\{(x^k\, \mu^k)\}$ is bounded. Write $r^k \equiv w^k + g(x^k)$ and

$s^k \equiv h(x^k)$. Since $\{r^k\}$ and $\{s^k\}$ are both bounded, let $c \in \mathbb{R}^m_+$ and $d \in \mathbb{R}_+$ be such that $c \geq r^k$ and $d \geq \|s^k\|$ for all k. Then $\{x^k\} \subset K_{c,d}$. By what has been proved above, it follows that $\{x^k\}$ is bounded. Hence by the boundedness of the sequence $\{\mathbf{L}(x^k, \mu^k, \lambda^k)\}$ and the linear independence of the constant vectors $\{\nabla h_j(x^k)\}$, it follows that $\{\mu^k\}$ is bounded. □

Finally, we examine the assumption $0 \in H_{++}$ (required in part (b) of Theorem **11.4.16**) for the function H under consideration. In essence, the following result shows that this assumption is satisfied under the assumptions of Lemma **11.4.22** and the generalized Slater CQ on the set K in (11.4.20). Recall from the discussion following Proposition **3.2.7** that this CQ implies the existence of KKT multipliers at all solutions of a VI defined by such a set.

11.4.23 Lemma. Suppose that the function $F : \mathbb{R}^n \to \mathbb{R}^n$ is continuous, $h : \mathbb{R}^n \to \mathbb{R}^\ell$ is affine, and each $g_i : \mathbb{R}^n \to \mathbb{R}$ is continuously differentiable and convex. Assume also that the set K defined in (11.4.20) is bounded and there is a vector $\tilde{x} \in K$ such that $g(\tilde{x}) < 0$. There exists $(\lambda^*, w^*, x^*, \mu^*)$ with $(\lambda^*, w^*) > 0$ satisfying $H(\lambda^*, w^*, x^*, \mu^*) = 0$, where H is given by (11.4.19).

Proof. Let $c \equiv \frac{1}{2} g(\tilde{x})$ and write $\tilde{K} \equiv K_{c,0}$. Note that $g(x) < 0$ for all x in $\tilde{K}$. By the first part of the proof of Lemma **11.4.22** (which incidentally does not require the rank assumption of Jh), it follows that $\tilde{K}$ is nonempty, compact and convex. Define the function

$$\tilde{F}(x) \equiv F(x) - \sum_{i=1}^{m} g_i(x)^{-1} \nabla g_i(x), \quad x \in \tilde{K},$$

which is well defined and continuous on $\tilde{K}$. It follows that the VI $(\tilde{K}, \tilde{F})$ has a solution, say x^*. Moreover, since $g(\tilde{x}) < c$ and $h(\tilde{x}) = 0$, the set $\tilde{K}$ satisfies the generalized Slater CQ. Hence there exists a multiplier $(\mu^*, \tilde{\lambda})$ with $\tilde{\lambda} \geq 0$ such that

$$\tilde{\mathbf{L}}(x^*, \mu^*, \tilde{\lambda}) = 0,$$

where

$$\tilde{\mathbf{L}}(x, \mu, \lambda) \equiv \tilde{F}(x) + \sum_{i=1}^{m} \lambda_i \nabla g_i(x) + \sum_{j=1}^{\ell} \mu_j \nabla h_j(x).$$

By defining $\lambda_i^* \equiv \tilde{\lambda}_i - 1/g_i(x^*)$ for each i and $w^* \equiv -g(x^*) > 0$, it is easily seen that $(\lambda^*, w^*, x^*, \eta^*)$ is a desired vector satisfying the requirements of the lemma. □

Combining the above lemmas, we obtain the following result for the function H given by (11.4.19) that arises from a variational inequality defined by a monotone mapping F. This corollary follows easily from Theorem **11.4.16** and Proposition **11.4.17** and requires no further proof.

11.4.24 Corollary. Suppose that the function $F : \mathbb{R}^n \to \mathbb{R}^n$ is continuous and monotone, each $g_i : \mathbb{R}^n \to \mathbb{R}$ is continuously differentiable and convex, and $h : \mathbb{R}^n \to \mathbb{R}^\ell$ is affine. Assume that either F is strictly monotone or one of the functions g_i is strictly convex. Assume also that the vectors $\{\nabla h_j(x) : j = 1, \ldots, \ell\}$ are linearly independent, that there exists a vector $\tilde{x} \in \mathbb{R}^n$ satisfying $g(\tilde{x}) < 0$ and $h(\tilde{x}) = 0$, and that the set K defined in (11.4.20) is bounded. The following four statements hold for the functions G and H defined in (11.1.2) and (11.4.19), respectively:

(a) G maps $\mathbb{R}^{2m}_{++} \times \mathbb{R}^{n+\ell}$ homeomorphically onto $\mathbb{R}^m_{++} \times H_{++}$;

(b) $\mathbb{R}^m_+ \times H_{++} \subseteq G(\mathbb{R}^{2m}_+ \times \mathbb{R}^{n+\ell})$;

(c) for every vector $a \in \mathbb{R}^m_+$, the system

$$H(\lambda, w, x, \mu) \;=\; 0, \quad w \circ \lambda = a$$

has a solution $(\lambda, w, x, \eta) \in \mathbb{R}^{2m}_+ \times \mathbb{R}^{n+\ell}$; and

(d) the set H_{++} is convex. □

All the results developed in this section can be extended to the CP in SPSD matrices. Basically, we only need to follow the same line of derivations and apply the main Theorem **11.2.1** or its corollaries, Theorems **11.2.2** or **11.2.4**. However, the development is very technical and requires careful verification of many matrix-theoretic details. See Section **11.10** for notes and comments.

11.5 IP Algorithms for the Implicit MiCP

Armed with the results established in the previous section, we apply Algorithm **11.3.2** to the implicit MiCP (11.1.3) based on its equivalent formulation as the CE (G, X) with $X \equiv \mathbb{R}^{2n}_+ \times \mathbb{R}^m$ and G given by (11.1.4). The blanket assumption we make on the mapping H is that it is continuously differentiable on $\operatorname{int} X$ and that the Jacobian matrix $JH(x, y, z)$ in partitioned form:

$$JH(x, y, z) \;=\; [\, J_x H(x, y, z) \quad J_y H(x, y, z) \quad J_z H(x, y, z) \,]$$

has the mixed P_0 property for every $(x, y, z) \in \operatorname{int} X$. Let $S \equiv \mathbb{R}^n_+ \times \mathbb{R}^{n+m}$ and $\zeta > n/2$. Define for every $(u, w) \in \mathbb{R}^n_{++} \times \mathbb{R}^{n+m}$,

$$p(u, w) \equiv \zeta \log(u^T u + w^T w) - \sum_{i=1}^{n} \log u_i. \tag{11.5.1}$$

Let $a \equiv (\mathbf{1}_n, 0) \in \mathbb{R}^n_{++} \times \mathbb{R}^{n+m}$ and $\bar{\sigma} \equiv 1$. By a demonstration similar to that in Subsections **11.2.1** and **11.3.1**, we can show that the conditions (IP1), (IP2), (IP3'), and (IP4)–(IP6) are all satisfied. Therefore Algorithm **11.3.2** and Theorem **11.3.4** are applicable to (11.1.3). In this application, let us examine the structure of the (modified) Newton equation (11.3.3) more closely. We have

$$X_{\mathcal{I}} = \operatorname{int} X = \mathbb{R}^{2n}_{++} \times \mathbb{R}^m.$$

Let $(x^k, y^k, z^k) \in \mathbb{R}^{2n}_{++} \times \mathbb{R}^m$ be a given iterate. The Newton equation (11.3.3) has the form:

$$\begin{bmatrix} \operatorname{diag}(y^k) & \operatorname{diag}(x^k) & 0 \\ J_x H(x^k, y^k, z^k) & J_y H(x^k, y^k, z^k) & J_z H(x^k, y^k, z^k) \end{bmatrix} \begin{pmatrix} dx^k \\ dy^k \\ dz^k \end{pmatrix}$$

$$= - \begin{pmatrix} x^k \circ y^k \\ H(x^k, y^k, z^k) \end{pmatrix} + \sigma_k \begin{pmatrix} \dfrac{(x^k)^T y^k}{n} \mathbf{1}_n \\ 0 \end{pmatrix}. \tag{11.5.2}$$

This system of linear equations is of order $(2n + m) \times (2n + m)$. Due to the special structure of the matrix on the left-hand side, the system can be reduced to one of the order $(n + m) \times (n + m)$ by eliminating either the dx^k or dy^k variable. Write $x^{-k} \equiv (x^k)^{-1}$ and $\varsigma_k \equiv (x^k)^T y^k / n$. From the first equation in (11.5.2), we have

$$dy^k = - \operatorname{diag}(x^{-k} \circ y^k) dx^k - y^k + \sigma_k \varsigma_k x^{-k}. \tag{11.5.3}$$

Substituting this into the second equation in (11.5.2), we obtain

$$M_k \begin{pmatrix} dx^k \\ dz^k \end{pmatrix} = r^k \tag{11.5.4}$$

where

$$M_k \equiv$$

$$\begin{bmatrix} J_x H(x^k, y^k, z^k) - J_y H(x^k, y^k, z^k) \operatorname{diag}(x^{-k} \circ y^k) & J_z H(x^k, y^k, z^k) \end{bmatrix}$$

is a nonsingular matrix of order $n + m$ and

$$r^k \equiv -H(x^k, y^k, z^k) + J_y H(x^k, y^k, z^k)\,(\,y^k - \sigma_k\,\varsigma_k\,x^{-k}\,).$$

Thus, to solve (11.5.2), we first solve the $(n+m)\times(n+m)$ system (11.5.4) and substitute the solution (dx^k, dz^k) into (11.5.3) to obtain dy^k. When the MiCP (11.1.3) corresponds to the CP (F, G) defined by a pair of functions F and G, the matrix M_k has further structure which can be profitably exploited when the system of linear equations (11.5.4) is solved. Consider for instance the NCP (F), which yields $m = 0$ and $H(x, y) \equiv F(x) - y$. In this case, we have $M_k = JF(x^k) + \mathrm{diag}(x^{-k} \circ y^k)$. For a monotone mapping F, so that $JF(x^k)$ is a positive semidefinite matrix, the matrix M_k is symmetric positive definite because x^k and y^k are positive vectors.

In general, the efficient solution of a system of linear equations of the form (11.5.4) is the most critical step in the successful application of an IP method for solving CPs. For standard monotone problems, the defining matrix M_k of such a system is symmetric positive definite. In this case, efficient linear solvers for positive definite systems can be applied, resulting in some highly effective practical methods for solving the CPs.

Summarizing the above discussion, we present the following IP method for solving the problem (11.1.3) under the blanket assumption set forth above. With the function $p(u, w)$ given by (11.5.1), the potential function in this algorithm is: for $(x, y, z) \in \mathbb{R}^{2n}_{++} \times \mathbb{R}^m$,

$$\psi(x, y, z) = \zeta\,\log(\|\,x \circ y\,\|^2 + \|\,H(x, y, z)\,\|^2) - \sum_{i=1}^{n} \log(x_i y_i).$$

A Potential Reduction Algorithm for the Implicit MiCP
(PRAIMiCP)

11.5.1 Algorithm.

Data: $\zeta > n/2$, $(x^0, y^0, z^0) \in \mathbb{R}^{2n}_{++} \times \mathbb{R}^m$, $\gamma \in (0, 1)$, and a sequence $\{\sigma_k\} \subset [0, 1)$.

Step 1: Set $k = 0$.

Step 2: If $H(x^k, y^k, z^k) = 0$ and $\varsigma_k \equiv (x^k)^T y^k/n = 0$, stop.

Step 3: Solve the system of linear equations (11.5.4) to obtain the pair (dx^k, dz^k) and substitute into (11.5.3) to obtain dy^k.

Step 4: Find the smallest nonnegative integer i_k such that with $i = i_k$,

$(x^k, y^k) + 2^{-i}(dx^k, dy^k) > 0$ and

$$\psi(x^k + 2^{-i}dx^k, y^k + 2^{-i}dy^k, z^k + 2^{-i}dz^k) - \psi(x^k, y^k, z^k)$$
$$\leq \gamma\, 2^{-i}\, \nabla\psi(x^k, y^k, z^k)^T \begin{pmatrix} dx^k \\ dy^k \\ dz^k \end{pmatrix};$$

set $\tau_k \equiv 2^{-i_k}$.

Step 5: Set $(x^{k+1}, y^{k+1}, z^{k+1}) \equiv (x^k, y^k, z^k) + \tau_k(dx^k, dy^k, dz^k)$ and let $k \leftarrow k + 1$; go to Step 2.

The convergence of the above algorithm is ensured by Theorem **11.3.4** and Corollary **11.3.5**. A precise statement of the specialized convergence result is as follows. No proof is required except for the assertion of solution existence in the case of an affine H in part (c) of the result.

11.5.2 Theorem. Let $H : \mathbb{R}^{2n}_{+} \times \mathbb{R}^m \times \mathbb{R}^{n+m}$ be continuous. Assume that H is continuously differentiable on $\mathbb{R}^{2n}_{++} \times \mathbb{R}^m$ and that the Jacobian matrix $JH(x, y, z)$ has the mixed P_0 property for every $(x, y, z) \in \mathbb{R}^{2n}_{++} \times \mathbb{R}^m$. Let $\{(x^k, y^k, z^k)\}$ be an infinite sequence generated by Algorithm **11.5.1** with

$$\sup_k \sigma_k < 1.$$

The following statements are valid.

(a) The sequences $\{x^k \circ y^k\}$ and $\{H(x^k, y^k, z^k)\}$ are bounded.

(b) Every accumulation point of $\{(x^k, y^k, z^k)\}$ is a solution of (11.1.3).

(c) If for every positive scalars δ and η, the set

$$\{\, (x, y, z) \in \mathbb{R}^{2n}_{++} \times \mathbb{R}^m : \delta \leq \| H(x, y, z) \| + x^T y \leq \eta \,\}$$

is bounded, then

$$\lim_{k\to\infty} x^k \circ y^k = 0 \quad \text{and} \quad \lim_{k\to\infty} H(x^k, y^k, z^k) = 0. \tag{11.5.5}$$

If in addition H is affine, then the problem (11.1.3) has a solution.

(d) If for every positive scalar η, the set

$$\{\, (x, y, z) \in \mathbb{R}^{2n}_{++} \times \mathbb{R}^m : \| H(x, y, z) \| \leq \eta,\ x \circ y \leq \eta\, \mathbf{1}_n \,\}$$

is bounded, then $\{(x^k, y^k, z^k)\}$ is bounded.

Proof. It remains to show that a solution to (11.1.3) exists if H is affine and the two limits in (11.5.5) hold. Write

$$H(x,y,z) \equiv q + Ax + By + Cz, \quad \forall\,(x,y,z) \in \mathbb{R}^{2n+m},$$

for some vector $q \in \mathbb{R}^{n+m}$, matrices A and B in $\mathbb{R}^{(n+m)\times n}$ and C in $\mathbb{R}^{(n+m)\times m}$. There exists a partition of the index set $\{1,\ldots,n\}$ into the disjoint union of two index subsets $\mathcal{I}$ and $\mathcal{J}$ such that for each $i \in \mathcal{I}$, the sequence $\{x_i^k\}$ is bounded and for each $j \in \mathcal{J}$, the sequence $\{x_j^k\}$ is unbounded. Without loss of generality, we may assume, by working with an appropriate subsequence, that for each $i \in \mathcal{I}$,

$$\lim_{k\to\infty} x_i^k = \bar{x}_i$$

for some $\bar{x}_i \geq 0$ and for each $j \in \mathcal{J}$,

$$\lim_{k\to\infty} x_j^k = \infty.$$

Since $\{x_i^k y_i^k\}$ converges to zero as $k \to \infty$ for all i, it follows that for each $j \in \mathcal{J}$,

$$\lim_{k\to\infty} y_j^k = 0.$$

We further partition $\mathcal{I}$ into the disjoint union of two index subsets:

$$\mathcal{I}_+ \equiv \{\, i \in \mathcal{I} : \bar{x}_i > 0 \,\} \quad \text{and} \quad \mathcal{I}_0 \equiv \{\, i \in \mathcal{I} : \bar{x}_i = 0 \,\}.$$

We also have for each $i \in \mathcal{I}_+$,

$$\lim_{k\to\infty} y_i^k = 0.$$

We may write

$$H(x^k,y^k,z^k) = q + Cz^k + \sum_{i\in\mathcal{I}} A_{\cdot i}x_i^k + \sum_{j\in\mathcal{J}} A_{\cdot j}x_j^k + \sum_{i\in\mathcal{I}_+} B_{\cdot i}y_i^k + \sum_{i\in\mathcal{I}_0} B_{\cdot i}y_i^k + \sum_{j\in\mathcal{J}} B_{\cdot j}y_j^k.$$

Since the left-hand side tends to zero as $k \to \infty$, by the fact the affine image of a polyhedron is a closed set, it follows that there exist $\bar{x}_j \geq 0$ for each $j \in \mathcal{J}$ and $\bar{y}_i \geq 0$ for each $i \in \mathcal{I}_0$, which need not be the limit of the corresponding sequences $\{x_j^k\}$ and $\{y_i^k\}$, and $\bar{z}$ such that

$$0 = q + \sum_{i\in\mathcal{I}_+} A_{\cdot i}\bar{x}_i \sum_{j\in\mathcal{J}} A_{\cdot j}\bar{x}_j + \sum_{i\in\mathcal{I}_0} B_{\cdot i}\bar{y}_i + C\bar{z}.$$

By letting $\bar{y}_i = 0$ for all $i \in \mathcal{I}_+ \cup \mathcal{J}$, it is easy to see that triple $(\bar{x},\bar{y},\bar{z})$ solves the problem (11.1.3). □

The sequence $\{(x^k, y^k, z^k)\}$ generated by Algorithm **11.5.1** is not necessarily feasible to the MiCP (11.1.3); that is, for each k, it is not necessary that $H(x^k, y^k, z^k) = 0$. In fact, when H is nonlinear, insisting that the iterates be feasible is a very difficult task. In contrary, if H is an affine mapping given by (11.4.9), maintaining the feasibility of the iterates is actually quite easy, provided that the starting iterate (x^0, y^0, z^0) satisfies $H(x^0, y^0, z^0) = 0$. Indeed if $H(x^k, y^k, z^k) = 0$, then, since H is affine, the Newton equation

$$H(x^k, y^k, z^k) + JH(x^k, y^k, z^k) \begin{pmatrix} dx^k \\ dy^k \\ dz^k \end{pmatrix} = 0$$

ensures that the next iterate

$$(\, x^{k+1}, y^{k+1}, z^{k+1}\,) \equiv (\, x^k, y^k, z^k\,) + \tau_k\, (\, dx^k, dy^k, dz^k\,),$$

where $\tau_k \in (0, 1]$, also satisfies $H(x^{k+1}, y^{k+1}, z^{k+1}) = 0$. The advantage of such a "feasible IP" method (as the sequence is feasible) for the affine, implicit, MiCP is that the so-generated sequence $\{(x^k, y^k, z^k)\}$ must be bounded under the equi-monotonicity property of the defining mapping and a strict feasibility condition of the problem.

11.5.3 Proposition. Let H be given by (11.4.9). Suppose that the matrix $[A \;\; B \;\; C]$ has the mixed P_0 property. Let $(x^0, y^0, z^0) \in \mathbb{R}^{2n}_{++} \times \mathbb{R}^m$ be a given iterate satisfying $H(x^0, y^0, z^0) = 0$. Let $\{(x^k, y^k, z^k)\}$ be an infinite sequence generated by Algorithm **11.5.1** initiated at such an iterate with

$$\sup_k \sigma_k < 1.$$

The following two statements are valid.

(a) If the implication (11.4.5) holds and C has full column rank, then $\{(x^k, y^k, z^k)\}$ is bounded.

(b) If for all positive scalars δ and η, the set

$$\{\, (\, x, y, z\,) \mathbb{R}^{2n}_{++} \times \mathbb{R}^m \,:\, H(x, y, z) = 0,\; \delta\, \mathbf{1}_n \le x \circ y \le \eta\, \mathbf{1}_n \,\}$$

is bounded, then the sequence $\{x^k \circ y^k\}$ converges to zero.

Proof. As noted above, we have $H(x^k, y^k, z^k) = 0$ for all k. Assume that the implication (11.4.5) holds and C has full column rank. By (11.4.5), we have

$$(\, x^k - x^0\,)^T (\, y^k - y^0\,) \ge 0.$$

By Theorem **11.3.4**(a), the sequence $\{(x^k)^T y^k\}$ is bounded. Since x^0 and y^0 are both positive, the above inequality easily implies that $\{(x^k, y^k)\}$ is bounded. Since C has full column rank and $q + Ax^k + By^k + Cz^k = 0$ for all k, it follows easily that $\{z^k\}$ is also bounded.

The proof of (b) is exactly the same as the proof of the first part of Corollary **11.3.5** by noting that $H(x^k, y^k, z^k) = 0$ for all k. □

Needless to say, in order to be able to initiate the algorithm as described in Proposition **11.5.3**, we need the problem (11.1.3) to have a strictly feasible vector (x^0, y^0, z^0). In general, it is not a trivial task to compute such a point (if it exists), even for H of the special form:

$$H(x, y) \equiv q + Mx - y$$

that corresponds to the LCP (q, M). For affine CPs where such a point is readily available, a feasible IP method is a viable alternative to an "infeasible IP" method.

The proof of part (a) in Proposition **11.5.3** shows that the assumption therein actually implies that the set

$$\{ (x, y, z) \in \mathbb{R}^{2n}_+ \times \mathbb{R}^m : H(x, y, z) = 0,\ x \circ y \leq \eta \mathbf{1}_n \}$$

is bounded for all $\eta > 0$. The latter property implies in turn that the problem (11.1.3) has a bounded solution set. In what follows, we give an example to show that the assumption in part (b) of Proposition **11.5.3**, which has to do with the boundedness of certain two-sided level sets, does not imply the boundedness of the above one-sided level sets.

11.5.4 Example. Let M be the 3×3 matrix:

$$M \equiv \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 1 & -1 & 0 \end{bmatrix}$$

and define $H(x, y) \equiv y - Mx$ for x and y in $\mathbb{R}^3$. It is easy to show that M is a P_0 matrix but not positive semidefinite. For every positive scalar $\varepsilon > 0$, let

$$x(\varepsilon) \equiv \begin{pmatrix} 2\varepsilon \\ \varepsilon \\ 1/\sqrt{\varepsilon} \end{pmatrix} \quad \text{and} \quad y(\varepsilon) \equiv \begin{pmatrix} 2\varepsilon \\ \varepsilon + 1/\sqrt{\varepsilon} \\ \varepsilon \end{pmatrix}.$$

It is trivial to verify that $H(x(\varepsilon), y(\varepsilon)) = 0$. For any given scalar $\eta > 0$, we clearly have $x(\varepsilon) \circ y(\varepsilon) \leq \eta \mathbf{1}_3$ for all $\varepsilon > 0$ sufficiently small. Yet, the pair $(x(\varepsilon), y(\varepsilon))$ is unbounded as $\varepsilon \to 0$.

Let $(x, y) > 0$ satisfy

$$y \,=\, Mx \quad \text{and} \quad \delta\, \mathbf{1}_3 \,\leq\, x \circ y \,\leq\, \eta\, \mathbf{1}_3$$

for a given pair of positive scalars δ and η. It is not difficult to deduce that

$$\sqrt{\delta} \,\leq x_1 \,\leq\, \sqrt{\eta} \quad \text{and} \quad x_2 \,\leq\, \sqrt{\eta}.$$

By an easy argument, we can show that x_3 must be bounded. Hence statement (c) of Proposition **11.5.3** is applicable to the LCP $(0, M)$ but statement (b) is not. In fact, this LCP has an unbounded solution ray; namely, $(0, 0, x_3)$ for any $x_3 \geq 0$. Consequently, for this example, the set

$$\{\, (\, x, y\,) \,:\, y - Mx \,=\, 0,\; \delta\, \mathbf{1} \,\leq\, x \circ y \,\leq\, \eta\, \mathbf{1}\,\}$$

is bounded for all positive scalars δ and η, whereas the larger set

$$\{\, (\, x, y\,) \,:\, y - Mx \,=\, 0,\; x \circ y \,\leq\, \eta\, \mathbf{1}\,\}$$

is unbounded for any positive η. □

11.5.1 The NCP and KKT system

We specialize further the previous developments to the monotone NCP (F), where $F : \mathbb{R}^n \to \mathbb{R}^n$ is continuous on $\mathbb{R}^n_+$ and continuously differentiable and monotone on $\mathbb{R}^n_{++}$. This implies that the mapping $H(x, y) \equiv y - F(x)$ is differentiably monotone on $\mathbb{R}^{2n}_{++}$; thus, the Jacobian matrix $JH(x, y)$ has the P_0 property for for all $(x, y) \in \mathbb{R}^{2n}_{++}$. With

$$G(x, y) \,\equiv\, \begin{pmatrix} x \circ y \\ y - F(x) \end{pmatrix}, \tag{11.5.6}$$

$$X \,\equiv\, \mathbb{R}^{2n}_+, \quad S \,\equiv\, \mathbb{R}^n_+ \times \mathbb{R}^n, \quad a \,\equiv\, \begin{pmatrix} \mathbf{1}_n \\ 0 \end{pmatrix} \,\in\, \mathbb{R}^{2n}, \quad \bar{\sigma} \,\equiv\, 1,$$

and

$$\psi(x, y) \,\equiv\, \zeta\, \log(\|\, x \circ y\,\|^2 + \|\, y - F(x)\,\|^2) - \sum_{i=1}^{n} \log(x_i y_i), \quad (\, x, y\,) \,\in\, \mathbb{R}^{2n}_{++},$$

where $\zeta > n/2$ is a given scalar, Algorithm **11.5.1** and Theorem **11.5.2** are applicable to the NCP (F). Due to the special structure of H, we can

establish a specialized convergence result under a further assumption on F. Specifically, extending the concept of an S matrix and restricting the class of S functions, we say that F is a *strongly S function* if, for every vector $q \in \mathbb{R}^n$, there exists a vector $x \geq 0$ such that $F(x) > q$. With F being continuous, this is equivalent to saying that for every vector $q \in \mathbb{R}^n$, there exists a vector $x' > 0$ such that $F(x') > q$.

11.5.5 Lemma. If $F : \mathbb{R}^n \to \mathbb{R}^n$ is a continuous, strongly S function that is monotone on $\mathbb{R}^n_{++}$, then for every nonnegative scalar η, the set

$$\{ (x, y) \in \mathbb{R}^{2n}_{++} : \| y - F(x) \| \leq \eta, \; x \circ y \leq \eta \mathbf{1}_n \}$$

is bounded.

Proof. Let a nonnegative scalar η be given. Choose a vector $\bar{x} > 0$ such that $\bar{y} \equiv F(\bar{x}) > \eta \mathbf{1}_n$. Suppose that there exists a sequence $\{(x^k, y^k)\}$ of vectors such that for every k,

$$(x^k, y^k) > 0, \quad \| y^k - F(x^k) \| \leq \eta, \quad x^k \circ y^k \leq \eta \mathbf{1}_n,$$

and

$$\lim_{k \to \infty} \| (x^k, y^k) \| = \infty.$$

Without loss of generality, we may assume that

$$\lim_{k \to \infty} \frac{(x^k, y^k)}{\| (x^k, y^k) \|} = (u^\infty, v^\infty)$$

for some nonzero pair of nonnegative vectors u^∞ and v^∞. For each k, let $q^k \equiv y^k - F(x^k)$. Note that $\|q^k\| \leq \eta$ for all k. Without loss of generality, we may assume that the sequence $\{q^k\}$ converges to a vector q^∞, which must satisfy $\|q^\infty\| \leq \eta$. By the monotonicity of F, we have, for each k,

$$0 \leq (x^k - \bar{x})^T (F(x^k) - F(\bar{x})) = (x^k - \bar{x})^T (y^k - \bar{y} - q^k),$$

which implies

$$(\bar{y} + q^k)^T x^k + \bar{x}^T y^k \leq (x^k)^T y^k + \bar{x}^T (\bar{y} + q^k).$$

Dividing by $\|(x^k, y^k)\|$ and letting k tend to infinity, we see that the right-hand side tends to zero because $(x^k)^T y^k \leq n\eta$ for all k. Thus we deduce

$$(\bar{y} + q^\infty)^T u^\infty + \bar{x}^T v^\infty \leq 0.$$

But this is impossible because $\bar{y} + q^\infty$ and $\bar{x}$ are positive vectors and the pair (u^∞, v^∞) is nonzero and nonnegative. □

The following convergence result is immediate and does not require a proof.

11.5.6 Corollary. Let $F : \mathbb{R}^n \to \mathbb{R}^n$ be a continuous, strongly S function. Suppose that F is continuously differentiable and monotone on $\mathbb{R}^n_{++}$. Every sequence $\{(x^k, y^k)\}$ produced by specializing Algorithm **11.5.1** to the NCP (F) as described above (and with $\sup \sigma_k < 1$) is bounded; moreover, for every accumulation point (x^∞, y^∞) of the sequence, the vector x^∞ solves the NCP (F). □

If F is a strongly S function, the NCP (F) is strictly feasible. It is therefore natural to ask whether the above result will remain valid for a monotone, strictly feasible NCP (F). After all, such an NCP must have a nonempty bounded solution set; see Theorem **2.4.4**. The critical issue here is the boundedness of the sequence $\{(x^k, y^k)\}$, which is in jeopardy without the strongly S property. In what follows, we introduce a variant of Algorithm **11.5.1** and show that every generated sequence must be bounded for a monotone, strictly feasible NCP. This variant takes full advantage of the special form of the function $H(x, y) \equiv y - F(x)$.

Specifically, let $F : \mathbb{R}^n \to \mathbb{R}^n$ be continuous on $\mathbb{R}^n_+$ and continuously differentiable and monotone on $\mathbb{R}^n_{++}$. Recall that the NCP (F) is (strictly) feasible if there exists a vector $\bar{x}$ $(>) \geq 0$ such that $F(\bar{x})$ $(>) \geq 0$. In terms of the function $H(x, y)$, feasibility means $0 \in H_+ \equiv \mathbb{R}^n_+ - F(\mathbb{R}^n_+)$ and strict feasibility means $0 \in H_{++} \equiv \mathbb{R}^n_{++} - F(\mathbb{R}^n_{++})$. Since F is monotone, by Proposition **11.4.18** and Lemma **11.4.14**, it follows that G is proper with respect to $\mathbb{R}^n_+ \times H_{++}$. The following simple lemma is easy to prove.

11.5.7 Lemma. Let $F : \mathbb{R}^n \to \mathbb{R}^n$ be any continuous function.

(a) If $0 \in H_+ \equiv \mathbb{R}^n_+ - F(\mathbb{R}^n_+)$, then $\mathbb{R}^{2n}_{++} \subseteq \mathbb{R}^n_+ \times H_{++}$.

(b) If $0 \in H_{++} \equiv \mathbb{R}^n_{++} - F(\mathbb{R}^n_{++})$, then $\mathbb{R}^{2n}_{+} \subseteq \mathbb{R}^n_+ \times H_{++}$.

Proof. We prove only (a); the proof of (b) is similar. Assume that $0 \in H_+$. It suffices to show that $\mathbb{R}^n_{++}$ is contained in H_{++}. Let q be an arbitrary vector in $\mathbb{R}^n_{++}$. Let $\bar{x} \geq 0$ be such that $F(\bar{x}) \geq 0$. By continuity of F, the vector $y \equiv q + F(\bar{x} + \varepsilon \mathbf{1}_n)$ is positive for all $\varepsilon > 0$ sufficiently small. Hence $q \in H_{++}$ as desired. □

An immediate consequence of the above lemma is that if the NCP (F) is (strictly) feasible, then G is proper with respect to the set ($\mathbb{R}^{2n}_+$, respectively) $\mathbb{R}^{2n}_{++}$, by Lemma **11.4.14**. This observation is useful subsequently. Let $S \equiv \mathbb{R}^{2n}_+$. Define the potential function for $\operatorname{int} S$:

$$p(u) \equiv \zeta \log u^T u - \sum_{i=1}^{2n} \log u_i, \quad u \in \mathbb{R}^{2n}_{++},$$

where $\zeta > n$ is a given scalar; also let the vector $a \equiv \mathbf{1}_{2n}$. With $X \equiv \mathbb{R}^{2n}_{+}$, it is easy to see that

$$X_{\mathcal{I}} \equiv G^{-1}(\operatorname{int} S) \cap \operatorname{int} X = \{ (x, y) \in \mathbb{R}^{2n}_{++} : y > F(x) \}.$$

In turn, the function p induces the potential function for this set $X_{\mathcal{I}}$: for $(x, y) \in X_{\mathcal{I}}$,

$$\psi(x, y) \equiv$$
$$\zeta \log(\| x \circ y \|^2 + \| y - F(x) \|^2) - \sum_{i=1}^{n} \log(x_i y_i) - \sum_{i=1}^{n} \log(y_i - F_i(x)),$$

We now have all the new ingredients to apply Algorithm **11.3.2**. Starting at a pair $(x^0, y^0) > 0$ satisfying $y^0 > F(x^0)$, the resulting modified IP algorithm generates a sequence $\{(x^k, y^k)\} \subset X_{\mathcal{I}}$ such that the sequence $\{\psi(x^k, y^k)\}$ is decreasing. Incidentally, the initial pair (x^0, y^0) is trivial to construct. By the properness of G with respect to either the entire set S or its interior, we have the following convergence result for the generated sequence $\{(x^k, y^k)\}$.

11.5.8 Proposition. Let $F : \mathbb{R}^n \to \mathbb{R}^n$ be a continuous function. Suppose that F is continuously differentiable and monotone on $\mathbb{R}^n_{++}$. Let $\{(x^k, y^k)\}$ be an infinite sequence generated by the modified IP algorithm as described above. For every accumulation point (x^∞, y^∞) of the sequence, the vector x^∞ solves the NCP (F). Furthermore, the following two statements hold.

(a) If the NCP (F) is feasible, then the sequences $\{x^k \circ y^k\}$ and $\{y^k - F(x^k)\}$ converge to zero. Thus

$$\lim_{k \to \infty} \min(x^k, F(x^k)) = 0.$$

(b) If the NCP (F) is strictly feasible, then the sequence $\{(x^k, y^k)\}$ is bounded.

Proof. Statement (b) and the first assertion in statement (a) follow easily from Theorem **11.3.4** by taking E to be S. It remains to show that $\{\min(x^k, y^k)\}$ converges to zero if both $\{x^k \circ y^k\}$ and $\{y^k - F(x^k)\}$ converge to zero. Recalling that $\{(x^k, y^k)\}$ is a positive sequence, assume that for an infinite subset $\kappa \subset \{1, 2, \ldots\}$,

$$\liminf_{k(\in\kappa) \to \infty} x^k_i > 0.$$

Since $\{x_i^k y_i^k\}$ converges to zero, we have

$$\lim_{k(\in\kappa)\to\infty} y_i^k \;=\; 0,$$

which implies that

$$\lim_{k(\in\kappa)\to\infty} F_i(x^k) \;=\; 0.$$

Thus

$$\lim_{k(\in\kappa)\to\infty} \min(\, x_i^k, F_i(x^k)\,) \;=\; 0.$$

Consequently, the entire sequence $\{\min(x^k, y^k)\}$ must converge to zero, as desired. □

Analytically, statement (a) of Proposition **11.5.8** is interesting in its own right. It asserts that every feasible, monotone NCP must have ε-solutions for all $\varepsilon > 0$. That is, for any $\varepsilon > 0$, there exists a (positive) vector x such that

$$\| \min(\, x, F(x)\,) \| \;\leq\; \varepsilon.$$

Moreover, such an approximate solution of the NCP (F) can be computed by the above modified version of the IP algorithm. The existence of such an ε-solution does not follow from the analytical results in Chapter **2**; it was alluded to in Example **2.4.6** for a simple monotone NCP that has no exact solution.

The two versions of the IP algorithm described above can be extended to the KKT system of a VI, which we restate below for ease of reference: find $(\lambda, w, x, \mu) \in \mathbb{R}^{2m}_+ \times \mathbb{R}^{n+\ell}$ such that

$$0 \;=\; G(\lambda, w, x, \mu) \;\equiv\; \begin{pmatrix} w \circ \lambda \\ w + g(x) \\ \mathbf{L}(x, \mu, \lambda) \\ h(x) \end{pmatrix} \;\in\; \mathbb{R}^{2m+n+\ell}. \tag{11.5.7}$$

The results obtained in Section **11.4** are highly relevant for establishing the convergence properties of the IP algorithms. In particular, the assumptions employed in the following analysis are variations of those in the previous section. Since the IP algorithms require the mapping H of (11.4.19), which we restate below:

$$H(\lambda, w, x, \mu) \;\equiv\; \begin{pmatrix} w + g(x) \\ \mathbf{L}(x, \mu, \lambda) \\ h(x) \end{pmatrix}, \tag{11.5.8}$$

to be continuously differentiable and its Jacobian matrix $JH(\lambda, w, x, \mu)$ to have the mixed P_0 property, we postulate some blanket assumptions on the triple (F, g, h). Specifically, we assume that the function $F : \mathbb{R}^n \to \mathbb{R}^n$ is continuously differentiable and $(g, h) : \mathbb{R}^n \to \mathbb{R}^{m+\ell}$ is twice continuously differentiable. We further assume the following two conditions:

(**L**inear **I**ndependence **(LI)**) for every $x \in \mathbb{R}^n$, the gradients

$$\{ \nabla h_j(x) : j = 1, \ldots, \ell \} \tag{11.5.9}$$

are linearly independent;

(**P**ositive **D**efiniteness **(PD)**) for every $(x, \lambda, \mu) \in \mathbb{R}^n \times \mathbb{R}^m_{++} \times \mathbb{R}^\ell$, the Jacobian matrix $J_x \mathbf{L}(x, \mu, \lambda)$ is (i) positive semidefnite on the null space of the gradient vectors (11.5.9), and (ii) positive definite on the null space of the gradient vectors (11.5.9) and $\{\nabla g_i(x) : i = 1, \ldots, m\}$; that is

$$[\, \nabla h_j(x)^T u = 0 \quad \forall\, i = 1, \ldots, \ell\,] \;\Rightarrow\; u^T J_x \mathbf{L}(x, \mu, \lambda) u \geq 0, \tag{11.5.10}$$

and

$$\left.\begin{array}{r} \nabla h_j(x)^T u = 0 \quad \forall\, j = 1, \ldots, \ell \\ \nabla g_i(x)^T u = 0 \quad \forall\, i = 1, \ldots, m \\ u \neq 0 \end{array}\right\} \;\Rightarrow\; u^T J_x \mathbf{L}(x, \mu, \lambda) u > 0; \tag{11.5.11}$$

(**B**oundedness **(BD)**) for every $(u, \eta) \in \mathbb{R}^n \times \mathbb{R}_+$, the set

$$K_{u,\eta} \equiv \{\, x \in \mathbb{R}^n : g(x) \leq u,\ \| h(x) \| \leq \eta \,\}$$

is bounded (possibly empty).

Notice that the monotonicity of F, the convexity of each g_i, and the affinity of h do not appear explicitly in the above assumptions. If F is monotone, each g_i is convex, and h is affine, then $J_x \mathbf{L}(x, \mu, \lambda)$ is a positive semidefinite matrix for all $(x, \lambda, \mu) \in \mathbb{R}^n \times \mathbb{R}^m_+ \times \mathbb{R}^\ell$; thus (11.5.10) holds trivially for a monotone VI (K, F), where $K \equiv K_{0,0}$. Moreover, by the proof of Lemma **11.4.22**, the boundedness assumption (BD) holds for a monotone VI (K, F) with a nonempty compact set K. The following result identifies several situations in which the implication (11.5.11) holds. In particular, part (a) is the differentiable version of the assumptions of Lemma **11.4.21**.

11.5.9 Proposition. Suppose that the function $F : \mathbb{R}^n \to \mathbb{R}^n$ is continuously differentiable and monotone, each $g_i : \mathbb{R}^n \to \mathbb{R}$ is twice continuously differentiable and convex, and $h : \mathbb{R}^n \to \mathbb{R}^\ell$ is affine. Condition (PD) holds under any one of the following three conditions:

(a) for every $x \in \mathbb{R}^n$, either $JF(x)$ or at least one of the matrices

$$\{ \nabla^2 g_i(x) : i = 1, \ldots, m \}$$

is positive definite;

(b) each g_i is quadratic and K is bounded;

(c) the polyhedron P consisting of all vectors x satisfying all the affine constraints among the constraints $g(x) \leq 0$ and $h(x) = 0$ has a vertex.

Proof. Condition (PD) is trivially valid when (a) holds. We next show that (b) implies (PD). It suffices to verify the implication (11.5.11) for (x, λ, μ) in $\mathbb{R}^n \times \mathbb{R}^m_{++} \times \mathbb{R}^\ell$. Assume for contradiction that (x, λ, μ) is such a triple for which the left-hand side of this implication holds but $u^T J_x \mathbf{L}(x, \mu, \lambda) u = 0$ By the monotonicity assumption of the VI (K, F) and the fact that $\lambda > 0$, it follows that $u^T \nabla^2 g_i(x) u = 0$ for all $i = 1, \ldots, m$. Since g_i is quadratic and $\nabla g_i(x)^T u = 0$, we deduce that g_i is a constant along the whole line $\{x + \tau u : \tau \in \mathbb{R}\}$. Since h is affine and $\nabla h_j(x)^T u = 0$, the same is true for each h_j for $j = 1, \ldots, \ell$. Hence the line $\{x + \tau u : \tau \in \mathbb{R}\}$ is contained in the set $K_{u,\eta}$ where $u \equiv g(x)$ and $\eta \equiv \|h(x)\|$. But this is impossible because each g_i is convex and h is affine and K is bounded; a contradiction. Thus (b) implies (PD). Finally, it remains to show that (11.5.11) holds under condition (c). It is well known that the polyhedron P has a vertex if and only if the lineality space of P is the singleton $\{0\}$. Let $\mathcal{I}$ be the index subset of $\{1, \ldots, m\}$ for which g_i is affine. The lineality space of P is equal to:

$$U \equiv \{ u \in \mathbb{R}^n : \nabla g_i(x)^T u = 0, \ \forall\, i \in \mathcal{I}; \ \nabla h_j(x)^T u = 0, \ \forall\, j \}$$

Hence $U = \{0\}$. Thus there exists no vector u satisfying the left-hand condition in (11.5.11). Hence this implication holds trivially. □

The next result is key to the applicability of the IP algorithms for solving the KKT system.

11.5.10 Lemma. Under conditions (PD) and (LI), the Jacobian matrix $JH(\lambda, w, x, \mu)$ in the partitioned form:

$$JH(\lambda, w, x, \mu) = \begin{bmatrix} J_\lambda H(\lambda, w, x, \mu) & J_w H(\lambda, w, x, \mu) & J_{x,\mu} H(\lambda, w, x, \mu) \end{bmatrix}$$

has the mixed P_0 property for all (λ, w, x, μ) in $\mathbb{R}^{2m}_{++} \times \mathbb{R}^{n+\ell}$.

Proof. It suffices to establish two things. One, the matrix

$$J_{x,\mu}H(\lambda,w,x,\mu) \;=\; \begin{bmatrix} Jg(x) & 0 \\ J_x\mathbf{L}(x,\mu,\lambda) & Jh(x)^T \\ Jh(x) & 0 \end{bmatrix}$$

has full column rank; two, the implication holds:

$$\left.\begin{aligned} J_x\mathbf{L}(x,\mu,\lambda)dx + \sum_{i=1}^{m} d\lambda_i\,\nabla g_i(x) + \sum_{j=1}^{\ell} d\mu_j\,\nabla h_j(x) \;&=\; 0 \\ dw_i + \nabla g_i(x)^T dx \;&=\; 0, \quad \forall\, i \;=\; 1,\ldots,m \\ \nabla h_j(x)^T dx \;&=\; 0, \quad \forall\, j \;=\; 1,\ldots,\ell \end{aligned}\right\}$$
$$\Rightarrow\; dw^T d\lambda \;\geq\; 0.$$

The latter implication follows easily because

$$dw^T d\lambda \;=\; dx\, J_x\mathbf{L}(x,\mu,\lambda)dx.$$

which is nonnegative by the condition (i) in (PD). The former full column rank condition follows from condition (ii) in (PD) and the condition (LI). □

The first IP algorithm for solving the KKT system is based on the formulation of this system as the CE (G, X), where G is given by (11.5.7) and $X \equiv \mathbb{R}^{2m}_{+} \times \mathbb{R}^{n+\ell}$. The algorithm is a specialization of Algorithm **11.3.2** with the following specifications:

$$S \;\equiv\; \mathbb{R}^{m}_{+} \times \mathbb{R}^{m+n+\ell}, \quad X_{\mathcal{I}} \;=\; \operatorname{int} X \;=\; \mathbb{R}^{2m}_{++} \times \mathbb{R}^{n+\ell},$$

$$a \;\equiv\; \begin{pmatrix} \mathbf{1}_m \\ 0 \end{pmatrix} \;\in\; \mathbb{R}^{m}_{+} \times \mathbb{R}^{m+n+\ell}, \quad \bar{\sigma} \;\equiv\; 1,$$

and for all $(\lambda, w, x, \mu) \in X_{\mathcal{I}}$,

$$\psi(\lambda,w,x,\mu) \;\equiv\; \zeta\,\log(\|\,\lambda \circ w\,\|^2 + \|\,H(\lambda,w,x,\mu)\,\|^2) - \sum_{i=1}^{m}\log(\lambda_i w_i)$$

where $\zeta > m/2$ is a given scalar. Initiated at a tuple $(\lambda^0, w^0, x^0, \mu^0)$ with $(\lambda^0, w^0) > 0$, the resulting algorithm generates a sequence of iterates $\{(\lambda^k, w^k, x^k\mu^k)\}$ with $(\lambda^k, w^k) > 0$ and

$$\psi(\lambda^{k+1}, w^{k+1}, x^{k+1}, \mu^{k+1}) \;<\; \psi(\lambda^k, w^k, x^k, \mu^k)$$

for every k. The following convergence result for this scheme is an immediate corollary of parts (a) and (b) of Theorem **11.3.4**.

11.5.11 Theorem. Let $F : \mathbb{R}^n \to \mathbb{R}^n$ be continuously differentiable and $(g, h) : \mathbb{R}^n \to \mathbb{R}^{m+\ell}$ be twice continuously differentiable. Suppose that (LI) and (PD) hold. Let $(\lambda^0, w^0, x^0, \mu^0) \in \mathbb{R}^{2m}_{++} \times \mathbb{R}^{n+\ell}$ be an arbitrary initial iterate. Let $\{(\lambda^k, w^k, x^k, \mu^k)\} \subset \mathbb{R}^{2m}_{++} \times \mathbb{R}^{n+\ell}$ be an infinite sequence generated by Algorithm **11.3.2** with the specifications given above and with

$$\sup_k \sigma_k < \bar{\sigma}.$$

The following statements are valid:

(a) the sequences $\{\lambda^k \circ w^k\}$ and $\{H(\lambda^k, w^k, x^k, \mu^k)\}$ are bounded;

(b) every accumulation point of $\{(\lambda^k, w^k, x^k, \mu^k)\}$ is a KKT tuple of the VI (K, F);

(c) if in addition (BD) holds, then the sequence $\{(x^k, w^k)\}$ is bounded.

Proof. It suffices to show (c). By (b), the sequences $\{w^k + g(x^k)\}$ and $\{h(x^k)\}$ are bounded. It then follows easily that the sequence $\{x^k\}$ is contained in the set $K_{u,\eta}$ for an appropriate pair (u, η). By (BD), $\{x^k\}$, and thus $\{w^k\}$ also, is bounded. □

The only restriction the starting tuple $(\lambda^0, w^0, x^0, \mu^0)$ in the above theorem is required to satisfy is the positivity of the pair (λ^0, w^0). In particular, the (primary) vector x^0 can be quite arbitrary. This generality has its drawback; namely, we can not conclude the boundedness of the multiplier sequence $\{(\lambda^k, \mu^k)\}$. Presumably, this drawback is compensated by the rather weak assumptions required. In order to ensure that a bounded sequence of tuples $\{(\lambda^k, w^k, x^k, \mu^k)\}$, including the multipliers, is generated, we can employ the modified version of the IP algorithm by requiring that $w^k + g(x^k) > 0$, among other things. For this application, we need to assume that each g_i is convex and h is affine. We also require that $h(x^k) = 0$ for all k. Specifically, we take

$$S \equiv \mathbb{R}^{2m}_{+} \times \mathbb{R}^{n+\ell}, \quad X_{\mathcal{I}} = \{\, (\lambda, w, x, \mu) \in \mathbb{R}^{2m}_{++} \times \mathbb{R}^{n+\ell} : w + g(x) > 0 \,\}$$

and for all $(\lambda, w, x, \mu) \in X_{\mathcal{I}}$,

$$\begin{aligned} \psi(\lambda, w, x, \mu) \equiv \; & \zeta \log(\| \lambda \circ w \|^2 + \| w + g(x) \|^2 + \| \mathbf{L}(x, \mu, \lambda) \|^2) \\ & - \sum_{i=1}^{m} [\log(w_i + g_i(x)) + \log(\lambda_i w_i)] \end{aligned}$$

where $\zeta > m$ is a given scalar. Initiated at a tuple $(\lambda^0, w^0, x^0, \mu^0)$ with $(\lambda^0, w^0) > 0$, $w^0 + g(x^0) > 0$, and $h(x^0) = 0$, Algorithm **11.3.2** generates a sequence of iterates $\{(\lambda^k, w^k, x^k \mu^k)\}$ satisfying

$$(\lambda^k, w^k) > 0, \quad w^k + g(x^k) > 0, \quad h(x^k) = 0,$$

and

$$\psi(\lambda^{k+1}, w^{k+1}, x^{k+1}, \mu^{k+1}) \,<\, \psi(\lambda^k, w^k, x^k, \mu^k)$$

for every k. We have the following convergence result for this special IP method.

11.5.12 Theorem. Let $F : \mathbb{R}^n \to \mathbb{R}^n$ be continuously differentiable and monotone, each $g_i : \mathbb{R}^n \to \mathbb{R}$ be twice continuously differentiable and convex, and $h : \mathbb{R}^n \to \mathbb{R}^\ell$ be affine satisfying (LI). Assume that there exists a vector $\hat{x}$ such that $g(\hat{x}) < 0$ and $h(\hat{x}) = 0$. Assume further that the set $K \equiv K_{0,0}$ is bounded and one of the three conditions (a), (b), and (c) of Proposition **11.5.9** holds. Let $(\lambda^0, w^0, x^0, \mu^0) \in \mathbb{R}^{2m}_{++} \times \mathbb{R}^{n+\ell}$ be an arbitrary initial iterate satisfying $w^0 + g(x^0) > 0$ and $h(x^0) = 0$. The sequence $\{(\lambda^k, w^k, x^k, \mu^k)\} \subset \mathbb{R}^{2m}_{++} \times \mathbb{R}^{n+\ell}$ generated by Algorithm **11.3.2** with the above specifications is bounded and every accumulation point is a KKT tuple of the VI (K, F); moreover, if $\bar{x}$ is any accumulation point of $\{x^k\}$, then x^k solves the VI (K, F).

Proof. It suffices to show that the sequence $\{(\lambda^k, \mu^k)\}$ is bounded. Assume for contradiction that there exists an infinite subset $\kappa \subset \{1, 2, \ldots\}$ such that

$$\lim_{k(\in\kappa)\to\infty} \| (\lambda^k, \mu^k) \| = \infty \quad \text{and} \quad \lim_{k(\in\kappa)\to\infty} \frac{(\lambda^k, \mu^k)}{\| (\lambda^k, \mu^k) \|} = (\bar{\lambda}, \bar{\mu})$$

for some nonzero vector $(\bar{\lambda}, \bar{\mu}) \in \mathbb{R}^{m+\ell}$ with $\bar{\lambda} \geq 0$. The sequences

$$\{ \mathbf{L}(x^k, \mu^k, \lambda^k) \}, \quad \{ w^k + g(x^k) \}, \quad \{ \lambda^k \circ w^k \} \quad \text{and} \quad \{ (x^k, w^k) \}$$

are bounded. Without loss of generality, we may assume that the sequence $\{(x^k, w^k)\}$ converges to some vector (x^∞, w^∞), which must satisfy

$$h(x^\infty) = 0, \quad w^\infty + g(x^\infty) \geq 0, \quad 0 \leq \bar{\lambda} \perp w^\infty \geq 0,$$

where $\bar{\lambda}^T w^\infty = 0$ is because $\{(\lambda^k)^T w^k\}$ is bounded. Hence it follows that

$$\{ i : \bar{\lambda}_i > 0 \} \subseteq \{ i : w_i^\infty = 0 \} \subseteq \{ i : g_i(x^\infty) \geq 0 \}.$$

We claim that $\bar{\lambda}$ is nonzero. We have

$$\mathbf{L}(x^k, \mu^k, \lambda^k) = F(x^k) + \sum_{i=1}^m \lambda_i^k \nabla g_i(x^k) + \sum_{j=1}^\ell \mu_j^k \nabla h_j(x^k);$$

normalizing by $\|(\lambda^k, \mu^k)\|$ and letting $k(\in \kappa) \to \infty$, using the boundedness of $\{\mathbf{L}(x^k, \mu^k, \lambda^k)\}$, we deduce

$$\sum_{i=1}^m \bar{\lambda}_i \nabla g_i(x^\infty) + \sum_{j=1}^\ell \bar{\mu}_j \nabla h_j(x^\infty) = 0. \tag{11.5.12}$$

Condition (LI) and the fact that $(\bar{\lambda}, \bar{\mu}) \neq 0$ therefore imply $\bar{\lambda} \neq 0$. Since $g(\hat{x}) < 0$ and $h(\hat{x}) = 0$, letting $d \equiv \hat{x} - x^\infty$, we have for all i such that $\bar{\lambda}_i > 0$

$$0 \leq g_i(x^\infty) \leq g_i(\hat{x}) - \nabla g_i(x^\infty)^T(\hat{x} - x^\infty),$$

which yields $\nabla g_i(x^\infty)^T d < 0$. Consequently, premultiplying (11.5.12) by d^T, we obtain, by the affinity of h, which implies $\nabla h_j(x^\infty)^T d = 0$,

$$\begin{aligned} 0 = & \sum_{i=1}^{m} \bar{\lambda}_i \nabla g_i(x^\infty)^T d + \sum_{j=1}^{\ell} \bar{\mu}_j \nabla h_j(x^\infty)^T d \\ = & \sum_{i:\bar{\lambda}_i > 0} \bar{\lambda}_i \nabla g_i(x^\infty)^T d < 0, \end{aligned}$$

where the last inequality follows because $\bar{\lambda} \neq 0$. This is a contradiction and the boundedness of the entire sequence $\{(\lambda^k, w^k, x^k, \mu^k)\}$ follows. □

11.6 The Ralph-Wright IP Approach

The IP algorithms presented so far are based on the reduction of a potential function. In this section, we present an alternative approach of designing IP algorithms that does not make explicit use of such a potential function. In addition, we wish to revisit an idea that plays a central role in the algorithms in Chapter **9**, which is very useful for inducing locally fast convergence. Namely, we should exploit a pure Newton direction as much as possible. In the context of the algorithms presented in this chapter for solving the CE (G, X), this means that we should set the scalar $\bar{\sigma} = 0$ and consider the Newton direction obtained by solving the familiar Newton equation:

$$G(x^k) + JG(x^k)d = 0.$$

As one could easily imagine, simply taking the solution of this equation, denoted d_N^k, would not be an effective way to deal with the presence of the constraint set X. Indeed, this was the motivation to introduce the scalar $\bar{\sigma} > 0$ in the first place to define a modified Newton equation whose solution facilitated the treatment of X. The downside of solving the modified Newton equation (11.3.3):

$$G(x^k) + JG(x^k)d = \sigma_k \frac{a^T G(x^k)}{\| a \|^2} a \tag{11.6.1}$$

is that if σ_k is not sufficient small, the resulting direction could be quite different from the Newton direction, thus jeopardizing the fast convergence of the overall method. To remedy the situation, we first use the Newton

direction d_N^k to see if it satisfies certain criteria to be specified; if so, use it to generate the next iterate x^{k+1}; otherwise resort to the IP direction d^k from (11.6.1). In what follows, to distinguish these two directions, we refer to the case of solving (11.6.1) with $\sigma_k = 0$ as the "fast" step and that with $\sigma_k \neq 0$ as the "safe" step. This approach has been used several times in the previous chapters, where the fall-back "safe" step was, for example, a gradient step (see, e.g., Algorithm **9.1.10**). However, we remark once again that the situation here is really different since we are not minimizing the potential function. The convergence of the algorithm is therefore based on the peculiar properties of the (biased) Newton direction under the assumed hypotheses on the problem.

We apply the above idea to the implicit MiCP (11.1.3) defined by the mapping $H : \mathbb{R}^{2n+m} \to \mathbb{R}^{n+m}$, which is continuous on $\mathbb{R}^{2n}_{+} \times \mathbb{R}^m$ and continuously differentiable on $\mathbb{R}^{2n}_{++} \times \mathbb{R}^m$. The corresponding function G is given by (11.1.4):

$$G(x,y,z) \equiv \begin{pmatrix} x \circ y \\ H(x,y,z) \end{pmatrix}, \quad (x,y,z) \in \mathbb{R}^{2n}_{+} \times \mathbb{R}^m.$$

We introduce several criteria for generating the iterates (x^k, y^k, z^k). These criteria do not make use of a potential function; instead they are based on the following considerations. Clearly, we want the residual of G to decrease; in fact, our goal is to drive both the (average) complementarity gap ς_k and the residual r_k to zero, where

$$\varsigma_k \equiv \frac{(x^k)^T y^k}{n} \quad \text{and} \quad r_k \equiv \left\| H(x^k, y^k, z^k) \right\|.$$

This goal has to be balanced with a strategy to maintain the positivity of the pair (x^k, y^k) in order to ensure that we stay "well inside" the interior of the set $\mathbb{R}^{2n}_{+} \times \mathbb{R}^m$. To accommodate this dual objective, we clearly do not want the complementarity gap ς_k to approach zero prematurely before the residual ρ_k becomes small enough. One way to ensure that this situation will not happen is to insist that the residual be bounded above by a positive multiple of the gap. This leads to the first criterion; namely, for some constant $\eta > 0$,

$$r_k \leq \eta \varsigma_k, \quad \forall\, k.$$

One important consequence of this criterion is that if ς_k approaches zero, then so does r_k. The other consideration is that we need to prevent an individual product $x_i^k y_i^k$ from approaching zero unless the gap ς_k, which

is an average of all the n products of this kind, itself tends to zero. One criterion that will satisfy this consideration is: for some constant δ in $(0,1)$,

$$x^k \circ y^k \;\geq\; \delta\, \varsigma_k\, \mathbf{1}_n, \quad \forall\, k.$$

The latter criterion is referred to as the *centrality condition*. It ensures that if any individual product $x_i^k y_i^k$ tends to zero, then the vector x^k is asymptotically complementary to y^k; that is the complementary gap ς_k also tends to zero. In turn, by the first criterion, this implies that r_k must converge to zero. Thus the sequence $\{(x^k, y^k)\}$ satisfies the MiCP conditions asymptotically.

Finally, we want the gap ς_k to decrease monotonically. One criterion that ensures this is a sufficient decrease condition, which is very common in a line search method. In the present context, this condition is as follows: for some constants γ and γ' in $(0,1)$,

$$\varsigma_{k+1} \;\leq\; \max(\,\gamma,\, 1 - \tau_k\,\gamma'\,)\,\varsigma_k, \quad \forall\, k,$$

which guarantees that for each k, ς_{k+1} decreases by at least either a constant fraction $(1-\gamma)$ of ς_k or a variable fraction $\tau_k\gamma'$ that depends on the step size τ_k at the current iteration. Typically, a decrease of the former kind is more stringent than the latter kind. The algorithm relies on the latter as a safety to fall upon in case the former decrease can not be realized.

Initiated at a positive iterate (x^0, y^0), the algorithm to be introduced generates a sequence $\{(x^k, y^k, z^k)\}$ of iterates that satisfy the above three criteria. Among other things, these criteria imply that (x^k, y^k) is positive for all k; moreover, the sequence $\{(x^k, y^k, z^k)\}$ is confined to the set:

$$\begin{aligned}\Omega \;\equiv\; \{\;\; (\,x, y, z\,) \;\in\; \mathbb{R}^{2n}_{++} \times \mathbb{R}^m \;:\\ x \circ y \;\geq\; \delta_{\min}\varsigma\, \mathbf{1}_n,\; \|\, H(x,y,z)\,\| \;\leq\; \eta_{\max}\varsigma\,\},\end{aligned}$$

where $\delta_{\min} \in (0, \frac{1}{2})$ and $\eta_{\max} > 0$ are certain scalars and $\varsigma \equiv x^T y/n$. Notice that all solutions of the MiCP (11.1.3) lie outside the set Ω.

In what follows, we establish two key lemmas that guarantee the well-definedness of the alternative IP algorithm. For a triple (x, y, z) in Ω and for a scalar σ in $[0,1)$, let (dx, dy, dz) satisfy the linear equation:

$$\begin{bmatrix} \mathrm{diag}(y) & \mathrm{diag}(x) & 0 \\ J_x H(x,y,z) & J_y H(x,y,z) & J_z H(x,y,z) \end{bmatrix} \begin{pmatrix} dx \\ dy \\ dz \end{pmatrix}$$

$$= \; -\begin{pmatrix} x \circ y \\ H(x,y,z) \end{pmatrix} + \sigma\varsigma \begin{pmatrix} \mathbf{1}_n \\ 0 \end{pmatrix}. \qquad (11.6.2)$$

For $\tau \in [0,1]$, let

$$\begin{pmatrix} x(\tau) \\ y(\tau) \\ z(\tau) \end{pmatrix} \equiv \begin{pmatrix} x \\ y \\ z \end{pmatrix} + \tau \begin{pmatrix} dx \\ dy \\ dz \end{pmatrix} \quad \text{and} \quad \varsigma(\tau) \equiv \frac{x(\tau)^T y(\tau)}{n}.$$

The following lemma deals with the case where the scalar σ is positive. This case pertains to the safe step of the algorithm.

11.6.1 Lemma. Let $\delta_{\min}$, $\eta_{\max}$, σ and γ be positive scalars with $\delta_{\min}$, σ and γ all less than 1. Let $(\hat{x}, \hat{y}, \hat{z})$ be a triple in Ω such that the matrix $JG(\hat{x}, \hat{y}, \hat{z})$ is nonsingular. There exist an open neighborhood $\mathcal{N}$ of $(\hat{x}, \hat{y}, \hat{z})$ and a scalar $\bar{\tau} \in (0,1]$ such that for all scalars $\delta \in (\delta_{\min}, 1)$, $\eta \in (0, \eta_{\max})$ and $\tau \in [0, \bar{\tau}]$ and all triples $(x, y, z) \in \mathcal{N}$ satisfying

$$x \circ y \geq \delta\, \varsigma \quad \text{and} \quad \| H(x,y,z) \| \leq \eta\, \varsigma, \tag{11.6.3}$$

it holds that

$$\begin{pmatrix} x(\tau) \\ y(\tau) \end{pmatrix} > 0, \quad x(\tau) \circ y(\tau) \geq \delta\, \varsigma(\tau)\, \mathbf{1}_n,$$

$$\| H(x(\tau), y(\tau), z(\tau)) \| \leq \eta\, \varsigma(\tau), \quad \text{and} \quad \varsigma(\tau) \leq [\, 1 - \tau\, \gamma\, (1 - \sigma\,)\,]\, \varsigma.$$

Proof. By the assumed properties on $(\hat{x}, \hat{y}, \hat{z})$, it follows that there exist an open neighborhood $\mathcal{N}$ of this triple and positive constant c_1 and c_2 such that for all triples (x, y, z) belonging to this neighborhood, the pair (x, y) remains positive, $x \circ y \geq c_1 \mathbf{1}_n$, the matrix $JG(x, y, z)$ remains nonsingular, and the vector (dx, dy, dz) satisfies

$$\| (\, dx, dy, dz\,) \| \leq c_2.$$

Hence by restricting $\mathcal{N}$ if necessary, we may assume without loss of generality that for all $(x, y, z) \in \mathcal{N}$, the pair $(x(\tau), y(\tau))$ is positive for all $\tau \in [0,1]$. Let $(x, y, z) \in \mathcal{N}$ satisfy (11.6.3). We have for all $\tau \in [0,1]$,

$$\begin{aligned} x(\tau) \circ y(\tau) &= (\, x + \tau dx\,) \circ (\, y + \tau dy\,) \\ &= (\, 1 - \tau\,)\, x \circ y + \tau\, \sigma\, \varsigma\, \mathbf{1}_n + \tau^2\, dx \circ dy \\ &\geq [\, (\, 1 - \tau\,)\, \delta + \tau\, \sigma\,]\, \varsigma\, \mathbf{1}_n + \tau^2\, dx \circ dy. \end{aligned}$$

Moreover,

$$\varsigma(\tau) = [\, (\, 1 - \tau\,) + \tau\, \sigma\,]\, \varsigma + \tau^2\, dx^T dy/n. \tag{11.6.4}$$

Since $\sigma > 0$, $\delta < 1$, $\varsigma \geq c_1$, and $\|(dx, dy)\| \leq c_2$, there must exist $\bar{\tau}_1 \in (0, 1]$ such that for all $\tau \in [0, \bar{\tau}_1]$,

$$x(\tau) \circ y(\tau) \;\geq\; \delta\, \varsigma(\tau);$$

moreover, $\bar{\tau}_1$ depends only on σ, δ, and the base vector $(\hat{x}, \hat{y}, \hat{z})$, and not on the particular triple (x, y, z). Next, we evaluate $H(x(\tau), y(\tau), z(\tau))$. We have

$$\begin{aligned} H(x(\tau), y(\tau), z(\tau)) \;&=\; H(x, y, z) + \\ &\quad \tau\,(\, J_x H(x, y, z)dx + J_y H(x, y, z)dy + J_z H(x, y, z)dz\,) + o(\tau) \\ &=\; (\,1 - \tau\,)\, H(x, y, z) + o(\tau). \end{aligned}$$

Taking norms on both sides, we deduce

$$\begin{aligned} \|\, H(x(\tau), y(\tau), z(\tau))\,\| \;&\leq\; (\,1 - \tau\,)\,\|\, H(x, y, z)\,\| + o(\tau) \\ &\leq\; (\,1 - \tau\,)\,\eta\,\varsigma + o(\tau). \end{aligned}$$

Comparing this expression with (11.6.4), we deduce that, since σ and η are both positive, $\varsigma \geq c_1$, and $\|(dx, dy)\| \leq c_2$, there must exist $\bar{\tau}_2 \in (0, 1]$, such that for all $\tau \in [0, \bar{\tau}_2]$,

$$\|\, H(x(\tau), y(\tau), z(\tau))\,\| \;\leq\; \eta\,\varsigma(\tau);$$

again $\bar{\tau}_2$ is independent of the triple (x, y, z). Finally, by (11.6.4) and a similar observation, it follows that there must exist $\bar{\tau}_3 \in (0, 1]$ such that for all $\tau \in [0, \bar{\tau}_3]$,

$$\varsigma(\tau) \;\leq\; [\,1 - \tau\,\gamma\,(\,1 - \sigma\,)\,]\,\varsigma.$$

Letting $\bar{\tau} \equiv \min(\bar{\tau}_1, \bar{\tau}_2, \bar{\tau}_3)$ completes the proof of the lemma. □

Part of the significance of the above lemma is that with a positive σ, if (x, y, z) is a triple satisfying (11.6.3) for a given pair of scalars δ and η, then the resulting triple $(x(\tau), y(\tau), z(\tau))$ obtained by taking a step size τ along the computed direction (dx, dy, dz) satisfies a similar condition with the same pair of scalars δ and η. The case $\sigma = 0$ is a bit more complicated. For one thing, the proof of Lemma **11.6.1** breaks down when $\sigma = 0$. However, by adjusting the pair of scalars δ and η, the resulting triple $(x(\tau), y(\tau), z(\tau))$ can still satisfy a set of relaxed criteria without the step size τ being too small to be effective. The following lemma is a formal statement of the case $\sigma = 0$, which pertains to the fast step of the algorithm.

11.6.2 Lemma. Let the triple $(x, y, z) \in \mathbb{R}^{2n}_{++} \times \mathbb{R}^m$ satisfy (11.6.3) for some positive scalars δ and η. For every pair of positive scalars $\delta' < \delta$ and $\eta' > \eta$, there exists a scalar $\bar{\tau} \in (0, 1]$ such that for all $\tau \in [0, \bar{\tau}]$, it holds that

$$\begin{pmatrix} x(\tau) \\ y(\tau) \end{pmatrix} > 0, \quad x(\tau) \circ y(\tau) \geq \delta' \varsigma(\tau) \mathbf{1}_n,$$

and $\|H(x(\tau), y(\tau), z(\tau))\| \leq \eta' \varsigma(\tau)$.

Proof. This is similar to the proof of Lemma **11.6.1**. The details are not repeated. □

With the above two lemmas, we are ready to present the Ralph-Wright IP algorithm for solving the implicit MiCP (11.1.3). A few words are in order to explain the steps of this algorithm. The algorithm requires as inputs two scalars $\delta_{\max} \in (0, \frac{1}{2})$ and $\eta_{\min} > 0$ and an initial iterate (x^0, y^0, z^0) in $\mathbb{R}^{2n}_{++} \times \mathbb{R}^m$ that satisifes

$$x^0 \circ y^0 \geq \delta_{\max} \varsigma_0 \mathbf{1}_n \quad \text{and} \quad \| H(x^0, y^0, z^0) \| \leq \eta_{\min} \varsigma_0, \tag{11.6.5}$$

where $\varsigma_0 \equiv (x^0)^T y^0 / n$. At a general iteration of the algorithm, the scalars δ_k and η_k (with $\delta_0 \equiv \delta_{\max}$ and $\eta_0 \equiv \eta_{\min}$) are adjusted downward and upward, respectively, in order to accommodate for the fast step as prescribed by Lemma **11.6.2**. The adjustment is made if such a fast step leads to a decrease of the complementarity gap by the constant fraction $1 - \gamma$, where $\gamma \in (0, 1)$. If the fast step fails, then a safe step is taken as prescribed by Lemma **11.6.1**. In the latter step, only a sufficient decrease of the complementarity gap is demanded. In either step, the new iterate $(x^{k+1}, y^{k+1}, z^{k+1})$ is required to satisfy a condition similar to (11.6.5) corresponding to suitable scalars δ_{k+1} and η_{k+1}. All iterates (x^k, y^k, z^k) lie in the set Ω, where the scalar $\delta_{\min}$ satisfies $0 < \delta_{\min} < \delta_{\max}$ and $\eta_{\max} \equiv \eta_{\min} \exp(3/2)$, with exp being the exponential function. The latter expression for $\eta_{\max}$ is dictated by the way the scalar η_k is adjusted at each fast step. The integer t_k in the algorithm counts the number of successful fast steps and is used in the adjustment of the scalars δ_k and η_k. The scalar $\bar{\delta} \in (0, \frac{1}{2})$ is the factor by which we adjust δ_k and η_k. The scalars γ, γ', and ρ are self-explanatory.

The system of linear equations (11.6.2) corresponding to the fast and safe step differs in the right-hand side, with the former step having $\sigma = 0$ and the latter step having $\sigma \neq 0$. Thus if this system is solved by a factorization procedure, one factorization of the defining matrix is sufficient for both steps.

The Ralph-Wright IP Algorithm (RWIPA)

11.6.3 Algorithm.

Data: $0 < \delta_{\min} < \delta_{\max} < \frac{1}{2}$, $\tilde{\sigma} \in (0, \frac{1}{2})$, $\bar{\delta} \in (0, \frac{1}{2})$, γ, γ' and ρ all in $(0,1)$, $\eta_{\min} > 0$, $\eta_{\max} \equiv \eta_{\min} \exp(3/2)$, and (x^0, y^0, z^0) in $\mathbb{R}^{2n}_{++} \times \mathbb{R}^m$ satisfying (11.6.5).

Step 1: Set $t_0 \equiv 0$, $\delta_0 \equiv \delta_{\max}$, and $\eta_0 \equiv \eta_{\min}$. Let $k = 0$.

Step 2: If $\varsigma_k = 0$, terminate.

Step 3: Solve the system of linear equations (11.6.2) with $\sigma = 0$. Set

$$\tilde{\delta} \equiv \delta_{\min} + \bar{\delta}^{t_k+1} (\delta_{\max} - \delta_{\min}) \quad \text{and} \quad \tilde{\eta} \equiv (1 + \bar{\delta}^{t_k+1}) \eta_{\min}.$$

If $\xi \equiv 1 - \sqrt{\varsigma_k}/\bar{\delta}^{t_k} \leq 0$, let $\tau_k = 0$ and go to Step 4. Otherwise, let τ_k be the first element τ in the sequence ξ, $\rho\xi$, $\rho^2\xi, \ldots$, such that

$$(x^k(\tau), y^k(\tau)) > 0, \quad x^k(\tau) \circ y^k(\tau) \geq \tilde{\delta}\, \varsigma_k(\tau)\, \mathbf{1}_n$$

and $\| H(x^k(\tau), y^k(\tau), z^k(\tau)) \| \leq \tilde{\eta} \varsigma_k(\tau)$.

Step 4: If $\varsigma_k(\tau_k) \leq \gamma\, \varsigma_k$, set $\delta_{k+1} \equiv \tilde{\delta}$, $\eta_{k+1} \equiv \tilde{\eta}$, $t_{k+1} \equiv t_k + 1$, and $(x^{k+1}, y^{k+1}, z^{k+1}) \equiv (x^k(\tau_k), y^k(\tau_k), z^k(\tau_k))$. Let $k \leftarrow k + 1$ and go to Step 2.

Step 5: If $\varsigma_k(\tau_k) > \gamma \varsigma_k$, solve the system of linear equations (11.6.2) with $\sigma = \tilde{\sigma}$. Let τ_k be the first element τ in the sequence 1, ρ, $\rho^2, \ldots$, such that

$$(x^k(\tau), y^k(\tau)) > 0, \quad x^k(\tau) \circ y^k(\tau) \geq \delta_k\, \varsigma_k(\tau)\, \mathbf{1}_n,$$

$$\| H(x^k(\tau), y^k(\tau), z^k(\tau)) \| \leq \eta_k\, \varsigma_k(\tau),$$

and

$$\varsigma_k(\tau) \leq [1 - \tau\, \gamma' (1 - \sigma)]\, \varsigma_k.$$

Set $\delta_{k+1} \equiv \delta_k$, $\eta_{k+1} \equiv \eta_k$, $t_{k+1} \equiv t_k$, and $(x^{k+1}, y^{k+1}, z^{k+1}) \equiv (x^k(\tau_k), y^k(\tau_k), z^k(\tau_k))$. Let $k \leftarrow k + 1$ and go to Step 2.

Notice that $\delta_k \geq \delta_{\min}$ for all k and

$$\begin{aligned} \eta_k &= \eta_{\min} \prod_{j=1}^{t_k} (1 + \bar{\delta}^j) \leq \eta_{\min} \prod_{j=1}^{\infty} (1 + 2^{-j}) \\ &\leq \eta_{\min} \exp(3/2) = \eta_{\max}. \end{aligned}$$

Hence all iterates (x^k, y^k, z^k) indeed belong to the set Ω. The convergence properties of such a sequence of iterates can be established fairly easily.

11.6.4 Theorem. Let the mapping $H : \mathbb{R}^{2n+m} \to \mathbb{R}^{n+m}$ be continuous on $\mathbb{R}^{2n}_{+} \times \mathbb{R}^m$ and continuously differentiable on $\mathbb{R}^{2n}_{++} \times \mathbb{R}^m$. Suppose that the Jacobian matrix $JH(x, y, z)$ has the mixed P_0 property for all $(x, y, z) \in \Omega$. Let $\{(x^k, y^k, z^k)\}$ be an infinite sequence generated by Algorithm **11.6.3**. The following two statements hold.

(a) Every accumulation point of $\{(x^k, y^k, z^k)\}$ is a solution of the implicit MiCP (11.1.3).

(b) If the set

$$\Omega \cap \{ (x, y, z) \in \mathbb{R}^{2n+m} : x^T y \leq (x^0)^T y^0 \}$$

is bounded, then the sequence $\{(x^k, y^k, z^k)\}$ is bounded.

Proof. It suffices to prove (a). Let $(x^\infty, y^\infty, z^\infty)$ be the limit of a subsequence $\{(x^k, y^k, z^k) : k \in \kappa\}$. Since the sequence $\{\varsigma_k\}$ is decreasing, it follows that $\{\varsigma_k\}$ converges. It suffices to show that

$$\varsigma_\infty \equiv \frac{(x^\infty)^T y^\infty}{n} = 0.$$

Assume for contradiction that ς_∞ is positive. It follows that $(x^\infty, y^\infty, z^\infty)$ belongs to the set Ω. Since $\varsigma_{k+1} \leq \gamma \varsigma_k$ if a fast step is taken at iteration k, it follows that only finitely many fast steps are taken. Hence for all but finitely many k, we have

$$\varsigma_{k+1} \leq [1 - \tau_k \gamma' (1 - \tilde{\sigma})] \varsigma_k.$$

By Lemma **11.6.1** and the definition of τ_k, there exists a scalar $\bar{\tau} > 0$ such that $\tau_k \geq \bar{\tau}$ for all $k \in \kappa$ sufficiently large. Since $\tilde{\sigma} < \frac{1}{2}$, we deduce

$$\varsigma_{k+1} \leq \varsigma_k - \frac{\bar{\tau}\gamma'}{2} \varsigma_k \leq \varsigma_k - \frac{\bar{\tau}\gamma'}{2} \varsigma_\infty,$$

which shows that ς_{k+1} decreases by a positive constant infinitely often. This contradicts the convergence of this sequence of scalars. □

Both Algorithm **11.6.3** and Theorem **11.6.4** can be specialized to the NCP and the KKT system of a VI. The details are omitted.

11.7 Path-Following Noninterior Methods

One advantage of the IP methods applied to the CE formulation of the implicit MiCP (11.1.3) is that the mapping H is required to be differentiable

only on $\mathbb{R}^{2n}_{++} \times \mathbb{R}^m$. To take advantage of such a favorable differentiability condition, the IP methods generate iterates in the domain $\mathbb{R}^{2n}_{++} \times \mathbb{R}^m$ by carefully modifying the Newton equation corresponding to the complementarity condition in order to maintain the positivity condition more effectively. For problems where the function H is everywhere differentiable on $\mathbb{R}^{2n+m}$, this advantage of the IP methods becomes less distinctive as the need to maintain the positivity of the variables diminishes. In this section, we describe a family of path-following, or homotopy methods for solving the implicit MiCP (11.1.3) that are based on an unconstrained equation formulation of the CP. Such a method employs a nonsmooth C-function that is made smooth by a homotopy parameter, which is being driven toward zero by an iterative descent procedure. Since the iterates generated are not required to be positive, the term "non-interior" is used in the literature to describe this family of methods in order to distinguish them from the IP methods presented in the previous sections. The methods described in this section also give an alternative interpretation to the central path and pave the way to the smoothing methods that are examined subsequently.

We use the FB C-function:

$$\psi_{\mathrm{FB}}(a, b) \;=\; \sqrt{a^2 + b^2} - a - b, \quad (\, a, b\,) \;\in\; \mathbb{R}^2$$

as the basis for the development in the section. The first thing we do is to introduce a smoothing of this function by ensuring the positivity of the term inside the square root. Specifically, for each scalar $\mu > 0$, consider the function:

$$\psi_{\mathrm{FB}_\mu}(a, b) \;\equiv\; \sqrt{a^2 + b^2 + 2\mu} - a - b, \quad (\, a, b\,) \;\in\; \mathbb{R}^2.$$

The function ψ_{FB_μ} is everywhere continuously differentiable on $\mathbb{R}^2$. Moreover,

$$\psi_{\mathrm{FB}_\mu}(a, b) \;=\; 0 \;\Leftrightarrow\; [\,(\, a, b\,) \;>\; 0 \text{ and } ab \;=\; \mu\,].$$

In Exercise **11.9.8**, the reader is asked to develop a similar non-interior method using a smoothing of the min function.

Based on the smoothed FB functional ψ_{FB_μ}, consider the equation:

$$0 \;=\; \mathbf{H}_{\mathrm{FB}_\mu}(x, y, z) \;\equiv\; \begin{pmatrix} \mathbf{\Psi}_{\mathrm{FB}_\mu}(x, y) \\ H(x, y, z) \end{pmatrix},$$

where $\mathbf{\Psi}_{\mathrm{FB}_\mu} : \mathbb{R}^{2n} \to \mathbb{R}^n$ is the separable function:

$$\mathbf{\Psi}_{\mathrm{FB}_\mu}(x, y) \;\equiv\; \begin{pmatrix} \psi_{\mathrm{FB}_\mu}(x_1, y_1) \\ \vdots \\ \psi_{\mathrm{FB}_\mu}(x_n, y_n) \end{pmatrix}, \quad \forall\, (\, x, y\,) \;\in\; \mathbb{R}^{2n}.$$

Trivially, if $\mu > 0$, a triple (x, y, z) such that $\mathbf{H}_{\mathrm{FB}_\mu}(x, y, z) = 0$ automatically satisfies the positivity condition: $(x, y) > 0$. More precisely, for every $\mu \geq 0$ it holds that

$$\mathbf{H}_{\mathrm{FB}_\mu}(x, y, z) \;=\; 0 \;\Leftrightarrow\; \left[\mathbf{H}_{\mathrm{IP}}(x, y, z) \;=\; \begin{pmatrix} \mu \mathbf{1}_n \\ 0 \end{pmatrix} \text{ and } (\, x, y\,) \;\geq\; 0\right].$$

This shows that the solutions of the system $\mathbf{H}_{\mathrm{FB}_\mu}(x, y, z) = 0$ for positive values of μ coincide with the central path (or one of its variants) provided that the latter path exists. The key point here, however, is that by using $\mathbf{H}_{\mathrm{FB}_\mu}$ to describe this path we no longer need the extra nonnegativity assumption on the variables x and y. Thus, by following the path of zeros of the family of functions $\mathbf{H}_{\mathrm{FB}_\mu}$ for $\mu > 0$, we can dispense with the nonnegativity of the iterates. Due to nonlinearity of $\mathbf{H}_{\mathrm{FB}_\mu}$, we can not in practice trace such a path exactly; even so, the positivity of the iterates (x^k, y^k) is no longer an issue in this approach.

In what follows, we gather some additional properties of the scalar function $\psi_{\mathrm{FB}_\mu}(a, b)$ that are useful subsequently.

11.7.1 Lemma. The following properties are valid for the function ψ_{FB_μ}.

(a) For every $\mu \geq 0$, $\psi^2_{\mathrm{FB}_\mu}$ is continuously differentiable on $\mathbb{R}^2$.

(b) For all $(a, b) \in \mathbb{R}^2$ and $\mu > 0$,

$$|\, x^T \nabla^2 \psi^2_{\mathrm{FB}_\mu}(a, b) x \,| \;\leq\; 2\,(\,9 + \sqrt{2}\,), \quad \forall\, x \;\in\; \mathbb{R}^2 \text{ satisfying } \|\, x \,\| \;=\; 1;$$

when $\mu = 0$, the above holds for all $(a, b) \in \mathbb{R}^2 \setminus \{(0, 0)\}$.

(c) For any $\mu_1 \geq 0$, $\mu_2 \geq 0$ and $(a, b) \in \mathbb{R}^2$,

$$|\, \psi^2_{\mathrm{FB}_{\mu_1}}(a, b) - \psi^2_{\mathrm{FB}_{\mu_2}}(a, b) \,| \;\leq\; (\,2 + \sqrt{2}\,)\,|\, \mu_1 - \mu_2 \,|.$$

(d) For all $\mu \geq 0$ and $(a, b) \in \mathbb{R}^2$,

$$\frac{2}{2 + \sqrt{2}}\,(\,\min(a, b)\,)^2 - (\,2 + \sqrt{2}\,)\,\mu \;\leq\; \psi^2_{\mathrm{FB}_\mu}(a, b),$$

$$(\,2 + \sqrt{2}\,)\,(\,\min(a, b)\,)^2 + (\,2 + \sqrt{2}\,)\,\mu \;\geq\; \psi^2_{\mathrm{FB}_\mu}(a, b).$$

(e) For every $\mu > 0$,

$$0 \;>\; \frac{\partial \psi_{\mathrm{FB}_\mu}(a, b)}{\partial a} \;>\; -2, \quad 0 \;>\; \frac{\partial \psi_{\mathrm{FB}_\mu}(a, b)}{\partial b} \;>\; -2,$$

and

$$\left(\frac{\partial \psi_{\mathrm{FB}_\mu}(a, b)}{\partial a}\right)^2 + \left(\frac{\partial \psi_{\mathrm{FB}_\mu}(a, b)}{\partial b}\right)^2 \;\geq\; 3 - 2\sqrt{2} \;>\; 0.$$

Proof. Part (a) is obvious because we know that the squared FB function ψ_{FB}^2 is continuously differentiable on $\mathbb{R}^2$. To prove part (b), we note that for $\mu > 0$,

$$\nabla^2 \psi^2_{\mathrm{FB}_\mu}(a,b) \;=\; 2\,\nabla\psi_{\mathrm{FB}_\mu}(a,b)\,\nabla\psi_{\mathrm{FB}_\mu}(a,b)^{\,T} + 2\psi_{\mathrm{FB}_\mu}(a,b)\,\nabla^2\psi_{\mathrm{FB}_\mu}(a,b);$$

hence for every x with $\|x\| = 1$,

$$\begin{aligned} &|\, x^{\,T}\nabla^2\psi^2_{\mathrm{FB}_\mu}(a,b)x\,| \\ &\quad\le\; 2\,\|\nabla\psi_{\mathrm{FB}_\mu}(a,b)\,\|^2 + 2\,|\,\psi_{\mathrm{FB}_\mu}(a,b)\,|\,|\,x^{\,T}\nabla^2\psi_{\mathrm{FB}_\mu}(a,b)x\,|. \end{aligned}$$

We have

$$\nabla^2\psi_{\mathrm{FB}_\mu}(a,b) \;=\; \begin{bmatrix} \dfrac{b^2+2\,\mu}{(\,a^2+b^2+2\,\mu\,)^{3/2}} & \dfrac{-ab}{(\,a^2+b^2+2\,\mu\,)^{3/2}} \\[2ex] \dfrac{-ab}{(\,a^2+b^2+2\,\mu\,)^{3/2}} & \dfrac{a^2+2\,\mu}{(\,a^2+b^2+2\,\mu\,)^{3/2}} \end{bmatrix}.$$

It is easy to verify that with $\|x\| = 1$,

$$|\,x^{\,T}\nabla^2\psi_{\mathrm{FB}_\mu}(a,b)x\,| \;\le\; \frac{1}{\sqrt{a^2+b^2+2\,\mu}},$$

and that

$$\|\nabla\psi_{\mathrm{FB}_\mu}(a,b)\,\|^2 \;\le\; 8.$$

Hence,

$$\begin{aligned} |\,x^{\,T}\nabla^2\psi^2_{\mathrm{FB}_\mu}(a,b)x\,| \;&\le\; 16 + 2\left|\,1 - \frac{a+b}{\sqrt{a^2+b^2+2\,\mu}}\,\right| \\ &\le\; 16 + 2\,(\,1+\sqrt{2}\,) \;=\; 2\,(\,9+\sqrt{2}\,). \end{aligned}$$

This establishes (b) when $\mu > 0$. The case when $\mu = 0$ and $(a,b) \neq 0$ can be proved by just substituting $\mu = 0$ in the above proof. To prove (c), we note that

$$\begin{aligned} &\psi^2_{\mathrm{FB}_{\mu_1}}(a,b) - \psi^2_{\mathrm{FB}_{\mu_2}}(a,b) \\ &\quad=\; 2\,(\mu_1-\mu_2\,) - 2\,(\,a+b\,)\,[\,\sqrt{a^2+b^2+2\,\mu_1} - \sqrt{a^2+b^2+2\,\mu_2}\,] \\ &\quad=\; 2\,(\mu_1-\mu_2\,) - 2\,(\,\mu_1-\mu_2\,)\,\frac{a+b}{\sqrt{a^2+b^2+2\,\mu_1}+\sqrt{a^2+b^2+2\,\mu_2}}. \end{aligned}$$

Hence,

$$|\,\psi^2_{\mathrm{FB}_{\mu_1}}(a,b) - \psi^2_{\mathrm{FB}_{\mu_2}}(a,b)\,| \;\le\; (\,2+2\sqrt{2}\,)\,|\,\mu_1-\mu_2\,|.$$

Part (d) follows easily from part (c) and Lemma **9.1.3**, which bounds $|\psi_{\rm FB}(a,b)|$ in terms of $|\min(a,b)|$.

To prove part (e), we note that

$$\nabla\psi_{{\rm FB}_\mu}(a,b) = \begin{pmatrix} \dfrac{\partial\psi_{{\rm FB}_\mu}(a,b)}{\partial a} \\ \\ \dfrac{\partial\psi_{{\rm FB}_\mu}(a,b)}{\partial b} \end{pmatrix} = \begin{pmatrix} \dfrac{a}{\sqrt{a^2+b^2+2\mu}} - 1 \\ \\ \dfrac{b}{\sqrt{a^2+b^2+2\mu}} - 1 \end{pmatrix}.$$

Thus both partial derivatives are negative and greater than -2. Moreover, if either a or b is nonpositive, then

$$\left(\frac{\partial\psi_{{\rm FB}_\mu}(a,b)}{\partial a}\right)^2 + \left(\frac{\partial\psi_{{\rm FB}_\mu}(a,b)}{\partial b}\right)^2 \geq 1 > 3 - 2\sqrt{2}.$$

Consider the case where both a and b are positive. Since $\mu > 0$, we have

$$\inf_{(a,b)\geq 0} \left\{ \left(\frac{a}{\sqrt{a^2+b^2+2\mu}} - 1\right)^2 + \left(\frac{b}{\sqrt{a^2+b^2+2\mu}} - 1\right)^2 \right\}$$

$$\geq \inf_{(a,b)\geq 0} \left\{ \left(\frac{a}{\sqrt{a^2+b^2}} - 1\right)^2 + \left(\frac{b}{\sqrt{a^2+b^2}} - 1\right)^2 \right\} \geq 3 - 2\sqrt{2},$$

where the last inequality follows from Proposition **9.1.6**. □

Part (c) of Lemma **11.7.1** says that the Hessian of $\psi^2_{{\rm FB}_\mu}(a,b)$ is bounded above for all (a,b) in $\mathbb{R}^2$ when $\mu > 0$ and for all nonzero (a,b) when $\mu = 0$. Part (d) implies that if $\{\mu_k\}$ is any positive sequence converging to zero, then

$$\lim_{k\to\infty} \psi_{{\rm FB}_{\mu_k}}(a_k,b_k) = 0 \Leftrightarrow \lim_{k\to\infty} \min(a_k,b_k) = 0$$

for any sequence $\{(a_k,b_k)\} \subset \mathbb{R}^2$. This observation is needed to show the convergence of the homotopy method to be presented momentarily.

Consistent with the IP methods, we introduce a set that contains the iterates to be generated by the homotopy algorithm. Specifically, given two positive scalars μ_0 and η and an initial triple (x^0,y^0,z^0), define

$$\begin{aligned} \mathcal{N}(\mu_0,\eta) \equiv \{ \quad & (x,y,z) \in \mathbb{R}^{2n+m} : \exists\, \mu \in (0,\mu_0] \text{ such that} \\ & \| \mathbf{H}_{{\rm FB}_\mu}(x,y,z) \|^2 \leq \| H(x^0,y^0,z^0) \|^2 + \eta\,\mu \} \end{aligned}$$

Initiated at the triple (x^0,y^0,z^0) that belongs to $\mathcal{N}(\mu_0,\eta)$ for some positive pair of scalars μ_0 and η, the homotopy algorithm generates a sequence of iterates $\{(x^k,y^k,z^k)\}$ that belongs to $\mathcal{N}(\mu_0,\eta)$. At each iteration k, an update is obtained by computing a Newton step (dx,dy,dz) at the current

triple (x^k, y^k, z^k), based on the equation $\mathbf{H}_{\mathrm{FB}_{\mu_k}}(x, y, z) = 0$ defined by the current parameter μ_k. A step size $\tau_k \in (0, 1]$ is computed and the parameter μ_k is adjusted, yielding the next triple $(x^{k+1}, y^{k+1}, z^{k+1})$ and a new parameter $\mu_{k+1} \in (0, \mu_k)$. This ends the current iteration. The algorithm continues until a prescribed termination rule is satisfied.

A Homotopy Method for the Implicit MiCP (HMIMiCP)

11.7.2 Algorithm.

Data: $(\mu_0, \eta) > 0$ and (x^0, y^0, z^0) satisfying

$$\| \mathbf{\Psi}_{\mu_0}(x^0, y^0, z^0) \|^2 \leq \eta \mu_0;$$

γ, γ', ρ, ρ' all in $(0, 1)$.

Step 1: Let $k = 0$.

Step 2: If $H(x^k, y^k, z^k) = 0$ and $\mathbf{\Psi}_{\mathrm{FB}}(x^k, y^k, z^k) = 0$, terminate.

Step 3: Solve the system of linear equations

$$\mathbf{H}_{\mathrm{FB}_{\mu_k}}(x^k, y^k, z^k) + J\mathbf{H}_{\mathrm{FB}_{\mu_k}}(x^k, y^k, z^k) \begin{pmatrix} dx \\ dy \\ dz \end{pmatrix} = 0.$$

Step 4: If $\mathbf{H}_{\mathrm{FB}_{\mu_k}}(x^k, y^k, z^k) = 0$, let $\tau_k \equiv 0$. Otherwise let τ_k be the first element in the sequence $\{1, \rho, \rho^2, \ldots\}$ such that

$$\begin{aligned} &\| \mathbf{H}_{\mathrm{FB}_{\mu_k}}(x^k + \tau_k dx, y^k + \tau_k dy, z^k + \tau_k dz) \|^2 \\ &\quad \leq (1 - \gamma \tau_k) \| \mathbf{H}_{\mathrm{FB}_{\mu_k}}(x^k, y^k, z^k) \|^2. \end{aligned}$$

Set $(x^{k+1}, y^{k+1}, z^{k+1}) \equiv (x^k, y^k, z^k) + \tau_k(dx, dy, dz)$.

Step 5: Let δ_k be the first element in the sequence $\{1, \rho', (\rho')^2, \ldots\}$ such that

$$\begin{aligned} &\| \mathbf{H}_{\mathrm{FB}_{(1-\gamma'\delta_k)\mu_k}}(x^{k+1}, y^{k+1}, z^{k+1}) \|^2 \\ &\quad \leq \| H(x^0, y^0, z^0) \|^2 + (1 - \gamma' \delta_k) \eta \mu_k; \end{aligned}$$

Set $\mu_{k+1} \equiv (1 - \gamma'\delta_k)\mu_k$; and $k \leftarrow k + 1$; go to Step 2.

We need a set of assumptions on the function H in order to ensure the well-definedness and convergence of the algorithm. This latter task requires the demonstration of several things. First and foremost is the existence

(and uniqueness) of the search direction (dx, dy, dz). This is dealt with by the following lemma.

11.7.3 Lemma. If the Jacobian matrix

$$JH(x,y,z) \;=\; [\;\; J_xH(x,y,z) \quad J_yH(x,y,z) \quad J_zH(x,y,z) \;\;]$$

has the mixed P_0 property, then the matrix $J\mathbf{H}_{\mathrm{FB}_\mu}(x,y,z)$ is nonsingular for every scalar $\mu > 0$.

Proof. It is easy to see that

$$J\mathbf{H}_{\mathrm{FB}_\mu}(x,y,z) \;=\; \begin{bmatrix} D_x & D_y & 0 \\ J_xH(x,y,z) & J_yH(x,y,z) & J_zH(x,y,z) \end{bmatrix},$$

where D_x and D_y are the diagonal matrices whose diagonal entries are, respectively,

$$(D_x)_{ii} \;=\; \frac{x_i}{\sqrt{x_i^2 + y_i^2 + 2\mu}} - 1 \quad \text{and} \quad (D_y)_{ii} \;=\; \frac{y_i}{\sqrt{x_i^2 + y_i^2 + 2\mu}} - 1$$

for $i = 1, \ldots, n$. Clearly, these diagonal entries are all negative. Hence the nonsingularity of $J\mathbf{H}_{\mathrm{FB}_\mu}(x,y,z)$ follows easily from Lemma **11.4.3**. □

Subsequently, we need a result that pertains to the nonsingularity of a limiting matrix of a converging sequence $\{J\mathbf{H}_{\mathrm{FB}_{\mu_k}}(x^k, y^k, z^k)\}$, where $\{\mu_k\}$ is a sequence of positive scalars converging to zero and $\{(x^k, y^k, z^k)\}$ is a converging sequence of tuples. In order to present this result, we need the concept of a reduced Schur complement of a complementary principal submatrix in a partitioned matrix, which we define below.

Let Q be given by (11.4.1), where A and B are of order $(n+m) \times n$ and C is of order $(n+m) \times m$. Given a subset $\mathcal{J} \equiv \mathcal{J}_A \cup \mathcal{J}_B$ of $\{1, \ldots, n\}$, where $\mathcal{J}_A$ and $\mathcal{J}_B$ are two disjoint subsets of $\{1, \ldots, n\}$, let

$$[\,A \;\; B\,]_{\mathcal{J}\mathcal{J}} \;\equiv\; [\,A_{\mathcal{J}\mathcal{J}_A} \;\; B_{\mathcal{J}\mathcal{J}_B}\,] \;\in\; \mathbb{R}^{|\mathcal{J}| \times |\mathcal{J}|}.$$

We call such a submatrix of $[A \;\; B]$ a *complementary principal submatrix*. For instance, if $\mathcal{J} = \{1, 3, 4\}$ with $\mathcal{J}_A = \{1\}$ and $\mathcal{J}_B = \{3, 4\}$, then

$$[A \quad B]_{\mathcal{J}\mathcal{J}} \;=\; \begin{bmatrix} a_{11} & b_{13} & b_{14} \\ a_{31} & b_{33} & b_{34} \\ a_{41} & b_{34} & b_{44} \end{bmatrix}. \tag{11.7.1}$$

In general, every nonsingular complementary principal submatrix T of $[A \quad B]$ corresponding to the index set $\mathcal{J} = \mathcal{J}_A \cup \mathcal{J}_B$ induces a *reduced*

Schur complement of the matrix Q in the following manner. We first form the Schur complement of T in Q in the usual way, obtaining $[A' \quad B' \quad C']$, and then we delete those columns in $A'(B')$ corresponding to the indices in $\mathcal{J}_B$ ($\mathcal{J}_A$, respectively). Such a reduced Schur complement is of the form $[\tilde{A} \quad \tilde{B} \quad C']$, where $\tilde{A}$ and $\tilde{B}$ are of order $(n - |\mathcal{J}| + m) \times (n - |\mathcal{J}|)$ and C' is of order $(n - |\mathcal{J}| + m) \times m$. With the above example and with $n = 4$ and $m = 0$, the reduced Schur complement of the complementary principal submatrix (11.7.1) in $[A \;\; B]$ is

$$\begin{bmatrix} a_{22} & b_{22} \end{bmatrix} - \begin{bmatrix} a_{21} & b_{23} & b_{24} \end{bmatrix} \begin{bmatrix} a_{11} & b_{13} & b_{14} \\ a_{31} & b_{33} & b_{34} \\ a_{41} & b_{34} & b_{44} \end{bmatrix}^{-1} \begin{bmatrix} a_{12} & b_{12} \\ a_{32} & b_{32} \\ a_{42} & b_{42} \end{bmatrix}.$$

Lemma **11.7.4** below is in the spirit of Theorem **9.1.23**. This lemma does not assume that the sequence $\{\mu_k\}$ of positive scalars converges. The application of the lemma in the proof of Theorem **11.7.6** pertains to the case where $\{\mu_k\}$ converges to zero.

11.7.4 Lemma. Let $H : \mathbb{R}^{2n+m} \to \mathbb{R}^{n+m}$ be continuously differentiable. Let $\{(x^k, y^k, z^k)\}$ be a sequence of vectors converging to $(x^\infty, y^\infty, z^\infty)$. Let $\{\mu_k\}$ be a sequence of positive scalars. Let D_x^k and D_y^k be diagonal matrices whose diagonal entries are defined by: for $i = 1, \ldots, n$,

$$(\,(D_x^k)_{ii}, (D_y^k)_{ii}\,) \;=\; \frac{(\,x_i^k, y_i^k\,)}{\sqrt{(x_i^k)^2 + (y_i^k)^2 + 2\mu_k}} - (\,1, 1\,).$$

Assume that

$$\lim_{k\to\infty} (\,D_x^k, D_y^k\,) \;=\; (\,D_x^\infty, D_y^\infty\,).$$

Let $\mathcal{J} \equiv \mathcal{J}_x \cup \mathcal{J}_y$, where

$$\mathcal{J}_x \;\equiv\; \{\, i \,:\, (\,D_x^\infty\,)_{ii} \;=\; 0\,\} \quad \text{and} \quad \mathcal{J}_y \;\equiv\; \{\, i \,:\, (\,D_y^\infty\,)_{ii} \;=\; 0\,\}.$$

Suppose that (a) the complementary principal submatrix

$$[\,J_x H^\infty \;\; J_y H^\infty\,]_{\mathcal{J}\mathcal{J}} \;\equiv\; \begin{bmatrix} (\,J_x H^\infty\,)_{\mathcal{J}_x\mathcal{J}_x} & (\,J_y H^\infty\,)_{\mathcal{J}_x\mathcal{J}_y} \\ (\,J_x H^\infty\,)_{\mathcal{J}_y\mathcal{J}_x} & (\,J_y H^\infty\,)_{\mathcal{J}_y\mathcal{J}_y} \end{bmatrix}$$

is nonsingular, where $J_s H^\infty \equiv J_s H(x^\infty, y^\infty, z^\infty)$ for $s = x, y, z$, and (b) the reduced Schur complement of the above matrix in JH^∞ has the mixed P_0 property. The limiting matrix

$$\begin{bmatrix} D_x^\infty & D_y^\infty & 0 \\ J_x H^\infty & J_y H^\infty & J_z H^\infty \end{bmatrix}$$

is nonsingular.

Proof. By part (e) of Lemma **11.7.1**, the two index sets $\mathcal{J}_x$ and $\mathcal{J}_y$ are disjoint subsets of $\{1,\ldots,n\}$. Let $\mathcal{K}$ be the complement of $\mathcal{J}_x \cup \mathcal{J}_y$ in $\{1,\ldots,n\}$. We can write

$$D_x^\infty = \begin{bmatrix} 0 & 0 & 0 \\ 0 & (D_x^\infty)_{\mathcal{J}_y\mathcal{J}_y} & 0 \\ 0 & 0 & (D_x^\infty)_{\mathcal{K}\mathcal{K}} \end{bmatrix},$$

and

$$D_y^\infty = \begin{bmatrix} (D_y^\infty)_{\mathcal{J}_x\mathcal{J}_x} 0 & & 0 \\ 0 & 0 & 0 \\ 0 & 0 & (D_y^\infty)_{\mathcal{K}\mathcal{K}} \end{bmatrix},$$

where $(D_x^\infty)_{\mathcal{K}\mathcal{K}}$ and $(D_y^\infty)_{\mathcal{K}\mathcal{K}}$ are both diagonal matrices with negative diagonal entries. The matrix

$$\begin{bmatrix} 0 & 0 & (D_y^\infty)_{\mathcal{J}_x\mathcal{J}_x} & 0 \\ 0 & (D_x^\infty)_{\mathcal{J}_y\mathcal{J}_y} & 0 & 0 \\ (J_xH^\infty)_{\mathcal{J}_x\mathcal{J}_x} & (J_xH^\infty)_{\mathcal{J}_x\mathcal{J}_y} & (J_yH^\infty)_{\mathcal{J}_x\mathcal{J}_x} & (J_yH^\infty)_{\mathcal{J}_x\mathcal{J}_y} \\ (J_xH^\infty)_{\mathcal{J}_y\mathcal{J}_x} & (J_xH^\infty)_{\mathcal{J}_y\mathcal{J}_y} & (J_yH^\infty)_{\mathcal{J}_y\mathcal{J}_x} & (J_yH^\infty)_{\mathcal{J}_y\mathcal{J}_y} \end{bmatrix}$$

is a nonsingular principal submatrix of the matrix

$$\begin{bmatrix} D_x^\infty & D_y^\infty & 0 \\ J_xH^\infty & J_yH^\infty & J_zH^\infty \end{bmatrix}.$$

Thus the latter matrix is nonsingular if the Schur complement of the former matrix in the latter matrix is nonsingular. It is not difficult to verify that such a Schur complement has the form:

$$\begin{bmatrix} (D_x^\infty)_{\mathcal{K}\mathcal{K}} & (D_y^\infty)_{\mathcal{K}\mathcal{K}} & 0 \\ \bar{A} & \bar{B} & \bar{C} \end{bmatrix}, \tag{11.7.2}$$

where the partitioned matrix

$$\begin{bmatrix} \bar{A} & \bar{B} & \bar{C} \end{bmatrix},$$

is the reduced Schur complement the matrix $[J_xH^\infty \;\; J_yH^\infty]_{\mathcal{J}\mathcal{J}}$ in JH^∞. By assumption, this Schur complement has the mixed P_0 property. The nonsingularity of (11.7.2) therefore follows from Lemma **11.4.3**. □

Returning to Algorithm **11.7.2**, we see that the equation

$$\mathbf{H}_{\mathrm{FB}_{\mu_k}}(x^k, y^k, z^k) + J\mathbf{H}_{\mathrm{FB}_{\mu_k}}(x^k, y^k, z^k) \begin{pmatrix} dx \\ dy \\ dz \end{pmatrix} = 0.$$

is the Newton equation of the system

$$\mathbf{H}_{\mathrm{FB}_{\mu_k}}(x, y, z) = 0$$

at the triple (x^k, y^k, z^k). Hence if the matrix $J\mathbf{H}_{\mathrm{FB}_{\mu_k}}(x^k, y^k, z^k)$ is nonsingular, then the step size τ_k in Step 4 of Algorithm **11.7.2** is well defined and can be determined in a finite number of trials, by the theory of a standard Newton method for a system of smooth equations. Next, we show that the scalar μ_{k+1} in Step 5 is well defined too; in doing so, we also obtain a lower bound for μ_{k+1} in terms of τ_k. Such a bound is essential in the convergence proof of the algorithm. In the following lemma, we use the fact that the iterate (x^k, y^k, z^k) satisfies

$$\| \mathbf{H}_{\mathrm{FB}_\mu}(x^k, y^k, z^k) \|^2 \leq \| H(x^0, y^0, z^0) \|^2 + \eta \mu_k,$$

which is satisfied for $k = 0$ and can be verified by an easy inductive argument for all $k \geq 1$.

11.7.5 Lemma. The scalar δ_k in Step 5 of Algorithm **11.7.2** satisfies

$$\frac{\gamma}{\gamma'} \frac{\eta \tau_k}{(2+\sqrt{2})n + \eta} \geq \delta_k \geq \rho' \min\left(1, \frac{\gamma}{\gamma'} \frac{\eta \tau_k}{(2+\sqrt{2})n + \eta}\right).$$

Proof. By the definition of δ_k, it suffices to establish that for all $\delta > 0$ such that

$$\delta \leq \frac{\gamma}{\gamma'} \frac{\eta \tau_k}{(2+\sqrt{2})n + \eta}, \tag{11.7.3}$$

we have

$$\begin{aligned} &\| \mathbf{H}_{\mathrm{FB}_{(1-\gamma'\delta)\mu_k}}(x^{k+1}, y^{k+1}, z^{k+1}) \|^2 \\ &\quad \leq \| H(x^0, y^0, z^0) \|^2 + (1 - \gamma' \delta) \eta \mu_k. \end{aligned} \tag{11.7.4}$$

Consider first the case where $\mathbf{H}_{\mathrm{FB}_{\mu_k}}(x^k, y^k, z^k)$ is nonzero. By part (c) of Lemma **11.7.1** and Step 4 of the algorithm, we deduce

$$\begin{aligned} &\| \mathbf{H}_{\mathrm{FB}_{(1-\gamma'\delta)\mu_k}}(x^{k+1}, y^{k+1}, z^{k+1}) \|^2 \\ &= \| \mathbf{H}_{\mathrm{FB}_{\mu_k}}(x^{k+1}, y^{k+1}, z^{k+1}) \|^2 + \\ &\quad \| \mathbf{\Psi}_{\mathrm{FB}_{(1-\gamma'\delta)\mu_k}}(x^{k+1}, y^{k+1}, z^{k+1}) \|^2 - \| \mathbf{\Psi}_{\mathrm{FB}_{\mu_k}}(x^{k+1}, y^{k+1}, z^{k+1}) \|^2 \\ &\leq (2+\sqrt{2}) n \gamma' \delta \mu_k + (1 - \gamma \tau_k) \| \mathbf{H}_{\mathrm{FB}_{\mu_k}}(x^k, y^k, z^k) \|^2 \\ &\leq (2+\sqrt{2}) n \gamma' \delta \mu_k + (1 - \gamma \tau_k) (\| H(x^0, y^0, z^0) \|^2 + \eta \mu_k). \end{aligned}$$

Hence in order for (11.7.4) to hold, it suffices that

$$(\,2+\sqrt{2}\,)\,n\,\gamma'\,\delta\,\mu_k + (\,1-\gamma\,\tau_k\,)\,\eta\,\mu_k \;\le\; (\,1-\gamma'\,\delta\,)\,\eta\,\mu_k.$$

A simple algebra shows that the latter inequality holds whenever (11.7.3) holds.

Suppose $\mathbf{H}_{\mathrm{FB}_{\mu_k}}(x^k, y^k, z^k) = 0$. We have

$$(\,x^{k+1}, y^{k+1}, z^{k+1}\,) \;=\; (\,x^k, y^k, z^k\,).$$

Hence

$$\|\,\mathbf{H}_{\mathrm{FB}_{(1-\gamma'\delta)\mu_k}}(x^{k+1}, y^{k+1}, z^{k+1})\,\|^2 \;\le\; (\,2+\sqrt{2}\,)\,n\,\gamma'\,\delta\,\mu_k.$$

Applying the last part of the above proof, we easily deduce if δ satisfies (11.7.3), then

$$\|\,\mathbf{H}_{\mathrm{FB}_{(1-\gamma'\delta)\mu_k}}(x^{k+1}, y^{k+1}, z^{k+1})\,\|^2 \;\le\; (\,1-\gamma'\,\delta\,)\,\eta\,\mu_k,$$

which clearly implies (11.7.4). □

Lemmas **11.7.3** and **11.7.5** ensure that Algorithm **11.7.2** generates a well-defined sequence of triples

$$\{\,(\,x^k, y^k, z^k\,)\,\} \;\subset\; \mathcal{N}(\mu_0, \eta)$$

along with a sequence of positive homotopy parameters $\{\mu_k\}$ such that for every k,

$$\|\,H(x^k, y^k, z^k)\,\|^2 \;\le\; \|\,H(x^0, y^0, z^0)\,\|^2 + \eta\,\mu_k$$

and

$$\mu_{k+1} \;=\; (\,1-\gamma'\,\delta_k\,)\,\mu_k.$$

The main convergence result for Algorithm **11.7.2** is as follows.

11.7.6 Theorem. Let the mapping $H : \mathbb{R}^{2n+m} \to \mathbb{R}^{n+m}$ be continuously differentiable on $\mathbb{R}^{2n}_{+} \times \mathbb{R}^m$. Suppose that the Jacobian matrix $JH(x, y, z)$ has the mixed P_0 property for all $(x, y, z) \in \mathcal{N}(\mu_0, \eta)$. Let $\{(x^k, y^k, z^k)\}$ be an infinite sequence generated by Algorithm **11.7.2**. The following two statements hold.

(a) Suppose that $(x^\infty, y^\infty, z^\infty)$ is the limit of a convergent subsequence $\{(x^k, y^k, z^k) : k \in \kappa\}$. If conditions (a) and (b) of Lemma **11.7.4** hold for every accumulation point (D_x^∞, D_y^∞) of the (bounded) sequence $\{(D_x^k, D_y^k) : k \in \kappa\}$, then $(x^\infty, y^\infty, z^\infty)$ is a solution of the implicit MiCP (11.1.3).

(b) If the set $\mathcal{N}(\mu_0, \eta)$ is bounded, then the sequence $\{(x^k, y^k, z^k)\}$ is bounded.

Proof. Suppose that

$$\liminf_{k(\in\kappa)\to\infty} \tau_k > 0.$$

By Lemma **11.7.5**, it follows that

$$\liminf_{k(\in\kappa)\to\infty} \mu_k > 0.$$

The sequence $\{\mu_k\}$ is monotonically decreasing; since each μ_k is positive, the sequence therefore converges. Hence

$$0 = \lim_{k\to\infty} (\mu_{k+1} - \mu_k),$$

which implies

$$\lim_{k\to\infty} \tau_k \mu_k = 0;$$

but this contradicts

$$\liminf_{k(\in\kappa)\to\infty} \mu_k \tau_k > 0.$$

Consequently, we must have

$$\lim_{k(\in\kappa)\to\infty} \tau_k = 0.$$

In turn, by Lemma **11.7.5**, it follows that

$$\lim_{k(\in\kappa)\to\infty} \mu_k = 0.$$

If $\mathbf{H}_{\mathrm{FB}_{\mu_k}}(x^k, y^k, z^k) = 0$ for infinitely many k in κ, then taking the limit on such a subsequence of iterates, we deduce that

$$\mathbf{H}_{\mathrm{FB}}(x^\infty, y^\infty, z^\infty) = 0,$$

which shows that the triple $(x^\infty, y^\infty, z^\infty)$ is a solution of the implicit MicCP (11.1.3). Hence, without loss of generality, we may assume that $\mathbf{H}_{\mathrm{FB}_{\mu_k}}(x^k, y^k, z^k)$ is nonzero for all but finitely many k in κ. Thus with $\tau_k' \equiv \rho^{-1}\tau_k$, we have

$$\frac{\| \mathbf{H}_{\mathrm{FB}_{\mu_k}}(x^k + \tau_k' dx, y^k + \tau_k' dy, z^k + \tau_k' dz) \|^2 - \| \mathbf{H}_{\mathrm{FB}_{\mu_k}}(x^k, y^k, z^k) \|^2}{\tau_k'}$$

$$> -\gamma \| \mathbf{H}_{\mathrm{FB}_{\mu_k}}(x^k, y^k, z^k) \|^2.$$

Substituting the definition of $\mathbf{H}_{\mathrm{FB}_{\mu_k}}$, we obtain

$$\frac{H_k + FB_k}{\tau_k'} \; > \; -\gamma \,\|\, \mathbf{H}_{\mathrm{FB}_{\mu_k}}(x^k, y^k, z^k) \,\|^2,$$

where

$$H_k \;\equiv\; \|\, H(x^k + \tau_k' dx, y^k + \tau_k' dy, z^k + \tau_k' dz) \,\|^2 - \|\, H(x^k, y^k, z^k) \,\|^2$$

and

$$FB_k \;\equiv$$
$$\|\, \mathbf{\Psi}_{\mathrm{FB}_{\mu_k}}(x^k + \tau_k' dx, y^k + \tau_k' dy, z^k + \tau_k' dz) \,\|^2 - \|\, \mathbf{\Psi}_{\mathrm{FB}_{\mu_k}}(x^k, y^k, z^k) \,\|^2.$$

The sequence of matrices $\{J\mathbf{H}_{\mathrm{FB}_{\mu_k}}(x^k, y^k, z^k) : k \in \kappa\}$ is bounded. Without loss of generality, we may assume that the sequence converges to a matrix $J\mathbf{H}^\infty$, which must be nonsingular by assumption and Lemma **11.7.4**. Hence the sequence of directions $\{(dx^k, dy^k, dz^k) \,:\, k \in \kappa\}$ is bounded. Without loss of generality, we may assume that the latter sequence converges to a limit $(dx^\infty, dy^\infty, dz^\infty)$. Hence the sequence

$$\{\, (\, x^k + \tau_k' dx^k, y^k + \tau_k' dy^k, z^k + \tau_k' dz^k \,) \,:\, k \in \kappa \,\}$$

also converges to $(x^\infty, y^\infty, z^\infty)$. By the definition of (dx^k, dy^k, dz^k) and the continuous differentiability of H, we can write

$$H_k \;=\; -\tau_k' \,\|\, H(x^k, y^k, z^k) \,\|^2 + o(\tau_k').$$

Similarly, by the mean-value theorem and part (b) of Lemma **11.7.1**, we have

$$FB_k \;=\; -\tau_k' \,\|\, \mathbf{\Psi}_{\mathrm{FB}_{\mu_k}}(x^k, y^k, z^k) \,\|^2 + O((\tau_k')^2).$$

Consequently, we deduce

$$-\|\, \mathbf{H}_{\mathrm{FB}}(x^\infty, y^\infty, z^\infty) \,\|^2 \;\geq\; -\gamma \,\|\, \mathbf{H}_{\mathrm{FB}}(x^\infty, y^\infty, z^\infty) \,\|^2,$$

which is a contradiction because $\gamma < 1$ and $\mathbf{H}_{\mathrm{FB}}(x^\infty, y^\infty, z^\infty)$ is nonzero. □

11.8 Smoothing Methods

The mapping $\mathbf{H}_{\mathrm{FB}_\mu}(x, y, z)$ derived from the nonsmooth FB C-function provides an example of a smoothing function. In this section, we formally define the latter concept (see Definition **11.8.1**) and present a family of methods for solving a nonsmooth equation based on it. In the previous

section we saw that "smoothing" methods and interior point methods have much in common. One of the main differences between the methods discussed so far and those we present in this section is that we no longer insist on using only some (biased) Newton direction but allow for the use of a wider range of options.

We are interested in solving the unconstrained system of nonlinear equations:

$$G(x) \;=\; 0 \tag{11.8.1}$$

where $G : \mathbb{R}^n \to \mathbb{R}^n$ is locally Lipschitz continuous but not assumed to be differentiable. The key to the developments in this section is the following definition.

11.8.1 Definition. Let $G : \mathbb{R}^n \to \mathbb{R}^n$ be a locally Lipschitz continuous function. We say that a family of functions

$$\{\, G_\varepsilon \,:\, \varepsilon \,\in\, \mathcal{P} \,\}, \tag{11.8.2}$$

where $\mathcal{P}$ is a subset of $\mathbb{R}^m$ containing 0 and a sequence of nonzero vectors converging to zero, is a *smoothing* of G in a domain $\mathcal{D} \subseteq \mathbb{R}^n$ if for every ε in $\mathcal{P}$, $G_\varepsilon : \mathbb{R}^n \to \mathbb{R}^n$ is continuously differentiable on $\mathcal{D}$, $G_\varepsilon(x)$ is continuous as a function of $(\varepsilon, x) \in \mathcal{P} \times \mathcal{D}$ and $G_0(x) = G(x)$ for all $x \in \mathcal{D}$. We call each member G_ε a smoothing function of G on $\mathcal{D}$. If $\mathcal{D}$ is the neighborhood of a given vector $\bar{x}$, we say that G_ε is a smoothing function of G near $\bar{x}$. □

Obviously, if (11.8.2) is a smoothing of G in $\mathcal{D}$, then

$$\lim_{\substack{z \to x \\ \varepsilon \downarrow 0}} G_\varepsilon(z) \;=\; G(x), \quad \forall\, x \,\in\, \mathcal{D}.$$

In the above definition we allow for a lot of freedom in the choice of the set $\mathcal{P}$, even if we remark that in most practical cases $m = 1$ and $\mathcal{P} = [0, \infty)$. With the aid of the family (11.8.2) of smoothing functions G_ε, the central idea of a smoothing method for computing a zero of G is very simple. Namely, we attempt to solve, possibly inexactly, a sequence of smooth equations

$$G_{\varepsilon_k}(x) = 0 \tag{11.8.3}$$

associated with a sequence $\{\varepsilon_k\}$ converging to zero. In principle we could try to follow the developments we made for IP methods and study the path described by the family of zeros of G_{ε_k}. This approach however, is likely to be profitable only under some kind of monotonicity assumption on the function G. The latter assumption is very natural in the context of interior point methods, because the (biased) Newton directions, which lie at the

heart of the IP methods, cannot be guaranteed to exist otherwise; furthermore, it is not easy to obtain meaningful, global convergence results in the absence of some sort of monotonicity (or P_0) property. As we said at the beginning of this section, when dealing with smoothing methods we want to go beyond the exclusive use of a (biased) Newton direction; therefore we intend not to assume any monotonicity condition on the function G. The price we pay for this is, obviously, that we are only able to guarantee convergence to stationary solutions of some merit function. From this point of view smoothing methods are close in spirit to the methods of Chapters **8** and **9**.

In principle, we can apply any convergent algorithm for solving (11.8.3). In order to focus the discussion, we intend to apply a standard line-search Newton method to (11.8.3). Although for each fixed ε_k such a method is locally, superlinearly convergent under a suitable nonsingular assumption on the Jacobian matrix, the overall convergence of a smoothing method for solving the original, nonsmooth equation (11.8.1) cannot be expected to be fast unless the smoothing satisfies a stronger property than Definition **11.8.1**. For this purpose, we introduce an important class of smoothing functions. For G_ε chosen within the latter class, and with a progressive accuracy in solving the smoothed equation (11.8.3), the superlinear convergence of a smoothing method can be accomplished easily; this is in contrast to a homotopy algorithm where superlinear convergence is somewhat more difficult to obtain. For the detailed description of a complete smoothing method, see Subsection **11.8.1**.

11.8.2 Definition. Let $G : \mathbb{R}^n \to \mathbb{R}^n$ be a locally Lipschitz continuous function and suppose that G_ε is a smoothing of G in a neighborhood of a vector $\bar{x}$. We say that G_ε is a *superlinear approximation* of G at $\bar{x}$ if there exist functions $\Delta_i : (0, \infty) \to [0, \infty)$ satisfying

$$\lim_{t \downarrow 0} \Delta_1(t) = 0 \quad \text{and} \quad \Delta_2(t) = O(t),$$

and a neighborhood $\mathcal{N}$ of $\bar{x}$ such that for every $x \in \mathcal{N}$,

$$\begin{aligned} &\| G(x) + JG_\varepsilon(x)(\bar{x} - x) - G(\bar{x}) \| \\ &\quad \leq \| x - \bar{x} \| \, \Delta_1(\| x - \bar{x} \|) + \Delta_2(\| \varepsilon \|). \end{aligned} \tag{11.8.4}$$

We say that G_ε is a *quadratic approximation* of G at $\bar{x}$ if a constant $c > 0$ exists such that, instead of (11.8.4),

$$\| G(x) + JG_\varepsilon(x)(\bar{x} - x) - G(\bar{x}) \| \leq c \| x - \bar{x} \|^2 + \Delta_2(\| \varepsilon \|). \tag{11.8.5}$$

Since by definition $G_\varepsilon(x)$ is continuous as a function of ε and x and therefore is also uniformly continuous on a compact set containing $(\varepsilon, \bar{x})$, we can write for some positive constant L and for any x sufficiently close to $\bar{x}$ and ε sufficiently close to zero,

$$\| G_\varepsilon(x) - G(x) \| \leq L\,\varepsilon.$$

Taking into account this fact, (11.8.4) and (11.8.5) can equivalently be rewritten as

$$\begin{aligned} &\| G_\varepsilon(x) + JG_\varepsilon(x)(\bar{x} - x) - G(\bar{x}) \| \\ &\qquad \leq \| x - \bar{x} \| \Delta_1(\| x - \bar{x} \|) + \Delta_2(\| \varepsilon \|) \end{aligned} \tag{11.8.6}$$

and

$$\| G_\varepsilon(x) + JG_\varepsilon(x)(\bar{x} - x) - G(\bar{x}) \| \leq c \| x - \bar{x} \|^2 + \Delta_2(\| \varepsilon \|). \tag{11.8.7}$$

Recalling Definition **7.2.2**, we see that if G has a Newton approximation $\mathcal{A}$ near $\bar{x}$ and if G_ε is a smoothing function of G near $\bar{x}$, then a sufficient condition for G_ε to be a superlinear approximation of G at $\bar{x}$ is that a neighborhood $\mathcal{N}$ of $\bar{x}$ exists such that for every $x \in \mathcal{N}$, there exists $A(x, \cdot)$ in $\mathcal{A}(x)$ satisfying

$$\| A(x, \bar{x} - x) - JG_\varepsilon(x)(\bar{x} - x) \| \leq \Delta_2(\varepsilon).$$

Moreover, if the Newton approximation $\mathcal{A}$ is strong, then the smoothing function G_ε is a quadratic approximation. Thus, roughly speaking, the smoothing function G_ε approximates G superlinearly (or quadratically) at $\bar{x}$ if $JG_\varepsilon(x)$ is an approximation of $A(x, \cdot)$ in the direction $\bar{x} - x$ of the order $\Delta_2(\| \varepsilon \|)$.

In general, if $\{x^k\}$ is a sequence converging to $\bar{x}$ and $\{\varepsilon_k\}$ is a sequence of positive numbers converging to zero, it is not necessarily true that every limit point of the sequence of Jacobian matrices $\{JG_{\varepsilon_k}(x^k)\}$ belongs to $\partial G(\bar{x})$, although this seems a rather natural property. A smoothing that possesses this property is very desirable for algorithmic reasons, even if it is not always easy to guarantee the property. This motivates the introduction of a definition. First we recall the notation from (7.5.18):

$$\partial_C G(x) \equiv (\, \partial G_1(x) \times \partial G_2(x) \times \cdots \times \partial G_n(x) \,)^T.$$

We know that $\partial_C G(x)$ contains $\partial G(x)$; moreover, $\partial_C G$ is an upper semi-continuous set-valued map.

11.8.3 Definition. Let $G : \mathbb{R}^n \to \mathbb{R}^n$ be a locally Lipschitz continuous function and suppose that G_ε is a smoothing of G in a neighborhood of a vector $\bar{x}$. We say that G_ε is a *weakly Jacobian consistent smoothing* of G at $\bar{x}$ if for every sequence $\{x^k\}$ converging to $\bar{x}$ and every sequence $\{\varepsilon_k\}$ in $\mathcal{P}$ converging to zero, the sequence of Jacobian matrices $\{JG_{\varepsilon_k}(x^k)\}$ has at least one accumulation point and every such point belongs to $\partial_C G(x)$. If every limit point actually belongs to the smaller set $\partial G(\bar{x})$, we say that G_ε is a *Jacobian consistent smoothing* of G at $\bar{x}$. □

The (weakly) Jacobian consistency property is important in establishing that limit points of sequences generated by smoothing methods are stationary point of a merit function. In fact, when we compute a zero of a nonsmooth function G via solving a sequence of smooth equations (11.8.3) corresponding to a sequence $\{\varepsilon_k\}$ converging to zero, we can expect to obtain a sequence of iterates $\{x^k\}$ that satisfies

$$\lim_{k\to\infty} \nabla\theta_{\varepsilon_k}(x^k) \;=\; 0, \tag{11.8.8}$$

where

$$\theta_{\varepsilon_k}(x) \;\equiv\; \tfrac{1}{2}\, G_{\varepsilon_k}(x)^T G_{\varepsilon_k}(x).$$

Under the Jacobian consistency property of the smoothing, we can deduce that every accumulation point x^∞ of the sequence $\{x^k\}$ is a C-stationary point of the limiting merit function $\theta \equiv \frac{1}{2}G^T G$, i.e., $0 \in \partial\theta(x^\infty)$, provided that G is locally Lipschitz continuous. If the weakly Jacobian consistency property holds instead, we can only show that $0 \in (\partial_C G(x^\infty))^T G(x^\infty)$. The latter condition can be regarded as a weak form of stationarity.

11.8.4 Proposition. Let G_ε be a smoothing of the locally Lipschitz function $G : \mathbb{R}^n \to \mathbb{R}^n$. Let $\{x^k\}$ be any sequence satisfying (11.8.8) and converging to x^∞. If $\{\varepsilon_k\}$ converges to zero, then the following two statements hold.

(a) If G_ε is weakly Jacobian consistent at x^∞, then

$$0 \;\in\; (\,\partial_C G(x^\infty)\,)^T G(x^\infty).$$

(b) If G_ε is Jacobian consistent at x^∞, x^∞ is a C-stationary point of θ.

Proof. We have

$$\partial\theta(x^\infty) \;=\; \partial G(x^\infty)^T G(x^\infty).$$

Assume that the smoothing enjoys the Jacobian consistency property. Since

$$\nabla\theta_{\varepsilon_k}(x^k) \;=\; JG_{\varepsilon_k}(x^k)^T G_{\varepsilon_k}(x^k),$$

a simple limiting argument easily establishes $0 \in \partial\theta(x^\infty)$. The proof for the weakly consistency case is analogous. □

We illustrate Definitions **11.8.2** and **11.8.3** with the smoothing of the FB reformulation of the implicit MiCP (11.1.3). In Subsection **11.8.2** we give other examples in a more systematic setting.

11.8.5 Example. Let $H : \mathbb{R}^{2n+m} \to \mathbb{R}^{n+m}$ be C^1. Let $\mathbf{H}_{\mathrm{FB}_\varepsilon}(x, y, z)$ be the smoothing of the function $\mathbf{H}_{\mathrm{FB}}(x, y, z)$; that is, for $\varepsilon \geq 0$,

$$\mathbf{H}_{\mathrm{FB}_\varepsilon}(x, y, z) \;=\; \begin{pmatrix} \psi_{\mathrm{FB}_\varepsilon}(x_1, y_1) \\ \vdots \\ \psi_{\mathrm{FB}_\varepsilon}(x_n, y_n) \\ H(x, y, z) \end{pmatrix},$$

where

$$\psi_{\mathrm{FB}_\varepsilon}(a, b) \;=\; \sqrt{a^2 + b^2 + 2\varepsilon} - a - b.$$

By Exercise **7.6.11**, we have

$$\partial H(x, y, z) \;\equiv\; \left\{ \begin{pmatrix} D_x & D_y & 0 \\ J_x H(x, y, z) & J_y H(x, y, z) & J_z H(x, y, z) \end{pmatrix} \right\},$$

where D_x and D_y are diagonal matrices whose diagonal entries are given by: for $i = 1, \ldots, n$,

$$(\,(D_x)_{ii}, (D_y)_{ii}\,) \begin{cases} = \dfrac{(x_i, y_i)}{\sqrt{x_i^2 + y_i^2}} - (1, 1) & \text{if } (x_i, y_i) \neq 0 \\ \in \mathrm{cl}\, \mathrm{I\!B}(0, 1) - (1, 1) & \text{otherwise.} \end{cases}$$

For every $\varepsilon > 0$, we have

$$J\mathbf{H}_{\mathrm{FB}_\varepsilon}(x, y, z) \;=\; \begin{pmatrix} D_x^\varepsilon & D_y^\varepsilon & 0 \\ J_x H(x, y, z) & J_y H(x, y, z) & J_z H(x, y, z) \end{pmatrix},$$

where D_x^ε and D_y^ε are diagonal matrices whose diagonal entries are given by: for $i = 1, \ldots, n$,

$$(\,(D_x^\varepsilon)_{ii}, (D_y^\varepsilon)_{ii}\,) \;=\; \frac{(x_i, y_i)}{\sqrt{x_i^2 + y_i^2 + 2\varepsilon}} - (1, 1).$$

From the expression of $J\mathbf{H}_{\mathrm{FB}_\varepsilon}(x, y, z)$, it is clear that $\mathbf{H}_{\mathrm{FB}_\varepsilon}$ is a Jacobian consistent smoothing of $\mathbf{H}_{\mathrm{FB}}$ at any triple $(\bar{x}, \bar{y}, \bar{z})$.

We next show that $\mathbf{H}_{\mathrm{FB}_\varepsilon}$ is a superlinear approximation of $\mathbf{H}_{\mathrm{FB}}$. Comparing $J\mathbf{H}_{\mathrm{FB}_\varepsilon}(x,y,z)$ with the corresponding matrix in $\partial H(x,y,z)$, we see that it suffices to show for every pair of scalars $(\bar{a},\bar{b})$, there exists a constant $\eta > 0$ such that for all $\varepsilon > 0$,

$$\left| \frac{a}{\sqrt{a^2+b^2}} - \frac{a}{\sqrt{a^2+b^2+2\varepsilon}} \right| |\,\bar{a}-a\,| \;\le\; \eta\sqrt{2\,\varepsilon}$$

for all nonzero (a,b) sufficiently close to $(\bar{a},\bar{b})$. The left-hand side of the above expression is equal to

$$\frac{|\,a\,|}{\sqrt{a^2+b^2}\sqrt{a^2+b^2+2\varepsilon}} \frac{2\,\varepsilon}{\sqrt{a^2+b^2}+\sqrt{a^2+b^2+2\varepsilon}} |\,\bar{a}-a\,|$$

$$= \sqrt{2\,\varepsilon}\,\frac{|\,a\,|}{\sqrt{a^2+b^2}}\,\frac{|\,\bar{a}-a\,|}{\sqrt{a^2+b^2+2\varepsilon}}\,\frac{\sqrt{2\,\varepsilon}}{\sqrt{a^2+b^2}+\sqrt{a^2+b^2+2\varepsilon}}.$$

The first and third fraction are both less than one; so is the middle fraction if $\bar{a} = 0$. If $\bar{a} \neq 0$, then provided that a is sufficiently close to $\bar{a}$, the denominator of the middle fraction is bounded away from zero and therefore the fraction is bounded by a constant. This establishes the claim. □

11.8.1 A Newton smoothing method

Based on a given family of smoothing functions $\{G_\varepsilon\}$ of the nonsmooth function G, we describe an iterative method for computing a zero of G. The iterations of the method are of two kinds: outer and inner iterations. At the beginning of each outer iteration, an iterate x^k is given and a smoothing parameter ε_k is determined. Starting at x^k, a globally convergent Newton method is applied to the smooth equation (11.8.3), generating a sequence of iterates $\{y^{j,k}\}$, where $y^{0,k} \equiv x^k$. The Newton iterations between two consecutive outer iterations constitute the inner iterations. A stopping criterion is employed to terminate the inner iterations. When such a criterion is satisfied, the inner Newton iterations within the current outer iteration are completed, yielding an updated outer iterate x^{k+1}. An outer stopping criterion is then tested for the termination of the overall iterative method. If x^{k+1} fails the latter criterion, a new smoothing parameter ε_{k+1} is chosen and a new cycle of inner iterations begins.

The stopping criterion for the inner iterations is of the standard type for a Newton method; that is

$$\|\,\nabla\theta_{\varepsilon_k}(y^{j,k})\,\| \;\le\; \text{tolerance.} \tag{11.8.9}$$

Placed in the context of a smoothing method, the above termination criterion of the inner iterations has an obvious deficiency. Indeed, suppose that

an iterate $y^{j,k}$ is generated such that $G(y^{j,k}) = 0$ but $\|\nabla\theta_{\varepsilon_k}(y^{j,k})\|$ exceeds the prescribed tolerance, i.e., the test (11.8.9) fails at $y^{j,k}$. In this event using the latter test alone for the inner iterations, we would throw away a solution of the system $G(x) = 0$! This is a real waste, the more so since we can easily detect such a favorable situation. In the inner iterations, we obviously desire progress in solving (11.8.3), but we should not ignore the possibility of detecting any point that satisfies the given termination criterion for the outer iterations, and thus for the overall method. Motivated by this consideration, we augment (11.8.9) by the following test:

$$\| G(y^{j,k}) \| \leq \tfrac{1}{2} \| G(x^k) \|, \tag{11.8.10}$$

which is used as an alternative to (11.8.9) for terminating the inner iterations. There is an obvious reason why (11.8.10) is a reasonable inner termination rule; indeed, if (11.8.10) holds, the norm of G is halved from its current value $\|G(x^k)\|$. Thus we could decrease the smoothing parameter ε_k and proceed to the next smooth equation $G_{\varepsilon_{k+1}}(x) = 0$.

The following is a detailed description of a smoothing method for approximating a zero of the function G. We let $\theta(x) \equiv \frac{1}{2}\|G(x)\|^2$.

A Line Search Smoothing Algorithm (LSSA)

11.8.6 Algorithm.

Data: $x^0 \in \mathbb{R}^n$, $\varepsilon_0 \in \mathcal{P}$, $\eta \in (0, \infty)$, $\gamma \in (0, 1/2)$, $p > 2$, $\rho > 0$. Set $k = 0$.

Step 1: Set $j = 0$ and $y^{0,k} = x^k$.

Step 2: If $G(y^{j,k}) = 0$ stop.

Step 3: Find a solution $d^{j,k}$ of the system

$$G_{\varepsilon_k}(y^{j,k}) + JG_{\varepsilon_k}(y^{j,k})d = 0. \tag{11.8.11}$$

If the system (11.8.11) is not solvable or if the condition

$$\nabla\theta_{\varepsilon_k}(y^{j,k})^T d^{j,k} \leq -\rho \| d^{j,k} \|^p \tag{11.8.12}$$

is not satisfied, set $d^{j,k} \equiv -\nabla\theta_{\varepsilon_k}(y^{j,k})$. Find the smallest integer $i_j = 0, 1, \ldots$ such that either

$$\theta_{\varepsilon_k}(y^{j,k} + 2^{-i_j} d^{j,k}) \leq \theta_{\varepsilon_k}(y^{j,k}) + \gamma\, 2^{-i_j} \nabla\theta_{\varepsilon_k}(y^{j,k})^T d^{j,k} \tag{11.8.13}$$

or

$$\| G(y^{j,k} + 2^{-i_j} d^{j,k}) \| \leq \tfrac{1}{2} \| G(x^k) \|. \tag{11.8.14}$$

Set $y^{j+1,k} \equiv y^{j,k} + 2^{-i_j} d^{j,k}$.

Step 4: If

$$\| \nabla\theta_{\varepsilon_k}(y^{j+1,k}) \| \leq \| \varepsilon_k \| \eta, \tag{11.8.15}$$

or

$$\| G(y^{j+1,k}) \| \leq \tfrac{1}{2} \| G(x^k) \|, \tag{11.8.16}$$

let $x^{k+1} \equiv y^{j+1,k}$. Select an $\varepsilon_{k+1} \in \mathcal{P}$ such that

$$0 < \| \varepsilon_{k+1} \| \leq \min \left\{ \tfrac{1}{2} \| \varepsilon_k \|, \theta(x^{k+1}) \right\}; \tag{11.8.17}$$

replace k by $k+1$, and go to Step 1. Otherwise, replace j by $j+1$ and go to Step 2.

In principle, two cases are possible in the above algorithm. Either an infinite sequence $\{x^k\}$ is generated or an index $\bar{k}$ exists such that the sequence $\{y^{j,\bar{k}}\}$ is infinite. This latter, undesirable case cannot occur if we can ensure that for every k

$$\liminf_{j \to \infty} \nabla\theta_{\varepsilon_k}(y^{j+1,k}) = 0. \tag{11.8.18}$$

Conditions guaranteeing the stronger condition

$$\lim_{j \to \infty} \nabla\theta_{\varepsilon_k}(y^{j+1,k}) = 0. \tag{11.8.19}$$

were discussed in Chapters **8** and **9**, but require some uniform continuity assumption. In the present setting a simple condition that guarantees (11.8.19) is that the level sets of θ_ε be bounded for every positive ε. We will come back to this point in the next subsection, when discussing concrete examples of smoothing.

Here instead, we take another route and show that by augmenting (11.8.12) by a further simple test we can always guarantee the satisfaction of (11.8.18) for every fixed k. Consider then substituting (11.8.12) by

$$\begin{aligned} &\nabla\theta_{\varepsilon_k}(y^{j,k})^T d^{j,k} \leq -\rho \| d^{j,k} \|^p \\ &\| d^{j,k} \| \geq \rho_1 \min \left\{ \| \nabla\theta_{\varepsilon_k}(y^{j,k}) \|, \| \nabla\theta_{\varepsilon_k}(y^{j,k}) \|^2 \right\}, \end{aligned} \tag{11.8.20}$$

where ρ_1 is a (very small) positive constant. Note that for a fixed k the additional test (11.8.20) does not change the convergence properties of the

algorithm. In fact, from the global convergence point of view, (11.8.20) can only imply that the gradient can be used more often, and this obviosuly does not affect global convergence. On the other hand, suppose the sequence $\{y^{j,k}\}$ is converging to a zero $x^*_{\varepsilon_k}$ of G_{ε_k} such that $JG_{\varepsilon_k}(x^*_{\varepsilon_k})$ is nonsingular. Multiplying the equation (11.8.11) by $JG_{\varepsilon_k}(y^{j,k})^T$, it is trivial to check that (11.8.20) is eventually always satisfied, so that also the superlinear convergence properties of the algorithm (for a fixed k) are not altered.

The benefits of using the additional test (11.8.20) are illustrated by the following proposition. As usual we assume that the stopping criterion at Step 2 is never satisfied.

11.8.7 Proposition. Let $G : \mathbb{R}^n \to \mathbb{R}^n$ be locally Lipschitz and let G_ε be a smoothing of G. Algorithm **11.8.6** as modified above generates an infinite sequence $\{x^k\}$.

Proof. Assume for the sake of contradiction that a certain $\bar{k}$ is reached for which the tests (11.8.15) and (11.8.16) are never satisfied. In what follows, we drop the index $\bar{k}$ and write ε, y^j, and d^j for $\varepsilon_{\bar{k}}$, $y^{j,\bar{k}}$, and $d^{j,\bar{k}}$, respectively. We also use several constants c_i that are all assumed to be positive.

Since (11.8.15) is never satisfied, we have

$$\| \nabla\theta_\varepsilon(y^j) \| \geq c_1, \qquad \forall\, j = 0, 1, \ldots. \tag{11.8.21}$$

By the additional test (11.8.20), we have that also d^j is bounded away from zero:

$$\| d^j \| \geq c_2, \qquad \forall\, j = 0, 1, \ldots. \tag{11.8.22}$$

Since ε is never updated, (11.8.14) is never satisfied, so that at each iteration the integer i_j is determined by the line-search procedure (11.8.13). For simplicity we set $\tau_j \equiv 2^{-i_j}$. By the line search, we see that at each iteration

$$\theta_\varepsilon(y^j + \tau_j d^j) - \theta_\varepsilon(y^j) \leq -\gamma\, \tau_j\, c_3 \min\left\{ \| d^j \|^2, \| d^j \|^p \right\},$$

where the two terms in the min correspond to the case in which the gradient or the solution of (11.8.11) have been used as search direction. Since θ_ε is bounded from below and the sequence $\{\theta_\varepsilon(y^j)\}$ is decreasing we get, recalling (11.8.22),

$$\sum_{j=0}^{\infty} \tau_j \| d^j \| < \infty. \tag{11.8.23}$$

From the definition of y^{j+1} we see that

$$\| y^{j+1} - y^j \| \leq \tau_j \| d^j \|,$$

from which, taking into account (11.8.23), we obtain

$$\sum_{j=0}^{\infty} \| y^{j+1} - y^j \| < \infty.$$

This implies that $\{y^j\}$ converges, say to y^∞. Note also that $\{d^j\}$ is bounded. In fact, if $d^j = -\nabla\theta_\varepsilon(y^j)$ the boundedness follows from the continuity of $\nabla\theta_\varepsilon(y^j)$; if d^j is the solution of (11.8.11) then the boundedness of $\{d^j\}$ follows from (11.8.12) and the boundedness of $\nabla\theta_\varepsilon(y^j)$. We assume then, without loss of generality, that $\{d^j\}$ converges to d^∞. Finally note that $\{\tau_j\}$ goes to zero by (11.8.23) and (11.8.22).

By the line search, we have

$$\theta_\varepsilon(y^j + 2\tau_j d^j) - \theta_\varepsilon(y^j) > \gamma 2\tau_j \nabla\theta_\varepsilon(y^j)^T d^j.$$

Dividing by $2\tau_j$ and passing to the limit, we therefore get

$$\nabla\theta_\varepsilon(y^\infty)^T d^\infty \geq \gamma \nabla\theta_\varepsilon(y^\infty)^T d^\infty,$$

which is only possible if $\nabla\theta_\varepsilon(y^\infty)^T d^\infty = 0$. Recalling that

$$d^\infty = -JG_\varepsilon(y^\infty)^{-1} G_\varepsilon(y^\infty), \quad \text{and} \quad \nabla\theta_\varepsilon(y^\infty) = JG_\varepsilon(y^\infty)^T G_\varepsilon(y^\infty),$$

we deduce $\|G_\varepsilon(y^\infty)\|^2 = 0$ and, therefore, $\nabla\theta_\varepsilon(y^\infty) = 0$. By continuity, the latter contradicts (11.8.21), so completing the proof. □

The next theorem establishes the convergence properties of the sequence $\{x^k\}$, assuming this sequence is infinite.

11.8.8 Theorem. Let $G : \mathbb{R}^n \to \mathbb{R}^n$ be locally Lipschitz continuous and let G_ε be a smoothing of G. Assume that Algorithm **11.8.6** produces an infinite sequence $\{x^k\}$ and suppose that G_ε is Jacobian consistent at every limit point of $\{x^k\}$. The following two statements hold.

(a) If x^∞ is an accumulation point of $\{x^k\}$, then x^∞ is a C-stationary point of θ.

(b) If in addition $G(x^\infty) = 0$, $\partial G(x^\infty)$ is nonsingular, G is semismooth (strongly semismooth) at x^∞ and G_ε approximates G superlinearly (quadratically) at x^∞, then $\{x^k\}$ converges to x^∞ and the convergence rate is Q-superlinear (Q-quadratic).

Proof. Suppose that the subsequence $\{x^k : k \in \kappa\}$ converges to x^∞. On the one hand, if (11.8.15) holds for infinitely many $k \in \kappa$, we have

$$\| \nabla\theta_{\varepsilon_{k-1}}(x^k) \| \leq \varepsilon_{k-1}\,\eta$$

for infinitely many $k \in \kappa$. By (11.8.17), $\{\varepsilon_{k-1}\}$ converges to zero. Hence Proposition **11.8.4** yields that x^∞ is a C-stationary point of θ. On the other hand, if (11.8.16) holds for all but finitely many iterations, we have that $\{\|G(x^k)\|\}$ converges to zero, hence $G(x^\infty) = 0$ and x^∞ is again a C-stationary point of θ. This complete the proof of (a).

To prove (b) assume now that $G(x^\infty) = 0$, $\partial G(x^\infty)$ is nonsingular, G is semismooth at x^∞ and G_ε approximates G superlinearly at x^∞. The Jacobian consistency property implies that for all $k \in \kappa$ sufficienlty large $JG_{\varepsilon_k}(x^k)^{-1}$ exists and $\{\|JG_{\varepsilon_k}(x^k)^{-1}\|\}$ is uniformly bounded from above and away from zero. This implies that for all $k \in \kappa$ sufficienlty large $d^{0,k} = -JG_{\varepsilon_k}(x^k)^{-1}G_{\varepsilon_k}(x^k)$ (by the discussion before Proposition **11.8.7**; this is true also if the additional test (**11.8.20**) is used). By (11.8.6) we then have

$$\begin{aligned}
\| x^k + d^{0,k} - x^\infty \| &= \| - JG_{\varepsilon_k}(x^k)^{-1}G_{\varepsilon_k}(x^k) + x^k - x^\infty \| \\
&\leq \| JG_{\varepsilon_k}(x^k) \| \, \| G_{\varepsilon_k}(x^k) + JG_{\varepsilon_k}(x^k)(x^\infty - x^k) \| \\
&= O(\| G_{\varepsilon_k}(x^k) - G(x^\infty) + JG_{\varepsilon_k}(x^k)(x^\infty - x^k) \|) \\
&= o(\| x^k - x^\infty \|) + O(\| \varepsilon_k \|) \\
&\leq o(\| x^k - x^\infty \|) + O(\| G(x^k) \|^2),
\end{aligned}$$

where the second inequality follows from (11.8.17). By the nonsingularity assumption on $\partial G(x^\infty)$ and Proposition **8.3.16**, we therefore obtain

$$\lim_{k\to\infty} \frac{x^k + d^{0,k} - x^\infty}{\| x^k - x^\infty \|} = 0. \tag{11.8.24}$$

Again by Proposition **8.3.16** and by Steps 3 and 4 of the algorithm we can then conclude that $x^{k+1} = x^k + d^{0,k}$. By using (11.8.12) and standard arguments we know that $\{d^{0,k} : k \in \kappa\}$ converges to zero. The vector x^∞ is an isolatd zero of G by the nonsingularity assumption on $\partial G(x^\infty)$. By the upper semicontinuity of the generalized Jacobian we then deduce that x^∞ is also an isolated stationary point of θ. Proposition **8.3.10** therefore implies that $\{x^k\}$ converges to x^∞. The superlinear convergence rate now follows from (11.8.24). The quadratic convergence rate result can be proved in a similar way. □

11.8.9 Remarks. We make several observations regarding the above theorem.

(a) Conditions ensuring that a stationary point of θ is a zero of G were extensively discussed in the previous chapters for all cases of interest.

(b) If G_ε is only weakly Jacobian consistent at every limit point x^∞ of $\{x^k\}$, a simple modification of the proof of the theorem shows that $0 \in \partial_C G(x^\infty)^T G(x^\infty)$.

(c) Under the assumptions of (b) in the theorem, inner and outer iterations coincide. □

Several variants of Algorithm **11.8.6** exist. One of the most interesting is known as the *Jacobian smoothing* method and is analyzed in Exercise **11.9.10**. Further variants are discussed in the Notes and Comments.

11.8.2 A class of smoothing functions

In this subsection, we provide several examples of smoothing families and give conditions under which the assumptions needed in the analysis of these methods are satisfied. Furthermore we also analyze some related issues relevant to the equation reformulations of VIs and CPs.

There exist general methods for building smoothing functions based on convolution and actually the methods we consider below are all in a direct or indirect way based on convolution. However, we refrain from considering such a general theory and prefer to study specific instances because the general approach is technically more demanding and is not easily applicable in practice, being based on the calculation of multivariate integrals. The reader can find suitable references to the general approach in the Notes and Comments.

We begin by providing a general approach for constructing smoothing functions of the scalar plus (i.e., max) function $a_+ \equiv \max(a, 0)$, where $a \in \mathbb{R}$. Since $\min(a, b) = b - (b - a)_+$, any smoothing function for the plus function easily yields a smoothing function for the min function, which in turn defines a smooth equation that provides an approximation of the NCP. Similarly, since $|a| = \max(0, a) + \max(0, -a)$, we can obtain a smoothing function for the absolute value function from any smoothing function for the plus function. At the end of this subsection, we also discuss an extension to the smoothing of the mid function introduced in Section **9.4**, which plays the analogous role of the min function for the box constrained VI.

To begin, we note that the plus function is the integral of the Heaviside unit step function; that is, $x_+ = \int_{-\infty}^{x} \sigma(t)dt$, where for $x \in \mathbb{R}$,

$$\sigma(x) \equiv \begin{cases} 1 & \text{if } x > 0 \\ 0 & \text{otherwise.} \end{cases}$$

The latter step function is in turn the integral of the Dirac delta function; that is, $\sigma(x) = \int_{-\infty}^{x} \delta(t)dt$, where $\delta(x)$ is the Dirac delta function satisfying:

$$\delta(x) \geq 0 \quad \text{and} \quad \int_{-\infty}^{\infty} \delta(t)dt = 1.$$

The latter properties of the Dirac delta function prompt us to use a probability density function as a smoothing function. Hence let $\rho(x)$ be any piecewise continuous function defined on the real line with finitely many discontinuous points such that

$$\rho(x) \geq 0 \quad \text{and} \quad \int_{-\infty}^{\infty} \rho(t)dt = 1.$$

A function satisfying the two relations above is usually called a probability density function. For practical purposes, we restrict $\rho(x)$ to be of finite absolute mean; that is,

$$\int_{-\infty}^{\infty} |\, t \,|\, \rho(t)dt < \infty. \tag{11.8.25}$$

With such a function $\rho(x)$, we define the smoothing function: for $\varepsilon > 0$,

$$p_\varepsilon(x) \equiv \varepsilon^{-1} \int_{-\infty}^{\infty} (\, x - t \,)_+ \, \rho(t/\varepsilon)dt, \quad x \in \mathbb{R}.$$

Equivalently, we have

$$p_\varepsilon(x) \equiv \int_{-\infty}^{\frac{x}{\varepsilon}} (\, x - \varepsilon\, t \,)\, \rho(t)dt, \quad x \in \mathbb{R}. \tag{11.8.26}$$

Proposition **11.8.10** below summarizes the essential properties of $p_\varepsilon(x)$. In particular, part (b) of this result shows that as ε tends to zero, $p_\varepsilon(x)$ approaches the plus function x_+. Part (c) shows that this smoothing is a quadratic approximation. Part (d) shows that $p_\varepsilon(x)$ preserves two essential properties of the plus function: convexity and monotonicity.

11.8.10 Proposition. Let ρ be a probability density function with finite absolute mean. Let $\varepsilon > 0$ be given. The following statements are valid.

(a) p_ε is twice continuously differentiable on $\mathbb{R}$; moreover

$$p'_\varepsilon(x) = \int_{-\infty}^{\frac{x}{\varepsilon}} \rho(t)dt \quad \text{and} \quad p''_\varepsilon(x) = \varepsilon^{-1}\, \rho(x/\varepsilon).$$

Thus $p'_\varepsilon(x) \in [0, 1]$ for all $x \in \mathbb{R}$; hence p_ε is nonexpansive.

(b) For every $x \in \mathbb{R}$,

$$-c_2\,\varepsilon \;\le\; p_\varepsilon(x) - x_+ \;\le\; c_1\,\varepsilon,$$

where

$$c_1 \;\equiv\; \int_{-\infty}^{0} |\,t\,|\,\rho(t)dt \quad \text{and} \quad c_2 \;\equiv\; \max\left\{ \int_{-\infty}^{\infty} t\,\rho(t)dt, 0 \right\}.$$

(c) For any scalar $\bar{x}$, there exist positive constants δ and $\bar{\varepsilon}$ and a function $\Delta_2 : (0, \bar{\varepsilon}] \to [0, \infty)$ satisfying $\Delta_2(\varepsilon) = O(\varepsilon)$ and

$$\left.\begin{array}{r} |\,x - \bar{x}\,| \;\le\; \delta \\ 0 \;<\; \varepsilon \;\le\; \bar{\varepsilon} \end{array}\right\} \;\Rightarrow\; |\,p_\varepsilon(x) + p'_\varepsilon(x)(\,\bar{x} - x\,) - \bar{x}_+\,| \;\le\; \Delta_2(\varepsilon);$$

thus p_ε is a quadratic approximation of the plus function.

(d) $p_\varepsilon(x)$ is nondecreasing and convex.

(e) If $\rho(t) > 0$ for all $t \in \mathbb{R}$, then

(i) $p_\varepsilon(x)$ is strictly increasing and strictly convex;

(ii) $p'_\varepsilon(x) \in (0, 1)$ for all $x \in \mathbb{R}$.

(f) If $\{\varepsilon_k\}$ is a sequence of positive scalars converging to 0 and $\{x^k\}$ is a sequence of scalars converging to $\bar{x}$, then

$$\lim_{k\to\infty} p'_{\varepsilon_k}(x^k) \;=\; \left\{ \begin{array}{rl} 1 & \text{if } \bar{x} > 0 \\ -1 & \text{if } \bar{x} < 0. \end{array} \right.$$

Proof. In view of the finiteness of the absolute mean of ρ, (11.8.26) shows that p_ε is everywhere continuously differentiable on the real line and its derivative $p'_\varepsilon(x)$ can be computed by the fundamental theorem of calculus and the product rule of differentiation. With $p'_\varepsilon(x)$ computed, another application of the fundamental theorem of calculus yields the second derivative $p''_\varepsilon(x)$. Thus (a) follows.

To prove (b), let $x \ge 0$. We have

$$\begin{aligned} p_\varepsilon(x) - x_+ \;&=\; \int_{-\infty}^{\frac{x}{\varepsilon}} (\,x - \varepsilon t\,)\,\rho(t)dt - x \\ &=\; x\left(\int_{-\infty}^{\frac{x}{\varepsilon}} \rho(t)dt - 1 \right) - \varepsilon \int_{-\infty}^{\frac{x}{\varepsilon}} t\,\rho(t)dt \\ &=\; \int_{\frac{x}{\varepsilon}}^{\infty} (\,\varepsilon\, t - x\,)\;\rho(t)dt - \varepsilon \int_{-\infty}^{\infty} t\,\rho(t)dt \\ &\ge\; -c_2\,\varepsilon. \end{aligned}$$

Furthermore, we also have

$$\begin{aligned} p_\varepsilon(x) - x_+ &\leq \varepsilon \int_{\frac{x}{\varepsilon}}^{\infty} t\,\rho(t)dt - \varepsilon \int_{-\infty}^{\infty} t\,\rho(t)dt \\ &\leq -\varepsilon \int_{-\infty}^{0} t\,\rho(t)dt = c_1\varepsilon. \end{aligned}$$

If $x < 0$, then

$$p_\varepsilon(x) - x_+ = p_\varepsilon(x) \geq 0 \geq -c_2\,\varepsilon.$$

Furthermore,

$$\begin{aligned} p_\varepsilon(x) - x_+ &= x \int_{-\infty}^{\frac{x}{\varepsilon}} \rho(t)dt + \varepsilon \int_{-\infty}^{\frac{x}{\varepsilon}} |\,t\,|\,\rho(t)dt \\ &\leq \varepsilon \int_{-\infty}^{0} |\,t\,|\,\rho(t)dt = c_1\,\varepsilon. \end{aligned}$$

This establishes (b). To prove (c), note that

$$p_\varepsilon(x) + p'_\varepsilon(x)(\,\bar{x} - x\,) - \bar{x}_+ = \int_{-\infty}^{\frac{x}{\varepsilon}} (\,\bar{x} - \varepsilon\,t\,)\,\rho(t)\,dt - \bar{x}_+.$$

If $\bar{x} = 0$, then

$$|\,p_\varepsilon(x) + p'_\varepsilon(x)(\,\bar{x} - x\,) - \bar{x}_+\,| \leq \varepsilon \int_{-\infty}^{\infty} |\,t\,|\,\rho(t)\,dt.$$

If $\bar{x} > 0$, then

$$p_\varepsilon(x) + p'_\varepsilon(x)(\,\bar{x} - x\,) - \bar{x}_+ = \int_{\frac{x}{\varepsilon}}^{\infty} (\,\varepsilon\,t - \bar{x}\,)\,\rho(t)dt - \varepsilon \int_{-\infty}^{\infty} t\,\rho(t)dt;$$

as in part (b), from this inequality, we can deduce

$$-c_2\,\varepsilon \leq p_\varepsilon(x) + p'_\varepsilon(x)(\,\bar{x} - x\,) - \bar{x}_+ \leq c_1\,\varepsilon,$$

with the same constants c_1 and c_2 as in the previous part. If $\bar{x} < 0$, then $x < 0$ for all x sufficiently close to $\bar{x}$. Hence,

$$p_\varepsilon(x) + p'_\varepsilon(x)(\,\bar{x} - x\,) - \bar{x}_+ = \bar{x} \int_{-\infty}^{\frac{x}{\varepsilon}} \rho(t)dt + \varepsilon \int_{-\infty}^{\frac{x}{\varepsilon}} |\,t\,|\,\rho(t)dt \leq c_1\,\varepsilon.$$

Furthermore,

$$p_\varepsilon(x) + p'_\varepsilon(x)(\,\bar{x} - x\,) - \bar{x}_+ \geq (\,\bar{x} - x\,) \int_{-\infty}^{\frac{x}{\varepsilon}} \rho(t)\,dt.$$

Hence, if $|x - \bar{x}| \le \delta$, we deduce

$$p_\varepsilon(x) + p'_\varepsilon(x)(\,\bar{x} - x\,) - \bar{x}_+ \;\ge\; -\delta \int_{-\infty}^{\frac{\bar{x}+\delta}{\varepsilon}} \rho(t)\,dt.$$

Since $\bar{x} < 0$, letting $\delta \equiv \frac{1}{2}|\bar{x}|$, we have $\bar{x} + \delta < 0$ and

$$\lim_{\varepsilon \downarrow 0} \int_{-\infty}^{\frac{\bar{x}+\delta}{\varepsilon}} \rho(t)\,dt \;=\; 0.$$

Hence defining

$$\Delta_2(\varepsilon) \;\equiv\; \max\left\{ \max\left(c_1, c_2, \int_{-\infty}^{\infty} |t|\rho(t)\,dt\right)\,\varepsilon,\; \int_{-\infty}^{\frac{\bar{x}}{2\varepsilon}} (\,\bar{x}_-\,)\,\rho(t)\,dt \right\}$$

and letting

$$\delta \;\equiv\; \begin{cases} \infty & \text{if } \bar{x} \ge 0 \\ \frac{1}{2}\,|\,\bar{x}\,| & \text{if } \bar{x} < 0, \end{cases}$$

we obtain the first assertion in part (c) if we show that for every negative $\bar{x}$

$$\int_{-\infty}^{\frac{\bar{x}}{2\varepsilon}} (-\bar{x})\,\rho(t)\,dt \;=\; O(\varepsilon).$$

To show this it suffices to note that, since

$$\int_0^{\infty} t^{-1}\,dt \;=\; \infty,$$

(11.8.25) implies

$$\lim_{|t|\to\infty} t^2\rho(t) \;=\; 0.$$

Therefore, a negative $\bar{t}$ exists such that if $t \in (-\infty, \bar{t}]$, then $\rho(t) < t^2$. This implies that for every ε such that $\bar{x}/(2\varepsilon) \le \bar{t}$ we can write

$$\int_{-\infty}^{\frac{\bar{x}}{2\varepsilon}} (-\bar{x})\,\rho(t)\,dt \;\le\; (-\bar{x}) \int_{-\infty}^{\frac{\bar{x}}{2\varepsilon}} \frac{1}{t^2}\,dt \;=\; 2\,\varepsilon.$$

By combining parts (b) and (c), the desired quadratic approximation property of p_ε follows easily. Part (d) is an immediate consequence of part (a) because the first and second derivative of $p_\varepsilon(x)$ are both nonnegative. If $\rho(t)$ is positive for all t, then the first and second derivative of $p_\varepsilon(x)$ are both positive; hence part (e) follows. Finally, part (f) is also an immediate consequence of part (a). □

Since $x = x_+ - (-x)_+$ for all $x \in \mathbb{R}$, it seems reasonable for this simple relation to be inherited by the smooth approximation. With the plus function replaced by its approximation, the above identity becomes

$$x = p_\varepsilon(x) - p_\varepsilon(-x). \tag{11.8.27}$$

Note that it is not reasonable to expect that

$$|\,x\,| = p_\varepsilon(x) + p_\varepsilon(-x)$$

because the left-hand side is not differentiable, whereas the right-hand side is. Differentiating (11.8.27) with respect to x, we obtain

$$1 = p'_\varepsilon(x) + p'_\varepsilon(-x).$$

By part (a) of Proposition **11.8.10**, it is easy to show that the above equality, and thus (11.8.27), will hold if the density function ρ is an even function, that is, $\rho(x) = \rho(-x)$ for all x.

In what follows, we present examples of density functions $\rho(x)$ and the corresponding smooth function $p_\varepsilon(x)$.

11.8.11 Example. (a) The neural network density function:

$$\rho(x) \equiv \frac{e^{-x}}{(\,1 + e^{-x}\,)^2}, \quad x \in \mathbb{R}.$$

We have

$$p_\varepsilon(x) = \varepsilon \log(\,1 + e^{x/\varepsilon}\,), \quad x \in \mathbb{R}.$$

(b) The CHKS density function:

$$\rho(x) \equiv \frac{2}{(\,x^2 + 4\,)^{\frac{3}{2}}}, \quad x \in \mathbb{R}.$$

We have

$$p_\varepsilon(x) \equiv \frac{\sqrt{x^2 + 4\varepsilon^2} + x}{2}, \quad x \in \mathbb{R}.$$

(c) The Huber-Pinar-Zenios density function:

$$\rho(x) \equiv \begin{cases} 1 & \text{if } 0 \le x \le 1 \\ 0 & \text{otherwise.} \end{cases}$$

We have

$$p_\varepsilon(x) = \begin{cases} 0 & \text{if } x < 0 \\ \dfrac{x^2}{2\,\varepsilon} & \text{if } 0 \le x \le \varepsilon \\ x - \varepsilon/2 & \text{if } x > \varepsilon. \end{cases}$$

(d) The Zang density function:

$$\rho(x) \equiv \begin{cases} 1 & \text{if } -\frac{1}{2} \le x \le \frac{1}{2} \\ 0 & \text{otherwise.} \end{cases}$$

We have

$$p_\varepsilon(x) = \begin{cases} 0 & \text{if } x < -\varepsilon/2 \\ \dfrac{1}{2\,\varepsilon}\,(x+\varepsilon/2)^2 & \text{if } -\varepsilon/2 \le x \le \varepsilon/2 \\ x & \text{if } x > \varepsilon/2. \end{cases}$$

Notice that the first two densities functions are positive on the real line, whereas the last two functions have compact support. Furthermore, the first and third density function $\rho(x)$ is not even, and thus the associated p_ε fails to satisfy (11.8.27); whereas the second and fourth density function $\rho(x)$ is even, and thus p_ε satisfies (11.8.27). □

With $p_\varepsilon(t)$ being a given smoothing function of the plus function, it is easy to obtain a smooth approximation of the min formulation of the implicit MiCP (11.1.3). Specifically, let $\varepsilon > 0$ be a given scalar; define

$$\mathbf{H}_{\varepsilon,\min}(x,y,z) \equiv \begin{pmatrix} x_1 - p_\varepsilon(x_1 - y_1) \\ \vdots \\ x_n - p_\varepsilon(x_n - y_n) \\ H(x,y,z) \end{pmatrix}, \quad \forall\,(x,y,z) \in \mathbb{R}^{2n+m}.$$

By part (b) of Proposition **11.8.10**, it is easy to show that there exists a constant $c > 0$, dependent on the density function ρ only, such that

$$\|\,\mathbf{H}_{\min}(x,y,z) - \mathbf{H}_{\varepsilon,\min}(x,y,z)\,\|_\infty \le c\,\varepsilon, \tag{11.8.28}$$

where

$$\mathbf{H}_{\min}(x,y,z) \equiv \begin{pmatrix} \min(x,y) \\ H(x,y,z) \end{pmatrix}.$$

Since p_ε is a quadratic approximation of the plus function, it follows that $\mathbf{H}_{\varepsilon,\min}$ is a quadratic approximation of $\mathbf{H}_{\min}$.

The inequality (11.8.28) has two consequences that are worth mentioning. First, this inequality implies

$$\|\,\mathbf{H}_{\min}(x,y,z)\,\|_\infty \le c\,\varepsilon + \|\,\mathbf{H}_{\varepsilon,\min}(x,y,z)\,\|_\infty, \quad \forall\,(x,y,z) \in \mathbb{R}^{2n+m}.$$

Consequently, by computing an approximate zero to the smooth function $\mathbf{H}_{\varepsilon,\min}$, say by Newton's method, we can obtain an approximate solution to the implicit MiCP (11.1.3) as accurately as we want by controlling the scalar ε and the accuracy of the smooth equation solver to ensure that $\|\mathbf{H}_{\varepsilon,\min}(x,y,z)\|$ is small enough. Second, the inequality (11.8.28) also implies that for every scalar $\eta > 0$,

$$\begin{aligned}&\{\,(x,y,z) \in \mathbb{R}^{2n+m} : \|\mathbf{H}_{\varepsilon,\min}(x,y,z)\|_\infty \le \eta\,\}\\ &\quad\subseteq \{\,(x,y,z) \in \mathbb{R}^{2n+m} : \|\mathbf{H}_{\min}(x,y,z)\|_\infty \le \eta + c\,\varepsilon\,\}.\end{aligned}$$

Consequently, if the right-hand set is bounded, then so is the left-hand set. Thus if $\|\mathbf{H}_{\min}\|$ has bounded level sets, then so does $\|\mathbf{H}_{\varepsilon,\min}\|$ (see the discussion after Algorithm **11.8.6**).

Since we are interested in solving the smoothed equation

$$\mathbf{H}_{\varepsilon,\min}(x,y,z) = 0$$

by Newton's method, the nonsingularity of the Jacobian matrix

$$J\mathbf{H}_{\varepsilon,\min}(x,y,z)$$

is an issue that needs to be addressed. It is easy to see that

$$J\mathbf{H}_{\varepsilon,\min}(x,y,z) = \begin{bmatrix} \mathbf{I}_n - D_\varepsilon(x,y) & D_\varepsilon(x,y) & 0 \\ J_xH(x,y,z) & J_yH(x,y,z) & J_zH(x,y,z) \end{bmatrix},$$

where $D_\varepsilon(x,y)$ is the diagonal matrix whose i-diagonal entry is equal to $p'_\varepsilon(x_i - y_i)$. In particular, if p_ε is a positive density function on $\mathbb{R}$, then these diagonal elements are positive and less than unity; hence $J\mathbf{H}_{\varepsilon,\min}(x,y,z)$ is of exactly the same structure as several of the matrices we have encountered throughout this chapter, such as $J\mathbf{H}_{\mathrm{FB}_\mu}(x,y,z)$ and $J\mathbf{H}_{\mathrm{IP}}(x,y,z)$, the latter with $(x,y) > 0$. Hence if $JH(x,y,z)$ has the mixed P_0-property, then $J\mathbf{H}_{\varepsilon,\min}(x,y,z)$ is nonsingular. Finally, it is easy to see that $\mathbf{H}_{\varepsilon,\min}$ is a Jacobian consistent smoothing of $\mathbf{H}_{\min}$.

All the considerations above apply in particular to the NCP (F) with $H(x,y,z) = y - F(x)$. For this special problem, a natural alternative is to use the min function directly:

$$\mathbf{F}_{\min}(x) = \min(x, F(x)),$$

without the extra variable y. The corresponding smoothing is given by

$$\mathbf{F}_{\varepsilon,\min}(x) \equiv \begin{pmatrix} x_1 - p_\varepsilon(x_1 - F_1(x)) \\ \vdots \\ x_n - p_\varepsilon(x_n - F_n(x)) \end{pmatrix}, \quad \forall\, x \in \mathbb{R}^2.$$

Similar to what is shown for $\mathbf{H}_{\varepsilon,\min}$, we can easily verify that $\mathbf{F}_{\varepsilon,\min}$ is a quadratic smoothing of $\mathbf{F}_{\min}$ and that for some positive constant c depending on the density function ρ only

$$\| \mathbf{F}_{\min}(x) - \mathbf{F}_{\varepsilon,\min}(x) \|_\infty \leq c\,\varepsilon.$$

The main difference is that, in this case, it can easily be verified that we can only prove $\mathbf{F}_{\varepsilon,\min}$ to be a weakly Jacobian consistent smoothing of $\mathbf{F}_{\min}$. Hence Theorem**11.8.8** only guarantees convergence to points x such that $0 \in \partial_C G(x)^T G(x)$ (see Remark **11.8.9** (b)).

It is straightforward to extend the smoothing function for the (scalar) plus function to the (scalar) mid function. Indeed, for any two scalars $a < b$, define

$$\begin{aligned} p_\varepsilon^{\mathrm{mid}}(x) &\equiv \int_{-\infty}^{\infty} \mathrm{mid}(a, b, x - \varepsilon t)\, \rho(t) dt, \quad \forall\, x \in \mathbb{R} \\ &= a \int_{\frac{x-a}{\varepsilon}}^{\infty} \rho(t) dt + \int_{\frac{x-b}{\varepsilon}}^{\frac{x-a}{\varepsilon}} (\, x - \varepsilon t\,)\, \rho(t) dt + b \int_{-\infty}^{\frac{x-b}{\varepsilon}} \rho(t) dt \end{aligned}$$

It is then easy to derive a result for the function $p_\varepsilon^{\mathrm{mid}}$ analogous to Proposition **11.8.10**. Omitting the details, we just give the following formula for the first derivative of $p_\varepsilon^{\mathrm{mid}}$:

$$(p_\varepsilon^{\mathrm{mid}})'(x) = \int_{\frac{x-b}{\varepsilon}}^{\frac{x-a}{\varepsilon}} \rho(t) dt.$$

11.9 Excercises

11.9.1 Let $p(u, v)$ be given by (11.3.2). Show that (IP4)–(IP6) hold with $\zeta > n_1/2$, $a \equiv (\mathbf{1}_{n_1}, 0)$ and $\bar{\sigma} = 1$. Show further that

$$\liminf_{(u \downarrow 0, v \to 0)} p(u, v) = -\infty \quad \text{and} \quad \limsup_{(u \downarrow 0, v \to 0)} p(u, v) = +\infty.$$

11.9.2 Prove Lemma **11.3.1**.

11.9.3 Let A, B, X, and Y be real square matrices of the same order. Show that if X and Y commute, then

$$\det \begin{bmatrix} A & B \\ X & Y \end{bmatrix} = \det(AY + BX).$$

Show further that if D_1 and D_2 are diagonal matrices, then

$$\det(AD_1 + BD_2) = \sum \det D \det E,$$

where E and D are column representative matrices of the pair (A, B) and (D_1, D_2), respectively, and the same indexed columns are selected to form D and E, and where the summation ranges over all such column representative matrices.

11.9.4 Let A and B be real square matrices of order n. Show that the following statements are equivalent.

(a) The pair (A, B) has the P_0 property.

(b) For arbitrary diagonal matrices D_1 and D_2 in $\mathbb{R}^{n\times n}$ with positive diagonal entries, $\det(AD_1 + BD_2)$ is nonzero.

(c) One of the following two statements holds:

  (i) every column representative matrix of the pair (A, B) has a nonnegative determinant and there is at least one such determinant that is positive;

  (ii) every column representative matrix of the pair (A, B) has a nonpositive determinant and there is at least one such determinant that is negative.

Historically, a matrix pair (A, B) satisfying (c) is said to have the *column* $\mathcal{W}_0$ property. Thus this exercise shows that (A, B) has the P_0 property if and only if (A, B) has the column $\mathcal{W}_0$ property. See Exercise **4.8.15** for the related column $\mathcal{W}$ property.

11.9.5 Let A and B be real square matrices of order n. Show that (A, B) is column monotone if and only if $A+B$ is nonsingular and AB^T is positive semidefinite. (Hint: for the "if" statement, show that there must exist a nonsingular column representative matrix of (A, B).) Use this characterization to show that the pair of matrices

$$A \equiv \begin{bmatrix} 3/2 & -1/2 \\ 1/2 & -1/2 \end{bmatrix} \quad \text{and} \quad B \equiv \begin{bmatrix} 2 & 1 \\ 1 & 0 \end{bmatrix}$$

is column monotone but not row monotone; i.e., (A^T, B^T) is not column monotone.

11.9.6 Verify the expression for $p_\varepsilon(x)$ in each of the four cases in Example **11.8.11**. In each case, verify part (a) of Proposition **11.8.10** and compute the two constants c_1 and c_2 in part (b) of this proposition.

11.9.7 This exercise concerns the vertical CP (F, G) solved by an IP method. Specifically, let $F, G : \mathbb{R}^n \to \mathbb{R}^n$ be a pair of continuous maps that are jointly monotone on $\mathbb{R}^n$. Let

$$H(u, v, x) \equiv \begin{pmatrix} u - F(x) \\ v - G(x) \end{pmatrix}.$$

(a) Assume that for every scalar $\eta > 0$ there exists $\bar{x}$ satisfying $F(\bar{x}) > \eta \mathbf{1}$ and $G(\bar{x}) > \eta \mathbf{1}$. Show that for every nonnegative scalar η, the set

$$\{ (u, v, x) \in \mathbb{R}^{2n}_{++} \times \mathbb{R}^n : \| H(u, v, x) \| \leq \eta, \ u \circ v \leq \eta \mathbf{1}_n \}$$

is bounded.

(b) Show that if F and G are continuously differentiable on $\mathbb{R}^n$, then $JF(x)^T JG(x)$ is positive semidefinite for all $x \in \mathbb{R}^n$ and H is differentiably monotone on $\mathbb{R}^{2n}$.

(c) Assume the conditions in (b) and (c). With

$$G(u, v, x) \equiv \begin{pmatrix} u \circ v \\ H(u, v, x) \end{pmatrix}, \quad X \equiv \mathbb{R}^{2n}_{+} \times \mathbb{R}^n,$$

$$S \equiv \mathbb{R}^n_{+} \times \mathbb{R}^{2n}, \quad a \equiv \begin{pmatrix} \mathbf{1}_n \\ 0 \end{pmatrix} \in \mathbb{R}^{3n}, \quad \bar{\sigma} \equiv 1,$$

and for all $(u, v, x) \in \mathbb{R}^{2n}_{++} \times \mathbb{R}^n$,

$$\psi(u, v, x) \equiv \zeta \log(\| u \circ v \|^2 + \| H(u, v, x) \|^2) - \sum_{i=1}^{n} \log(u_i v_i),$$

where $\zeta > n/2$ is a given scalar, let $\{(u^k, v^k, x^k)\}$ be a sequence of triples generated by Algorithm **11.5.1**. Show that such a sequence must be bounded and if $(u^\infty, v^\infty, x^\infty)$ is any accumulation point of the sequence, then x^∞ solves the vertical CP (F, G).

(d) Derive a modified IP method and an extension of Proposition **11.5.8** to the vertical CP (F, G). Deduce that every feasible vertical CP with a jointly monotone pair of C^1 functions must have ε-solutions.

11.9.8 In Section **11.7** we showed the map ψ_{FB_μ} can be used to define a certain "central path". In this exercise, we ask the reader to extend this kind of analysis by using the CHKS smoothing of the min function. Specifically, for each $\varepsilon \geq 0$, we define

$$\psi_{\mathrm{CHKS}_\varepsilon}(a, b) \equiv \sqrt{(a - b)^2 + 4\varepsilon} - (a + b), \quad (a, b) \in \mathbb{R}^2.$$

Note that for $\varepsilon > 0$, $\psi_{\mathrm{CHKS}_\varepsilon}(a,b) = -2(a - p_\varepsilon(a-b))$, where p_ε is the smoothing of the max function obtained by using the CHKS density function; this justifies calling $\psi_{\mathrm{CHKS}_\varepsilon}$ the CHKS smoothing of the min function. The factor -2 is introduced just for simplicity and could be omitted.

Given $H : \mathbb{R}^{2n+m} \to \mathbb{R}^{n+m}$, define $\mathbf{H}_{\mathrm{CHKS}} : \mathbb{R}^n_+ \times \mathbb{R}^{2n+m} \to \mathbb{R}^{3n+m}$ by

$$\mathbf{H}_{\mathrm{CHKS}}(u,x,y,z) \equiv \begin{pmatrix} u \\ \mathbf{\Psi}_{\mathrm{CHKS}_u}(x,y) \\ H(x,y,z) \end{pmatrix}, \quad \forall\,(u,x,y,z) \in \mathbb{R}^n_+ \times \mathbb{R}^{2n+m},$$

where $\mathbf{\Psi}_{\mathrm{CHKS}_u} : \mathbb{R}^{2n} \to \mathbb{R}^n$ is the separable function:

$$\mathbf{\Psi}_{\mathrm{CHKS}_u}(x,y) \equiv \begin{pmatrix} \psi_{\mathrm{CHKS}_{u_1}}(x_1,y_1) \\ \vdots \\ \psi_{\mathrm{CHKS}_{u_n}}(x_n,y_n) \end{pmatrix}, \quad \forall\,(x,y) \in \mathbb{R}^{2n}.$$

It is clear that if $\mathbf{H}_{\mathrm{CHKS}}(u,x,y,z) = 0$, then (x,y,z) is a solution of (11.1.3). Let

$$\Omega_{++} \equiv W(\mathbb{R}^n_{++} \times \mathbb{R}^{2n+m})$$

where

$$W(u,x,y,z) \equiv \begin{pmatrix} \mathbf{\Psi}_{\mathrm{CHKS}_u}(x,y) \\ H(x,y,z) \end{pmatrix}, \quad \forall\,(u,x,y,z) \in \mathbb{R}^n_+ \times \mathbb{R}^{2n+m}.$$

(a) Show that for every scalar $\varepsilon \geq 0$, $\psi_{\mathrm{CHKS}_\varepsilon}(a,b) = c$ if and only if

$$\varepsilon = (a + c/2)(b + c/2) \quad \text{and} \quad (a + c/2, b + c/2) \geq 0.$$

(Hint: Square $\sqrt{(a-b)^2 + 4\varepsilon} = a + b + c$ and do some algebraic manipulations).

(b) Suppose that the mapping $H : \mathbb{R}^{2n+m} \to \mathbb{R}^{n+m}$ is continuous, (x,y) equi-monotone on $\mathbb{R}^{2n}$, z-injective on $\mathbb{R}^{2n}$, and z-coercive on $\mathbb{R}^{2n}$. Prove that the map $\mathbf{H}_{\mathrm{CHKS}} : \mathbb{R}^n_+ \times \mathbb{R}^{2n+m} \to \mathbb{R}^n_+ \times \mathbb{R}^{2n+m}$ is injective on $\mathbb{R}^n_{++} \times \mathbb{R}^{2n+m}$ and proper with respect to $\mathbb{R}^n_+ \times \Omega_{++}$. (Hint: to prove the injectivity argue by contradiction and suppose that $\mathbf{H}_{\mathrm{CHKS}}(u^1,x^1,y^1,z^1) = \mathbf{H}_{\mathrm{CHKS}}(u^2,x^2,y^2,z^2)$ for two tuples (u^i,x^i,y^i,z^i), $i = 1,2$, with $u^i > 0$. Show first that $u^1 = u^2$. By using (a), the equi-monotonicity and Lemma **11.4.13** prove that $x^1 = x^2$ and $y^1 = y^2$. Next use the z-injectivity of H to conclude that $z^1 = z^2$. The proof that $\mathbf{H}_{\mathrm{CHKS}}$ is proper with respect to $\mathbb{R}^n_+ \times \Omega_{++}$ parallels that of Lemma **11.4.14**)

(c) Under the same assumptions in (b), show further that $\mathbf{H}_{\mathrm{CHKS}}$ maps $\mathbb{R}^n_{++} \times \mathbb{R}^{2n+m}$ onto $\mathbb{R}^n_{++} \times \Omega_{++}$ homeomorphically.

11.9.9 For any scalars a, b, ε, and μ with ε and μ positive, show that

$$a \geq 0, \quad b + \varepsilon a \geq 0, \quad a(b + \varepsilon a) = \mu$$

if and only if

$$2\varepsilon a + b - \sqrt{b^2 + 4\varepsilon\mu} = 0.$$

Given a mapping $H : \mathbb{R}^{2n+m} \to \mathbb{R}^{n+m}$ and positive scalars ε and μ, define the mapping $\mathbf{H}_{\varepsilon,\mu} : \mathbb{R}^{2n+m} \to \mathbb{R}^{2n+m}$ as follows:

$$\mathbf{H}_{\varepsilon,\mu}(x,y,z) \equiv \begin{pmatrix} 2\varepsilon x + y - \sqrt{y \circ y + 4\varepsilon\mu \mathbf{1}} \\ H(x,y,z) \end{pmatrix}, \quad \forall (x,y,z) \in \mathbb{R}^{2n+m}.$$

Show that $\mathbf{H}_{\varepsilon,\mu}(x,y,z) = (a,b)$ if and only if

$$\begin{pmatrix} u \circ v \\ H(u + a/(2\varepsilon), v - \varepsilon u, z) \end{pmatrix} = \begin{pmatrix} \mu \mathbf{1} \\ b \end{pmatrix}$$

and

$$\begin{pmatrix} u \\ v \end{pmatrix} \equiv \begin{pmatrix} x - a/(2\varepsilon) \\ y + \varepsilon x - a/2 \end{pmatrix} \geq 0.$$

Deduce that if $m = 0$ and $H(x,y) \equiv y - F(x)$, where F is a P_0 function, then $\mathbf{H}_{\varepsilon,\mu}$ is a global homeomorphism on $\mathbb{R}^{2n}$.

11.9.10 The Jacobian smoothing method is identical to Algorithm **11.8.6** except that at Step 3 (11.8.11) is substituted by

$$G(y^{j,k}) + JG_{\varepsilon_k}(y^{j,k})d = 0. \tag{11.9.1}$$

This modification seems very natural because, in the end, we want find a zero of G and not of G_{ε_k}. Roughly speaking this corresponds to viewing the smoothing process simply as a way of constructing a Newton scheme for G. Prove that for the modified Algorithm **11.8.6** Theorem **11.8.8** still holds. (Hint: point (a) needs no modifications. To prove (b) use the fact that eventually, for every $k \in \kappa$, the difference between the solution of (11.8.11) and (11.9.1) is in the order of $\|\varepsilon_k\| \leq O(\|x^k - x^\infty\|^2$.)

11.9.11 Consider the function $\rho(t) \equiv \frac{1}{2}e^{-|t|}$.

(a) Show that ρ is a probability density function with finite absolute mean.

(b) Use ρ to define an alternative smoothing of the min function as follows: $\min_\varepsilon(a,b) \equiv a - p_\varepsilon(a-b)$. Show that

$$\min_\varepsilon(a,b) = \begin{cases} a - \dfrac{\varepsilon}{2}\, e^{(a-b)/\varepsilon} & \text{if } a \leq b \\ b - \dfrac{\varepsilon}{2}\, e^{(b-a)/\varepsilon} & \text{if } a > b. \end{cases}$$

11.9.12 Let $h : \mathbb{R}^{m+n} \to \mathbb{R}^m$ and $g : \mathbb{R}^{m+n} \to \mathbb{R}^n$ be continuously differentiable. Consider the MiCP in the variables $(x,y) \in \mathbb{R}^{m+n}$:

$$\begin{aligned} 0 &= h(x,y) \\ 0 \leq y &\perp g(x,y) \geq 0 \end{aligned}$$

and the homogenized problem in the variables (x,y,τ):

$$0 \leq \begin{pmatrix} 0 \\ y \\ \tau \end{pmatrix} \perp F(x,y,\tau) \equiv \begin{pmatrix} \tau\, h(x/\tau, y/\tau) \\ \tau\, g(x/\tau, y/\tau) \\ -x^T h(x/\tau, y/\tau) - y^T g(x/\tau, y/\tau) \end{pmatrix} \geq 0.$$

Let $f(x,y) \equiv (h(x,y), g(x,y))$. Show that

(a) if Jf is positive semidefinite on $\mathbb{R}^m \times \mathbb{R}^n_+$, then JF is positive semidefinite on $\mathbb{R}^m \times \mathbb{R}^{n+1}_{++}$;

(b) the original MiCP has a solution if and only if the homogenized MiCP has a solution with $\tau > 0$.

11.10 Notes and Comments

As a result of the famous example of Klee and Minty [332] that demonstrated the exponential behavior of the simplex method for linear programming, there has been a sustained interest in the mathematical programming community to design iterative methods for solving LPs and extended problems that are practically efficient and have polynomially bounded computational complexity. Interior point methods were born as part of the pursuit of this goal. While Khachiyan [322] proposed the first polynomial algorithm for linear programs and while there even were important preceding developments, it was Karmarkar's IP algorithm for a linear program [320] that inspired the contemporary advances in this family of iterative methods. Today, there is a huge literature on interior point methods for solving linear and nonlinear programs as well as their extensions, with many published articles, books and special volumes, a small sample of which includes

[421, 463, 511, 519, 599, 600, 612, 644]. The discussion in this section focuses on the literature that is directly related to the VIs, CPs, and CEs. See [252, 568] for specialized IP methods for solving saddle problems.

While IP methods for constrained optimization problems can be traced to the pioneering Sequential Unconstrained Minimization Technique of Fiacco and McCormick [197], IP methods for solving CPs were first studied by McLinden [398, 399] in the broad context of a set-valued problem defined as follows: For a given maximal monotone multifunction $F : \mathbb{R}^n \to \mathbb{R}^n$, find a pair $(x, y) \in \operatorname{gph} F$ such that

$$0 \leq x \perp y \geq 0.$$

The notion of the central path for a CP was formally born in McLinden's papers, which are purely theoretical in nature and pay no attention to practical algorithms. Fueled by the interest in solving linear programs, Megiddo [402] proposed the first IP path-following method based on the primal-dual (LCP) formulation of a pair of linear programs, thereby putting McLinden's conceptual scheme into practice.

Kojima, Megiddo, Noma, and Yoshise [339] presented a comprehensive study of IP methods for solving the LCP with a P_* matrix. Together with Mizuno, these authors [337, 338, 341, 342, 345] began an intensive investigation of IP methods for solving NCPs of the monotone class and of the uniformly P type. Among their many contributions are a careful investigation of the central path associated with the NCP of these classes and the associate path-following algorithms as well as many fundamental results for the system of parameterized equations:

$$\begin{pmatrix} y - F(x) \\ x \circ y \end{pmatrix} = t \begin{pmatrix} a \\ b \end{pmatrix}, \quad (x, y) \geq 0.$$

In particular, these authors were the first to study the above system with F being a P_0 function. Proposition **11.4.17** generalizes a convexity property of the monotone NCP established in the paper [344]. Proposed originally for a pair of primal-dual linear programs by Todd and Ye [582], the potential function (11.1.7) was extended to the LCP in [340, 343, 641].

Harker and Xiao [259] discussed a polynomial-time algorithm for solving a monotone AVI. Cao and Ferris [77] studied IP methods for a monotone AVI based on a reduction and a path-following algorithm applied to the reduced problem. Tseng [589] proposed an infeasible path-following method for monotone CPs.

Expanding the initial work of McLinden and Kojima and his group, several authors have studied the existence and limiting property of the

various trajectories (including the central trajectory) to the NCP using different approaches. Güler [246] considered this issue in the context of maximal monotone multifunctions. Facchinei [167], Facchinei and Kanzow [175], Gowda and Tawhid [242], and Zhao and Li [653] all treated this issue for the NCP with a P_0 function. Monteiro and Tsuchiya [411] studied the limiting behavior of the derivatives of certain trajectories associated with a monotone horizontal LCP.

Armed with his strong expertise with IP methods for linear and convex quadratic programs, Monteiro teamed up with Pang to begin an in-depth study of IP methods for solving CPs. Part of their initial ambition was to extend the IP methods beyond monotone problems and those of the P type by applying these methods to the min formulation of the NCP. Together with T. Wang, their effort led to the development of the "positive algorithm" [410], which, to their disappointment, lacked robustness and other desirable properties compared to the semismooth Newton methods described in Chapter **9**. Subsequently, these authors [603] abandoned the nonsmooth formulation and instead used a smooth CE formulation. The latter reference marked the dawn of the CE framework for the unified study of IP methods for solving CPs. This framework was further expanded in three additional papers [407, 408, 409]. The materials in Sections **11.2** through **11.5** are based on the collective work in these references. In particular, the article [409] develops an extensive theory for several interior point mappings for CPs in SPSD matrices.

We refer the reader to the monograph [11] by Ambrosetti and Prodi for a contemporary exposition of the theory of proper local homeomorphisms on metric spaces. The above-mentioned work of Monteiro and Pang is built entirely on this theory. In particular, the proof of Theorem **11.2.1**, which can be found in [407], is based on the work of Ambrosetti and Prodi.

The use of the horizonal LCP as a unified framework for the analysis of IP methods for linear and convex quadratic programs was popularized by Zhang in the paper [648], in which the polynomial complexity of a long-step path-following infeasible IP method was established. See [612, Chapter 6] for a brief historical account of this class of IP methods. In particular, while infeasible IP methods were proposed and implemented for LPs by Lustig [378], it was Zhang who first provided a convergence analysis of such methods. Extending Zhang's analysis, Kojima, Noma, and Yoshise [345] developed a generic class of infeasible IP methods for monotone NCPs and established that each algorithm in the class either generates an approximate solution with a given accuracy or provides the information that the complementarity problem has no solution in a given bounded set. Bellavia

and Macconi [44] show that the analysis in [345] can be extended to an inexact IP method. Billups and Ferris [52] refined Zhang's original proof of global convergence by dropping a requirement imposed on the starting point, which is required by Zhang to be (componentwise) greater than or equal to a certain vector that satisfies the equality constraints.

As pointed out in [648, 247] and utilized extensively in many papers including [60, 22], the horizontal LCP framework is better suited for the convergence analysis of the IP methods for LPs and convex QPs, especially when it comes to the issue of polynomial-time complexity and superlinear rate of convergence. Ye [642] developed a fully polynomial-time approximation algorithm for computing a stationary point of the general horizontal LCP. These prior studies provided the motivation for Monteiro and Pang to develop their theory for the nonlinear, implicit MiCP (11.1.3).

Generalizing a P_0 matrix, Wilson [608, 609] introduced the $\mathcal{W}$ and $\mathcal{W}_0$ properties for a pair of real square matrices; see Exercises **4.8.15** and **11.9.4**, respectively. Motivated by their study of the generalized order LCP [569], Sznajder and Gowda [570] extended the $\mathcal{W}$ properties to finitely many matrices of the same order. The counterexamples in Exercises **4.8.16** and **11.9.5** are due to Sznajder and Gowda, who also establish the equivalence of (b) and (c) in Exercise **11.9.4**. While the mixed P_0 property was introduced by Monteiro and Pang [407], the equivalence of the P_0 property and the column $\mathcal{W}$ property has not appeared in the literature. The column monotoncity property of a pair of real square matrices was used extensively in the IP literature. The characterization of this property in Exercise **11.9.5** was obtained in [247]. The equivalence of the horizontal LCP with a column monotone pair and a monotone LCP was first established in [570, 581]; see also [614]. In [21], the authors define the $P_*(\sigma)$ property of a pair of matrices in a horizontal LCP and show that this property is invariant when the latter LCP is transformed into an equivalent standard LCP. The authors of the latter paper use the term "geometric LCP" to describe a horizontal LCP.

Based on Ye's homogenization formulation (Exercise **11.9.12**), Lesaja [365, 366] studies IP methods for solving NCPs defined by P_* functions. Under several mild conditions, Lesaja establishes that the algorithm achieves linear global convergence and quadratic local convergence. Peng, Roos, Terlaky, and Yoshise [455] also study this class of NCPs using their "self-regularity theory", which originated from Peng's Ph.D. thesis [450].

Example **11.5.4**, which is the motivation to consider the boundedness of the two-sided level set in part (b) of Proposition **11.5.3**, appeared in [583]. This LCP is derived from a planar frictional contact problem.

The source of Section **11.6** is the article by Ralph and Wright [489], which in turn is an extension of the prior work of Wright [611] that pertains to the horizontal LCP and of Wright and Ralph [615] that pertains to the monotone NCP. These articles also establish the superlinear convergence of Algorithm **11.6.3** under various assumptions, which include the existence of a strictly complementary solution and the CRCQ. In the recent article [490], superlinear convergence is established when the CRCQ does not hold. Collectively, these results are very significant, because they show that IP methods for nonlinear, monotone VIs can converge superlinearly when multipliers are not unique and when active constraints are not linearly independent. The existence of a nondegenerate solution is an assumption that is often required for the superlinear convergence of IP methods. See the paper [412] for an analysis of the local convergence properties of feasible and infeasible interior point algorithms without this assumption. The latter paper shows in particular that for a broad class of such algorithms, superlinear convergence cannot be achieved, due to the non-existence of nondegenerate solutions.

There are several interesting IP topics for CPs that we have not covered in this chapter. Güler and Ye [248] show that for monotone LCPs, most IP methods generate a sequence in which every limit point is a solution with the maximal number of nonzero components among all solutions; such a solution is called *maximally complementary*. For an LCP with at least one nondegenerate solution (like that derived from the primal-dual pair of linear programs), a complementary solution is maximal if and only if it is nondegenerate. Potra and Ye [464] presented a potential reduction IP algorithm for monotone NCPs, which computes an approximate solution to a system of nonlinear equations at each iteration; they discussed the convergence of the algorithm toward a maximally complementary solution.

Extending an earlier algorithm of Ye [643] for solving the monotone LCP, the homogeneous algorithm of Anderson and Ye [12] solves an MiCP by solving a homogenized problem; Exercise **11.9.12** is drawn from this reference. There is a large literature on complexity analysis of IP methods for LPs, QPs, and monotone LCPs. This kind of analysis and that of superlinear convergence are often highly technical and involve a lot of detailed derivations. Many of the aforementioned references are devoted to such analyses; see also the series of papers by Sun and Zhao [565, 566, 567] where quadratically convergent IP methods for monotone VIs and CPs can be found. Stoer and Wechs [544] extend the results in [411] and establish the analyticity of interior point paths as a function of the barrier parameter. Such an analyticity property is used in several subsequent papers

[545, 546, 547] to design high-order IP methods for solving the sufficient horizontal LCP, even for degenerate problems with no strictly complementary solutions; see also [649] for related work on such high-order IP methods for solving LCPs.

Another significant development of the family of IP methods in recent years pertains to the class of so-called "conic optimization problems", which was the motivation for Kojima and his colleagues to introduce the class of CPs in SPSD matrices. We refer the reader to the two monographs [61, 421] for an introduction to the conic optimization problems, including many applications. A yet even more recent development is the application of IP methods for solving the class of "robust optimization problems"; see [46]. In essence, the robustness idea provides one way of dealing with uncertain data; this is done via a conservative approach in which a data "uncertainty set" is postulated and constraints involving all members in this set are imposed. For several important types of uncertain data sets, the resulting optimization problems are of the conic type to which the IP methods are applicable.

The extension of the robustness idea to equilibrium problems has yet to be explored. (This is not to be confused with the concepts of robust solutions to a VI considered in [442, 598], which are aimed at identifying a particular type of solution to such a VI.) In this regard, being very important for practical reasons, the entire subject of VIs/CPs with uncertain data deserves a careful investigation. To date, there exist only two studies (not related to IP methods) that address the class of "stochastic VIs" [43, 249].

Smoothing a nonsmooth function is a very natural approach in mathematics that can be traced back at least to the work of Steklov and Sobolev at the beginning of the last century. In optimization, smoothing is almost part of the folklore; early developments were made in the 1970s by many Russian mathematicians, one of the earliest references being [321]. The smoothing of the plus function $\max(0, t)$ is particularly relevant to the development of algorithms for complementarity problems. One of the first papers to deal with the smoothing of the plus function is [645]. Two sources that give general approaches to smoothing and provide a good overview of results up to the early 1990s are [164, 351]. From another angle, smoothing can also be linked to the so-called "splitting" methods for the solution of a nonsmooth system of equations. Many authors have proposed Newton-type methods for the solution of a nonsmooth system of equations $G(x) = 0$ in the following way. Suppose that we can write $G = S + N$, where S is smooth and N nonsmooth (this is obviously always possible by choosing

any smooth S and setting $N = G - S$). Define then an iterative process by setting $x^{k+1} = x^k - JS(x^k)^{-1}G(x^k)$. Obviously, we can view S as a smoothing of G. This approach was initiated by Zincenko [657], who credited Krasnosleskii for first suggesting the method. Further references to the construction of local Newton type algorithms based on this splitting idea can be found in [626].

However, the modern chapter of smoothing methods for VIs and CPs was initiated by Chen and Mangasarian [92, 93]. In particular, the latter of these two papers is entirely devoted to the development of algorithms for the solution of NCPs and box constrained VIs based on the smoothing of the plus function. In [93] it is also recognized that the solutions of the smoothed problems form (interior) trajectories that may or may not coincide with the central path according to the chosen type of smoothing. The good numerical results reported in this reference and also in [51], and the sound theoretical extensions given by Gabriel and Moré [231] for VIs on rectangles, prompted many researchers to take up the study of smoothing methods. Recent years have witnessed a flourishing of papers exploring the many facets of the approach. Attempts to summarize the results in this field are made in [481, 96].

Qi and colleagues made several contributions to the family of smoothing methods. In [478, 479], globally convergent versions of splitting methods related to those of Zincenko discussed above, and with S and N possibly changing at each iteration, are studied. These methods are also known in the literature as Jacobian smoothing or smoothing Newton methods. An important contribution along this line was given by Chen, Qi, and Sun [99] who, for the first time, proved a superlinear convergence rate for a smoothing method. This kind of splitting approach inspired Exercise **11.9.10**. Superlinear convergence was also proved by Gabriel [229], who, however, defined a hybrid algorithm where the smoothing method is used only in a first phase in order to get close to a solution. Modifications of the algorithm in [99] are given in [102, 308]. A trust region version of a smoothing Newton method is developed in [474]. Other algorithms using smoothing approaches as well as extensions and applications are analyzed in [98, 100, 103, 224, 367, 368, 468, 469, 483, 484, 487, 561, 564].

Algorithms in another series of papers, which employ the idea of tracing the solution path of a smoothed equation, are more akin to the IP methods. Specifically, in these papers, suitable neighborhoods of the followed path are introduced and used in the definition and analysis of the corresponding algorithms. Usually, these methods do not maintain the nonnegativity constraints and are therefore termed non-interior methods. The idea of

non-interior path-following for solving CPs originated from the Ph.D. research of B.T. Chen under the supervision of Harker; see the published papers [87, 88, 89]. Exercise **11.9.9** is based on Chen's thesis work. The original paper [87] was based on the smoothing of the min function (this is usually called the Chen-Harker-Kanzow-Smale smoothing function in the literature). Kanzow [298] proposed similar developments using a smoothed Fischer-Burmeister function and made the interesting observation that the path corresponding to this function and to the CHKS smoothing function is exactly the central path used in the interior point literature. The smoothed FB C-function ψ_{FB_μ} is used in an SQP algorithm for solving an MPEC by Fukushima, Luo, and Pang [223].

An extension of the approach in the previous papers to general VIs is given in [305]. The first complexity result for this kind of non-interior method was obtained by Xu and Burke [624]. This paper was immediately followed by a flurry of activity on rate of convergence results. All these results hinge on some new neighborhood concept for the smoothing paths that are not necessarily positive [73, 274]. Global linear convergence for a non-interior path following method was proved for the first time by Burke and Xu [73]; these results were then extended to nonlinear NCPs by Xu [623]. In [84, 85, 90, 91, 453, 592], the previous results are further extended to a wider class of smoothing functions; furthermore, superlinear or quadratic convergence is also shown to hold. Qi and Sun [482] proved analogous results for a method that extends the approach of [274], while in [76, 273] more complexity results are obtained for the LCP under different assumptions. Finally, based on the CHKS smoothing technique, Burke and Xu [74, 75] analyze the first instance of a non-interior predictor-corrector algorithm for monotone LCPs.

Our definitions of smoothing and superlinear or quadratic approximation are variants of the respective definitions in [483]. The definition of (weakly) Jacobian consistent smoothing is derived from that of Jacobian consistent smoothing introduced in [99]. The smoothing of the FB function used in Example **11.8.5** was proposed in [298]. The Line Search Smoothing Algorithm is drawn from [483], but has some peculiar characteristics of its own. In particular, this is one of the very few smoothing methods for which convergence to stationary points of the merit function can be proved under no monotonicity-type assumptions (see [308, 469] for other results of this type). This allows us to distinguish clearly between the assumptions needed to "make the algorithm work" and those guaranteeing that stationary points of the merit function are solutions of the system. We believe that this is very desirable. The convergence analysis and the results on the

convergence rate also parallel those in [483], but again with differences in the assumptions made. Proposition **11.8.7** and the related discussion are based on [606].

The smoothing of the plus function is basically the one introduced by Chen and Mangasarian [93]. In stating and proving Proposition **11.8.10** we took into account results in [93] and [483]. The examples of the density functions discussed in Example **11.8.11** were presented in [93]. The neural network density function was first used for smoothing purposes in [92]. The CHKS density function was introduced, independently, by Chen and Harker [87], Kanzow [298] and Smale [524]. The Huber-Pinar-Zenios density function is from [459], while the Zang function goes back to [645]. Pinar and Chen [458] have used related smoothing ideas to solve ℓ_1 formulations of linear inequalities. The extension to the mid function discussed at the end of Section **11.8.2** is carried out in detail in [231].

Chapter 12

Methods for Monotone Problems

Complementary to Chapter **11**, the present chapter is devoted to the study of solution methods for VIs that are of the monotone type and also for NCPs of the P_0 type. We already saw in Chapters **9** and **10** that the presence of some kind of monotonicity in a problem often leads to improved results for general algorithms capable of dealing with non-monotone problems. The methods studied in this chapter are distinctively different from those in previous chapters and strongly dependent on some kind of monotonicity of the problem; in fact, the methods presented herein usually cannot even be considered for problems lacking this particular property.

12.1 Projection Methods

Projection methods are conceptually simple methods for the solution of a monotone VI (K, F). There are two features common to all the methods that fall in this class. The first is that their implementation requires the ability to efficiently calculate the projection onto the closed convex set K; this feature certainly limits the applicability of the methods, especially when such projection is computationally expansive and/or difficult. The second characteristic is that the methods do not require the use of the derivatives of F and do not involve any complex computation besides the projection on K. On the one hand the latter feature makes these methods, when the projection is easily computable, extremely simple and cheap; on the other hand the use of no derivative information prevents these methods from being fast.

As a result, for sets K such that the projection Π_K can be easily carried out, projection methods can be applied to the solution of very large problems because of their simplicity; they can also be used in a two-phase

scheme in which a cheap projection method is first applied in order to move towards a "promising" region where a switch is then made to a more expansive but faster and more accurate method. In the next subsection we introduce and analyze several variants of a basic projection method.

12.1.1 Basic fixed-point iteration

We start with the most basic projection method, based on an application of the Banach fixed-point Theorem **2.1.21**. This simple method has a limited scope in practical applications, but it is the prototype for more advanced methods and it gives a clear illustration of the fixed-point perspective which underlies all projection-type methods that we examine in this section.

Throughout this section, K is a closed convex subset of $\mathbb{R}^n$ and F is a continuous mapping from K into $\mathbb{R}^n$. Let D be an $n \times n$ symmetric positive definite matrix. The solutions of the VI (K, F) are the solutions of the skewed natural equation

$$\mathbf{F}^{\mathrm{nat}}_{K,D}(x) \equiv x - \Pi_{K,D}(x - D^{-1}F(x)) = 0,$$

and vice versa, where $\Pi_{K,D}$ is the skewed projector onto K defined by the matrix D; that is, for every $x \in \mathbb{R}^n$, $\Pi_{K,D}(x)$ is the solution of the following convex minimization problem in the variable y:

$$\begin{array}{ll} \text{minimize} & \frac{1}{2}(y - x)^T D(y - x) \\ \text{subject to} & y \in K \end{array}$$

We recall that this projector is nonexpansive under the D-norm; see Exercise **1.8.16**. Therefore fixed points of the mapping

$$x \mapsto \Pi_{K,D}(x - D^{-1}F(x))$$

are solutions of the VI (K, F) and vice versa. Hence, if the latter map is a contraction we can use Algorithm **2.1.20** to calculate a solution of the VI (K, F). Below we rephrase Algorithm **2.1.20** in this context.

Basic Projection Algorithm (BPA)

12.1.1 Algorithm.

Data: $x^0 \in K$ and a symmetric positive definite $n \times n$ matrix D.

Step 0: Set $k = 0$.

Step 1: If $x^k = \Pi_{K,D}(x^k - D^{-1}F(x^k))$ stop.

Step 2: Set $x^{k+1} \equiv \Pi_{K,D}(x^k - D^{-1}F(x^k))$ and $k \leftarrow k + 1$; go to Step 1.

The following result gives sufficient conditions on the mapping F and the matrix D to ensure the convergence of the above algorithm.

12.1.2 Theorem. Let $F : K \to \mathbb{R}^n$, where K be a closed convex subset of $\mathbb{R}^n$. Suppose L and μ are such that for any x and y in K,

$$(\, F(x) - F(y)\,)^T (\, x - y\,) \;\geq\; \mu \,\|\, x - y \,\|_2^2 \tag{12.1.1}$$

and

$$\|\, F(x) - F(y) \,\|_2 \;\leq\; L \,\|\, x - y \,\|_2.$$

If

$$L^2 \,\lambda_{\max}(D) \;<\; 2\,\mu\,\lambda_{\min}^2(D), \tag{12.1.2}$$

the mapping $\Pi_{K,D}(x - D^{-1}F(x))$ is a contraction from K to K with respect to the norm $\|\cdot\|_D$; therefore every sequence $\{x^k\}$ produced by Algorithm **12.1.1** converges to the unique solution of the VI (K, F).

Proof. We note that, for any symmetric positive definite D, we have for any two vectors x and y in $\mathbb{R}^n$,

$$\lambda_{\min}(D) \,\|\, x - y \,\|_2^2 \;\leq\; \|\, x - y \,\|_D^2 \;\leq\; \lambda_{\max}(D) \,\|\, x - y \,\|_2^2.$$

Thus, if x and y are any two vectors in K, we can write

$$\begin{aligned}
&\|\, \Pi_{K,D}(x - D^{-1}F(x)) - \Pi_{K,D}(y - D^{-1}F(y)) \,\|_D^2 \\
&\leq\; \|\, (x - y) + D^{-1}(F(y) - F(x)) \,\|_D^2 \\
&=\; \|\, F(x) - F(y) \,\|_{D^{-1}}^2 + \|\, x - y \,\|_D^2 - 2\,(\, F(x) - F(y)\,)^T (\, x - y\,) \\
&\leq\; \left(1 + \frac{L^2}{\lambda_{\min}^2(D)} - \frac{2\,\mu}{\lambda_{\max}(D)} \right) \|\, x - y \,\|_D^2,
\end{aligned}$$

where the first inequality is due to Exercise **1.8.16**. Therefore if (12.1.2) holds, then $\Pi_{K,D}(x - D^{-1}F(x))$ is a contraction. □

If D is a positive multiple τ of the identity matrix, condition (12.1.2) becomes

$$\tau \;>\; \frac{L^2}{2\,\mu}.$$

Consequently, for a given VI (K, F) with a strongly monotone and Lipschitz continuous mapping F, provided that the constants L and μ are available and τ is chosen as described, the sequence $\{x^k\}$ defined iteratively by

$$x^{k+1} \;\equiv\; \Pi_K(x^k - \tau^{-1}F(x^k)), \quad k \;=\; 0, 1, 2, \ldots, \tag{12.1.3}$$

converges to the unique solution of the VI (K, F), for any x^0 in K. The iteration (12.1.3) is illustrated in Figure **12.1**.

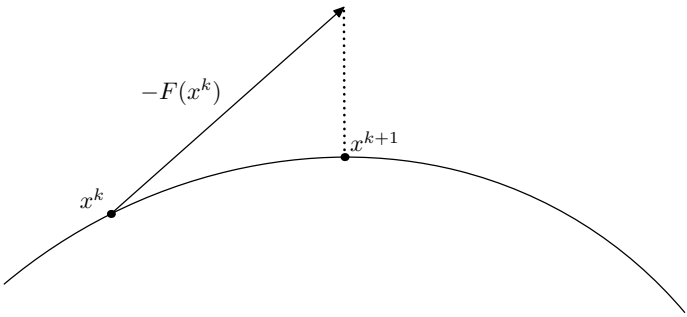


Figure 12.1: One iteration of the Basic Projection Algorithm.

If F is the gradient of a convex function θ with a nonempty set of constrained minima in K, it is known that the above method converges for suitably large values of τ (see Exercise **12.8.1**). This may lead one to suppose that it should be possible to prove convergence of Algorithm **12.1.1** under a mere monotonicity assumption on F. The following example shows that this conjecture is false.

12.1.3 Example. Let $F(x) \equiv Ax$, where

$$A \equiv \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}.$$

Note that the Jacobian of F is equal to the asymmetric matrix A, so that F is not a gradient map. It is immediate to see that the unique solution of the VI $(F, \mathbb{R}^2)$ is $x^* = 0$. However, direct calculation shows that for every $x^k \neq 0$ and $\tau > 0$ we have

$$\| x^{k+1} - x^* \| = \| x^k - \tau^{-1} F(x^k) \| = \sqrt{1 + \tau^{-2}} \, \| x^k \| > \| x^k \|,$$

so that it is clear that Algorithm **12.1.1** cannot converge to x^*. □

One of our aims in the remaining part of this section is therefore to develop projection algorithms which do not require the strong monotonicity of F. Another drawback of the Basic Projection Algorithm is that it requires the knowledge of the constants L and μ, which are typically not known in practice. Thus for practical reasons, it is desirable to have a projection scheme which does not require *a priori* such constants. Without some kind of a line search, such a scheme is not known to exist to date.

In Algorithm **12.1.4** below, the two constants L and μ are substituted by a co-coercivity constant of F; see Theorem **12.1.8**. In Subsection **12.1.2**, we present a projection-type algorithm applicable to a pseudo

monotone VI, which requires only the knowledge of the Lipschitz constant L of the function F. In Subsection **12.1.3**, we present a projection-type algorithm that uses a simple line search and which is applicable to a pseudo monotone VI (K, F) where F is not required to be Lipschitz continuous.

In the remainder of this subsection, we present a refined analysis of Algorithm **12.1.1** that is not based on the fixed-point argument but which yields an improved convergence result; see Theorem **12.1.8**. We apply this analysis to a variable-step projection scheme in which the step size τ is allowed to vary from one iteration to the next. This extends the iteration (12.1.3) which is a constant-step projection scheme. The resulting variable-step algorithm is not a line-search method, however, because the step size τ_k is not determined by a line search routine. (A word of caution: in Step 2 of the algorithm below, we use τ_k instead of τ_k^{-1} in the iterative formula; this is a notational but not substantial deviation from (12.1.3).)

Projection Algorithm with Variable Steps (PAVS)

12.1.4 Algorithm.

Data: $x^0 \in K$.

Step 0: Set $k = 0$.

Step 1: If $x^k = \Pi_K(x^k - F(x^k))$ stop.

Step 2: Choose $\tau_k > 0$. Set $x^{k+1} \equiv \Pi_K(x^k - \tau_k F(x^k))$ and $k \leftarrow k + 1$; go to Step 1.

The choice of the sequence of scalars $\{\tau_k\}$ is crucial for the success of Algorithm **12.1.4**. In what follows, we show that if F is co-coercive with constant $c > 0$, i.e.,

$$(\, F(x) - F(y)\,)^T (\, x - y\,) \;\geq\; c \,\|\, F(x) - F(y)\,\|_2^2, \quad \forall\, x, y \,\in\, K,$$

then $\{\tau_k\}$ can be chosen so that the sequence $\{x^k\}$ converges to a solution of the VI (K, F); see Theorem **12.1.8**. Since every strongly monotone, Lipschitz continuous function is co-coercive but not vice versa, we have therefore successfully extended the convergence of the projection algorithm to a broader class of VIs.

To accomplish our goal we need two preliminary results, the first of which is of independent interest. Specifically, we consider the application of Algorithm **12.1.4** to the case where $K = \mathbb{R}^n$, that is, to the solution of the system of equations $F(x) = 0$. In this case Algorithm **12.1.4** reduces

to the iteration

$$x^{k+1} \equiv x^k - \tau_k F(x^k), \quad k = 0, 1, 2, \ldots.$$

Actually, we consider a more general situation where the correction term $\tau_k F(x^k)$ is replaced by $F^k(x^k)$, where all the functions F^k have the same set of zeros. This generalized iteration is as follows:

$$x^{k+1} = x^k - F^k(x^k), \quad k = 0, 1, 2, \ldots. \tag{12.1.4}$$

The next result establishes the convergence of the generated sequence $\{x^k\}$.

12.1.5 Lemma. Let $F^k : \mathbb{R}^n \to \mathbb{R}^n$ for $k = 0, 1, 2, \ldots$ be co-coercive functions on $\mathbb{R}^n$ with modulus c_k such that

$$\rho \equiv \inf_k c_k > \tfrac{1}{2}.$$

If all the functions F^k have the same nonempty set S of zeros and

$$\inf_k \| F^k(x) \| > 0, \quad \forall x \notin S, \tag{12.1.5}$$

then the sequence $\{x^k\}$ produced by the iteration (12.1.4) converges to a point x^* in S.

Proof. Let $\{x^k\}$ and $\{y^k\}$ be two sequences produced by the iteration (12.1.4) starting from x^0 and y^0, respectively. We first show that

$$\lim_{k\to\infty} \| x^k - y^k \| \equiv \sigma < \infty; \tag{12.1.6}$$

$$\sum_{k=0}^{\infty} \| F^k(x^k) - F^k(y^k) \|^2 < \infty. \tag{12.1.7}$$

Taking into account the co-coercivity assumption, a direct calculation yields

$$\begin{aligned} \| x^{k+1} - y^{k+1} \|^2 &\leq \| x^k - y^k \|^2 - (2c_k - 1) \| F^k(x^k) - F^k(y^k) \|^2 \\ &\leq \| x^k - y^k \|^2 - (2\rho - 1) \| F^k(x^k) - F^k(y^k) \|^2. \end{aligned}$$

This show that the sequence $\{\|x^k - y^k\|\}$ is nonincreasing, and thus has a limit which we call σ; thus (12.1.6) follows. To show (12.1.7) it suffices to observe that a simple recursion of the above inequality leads to

$$\| x^{k+1} - y^{k+1} \|^2 \leq \| x^0 - y^0 \|^2 - (2\rho - 1) \sum_{i=0}^{k-1} \| F^i(x^i) - F^i(y^i) \|^2.$$

From this and (12.1.6), (12.1.7) follows.

Let $\bar{x}$ be any element in S. If we start the iterative at $y^0 = \bar{x}$ we have $y^k = \bar{x}$ for every k. By (12.1.7) we have

$$\sum_{k=0}^{\infty} \| F^k(x^k) \|^2 < \infty, \tag{12.1.8}$$

while (12.1.6) proves that

$$\lim_{k\to\infty} \| x^k - \bar{x} \| \equiv \sigma \tag{12.1.9}$$

exists. It follows that the sequence $\{x^k\}$ is bounded. Let x^* be the limit of the subsequence $\{x^k : k \in \kappa\}$. We show that $x^* \in S$. Since F^k is co-coercive with constant c_k, F^k is Lipschitz continuous with constant $1/c_k$. Therefore

$$\| F^k(x^k) - F^k(x^*) \| \leq \frac{1}{\rho} \| x^k - x^* \|,$$

from which we obtain

$$\lim_{k(\in\kappa)\to\infty} \| F^k(x^k) - F^k(x^*) \| = 0.$$

By (12.1.8), we also have

$$\lim_{k\to\infty} \| F^k(x^k) \| = 0.$$

Combining the last two limits, we deduce

$$\lim_{k(\in\kappa)\to\infty} \| F^k(x^*) \| = 0$$

From this limit and (12.1.5) we conclude that $x^* \in S$. By (12.1.9) with $\bar{x} = x^*$, it follows that $\{x^k\}$ converges to x^*. □

12.1.6 Remark. The assumption that each F^k is co-coercive on $\mathbb{R}^n$ was made for simplifying the above proof somewhat. The reader can readily see that it is sufficient to assume that each F^k is co-coercive on a closed set containing the sequence $\{x^k\}$ and S. □

Let $\mathbf{F}^{\mathrm{nat}}_{K,\tau}$ denote the natural map of the VI $(K, \tau F)$; this map was first introduced in Exercise **4.8.4**. The next lemma shows that $\mathbf{F}^{\mathrm{nat}}_{K,\tau}$ is a co-coercive map on K; this result is key to the application of Lemma **12.1.5** to analyze Algorithm **12.1.4**.

12.1.7 Lemma. Suppose that a function $F : K \subseteq \mathbb{R}^n \to \mathbb{R}^n$, with K convex and nonempty, is co-coercive on K with constant c. If $\tau \in (0, 4c)$, then $\mathbf{F}^{\mathrm{nat}}_{K,\tau}$ is co-coercive on K with constant $1 - \tau/4c$.

Proof. Since the projection is co-coercive with constant 1, it follows that, for any two vectors x and y in K,

$$[\,\Pi_K(x-\tau F(x))-\Pi_K(y-\tau F(y))\,]^T[\,(x-\tau F(x))-(y-\tau F(y))\,]$$
$$\geq \|\,\Pi_K(x-\tau F(x))-\Pi_K(y-\tau F(y))\,\|^2.$$

By the above inequality and an easy manipulation, we deduce

$$(\mathbf{F}^{\rm nat}_{K,\tau}(x)-\mathbf{F}^{\rm nat}_{K,\tau}(y))^T(x-y) \geq \|\,\mathbf{F}^{\rm nat}_{K,\tau}(x)-\mathbf{F}^{\rm nat}_{K,\tau}(y)\,\|^2$$
$$+\tau\,(F(x)-F(y))^T(x-y)-\tau\,(\mathbf{F}^{\rm nat}_{K,\tau}(x)-\mathbf{F}^{\rm nat}_{K,\tau}(y))^T(F(x)-F(y)).$$

Using the co-coercivity of F, the above inequality implies:

$$\begin{aligned}
&(\mathbf{F}^{\rm nat}_{K,\tau}(x)-\mathbf{F}^{\rm nat}_{K,\tau}(y))^T(x-y)\\
&\quad\geq \|\,\mathbf{F}^{\rm nat}_{K,\tau}(x)-\mathbf{F}^{\rm nat}_{K,\tau}(y)\,\|^2+\tau\,c\,\|\,F(x)-F(y)\,\|^2\\
&\qquad\quad -\tau\,(\mathbf{F}^{\rm nat}_{K,\tau}(x)-\mathbf{F}^{\rm nat}_{K,\tau}(y))^T(F(x)-F(y))\\
&\quad= \left(1-\frac{\tau}{4c}\right)\|\,\mathbf{F}^{\rm nat}_{K,\tau}(x)-\mathbf{F}^{\rm nat}_{K,\tau}(y)\,\|^2\\
&\qquad\quad +\left\|\,\sqrt{\frac{\tau}{4c}}\,(\mathbf{F}^{\rm nat}_{K,\tau}(x)-\mathbf{F}^{\rm nat}_{K,\tau}(y))-\sqrt{\tau c}\,(F(x)-F(y))\,\right\|^2\\
&\quad\geq \left(1-\frac{\tau}{4c}\right)\|\,\mathbf{F}^{\rm nat}_{K,\tau}(x)-\mathbf{F}^{\rm nat}_{K,\tau}(y)\,\|^2,
\end{aligned}$$

which establishes the lemma. □

Combining the above two lemmas, we obtain the convergence of Algorithm **12.1.4** for the class of solvable, co-coercive VIs.

12.1.8 Theorem. Let K be a closed convex subset of $\mathbb{R}^n$ and F be a mapping from K into $\mathbb{R}^n$ that is co-coercive on K with constant c. Suppose that $\mathrm{SOL}(K,F)\neq\emptyset$. If

$$0 < \inf_k \tau_k \leq \sup_k \tau_k < 2\,c,$$

Algorithm **12.1.4** produces a sequence $\{x^k\}$ converging to a solution of the VI (K,F).

Proof. Algorithm **12.1.4** can be seen to be an application of iteration (12.1.4) with $F^k(x)\equiv \mathbf{F}^{\rm nat}_{K,\tau_k}(x)$, whose set of zeros clearly coincides with $\mathrm{SOL}(K,F)$. By Lemma **12.1.7**, each F^k is co-coercive on K, which is a closed set containing the sequence $\{x^k\}$ and $\mathrm{SOL}(K,F)$, with modulus $1-\tau_k/4c$. Moreover, a simple argument shows that, for every x that is not a solution of the VI (K,F),

$$\inf_k \|\,F^k(x)\,\| > 0.$$

Therefore, according to Remark **12.1.6**, we may apply Lemma **12.1.5** to deduce that the sequence $\{x^k\}$ converges to a solution of the VI (K, F). □

12.1.2 Extragradient method

Theorem **12.1.8** is certainly an improvement over Theorem **12.1.2** in that the strong monotonicity of F is replaced by its co-coerciveness. In this subsection we introduce a projection algorithm that executes two projections per iteration. Although this undoubtedly requires twice the amount of computations, the benefit is significant because the resulting algorithm is applicable to the class of pseudo monotone VIs. However, the function F is still required to be Lipschitz continuos and an estimate of its Lipschitz constant is needed. The extragradient method presented below takes its name from the extra evaluation of F (and the extra projection) that is called for in each iteration. The name originates from the case of a symmetric VI. In this case, the VI represents the optimality condition of a differentiable optimization problem, the extra evaluation of F corresponds to an extra evaluation of the gradient of the objective function, thus extragradient.

Extragradient Algorithm (EgA)

12.1.9 Algorithm.

Data: $x^0 \in K$ and $\tau > 0$.

Step 0: Set $k = 0$.

Step 1: If $x^k \in \mathrm{SOL}(K, F)$, stop.

Step 2: Compute

$$\begin{aligned} x^{k+1/2} &\equiv \Pi_K(x^k - \tau F(x^k)), \\ x^{k+1} &\equiv \Pi_K(x^k - \tau F(x^{k+1/2})); \end{aligned}$$

set $k \leftarrow k + 1$ and go to Step 1.

The generation of $x^{k+1/2}$ and x^{k+1} at Step 2 is illustrated in Figure **12.2**. In order to give some geometric motivation for this algorithm, we consider its application to the solution of a system of monotone equations ($K = \mathbb{R}^n$); in this case, Step 2 becomes

$$\begin{aligned} x^{k+1/2} &\equiv x^k - \tau F(x^k), \\ x^{k+1} &\equiv x^k - \tau F(x^{k+1/2}), \end{aligned}$$

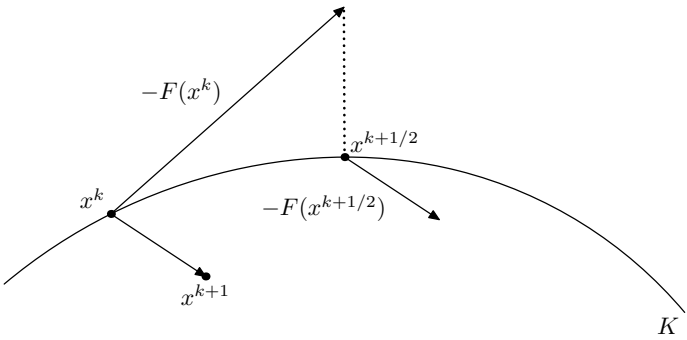


Figure 12.2: One iteration of the Extragradient Algorithm.

Consider the hyperplane $H^k \equiv \{x \in \mathbb{R}^n : F(x^{k+1/2})^T(x - x^{k+1/2}) = 0\}$ and let x^* be a solution of the system of equations $F(x) = 0$. By the monotonicity of F we have $F(x^{k+1/2})^T(x^* - x^{k+1/2}) \leq 0$. On the other hand, $F(x^{k+1/2})^T(x^k - x^{k+1/2}) = \tau F(x^{k+1/2})^T F(x^k)$. When τ converges to zero, $x^{k+1/2}$ converges to x^k and therefore, for τ small enough we have $F(x^{k+1/2})^T F(x^k) > 0$ (it cannot be that $F(x^k)$ is zero, since otherwise we would have stopped at Step 1). Therefore, for τ small enough, $F(x^{k+1/2})^T(x^k - x^{k+1/2}) > 0$ and H^k separates x^k from any solution x^*. As a consequence of this fact we see that x^{k+1} is a move from x^k in the direction of the projection of this latter point onto the separating hyperplane H^k so that, for τ small enough, x^{k+1} is closer than x^k to any solution of the system of equations. This property (which is called Fejér monotonicity of $\{x^k\}$ with respect to the solution set, see Exercise **12.8.3**) is the basis for the convergence analysis. If $K \neq \mathbb{R}^n$, the Fejér monotonicity of $\{x^k\}$ with respect to the solution set can still be shown, thanks to the properties of the projector.

In the previous discussion the monotonicity of F was used only to show that $F(x^{k+1/2})^T(x^* - x^{k+1/2}) \leq 0$ for any solution x^*. This motivates the following definition. We say that the function $F : K \to \mathbb{R}^n$ is pseudo monotone on K with respect to $\mathrm{SOL}(K, F)$ if the latter set is nonempty and for every $x^* \in \mathrm{SOL}(K, F)$ it holds that

$$F(x)^T(x - x^*) \geq 0, \quad \forall x \in K.$$

This concept is a weakening of pseudo monotonicity of F on K and plays a central role in the convergence of Algorithm **12.1.9**. The key lemma below makes the previous discussion on the algorithm precise and extends it to the case $K \neq \mathbb{R}^n$.

12.1.10 Lemma. Let K be a closed convex set in $\mathbb{R}^n$ and let F be a mapping from K into $\mathbb{R}^n$ that is pseudo monotone on K with respect to $\mathrm{SOL}(K,F)$ and Lipschitz continuous on K with constant $L>0$. Let x^* be any solution of the VI (K,F). For every k it holds that

$$\| x^{k+1} - x^* \|^2 \leq \| x^k - x^* \|^2 - (1 - \tau^2 L^2) \| x^{k+1/2} - x^k \|^2.$$

Proof. Since $x^* \in \mathrm{SOL}(K,F)$, $x^{k+1/2} \in K$, and F is pseudo monotone with respect to $\mathrm{SOL}(K,F)$, we have $F(x^{k+1/2})^T(x^{k+1/2} - x^*) \geq 0$ for all k. This implies

$$F(x^{k+1/2})^T(x^* - x^{k+1}) \leq F(x^{k+1/2})^T(x^{k+1/2} - x^{k+1}).$$

By the variational characterization of the projection, we have

$$\begin{aligned}
&(x^{k+1} - x^{k+1/2})^T(x^k - \tau F(x^{k+1/2}) - x^{k+1/2}) \\
&= (x^{k+1} - x^{k+1/2})^T(x^k - \tau F(x^k) - x^{k+1/2}) \\
&\quad +\tau\, (x^{k+1} - x^{k+1/2})^T(F(x^k) - F(x^{k+1/2})) \\
&\leq (x^{k+1} - \Pi_K(x^k - \tau F(x^k)))^T(x^k - \tau F(x^k) - \Pi_K(x^k - \tau F(x^k))) \\
&\quad +\tau\, (x^{k+1} - x^{k+1/2})^T(F(x^k) - F(x^{k+1/2})) \\
&\leq \tau\, (x^{k+1} - x^{k+1/2})^T(F(x^k) - F(x^{k+1/2})).
\end{aligned}$$

By letting $y^k \equiv x^k - \tau F(x^{k+1/2})$ for simplicity and by using the above inequalities, we obtain

$$\begin{aligned}
&\| x^{k+1} - x^* \|^2 = \| \Pi_K(y^k) - x^* \|^2 \\
&= \| y^k - x^* \|^2 + \| y^k - \Pi_K(y^k) \|^2 + 2\, (\Pi_K(y^k) - y^k)^T (y^k - x^*) \\
&\leq \| y^k - x^* \|^2 - \| y^k - \Pi_K(y^k) \|^2 \\
&= \| x^k - x^* - \tau F(x^{k+1/2}) \|^2 - \| x^k - x^{k+1} - \tau F(x^{k+1/2}) \|^2 \\
&= \| x^k - x^* \|^2 - \| x^k - x^{k+1} \|^2 + 2\tau (x^* - x^{k+1})^T F(x^{k+1/2}) \\
&\leq \| x^k - x^* \|^2 - \| x^k - x^{k+1} \|^2 + 2\tau (x^{k+1/2} - x^{k+1})^T F(x^{k+1/2}) \\
&= \| x^k - x^* \|^2 - \| x^k - x^{k+1/2} \|^2 - \| x^{k+1/2} - x^{k+1} \|^2 \\
&\qquad +2(x^{k+1} - x^{k+1/2})^T (x^k - \tau F(x^{k+1/2}) - x^{k+1/2}) \\
&\leq \|x^k - x^*\|^2 - \|x^k - x^{k+1/2}\|^2 - \|x^{k+1/2} - x^{k+1}\|^2 \\
&\quad +2\,\tau\, L \| x^{k+1} - x^{k+1/2} \| \| x^k - x^{k+1/2} \| \\
&\leq \| x^k - x^* \|^2 - (1 - \tau^2 L^2) \| x^k - x^{k+1/2} \|^2,
\end{aligned}$$

which completes the proof of the lemma. □

Based on the above lemma, we can establish the convergence of Algorithm **12.1.9**. Notice that the Lipschitz constant L of F still plays a role in controlling the step τ in the algorithm.

12.1.11 Theorem. Let K be a closed convex set in $\mathbb{R}^n$ and let F be a mapping from K into $\mathbb{R}^n$ that is pseudo monotone on K with respect to $\mathrm{SOL}(K, F)$ and Lipschitz continuous on K with constant L. If $\tau < 1/L$, then the sequence $\{x^k\}$ generated by Algorithm **12.1.9** converges to a solution of the VI (K, F).

Proof. Let x^* be an arbitrary element of $\mathrm{SOL}(K, F)$. Let $\rho \equiv 1-\tau^2 L^2$. By the assumption on τ, we have $\rho \in (0, 1)$. By Lemma **12.1.10** the sequence $\{x^k\}$ is bounded so that it has at least one limit point $\bar{x}$ that must belong to K. We claim that $\bar{x} \in \mathrm{SOL}(K, F)$. From Lemma **12.1.10** and the fact that $\rho \in (0, 1)$, we get

$$\rho \sum_{k=0}^{\infty} \| x^k - x^{k+1/2} \|^2 \leq \| x^0 - x^* \|^2.$$

This implies

$$\lim_{k\to\infty} \| x^k - x^{k+1/2} \| = 0.$$

If $\bar{x}$ is the limit of the subsequence $\{x^k : k \in \kappa\}$, then we must have

$$\lim_{k(\in\kappa)\to\infty} x^{k+1/2} = \bar{x}.$$

By the definition of $x^{k+1/2}$ at Step 2 of Algorithm **12.1.9** and by the continuity of F and of the projector, we see that

$$\begin{aligned} \bar{x} &= \lim_{k(\in\kappa)\to\infty} x^{k+1/2} \\ &= \lim_{k(\in\kappa)\to\infty} \Pi_K(x^k - \tau F(x^k)) \\ &= \Pi_K(\bar{x} - \tau F(\bar{x})), \end{aligned}$$

which shows that $\bar{x} \in \mathrm{SOL}(K, F)$.

It remains to show that the whole sequence $\{x^k\}$ converges to $\bar{x}$. To this end it suffices to apply Lemma **12.1.10** with $x^* = \bar{x}$ to deduce that the sequence $\{\| x^k - \bar{x} \|\}$ is monotonically decreasing and therefore converges. Since

$$\lim_{k\to\infty} \| x^k - \bar{x} \| = \lim_{k(\in\kappa)\to\infty} \| x^k - \bar{x} \| = 0,$$

$\{x^k\}$ converges to $\bar{x}$ as desired. □

12.1.3 Hyperplane projection method

As noted before, the extragradient method still requires the knowledge of the Lipschitz constant L of F, which is usually not known. In this subsection we present an enhanced extragradient-like method that neither requires F to be Lipschitz continuous nor calls for the knowledge of potentially unknown constants.

Let K be a closed convex set in $\mathbb{R}^n$ and F be a continuous mapping from K into $\mathbb{R}^n$ that is pseudo monotone on K with respect to $\mathrm{SOL}(K, F)$. Let $\tau > 0$ be a fixed scalar. The algorithm can be described geometrically as follows. Let $x^k \in K$ be given. First compute the point $\Pi_K(x^k - \tau F(x^k))$. Next search the line segment joining x^k and $\Pi_K(x^k - \tau F(x^k))$, by a simple Armijo-type line search routine, for a point z^k such that the hyperplane

$$H^k \equiv \{ x \in \mathbb{R}^n : F(z^k)^T (x - z^k) = 0 \}$$

strictly separates x^k from $\mathrm{SOL}(K, F)$. We then project x^k onto H^k and the resulting point onto K, obtaining x^{k+1}. It can be shown that x^{k+1} is closer to $\mathrm{SOL}(K, F)$ than x^k. If we refer to the discussion made after the statement of the Extragradient Algorithm for the case $K = \mathbb{R}^n$, we can easily see what the main changes are. First, since we can no longer use the information on the Lipschitz constant L to determine a shift τ along $-F(x^k)$ which gives a suitable separating hyperplane H^k, we perform a line search along the direction $-F(x^k)$ to find such a hyperplane (see (12.1.10)). Second, instead of moving from x^k *along* the direction to the projection of x^k onto H^k, we take x^{k+1} to be exactly such a projection. The resulting method requires three projections per iteration, two onto K and one onto H^k. The latter projection is easily performed and is given by an explicit formula.

Hyperplane Projection Algorithm (HPA)

12.1.12 Algorithm.

Data: $x^0 \in K$, $\tau > 0$, and $\sigma \in (0, 1)$.

Step 0: Set $k = 0$.

Step 1: If $x^k \in \mathrm{SOL}(K, F)$, stop.

Step 2: Compute

$$y^k \equiv \Pi_K(x^k - \tau F(x^k)),$$

and find the smallest nonnegative integer i_k such that with $i = i_k$,

$$F(2^{-i}y^k + (1-2^{-i})x^k)^T(x^k - y^k) \geq \frac{\sigma}{\tau}\| x^k - y^k \|^2. \quad (12.1.10)$$

Step 3: Set

$$z^k \equiv 2^{-i_k} y^k + (1-2^{-i_k})\, x^k,$$

and

$$w^k \equiv \Pi_{H^k}(x^k) = x^k - \frac{F(z^k)^T(x^k - z^k)}{\| F(z^k) \|^2} F(z^k),$$

Step 4: Set $x^{k+1} \equiv \Pi_K(w^k)$ and $k \leftarrow k+1$; go to Step 1.

Noting that $z^k = \tau_k y^k + (1-\tau_k)x^k$, where $\tau_k \equiv 2^{-i_k}$, we have

$$w^k = x^k - \tilde{\tau}_k F(z^k)$$

where

$$\tilde{\tau}_k \equiv \tau_k \frac{F(z^k)^T(x^k - y^k)}{\| F(z^k) \|^2}.$$

Hence

$$x^{k+1} = \Pi_K(x^k - \tilde{\tau}_k F(\tau_k y^k + (1-\tau_k)x^k)).$$

The above expressions shows, once again, that the hyperplane projection method is closely related to the extragradient method. Indeed, when $\tau_k = 1$, we get $z^k = y^k$ and

$$x^{k+1} = \Pi_K(x^k - \tilde{\tau}_k F(y^k));$$

this iteration formula differs from the iteration formula in the extragradient method in that the variable step size $\tilde{\tau}_k$ is employed instead of the fixed step size τ in the second projection.

The iterations of the Hyperplane Projection Algorithm may seem more complicated than those of the projection algorithms considered so far. Figure **12.3** shows that the complication is more superficial than real.

To prove the convergence of Algorithm **12.1.12**, we need three preparatory results. The first result formulates two known properties of the Euclidean projector in a form more convenient for subsequence use.

12.1.13 Lemma. Let K a nonempty closed convex set. The following statements hold.

(a) For all x, y in $\mathbb{R}^n$,

$$\| \Pi_K(x) - \Pi_K(y) \|^2 \leq \| x - y \|^2 - \| \Pi_K(x) - x + y - \Pi_K(y) \|^2.$$

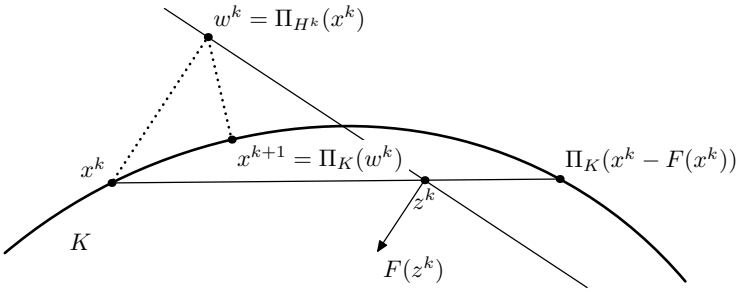


Figure 12.3: One iteration of the Hyperplane Projection Algorithm.

(b) For all $x \in K$ and for all $y \in \mathbb{R}^n$,

$$(\, x - y \,)^T (\, x - \Pi_K(y) \,) \;\geq\; \| \, x - \Pi_K(y) \, \|^2.$$

Proof. To prove (a) observe that

$$\begin{aligned} \| \, \Pi_K(x) - x + y - \Pi_K(y) \, \|^2 \;&=\; \| \, \Pi_K(x) - \Pi_K(y) \, \|^2 + \| \, x - y \, \|^2 \\ &\quad -2(x - y)^T (\Pi_K(x) - \Pi_K(y)) \\ &\leq\; -\| \, \Pi_K(x) - \Pi_K(y) \, \|^2 + \| \, x - y \, \|^2, \end{aligned}$$

where the inequality follows from the co-coercivity of the projector; see Theorem **1.5.5** (c). This property of the projector also yields (b) because $x = \Pi_K(x)$ for $x \in K$. □

The next lemma shows, among other things, that the line search at Step 2 terminates finitely, so that the Hyperplane Projection Method is well defined.

12.1.14 Lemma. Suppose that F is continuous on K and let x^k be a point in K that is not a solution of the VI (K, F). It holds that

(a) a finite integer $i_k \geq 0$ exists such that (12.1.10) holds;

(b) $F(z^k)^T (x^k - z^k) > 0$.

Proof. Assume for contradiction that for every nonnegative integer i we have

$$F\left(2^{-i} y^k + (1 - 2^{-i}) x^k\right)^T (\, x^k - y^k \,) \;<\; \frac{\sigma}{\tau} \, \| \, x^k - y^k \, \|^2.$$

Multiplying by τ and passing to the limit $i \to \infty$, we obtain,

$$\tau \, F(x^k)^T (\, x^k - y^k \,) \;\leq\; \sigma \, \| \, x^k - y^k \, \|^2.$$

Recalling the definition of y^k at Step 2 and Lemma **12.1.13** (b), we get

$$\begin{aligned}\sigma \| x^k - y^k \|^2 &\geq \tau F(x^k)^T (x^k - y^k) \\ &= [x^k - (x^k - \tau F(x^k))]^T [x^k - \Pi_K(x^k - \tau F(x^k))] \\ &\geq \| x^k - y^k \|^2.\end{aligned}$$

Since $\sigma \in (0,1)$, this implies $x^k = y^k = \Pi_K(x^k - \tau F(x^k))$. But this contradicts the assumption that $x^k \notin \mathrm{SOL}(K,F)$. □

If F is pseudo monotone on K with respect to $\mathrm{SOL}(K,F)$, part (b) of the above lemma shows that the hyperplane H^k strictly separates x^k from $\mathrm{SOL}(K,F)$. Indeed, since z^k belongs to K we have, for any solution x^* of the VI (K,F),

$$F(z^k)^T (z^k - x^*) \geq 0 > F(z^k)^T (z^k - x^k).$$

Hence $\mathrm{SOL}(K,F)$ and x^k lie in opposite sides of the hyperplane H^k. This property lies at the heart of Algorithm **12.1.12** and is where the pseudo monotonicity property of the pair (K,F) is needed.

We come to the third and last preparatory result before proving the main convergence theorem of Algorithm **12.1.12**.

12.1.15 Proposition. Let K be a closed convex set in $\mathbb{R}^n$ and let F be a continuous mapping from K into $\mathbb{R}^n$ that is pseudo monotone on K with respect to $\mathrm{SOL}(K,F)$. Let x^k be a sequence produced by Algorithm **12.1.12**. The following four statements hold.

(a) For every solution x^*, the sequence $\{\|x^k - x^*\|\}$ is nonincreasing, and therefore convergent.

(b) The sequence $\{x^k\}$ is bounded.

(c) $\lim_{k\to\infty} F(z^k)^T (x^k - z^k) = 0$.

(d) If an accumulation point of $\{x^k\}$ is a solution of VI (K,F), the whole sequence converges to this solution.

Proof. Only (a) and (c) need a proof. Let

$$H^k_{\leq} \equiv \{ x \in \mathbb{R}^n : F(z^k)^T (x - z^k) \leq 0 \}.$$

We have

$$\mathrm{SOL}(K,F) \subseteq H^k_{\leq} \not\ni x^k.$$

Thus

$$x^* = \Pi_K(x^*) = \Pi_{H^k_{\leq}}(x^*)$$

while w^k is the projection of x^k on $H^k_{\leq}$. Therefore, by Lemma **12.1.13** (a) and the definition of x^{k+1}, we have, for any $x^* \in \mathrm{SOL}(K, F)$,

$$\begin{aligned}
\| x^{k+1} - x^* \|^2 &= \| \Pi_K(w^k) - \Pi_K(x^*) \|^2 \\
&\leq \| w^k - x^* \|^2 - \| \Pi_K(w^k) - w^k \|^2 \\
&= \| \Pi_{H^k_{\leq}}(x^k) - \Pi_{H^k_{\leq}}(x^*) \|^2 - \| \Pi_K(w^k) - w^k \|^2 \\
&\leq \| x^k - x^* \|^2 - \| \Pi_{H^k_{\leq}}(x^k) - x^k \|^2 - \| \Pi_K(w^k) - w^k \|^2 \\
&= \| x^k - x^* \|^2 - \| w^k - x^k \|^2 - \| \Pi_K(w^k) - w^k \|^2 \\
&\leq \| x^k - x^* \|^2.
\end{aligned}$$

This establishes part (a). By looking at the first and next to last inequality in the above chain of inequalities, we see that

$$\begin{aligned}
0 &\leq \| w^k - x^k \|^2 \\
&\leq \| x^k - x^* \|^2 - \| x^{k+1} - x^* \| - \| \Pi_K(w^k) - w^k \|^2 \\
&\leq \| x^k - x^* \|^2 - \| x^{k+1} - x^* \|.
\end{aligned}$$

By part (a), so that we deduce

$$0 = \lim_{k\to\infty} \| w^k - x^k \| = \lim_{k\to\infty} \frac{F(z^k)^T(x^k - z^k)}{\| F(z^k) \|}.$$

Part (b) clearly implies the boundedness of $\{z^k\}$. Hence we get

$$\lim_{k\to\infty} F(z^k)^T(x^k - z^k) = 0,$$

which is part (c). □

We are ready to state and prove the promised convergence result of Algorithm **12.1.12**.

12.1.16 Theorem. Let K be a closed convex set in $\mathbb{R}^n$ and let F be a continuous mapping from K into $\mathbb{R}^n$ that is pseudo monotone on K with respect to $\mathrm{SOL}(K, F)$. If $\{x^k\}$ is a sequence produced by Algorithm **12.1.12**, then $\{x^k\}$ converges to a solution of the VI (K, F).

Proof. Let us set, for simplicity, $t_k = 2^{-i_k}$. By Lemma **12.1.15** (c),

$$\lim_{k\to\infty} F(z^k)^T(x^k - z^k) = 0 = \lim_{k\to\infty} t_k F(z^k)^T(x^k - y^k). \tag{12.1.11}$$

We consider two cases.

Case 1: $\limsup_{k\to\infty} t_k > 0$. There exist $\bar{t} > 0$ and a subsequence κ such that $t_k \geq \bar{t}$ for every $k \in \kappa$. From (12.1.11) we then get

$$\lim_{k(\in\kappa)\to\infty} F(z^k)^T(x^k - y^k) = 0.$$

By (12.1.10) and the definition of z^k, the above limit implies

$$\lim_{k(\in\kappa)\to\infty} \| x^k - y^k \| = 0 = \lim_{k(\in\kappa)\to\infty} \| x^k - \Pi_K(x^k - \tau F(x^k)) \|.$$

Since $\{x^k\}$ is bounded by Lemma **12.1.15** (b), we may assume without loss of generality that $\{x^k : k \in \kappa\}$ converges to x^∞. Hence we get

$$\| x^\infty - \Pi_K(x^\infty - \tau F(x^\infty)) \| = 0.$$

This shows that x^∞ is a solution of the VI (K, F). By Lemma **12.1.15** (d) we therefore conclude that the whole sequence $\{x^k\}$ converges to x^∞ and thus completing the proof in this case.

Case 2: $\lim_{k\to\infty} t_k = 0$. Let us set

$$\bar{z}^k \equiv 2^{-i_k+1} y^k + (1 - 2^{-i_k+1}) x^k.$$

As before, we may assume without loss of generality that some subsequence $\{x^k : k \in \kappa\}$ converges to a limit x^∞. Since $\{t^k\}$ converges to zero, we have

$$\lim_{k(\in\kappa)\to\infty} \bar{z}^k = x^\infty.$$

By the rules of Step 2 of Algorithm **12.1.12**, and since $\{t_k\} \to 0$, we also have

$$F(\bar{z}^k)^T(x^k - y^k) < \frac{\sigma}{\tau} \| x^k - y^k \|^2.$$

Passing to the limit $k(\in \kappa) \to \infty$, we obtain

$$F(x^\infty)^T(x^\infty - y^\infty) \leq \frac{\sigma}{\tau} \| x^\infty - y^\infty \|^2,$$

where $y^\infty \equiv \Pi_K(x^\infty - \tau F(x^\infty))$. By Lemma **12.1.13** (b), it follows that

$$\begin{aligned} \sigma \| x^\infty - y^\infty \|^2 &\geq \tau F(x^\infty)^T(x^\infty - y^\infty) \\ &= [x^\infty - \Pi_K(x^\infty - \tau F(x^\infty))]^T(x^\infty - y^\infty) \\ &\geq \| x^\infty - y^\infty \|. \end{aligned}$$

Since $\sigma \in (0, 1)$, this implies $x^\infty = \Pi_K(x^\infty - \tau F(x^\infty))$ so that x^∞ is a solution of the VI (K, F). The whole sequence $\{x^k\}$ then converges to x^∞ by Lemma **12.1.15** (d). □

There is a simple variant of the Hyperplane Projection Algorithm which may be of interest. We note that at Step 3 of this algorithm we obtain x^{k+1} by projecting x^k on H^k first, obtaining w^k, and then projecting w^k back onto K. It is possible to show, by simple modifications of the proofs, that Theorem **12.1.16** still holds if we calculate x^{k+1} as the projection of x^k onto $H^k_{\leq} \cap K$. This variant is illustrated in Figure **12.4**. Some obvious

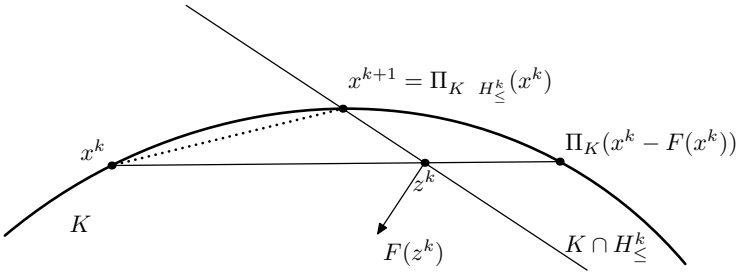


Figure 12.4: One iteration of the variant of Hyperplane Projection Algorithm.

geometric considerations show that, given x^k, the point x^{k+1} calculated according to this modified procedure is closer to $\mathrm{SOL}(K, F)$ than the point x^{k+1} calculated as in Step 3 of Algorithm **12.1.12**. This suggests that the variant could be faster than the original scheme. This advantage has to be weighted against the fact that projecting onto $H^k_{\leq} \cap K$ may be more difficult than projecting onto H^k and K separately. For example, suppose that K is a rectangle or a sphere. The respective projection onto H^k and K can be computed analytically. However the projection onto $H^k_{\leq} \cap K$ entails the solution of a nontrivial convex minimization problem.

12.2 Tikhonov Regularization

One of the basic ideas that is often exploited in the solution of VIs is the substitution of the original problem by a sequence of problems that are, in some sense, better behaved. In this section we present one of the classical and simplest realizations of this idea: the *Tikhonov regularization*.

The process of "regularizing" the VI (K, F) in terms of the family of perturbed VIs (K, F_ε), where $F_\varepsilon \equiv F + \varepsilon \mathbf{I}$ and ε is a positive parameter, was historically defined for monotone VIs. The motivation is that a monotone problem generally lacks the kind of strong stability properties that are present in a strongly monotone problem. In this section we broaden this historical consideration by studying a VI of the P_0 type; specialization of the results to a monotone VI yields stronger conclusions.

The setting herein is that of Section **3.5**; see in particular Subsection **3.5.2**. We consider a set $K \subseteq \mathbb{R}^n$ given by:

$$K = \prod_{\nu=1}^{N} K_\nu, \tag{12.2.1}$$

where N is a positive integer and each K_ν is a subset of $\mathbb{R}^{n_\nu}$ with

$$\sum_{\nu=1}^{N} n_\nu = n.$$

We assume throughout the section that F is continuous P_0 function on K, unless otherwise stated. For each $\varepsilon > 0$, let $x(\varepsilon)$ be the unique solution of the VI (K, F_ε); see Theorem **3.5.15**. The family of solutions

$$\{\, x(\varepsilon) \,:\, \varepsilon > 0 \,\} \tag{12.2.2}$$

is called the *Tikhonov trajectory* of the VI (K, F). Our first goal is to establish various properties of this trajectory. A preliminary result is the boundedness and continuity of this trajectory for ε restricted to a bounded interval which does not contain zero.

12.2.1 Proposition. Let K be given by (12.2.1) where each K_ν is closed convex. Let $F : K \to \mathbb{R}^n$ be a continuous P_0 function on K. For any $\overline{\varepsilon} > \underline{\varepsilon} > 0$, the family

$$\{\, x(\varepsilon) \,:\, \varepsilon \in [\,\underline{\varepsilon}, \overline{\varepsilon}\,] \,\} \tag{12.2.3}$$

is bounded. Consequently, $x(\varepsilon)$ is continuous at every positive ε.

Proof. The proof of the boundedness of the solutions (12.2.3) is similar to the proof of Theorem **3.5.15**. Indeed suppose that there is a sequence of scalars $\{\varepsilon_k\} \subset [\underline{\varepsilon}, \overline{\varepsilon}]$ such that with $x^k \equiv x(\varepsilon_k)$,

$$\lim_{k\to\infty} \| x^k \| = \infty.$$

Similar to the proof of Theorem **3.5.15**, we may deduce the existence of an index $\nu \in \{1, \ldots, N\}$ and a bounded sequence of vectors $\{y^k\}$ such that

$$\lim_{k\to\infty} \| x^k_\nu \| = \infty$$

and for infinitely many k's,

$$(\, y^k_\nu - x^k_\nu \,)^T [\, F_\nu(y^k) + \varepsilon_k\, x^k_\nu \,] \geq 0.$$

This is a contradiction because $\varepsilon_k \geq \underline{\varepsilon} > 0$ for all k. Thus the family (12.2.3) is bounded.

To show the continuity of $x(\varepsilon)$ at an arbitrary $\varepsilon > 0$, fix such an ε and let $\{\varepsilon_k\}$ be a sequence approaching ε. By what has just been proved, the sequence $\{x(\varepsilon_k)\}$ is bounded. Moreover, every accumulation point of this sequence (and there must be at least one such point) must be a solution of the VI (K, F_ε), by simple continuity arguments. Since the latter VI has a unique solution, it follows that $\{x(\varepsilon_k)\}$ has a unique accumulation point; therefore this sequence converges and its limit must be $x(\varepsilon)$. □

As a way of regularizing the VI (K, F), we wish to investigate the limit behavior of the trajectory $\{x(\varepsilon)\}$ as $\varepsilon \downarrow 0$. The above proposition does not address this issue. One complication is that unlike the VI (K, F_ε) with a positive ε, the original VI (K, F) may have multiple or no solutions. Thus the limit

$$\lim_{\varepsilon \downarrow 0} x(\varepsilon) \tag{12.2.4}$$

does not always exist. Indeed the nonemptiness of $\mathrm{SOL}(K, F)$ is a necessary condition for this limit to exist because such a limit must be a solution of the VI (K, F). But this condition is in general not sufficient. The following example illustrates the unboundedness of the Tikhonov trajectory in the case of a simple LCP.

12.2.2 Example. Consider the LCP (q, M) where

$$M \equiv \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \quad \text{and} \quad q \equiv \begin{pmatrix} -1 \\ 0 \end{pmatrix}.$$

Clearly M is a P_0 matrix and the solution set of the LCP (q, M) consists of two unbounded rays:

$$\mathrm{SOL}(q, M) = \{ (x_1, 1) : x_1 \geq 0 \} \cup \{ (0, x_2) : x_2 \geq 1 \}.$$

It is easy to check that the regularized LCP $(q, M + \varepsilon \mathbf{I}_2)$ has a unique solution given by $x(\varepsilon) = (1/\varepsilon, 0)$ for every $\varepsilon > 0$. The latter solution is clearly unbounded as ε tends to zero. More interestingly, the distance of the regularized solution $x(\varepsilon)$ to the set $\mathrm{SOL}(q, M)$ is equal to unity for all $\varepsilon > 0$. Notice that for this example, $\mathrm{SOL}(q, M)$ is nonconvex. This is illustrated in Figure **12.5**. □

The above example suggests that the boundedness of $\mathrm{SOL}(K, F)$ may have something to do with the convergence of the Tikhonov trajectory as the perturbation parameter tends to zero. A full treatment of this issue for the general P_0 VI requires the theory of weakly univalent functions developed in Section **3.6**; see Theorem **12.2.7** below. In what follows,

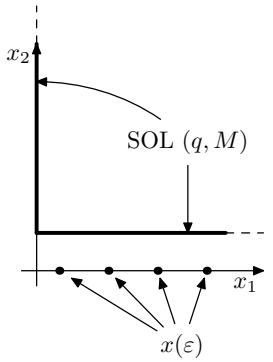


Figure 12.5: Solution set and Tikhonov trajectory.

we discuss first the special case of a monotone VI on a closed convex set that is not necessarily of the Cartesian form (3.5.1). For this case, the nonemptiness of $\mathrm{SOL}(K, F)$ is both necessary and sufficient for the limit (12.2.4) to exist. The key to the proof of this result is the convexity of the solution set of such a VI (see Theorem **2.3.5**). A consequence of this convexity property is that if the VI (K, F), where K is closed convex and F is continuous and monotone on K, has a solution, then it must have a unique solution with least Euclidean norm. We call this the *least-norm solution* of the monotone VI. This solution turns out to be the limit of the Tikhonov trajectory as the perturbation tends to zero.

12.2.3 Theorem. Let $K \subseteq \mathbb{R}^n$ be closed convex and $F : K \to \mathbb{R}^n$ be continuous and monotone on K. Let (12.2.2) be the Tikhonov trajectory. The following three statements are equivalent:

(a) the limit (12.2.4) exists;

(b) $\limsup_{\varepsilon \downarrow 0} \|x(\varepsilon)\| < \infty$;

(c) $\mathrm{SOL}(K, F) \neq \emptyset$.

Moreover, if any one of these statements holds, the limit (12.2.4) is equal to the least-norm solution of the VI (K, F).

Proof. (a) $\Rightarrow$ (b). This is obvious.

(b) $\Rightarrow$ (c). The boundedness of $\{x(\varepsilon)\}$ as $\varepsilon \downarrow 0$ implies that for every sequence of positive scalars $\{\varepsilon_k\}$ converging to zero, the associated sequence of solutions $\{x(\varepsilon_k)\}$ must have at least one accumulation point. It is clear that any such point must be a solution of the VI (K, F).

(c) $\Rightarrow$ (a). This is the nontrivial part of the proposition. Let $\bar{x}$ denote the least-norm solution of $\mathrm{SOL}(K, F)$. Let $\{\varepsilon_k\}$ be an arbitrary sequence

of positive scalars converging to zero. Write $x^k \equiv x(\varepsilon_k)$. It suffices to show that

$$\lim_{k\to\infty} x^k \;=\; \bar{x}.$$

This limit also establishes the last assertion of the proposition. For each k we have

$$(\,\bar{x} - x^k\,)^T[\,F(x^k) + \varepsilon_k\, x^k\,] \;\geq\; 0;$$

and

$$(\,x^k - \bar{x}\,)^T F(\bar{x}) \;\geq\; 0.$$

By the monotonicity of F on K, the latter inequality implies

$$(\,x^k - \bar{x}\,)^T F(x^k) \;\geq\; 0.$$

Therefore we obtain

$$\varepsilon_k\,(\,\bar{x} - x^k\,)^T x^k \;\geq\; 0$$

which implies

$$\bar{x}^T x^k \;\geq\; \|\,x^k\,\|_2^2.$$

By the Cauchy-Schwarz inequality applied to the left-hand inner product, we deduce

$$\|\,x^k\,\|_2 \;\leq\; \|\,\bar{x}\,\|_2.$$

Thus the sequence $\{x^k\}$ is bounded; moreover the Euclidean norm of every accumulation point of this sequence is bounded above by the Euclidean norm of the least-norm solution. This shows that the sequence $\{x^k\}$ must converge to the least-norm solution $\bar{x}$ as claimed. □

12.2.4 Remark. It is not clear if Theorem **12.2.3** will remain valid if F is pseudo monotone on K. In this case, although the VI (K, F) still possesses a unique least-norm solution, due to the convexity of $\mathrm{SOL}(K, F)$, the existence and uniqueness of the Tikhonov trajectory is in jeopardy. We leave this as an unresolved question. □

Theorem **12.2.3** can be generalized by considering a nonlinear regularization of the monotone VI. Instead of the Tikhonov regularization, consider the family:

$$\{\;\mathrm{VI}\;(K, F + \varepsilon\, G)\;:\;\varepsilon\;>\;0\,\}$$

where $G : \mathbb{R}^n \to \mathbb{R}^n$ is an arbitrary mapping that is strongly monotone on K. When G is the identity map, we recover the Tikhonov regularization. Since F is assumed monotone on K, it follows that $F + \varepsilon G$ is strongly monotone on K for all $\varepsilon > 0$. Consequently, for every $\varepsilon > 0$, there exists

a unique vector $x(\varepsilon)$ that solves the VI $(K, F + \varepsilon G)$. We call this family of solutions $\{x(\varepsilon) : \varepsilon > 0\}$ the *G-trajectory*. The following result concerns the limiting behavior of the G-trajectory as ε tends to zero.

12.2.5 Theorem. Let $K \subseteq \mathbb{R}^n$ be closed convex and $F : K \to \mathbb{R}^n$ be continuous and monotone on K. Let $G : K \to \mathbb{R}^n$ be continuous and strongly monotone on K. The three statements (a), (b), and (c) in Theorem **12.2.3** remain equivalent for the G-trajectory. Moreover, if any one of these statements holds, the limit (12.2.4) is the unique solution of the VI $(\bar{K}, G)$, where $\bar{K} \equiv \mathrm{SOL}(K, F)$.

Proof. It suffices to show (c) $\Rightarrow$ (a) and the last statement about the limit (12.2.4). We proceed as in the proof of Theorem **12.2.3**. Assume that $\mathrm{SOL}(K, F)$ is nonempty; let $\bar{x}$ be an arbitrary element of this set. Let $\{\varepsilon_k\}$ be an arbitrary sequence of positive scalars converging to zero. Write $x^k \equiv x(\varepsilon_k)$. For each k we have

$$(\,\bar{x} - x^k\,)^T[\,F(x^k) + \varepsilon_k\, G(x^k)\,] \;\geq\; 0; \tag{12.2.5}$$

and

$$(\,x^k - \bar{x}\,)^T F(\bar{x}) \;\geq\; 0.$$

By the monotonicity of F on K, the latter inequality implies

$$(\,x^k - \bar{x}\,)^T F(x^k) \;\geq\; 0.$$

Adding this inequality to (12.2.5), we deduce

$$(\,\bar{x} - x^k\,)^T G(x^k) \;\geq\; 0. \tag{12.2.6}$$

By the strong monotonicity of G, there exists a scalar $c > 0$ such that for all k,

$$(\,\bar{x} - x^k\,)^T(\,G(\bar{x}) - G(x^k)\,) \;\geq\; c\,\|\,\bar{x} - x^k\,\|_2^2.$$

Adding the last two inequalities, we deduce

$$(\,\bar{x} - x^k\,)^T G(\bar{x}) \;\geq\; c\,\|\,\bar{x} - x^k\,\|_2^2.$$

By the Cauchy-Schwarz inequality, we obtain

$$\|\,\bar{x} - x^k\,\|_2 \;\leq\; c^{-1}\,\|\,G(\bar{x})\,\|_2.$$

Hence the sequence $\{x^k\}$ is bounded. Let x^∞ be an accumulation point of this sequence. Clearly, x^∞ belongs to $\mathrm{SOL}(K, F)$. Moreover, from (12.2.6), we easily obtain

$$(\,\bar{x} - x^\infty\,)^T G(x^\infty) \;\geq\; 0.$$

Since $\bar{x}$ is an arbitrary element of the set $\mathrm{SOL}(K, F)$, it follows that x^∞ is a solution of the VI $(\bar{K}, G)$. Since G is strongly monotone on K which contains $\bar{K}$, the latter VI has a unique solution. We have therefore shown that the sequence $\{x^k\}$ is bounded and it has a unique accumulation point that is the solution of the VI $(\bar{K}, G)$. Therefore the sequence $\{x^k\}$ converges with the desired limit. □

In what follows, we establish a generalization of Theorem **12.2.3** by weakening the monotonicity of F. Specifically, we consider a partitioned VI (K, F), where K is the Cartesian product of finitely many closed convex sets in lower-dimensional spaces and F is a $\mathrm{P}_*(\sigma)$ function on K for some $\sigma > 0$; see Definition **3.5.8**.

12.2.6 Theorem. Let K be given by (12.2.1) where each K_ν is closed convex. Let $F : K \to \mathbb{R}^n$ be a continuous $\mathrm{P}_*(\sigma)$ function on K for some $\sigma > 0$. For each $\varepsilon > 0$, let $x(\varepsilon)$ denote the unique solution of the VI $(K, F + \varepsilon \mathbf{I})$. The following two statements are equivalent:

(a) $\limsup_{\varepsilon \downarrow 0} \| x(\varepsilon) \| < \infty$;

(b) the VI (K, F) has a solution.

Proof. It suffices to show that (b) implies (a) because the other implication is obvious. Let $\bar{x} \in \mathrm{SOL}(K, F)$ be arbitrary. For each $\varepsilon > 0$ and ν, we have

$$(F_\nu(x(\varepsilon)) + \varepsilon\, x_\nu(\varepsilon))^T (\bar{x}_\nu - x_\nu(\varepsilon)) \geq 0 \quad \text{and} \quad F_\nu(\bar{x})^T (x_\nu(\varepsilon) - \bar{x}_\nu) \geq 0.$$

Adding and rearranging terms, we obtain

$$0 \geq (F_\nu(x(\varepsilon)) - F_\nu(\bar{x}) + \varepsilon\, x_\nu(\varepsilon))^T (x_\nu(\varepsilon) - \bar{x}_\nu),$$

which implies, with $I_+(\varepsilon) \equiv I_+(x(\varepsilon), \bar{x})$

$$\begin{aligned} 0 &\geq (F(x(\varepsilon)) - F(\bar{x}) + \varepsilon\, x(\varepsilon))^T (x(\varepsilon) - \bar{x}) \\ &\geq -\sigma \sum_{\nu \in I_+(\varepsilon)} (F_\nu(x(\varepsilon)) - F_\nu(\bar{x}))^T (x_\nu(\varepsilon) - \bar{x}_\nu) + \varepsilon\, x(\varepsilon)^T (x(\varepsilon) - \bar{x}) \\ &\geq \sigma\, \varepsilon \sum_{\nu \in I_+(\varepsilon)} x_\nu(\varepsilon)^T (x_\nu(\varepsilon) - \bar{x}_\nu) + \varepsilon\, x(\varepsilon)^T (x(\varepsilon) - \bar{x}). \end{aligned}$$

Thus

$$0 \geq \sigma \sum_{\nu \in I_+(\varepsilon)} x_\nu(\varepsilon)^T (x_\nu(\varepsilon) - \bar{x}_\nu) + x(\varepsilon)^T (x(\varepsilon) - \bar{x}). \tag{12.2.7}$$

For each $\nu \in I_+(\varepsilon)$, we have

$$0 \;\geq\; \varepsilon\, x_\nu(\varepsilon)^T (\, x_\nu(\varepsilon) - \bar{x}_\nu \,), \tag{12.2.8}$$

which implies $\|x_\nu(\varepsilon)\| \leq \|\bar{x}_\nu\|$. Consequently, (12.2.7) implies that $\{x_\nu(\varepsilon)\}$ is bounded also for every $\nu \notin I_+(\varepsilon)$. □

Having dealt with the monotone case and the $\mathrm{P}_*(\sigma)$ generalization, we consider the Tikhonov trajectory associated with a VI of the P_0-type. The following result is a consequence of Corollary **3.6.5**.

12.2.7 Theorem. Let K be given by (12.2.1) where each K_ν is closed convex. Let $F : K \to \mathbb{R}^n$ be a continuous P_0 function on K. Let $\{x(\varepsilon)\}$ be the Tikhonov trajectory of the VI (K, F). If $\mathrm{SOL}(K, F)$ is nonempty and bounded, then

$$\limsup_{\varepsilon \downarrow 0} \,\| \, x(\varepsilon) \, \| \;<\; \infty.$$

Proof. For $\varepsilon > 0$, with $\mathbf{F}^{\mathrm{nat}}_{\varepsilon,K}$ denoting the natural map of the perturbed VI $(K, F + \varepsilon \mathbf{I})$, we have

$$\| \, \mathbf{F}^{\mathrm{nat}}_K(x) - \mathbf{F}^{\mathrm{nat}}_{\varepsilon,K}(x) \, \| \;\leq\; \varepsilon \, \| \, x \, \|.$$

The boundedness of $\{x(\varepsilon)\}$ as $\varepsilon \downarrow 0$ follows readily from part (c) of Corollary **3.6.5**. □

In the P_0 case, it is not easy to prove the convergence of the Tikhonov trajectory when ε goes to zero, even under the assumption that $\mathrm{SOL}(K, F)$ is nonempty and bounded. To establish such a convergence result we need an additional subanalytic assumption on the pair (K, F). The proof of the following theorem makes use of the subanalyticity properties (p1)–(p6) in Section **6.6**.

12.2.8 Theorem. Let K be given by (12.2.1) where each K_ν is a closed convex subanalytic subset of $\mathbb{R}^\nu$. Let $F : \mathbb{R}^n \to \mathbb{R}^n$ be a continuous P_0 subanalytic function. Let $\{x(\varepsilon)\}$ be the Tikhonov trajectory of the VI (K, F). It holds that

$$\left[\limsup_{\varepsilon \downarrow 0} \,\| \, x(\varepsilon) \, \| \;<\; \infty \right] \;\Leftrightarrow\; \left[\lim_{\varepsilon \downarrow 0} x(\varepsilon) \text{ exists } \right].$$

Thus, if $\mathrm{SOL}(K, F)$ is nonempty and bounded, then the Tikhonov trajectory $\{x(\varepsilon)\}$ converges to a solution of the VI (K, F) as $\varepsilon \downarrow 0$.

Proof. It suffices to prove the "⇒" implication. Let us consider the natural map of the VI $(K, F + \varepsilon\mathbf{I})$, as a function of both ε and x:

$$G(\varepsilon, x) \equiv x - \Pi_K(x - (F(x) + \varepsilon x)).$$

Since F is subanalytic and continuous by assumption and the projection map is subanalytic, it follows that $G(\varepsilon, x)$ is subanalytic by Lemma **6.6.2** and (p5). Therefore, by (p4), $G^{-1}(0)$ is subanalytic. It is easy to check that the graph of the function $x(\varepsilon)$ for $\varepsilon > 0$ is given by

$$\mathcal{X} = G^{-1}(0) \cap ((0, +\infty) \times \mathbb{R}^n);$$

by (p1), $\mathcal{X}$ is a subanalytic set.

By the boundedness assumption, $x(\varepsilon)$ has at least one limit point, say $\bar{x}$, when ε tends to 0. Suppose for the sake of contradiction that there exists another accumulation point $\tilde{x}$ with $\tilde{x} \neq \bar{x}$. By definition $(0, \bar{x})$ belongs to the closure of $\mathcal{X}$. Therefore, by (p6), there exists a continuous function $\bar{g} : [0, 1] \to \mathbb{R}^{1+n}$ such that $\bar{g}(0) = (0, \bar{x})$ and for $s \in (0, 1]$, $\bar{g}(s) = (\bar{\varepsilon}(s), x(\bar{\varepsilon}(s)))$, with $\bar{\varepsilon}(s) > 0$. Let $\{s_k\}$ be any sequence of positive scalars converging to 0. We have, by continuity,

$$\lim_{k\to\infty} \bar{g}(s_k) = \lim_{k\to\infty} (\varepsilon_k, x(\varepsilon_k)) = (0, \bar{x}) \tag{12.2.9}$$

(where we set $\varepsilon_k \equiv \bar{\varepsilon}(s_k)$). Since $(0, \tilde{x})$ also belongs to the closure of $\mathcal{X}$ we can reason in a similar way to deduce the existence of a continuous function $\tilde{g} : [0, 1] \to \mathbb{R}^{1+n}$ such that $\tilde{g}(0) = (0, \tilde{x})$ and, for $s \in (0, 1]$, $\tilde{g}(s) = (\tilde{\varepsilon}(s), x(\tilde{\varepsilon}(s)))$, with $\tilde{\varepsilon}(s) > 0$. By using continuity and the fact that $\tilde{g}_1$, the first component of $\tilde{g}$, is positive for $s > 0$, it is easy to check that the sequence $\{\tilde{s}_k\}$, where for each k,

$$\tilde{s}_k \equiv \inf\{ s \in [0, 1] : \tilde{g}_1(s) \geq \varepsilon_k \}$$

which is well defined for ε_k sufficiently small, converges to 0 as $k \to \infty$; moreover, $\tilde{g}_1(\tilde{s}_k) = \varepsilon_k$. By the continuity of $\tilde{g}$, we deduce

$$\lim_{k\to\infty} \tilde{g}(\tilde{s}_k) = \lim_{k\to\infty} (\varepsilon_k, x(\varepsilon_k)) = (0, \tilde{x}),$$

thus contradicting (12.2.9). The last statement of the theorem follows from Theorem **12.2.7**. □

12.2.1 A regularization algorithm

The Tikhonov regularization leads to an iterative algorithm for approximating a solution of a VI with a P_0 function, provided that such a solution

exists. Such an algorithm, whose practical implementation takes different forms, involves the solution of a sequence of sub-VIs each of the P type. In the most basic form, we fix a decreasing sequence $\{\varepsilon_k\}$ of positive numbers converging to zero. For each k, we in turn use an appropriate algorithm, which usually is another iterative process, to compute $x(\varepsilon_{k+1})$, possibly inexactly. For the latter computation, it is advisable to initiate the (inner) algorithm for solving the sub-VI $(K, F+\varepsilon_k \mathbf{I})$ at the previous iterate $x(\varepsilon_k)$; the reason is that eventually $x(\varepsilon_k)$ is close to $x(\varepsilon_{k+1})$ if the Tikhonov trajectory is convergent. In what follows, we present a version of the Tikhonov regularization algorithm in which we use $\|\mathbf{F}^{\mathrm{nat}}_{\varepsilon_k,K}\|$ to gauge the suitability of an iterate to be a good approximation to $x(\varepsilon_{k+1})$. An equivalent residual of the same growth order as the natural residual can also be used.

Tikhonov Regularization Algorithm (TiRA)

12.2.9 Algorithm.

Data: $x^0 \in \mathbb{R}^n$, $\{\rho_k\} \downarrow 0$, and $\{\varepsilon_k\} \downarrow 0$

Step 0: Set $k = 0$.

Step 1: If x^k solves the VI (K, F), stop.

Step 2: Calculate a point x^{k+1} such that

$$\| \mathbf{F}^{\mathrm{nat}}_{\varepsilon_k,K}(x^{k+1}) \| \leq \rho_k.$$

Set $k \leftarrow k+1$ and go to Step 1.

The following theorem, which yields the subsequential convergence of the Tikhonov regularization Algorithm, is easy to show.

12.2.10 Theorem. Let K be given by (12.2.1), where each K_ν is closed convex. Let $F : K \to \mathbb{R}^n$ be a continuous P_0 function on K and suppose $\mathrm{SOL}(K, F)$ is nonempty and bounded. Let $\{x^k\}$ be a sequence produced by Algorithm **12.2.9**. The sequence $\{x^k\}$ is bounded and each of its limit points belongs to $\mathrm{SOL}(K, F)$.

Proof. A simple continuity argument shows that every accumulation point of $\{x^k\}$ is a solution of the VI (K, F). Hence we only have to show that $\{x^k\}$ is bounded. But this easily follows from Corollary **3.6.5**. □

Algorithm **12.2.9** provides a broad framework within which many different schemes can be realized. These schemes vary with (a) the process to compute $x(\varepsilon_{k+1})$ and (b) the choice of the sequences $\{\rho_k\}$ and $\{\varepsilon_k\}$.

Only a good combination of these two elements can give rise to efficient algorithms. We refer the readers to Section **12.9** for some more detail on this point.

12.3 Proximal Point Methods

The Tikhonov regularization method suffers a potential computational drawback. Namely, when ε_k goes to zero the perturbed problems approach the original problem and thus it may become more and more difficult to solve them, even inexactly. Proximal point methods provide a way of alleviating such difficulty. Roughly speaking, in proximal point methods, similar to Tikhonov regularization, we (approximately) solve a sequence of subproblems, but at step k the perturbing function is given by $c_k(x-x^{k-1})$, for some $c_k > 0$, instead of $\varepsilon_k x$. Intuitively, the benefit of the former perturbation is that if the sequence $\{x^k\}$ converges, the term $c_k(x^k - x^{k-1})$ approaches zero provided that c_k remains bounded; thus c_k does not need to go to zero. As a result, we can maintain the "uniformly strong monotonicity" of the perturbed functions $F + c_k(\mathbf{I} - x^k)$ for all k even when F is just monotone. It turns out that the proximal idea has far reaching applications and leads to developments that are much more than just a modification of the Tikhonov regularization method. These developments are described in this and in the next section. The natural setting for these developments is the framework of "maximal monotone" maps. Although this framework is introduced only at this late stage of the book and probably can be avoided for some of the simpler results, maximal monotone maps provide a powerful setting for the derivation of the advanced results which would otherwise not be easy to establish.

12.3.1 Maximal monotone maps

Set-valued maps were introduced in Subsection **2.1.3**; here we start by giving some additional definitions and results. Recall the definition of a (strongly) monotone set-valued map, first encountered in Exercise **2.9.15**: A set-valued map $\Phi : \mathbb{R}^n \to \mathbb{R}^n$ is (strongly) monotone if there exists a constant $c(>) \geq 0$ such that

$$(x - y)^T(u - v) \geq c\,\|x - y\|^2$$

for all x and y in $\operatorname{dom}\Phi$, and all u in $\Phi(x)$ and v in $\Phi(y)$. We regard that Φ, as all set-valued maps we encounter in this chapter, is defined on the whole space $\mathbb{R}^n$; the set of points x where $\Phi(x)$ is nonempty is the domain $\operatorname{dom}\Phi$

of Φ. (Strong) monotonicity of a set-valued map is basically a property of the map's graph. Sometimes we write $\Phi : D \subseteq \mathbb{R}^n \to R \subseteq \mathbb{R}^n$ to mean that $\operatorname{dom}\Phi = D$ and that $\Phi(x)$ is contained in R for all $x \in \operatorname{dom}\Phi$. A classical example of a monotone set-valued map is the subdifferential of a convex function (see Exercise **12.8.8**). Monotone maps can also be generated from other monotone maps. For example, it is easy to see that if Φ is monotone, then so are every positive multiple of Φ and the following derived maps of Φ: the closure map $\operatorname{cl}\Phi$, where $(\operatorname{cl}\Phi)(x) \equiv \operatorname{cl}\Phi(x)$ for all x, the convex hull map $\operatorname{conv}\Phi$, where $(\operatorname{conv}\Phi)(x) \equiv \operatorname{conv}\Phi(x)$ for all x; and the inverse map Φ^{-1}, where $\Phi^{-1}(x) \equiv \{y : x \in \Phi(y)\}$ for all x. Notice that the domain of Φ^{-1} is the range of Φ and $\operatorname{gph}\Phi$ and $\operatorname{gph}\Phi^{-1}$ are related by

$$(x, y) \in \operatorname{gph}\Phi \;\Leftrightarrow\; (y, x) \in \operatorname{gph}\Phi^{-1}.$$

Based on this relation, it is easy to see that the monotonicity of Φ^{-1} is equivalent to the monotonicity of Φ.

Proposition **12.3.1** below provides a simple technical characterization of monotone maps. To this end, it would be convenient for us to call a single-valued map F *1-co-coercive* if it is co-coercive with constant 1; i.e.,

$$(F(x) - F(y))^T(x - y) \geq \|F(x) - F(y)\|^2, \quad \forall\, x, y \in \operatorname{dom} F.$$

12.3.1 Proposition. Let a set-valued map $\Phi : \mathbb{R}^n \to \mathbb{R}^n$ be given. The following statements are equivalent.

(a) Φ is monotone;

(b) For every positive constant c, $(\mathbf{I} + c\Phi)^{-1}$ is a singled-valued, nonexpansive map, that is

$$\|x - y\| \leq \|(x - y) + c(u - v)\|, \quad \forall\, (x, u), (y, v) \in \operatorname{gph}\Phi.$$

(c) For every positive constant c, $(\mathbf{I}+c\Phi)^{-1}$ is a single-valued, 1-co-coercive map, that is

$$((x-y)+c(u-v))^T(y-x) \geq \|y-x\|^2, \quad \forall\, (x, u), (y, v) \in \operatorname{gph}\Phi.$$

Proof. Suppose Φ is monotone. For all (x, u) and (y, v) belonging to $\operatorname{gph}\Phi$ we have

$$\begin{aligned} \|(x - y) + c(u - v)\|^2 &= \|x - y\|^2 + 2c(u - v)^T(x - y) + c^2\|u - v\|^2 \\ &\geq \|x - y\|^2, \end{aligned}$$

which shows that $(\mathbf{I} + c\Phi)^{-1}$ is single-valued and nonexpansive. The converse is also true. In fact, if $(\mathbf{I} + c\Phi)^{-1}$ is single-valued and nonexpansive,

the above equality and inequality hold, so that we get

$$2\,c\,(\,u-v\,)^T(\,x-y\,)+c^2\,\|\,u-v\,\|^2\;\geq\;0.$$

Dividing by c and passing to the limit $c\downarrow 0$, we obtain the monotonicity of Φ. Consequently (a) and (b) are equivalent.

We next prove that (a) is equivalent to (c). It suffices to observe that

$$(\,x,u\,)\;\in\;\mathrm{gph}\,\Phi\;\Leftrightarrow\;(\,x+cu,x\,)\;\in\;\mathrm{gph}(\mathbf{I}+c\Phi)^{-1}. \tag{12.3.1}$$

From this observation, we deduce, for every positive c:

$$\begin{aligned} T \text{ monotone} \;&\Leftrightarrow\; (\,y-x\,)^T(\,v-u\,)\;\geq\;0,\quad \forall\,(\,x,u\,),(\,y,v\,)\;\in\;\mathrm{gph}\,\Phi\\ &\Leftrightarrow\; [(\,y-x\,)+c\,(\,v-u\,)]^T\,(\,y-x\,)\;\geq\;\|\,y-x\,\|^2,\\ &\qquad\qquad \forall\,(\,x,u\,),(\,y,v\,)\;\in\;\mathrm{gph}\,\Phi\\ &\Leftrightarrow\; (\,\mathbf{I}+c\,\Phi\,)^{-1}\text{ 1-co-coercive,} \end{aligned}$$

thus concluding the proof. □

We introduce the important notion of a "maximal monotone" map. Such is a monotone map that enjoys an additional maximality property.

12.3.2 Definition. A monotone map Φ is *maximal monotone* if no monotone map Ψ exists such that $\mathrm{gph}\,\Phi\subset\mathrm{gph}\,\Psi$. □

If we identify the map Φ with its graph, maximal monotone maps are monotone maps that are maximal with respect to set inclusion. Another rephrasing of maximal monotonicity is the following: A monotone map Φ is maximal monotone if and only if every solution $(y,v)\in\mathbb{R}^n\times\mathbb{R}^n$ of the system of inequalities

$$(\,v-u\,)^T(\,y-x\,)\;\geq\;0,\quad\forall\,(\,x,u\,)\;\in\;\mathrm{gph}\,\Phi,$$

belongs to $\mathrm{gph}\,\Phi$.

Our next task is to give a useful characterization of maximal monotone maps, showing their rich structure. This is the main result of this subsection, which is basically due to Minty and referred to as Minty's Theorem.

12.3.3 Theorem. Let a set-valued map $\Phi:\mathbb{R}^n\to\mathbb{R}^n$ be given. The following statements are equivalent.

(a) Φ is maximal monotone.

(b) Φ is monotone and $\mathrm{ran}(\mathbf{I}+\Phi)=\mathbb{R}^n$.

(c) For any positive c, $(\mathbf{I}+c\Phi)^{-1}$ is nonexpansive and $\mathrm{dom}(\mathbf{I}+c\Phi)^{-1} = \mathbb{R}^n$.

(d) For any positive c, $(\mathbf{I}+c\Phi)^{-1}$ is 1-co-coercive and $\mathrm{dom}(\mathbf{I}+c\Phi)^{-1} = \mathbb{R}^n$.

Proof. The equivalence of (c) and (d) and the implication (c) $\Rightarrow$ (b) are an immediate consequence of Proposition **12.3.1**. Suppose that (b) holds but Φ is not maximal. A monotone map Φ' exists such that $\mathrm{gph}\,\Phi \subset \mathrm{gph}\,\Phi'$. By (b), if u belongs to $\Phi'(x)$, a $(y, v) \in \mathrm{gph}\,\Phi$ exists such that $x+u = y+v$. But (y, v) also belong to $\mathrm{gph}\,\Phi'$ and therefore, again by Proposition **12.3.1** with $c = 1$, we see that $x = y$. Therefore $u = v$ and so $(x, u) \in \mathrm{gph}\,\Phi$ thus showing that Φ is maximal monotone and (b) $\Rightarrow$ (a). The rest of the proof is devoted to the somewhat more complex implication (a) $\Rightarrow$ (c). Again because of Proposition **12.3.1**, it suffices to show that $\mathrm{ran}(\mathbf{I} + c\Phi) = \mathbb{R}^n$ for every positive c.

We need the following intermediate result: For every monotone map Φ, and for every $y \in \mathbb{R}^n$ an $x \in \mathbb{R}^n$ exists such that

$$(\,w + x\,)^T(\,z - x\,) \;\geq\; y^T(\,z - x\,), \qquad \forall\,(\,z, w\,) \;\in\; \mathrm{gph}\,\Phi. \tag{12.3.2}$$

Suppose for a moment that (12.3.2) is true. For any $y \in \mathbb{R}^n$ we can rewrite (12.3.2) as

$$(\,(y - x) - w\,)^T(\,x - z\,) \;\geq\; 0, \qquad \forall\,(\,z, w\,) \;\in\; \mathrm{gph}\,\Phi.$$

By the maximal monotonicity of Φ it follows that $y - x \in \Phi(x)$, which shows that $\mathrm{ran}(\mathbf{I} + \Phi) = \mathbb{R}^n$. Since Φ is monotone if and only if $c\Phi$ is monotone for every positive c, this would complete the proof.

Let us show, then, that (12.3.2) holds. We may assume, without loss of generality, that $y = 0$. For every $(z, w) \in \mathrm{gph}\,\Phi$ we set

$$S(z, w) \;\equiv\; \{\,x \;\in\; \mathbb{R}^n \;:\; (\,w + x\,)^T(\,z - x\,) \;\geq\; 0\,\}.$$

It is clear that $S(z, w)$ is convex and compact. We only need to show that

$$\bigcap_{(z,w)\in\mathrm{gph}\,\Phi} S(z, w) \;\neq\; \emptyset.$$

Since all the sets $S(z, w)$ are compact, this is in turn equivalent to showing that the intersection of every finite family of sets $S(z^i, w^i)$, $i = 1, \ldots, r$, is nonempty, that is

$$\bigcap_{i=1}^{r} S(z^i, w^i) \;\neq\; \emptyset.$$

Set

$$K \;\equiv\; \left\{\,\lambda \;\in\; \mathbb{R}^r \;:\; \lambda \;\geq\; 0,\, \sum_{i=1}^{r} \lambda_i \;=\; 1\,\right\},$$

and define $L : K \times K \to \mathbb{R}$ by

$$L(\lambda, \mu) \equiv \sum_{i=1}^{r} \mu_i \, (\, x(\lambda) + w^i \,)^T (\, x(\lambda) - z^i \,),$$

where $x(\lambda) \equiv \sum_{j=1}^{r} \lambda_j z^j$. The function L is continuously differentiable, convex in λ and linear in μ; therefore, by Corollary **2.2.10** a saddle point $(\bar{\lambda}, \bar{\mu})$ of the triple (L, K, K) exists. In particular, by Theorem **1.4.1** we can write, for every $\mu \in K$,

$$\begin{aligned} L(\bar{\lambda}, \mu) &\leq \max_{\lambda \in K} L(\lambda, \lambda) \\ &= \max_{\lambda \in K} \left[\sum_{i=1}^{r} \sum_{j=1}^{r} \lambda_i \, \lambda_j \, (\, w^i \,)^T (\, z^j - z^i \,) \right] \\ &= \max_{\lambda \in K} \tfrac{1}{2} \left[\sum_{i=1}^{r} \sum_{j=1}^{r} \lambda_i \, \lambda_j \, (\, w^i - w^j \,)^T (\, z^j - z^i \,) \right] \leq 0. \end{aligned}$$

But this shows that, for every $\mu \in K$,

$$\sum_{i=1}^{r} \mu_i \, (\, x(\bar{\lambda}) + w^i \,)^T (\, x(\bar{\lambda}) - z^i \,) \leq 0,$$

so that

$$x(\bar{\lambda}) \in \bigcap_{i=1}^{r} S(z^i, w^i),$$

establishing the finite intersection property of the sets $S(z, w)$. □.

Comparing Theorem **12.3.3** and Proposition **12.3.1**, we see that they are similar; basically, what maximality gives in addition to monotonicity is the surjectivity of $\mathbf{I} + c\Phi$. From the point of view of the solution of certain kind of inclusion, Proposition **12.3.1** tells us that for every $y \in \mathbb{R}^n$ the system

$$y = x + c\,u, \quad u \in \Phi(x)$$

has at most one solution x when Φ is monotone; if in addition Φ is maximal then the same system always has one (and only one) solution x. Thus maximal monotonicity can be viewed as a sufficient condition for the solvability of a special system. However, it should not be overlooked that maximal monotonicity also gives us many additional properties besides solvability of an equation.

The map $(\mathbf{I}+c\Phi)^{-1}$, which plays a central role in Proposition **12.3.1** and Theorem **12.3.3** and in fact throughout the developments herein, is called the *resolvent* of Φ (with constant c) and is denoted by $J_{c\Phi}$. This resolvent is a single-valued, nonexpansive map for a monotone Φ; the domain of $J_{c\Phi}$ is equal to $\mathbb{R}^n$ if Φ is maximal monotone. These facts can be restated in an alternative form that facilitate their use.

12.3.4 Proposition. Let Φ be a monotone map and c a positive constant. Every vector $z \in \mathbb{R}^n$ can be written in at most one way as $x + cu$ where $u \in \Phi(x)$. If Φ is maximal monotone, every vector $z \in \mathbb{R}^n$ can be written in exactly one way as $x + cu$ where $u \in \Phi(x)$.

Proof. Suppose we can write

$$z \;=\; x + c\,u \;=\; y + c\,v,$$

for some pairs (x, u) and (y, v) in $\operatorname{gph}\Phi$. This implies (recall (12.3.1)),

$$x \;=\; J_{c\Phi}(x + cu) \;=\; J_{c\Phi}(z) \;=\; J_{c\Phi}(y + cv) \;=\; y,$$

and consequently $u = v$. The second assertion of the proposition is obvious. □

There exists another important relation between a map Φ and its resolvent. Analogously to a single-valued map, a zero of a set-valued map Φ is a point x such that $0 \in \Phi(x)$. In fact, it turns out that the zeros of a maximal monotone map coincide with the fixed points of its resolvent.

12.3.5 Proposition. Let Φ be a maximal monotone map. For every positive c and $x \in \mathbb{R}^n$, we have $0 \in \Phi(x)$ if and only if $J_{c\Phi}(x) = x$.

Proof. We know that $\operatorname{gph} J_{c\Phi} = \{(x + cu, x) : (x, u) \in \operatorname{gph}\Phi\}$. (We have already used this fact several times.) Therefore

$$0 \;\in\; \Phi(x) \;\Leftrightarrow\; (\,x, 0\,) \;\in\; \operatorname{gph}\Phi \;\Leftrightarrow\; (\,x, x\,) \;\in\; \operatorname{gph} J_{c\Phi}.$$

Since Φ is maximal monotone, its resolvent is single-valued, and this completes the proof. □

Our main interest is in applying the machinery developed so far to the solution of variational inequalities. Let then a VI (K, F) be given. We can rewrite it as the problem of finding a zero of the set-valued map $T \;\equiv\; F + \mathcal{N}(\cdot; K)$, that is the problem of finding a point x such that $0 \in T(x)$. In this context it is interesting to understand when T is maximal monotone and what the resolvent of T is. These issues are analyzed in the next proposition.

12.3.6 Proposition. Let $K \subseteq \mathbb{R}^n$ be nonempty closed and convex and $F : K \to \mathbb{R}^n$ continuous. The following three properties hold for the map $T \equiv F + \mathcal{N}(\cdot; K)$.

(a) $J_{cT}(x) = \mathrm{SOL}(K, F_{c,x})$, where $F_{c,x}(y) \equiv y - x + cF(y)$.

(b) $\mathcal{N}(\cdot; K)$ is maximal monotone.

(c) If F is monotone then T is maximal monotone.

Proof. We have

$$\begin{aligned} y \in J_{cT}(x) = (\mathbf{I} + cT)^{-1}(x) \quad &\Leftrightarrow \quad x \in (\mathbf{I} + cF + c\mathcal{N}(\cdot; K))(y) \\ &\Leftrightarrow \quad 0 \in y - x + cF(y) + \mathcal{N}(y; K); \end{aligned}$$

this establishes (a). To prove (b) take F to be identically zero. We get $T = \mathcal{N}(\cdot; K)$ and, by using (a),

$$J_{cT}(x) = \mathrm{SOL}(K, \mathbf{I} - x) = \Pi_K(x).$$

Recalling the discussion after Theorem **1.5.5**, we know that the projector Π_K is defined everywhere and 1-co-coercive, Minty's Theorem then yields that $\mathcal{N}(\cdot; K)$ is maximal monotone. Assume that F is monotone on K. By (b) it is immediate to show that T, being the sum of two monotone maps, is also monotone. By (a), the resolvent $J_{cT}(x)$ is equal to the solution set of the VI $(K, F_{c,x})$. Since the map $F_{c,x}$ is clearly strongly monotone, it follows that the latter solution set is a singleton. Again by Minty's Theorem (c) follows. □

Part (a) of Proposition **12.3.6** provides the basis for the application of the proximal point method presented in the next subsection to the solution of a monotone VI. It should be noted that a map F defined and monotone on K does not imply that F is maximal monotone, even if the sum of F and the normal cone $\mathcal{N}(\cdot; K)$ is actually maximal monotone. A sufficient condition for a single-valued monotone map to be maximal monotone is that its domain is the entire space $\mathbb{R}^n$; see Exercise **12.8.11**.

12.3.2 The proximal point algorithm

We introduce an iterative algorithm for the solution of the inclusion

$$0 \in T(x),$$

where T is a maximal monotone operator. The key to the algorithm is Proposition **12.3.5** which shows that the zeros of T are precisely the fixed

points of its resolvent. Since the latter resolvent is nonexpansive, by Minty's Theorem, it is natural to consider the recursion $x^{k+1} \equiv J_{cT}(x^k)$ hoping it will converge to a zero of T. Actually this turns out to be true provided that T has a zero. However our goal is more ambitious and we want to analyze a more flexible algorithm. First of all we want to allow the inexact calculation of the resolvent. This is useful because invariably every evaluation of the resolvent involves solving a nontrivial subproblem. For instance, in the case where $T \equiv F + \mathcal{N}(\cdot; K)$, computing $J_{cT}(x)$ amounts to solving the VI $(K, \mathbf{I} - x + cF)$. Secondly we want to allow the coefficient c to change from one iteration to the next. Finally we also want to consider an iteration of the following form:

$$x^{k+1} \;=\; (\,1 - \rho_k\,)\, x^k + \rho_k\, J_{c_k T}(x^k),$$

where $\rho_k \in (0, 2)$ is a relaxation factor. The case $\rho_k \in (0, 1)$ is usually referred to as *under-relaxation* and the case $\rho_k \in (1, 2)$ *over-relaxation*.

The following theorem is the main result of the subsection and fully describes the convergence of the proximal point method for finding the zero of a maximal monotone set-valued map.

12.3.7 Theorem. Let $T : \mathbb{R}^n \to \mathbb{R}^n$ be a maximal monotone map and let $x^0 \in \mathbb{R}^n$ be given. Define the sequence $\{x^k\}$ by setting

$$x^{k+1} \;\equiv\; (\,1 - \rho_k\,)\, x^k + \rho_k\, w^k,$$

where, for every k,

$$\| \, w^k - J_{c_k T}(x^k) \, \| \;\leq\; \varepsilon_k$$

and $\{\varepsilon_k\} \subset [0, \infty)$ satisfies $E \equiv \sum_{k=1}^{\infty} \varepsilon_k < \infty$, $\{\rho_k\} \subset [R_m, R_M]$, where $0 < R_m \leq R_M < 2$, and $\{c_k\} \subset (C_m, \infty)$, where $C_m > 0$. If T has a zero the sequence $\{x^k\}$ converges to a zero of T.

Proof. For the sake of notational simplicity we introduce a new map:

$$Q^k \;\equiv\; \mathbf{I} - J_{c_k T}.$$

Since $J_{c_k T}$ is 1-co-coercive it is simple to verify that Q^k is also 1-co-coercive (the reader is asked to prove this fact in Exercise **12.8.9**). Furthermore, any zero of T, being a fixed point of its resolvent, is also a zero of Q^k. For all k we denote by $\bar{x}^k$ the point that would be obtained by the proximal point iteration were the resolvent computed exactly; that is,

$$\bar{x}^{k+1} \;\equiv\; (\,1 - \rho_k\,)\, x^k + \rho_k\, J_{c_k T}(x^k).$$

For every zero x^* of T, we can write:

$$\begin{aligned}
\|\bar{x}^{k+1} - x^*\|^2 &= \|x^k - \rho_k Q^k(x^k) - x^*\|^2 \\
&= \|x^k - x^*\|^2 - 2\rho_k (x^k - x^*)^T Q^k(x^k) + \rho_k^2 \|Q^k(x^k)\|^2 \\
&= \|x^k - x^*\|^2 - 2\rho_k (x^k - x^*)^T (Q^k(x^k) - Q^k(x^*)) \\
&\quad + \rho_k^2 \|Q^k(x^k)\|^2 \\
&\leq \|x^k - x^*\|^2 - \rho_k(2 - \rho_k)\|Q^k(x^k)\|^2 \\
&\leq \|x^k - x^*\|^2 - R_m(2 - R_M)\|Q^k(x^k)\|^2 \\
&\leq \|x^k - x^*\|^2.
\end{aligned}$$

Since $\|x^k - \bar{x}^k\| \leq \rho_k \varepsilon_k$, we get

$$\|x^{k+1} - x^*\| \leq \|\bar{x}^{k+1} - x^*\| + \|x^{k+1} - \bar{x}^{k+1}\| \leq \|x^k - x^*\| + \rho_k \varepsilon_k.$$

From this we get

$$\|x^{k+1} - x^*\| \leq \|x^0 - x^*\| + \sum_{i=0}^{k} \rho_i \varepsilon_i \leq \|x^0 - x^*\| + 2E,$$

so that the sequence $\{x^k\}$ is bounded. We can also write:

$$\begin{aligned}
\|x^{k+1} - x^*\|^2 &= \|\bar{x}^{k+1} - x^* + (x^{k+1} - \bar{x}^{k+1})\|^2 \\
&= \|\bar{x}^{k+1} - x^*\|^2 + 2(\bar{x}^{k+1} - x^*)^T (x^{k+1} - \bar{x}^{k+1}) \\
&\quad + \|x^{k+1} - \bar{x}^{k+1}\|^2 \\
&\leq \|\bar{x}^{k+1} - x^*\|^2 + 2\|\bar{x}^{k+1} - x^*\| \|x^{k+1} - \bar{x}^{k+1}\| \\
&\quad + \|x^{k+1} - \bar{x}^{k+1}\|^2 \\
&\leq \|x^k - x^*\|^2 - R_m(2 - R_M)\|Q^k(x^k)\|^2 \\
&\quad + 2\rho_k \varepsilon_k (\|x^0 - x^*\| + 2E) + \rho_k^2 \varepsilon_k^2.
\end{aligned}$$

Letting $E_2 \equiv \sum_{k=0}^{\infty} \varepsilon_k^2 < \infty$, we have for every k

$$\begin{aligned}
\|x^{k+1} - x^*\|^2 &\leq \|x^0 - x^*\|^2 + 4E(\|x^0 - x^*\| + 2E) \\
&\quad + 4E_2 - R_m(2 - R_M) \sum_{i=0}^{k} \|Q^k(x^k)\|^2.
\end{aligned}$$

Passing to the limit $k \to \infty$, we deduce

$$\sum_{i=0}^{\infty} \| Q^k(x^k) \|^2 < \infty \quad \Rightarrow \quad \lim_{k\to\infty} Q^k(x^k) = 0.$$

By Proposition **12.3.4**, for every k there exists a unique pair (y^k, v^k) in $\operatorname{gph} T$ such that $x^k = y^k + c_k v^k$. Then $J_{c_k T}(x^k) = y^k$ so that $Q^k(x^k) \to 0$ implies $(x^k - y^k) \to 0$. Since c_k is bounded away from zero, it follows that $c_k^{-1} Q^k(x^k) = v^k \to 0$. Since $\{x^k\}$ is bounded it has at least a limit point. Let x^∞ be such a limit point and assume that the subsequence $\{x^k : k \in \kappa\}$ converges to x^∞. It follows that $\{y^k : k \in \kappa\}$ also converges to x^∞. For every (y, v) in $\operatorname{gph} T$ we have, by the monotonicity of T,

$$(y - y^k)^T (v - v^k) \geq 0.$$

Passing to the limit $k(\in \kappa) \to \infty$ we get, $(y - x^\infty)^T (v - 0) \geq 0$. Since T is maximal monotone this implies $(x^\infty, 0) \in \operatorname{gph} T$, that is $0 \in T(x^\infty)$.

To complete the proof it remains to show that x^∞ is the unique limit point of $\{x^k\}$. We have just seen that for any zero of T and therefore, in particular, for x^∞, we can write

$$\begin{aligned} \| x^{k+1} - x^\infty \|^2 \quad &\leq \quad \| x^k - x^\infty \|^2 - R_m(2 - R_M) \| Q^k(x^k) \|^2 \\ &\quad + 2\rho_k \varepsilon_k (\| x^0 - x^\infty \| + 2E) + \rho_k^2 \varepsilon_k^2. \end{aligned}$$

Since

$$\sum_{k=0}^{\infty} \left(-R_m(2 - R_M) \| Q^k(x^k) \|^2 + 2\rho_k \varepsilon_k (\| x^0 - x^\infty \| + 2E) + \rho_k^2 \varepsilon_k^2 \right) < \infty,$$

this shows that $\|x^k - x^\infty\|$ converges. Since $\{\|x^k - x^\infty\| : k \in \kappa\} \to 0$, the whole sequence $\{x^k\}$ must converge to x^∞. □

The possibility of an inaccurate evaluation of the resolvent is of obvious importance, as is the use of a varying c_k. Here we give a hint to why considering over/under-relaxations may also be of practical relevance. Suppose that T is strongly monotone with constant m and Lipschitz continuous, and therefore single-valued, with constant L (which can be chosen to satisfy $L > m$). By the strong monotonicity assumption we have (see Exercise **12.8.13**), that the resolvent of J_{cT} of T is a contraction with constant $(1 + cm)^{-1}$. Hence if x^* is the unique solution of the equation $0 = T(x)$, one can write

$$\| \bar{x}^{k+1} - x^* \| \leq (1 + c_k m)^{-1} \| x^k - x^* \|, \tag{12.3.3}$$

where $\bar{x}^{k+1} \equiv J_{c_k T}(x^k)$, that is $\bar{x}^{k+1}$ corresponds to $\rho_k = 1$, so that no over/under-relaxation is used. If we use over/under-relaxation we have, instead, by using the identity $x^k = \bar{x}^{k+1} + c_k T(\bar{x}^{k+1})$,

$$x^{k+1} \;=\; (1-\rho_k)x^k + \rho_k \bar{x}^{k+1} \;=\; \bar{x}^{k+1} + c_k(1-\rho_k)T(\bar{x}^{k+1})$$

so that, assuming $\rho_k > 1$,

$$\| x^{k+1} - x^* \|^2 \;\leq\; \left[\, 1 + c_k^2 L^2 (\rho_k - 1)^2 - 2mc_k(\rho_k - 1)) \,\right] \| \bar{x}^{k+1} - x^* \|^2.$$

For $\rho_k^m = 1 + m/(c_k L^2)$ the coefficient on the right-hand side reaches its minimum; furthermore $\rho_k^m < 2$ if $c_k > m/(L^2)$. In this situation, we can then write

$$\| x^{k+1} - x^* \| \;\leq\; \left(1 - \frac{m^2}{L^2}\right)^{1/2} (1 + c_k m)^{-1} \| x^k - x^* \|,$$

which compares favorably to (12.3.3) because of the first multiplicative factor that is less than 1.

Theorem **12.3.7** will be useful for the study of splitting methods in the next section. For now, we specialize and refine the theorem in the case $T \equiv F + \mathcal{N}(\cdot; K)$, that is derived from the monotone VI (K, F). We present the complete proximal point algorithm for solving the latter problem based on the map T.

Generalized Proximal Point Algorithm (GPPA)

12.3.8 Algorithm.

Data: $x^0 \in \mathbb{R}^n$, $c_0 > 0$, $\varepsilon_0 \geq 0$, and $\rho_0 \in (0, 2)$.

Step 1: Set $k = 0$.

Step 2: If x^k is a solution of VI (K, F), stop.

Step 3: Find w^k such that $\|w^k - J_{c_k T}(x^k)\| \leq \varepsilon_k$.

Step 4: Set $x^{k+1} \equiv (1-\rho_k)x^k + \rho_k w^k$ and select c_{k+1}, ε_{k+1}, and ρ_{k+1}. Set $k \leftarrow k+1$ and go to Step 1.

The core step in this algorithm is the calculation of w^k at Step 3. As we have seen, if $\varepsilon_k = 0$ this amounts to the exact solution of the strongly monotone VI (K, F^k), where

$$F^k(x) \;\equiv\; x - x^k + c_k\, F(x),$$

while, if $\varepsilon_k > 0$, w^k can be computed as follows. Note that $J_{c_k T}(x^k)$ is the unique solution of the VI (K, F^k). Hence, w^k is an inexact solution of the latter VI satisfying

$$\operatorname{dist}(w^k, \operatorname{SOL}(K, F^k)) \leq \varepsilon_k.$$

Assume that F is Lipschitz continuous with constant $L > 0$. Since F^k is strongly monotone with constant $1 + c_k$ and Lipschitz continuous with constant $1 + L$, Proposition **6.3.1** implies

$$\operatorname{dist}(w^k, \operatorname{SOL}(K, F^k)) \leq \frac{2 + L}{1 + c_k} \| (\mathbf{F}^k)_K^{\mathrm{nat}}(w^k) \|,$$

where $(\mathbf{F}^k)_K^{\mathrm{nat}}$ is the natural map associated with the VI (K, F^k). Consequently, the computation of w^k can be accomplished by obtaining an inexact solution of the latter VI satisfying the residual rule

$$\| (\mathbf{F}_k)_K^{\mathrm{nat}}(w^k) \| \leq \frac{1 + c_k}{2 + L} \varepsilon_k.$$

If F is not Lipschitz continuous, we can use the results of Subsection **6.3.1** to obtain alternative inexact residual rules that w^k has to satisfy. For example, we can stipulate that w^k belongs to K and apply Proposition **6.3.7**.

As the following theorem shows, the constant c_k^{-1} does not need to tend to zero. Thus F^k can remain uniformly well behaved throughout the iterations. Unlike the Tikhonov regularization, the small price we pay for this is that we can no longer deduce that the solution to which the sequence $\{x^k\}$ converges to is the least-norm solution of the VI (K, F); cf. Theorem **12.2.3**.

12.3.9 Theorem. Let $K \subseteq \mathbb{R}^n$ be closed convex and $F : K \to \mathbb{R}^n$ be continuous and monotone. Let $x^0 \in \mathbb{R}^n$ be given. Let $\{\varepsilon_k\} \subset [0, \infty)$ satisfy $E \equiv \sum_{k=1}^{\infty} \varepsilon_k < \infty$, $\{c_k\} \subset (C_m, \infty)$, where $C_m > 0$, and $\{\rho_k\} \subset [R_m, R_M]$, where $0 < R_m \leq R_M < 2$. Let $\{x^k\}$ be generated by Algorithm **12.3.8**.

(a) If VI (K, F) has a solution, the sequence $\{x^k\}$ converges to a point in $\operatorname{SOL}(K, F)$.

(b) If VI (K, F) has no solutions the sequence $\{x^k\}$ is unbounded.

Proof. By Theorem **12.3.7**, we only need to show (b). Suppose that the solution set of the VI is empty and assume for contradiction that the sequence $\{x^k\}$ is bounded, so that a constant M exists such that $\|x^k\| \leq \frac{1}{2}M$ for every k. Assume, without loss of generality, that $\varepsilon_k < (1/2)M$ for

every k. The sequence $\{x^k\}$ can also be seen as produced by the application of Algorithm **12.3.8** to the VI $(K \cap \mathrm{cl}\, \mathrm{I\!B}(0, M), F)$. But this VI, being defined on a compact set, has a solution. Therefore, by (a) and the boundedness of $\{x^k\}$, the sequence $\{x^k\}$ converges to a solution x^* of the VI $(K \cap \mathrm{cl}\, \mathrm{I\!B}(0, M), F)$ such that $\|x^*\| \leq \frac{1}{2}M$. By Proposition **2.2.8** x^* is also a solution of the VI (K, F). This is a contradiction. □

We remark that a result similar to part (b) of the above theorem can be obtained also in the more general setting of Theorem **12.3.7**. The proof of such an extended result is, however, much more technical. Since we are mainly interested in the application of the proximal point algorithm to VIs, we do not discuss further the generalized result.

12.4 Splitting Methods

The fundamental idea underlying splitting methods is that in many situations we want to find a zero of a monotone map T that is given by the sum of two other maps: $T \equiv A + B$ and whose resolvent $(\mathbf{I} + cT)^{-1}$ is difficult to evaluate. However, if the resolvents of A and B are easier to evaluate, one may wish to develop solution procedures that employ these resolvents only (or maybe just the resolvent of A or B) to solve the inclusion $0 \in T(x)$. Many splitting procedures exist, in this section we concentrate on two methods only, the Douglas-Rachford splitting procedure and the forward-backward splitting procedure. These two are certainly the most interesting among the splitting procedures and their application to the solution of VIs is illustrated in the next section.

12.4.1 Douglas-Rachford splitting method

Let T be a set-valued operator, and suppose that $T = A + B$, where A and B are set-valued and maximal monotone (T itself is not required to be maximal monotone even though it is obviously monotone). A possible way of solving the inclusion $0 \in T(x)$ is to apply the proximal point algorithm studied in the previous section. The main computational burden in this approach is obviously the evaluation of the resolvent $J_{cT}(x)$. In this context, splitting methods try to alleviate this burden by evaluating J_{cA} and J_{cB} only. If these two resolvents are easier to evaluate than the original one, then this may be a viable approach.

For a fixed $c > 0$, we say that the sequence $\{x^k\}$ obeys the (exact) Douglas-Rachford recursion, if

$$x^{k+1} = J_{cA}((2J_{cB} - \mathbf{I})(x^k)) + (\mathbf{I} - J_{cB})(x^k) \equiv L_{c,A,B}(x^k). \quad (12.4.1)$$

Note that the resolvent J_{cT} is not involved in the computation of x^{k+1}. Our aim is to show that if the inclusion $0 \in T(x)$ has a solution, then the sequence $\{J_{cB}(x^k)\}$ converges to a zero of T. To do so we first show that the sequence $\{x^k\}$ produced by (12.4.1) can also be seen as produced by the proximal point algorithm applied to an (essentially set-valued) map that has a strong relation to the zeros of T; then we apply Theorem **12.3.7**.

We need some preliminary considerations. Suppose that the sequence $\{x^k\}$ obeys the Douglas-Rachford recursion. For every k, let (y^k, b^k) the unique element in $\operatorname{gph} B$ such that $y^k + cb^k = x^k$ (use Proposition **12.3.4**). It is immediate to see that

$$\begin{aligned}(\,\mathbf{I} - J_{cB}\,)(x^k) &= y^k + cb^k - y^k \;=\; cb^k\\ (\,2J_{cB} - \mathbf{I}\,)(x^k) &= 2y^k - (y^k + cb^k) \;=\; y^k - cb^k.\end{aligned}$$

Similarly, if (z^k, a^k) belongs to $\operatorname{gph} A$, we have

$$J_{cA}(z^k + ca^k) \;=\; z^k.$$

Using these three identities, we can calculate x^{k+1} from $x^k = y^k + cb^k$ in the following way:

(a) Find the unique $(z^{k+1}, a^{k+1}) \in \operatorname{gph} A$ such that $z^{k+1} + ca^{k+1} = y^k - cb^k$.

(b) Set $x^{k+1} \equiv z^{k+1} + cb^k$.

(c) Find the unique $(y^{k+1}, b^{k+1}) \in \operatorname{gph} B$ such that $x^{k+1} = y^{k+1} + cb^{k+1}$. (This step is needed for the next iteration.)

Consider the mapping

$$M_{c,A,B} \;\equiv\; (L_{c,A,B}(x^k))^{-1} - \mathbf{I}.$$

By (a)–(c) above it is easy to see that

$$\begin{aligned}&\operatorname{gph} L_{c,A,B} \;=\\ &\{\,(u + cb, v + cb) \,:\, (u,b) \,\in\, \operatorname{gph} B,\; (v,a) \,\in\, \operatorname{gph} A,\; v + ca \,=\, u - cb\,\},\end{aligned}$$

from which we get

$$\begin{aligned}&\operatorname{gph} M_{c,A,B} \;=\\ &\{\,(v + cb, u - v) \,:\, (u,b) \,\in\, \operatorname{gph} B,\; (v,a) \,\in\, \operatorname{gph} A,\; v + ca \,=\, u - cb\,\}.\end{aligned}$$

The following proposition establishes some crucial properties of $M_{c,A,B}$.

12.4.1 Proposition. Suppose that A and B are maximal monotone maps and $T = A + B$.

(a) The map $M_{c,A,B}$ is maximal monotone.

(b) Let S be the set of zeros of the map $M_{c,A,B}$; then

$$\begin{aligned} S &= \{\, u + cb \,:\, b \in B(u),\ -b \in A(u) \} \\ &\subseteq \{\, u + cb \,:\, 0 \in T(u),\ b \in B(u) \,\} \end{aligned}$$

(c) If $x^{k+1} = L_{c,A,B}(x^k)$, then x^{k+1} can also be obtained as an application of Algorithm **12.3.8** to the map $M_{c,A,B}$ with $c_k = 1$, $\rho_k = 1$ and exact evaluation of the resolvent (that is $\varepsilon_k = 0$).

Proof. We first show that $M_{c,A,B}$ is monotone. To this end, let (u^i, b^i), and (v^i, a^i) for $i = 1, 2$ be such that (u^1, b^1) and (u^2, b^2) in $\operatorname{gph} B$, (v^1, a^1) and (v^2, a^2) in $\operatorname{gph} A$, $v^1 + ca^1 = u^1 - cb^1$, and $v^2 + ca^2 = u^2 - cb^2$. We have

$$a^1 \,=\, c^{-1}(u^1 - v^1) - b^1, \qquad a^2 \,=\, c^{-1}(u^2 - v^2) - b^2,$$

and

$$\begin{aligned} &\big((v^2 + cb^2) - (v^1 + cb^1)\big)^{\,T} \big((u^2 - v^2) - (u^1 - v^1)\big) \\ &\quad = \; c(v^2 + cb^2)^{\,T} \big(c^{-1}(u^2 - v^2) - b^2 - c^{-1}(u^1 - v^1) + b^1\big) \\ &\qquad\; -c(v^1 + cb^1)^{\,T} \big(c^{-1}(u^2 - v^2) - b^2 - c^{-1}(u^1 - v^1) + b^1\big) \\ &\qquad\; +c\,\big((v^2 + cb^2) - (v^1 + cb^1)\big)^{\,T} \big(b^2 - b^1\big) \\ &\quad = \; c(v^2 - v^1)^{\,T} \big(c^{-1}(u^2 - v^2) - b^2 - c^{-1}(u^1 - v^1) + b^1\big) \\ &\qquad\; +c^2(b^2 - b^1)^{\,T} \big(c^{-1}(u^2 - v^2) - b^2 - c^{-1}(u^1 - v^1) + b^1\big) \\ &\qquad\; +c(v^2 - v^1)^{\,T}(b^2 - b^1) + c^2(b^2 - b^1)^{\,T}(b^2 - b^1) \\ &\quad = \; c(v^2 - v^1)^{\,T}(a^2 - a^1) + c(b^2 - b^1)^{\,T}(u^2 - u^1) \\ &\qquad\; -c(b^2 - b^1)^{\,T}(v^2 - v^1) - c^2(b^2 - b^1)^{\,T}(b^2 - b^1) \\ &\qquad\; +c(v^2 - v^1)^{\,T}(b^2 - b^1) + c^2(b^2 - b^1)^{\,T}(b^2 - b^1) \\ &\quad = \; c(v^2 - v^1)^{\,T}(a^2 - a^1) + c(b^2 - b^1)^{\,T}(u^2 - u^1) \;\geq\; 0, \end{aligned}$$

where the last inequality follows from the monotonicity of A and B. The above chain of equalities shows that $M_{c,A,B}$ is monotone. Since

$$L_{c,A,B} \,=\, J_{cA} \circ (\, 2J_{cB} - \mathbf{I}\,) + (\, \mathbf{I} - J_{cB}\,)$$

and $\operatorname{dom} J_{cA} = \operatorname{dom} J_{cB} = \mathbb{R}^n$, we have

$$\operatorname{ran}(\mathbf{I} + M_{c,A,B}) \,=\, \operatorname{dom}(\mathbf{I} + M_{c,A,B})^{-1} \,=\, \operatorname{dom} L_{c,A,B} \,=\, \mathbb{R}^n.$$

The maximal monotonicity of $M_{c,A,B}$ then follows from Theorem **12.3.3**.

To show (b), let s be a point in S. By the explicit expression for the graph of $M_{c,A,B}$ given just before this proposition, we see that there must exist u, b, v, a in $\mathbb{R}^n$ such that $v + cb = s$, $u - v = 0$, $(u, b) \in \operatorname{gph} B$ and $(v, a) \in \operatorname{gph} A$, so that

$$a = -b, \quad u + cb = s, \quad (u, b) \in \operatorname{gph} B, \quad (u, -b) \in \operatorname{gph} A.$$

From these relations, $s \in \{u + cb : b \in B(u), -b \in A(u)\}$ follows. Conversely, if $s \in \{u + cb : b \in B(u), -b \in A(u)\}$, then $s = u + cb$, with $b \in B(u)$ and $-b \in A(u)$. Therefore $(s, 0) \in \operatorname{gph} M_{c,A,B}$. To show the inclusion claimed in (b) just note that $b \in B(u)$ and $-b \in A(u)$ imply that u is a zero of T.

Finally, $x^{k+1} = L_{c,A,B}(x^k)$ can be rewritten as

$$x^{k+1} = (\mathbf{I} + M_{c,A,B})^{-1}(x^k),$$

which proves (c). □

The upshot of the previous proposition is clear: a sequence $\{x^k\}$ obeying the Douglas-Rachford recursion can also be viewed as generated by the proximal point method applied to $M_{c,A,B}$. If the inclusion $0 \in M_{c,A,B}(x)$ has a solution, then $\{x^k\}$ converges to a zero x^* of $M_{c,A,B}$ and $J_{cB}(x^*)$ is a zero of T. In turn, by part (b) of the proposition, it is clear that $M_{c,A,B}$ has a zero if and only if T has a zero.

Douglas-Rachford Splitting Algorithm (DRSA)

12.4.2 Algorithm.

Data: $x^0 \in \mathbb{R}^n$ and $T = A + B$.

Step 0: Set $k = 0$.

Step 1: Calculate $u^k \equiv J_{cB}(x^k)$.

Step 2: If u^k solves the inclusion $0 \in T(u^k)$, stop.

Step 3: Calculate $v^k \equiv J_{cA}(2u^k - x^k)$.

Step 4: Set $x^{k+1} \equiv v^k + x^k - u^k$ and $k \leftarrow k + 1$; go to Step 1.

Note that in this algorithm nowhere is the resolvent of T used. Instead, the resolvents of A and B are used at each iteration. In view of the above discussion and of the continuity of J_{cB}, the following theorem needs no proof.

12.4.3 Theorem. Suppose that A and B are maximal monotone and that the inclusion $0 \in (A+B)(x)$ has a solution. The Douglas-Rachford Splitting Algorithm produces a sequence $\{x^k\}$ converging to a zero x^* of $M_{c,A,B}$ and a sequence $\{u^k\}$ converging to a zero u^* of $A + B$. □

As we already remarked in Proposition **12.4.1**, Algorithm **12.4.2** corresponds to the application of the proximal point Algorithm **12.3.8** to $M_{c,A,B}$, with $c_k = 1$, $\rho_k = 1$ and $\varepsilon_k = 0$ for all k. In view of Proposition **12.4.1** (b), a generalization of the Douglas-Rachford method can be easily defined by taking different values for these parameters, in accordance with Theorem **12.3.9**. Before following this idea we note however that, while varying the value of ρ_k and ε_k turns out to be a sensible thing to do, it does not seem practical to take a c_k different from 1. In fact, consider trying to compute $(\mathbf{I} + c_1 M_{c,A,B})^{-1}(x)$. We have

$$\begin{aligned}\operatorname{gph}(\mathbf{I} + c_1 M_{c,A,B})^{-1} = \{ \, (\, (1-c_1)v + c_1 u + cb, v + cb \,) \, : \\ (u,b) \in \operatorname{gph} B, \ (v,a) \in \operatorname{gph} A, \ v + ca = u - cb \, \}.\end{aligned}$$

Therefore, to compute $(\mathbf{I} + c_1 M_{c,A,B})^{-1}(x)$, one must find $(u,b) \in \operatorname{gph} B$ and $(v,a) \in \operatorname{gph} A$ such that

$$x = (1-c_1)v + c_1 u + cb, \qquad v + ca = u - cb$$

which, unless $c_1 = 1$, does not appear to be easier than computing the resolvent of T. When $c_1 = 1$, instead, the above formulas reduce to

$$x = u + cb, \qquad v + ca = u - cb,$$

so that we see that A and B are used separately and the problem has been decomposed. Therefore, keeping $c_k = 1$ in applying the proximal method to $M_{c,A,B}$ seems essential if one wants to achieve some kind of decomposition. However, the option of taking $\rho_k \neq 1$ and $\varepsilon_k \neq 0$ is certainly feasible and gives rise to over/under-relaxed versions and inexact versions of the Douglas-Rachford splitting method. We summarize these extensions in the algorithm below.

Inexact Douglas-Rachford Splitting Algorithm (IDRSA)

12.4.4 Algorithm.

Data: $x^0 \in \mathbb{R}^n$ and $T = A + B$.

Step 0: Set $k = 0$.

Step 1: Calculate u^k such that $\|u^k - J_{cB}(x^k)\| \le \varepsilon'_k$.

Step 2: If u^k solves the inclusion $0 \in T(u^k)$, stop.

Step 3: Calculate v^k such that $\|v^k - J_{cA}(2u^k - x^k)\| \le \varepsilon''_k$.

Step 4: Set $x^{k+1} \equiv \rho_k(v^k + x^k - u^k) + (1 - \rho_k)x^k$, and $k \leftarrow k + 1$; go to Step 1.

The convergence properties of this algorithm follow readily from Theorem **12.3.7** and the above considerations.

12.4.5 Theorem. Suppose that A and B are maximal monotone and that the inclusion $0 \in (A + B)(x)$ has a solution. Suppose that in the Inexact Douglas-Rachford Splitting Algorithm, $\varepsilon'_k > 0$ and $\varepsilon''_k > 0$ for every k,

$$\sum_{k=0}^{\infty} \varepsilon'_k < \infty, \quad \sum_{k=0}^{\infty} \varepsilon''_k < \infty,$$

and

$$0 < \inf_k \rho_k \le \sup_k \rho_k < 2.$$

The sequence $\{x^k\}$ converges to a zero x^* of $M_{c,A,B}$, and $\{u^k\}$ converges to a zero $u^* = J_{cB}(x^*)$ of $A + B$.

Proof. The only thing to show is that for every k

$$\| (v^k + x^k - u^k) - J_{M_{c,A,B}}(x^k) \| \le \varepsilon_k$$

for some nonnegative ε_k such that

$$\sum_{k=0}^{\infty} \varepsilon_k < \infty.$$

Since $\|u^k - J_{cB}(x^k)\| \le \varepsilon'_k$ we also have

$$\| (2u^k - x^k) - (2J_{cB}(x^k) - x^k) \| \le 2\varepsilon'_k.$$

Since the resolvent is a nonexpansive map, the above implies

$$\| J_{cA}(2u^k - x^k) - J_{cA}(2J_{cB}(x^k) - x^k) \| \le 2\varepsilon'_k;$$

therefore

$$\| v^k - J_{cA}(2J_{cB}(x^k) - x^k) \| \le 2\varepsilon'_k + \varepsilon''_k.$$

From this last inequality, we finally have

$$\begin{aligned}
&\| \, (\, v^k + x^k - u^k \,) - J_{M_{c,A,B}}(x^k) \, \| \\
&\quad = \| \, (\, v^k + x^k - u^k \,) - J_{cA}(2J_{cB}(x^k - x^k)) + (\, x^k - J_{cB}(x^k) \,) \, \| \\
&\quad \leq 3\varepsilon_k' + \varepsilon_k''.
\end{aligned}$$

Setting $\varepsilon_k \equiv 3\varepsilon_k' + \varepsilon_k''$ then concludes the proof. □

12.4.2 The forward-backward splitting method

In this subsection we consider another interesting splitting method, the so-called forward-backward splitting method. Let $T : \mathbb{R}^n \to \mathbb{R}^n$ be a set-valued map with a nonempty, closed convex domain. Suppose that

$$T \, = \, A + B,$$

where A is maximal monotone and B is a single-valued function from $\mathrm{dom}\, A$ into $\mathbb{R}^n$. We want to solve the inclusion $0 \in T(x)$ via the following iteration: with $x^k \in \mathrm{dom}\, A$,

$$x^{k+1} \, = \, J_{c_k A}((\mathbf{I} - c_k B)(x^k)). \tag{12.4.2}$$

The iteration can be motivated as follows. Note that $0 \in T(x)$ if and only if

$$\begin{aligned}
-c\, B(x) \, \in \, c\, A(x) \quad &\Leftrightarrow \quad (\, \mathbf{I} - c\, B \,)(x) \, \in \, (\, \mathbf{I} + c\, A \,)(x) \\
&\Leftrightarrow \quad x \, = \, J_{cA}((\mathbf{I} - cB)(x)).
\end{aligned}$$

Thus (12.4.2) is simply a fixed-point iteration of the latter equation with a varying multiplier c_k at each iteration.

Note that $J_{c_k A}$ is a single-valued function from $\mathbb{R}^n$ to $\mathrm{dom}\, A$, thus x^{k+1} belongs to $\mathrm{dom}\, A$ and the iteration is well defined. As in the Douglas-Rachford splitting, also in this case we never use T explicitly, but rather alternate between a very easy "forward" step involving only B and a "backward" step involving the resolvent of A.

At first sight, the iteration (12.4.2) may appear somewhat restrictive. For instance, suppose we consider the following seemingly more general iterative algorithm. Let $\tilde{A}$ be a maximal, strongly monotone set-valued map and let $\tilde{B}$ be a single-valued map such that $T = \tilde{A} + \tilde{B}$. Consider the iteration: given x^k, let x^{k+1} be the unique vector satisfying:

$$0 \, \in \, \tilde{A}(x^{k+1}) + \tilde{B}(x^k). \tag{12.4.3}$$

By Proposition **12.4.14**, x^{k+1} is well defined. The iteration (12.4.2) can be seen to be a special realization of (12.4.3) with $\tilde{A} \equiv \mathbf{I} + cA$ and $\tilde{B} \equiv cB - \mathbf{I}$.

Notice that if A is monotone, then $\tilde{A}$ is strongly monotone. The converse is also true too; in other words, (12.4.2) includes (12.4.3) as a special case. Indeed, for any $\tau > 0$, (12.4.3) is equivalent to

$$0 \in \tau^{-1}\tilde{A}(x^{k+1}) + \tau^{-1}\tilde{B}(x^k).$$

Thus by letting $c \equiv 1$, $A \equiv \tau^{-1}\tilde{A} - \mathbf{I}$, and $B \equiv \tau^{-1}\tilde{B} + \mathbf{I}$, we deduce that the iterations (12.4.3) and (12.4.2) will produce the same sequence of iterates. The reason for scaling the iteration (12.4.3) by τ^{-1} before identifying it as a special case of (12.4.2) is that with $\tilde{A}$ strongly monotone, we can choose τ so that the resulting A is monotone; the latter monotonicity is needed in the setting of the forward-backward splitting method. See Exercise **12.8.19** for an application of this discussion.

The next theorem gives the main convergence result for the forward-backward splitting iteration (12.4.2). As in the Douglas-Rachford splitting, we allow the possibility of inaccurate evaluation of the resolvent of A (but assume exact evaluation of B).

12.4.6 Theorem. Let T be a set-valued map with a nonempty, closed convex domain and at least one zero. Assume that $T = A + B$, where A is maximal monotone and B is a single-valued function from $\operatorname{dom} A$ into $\mathbb{R}^n$. Suppose that x^0 belongs to $\operatorname{dom} A$ and let $\{x^k\}$ be a sequence such that

(a) $\|x^{k+1} - J_{c_k A}(y^k)\| \le \varepsilon_k$, where $y^k \equiv (\mathbf{I} - c_k B)(x^k)$.

(b) $x^{k+1} \in \operatorname{dom} A$;

and suppose that $\sum_{k=0}^{\infty} \varepsilon_k < \infty$. If B is co-coercive with modulus σ, and

$$0 < m \le c_k \le M < 2\sigma, \qquad \forall\, k,$$

then $\{x^k\}$ converges to a point x^* such that $0 \in T(x^*)$.

Proof. We prove first the theorem for the case of exact evaluation of the resolvent of A, that is $\varepsilon_k = 0$ for all k. In this case x^{k+1} is given by (12.4.2). Given x^k, we can calculate x^{k+1} in the following way:

(a) find a $z^k \in A(x^k - c_k(B(x^k) + z^k))$;

(b) set $x^{k+1} = x^k - c_k(B(x^k) + z^k)$.

This can be verified by observing that (a) and (b) give

$$x^k - x^{k+1} - c_k B(x^k) \in c_k A(x^{k+1})$$

so that solving with respect to x^{k+1} we get (12.4.2). Let x^* be a solution of $0 \in T(x)$, so that $0 = z^* + B(x^*)$ for some z^* belonging to $A(x^*)$. By the co-coerciveness of B we can write

$$\begin{aligned} 0 &= (x^k - x^*)^T(B(x^k) - B(x^*)) - (x^k - x^*)^T(B(x^k) - B(x^*)) \\ &\geq \sigma \| B(x^k) - B(x^*) \|^2 - (x^k - x^*)^T(B(x^k) - B(x^*)) \\ &= c_k \| B(x^k) - B(x^*) \|^2 - (x^k - x^*)^T(B(x^k) - B(x^*)) + \Delta_k \end{aligned}$$

where $\Delta_k \equiv (\sigma - c_k)\|B(x^k) - B(x^*)\|^2$. Substituting $B(x^*) = -z^*$ into the above inequality, we obtain

$$\begin{aligned} 0 &\geq (-x^k + x^* + c_k(B(x^k) + z^k))^T(B(x^k) - B(x^*)) \\ &\quad -c_k (z^k - z^*)^T(B(x^k) - B(x^*)) + \Delta_k \\ &= -(x^{k+1})^T(B(x^k) - B(x^*)) - c_k(z^k - z^*)^T(B(x^k) - B(x^*)) + \Delta_k, \end{aligned}$$

where the last equality follows from (b). Similarly, using $z^* \in A(x^*)$ and $z^k \in A(x^{k+1})$ by (a) and (b), and the monotonicity of A, we get

$$\begin{aligned} 0 &= (x^{k+1} - x^*)^T(z^k - z^*) - (x^{k+1} - x^*)^T(z^k - z^*) \\ &\geq -(x^{k+1} - x^*)^T(z^k - z^*). \end{aligned}$$

Summing the last two relations, recalling that $z^* + B(x^*) = 0$, and using (b), we obtain

$$\begin{aligned} 0 &\geq -(x^{k+1} - x^*)^T(B(x^k) + z^k) - c_k(x^k - x^*)^T(B(x^k) - B(x^*)) + \Delta_k \\ &= (x^{k+1} - x^*)^T \left(\frac{(x^{k+1} - x^*) - (x^k - x^*)}{c_k} \right) \\ &\quad -c_k (z^k - z^*)^T(B(x^k) - B(x^*)) + \Delta_k. \end{aligned}$$

In what follows we set, for brevity, $\hat{x}^k \equiv x^k - x^*$, $\hat{x}^{k+1} \equiv x^{k+1} - x^*$, $\hat{z}^k \equiv z^k - z^*$, and $\hat{B}(x^k) \equiv B(x^k) - B(x^*)$. The last relation then, by using the following two identities,

$$\begin{aligned} 2(\hat{x}^{k+1})^T(\hat{x}^{k+1} - \hat{x}^k) &= \| \hat{x}^{k+1} - \hat{x}^k \|^2 + \| \hat{x}^{k+1} \|^2 - \| \hat{x}^k \|^2, \\ \| \hat{x}^{k+1} - \hat{x}^k \|^2 &= c_k^2 \| \hat{B}(x^k) \|^2 + c_k^2 \| \hat{z}^k \|^2 + 2c_k^2 (\hat{z}^k)^T \hat{B}(x^k) \end{aligned}$$

implies

$$0 \geq \| \hat{x}^{k+1} \|^2 - \| \hat{x}^k \|^2 + c_k^2 \| \hat{B}(x^k) \|^2 + c_k^2 \| \hat{z}^k \|^2 + 2c_k \Delta_k.$$

so that, by substituting the definition of Δ_k, we get

$$\begin{aligned} \| \hat{x}^k \|^2 &= \| \hat{x}^{k+1} \|^2 + c_k(2\sigma - c_k) \| \hat{B}(x^k) \|^2 + c_k^2 \| \hat{z}^k \|^2 \\ &\geq \| \hat{x}^{k+1} \|^2 + m(2\sigma - M) \| \hat{B}(x^k) \|^2 + m^2 \| \hat{z}^k \|^2, \end{aligned}$$

where m and $(2\sigma - M)$ are positive numbers by assumption. Substituting back in this expression the definitions of $\hat{x}$, $\hat{z}$ and $\hat{B}$, we get

$$\| x^{k+1} - x^* \|^2 \leq \| x^k - x^* \|^2 - m\,(2\sigma - M) \| B(x^k) - B(x^*) \|^2 - m^2 \| z^k + B(x^*) \|^2. \tag{12.4.4}$$

This shows that $\{x^k\}$ is bounded,

$$\lim_{k\to\infty} B(x^k) = B(x^*) \qquad \text{and} \qquad \lim_{k\to\infty} z^k = -B(x^*). \tag{12.4.5}$$

Since $\{x^k\}$ is bounded it has at least one limit point. Let $\{x^k : k \in \kappa\}$ converge to x^∞. We want to show that $0 \in A(x^\infty) + B(x^\infty)$. Since B is co-coercive, it is therefore continuous, thus, by (12.4.5) we have

$$B(x^\infty) = B(x^*). \tag{12.4.6}$$

We claim that

$$-B(x^*) \in A(x^\infty) \tag{12.4.7}$$

In fact, for any $x \in \operatorname{dom} A$ and any $z \in A(x)$ we have

$$\begin{aligned} 0 &\leq (z - z^{k-1})^T (x - x^k) \\ &= (z - z^{k-1})^T (x - x^\infty) + (z - z^{k-1})^T (x^\infty - x^k). \end{aligned}$$

Since the left-hand side tends to $(z - (-B(x^*)))^T (x - x^\infty)$, by the maximal monotonicity of A, (12.4.7) follows. But (12.4.6) and (12.4.7) together obviously show that x^∞ is a solution of the inclusion $0 \in T(x)$. It remains to show that x^∞ is the only limit point of the sequence $\{x^k\}$. But by (12.4.4), with $x^* = x^\infty$, we see that the entire sequence $\{\|x^k - x^\infty\|\}$ is nonincreasing and therefore convergent. Since the subsequence $\{x^k : k \in \kappa\}$ converges to x^∞, it follows that the entire sequence $\{x^k\}$ converges to x^∞.

We next consider the case of inaccurate evaluation of the resolvent. Let us write $\bar{x}^{k+1} = J_{c_k A}((\mathbf{I} - c_k B)(x^k))$. Inequality (12.4.4) gives

$$\| \bar{x}^{k+1} - x^* \| \leq \| x^k - x^* \|.$$

Since $\|x^{k+1} - \bar{x}^{k+1}\| \leq \varepsilon_k$, we get

$$\| x^{k+1} - x^* \| \leq \| \bar{x}^{k+1} - x^* \| + \| x^{k+1} - \bar{x}^{k+1} \| \leq \| x^k - x^* \| + \varepsilon_k.$$

From this we get

$$\| x^{k+1} - x^* \| \leq \| x^0 - x^* \| + \sum_{i=0}^{k} \varepsilon_i \leq \| x^0 - x^* \| + 2E,$$

where we set $E \equiv \sum_{i=0}^{\infty} \varepsilon_i$. This shows that the sequence $\{x^k\}$ is bounded. We can also write:

$$\begin{aligned}
\| x^{k+1} - x^* \|^2 &= \| \bar{x}^{k+1} - x^* + (x^{k+1} - \bar{x}^{k+1}) \|^2 \\
&= \| \bar{x}^{k+1} - x^* \|^2 + 2(\bar{x}^{k+1} - x^*)^T (x^{k+1} - \bar{x}^{k+1}) \\
&\quad + \| x^{k+1} - \bar{x}^{k+1} \|^2 \\
&\leq \| \bar{x}^{k+1} - x^* \|^2 + 2\| \bar{x}^{k+1} - x^* \| \, \| x^{k+1} - \bar{x}^{k+1} \| \\
&\quad + \| x^{k+1} - \bar{x}^{k+1} \|^2 \\
&\leq \| x^k - x^* \|^2 - m(2\sigma - M)\| B(x^k) - B(x^*) \|^2 \\
&\quad -m^2 \| z^k + B(x^*) \|^2 + 2\varepsilon_k (\| x^0 - x^* \| + 2E) + \varepsilon_k^2.
\end{aligned}$$

If we set $E_2 \equiv \sum_{k=0}^{\infty} \varepsilon_k^2 < \infty$, we can write for every k

$$\| x^{k+1} - x^* \|^2 \leq \| x^0 - x^* \|^2 + 4E(\| x^0 - x^* \| + 2E) + E_2$$
$$-\sum_{k=0}^{k} \left[m(2\sigma - M)\| B(x^k) - B(x^*) \|^2 + m^2 \| z^k + B(x^*) \|^2 \right].$$

This implies (12.4.5) and the proof can be easily completed as in the case of exact evaluation of the resolvent. □

12.4.7 Remark. In the proof of the above theorem, when we showed that (12.4.7) holds, we have actually showed that any maximal monotone map is closed according to Definition **2.1.16**. This fact will be used again later in this section. □

If either A or B is strongly monotone, we can establish a R-linear convergence rate for the sequence $\{x^k\}$ produced by the forward-backward splitting method. This rate result asserts that $\{x^k\}$ converges to x^* at least R-linearly; that is, there exist positive constants c and η with $\eta < 1$ such that $\|x^{k+1} - x^*\| \leq c\eta^k$ for all k sufficiently large. See Subsection **12.6.2** for an improved rate analysis.

12.4.8 Corollary. Assume the same setting of Theorem **12.4.6**. If in addition either A or B are strongly monotone, the sequence $\{x^k\}$ converges at least R-linearly.

Proof. Let x^* be the unique solution of $0 \in T(x)$ and let η_A and η_B be the moduli of (strong) monotonicity of A and B respectively. Note that

either η_A or η_B is positive. Using the same notation as in the proof of Theorem **12.4.6**, we have

$$\begin{aligned}(z^k + B(x^*))^T(x^{k+1} - x^*) &\geq \eta_A \| x^{k+1} - x^* \|^2 \\ (B(x^k) - B(x^*))^T(x^k - x^*) &\geq \eta_B \| x^k - x^* \|^2,\end{aligned}$$

so that

$$\begin{aligned}\| z^k + B(x^*) \| &\geq \eta_A \| x^{k+1} - x^* \| \\ \| B(x^k) - B(x^*) \| &\geq \eta_B \| x^k - x^* \|.\end{aligned}$$

Together with (12.4.4) this yields

$$\begin{aligned}\| x^{k+1} - x^* \|^2 \leq \; & \| x^k - x^* \|^2 - m(2\sigma - M)\eta_B \| x^k - x^* \|^2 \\ & -m^2\eta_A \| x^k - x^* \|^2.\end{aligned}$$

Since either η_A or η_B is positive, this concludes the proof. □

Noting that we can write $T = T + 0$ and that the zero mapping is cocoercive with a modulus that can be taken to be any positive number, we get the corollary below on the proximal point method with exact calculation of the resolvent and without relaxation. No proof of the corollary is necessary.

12.4.9 Corollary. Let T be a maximal monotone set-valued map with $T^{-1}(0) \neq \emptyset$. If

$$0 < m \leq c_k \leq M \qquad \forall\, k, \tag{12.4.8}$$

for some positive m and M, the sequence $\{x^k\}$ produced by the iteration:

$$x^{k+1} \equiv J_{c_k T}(x^k), \tag{12.4.9}$$

converges to a zero x^* of T. Furthermore, if T is strongly monotone, $\{x^k\}$ converges at least R-linearly. □

In order to enhance its convergence properties, the forward-backward splitting method can be modified in a way that is similar to the extragradient algorithm. This will require, however, a larger computational cost per iteration. More specifically, in each iteration of the modified method, we require a closed convex set X^k satisfying

$$X^k \subseteq \operatorname{dom} B \quad \text{and} \quad X^k \cap T^{-1}(0) \neq \emptyset,$$

and we perform an extra projection onto X^k to obtain the next iterate.

Modified Forward-Backward Algorithm (MFBA)

12.4.10 Algorithm.

Data: $T = A + B$, $X^0 \subseteq \operatorname{dom} B$, $X^0 \cap T^{-1}(0) \neq \emptyset$, and $x^0 \in X^0$.

Step 0: Set $k = 0$.

Step 1: If $0 \in T(x^k)$ stop.

Step 2: Select a $c_k > 0$ and a closed convex set $X^k \subseteq \operatorname{dom} B$ such that $X^k \cap T^{-1}(0) \neq \emptyset$. Set

$$\begin{aligned} x^{k+1/2} &\equiv J_{c_k A}((\mathbf{I} - c_k B)(x^k)) \\ x^{k+1} &\equiv \Pi_{X^k}\left(x^{k+1/2} - c_k(B(x^{k+1/2}) - B(x^k))\right) \end{aligned}$$

Step 3: Set $k \leftarrow k + 1$ and go to Step 1.

To analyze this algorithm, we assume that $\operatorname{dom} A \subseteq \operatorname{dom} B$. Since x^0 belongs to $X^0 \subseteq \operatorname{dom} B$, a simple inductive argument shows that the algorithm is well defined. There are many possible choices for X^k, we will return to this point later. For the time being we only require that X^k satisfies the conditions in Step 2. A crucial step in Algorithm **12.4.10** is the choice of c_k. We choose it by a line-search procedure as follows. For any constant γ in $(0, 1)$, we set $c_k \equiv 2^{-i_k}$, where i_k is the smallest i in $\{0, 1, 2, \ldots\}$ such that

$$\begin{aligned} &2^{-i} \,\|\, B(J_{2^{-i}A}((\mathbf{I} - 2^{-i}B)(x^k)) - B(x^k) \,\| \\ &\quad \leq \gamma \,\|\, J_{2^{-i}A}((\mathbf{I} - 2^{-i}B)(x^k) - x^k \,\|. \end{aligned} \tag{12.4.10}$$

Thus we have

$$c_k \,\|\, B(x^{k+1/2}) - B(x^k) \,\| \leq \gamma \,\|\, x^{k+1/2} - x^k \,\|. \tag{12.4.11}$$

12.4.11 Lemma. Let A and B be two maximal monotone maps from $\mathbb{R}^n$ into $\mathbb{R}^n$ with $\operatorname{dom} A \subseteq \operatorname{dom} B$ and B single-valued. Let $X \subseteq \operatorname{dom} B$ be a closed convex set such that $X \cap (A + B)^{-1}(0) \neq \emptyset$. For any positive c and any $x \in \operatorname{dom} B$ let $x^+ \equiv J_{cA}((\mathbf{I} - cB)(x))$ and $z \equiv x^+ - c(B(x^+) - B(x))$. For every $x^* \in X \cap (A + B)^{-1}(0)$,

$$\begin{aligned} \|\, \Pi_X(z) - x^* \,\|^2 &\leq \|\, z - x^* \,\|^2 \\ &= \|\, x - x^* \,\|^2 + c^2 \|\, B(x^+) - B(x) \,\|^2 \\ &\quad - \|\, x^+ - x \,\|^2 - 2c\,\eta, \end{aligned} \tag{12.4.12}$$

where $\eta \geq 0$ has the property that, if $A+B$ is strongly monotone on $\operatorname{dom} A$ with modulus $\sigma > 0$, then $\eta \geq \sigma \|x^+ - x^*\|^2$.

Proof. The inequality in (12.4.12) follows from $x^* = \Pi_X(x^*)$ and the nonexpansiveness of the projector. It remains to show the equality in (12.4.12). From the definition of x^+ and z we have

$$\begin{aligned} &x^+ + cu^+ \;=\; x - cv, \qquad z \;=\; x^+ - c(v^+ - v), \\ &v \;=\; B(x), \qquad v^+ \;=\; B(x^+), \qquad u^+ \;\in\; A(x^+). \end{aligned} \tag{12.4.13}$$

Since $0 \in A(x^*) + B(x^*)$, there exist $u^* \in A(x^*)$ and $v^* \in B(x^*)$ satisfying

$$u^* + v^* \;=\; 0. \tag{12.4.14}$$

Therefore

$$\begin{aligned} \| x - x^* \|^2 &= \| x - x^+ + x^+ - z + z - x^* \|^2 \\ &= \| x - x^+ \|^2 + \| x^+ - z \|^2 + \| z - x^* \|^2 \\ &\quad +2(x - x^+)^T(x^+ - x^*) + 2(x^+ - z)^T(z - x^*) \\ &= \| x - x^+ \|^2 - \| x^+ - z \|^2 + \| z - x^* \|^2 \\ &\quad +2(x - z)^T(x^+ - x^*) \\ &= \| x - x^+ \|^2 - c^2 \| B(x^+) - B(x) \|^2 \\ &\quad +\| z - x^* \|^2 + 2c(v^+ + u^+)^T(x^+ - x^*) \\ &= \| x - x^+ \|^2 - c^2 \| B(x^+) - B(x) \|^2 \\ &\quad +\| z - x^* \|^2 + 2c(v^+ - v^* + u^+ - u^*)^T(x^+ - x^*), \end{aligned}$$

where the fourth equality follows from (12.4.13) and the fifth from (12.4.14). This proves the equality in (12.4.12) with

$$\begin{aligned} \eta &\equiv (\, v^+ - v^* + u^+ - u^* \,)^T (\, x^+ - x^* \,) \\ &= (\, v^+ + u^+ \,)^T (\, x^+ - x^* \,). \end{aligned}$$

Since A and B are monotone, it follows from (12.4.13) and (12.4.14) that η is nonnegative. If in addition $A+B$ is strongly monotone on $\operatorname{dom} A$ with modulus $\sigma > 0$, then $\eta \geq \sigma \|x^+ - x^*\|^2$. □

12.4.12 Lemma. Let A and B be two maximal monotone maps from $\mathbb{R}^n$ into $\mathbb{R}^n$ with $\operatorname{dom} A \subseteq \operatorname{dom} B$ and B single-valued. For any positive c and any $x \in \operatorname{dom} B$, with $x^+ \equiv J_{cA}((\mathbf{I} - cB)(x))$,

$$c^{-1} \| x^+ - x \| \;\leq\; \min_{w \in A(x) + B(x)} \| w \|. \tag{12.4.15}$$

Proof. Note that $A(x)$ is closed and convex (see Exercise **12.8.10**) and therefore the minimum in (12.4.15) is well defined. By the definition of x^+ we have $c^{-1}(x - x^+) \in A(x^+) + B(x)$, so that, for any $w \in A(x) + B(x)$ the monotonicity of B implies

$$\big((c^{-1}(x - x^+) - B(x)) - (w - B(x))\big)^T (x^+ - x) \geq 0.$$

Simple manipulations then show that

$$c^{-1} \| x^+ - x \|^2 \leq w^T(x - x^+) \leq \| w \| \|x - x^+ \|,$$

which easily yields (12.4.15). □

The theorem below contains the main convergence properties of the Modified Forward-Backward algorithm.

12.4.13 Theorem. Let A and B be two maximal monotone map from $\mathbb{R}^n$ to $\mathbb{R}^n$ with $\operatorname{dom} A \subseteq \operatorname{dom} B$ and B single-valued and continuous on $\operatorname{dom} A$. Suppose that $T \equiv A + B$ is maximal monotone. For every k, let X^k be a closed convex set such that $X^k \subseteq \operatorname{dom} B$. Assume that

$$\Xi \equiv \left(\bigcap_{k=0}^{\infty} X^k \right) \bigcap (A + B)^{-1}(0) \neq \emptyset.$$

Let $\{x^k\}$ and $\{x^{k+1/2}\}$ be the sequences produced by the Modified Forward-Backward Algorithm with c_k determined by the procedure (12.4.10). The following four statements hold.

(a) The Modified Forward-Backward Algorithm is well defined.

(b) For every $x^* \in \Xi$ and for every k, we have

$$\begin{aligned} &\| x^{k+1} - x^* \|^2 \leq \\ &\| x^k - x^* \|^2 - (1 - \gamma^2) \| x^{k+1/2} - x^k \|^2 - 2c_k \eta_k, \end{aligned} \tag{12.4.16}$$

for some nonnegative η_k with the property that if T is strongly monotone on $\operatorname{dom} A$ with modulus $\sigma > 0$, then $\eta_k \geq \sigma \|x^{k+1/2} - x^*\|^2$.

(c) If the function $x \mapsto \min\{\|w\| : w \in T(x)\}$ is locally bounded on $\operatorname{dom} A$, the sequence $\{x^k\}$ converges to a zero of T.

(d) If T is strongly monotone on $\operatorname{dom} A$ with modulus $\sigma > 0$, then

$$\| x^{k+1} - x^* \|^2 \leq [\, 1 - \min(1 - \gamma^2, 2c_k\sigma) \,] \| x^k - x^* \|^2$$

for every k and where x^* is the unique element in $T^{-1}(0)$.

Proof. In view of the discussion after the presentation of the algorithm, to show that the Modified Forward-Backward Algorithm is well defined we only have to show that c_k can always be determined in a finite number of trials according to the line search (12.4.10). Let x^k be given. By Step 1 in the algorithm, it holds that $x^k \notin \Xi$. Since $x^k \in X^k \subseteq \operatorname{dom} A$, applying Lemma **12.4.12** with $x = x^k$ and $x^{k,+}(c) \equiv J_{cA}(\mathbf{I} - cB)(x^k)$, we get

$$c^{-1} \| x^+(c) - x^k \| \leq \min_{w \in T(x^k)} \| w \|,$$

which shows that $x^{k,+}(c)$ converges to x^k as c goes to zero. Notice that $x^{k,+}(c_k) = x^{k+1/2}$. Since B is continuous, this implies

$$\lim_{c \downarrow 0} B(x^{k,+}(c)) = B(x^k). \tag{12.4.17}$$

By the definition of $x^{k,+}(c)$, we have

$$c^{-1} (x^k - x^{k,+}(c)) \in A(x^{k,+}(c)) + B(x^k).$$

There are two possibilities. If

$$\liminf_{c \downarrow 0} c^{-1} (x^k - x^{k,+}(c)) = 0,$$

then by Remark **12.4.7** it follows that x^k is a zero of T, a contradiction. Therefore,

$$\liminf_{c \downarrow 0} c^{-1} \| x^k - x^{k,+}(c) \| > 0. \tag{12.4.18}$$

But (12.4.17) and (12.4.18) show that for any $\gamma > 0$,

$$\| B(x^{k,+}(c)) - B(x^k) \| \leq c^{-1} \| x^{k,+}(c) - x^k \|$$

for all $c > 0$ sufficiently small. Consequently, statement (a) holds.

Let $x^* \in \Xi$. A direct application of Lemma **12.4.11** and (12.4.11) shows that (b) holds. We next consider (c). By (12.4.16) we see that the sequence $\{x^k\}$ is bounded and therefore has at least one limit point, say x^∞. We want to show that x^∞ is a zero of T. Assume that $\{x^k : k \in \kappa\}$ converges to x^∞. We consider two cases. Suppose first that $\{c_k : k \in \kappa\}$ contains a subsequence $\{c_k : k \in \kappa'\}$ bounded away from zero, where κ' is an infinite subset of κ. Inequality (12.4.16) implies that $\|x^{k+1/2} - x^k\|$ tends to zero as $k(\in \kappa') \to \infty$ and this together with (12.4.10) shows that $\{\|B(x^{k+1/2}) - B(x^k)\| : k \in \kappa'\}$ goes to zero. The definition of $x^{k+1/2}$ implies

$$c_k^{-1}(x^k - x^{k+1/2}) + B(x^{k+1/2}) - B(x^k) \in A(x^{k+1/2}) + B(x^{k+1/2})$$

for all $k \in \kappa'$. The left-hand side in the above inclusion goes to zero as $k(\in \kappa') \to \infty$ so that, again by Remark **12.4.7**, x^∞ is a zero of T.

Suppose next that

$$\lim_{k(\in\kappa)\to\infty} c_k \;=\; 0.$$

This implies that eventually (on the subsequence κ) $c_k < 1$, so that

$$\frac{\gamma}{2c_k}\| x^{k,+}(2c_k) - x^k \| \;<\; \| B(x^{k,+}(2c_k)) - B(x^k) \|. \tag{12.4.19}$$

We can apply Lemma **12.4.12** with $x = x^k$ and $c = c_k$, obtaining

$$(2c_k)^{-1} \| x^{k,+}(2c_k) - x^k \| \;\le\; \min_{w\in T(x^k)} \| w \|, \quad \forall\, k \in \kappa.$$

Since $\{x^k : k \in \kappa\}$ converges to x^∞, by assumption (c), it follows that $\{x^{k,+}(2c_k) : k \in \kappa\}$ also converges to x^∞. By the definition of $x^{k,+}(2c_k)$ we have

$$w^k \;\equiv\; (2c_k)^{-1}(x^k - x^{k,+}(2c_k)) + B(x^{k,+}(2c_k)) - B(x^k) \;\in\; T(x^{k,+}(2c_k)).$$

If $\liminf_{k(\in\kappa)\to\infty} \|w^k\| = 0$, then $0 \in T(x^\infty)$. If $\liminf_{k(\in\kappa)\to\infty} \|w^k\| > 0$, then

$$\liminf_{k(\in\kappa)\to\infty} \frac{\| x^k - x^{k,+}(2c_k) \|}{2c_k} \;>\; 0.$$

But this contradicts (12.4.19). To complete the proof of (c), we only have to show that x^∞ is the only accumulation point of the sequence $\{x^k\}$. Using (12.4.16) we may proceed as in the corresponding parts of the proofs of Theorems **12.3.7** and **12.4.6**; the details are omitted.

Suppose finally that T is strongly monotone on $\operatorname{dom} A$ with modulus $\sigma > 0$. By (b) we know that (12.4.16) holds with $\eta_k \ge \sigma\| x^{k+1/2} - x^* \|^2$. Noting that

$$\| x^k - x^* \|^2 \;\le\; 2\,(\, \| x^{k+1/2} - x^k \|^2 + \| x^{k+1/2} - x^* \|^2\,).$$

we have

$$\begin{aligned}
&\| x^{k+1} - x^* \|^2 \\
&\le\; \|x^k - x^* \|^2 - (1-\gamma^2) \| x^{k+1/2} - x^k \|^2 - 2c_k\sigma \| x^{k+1/2} - x^* \|^2 \\
&\le\; \|x^k - x^*\|^2 - \min(1-\gamma^2, 2c_k\sigma)(\, \| x^{k+1/2} - x^k \|^2 + \| x^{k+1/2} - x^* \|^2\,) \\
&\le\; \|x^k - x^* \|^2 - \tfrac{1}{2}\, \min(1-\gamma^2, 2c_k\sigma\,) \| x^k - x^* \|^2.
\end{aligned}$$

Thus (d) holds. □

The choice of the set X^k is highly dependent on the problem at hand. If dom B is the whole space then we can obviously take X^k to be the whole space; in this case, the projection on X^k is trivial. When applying the method to the solution of a VI (K, F) a suitable choice is to take X^k to be K. In some cases, strategies analogous the one adopted in the Hyperplane Projection Algorithm may allow to find at each iteration a halfspace X^k that contains all the solutions of the problem.

For a given $x \in \mathbb{R}^n$, the two quantities

$$\| x - J_{cA}(\mathbf{I} - cB)(x) \| \quad \text{and} \quad \min_{w \in A(x)+B(x)} \| w \|$$

provide two residuals for the inclusion $0 \in T(x)$. Indeed, it is easy to see that either of the above quantities is equal to zero if and only if x is a solution of the inclusion. If T is strongly monotone, we have the following error bound in terms of the second residual.

12.4.14 Proposition. Let $T : \mathbb{R}^n \to \mathbb{R}^n$ be a maximal, strongly monotone set-valued map. The inclusion $0 \in T(x)$ has a unique solution, say x^*. Moreover, there exists a constant $c > 0$ such that

$$\| x - x^* \| \leq c \min_{w \in T(x)} \| w \|, \quad \forall\, x \in \mathbb{R}^n.$$

Proof. The strong monotonicity of T with constant $\sigma > 0$ is equivalent to the monotonicity of $T_0 \equiv T - \sigma\mathbf{I}$, and we have $T^{-1} = (T_0 + \sigma\mathbf{I})^{-1}$. Since T is maximal, so is T_0. Thus T^{-1} is a single-valued, globally Lipschitz continuous function with domain $\mathbb{R}^n$. This establishes that $0 \in T(x)$ has a unique solution, which we denote x^*. If $w \in T(x)$, the strong monotonicity of T yields

$$w^T(\, x - x^*\,) \geq \sigma \| x - x^* \|^2$$

which implies

$$\| x - x^* \| \leq \sigma^{-1} \| w \|.$$

Since w is an arbitrary vector in $T(x)$, the desired error bound follows. □

12.5 Applications of Splitting Algorithms

The splitting methods presented in the previous section can be used in several ways to solve VIs. We have studied inexact versions of these methods and, in the case of the Douglas-Rachford method also the use of relaxations. For simplicity, in this section we only present results for the exact versions of the splitting methods without relaxation. The interested reader will have

no difficulty in extending the results by considering inexact evaluation of the resolvents and over/under-relaxations, if needed.

Basically, the cornerstone of all applications of the splitting methods to a VI (K, F) is the reformulation of the VI as the zero finding problem $0 \in F(x) + \mathcal{N}(x; K)$. For our purpose here, we assume that $F \equiv G + H$ for some mappings G and H. In this case the VI $(K, G + H)$ is equivalent to finding a zero of the inclusion

$$0 \in G(x) + H(x) + \mathcal{N}(x; K).$$

Assuming appropriate monotonicity assumptions are satisfied, Proposition **12.3.6** tells us that we can apply all of the splitting methods considered in the previous section. Obviously various choices are possible. For example, we may take $A = \mathcal{N}(x; K)$ and $B = G + H$ and then apply either the Douglas-Rachford splitting or the forward-backward method. Or we can take $A = G + \mathcal{N}(x; K)$ and $B = H$ and apply again one of the methods considered previously. An important point of interest is then to understand what the resolvents of such A or B become and how they can be computed. This should make clear that there are many possibilities to analyze. The aim of this section is merely to illustrate the potential of this approach by giving some selected examples of interesting splittings.

12.5.1 Projection algorithms revisited

Splitting methods can be used to recover some of the results presented in Section **12.1** in a different and elegant way, shedding new light on projection methods and also leading to new variants. Suppose we want to solve the VI (K, F), where F is continuous on K. We can reformulate this problem as the zero finding problem

$$0 \in F(x) + \mathcal{N}(x; K).$$

We want to apply the forward-backward splitting method. By Proposition **12.3.6** we know that $\mathcal{N}(\cdot; K)$ is maximal monotone. Take then $A = \mathcal{N}(\cdot; K)$ and $B = F$, so that the iteration (12.4.2) becomes

$$x^{k+1} \equiv J_A((\mathbf{I} - F)(x^k)) = \Pi_K(x^k - F(x^k)), \tag{12.5.1}$$

where we have used the fact, already encountered in the proof of Proposition **12.3.6**, that the resolvent of the normal cone map is the Euclidean projector. Iteration (12.5.1) converges if F is co-coercive with modulus greater than $1/2$. We have thus recovered a particular case of Theorem **12.1.8**. Furthermore, if F is strongly monotone, Corollary **12.4.8** shows that this

projection method converges to the unique solution of the problem at a linear rate (at least). Using the results on the inexact evaluation of the resolvent (that is of the projection, in this case) that we obtained for the forward-backward method, we could easily design inexact versions of the basic projection method.

Asymmetric projection algorithm

If F is a monotone function and D is any positive definite matrix, for every vector $x^k \in \operatorname{dom} F$, the map

$$F_D^k(x) \equiv F(x^k) + D(x - x^k), \quad \forall\, x \in \mathbb{R}^n$$

is strongly monotone; hence the VI (K, F_D^k) has a unique solution. When D is in addition symmetric, the unique solution of the latter VI is equal to the skewed projection $\Pi_{K,D}(x^k - D^{-1}F(x^k))$. In what follows, we extend the latter notation to the case of an asymmetric positive definite matrix D, with the understanding that $\Pi_{K,D}(x^k - D^{-1}F(x^k))$ denotes the unique solution of the VI (K, F_D^k). We call the operator $\Pi_{K,D}$ an asymmetric projector and consider the following variant of the Basic Projection Algorithm **12.1.1**.

Asymmetric Projection Algorithm (APA)

12.5.1 Algorithm.

Data: $x^0 \in K$ and a positive definite $n \times n$ matrix D.

Step 0: Set $k = 0$.

Step 1: If $x^k = \Pi_{K,D}(x^k - D^{-1}F(x^k))$ stop.

Step 2: Set $x^{k+1} \equiv \Pi_{K,D}(x^k - D^{-1}F(x^k))$ and $k \leftarrow k + 1$; go to Step 1.

The only difference between this method and the Basic Projection Algorithm **12.1.1** is that the matrix D is not required to be symmetric. Due to the asymmetry of D, we can no longer use the D-norm to establish the convergence of Algorithm **12.5.1** (cf. Theorem **12.1.2**). Instead, we show that this extended algorithm is a special case of the forward-backward splitting method and use this fact to establish properties of Algorithm **12.5.1** that are stronger than the ones established for the former Basic Projection Algorithm. To simplify the notation, we set

$$C \;=\; D_s^{-1/2}\,(\,D - D_s\,)D_s^{-1/2},$$

where $D_s \equiv \frac{1}{2}(D + D^T)$ is the symmetric part of D. Note that C is skew

symmetric. Furthermore, we also let

$$Y \equiv D_s^{1/2}K = \{ D_s^{1/2}x : x \in K \}$$

and define the function $\tilde{F}_D : Y \to \mathbb{R}^n$ as

$$\tilde{F}_D(y) \equiv D_s^{-1/2}F(D_s^{-1/2}y), \quad \forall\, y \in Y.$$

The following is the main convergence theorem for the APA.

12.5.2 Theorem. Suppose that $\tilde{F}_D - C$ is co-coercive on Y with modulus greater than $1/2$. The sequence $\{x^k\}$ produced by the Asymmetric Projection Algorithm converges to a solution of the VI (K, F).

Proof. By the definition of x^{k+1}, we have, for all $x \in K$,

$$(\, x - x^{k+1}\,)^T\,(\, F(x^k) + D(x^{k+1} - x^k)\,) \geq 0;$$

the left-hand expression is equal to

$$(\, D_s^{1/2}x - D_s^{1/2}x^{k+1}\,)^T\,[\, D_s^{-1/2}F(D_s^{-1/2}D_s^{1/2}x^k\,)$$
$$+(\, D_s^{-1/2}(D - D_s)D_s^{-1/2} + \mathbf{I}\,)\,(\, D_s^{1/2}x^{k+1} - D_s^{1/2}x^k\,)\,]$$

which upon the substitution of variables:

$$y \equiv D_s^{1/2}x, \quad y^{k+1} \equiv D_s^{1/2}x^{k+1}, \quad \text{and} \quad y^k \equiv D_s^{1/2}x^k$$

becomes

$$(\, y - y^{k+1}\,)^T[\, \tilde{F}_D(y^k) + (\,\mathbf{I} + C\,)(y^{k+1} - y^k\,)\,].$$

Hence, we obtain

$$0 \in (\,\mathbf{I} + C\,)\,(\, y^{k+1} - y^k\,) + \tilde{F}_D(y^k) + \mathcal{N}_Y(y^{k+1}),$$

which, in turn, is equivalent to

$$y^{k+1} = J_A((\mathbf{I} - B)(y^k)),$$

where

$$A \equiv C + \mathcal{N}_Y \quad \text{and} \quad B = \tilde{F}_D - C.$$

Since C is skew symmetric, thus positive semidefinite, it follows from Proposition **12.3.6** that A is maximal monotone with $\operatorname{dom} A = Y$. Thus, the sequence $\{y^k\}$ can be viewed as generated by the forward-backward splitting method. Thus, by Theorem **12.4.6**, the sequence $\{y^k\}$ converges to a point y^* satisfying

$$0 \in \tilde{F}_D(y^*) + \mathcal{N}_Y(y^*),$$

which is equivalent to saying that y^* is a solution of the VI $(Y, \tilde{F}_D)$. But then it easily follows that $x^* \equiv D_s^{-1/2} y^*$ is a solution of the VI (K, F). □

Since C is skew symmetric, a necessary condition for $\tilde{F}_D - C$ to be co-coercive on Y is that F is monotone on K. Thus although not explicitly stated in Theorem **12.5.2**, the Asymmetric Projection Algorithm is applicable only to monotone VIs. The proposition below gives a sufficient condition for $\tilde{F}_D - C$ to be co-coercive on Y with modulus exceeding $1/2$.

12.5.3 Proposition. Under either one of the following two conditions:

(a) F is Lipschitz continuous on K with modulus L and a $\sigma \in (0,1)$ exists such that, for all y^1 and $y^2 \in Y$,

$$\begin{aligned} &\| \tilde{F}_D(y^2) - \tilde{F}_D(y^1) - D_s^{-1/2} D D_s^{-1/2}(y^2 - y^1) \| \\ &\quad \leq \sigma \| y^2 - y^1 \|; \end{aligned} \tag{12.5.2}$$

(b) $F(x) = Mx + q$, for some positive semidefinite matrix $M \equiv D + E$ with D positive definite and E symmetric, and $0 < \|\mathbf{I} + H\| < 2$, where $H \equiv D_s^{-1/2} E D_s^{-1/2}$;

the map $\tilde{F}_D - C$ is co-coercive on Y with modulus greater than $1/2$.

Proof. To simplify the notation somewhat, we write F_D for $\tilde{F}_D$. For part (a), we need to prove that for all y^1 and y^2 belonging to Y,

$$\begin{aligned} &\big(F_D(y^2) - Cy^2 - (F_D(y^1) - Cy^1)\big)^T (y^2 - y^1) \\ &\qquad \geq c \| F_D(y^2) - Cy^2 - (F_D(y^1) - Cy^1) \|^2 \end{aligned} \tag{12.5.3}$$

for some $c > 1/2$. By (12.5.2) and by the definition of C we can write:

$$\begin{aligned} &\| F_D(y^2) - Cy^2 - (F_D(y^1) - Cy^1) \|^2 \\ &\quad = \| (y^2 - y^1) + (F_D(y^2) - F_D(y^1)) - D_s^{-1/2} D D_s^{-1/2}(y^2 - y^1) \|^2 \\ &\quad = 2(y^2 - y^1)^T (F_D(y^2) - F_D(y^1) - D_s^{-1/2} D D_s^{-1/2}(y^2 - y^1)) \\ &\qquad + \| y^2 - y^1 \|^2 + \| F_D(y^2) - F_D(y^1) - D_s^{-1/2} D D_s^{-1/2}(y^2 - y^1) \|^2 \\ &\quad \leq 2(y^2 - y^1)^T (F_D(y^2) - F_D(y^1) - D_s^{-1/2} D D_s^{-1/2}(y^2 - y^1)) \\ &\qquad + (1 + \sigma^2) \| y^2 - y^1 \|^2 \\ &\quad = 2(y^2 - y^1)^T (F_D(y^2) - Cy^2 - (F_D(y^1) - Cy^1)) \\ &\qquad - (1 - \sigma^2) \| y^2 - y^1 \|^2. \end{aligned}$$

By the Lipschitz continuity of F we have

$$\| F_D(y^2) - Cy^2 - (F_D(y^1) - Cy^1) \| \leq (\| C \| + L \| D_s^{-1} \|) \| y^2 - y^1 \|;$$

hence (12.5.3) holds with

$$c \;=\; \tfrac{1}{2}\left(1+\frac{1-\sigma^2}{\|C\|+L\|D_s^{-1}\|}\right) \;>\; \tfrac{1}{2},$$

and thus concluding the proof of (a).

(b) It is easy to see that in this case $F_D - C$ is affine with

$$F_D(y) - Cy \;=\; (\mathbf{I}+H)y + D_s^{-1}q.$$

Since M is positive semidefinite, so is the matrix

$$\mathbf{I}+H \;=\; D_s^{-1/2}(\,D_s+E\,)\,D_s^{-1/2}.$$

Moreover, we have $\|\mathbf{I}+H\| = \lambda_{\max}(\mathbf{I}+H)$. Therefore, for all vectors y,

$$y^{\,T}(\,\mathbf{I}+H\,)y \;\geq\; \frac{1}{\|\,\mathbf{I}+H\,\|}\,\|\,(\,\mathbf{I}+H\,)y\,\|^2.$$

This shows that $F_D - C$ is co-coercive with constant $1/(\|\mathbf{I}+H\|) > 1/2$. □

If F is affine as in (b) of the above proposition, (12.5.2) is easily seen to imply $\|H\| < 1$, which is more restrictive than $0 < \|\mathbf{I}+H\| < 2$ assumed in (b). Therefore the conditions for the affine case considered in (b) are a weakening of (12.5.2) specialized to an affine F. The following proposition shows that in the affine case we can always easily find a D that makes the condition $0 < \|\mathbf{I}+H\| < 2$ satisfied.

12.5.4 Proposition. Let $M = D + E$ be an $n \times n$ positive semi-definite matrix and set $D = \rho\mathbf{I} - G$, where G is any $n \times n$ matrix. For every $\rho > 0$ sufficiently large, D is positive definite and $0 < \|\mathbf{I}+H\| < 2$.

Proof. It is easy to see that $D_s = \rho\mathbf{I} - G_s$ so that, if $\rho > \|G_s\|$, D_s is positive definite and we can write

$$\begin{aligned} D_s^{-1/2} &= \rho^{-1/2}\left(\mathbf{I}-\frac{1}{\rho}G_s\right)^{-1/2} \\ &= \rho^{-1/2}\left[\mathbf{I}+\sum_{i=1}^{\infty}\frac{1\times 3\times\cdots\times(2i-1)}{i!\,2^i}\left(\frac{G_s}{\rho}\right)^i\right]. \end{aligned}$$

Since all the coefficients in the power expansion are between 0 and 1/2, we have

$$D_s^{-1/2} \;=\; \rho^{-1/2}\left(\mathbf{I}+\frac{1}{2\rho}G_s+\mathrm{O}(1/\rho^2)\right),$$

where $\mathrm{O}(\delta)$ denotes a matrix with norm bounded above by δ. Therefore we get

$$\begin{aligned} & D_s^{-1/2} E D_s^{-1/2} \\ &= \frac{1}{\rho}\left(\mathbf{I} + \frac{1}{2\rho}G_s + \mathrm{O}(1/\rho^2)\right)(M + G - \rho\mathbf{I})\left(\mathbf{I} + \frac{1}{2\rho}G_s + \mathrm{O}(1/\rho^2)\right) \\ &= -\mathbf{I} + \mathrm{O}(1/\rho), \end{aligned}$$

which shows that for all ρ large enough, $0 < \|\mathbf{I} + D_s^{-1/2} E D_s^{-1/2}\| < 2$. □

The way to apply the above result to an arbitrary matrix M is as follows. Choose a matrix G so that $M + G$ is symmetric. Choose a scalar $\rho > \lambda_{\max}(G_s) = \|G_s\|$. Define $D \equiv \rho\mathbf{I} - G$. We then have

$$E \equiv M - D = M + G - \rho\,\mathbf{I},$$

which is clearly a symmetric matrix by the choice of G. We illustrate this application by considering a VI (K, q, M) that corresponds to a saddle problem (L, X, Y) with a quadratic saddle function L; see Subsection **1.4.1**. Specifically, let

$$M = \begin{pmatrix} P & R \\ -R^T & Q \end{pmatrix}, \qquad q = \begin{pmatrix} p \\ r \end{pmatrix}, \qquad K = X \times Y, \tag{12.5.4}$$

where $P \in \mathbb{R}^{n\times n}$ and $Q \in \mathbb{R}^{m\times m}$ are symmetric positive semidefinite, $R \in \mathbb{R}^{n\times m}$, $p \in \mathbb{R}^n$, $r \in \mathbb{R}^m$, and $X \subseteq \mathbb{R}^n$ and $Y \subseteq \mathbb{R}^m$ are nonempty closed convex sets. Note that M is positive semidefinite but not necessarily positive definite. The corresponding saddle function is

$$L(x, y) = p^T x + q^T y + \tfrac{1}{2}\, x^T P x + x^T R y - \tfrac{1}{2}\, y^T Q y, \quad (x, y) \in \mathbb{R}^{n+m}.$$

We can now apply the Asymmetric Projection Algorithm by first taking $G \in \mathbb{R}^{(n+m)\times(n+m)}$ so that $M+G$ is symmetric. Obviously there are many choices. Two such choices are

$$G_1 \equiv \begin{pmatrix} 0 & 0 \\ 2\,R^T & 0 \end{pmatrix} \quad \text{and} \quad G_2 \equiv \begin{pmatrix} 0 & -R \\ R^T & 0 \end{pmatrix}$$

With G chosen, let $\rho > \|G_s\|$ and set $D \equiv \rho\mathbf{I} - G$. Note that the symmetric part of G_2 is the zero matrix. Thus ρ can be any positive scalar with this choice of G. Applying the Asymmetric Projection Algorithm to the choice G_1 results in the following iterative process for solving the saddle problem

(L, X, Y):

$$x^{k+1} \equiv \operatorname*{argmin}_{x \in X} \left\{ \frac{\rho}{2} \| x - x^k \|^2 + (Px^k + p + Ry^k)^T x \right\}$$

$$y^{k+1} \equiv \operatorname*{argmin}_{y \in Y} \left\{ \frac{\rho}{2} \| y - y^k \|^2 + (Qy^k + r - 2R^T x^{k+1} + R^T x^k)^T y \right\}.$$

By Propositions **12.5.2**, **12.5.3** and **12.5.4** it follows that if ρ is sufficiently large, the sequence $\{(x^k, y^k)\}$ so generated converges to a saddle point of the triple (L, X, Y), provided that this saddle problem has a solution. It is important to note that both x^{k+1} and y^{k+1} are obtained by minimizing strictly convex, separable quadratic functions over X and Y respectively. (Indeed, these minimization problems are Euclidean projections.) This is obviously very convenient from the computational point of view because we can take full advantage of the separability of the sets X and Y. In contrast, such a separation cannot be accomplished with the choice of the matrix G_2.

12.5.2 Applications of the Douglas-Rachford splitting

The Doulgas-Rachford splitting is interesting because its convergence can be obtained under very mild assumptions. In this subsection we consider the direct application of the Douglas-Rachford splitting method to the VI (K, F), where F is the composite mapping: $F \equiv M^T \circ G \circ M$ with M being an $s \times n$ matrix and $G : \mathbb{R}^s \to \mathbb{R}^s$. When G is (strongly) monotone, F is a monotone composite map. As usual we rewrite the VI (K, F) as the zero finding problem

$$0 \in M^T \circ G \circ M(x) + \mathcal{N}(x; K). \tag{12.5.5}$$

Notice that this inclusion is well defined for a set-valued map G. Thus throughout the remainder of this subsection, we let G be a set-valued map. The problem (12.5.5) is an instance of the generalized VI (K, F), where $F \equiv M^T \circ G \circ M$ is a multifunction. A solution of this problem is an element x of K such that there exists $v \in G(Mx)$ satisfying

$$(y - x)^T M^T v \geq 0, \quad \forall y \in K.$$

To apply the Douglas-Rachford splitting method, our first order of business is to establish when $M^T \circ G \circ M$ is maximal monotone. The result below deals with this issue.

12.5.5 Proposition. Suppose that $G : \mathbb{R}^n \to \mathbb{R}^n$ is a maximal monotone (set-valued) map, and that the $s \times n$ matrix M has full row rank. The (set-valued) map $M^T \circ G \circ M$ is maximal monotone.

Proof. It is easy to see that $F \equiv M^T \circ G \circ M$ is monotone on its domain, which is given by

$$\operatorname{dom} F = \{ x \in \mathbb{R}^n : Mx \in \operatorname{dom} G \}.$$

To prove the maximal monotonicity of F, it is sufficient to show, by Theorem **12.3.3**, that the resolvent of F has full range, that is, for any $y \in \mathbb{R}^n$ there exists an $x \in \mathbb{R}^n$ such that

$$y \in x + M^T G(Mx). \tag{12.5.6}$$

We claim that, given y,

$$x \equiv y - M^T (G^{-1} + M M^T)^{-1}(My)$$

satisfies (12.5.6). First of all note that since M has full row rank, the map $G^{-1} + MM^T$ is strongly monotone and therefore surjective. Hence $(G^{-1} + MM^T)^{-1}$ is well defined and single-valued. We can rewrite x as

$$x = y - M^T z, \tag{12.5.7}$$

where

$$z = (G^{-1} + MM^T)^{-1}(My) \tag{12.5.8}$$

or, equivalently,

$$My \in (G^{-1} + MM^T)(z).$$

Therefore

$$M(y - M^T z) \in G^{-1}(z),$$

that, together with (12.5.7) gives

$$z \in G(Mx) \tag{12.5.9}$$

From this and from (12.5.7) we see that x satisfies (12.5.6). □

We can now apply the Douglas-Rachford splitting method to the solution of (12.5.5). With reference to the terminology employed in Subsection **12.4.1** we have to decide the choice of A and B. There are at least two obvious choices. The first is to set $A \equiv M^T \circ G \circ M$ and $B \equiv \mathcal{N}(\cdot; K)$; the second is to reverse the roles of $\mathcal{N}(\cdot; K)$ and $M^T \circ G \circ M$. We consider only the former option and leave to the reader to work out the details of the latter choice. Actually, there is not much to say, we only need to observe that the resolvent of B is the Euclidean projector onto K while calculating $J_{M^T \circ G \circ M}(w)$ simply amounts to finding the unique solution z

of the equation $z + M^T G(Mz) = w$. In this way we obtain the following algorithm for solving the inclusion (12.5.5).

DR Splitting Algorithm for a Multi-Valued VI (DRSAMVI)

12.5.6 Algorithm.

Data: $x^0 \in \mathbb{R}^n$.

Step 0: Set $k = 0$.

Step 1: Calculate $y^k = \Pi_K(x^k)$.

Step 2: If $0 \in M^T \circ G \circ M(y^k) + \mathcal{N}(y^k; K)$, stop.

Step 3: Calculate v^k as the unique solution of the equation

$$v^k + M^T G(Mv^k) \;=\; 2y^k - x^k.$$

Step 4: Set $x^{k+1} \equiv v^k + x^k - y^k$ and $k \leftarrow k + 1$; go to Step 1.

The algorithm calls for the solution of a system of equations involving G (Step 3) and for a projection onto K (Step 1) at each iteration. It may be interesting to give an alternative way to execute Step 3 that involves the function G only and not the function $M^T \circ G \circ M$. As we see later, this is useful in some applications. In the proof of Proposition **12.5.5** we have shown that

$$J_{M^T \circ G \circ M}(y) \;=\; y - M^T(G^{-1} + MM^T)^{-1}(My),$$

Let z be defined by (12.5.8), and let

$$r \;\equiv\; My - MM^T z.$$

This and (12.5.7) imply

$$r \;=\; Mx. \tag{12.5.10}$$

Since $(MM^T)^{-1}$ exists by the full row rank assumption on M, the definition of r yields

$$z \;=\; (MM^T)^{-1}(My - r). \tag{12.5.11}$$

By (12.5.9), (12.5.10) and (12.5.11) we get

$$0 \;\in\; (MM^T)^{-1}(r - My) + G(r).$$

We see then that we can give the following recipe for calculating the resolvent of $M^T \circ G \circ M$.

Algorithm for Evaluating $J_{M^T \circ G \circ M}(u)$

12.5.7 Algorithm.

Data: $u \in \mathbb{R}^n$.

Step 1: Calculate the unique solution r of the inclusion

$$0 \in (r - Mu) + MM^T G(r).$$

Step 2: Solve the system of linear equations for z:

$$MM^T z = Mu - r.$$

Step 3: Set $J_{M^T \circ G \circ M}(u) = u - M^T z$.

Observe that if G is single-valued, the inclusion in Step 1 is an equation. The convergence properties of Algorithm **12.5.6** follow readily from Theorem **12.4.3** and require no proof.

12.5.8 Theorem. Suppose that $G : \mathbb{R}^n \to \mathbb{R}^n$ is maximal monotone and that $M \in \mathbb{R}^{s \times n}$ has full row rank. Let K be a closed convex subset of $\mathbb{R}^n$. Suppose that the inclusion (12.5.5) has a solution. Algorithm **12.5.6** produces a sequence $\{y^k\}$ converging to a solution of (12.5.5). □

An application to traffic equilibrium problems

We can apply the method just described to the solution of traffic equilibrium problems. In particular, we consider the static traffic equilibrium problem with fixed demand considered in Subsection **1.4.5**. For convenience, we briefly recall the problem and take this as an opportunity to present it in a way that is more amenable to the application of the Douglas-Rachford method.

We have a traffic network represented by a graph with node set $\mathcal{N}$ and arc set $\mathcal{A}$. The cost of travel along an arc $a \in \mathcal{A}$ is a nonlinear function $c_a(f)$ of the total flow vector f with components f_b for all $b \in \mathcal{A}$. We denote by $c(f)$ the vector with components $c_a(f)$, $a \in \mathcal{A}$. There are two subsets of $\mathcal{N}$ that represent the set of origin nodes $\mathcal{O}$ and destination nodes $\mathcal{D}$, respectively. The set of origin-destination (OD) pairs is a given subset $\mathcal{W}$ of $\mathcal{O} \times \mathcal{D}$. For a given $w = (w_o, w_d) \in \mathcal{W}$ we have a flow vector $x_w \in \mathbb{R}^{|\mathcal{A}|}$ which satisfies the conservation of flows and is nonnegative:

$$Ex_w = g_w, \qquad x_w \geq 0,$$

where E is the node-arc incidence matrix of the network and $g_w \in \mathbb{R}^{|\mathcal{N}|}$ is the demand vector for the OD pair w whose components are all zero except for the two entries g_{w_o} and g_{w_d} that correspond to the origin and destination nodes of the pair $w = (w_o, w_d)$ and that are equal to d_w and $-d_w$, respectively. Therefore, for a given w, x_w "brings" a flow of d_w from the origin w_o to the destination w_d. The set of all feasible arc flows f is therefore given by the sum of all possible feasible vectors x_w and can be written as

$$\mathcal{F} \equiv \left\{ f : f = \sum_{w \in \mathcal{W}} x_w \text{ for some } x_w \geq 0 \text{ satisfying } E x_w = g_w \, \forall \, w \in \mathcal{W} \right\}.$$

This representation of the set $\mathcal{F}$ is equivalent to the one given before Proposition **1.4.8**. Under a nonnegativity and additivity assumption on the cost function c, Proposition **1.4.8** shows that a feasible flow $f \in \mathcal{F}$ induces a user equilibrium if and only if f is a solution of the VI $(\mathcal{F}, c)$.

Introduce the $|\mathcal{A}| \times (|\mathcal{W}||\mathcal{A}|)$ matrix

$$M \equiv [\ \mathbf{I} \quad \mathbf{I} \quad \cdots \quad \mathbf{I}\ ],$$

where $\mathbf{I}$ is the $|\mathcal{A}| \times |\mathcal{A}|$ identity matrix. Clearly M has full row rank. Let us denote by $x \in \mathbb{R}^{|\mathcal{W}||\mathcal{A}|}$ the vector that has the x_w as subvectors; also let K be defined by

$$K \equiv \{\, x \in \mathbb{R}^{|\mathcal{W}||\mathcal{A}|} : E x_w = g_w,\ x_w \geq 0, \forall \, w \in \mathcal{W} \,\}.$$

Notice that K has the Cartesian product structure:

$$K = \prod_{w \in \mathcal{W}} K_w,$$

where

$$K_w \equiv \{\, x_w \in \mathbb{R}^{|\mathcal{A}|} : E x_w = g_w,\ x_w \geq 0, \,\}, \quad w \in \mathcal{W}.$$

The VI $(\mathcal{F}, c)$ can be written as

$$(\, Mz - Mx \,)^T c(Mx) \geq 0, \quad \forall \, z \in K$$

or, equivalently, as

$$(\, z - x \,)^T M^T c(Mx) \geq 0, \quad \forall \, z \in K,$$

which clearly falls in the framework considered in this subsection. The structure of the matrix M allows us to easily compute $(MM^T)^{-1}$, which is

equal to $(1/|\mathcal{W}|)\mathbf{I}$. Furthermore, due to the Cartesian product structure of K, the projection onto K naturally decomposes into $|\mathcal{W}|$ independent projections onto the sets K_w each in a lower-dimensional space than K. Taking advantage of these simplifications and using Algorithm **12.5.7**, we present the specialization of the Douglas Rachford Splitting Algorithm **12.5.6** to the traffic equilibrium problem as follows.

DR Splitting Algorithm for the Traffic Equilibrium Problem
(DRSATEP)

12.5.9 Algorithm.

Data: $x^0 \in \mathbb{R}^n$, $c > 0$.

Step 0: Set $k = 0$.

Step 1: For every $w \in \mathcal{W}$, calculate $y_w^k \equiv \Pi_{K_w}(x_w^k)$.

Step 2: If $y^k \equiv (y_w^k)$ solves the VI $(K, M^T \circ c \circ M)$, stop.

Step 3: Calculate r^k as the unique solution of the equation

$$\frac{1}{c\,|\mathcal{W}|}\left(r - \sum_{w\in\mathcal{W}}(2y_w^k - x_w^k)\right) + c(r) \;=\; 0$$

and set

$$v^k \;\equiv\; \frac{1}{c\,|\mathcal{W}|}\left(\sum_{w\in W}(2y_w^k - x_w^k) - r^k\right).$$

Step 4: Set $x^{k+1} \equiv v^k + x^k - y^k$ and $k \leftarrow k+1$; go to Step 1.

Note that each projection in Step 1 can be calculated by solving a strictly convex quadratic program and all the required projections can be executed in parallel.

12.6 Rate of Convergence Analysis

Several of the algorithms presented in the previous sections are fixed-point iterations of the form:

$$x^{k+1} \;\equiv\; \Phi(x^k), \quad k \;=\; 0, 1, 2, \ldots, \tag{12.6.1}$$

for a certain (single-valued) mapping $\Phi : \operatorname{dom}\Phi \subseteq \mathbb{R}^n \to \mathbb{R}^n$. For the forward-backward splitting iteration (12.4.2), we have shown a R-linear convergence rate of the iterates $\{x^k\}$ under a strong monotonicity assumption; see Corollary **12.4.8**. In this section, we derive a general rate of

convergence theory for the iteration (12.6.1) under the assumption of the existence of an appropriate residual function for the fixed-point problem $x = \Phi(x)$ that satisfies two key properties. We then apply this theory to several iterative algorithms for solving the (pseudo) monotone VI.

Let X^* denote the set of fixed points of Φ, which we assume is nonempty. Let $\psi : \operatorname{dom}\Phi \to \mathbb{R}_+$ be a residual function of X^*; that is, ψ is continuous and $\psi(x) = 0$ if and only if $x \in X^*$. Suppose that there exist three positive constants η_1, η_2, and δ satisfying

$$\psi(x) - \psi(\Phi(x)) \geq \eta_1 \| x - \Phi(x) \|^2, \quad \forall\, x \in \operatorname{dom}\Phi, \tag{12.6.2}$$

and

$$\begin{aligned} &\min(\, \psi(x), \psi(\Phi(x)) \,) \\ &\quad \leq \eta_2 \| x - \Phi(x) \|^2, \quad \forall\, x \in \operatorname{dom}\Phi \text{ with } \| x - \Phi(x) \| \leq \delta. \end{aligned} \tag{12.6.3}$$

The first condition (12.6.2) implies

$$\psi(x^k) - \psi(x^{k+1}) \geq \eta_1 \| x^k - x^{k+1} \|^2 \tag{12.6.4}$$

so that there is a sufficient decrease in the value $\psi(x^{k+1})$ from the previous value $\psi(x^k)$. In this sense, ψ is a merit function for the iteration (12.6.1). Conditions (12.6.2) and (12.6.3) together imply

$$\psi(x^{k+1}) \leq \eta_2 \| x^k - x^{k+1} \|^2, \tag{12.6.5}$$

if $\|x^k - x^{k+1}\|$ is sufficiently small. Roughly speaking, (12.6.3) ensures that ψ does not grow too fast near X^*. In the context of VIs, there is a close connection between the latter condition and the local error bounds. Before revealing this connection and presenting how conditions (12.6.2) and (12.6.3) apply to the iterative methods in the previous sections, we state and prove a rate result for the iteration (12.6.1) under these two conditions.

12.6.1 Theorem. Let $\Phi : \operatorname{dom}\Phi \subseteq \mathbb{R}^n \to \mathbb{R}^n$ be a continuous function with a nonempty set X^* of fixed points. Let $\{x^k\}$ be defined by the iteration (12.6.1). If there exists a continuous residual function $\psi : \operatorname{dom}\Phi \to \mathbb{R}_+$ for X^* satisfying (12.6.2) and (12.6.3), then $\{\psi(x^k)\}$ converges to zero at least R-linearly, and the sequence $\{x^k\}$ converges to an element of X^* at least R-linearly.

Proof. By (12.6.4), the nonnegative sequence $\{\psi(x^k)\}$ of scalars is monotonically decreasing and therefore converges. This implies that

$$\lim_{k\to\infty} \| x^k - x^{k+1} \| = 0.$$

In turn, (12.6.5) implies that $\{\psi(x^k)\}$ converges to zero. The two inequalities (12.6.4) and (12.6.5) together yield

$$\psi(x^{k+1}) \;\leq\; \frac{\eta_2}{\eta_1+\eta_2}\,\psi(x^k)$$

for all k sufficiently large. Iterating this inequality easily establishes the R-linear rate of convergence of $\{\psi(x^k)\}$. From (12.6.4), we get

$$\eta_1 \,\|\, x^k - x^{k+1} \,\|^2 \;\leq\; \psi(x^k) \;\leq\; \left(\frac{\eta_2}{\eta_1+\eta_2}\right)^k \psi(x^0),$$

which yields,

$$\|\, x^k - x^{k+1} \,\| \;\leq\; \sqrt{\frac{\psi(x^0)}{\eta_1}}\left(\sqrt{\frac{\eta_2}{\eta_1+\eta_2}}\right)^k.$$

Therefore,

$$\|\, x^k - x^{k+m} \,\| \;\leq\; \sqrt{\frac{\psi(x^0)}{\eta_1}}\sum_{j=k}^{k+m-1}\left(\sqrt{\frac{\eta_2}{\eta_1+\eta_2}}\right)^j.$$

Thus $\{x^k\}$ is a Cauchy sequence and hence converges to a limit, say x^∞. By the continuity of Φ, x^∞ belongs to X^*; moreover, we have

$$\|\, x^k - x^\infty \,\| \;\leq\; \sqrt{\frac{\psi(x^0)}{\eta_1}}\,\frac{1}{1-\sqrt{\dfrac{\eta_2}{\eta_1+\eta_2}}}\left(\sqrt{\frac{\eta_2}{\eta_1+\eta_2}}\right)^k,$$

showing that $\{x^k\}$ converges to x^∞ at least R-linearly. □

12.6.1 Extragradient method

The sequence $\{x^k\}$ produced by the extragradient Algorithm **12.1.9** for solving the VI (K, F) is a special case of the fixed-point iteration (12.6.1) with

$$\Phi_{\rm ex}(x) \;\equiv\; x - \Pi_K(x - \tau F(x - \Pi_K(x - \tau F(x)))), \quad x \in K.$$

We recall from Exercise **1.8.29** that a fixed point of $\Phi_{\rm ex}$ is indeed a solution of the VI (K, F). We show that condition (12.6.2) holds with

$$\psi(x) \;\equiv\; \mathrm{dist}(x, \mathrm{SOL}(K, F))^2.$$

12.6.2 Proposition. Let K be a closed convex set in $\mathbb{R}^n$ and let F be a mapping from K into $\mathbb{R}^n$ that is pseudo monotone on K with respect to

$\mathrm{SOL}(K,F)$ and Lipschitz continuous on K with constant $L > 0$. For any $\tau \in (0, 1/L)$ and any $x \in K$,

$$\begin{aligned}&\mathrm{dist}(x, \mathrm{SOL}(K,F))^2 - \mathrm{dist}(\Phi_{\mathrm{ex}}(x), \mathrm{SOL}(K,F))^2 \\ &\quad \geq \tfrac{1}{2}\,(1 - \tau L)\,\|\, x - \Phi_{\mathrm{ex}}(x)\,\|^2.\end{aligned}$$

Proof. Let $x^* \in \mathrm{SOL}(K,F)$ be such that $\|x - x^*\| = \mathrm{dist}(x, \mathrm{SOL}(K,F))$. Let $y \equiv \Pi_K(x - \tau F(x))$. From the proof of Lemma **12.1.10**, we obtain the first of the following three inequalities:

$$\begin{aligned}&\mathrm{dist}(x, \mathrm{SOL}(K,F))^2 - \mathrm{dist}(\Phi_{\mathrm{ex}}(x), \mathrm{SOL}(K,F))^2 \\ &\quad \geq \|\, x - y\,\|^2 + \|\, y - \Phi_{\mathrm{ex}}(x)\,\|^2 - 2\,\tau\, L\,\|\,\Phi_{\mathrm{ex}}(x) - y\,\|\,\|\, y - x\,\| \\ &\quad \geq (1 - \tau L)\,[\,\|\, x - y\,\|^2 + \|\, y - \Phi_{\mathrm{ex}}(x)\,\|^2\,] \\ &\quad \geq \tfrac{1}{2}\,(1 - \tau L)\,\|\, x - \Phi_{\mathrm{ex}}(x)\,\|^2,\end{aligned}$$

where the second inequality follows from the assumption that $1 > \tau L$ and the third inequality follows from the triangle inequality and the Cauchy-Schwarz inequality. □

Under condition (12.6.2), condition (12.6.3) holds if we can show that there exist positive constants η and δ such that

$$\mathrm{dist}(x, \mathrm{SOL}(K,F)) \;\leq\; \eta\,\|\, x - \Phi_{\mathrm{ex}}(x)\,\|,$$

for all $x \in K$ satisfying $\|x - \Phi_{\mathrm{ex}}(x)\| \leq \delta$. We recognize the latter as a local error bound for $\mathrm{SOL}(K,F)$ with $r_{\mathrm{ex}}(x) \equiv \|x - \Phi_{\mathrm{ex}}(x)\|$ as the residual. In what follows, we state and prove a result which shows that the extragradient residual $r_{\mathrm{ex}}(x)$ is equivalent to the residual $\|\mathbf{F}^{\mathrm{nat}}_{K,\tau}(x)\|$. In turn, we recall from Exercise **4.8.4** that the latter residual is equivalent to the natural residual $\|\mathbf{F}^{\mathrm{nat}}_{K}(x)\|$. The following proposition does not require F to be pseudo monotone.

12.6.3 Proposition. Let F be Lipschitz continuous with constant $L > 0$ on the closed convex set $K \subseteq \mathbb{R}^n$. For any $\tau > 0$ and any $x \in K$,

$$(1 - \tau L)\,\|\,\mathbf{F}^{\mathrm{nat}}_{K,\tau}(x)\,\| \;\leq\; r_{\mathrm{ex}}(x) \;\leq\; (1 + \tau L)\,\|\,\mathbf{F}^{\mathrm{nat}}_{K,\tau}(x)\,\|.$$

Proof. Let $x \in K$ be arbitrary. Let $y \equiv \Pi_K(x - \tau F(x))$. By the triangle inequality, we obtain the first of the following three inequalities:

$$\begin{aligned}r_{\mathrm{ex}}(x) \;&\geq\; \|\, x - y\,\| - \|\, y - \Phi_{\mathrm{ex}}(x)\| \\ &\geq\; \|\,\mathbf{F}^{\mathrm{nat}}_{K,\tau}(x)\,\| - \tau\,\|\, F(y) - F(x)\,\| \;\geq\; (1 - \tau L)\,\|\,\mathbf{F}^{\mathrm{nat}}_{K,\tau}(x)\,\|,\end{aligned}$$

where the second inequality follows from the nonexpansiveness of Π_K and the third inequality follows from the Lipschitz continuity of F. An analogous argument proves the right-hand inequality in the lemma. □

Combining Propositions **12.6.2** and **12.6.3**, Exercise **4.8.4**, and Theorem **12.6.1**, we obtain the following result that gives a R-linear rate of convergence of the extragradient method for solving a pseudo monotone VI, under a local error bound for the solution set of the latter problem. No proof is needed.

12.6.4 Theorem. Let K be a closed convex set in $\mathbb{R}^n$ and let F be a mapping from K into $\mathbb{R}^n$ that is pseudo monotone on K with respect to $\mathrm{SOL}(K, F)$ and Lipschitz continuous on K with constant $L > 0$. Suppose that $\mathrm{SOL}(K, F)$ admits a local error bound on K with $\|\mathbf{F}_K^{\mathrm{nat}}(x)\|$ as the residual. For any $\tau \in (0, 1/L)$ and any $x^0 \in K$, the sequence $\{x^k\}$ generated by the extragradient Algorithm **12.1.9** converges to a solution of the VI (K, F) at least R-linearly. □

The above analysis provides a simple recipe for applying the general Theorem **12.6.1** to obtain the R-linear convergence of fixed-point iterations for solving the VI. First, we need to ensure that each fixed-point step provides a sufficient decrease for the distance function to the solution set of the VI (condition (12.6.2)). Second, we need to demonstrate that the fixed-point residual is equivalent to the natural residual. With these two requirements met, the R-linear rate then follows readily, provided that a local error bound holds for the solution set of the VI with the natural residual. See Section **6.2** for discussion of such error bounds.

12.6.2 The forward-backward splitting method

The forward-backward splitting iteration (12.4.2) is a special case of the iteration (12.6.1) with $\Phi_{\mathrm{fb}}(x) \equiv J_{cA}((\mathbf{I} - cB)(x))$. In this analysis, the (positive) constant c is fixed throughout the iterations and satisfies $c < 2\sigma$, where σ is the co-coercivity constant of B. From (12.4.4), we deduce that for all $x \in \mathrm{dom}\, A$, with $\Phi_{\mathrm{fb}}(x) = x - c(B(x) + z)$,

$$
\begin{aligned}
&\mathrm{dist}(\Phi_{\mathrm{fb}}(x), T^{-1}(0))^2 - \mathrm{dist}(x, T^{-1}(0))^2 \\
&\leq -c\,(2\sigma - c)\,\|\,B(x) - B(x^*)\,\|^2 - c^2\,\|\,z + B(x^*)\,\|^2 \\
&\leq -c\,\min(c, 2\sigma - c)\,[\,\|\,B(x) - B(x^*)\,\|^2 + \|\,z + B(x^*)\,\|^2\,] \\
&\leq -\frac{c\,\min(c, 2\sigma - c)}{2}\,\|\,B(x) + z\,\|^2 \;=\; -\frac{\min(\,c, 2\sigma - c\,)}{2\,c}\,\|x - \Phi_{\mathrm{fb}}(x)\,\|^2.
\end{aligned}
$$

Hence condition (12.6.2) holds with $\psi(x) \equiv \operatorname{dist}(x, T^{-1}(0))^2$. Under the identification: $T \equiv F + \mathcal{N}(\cdot; K)$, $F \equiv G + H$, $A \equiv G + \mathcal{N}(\cdot; K)$ and $B \equiv H$, we establish below the equivalence between the fixed-point residual $\|x - \Phi_{\rm fb}(x)\|$ and the scaled residual $\|\mathbf{F}^{\rm nat}_{K,c}(x)\|$ of the VI (K, F)

12.6.5 Proposition. Let K be a closed convex set in $\mathbb{R}^n$ and $F \equiv G + H$ be a mapping from K into $\mathbb{R}^n$ with G monotone and Lipschitz continuous on K with constant $L > 0$. For all $c \in (0, 1/L)$ and $x \in K$,

$$(1 + c\,L)\,\|\,\mathbf{F}^{\rm nat}_{K,c}(x)\,\| \;\geq\; \|\,x - J_{cA}((\mathbf{I} - cH)(x))\,\| \;\geq\; (1 - c\,L)\,\|\,\mathbf{F}^{\rm nat}_{K,c}(x)\,\|,$$

where $A \equiv G + \mathcal{N}(\cdot; K)$.

Proof. Let

$$r \equiv x - J_{cA}((\mathbf{I} - cH)(x)) \quad \text{and} \quad s \equiv \mathbf{F}^{\rm nat}_{K,c}(x).$$

By Proposition **12.3.6**(a), it follows that vector $x - r$ is a solution of the VI $(K, cG + \mathbf{I} - x + cH(x))$. Hence $x - r$ belongs to K and

$$(\,v - x + r\,)^T[\,c\,G(x - r) + c\,H(x) - r\,] \;\geq\; 0, \quad \forall\, v \;\in\; K.$$

By the definition of s, we have $x - s$ belongs to K and

$$(\,v - x + s\,)^T[\,c\,F(x) - s\,] \;\geq\; 0, \quad \forall\, v \;\in\; K.$$

Substituting $v = x - s$ into the former inequality and $v = x - r$ into the latter inequality and adding, we deduce

$$(\,r - s\,)^T[\,c\,(\,G(x - r) - G(x)\,) - r + s\,] \;\geq\; 0.$$

Rearranging terms, we obtain

$$c\,(\,r - s\,)^T(\,G(x - r) - G(x - s)\,) + c\,(\,r - s\,)^T(\,G(x - s) - G(x)\,) \;\geq\; \|\,r - s\,\|^2.$$

By the monotonicity and the Lipschitz continuity of G and the Cauchy-Schwarz inequality, we further deduce

$$c\,L\,\|\,s\,\| \;\geq\; \|\,r - s\,\|.$$

By the triangle inequality, we easily obtain the desired inequalities of the proposition. □

Employing the results in this subsection, we obtain the R-linear convergence of several iterative methods for solving the VI (K, F).

12.6.6 Theorem. Let K be a closed convex set in $\mathbb{R}^n$ and $F : K \to \mathbb{R}^n$ be such that $\mathrm{SOL}(K, F)$ is nonempty and has a local error bound with the natural residual.

(a) If F is co-coercive on K with modulus greater than $1/2$, then for any $x^0 \in K$, the sequence $\{x^k\}$ generated by the projection iteration (12.5.1)

$$x^{k+1} = \Pi_K(x^k - F(x^k)), \quad k = 0, 1, 2, \ldots,$$

converges to an element of $\mathrm{SOL}(K, F)$ at least R-linearly.

(b) If F is monotone and Lipschitz continuous on K with constant $L > 0$, then for any $c \in (0, 1/L)$ and $x^0 \in K$, the sequence $\{x^k\}$ generated by the proximal point iteration (12.4.9):

$$x^{k+1} \equiv \text{ unique solution of VI } (K, \mathbf{I} - x^k + cF), \quad k = 0, 1, 2, \ldots,$$

converges to an element of $\mathrm{SOL}(K, F)$ at least R-linearly.

(c) Let D be a positive definite matrix. Suppose that $\tilde{F}_D - C$ is co-coercive on $Y \equiv D_s^{1/2} K$ with modulus greater than $1/2$, where

$$C = D_s^{-1/2} (D - D_s) D_s^{-1/2},$$

and

$$\tilde{F}_D(y) \equiv D_s^{-1/2} F(D_s^{-1/2} y) \quad \forall\, y \in Y.$$

Assume further that $\|C\| < 1$. For any $x^0 \in K$, the sequence $\{x^k\}$ generated by the asymmetric projection iteration:

$$x^{k+1} \equiv \Pi_{K,D}(x^k - D^{-1} F(x^k)), \quad k = 0, 1, 2, \ldots,$$

converges to an element of $\mathrm{SOL}(K, F)$ at least R-linearly.

Proof. The first two iterative schemes in (a) and (b) are special cases of the forward-backing splitting iteration $x^{k+1} = \Phi_{\mathrm{fb}}(x^k)$ applied to the inclusion $0 \in F(x) + \mathcal{N}(x; K)$. For each of these two cases, it suffices to identify the splitting $F = G + H$ and the constant c and to observe that $B \equiv H$ is co-coercive with modulus greater than $\frac{1}{2}c$ and that G is monotone and Lipschitz continuous on K with constant less than c. For (a), we have $c \equiv 1$, $G \equiv 0$ and $H \equiv F$. For (b), we have $G \equiv F$ and $H \equiv 0$. For case (c), the proof of Theorem **12.5.2** shows that $x^{k+1} \equiv D_s^{-1/2} y^{k+1}$ where the sequence $\{y^k\}$ is obtained from the forward-backward splitting iteration with $c \equiv 1$ applied to the inclusion $0 \in \tilde{F}_D(y) + \mathcal{N}_Y(y)$ with the splitting

$\tilde{F}_D \equiv G + H$, where $G \equiv C$, $H \equiv \tilde{F}_D - C$. Consequently, $\{y^k\}$ converges to a solution of the latter inclusion at least R-linearly. It then follows easily that $\{x^k\}$ converges to an element of $\mathrm{SOL}(K, F)$ at least R-linearly. □

If $F(x) \equiv q + Mx$, we may choose a matrix $D \equiv \rho \mathbf{I} - G$ with $\rho > 0$ sufficiently large such that $\|C\| < 1$; indeed with the so-chosen D, we have

$$C \;=\; D_s^{-1/2}\,(\,G_s - G\,)D_s^{-1/2}.$$

This expression, together with the proof of Proposition **12.5.4**, shows that $\|C\|$ can be made arbitrarily small with $\rho > 0$ sufficiently large. Thus the conditions in part (c) of Theorem **12.6.6** are satisfied.

12.7 Equation Reduction Methods

In the previous sections we presented and analyzed Tikhonov regularization and proximal point methods. The central idea in these methods is to substitute the original VI by a sequence of better-behaved VIs. We saw that this could bring many advantages and yet, the subproblems that replace the original problem are structurally similar to the original one. In this section we take a step further and consider algorithms that call for the solution of a sequence of essentially unconstrained systems of (smooth) equations. Since solving systems of equations is conceptually simpler than solving a VI, these methods are clearly attractive. On the other hand, invariably the convergence of these methods requires assumptions that are stronger than the ones considered, for example, for proximal point algorithms.

The analysis of the algorithms in this section requires some results about recession and conjugate functions that we summarize in the next subsection. This summary is followed by two subsections where we consider algorithms that can be viewed as nonlinear extensions of the basic proximal point algorithm. In the fourth and last subsection we consider algorithms akin to the classical (interior and exterior) barrier methods for constrained optimization. All the algorithms presented in this section (except for the exterior barrier method in Subsection **12.7.4**) generate iterates that are contained in the interior of the closed convex set defining the VI; in this regard, they share many common features with the methods of Chapter **11**. Nevertheless, a key difference between the IP methods in the previous chapter and the algorithms in the present chapter is that the former methods involve solving linear equations that are the result of the linearization of some nonlinear equations whereas the latter methods involve solving nonlinear equations of a different kind by unspecified methods. The question of whether refinements of the algorithms developed in this section can be

designed so that the resulting refined algorithms will parallel more closely the IP methods in Chapter **11** remains open and deserves to be investigated. Such refinements will most likely be easiest to obtain in the case of CPs or VIs with special structures, such as VIs on rectangles.

12.7.1 Recession and conjugate functions

All the facts we report here, with the exception of Proposition **12.7.1** which we give a proof for, are standard; see Section **12.9** for relevant references. We start with some facts about recession cones that complement those already presented in Section **2.3**.

Let X be a nonempty closed convex set in $\mathbb{R}^n$. The recession cone of X is defined by

$$X_\infty \equiv \{ d \in \mathbb{R}^n : x + \tau d \in X, \ \forall x \in X, \ \forall \tau \geq 0 \}.$$

It is known that a vector d is a recession direction of X if and only if there exists a sequence of scalars $\{\tau_k\}$ tending to infinity and a sequence of vectors $\{x^k\} \subset X$ such that

$$d = \lim_{k \to \infty} \tau_k^{-1} x^k.$$

This characterization is useful to understand some properties of "recession functions" that we introduce next.

Let $\varphi : \mathbb{R}^n \to \mathbb{R} \cup \{+\infty\}$ be an extended-valued convex function. In order to avoid lengthy specifications about the domains of functions in this section, unless otherwise specified, all convex functions are considered to be extended-valued defined on the whole space $\mathbb{R}^n$. Adopting standard notation in convex analysis, we denote by $\operatorname{dom} \varphi$ the set of points where φ takes finite values:

$$\operatorname{dom} \varphi \equiv \{ x \in \mathbb{R}^n : \varphi(x) < +\infty \}.$$

The convex function φ is said to be *proper* if it has a nonempty domain. The *epigraph* of the convex function φ is the convex set defined by

$$\operatorname{epi} \varphi \equiv \{ (x, \eta) \in (\operatorname{dom} \varphi) \times \mathbb{R} : \varphi(x) \leq \eta \}.$$

The function φ is said to be *closed* if its epigraph is closed. It is not difficult to see that φ is closed if and only if it is lower semicontinuous. The epigraph of a closed proper convex function c contains all the information about the function, in particular

$$\varphi(x) = \inf \{ y \in \mathbb{R} : (x, y) \in \operatorname{epi} \varphi \},$$

while $\operatorname{dom}\varphi$ is just the canonical projection on the first n components of the epigraph of φ.

To each closed proper convex function φ we can associate a convex function φ_∞, which is called the *recession function* of φ and whose definition in terms of its epigraph is given by

$$\operatorname{epi}(\varphi_\infty) \equiv (\operatorname{epi}\varphi)_\infty.$$

The recession function φ_∞ in some sense characterizes the behavior at infinity of φ in the same way as the recession cone of a convex set K does the behavior of K at infinity. Since the recession cone of a convex set is always a closed set, recession functions are always closed or, equivalently, lower semicontinuous. From its definition, we can see that

$$\varphi_\infty(d) = \inf\left\{ \liminf_{k\to\infty} \frac{\varphi(\tau_k d^k)}{\tau_k} : \{\tau_k\} \to \infty, \{d^k\} \to d \right\}. \tag{12.7.1}$$

It is known that, for any convex function φ and $x \in \operatorname{dom}\varphi$,

$$\varphi_\infty(d) = \lim_{\tau\to\infty} \frac{\varphi(x+\tau d) - \varphi(x)}{\tau}, \quad \forall\, d \in \mathbb{R}^n.$$

This relation clearly shows that $\varphi_\infty(d)$ describes the "far behavior" of φ in the direction d. Furthermore it also reveals that the recession function is positively homogeneous. Since the quotient in the right-hand side of the above limit is a nondecreasing function of τ, we have

$$\varphi_\infty(d) = \sup_{\tau>0} \frac{\varphi(x+\tau d) - \varphi(x)}{\tau}.$$

If K is a nonempty, closed convex, finitely representable set, that is if

$$K \equiv \{\, x \in \mathbb{R}^n : \varphi_i(x) \le 0,\ i = 1, \ldots, m \,\},$$

where each φ_i is a closed proper convex function, then the recession cone of K can be described by using the recession functions of the constraint functions φ_i:

$$K_\infty = \{\, d \in \mathbb{R}^n : (\varphi_i)_\infty(d) \le 0,\ i = 1, \ldots, m \,\}. \tag{12.7.2}$$

We next give a formula for the calculation of the recession function of the composition of two convex functions that we will use in Subsection **12.7.4**.

12.7.1 Proposition. Let $r : \mathbb{R} \to \mathbb{R} \cup \{\infty\}$ be a convex nondecreasing function with $\operatorname{dom} r = (-\infty, \eta)$ for some $\eta \in [0, \infty]$ and with $r_\infty(1) = \infty$.

Let $\varphi : \mathbb{R}^n \to \mathbb{R} \cup \{\infty\}$ be a closed proper convex function such that $\varphi(\mathbb{R}^n) \cap \operatorname{dom} r \neq \emptyset$. Consider the composite function

$$g(x) \equiv \begin{cases} r(c(x)) & \text{if } x \in \operatorname{dom} \varphi \\ +\infty & \text{otherwise.} \end{cases}$$

The function g is closed proper and convex and

$$g_\infty(d) = \begin{cases} r_\infty(\varphi_\infty(d)) & \text{if } d \in \operatorname{dom}(\varphi_\infty) \\ +\infty & \text{otherwise.} \end{cases}$$

Proof. It is easy to see that g is closed proper and convex. Let x belong to $\operatorname{dom} \varphi$ be such that $\varphi(x) \in \operatorname{dom} r$. For every $a < \varphi_\infty(d)$ we have

$$\varphi(x + \tau d) \geq \varphi(x) + \tau\, a$$

if $\tau > 0$ is sufficiently large. Therefore, since r is nondecreasing, we get

$$\frac{g(x + \tau d) - g(x)}{\tau} \geq \frac{r(\varphi(x) + \tau a) - r(\varphi(x))}{\tau}$$

and, passing to the limit $\tau \to \infty$, we deduce $g_\infty(d) \geq r_\infty(a)$.

If $\varphi_\infty(d) = \infty$, we can take $a = 1$, so that since $r_\infty(1) = \infty$, we get $g_\infty(d) = \infty$. If $\varphi_\infty(d) < \infty$, then since r_∞ is lower semicontinuous, by letting $a \to \varphi_\infty(d)$, we get $g_\infty(d) \geq r_\infty(\varphi_\infty(d))$. But using the monotonicity of r and recalling that $\varphi(x + \tau d) \leq \varphi(x) + \tau \varphi_\infty(d)$, we also get

$$\begin{aligned} g_\infty(d) &= \lim_{\tau \to \infty} \frac{g(x + \tau d) - g(x)}{\tau} \\ &\leq \lim_{\tau \to \infty} \frac{r(\varphi(x) + \tau \varphi_\infty(d)) - r(\varphi(x))}{\tau} \\ &= r_\infty(\varphi_\infty(d)), \end{aligned}$$

thus equality must hold between $g_\infty(d)$ and $r_\infty(\varphi_\infty(d))$. □

We review a few facts about "conjugate functions" that are used in the proof of Proposition **12.7.3**. Let $\varphi : \mathbb{R}^n \to \mathbb{R} \cup \{\infty\}$ be a convex function; its *conjugate*, denoted by φ^*, is defined by

$$\varphi^*(y) \equiv \sup_{x \in \mathbb{R}^n} \{\, y^T x - \varphi(x) \,\}.$$

It is not difficult to show that φ^* is a closed convex function. Conjugate functions play an important role in deriving duality schemes and optimality conditions, but for our subsequence purposes, conjugate functions allow us to simplify certain calculations. A natural question is whether $\varphi^*(y)$ is

finite. There are many results in this vein, the one we need is the following. Suppose that φ is additionally closed and proper, then

$$\operatorname{int}\operatorname{dom}(\varphi^*) = \{\, y \in \mathbb{R}^n : y^T d < \varphi_\infty(d),\ \forall\, d \neq 0 \,\}.$$

In view of the definition of conjugate and recession function, the above expression is not surprising. Since it can be shown that if φ is closed proper and convex then $(\varphi^*)^* = \varphi$ holds, we can easily derive from the previous formula that

$$\operatorname{int}\operatorname{dom}\varphi = \{\, x \in \mathbb{R}^n : x^T d < (\varphi^*)_\infty(d),\ \forall\, d \neq 0 \,\}. \tag{12.7.3}$$

A final technical result we need is that if φ is a closed convex function with $\operatorname{int}\operatorname{dom}\varphi \neq \emptyset$ and φ is differentiable in this interior, then for every x in $\operatorname{int}\operatorname{dom}\varphi$, $\varphi^*(\nabla\varphi(x))$ is finite and

$$\varphi(x) + \varphi^*(\nabla\varphi(x)) = \nabla\varphi(x)^T x. \tag{12.7.4}$$

12.7.2 Bregman-based methods

The class of methods we consider herein is based on "Bregman functions and Bregman distances". To motivate the approach recall the proximal point iteration (with exact evaluation of the resolvent) for solving a VI. Given the current iterate x^k, x^{k+1} is the solution of the VI $(K, c_k F + \mathbf{I} - x^k)$. The idea behind the methods considered in this and in the next section is to substitute the simple function $\mathbf{I} - x^k$ by an appropriate nonlinear term. In particular we may calculate x^{k+1} as the solution of the sub-VI $(K, c_k F + \nabla f - \nabla f(x^k))$, where f is a suitable real-valued function. The key thing we want to accomplish is that by using a suitable f the VI $(K, c_k F + \nabla f - \nabla f(x^k))$ reduces to an essentially unconstrained system of nonlinear equations defined on $\operatorname{int} K$, which we assume to be nonempty.

Bregman functions

In order to better understand the properties we need to place on f, we first observe that by setting $f \equiv (1/2)\|\cdot\|^2$ we can write

$$\tfrac{1}{2}\ \operatorname{dist}(x,y)^2 = f(x) - f(y) - \nabla f(y)^T(\,x - y\,)$$

for any x and y in $\mathbb{R}^n$. This identity suggests to define a sort of "generalized distance" by setting

$$D_f(x,y) \equiv f(x) - f(y) - \nabla f(y)^T(\,x - y\,),$$

where f is yet to be specified. See Figure 12.6 for an illustration. Since $D_f(x,y) \neq D_f(y,x)$ in general, the resulting function D_f is not a true distance function. (For a stronger reason why D_f is not a true distance function, see the comment following Definition **12.7.2**.)

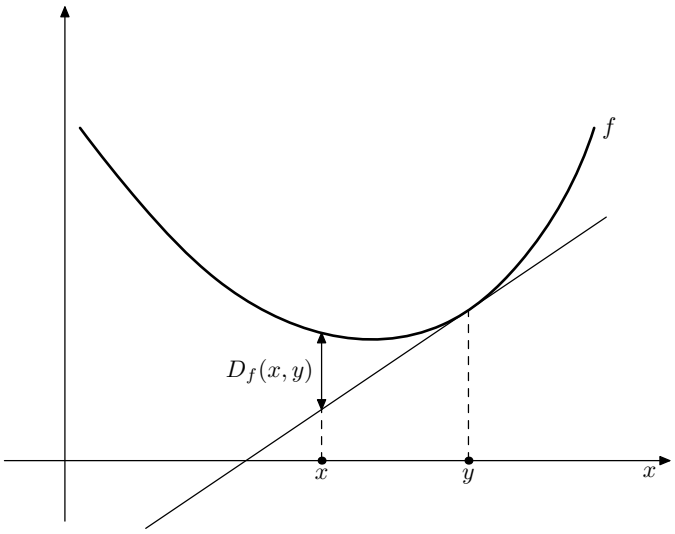


Figure 12.6: The definition of $D_f(x,y)$.

Let K be a solid closed convex set in $\mathbb{R}^n$ and let f be a strictly convex continuous function defined on K which is continuously differentiable on $\operatorname{int} K$. The function D_f can be given a simple geometric interpretation: $D_f(x,y)$ is the difference between $f(x)$ and the value at x of the linearized approximation at y of $f(x)$. By the strict convexity of f it follows that, for any $x \in K$ and $y \in \operatorname{int} K$, the function $D_f(x,y)$ is nonnegative and it is zero if and only if $x = y$. In essence, we want to impose some additional properties on f restricted to $\operatorname{int} K$ so that $D_f(x,y)$ behaves on $\operatorname{int} K$ somewhat like $\frac{1}{2}\operatorname{dist}(x,y)^2$ does on $\mathbb{R}^n$. (For example, we want for x and y in $\operatorname{int} K$, $D_f(x,y) = 0 \Leftrightarrow x = y$; see Corollary **12.7.7**.)

With the above introduction, we give the formal definition of a Bregman function.

12.7.2 Definition. Let K be a solid closed convex set in $\mathbb{R}^n$. A function $f : K \to \mathbb{R}$ is a *Bregman function* with zone K if

(a) f is strictly convex and continuous on K;

(b) f is continuously differentiable on $\operatorname{int} K$;

(c) for all $x \in K$ and all constants η, the set

$$\{\, y \in \operatorname{int} K \,:\, D_f(x,y) \le \eta \,\}$$

is bounded;

(d) if $\{x^k\}$ is a sequence of points in $\operatorname{int} K$ converging to x, then

$$\lim_{k\to\infty} D_f(x, x^k) = 0.$$

If f is a Bregman function, we say that D_f is a *Bregman distance*. If in addition f also satisfies the following condition:

(e) $\nabla f(\operatorname{int} K) = \mathbb{R}^n$;

then we say that f is a *full range Bregman function*. □

While conditions (a)–(d) are usually rather easy to satisfy and check, condition (e) is needed in order to establish the existence of the solution in $\operatorname{int} K$ of the sub-VI $(K, c_k F + \nabla f - \nabla f(x^k))$. Notice that whereas $D_f(x,y)$ is well defined for any $x \in K$ and $y \in \operatorname{int} K$, $D_f(y,x)$ is not defined unless x also belongs to $\operatorname{int} K$ because ∇f is only well defined on $\operatorname{int} K$. This is a stronger reason why the Bregman distance is not a true distance.

It is easy to check that $f(x) = (1/2)\|x\|^2$ is a full range Bregman function with zone $\mathbb{R}^n$. Before considering other examples it may be interesting to discuss further the key condition (e). Roughly speaking this condition requires that ∇f "stretches" the interior of K onto $\mathbb{R}^n$, thus making the solution of the sub-VIs essentially unconstrained. The following proposition not only sheds some light on this condition, but may be useful in verifying the satisfaction of (e). It also makes clear some consequences of this condition that will be used subsequently.

12.7.3 Proposition. Let f be a Bregman function with zone $K \subseteq \mathbb{R}^n$. Consider the following statements:

(a) $\nabla f(\operatorname{int} K) = \mathbb{R}^n$;

(b) the level sets

$$L(\eta) \equiv \{\, x \in \operatorname{int} K \,:\, \|\nabla f(x)\| \le \eta \,\} \tag{12.7.5}$$

are compact for all $\eta \in \mathbb{R}$.

(c) for all $x \in \operatorname{int} K$, if $\{y^k\}$ is a sequence of points in $\operatorname{int} K$ converging to a boundary vector of K, then

$$\lim_{k\to\infty} \nabla f(y^k)^T (\, y^k - x \,) = \infty.$$

It holds that (a) $\Rightarrow$ (b) $\Rightarrow$ (c). If f is twice continuously differentiable on $\operatorname{int} K$ with a nonsingular Hessian at all points in $\operatorname{int} K$, then (a) and (b) are equivalent.

Proof. As a preliminary observation we show that if u and v belong to $\operatorname{int} K$ and $w \equiv (1-\tau)u + \tau v$, with $\tau \in (0,1)$, then

$$\begin{aligned} 0 &\leq (\nabla f(u) - \nabla f(w))^T (u - w) \\ &\leq \tau (\nabla f(u) - \nabla f(v))^T (u - v). \end{aligned} \tag{12.7.6}$$

We only need to show the second inequality, since the first follows from the monotonicity of ∇f on $\operatorname{int} K$. Note that

$$u - w = \tau (u - v) \quad \text{and} \quad w - v = (1 - \tau)(u - v).$$

Therefore we can write

$$\begin{aligned} &\tau (\nabla f(u) - \nabla f(v))^T (u - v) \\ &\quad = \tau (\nabla f(u) - \nabla f(w))^T (u - v) + \tau (\nabla f(w) - \nabla f(v))^T (u - v) \\ &\quad = (\nabla f(u) - \nabla f(w))^T (u - w) + \frac{\tau}{1-\tau} (\nabla f(w) - \nabla f(v))^T (w - v) \\ &\quad \geq (\nabla f(u) - \nabla f(w))^T (u - w), \end{aligned}$$

where the last inequality follows from the monotonicity of ∇f and the fact that $\tau \in (0,1)$.

Suppose that (a) holds and assume for the sake of contradiction that (b) does not hold. A sequence $\{y^k\} \subset \operatorname{int} K$ and an $\eta \in \mathbb{R}$ exist such that, for all k, $\|\nabla f(y^k)\| \leq \eta$ and either

$$\lim_{k\to\infty} \|y^k\| = \infty \quad \text{or} \quad \lim_{k\to\infty} y^k = y^\infty \in \operatorname{bd} K.$$

Since $\{\nabla f(y^k)\}$ is bounded we may assume without loss of generality that $\{\nabla f(y^k)\}$ converges to a vector $z \in \mathbb{R}^n$. By (a), a $\bar{y} \in \operatorname{int} K$ exists such that $\nabla f(\bar{y}) = z$. Set, for $\tau_k \in (0,1)$, $\tilde{y}^k \equiv (1-\tau_k)\bar{y} + \tau_k y^k$. By (12.7.6) we get

$$\begin{aligned} 0 &\leq (\nabla f(\bar{y}) - \nabla f(\tilde{y}^k))^T (\bar{y} - \tilde{y}^k) \\ &\leq \tau_k \nabla f(\bar{y}) - \nabla f(y^k))^T (\bar{y} - y^k). \end{aligned} \tag{12.7.7}$$

If $\{y^k\}$ tends to $y^\infty \in \operatorname{bd} K$, take $\tau_k \equiv 1/2$ for all k, so that

$$\lim_{k\to\infty} \tilde{y}^k \equiv \tilde{y} = \lim_{k\to\infty} \tfrac{1}{2}(y^\infty + \bar{y}),$$

and $\tilde{y} \in \operatorname{int} K$. Passing to the limit in (12.7.7) we therefore get

$$(\nabla f(\bar{y}) - \nabla f(\tilde{y}))^T (\bar{y} - \tilde{y}) = 0,$$

which implies, by the strict monotonicity of ∇f, $\bar{y} = \tilde{y} = \frac{1}{2}(y^\infty + \tilde{y})$. The latter relation implies, in turn, $\bar{y} = y^\infty$, which is a contradiction because $y^\infty \in \operatorname{bd} K$ while $\bar{y} \in \operatorname{int} K$.

Assume then that $\lim_{k\to\infty} \|y^k\| = \infty$. In this case we take $\tau_k \equiv \|y^k - \bar{y}\|^{-1}$ which, for k large enough, is in $(0,1)$. With this choice we have $\|\tilde{y}^k - \bar{y}\| = 1$ for every k, so that $\{\tilde{y}^k\}$ is bounded and we may assume, without loss of generality, that $\{\tilde{y}^k\}$ converges to $\tilde{y}$. Take limits in (12.7.7) and note that the rightmost-hand side converges to zero because

$$\lim_{k\to\infty} \nabla f(y^k) \;=\; \nabla f(\bar{y}) \quad \text{and} \quad \|\, \tau_k\, (\, \bar{y} - y^k\,)\,\| \;=\; 1.$$

Similar to the previous case, we deduce

$$(\, \nabla f(\bar{y}) - \nabla f(\tilde{y})\,)^T (\, \bar{y} - \tilde{y}\,) \;=\; 0,$$

which implies $\bar{y} = \tilde{y}$. This is impossible because $\|\bar{y} - \tilde{y}\| = 1$.

To derive the implication (b) $\Rightarrow$ (c), let $\{y^k\}$ be a sequence of points in $\operatorname{int} K$ converging to a vector in $\operatorname{bd} K$. By (12.7.4) we can write

$$f(y^k) + f^*(\nabla f(y^k)) \;=\; \nabla f(y^k)^T y^k$$

so that

$$D_f(x, y^k) \;=\; f(x) + f^*(\nabla f(y^k)) - \nabla f(y^k)^T x. \tag{12.7.8}$$

We claim that

$$\lim_{k\to\infty} D_f(x, y^k) \;=\; +\infty. \tag{12.7.9}$$

Assume for contradiction the contrary; that is assume, without loss of generality, that

$$\limsup_{k\to\infty} D_f(x, y^k) \;<\; +\infty.$$

By (b), $\{\nabla f(y^k)\}$ is unbounded. Without loss of generality, assume that

$$\lim_{k\to\infty} \|\, \nabla f(y^k)\,\| \;=\; +\infty \quad \text{and} \quad \lim_{k\to\infty} \frac{\nabla f(y^k)}{\|\, \nabla f(y^k)\,\|} \;=\; d \;\neq\; 0.$$

Dividing (12.7.8) by $\|\nabla f(y^k)\|$ and passing to the limit we get

$$\lim_{k\to\infty} \frac{f^*(\nabla f(y^k))}{\|\, \nabla f(y^k)\,\|} \;=\; d^T x$$

which, by (12.7.1), implies

$$d^T x \;\geq\; (f^*)_\infty(d).$$

But by (12.7.3) this implies $x \notin \operatorname{int} K$, a contradiction. Therefore (12.7.9) holds. But this, in turn, easily implies (c).

Finally, assume that f is twice continuously differentiable on $\operatorname{int} K$ with a nonsingular Hessian at all points in $\operatorname{int} K$. Since $\nabla f(\operatorname{int} K)$ is nonempty, it suffices to show that $\nabla f(\operatorname{int} K)$ is both open and closed. The fact that $\nabla f(\operatorname{int} K)$ is open follows from the nonsingularity of $\nabla^2 f$ and the open mapping theorem. In fact, the nonsingularity of $\nabla^2 f$ ensures that ∇f is a local homeomorphism in some open neighborhood of any point in $\operatorname{int} K$. It remains to show that $\nabla f(\operatorname{int} K)$ is closed. Let $\{z^k\}$ be a sequence in $\nabla f(\operatorname{int} K)$ converging to z^∞; we want to show that z^∞ belongs to $\nabla f(\operatorname{int} K)$. Consider the level set $L(2\|z^\infty\|)$. For all k there exists an $x^k \in \operatorname{int} K$ such that $\nabla f(x^k) = z^k$. By continuity, x^k belongs to $L(2\|z^\infty\|)$ for all k large enough. By (b) $L(2\|z^\infty\|)$ is compact and therefore we may assume without loss of generality, that $\{x^k\}$ converges to a vector $x^\infty \in L(2\|z^\infty\|)$. Again by continuity, we therefore have $\nabla f(x^\infty) = z^\infty$, showing that $\nabla f(\operatorname{int} K)$ is closed. □

In what follows, we give two nontrivial examples of Bregman functions. The first one concerns the nonnegative orthant and is therefore relevant to the nonlinear complementarity problem. Figure 12.7 illustrates a Bregman function (on the left) and a full range Bregman function (on the right).

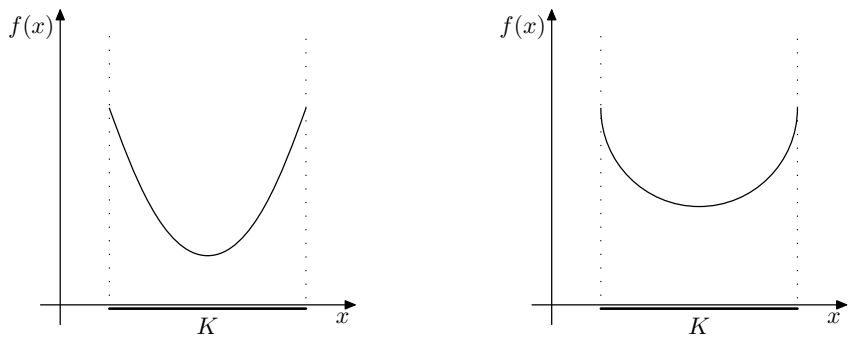


Figure 12.7: More Bregman functions.

12.7.4 Example. Set $K = \mathbb{R}^n_+$ and consider the function $f : K \to \mathbb{R}$ defined by

$$f(x) \equiv \sum_{i=1}^{n} x_i \log x_i,$$

with the convention that $0 \log 0 = 0$ (this convention is assumed to hold throughout this example). It is immediate to check that conditions (a) and

(b) of Definition **12.7.2** are met. Direct calculations yield

$$D_f(x,y) \;=\; \sum_{i=1}^{n} \left(x_i \log \frac{x_i}{y_i} + y_i - x_i \right),$$

from which we see that also conditions (c) and (d) in the definition hold. Therefore f is a Bregman function with zone $\mathbb{R}^n_+$. We show that f is a full range Bregman function. In fact

$$\frac{\partial f(x)}{\partial x_i} \;=\; 1 + \log x_i \,;$$

it is therefore clear that $\nabla f(\mathbb{R}^n_{++}) = \mathbb{R}^n$. □

Another example of a Bregman function for the nonnegative orthant is given in Exercise **12.8.22**. We next consider a simple extension in which K is a bounded rectangle given by (1.1.7).

12.7.5 Example. Suppose that

$$K \;\equiv\; \{\, x \in \mathbb{R}^n \,:\, a_i \le x_i \le b_i,\; i = 1, \ldots n \,\},$$

where for each i, $-\infty < a_i < b_i < \infty$. Consider the function $f : K \to \mathbb{R}$ defined by

$$f(x) \;\equiv\; \sum_{i=1}^{n} \left[(x_i - a_i) \log(x_i - a_i) + (b_i - x_i) \log(b_i - x_i) \right]$$

where, again, we make the convention that $0 \log 0 = 0$. An easy calculation shows that

$$D_f(x,y) \;=\; \sum_{i=1}^{n} \left[(x_i - a_i) \log \left(\frac{x_i - a_i}{y_i - a_i} \right) + (b_i - x_i) \log \left(\frac{b_i - x_i}{b_i - y_i} \right) \right],$$

also in this case there are no difficulties in verifying that f is a full range Bregman function. □

Combining the previous two examples, it is possible to define easily a full range Bregman function for an unbounded rectangle, where some of the variables may have unbounded lower or upper bounds. In Exercise **12.8.22** the reader is asked to analyze a different Bregman function for rectangles and also to extend the approach of the previous two examples to a general polyhedral set.

We conclude the analysis of Bregman functions by proving some technical properties that will be used in the convergence analysis of the Bregman proximal point algorithm. The identity in part (a) of the following proposition is called the *three-point formula* for the Bregman function.

12.7.6 Proposition. Let f be a Bregman function with zone K. The following statements hold.

(a) For every y and z in $\operatorname{int} K$ and for every x in K, it holds that

$$D_f(x,y) \;=\; D_f(x,z) + D_f(z,y) + (\,\nabla f(y) - \nabla f(z)\,)^T(\,z - x\,).$$

(b) Let $\{x^k\}$ and $\{y^k\}$ be sequences in K and in $\operatorname{int} K$, respectively. If

$$\lim_{k\to\infty} D_f(x^k, y^k) \;=\; 0$$

and either $\{x^k\}$ or $\{y^k\}$ converges, the other sequence also converges and to the same limit.

Proof. (a) By the definition of D_f we have

$$\begin{aligned}
\nabla f(z)^T(\,x - z\,) &= f(x) - f(z) - D_f(x,z),\\
\nabla f(y)^T(\,z - y\,) &= f(z) - f(y) - D_f(z,y),\\
\nabla f(y)^T(\,x - y\,) &= f(x) - f(y) - D_f(x,y).
\end{aligned}$$

Subtracting the first two equalities from the third gives the desired result.

The proof of (b) is carried out in several steps. We first show that if x belongs to K, y to $\operatorname{int} K$, and $w = y + (1-\rho)(x-y)$ with $\rho \in (0,1)$, then

$$D_f(x,w) + D_f(w,y) \;\le\; D_f(x,y). \tag{12.7.10}$$

Clearly w belongs to $\operatorname{int} K$. Substituting $z = w$ into the three-point formula in (a), we have

$$D_f(x,y) \;=\; D_f(x,w) + D_f(w,y) + (\,\nabla f(y) - \nabla f(w)\,)^T(\,w - x\,).$$

Furthermore, since

$$w - x \;=\; \rho\,(\,y - x\,) \;=\; \frac{\rho}{1-\rho}\,(\,y - w\,),$$

(12.7.10) follows readily by the monotonicity of f on K. We next prove that if $\{x^k\}$ is a sequence of points in K converging to x and if $\{y^k\}$ is a sequence in $\operatorname{int} K$ converging to $y \neq x$, then

$$\liminf_{k\to\infty} D_f(x^k, y^k) \;>\; 0.$$

Set $z^k \equiv \frac{1}{2}(x^k + y^k)$. The sequence $\{z^k\}$ is in $\operatorname{int} K$ and converges to $z \equiv \frac{1}{2}(x+y) \in K$. By the convexity of f we can write

$$\begin{aligned}
f(z^k) &\ge f(y^k) + \nabla f(y^k)^T(z^k - y^k)\\
&= f(y^k) + \tfrac{1}{2}\,\nabla f(y^k)^T(x^k - y^k).
\end{aligned}$$

Therefore

$$\begin{aligned}\frac{f(x^k)-f(y^k)}{2}-f(z^k) &\leq \frac{f(x^k)-f(y^k)}{2}\\ &\quad -f(y^k)-\tfrac{1}{2}\,\nabla f(y^k)^T(\,x^k-y^k\,)\\ &= \tfrac{1}{2}\,D_f(x^k,y^k).\end{aligned}$$

Passing to the limit $k\to\infty$, we get

$$\frac{f(x)-f(y)}{2}-f(z)\;\leq\;\tfrac{1}{2}\liminf_{k\to\infty} D_f(x^k,y^k).$$

By the strict convexity of f and the fact that $x\neq y$, the left-hand side, and thus the right-hand side, is positive.

We can now complete the proof of (b). Suppose for the sake of contradiction that one of the sequences converges and the other does not converge or converges to a different limit. There exists a subsequence κ and a positive ε for which

$$\|\,x^k-y^k\,\|\;>\;\varepsilon,\qquad \forall\,k\;\in\;\kappa.$$

Suppose first that $\{y^k\}$ converges to y. Set

$$w^k\;\equiv\;y^k+\frac{\varepsilon}{\|\,x^k-y^k\,\|}\,(\,x^k-y^k\,).$$

By (12.7.10), it holds that for every $k\in\kappa$, $D_f(w^k,y^k)\leq D_f(x^k,y^k)$; thus

$$\lim_{k(\in\kappa)\to\infty} D_f(w^k,y^k)\;=\;0.$$

Since $\|w^k-y^k\|=\varepsilon$ for every $k\in\kappa$ and $\{y^k\}$ converges, it follows that $\{w^k:k\in\kappa\}$ is bounded. Without loss of generality, we may assume that $\{w^k:k\in\kappa\}$ converges to a w. We have

$$\lim_{k(\in\kappa)\to\infty} D_f(w^k,y^k)\;=\;0,\qquad \lim_{k(\in\kappa)\to\infty} y^k\;=\;y^\infty$$

$$\lim_{k(\in\kappa)\to\infty} w^k\;=\;w,\qquad \|\,w-y^\infty\,\|\;=\;\varepsilon,$$

which is a contradiction. The case in which $\{x^k\}$ converges can be analyzed similarly by exchanging the roles of $\{x^k\}$ and $\{y^k\}$. □

Part (b) of Proposition **12.7.6** yields the following consequence, which implies that the converse of part (d) of Definition **12.7.2** is valid.

12.7.7 Corollary. Let f be a Bregman function with zone K. A sequence $\{x^k\}\subset\operatorname{int}K$ converges to a limit x^∞ if and only if

$$\lim_{k\to\infty} D_f(x^\infty,x^k)\;=\;0.$$

Proof. The "only if" statement follows from part (d) of Definition **12.7.2**; the "if" statement follows from part (b) of Proposition **12.7.6**. □

Bregman proximal point algorithms

We apply the previous results to the development of a Bregman proximal point method for solving a VI (K, F) with F monotone and K a solid closed convex set. The algorithm has already been outlined in the introduction to this section. Below we formally give the complete statement of the algorithm which includes the possibility of inexact solution of the subproblems.

Bregman Proximal Point Algorithm (BPPA)

12.7.8 Algorithm.

Data: $x^0 \in \operatorname{int} K$, $c_0 > 0$, $\varepsilon_0 \geq 0$, and a Bregman function f with zone K.

Step 1: Set $k = 0$.

Step 2: If x^k is a solution of VI (K, F), stop.

Step 3: Find $x^{k+1} \in \operatorname{int} K$ such that $\|e^{k+1}\| \leq \varepsilon_k$, where

$$e^{k+1} \equiv c_k F(x^{k+1}) + \nabla f(x^{k+1}) - \nabla f(x^k).$$

Step 4 Select c_{k+1} and ε_{k+1}. Set $k \leftarrow k+1$ and go to Step 2.

In Step 3, we are in essence solving the (nonlinear) equation

$$c_k F(x) + \nabla f(x) - \nabla f(x^k) = 0 \tag{12.7.11}$$

inexactly; we accept an approximate solution of the equation as the next iterate x^{k+1} if the residual $\|e^{k+1}\|$ is less than the prescribed tolerance ε_k and x^{k+1} is an interior point of K. A key step in the analysis of Algorithm **12.7.8** is the demonstration that such a vector x^{k+1} is well defined. This is accomplished via Theorem **11.2.1**, which is the basis for proving the following preliminary result.

12.7.9 Lemma. Let K be a solid, closed convex subset of $\mathbb{R}^n$. Suppose that $F : K \to \mathbb{R}^n$ is continuous and strongly monotone on K and that $H : \operatorname{int} K \to \mathbb{R}^n$ is continuous, surjective, and strictly monotone on $\operatorname{int} K$. The sum $F + H$ maps $\operatorname{int} K$ homeomorphically onto $\mathbb{R}^n$.

Proof. It is clear that $G \equiv F + H$ is strictly monotone and thus injective. Consequently, G is a local homeomorphism mapping $\operatorname{int} K$ into $\mathbb{R}^n$. With $M_0 = M = \operatorname{int} K$ and $N_0 = E = N = \mathbb{R}^n$, it follows from Theorem **11.2.1** that if $G : \operatorname{int} K \to \mathbb{R}^n$ is proper with respect to $\mathbb{R}^n$, then $G(\operatorname{int} K) = \mathbb{R}^n$ and the proof is complete. To establish the properness requirement, let S be a compact set in $\mathbb{R}^n$, it suffices to show that the set

$$G^{-1}(S) \equiv \{\, x \in \operatorname{int} K \,:\, F(x) + H(x) \in S \,\}$$

is compact. Let $\{x^k\}$ be an arbitrary sequence of vectors in $G^{-1}(S)$. We first show that $\{x^k\}$ is bounded. Since $\{F(x^k) + H(x^k)\} \subset S$ and S is compact, we may assume without loss of generality that

$$\lim_{k\to\infty} (\, F(x^k) + H(x^k) \,) = q^\infty$$

exists. By the surjectivity of H, it follows that there exists a vector x^∞ in $\operatorname{int} K$ such that $H(x^\infty) = q^\infty$. Write

$$r^k \equiv F(x^k) + H(x^k) - H(x^\infty);$$

then

$$\lim_{k\to\infty} r^k = 0.$$

Let $c > 0$ be a constant of strong monotonicity of F; thus

$$(\, x^k - x^\infty \,)^T (\, F(x^k) - F(x^\infty) \,) \geq c \,\| x^k - x^\infty \|^2.$$

We have

$$\begin{aligned}
&(\, x^k - x^\infty \,)^T r^k \\
&\quad = (\, x^k - x^\infty \,)^T (\, F(x^k) - F(x^\infty) \,) + (\, x^k - x^\infty \,)^T F(x^\infty) \\
&\qquad + (\, x^k - x^\infty \,)^T (\, H(x^k) - H(x^\infty) \,) \\
&\quad \geq c \,\| x^k - x^\infty \|^2 + (\, x^k - x^\infty \,)^T F(x^\infty).
\end{aligned}$$

Since $(x^k - x^\infty)^T r^k$ is of order $o(\|x^k - x^\infty\|)$, the boundedness of $\{x^k\}$ follows readily.

To complete the proof, we need to show that $G^{-1}(S)$ is closed. Let $\{x^k\} \subset G^{-1}(S)$ be a sequence converging to a vector $\bar{x}$. By the continuity of G on its domain, it suffices to show that $\bar{x}$ belongs to $\operatorname{int} K$. As above, we may assume without loss of generality that $\{F(x^k) + H(x^k)\}$ converges to a vector q^∞. The sequence $\{H(x^k)\}$ converges to $q^\infty - F(\bar{x})$. By the surjectivity of H, there exists a vector x^∞ in $\operatorname{int} K$ such that

$$H(x^\infty) = q^\infty - F(\bar{x}).$$

Let $\varepsilon > 0$ be such that the closed neighborhood $\operatorname{cl} \mathbb{B}(x^\infty, \varepsilon)$ is contained in $\operatorname{int} K$. Since H is injective and continuous, by the open mapping theorem, the image $H(\mathbb{B}(x^\infty, \varepsilon))$ is an open set containing $H(x^\infty)$, which is the limit of $\{H(x^k)\}$. Hence $H(x^k)$ belongs to $H(\mathbb{B}(x^\infty, \varepsilon))$ for all k sufficiently large. By the injectivity of H, it follows that x^k belongs to $\mathbb{B}(x^\infty, \varepsilon)$ for all k sufficiently large. Consequently, the limit of $\{x^k\}$, which is $\bar{x}$, must belong to the closure of $\mathbb{B}(\bar{x}, \varepsilon)$, which is contained in $\operatorname{int} K$. Therefore $\bar{x}$ belongs to $\operatorname{int} K$ as desired. □

The above lemma concerns a sum $F + H$, where F is assumed strongly monotone and H is continuous, strictly monotone, and maps $\operatorname{int} K$ surjectively onto $\mathbb{R}^n$. In the next theorem, we assume that F is monotone but restrict H to be the gradient map of a strictly convex function.

12.7.10 Theorem. Let $K \subseteq \mathbb{R}^n$ be a solid closed convex set and let F be a monotone function on K. Suppose that $f : \operatorname{int} K \to \mathbb{R}$ is a continuously differentiable strictly convex function such that $\nabla f(\operatorname{int} K) = \mathbb{R}^n$. The sum $F + \nabla f$ maps $\operatorname{int} K$ homeomorphically onto $\mathbb{R}^n$.

Proof. We divide the proof into three steps.

Step 1. We prove that, for every $w \in \mathbb{R}^n$ and for every $y \in \operatorname{int} K$,

$$\sup_{z \in \operatorname{int} K} (\nabla f(z) - w)^T (y - z) < \infty.$$

Since $\nabla f(\operatorname{int} K) = \mathbb{R}^n$ a vector $x \in \operatorname{int} K$ exists such that $\nabla f(x) = w$. By the convexity of f we can write, for any $z \in \operatorname{int} K$,

$$\begin{aligned} f(y) - f(z) &\geq \nabla f(z)^T (y - z) \\ &= (\nabla f(z) - \nabla f(x))^T (y - z) + \nabla f(x)^T (y - z). \end{aligned}$$

This implies

$$\begin{aligned} (\nabla f(z) - w)^T (y - z) &\leq f(y) - f(z) - \nabla f(x)^T y + \nabla f(x)^T z \\ &\leq f(y) - \nabla f(x)^T y + (\nabla f(x)^T x - f(x)) \\ &\equiv a(w, y) \end{aligned}$$

where the second inequality holds because f is convex and thus we have $f(z) \geq f(x) + \nabla f(x)^T (z - x)$, so that $\nabla f(x)^T z - f(z) \leq \nabla f(x)^T x - f(x)$. It follows that

$$\sup_{z \in \operatorname{int} K} (\nabla f(z) - w)^T (y - z) \leq a(w, y).$$

Step 2. Set $G \equiv F + \nabla f$. We claim that, for every $w \in \mathbb{R}^n$ and for every $y \in \operatorname{int} K$,

$$\sup_{z \in \operatorname{int} K} (\, G(z) - w \,)^T (\, y - z \,) \, < \, \infty.$$

With $\tilde{w} \equiv w - F(y)$, Step 1 implies the existence of a vector $a(\tilde{w}, y)$ such that

$$\sup_{z \in \operatorname{int} K} (\, \nabla f(z) - \tilde{w} \,)^T (\, y - \, z) \, \leq \, a(\tilde{w}, y).$$

By the monotonicity of F we also have $F(z)^T(y - z) \, \leq \, F(y)^T(y - z)$; therefore

$$\sup_{z \in \operatorname{int} K} (\, F(z) + \nabla f(z) - w \,)^T (\, y - z \,) \, \leq \, \tilde{a}(w, y),$$

where $\tilde{a}(w, y) \equiv a(\tilde{w}, y)$.

Step 3. We are ready to prove that $G \equiv F + \nabla f$ maps $\operatorname{int} K$ homeomorphically onto $\mathbb{R}^n$. Since G is obviously strictly monotone, it suffices to show that for any $w \in \mathbb{R}^n$ we can find an $x \in \operatorname{int} K$ such that $w = F(x) + \nabla f(x)$. Consider the perturbed equation $G(z) + \varepsilon z = w$, where ε is a positive constant. By Lemma **12.7.9** the latter equation has one and only one solution belonging to $\operatorname{int} K$. We denote such solution by $x(\varepsilon)$.

We show that $x(\varepsilon)$ is bounded as ε goes to zero. Let y be any vector in $\operatorname{int} K$ and w' any vector in $\mathbb{R}^n$. By Step 2 of the proof, we can write

$$(\, G(z) - (w + w') \,)^T (\, y - z \,) \, \leq \, \tilde{a}(w + w', y)$$

for all $z \in \operatorname{int} K$. Taking $z = x(\varepsilon)$ we get

$$-(\, \varepsilon \, x(\varepsilon) + w' \,)^T (\, y - x(\varepsilon) \,) \, \leq \, \tilde{a}(w + w', y),$$

which implies

$$x(\varepsilon)^T w' \, \leq \, \tilde{a}(w + w', y) + y^T w' + \varepsilon \, x(\varepsilon)^T (\, y - x(\varepsilon) \,).$$

This last inequality holds for any w' and all $\varepsilon > 0$. Thus it follows that $x(\varepsilon)$ is bounded as ε goes to zero. Assume, without loss of generality, that

$$\lim_{\varepsilon \downarrow 0} x(\varepsilon) = \, x^\infty.$$

Obviously x^∞ belongs to K. We claim that x^∞ actually belongs to $\operatorname{int} K$. By definition we have, for every positive ε, $G(x(\varepsilon)) + \varepsilon x(\varepsilon) = w$, that is

$$F(x(\varepsilon)) + \nabla f(x(\varepsilon)) + \varepsilon \, x(\varepsilon)) \, = \, w, \qquad \forall \, \varepsilon \, > \, 0. \tag{12.7.12}$$

Multiplying both sides by $z - x(\varepsilon)$, where z is any point in $\operatorname{int} K$, we get

$$F(x(\varepsilon))^T(\,z - x(\varepsilon)\,) + \nabla f(x(\varepsilon))^T(\,z - x(\varepsilon)\,) + \varepsilon\, x(\varepsilon)^T(\,z - x(\varepsilon)\,) \;=\; w.$$

By part (c) of Proposition **12.7.3**, it follows that x^∞ cannot belong to $\operatorname{bd} K$. Therefore, we may pass to the limit $\varepsilon \downarrow 0$ in (12.7.12), obtaining, by continuity, $F(x^\infty) + \nabla f(x^\infty) = w$, with $x^\infty \in \operatorname{int} K$. □

As a consequence of Theorem **12.7.10**, it follows that equation (12.7.11) always has a unique solution in $\operatorname{int} K$. In addition to the possibility of letting x^{k+1} be the exact solution of (12.7.11) in $\operatorname{int} K$, we can obtain x^{k+1}, conceptually, as the exact solution in $\operatorname{int} K$ of the equation

$$e^{k+1} \;\equiv\; c_k\, F(x^{k+1}) + \nabla f(x^{k+1}) - \nabla f(x^k)$$

for a given vector e^{k+1} satisfying $\|e^{k+1}\| \le \varepsilon_k$. The theorem below guarantees that such a x^{k+1} exists for every given e^{k+1}.

To prepare for the analysis of Algorithm **12.7.8**, we establish two technical results.

12.7.11 Lemma. Let K be a closed convex subset of $\mathbb{R}^n$ and $F : K \to \mathbb{R}^n$ be continuous and monotone. Let $\{x^k\}$ be generated by Algorithm **12.7.8** with f being a Bregman function with zone K. Let $x^* \in \operatorname{SOL}(K, F)$ be arbitrary. For all k it holds that

$$\begin{aligned} D_f(x^*, x^{k+1}) \;=\; & D_f(x^*, x^k) - D_f(x^{k+1}, x^k) \\ & -(\,\nabla f(x^k) - \nabla f(x^{k+1})\,)^T(\,x^{k+1} - x^*\,). \end{aligned} \tag{12.7.13}$$

Consequently,

$$\begin{aligned} & D_f(x^*, x^{k+1}) \;\le \\ & D_f(x^*, x^k) - D_f(x^{k+1}, x^k) + (\,x^{k+1} - x^*\,)^T e^{k+1}. \end{aligned} \tag{12.7.14}$$

Proof. The expression (12.7.13) follows easily from the three-point formula of the Bregman function. Specifically, it suffices to substitute $x \equiv x^*$, $y \equiv x^k$ and $z \equiv x^{k+1}$ in Proposition **12.7.6**(a). By the definition of e^{k+1}, we have

$$\frac{1}{c_k}\,(\,\nabla f(x^k) - \nabla f(x^{k+1}) + e^{k+1}\,) \;=\; F(x^{k+1});$$

which implies

$$\begin{aligned} & \frac{1}{c_k}\,(\,\nabla f(x^k) - \nabla f(x^{k+1}) + e^{k+1}\,)^T(\,x^{k+1} - x^*) \\ & \quad =\; F(x^{k+1})^T(\,x^{k+1} - x^*\,) \\ & \quad =\; (\,F(x^{k+1}) - F(x^*)\,)^T(\,x^{k+1} - x^*\,) + F(x^*)^T(\,x^{k+1} - x^*\,) \\ & \quad \ge\; 0, \end{aligned}$$

where the last inequality follows from the monotonicity of F and the definition of x^*. The inequality (12.7.14) now follows readily. □

12.7.12 Lemma. Assume the same setting as in Lemma **12.7.11**. If

$$\sum_{k=0}^{\infty} (x^{k+1} - x^*)^T e^{k+1} < \infty,$$

then the sequence $\{x^k\}$ is bounded and $\{D_f(x^{k+1}, x^k)\}$ converges to zero.

Proof. By (12.7.14) and an inductive argument we deduce that

$$\begin{aligned} &D_f(x^*, x^r) \leq \\ &D_f(x^*, x^0) - \sum_{k=0}^{r-1} D_f(x^{k+1}, x^k) + \sum_{k=0}^{r-1} (x^{k+1} - x^*)^T e^{k+1} \end{aligned} \tag{12.7.15}$$

for every $r \geq 0$. By the summability of $\{(x^{k+1} - x^*)^T e^{k+1}\}$, it follows that

$$\sigma \equiv \sup_{r \geq 0} \sum_{k=0}^{r-1} (x^{k+1} - x^*)^T e^{k+1} < \infty.$$

Using the nonnegativity of $D_f(x^{k+1}, x^k)$, we have

$$D_f(x^*, x^r) \leq D_f(x^*, x^0) + \sigma < \infty$$

for every $r \geq 0$. By property (c) in Definition **12.7.2**, it follows that $\{x^k\}$ must be bounded. From (12.7.15), we have

$$D_f(x^*, x^r) \leq D_f(x^*, x^0) + \sigma - \sum_{k=0}^{r-1} D_f(x^{k+1}, x^k), \quad \forall\, r \geq 0.$$

By the nonnegativity of the function D_f again, we finally get

$$0 \leq \sum_{k=0}^{\infty} D_f(x^{k+1}, x^k) \leq D_f(x^*, x^0) + \sigma < \infty,$$

which implies that $\{D_f(x^{k+1}, x^k)\}$ converges to zero. □

In order to establish the convergence of Algorithm **12.7.8**, we need to restrict the class of monotone functions F with an additional property that we motivate as follows. It is clear that the inclusion

$$\mathrm{SOL}(K, F) \subseteq \{ x \in K : F(x)^T (x^* - x) \geq 0 \}$$

holds for any $x^* \in K$. Moreover, if F is pseudo monotone on K and if $x^* \in \mathrm{SOL}(K, F)$, then

$$\{ x \in K : F(x)^T (x^* - x) \geq 0 \} \subseteq \{ x \in K : F(x^*)^T (x^* - x) = 0 \}.$$

Recalling the minimum principle sufficiency for an optimization problem introduced in the discussion following Lemma **6.4.3**, we may define the following class of VIs.

12.7.13 Definition. We say that the VI (K, F) satisfies the

(a) *minimum principle sufficiency* (MPS) if an $x^* \in \mathrm{SOL}(K, F)$ exists such that

$$\mathrm{SOL}(K, F) = \{ x \in K : F(x^*)^T (x^* - x) = 0 \};$$

(b) *weak minimum principle sufficiency* (WMPS) if an $x^* \in \mathrm{SOL}(K, F)$ exists such that

$$\mathrm{SOL}(K, F) = \{ x \in K : F(x)^T (x^* - x) \geq 0 \}. \tag{12.7.16}$$

According the preceding discussion, it is clear that if F is pseudo monotone on K, then MPS implies WMPS. In Exercise **12.8.23**, we ask the reader to show that every monotone function F that is pseudo monotone plus on K must satsify the WMPS if $\mathrm{SOL}(K, F)$ is nonempty. This exercise also gives an example of a monotone LCP that satisfies the WMPS but the defining matrix is not positive semidefinite plus; furthermore, the example shows that it is possible for (12.7.16) to hold for some but not all $x^* \in \mathrm{SOL}(K, F)$.

Since in general the class of pseudo monotone plus functions includes the class of monotone plus functions, which in turn includes the strictly monotone composite functions and the symmetric monotone functions, it follows that the class of monotone VIs with the WMPS property is rather broad. In particular, if F is the gradient of a C^1 convex function, then the WMPS holds for the VI (K, F), provided that the latter problem is solvable. Consequently, the Bregman proximal point algorithm is always applicable to convex differentiable minimization problems and to strictly monotone composite VIs, provided that they are solvable.

12.7.14 Theorem. Let K be a closed convex set in $\mathbb{R}^n$ and F be a continuous and monotone mapping on K. Let $\{x^k\}$ be generated by Algorithm **12.7.8** with f being a Bregman function with zone K and with $\{c_k\}$ bounded away from zero. Assume that

$$\sum_{k=0}^{\infty} \| e^k \| < \infty \tag{12.7.17}$$

$$\sum_{k=0}^{\infty} (e^k)^T x^k \quad \text{exists and is finite.} \tag{12.7.18}$$

and that the VI (K, F) satisfies the WMPS. The sequence $\{x^k\}$ converges to a solution of VI (K, F).

Proof. Let x^* be a solution of VI (K, F) such that the WMPS holds. Trivially

$$(\, x^{k+1} - x^* \,)^T e^{k+1} \;=\; (\, x^{k+1} \,)^T e^{k+1} - (\, x^* \,)^T e^{k+1}.$$

The assumption (12.7.17) implies

$$\lim_{k \to \infty} e^k \;=\; 0$$

and

$$0 \;\leq\; \sum_{k=0}^{\infty} |\, (x^*)^T e^{k+1} \,| \;\leq\; \| \, x^* \, \| \sum_{k=0}^{\infty} \| \, e^{k+1} \, \| \;<\; \infty,$$

which, in turn, implies that $\sum_{k=0}^{\infty} (x^*)^T e^{k+1}$ is convergent. By (12.7.18), the infinite sum $\sum_{k=0}^{\infty} (x^{k+1} - x^*)^T e^{k+1}$ is finite; hence

$$\lim_{k \to \infty} (\, x^{k+1} - x^* \,)^T e^{k+1} \;=\; 0. \tag{12.7.19}$$

All the assumptions of Lemmas **12.7.11** and **12.7.12** are satisfied. As a first consequence we get that $\{x^k\}$ is bounded. Let x^∞ be any limit point of $\{x^k\}$, and assume that $\{x^k : k \in \kappa\}$ converges to x^∞. By Lemma **12.7.12** it follows that $\{D_f(x^{k+1}, x^k) : k \in \kappa\}$ converges to zero. By Proposition **12.7.6**(b) this implies that $\{x^{k+1} : k \in \kappa\}$ also converges to x^∞. By (12.7.14) we get

$$D_f(x^*, x^{k+1}) \;\leq\; D_f(x^*, x^k) + (\, x^{k+1} - x^* \,)^T e^{k+1}.$$

Since the infinite sum $\sum_{k=0}^{\infty} (x^{k+1} - x^*)^T e^{k+1}$ is finite, by Lemma **10.4.21**, the sequence $\{D_f(x^*, x^{k+1})\}$ converges to some nonnegative scalar. By (12.7.13) and Step 3 of the algorithm, we can write

$$\begin{aligned} & c_k \, F(x^{k+1})^T (\, x^{k+1} - x^* \,) \\ & \quad = \; D_f(x^*, x^k) - D_f(x^*, x^{k+1}) - D_f(x^{k+1}, x^k) + (\, x^{k+1} - x^* \,)^T e^{k+1}. \end{aligned}$$

Passing to the limit, we get

$$\lim_{k(\in \kappa) \to \infty} c_k \, F(x^{k+1})^T (\, x^{k+1} - x^* \,) \;=\; 0.$$

Since $\{c_k\}$ is bounded away from zero, it follows that

$$\lim_{k(\in\kappa)\to\infty} F(x^{k+1})^T(x^{k+1} - x^*) = 0.$$

By the continuity of F we obtain $F(x^\infty)^T(x^\infty - x^*) = 0$. By the WMPS of the VI (K, F), we deduce that $x^\infty \in \mathrm{SOL}(K, F)$.

To conclude the proof it remains to show that x^∞ is the only limit point of $\{x^k\}$. By property (d) of Definition **12.7.2**, we deduce

$$\lim_{k(\in\kappa)\to\infty} D_f(x^\infty, x^k) = 0.$$

But taking x^* to be x^∞ in the above proof, we conclude that the sequence $\{D_f(x^\infty, x^k)\}$ converges. Consequently,

$$\lim_{k\to\infty} D_f(x^\infty, x^k) = 0.$$

The convergence of $\{x^k\}$ to x^∞ follows from Corollary **12.7.7**. □

We conclude this subsection with some comments on the inexact criteria (12.7.17) and (12.7.18). The former criterion is similar (although simpler) than the one for the standard proximal point algorithm, while the latter is an additional requirement that is needed because we are using Bregman functions to define the subproblems. Condition (12.7.18) is rather difficult to enforce as it is, but it is implied in particular by the easily implementable condition

$$\sum_{k=0}^{\infty} (e^k)^T x^k = 0.$$

With this observation in mind, a way to enforce (12.7.17) and (12.7.18) is simply to require at each iteration

$$\max(\|e^k\|, (e^k)^T x^k) \le c\,\rho^k,$$

where c is any nonnegative constant and $\rho \in (0, 1)$.

12.7.3 Linearly constrained VIs

When K is polyhedral, we can design an equation-reduction proximal-like algorithm for solving the linearly constrained, monotone VI (K, F) without imposing the WMPS. This subsection is devoted to the discussion of such an algorithm and its convergence analysis. Let

$$K = \{x \in \mathbb{R}^n : Ax \le b\}, \tag{12.7.20}$$

where A is an $m \times n$ matrix with full column rank (therefore $m \geq n$). This assumption is certainly satisfied if all the variables have lower or upper bounds. We assume that the Slater constraint qualification holds, that is,

$$\operatorname{int} K \;=\; \{\, x \in \mathbb{R}^n \;:\; Ax \;<\; b \,\} \;\neq\; \emptyset,$$

For brevity, we also introduce the notation $l_i(x) \equiv b_i - A_{i\cdot}x$, where $A_{i\cdot}$ is the i-th row of A. The vector $l(x)$ is the m-vector whose components are $l_i(x)$ for $i = 1, \ldots, m$. Note that $K = \{x \in \mathbb{R}^n : l(x) \geq 0\}$.

For any u and v in $\mathbb{R}^m_{++}$ we define the function

$$p(u,v) \;\equiv\; \sum_{i=1}^{m} \left(u_i^2 - u_i v_i - v_i^2 \log \frac{u_i}{v_i} \right).$$

Associated with this function we introduce, for any x and $y \in \operatorname{int} K$, the function

$$P(x,y) \;\equiv\; p(l(x), l(y)).$$

It is easy to see that P is (infinitely) differentiable on $\operatorname{int} K \times \operatorname{int} K$ and that

$$\nabla_x P(x,y) \;\equiv\; -\sum_{i=1}^{m} \left(2l_i(x) - l_i(y) - \frac{l_i(y)^2}{l_i(x)} \right) (\, A_{i\cdot}\,)^T.$$

The following is the special proximal-like algorithm for solving the linearly constrained VI.

Proximal-like Algorithm for a Linearly Constrained VI
(PALCVI)

12.7.15 Algorithm.

Data: $x^0 \in \operatorname{int} K$, $c_0 > 0$, and $\varepsilon_0 \geq 0$.

Step 1: Set $k = 0$.

Step 2: If x^k is a solution of VI (K, F), stop.

Step 3: Find $x^{k+1} \in \operatorname{int} K$ such that $\|e^{k+1}\| \leq \varepsilon_k$, where

$$e^{k+1} \;\equiv\; c_k\, F(x^{k+1}) + \nabla_x P(x^{k+1}, x^k).$$

Step 4: Select c_{k+1} and ε_{k+1}. Set $k \leftarrow k+1$ and go to Step 2.

The similarities of this algorithm with Algorithm **12.7.8** are obvious; note however that the term $\nabla_x P(x^{k+1}, x^k)$ cannot be recovered from a Bregman function, and is specially tailored to the polyhedral structure of

K. As in the Bregman algorithm, our first order of business is to show that Algorithm **12.7.15** is well defined; this amounts to showing that the iterate x^{k+1} in Step 3 exists.

12.7.16 Theorem. In the above setting, $F + \nabla_x P(\cdot, x^k)$ maps $\operatorname{int} K$ onto $\mathbb{R}^n$.

Proof. We apply Theorem **12.7.10**. For a fixed $x^k \in \operatorname{int} K$ consider the function

$$\tilde{p}(x) \equiv P(x, x^k), \quad x \in \operatorname{int} K.$$

Note that $\tilde{p}$ is strictly convex and continuously differentiable on $\operatorname{int} K$ with

$$\nabla \tilde{p}(x) = \nabla_x P(x, x^k).$$

We need to show that $\nabla \tilde{p}(\operatorname{int} K) = \mathbb{R}^n$. Let $y \in \mathbb{R}^n$ be a given point and consider the function

$$p_y(x) \equiv \tilde{p}(x) - y^T x.$$

This function is well defined, convex, and continuously differentiable on $\operatorname{int} K$. We claim that the level sets

$$L_y(\eta) \equiv \{ x \in \operatorname{int} K : p_y(x) \leq \eta \}$$

are compact. If this is not true, we can find a constant η and a sequence $\{x^i\}$ of points in $\operatorname{int} K$ such that $p_y(x^i) \leq \eta$ and either of the following conditions hold:

(a) $\lim_{i \to \infty} \|x^i\| = \infty$; or

(b) $\lim_{i \to \infty} x^i = \bar{x} \in \operatorname{bd} K$.

But looking at the structure of p_y, we see that if (a) or (b) holds, then $p_y(x^i)$ tends to infinity. Therefore since $\operatorname{int} K \neq \emptyset$ by assumption, the convex function p_y has an unconstrained minimum point, let us say z, in $\operatorname{int} K$. It follows then that

$$0 = \nabla p_y(z) - y = \nabla \tilde{p}(z) - y.$$

Since y was arbitrary in $\mathbb{R}^n$, $\nabla \tilde{p}$ therefore maps $\operatorname{int} K$ onto $\mathbb{R}^n$. □

In order to study the convergence properties of Algorithm **12.7.15**, we still need a preliminary lemma.

12.7.17 Lemma. For any scalars s, t and u, with s and t positive and u nonnegative,

(a) $(t-u)\left(2t-s-\dfrac{s^2}{t}\right) \geq (s-t)(3u-2t-s)$,

(b) $(t-u)\left(2t-s-\dfrac{s^2}{t}\right) \geq \frac{3}{2}((u-t)^2-(u-s)^2)+\frac{1}{2}(t-s)^2$.

Proof. By simple manipulations, we can write:

$$\begin{aligned}(t-u)\left(2t-s-\frac{s^2}{t}\right) &= 2t^2-st-s^2-u(2t-s)+u\frac{s^2}{t}\\ &\geq 2t^2-st-s^2-u(2t-s)+u(2s-t).\end{aligned}$$

Simple calculations now yield:

$$\begin{aligned}(t-u)\left(2t-s-\frac{s^2}{t}\right) &\geq (2t+s)(t-s)-3u(t-s)\\ &= (s-t)(3u-2t-s)\\ &= (t-s)(s-u)+2(t-s)(t-u).\end{aligned}$$

The first equality shows (a). Furthermore, substituting the identities

$$\begin{aligned}(t-s)(s-u) &= \tfrac{1}{2}\left((u-t)^2-(u-s)^2-(t-s)^2\right)\\ 2(t-s)(t-u) &= (u-t)^2-(u-s)^2+(t-s)^2,\end{aligned}$$

in the second equality completes the proof of (b). □

We can state and prove the promised convergence properties of Algorithm **12.7.15**.

12.7.18 Theorem. Let K be given by (12.7.20) with A having full column rank. Let F be continuous and monotone on K. Let $\{x^k\}$ be generated by Algorithm **12.7.15**, where $\{c_k\}$ is bounded away from zero. Assume (12.7.17) and (12.7.18) and that the VI (K,F) has at least one solution. The sequence $\{x^k\}$ converges to a solution of the VI (K,F).

Proof. By Step 3 of the algorithm, taking into account the expression of $\nabla_x P(x^{k+1},x^k)$, we have

$$\begin{aligned}F(x^{k+1}) &= (c_k)^{-1}\,e^{k+1}\\ &\quad +(c_k)^{-1}\sum_{i=1}^{m}\left(2l_i(x^{k+1})-l_i(x^k)-\frac{l_i(x^k)^2}{l_i(x^{k+1})}\right)(\,A_{i\cdot}\,)^T.\end{aligned}$$

Furthermore, if $x^*\in \mathrm{SOL}(K,F)$, then

$$\begin{aligned}&F(x^{k+1})^T(\,x^{k+1}-x^*\,)\\ &\quad = F(x^*)^T(\,x^{k+1}-x^*\,)+(\,F(x^{k+1})-F(x^*)\,)^T(\,x^{k+1}-x^*\,);\end{aligned}$$

since both terms on the right-hand side are nonnegative, so is the left-hand product. Hence, by Lemma **12.7.17**(b) we obtain

$$\begin{aligned}
&(x^* - x^{k+1})^T(-e^{k+1}) \\
&\geq \tfrac{3}{2}\left(\| A(x^* - x^{k+1})\|^2 - \| A(x^* - x^k)\|^2\right) + \tfrac{1}{2}\| A(x^{k+1} - x^k)\|^2 \\
&\geq \tfrac{3}{2}\left(\| A(x^* - x^{k+1})\|^2 - \| A(x^* - x^k)\|^2\right).
\end{aligned}$$

By (12.7.17) and (12.7.18), it follows that $\{\|A(x^* - x^{k+1})\|\}$ converges. Since A has full column rank, $\{\|x^* - x^{k+1}\|\}$ converges; hence $\{x^k\}$ is bounded. Using the first of the above two inequalities, we then see that $\{\|A(x^{k+1} - x^k)\|\}$ converges to zero; thus

$$\lim_{k\to\infty} \| x^{k+1} - x^k \| = 0. \tag{12.7.21}$$

Set now, for simplicity,

$$q_{i,k}(x) \equiv \left(l_i(x^{k+1}) - l_i(x)\right)\left(2l_i(x^{k+1}) - l_i(x^k) - \frac{l_i(x^k)^2}{l_i(x^{k+1})}\right),$$

and note that by Lemma **12.7.17**(a) we have

$$q_{i,k}(x) \geq (l_i(x^k) - l_i(x^{k+1}))(3l_i(x) - l_i(x^k) - 2l_i(x^{k+1})).$$

Since $\{x^k\}$ is bounded and (12.7.21) holds, this shows that

$$\liminf_{k\to\infty} q_{i,k}(x) \geq 0. \tag{12.7.22}$$

Consider now any fixed $x \in K$. Again by Step 3 of the algorithm we can write

$$c_k\, F(x^{k+1})^T(\, x - x^{k+1}\,) = (\, e^{k+1}\,)^T(\, x - x^{k+1}\,) - q_{i,k}(x).$$

Assume, without loss of generality, that $\{x^{k+1} : k \in \kappa\}$ converges to x^∞. Passing to the limit $k(\in \kappa) \to \infty$ in the above expression, we get

$$F(x^\infty)^T(\, x - x^\infty\,) \geq 0,$$

where we have used the fact that c_k is bounded away from zero, the fact that $\{e^k\}$ converges to zero, and (12.7.22). This shows that $x^\infty \in \mathrm{SOL}(K, F)$. Letting $x^* = x^\infty$ in the above developement, we see that $\{\|x^k - x^\infty\|\}$ converges. Hence the whole sequence $\{x^k\}$ converges to x^∞. □

12.7.4 Interior and exterior barrier methods

In this final subsection we consider equation-reduction methods that are the generalization of the classical interior and exterior barrier methods for constrained optimization. The distinctive feature of the methods to be presented here is that the subproblems are basically unconstrained systems of equations and that the points produced could lie outside the set K. We consider two classes of algorithms: interior and exterior methods. Interior algorithms are applicable to problems where the set K has a nonempty interior; these methods generate only points that belong to the interior of K. Exterior algorithms are applicable to problems where $\operatorname{int} K$ can be empty; these methods generate points that may lie outside K.

Throughout this subsection we assume that the set K is nonempty and finitely representable in the form:

$$K \equiv \{\, x \in \mathbb{R}^n : g(x) \leq 0 \,\}, \tag{12.7.23}$$

where $g : \mathbb{R}^n \to \mathbb{R}^m$ and each g_i is convex. Let F be a continuous, monotone function on K. We assume that the VI (K, F) has a nonempty and compact solution set. We consider an algorithm which involves the solution of a sequence of equations each of the form

$$F(x) + \varepsilon\, G_\varepsilon(x) = 0, \tag{12.7.24}$$

where ε is a positive parameter that eventually goes to zero. The function G_ε is defined by

$$G_\varepsilon(x) \equiv \nabla g_\varepsilon(x), \qquad g_\varepsilon(x) \equiv \sum_{i=1}^{m} r(\varepsilon^{-1} g_i(x)),$$

where $r : \mathbb{R} \to \mathbb{R} \cup \{\infty\}$ is a convex nondecreasing function satisfying the following conditions:

(r1) $\operatorname{dom} r = (-\infty, \eta)$ with $\eta \in [0, \infty]$;

(r2) r is continuously differentiable on its domain;

(r3) $\lim_{t \uparrow \eta} r(t) = +\infty$;

(r4) $r_\infty(-1) = 0$

(r5) $r_\infty(1) = +\infty$.

Condition (r5) is automatically satisfied if η is finite. In the case $\eta = +\infty$, (r5) says that when t grows to infinity, r goes to infinity "faster than

linearly". On the other hand, condition (r4) just requires r to grow "slower than linearly" when t goes to $-\infty$. Since r_∞ is positively homogeneous, (r4) and (r5) imply

$$r_\infty(-t) = 0 \quad \text{and} \quad r_\infty(t) = \infty, \quad \forall\, t > 0.$$

The choice of the function r depends on the given set K. If $\operatorname{int} K$ is nonempty, we may choose a function r with $\eta = 0$. If $\operatorname{int} K$ is empty, we choose a function r with $\eta > 0$. In the former case, the method we are about to present behaves like an interior point method and generates iterates in the interior of K. In the latter case, the method generates points that may lie outside the set K. In both cases we show that (12.7.24) always has solutions.

To simplify the presentation we assume that the functions g_i in (12.7.23) are continuously differentiable on the whole space. Under this assumption the function g_ε is convex and its domain is the open set

$$\{\, x \in \mathbb{R}^n : g_i(x) < \eta,\ i = 1, 2, \ldots, m \,\};$$

furthermore g_ε is continuously differentiable on its domain. Assuming that F is defined on a set containing $\operatorname{dom} g_\varepsilon$, solving the equation (12.7.24) is therefore a well-posed problem and consists of finding a zero of a function defined on an open set. Note finally that if $\eta > 0$ then $\operatorname{dom} g_\varepsilon$ is certainly nonempty. If $\eta = 0$ instead, the nonemptiness of $\operatorname{dom} g_\varepsilon$ is equivalent to Slater's constraint qualification for K.

We begin by giving some examples of suitable functions r which show that a wide range of options is available. In Exercise **12.8.24** we outline a systematic way of generating functions r satifying the conditions (r1)–(r5). Below is a list of functions r satisfying (r1)–(r5) along with the corresponding scalars η:

1. $r(t) = e^t, \quad (\eta = +\infty);$

2. $r(t) = \begin{cases} t + \frac{1}{2}\, t^2 & \text{if } t \geq -\frac{1}{2} \\ -\frac{1}{4}\, \log(-2t) - \frac{1}{8} & \text{if } t \leq -\frac{1}{2}; \end{cases} \qquad (\eta = +\infty);$

3. $r(t) = -\log(1 - t), \quad (\eta = 1);$

4. $r(t) = t/(1 - t), \quad (\eta = 1);$

5. $r(t) = -\log(-t), \quad (\eta = 0);$

6. $r(t) = -1/t, \quad (\eta = 0).$

It is a routine exercise to check that all these functions satisfy conditions (r1)–(r5) and we leave this task to the reader.

As before, our next step is to show that the equation (12.7.24) has at least one solution. This issue is analyzed in the next theorem.

12.7.19 Theorem. Let K be defined by (12.7.23) with all the functions g_i being convex and continuously differentiable on the whole space $\mathbb{R}^n$, and let $F : \mathbb{R}^n \to \mathbb{R}^n$ be continuous and monotone on K. Assume that the solution set of the VI (K, F) is nonempty and compact. Suppose further that

(a) if $\eta = 0$ then Slater's constraint qualification holds for K;

(b) if $\eta > 0$ then F is continuous and monotone on $\mathbb{R}^n$.

For each positive ε, equation (12.7.24) has at least one solution and every such solution belongs to $\operatorname{dom} g_\varepsilon$.

Proof. For each $k = 1, 2, \ldots$, let

$$C^k \equiv \{\, x \in \operatorname{cl} \mathbb{B}(0,k) : g_i(x) \leq \eta - 2^{-k},\ i = 1, 2, \ldots \}.$$

For every k large enough the set C^k is nonempty, convex, and compact. Moreover, C^k is contained in the domain of g_ε; thus $F + \varepsilon G_\varepsilon$ is well-defined and continuous on C^k. Therefore the VI $(C^k, F + \varepsilon G_\varepsilon)$ has at least one solution, which we denote by x^k. We claim that the sequence $\{x^k\}$ is bounded and that if x^∞ is a limit point of $\{x^k\}$ then x^∞ solves the equation (12.7.24).

By definition we have

$$F(x^k)^T(\, x - x^k\,) + \varepsilon\, G_\varepsilon(x^k)^T(\, x - x^k\,) \geq 0, \qquad \forall\, x \in C^k$$

and therefore, taking into account the fact that G_ε is the gradient of the convex function g_ε, we get

$$F(x^k)^T(\, x - x^k\,) + \varepsilon\, g_\varepsilon(x) \geq \varepsilon\, g_\varepsilon(x^k), \qquad \forall\, x \in C^k. \tag{12.7.25}$$

By the monotonicity of F we then get

$$F(x)^T x + \varepsilon g_\varepsilon(x) \geq F(x)^T x^k + \varepsilon g_\varepsilon(x^k), \qquad \forall\, x \in C^k. \tag{12.7.26}$$

Suppose for the sake of contradiction that $\{x^k\}$ is unbounded. Without loss of generality we may assume that

$$\lim_{k\to\infty} \|\, x^k \,\| = \infty \quad \text{and} \quad \lim_{k\to\infty} \frac{x^k}{\|\, x^k \,\|} = d \neq 0.$$

Let y be an arbitrary vector in K if $\eta > 0$ and in $\operatorname{int} K$ if $\eta = 0$; in both cases y belongs to $C^k \cap \operatorname{dom} g_\varepsilon$ for every k large enough. Dividing (12.7.26) by $\|x^k\|$ and passing to the limit, we get, taking also into account (12.7.1),

$$F(y)^T d + \varepsilon\, (\, g_\varepsilon\,)_\infty(d) \;\leq\; 0.$$

By Proposition **12.7.1** and conditions (r4) and (r5), it follows that

$$(g_i)_\infty(d) \;\leq\; 0, \quad i \;=\; 1, 2, \ldots, m, \quad \text{and} \quad F(y)^T d \;\leq\; 0,$$

or, equivalently,

$$d \;\in\; K_\infty, \quad \text{and} \quad F(y)^T d \;\leq\; 0. \tag{12.7.27}$$

If $\eta > 0$, (12.7.27) holds for every $y \in K$ and therefore we get a contradiction to Theorem **2.3.16**. If $\eta = 0$, $F(y)^T d \leq 0$ holds for every $y \in \operatorname{int} K$ and therefore, by a simple continuity argument, for every $y \in K$. We obtain the same contradiction. Therefore the sequence $\{x^k\}$ is bounded.

Assume, without loss of generality, that $\{x^k\}$ converges to x^∞. First observe that $g_i(x^\infty) < \eta$ for all i; for otherwise $g_\varepsilon(x^k)$ would tend to infinity and we would get a contradiction to (12.7.25). Therefore $x^\infty \in \operatorname{dom} g_\varepsilon$. Let Ω be an open neighborhood of x^∞; if Ω is sufficiently small it holds that $\Omega \subseteq C^k$ for every k sufficiently large. But then, passing to the limit in (12.7.25) we obtain, by continuity (recall also assumptions (a) and (b)),

$$F(x^\infty)^T (\, x - x^\infty\,) + \varepsilon\, g_\varepsilon(x) \;\geq\; \varepsilon\, g_\varepsilon(x^\infty), \qquad \forall\, x \;\in\; \Omega.$$

This shows that x^∞ is a local minimizer of the differentiable convex function

$$h(x) \;\equiv\; F(x^\infty)^T (\, x - x^\infty\,) + \varepsilon\, g_\varepsilon(x)$$

so that

$$0 \;=\; F(x^\infty) + \varepsilon\, \nabla g_\varepsilon(x^\infty) \;=\; F(x^\infty) + \varepsilon\, G_\varepsilon(x^\infty),$$

and thus concluding the proof. □

We are ready to state and analyze an algorithm based on the subproblem (12.7.24). As usual, we also consider the possibility of solving inexactly the subproblems, which is an essential feature for practical efficiency.

Interior/Barrier Algorithm (IBA)

12.7.20 Algorithm.

Data: $x^0 \in \mathbb{R}^n$, $\rho_0 > 0$, and $\varepsilon_0 > 0$.

Step 1: Set $k = 0$.

Step 2: If x^k is a solution of VI (K, F), stop.

Step 3: Find $x^{k+1} \in \operatorname{dom} g_{\varepsilon_k}$ such that $\|e^{k+1}\| \leq \rho_k$, where

$$e^{k+1} \equiv F(x^{k+1}) + \varepsilon_k G_{\varepsilon_k}(x^{k+1}).$$

Step 4: Select positive $\rho_{k+1} < \rho_k$ and $\varepsilon_{k+1} < \varepsilon_k$ and set $k \leftarrow k+1$; go to Step 2.

The convergence of the algorithm is asserted by the following result.

12.7.21 Theorem. Assume the same setting of Theorem **12.7.19** and let $\{x^k\}$ be a sequence generated by Algorithm **12.7.20**. Suppose further that the sequences $\{\rho_k\}$ and $\{\varepsilon_k\}$ both converge to zero. The sequence $\{x^k\}$ is bounded and each of its limit points is a solution of the VI (K, F).

Proof. By the definition fo e^k we have, for every $k \geq 1$ and every $x \in \mathbb{R}^n$,

$$-(e^k)^T(x - x^k) + F(x^k)^T(x - x^k) + \varepsilon_{k-1} G_{\varepsilon_{k-1}}(x^k)^T(x - x^k) = 0.$$

Similar to the proof of Theorem **12.7.19**, the monotonicity of F and the convexity of g imply then, for every $x \in \operatorname{dom} g_{\varepsilon_{k-1}}$,

$$\begin{aligned} &-(e^k)^T x + F(x)^T x + \varepsilon_{k-1}\, g_{\varepsilon_{k-1}}(x) \\ &\quad \geq -(e^k)^T x^k + F(x)^T x^k + \varepsilon_{k-1}\, g_{\varepsilon_{k-1}}(x^k). \end{aligned} \tag{12.7.28}$$

We first show that $\{x^k\}$ is bounded. Suppose for the sake of contradiction that this is not true. Without loss of generality, we may assume that

$$\lim_{k\to\infty} \|x^k\| = \infty \quad \text{and} \quad \lim_{k\to\infty} \frac{x^k}{\|x^k\|} = d^\infty \neq 0.$$

Let y be a point in K if $\eta > 0$ and in $\operatorname{int} K$ if $\eta = 0$; note that y belongs to $\operatorname{dom} g_{\varepsilon_{k-1}}$ for every k. Set

$$s(y) \equiv \max\{\, g_i(y) : i = 1, 2, \ldots, m \,\}.$$

Since r is nondecreasing we see from (12.7.28) that

$$\begin{aligned} &-(e^k)^T y + F(y)^T y + m\,\varepsilon_{k-1}\, r(\varepsilon_{k-1}^{-1}\, s(y)) \\ &\quad \geq -(e^k)^T x^k + F(y)^T x^k + \varepsilon_{k-1} \sum_{i=1}^m r(\varepsilon_{k-1}^{-1}\, g_i(x^k)). \end{aligned} \tag{12.7.29}$$

Let σ_i be any number such that $\sigma_i < (g_i)_\infty(d^\infty)$ for $i = 1, 2, \ldots, m$. By (12.7.1) we know that, for every k sufficiently large, $g_i(x^k) \geq \sigma_i \|x^k\|$. Since

r is nondecreasing, we deduce from the above inequality that

$$-(\,e^k\,)^T\frac{y}{\|x^k\|}+F(y)^T\frac{y}{\|x^k\|}+\frac{m\varepsilon_{k-1}}{\|x^k\|}\,r(\varepsilon_{k-1}^{-1}\,s(y))$$
$$\geq\ -(\,e^k\,)^Td^k+F(y)^Td^k+\sum_{i=1}^{m}\frac{\varepsilon_{k-1}}{\|x^k\|}\,r\left(\sigma_i\,\frac{\|x^k\|}{\varepsilon_{k-1}}\right),$$

where we set $d^k\equiv x^k/\|x^k\|$. Since $s(y)<0$ if $\eta=0$ and $s(y)\leq 0$ otherwise, passing to the limit and taking into account (r4), we get

$$F(y)^Td^\infty+\sum_{i=1}^{m}r_\infty(\sigma_i)\ \leq\ 0. \tag{12.7.30}$$

By (r5) this implies $\sigma_i\leq 0$. Since σ_i is arbitrary, we deduce $(g_i)_\infty(d^\infty)\leq 0$ for all i. This observation, (r4) and (12.7.30) then imply

$$d^\infty\ \in\ K_\infty\quad\text{ and }\quad F(y)^Td^\infty\ \leq\ 0.$$

Reasoning as in the proof of Theorem **12.7.19** we can derive a contradiction, thus proving that $\{x^k\}$ is bounded.

Assume without loss of generality that $\{x^k\}$ converges to a limit x^*. We show that $x^*\in\mathrm{SOL}(K,F)$. First of all note that $x^*\in K$. In fact $x^k\in\operatorname{dom}g_{\varepsilon_{k-1}}$ by Step 3, if x^* were not in K, a simple observation on the structure of the sets $\operatorname{dom}g_{\varepsilon_{k-1}}$ and an elementary continuity argument would show that $x^k\notin\operatorname{dom}g_{\varepsilon_{k-1}}$ eventually. Let y and $s(y)$ be defined as before, and let δ_i be numbers such that $\delta_i<g_i(x^*)$ for $i=1,2,\ldots,m$. By (12.7.29) and the fact that r is nondecreasing we get

$$-(\,e^k\,)^Ty+F(y)^Ty+m\,\varepsilon_{k-1}\,r(\varepsilon_{k-1}^{-1}\,s(y))$$
$$\geq\ -(\,e^k\,)^Tx^k+F(y)^Tx^k+\varepsilon_{k-1}\sum_{i=1}^{m}r(\varepsilon_{k-1}^{-1}\,\delta_i).$$

Since $\delta_i<0$, $s(y)<0$ if $\eta=0$ and $s(y)\leq 0$ otherwise, passing to the limit we get

$$F(y)^T(\,y-x^*\,)\ \geq\ 0.$$

This inequality holds for all $y\in K$ if $\eta>0$ and for all $y\in\operatorname{int}K$ if $\eta=0$. But in the latter case we can use a simple continuity argument to show that $F(y)^T(y-x^*)$ is nonnegative for any y in K. By the proof of Theorem **2.3.5**, we conclude that x^* is a solution of the VI (K,F). □

12.8 Exercises

12.8.1 This exercise shows that if F is a gradient map, then the Basic Projection Algorithm enjoys some enhanced convergence properties. Let

$F = \nabla\theta$ for some continuously differentiable convex function $\theta : K \to \mathbb{R}$, with $K \subseteq \mathbb{R}^n$ closed and convex. Suppose that F is Lipschitz continuous on K with constant L. Consider the sequence $\{x^k\}$ generated by the iteration

$$x^{k+1} \;=\; x^k - \tau^{-1} F(x^k)$$

starting with $x^0 \in K$. Show that if $\tau > L/2$ then every limit point of $\{x^k\}$ is a solution of the VI (K, F). (Hint: by using the properties of the projection prove first that

$$F(x^k)^T (x^{k+1} - x^k) \;\leq\; \tau \| \, x^{k+1} - x^k \, \|^2.$$

Using this fact and the Taylor expansion of the function θ show that

$$\theta(x^{k+1}) - \theta(x^k) \;\leq\; \left(\frac{L}{2} - \tau \right) \| \, x^{k+1} - x^k \, \|^2.$$

Complete the proof using the last inequality.)

12.8.2 In this exercise we consider a simple subgradient type algorithm for the solution of the VI (K, F). Suppose that $F : K \to \mathbb{R}^n$ is strongly monotone and Lipschitz continuous on K. Consider the iteration

$$x^{k+1} \;=\; \Pi_K(x^k - \tau_k F(x^k)).$$

Show that if

$$\tau_k \;\geq\; 0, \quad \lim_{k \to \infty} \tau_k \;=\; 0, \quad \text{and} \quad \sum_{k=0}^{\infty} \tau_k \;=\; \infty,$$

then the sequence $\{x^k\}$ converges to the unique solution of VI (K, F). Note that even if the assumptions on the problem are the same used in the Basic Projection Algorithm **12.1.1**, no knowledge of the Lipschitz constant L or of the modulus of strong monotonicity μ is necessary. (Hint: use the fixed-point charecterization of the (unique) solution x^* to the VI (K, F) to show that $\|x^{k+1} - x^*\|^2 \leq (1 - 2\mu\tau_k + L^2\tau_k^2)\|x^k - x^*\|^2$ and then conclude.)

12.8.3 In this exercise we explore some properties of Fejér sequences. Let $C \subseteq \mathbb{R}^n$ be a closed, convex and nonempty set. A sequence $\{x^k\}$ in $\mathbb{R}^n$ is *Fejér monotone* with respect to C if for all k

$$\| \, x^{k+1} - x \, \| \;\leq\; \| \, x^k - x \, \|, \quad \forall \, x \in C.$$

Prove that if $\{x^k\}$ in $\mathbb{R}^n$ is Fejér monotone with respect to C the following assertions hold.

(a) $\{x^k\}$ is bounded and $\text{dist}(x^{k+1}; C) \leq \text{dist}(x^k; C)$.

(b) $\{x^k\}$ has at most one limit point in C.

(c) The sequence $\{\Pi_C(x^k)\}$ converges. (Hint: use the identity

$$\| a - b \|^2 = 2 \| a \|^2 + 2 \| b \|^2 - \| a + b \|^2$$

with $a \equiv \Pi_C(x^{k+i})$ and $b = \Pi_C(x^k)$ to show that $\{\Pi_C(x^k)\}$ is a Cauchy sequence.

(d) The following facts are equivalent.

1. $\{x^k\}$ converges to a point in C.
2. $\{\text{dist}(x^k; C)\}$ converges to zero.
3. $\{x^k - \Pi_C(x^k)\}$ converges to zero.

12.8.4 Consider a monotone VI (K, F) with F continuous, and the solution $x(\varepsilon)$ of its Tikhonov regularization. Show that $\|x(\varepsilon_1)\| \leq \|x(\varepsilon_2)\|$ if $\varepsilon_1 \leq \varepsilon_2$.

12.8.5 This exercise concerns further properties of the Tikhonov trajectory of the NCP (F), where $F : \mathbb{R}^n \to \mathbb{R}^n$ is a continuous P_0 function. Let $x(\varepsilon)$ denote the unique solution of the regularized NCP $(F + \varepsilon \mathbf{I})$ for $\varepsilon > 0$.

(a) Show that if x^∞ is an accumulation point of the Tikhonov trajectory at $\varepsilon = 0$, i.e., if there exists a sequence of positive scalars $\{\varepsilon_k\}$ converging to zero such that $\{x(\varepsilon_k)\}$ converges to x^∞, then x^∞ is a *weak Pareto minimal solution* of the NCP (F); that is, x^∞ solves the NCP (F) and there exists no solution $\bar{x}$ such that $\bar{x} < x^\infty$.

(b) Show that if in addition F is a Z function (see Exercise **3.7.21**), then $x(\varepsilon) \leq x(\varepsilon')$ if $0 < \varepsilon' < \varepsilon$. Moreover the following statements are equivalent:

(i) the NCP (F) is feasible;

(ii) the trajectory $\{x(\varepsilon)\}$ as $\varepsilon \downarrow 0$ is bounded;

(iii) the limit $\{x(\varepsilon)\}$ as $\varepsilon \downarrow 0$ exists.

Finally, if any one of the above three statements holds, then $\{x(\varepsilon)\}$ converges to the least-element solution of the NCP (F) as $\varepsilon \downarrow 0$.

12.8.6 Let $F \equiv G + H$ be a splitting of the function F, where G is a Z function and H is antitone (i.e., $x \leq y \Rightarrow H(y) \geq H(x)$).

(a) Show that F is a Z function.

(b) Consider a sequence $\{x^k\}$ generated as follows. The starting iterate x^0 is any feasible vector of the NCP (F) and for $k = 0, 1, 2, \ldots$, x^{k+1} is the least element solution of the NCP $(G + H(x^k))$. Show that the sequence $\{x^k\}$ is well defined and converges monotonically to a solution of the NCP (F), provided that x^0 exists.

(c) Suppose that F is in addition a P_0 function. Fix a scalar $c > 0$. Let x^0 be any feasible vector of the NCP (F). Let $\{x^k\}$ be the sequence generated by the exact proximal point method without relaxation; i.e., for each k, x^{k+1} is the (unique) solution of the NCP $(F + c(\mathbf{I} - x^k))$. Use part (b) to show that the sequence $\{x^k\}$ converges monotonically to a solution of the NCP (F).

12.8.7 Let ϕ be an increasing function from $\mathbb{R}$ to $\mathbb{R}$. Define a set-valued map $\Phi : \mathbb{R} \to \mathbb{R}$ by setting

$$\Phi(x) \equiv \left[\lim_{y \uparrow x} \phi(y), \lim_{y \downarrow x} \phi(y) \right].$$

Show that Φ is monotone.

12.8.8 Let $\phi : \mathbb{R}^n \to \mathbb{R}$ be a convex function. Show that $\partial\phi$, the subdifferential of ϕ, is a monotone set-valued map. (We have assumed for simplicity that ϕ is finite-valued everywhere, even though this is not necessary).

12.8.9 Show that a set-valued map Φ is 1-co-coercive if and only if $\mathbf{I} - \Phi$ is 1-co-coercive.

12.8.10 Let $\Phi : \mathbb{R}^n \to \mathbb{R}^n$ be a maximal monotone set-valued map. Show that

(a) $\operatorname{gph} \Phi$ is closed;

(b) $\Phi(x)$ is closed and convex for every $x \in \mathbb{R}^n$. (Hint: use the maximal monotonicity to show that $\Phi(x)$ is the intersection of closed half-spaces.)

12.8.11 In the case of a single-valued function, maximal monotonicity can be written as

$$[\, (v - F(x))^T (y - x) \geq 0 \quad \forall\, x \in \mathbb{R}^n \,] \quad \Rightarrow \quad v = F(y).$$

Use this fact to show if $F : \mathbb{R}^n \to \mathbb{R}^n$ is a single-valued monotone function defined on the whole space, then F is maximal monotone.

12.8.12 Consider the proximal point algorithm for solving a monotone VI (K, F) without relaxation ($\rho_k = 1$) and with exact evaluation of the resolvent ($\varepsilon_k = 0$). Give an alternative, direct proof of the convergence of the method. (Hint: Use the definition of monotonocity, that of the iterates, and some algebraic manipulations to show that sequence $\{x^k\}$ is bounded; then argue using continuity.)

12.8.13 Let $T : \mathbb{R}^n \to \mathbb{R}^n$ be a strongly monotone set-valued map. Consider the proximal point algorithm for solving the inclusion $0 \in T(x)$ without relaxation and with exact evaluation of the resolvent. Show that if $\lim_{k\to\infty} c_k = \infty$, the sequence $\{x^k\}$ converges Q-superlinearly to the unique solution of the inclusion. (Hint: Show preliminarily that if $m > 0$ is a constant of strong monotonicity, then the resolvent is a contraction with modulus $(1 + mc_k)^{-1}$. This can be seen by considering the monotone map $T' \equiv T - m\mathbf{I}$, for which $J_{c_k T} = J_{c_k T'}((1 + mc_k)^{-1}$, and then by using the nonexpansiveness of T'.)

12.8.14 Suppose that T is a maximal monotone map. Show that

$$J_{T^{-1}} = \mathbf{I} - J_T.$$

12.8.15 Suppose that the inclusion $0 \in T(x)$, with T maximal monotone, has a solution x^* such that $0 \in \operatorname{int} T(x^*)$. Consider the proximal point algorithm for solving the inclusion $0 \in T(x)$, assuming exact evaluation of the resolvent and no relaxation. Show that $x^k = x^*$ for some finite k. (Hint: Argue first that T^{-1} is single-valued and constant in a neighborhood of 0. Then go through the proof of Theorem **12.3.7** taking into account that $J_{c_k T}(x^k) \in T^{-1}(c_k^{-1} Q^k(x^k))$.)

12.8.16 Let E be an $n \times n$ symmetric matrix and D be an $n \times n$ positive definite matrix such that $D + E$ is positive definite. Show that $\|D_s^{-1/2} E D_s^{-1/2}\| < 1$ if and only if $D - E$ is positive definite. This shows how condition $\|H\| < 1$ mentioned after Proposition **12.5.3** can be verified easily.

12.8.17 Assume that the following standard linear program (LP)

$$\begin{aligned} \text{minimize} \quad & c^T x \\ \text{subject to} \quad & Ax = b \\ & x \geq 0 \end{aligned}$$

has a solution and denote by y the dual variables. Let the sequence $\{(x^k, y^k)\}$ be generated according to the iteration

$$\begin{aligned} y^{k+1} &= y^k + \rho^{-1}\,(\,b - Ax^k\,) \\ x^{k+1} &= \Pi_{\mathbb{R}^n_+}(x^k - \rho^{-1}(c - 2A^T y^{k+1} + A^T y^k)), \end{aligned}$$

where (x^0, y^0) is any primal-dual pair and ρ is a positive constant. Show that for every $\rho > 0$ sufficiently large the sequence $\{(x^k, y^k)\}$ converges to a pair (x^*, y^*) with x^* an optimal solution of the LP and y^* an optimal solution of the dual.

12.8.18 Suppose a nonempty polyhedron P is given by the intersection of m half-spaces $H^\nu \equiv \{x \in \mathbb{R}^n : a_\nu^T x \le 0\}$, where $a_\nu \in \mathbb{R}^n$. Consider the following iterative process:

$$\begin{aligned} y_\nu^{k+1} &= y_\nu^k + \rho^{-1}(x_\nu^k - x_1^k), && \nu = 2, \ldots, m \\ x_\nu^{k+1} &= \Pi_{H^\nu}\left[x_\nu^k - \rho^{-1}(x_\nu^k - \bar{x} + 2y_\nu^{k+1} - y_\nu^k)\right], && \nu = 2, \ldots, m \\ x_1^{k+1} &= \Pi_{H^1}\left[x_1^k - \rho^{-1}\left(x_1^k - \bar{x} - \sum_{\nu=2}^m (2y_\nu^{k+1} - y_\nu^k)\right)\right], \end{aligned}$$

where $\bar{x} \in \mathbb{R}^n$ is a given vector and for all ν the x_ν and y_ν belong to $\mathbb{R}^n$. Show that for every ρ sufficiently large the sequence $\{x_1^k\}$ converges to the projection x^* of $\bar{x}$ onto P. (Hint: Use the fact that x^* is the solution of

$$\begin{aligned} \text{minimize} \quad & \sum_{\nu=1}^m \| x_\nu - \bar{x} \|^2 \\ \text{subject to} \quad & x_1 = x_\nu, \qquad \nu = 2, \ldots, m \\ & x_\nu \in H^\nu, \qquad \nu = 2, \ldots, m, \end{aligned}$$

and apply the asymmetric projection algorithm.)

12.8.19 Another noteworthy special case of the forward-backward splitting method is the *auxiliary problem principle*. In essence, this "principle" states that a solution to a VI can be obtained as a fixed point of a mapping defined by an auxiliary minimization problem. In turn, such a mapping leads to a fixed-point method for solving the original VI, which may be stated as follows. Let $\tau > 0$ be a given scalar and $\theta : \mathbb{R}^n \to \mathbb{R}$ be a strongly convex function on K that is continuously differentiable on an open set containing K. With $x^k \in K$ given, the iterate x^{k+1} is obtained by solving the auxiliary minimization problem:

$$\begin{aligned} \text{minimize} \quad & \theta(x) + (\,\tau\, F(x^k) - \nabla\theta(x^k)\,)^T x \\ \text{subject to} \quad & x \in K. \end{aligned}$$

Show that

$$x^{k+1} \;=\; J_A((\mathbf{I}-B)(x^k)),$$

where $A \equiv \tau^{-1}\nabla\theta - \mathbf{I} + \mathcal{N}(\cdot;K)$ and $B \equiv F - \tau^{-1}\nabla\theta + \mathbf{I}$. Suppose that F is monotone and the VI (K,F) has a solution. Give sufficient conditions on τ and θ so that the generated sequence $\{x^k\}$ converges to an element of $\mathrm{SOL}(K,F)$ at least R-linearly.

12.8.20 The auxiliary principle can be applied to the hemivariational inequality of finding a vector $x \in K$ satisfying

$$F(x)^T(\,y-x\,)+\varphi(y)-\varphi(x) \;\geq\; 0, \quad \forall\, y \;\in\; K.$$

Let $\theta : \mathbb{R}^n \to \mathbb{R}$ be a strongly convex function on K that is continuously differentiable on an open set containing K. With $x^k \in K$ given, the iterate x^{k+1} is obtained by solving the auxiliary minimization problem:

$$\begin{array}{ll} \text{minimize} & \theta(x)+\varphi(x)+(\,F(x^k)-\nabla\theta(x^k)\,)^T x \\ \text{subject to} & x \;\in\; K. \end{array}$$

Establish the convergence of this iterative method by showing that it is a realization of the forward-backward splitting method.

12.8.21 Consider the zero finding problem

$$0 \;\in\; A(x)+M^T B(Mx), \tag{12.8.1}$$

where $A : \mathbb{R}^n \to \mathbb{R}^n$ and $B : \mathbb{R}^s \to \mathbb{R}^s$ are point-to-set maps and M is an $s \times n$ matrix. This setting generalizes the one considered in (12.5.5). Consider the "dual" problem

$$0 \;\in\; A^{-1}(y) - MB^{-1}(-M^T y). \tag{12.8.2}$$

Suppose y^* is a zero of (12.8.2). Show that an $x^* \in B^{-1}(-M^T y^*)$ exists that solves (12.8.1).

12.8.22 (a) Consider the function

$$f(x) \;\equiv\; \sum_{i=1}^{n}(\,x_i^{\rho}-x_i^{\theta}\,), \quad \forall\, x \;\geq\; 0,$$

where ρ and θ are positive constants such that $\rho \geq 1$ and $\theta \in (0,1)$. Show that f is a Bregman function with zone $\mathbb{R}^n_+$ and that if $\rho > 1$ then f is also full range.

(b) Consider the set $K = \{x \in \mathbb{R}^n : -1 \leq x_i \leq 1,\ i = 1, 2, \ldots, m\}$. Show that

$$f(x) \equiv -\sum_{i=1}^{n} \sqrt{1 - x_i^2}.$$

is a full range Bregman function with zone K. Extend the above function to the case of a general bounded rectangle.

(c) Suppose that $K \equiv \{x \in \mathbb{R}^n : Ax \leq b\}$ has a nonempty interior. Show that the function

$$f(x) \equiv \tfrac{1}{2} \| x \|^2 + \sum_{i=1}^{n} (b_i - A_i.x) \log(b_i - A_i.x)$$

is a full range Bregman function with zone K. (Hint: To show ∇f maps $\operatorname{int} K$ onto $\mathbb{R}^n$ use a technique similar to the one in Subsection **12.7.3**.)

12.8.23 Show that if the function $F : K \to \mathbb{R}^n$ is monotone and pseudo monotone plus on the closed convex set K, then the VI (K, F) satisfies the WMPS if it is solvable. Consequently, if θ is a convex C^1 function, then the convex program

$$\begin{array}{ll} \text{minimize} & \theta(x) \\ \text{subject to} & x \in K \end{array}$$

must satisfy the WMPS. Of course, the MPS does not necessarily hold. Let

$$M \equiv \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix},$$

which is positive semidefinite but not positive semidefinite plus. Show that the homogenous LCP $(0, M)$ satisfies the WMPS; nevertheless, with $\tilde{x} = 0$ $\mathrm{SOL}(0, M)$ is a proper subset of $\{x \geq 0 : F(x)^T(\tilde{x} - x) = 0\}$, where $F(x) \equiv Mx$. Furthermore, show that F is not pseudo monotone plus on $\mathbb{R}^2_+$.

12.8.24 Consider the construction we gave in Subsection **11.8.2** for generating smoothing functions of the scalar plus function. Specifically, let p_ε be the function given by (11.8.26). Show that $(p_1)^2$ and any primitive of p_1 satisfy (r1)-(r5) (Hint: Use De l'Hospital rule to verify (r4) and (r5).)

12.8.25 Consider Algorithm **12.7.20** and assume the same setting of Theorem **12.7.19**. Suppose additionally that the Slater CQ holds also in case (b) of the theorem and that $\rho_k = 0$ for all k (i.e. that the subproblems are solved exactly at every iteration). Consider the sequence $\{\lambda^k\}$ defined by

$$\lambda_i^k \equiv r'(\varepsilon^{-1} g_i(x^k)), \quad \forall\, i = 1, \ldots, m.$$

Assume, without loss of generality, that the sequence $\{x^k\}$ produced by the Algorithm **12.7.20** converges to a solution x^* of the VI (K, F). Show that $\{\lambda^k\}$ is bounded and that each of its limit points belongs to $\mathcal{M}(x^*)$. (Hint: Show preliminarily, by using (r4), that $\lim_{t\to-\infty} r'(t) = 0$. Use this fact, a normalization argument, $F(x^k) + \varepsilon_k G(x^k) = 0$ and the Slater CQ to prove the boundedness of $\{\lambda^k\}$. Finally employ a standard limiting argument on $F(x^k) + \varepsilon_k G(x^k) = 0$ to conclude the exercise.)

12.9 Notes and Comments

More so than in the other parts of this book, the results of this chapter, which are restricted to the finite-dimensional case, are often analyzed and discussed in an infinite-dimensional framework in the literature. In order to avoid lengthy distinctions that would make the exposition heavy, we do not discuss the setting of every single paper. Occasionally, we do, however, go into greater detail, especially when a precise specification is relevant to the discussion.

Projection methods for the solution of the VI (K, F) were first proposed in an infinite-dimensional setting by Sibony [522], building on the previous work [65], where projection-type methods were considered for the solution of monotone equations. The approach adopted in the former reference is motivated by a fixed-point argument identical to the one that we adopt in Subsection **12.1.1**. In [522] the use of skewed projections and of variable steps was also advocated. The approach used here to analyze projection algorithms with variable steps (see Algorithm **12.1.4**) is based on the book by Golshtein and Tretyakov [241]; very closely related results were also obtained independently in [391]. In [241] inexact evaluation of the function F and of the projection are also considered, and it is shown that Algorithm **12.1.4** still converges to a solution of the VI if the errors in both inexact calculations have a finite sum.

Basically, all developments subsequent to [522] represent an attempt to design a projection algorithm for the widest class of problems and without the *a priori* knowledge of unknown constants. Early references include [25, 39, 68, 69, 239], where subgradient-like methods were proposed. In the latter methods, the iteration has the form $x^{k+1} \equiv \Pi_K(x^k - \tau_k^{-1} F(x^k))$, but the step size τ_k tends to zero in a suitable, controlled way. These methods extend Shor's famous subgradient algorithms for the minimization of a convex function. A simple example of these methods is given in Exercise **12.8.2**. A more recent, good exposition on subgradient-like methods is given in Chapter 5 of [241], where convergence of a subgradient method is

proved under the assumptions that (a) the solution set of the VI (K, F) is nonempty and bounded; (b) for every solution x^* of the VI (K, F) and for all x in K different from x^*, $F(x)^T(x - x^*) > 0$; and (c) the step size τ_k is always smaller than a threshold value that depends on the starting point. In the same reference, it is shown that stronger results can be obtained if one considers *ergodic* convergence, i.e. convergence not of the sequence $\{x^k\}$ generated by the subgradient method, but of an auxiliary sequence $\{y^k\}$ obtained by "averaging" the iterates:

$$y^k \equiv \left(\sum_{i=0}^{k} \tau_i \right)^{-1} \sum_{i=0}^{k} \tau_i \, x^i.$$

Ergodic convergence was used for the first time in the VI context in [69]. Another interesting work in this area is [219], where a subgradient-type projection method is studied in which projections are not performed on the original set K defining the variational inequality but on a halfspace containing K. The computational advantages of the approach are evident but should be weighed against the rather strong assumptions needed for convergence.

The extragradient method was first proposed in [350], and it generated much interest, since it could be proved to be convergent for problems that are just monotone, even though the step size is still required to be smaller than a quantity dependent on the Lipschitz constant of the function F. Many subsequent papers [264, 323, 388, 538, 554, 555, 556, 621] aimed at improving the extragradient method, the main concern being how to avoid the prior knowledge of the Lipschitz constant, or even better yet, to avoid the Lipschitz assumption altogether. With the exception of [538], whose main aim is to dispense with the extra projection but still requires the knowledge of the Lipschitz constant of F, a common feature in all these methods is that at each iteration some kind of "line search" is performed, which invariably involves further projections on the set K and thus adds to the cost of the overall computations. Some improved convergence results can be obtained in the case of linearly constrained VIs [116, 261, 262, 263, 265, 538, 646].

The drawbacks of the extragradient method and its variations did not get removed until the arrival of the hyperplane projection approach. Apparently, the idea of using the projection on a separating hyperplane determined by performing a line-search-like step satisfying (12.1.10) was first proposed by Konnov [347] for a monotone VI with a Lipschitz continuous function F. Actually, Konnov considered a more general algorithmic scheme and subsequently developed many refinements, all of which are sum-

marized in his interesting monograph [349]. In [348], convergence and rate of convergence are proved in a setting that includes Algorithm **12.1.12**, under the assumption that F is pseudo monotone with respect to the solution set. Our exposition follows the approach of Iusem and Svaiter in [280], who, considering the monotone problems only, independently proposed the Hyperplane Projection Algorithm. The observation made after Theorem **12.1.16** that at Step 3, x^{k+1} can be calculated as the projection of x^k onto $H^k_{\leq} \cap K$, is from [531], where some other algorithmic variants can be found. The trick of adding a projection on a suitably defined separating hyperplane turns out to be powerful and enhances the properties of several algorithms, besides the projection ones, under simple modifications. See [349] and [530] for details.

Motivated by the need to solve ill-posed inverse problems in mathematical physics, Tikhonov introduced his renowned regularization technique in [577]. In the special case of optimization problems, the idea of regularizing a problem by adding a strongly convex term to the objective function can actually be traced back at least to the work of Karlin [319]. The regularization technique proved to be an invaluable tool in the solution of ill-posed problems, and an enormous amount of work has been devoted to its study. The classical reference for Tikhonov regularization is the book written by Tikhonov and Arsenin [578] in 1977. A more recent overview where further general references can be found is [420]. Tseng [591] also contains many bibliographical references. The application of regularization techniques to monotone VIs was investigated by Browder in the early days of the study of variational inequalities. In particular, Browder [67] proved a result implying that if VI (K, F) is solvable and monotone and G is continuous and strongly monotone, the G-trajectory converges to the unique solution of VI $(\mathrm{SOL}(K, F), G)$ (cf. the last part of Theorem **12.2.5**). The motivation for Browder to consider such a result was to prove solution existence for some classes of VIs. For a recent work on the regularization of a monotone VI via its KKT reformulation, see [18]. Brezis [63] might have been the first to observe explicitly that (a) implies (c) in Theorem **12.2.3**, thus showing the equivalence, in the monotone case, of the boundedness of the Tikhonov trajectory and the nonemptiness of the solution set of the underlying VI. The continuity of the Tikhonov trajectory in the monotone case follows easily from stability results. Subramanian [551] considered this issue in the case of monotone complementarity problems and gave an explicit estimate of $\|x(\varepsilon_1) - x(\varepsilon_2)\|$ from which Proposition **12.2.1** follows readily.

The extension of the results discussed so far to a class of problems larger than the monotone one was initiated already by Browder [67], who proved

the results in his paper for a class of functions he called "quasi monotone" (distinct from the one referred to in Section **2.10**) that however was not used much afterwards. Qi [467] considered quasi monotone VIs in the sense of [318] and showed that if an NCP is quasi monotone and strictly feasible then the Tikhonov trajectory is bounded and $\{x(\varepsilon)\}$ is a minimizing curve for the D-gap function. In the case of the LCP (q, M) some partial results were obtained in [114], where it was shown that if M is a P_0 and R_0 matrix, every sequence $\{x(\varepsilon_k)\}$ is bounded if $\{\varepsilon_k\}$ goes to zero. Venkateswaran [601], under a condition that essentially says that the problem has a unique solution, proves convergence of the entire Tikhonov trajectory in the case of a P_0 LCP. The approach we take here for P_0 problems was started by Facchinei and Kanzow [175] in the NCP case. They showed that the Tikhonov trajectory is well defined for an NCP of the P_0 kind; moreover, by applying the mountain pass theorem to the Fischer-Burmeister equation reformulation of the problem, they proved the continuity of such a trajectory and its boundedness when the solution set of the NCP is bounded. The results in [175] required the continuous differentiability of F. By using the theory of weakly univalent functions, Gowda and Ravindran [491] extended the results in [175] to the case of box constrained VIs while only requiring F to be continuous; see also [242]. The "intermediate" case of a VI defined by a $P_*(\sigma)$ function analyzed in Theorem **12.2.6** was inspired by results in [651, 653] for the NCP.

Proposition **12.2.1** and Theorem **12.2.7** are taken from [178], where the authors borrow the techniques of [491] and extend the results of [175] to general VIs. Proving convergence of the Tikhonov trajectory in the general P_0 case is difficult, and only some partial results are known. Theorem **12.2.8** is due to Facchinei and Pang [178], who generalized earlier results in [571, 242], where the theory of semi-algebraic sets was used to establish the convergence of Tikhonov trajectories for some classes of NCP (F) that include the case where F is a polynomial P_0 function. To date it is not known whether the Tikhonov trajectory relative to a VI (K, F), where F is a continuous (or even continuously differentiable) P_0 function, can have more than one limit point.

A related problem that has attracted the attention of researchers is the characterization of the limiting point of the Tikhonov trajectory when such a point exists. For monotone problems, Theorem **12.2.5** settles the issue completely. In the case of NCPs, Tseng [591] showed that if F is pseudo monotone, a limit point of the Tikhonov G-trajectory is still a solution of the NCP $(F+G)$ as in Theorem **12.2.5**. If the function F is P_0, much less is known. In [571] the authors established that in the case of a P_0 LCP, such a

limit point is a weak Pareto minimal point of the solution set, i.e. a solution that is not strictly greater than any other solution; see Exercise **12.8.5**. Such information is rather minimal, however, since any solution with at least a zero component is weak Pareto minimal and examples can be built where every solution is weak Pareto minimal. The least-element portion in the mentioned exercise is discussed in [651].

A more substantial issue we have not touched upon is the study of the "convergence rate" of $x(\varepsilon)$. Specifically, let $x(\varepsilon)$ be a Tikhonov trajectory converging to a point x^*. The issue is to relate the error $\|x(\varepsilon) - x^*\|$ to the perturbation ε. This problem has been considered mainly in the monotone case. Doktor and Kucera [141] gave some results when F is strongly monotone. Although interesting per se, these results refer to a case where regularization is very unlikely to be used. Liu and Nashed [374, 420] considered the problem under sets of assumptions that always include the co-coercivity of F and showed that $\|x(\varepsilon) - x^*\| = O(\varepsilon^{1/3})$ or $\|x(\varepsilon) - x^*\| = O(\varepsilon^{1/2})$ depending on the assumptions made. In the context of the NCP, Tseng [591] made a very detailed study of the behavior of various "trajectories". Among other results he established that if F is analytic and $\{\varepsilon_k\}$ is a sequence of positive scalars converging to zero, there exist positive constants c and γ such that the distance of $x(\varepsilon_k)$ to the solution set of the unperturbed NCP is bounded by $c\varepsilon_k\|x(\varepsilon_k)\|^\gamma$, with $\gamma = 1$ if F is affine. It is interesting to note that in this result F is not assumed to be monotone; it is required only that the NCP $(F + \varepsilon_k \mathbf{I})$ have a nonempty solution set, that $x(\varepsilon_k)$ belong to the latter solution set, and that $\{x(\varepsilon_k)\}$ converge to some point x^*. Therefore, the result applies, in particular, to an analytic P_0 NCP with a nonempty bounded solution set. A similar result was proved in the case of affine monotone problems by Robinson [495]. Other results in this vein are given by Fischer [204].

There are two important considerations in the practical implementation of the Tikhonov regularization idea: One is the fact that the regularized subproblems should not (or in some cases, cannot) be solved exactly; two is the criterion for reducing ε even in the inexact solution of the subproblems. These issues have been clearly understood since the beginning, as shown by the works of Bakushinskii and Poljak [37, 38, 39]. In [39] the authors considered the solution of the regularized subproblems of a monotone VI by means of projection methods and gave specific rules to update ε. A similar approach is described in [37] that combines the Josephy-Newton method with regularization; more general strategies are discussed in [38].

The broad algorithmic scheme presented in Subsection **12.2.1** extends the work of Facchinei and Kanzow [175] for NCPs, where the Fischer-

Burmeister merit function is used instead of the natural residual. This turns out to be the first provably convergent algorithm for P_0 NCPs with bounded solution sets. An important advantage of Algorithm **12.2.9** is its flexibility; specifically, it allows any (iterative) algorithm to be used for computing an approximate solution of each perturbed problem; see [477, 557] for elaborations. Akin to the methods described so far, El Farouq and Cohen [163] proposed a method for the solution of monotone problems whereby regularization is embedded in a scheme based on Cohen's auxiliary principle; see also the related paper [162] for the convergence analysis of an iterative method based on the latter principle.

Maximal monotone set-valued maps have been and still are intensively investigated and used, since they provide a powerful unified framework for many problems in nonlinear analysis. These maps were introduced by Minty [406]. A classical reference on the subject is [63]. All the results in Subsection **12.3.1** are standard; our exposition is based on [63, 157, 241].

The proximal point algorithm for VIs was first proposed by Martinet [392] in 1970 in an attempt to alleviate the difficulties in a Tikhonov regularization method when one has to solve a progressively more and more ill-conditioned problem. Martinet was motivated by a similar approach used in [45] in the case of convex quadratic minimization problems. In the finite-dimensional case we are interested in, the main difference between Tikhonov regularization and the proximal point method is that we are not able to predict to which point the sequence produced by the latter method converges. We note, however, that in the infinite-dimensional setting, which is considered by most authors dealing with these issues, there is a further important difference: For Tikhonov regularization methods strong convergence to a solution of the original problem can be proved, while for the proximal point method only weak convergence can be obtained unless further strong assumptions are made. Güler [245] gave an example to show the lack of strong convergence of the method.

Rockafellar's paper [508] was an important step toward a wider appreciation of the importance of the proximal point method. He analyzed the proximal point algorithm for the problem of finding a zero of a maximal monotone map, which obviously includes that of solving a VI. Actually, this general framework persists in most subsequent papers dealing with proximal point algorithms. In the mentioned paper, Rockafellar presented several significant results. First he allowed the coefficient c to vary from iteration to iteration (while it was fixed in [392]); second, and more importantly, he allowed for the inexact solution of the perturbed subproblems. He also gave some "convergence rate results". Specifically, he proved that

if x^* is the limit point produced by the method (with no relaxation, i.e., $\rho_k = 0$ in Algorithm **12.3.8**, and with a certain inexactness rule) and under the additional assumption that T^{-1} satisfies a certain Lipschitz property at 0, then

$$\| x^{k+1} - x^* \| \leq \eta_k \| x^k - x^* \|,$$

where $\{\eta_k\}$ is a bounded sequence of positive scalars that goes to zero if c_k goes to infinity. In [508] it is also shown that if $0 \in \operatorname{int} T^{-1}(x^*)$ and $\varepsilon_k = 0$ for each k in the Generalized Proximal Point Algorithm (i.e. exact evaluation of the resolvent), then $x^k = x^*$ for some k. Burachik, Iusem, and Svaiter [72] introduced an enlargement of monotone operators and use it to define a family of inexact proximal point algorithms for VIs. Ferris [181] showed finite termination by assuming "weak sharpness" of the solution. For an extensive discussion on the finite termination property of the proximal point method, in the case of both optimization problems and VIs, we refer the reader to [440].

Around the same time, Rockafellar also gave another important contribution by showing that the Hestenes-Powell method of multipliers for nonlinear programming is nothing else than the proximal point algorithm applied to the dual optimization problem [509] (see also [507]), and that new methods of the multipliers can be developed based on this observation. The idea of using under- and overrelaxations was put forward by Golshtein and Tretyakov [240], who proposed a scheme similar to the Generalized Proximal Point Algorithm, where, however, c_k is not allowed to vary from iteration to iteration. Our analysis of the algorithm is based on the work of Eckstein and Bertsekas [157]. The example before Algorithm **12.3.8**, showing the utility of overrelaxation, is from [227].

There are several other aspects of the proximal point algorithm that have been intensively investigated. The first is that of the convergence rate that we already mentioned when discussing [508]. Luque [377] extended Rockafellar's analysis. By assuming various conditions on the growth properties of T^{-1} in a neighborhood of 0 and by considering broad inexactness rules, Luque studied in detail when the sequence $\{x^k\}$ generated by the Generalized Proximal Point Algorithm converges sublinearly, linearly, or superlinearly. A further contribution of Luque is that, in contrast to the analysis in [508], the convergence rate results are obtained without assuming uniqueness of the solution of the unperturbed problem.

The convergence rates alluded to above deserve an elaboration. These rates refer to the outer sequence $\{x^k\}$ produced by the algorithm. However, in order to generate these iterates, an iterative "inner" method has to be used in order to approximatively calculate the resolvent; the latter typically

requires multiple iterations. Therefore, the calculation of x^k can in general require a possibly huge computational effort. This issue has recently been addressed by Yamashita and Fukushima [634], where the proximal point algorithm is considered for the solution of an NCP. By using the Newton method applied to the FB equation reformulation of the subproblems, the authors give conditions ensuring that one step of the inner Newton method is eventually sufficient to generate a suitable outer x^k, thus ensuring a *genuine* superlinear convergence rate. A key assumption in this reference is that the point to which the method converges is nondegenerate. This rather strong assumption is eliminated in [121], by suitably using a technique proposed in [629] that identifies the zero variable(s) at a solution.

Another problem that has been investigated in relation to proximal point algorithms is its applicability to a class of problems wider than the monotone one. Extensions to pseudo monotone VIs are analyzed in [161], while applications of the proximal point method to P_0 NCPs is considered in [636]. Solodov and Svaiter have also been very active in the definition of new variants of the proximal point algorithms, see [532, 533, 537], their main concern being the use of some new and more practical and/or effective tolerance requirements on the solution of the subproblems. One of their noteworthy contributions is [536], where they modify the proximal point algorithm by using a technique similar to the one used in the Hyperplane Projection Algorithm to enforce strong convergence of the method in a Hilbert space. The same result was achieved in [361] by combining the proximal algorithm with Tikhonov regularization.

Operator splitting has long been a known technique in numerical linear algebra and in partial differential equations. However, in these fields the operators are usually linear and single-valued. Ample accounts of these methods along with extensive bibliographical references can be found in [196, 386]. Before plunging into a detailed discussion on splitting methods for monotone maps, let us mention that Eckstein's Ph.D. thesis [154] is a good source of information up to 1989. This reference, along with [584, 585, 593], provides an entry point to the literature on the application of splitting methods to optimization problems. Further general references on splitting methods and on its applications are [158, 215, 237, 418].

The idea of generalizing splitting methods to the solution of inclusion problems defined by nonlinear, monotone, set-valued maps first appeared in the seminal paper by Lions and Mercier [372]. Since then, splitting algorithms have been extensively studied, mainly in the French mathematical community. Besides the splitting methods discussed in the text, which are certainly the most useful for our purposes, several other methods have been

discussed in the literature. In order to place the results discussed herein in the right perspective, let us first briefly comment on two of the most often encountered of these other methods.

Let T be a set-valued monotone map and suppose that $T = A + B$. The double-backward splitting method chooses $x^{k+1} \in J_{c_k A}(J_{c_k B}(x^k))$ at each iteration. This method was discussed by Passty [437] and, in the special case in which B is a normal cone to a convex set, by Lions [371]. Assuming that A and B are maximal monotone and supposing that the sequence $\{c_k\}$ satisfies some conditions that imply in particular its convergence to zero, Passty proved ergodic convergence of the method. It is clear that both because of the requirements on $\{c_k\}$ and because of the weak convergence results, the method is not very appealing in the general case.

Another method is a forward-backward splitting method in which the roles of A and B are exchanged in each iteration. This results in the scheme

$$x^{k+1/2} \in J_{c_k A}((\mathbf{I} - c_k B)(x^k)) \quad \text{and} \quad x^{k+1} \in J_{c_k B}((\mathbf{I} - c_k A)(x^{k+1/2}).$$

This kind of scheme originates from the solution method proposed by Peaceman and Rachford [443] for the numerical solution of partial differential equations. It was generalized to set-valued monotone maps by Lions and Mercier [372]. The convergence results of this method do not place any particular restriction on the choice of c_k but do place some restrictions on the properties of A or B.

The Peaceman-Rachford algorithm should be compared to the forward-backward iteration (12.4.2). Although the latter algorithm requires c_k to satisfy appropriate conditions, it is simpler and different from the former algorithm. When specialized to the VI, we have seen that (12.4.2) leads to some interesting methods for solving the problem.

In contrast to the above methods, the Douglas-Rachford splitting Algorithm **12.4.2** does not impose any restrictions on the splitting $A+B$ and on the choice of the parameter c. In this sense it is the most widely applicable splitting method. Its origin goes back to the paper by Douglas and Rachford [146] on the numerical solution of the heat equation. Once again, Lions and Mercier [372] extended to the general setting of set-valued monotone maps. The Douglas-Rachford splitting method has important ramifications in the case of nonlinear programs, as pointed out already in [372]. Gabay [227] showed that the alternating direction method of multipliers (a variant of the method of multipliers for nonlinear problems that is suitable to decomposition) is just a special case of the Douglas-Rachford method applied to some kind of "dual" problem and extended the former method to VIs (see also [47]). The exposition in Subsection **12.4.1** is drawn from Eck-

stein and Bertsekas [157], who showed that the Douglas-Rachford splitting method is just an application of the proximal point algorithm to a suitably defined maximal monotone map. Based on this observation, they were able to generalize the classical Douglas-Rachford method to include inexact evaluation of resolvents and over- and under-relaxations, as reported in Subsection **12.4.1**. The relation between the proximal point method and the Douglas-Rachford method was also highlighted, using a different point of view, by Lawrence and Spingarn [360]. In turn, Spingarn's method of "partial inverses" [540], which has several applications to parallel optimization [541] (see [362] for a succinct introduction to this topic), can be shown to be a particular case of the Douglas-Rachford splitting method [154].

Subsection **12.5.2** is based on [222] and illustrates the application of the Douglas-Rachford splitting method to a "primal" VI instead of a "dual" VI as in the paper of Gabay [227]. This application evolved from a previous work along the same lines for convex optimization [160]. For other decomposition methods applied to the traffic equilibrium problem, see [266, 359, 617].

As made clear in the beginning of Subsection **12.5.1**, forward-backward splitting methods have their roots in projection methods. This simple case suggests that the forward-backward algorithm may require some strong assumptions for convergence. In fact, in the case where $A = \mathcal{N}(\cdot; K)$ and $B = \nabla\theta$, with θ being a differentiable convex function, we need the Lipschitz continuity (or equivalently, the co-coerciveness, see Subsection **2.3.1**) of $\nabla\theta$ to prove convergence. In fact, if one considers a decomposition $T = A + B$ with A and B maximal monotone and no further structure, the best known result is an ergodic convergence result requiring c_k to converge to zero and some boundedness condition, see [437]. Gabay [227] showed that if B is co-coercive and c_k is a fixed positive number smaller than twice the co-coercivity constant of B, then the forward-backward algorithm converges. This result was extended to non-constant c_k in [585]. Theorem **12.4.6** is a simplified version of the results in [585]; the convergence rate result reported in Corollary **12.4.8** is also from the same source. A related result is given by Chen and Rockafellar [95], who show that if B is Lipschitz continuous and T is strongly monotone, the method converges linearly to the unique solution, provided c_k is smaller than a constant dependent on the Lipschitz and strong monotonicity constants. The Modified Forward-Backward Algorithm, whereby neither the co-coercivity of B nor the strong monotonicity of T are required, is due to Tseng [593].

Asymmetric projection methods have long existed for solving complementarity problems; see, for example, [114], where these methods are de-

rived from simple matrix splitting schemes for solving LCPs. Pang and Chan [81] and Dafermos [119] gave general algorithmic schemes for VIs that encompass asymmetric projection. A major drawback in these iterative schemes is that the conditions imposed on F for convergence invariably imply that F is at least strictly monotone, thereby ruling out many important applications. All of Subsection **12.5.1** is based on the work of Tseng [584], who also coined the name "asymmetric projection" to describe this class of methods. The main advantage of the adopted approach is that it is possible to relax the assumptions in [81, 119] to obtain convergence. Convergence of Algorithm **12.5.1** under the conditions in part (a) of Proposition **12.5.3** was proved in [81]. The latter reference also discussed the nonlinear Jacobi method for solving partitioned VIs, which was the generalization of the original PIES algorithm [5, 6, 7]. See [255] for an early proposal to accelerate the projection method for VIs.

As mentioned in Section **6.10**, Luo and Tseng are the champions of using error bounds to obtain rates of convergence of iterative algorithms for a host of mathematical programming problems. Section **12.6** is based on the work of Tseng [586]. As one can see from the results therein, the existence of local error bounds allows one to establish rates of convergence under weak conditions. A recent work of Tseng [594] applies error bounds to establish the superlinear convergence of some Newton-type methods in optimization. Luo [375] gives further applications of error bounds to convergence analysis of iterative algorithms.

Recession and conjugate functions are powerful tools in convex analysis. With the exception of Proposition **12.7.1**, which is from [28], all the results in Subsection **12.7.1** can be found in classical references such as [269, 505]. See the most recent paper by Auslender [27] and the monograph by Auslender and Teboulle [32] for an extensive, contemporary study of these functions and their role in the convergence of algorithms without boundedness.

The standard, formal definition of Bregman function was given by Censor and Lent [79] and is based on the work of Bregman [62] on the generalization of the cyclic projection method for finding a point in the intersection of a finite number of closed convex sets. As observed in [80], a good source of bibliographical references, there is no universal agreement on the definition of a Bregman function, and different authors often give different, if obviously related, definitions in order to achieve their particular goals. Definition **12.7.2** is employed by Solodov and Svaiter [534]; as discussed in the reference, the imposed conditions appear to be "minimal" relative to those used in the literature.

An in-depth study of properties of Bregman functions is [41]. What we called a full range Bregman function is usually termed a "zone coercive" Bregman function; this property is crucial in the definition of algorithms. Proposition **12.7.3** is based on results in [41] and [78]. Condition (c) in this proposition is also known as "boundary coerciveness" and is equivalent to f being essentially smooth on $\operatorname{int} K$. Example **12.7.4** is from Bregman's original paper [62]; Example **12.7.5** is given, for example, in [78]. The latter reference also mentions the Bregman functions in (a) and (c) of Exercise **12.8.22**, while the example in (b) is suggested in [155] among others. The three-point formula in Proposition **12.7.6** (a) is from [94], while part (b) was first proved in [534].

As we have mentioned, Bregman functions were introduced to solve feasibility problems. Subsequently they were extended to the definition of proximal-like algorithms for minimization problems. We refer the interested reader to [41, 94, 80] for accounts on these issues. The use of Bregman functions to define new proximal point algorithms for maximal monotone inclusions was initiated by Eckstein [155]. Further contributions include [70, 71, 78, 156, 279, 534].

A crucial issue in the analysis of a Bregman proximal point method for a VI (or for a maximal monotone inclusion) is the solvability of the subproblems. This difficult issue is tackled by Eckstein [155] by invoking some deep and advanced results on the image of the sum of two monotone operators established by Brézis and Haraux [64]. Burachik and Iusem [70] follow the same path, while in [78, 279, 534] either reference is made to [70, 156] or solvability is just assumed to begin with. The route we take in Theorem **12.7.10** is different and is based on Theorem **11.2.1** via Lemma **12.7.9**; some details are nevertheless inspired by the proof techniques of [64].

The WMPS is an essential requirement in the convergence analysis of Bregman proximal point algorithms for solving the monotone VI (K, F). The usual way such algorithms are derived in the literature begins with the conversion of this problem into the inclusion problem $0 \in F(x) + \mathcal{N}(x; K)$, which is a special instance of the general problem $0 \in \Phi(x)$. For the latter problem, a key "paramonotonicity" property on Φ is imposed; see [68, 70, 78, 277, 278]. Specifically, a set-valued map $\Phi : \mathbb{R}^n \to \mathbb{R}^n$ is said to be *paramonotone* if it is monotone and for any two pairs (x, u) and (y, v) in $\operatorname{gph} \Phi$,

$$\langle u - v, x - y \rangle = 0 \;\Rightarrow\; (u, v) \in \Phi(y) \times \Phi(x).$$

Clearly, when Φ is single-valued, this is exactly the monotone plus property in part (b) of Definition **2.3.9**. In the application to the VI, Φ is

never single-valued because it is equal to $F + \mathcal{N}(\cdot; K)$. Nevertheless, it has been shown by the authors in the cited references that if F is monotone plus on K, then the Bregman proximal point algorithms can be applied to the VI (K, F) and its convergence can be established. As seen from Exercise **12.8.23**, the WMPS of the monotone VI (K, F) is weaker than the monotonicity plus property, or equivalently, paramonotonicity of F on K. Therefore Theorem **12.7.14** is an improvement of the convergence results in the literature in the context of variational inequalities.

In all papers dealing with Bregman proximal algorithms for VIs (or maximal monotone inclusions) a further assumption is usually required that is often called "quasimonotonicity". The only exception is [534], where the authors dispense with this assumption thanks to a more refined analysis that hinges on Proposition **12.7.6**(b). The Bregman Proximal Point Algorithm and its convergence analysis are essentially based on [156]; one difference is that we use the technique of [534] to avoid the "quasimonotonicity" assumption. Unlike [534], we employ a different inexactness criterion for the resolution of the subproblems.

When applied to a VI, the classical proximal algorithm (in its exact version) calculates x^{k+1} as the solution of the VI $(K, c_k F + (\cdot - x^k))$, while in the Bregman Proximal Point Algorithm x^{k+1} solves the equation $c_k F(x) + \nabla f(x) - \nabla f(x^k) = 0$. Prompted by this observation, one is tempted to consider still further methods where x^{k+1} is the solution of an equation of the type $c_k F(x) + \phi(x, x^k) = 0$, for some appropriate function ϕ. Indeed, this approach is viable and has been pursued; but there are to date not too many results. The usefulness of these alternative approaches is well illustrated by the algorithm for linearly constrained VIs in Subsection **12.7.3**, which is adopted from [34]. Other references on these generalized proximal point methods and on the applications of non-quadratic (such as logarithmic) proximal point algorithms to decomposition include [30, 31, 33, 281, 357, 576].

The methods in Subsection **12.7.4** generalize the classical interior and exterior barrier methods for NLPs [197]. The section is based on the work of Auslender [26]. On the one hand, we have simplified the presentation by restricting the setting; on the other hand, we introduce the possibility of the inexact resolution of the subproblems, a feature that is absent in the reference. Auslender's paper generalizes the prior work [28, 29] for convex programs. In [28] the relations to the methods in [197] are apparent and discussed in detail. Exercises **12.8.24** and **12.8.25** are based on the results in [26].

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